Dirac, Feynman, Penrose

An Entangled Golden Braid

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Prelude on a Blank Sheet

行く秋や
仕上がりし絵の
白きまま

yuku aki ya
shiagarishi e no
shiroki mama

Departing autumn —
the picture is completed,
and it is still white.

[*Achilles's table, late on a still autumn afternoon. A sheet of paper lies before him, blank on both sides. The Tortoise lets herself in with another under her arm.*]

[*Outside, the lane is so still that the last leaves hang on the trees as if someone had set them there, and in the yard the water-butt stands full and dark, its surface unmoving. The afternoon light is already going, grey and even, lying on the room without falling anywhere in particular. The grate is swept and cold; no fire has been laid in it. The walls are bare, and the table is bare but for the teapot, the cups and the paper; there is nothing on it to write with, and nothing in the room that anyone has begun. A shawl hangs over the back of Achilles's chair.*]

Tortoise: Not a breath of wind in the lane. Every leaf that is left is staying put.

Achilles: Then the leaves have more resolve than I have. Sit; the tea is still

warm. I have sat here since lunch, meaning to begin something, and I have not begun.

Tortoise: [*She takes the chair beside him, and the cup he pours, which she sets down by the pot.*] I have been luckier. [*She holds her sheet flat for him to see.*] This is a picture I made this morning, and it is finished.

Achilles: There is nothing on it.

Tortoise: Nothing at all. That is what it is a picture of.

Achilles: Then it is worth nothing. Zero.

[*The Tortoise says nothing. His sheet lies blank before him, alone on the table. She lowers hers and slides it across until its edge meets the edge of his, then takes her hand away; his does not stir. Where one blank sheet lay, two lie edge to edge. Nothing has been added to his and nothing taken from it; hers is as blank as when it came in; the pot and cups stand as they stood. The one difference in the room is that hers now lies beside his.*]

[*Beyond the window a leaf lets go of its tree at last, drifts down through the windless air, and comes to rest on the water in the butt. It lies on the surface, and the surface is as it was: not a ripple goes out from it. Achilles looks at the table, then at his cup, as if one of them might have moved.*]

Tortoise: What has it done to yours?

Achilles: Nothing. But that proves my point: adding zero does nothing either.

Tortoise: Did I add it? I put it after yours, as one thing is done after another. Say yours makes something twice as much. Twice as much, and then my picture: how much?

Achilles: Twice as much, and then nothing done: still twice. Twice, and then three times, would have been six times. So your picture is the one that leaves it as it was: once. One, not zero. ...Oh.

[*For a while neither of them says anything. The light has gone down another shade, and the room has grown quieter with it, the way a room does when nobody needs to*

move. Achilles sits back, pulls the shawl round his shoulders, and warms his hands on his cup.]

Tortoise: [*She leaves her sheet where it lies.*] Keep it. It takes up no room at all.

Achilles: [*He looks from one blank sheet to the other, and touches neither.*] I shall. But why should a picture of nothing come to one, and not to nothing?

Tortoise: Drink your tea; the light is going. The question will keep, and so will the sheets.

[*Achilles drinks his tea. Outside, the leaf still lies on the water in the butt, and the water has kept its stillness round it; the last leaves are still on the trees. The light has gone low and grey, and settles on the bare walls, the cold grate, the pot and the cups, and the sheets where they lie, and on nothing else, since there is nothing else. Neither of them moves; there is no hurry.*]

Chapter 0: The Empty Diagram

— Prelude on a Blank Sheet —

What is a picture with nothing in it worth?

Nothing, surely: an empty purse holds no money, an empty page tells no story. But a picture can be put together with other pictures, and then it must be judged by what it does to them. Judged that way, the empty page is not worth nothing.

§0.1 The Empty Diagram

Here is the picture, to the left of the sign. There is nothing in it.

The Empty Diagram.

This is not a printing error. The space to the left of the sign *is* the picture: a page on which nothing has been drawn. We call it **the empty diagram**, and it is the number 1. Not 0, but 1.

§0.2 The Theorem, Twice

First an example: seven times one is seven. Here is the theorem.

Theorem 0.1 (The empty diagram is one). The empty diagram is the number 1: composing anything with it changes nothing, exactly.

Dirac's Blackboard

Composing is doing one thing and then another, and for numbers that is multiplying: twice, then three times, is six times. So the number of nothing is the number that changes nothing under multiplying, and that is 1 — not 0, which changes nothing only under adding.

The number of nothing is the product of no numbers. Appending a number a to a product multiplies the product by a, and appending it to no numbers gives the product of a alone, which is a. So

$$a \;=\; (\text{the product of no numbers}) \cdot a \qquad \text{for every number } a,$$

and at $a = 1$ this says the product of no numbers is 1. No other number serves: a number that leaves every number as it was leaves 1 as it was, and a number times 1 is itself, so it is 1. And 0 fails: nought times seven is nought, not seven. The empty diagram is nothing drawn, so it is the product of no numbers: 1. ■

Penrose's Sketchbook

Here the proof is to look. Two empty diagrams beside each other, to the left of the sign; one, to the right:

=

Nobody could tell the two sides apart: they are the same page. Beside any picture the same holds: nothing has been drawn, so nothing has been added. The empty diagram is the picture that changes nothing when anything is composed with it, which is what "the number 1" says in pictures; the display of §0.1 gives it that name. No rule was used, and none was needed. ■

The verdict

A tie, honestly. Each proof is a few sentences. The picture needs no rule, and sees "changes nothing" for free. But it cannot say *which* number that is. The name 1 comes from the blackboard, which needs one fact, that composing is multiplying, and one sentence more to rule out every other number. Each proves the theorem in its own language, as every theorem in this book will be proved: twice.

That is the whole Prelude, and the page is still blank. From the next chapter on, something is drawn. The book returns to such a page only at its end, and finds it worth just what it is worth now.

Overture for a Single String

木枯や
裏返りまた
裏返る

kogarashi ya
uragaeri mata
uragaeru

The withering wind —
something turns over, and then
turns over again.

[*Achilles's table, after supper. He is bent over a coin and a much-scribbled sheet; the Tortoise has just let herself in.*]

[*Outside, the first hard wind of the year is stripping the lane of its last leaves and worrying at the window, which has gone black and cold; a little of the draught finds its way in under the door. The fire has not long been lit and has not yet made up its mind — it catches, gutters, catches again, and throws a light that will not sit still on anything. On the wall above the hearth hangs a woodcut, Escher's* Day and Night*: the chequered fields between two towns, daylight lying over the one side and night over the other. Supper is cleared; a pot of tea stands cooling at the far corner of the table, and beside it the coin, the pencil and the much-scribbled sheet are Achilles's.*]

Tortoise: You started without me.

Achilles: Only fidgeting. Come round to this side and sit — good, now the table reads the same way for us both. Was the road very cold?

Tortoise: Cold enough that I'm grateful for the chair, and the tea, and a table indoors. Don't mind me; you have the face of a man in the middle of something.

Achilles: In the middle of it, and quite turned around in the middle of it. I'm glad you've come.

[*The Tortoise, settled beside him, holds her hands a moment toward the grate, where the flames are still finding their feet. Achilles reaches across for the cooling pot and pours. For a while there is only the wind at the glass and the small business of getting warm.*]

First Challenge — *A Coin on a String*

[*Achilles's table. A coin, a pencil, and a sheet of paper crowded with his marks. The Tortoise sits beside him, on the same side of the table.*]

Achilles: Over. Over again. Then I laid it heads up — or was that before the second turn? [*He turns the coin over once more and makes another mark.*] Tortoise, I have been turning this coin over since supper, I have written down everything I did to it, and I can no longer tell from my own notes which face ought to be showing.

Tortoise: Which face is showing?

Achilles: Tails: the wreath. Heads is the head in profile, on the other side. But is it *right* that it's tails? That is what the paper was for.

Tortoise: Before you cross out one more line — may I? I keep a length of string in my shell for exactly this kind of muddle. Laid flat along the table, it holds things in the order they happen, which is the one thing your paper won't do.

Achilles: By all means. My notes have not covered themselves in glory tonight.

Tortoise: Your paper says what you did. It doesn't say in what order, or to what. [*From her shell she draws a length of string; she lays it across the table in front of them, left to right, and pulls it taut between her hands until the last of the slack is gone and it lies dead straight. She pins down its two ends, and plucks it once: a single clear note sounds and dies away, and the string settles into stillness.*] Let us keep the record somewhere it can't get out of order.

Achilles: A string. There's nothing on it.

Tortoise: Then so far nothing has been done. [*She writes "heads up" on a slip of paper, pencils a small head in profile beside the words, so that a glance tells the start, and ties it to the string's left end.*] The end tag: how the traveller starts. Nothing comes before it. [*She writes "turn it over" on another slip — words only this time, and no face, for it names none — folds it into a little tent, and sets it astride the string halfway along, like a saddle.*] And a tag: what is done on the way. Set the coin as the end tag says, and walk it along.

[*Achilles lays the coin heads up at the right-hand end of the string, where his hand is.*]

Tortoise: Read it from the left. The end tag is where it starts.

Achilles: [*The coin sits heads up at the right-hand end, where his hand is. He carries it round to the left end, under the "heads up" end tag, and slides it rightward along the table beside the string — the string and its one saddled tag holding still — and where the tag hangs he stops and turns the coin over in his fingers, the head going under and the wreath coming up, so that the face which was heads is now tails; it is the same coin, only the other way up. Then he carries it on to the right-hand end.*] Tails up. Well, of course it's tails up.

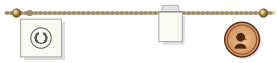


from heads up

[*Under the door the draught comes again, harder than before, and lays the hearth-flames flat over to one side. They burn that way a moment, all of them leaning the one direction.*]

Tortoise: And if it had started the other way? [*She unties the end tag and ties on another: "tails up", with a small wreath pencilled beside the words.*]

Achilles: [*setting the coin tails up at the left end and walking it along, turning it over at the tag*] Heads. So what you're saying is, the string remembers what I did to the coin.



from tails up

Tortoise: The string says what is *to be* done, and in what order, to whatever the end tag sends along. It would say so if you never walked a coin past it at all.

Achilles: Like a recipe, then: it says what to do to whatever goes into the bowl, and says it before anything is in the bowl. Then it can say more than I've done. [*He writes "turn it over" on a fresh slip, folds it, and sets it on the string beside the first.*] And two of them?

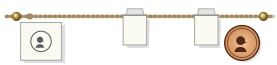
Second Challenge — *There and Back*

Tortoise: What do you expect?

Achilles: One tag turns the coin over. Two tags must do twice as much.

Tortoise: Twice as much as turning over. And how will the coin look, when it has been turned over twice as much?

Achilles: Very... thoroughly tails. No — I had better walk it. [*The Tortoise ties the "heads up" end tag on again. He sets the coin heads up at the left end and walks it along: over at the first tag, over at the second.*] Tails, then heads. It arrives as it left.



from heads up

[*As he says it, the gust falls away outside. The flames right themselves and stand up where they had stood before, unhurried, come back to exactly where they began.*]

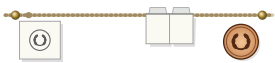
Tortoise: From that start.

Achilles: [*The end tag is changed to "tails up"; he walks the coin again.*] As it left, again — like a switch pressed twice: whichever way the lamp was, that way it is again. And a coin has only the two ways of starting, so I've walked them all.



from tails up

[*Two "turn it over" tags hang astride the string a little apart, the end tag at the left. The Tortoise slides them along the string toward each other until they meet, and lifts the touching pair straight up and off. The string stays pinned and taut and the end tag stays at its knot; only the pair is gone, and where they hung there is now nothing to be done. The string is bare but for its end tag.*]



slid together



lifted off

Tortoise: Walk it again.

Achilles: Past what? [*He slides the coin from the left end to the right without stopping.*] There was nothing to do. It arrives as it left — which is exactly what it did a moment ago, with both tags on. ...Oh. I needn't have walked it. Two turn-overs side by side are so much bare string: wherever I see the pair, I may lift it off.

Tortoise: And wherever the string is bare, you may set a pair on.

Achilles: Whyever would I want to?

Tortoise: Another evening.

Achilles: Then any tag, hung twice, comes off?

Third Challenge — *Which Tag First?*

Tortoise: [*writing on a fresh slip, and pencilling the little head beside the words, as on the "heads up" end tag*] "Heads up, whatever it was." [*She writes a second the same, head and all, folds both into tents, and sets them astride the string side by side — saddled along the string like the turn-over tags, not knotted at its end like the end tag.*] Any tag, you said.

Achilles: Twins. Off they come.

Tortoise: Walk it first. The end tag still says tails.

Achilles: [*walking the coin from its tails start: at the first tag he lays it heads up, the wreath going under; at the second it is heads up already, and he leaves it*] It arrives heads — but along a bare string it would have arrived tails! The pair can't come off.



twins

Tortoise: [*lifting one of the two off*] And now?

Achilles: [*walking it again*] Heads, just the same. So one of the twins comes off, and never both. Hung twice, this tag does what it does once.

Tortoise: So not every tag.

Achilles: Very well: *some* tags come off in pairs. But the order along the string can't matter, at least. It didn't for the two turn-overs.

Tortoise: The two turn-overs were the same tag. [*She sets a "turn it over" tag on the string to the left of the "heads up, whatever it was" tag, and changes the end tag to "heads up".*] Walk it.

Achilles: [*walking it from heads: over at the first tag, and the wreath comes up; heads up at the second, and the head is back*] It arrives heads.



as hung

[*On the string, "turn it over" hangs at the left and "heads up, whatever it was" at the right. The Tortoise lifts both clear of the string and sets them down again with their places swapped — "heads up, whatever it was" now at the left, "turn it over" now at the right. They are the very same two tags, and the coin's start is the very same start; what has changed is only their order along the string.*]

Tortoise: Same tags. Walk it again.

Achilles: [*walking it from heads: at the first tag it is heads up already, and he leaves it; over at the second, and the wreath comes up*] It arrives *tails*. Same coin, same start, the same two tags, and the other face — so the order along the string is part of what the string says. Like washing your hands and drying them: the same two jobs, and one order leaves you dry, the other wet.



swapped

[*The tea has stood long enough to want reviving. Achilles pours a little milk into his cup, where it blooms and greys the dark brew; the Tortoise takes hers the other way about, the tea let down onto the milk already in the cup, and the two settle to different colours.*]

Tortoise: Change the end tag, and walk it once more.

Achilles: [*with "tails up" tied on, walking it: heads up at the first tag, the wreath going under; over at the second, and the wreath comes up again*] Both starts arrive alike! Once the coin has passed that "heads up" tag, nothing further along the string can tell how it started, and nothing I could hang there would bring it back — not even its twin. The turn-over tag never loses anything, and this one does: that is why a second turn-over undoes the first, and a second of these undoes nothing. Is there a rule for which tags lose things?



from tails up

[*He tops up both cups from the jug while he thinks, milk into each until the two are of a colour. Whatever they had been — the strong cup and the weak — they are the same pale tea now, and no amount of looking into either would tell you which had been which.*]

Tortoise: Another evening. [*She lifts both tags off. The "heads up, whatever it was" tags go away into her shell; the two "turn it over" tags stay on the table.*] Every walk so far began with a coin you could see. [*She takes the coin and a die behind her shell, where the die rattles once and settles; then, by the rule she has set herself, she sets the coin down just short of the string's left end, out of his sight, and lays her foot over it. It is one coin still, with one true face — she knows which, and will not say. She unties the end tag; her foot does not lift again all evening.*] How does this one start?

Fourth Challenge — *The Covered Coin*

Achilles: Heads up or tails up. It must be one of them; coins don't lie on their edges under tortoises. What did the die say?

Tortoise: Heads, unless the die showed a six. I shan't tell you what it showed.

Achilles: Then lift your foot.

Tortoise: That would tell us how the coin lies. I asked what we may honestly tie to the string, now, with my foot where it is. "Heads up"?

Achilles: I couldn't swear to it. Nor to "tails up". Neither tag is honest, and yet the coin *is* one or the other. Let me take a plainer case first: a coin tossed and covered. Evens.

Tortoise: By all means. [*She cuts a strip of paper and folds it in six.*] An end tag for a coin under a foot. The strip is tied on by one end. From the knot

out to where the shading changes is heads' share; from there out to the free end is tails'. The shading only marks the place: it is how far from the knot that counts, not which part is dark.

Achilles: [*shading three parts of the six from one end, and tying the strip to the string's left end by its dark end, so that it lies flat on the near side of the string, toward the two of them*] From the knot, three parts, and then the change: heads, three parts in six. Tails, the other three. A tossed coin.



a tossed coin

Tortoise: [*setting one "turn it over" tag on the string*] Walk it.

Achilles: The coin is under your foot.

Tortoise: Walk what you have.

Achilles: [*untying the strip and carrying it along the string as far as the tag*] And how does one turn over a wager?

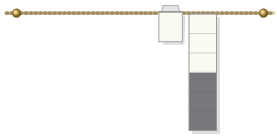
Tortoise: If I turned the coin over where it lies, unseen, what would become of heads' share?

Achilles: It would be tails' share, and tails' would be heads'.

Tortoise: Then turn the strip end for end, so that it will be tied on by its other end.

Achilles: [*The strip is in his hands at the tag, untied; the end that had sat at the knot is the dark one, and from it three parts run dark to the change of shade — heads' share — then three parts white to the free end. He turns it end for end and holds its white end to the string. What has moved is which end is held to the string; what has not moved is the change of shade, still three parts in from the held end, for on this strip it sits dead in the middle.*] Tied on by this end: three in six. [*He turns it back, and round again: three parts from either end, every time.*] It reads

the same from either end. Then for a coin nobody has seen, this tag does nothing! It needn't wait for a twin; it may come off alone.

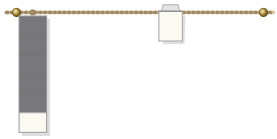


turned at the tag

Tortoise: That is your tossed coin. Mine is under my foot. [*Achilles lays his strip aside. She folds a second strip in six, shades five parts from one end, leaves the sixth white, and ties it to the string's left end by its dark end.*]

[*The fire has burned down while they worked, and sits now as one low bed of embers: most of it still redly alight, and over in one corner a patch already fallen to grey ash. It gives less light than it did, and the room has drawn in a little closer around the table.*]

Achilles: Heads, five parts in six. That's your die.



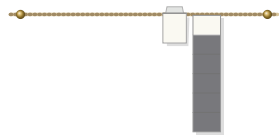
the Tortoise's strip

Tortoise: Walk it.

[*The fire drops with a soft collapse of coals, and for a breath the room holds still — only the wind. In the lower light the woodcut on the wall seems to come forward off its nail, nearer than it had hung all evening.*]

Achilles: [*This strip hangs by its dark end: from the knot, five parts dark to the change — heads' share, five in six — then one white part to the free end. He unties it, carries it along to the tag, turns it end for end, and holds its white end to the string. The strip is the same length it was; but the change of shade now sits only one part in from the held end, where a moment ago it sat five parts in, and the two shares have traded sizes.*] Tied on by this end: from the knot, one part — and

then the change. It moved! [*He looks up, the strip still in his hands, dark end and white end, at the print on the wall — Escher's* Day and Night.] Black birds flying one way, white birds the other, over fields that are neither yet. And it is all one sheet of paper.



turned at the tag

Tortoise: And the tag?

Achilles: At work all along. My strip couldn't show it: turned end for end, half-and-half is half-and-half. Put your socks on the wrong feet and nobody could tell; do it with your shoes and anybody could.

Tortoise: One start was not enough to tell a tag from no tag. [*She sets the second "turn it over" tag on the string beside the first.*] Go on.

Achilles: [*carrying the strip on to the second tag and turning it end for end again*] Tied on by its dark end again: from the knot, five parts, and then the change — heads, five in six, as it began. Turned end for end, it is still one strip — the same length however it's held, and the two shares together always the whole of it. [*He carries it back and ties it to the string's left end by its dark end.*]

[*The Tortoise slides the two tags together and lifts them both off. The strip hangs at the left end of a bare string, as she first tied it.*]



lifted off

Achilles: As before. The pair comes off for the strip as it did for the coin, and the coin itself never stirred. So a start is either a fact or a wager. There is nothing else it could be.

[*The bed of embers has settled again while they talked: most of the live red gathered at one end of the grate now, the barest coal at the other, and more grey between them than there was. The wind takes up and dies and takes up again beyond the glass.*]

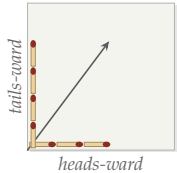
[*The Tortoise reaches into her shell and brings out a square of tracing paper.*]

Fifth Challenge — *The Arrow*

Tortoise: Tracing paper this time, and a handful of matchsticks. I choose tracing paper because a mark on it shows through from behind, so we may turn it over and still follow the very same mark; the matches are nothing but a small ruler — set end to end, they say how far a thing reaches.

Achilles: Matchsticks for a ruler. Very well — show me a start.

Tortoise: [*She lays the square flat on the table. From its near left-hand corner she lays three matchsticks end to end along the bottom edge, and four up the side.*] Along the table, the way the string runs: call that heads-ward. Away from us: tails-ward. [*With her pencil she draws an arrow from the corner to the spot three matches along and four matches up.*] A start.



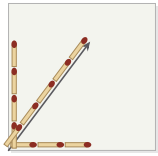
a start

Achilles: A strip with a corner in it! Three parts heads-ward and four tails-ward: seven parts in all. Your arrow is seven matches long.

Tortoise: Measure it.

Achilles: [*laying matchsticks end to end along the arrow, from the corner to its tip*] Five. Not seven. But nine and sixteen make twenty-five, and five fives are twenty-five: it's the squares on the two reaches that make up the square

on the arrow, not the reaches themselves. Then these aren't shares — three and four overshoot the whole. As a wager it's nonsense.



five, not seven

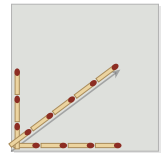
[*The teapot has sat cooling at the far corner of the table this hour past. Achilles reaches for it now without rising — straight across the corner, the short way, a shorter reach than following the two edges round — finds it barely warm, and pours anyway.*]

Tortoise: As a wager, yes.

Achilles: And it's no fact either: it lies along neither edge. It cuts across the corner, as my hand did just now for the teapot, and that is shorter than going round by the two edges. So there was something else a start could be. What does it mean, to reach three heads-ward?

Tortoise: Another evening. Tonight, only what a tag does to it. [*The arrow reaches three matches heads-ward and four tails-ward, and five along its own length. She clears the matches away and turns the square face-down about the line running from the arrow's corner to the far corner. The corner stays pinned where it was and the line she turns it about stays put; what swings across is the rest of the square, so that the heads-ward edge comes to lie where the tails-ward edge was, and the reverse — and through the paper the arrow shows, as long as ever.*] Turned over.

Achilles: [*laying matches again: along the bottom, up the side, along the arrow*] The reaches have changed places, as the shares did. And along the arrow, five still: turned over, it is still one arrow.

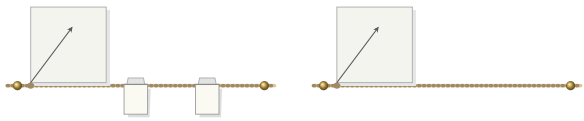


turned over

Tortoise: And home again?

Achilles: [*He clears the matches and turns the square over again about the same line.*] Three and four. Home.

[*He unties the strip and ties the square to the string's left end by the arrow's corner, so that the square lies on the far side of the string, its tails-ward edge running away from them. He sets the two "turn it over" tags on the string side by side. His hand goes to the knot, to untie the square for its walk — and stops. He slides the two tags together and lifts them off.*]



tied on *lifted off*

Achilles: ...Oh. I needn't walk anything; I never need have. The string and its tags never once asked what was travelling — a coin, a strip, an arrow. Each has its own way of being turned over, and the tag doesn't care which. Two of them side by side come off for all three.

Tortoise: They always did.

Achilles: But which ways may an arrow lie? And when it comes to the right-hand end of the string — how should I find out what had arrived?

Tortoise: Another evening.

Achilles: You say that often.

[*The wind has dropped to a long low sound at the glass, gone from worrying at the house to something almost companionable. The fire is a clear bed of coals now with*

hardly a flame left on it, warming without lighting; the hour has got late without either of them marking it.]

Tortoise: It is a thrifty phrase. It puts nothing away for good.

Achilles: [*He gathers the loose tags into a small pile and slides his crowded, useless sheet to the far edge of the table.*] Then I'll leave off crossing things out. The string kept the better record of the two.

[*She unties the square and puts it away. The string lies across the table, pinned and straight, with nothing on it, as it did at the start. Her foot has not moved from the coin. Achilles plucks the string once: the same note.*]

Achilles: But what does travel along it, then? A coin?

Tortoise: Not just a coin.

[*The fire is banked low and steady, a clear bed of coals that will keep till morning, and the wind has all but given over. Above the hearth the woodcut has gone quiet in the dim — the chequered fields and the two towns settled back into the wall. Neither of them moves to clear anything away just yet.*]

Chapter 1: Not Just a Bit

— Overture for a Single String —

Part I: The Single Wire

Escher: "Day and Night" — white birds fly one way into the night, black birds the other way into the day, and both grow out of the fields between

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When something carries the answer to a yes-or-no question, what exactly is it carrying?

A light switch is up or down. A coin on a table shows heads or tails. A door is open or shut. Each of these settles one question, and it seems that nothing could be simpler: there are two possibilities, the world has picked one, and the thing in front of us is the record of the pick. Every message ever sent, every photograph stored in a telephone, every sum done by a machine is built from records of this kind, a great many of them, and the theory of computing begins by taking one of them seriously.

But put a hand over the coin before anybody has looked. There are still two possibilities, and the world has still picked one, yet the honest description of what is under the hand is no longer "heads" or "tails". It is something like "heads, five times in six". That description is not a face of the coin. It is a pair of numbers, and it has to be carried from place to place, and changed when the coin is turned over, just as a face would be. So already there are two different things that can travel down the same line: a fact, and a wager.

This chapter is about a third. It can be written in the same place as the other two, it is changed by the same operations, and it is neither a fact nor a wager. We shall define it carefully and say honestly how little we yet know about it.

Along the way we need a language for things that travel and things that are done to them, and we shall build two: one of columns and matrices, written on a blackboard, and one of lines and boxes, drawn in a sketchbook. They will say the same things. The interest of the book is in how differently they say them.

The smallest piece of all this is a single line with something travelling along it. What travels? A coin, to begin with. Not just a coin.

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§1.1 A Coin on a String

Start with the plainest case. A **bit** is a thing with two possible values, which we call 0 and 1. A coin lying on a table is a bit: say heads is 0 and tails is 1. We want to describe two matters: how a bit starts, and what is done to it.

On the blackboard. We could write the value of a bit as the digit 0 or the digit 1 and be done. Instead we write it as a column of two numbers, with a 1 in the place that belongs to the value and a 0 in the other place:

$$\text{the state } 0 \;=\; \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad\qquad \text{the state } 1 \;=\; \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

The convention, fixed for the whole book, is that **the first (top) entry belongs to** 0 **and the second to** 1. Read a column as two answers, to the questions "is it 0?" and "is it 1?"; the state 0 answers *yes, no*.

Something done to a bit is called a **process**, and on the blackboard a process is a 2×2 matrix, a square of four numbers. A matrix acts on a column by the usual rule,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \;=\; \begin{pmatrix} ax + by \\ cx + dy \end{pmatrix}.$$

Try the rule on our two columns. With $x = 1, y = 0$ the result is $\begin{pmatrix} a \\ c \end{pmatrix}$, the

first column of the matrix; with $x = 0, y = 1$ it is $\begin{pmatrix} b \\ d \end{pmatrix}$, the second. So a matrix wears its behaviour on its face: **its first column is what it does to the state** 0**, and its second column is what it does to the state** 1**.** We shall use this remark more than once.

The process we care about most in this chapter turns the coin over. It is called **NOT**, and we write X for its matrix. It must send the state 0 to the state 1 and the state 1 to the state 0, so by the remark just made its columns are forced:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Check it by the rule:

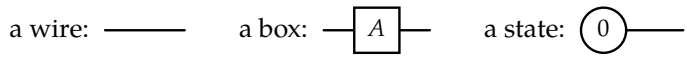
$$X\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \qquad X\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \cdot 0 + 1 \cdot 1 \\ 1 \cdot 0 + 0 \cdot 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

In the sketchbook. We now draw the same things, and since these are the first pictures of the book, each element is described before it appears.

A **wire** is a horizontal line. It stands for the system: the thing that has a value, here our one coin. A wire is not a moment and not a place; it is the coin's whole career, drawn from left to right.

A **box** is a rectangle sitting on the wire, with the wire entering on its left and leaving on its right, and a name written inside. It stands for a process: something done to the system. What enters on the left is what the process is given; what leaves on the right is what it hands on.

A **state** is a box with *no wire coming in*: nothing enters it, and one wire leaves it on the right. It stands for a way of starting. To tell states from processes at a glance we draw a state with rounded ends. The two states we have so far are labelled 0 and 1.



One more element, and it is not a shape but a way of placing shapes. Several things may sit **in a row** on the same wire, and a row is read **from left to right: first this, then that.** A state at the left end followed by a box means "start like this, then do that". Whatever is further to the left happens earlier.

With these four elements the two blackboard calculations above become two picture equations. Start in the state 0, then apply NOT: the result is the state 1. And the other way about:



We shall need to refer to these two equations by name. The left one is ***NOT on 0***; the right one is ***NOT on 1***. Each side of each equation is a state: no wire comes in, one wire goes out. An equals sign between two pictures means that they stand for the same column or the same matrix, entry for entry. In this chapter every picture equation is exact in that sense. See `Code/Ch1/Flip.py`.

Why this definition. Why write a bit as a column of two numbers, when one digit would do? For two reasons, and both are investments. First, once a start is a column, a process can be a matrix, and everything that matrices know how to do — above all, to be multiplied — becomes available. Second, a column has room in it. The two columns above use only the numbers 0 and 1; the places where they sit could hold other numbers. In §1.4 and §1.5 we shall change what is written in the column *without changing a single matrix or a single picture*, and that is the whole plot of the chapter.

Feynman's Notebook.

What a bit is made of. No machine has ever held a 0 or a 1. What a machine holds is some physical thing that can be put into either of two conditions and found, later, reliably in the one it was put in: a voltage that is high or low, a patch of magnetic material

lying one way or the other, a hole punched or not punched in a card. The 0 and the 1 are names we give to the two conditions. The coin on the table is as good a bit as any of these, only slower.

This is why the theory can afford to be abstract. Nielsen and Chuang, in the textbook on which the classical half of this book leans (*Quantum Computation and Quantum Information*, §1.2), introduce the bit as the fundamental concept of classical computation and classical information, and then describe its quantum counterpart deliberately as a mathematical object, so that what is proved does not depend on any particular way of building one. Our wire is drawn in the same spirit. It does not say what the system is made of. It says only that there is one.

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§1.2 There and Back

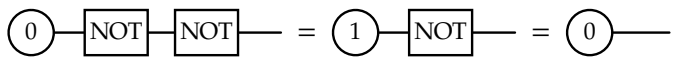
Turn the coin over, and then turn it over again. Anyone can say what happens. The point of this section is to say it in our two languages and to notice how much, or how little, each has really said.

On the blackboard. Start in the state 0 and apply X twice, one step at a time, using the two calculations of §1.1:

$$\begin{pmatrix}1\\0\end{pmatrix} \xrightarrow{X} \begin{pmatrix}0\\1\end{pmatrix} \xrightarrow{X} \begin{pmatrix}1\\0\end{pmatrix}, \qquad \begin{pmatrix}0\\1\end{pmatrix} \xrightarrow{X} \begin{pmatrix}1\\0\end{pmatrix} \xrightarrow{X} \begin{pmatrix}0\\1\end{pmatrix}.$$

Each start comes home.

In the sketchbook. Put two NOT boxes in a row after the state 0. Reading from the left, the state 0 followed by the first NOT is, by *NOT on 0*, the state 1; and the state 1 followed by the remaining NOT is, by *NOT on 1*, the state 0:



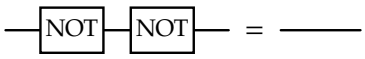
The same two steps in the other order bring the state 1 home.

So far this is an *example*: two particular starts, walked through two particular boxes. It says that for a coin we can see, two turns are as good as none. It would be more useful to have a statement about the two boxes themselves, with no start mentioned — a statement one could apply in the middle of a long row without asking what had come in from the left. Such a statement needs a picture of "no boxes at all". That is the **bare wire**: a wire with nothing on it, standing for the process of doing nothing. On the blackboard doing nothing is the **identity matrix**

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

which by the rule of §1.1 returns every column unchanged.

> **Theorem 1.1 (NOT twice is nothing).** $XX = I$, exactly. In pictures: two NOT boxes in a row are the bare wire.



Proof idea. A matrix is known once its two columns are known, and its columns are what it does to the states 0 and 1. The example has just shown that "NOT, then NOT" sends each of those two states where doing nothing sends it. So the two processes have the same columns.

Proof sketch. The matrix of "NOT, then NOT" is the product XX (why a row of boxes is a product is the business of §1.3). Its first column is $X(X\begin{pmatrix}1\\0\end{pmatrix}) = \begin{pmatrix}1\\0\end{pmatrix}$ and its second is $X(X\begin{pmatrix}0\\1\end{pmatrix}) = \begin{pmatrix}0\\1\end{pmatrix}$; those are the columns of I. ■ See `Code/Ch1/TwiceOver.py`.

This is the headline theorem of the chapter, and §1.7 proves it twice, in full. It may seem a great deal of ceremony for a coin turned over twice. The reason for the ceremony is that the theorem is about the *matrix*, not about the coin. Before the chapter is over the wire will be carrying things that are not coins, and the theorem will hold for them without a word being

changed.

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§1.3 Boxes in a Row

Two copies of the same box in a row cannot show us everything about rows. We need a second kind of box.

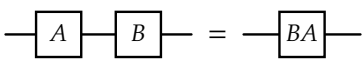
On the blackboard. How many processes are there that take a bit and return a bit? A process must say where 0 goes and where 1 goes, and each has two places to go, so there are exactly four. Writing each as the matrix whose columns are the two results:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad S_0 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad S_1 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

The first does nothing and the second is NOT. The third, **SET-0**, sends both states to the state 0 — "heads up, whatever it was" — and the fourth, **SET-1**, sends both to the state 1.

Now do one process and then another. If a column v is first given to a matrix A, the result is Av; if that is then given to B, the result is $B(Av)$. Matrix multiplication is defined precisely so that this equals $(BA)v$: the product BA is the single matrix that does the work of "A, then B". Notice where the letters have ended up.

In the sketchbook. Two boxes in a row, A on the left and B on the right, are read as always from left to right: first A, then B. Draw an outline around the two of them and what is inside has one wire in and one wire out: **a row of boxes is itself a box**. Its matrix is BA.



This is the one convention in the book most worth saying loudly, so here it

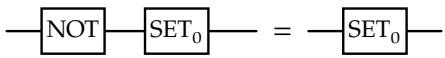
is, loudly. **Pictures are read from left to right. Matrix products are read from right to left. The row "A, then B" is the product BA.** The reason is nothing deeper than the habit of writing a matrix to the *left* of the column it acts on, so that the matrix nearest the column acts first. Neither convention will change; the reader who remembers that they run opposite ways will be spared a great many sign errors and transposed answers later in the book.

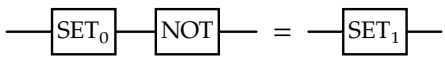
With NOT alone the convention was invisible, since XX reads the same in both directions. With two *different* boxes it shows. Take NOT and SET-0, in both orders:

$$\text{NOT, then SET-0:}\quad S_0X = \begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix}\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} = \begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix} = S_0,$$

$$\text{SET-0, then NOT:}\quad XS_0 = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}\begin{pmatrix}1 & 1\\ 0 & 0\end{pmatrix} = \begin{pmatrix}0 & 0\\ 1 & 1\end{pmatrix} = S_1.$$

In words: turn the coin over and then lay it heads up, and it is heads up; lay it heads up and then turn it over, and it is tails up. The same two boxes, in the two orders, are two different processes:





Order matters. See `Code/Ch1/OrderMatters.py`, which also checks the left-to-right, right-to-left convention on a pair of matrices with no helpful symmetries at all.

The bare wire of §1.2 now has a second description: it is the *empty row*, the row with no boxes on it, and its matrix is I. Putting a bare stretch of wire before or after a box changes nothing, which on the blackboard reads $AI = A = IA$.

Two further lessons can be read from SET-0. First, Theorem 1.1 is a fact

about NOT and not a fact about rows: two SET-0 boxes in a row are *not* the bare wire, since $S_0 S_0 = S_0$. Doing it twice is the same as doing it once. Second, nothing placed after SET-0 can repair it. For any matrix B whatever,

$$B\,S_0 \;=\; \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \;=\; \begin{pmatrix} b_{11} & b_{11} \\ b_{21} & b_{21} \end{pmatrix},$$

a matrix with two equal columns, and I does not have two equal columns. Once the coin has been laid heads up, nothing further along the wire can tell how it started. We record this as a fact about this one matrix. What distinguishes the boxes that can be undone from those that cannot is a question for Chapter 3.

Dirac's Blackboard.

All sixteen rows of two. Four boxes, taken two at a time with order respected, make sixteen rows. Here are all of them. The entry in the row labelled A and the column labelled B is the process "A, then B", which is the matrix product BA — the *column* label is written on the *left* of the product.

first\then	I	X	S_0	S_1
I	I	X	S_0	S_1
X	X	I	S_0	S_1
S_0	S_0	S_1	S_0	S_1
S_1	S_1	S_0	S_0	S_1

Three things to read off. Every entry is again one of the four, as it must be, since a bit-process followed by a bit-process is a bit-process. The table is not symmetric about its diagonal: compare the entry for X, then S_0 (which is S_0) with the entry for S_0, then X (which is S_1). And the symbol I appears on the diagonal exactly twice: only doing nothing, and NOT, are undone by a second copy of themselves.

Why this definition. Why let a row of boxes *be* a box, rather than keep rows and boxes as two kinds of thing? Because then every rule about boxes

applies to rows for nothing, and an equation such as Theorem 1.1, which says that a certain row equals a certain simpler row, may be used anywhere inside a longer one.

What we have drawn in this section are the first **circuits** of the book: boxes on a wire, doing to a bit the things a classical computer does. They are, admittedly, the circuits of a computer with a single bit in it. Real circuits have many wires side by side and boxes that join them, and those need one new idea — how to draw, and how to calculate, two systems at once — which belongs to Chapter 4.

Penrose's Sketchbook.

Where the pictures come from, and what they do not say. Drawing a calculation is an old idea. In 1971 Roger Penrose published "Applications of negative dimensional tensors"; as Peter Selinger's survey of graphical languages tells it, Penrose proposed drawing each map as a box, with a wire for each of its indices, and joining the wires of indices that are paired. In 2004 Samson Abramsky and Bob Coecke set out to replace ad hoc calculations with matrices and normalizing constants by conceptual definitions and proofs; of their networks of boxes they observe (in their §4) that the quantum information seems to flow along the line. Our reading of a wire as a system and a box as a process descends from that work.

Two cautions, to carry through the book. A picture does not say what its wire carries; the next two sections lean on that. And at this stage a picture has no rules of its own: every equation between pictures in this chapter is a matrix equation underneath, and is believed for that reason and no other.

§1.4 The Covered Coin

Now roll a die out of sight. If it shows a six, lay the coin tails up; otherwise lay it heads up. Cover the coin. How does it start?

Neither of our two state boxes is an honest answer. The coin under the hand is heads or it is tails, but *we* can say only that it is heads in five cases out of six.

On the blackboard. A **probabilistic bit** is described by a column

$$\begin{pmatrix} p \\ 1-p \end{pmatrix}, \qquad 0 \le p \le 1,$$

in which the first entry is the probability that the value is 0 and the second the probability that it is 1. Such a column — entries that are real and not negative, and that **sum to** 1 — is called a **stochastic vector**. The covered coin is $\begin{pmatrix} 5/6 \\ 1/6 \end{pmatrix}$. The two columns of §1.1 are stochastic vectors too, the ones in which nothing is in doubt.

What should a process do to such a column? Suppose the process is the matrix A. With probability p the coin is really in the state 0, and then the result is the first column of A; with probability $1-p$ it is in the state 1, and the result is the second column. The honest description of the result is the two mixed in those proportions:

$$p \cdot (\text{first column of } A) + (1-p) \cdot (\text{second column of } A) \;=\; A \begin{pmatrix} p \\ 1-p \end{pmatrix}.$$

The right-hand side is the rule of §1.1 read backwards. So **the same matrices act, in the same way**; nothing needs to be redefined. For NOT,

$$X \begin{pmatrix} p \\ 1-p \end{pmatrix} = \begin{pmatrix} 1-p \\ p \end{pmatrix}, \qquad \text{for instance} \qquad X \begin{pmatrix} 5/6 \\ 1/6 \end{pmatrix} = \begin{pmatrix} 1/6 \\ 5/6 \end{pmatrix}:$$

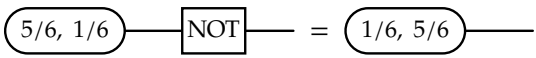
the two shares change places. Nobody has touched the coin's description by hand; the matrix did it.

Does a process always return a legal description? Add the two entries of Av using the rule of §1.1: the sum is $(a + c)\,x + (b + d)\,y$. If each *column* of A sums to 1, that is $x + y$, and the total is kept. All four matrices of §1.3 have entries that are not negative and columns that sum to 1 (such matrices are called **stochastic matrices**), and that is exactly why they turn stochastic vectors into stochastic vectors.

And two NOTs? There is nothing to prove. Theorem 1.1 says $XX = I$, and I returns every column unchanged, whatever is written in it:

$$X\Big(X\begin{pmatrix}5/6\\1/6\end{pmatrix}\Big) = X\begin{pmatrix}1/6\\5/6\end{pmatrix} = \begin{pmatrix}5/6\\1/6\end{pmatrix}.$$

In the sketchbook. There is no new element. The wire is the same wire and the boxes are the same boxes; only the label inside the state box has changed. We write the two entries of the column in it, first entry first:



Every picture of §1.1–§1.3 stands exactly as it was drawn. None of them ever said what the wire carries.

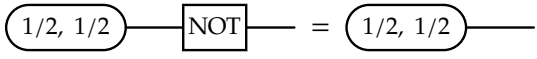
A trap

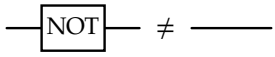
Here is a tempting experiment that goes wrong. Take the fairest covered coin, heads and tails in equal shares, and apply NOT:

$$X\begin{pmatrix}1/2\\1/2\end{pmatrix} = \begin{pmatrix}1/2\\1/2\end{pmatrix}.$$

The column has not moved. One might conclude that for a coin nobody has seen, NOT does nothing, and might as well be taken off the wire *by itself*. In

pictures the tempting inference is from the first line below to the denial of the second:





Both lines are true. NOT leaves the fair coin's description alone and is nevertheless not the bare wire, as the five-in-six coin showed a moment ago. The moral is worth a sentence of its own: **one state is not enough to tell two boxes apart.** Agreement on a single start, however natural a start it is, proves nothing about the boxes. How many starts *are* enough is a question we shall have to answer in §1.7, since the sketchbook proof of Theorem 1.1 depends on it. See `Code/Ch1/HiddenCoin.py`.

Why this definition. Why describe ignorance by a column of two numbers summing to 1, rather than by a single number p? Because the column was already there. In §1.1 we chose to write a bit in two places precisely so that something other than 0 and 1 could later be written in them, and so that the matrices would not have to be told. A single number p would have needed a new rule for each process; the column needs none.

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§1.5 The Arrow

So a start is either a fact or a wager? It is natural to think that there is nothing else it could be. The columns say otherwise. A column has two places, and we have filled them in two ways: with a 1 and a 0; and with two shares of a whole. Here is a third way.

On the blackboard. A **qubit** is a system whose state is a column of two *complex* numbers

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \qquad \text{with} \qquad |\alpha|^2 + |\beta|^2 = 1.$$

Here $|z|$ is the modulus of a complex number: for $z = a + bi$ with a, b real, $|z|^2 = a^2 + b^2$. The set of all columns of two complex numbers is written $\mathbb{C}^2$, so a qubit state is a column in $\mathbb{C}^2$ whose **squared moduli sum to** 1. The entries need not be shares of anything: being complex numbers, they may be negative, or not real at all.

Put the two rules side by side on one example:

$$\begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}: \qquad \left(\tfrac{3}{5}\right)^2 + \left(\tfrac{4}{5}\right)^2 = \tfrac{9}{25} + \tfrac{16}{25} = 1, \qquad \text{but} \qquad \tfrac{3}{5} + \tfrac{4}{5} = \tfrac{7}{5}.$$

This column is a perfectly good qubit state and is *not* a stochastic vector. Going the other way, the covered coin $\begin{pmatrix} 5/6 \\ 1/6 \end{pmatrix}$ is a stochastic vector and not a qubit state, since $\frac{25}{36} + \frac{1}{36} = \frac{26}{36}$. The two rules are different rules, and the two kinds of column are different kinds of thing. Only the states 0 and 1 are legal under both — and as bits besides. Another legal qubit state, to show an entry that is not real, is $\begin{pmatrix} 3/5 \\ 4i/5 \end{pmatrix}$, since $|4i/5|^2 = \frac{16}{25}$.

When both entries are real, there is a picture of the rule that everyone knows. Draw an arrow in the plane, starting at the origin, reaching α to the right and β upward. By Pythagoras its length squared is $\alpha^2 + \beta^2$. A real qubit state is an arrow **of length** 1; our example reaches three fifths one way and four fifths the other, and the old triangle with sides three, four and five does the rest. The dialogue draws its arrow this way, on paper. It is an honest picture of a slice of the truth: the full space $\mathbb{C}^2$ has complex entries and does not fit on a sheet of paper.

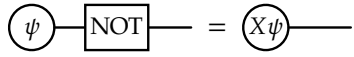
The matrices act as before, by the rule of §1.1, which never asked what kind of numbers it was multiplying. NOT exchanges the entries,

$$X\begin{pmatrix}\alpha\\ \beta\end{pmatrix} = \begin{pmatrix}\beta\\ \alpha\end{pmatrix}, \qquad \text{for instance} \qquad X\begin{pmatrix}3/5\\ 4/5\end{pmatrix} = \begin{pmatrix}4/5\\ 3/5\end{pmatrix},$$

and since $|\beta|^2 + |\alpha|^2 = |\alpha|^2 + |\beta|^2$, NOT turns a qubit state into a qubit state. Two NOTs bring it home, by Theorem 1.1, with nothing new to prove. See `Code/Ch1/Arrow.py`.

Not every box of §1.3 is so well behaved with the new passenger. SET-0 keeps sums but not sums of squares: $S_0\begin{pmatrix}3/5\\ 4/5\end{pmatrix} = \begin{pmatrix}7/5\\ 0\end{pmatrix}$, which is no qubit state. Which boxes may sit on a wire that carries a qubit is, again, Chapter 3's question.

In the sketchbook. Once more, no new element. The wire now stands for $\mathbb{C}^2$; a general state box is labelled ψ, the Greek letter psi, standing for a column $\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$; and NOT acts on it as the blackboard says:



The three passengers the wire has carried in this chapter, in one table:

passenger	its column	the rule
a bit	$\begin{pmatrix}1\\ 0\end{pmatrix}$ or $\begin{pmatrix}0\\ 1\end{pmatrix}$	one entry is 1, the other 0
a probabilistic bit	$\begin{pmatrix}p\\ 1-p\end{pmatrix}$	real, not negative, **entries sum to** 1
a qubit	$\begin{pmatrix}\alpha\\ \beta\end{pmatrix}$	complex, **squared moduli sum to** 1

Why this definition. Here we must be candid: this chapter does not say. We have *defined* the qubit's rule; we have not explained it. Why squares, and not the entries themselves? What do α and β have to do with what anyone

would see who looked? If they are not shares, what are they? Those are the questions of Chapter 2, and the possibility of negative entries, which we have mentioned and then carefully not used, is where that chapter begins. What this chapter can say is that the definition is *not* a relabelling of the covered coin: the column $\begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}$ is legal under one rule and illegal under the other, so whatever a qubit state is, it is not a wager about a hidden bit written in unusual units.

Feynman's Notebook.

Two levels of anything. What in the world is described by such a column? Nielsen and Chuang give a short list (*Quantum Computation and Quantum Information*, §1.2): the two polarizations of a photon; the alignment of a nuclear spin in a uniform magnetic field; two states of an electron orbiting a single atom, the ground state and an excited one. As with the bit, the two labels 0 and 1 name two conditions of a physical thing that can be reliably told apart. What is new is what else the thing can do.

The idea of computing with such systems is usually traced to Richard Feynman's "Simulating Physics with Computers", published in the *International Journal of Theoretical Physics* in 1982. As Nielsen and Chuang tell it (§1.1), Feynman pointed to the difficulty of simulating quantum mechanical systems on ordinary computers and suggested that computers built on quantum mechanical principles would avoid it. The word "qubit" is younger: Nielsen and Chuang credit it to Benjamin Schumacher, in 1995.

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§1.6 The Rosetta Table

This chapter seeds the book's running dictionary between the blackboard and the sketchbook. Five of the seven rows introduce picture elements — the wire, the box, the state, the row, and the bare wire; one names the two states we use most; and the last is the book's first rewrite rule.

#	Algebra (Dirac)	Picture (Penrose)
1	a two-valued system, and the space its columns live in: $\{0, 1\}$, then stochastic vectors, then $\mathbb{C}^2$	a **wire** — *new element*
2	a 2×2 matrix A	a **box** labelled A on the wire, input on the left, output on the right — *new element*
3	a column $\begin{pmatrix} \alpha \\ \beta \end{pmatrix}$	a **state**: a box with no input and one wire out — *new element*
4	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$	the state boxes labelled 0 and 1
5	the product BA (first A, then B)	A and B **in a row** on the wire, A on the left: read the picture left to right, the product right to left — *new element*
6	the identity matrix I	the **bare wire** — *new element*
7	$XX = I$	two NOT boxes in a row = the bare wire — the first **rewrite rule**

All seven rows are exact: the same numbers on both sides, entry for entry.

§1.7 The Theorem, Twice

Theorem 1.1 (NOT twice is nothing). Let X be the matrix of NOT and I the identity matrix. Then

$$XX = I, \qquad \text{that is,} \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

exactly. In pictures: two NOT boxes in a row are the bare wire.

The example of §1.2 walked two starts through the two boxes. Here are two proofs of the statement about the boxes themselves.

Dirac's Blackboard

The entry of a product BA in a given row and column is found by running along that row of B and down that column of A, multiplying in pairs and adding. For XX all four entries are:

$$\begin{aligned} \text{row 1, column 1:} \quad & 0 \cdot 0 + 1 \cdot 1 = 1, \\ \text{row 1, column 2:} \quad & 0 \cdot 1 + 1 \cdot 0 = 0, \\ \text{row 2, column 1:} \quad & 1 \cdot 0 + 0 \cdot 1 = 0, \\ \text{row 2, column 2:} \quad & 1 \cdot 1 + 0 \cdot 0 = 1. \end{aligned}$$

Therefore

$$XX = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.$$

And so, for *every* column v of two numbers — be they 0 and 1, shares of a whole, or complex —

$$X(Xv) = (XX)\,v = Iv = v,$$

where the first equality is the defining property of the matrix product (§1.3)

and the last is the defining property of I. Four entries, eight multiplications, and nothing assumed about v. ■

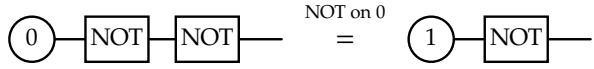
Penrose's Sketchbook

The sketchbook owns, at this stage, exactly two equations that mention NOT: *NOT on 0* and *NOT on 1*, from §1.1. Both have a state box in them. The theorem has none. To get from equations about states to an equation about boxes we need a principle, and we state it, and its source, before using it.

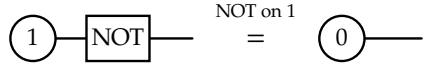
Two states suffice. *If two boxes (or two rows of boxes) give equal pictures when the state* 0 *is placed before them, and equal pictures when the state* 1 *is placed before them, then they are equal.*

This is not a fact the sketchbook has found for itself. It is lent by the blackboard, where it is the remark of §1.1: the first column of a matrix is what it does to the state 0, the second is what it does to the state 1, and a matrix is nothing more than its two columns. It is the other side of the trap of §1.4: one state was not enough to tell two boxes apart; these two are. We also use, without further comment, that a piece of a picture may be replaced by an equal piece — equal factors give equal products.

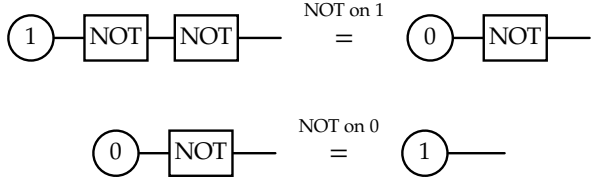
Step 1 — *NOT on 0.* Place the state 0 before the two NOT boxes. The state and the first box together are the left side of *NOT on 0*; replace them by its right side, the state 1.



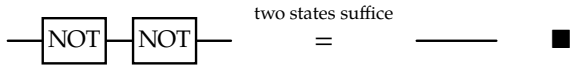
Step 2 — *NOT on 1.* What is left is the left side of *NOT on 1*; replace it by the state 0. A state followed by a bare stretch of wire is that state, so this is also the state 0 placed before the bare wire.



Steps 3 and 4 — *NOT on 1,* then *NOT on 0.* Place the state 1 before the two NOT boxes and use the same two equations in the other order.



Step 5 — *two states suffice*. The row of two NOT boxes and the bare wire give equal pictures after the state 0 (Steps 1–2) and equal pictures after the state 1 (Steps 3–4). By the principle, they are equal:



The verdict

The algebra wins, plainly. The blackboard proof is one matrix product, four entries, and it owes nothing to anyone. The sketchbook proof is five steps, and its last step is not a picture-move at all: *two states suffice* is a fact about matrices and their columns, borrowed from the blackboard. At this stage the sketchbook has no rule of its own that could make two boxes vanish, and it would be dishonest to pretend otherwise.

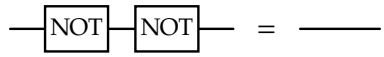
What the picture gives in return is not a shorter proof but a different kind of statement. The equation between the two pictures never mentions what the wire carries; and from this page on it is a **rewrite rule** — wherever two NOT boxes stand side by side in a row, they may be lifted off, and wherever a wire is bare, two may be hung on it — which is how every later picture proof in this book will work. That is a promise about the future, not a victory in the present. It proves an equality of processes, and nothing else.

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§1.8 The Round Trip

Here is the chapter's master diagram. We call it **The Round Trip**: a wire, open at both ends, running from left to right, with two NOT boxes in a row

on it. Out, and home again. Its evaluation is Theorem 1.1:



On the blackboard, $XX = I$; the equation is exact. We do not prove it again. Instead we *read* it, three times, by placing a state box at the open left end. The picture is the same each time, and only the label in the state box changes.

First reading: a bit. The passenger is a coin we can see, say the state 0.



$$XX\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$$

Second reading: a probabilistic bit. The passenger is a coin under a hand; the state box is labelled p, standing for the column $\begin{pmatrix}p\\1-p\end{pmatrix}$.



$$XX\begin{pmatrix}p\\1-p\end{pmatrix} = \begin{pmatrix}p\\1-p\end{pmatrix}$$

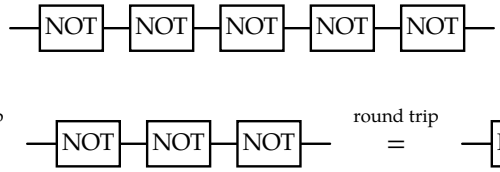
Third reading: a qubit. The passenger is a column $\psi = \begin{pmatrix}\alpha\\\beta\end{pmatrix}$ of $\mathbb{C}^2$ with $|\alpha|^2 + |\beta|^2 = 1$.



$$XX\begin{pmatrix}\alpha\\\beta\end{pmatrix} = \begin{pmatrix}\alpha\\\beta\end{pmatrix}$$

One picture, three passengers, one answer. In each reading the passenger really does make the trip — the state 0 becomes the state 1, the shares change places, the entries α and β are exchanged — and in each it comes home exactly as it left. The equation did not need to be told which reading was intended, because it was never about the passenger. See `Code/Ch1/TwiceOver.py` for the equation and for the first two passengers, and `Code/Ch1/Arrow.py` for the third.

Let us also use the equation once as what §1.7 said it had become, a rewrite rule. Here is a row of five NOT boxes. Lift off one adjacent pair, then another:



No state was plugged in and no matrix was multiplied: $X^5 = X$, by lifting boxes off a wire. It is a small thing. It is also the first calculation in this book done entirely in the sketchbook, on credit from the blackboard, and the later chapters will run up a considerable account.

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§1.9 The Other End of the Wire

Let us gather what we have. A system is a wire; a process is a box, which is a 2×2 matrix; a way of starting is a state, which is a column (§1.1). Two NOTs in a row are the bare wire, which we proved twice, and the algebra won (§1.2, §1.7). Boxes in a row are a matrix product written backwards, and their order matters (§1.3). The column can hold a fact; it can hold a wager, with entries that sum to 1 (§1.4); and it can hold something that is neither, with complex entries whose squared moduli sum to 1 (§1.5). Through all three the pictures did not change, and *The Round Trip* came out the same (§1.8).

Now look at the pictures once more and notice what none of them has. Every wire in this chapter starts somewhere — we have a box with no input, and we have used it on every page — but no wire *ends*. Each one runs off the right-hand side of the picture, still carrying its passenger, and nobody has asked the passenger anything. For a coin that hardly mattered: one looks, and sees a face. For a covered coin one lifts the hand, and the wager is settled by a fact. But for the third passenger we have not said what looking

would be, or what one would see, or how often. We have a column of two complex numbers and a rule about their squared moduli, and no connection yet between those numbers and anything that happens.

That is the business of the next chapter. It will draw the thing that closes the right-hand end of a wire, and find that a wire closed at both ends is no longer a process at all but a single number. It will say what that number has to do with what is seen, and why the rule must use squares. And it will finally use the fact we mentioned in §1.5 and then set carefully aside: that the entries of a qubit's column, unlike any wager's, are allowed to be negative.

Nocturne in Negative Numbers

水底に
もう一つある
雪催

minasoko ni
mō hitotsu aru
yukimoyoi

Under the water
there is another one, too —
sky heavy with snow.

[*The snow has not come yet but means to: all evening the sky has hung low and close, the loaded yellow-grey that stands before a heavy fall, and the shutters are drawn against it. Above the hearth, where the firelight can reach it, hangs a print Achilles had framed in the spring — Escher's* Sky and Water I*: dark birds ranked across the top, pale fish across the bottom, and a broad middle band where the two come to the same size and fit close, wing against fin. As the coals shift, the light picks out the birds and loses the fish, then loses the birds and finds the fish. The table stands bare and scrubbed, waiting.*]

[*Evening; a low fire burning down. The Tortoise sits exactly where she sat last evening, one foot planted just so — she has not shifted it since. Achilles comes back from the cold hall with a fresh pot and two cups.*]

Achilles: Still there. You've not budged an inch since last evening — I do believe you've taken root in my chair.

Tortoise: I have a coin to mind. [*She settles more comfortably, without moving the foot.*] Some things keep better left exactly as they are.

Achilles: [*pouring*] Then keep it. It's a filthy night for anything but the fire and a warm cup. [*He hands her one.*] That overture of last evening is still going round in my head, mind — I hummed it half the day and got the ending wrong every time.

Tortoise: The good ones do that. [*warming her hands round the cup*] There is nothing to be done on a night like this but sit close to the fire and think slowly.

Achilles: Slowly suits me. Though I'll warn you — there's a question left over from last evening that has kept me from my sleep.

Tortoise: They always are. Let me get my strings out of my shell, and we shall see what can be done about it.

First Challenge — *The Right-Hand End*

[*Evening; Achilles's table. The Tortoise draws a pale string out of her shell and a dark one after it. She pins the pale string taut and straight, left to right, on the left half of the table; she pins the dark string on the same line after it, a hand's breadth of bare table between. She plucks the pale string once: one note. Under her foot is a coin that was covered on an earlier evening; nobody has looked at it since.*]

Achilles: You never answered me. An arrow sets out from the left-hand end. When it comes to the right-hand end, how do I find out what has arrived?

Tortoise: [*She rests a fingertip on the pale string, then on the dark one lying after it.*] Gently, now. This pale string is the one we shall work along: your arrow sets out from its left end, and your question waits at the right. The dark one lies after it, a good hand's breadth of bare table between, so they never muddle — it will have its turn. And everything I carry about tonight I'll draw on one of these little squares of tracing paper: an arrow inked in from a corner, stiff enough to knot on by that corner. Tracing paper, mind, so that

when I want the same arrow in two places at once I can copy it rather than move it.

Tortoise: [*She brings out a square of tracing paper with an arrow drawn on it from the near left-hand corner, and ties it by that corner to the pale string's left end.*] Last evening's arrow: three matches heads-ward, four tails-ward. I called this the end tag, then. Tonight a string has an end to spare, so call it the start tag.

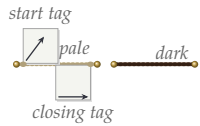
Achilles: Five matches long, I remember. And the dark string?

Tortoise: Later. [*She draws an arrow on a second square, from the near left-hand corner straight along the bottom edge, five matches long.*] Heads-ward, and nothing else. It would do for a start tag.

Achilles: A dull one. A coin lying heads up.

Tortoise: Dull or not, watch what I do with it. A string has an end to spare tonight, and a tag can be tied on for either errand: to send an arrow off from the near end, or to sit at the far end and wait for whatever comes along the string to it.

Tortoise: [*The start tag at the left end is set to send its reach off down the string. She takes a fresh square drawn the very same way and carries it along the pale string to the right-hand end as it lies — not turned round on the table, not turned over — so the arrow keeps pointing exactly as it did; only she ties it on now by the far right-hand corner, the corner opposite, so that the knot, and not the arrow, faces back along the string toward the start, and the square hangs down on the near side of the string, toward the two of them, where the start tag stands off beyond it.*] The arrow points as it pointed. Only the knot has changed its mind: this tag doesn't send anything along the string. It waits for what comes. A closing tag.



a start, and
a closing tag

Achilles: Then hang something beyond it, and let us see what falls off the end.

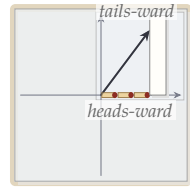
Tortoise: Where?

Achilles: [*His hand goes to the right of the closing tag, and finds bare table.*] There's no string left to hang it on. So what you're saying is, a start at one end, a question at the other, and the string hands over a coin. Heads, I expect.

Tortoise: Then let me show you how we put a question to the string at all. I have another sheet, ruled with two lines that cross square, an arrowhead on each — picture the crossing as the question itself, and each line as one of the answers it might be given, the one way or the other. Laid over an arrow, it lets us read how far the arrow reaches toward each answer. For measuring that reach I keep this square-cornered card, whose true corner drops a straight line from the arrow's tip down onto a ruled line, and matchsticks all cut to one length, for a ruler — so every reach we find is spoken in matches.

Tortoise: Let us find what it hands over. [*She brings out a sheet of tracing paper ruled with two lines crossing at right angles, an arrowhead on each. She lays it over the closing tag with one line along the tag's arrow, arrowheads agreeing; then she slides it back along the pale string, as it lies, and lets it rest on the start tag with the crossing on the arrow's tail.*] A question with two answers: heads-ward along this line, tails-ward along that. [*She hands him a square-cornered card.*] From the arrow's tip, straight down to the heads-ward line. Then matches, from the crossing to the foot of the card.

Achilles: [*He stands the square-cornered card on its true corner at the arrow's tip and lets its edge fall straight down onto the heads-ward line; where the corner meets the line is the foot. From the crossing out to that foot he lays the matches end to end.*] Three. And the arrow is five. Three parts in five.



three in five

Tortoise: [*She unties the closing tag, and ties on in its place, the same way, a square with an arrow five matches long up its left-hand side.*] Tails-ward, and nothing else.

Achilles: [*The card drops from the tip to the tails-ward line this time; the foot comes down further out along it.*] Four. Four parts in five. But that's no coin. A string tagged at both ends hands over a *number*?

Tortoise: One number. It has nothing left to travel to. Lean a stick in the noon sun and ask the ground about it, and the ground answers with a shadow: a length, not a stick.

[*Somewhere in the shutter's frame the wind has found a gap and settled into it, and out of that one closed slot it draws a single steady note, the same however hard or soft it blows. Achilles tips his head at the sound, then lets it go. Against the shutter the first snow ticks — a few grains, no more.*]

Achilles: Then those are the shares! Heads three parts in five, tails four. Which makes seven parts in — five. They overshoot the whole, exactly as they did last evening. [*He looks along the table.*] And the dark one?

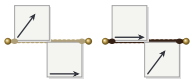
Tortoise: Now.

Second Challenge — *Why Squares*

Tortoise: The same tags, each at the other end. Trace them; a square can't hang in two places.

[*First the Tortoise unties the tails-ward square from the pale string's right-hand end and ties the heads-ward square back on there as its closing tag, knot facing*

back, where it was. Achilles lays a fresh square over the heads-ward arrow and, the tracing paper letting the shape through, inks its copy; he traces last evening's three-and-four arrow the same way. Now the Tortoise gives the dark string the two squares the pale string had — but each at the end the pale string did not use. The heads-ward tracing she ties to the dark string's left end, where a start tag sends its reach off; the three-and-four tracing she carries to the right-hand end as it lies and ties with the knot facing back, to wait. The arrows are the very ones from the pale string; only their errands have swapped ends — the arrow that asked now waits, and the arrow that waited now asks.]

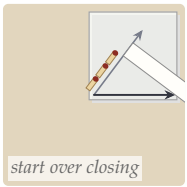


each at the other end

Achilles: The question sets out, and my arrow waits for it. That's the wrong way about.

Tortoise: Find its number. No ruled sheet this time: lay its start over its closing tag, tail on tail, as they lie. Drop the card from the tip of the heads-ward arrow onto the line of the other.

Achilles: [*He slides the heads-ward start square, its arrow drawn five matches long, over the three-and-four square until tail sits on tail, the two arrows springing from one point; the card drops from the heads-ward tip onto the three-and-four arrow's line, and from that arrow's tail out to the foot he lays the matches.*] Three parts in five — the pale string's number, arriving from the other end.



from the other end

Tortoise: And tails-ward?

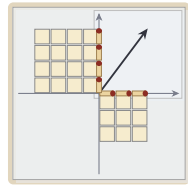
Achilles: [*He traces the tails-ward arrow onto a fresh square and swaps it in for*

the heads-ward start, and the tails-ward square goes back in at the pale string's closing end; he lays the new start over the three-and-four closing, tail on tail as before, and drops the card from its tip — and now the foot lands further along the line, and the matches reach four.] Four in five on the pale string. Four in five on the dark. Like an echo off the yard wall: it only repeats what it was told!

Tortoise: Good — keep hold of that. Those little squares in the dish, each a single matchstick to a side, are for laying a shape out flat where you can count it — nothing cleverer than that. Reach for a handful when you want them.

Tortoise: Then you have two equal sides. [*She tips a heap of little paper squares onto the table, well clear of the pale string and the dark, each one match on a side.*] One side from the pale string, one from the dark. Fill it in.

Achilles: [*The heads-ward reach lies bare along its line, three matches from the crossing. He sets the little squares out along it and stacks equal rows out from it, on the side away from the arrow, as many deep as the reach is long, until the reach has grown into a solid block; then a second block the same way on the tails-ward reach, on its own side away from the arrow. The matches do not move; only paper has been added.*] Three by three on the heads-ward reach: nine. Four by four on the tails-ward: sixteen. Nine and sixteen. Twenty-five in all.



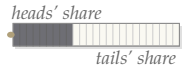
nine and sixteen

Tortoise: And to hold the answer, reach again for the strip from last evening — the paper that stood in for the covered coin's wager.

Tortoise: Now a strip. Fold it in twenty-five. From the knot end out to the change of shade is heads' share, as it was for the covered coin.

Achilles: [*He folds the strip across into twenty-five equal parts, then shades it from the knot end inward, part by part, and stops after the ninth; from there to the far end the strip stays pale.*] Heads, nine in twenty-five. And those *do* make

one whole. But this is a wager — a coin under a foot! There's no foot here, only an arrow and a question.



heads, nine
in twenty-five

[*On the sill beyond the shutter the snow has begun to lie, and it only ever adds — a grain, another grain, never one taken back — so that the few grains of a moment ago have become a covering, and the covering deepens and evens until the whole ledge carries the same soft weight from end to end. Achilles fills both cups again from the pot without looking up; the tea has gone from scalding to merely warm.*]

Tortoise: That is the recipe. An arrow, a question, the reach on the pale string, the same on the dark, the square on the two. If ever that question were put to that arrow, the strip says how often each answer would come.

Achilles: But why should the world go by the squares?

Tortoise: That is how often. Nobody has caught it doing otherwise. [*She takes a fresh square of tracing paper and draws an arrow from its far left-hand corner: one match heads-ward, and one match towards herself.*] One heads-ward. One tails-ward, backwards.

Achilles: A reach of less than nothing?

Third Challenge — *A Reach Backwards*

Tortoise: First the honest one. [*She draws another from the near left-hand corner: one match heads-ward, one tails-ward.*] The slant. Tie it on and ask it.

[*The slant is tied to the pale string's left end. The Tortoise marks its length along the card's edge and draws the closing arrows, heads-ward and tails-ward, exactly so long. Achilles traces; the dark string is tagged the other way about.*]

Achilles: [*He lays the ruled sheet square over the slant, crossing on the tail, and drops the card from the arrow's tip to each line in turn — to the heads-ward line,*

then to the tails-ward — laying matches from the crossing to each foot.] Heads-ward, one match. Tails-ward, one match. And along the slant itself — [*He turns the card to the slant's own arrow and lays matches up it from the crossing.*] — one and a bit. Not two. One part in... it doesn't come out in matches.

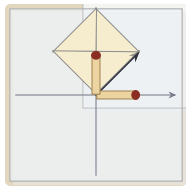
Tortoise: Then don't say it in matches. And the dark one?

Achilles: One match, and one match. The same.

Tortoise: Squares.

Achilles: One little square on each reach. But the square on the slant?

Tortoise: [*She takes the two little squares — the one from the heads-ward reach, the one from the tails-ward — and cuts each along its diagonal, corner to corner; the four half-squares she fits point to point around the slant until they close into one square standing squarely on it, on its tails-ward side, away from the heads-ward line. The paper is all there: none left over, none wanting.*] Made of those two, and nothing over.



nothing over

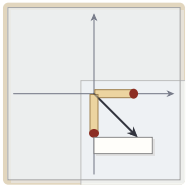
Achilles: Then each reach has one square of the two. Heads, one in two. [*He folds a strip in two and shades one part.*] Evens: my half-and-half strip again. Now your back-slant.

[*The back-slant is tied on as the pale string's start; its tracing becomes the dark string's closing tag.*]

[*He lays the sheet square over the back-slant and drops the card from its tip to each line. The heads-ward reach lands out along its line, honest enough; but the tails-ward stroke of this arrow leans back past its own tail, and the card comes down where no reach has come down before.*]

Achilles: Tails-ward — the foot of the card falls *behind* the crossing. One

match, backwards.



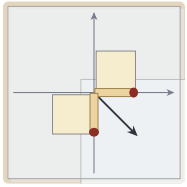
one match, backwards

Tortoise: And the dark one?

Achilles: [*He lays the tails-ward start over the back-slant. The card's foot misses the arrow; he produces the arrow's line backwards through its tail with the card's edge, and lays a match there.*] One match, backwards. Backwards, and backwards again. But can one lay paper on a reach of less than nothing?

Tortoise: [*She lays a little square on the heads-ward reach, as before; then another on the two backwards sides, on the far side of the crossing, where no square has lain tonight.*] Pick it up. Is there less of it than of the others?

Achilles: It's a square like any other. A shilling lent or a shilling borrowed is a shilling either way, if all you ask is how much. [*He holds the strip against the back-slant, and then against the slant.*] So the backwards reach is bookkeeping! A mark for the paper, and nothing to the world. Same strip, same start. [*He ties the one strip on at the pale string's left end, beside the square.*] One strip will do for both arrows.



a square like any other

[*Outside, the wind has quietly gone round to blow from the other side of the house — the chimney, which had been steady, takes up a low mutter, and the draught that had been coming in at the door slackens and dies. The fire does not notice; it burns*

exactly as it burned, and the one note in the shutter holds its pitch without a waver. Achilles sits back, satisfied, and warms his hands.]

Tortoise: [*letting him*] You have asked it one question.

[*She sets a fingertip on the crossing of the ruled sheet, holding it fast to the arrow's tail, and swings the sheet about that one point, the heads-ward line coming round toward the two of them. The crossing does not stir; the two ruled lines wheel away from heads-ward and tails-ward.*]

Fourth Challenge — *The Turned Question*

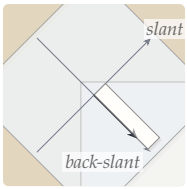
[*The ruled sheet has come to rest a half-right-angle round, the crossing still pinned to the arrow's tail. The line that was heads-ward now runs along the back-slant; the line that was tails-ward, along the slant.*]

[*The wind veers again — not round to the far side now but a half-turn, crosswise — and the note that the shutter had held falls to a low nothing, while a chink somewhere that had been silent all evening begins to whistle. The house sounds like another house.*]

Achilles: That's no question. Neither line is heads-ward or tails-ward.

Tortoise: Two lines at right angles, two answers. It needs closing tags, that is all: an arrow along each line, as long as the start. You have drawn them already.

Achilles: The slant and the back-slant themselves! [*He traces them. The back-slant is the pale string's start; a tracing of the slant is carried across as it lies and tied on, knot facing back.*] Back-slant, asked along the slant's line. [*He drops the card. Its foot falls on the crossing.*] No matches at all. It has no reach along that line.



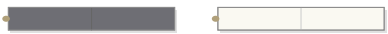
no reach

Tortoise: And along the other?

Achilles: It *lies* along the other. The whole arrow, and nothing left over.

Tortoise: And the dark one?

Achilles: [*laying one tracing over the other*] Nothing; and then the whole arrow. The same. And with the slant for a start it is the other way about: the whole arrow along its own line, none along the back-slant's. So the strips — [*Two fresh strips lie pale on the table. He shades the back-slant's strip from knot to far end — under the turned sheet its share from the knot goes to the line that heads-ward swung round to, the back-slant's own — and leaves the slant's strip white from end to end.*] All dark. All white.



the back-slant *the slant*

[*He unties the half-and-half strip from beside the start tag. Then he swings the ruled sheet back until it sits square, heads-ward and tails-ward; then round again until it lies along the slant and the back-slant; then back to square — rocking it between the two, the same two lines telling him first that the arrows are alike and then that they are utterly apart. He looks up at the print on the wall — Escher's* Sky and Water I.]

[*The fire chooses that moment to settle, and for a breath the flames stand still; the light lies flat and even across the whole print, birds and fish alike, neither nearer than the other. Then a coal drops, and the shadows come back.*]

Achilles: The fish are only what is left between the birds. And the birds are what is left between the fish.

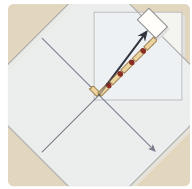
Tortoise: And the sheet?

Achilles: ...Oh. They were never the same start. One question hands me the same strip for both, and the next tells them apart every time — they stand at right angles, as far apart as heads-ward from tails-ward. The backwards reach was the whole difference. Two parcels of one weight: the scales can't tell them apart, and one good shake does.

Tortoise: Last evening's arrow, under the turned sheet.

[*The three-and-four arrow is tied on again. The Tortoise lays five matches along each turned line and draws the closing arrows so; Achilles traces, and the dark string is tagged the other way about.*]

Achilles: Along the slant's line: just short of five matches. Along the back-slant's: short of one match, and backwards.



all but a sliver

Tortoise: And the dark one?

Achilles: Just short of five; short of one, backwards. So the shares: all but a sliver — and the sliver comes from a backwards reach. Then every backwards reach can be caught, if one turns the question far enough!

Fifth Challenge — *The Arrow Reversed*

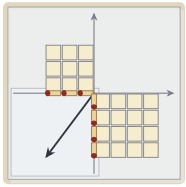
Tortoise: [*She unties the pale string's start tag and turns the square end for end on the table — the tip swung down to where the tail was, the ink not redrawn and its length unchanged, only the arrow pointing now the opposite way — and ties it on again by what had been its far corner.*] Every reach backwards at once.

Achilles: Then it will be caught at once. [*He ties the five-long heads-ward square she drew at the start of the evening on again as the pale string's closing*

tag, knot facing back; then he lays the sheet square and drops the card from the reversed arrow's tip to each line; both feet come down behind the crossing now, where before they fell in front of it.] Heads-ward: three, backwards. Tails-ward: four, backwards.

Tortoise: And the dark one? Its closing tag is a tracing of this. [*She turns that square end for end likewise, and ties it on, knot facing back; the heads-ward tracing is the dark string's start again.*]

Achilles: [*producing the reversed arrow's line back through its tail*] Three, backwards. Four, backwards. [*He builds the two blocks again, but on the far side of the crossing now, each behind its backwards reach; the blocks are the old three-by-three and four-by-four, only their side has changed.*] Nine and sixteen. My first strip.



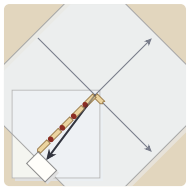
backwards, nine and sixteen

Tortoise: Don't tie it on yet.

Achilles: Well — the square sheet missed the back-slant too. [*He turns the sheet as she turned it; the five-long arrows along its lines go back on as the pale string's closing tag, their tracings as the dark string's start.*] Just short of five, backwards — and the sliver comes forwards now. And the dark one?

Tortoise: That was my line.

Achilles: The same, reach for reach. All but a sliver, as before.

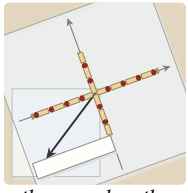


just short of five, backwards

Tortoise: A different arrow. Find me the question that knows it.

[*Achilles turns the ruled sheet to a position of his own choosing — the square sheet swung only a little way round, the heads-ward line tipped a little away from them — and holds it there. He lays matches along its lines for the arrow reversed. Then, the sheet held still, he turns the square end for end beneath it, lays a second row of matches beside the first, and turns the square back.*]

Achilles: The same lengths. Every one on the other side of the crossing. [*The Tortoise lays five matches along one of his lines and draws the closing arrow; he traces, and lays the dark string's squares one over the other.*] And whatever turns backwards on the pale string turns backwards on the dark. Two sides, a paper square, the same square. I can't catch it!



the same lengths

[*At the height of it the storm comes round the whole way — the wind that had been crosswise swings until it stands head-on against the shutters, and in one breath every draught in the room trades places, the cold from the door and the cold from the chimney changing ends. The fire burns on as it burned; the tea sits cooling in the cups. Every draught is reversed, and nothing in the room can be seen to have moved.*]

Tortoise: One reach turned backwards?

Achilles: A different start, and some question knows it every time. Every reach turned backwards: nothing to catch. Every clock in town an hour fast, and nobody misses a train; only the station clock fast, and everybody does.

Tortoise: A different arrow on the paper, even so. And no question we can tie on tells them apart.

Achilles: Then what *is* fixed? Five of what? Twenty-five of what?

Sixth Challenge — *The Whole Square*

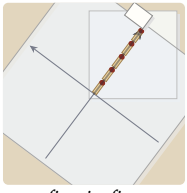
Tortoise: [*She turns the reversed squares end for end again: last evening's arrow, as it was drawn. One she ties to the pale string's left end.*] Trace it.

[*Achilles traces it. The Tortoise carries the tracing to the pale string's right-hand end as it lies, and ties it on with the knot facing back.*]

Achilles: The arrow, asked about — itself?

Tortoise: [*She turns the ruled sheet until one line lies along the arrow.*] Ask.

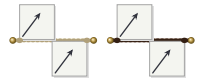
Achilles: [*The card's foot falls on the arrow's own tip.*] Five. Five parts in five: all of it. And along the other line, no reach at all.



five in five

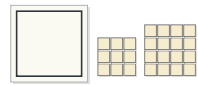
Tortoise: And the dark one?

Achilles: [*He traces the arrow twice more, ties one copy to the dark string's left end and carries the other to its right, and lays the start square over the closing, tail on tail; the card drops from the tip clean onto the arrow's own tip.*] Five in five. What else could it say?



asked about itself

Tortoise: [*She draws on a sheet of paper a square five matches on a side.*] The square on it. Fill it in.



the outline

the square on it

Achilles: I have no more little squares. Only the nine and the sixteen from — [*He breaks the three-by-three block into its nine loose squares and the four-by-four into its sixteen, and sets them one by one into her five-by-five outline, row filling into row, until the outline is a full field of squares, and nothing is left of either block.*] It fills. Slice a loaf any way you like, and the slices still come back to the one loaf.



it fills

Tortoise: Exactly?

Achilles: None over, none short. ...Oh. A string tagged at both ends is a number, and the number may be backwards. The number with its twin from the dark string is a square of paper, and that never is. And the squares, whatever the question, make up the square on the arrow asked about itself: one whole.

[*The snow has stopped. Where the shutter stands a little open, the world outside has gone to one unbroken white, sill and yard and low roofs filled level and made all of a piece. The wind has dropped away to nothing, and the house has stopped sounding altogether.*]

Tortoise: Twenty-five of that.

[*She unties every tag and puts the squares away. The pale string and the dark string lie bare, end to end, a hand's breadth of table between them, as they lay at the start. Her foot has not moved from the coin. Achilles plucks the pale string: one note. He plucks the dark string: the same note, after it.*]

Achilles: [*He warms his hands at what is left of the fire.*] The embers are nearly out, and the tea went cold an age ago. A good hour to stop, this.

Tortoise: [*not moving*] A very good hour.

Achilles: But the reaches themselves — three in five, one match backwards. Not shares. Not facts. What *are* they?

Tortoise: They have a name.

Achilles: Tell me.

Tortoise: It will keep till morning.

[*The fire is a low red breathing under grey. Beyond the shutter the white field keeps its stillness, and will keep it till morning. Above the hearth the print has gone back to being a print — birds along the top, fish along the bottom, dim now and part of the wall. The pale string and the dark lie bare where she left them, and under the Tortoise's foot the coin lies covered, as it has lain all night, and nobody has looked.*]

Chapter 2: Amplitudes

— Nocturne in Negative Numbers —

Part I: The Single Wire

Escher: "Sky and Water I" — birds above and fish below, and in the band between them each is only the space the other leaves

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How can a number that is allowed to be less than nothing tell us how often something will happen?

For the covered coin of the last chapter there was no such puzzle. The two numbers in its column *were* the chances: heads five times in six, tails once in six. They were never negative, they made up one whole, and lifting the hand settled the wager with a fact. Whatever the column said, it said about what would be seen.

Then we wrote a third kind of thing in the same two places. Its entries reach three fifths one way and four fifths the other, and three fifths and four fifths make seven fifths, which is no whole at all. Its entries may be negative. They may fail to be real numbers. Nobody's chances look like that. And yet when a qubit is finally asked a question with two answers, an answer comes; and when the same question is put to a great many qubits prepared in the same way, the answers come in steady proportions, as reliable as a die. Somewhere between the column and the counting there must be a rule that turns the one into the other.

This chapter finds the place where that rule lives. We left every wire of Chapter 1 running off the right-hand side of the page with nobody asking its passenger anything. Here we close that end. What comes out of a wire

closed at both ends is a single number, and it is not yet a chance: it has a sign, or worse. The chance appears only when the number is heard together with its mirror twin. So this is a chapter about numbers that are never themselves seen — it is night, the coin stays covered, nothing is looked at — and about the one way they have of showing in the morning.

Those numbers have a name. It is the title of the chapter.

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§2.1 A Name for a Column

Before we close the wire we need a better way of writing what travels on it. Chapter 1 wrote every state out as a column, in full, every time. That is honest and it is slow, and it has a deeper fault: it gives us no way of *naming* a column, so that we could speak of the same thing lying on its side.

On the blackboard. A **ket** is a name for a column, written between a vertical bar and an angle bracket: $|\psi\rangle$, read "ket psi". What stands inside is a label and nothing more. We fix two labels for good:

$$|0\rangle \;=\; \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad\qquad |1\rangle \;=\; \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These are the states 0 and 1 of §1.1, with the first (top) entry belonging to 0 as always. A word of warning that every beginner needs once: $|0\rangle$ **is not the zero column.** The 0 inside the ket is a name, as "heads" would be; the column it names has a 1 in it. The column with two zero entries is not a qubit state at all, since its squared moduli sum to 0, and we shall never give it a ket.

Columns can be multiplied by numbers, entry by entry, and added, entry by entry. Doing so with our two named columns recovers every column there is:

$$\alpha\,|0\rangle + \beta\,|1\rangle \;=\; \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \;=\; \begin{pmatrix} \alpha \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \beta \end{pmatrix} \;=\; \begin{pmatrix} \alpha \\ \beta \end{pmatrix}.$$

So the general qubit state of §1.5 has the ket spelling

$$|\psi\rangle \;=\; \alpha\,|0\rangle + \beta\,|1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1,$$

and Chapter 1's arrow, which we shall use all through this chapter and call T for its triangle, is

$$|T\rangle \;=\; \tfrac{3}{5}\,|0\rangle + \tfrac{4}{5}\,|1\rangle \;=\; \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}.$$

Nothing has been added to Chapter 1 except a way of writing. The definition of a qubit state is unchanged, and so is every matrix.

In the sketchbook. There is no new element, and nothing is redrawn. The state boxes of Chapter 1 were already labelled 0, 1 and ψ; the ket is the blackboard catching up with a habit the sketchbook had from the start, which is to put a name in a box and let the box's *shape* say what kind of thing is named. On the blackboard the shape is the bar and the bracket; in the sketchbook it is the rounded box with a wire leaving on its right. The state box labelled ψ stands for $|\psi\rangle$, and the state boxes labelled 0 and 1 stand for $|0\rangle$ and $|1\rangle$.

Why this definition. Why wrap a label in a bar and a bracket, when a plain letter ψ would name a column just as well? Because the wrapping says *which way up* the thing is. A ket is a column. In the next section we shall need the same label on a row, and the notation will do it by turning the wrapping round. A plain letter cannot be turned round.

Feynman's Notebook.

Where the ket comes from. The notation is Dirac's. It was introduced in a paper whose title says exactly what it contains, "A new notation for quantum mechanics", published in 1939 in the journal of the Cambridge Philosophical Society, and it is the reason the algebraic strand of this book is called *Dirac's Blackboard*. It has long since stopped being new. Nielsen and Chuang's *Quantum Computation and Quantum Information*, the textbook

> our blackboard leans on, develops in its second chapter the linear algebra a quantum computer scientist needs, and writes it in kets throughout; so does nearly everything else the reader will meet.
>
> It is also sturdier than a mere abbreviation. When Abramsky and Coecke rebuilt the formalism of quantum mechanics from abstract parts in 2004, one of the things they checked (in their §7.2, "Bras and kets") was that Dirac's notation could be interpreted in the new setting. A notation that survives having its foundations replaced was well chosen.

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§2.2 The Other End of the Wire

Now to the debt of §1.9: something that closes the *right-hand* end of a wire.

Begin with the plainest case. In §1.1 we read a column as two answers, to the questions "is it 0?" and "is it 1?". Picking out the first answer is itself something a matrix can do — not a square matrix this time, but a **row**, a matrix with one row and two columns. A row acts on a column by the same rule as any matrix product, along the row and down the column, multiplying in pairs and adding; only there is one row to run along, so the result has a single entry:

$$\begin{pmatrix} c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = cx + dy.$$

A matrix with one row and one column is a number, and we shall treat it as one. Now ask the general state the first question:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 1 \cdot \alpha + 0 \cdot \beta = \alpha, \qquad \begin{pmatrix} 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \beta.$$

For the arrow T the two results are $\frac{3}{5}$ and $\frac{4}{5}$. The state went in; a number

came out; and there is nothing left for any further matrix to act on. That is what closing the right-hand end means.

The rows $\begin{pmatrix}1 & 0\end{pmatrix}$ and $\begin{pmatrix}0 & 1\end{pmatrix}$ are the columns $|0\rangle$ and $|1\rangle$ laid on their sides. Could *any* state be laid on its side and used as a question — "how much of you lies along ϕ?" Let us try it, and test the attempt in the one case where we know what the answer ought to be. A state asked about *itself* should answer as emphatically as anything can. For T, laid on its side and applied to itself:

$$\begin{pmatrix}\frac{3}{5} & \frac{4}{5}\end{pmatrix}\begin{pmatrix}3/5\\4/5\end{pmatrix} = \tfrac{9}{25} + \tfrac{16}{25} = 1.$$

Good. But Chapter 1 planted a second example for exactly this moment, the state $\begin{pmatrix}3/5\\4i/5\end{pmatrix}$ with an entry that is not real. Laid on its side and applied to itself it gives $\frac{9}{25} + (\frac{4i}{5})^2 = \frac{9}{25} - \frac{16}{25} = -\frac{7}{25}$. A legal state, asked about itself, answers with a negative number smaller than one. Something is wrong with the recipe, and Chapter 1's rule says what: the rule was never about the squares of the entries. It was about their squared *moduli*.

The repair needs one piece of school algebra. The **complex conjugate** of $z = a + bi$ (with a, b real) is $\bar{z} = a - bi$. A real number is its own conjugate. Conjugating twice does nothing; conjugation passes through sums and products, $\overline{z+w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z}\,\bar{w}$; and, the fact that matters most in this chapter,

$$\bar{z}\,z \;=\; (a-bi)(a+bi) \;=\; a^2 + b^2 \;=\; |z|^2.$$

A number times its conjugate is its squared modulus: real, and never negative.

On the blackboard. For a state $|\phi\rangle = \begin{pmatrix}\gamma\\\delta\end{pmatrix}$, the **bra** $\langle\phi|$, read "bra phi", is the column laid on its side *with every entry conjugated*:

$$\langle\phi| \;=\; \begin{pmatrix}\bar\gamma & \bar\delta\end{pmatrix}.$$

The label is the same; the wrapping has been turned round, angle bracket on the left and bar on the right, and that is how the notation says "row". For real columns the conjugation is invisible: $\langle 0| = \begin{pmatrix}1 & 0\end{pmatrix}$ and $\langle 1| = \begin{pmatrix}0 & 1\end{pmatrix}$, the two rows we began with.

A bra applied to a ket is written with the two bars merged into one, $\langle\phi|\psi\rangle$, and called the **inner product** of ϕ and ψ, or simply the **bracket**:

$$\langle\phi|\psi\rangle \;=\; \begin{pmatrix}\bar\gamma & \bar\delta\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix} \;=\; \bar\gamma\,\alpha + \bar\delta\,\beta.$$

It is a number, in general a complex one. We call it the **amplitude** of ψ for ϕ: what the state ψ answers when asked ϕ. In particular

$$\langle 0|\psi\rangle = \alpha, \qquad\qquad \langle 1|\psi\rangle = \beta,$$

so the entries of a column, which Chapter 1 could not name, are its amplitudes for 0 and for 1.

Exchange the two labels and the bracket is conjugated:

$$\langle\psi|\phi\rangle \;=\; \bar\alpha\gamma + \bar\beta\delta \;=\; \overline{\alpha\bar\gamma + \beta\bar\delta} \;=\; \overline{\langle\phi|\psi\rangle}.$$

Asking ψ about ϕ and asking ϕ about ψ give mirror-twin numbers. We shall use this on almost every page.

Dirac's Blackboard.

Why the conjugate. Take $|\chi\rangle = \begin{pmatrix}3/5\\ 4i/5\end{pmatrix}$. It is a state: $|3/5|^2 + |4i/5|^2 = \frac{9}{25} + \frac{16}{25} = 1$. Its bra is $\langle\chi| = \begin{pmatrix}\frac{3}{5} & -\frac{4i}{5}\end{pmatrix}$, and

$$\langle\chi|\chi\rangle = \tfrac{3}{5}\cdot\tfrac{3}{5} + \left(-\tfrac{4i}{5}\right)\left(\tfrac{4i}{5}\right) = \tfrac{9}{25} - \tfrac{16\,i^2}{25} = \tfrac{9}{25} + \tfrac{16}{25} = 1.$$

Without the conjugate the second term is $(\frac{4i}{5})^2 = -\frac{16}{25}$ and the total is $-\frac{7}{25}$. The plain row is not wrong as a matrix; it is the wrong question.

The twin rule, on numbers. With $|T\rangle = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}$:

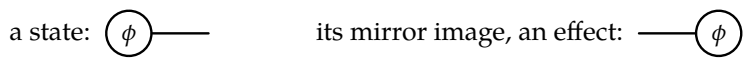
$$\langle\chi|T\rangle = \tfrac{3}{5}\cdot\tfrac{3}{5} + \left(-\tfrac{4i}{5}\right)\tfrac{4}{5} = \tfrac{9-16i}{25}, \qquad \langle T|\chi\rangle = \tfrac{3}{5}\cdot\tfrac{3}{5} + \tfrac{4}{5}\cdot\tfrac{4i}{5} = \tfrac{9+16i}{25}.$$

Conjugates, as promised: the real parts agree and the imaginary parts are opposite, which is all that conjugation ever does. Their product is $\frac{81+256}{625} = \frac{337}{625}$: real, and not negative. Keep an eye on products of that kind.

In the sketchbook. One new element, described before it is drawn.

An **effect** is a box with one wire coming *in*, on its left, and **no wire going out**. It is the mirror image of a state. Hold a mirror upright at the right of a state box: the box that had nothing on its left and a wire leaving on its right becomes a box with a wire arriving on its left and nothing on its right. A state is a way of starting; an effect is a way of ending — a question put to whatever arrives.

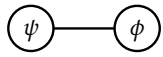
We draw an effect with the same rounded ends as a state, and the reader should be warned how to tell them apart, because the outline of the box itself is symmetric and looks the same in the mirror. **Look for the wire.** A rounded box with its wire on the *right* is a state; a rounded box with its wire on the *left* is an effect. It is the same box, met from the other side.



The effect labelled ϕ stands for the bra $\langle\phi|$. So the mirror does one thing that

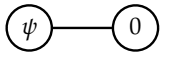
cannot be seen in the drawing: **the mirror conjugates.** The state ϕ carries the entries γ, δ in a column; its mirror image carries $\bar{\gamma}, \bar{\delta}$ in a row.

Now put a state at the left end of a wire and an effect at the right end. Nothing comes in and nothing goes out: the picture has no open end anywhere. We call it a **closed diagram**. By the rule of §1.3 a row of things on a wire is the matrix product of those things, written in the opposite order; here the things are a column and a row, and their product is a number. **A closed diagram is a number:**

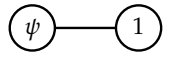


$$= \langle \phi | \psi \rangle .$$

Here is Chapter 1's loud convention again, and it deserves to be loud again. **In the picture the effect is on the right. In the bracket the bra is on the left.** The picture is read left to right — first the state ψ, then the question ϕ — and the bracket $\langle \phi | \psi \rangle$ is a matrix product, read right to left. They say the same thing. The two calculations that opened this section are now two picture equations:

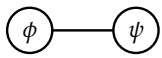


$$= \alpha$$



$$= \beta$$

And the twin rule $\langle \psi | \phi \rangle = \overline{\langle \phi | \psi \rangle}$ is what happens when the mirror is turned on a whole closed diagram: the state ψ on the left becomes the effect ψ on the right, the effect ϕ on the right becomes the state ϕ on the left, and the number is conjugated.



$$= \langle \psi | \phi \rangle = \overline{\langle \phi | \psi \rangle}$$

Every picture equation in this chapter is exact: the same number on both sides, with nothing swept aside.

Penrose's Sketchbook.

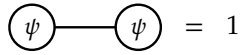
The mirror, stated carefully. Three things the mirror does, and one it does not yet do. It turns a state into an effect with the same

label, and an effect back into the state: two reflections are none, as two conjugations are none. It conjugates the entries, which no drawing shows; the label ϕ on an effect always names the *state* whose mirror image it is, so its row is $\bar{\gamma}, \bar{\delta}$ and not γ, δ. And turned on a closed diagram made of one state, one wire and one effect, it reverses the order, mirrors each end, and conjugates the number.

What it does not do, in this chapter, is reflect a *box*. We have said nothing about the mirror image of NOT, or of any process with a wire on both sides, and nothing here should be read as a rule for reflecting pictures in general. That is Chapter 3's subject.

The test that broke the plain row now passes, and not only for our two examples. It is the headline theorem of the chapter:

Theorem 2.1 (a state meets its own mirror image in the number 1). For every qubit state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $\langle\psi|\psi\rangle = 1$, exactly. In pictures: the state ψ, a wire, and the effect ψ make a closed diagram, and that closed diagram is the number 1.



Proof idea. The bra was built so that each entry meets its own conjugate, and an entry times its conjugate is its squared modulus; the squared moduli of a qubit state sum to 1 by definition. §2.8 proves it twice, in full, and draws from it a consequence that is not a restated definition. See `Code/Ch2/Closing.py` for the bra, the effect and the closed diagram, including the check that the *un*conjugated mirror fails.

Why this definition. Why should the mirror conjugate, when simply laying the column on its side is simpler? Because only the conjugated row makes a state meet itself in $|\alpha|^2 + |\beta|^2$, the one quantity Chapter 1's rule speaks about. With the plain row, the state $|\chi\rangle$ of the sidebar meets itself in $-\frac{7}{25}$, and the closed diagram of Theorem 2.1 would be 1 for some states and not for others.

§2.3 Why Squares

We have a number, and we must now say what it has to do with anything that happens.

The first guess is the natural one: the amplitudes are the chances. The arrow T answers $\frac{3}{5}$ when asked 0 and $\frac{4}{5}$ when asked 1; perhaps those are the shares of heads and tails. They cannot be. A pair of shares must make one whole, and $\frac{3}{5} + \frac{4}{5} = \frac{7}{5}$. Shares are never negative, and amplitudes may be (§2.4 has one). Shares are real numbers, and the amplitudes of $|\chi\rangle$ are not. An amplitude is not a chance of anything.

But we met, a page ago, a way of making a real, non-negative number out of any complex one: multiply it by its conjugate. And Chapter 1's rule says that for the two amplitudes of a state, the numbers made this way sum to exactly 1. They are real, they are not negative, they make one whole. They have every property a pair of shares must have.

On the blackboard. That the world actually uses them so is not something we can prove. It is a postulate — in Nielsen and Chuang's presentation (*Quantum Computation and Quantum Information*, ch. 2) it stands among the postulates of quantum mechanics — and it is called the **Born rule**:

> **The Born rule (postulate).** When a qubit in the state $|\psi\rangle$ is asked a question one of whose possible answers is b, the answer b comes with probability
>
> $$p(b) \;=\; |\langle b|\psi\rangle|^2.$$

For the moment "a question" means the one question we have, with the two answers 0 and 1; what other questions there are is the business of §2.5. Since $|z|^2 = \bar{z}z$, and since the conjugate of a bracket is the bracket with its labels exchanged (§2.2), the rule has a second spelling that we shall find more useful than the first:

$$p(b) \;=\; \overline{\langle b|\psi\rangle}\;\langle b|\psi\rangle \;=\; \langle\psi|b\rangle\;\langle b|\psi\rangle\,.$$

A probability is an amplitude times its mirror twin. For the general state the two values are $|\alpha|^2$ and $|\beta|^2$. For the arrow:

$$p(0) = \left(\tfrac{3}{5}\right)^2 = \tfrac{9}{25}, \qquad p(1) = \left(\tfrac{4}{5}\right)^2 = \tfrac{16}{25}, \qquad \tfrac{9}{25} + \tfrac{16}{25} = 1.$$

Look at what has been produced. The column $\begin{pmatrix} 9/25 \\ 16/25 \end{pmatrix}$ has real entries, not negative, summing to 1: it is a *stochastic vector*, a covered coin in the exact sense of §1.4. So the Born rule is a bridge between the two rules of Chapter 1's table. It takes a qubit state **and a question**, and hands back a wager. The normalization $|\alpha|^2 + |\beta|^2 = 1$, which Chapter 1 could only state, now reads: *the chances make one whole.*

In the sketchbook. The second spelling of the rule is a product of two brackets, and each bracket is a closed diagram. We need to say what two closed diagrams *in a row* mean, and the answer is already in our hands. A row is a matrix product whatever the shapes of the matrices in it (§1.3); in the row "state ψ, effect b, state b, effect ψ" the product, read right to left, is $\langle\psi|\;|b\rangle\;\langle b|\;|\psi\rangle$, and since matrix multiplication may be grouped as we please, this is $(\langle\psi|\,|b\rangle)(\langle b|\,|\psi\rangle)$: the product of two numbers. So **closed diagrams in a row multiply.** This is a consequence of the row rule, worked out on the blackboard, and not a new rule of drawing; in particular, where an effect is followed by a state, *no wire joins them*. The first wire has ended. The second has not yet begun.

Here, then, is the Born rule as a picture. We shall meet it on most of the remaining pages, so it gets a name: ***The Echo***.



$$=\; \langle\psi|b\rangle\;\langle b|\psi\rangle \;=\; |\,\langle b|\psi\rangle\,|^2$$

Read it from the left, looking for the wires as §2.2 advised. The state ψ, with its wire on the right; that wire ends in the effect b. Then a gap: bare paper, no wire. Then the state b, whose wire ends in the effect ψ. The left half says

"ψ, asked b", and its number is the amplitude $\langle b|\psi\rangle$. The right half is the left half seen in the mirror, every piece reflected and the order reversed, and its number is the conjugate, $\langle\psi|b\rangle$. The call and its echo; and the chance is the two together.

This is why a Born value is real and never negative, in the sketchbook's terms: it is a closed diagram followed by its own mirror image, a number next to its twin. The amplitude alone may have a sign, or an i in it. The amplitude *with its echo* cannot.

Let us be plain about what has not been claimed. The picture does not explain why nature computes chances this way, and it does not derive the rule; the rule is a postulate, on the blackboard and in the sketchbook alike. What the picture does is *locate* the square. The question "why squares?" with which Chapter 1 ended has become a question one can point at: *why is there a second wire?* We shall say in §2.8 how much of an answer this chapter can honestly give. See `Code/Ch2/Squares.py`.

Why this definition. Why $\bar{z}z$, and not the amplitude itself, or its plain square z^2? The amplitude itself fails every test for a share, as we saw. The plain square fails for the same reason the plain row failed in §2.2: for $|\chi\rangle$ it would give $p(1) = (\frac{4i}{5})^2 = -\frac{16}{25}$, a negative chance. The product of a number with its conjugate is the simplest expression that is real and non-negative for *every* amplitude.

> **Feynman's Notebook.**
>
> *How a probability is met in a laboratory.* Nobody has ever read an amplitude off an instrument. What a laboratory can do is prepare a system in the same way a great many times, put the same two-answer question to it each time — Nielsen and Chuang call this step a measurement, and give it a postulate of its own in their second chapter — and count: so many runs answered one way, so many the other. A single run yields a single answer, never a number like $\frac{3}{5}$, and certainly never one with a minus sign or an i in it.
>
> The Born rule is a claim about the counts: over a long series

> of runs, the fraction answering b settles near $|\langle b|\psi\rangle|^2$. It is a postulate, not a deduction from anything more primitive, and it is kept because the counting agrees with it. That is the whole of its authority, and it is a great deal. The amplitudes belong to the night, when nothing is looked at; the counts are what is there in the morning.

---

§2.4 A Sign You Cannot See

Now, at last, the negative entry that Chapter 1 mentioned and set aside.

On the blackboard. Two states get names of their own, a plus sign and a minus sign inside a ket:

$$|+\rangle \;=\; \tfrac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad\qquad |-\rangle \;=\; \tfrac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}.$$

Both are legal: $(\frac{1}{\sqrt{2}})^2 + (\pm\frac{1}{\sqrt{2}})^2 = \frac{1}{2} + \frac{1}{2} = 1$. As arrows on paper, $|+\rangle$ reaches equally toward 0 and toward 1; $|-\rangle$ reaches toward 0 and *backwards*, away from 1. The amplitude $\langle 1|-\rangle = -\frac{1}{\sqrt{2}}$ is a reach of less than nothing.

Ask each of them the standard question:

$$|+\rangle: \qquad p(0) = |\tfrac{1}{\sqrt{2}}|^2 = \tfrac{1}{2}, \qquad p(1) = |\tfrac{1}{\sqrt{2}}|^2 = \tfrac{1}{2};$$

$$|-\rangle: \qquad p(0) = |\tfrac{1}{\sqrt{2}}|^2 = \tfrac{1}{2}, \qquad p(1) = |-\tfrac{1}{\sqrt{2}}|^2 = \tfrac{1}{2}.$$

The same wager from both: the fair covered coin of §1.4. The square has eaten the sign. And this is general. The states $\alpha|0\rangle + \beta|1\rangle$ and $\alpha|0\rangle - \beta|1\rangle$ give the standard Born values $|\alpha|^2$ and $|\pm\beta|^2 = |\beta|^2$ alike.

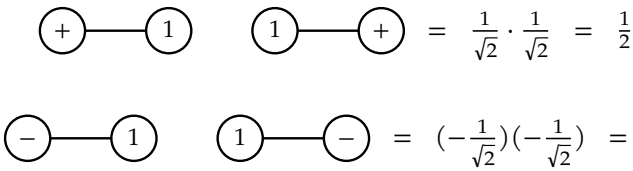
Yet they are different columns — and not slightly different. Ask one about the other:

$$\langle +|-\rangle \;=\; \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}(1 \cdot 1 + 1 \cdot (-1)) \;=\; \tfrac{1}{2}\,(1-1) \;=\; 0,$$

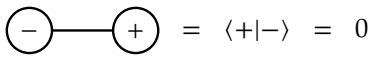
whereas $\langle +|+\rangle = \frac{1}{2}\,(1+1) = 1$.

For real arrows a bracket equal to 0 has a familiar meaning: the arrows are **perpendicular**; neither has any reach along the other. Draw $(1, 1)$ and $(1, -1)$ on squared paper and they stand at right angles. We take the word over for all states: $|\phi\rangle$ and $|\psi\rangle$ are perpendicular when $\langle\phi|\psi\rangle = 0$. By that test $|+\rangle$ and $|-\rangle$ are as different as two states can be, exactly as different as $|0\rangle$ and $|1\rangle$, for which $\langle 0|1\rangle = 1 \cdot 0 + 0 \cdot 1 = 0$ too. (Chapter 5 will read a zero like this one in quite another way; here it is plain geometry.)

In the sketchbook. No new element. Two *Echo* pictures with different labels in their outer boxes, and the same number:



In the second picture each half is negative, and the echo repeats the sign faithfully; the product cannot show it. And one closed diagram whose number is zero:



A trap

Here is the tempting conclusion. Everything that anyone will ever *count* comes through the Born rule; the Born rule gives $|+\rangle$ and $|-\rangle$ the same wager; so the minus sign is bookkeeping, a mark on the blackboard with no echo in the world, and the two states might as well be called one.

Every step of that is sound except the word "everything". We have put *one question* to these two states. §1.4 taught that one state is not enough to tell

two boxes apart. It would be rash to assume that one question is enough to tell two states apart — and the bracket $\langle +|-\rangle = 0$ is a warning that it is not. See `Code/Ch2/HiddenSign.py`.

---

§2.5 The Turned Question

What would another question be? The standard question has two answers, 0 and 1, and each answer is an effect: the mirror image of a state. Nothing in the Born rule singles those two states out. What we need is to say which *pairs* of states can serve as the two answers of a question.

On the blackboard. An **orthonormal basis** is a pair of states $|b_0\rangle, |b_1\rangle$ that are perpendicular to one another:

$$\langle b_0|b_0\rangle = 1, \qquad \langle b_1|b_1\rangle = 1, \qquad \langle b_0|b_1\rangle = 0$$

(and then $\langle b_1|b_0\rangle = \bar{0} = 0$ as well). The first two conditions say only that both are states; the third is the new one. The pair $|0\rangle, |1\rangle$ is an orthonormal basis, the **standard basis**. By the calculation of §2.4, so is the pair $|+\rangle, |-\rangle$. On paper it is the standard pair of directions turned through half a right angle.

From now on **a question is an orthonormal basis**, its two answers are the two members, and the Born rule of §2.3 is complete as it stands: asked the question $\{b_0, b_1\}$, the state $|\psi\rangle$ answers b_0 with probability $|\langle b_0|\psi\rangle|^2$ and b_1 with probability $|\langle b_1|\psi\rangle|^2$.

Put the turned question to the two states of the trap:

$$|+\rangle: \qquad p(+) = |\langle +|+\rangle|^2 = 1, \qquad p(-) = |\langle -|+\rangle|^2 = 0;$$

$$|-\rangle: \qquad p(+) = |\langle +|-\rangle|^2 = 0, \qquad p(-) = |\langle -|-\rangle|^2 = 1.$$

Certainty, and opposite certainties. The two states that one question could not tell apart at all, another tells apart every single time. The sign was never bookkeeping. It was the whole difference, waiting for the right question.

The arrow T under the turned question gives two less tidy numbers, worth having:

$$\langle +|T\rangle = \tfrac{1}{\sqrt{2}}(\tfrac{3}{5}+\tfrac{4}{5}) = \tfrac{7}{5\sqrt{2}}, \qquad \langle -|T\rangle = \tfrac{1}{\sqrt{2}}(\tfrac{3}{5}-\tfrac{4}{5}) = -\tfrac{1}{5\sqrt{2}},$$

$$p(+) = \tfrac{49}{50}, \qquad p(-) = \tfrac{1}{50}.$$

All but a sliver — and the sliver comes from a negative amplitude. The conjugate has its turned question too: T and the non-real state $|\chi\rangle$ of §2.2 give the same standard values $\frac{9}{25}, \frac{16}{25}$, but $\langle +|\chi\rangle = \frac{3+4i}{5\sqrt{2}}$, whose squared modulus is $\frac{9+16}{50} = \frac{1}{2}$, against T's $\frac{49}{50}$.

The word *basis* makes a promise, which we now keep: every state can be written in terms of the two members, and the coefficients are its amplitudes.

> **Lemma 2.3 (expansion in an orthonormal basis).** If $|b_0\rangle, |b_1\rangle$ is an orthonormal basis, then for every state $|\psi\rangle$
>
> $$|\psi\rangle \;=\; \langle b_0|\psi\rangle\,|b_0\rangle + \langle b_1|\psi\rangle\,|b_1\rangle\,.$$

For the standard basis this is the first display of §2.1, since $\langle 0|\psi\rangle = \alpha$ and $\langle 1|\psi\rangle = \beta$. For the turned basis and the arrow T it is checked by hand in the sidebar below. *Proof idea:* subtract from $|\psi\rangle$ its reach along each member; what is left over has no reach along either, and in a space with only two places that leaves nothing. *Proof sketch.* Write $c_0 = \langle b_0|\psi\rangle$, $c_1 = \langle b_1|\psi\rangle$, and let r be the leftover column $|\psi\rangle - c_0\,|b_0\rangle - c_1\,|b_1\rangle$. A row times a sum of columns is the sum of the products, so $\langle b_0|$ applied to r gives $c_0 - c_0\cdot 1 - c_1\cdot 0 = 0$, and likewise $\langle b_1|$ gives 0. Stack the two bras into a 2×2 matrix B, and set the two kets side by side as the columns of a 2×2 matrix K. The four entries of BK are the four brackets $\langle b_i|b_j\rangle$, so orthonormality says precisely $BK = I$. For square matrices

a one-sided inverse is two-sided, so $KB = I$ too, and $r = K(Br) = K\begin{pmatrix}0\\0\end{pmatrix}$ is the zero column. ■ See `Code/Ch2/TurnedQuestion.py`.

Dirac's Blackboard.

The turned basis, by brute force. Orthonormal: $\langle +|+\rangle = \frac{1}{2}(1+1) = 1$, $\langle -|-\rangle = \frac{1}{2}(1+1) = 1$, $\langle +|-\rangle = \frac{1}{2}(1-1) = 0$.

The expansion of T. With $c_+ = \frac{7}{5\sqrt{2}}$ and $c_- = -\frac{1}{5\sqrt{2}}$,

$$c_+ |+\rangle + c_- |-\rangle = \tfrac{7}{10}\begin{pmatrix}1\\1\end{pmatrix} - \tfrac{1}{10}\begin{pmatrix}1\\-1\end{pmatrix} = \tfrac{1}{10}\begin{pmatrix}6\\8\end{pmatrix} = \begin{pmatrix}3/5\\4/5\end{pmatrix}.$$

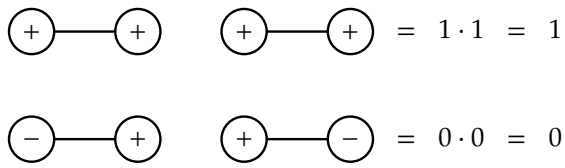
The Born values. $c_+^2 = \frac{49}{50}$, $c_-^2 = \frac{1}{50}$, and

$$\tfrac{49}{50} + \tfrac{1}{50} = 1.$$

Both amplitudes are real, so each squared modulus is a plain square, and the minus sign on c_- disappears in it.

Compare the standard question: $\frac{9}{25} + \frac{16}{25} = 1$. Two different wagers from one state, and each makes one whole. Nothing in the definition of a state mentioned the turned basis. Why should the total come out right there as well? That is Corollary 2.2, in §2.8.

In the sketchbook. No new element, and that is the point worth drawing. The effects labelled + and − are mirrored states like any others, and *The Echo* does not care which labels it carries:



Set these beside the two pictures of §2.4. There the outer labels + and − made no difference to the number; here they make all the difference there can be.

The expansion of Lemma 2.3 stays on the blackboard. It is a *sum* of two columns, and the sketchbook does not yet have a way to add two pictures that have open wires; we shall meet that tool in Chapter 5. What the sketchbook may do is add *numbers*, and a closed diagram is a number. We shall use that liberty once, in §2.8, and say so when we do.

Why this definition. Why must the answers of a question be perpendicular states, rather than any two different states? Because answers must not be confusable. If the qubit *is* in the state $|b_0\rangle$, the question ought to return b_0 with certainty and b_1 never, and the Born rule makes those two demands into $|\langle b_0|b_0\rangle|^2 = 1$ and $|\langle b_1|b_0\rangle|^2 = 0$. That is orthonormality, no more and no less.

---

§2.6 The Arrow Reversed

One might now over-learn the lesson: every sign is caught by *some* question, if the question is turned far enough. Let us test that on the boldest sign of all, the one in front of the whole column.

Reverse the arrow end for end: $-|T\rangle = \begin{pmatrix} -3/5 \\ -4/5 \end{pmatrix}$. It is a state, since squaring removes the signs, and it is a different column from $|T\rangle$. Every amplitude changes sign: $-\frac{3}{5}$ and $-\frac{4}{5}$ for the standard question, $-\frac{7}{5\sqrt{2}}$ and $+\frac{1}{5\sqrt{2}}$ for the turned one. And every Born value stays where it was: $\frac{9}{25}, \frac{16}{25}; \frac{49}{50}, \frac{1}{50}$. Turn the question to any position you like, and the same will happen, because the bracket of *any* bra with $-|T\rangle$ is minus its bracket with $|T\rangle$, and the square does not see the minus.

On the blackboard. The general form of this needs the complex numbers of modulus 1. For a real number θ, write $e^{i\theta} = \cos\theta + i\sin\theta$. Its squared modulus is $\cos^2\theta + \sin^2\theta = 1$; its conjugate is $\cos\theta - i\sin\theta = e^{-i\theta}$; and

$e^{i\theta}e^{-i\theta} = 1$. With $\theta = \pi$ it is -1. Such a number is called a **phase**, and when it multiplies a whole column, a **global phase**.

> **Proposition 2.4 (a global phase is seen by no question).** Let $|\psi\rangle$ be a state, θ real, and $|\psi'\rangle = e^{i\theta}|\psi\rangle$. Then $|\psi'\rangle$ is a state, it is a different column from $|\psi\rangle$ whenever $e^{i\theta} \neq 1$, and for **every** state $|b\rangle$
>
> $$|\langle b|\psi'\rangle|^2 \;=\; |\langle b|\psi\rangle|^2.$$

Proof idea. The factor comes out of the amplitude as $e^{i\theta}$ and out of its mirror twin as $e^{-i\theta}$, because the mirror conjugates; and the two meet in the product.
Proof sketch. The entries of $|\psi'\rangle$ are $e^{i\theta}\alpha$ and $e^{i\theta}\beta$, with squared moduli $|\alpha|^2$ and $|\beta|^2$, so $|\psi'\rangle$ is a state. A row applied to a multiple of a column is the multiple of the result: $\langle b|\psi'\rangle = e^{i\theta}\langle b|\psi\rangle$. The bra of $|\psi'\rangle$ has the conjugated entries $e^{-i\theta}\bar{\alpha}$, $e^{-i\theta}\bar{\beta}$, that is, $\langle\psi'| = e^{-i\theta}\langle\psi|$, so $\langle\psi'|b\rangle = e^{-i\theta}\langle\psi|b\rangle$. Therefore

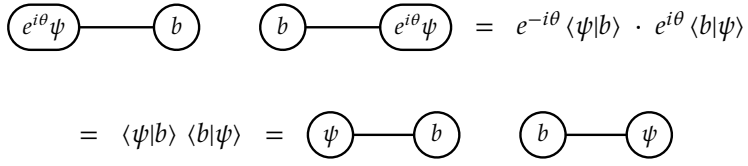
$$\langle\psi'|b\rangle\,\langle b|\psi'\rangle \;=\; e^{-i\theta}\,e^{i\theta}\;\langle\psi|b\rangle\,\langle b|\psi\rangle \;=\; \langle\psi|b\rangle\,\langle b|\psi\rangle\,. \qquad \blacksquare$$

See `Code/Ch2/Reversed.py`, which tries effects drawn at random as well as our own.

In the sketchbook. A number may multiply a state picture, as it may multiply a column: $e^{i\theta}$ times the state ψ is the state box labelled $e^{i\theta}\psi$, and its mirror image is the effect with the same label, whose row is $e^{-i\theta}\langle\psi|$. The two state pictures are different,



and their two *Echo* pictures are the same number, for every label b:



The factor leaves the left half as $e^{i\theta}$ and the right half as $e^{-i\theta}$. Whatever the call gains, the echo gives back.

We must be careful about what this says. It does **not** say that $e^{i\theta}|\psi\rangle$ and $|\psi\rangle$ are the same state, or the same picture, in some loosened sense of "same". They are different columns and different pictures; the closed diagram of one with an effect b is a different number from that of the other. What the proposition says is that their *Born values* agree for every effect, so that no question of the kind we have defined can tell them apart.

Set this beside §2.4. A sign — or any phase other than 1 — on *one* entry of a column whose entries are both non-zero makes a different state, and some question gives the two different chances (for $|+\rangle$ and $|-\rangle$, opposite certainties). The same phase on *both* entries makes a different column that no question detects at all. The difference between those two cases is the most important thing to carry out of this chapter. (We have shown it for every question asked directly of the state. That it survives whatever boxes are placed on the wire first is true as well, and belongs to Chapter 3.)

---

§2.7 The Rosetta Table

This chapter adds eight rows to the running dictionary, numbered on from Chapter 1's seven. Rows 8 and 9 are new spellings, on the algebra side only, of Chapter 1's third and fourth rows. Rows 10 and 11 introduce the chapter's two new picture elements, the effect and the closed diagram. The remaining four are things those elements say.

#	Algebra (Dirac)	Picture (Penrose)
8	$\|\psi\rangle = \alpha\|0\rangle + \beta\|1\rangle$, the ket spelling of a column	the state box labelled ψ (no new element)
9	$\|0\rangle$, $\|1\rangle$	the state boxes labelled 0 and 1 (no new element)

#	Algebra (Dirac)	Picture (Penrose)
10	the bra $\langle\phi\vert = \begin{pmatrix}\bar{\gamma} & \bar{\delta}\end{pmatrix}$	an **effect**: a rounded box with its wire on the left and no output, the mirror image of the state ϕ (the mirror conjugates); *new element*
11	the bracket $\langle\phi\vert\psi\rangle$, bra on the left	state ψ, wire, effect ϕ, effect on the right: a **closed diagram**, which is a number — *new element*
12	a product of numbers $\langle\chi\vert\omega\rangle\ \langle\phi\vert\psi\rangle$	closed diagrams in a row (the row rule, with no open end)
13	the Born rule $\vert\langle b\vert\psi\rangle\vert^2 = \langle\psi\vert b\rangle\ \langle b\vert\psi\rangle$	***The Echo***: a closed diagram followed by its mirror image
14	$\langle\psi\vert\psi\rangle = 1$	a state meeting its own mirror image = 1
15	$e^{i\theta}\vert\psi\rangle \neq \vert\psi\rangle$, yet $\vert\langle b\vert e^{i\theta}\psi\rangle\vert^2 = \vert\langle b\vert\psi\rangle\vert^2$ for every b	two different state pictures; the same *Echo* number for every effect

All eight rows are exact. In row 12 the labels χ, ω stand for any two further states; in row 15 the label $e^{i\theta}\psi$ inside a bracket stands for the state $e^{i\theta}\vert\psi\rangle$.

---

§2.8 The Theorem, Twice

We have met the theorem in examples three times: $\vert T\rangle$ met itself in $\frac{9}{25} + \frac{16}{25} = 1$ (§2.2); so did the non-real $\vert\chi\rangle$, once its row was conjugated; so did $\vert+\rangle$ in §2.4. Here is the statement for all states at once.

Theorem 2.1 (a state meets its own mirror image in the number 1). Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ be a qubit state: α, β complex with $|\alpha|^2 + |\beta|^2 = 1$. Let $\langle\psi| = \begin{pmatrix}\bar{\alpha} & \bar{\beta}\end{pmatrix}$ be its bra, the conjugated row. Then

$$\langle\psi|\psi\rangle \;=\; 1,$$

exactly. In pictures: the state ψ, a wire, and the effect ψ form a closed diagram, and that closed diagram is the number 1.

Dirac's Blackboard

Write out the row and the column and multiply, along the row and down the column:

$$\langle\psi|\psi\rangle \;=\; \begin{pmatrix}\bar{\alpha} & \bar{\beta}\end{pmatrix}\begin{pmatrix}\alpha \\ \beta\end{pmatrix} \;=\; \bar{\alpha}\,\alpha + \bar{\beta}\,\beta.$$

For any complex number $z = a + bi$, $\bar{z}z = (a - bi)(a + bi) = a^2 + b^2 = |z|^2$ (§2.2). Apply this to each term:

$$\bar{\alpha}\,\alpha + \bar{\beta}\,\beta \;=\; |\alpha|^2 + |\beta|^2.$$

And $|\psi\rangle$ is a qubit state, which by the definition of §1.5 means

$$|\alpha|^2 + |\beta|^2 \;=\; 1.$$

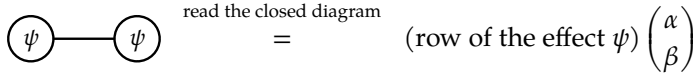
One row times one column, two products, one appeal to the definition. ■

Penrose's Sketchbook

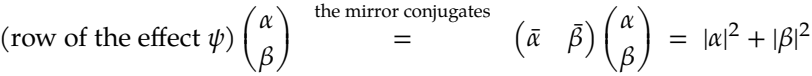
The sketchbook owns three things that bear on this picture, and the proof uses each once. We name them. *Read the closed diagram* (row 11 of the table): a state, a wire and an effect are the bracket of the effect's row with the state's

column. *The mirror conjugates* (row 10): the effect labelled ψ carries the row $\bar{\alpha}, \bar{\beta}$. *Chapter 1's rule*: the label ψ in a state box stands for a column whose squared moduli sum to 1.

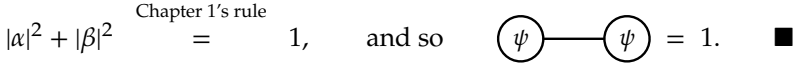
Step 1 — *read the closed diagram*. The picture has no open end, so it is a number: the row of the effect ψ times the column of the state ψ.



Step 2 — *the mirror conjugates*. The effect ψ is the mirror image of the state ψ, so its row is the conjugated one.



Step 3 — *Chapter 1's rule*. The box on the left is a state box, so the sum of squared moduli is 1.



■

Notice what Step 2 guards against. Had the mirror *not* conjugated, the middle expression would have been $\alpha^2 + \beta^2$, and for $|\chi\rangle$ the picture would have come to $-\frac{7}{25}$.

What the theorem is for

Taken alone, Theorem 2.1 is Chapter 1's definition in a new dress, and we should not pretend otherwise. Its worth is in what follows from it. The sidebar of §2.5 found that the arrow's Born values make one whole under the standard question, $\frac{9}{25} + \frac{16}{25}$, and again under the turned one, $\frac{49}{50} + \frac{1}{50}$ — though nothing in the definition of a state mentions the turned basis. That cannot be left as a coincidence. If the Born rule is to hand back an honest

wager for *every* question, the total must be 1 for every orthonormal basis, and it is.

> **Corollary 2.2 (the chances make one whole, for every question).** For every state $|\psi\rangle$ and every orthonormal basis $|b_0\rangle, |b_1\rangle$,
>
> $$|\langle b_0|\psi\rangle|^2 + |\langle b_1|\psi\rangle|^2 \;=\; \langle\psi|\psi\rangle \;=\; 1.$$

Proof idea. Expand the ket inside $\langle\psi|\psi\rangle$ in the basis, by Lemma 2.3; the bracket breaks into two terms, and each term is an amplitude times its mirror twin. *Proof.* By Lemma 2.3, $|\psi\rangle = \langle b_0|\psi\rangle\,|b_0\rangle + \langle b_1|\psi\rangle\,|b_1\rangle$. Apply the row $\langle\psi|$ to both sides; a row times a sum of columns is the sum of the products, and numbers may be moved to the front:

$$\langle\psi|\psi\rangle \;=\; \langle b_0|\psi\rangle\,\langle\psi|b_0\rangle + \langle b_1|\psi\rangle\,\langle\psi|b_1\rangle \;=\; |\langle b_0|\psi\rangle|^2 + |\langle b_1|\psi\rangle|^2,$$

using $\langle\psi|b\rangle = \overline{\langle b|\psi\rangle}$ in the last step. The left side is 1 by Theorem 2.1. ■ See `Code/Ch2/WholeSquare.py`, which checks the theorem, the failure of the unconjugated mirror, and the corollary for the standard basis, the turned basis and a basis drawn at random.

This proof is the blackboard's alone, and we say so plainly. Its one real step is the expansion, a sum of two kets, which the sketchbook cannot yet draw. What the sketchbook can do is state the result, as arithmetic on pictures already evaluated: the number of *The Echo* with $b = b_0$, plus the number of *The Echo* with $b = b_1$, equals the number of the closed diagram of Theorem 2.1. Three pictures, three numbers, and one sum of numbers borrowed from the blackboard. It is not a rewrite.

Here, too, is the honest answer to the question Chapter 1 left us with, *why the rule must use squares*. We have not shown that nature had no other choice; the Born rule is a postulate. What we have shown is narrower and still worth having. The amplitudes themselves cannot be shares (§2.3). Their plain squares cannot either. The products $\bar{z}z$ can: they are real and never negative, and — this is the corollary — they make exactly one whole not for one favoured question but for every question there is, and for no deeper

reason than the rule $|\alpha|^2 + |\beta|^2 = 1$ by which a qubit state was defined. The definition of Chapter 1 and the postulate of this chapter were made for each other. That is the sense, and the only sense, in which the rule "must" use squares.

The verdict

A draw, and a modest one; if anything, the algebra has it. Both proofs unfold the same definition. The blackboard does so in one line. The sketchbook's three steps are that same line, reached by first reading the picture; it owns no rewrite rule that *produces* the number 1, and its Step 3 is Chapter 1's blackboard rule under another name.

What the picture gives is not a proof but a place. On the blackboard the normalization is a rule about the two entries of a column, and so about the standard basis. In the sketchbook it is one closed picture with a ψ at each end and **no basis in it anywhere** — which is the right way to hold it if one wants to see why, by Corollary 2.2, the total is one whole for every question. The picture locates the square in the second wire of *The Echo*. It does not explain the square, and it does not derive the Born rule.

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§2.9 The Echo

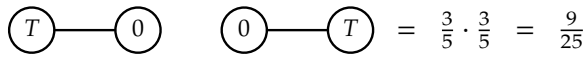
Here is the chapter's master diagram, first drawn in §2.3. We call it **The Echo**. It is a row of four rounded boxes on one line: the state ψ, whose wire runs to the effect b; then bare paper; then the state b, whose wire runs to the effect ψ. Each of its wires is closed at both ends, and the right half is the left half in the mirror. Its evaluation is the Born rule:



The equation is exact. The wire on the left contributes the amplitude $\langle b|\psi\rangle$; the wire on the right contributes its conjugate $\langle\psi|b\rangle$; nothing joins the effect b to the state b, and the two numbers multiply. We prove nothing again.

Instead, as with Chapter 1's master diagram, we *read* the one picture three times, changing only the labels.

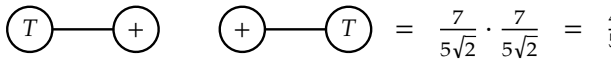
First reading: the standard question. Take $\psi = T$ and $b = 0$. Each half is $\frac{3}{5}$.



$$= \frac{3}{5} \cdot \frac{3}{5} = \frac{9}{25}$$

With $b = 1$ it is $\frac{4}{5} \cdot \frac{4}{5} = \frac{16}{25}$, and the two numbers are the covered coin $\begin{pmatrix} 9/25 \\ 16/25 \end{pmatrix}$ of §2.3.

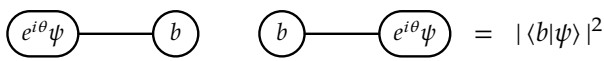
Second reading: the turned question. The same state, with $b = +$. Each half is $\frac{7}{5\sqrt{2}}$.



$$= \frac{7}{5\sqrt{2}} \cdot \frac{7}{5\sqrt{2}} = \frac{49}{50}$$

With $b = -$ each half is $-\frac{1}{5\sqrt{2}}$, negative on both wires, and the product is $\frac{1}{50}$. One state, a different question, a different coin: $\begin{pmatrix} 49/50 \\ 1/50 \end{pmatrix}$. For real states such as T the two halves are equal, sign and all. For a state with non-real entries they are conjugates and in general not equal: with $\psi = \chi$ and $b = +$ the left half is $\frac{3+4i}{5\sqrt{2}}$, the right half $\frac{3-4i}{5\sqrt{2}}$, and the product $\frac{25}{50} = \frac{1}{2}$.

Third reading: the arrow reversed. Replace ψ by $e^{i\theta}\psi$ and leave b alone.



$$= |\langle b|\psi\rangle|^2$$

A different picture from the one at the head of the section — two of its four labels have changed — and the same number, for every b (Proposition 2.4). With $\theta = \pi$ and $\psi = T$ this is the reversed arrow of §2.6: each half of the first reading becomes $-\frac{3}{5}$, and the product is still $\frac{9}{25}$.

There is a fourth reading, which we have already had: put $b = \psi$, and each half is the closed diagram of Theorem 2.1, so that *The Echo* is $1 \cdot 1 = 1$. A state asked whether it is itself says yes, every time.

One picture; the question changed, the state changed, and each time the square was found in the same place. See `Code/Ch2/Squares.py` for the picture and its first reading, `Code/Ch2/TurnedQuestion.py` for the second, and `Code/Ch2/Reversed.py` for the third.

Penrose's Sketchbook.

What the picture does not say. The Echo says where the square is. It does not say why the world counts that way, and no amount of looking at it will.

The idea that a picture with no open wire is a number has a precise home. Abramsky and Coecke (2004, §6) call the maps from the trivial system to itself *scalars*, and note that for finite-dimensional vector spaces these are just the elements of the field; "no open wire" is our gloss on that, not their wording. They go on to interpret Dirac's notation in their setting (§7.2), to observe that on the scalars of finite-dimensional Hilbert spaces the operation that turns a map round is complex conjugation (§7.3), and to write the probability of an outcome as a scalar composed with its turned-round twin, a scalar they say can be thought of as the "probability amplitude" (§8). In their framework the Born rule emerges from the axioms. In this chapter it is a postulate, and the picture is a place to keep it.

---

§2.10 What May Sit on the Wire

Let us gather what we have. A ket is a name for a column, and a bra is the same name on the conjugated row (§2.1, §2.2). In the sketchbook the bra is an effect, the mirror image of a state, and the mirror conjugates; a state and an effect on one wire make a closed diagram, which is a number, an amplitude (§2.2). An amplitude is not a chance. The chance is the amplitude times its mirror twin, by a postulate that the picture locates and does not explain (§2.3). A sign on one entry of a column is invisible to one question and visible to another — decisive, for $|+\rangle$ against $|-\rangle$ (§2.4, §2.5);

a phase on the whole column is a different column that no question can detect (§2.6). And a state meets its own mirror image in the number 1, from which it follows that the chances make one whole whatever is asked (§2.8). Through all of it, one picture with two labels to change (§2.9).

Now look at the pictures once more and notice what none of them has. Every wire in this chapter runs *bare* from its state to its effect. Chapter 1 hung boxes on wires and never closed them; this chapter closed wires and never hung a box on them. Put the two together and a question presents itself at once. Place a box between the state ψ and the effect ψ, and then the same between the pieces of the echo: is the closed picture still 1? Do the chances still make one whole? For NOT, which only exchanges the two entries, one can check that they do. For SET-0 we already know they do not: §1.5 watched it turn the arrow into $\begin{pmatrix} 7/5 \\ 0 \end{pmatrix}$, which meets its own mirror image in $\frac{49}{25}$.

So some boxes keep the promise of Theorem 2.1 and some break it, and a wire that carries a qubit can bear only the first kind. To say which they are we shall need something this chapter was careful not to provide: the mirror image of a *box*. We have reflected states into effects and left every process alone. What is the mirror image of a box, and which boxes keep a closed picture equal to 1? We leave both questions open here, with the wire bare and the coin still covered. The next chapter turns the mirror on the boxes themselves.

Ländler on a Sphere

足跡を
踏んで戻るや
深雪道

ashiato o
funde modoru ya
miyukimichi

Deep snow on the path —
going back, a foot set down
in each waiting print.

[*The valley has been white for a day and a night, and the snow lies now, folded in drifts over the one narrow path to the Crab's door. It is late; the cold has teeth. A single track of prints goes up that path — the Tortoise ahead, Achilles set down in her prints behind — and nothing else has passed all evening. At the end of it, one curtained window holds a low, unhurried glow, the only warm thing in all the dark.*]

[*Deep winter; a lamp-lit evening; the snow lies. Achilles and the Tortoise stand on the Crab's doorstep. The door-knob turns.*]

Achilles: When one watches a turn from the other side, which way round does it go?

Tortoise: You are looking at the door-knob, Achilles.

Achilles: It turned by itself. Someone has hold of it on the other side.

[*The door stands open, and the Crab in it.*]

Crab: Good evening, good evening! Mind the step; it is the same step either way.

Tortoise: Good evening, Mr Crab.

Crab: One set of footprints between you, I hope. My path is narrow.

Achilles: The Tortoise goes first and I go in her prints. It is slow, but it is tidy.

Tortoise: Shall we, Achilles?

[*Over the threshold.*]

Crab: In, in, both of you, and let the warm keep what it has. Coats on the hook that isn't the clock — everyone tries the clock, and then the hour is wrong until midnight.

[*They come in stamping, and the cold comes off them in a grey breath the room swallows. The Tortoise lays her wet mittens on the fender to steam, the one pulled off last dropped on top of the other. In the grate a new fire has only just caught, spitting and ducking, not yet sure of its draught; and over it all the clock knocks out its patient three.*]

Achilles: [*warming his hands*] I had forgotten there was such a thing as warm. The path did its best to keep us out of doors.

Tortoise: A short path, well kept. Is that your grandmother's clock still, Mr Crab, keeping its three-time?

Crab: It ticks in threes and always has; I gave up correcting it years ago. There is tea in the brown pot, and something warmer in the other if the evening runs long.

Achilles: Tea, before anything. My hands have stopped believing in themselves.

[*The room is low and warm and full of the fire's moving light. On the wall beside the dresser hangs a print the Crab has had for years — Escher's* Hand with Reflecting Sphere*: a hand holds up a mirrored ball, and in the ball the man himself, and the*

whole curved room bending round him to the corners. The lamp lays a soft sheen along it.]

First Challenge — *One Turn and Its Mirror*

[*Indoors. On the Crab's table, on a felt-lined ring on a low pedestal, rests a ball the size of an orange, of dull lamp-blacked brass.*]

Crab: There. From a binnacle-maker's widow in Bergen, who had no notion what she was selling. Go on, Achilles: ask me why it doesn't shine.

Achilles: Why doesn't it shine? A brass ball ought to.

Crab: Lamp-black, and meant. It is not there to be looked at; it is there to be turned. Now — [*he touches a small bead on top of the ball*] — the glass needle. It runs right through and out underneath, down the hollow of the stand, with a bead on this end only. At rest it stands upright, bead on top.

Achilles: And that one, pointing at my waistcoat?

Crab: The steel needle. Level, through the middle, bead towards the company — by which I mean the two of you; I am furniture. It passes the glass one by a hair's breadth and touches nothing. I have lain awake over that hair's breadth.

Tortoise: And the one between, Mr Crab?

Crab: The brass needle, exactly midway between the others: bead up and towards the company. We shan't want it yet. Glass bead up, steel bead towards you — that is *home*, and the brass one follows of itself.

Achilles: And inside?

Crab: Something inside keeps its bearing, however the case is turned — as a compass card does in a turned binnacle. The beads tell only how the *case* has been turned.

Achilles: Then open it, and see how the inside stands!

Crab: [*a claw over a small shutter in the ball's side*] There is a shutter. Open it once and look, and the instrument is spoiled for good. I have never opened

it. Eleven winters, Achilles.

Tortoise: How does one turn it, Mr Crab?

Crab: One way only. Lift the ball clear of the stand; take a needle by its two ends and hold it still; and with your thumbs turn the ball about the needle, as a bead turns on a wire. The other needles go round with the ball. My grandmother danced the ländler so: the couple turn, and nobody lets go.

Achilles: How far?

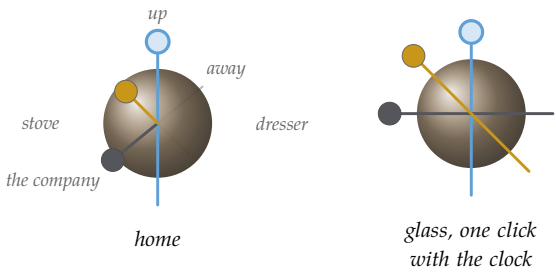
Crab: The collar where each needle goes in has four notches: one click is a quarter of the way round, two clicks half. Gently!

Achilles: [*taking the glass needle by its two ends*] Glass, one click, with the clock.

Crab: With the clock as seen from that needle's bead, mind. For the glass, that is looking down on the top of the ball.

[*Achilles lifts the ball just clear of its stand and holds the glass needle fast at both ends. The glass bead is the pin the turn is made about: it does not stir. A quarter-turn with the clock carries the steel and brass needles round with the ball, and the steel bead, that had pointed at the company, swings a quarter of the way round. A click; the Crab winces.*]

Achilles: The steel bead lies level and points at the stove.



Crab: Stove on the one hand, dresser on the other. In this room that is all the sideways there is.

Tortoise: [*She has brought tracing paper, as usual, and no string. On a strip of it*

she draws a curved arrow running with the clock, and beside it a dot of pale blue.] Pale blue for glass, and one arrowhead for one click. If the arrow runs with the clock on the strip as it lies before the player, the turn is with the clock as seen from the bead.

Achilles: A score! For one note.

[*She holds the strip by its short upright edge and turns it over towards her, as one turns a page. It is the same strip — the blue dot has not moved, the length has not changed — but the ink now shows through from the back of the paper.*]



as drawn



like a page

Tortoise: I turn it like a page. Seen through the paper, the last arrow would come first, and every arrow runs the other way round. Now I play what I see. [*She lifts the ball and holds the glass needle fast: a quarter-turn against the clock. The glass bead keeps the top; the steel bead swings back the quarter it had gone.*]

Achilles: Home.

Crab: Thank heaven.

Achilles: One note is nothing. Let me write a real one. [*He draws two arrows on a fresh strip, each with two heads: a blue dot, then a grey.*] Grey for steel, two heads for two clicks, and we read from the left. Glass, two clicks; then steel, two clicks.

[*He lifts the ball. The glass needle held: the steel bead is carried the whole half-way round, off the company and out the far side. Then the steel needle, held where it now lies: the glass bead is carried the half-way round, from the top down into the hollow of the stand. The Crab winces four times.*]

Achilles: Glass bead straight down, in the hollow of the stand; steel away, towards Mr Crab.

Tortoise: And to undo it?

[*Achilles lays his strip flat and turns it half-way round on the table without lifting it: the paper stays face up, and each arrow is carried round with it, its curl unchanged.*]



as drawn *round on the table*

Achilles: I turn the strip round on the table — flat, half-way round, not turned over. The last arrow comes first, and each runs as it was drawn. Steel, two clicks; glass, two clicks. [*He plays it: the steel needle held, then the glass, each carrying the ball half-way back.*] Home!

Tortoise: Is that the only way to read a strip backwards?

Achilles: There's another. Flip it top over bottom, about its long edge: the order stays, and every arrow runs the other way round.

[*Twice more the score is played from home, and twice more undone: once from the strip flipped top over bottom, once from the strip turned like a page. Home; and home. Then Achilles plays it steel first, glass second, and the ball comes to the very place the first order left it; he plays the same again, and it is home.*]



glass, then steel, two clicks each *steel, then glass*

[*The Crab has crossed to the dresser for the other pot and comes back the way he went, sidelong, the same step carrying him out and in — a crab is at no more trouble to go one way than the other. He tops up the cups without being asked and settles again into his chair.*]

Achilles: The same place, whichever goes first. So what you're saying is: to undo a score, play it again from the other end — any other end — and the order never mattered anyway.

Tortoise: I said nothing. But it is a handsome rule, and it has not failed yet.

Achilles: Nor will it.

Tortoise: Then one click each will be no harder.

Second Challenge — *The Score Read Backwards*

Tortoise: [*On Achilles's strip she rubs out one head of each arrow.*] Glass, one click; then steel, one click. Both with the clock, as you drew them. Mr Crab?

Crab: [*He plays it, wincing. Glass, a quarter: the glass keeps the top while the steel leaves the company for the stove. Then the steel needle, held where it now lies at the stove: it keeps its place while the glass bead is carried a quarter down off the top, to face the company.*] If I must.

Achilles: Now, my rule. The strip round on the table: steel one click, then glass one click, each with the clock as drawn. [*He plays it from where the ball stands.*] Glass bead… straight down. The steel is home and the glass is standing on its head!

Crab: Allow me. [*He backs out of Achilles's clicks, last click first and each the other way.*] As it was before you began.

Achilles: Then my other rule. The strip top over bottom: glass first, then steel, each against the clock. [*He plays it: the glass held while the steel swings a quarter the other way, then the steel held while the glass swings the other way too.*] Now the steel is on its head, and the glass lies at the stove. That's not home either. It isn't even the same mistake!

Crab: Allow me. [*He backs out again, last click first, each the other way.*] As it was.



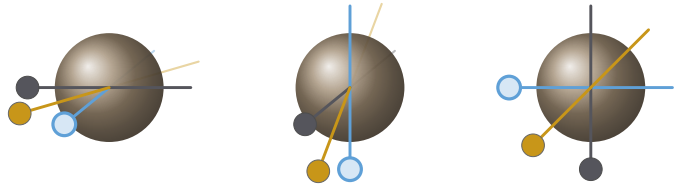
as drawn



round on the table



top over bottom



as played *round on the table* *top over bottom*

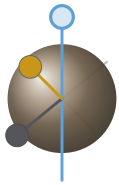
Achilles: But both rules worked. Every time!

Tortoise: On arrows with two heads. [*She turns the strip over about its short edge, as she turned her own; nothing is rubbed out, nothing redrawn.*] The page. Read me what you see.



the page

Achilles: Steel first — that's my first rule. Each against the clock — that's my second. [*He plays it from where the ball stands: the steel needle held while the glass bead swings a quarter back up to the top; then the glass needle held while the steel bead swings a quarter back to the company.*] Home. ...Oh. Both at once. Last arrow first *and* every arrow the other way round, and neither half will do alone.



the page: home

[*On the fender the mittens have gone quiet, done steaming. The one pulled off last lies on top of the other and inside-out, as it left the hand.*]

Tortoise: And the two-headed arrows?

Achilles: They hid both halves. Two clicks have no with or against, and

those two turns didn't mind which went first. And Mr Crab has been doing it under my nose. Twice!

Crab: Doing what? I went out the way you came in. I always leave a room by the way I came in, only backwards. I see nothing remarkable in it.

Achilles: It is what anyone does at a door, and I never saw it. What went in last comes out first — and it comes *out*; it does not go in again.

Tortoise: Are there scores that the page does not change?

Third Challenge — *The Sandwich*

[*The fire has its draught now and burns tall and even, and the room has stopped feeling like a place one has just come in from the cold. The clock ticks on over everything, unhurried, as it has all evening.*]

Tortoise: [*On a fresh strip she draws a two-headed arrow with a yellow dot, a two-headed arrow with a blue dot, a two-headed arrow with a yellow dot.*] Yellow for brass. Brass, two clicks. Glass, two clicks. Brass, two clicks. [*She turns the strip over about its short edge, and back again. Through the paper every arrow curls the other way; but each is a half-turn, and a half-turn has no other way round.*] The same score, either side of the paper.



as drawn *like a page*

Crab: The brass needle at last.

Achilles: Before anyone touches it — Mr Crab, the maker's card that came in the case. May I?

Crab: For each needle's turn a little square of four numbers. I never could read it.

Achilles: I can, and I needn't even multiply. Brass twice over is no turn at all: two half-turns. So the brass arrows cancel, near enough, and what

is left is the glass half-turn in the middle. Glass bead up, steel bead away. Nearly home.

Tortoise: There is a glass turn standing between your two brass ones.

Achilles: With two-headed arrows the order never mattered. We tried it! Nearly home, I say.

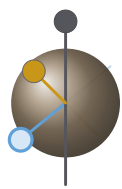
Crab: [*He takes the brass needle by its ends: two clicks. The brass bead is the pin now — it holds fast while the glass and steel needles swing a half-turn right past one another.*] Brass.

Tortoise: Look where the glass needle lies now.

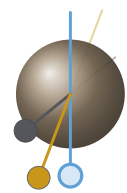
Achilles: Level, bead at the company — where the steel one lay. And the steel one stands upright, bead on top, in the glass one's place. They have changed places, bead for bead.

Crab: [*The glass needle, as it now lies, level at the company: two clicks. The glass bead holds at the company while the steel swings from the top round to the bottom. Then the brass again: two clicks, and glass and steel trade ends once more.*] Glass. Brass.

Achilles: Glass straight down, steel towards the company. That isn't nearly home. I said glass up and steel away, and it is exactly the other thing!



after one brass turn



brass, glass, brass

Tortoise: [*She turns the strip like a page once more.*] The page, Mr Crab.

[*The Crab plays the same score again, the ball lifted clear in his claw — brass, glass, brass — and the needles walk back the way they came: home. With the dull ball still held up before him he glances at the print on the wall — Escher's* Hand with Reflecting Sphere.]

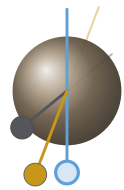
[*The clock knocks out its three, and the room goes still with it. Held up so, between the lamp and the wall, the ball throws a soft round shadow across the print, and for a moment the little curved room inside the mirrored sphere seems to lean out toward the lamplight.*]

Crab: That fellow holds up his ball and it shows him himself, and the whole room behind him. Mine shows me nothing at all — which is the whole point of it.

Tortoise: Home, Achilles. Remember where the beads were.

Crab: [*He sets the ball down, takes it up again by the two ends of the steel needle: two clicks. The steel needle holds fast while everything else swings a half-turn about it — the steel bead does not stir, and the glass and brass beads come round to the far side.*] One turn of the steel — and there they all are again.

Achilles: Bead for bead the same!



one steel turn, from home

Tortoise: And your numbers?

Achilles: Honestly, this time. [*He takes the maker's card and a scrap of paper.*] The brass square, then the glass square, then the brass square: row against column, twice over. The halves double up into wholes, the signs fall where they should — and what is left is the steel needle's square, number for number. And it was quicker than Mr Crab's six clicks.

[*He sets the scrap down and sits back, pleased with it, and for a moment only the fire and the clock are working.*]

Tortoise: It was. Did it tell you why?

Achilles: No. Only that it is so. But after the first brass turn the glass needle

lay where the steel one had lain — so the glass turn in the middle was made about the steel needle's place. And the last brass turn only put them back. ...Oh. That is why it comes out a steel turn.

Tortoise: A turn, seen through a swap.

Crab: [*The steel needle again: two clicks.*] And two clicks more of the steel, and we are home.

Achilles: But the brass needle is a luxury, surely — could one do without it?

Fourth Challenge — *Three Turns Make Any Turn*

Tortoise: Glass, steel, glass. One click each, with the clock.

Achilles: [*He plays it: glass, steel, glass, a quarter each, and each time the held needle's bead stands still while the others swing.*] The glass at the company. The steel on top. But that is where the first brass turn left them!

Tortoise: Now say the score from the other end.

Achilles: Glass, steel, glass. It is the same from both ends. But on the page each arrow would run against the clock, so that is its undoing. [*He plays it, every quarter-turn walking back the way it came.*] Home, as the page promised.

Tortoise: Again. The page again, from home.

Achilles: From *home*? [*He plays it: glass, steel, glass, each against the clock.*] The glass at the company, the steel on top — the same place! The score and its page go to the same place.

Tortoise: As a brass turn should.

Crab: [*He takes the brass needle by its ends: two clicks, and glass and steel trade ends.*] Then this will take it back. Home.

Achilles: So the brass needle *is* a luxury. And the others? Could one get anywhere at all without it?

Tortoise: Mr Crab, would you shut your eyes and tumble it?

Crab: Tumble it! [*Eyes shut, he turns the ball over and over in his claws and sets it down anyhow.*] It can be coaxed to rest between the notches of a collar, if one is gentle. Which I beg you to be.

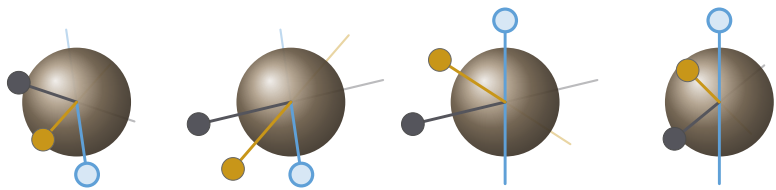
Achilles: The glass low, leaning towards us and the dresser. The steel high, leaning towards us and a little to the stove. Nowhere with a name.

Crab: Nowhere I should care to leave it.

Achilles: I'll take the glass needle. [*He holds it by its ends and turns the ball slowly about it.*] The steel goes round and round. The glass bead doesn't stir. Of course it doesn't: I am holding it.

Tortoise: Glass, until the steel needle lies level. Stop. Now steel, until the glass bead is on top. Stop. Now glass again, until the steel bead points at the company.

Achilles: [*He does each as she says it: the glass needle held fast while the steel swings round until it lies level; then the steel needle held while the glass climbs to the top; then the glass needle held again while the steel comes round to the company — each time the held bead standing still, and the others travelling only as far as she says, then stopping.*] Level. On top. At the company. Home, from nowhere.



tumbled | *glass: the steel lies level* | *steel: the glass on top* | *glass: home*

Tortoise: Glass, steel, glass. It is always enough.

[*In the grate a log that has burned through at the middle gives way, drops, and rolls a little until it finds a flat seat among the coals and holds there — settled, from wherever it happened to fall.*]

Achilles: I see why. The first two turns have one job only: to stand the glass

bead on top. After that the ball can be wrong in one way only, spun about the glass needle — and the last glass turn spins it right.

Crab: As one puts a jar back on the shelf: stand it upright, then turn its label to the front. And now nothing sits in its notch. Hold the ball still on its ring, Achilles — still! — while I roll each needle alone until it clicks into its collar. [*He rolls the glass needle between his claws: a click. The steel: a click. The brass: a click.*] There. The ball has not moved, and a click is a quarter of the way round again.

Achilles: So everything can be undone, if one is patient.

Tortoise: Everything on the strip.

Fifth Challenge — *What Has No Mirror*

Achilles: What else is there? [*He looks over the stand.*] This little brass catch on the pedestal?

Crab: Ah. Put the ball somewhere first, and tell us where.

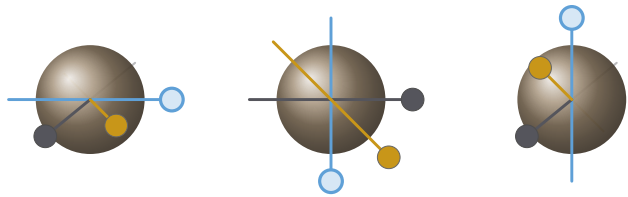
Achilles: Steel, one click with the clock. [*He plays it: the steel needle held, and the ball tips over sideways about it.*] The glass lies over at the dresser.

Crab: That little catch is the ball's valet — when a guest leaves it in a muddle, one press sets it to rights: home again, however it was left.

Crab: [*He presses the catch. Wherever the beads stood, they are gone from there in the one instant: glass up, steel at the company — home, as though no turn had been made at all.*] Home.

Achilles: So it is! Again. Glass, one click against the clock; then steel, two clicks. [*He plays it.*] Glass straight down, steel at the dresser — nowhere near the first place.

Crab: [*He presses the catch, and from here too the beads leap to the same home.*] Home.



one place *another place* *the catch, from either: home*

Achilles: From there as well! How does it do it?

Crab: A weight inside, I dare say. Will anyone take more tea?

[*He pours, and tips a little milk after it; the pale cloud turns over once in the cup and thins away to one even brown. Achilles takes the cup and warms both hands round it.*]

Tortoise: [*She hands Achilles a blank strip.*] Draw the catch.

Achilles: One arrow... From the first place it did what steel, one click against the clock, would have done. From the second it did something else altogether. It isn't one turn. It is whichever turn happens to be wanted.

Tortoise: Draw it anyway. Then turn the page.

Achilles: The page would take the ball from home back to where it was. But it was in two different places, and the catch took both to the same one. The page would have to lead to two places at once. ...Oh. It can't be drawn.



the catch

Tortoise: And what can?

Achilles: What the page can undo — a turn, and nothing but a turn. The catch is not on the strip. It is like the milk in this tea: pour it how you like, it comes to the one brown, and no stirring brings the white back.

Tortoise: [*She takes the evening's strips — her one blue arrow; Achilles's strip with the heads rubbed out; brass, glass, brass — turns each face up again, lays them*

end to end in that order, and gums them into one.] Then let us see how much the page can undo. From home, Mr Crab, left to right.

Crab: [*He plays the long score from home, left to right, wincing at every click.*] All of it. My poor collars.

Achilles: The glass at the dresser, the steel away, everything lying level. Nowhere near home.

Tortoise: [*She turns the whole long strip over like a page.*] Read it to him.



the evening's strips, gummed

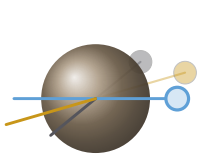


like a page

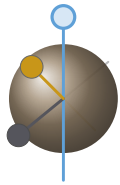
Achilles: Your sandwich first, Mr Crab, just as it was: brass, glass, brass. Then steel, glass and glass, one click each, and every one against the clock.

Crab: [*He plays it as it is read.*] And the last.

Achilles: Home — by turns alone, to the last click.



the evening's score



its page: home

Crab: Home, and nothing owing. Every click of the evening has been paid back. [*His claw rests on the shutter.*] There is one thing in this house for which nobody can write a page.

Tortoise: No, Mr Crab. Nobody can.

Crab: Eleven winters. I find I should rather know.

Achilles: Whatever it shows you, Mr Crab, we are glad we came.

[*The Crab opens the shutter, and looks.*]

[*On the wall the print is only a print again: the mirrored sphere holds its same curved room and nothing that has happened tonight. The fire has burned down to a bed of coals with one red eye still open in the grey, and the clock ticks on, softer now.*]

[*For a moment no one speaks.*]

Tortoise: Shall we, Achilles?

[*The doorstep. The snow lies; one set of footprints runs down the path, and every print points at the house.*]

Achilles: The Tortoise goes first and I go in her prints. It is slow, but it is tidy.

Crab: One set of footprints between you, I hope. My path is narrow.

Tortoise: Good evening, Mr Crab.

Crab: Good evening, good evening! Mind the step; it is the same step either way.

[*The door closes. The door-knob turns.*]

Achilles: It turned by itself. Someone has hold of it on the other side.

Tortoise: You are looking at the door-knob, Achilles.

Achilles: When one watches a turn from the other side, which way round does it go?

[*Outside, the one track stays as it was — the Tortoise's prints and Achilles's set in them — and the frost is setting the snow hard about their edges, to keep them till morning. The window holds its low glow behind the drawn curtain; behind that, a banked fire the Crab will build up again once the cold has the house to itself.*]

Chapter 3: Turning the Sphere

— Ländler on a Sphere —

Part I: The Single Wire

Escher: "Hand with Reflecting Sphere" — a hand holds up a mirrored ball, and in the ball are the man who holds it and the whole room behind him

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When you watch a turn from the other side, which way round does it go — and why should that be the way to undo it?

Stand in front of a shop window while someone inside turns a globe on its stand. From where they stand it goes with the clock. From where you stand, looking through the glass, it goes against. Nothing about the globe is in dispute between you; you are watching one turn from its two sides. Now suppose you wanted to put the globe back as it was. You would do what *you* saw, from where *they* stand. The undoing of a turn is the same turn, seen from behind.

That is easy enough for one turn. It becomes a real question when there are several, one after another, about different axes. Do you undo them in the order they were made, or in the other order? Do you reverse each of them, or only the whole? People who have packed a suitcase know the answer in their hands — last in, first out, and every movement backwards — and still get it wrong on paper.

For us the question has an edge to it. Two chapters have now said that a qubit is a thing nobody looks at: a covered coin, an arrow in the dark. Chapter 1 hung boxes on its wire and never closed the wire; Chapter 2 closed the wire and never hung a box on it. If a box is to act on something

that may not be inspected before, during or after, then there had better be a guarantee, written into the box itself, that it has not spoiled what it carries. We shall find that the guarantee and the view from the other side of the glass are the same thing. The boxes that may sit on a qubit's wire are exactly the ones undone by their own mirror image. And when we ask what those boxes *do*, the answer will be the plainest picture in this book's subject: they turn a sphere.

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§3.1 A Box on a Closed Wire

Chapter 2 ended (§2.10) with an instruction and two questions: *place a box between the state ψ and the effect ψ, and then the same between the pieces of the echo: is the closed picture still* 1*? Do the chances still make one whole?* Before we can answer we must learn to read a closed picture that has a box on it, and then we must be careful about which closed picture the question means.

One change of name first. Chapter 2 called Chapter 1's arrow T, for its triangle. This chapter needs the letter T for a box that the whole literature calls by that name (§3.4), so here the arrow is $|a\rangle$:

$$|a\rangle \;=\; \tfrac{3}{5}\,|0\rangle + \tfrac{4}{5}\,|1\rangle \;=\; \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}.$$

And the box NOT will from now on carry the name of its matrix, X.

On the blackboard. A bra is a row, a 2×2 matrix is a matrix, a ket is a column, and the product of the three, in that order, is a row times a column: a number. We write it with the matrix between the bars,

$$\langle \phi | \, A \, | \psi \rangle \,,$$

and compute it in whichever grouping is convenient, since matrix products may be grouped as we please: either the row $\langle \phi |$ applied to the column $A\,|\psi\rangle$, or the row $\langle \phi | \, A$ applied to the column $|\psi\rangle$.

The plainest case tells us what these numbers are. Number the entries of a matrix from zero, row first,

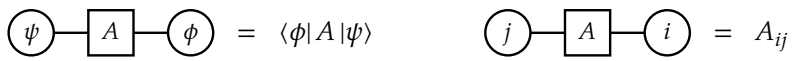
$$A \;=\; \begin{pmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{pmatrix}.$$

The column $A\,|j\rangle$ is the j-th column of A (§1.1), and the row $\langle i|$ picks out the i-th entry of whatever column it meets (§2.2). So, for i and j each 0 or 1,

$$\langle i|\,A\,|j\rangle \;=\; A_{ij}.$$

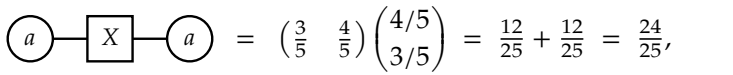
The four numbers in a matrix are its four closed pictures on the standard states. For NOT, $\langle 1|\,X\,|0\rangle = 1$ and $\langle 0|\,X\,|0\rangle = 0$: the state 0, after NOT, answers "yes" to the question 1 and "no" to the question 0.

In the sketchbook. No new element yet. A state, then a box, then an effect, all on one wire, is a row of three things with no open end, so by the row rule (§1.3) and the reading of closed diagrams (§2.2) it is a number, and the number is the product just written. As always the picture runs left to right and the product right to left:



Every picture equation in this chapter is exact: the same number, or the same matrix, on both sides, with no factor swept aside.

Now the question of §2.10, and a warning about how to read it. Taken literally, "a box between the state ψ and the effect ψ" is the picture on the left above with $\phi = \psi$. It is a perfectly good number. For the arrow and NOT,



which is not 1, and nothing has gone wrong: the arrow was turned over and then asked whether it was still the *old* arrow. (Keep the number $\frac{24}{25}$; it returns in §3.5 with a new meaning.) That is not the promise of Theorem 2.1. The promise was that a state meets *its own* mirror image in 1, and the state on the wire after the box is no longer $|a\rangle$; it is $X|a\rangle$. What §2.10 computed for SET-0, and what we must ask of every box, is whether the state *after the box* still meets its own mirror image in 1:

$$X|a\rangle = \begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix} : \quad \tfrac{16}{25} + \tfrac{9}{25} = 1;$$

$$S_0|a\rangle = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} = \begin{pmatrix} 7/5 \\ 0 \end{pmatrix} : \quad \tfrac{49}{25}.$$

NOT keeps the promise for the arrow; SET-0 breaks it, as §1.5 foresaw. See `Code/Ch3/NoMirror.py`.

Here the sketchbook is stuck, and the place where it sticks is the subject of the chapter. We can draw the state "a, then X". To draw *its mirror image* we could cheat, work out the column on the blackboard and put the answer in a rounded box. But a mirror ought to work on the picture as it stands, piece by piece. Chapter 2 taught the mirror to reflect a state into an effect. It was careful to say (§2.2) that it had not taught it to reflect a box.

Why this definition. Why read §2.10's question as "the state after the box meets its own mirror image", and not as the literal picture with ψ at both ends? Because only the first is the promise of Theorem 2.1, on which the Born rule's "one whole" rests (Corollary 2.2). The literal picture compares the new state with the old one, and a box that does anything at all is entitled to fail that comparison.

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§3.2 The Mirror of a Box

We do not get to choose what the mirror image of a box is. Chapter 2 has already decided it.

Hold the mirror up to the closed picture "state j, box A, effect i", whose number is A_{ij}. §2.2 told us three things about what we must see. The state j on the left becomes the effect j on the right. The effect i on the right becomes the state i on the left. And the number is conjugated. In between there is some box, the reflection of A; call it A' for a moment. The mirrored picture reads "state i, box A', effect j", which is the entry A'_{ji}. So

$$A'_{ji} \;=\; \overline{A_{ij}} \qquad \text{for all four choices of } i \text{ and } j,$$

and four entries are a whole matrix. There is nothing left to decide.

On the blackboard. The **conjugate transpose** of a matrix A, written $A^\dagger$ and read "A dagger", is the matrix whose entry in row j and column i is the conjugate of the entry of A in row i and column j:

$$(A^\dagger)_{ji} \;=\; \overline{A_{ij}}, \qquad\qquad \begin{pmatrix} p & q \\ r & s \end{pmatrix}^\dagger \;=\; \begin{pmatrix} \bar{p} & \bar{r} \\ \bar{q} & \bar{s} \end{pmatrix}.$$

Two things happen at once. The matrix is flipped about its diagonal, rows becoming columns: that alone is the **transpose**, A^T. And every entry is conjugated: that alone is the **conjugate**, $\overline{A}$. The dagger is both. Done twice it does nothing, $(A^\dagger)^\dagger = A$, since two flips are none and two conjugations are none; and $I^\dagger = I$.

The same recipe applies to matrices of any shape, and we have been using it since §2.2 without the name: the conjugate transpose of a column is the conjugated row. **The bra is the dagger of the ket**, $\langle\psi| = (|\psi\rangle)^\dagger$, and the ket is the dagger of the bra.

To see that the three operations really differ we need a matrix that is neither real nor symmetric about its diagonal; for real matrices the conjugate is

invisible, for symmetric ones the transpose is. Here are two matrices that §3.4 will introduce properly. For now they are only arrays of numbers:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad V = \tfrac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix},$$

$$U \;=\; VS \;=\; \tfrac{1}{2}\begin{pmatrix} 1+i & 1+i \\ 1-i & -1+i \end{pmatrix}.$$

(Multiplying on the right by S multiplies the second column by i, and $i(1-i) = 1+i$, $i(1+i) = -1+i$.) The matrix U is the row "S, then V". It is not real, and it is not symmetric: its off-diagonal entries are $\frac{1+i}{2}$ and $\frac{1-i}{2}$. It will be our test piece for the whole chapter. Side by side:

$$U^T = \tfrac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1+i & -1+i \end{pmatrix}, \qquad \overline{U} = \tfrac{1}{2}\begin{pmatrix} 1-i & 1-i \\ 1+i & -1-i \end{pmatrix},$$

$$U^\dagger = \tfrac{1}{2}\begin{pmatrix} 1-i & 1+i \\ 1-i & -1-i \end{pmatrix}.$$

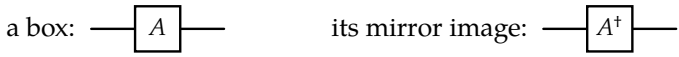
Three different matrices, and none of them is U.

In the sketchbook. One new element, described before it is drawn; it is the only new element of the chapter.

The **mirror image of a box** A is the box seen in a mirror held upright across the wire. The wire that came in on the left now leaves on the right, and the wire that left on the right now comes in on the left: *input and output change places*. And, as for states in §2.2, **the mirror conjugates.** The reflected box stands for the matrix $A^\dagger$.

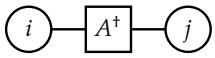
We have to be candid about the drawing. A box on one wire has one wire on each side, and its outline is a plain rectangle, so the reflected box has exactly the outline of the original. With a state, §2.2 could say "look for the wire"; here the wire is on both sides, before the mirror and after. **On a single wire the mirror shows in only two places: in the label, which we**

write $A^\dagger$, and in the order of a row (below). So we draw it like this:



Those who draw such pictures for a living meet the same difficulty, and solve it by putting a mark in one corner of every box, so that the reflection can be told from the original by where the mark has gone (the *Feynman's Notebook* of §3.8 has the reference). Our mark is the dagger on the label.

With the new element the opening argument of this section becomes a picture equation, true for all four choices of i and j:



$$= \langle j|\, A^\dagger\, |i\rangle = \overline{A_{ij}}.$$

Why this definition. Why both the flip and the conjugation, when either alone is simpler? Because Chapter 2 left no room. The mirror exchanges the two ends of a closed picture, and that is the flip, ij becoming ji; and the mirror conjugates the number, and that is the bar. A reflection that did only one of the two would disagree with §2.2 about the four closed pictures that make up the box.

The mirror of a row

Our test piece U is a row of two boxes, S then V. So there are two ways to find its mirror image: reflect the matrix U as a whole, which we have done; or hold the mirror up to the row. A mirror held up to a row of things reverses their order: what was leftmost is rightmost in the glass. So the mirror image of "S, then V" should be "$V^\dagger$, then $S^\dagger$", whose matrix (rightmost box first in the product, as always) is $S^\dagger V^\dagger$. Let us see. Both S and V happen to be symmetric, so their daggers are their conjugates:

$$S^\dagger V^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} \cdot \tfrac{1}{2} \begin{pmatrix} 1-i & 1+i \\ 1+i & 1-i \end{pmatrix} = \tfrac{1}{2} \begin{pmatrix} 1-i & 1+i \\ 1-i & -1-i \end{pmatrix} = U^\dagger.$$

(Multiplying on the left by $S^\dagger$ multiplies the second row by $-i$.) It is. And the two cheaper recipes fail. Reflect each piece but keep the order, "$S^\dagger$, then $V^\dagger$": the matrix is $V^\dagger S^\dagger = \overline{V}\,\overline{S} = \overline{U}$, which is not $U^\dagger$. Reverse the order but do not reflect the pieces, "V, then S": the matrix is $SV = S^T V^T = (VS)^T = U^T$, which is not $U^\dagger$ either. (That these two wrong answers are exactly the conjugate and the transpose of U is an accident of our having chosen symmetric pieces; that they are *wrong* is not.)

> **Theorem 3.1 (The Mirror of a Row).** For matrices A and B whose product BA is defined, of any shapes,
>
> $$(BA)^\dagger \;=\; A^\dagger B^\dagger.$$
>
> In particular, for every 2×2 matrix A and all states $|\psi\rangle$, $|\phi\rangle$:
>
> $$(A\,|\psi\rangle)^\dagger = \langle\psi|\,A^\dagger, \qquad\qquad \overline{\langle\phi|\,A\,|\psi\rangle} \;=\; \langle\psi|\,A^\dagger\,|\phi\rangle\,.$$
>
> In pictures: the mirror image of a row is the row of the mirror images, **in the reverse order**; and the mirror image of a closed picture is the conjugate number.

Proof idea. An entry of a product is a sum of products of entries, one from each factor. The flip exchanges the roles of rows and columns in both factors, which is what reverses their order; the conjugation passes through sums and products (§2.2) and so can be done entry by entry. *Proof sketch.* The index calculation is two lines and is carried out in the sidebar. The first special case is the theorem with the column $|\psi\rangle$ in the place of A, and A in the place of B. For the second, a number is a matrix with one row and one column, and its dagger is its conjugate; apply the theorem twice to the product of three, $(\langle\phi|\,A\,|\psi\rangle)^\dagger = (|\psi\rangle)^\dagger A^\dagger (\langle\phi|)^\dagger = \langle\psi|\,A^\dagger\,|\phi\rangle$. ■ See `Code/Ch3/Backwards.py`, which checks the theorem on matrices drawn at random and checks that the two cheaper recipes fail; and `Code/Ch3/Mirror.py` for the mirror of a single box.

Dirac's Blackboard.

The mirror of a product, entry by entry. Let A and B be 2×2; other

shapes change only the ranges of the indices. The entry of BA in row i, column j is, by the rule "along the row and down the column",

$$(BA)_{ij} \;=\; B_{i0}A_{0j} + B_{i1}A_{1j}.$$

By definition the entry of $(BA)^\dagger$ in row j, column i is the conjugate of this, and conjugation passes through sums and products:

$$\left((BA)^\dagger\right)_{ji} \;=\; \overline{B_{i0}}\,\overline{A_{0j}} + \overline{B_{i1}}\,\overline{A_{1j}}.$$

Now read the right side again. $\overline{A_{kj}} = (A^\dagger)_{jk}$ and $\overline{B_{ik}} = (B^\dagger)_{ki}$, so it is

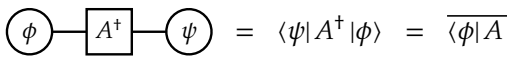
$$(A^\dagger)_{j0}(B^\dagger)_{0i} + (A^\dagger)_{j1}(B^\dagger)_{1i} \;=\; (A^\dagger B^\dagger)_{ji}.$$

Numbers may be multiplied in either order; matrices may not. The index k that joined the *back* of B to the *front* of A now joins the back of $A^\dagger$ to the front of $B^\dagger$, and that is the whole reason for the reversal. ■

In the sketchbook the theorem is the rule for reflecting any picture that this book has so far drawn. **Mirror every piece, and reverse the order.** A state becomes an effect (§2.2), an effect a state, a box its mirror image, and what was first is last. Here it is for a state followed by two boxes,



(on the blackboard: $(BA\,|\psi\rangle)^\dagger = \langle\psi|\,A^\dagger B^\dagger)$, and for a closed picture:



$$= \langle\psi|\,A^\dagger\,|\phi\rangle \;=\; \overline{\langle\phi|\,A\,|\psi\rangle}.$$

A worked instance, with a number that is not real so that the bar can be seen.

Take the arrow, the test piece and the effect $+$. First $U|a\rangle = \frac{1}{10}\begin{pmatrix} 7+7i \\ -1+i \end{pmatrix}$, and then

$$\langle +|\, U\,|a\rangle \;=\; \frac{1}{\sqrt{2}} \cdot \frac{1}{10}((7+7i)+(-1+i)) \;=\; \frac{6+8i}{10\sqrt{2}} \;=\; \frac{3+4i}{5\sqrt{2}}.$$

The mirrored picture is "state $+$, box $U^\dagger$, effect a". From the matrix $U^\dagger$ above, $U^\dagger|+\rangle = \frac{1}{2\sqrt{2}}\begin{pmatrix} 2 \\ -2i \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$, and

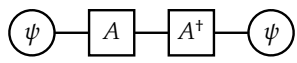
$$\langle a|\, U^\dagger\,|+\rangle \;=\; \frac{1}{\sqrt{2}}\left(\tfrac{3}{5} - \tfrac{4}{5}i\right) \;=\; \frac{3-4i}{5\sqrt{2}}.$$

Conjugates, as the theorem requires. With U^T or $\overline{U}$ in the mirrored picture they would not be.

---

§3.3 Boxes That Keep the Promise

Now the sketchbook of §3.1 is no longer stuck. The state "ψ, then A" has the mirror image "$A^\dagger$, then the effect ψ", and a state meeting its own mirror image is one closed picture with two boxes on it:



$$= \langle\psi|\, A^\dagger A\,|\psi\rangle\,.$$

This is the picture §2.10 was asking about. If it is 1 for every state ψ, the box A **keeps the promise** of Theorem 2.1: whatever state goes in, a state comes out.

Try the two boxes we know. For NOT: X is real and symmetric, so $X^\dagger = X$, and $X^\dagger X = XX = I$ by Theorem 1.1. The two boxes in the middle of the picture are a bare wire, and the picture is $\langle\psi|\psi\rangle = 1$ for every ψ at once, not only for the arrow. For SET-0:

$$S_0^\dagger S_0 \;=\; \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \;=\; \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \langle a|\, S_0^\dagger S_0\, |a\rangle \;=\; \left(\tfrac{3}{5}+\tfrac{4}{5}\right)^2 \;=\; \tfrac{49}{25}.$$

And for the test piece, which is neither real nor symmetric, multiply out $U^\dagger U$ with the matrices of §3.2. The four entries are

$$\tfrac{1}{4}\big((1-i)(1+i)+(1+i)(1-i)\big) = 1,$$

$$\tfrac{1}{4}\big((1-i)(1+i)+(1+i)(-1+i)\big) = \tfrac{1}{4}(2-2) = 0,$$

and likewise 0 and 1 in the second row: $U^\dagger U = I$. The cheaper reflections do *not* give the bare wire:

$$U^T U \;=\; \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad \overline{U}\,U \;=\; \tfrac{1}{2}\begin{pmatrix} 1-i & 1+i \\ -1+i & 1+i \end{pmatrix}.$$

In the examples that work, the promise is kept for the strongest possible reason: the box followed by its mirror image *is* the bare wire. Could a box keep the promise for some weaker reason? It could not.

On the blackboard. A 2×2 matrix U is **unitary** when $U^\dagger U = I$.

> **Theorem 3.2 (Boxes That Keep the Promise).** For a 2×2 complex matrix U the following are equivalent:
>
> 1. $U^\dagger U = I$ (the box, then its mirror image, is the bare wire);
> 2. for every state $|\psi\rangle$, the column $U|\psi\rangle$ is a state: $\langle\psi|\, U^\dagger U\,|\psi\rangle = 1$.
>
> When they hold, also $UU^\dagger = I$ (the mirror image, then the box, is the bare wire); the two columns of U are an orthonormal basis; and U preserves every bracket: for all states $|\phi\rangle$, $|\psi\rangle$, the bracket of $U|\phi\rangle$ with $U|\psi\rangle$ is $\langle\phi|\psi\rangle$.

Proof idea. One direction is a single application of *The Mirror of a Row*. For the other, a matrix is pinned down by a handful of its closed pictures: two standard states fix the diagonal of $U^\dagger U$, and two further states, one real and one not, fix the real and imaginary parts of what is off the diagonal.

Proof sketch. ($1 \Rightarrow 2$, and the bracket.) By Theorem 3.1 the bra of $U|\phi\rangle$ is $\langle\phi|U^\dagger$, so the bracket of $U|\phi\rangle$ with $U|\psi\rangle$ is $\langle\phi|U^\dagger U|\psi\rangle = \langle\phi|I|\psi\rangle = \langle\phi|\psi\rangle$. With $\phi = \psi$ this is statement 2.

($2 \Rightarrow 1$.) Write $M = U^\dagger U$ with entries m_{ij}, and put four states into $\langle\psi|M|\psi\rangle = 1$. The states $|0\rangle$ and $|1\rangle$ give $m_{00} = 1$ and $m_{11} = 1$. The state $|+\rangle$ gives $\frac{1}{2}(m_{00} + m_{01} + m_{10} + m_{11}) = 1$, hence $m_{01} + m_{10} = 0$. The state $|\omega\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, whose bra is $\frac{1}{\sqrt{2}}\begin{pmatrix}1 & -i\end{pmatrix}$, gives $\frac{1}{2}(m_{00} + i\,m_{01} - i\,m_{10} + m_{11}) = 1$, hence $m_{01} - m_{10} = 0$. So $m_{01} = m_{10} = 0$ and $M = I$.

(The rest.) For square matrices a one-sided inverse is two-sided (we used this in Lemma 2.3), so $U^\dagger U = I$ gives $UU^\dagger = I$. And the entry m_{ij} of $U^\dagger U$ is $\langle i|U^\dagger U|j\rangle$, the bracket of the i-th column of U with the j-th; $M = I$ says these brackets are $1, 0, 0, 1$, which is orthonormality (§2.5). ■

See `Code/Ch3/Mirror.py`, which checks all of this on a unitary drawn at random and checks that the transpose and the conjugate are *not* its undoing, and `Code/Ch3/NoMirror.py` for SET-0.

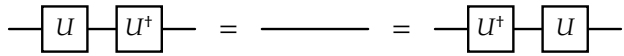
For the test piece, take the factor $\frac{1+i}{\sqrt{2}} = e^{i\pi/4}$ out of each column of U, using $\frac{1-i}{1+i} = -i$ and $\frac{-1+i}{1+i} = i$: the columns are $e^{i\pi/4} \cdot \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$ and $e^{i\pi/4} \cdot \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle) = e^{i\pi/4}|\omega\rangle$. Their bracket is $\frac{1}{2}(1 + (i)(i)) = 0$: an orthonormal basis, and not one we have met before, each member carrying a global phase.

Two remarks keep the theorem honest. First, "has an undoing" is weaker than "is undone by its mirror image". The matrix $D = \begin{pmatrix}1 & 0\\ 0 & 2\end{pmatrix}$ is undone by $\begin{pmatrix}1 & 0\\ 0 & 1/2\end{pmatrix}$, but $D^\dagger D = \begin{pmatrix}1 & 0\\ 0 & 4\end{pmatrix}$, and $\langle a|D^\dagger D|a\rangle = \frac{9}{25} + \frac{64}{25} = \frac{73}{25}$. It breaks the promise, and may not sit on a qubit's wire. SET-0 is worse: it sends both $|0\rangle$ and $|1\rangle$ to $|0\rangle$, so nothing placed after it can recover which one went in

(§1.3), and it has no undoing of any kind.

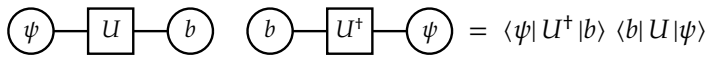
Second, §2.6 left a debt. It showed that a global phase is seen by no question asked directly of the state, and said that this survives whatever boxes are placed on the wire first. It does. Matrices act on a multiple of a column by giving the same multiple of the result, so $U(e^{i\theta}|\psi\rangle) = e^{i\theta}U|\psi\rangle$; if U is unitary then $U|\psi\rangle$ is a state, and Proposition 2.4, applied to *that* state, says no question separates it from its multiple. The same argument covers a phase on a box: $e^{i\theta}U$ is a different matrix from U when $e^{i\theta} \neq 1$, it is again unitary, and it sends each state to $e^{i\theta}$ times where U sends it. Different boxes; and no question put afterwards can tell which of them was on the wire.

In the sketchbook. Unitarity is a rewrite rule, the first this book has had since Theorem 1.1, and it contains that theorem as a special case:



A box meeting its reflection, in either order, leaves a bare wire. We call the rule ***a box meets its mirror***. It is licensed by Theorem 3.2 and by nothing else, and it may be used only for a box known to be unitary.

The second half of §2.10's question was about *The Echo*. Put the box between the pieces of each half, remembering that the right half is the mirror image of the left, so that by *The Mirror of a Row* it carries $U^\dagger$, on the other side of the gap:

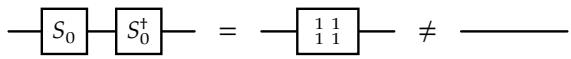


The number is $|\langle b|\,U\,|\psi\rangle|^2$: by the Born rule, the chance of the answer b when the box has acted first. Do the chances still make one whole? Add the numbers for $b = 0$ and $b = 1$ (a sum of *numbers*, the liberty §2.5 allowed). They are the squared moduli of the two entries of the column $U|\psi\rangle$, and their sum is that column meeting its own mirror image: $\langle\psi|\,U^\dagger U\,|\psi\rangle$, the closed picture at the head of this section. For a unitary box that is 1, and then $U|\psi\rangle$ is a state and Corollary 2.2 makes the total 1 for every other question too. For SET-0 and the arrow the two numbers are $\frac{49}{25}$ and 0. "Chances" that

come to nearly two are not chances.

Penrose's Sketchbook.

A reflection that is not an undoing. The rule *a box meets its mirror* is not a fact about mirrors. Every box has a mirror image; only some are undone by it. Here is SET-0 meeting its reflection:



The middle picture is a box whose matrix has all four entries 1. In the other order, $S_0^\dagger$ then S_0, the matrix is $\begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$: not a bare wire either, and not even the same failure. So when a row contains a box and its reflection side by side, do not cancel them by eye. Cancel them because the box is unitary, and say so. The sketchbook cannot tell a unitary box from any other by its outline; that knowledge comes from the blackboard, through Theorem 3.2.

Why this definition. Why demand $U^\dagger U = I$, when the requirement we actually care about is only that states go to states? Because they are the same requirement (Theorem 3.2), and the first can be checked by one matrix product, while the second speaks of infinitely many states. And why the mirror image, not merely *some* undoing? Because D above has an undoing and still turns the arrow into a column of squared length $\frac{73}{25}$.

---

§3.4 Six Boxes

It is time to meet the boxes that the rest of the book is built from. There are six, and we already know the first.

On the blackboard.

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$H = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix},$$

where $e^{i\pi/4} = \frac{1+i}{\sqrt{2}}$.

The first three are called the **Pauli matrices**; H is the **Hadamard** box; S and T are the **phase** boxes. (S is the S of §3.2, and the matrix V there was HSH, as the reader can now check.)

Four are their own mirror images. X, Z and H are real and symmetric, so neither the bar nor the flip changes them. Y is more interesting: it is *not* real and *not* symmetric. Its transpose is $-Y$ and its conjugate is $-Y$; the two signs cancel, and $Y^\dagger = Y$. Each of the four is unitary, and a unitary box that is its own mirror image is its own undoing:

$$XX = I, \qquad YY = \begin{pmatrix} (-i)(i) & 0 \\ 0 & (i)(-i) \end{pmatrix} = I, \qquad ZZ = I,$$

$$HH = \tfrac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = I.$$

Two are not. $S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix} \neq S$ and $T^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-i\pi/4} \end{pmatrix} \neq T$. Both are unitary, since $\bar{\imath}\, i = 1$ and $e^{-i\pi/4} e^{i\pi/4} = 1$. But done twice they are not the bare wire. They climb a ladder instead:

$$T^2 = S, \qquad S^2 = Z, \qquad Z^2 = I.$$

They are related. Multiply out: $XZ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $ZX = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so

$$XZ \;=\; -ZX, \qquad\qquad Y \;=\; iXZ.$$

The row "Z, then X" and the row "X, then Z" are different boxes; they differ by a sign on the whole box. By the second remark of §3.3, no question asked afterwards can tell them apart.

What H and Z do to the four states we have names for. These eight small facts are used constantly, and the sketchbook proof of §3.8 will borrow them:

$$H|0\rangle = \tfrac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = |+\rangle\,, \qquad H|1\rangle = \tfrac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix} = |-\rangle\,,$$

$$Z|+\rangle = |-\rangle\,, \qquad Z|-\rangle = |+\rangle\,,$$

the first two because $H|j\rangle$ is the j-th column of H, the last two because Z changes the sign of the second entry. Apply H to the first two and use $HH = I$: $H|+\rangle = |0\rangle$ and $H|-\rangle = |1\rangle$. And $Z|0\rangle = |0\rangle$, $Z|1\rangle = -|1\rangle$. So H carries the standard basis to the turned basis of §2.5 and back; Z **does to the turned basis what NOT does to the standard one**; and to the standard basis Z does nothing that a standard question can see, which is §2.4's hidden sign, now in a box.

Take the dagger of $H|0\rangle = |+\rangle$. By *The Mirror of a Row* and $H^\dagger = H$:

$$\langle 0|H = \langle +|\,, \qquad\qquad \langle 1|H = \langle -|\,.$$

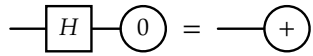
In the sketchbook. No new element: six labelled boxes. Four of them are their own mirror images ($X^\dagger$ *is* the box X, and we shall never write the dagger on it) and two are not. The facts just computed are picture equations between states, in the manner of Chapter 1's *NOT on 0*, and we name them the same way. ***H on 0*** and ***H on 1***:



Z on + and ***Z on −***:



And the mirror image of *H on 0*, every piece reflected and the order reversed, is an equation between effects, with *H* its own reflection:



This last equation says what *H* is *for*. To ask the turned question of §2.5, put an *H* on the wire and then ask the standard one.

Why this definition. Why these six, out of all the unitary matrices there are? For this chapter the answer is modest: *X*, *Y* and *Z* will be the three axes of the sphere (§3.5); *H* passes between two of them; *S* and *T* are the simplest boxes that are *not* their own mirror images, without which nothing in this chapter could be tested. That a great deal can be built from them is a claim for later chapters, and we do not make it here.

Feynman's Notebook.

Names that everybody uses. The letters in this section are not ours. Van de Wetering's survey, written for working quantum computer scientists, opens its account of circuits (§2.1) by recording that the community has mostly agreed on its naming conventions: *X*, *Y* and *Z* are the Pauli matrices; the *S* gate and the *T* gate are the quarter- and eighth-turn members of the family of *Z*-rotations (§3.6 below says what the turning is); *H* is the Hadamard gate; and a dagger on a gate's name means the inverse gate. That last convention is Theorem 3.2 in everyday use: nobody who writes $S^\dagger$ in a circuit stops to say whether they mean the mirror image or the undoing, because for a unitary box there is no difference. "Gate" is the engineers' word for what we call a box. The textbook treatment of the Pauli matrices, of *H* and of the sphere they turn is Nielsen & Chuang, ch. 4.

A trap

Here are two tempting rules. *To undo a row, play it again from the other end. And the order of two boxes never really matters.*

Both are true of everything a cautious person might try first. Take any row made of X, Z and H — say "H, then Z". Its mirror image is "$Z^\dagger$, then $H^\dagger$", which is "Z, then H": the same boxes from the other end, because each is its own reflection. And X with Z in the two orders differ only by a sign on the box, which no question sees.

Now try S. "Play it again" gives S, then S, which is Z and not the bare wire. Play our test row "S, then V" again from the other end, "V, then S": the four boxes multiply to $SVVS = U^TU = iX$ (§3.3), nowhere near a bare wire. There is a subtler version. Wrap X between S and its mirror image, and between S and itself:

$$SXS^\dagger = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = Y, \qquad SXS = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = iX.$$

The first is a new box; the second is X again with a phase on it. Someone who had only ever wrapped things in H, where $H^\dagger = H$ hides the distinction, would not know that there was a choice to make.

As for order: $HZ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ and $ZH = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, and no sign or phase turns one into the other, since their top-left entries agree and their top-right entries do not. The moral is the moral of §1.4 over again. X, Z and H are real and symmetric, and **a real symmetric box can never test a claim about mirrors.** See `Code/Ch3/Mirror.py` and `Code/Ch3/Backwards.py`.

---

§3.5 The Sphere

A qubit state is two complex numbers, which is four real ones; the rule $|\alpha|^2 + |\beta|^2 = 1$ removes one, and the global phase, which no question sees, removes another. Two real numbers are left. That is as many as it takes

to say where you are on the surface of the Earth. This section shows that the resemblance is exact. It is a blackboard section: the sphere is a way of reading numbers, and we add no sphere to the sketchbook's elements.

On the blackboard. For a state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ define three numbers, one for each Pauli matrix:

$$x = \langle\psi|\,X\,|\psi\rangle\,, \qquad y = \langle\psi|\,Y\,|\psi\rangle\,, \qquad z = \langle\psi|\,Z\,|\psi\rangle\,.$$

Multiplying out,

$$x = \bar{\alpha}\beta + \bar{\beta}\alpha, \qquad y = -i\,\bar{\alpha}\beta + i\,\bar{\beta}\alpha, \qquad z = |\alpha|^2 - |\beta|^2.$$

Write $w = \bar{\alpha}\beta$. Then $\bar{\beta}\alpha = \bar{w}$, and a number plus its conjugate is twice its real part, while $-i(w - \bar{w})$ is twice its imaginary part. So

$$x = 2\,\mathrm{Re}(\bar{\alpha}\beta), \qquad y = 2\,\mathrm{Im}(\bar{\alpha}\beta), \qquad z = |\alpha|^2 - |\beta|^2.$$

Proposition 3.3 (the point of a state). For every state $|\psi\rangle$ the numbers x, y, z are real, and

$$x^2 + y^2 + z^2 \;=\; 1.$$

The states $|\psi\rangle$ and $e^{i\theta}\,|\psi\rangle$ have the same three numbers. And every point (x, y, z) with $x^2 + y^2 + z^2 = 1$ belongs to some state.

So each state marks a point on the sphere of radius 1 about the origin — it is called the **Bloch sphere** (Nielsen & Chuang, ch. 4) — and the point does not notice a global phase.

Proof idea. Reality is a mirror argument; the sum of squares is the identity $(p-q)^2 + 4pq = (p+q)^2$. *Proof sketch.* The formulas above show x, y, z real. Next, $x^2 + y^2 = 4(\mathrm{Re}(w)^2 + \mathrm{Im}(w)^2) = 4|w|^2 = 4|\alpha|^2|\beta|^2$, so

$$x^2 + y^2 + z^2 \;=\; 4|\alpha|^2|\beta|^2 + \left(|\alpha|^2 - |\beta|^2\right)^2 \;=\; \left(|\alpha|^2 + |\beta|^2\right)^2 \;=\; 1.$$

A global phase multiplies α and β by $e^{i\theta}$ and their conjugates by $e^{-i\theta}$, and in each of $\bar{\alpha}\beta$, $|\alpha|^2$, $|\beta|^2$ the two factors cancel. Last, given a point of the sphere, write it as $(\sin\vartheta\cos\varphi,\ \sin\vartheta\sin\varphi,\ \cos\vartheta)$, as one does with latitude and longitude, and take

$$|\psi\rangle \;=\; \cos\tfrac{\vartheta}{2}\,|0\rangle + e^{i\varphi}\sin\tfrac{\vartheta}{2}\,|1\rangle\,.$$

Then $\bar{\alpha}\beta = \cos\frac{\vartheta}{2}\sin\frac{\vartheta}{2}\,e^{i\varphi} = \frac{1}{2}\sin\vartheta\,e^{i\varphi}$ and $z = \cos^2\frac{\vartheta}{2} - \sin^2\frac{\vartheta}{2} = \cos\vartheta$, which is the given point. ■ See `Code/Ch3/Sandwich.py`, which checks the proposition on states drawn at random.

Where do the states we know sit?

$$|0\rangle:\ (0,0,1), \qquad |1\rangle:\ (0,0,-1),$$

$$|+\rangle:\ (1,0,0), \qquad |-\rangle:\ (-1,0,0), \qquad |\omega\rangle:\ (0,1,0),$$

where $|\omega\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ is the fourth test state of §3.3. The arrow has $\bar{\alpha}\beta = \frac{12}{25}$ and $|\alpha|^2 - |\beta|^2 = -\frac{7}{25}$:

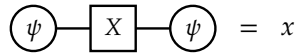
$$|a\rangle:\ (\tfrac{24}{25},\ 0,\ -\tfrac{7}{25}).$$

There is the $\frac{24}{25}$ of §3.1. The "literal" picture with the arrow at both ends and X between was never a broken promise; it was the arrow's x-coordinate. Chapter 2's non-real state $|\chi\rangle = \begin{pmatrix} 3/5 \\ 4i/5 \end{pmatrix}$ has $\bar{\alpha}\beta = \frac{12i}{25}$ and sits at $(0, \frac{24}{25}, -\frac{7}{25})$: the same latitude as the arrow, a quarter of the way round. The real states all lie on the one great circle $y = 0$; the i in an amplitude is what takes a state off it.

An honest word about Chapter 2's paper arrows. On squared paper, $|0\rangle$ and $|1\rangle$ were perpendicular, and so were $|+\rangle$ and $|-\rangle$. On the sphere each pair sits at *opposite poles*. The arrows $|0\rangle$ and $|+\rangle$ were half a right angle apart on paper; their points $(0,0,1)$ and $(1,0,0)$ are a full right angle apart. The

sphere doubles every angle of the paper picture. That is the half-angle $\frac{\vartheta}{2}$ in the formula above, and it is why perpendicular states (the two answers of a question, §2.5) are opposite points. Neither picture is wrong. The paper arrow kept the sign of the whole column, which no question sees, and could show only real states. The sphere shows every state and nothing that cannot be seen.

In the sketchbook. Each of the three numbers is a closed picture of the kind §3.1 taught us to read:



and the same with Y for y, and with Z for z. The sketchbook can say, in its own terms, why they are real. Hold the mirror up to the picture above. The state ψ and the effect ψ change places, and X is its own mirror image: the reflected picture *is the picture we started with*. But the number of a reflected picture is the conjugate (Theorem 3.1). A number equal to its own conjugate is real. The same argument serves for any box that is its own reflection, and it is a better reason than the formulas, which only happen to come out real.

One caution. The three numbers are three separate pictures. We do not set them side by side in a row, because a row of closed pictures is a *product* (§2.3), and the product xyz means nothing here. The sum of their squares is 1; that is arithmetic on three numbers, done on the blackboard.

Why this definition. Why these three numbers and not, say, the two amplitudes themselves? Because the amplitudes change with the global phase and these do not: each is a closed picture with the state at one end and its mirror image at the other, and whatever phase the state gains its reflection gives back (§2.6). The point on the sphere is the state with exactly the unseeable part removed.

---

§3.6 Turning

What does a box do to the point?

Start with an example. The arrow sits at $(\frac{24}{25}, 0, -\frac{7}{25})$. Send it through H: §2.5 already computed the amplitudes, $H|a\rangle = \frac{1}{5\sqrt{2}}\begin{pmatrix} 7 \\ -1 \end{pmatrix}$, so $\bar{\alpha}\beta = -\frac{7}{50}$ and $|\alpha|^2 - |\beta|^2 = \frac{49}{50} - \frac{1}{50}$, and the new point is

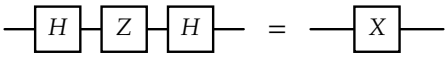
$$H|a\rangle : \ (-\tfrac{7}{25},\ 0,\ \tfrac{24}{25}).$$

The first and third coordinates have changed places. That is too tidy to be luck, and the reason can be found without choosing a state at all. The new x-coordinate is, by *The Mirror of a Row* and $H^\dagger = H$,

$$(\langle\psi|\, H)\, X\, (H\,|\psi\rangle) \ = \ \langle\psi|\, HXH\, |\psi\rangle\,,$$

the closed picture of the *old* state with the row H, X, H in the middle. If the new x is always the old z, the row H, X, H must be the box Z; and if the new z is the old x, the row H, Z, H must be the box X.

> **Theorem 3.4 (The Sandwich).** $HZH = X$, exactly. In pictures: the row of boxes H, Z, H on one wire is the box X.



Proof idea. H turns the standard question into the turned one and back again; in between, Z does to the turned basis what NOT does to the standard one (§3.4). This is the headline theorem of the chapter, and §3.8 proves it twice in full.

> **Corollary 3.5 (the other two sandwiches).** $HXH = Z$ and $HYH = -Y$, exactly.

Proof. Wrap both sides of Theorem 3.4 in H and use $HH = I$: $Z = H(HZH)H = HXH$. For $Y = iXZ$, slip $HH = I$ between the factors: $HYH = i\,(HXH)(HZH) = i\,ZX = -i\,XZ = -Y$, using $ZX = -XZ$. ■

On the blackboard. Now we can say what several boxes do to the point of

every state. Two families of boxes are needed, with a real number θ for a dial:

$$R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix},$$

$$R_x(\theta) = H\,R_z(\theta)\,H = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}.$$

Both are unitary: $R_z(\theta)^\dagger = R_z(-\theta)$ undoes $R_z(\theta)$, and the mirror image of the row H, $R_z(\theta)$, H is the row H, $R_z(-\theta)$, H, which undoes it box by box.

Proposition 3.6 (turns). Let $|\psi\rangle$ be any state, with point (x, y, z).

1. The point of $H|\psi\rangle$ is $(z, -y, x)$.
2. The point of $R_z(\theta)|\psi\rangle$ is $(x\cos\theta - y\sin\theta,\quad x\sin\theta + y\cos\theta,\quad z)$: the sphere is turned through the angle θ about the z-axis.
3. The point of $R_x(\theta)|\psi\rangle$ is $(x,\quad y\cos\theta - z\sin\theta,\quad y\sin\theta + z\cos\theta)$: a turn through θ about the x-axis.
4. If U is unitary, $e^{i\delta}U$ moves every point exactly as U does.

Proof idea. The new coordinate for the Pauli matrix P is $\langle\psi|\,U^\dagger P U\,|\psi\rangle$: wrap each Pauli matrix in the box and its mirror image, and read off the result in terms of the old Pauli matrices. *Proof sketch.* (1) is Theorem 3.4 and Corollary 3.5. (2) $R_z(\theta)^\dagger X R_z(\theta)$ has zero diagonal and off-diagonal entries $e^{i\theta}$ (top right) and $e^{-i\theta}$ (bottom left); so has $\cos\theta\,X - \sin\theta\,Y$, since $\cos\theta + i\sin\theta = e^{i\theta}$. Hence the new x is $x\cos\theta - y\sin\theta$. In the same way $R_z(\theta)^\dagger Y R_z(\theta) = \sin\theta\,X + \cos\theta\,Y$; and Z, being diagonal like $R_z(\theta)$, is left alone. (3) $R_x(\theta)$ is the row H, $R_z(\theta)$, H, so apply (1), then (2), then (1): (x, y, z) goes to $(z, -y, x)$, then to $(z\cos\theta + y\sin\theta,\ z\sin\theta - y\cos\theta,\ x)$, then to $(x,\ y\cos\theta - z\sin\theta,\ y\sin\theta + z\cos\theta)$. (4) In $(e^{i\delta}U)^\dagger P (e^{i\delta}U)$ the factors $e^{-i\delta}$ and $e^{i\delta}$ cancel. ■ See `Code/Ch3/Sandwich.py`.

So H is a **half-turn about the direction midway between the x- and z-axes**: that half-turn exchanges the two axes and reverses the one perpendicular to both, which is what $(z, -y, x)$ says. And our diagonal boxes are z-turns

with a phase in front:

$$Z = e^{i\pi/2}R_z(\pi), \qquad S = e^{i\pi/4}R_z(\tfrac{\pi}{2}), \qquad T = e^{i\pi/8}R_z(\tfrac{\pi}{4}),$$

$$X = e^{i\pi/2}R_x(\pi), \qquad V = HSH = e^{i\pi/4}R_x(\tfrac{\pi}{2}).$$

Z is a half-turn about the z-axis, S a quarter-turn, T an eighth; X is a half-turn about the x-axis and V a quarter-turn about it. The ladder $T^2 = S$, $S^2 = Z$ of §3.4 is an eighth and an eighth making a quarter, a quarter and a quarter making a half. The mirror image $S^\dagger$ is the quarter-turn the other way — the turn seen through the shop window. For a half-turn the two senses agree, and that is how X, Z and H come to be their own mirror images, and why they could never test a claim about mirrors.

The Sandwich can now be read on the sphere. A half-turn about z, performed between two exchanges of the x- and z-axes, is a half-turn about x: follow the point, $(x, y, z) \mapsto (z, -y, x) \mapsto (-z, y, x) \mapsto (x, -y, -z)$, and the last is what X does. And the sign in $XZ = -ZX$ is explained: a half-turn about x and a half-turn about z, in either order, are the same turn of the sphere, $(x, y, z) \mapsto (-x, y, -z)$, the half-turn about y; the two orders give boxes that differ only by the phase -1. For H and Z the two orders are different turns: ZH sends the point of $|0\rangle$ to $(-1, 0, 0)$, and HZ sends it to $(1, 0, 0)$.

One honest sentence about the half-angle. $R_z(2\pi) = -I$: a full turn of the sphere brings every point home, and the box that performs it is not the bare wire but the bare wire with a sign; it takes two full turns to get the box itself back. The sphere cannot see the difference, by part 4. Later chapters, with more than one wire, will.

Three turns

How much can be done with z-turns alone? Not much. $R_z(\theta)\,|0\rangle = e^{-i\theta/2}\,|0\rangle$: the point $(0, 0, 1)$ never moves. With x-turns alone, $(1, 0, 0)$ never moves. But take one of each kind, a quarter-turn apiece, in the row S, V, S. The first two boxes are our test piece, $VS = U$, so the product is SU; multiplying on

the left by S multiplies the second row by i:

$$SVS \;=\; SU \;=\; \tfrac{1}{2}\begin{pmatrix} 1+i & 1+i \\ i(1-i) & i(-1+i) \end{pmatrix} \;=\; \tfrac{1+i}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \;=\; e^{i\pi/4}\,H.$$

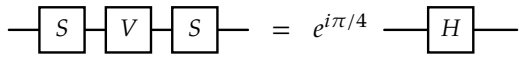
A quarter-turn about z, a quarter-turn about x, a quarter-turn about z, is the half-turn H — exactly, with the scalar $e^{i\pi/4}$ written out. Removing the phases carried by S and V, the same equation reads $H = e^{i\pi/2}R_z(\frac{\pi}{2})R_x(\frac{\pi}{2})R_z(\frac{\pi}{2})$. Reflect the whole equation: $S^\dagger V^\dagger S^\dagger = e^{-i\pi/4}H$, three quarter-turns the other way, and the same half-turn. So a box about a slanting axis can be had from boxes about the two upright ones. It needs three, and the row reads the same from either end. The general fact is this.

> **Theorem 3.7 (Three Turns; cited, not proved here).** For every 2×2 unitary matrix U there are real numbers $\alpha, \beta, \gamma, \delta$ with
>
> $$U \;=\; e^{i\delta}\,R_z(\alpha)\,R_x(\beta)\,R_z(\gamma).$$

We prove only the instance above. The general statement is the decomposition of a rotation into *Euler angles*; it is standard textbook material (Nielsen & Chuang, ch. 4), and the z–x–z form we use, together with the instance $U = H$ with all three angles equal to $\frac{\pi}{2}$, is stated in van de Wetering's survey (§3.6). The companion code does not prove it either, but it does test it: `Code/Ch3/ThreeTurns.py` computes the four angles from the four entries of a hundred unitaries drawn at random and checks the equation each time.

Granting the theorem, Proposition 3.6 says what every unitary box does to the sphere: three turns, one after another. Turning is all that a box on a single qubit's wire can do.

In the sketchbook. The instance is a picture equation with a scalar in it. A number beside a picture multiplies it, as in §2.6:



The scalar is part of the equation. Without it the two sides are different boxes that turn the sphere alike, and "turn the sphere alike" is a sentence about the sphere, not an equals sign between pictures. The sketchbook has, at this stage, no rewrite that would *find* such an equation; it records what the blackboard has multiplied out.

Why this definition. Why write $e^{-i\theta/2}$ and $e^{i\theta/2}$ in $R_z(\theta)$, when $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$ is simpler and turns the sphere identically? For the symmetry: with the half-angles shared out evenly, $R_x(\theta) = HR_z(\theta)H$ comes out with plain cosines and sines, and the mirror image is simply $\theta \mapsto -\theta$. The simpler matrix is the one the phase boxes S and T use, and the price is the scalars $e^{i\pi/4}$ and $e^{i\pi/8}$ above. Both conventions are in use; one must say which, every time.

---

§3.7 The Rosetta Table

This chapter adds seven rows to the running dictionary, numbered on from Chapter 2's fifteen. Row 16 is the chapter's one new picture element. Row 17 is the rule for reflecting a whole picture. Rows 18 to 21 are things that can now be said without any new element, and row 22 is the headline theorem.

#	Algebra (Dirac)	Picture (Penrose)
16	$A^\dagger$, the conjugate transpose	the **mirror image of the box** A, labelled $A^\dagger$: input and output change places, and the mirror conjugates — *new element*
17	$(BA)^\dagger = A^\dagger B^\dagger$; $(A\,\lvert\psi\rangle)^\dagger = \langle\psi\rvert\, A^\dagger$	the mirror image of a row: every piece mirrored, the order reversed
18	$\langle i\rvert\, A\,\lvert j\rangle = A_{ij}$	state j, box A, effect i: a closed picture, a number

#	Algebra (Dirac)	Picture (Penrose)
19	$U^\dagger U = I = UU^\dagger$ (unitary)	***a box meets its mirror***: in either order, a bare wire
20	$A^\dagger = A$ (here X, Y, Z, H)	a box that is its own mirror image
21	$H\|0\rangle = \|+\rangle$, $H\|1\rangle = \|-\rangle$; $\langle 0\|H = \langle +\|$, $\langle 1\|H = \langle -\|$	state 0, then H = state $+$; H, then effect 0 = effect $+$ (likewise 1 and $-$)
22	$HZH = X$	the row H, Z, H = the box X

All seven rows are exact. There is no row for the sphere: it is a way of reading three closed pictures of the kind in row 18, and not a picture element.

---

§3.8 The Theorem, Twice

We have met the theorem through its consequence: the arrow's first and third coordinates changed places under H (§3.6). Here is the statement.

Theorem 3.4 (The Sandwich). Let $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then

$$HZH = X,$$

exactly, scalar included. In pictures: the row of boxes H, Z, H on one wire is the box X.

Dirac's Blackboard

Two matrix products. First ZH; multiplying on the left by Z changes the sign of the second row:

$$ZH \;=\; \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \;=\; \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

Then H times that, along the rows and down the columns, with $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$ in front:

$$H(ZH) \;=\; \tfrac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

$$= \tfrac{1}{2} \begin{pmatrix} 1\cdot 1 + 1\cdot(-1) & 1\cdot 1 + 1\cdot 1 \\ 1\cdot 1 + (-1)(-1) & 1\cdot 1 + (-1)\cdot 1 \end{pmatrix} \;=\; \tfrac{1}{2} \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix} \;=\; X.$$

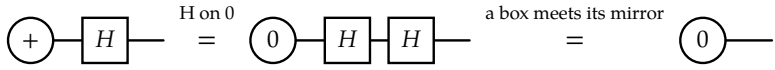
Two products, eight multiplications in the second, and the two factors $\frac{1}{\sqrt{2}}$ meet to cancel the 2. ■

Penrose's Sketchbook

The sketchbook owns no rewrite that turns three boxes into one. What it owns, and where each thing comes from, is this. *H on 0*, *H on 1*, *Z on +* and *Z on −* (§3.4) are **lent by the blackboard**: each is a column computed there and recorded as a picture. *A box meets its mirror* (row 19) is this chapter's own rule, and with row 20 (H is its own mirror image) it says that two H boxes in a row are a bare wire. From Chapter 1 we have *NOT on 0* and *NOT on 1* (§1.1), which with NOT labelled X read "state 0, then X, is the state 1" and "state 1, then X, is the state 0"; and the principle ***two states suffice*** (§1.7), also lent by the blackboard: two rows of boxes that agree after the state 0 and after the state 1 are equal.

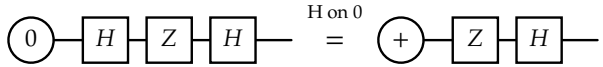
Step 0 — *H on +*. One more small equation is needed, and the chapter's own rule supplies it. Read *H on 0* backwards to replace the state +; then

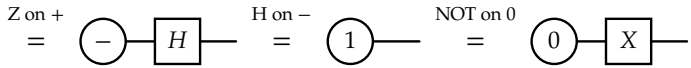
two H boxes are a bare wire.



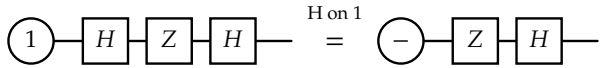
In the same way, from *H on 1*: the state $-$, then H, is the state 1. Call these ***H on +*** and ***H on −***.

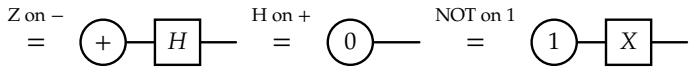
Step 1 — follow the state 0**.** Place the state 0 before the row and replace, three times, a state and the box after it by a single state.



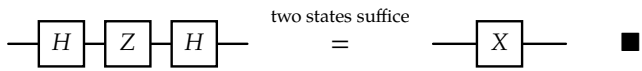


Step 2 — follow the state 1**.** The same, with the signs the other way round.





Step 3 — *two states suffice*. The row H, Z, H and the box X give equal pictures after the state 0 (Step 1) and after the state 1 (Step 2). So they are equal:



■

Every equals sign above is exact. The states passed through were $0, +, -, 1$ in Step 1 and $1, -, +, 0$ in Step 2, the same four in the opposite order. §3.9 draws that.

The verdict

The algebra wins, and it is not close. The blackboard needs two products of 2×2 matrices. The sketchbook needs eight replacements and a closing principle, and of the facts it uses only one, *a box meets its mirror*, is its own; the columns of H, the action of Z on $\pm$, and *two states suffice* are all on loan from the blackboard. At this stage the sketchbook has no rule that acts on H, Z, H as boxes. A later chapter brings a notation in which this equation is a single move; we do not have it yet, and we shall not pretend to.

What the longer proof gives is a reason, which the matrix product does not. The first H turns the standard question into the turned one. In the turned basis, Z is a NOT. The second H turns back. So "NOT" is not the name of a matrix but of a job, exchanging the two answers of a question, and Z and X are the same job done for two different questions. The blackboard proves the theorem and shows none of this.

Feynman's Notebook.

The adjoint as a mirror image. That the dagger of a picture is its reflection is not this book's conceit. Selinger's survey of graphical languages (§7) sets out the general notion, a *dagger category*, in which every map has an adjoint going the other way, with exactly the three laws of our §3.2: the adjoint of doing nothing is doing nothing, the adjoint of a composite is the composite of the adjoints in the reverse order, and the adjoint of an adjoint is the original. He states the graphical rule in one line: the adjoint of any diagram is its mirror image. He also notes the difficulty we met, that a box's reflection can be told from the box only because the boxes carry a mark in one corner. The same section defines a unitary map as an invertible one whose inverse is its adjoint. Abramsky and Coecke (2004, §7.1) have the same definition and prove from it, in their abstract setting, that unitary maps preserve the inner product: our Theorem 3.2, without a matrix in sight.

§3.9 The Turn in Three Steps

Here is the chapter's master diagram. We call it **The Turn in Three Steps**. It is Step 1 of the sketchbook proof with every intermediate state *asked whether it is what we said it was*. On one row, from the left: the state 0, the box H, the effect $+$; bare paper; the state $+$, the box Z, the effect $-$; bare paper; the state $-$, the box H, the effect 1. Each wire is closed at both ends and carries one box, and nothing joins an effect to the state that follows it.



$$= \quad 1 \cdot 1 \cdot 1 \;=\; 1$$

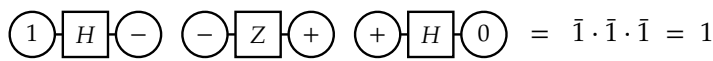
The equation is exact. Closed pictures in a row multiply (§2.3), and the three factors are

$$\langle +|H|0\rangle = \langle +|+\rangle = 1, \qquad \langle -|Z|+\rangle = \langle -|-\rangle = 1, \qquad \langle 1|H|-\rangle = \langle 1|1\rangle = 1,$$

by *H on 0*, *Z on* $+$ and *H on* $-$ followed each time by Theorem 2.1. Each 1 is a Born certainty: the state 0, after H, is $+$ for sure; $+$, after Z, is $-$ for sure; $-$, after H, is 1 for sure. The number 1 is also the entry $\langle 1|X|0\rangle$ of the box X, as it must be by *The Sandwich*. But we draw no equals sign between this row and the closed picture with all three boxes on a single wire between the state 0 and the effect 1. That is a different picture, and at this stage only the blackboard can say why it has the same value: between its boxes it carries whatever state there is, and here we happen to have named each one. As with *The Echo*, we prove nothing again, and read the picture three times.

First reading: as drawn. 0 to $+$ to $-$ to 1: the first column of X, in three certainties. Value 1.

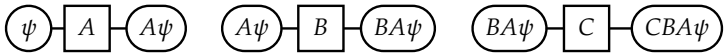
Second reading: in the mirror. Reflect the whole row by the rule of §3.2: every piece mirrored, the order reversed. States become effects and effects states; H and Z are their own mirror images; the last wire becomes the first.



$$= \quad \bar{1} \cdot \bar{1} \cdot \bar{1} \;=\; 1$$

Its value is the conjugate, which for the number 1 is 1 again. And look at what it says: 1 to − to + to 0. That is Step 2 of the sketchbook proof, the *other* column of X. The picture and its reflection together are the whole proof of *The Sandwich*. The theorem is a row that reads the same from either end, and so is its proof.

Third reading: any three boxes that keep the promise. Replace 0 by any state ψ, and H, Z, H by any three unitary boxes A, B, C; label each effect with the state that ought to arrive there.



$$= \quad 1 \cdot 1 \cdot 1 \quad = \quad 1$$

Here a rounded box labelled $A\psi$ stands for the state $A\,|\psi\rangle$, or for its mirror image when the wire is on its left. Why is the first factor 1? The effect labelled $A\psi$ is the mirror image of "state ψ, then A", which by *The Mirror of a Row* is "$A^\dagger$, then the effect ψ". So the first wire is the closed picture at the head of §3.3,

$$\langle\psi|\, A^\dagger A\,|\psi\rangle \;=\; 1,$$

by *a box meets its mirror* and Theorem 2.1. The other two wires are the same argument with $A\,|\psi\rangle$ and $BA\,|\psi\rangle$ in the place of $|\psi\rangle$; they are states because A and B keep the promise. To make the reading an honest test of the dagger, take boxes that are not their own reflections: $A = S$, $B = U$ (our test piece, neither real nor symmetric), $C = T$, and ψ any state at all. Every factor is still 1. A cheaper reflection fails at the first wire: with the transpose in the place of the mirror, and the arrow for ψ, the first factor would be $\langle a|\, S^T S\,|a\rangle = \langle a|\, Z\,|a\rangle = -\frac{7}{25}$.

Now take a box that does not keep the promise. With $A = S_0$ and the arrow, the first wire is "state a, box S_0, effect $S_0 a$", and its value is $\frac{49}{25}$: not a certainty, and not a chance of any kind. This third reading is the answer to §2.10.

The boxes that keep every wire of this picture equal to 1 *are the unitary ones.* See `Code/Ch3/Sandwich.py`, which evaluates all three readings.

Penrose's Sketchbook.

What the gaps are. The bare paper between an effect and the next state is part of the picture. Left of a gap, a wire has ended in a question; right of it, a new wire begins in a state. That the new state has the same label as the old question is our choice, made because the answer was certain. It is not something the picture enforces. Nothing flows across a gap.

This is why the row is not the sandwich itself. Join the wires into one by erasing the inner effects and states, and you have the closed picture "state 0, then H, Z, H, then effect 1", in which the boxes pass along whatever is on the wire, certain or not. For these labels, and for the third reading, the two pictures have the same number. For boxes in general, with intermediate questions whose answers are *not* certain, cutting a wire and asking changes the number. What cutting and asking does to a wire is a later chapter's subject. We have been careful to cut only where the answer was already sure.

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§3.10 A Second Wire

Let us gather what we have. A box on a closed wire is a number, and four such numbers are the box (§3.1). The mirror image of a box is forced by what Chapter 2 said about mirrors: the conjugate transpose, and neither of the two cheaper things (§3.2). The mirror image of a row reverses the order *and* reflects each piece (§3.2). The boxes that send every state to a state are exactly those that, meeting their reflection, leave a bare wire (§3.3). Six such boxes have names; four are their own reflections, and for that very reason can test nothing about reflections (§3.4). A state, stripped of the phase no question sees, is a point on a sphere (§3.5). Each of our boxes turns that sphere; H exchanges two of its axes; and, granting one cited theorem, every

box on a qubit's wire is three turns (§3.6). The row H, Z, H is the box X; the blackboard proves it in two products, and the sketchbook, with borrowed facts, in three steps and their reflection (§3.8, §3.9).

All of this comes to one sentence, and it answers the question with which Chapter 2 closed. **A box may sit on a qubit's wire exactly when its mirror image is its undoing.** Every such box can be run backwards by whoever stands on the other side of the glass. SET-0 is not such a box. It sends many states to one, and its reflection would have to send that one state back to many. We said in Chapter 1 that a qubit is a thing not looked at, and we can now add that everything done to it on the wire is a thing that could be undone. Whatever looking is — and we have still not said — it is not a box of this kind, for no one has ever unlooked.

Now notice what none of our pictures has. So far every diagram in this book has lain along a single line. When we needed more than one wire, in §3.9, we set them end to end, with bare paper between, and their numbers multiplied; nothing passed from one to the next. But a coin is seldom alone. Put a second coin on the table beside the first. Is its wire drawn beside the first wire — above it, below it? What is the column for two coins: two entries and two more, or something else? If a box sits on one wire and nothing on the other, what is the matrix of the pair? And if two boxes sit on two wires, does it matter which we think of as acting first?

One wire, we have learned, is a sphere, and all that can be done to it is to turn it. We leave the sphere turning. The next chapter lays a second wire beside the first.

Yoked Inventions

糸と糸
並べて張るや
春隣

ito to ito
narabete haru ya
haru tonari

A string, and a string,
stretched taut and laid side by side —
spring, one door away.

[*The snow is going. It has drawn back from the foot of the Tortoise's wall and from the sunny side of every post, and what is left of it in the yard is grey and full of holes; the path to the gate is wet earth. Along the porch eave hangs a row of icicles, and at the end the sun reached first one of them has begun to drip — a drop, a long wait, another drop — as though the thaw had not yet made up its mind. The rest hang dry. Out in the yard the old pear tree is bare; across the yard the woodshed roof, and the roof of the lean-to under it, still carry their snow.*]

[*Achilles comes up the porch steps; the Tortoise is unfolding a wooden frame onto the table.*]

Achilles: The whole hillside is running water this morning — your snow has gone to soup, and I came the last stretch more sliding than walking.

Tortoise: It does that, about now; by noon the path will be firmer than it

was at dawn. Sit and warm your hands, the kettle is hot. Though I did hope you might come early enough to be useful.

Achilles: Useful is never a restful word from you. What is this you are unfolding?

Tortoise: A frame for stretching strings across — two uprights, a row of hooks down each, and a string with a loop at either end to go between a facing pair. It lets us set one thing beside another and follow them together, in place of one at a time. And here beside it a stiff card, to stand a coin behind, so that we speak of its honest shares and not of the face it happens to show.

Achilles: One string and one coin I can manage before the tea has cooled. Hook it up, and let me begin on ground I know.

First Challenge — *A String Below*

[*The porch is open to the yard along its front and has the house wall at its back. The door has a window in it, a bar across and a bar down, all clear glass but the upper pane on the hinge side, which is old green bottle-glass; where Achilles stamped his boots before it there is a small puddle on the boards. On the wall beside the door hangs Escher's* Three Spheres II *in a narrow frame: three balls in a row on a table top, the left-hand one letting the light straight through it, the right-hand one giving none of it back. The shadow of the corner post lies across it just now. Against the end rail leans a glazed light off the cold frame, waiting for warmer nights, and on the rail itself stands a seed tray parted into cells by slats. The table is at the front, in the sun.*]

[*Late winter, the first thaw; a clear morning; the Tortoise's porch. On the table stands a small folding wooden frame out of her shell: two uprights with a column of hooks down each, and a silk string, a loop at either end, stretched left to right between the top hooks. Achilles walks a coin along the string, from the left upright to the right.*]

[*The frame does not stand bolt upright. A strut behind lets it lean far back, as a*

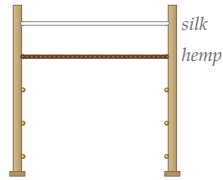
reading-slope leans, so that a coin on the table lies under its string. Its left upright is the one towards the porch steps; Achilles sits on the side of the top hooks and the Tortoise on the side of the bottom ones, and there are more hooks down each upright than the morning looks likely to want. The silk is the finest thing on the porch: drawn tight, it takes the sun along its whole length and shows as hardly more than a line of light.]

Achilles: Heads at the left, heads at the right. A bare string says nothing; I remember that much. One string, one coin, and a list of two lines for it — a line for heads, a line for tails. I could do this in my sleep.

[*Achilles settles back, both hands round the hot cup. The kettle ticks as it cools on its trivet.*]

Tortoise: [*She loops another string over the next pair of hooks down, sets up a card on edge as a screen, takes Achilles's coin behind it, and there tosses it, and two pennies after it.*] A string below: hemp. A coin on it, behind the card: tails, if both my pennies fell tails, and heads otherwise. And what may honestly be said of that coin: heads, three times in four.

Achilles: [*She has laid the tossed coin behind the card at the silk string's left end.*] A string above: silk. A coin on it, behind the card: as it fell. And what may honestly be said of that coin: heads, one time in two.



a string below

[*The hemp is the colour of a doormat, and hairy where the silk is smooth; the sun makes nothing of it. It runs a hand's breadth below the silk from hook to hook and touches it nowhere.*]

Tortoise: Write down what you know of the pair.

Achilles: [*writing on two slips of paper*] Two lines for the upper coin: a half, a half. Two lines for the lower: three-quarters, a quarter. I keep two little

lists, one under the other, as the strings are.

Tortoise: Then read me off the share of heads above and tails below, together.

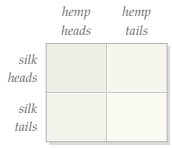
Achilles: Together. Well — a half from the upper list, a quarter from the lower: three-quarters?

Tortoise: And heads above with heads below?

Achilles: A half and three-quarters: one and a quarter. More than the whole morning. No. My two lists have no line for it, and adding won't make one. A list of my shirts and a list of my hats will never tell you how often the blue shirt goes out with the straw hat.

Tortoise: Then we want something with a place for every such pairing at once. This plain sheet will serve: folded across and across, it falls into panels, and we can give one panel to each way the two coins might come up.

Tortoise: [*She takes the blank sheet — one whole face — and folds it once across the middle; the sheet is no smaller folded, but the crease has scored it into an upper half and a lower half.*] For the silk: above the crease it shows heads, below it tails. [*She folds it again, this time end to end, and opens it flat: the outline is unchanged, yet the two creases have quartered the single face into four panels.*] For the hemp: left of the crease heads, right of it tails. Four panels. Fill them.



a panel for each pairing

Achilles: Upper right is the one you asked for. In half of all mornings the silk coin shows heads; in a quarter of *those* the hemp shows tails. A half of a quarter: one-eighth.

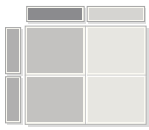
Tortoise: And its neighbours?

Achilles: [*writing in each panel, and pencilling it in as dark as its share*] Upper

left, a half of three-quarters: three-eighths. Below them the same again: three-eighths, one-eighth. Eight eighths. They make one whole, which my two lists never did!

Tortoise: What would you call it?

Achilles: [*He lays the silk's slip down the left edge of the sheet and the hemp's along the top; the panels stay as they are, and each now has its row's line beside it and its column's line above.*] A patchwork. Every panel is its row's share times its column's share, and the two little lists sit along the edges like a border.



a patchwork

[*On the rail the seed tray stands in the sun. The Tortoise has stuck a label in the earth at the head of each column, for what was sown there, and one at the end of each row, for the week it was sown in. The labels go round two edges and are soon read; the cells fill the tray. Nothing is up in any of them yet.*]

Tortoise: And had I given you reaches, not shares?

Achilles: Reach times reach in every panel, and the squaring after. It comes to the same patchwork.

[*The Tortoise takes down the card. The silk coin shows tails; the hemp coin, heads. Achilles puts a finger on the lower left panel.*]

Achilles: So what you're saying is, the pair wants a line for every panel, and the panels go by strings. Two and two — four. So a third string makes six.

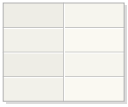
Tortoise: Fold it.

Second Challenge — *Sixteen Panels*

Achilles: [*The Tortoise hands him a string out of her shell; he loops it over the next hooks down, as she did.*] A string below: gut. And a fold for it. [*He takes a fresh sheet and folds it as he speaks, the last crease going across again and turning the whole folded square back over on itself.*] Once across for the silk. Once the other way for the hemp. Once more for the gut. Six panels. [*He opens it flat and runs a finger down the rows of panels, two to a row, counting.*] Two, four, six... eight.

Tortoise: Count them again, by all means.

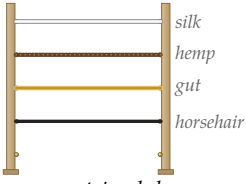
Achilles: Eight. ...Oh. The new fold didn't add two panels at the edge. It went through every panel that was already there.



eight, not six

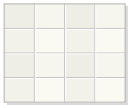
[*Out in the yard the pear tree is bare, and with the snow off it one can see how the Tortoise has pruned it all these years. The trunk parts in two at the height of the wall; each limb parts in two above that, and each branch again, and again. Nobody has ever thought of counting the twigs.*]

Tortoise: [*She loops another string over the next free hooks; it settles straight beneath the others, and not one of the strings already hooked so much as stirs.*] A string below: horsehair. And a fold for it.



a string below:
horsehair

Achilles: [*folding the sheet over on itself again, and once more, end to end this time; it is growing thick, and each fold splits every panel there is anew*] A fourth fold. [*He opens it out and counts along the rows.*] Sixteen. And a fifth would be thirty-two, and I can no longer crease the paper. Not two more for each string, then: twice as many for each string. It's a row of lamps, each lit or dark: put one more lamp on the end, and every way the row could look splits in two.



sixteen

[*The gut is yellow and hard, a little clear at the edges like a sliver of horn; it gave a small dry note when Achilles's thumbnail caught it in the hooking. The horsehair under it is dark and dull and gives no note at all. Below the horsehair the hooks go on down the uprights, empty.*]

Tortoise: And if the coins are separate, as ours were — each tossed by itself behind the card?

Achilles: Then a little list for each would do, along the borders. Two lines for the silk coin, two for the hemp, two for the gut, two for the horsehair: eight lines, and I multiply across to fill the panels.

Tortoise: Eight lines in all. The sheet has sixteen panels. And not every sheet is a patchwork.

Achilles: What else could a sheet be? You folded it; I multiplied.

[*The Tortoise does not answer. She puts the blank sheet by, keeps the sheet of the first folding at hand, and takes from her shell a small box of wooden clips. Each has a rule written on it; the two that say turn it over are of plain pale wood, the two that say heads up are stained dark, a small head painted on each.*]

Tortoise: You have met these — the little clips, each with its rule. This one says turn the coin over; that one says set it heads, whatever it showed. Pinch

a clip onto a string, and it does its rule to that coin as it passes, and to no other coin at all.

[*The sun has come along the eave. Where one icicle dripped there are now a dozen at it, each in its own time — quick ones, slow ones, and one that only gathers and gathers — and together they make a patter that nobody could beat time to. Under them the snow along the foot of the porch is beginning to be pitted.*]

Achilles: And a clip? A clip knows nothing of panels. It sits on one string.

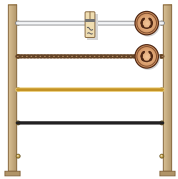
Third Challenge — *One Clip, Nothing on the Other*

Tortoise: [*She takes one clip and pinches it onto the silk string, halfway along; the hemp string below it is left bare.*] A clip on the silk: "turn it over". Nothing on the hemp. What becomes of the sheet?

Achilles: The hemp coin is untouched, so half the sheet is untouched. And better: the silk coin was half-and-half, and turned over it is half-and-half still. Three-eighths, one-eighth above; three-eighths, one-eighth below. The whole sheet is untouched!

Tortoise: The numbers are alike. Are the panels? Lay the coins where you can see them, and walk them. [*She gives him a blue pencil.*] Mark where each panel goes.

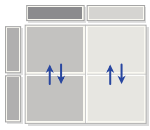
Achilles: Heads above, heads below. [*He sets a coin on each string and walks the pair along together, level, from the left upright; at the clip the silk coin is flipped, head going under and wreath coming up and the hemp coin below passes it untouched.*] Upper left goes to lower left. [*He draws a blue arrow on the sheet.*] Heads above, tails below: arrives tails above, tails below. Upper right goes to lower right.



over on the silk

Tortoise: And from the lower panels?

Achilles: Up. Each to the panel above it. [*Two more walks, two more arrows; and every arrow he has drawn runs straight up or straight down, not one of them sideways.*] Every panel has moved! The halves made by the first fold have changed places, upper for lower. The shares looked the same only because the two halves were twins.



blue: over
the first fold

[*By the door, where Achilles stamped his boots, the puddle has the door's window in it, standing on its head. The green pane, which is at the top on the hinge side, lies at the bottom on the hinge side, and every clear pane has gone over the bar across with it. A drop blown in from the eave shakes the whole window to pieces, and it comes back the same.*]

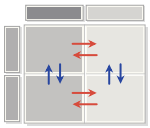
Tortoise: Your turn to say it.

Achilles: A clip on the hemp: "turn it over". Nothing on the silk.

[*The Tortoise lifts the clip off the silk and pinches it onto the hemp instead; now it is the lower coin the rule will turn. She gives him a red pencil, and he walks the coins from each panel in turn.*]

Achilles: [*At the clip it is now the hemp coin that turns, and the silk coin above passes it untouched; he draws a red arrow for each walk, and every red arrow runs*

sideways, from a panel to its neighbour in the same row.] Upper left and upper right trade places; below the crease, the same. Every panel moves again — but this time nothing crosses the first fold. The panels change places *inside* each half.



red: inside each half

[*The glazed light leaning on the end rail has the same window in it, right way up this time: the green pane at the top still, and on the latch side. In the puddle every pane went over the bar across; in the glass every pane has gone over the bar down.*]

Tortoise: One clip; one bare string; and the sheet?

Achilles: Moved all over, both times, and differently. There isn't a panel anywhere that belongs to one coin. A panel is like a seat in a theatre, a row and a number: move everyone to another row, and every seat is another seat, though each keeps its number. [*The Tortoise moves the clip back to the silk string.*] Then put one on each. — Which goes first?

Fourth Challenge — *Which First?*

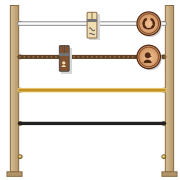
Tortoise: [*She clips a second clip to the hemp string: "heads up, whatever it was".*] One on each.

Achilles: But which do the coins meet first? It makes a difference; I have it in my notes. On one string, "turn it over" and then "heads up, whatever it was" arrives heads; the same two in the other order arrive tails. I shall write out both orders. [*He begins a column on the back of the blank sheet.*]

Tortoise: [*She slides her clip along the hemp string until it stands well to the left of his; it runs freely the whole way, for his clip rides the silk above and there is*

nothing on the hemp to check it.] Walk them. Heads above, tails below.

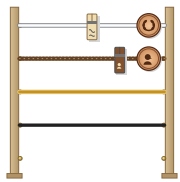
Achilles: [*walking the coins, level*] The hemp coin meets its clip first: heads up. Then the silk coin meets its own: over. They arrive tails above, heads below.



hers to the left

Tortoise: [*She slides her clip rightward and it passes clean under his — the two never so much as graze, being on strings one above the other — and comes to rest beyond it.*] I slide it past yours. Again.

Achilles: [*walking the coins, level, from heads above and tails below: now the silk coin reaches its clip and is turned before the hemp coin reaches hers and is laid heads up; the clips have not changed, only which one a coin meets first*] The silk coin first, this time. The same!



slid past his

Tortoise: [*She slides her clip back until it stands exactly under his.*] And level.

Achilles: Both at once. Tails above, heads below.

[*He walks the coins from every other pair of starts, with her clip left, right and level. The same faces arrive each time. His column of orders is left unfinished.*]

[*At the corner of the eave two icicles hang side by side, and each has worn its own hollow in the snow beneath. Now the left one lets go a drop first, now the right, now the two together, and there is no telling which it will be; the hollows deepen*

alike. All along the foot of the porch the pitting has become a row of such hollows, one to an icicle.]

Achilles: Those were single clips. With more of them there must be an order somewhere.

Tortoise: [*She adds a second clip to each string, to the right of the first: on the silk "heads up, whatever it was", on the hemp "turn it over". The first clips stand level, and the second clips stand level.*] Take the silk; I shall take the hemp. We walk together, and each of us says what is done.

Achilles: Silk. The coin starts tails up.

Tortoise: Hemp. The coin starts heads up.

Achilles: Silk, the first clip: turn it over. Heads.

Tortoise: Hemp, the first clip: heads up, whatever it was. Heads.

Achilles: Silk, the second clip: heads up, whatever it was. Heads still.

Tortoise: Hemp, the second clip: turn it over. Tails.

Achilles: Silk arrives heads.

Tortoise: Hemp arrives tails.

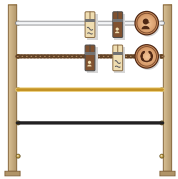
Achilles: Both first clips, and then both second clips. So there *was* an order, after all.

Tortoise: Set the coins back. Now walk all of the silk before I stir.

Achilles: Tails; over, heads; heads up, heads. Heads, as before.

Tortoise: Heads; heads up, heads; over, tails. Tails, as before.

Achilles: The same again. First both firsts and then both seconds; or all of the silk and then all of the hemp. But between the two walks nobody touched a clip. ...Oh. Nothing was moved. It was one frame, and I read it two ways.



nothing moved

[*He looks up from the frame. On the wall beside the door hangs a print — Escher's* Three Spheres II.]

[*The shadow of the corner post, which lay over the print when they sat down, has moved off along the wall, and the sun is full on it. For the length of his look no drop falls anywhere along the eave.*]

Achilles: Your three balls stand side by side on their table just so, and nobody asks which of them comes first. Though the middle one has both its neighbours in its shine.

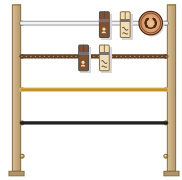
Tortoise: [*On the silk string she pushes the "turn it over" clip to the right, towards its neighbour.*] Now I slide this one past that one.

Achilles: It won't go. They knock together. They are on the same string!

[*Down any one icicle the water goes in file. The drop behind cannot get by the drop in front, and they come to the tip in the order they set out.*]

Tortoise: [*Because both clips share the one silk string, neither can pass the other; she lifts the "turn it over" clip clean off the silk and pinches it on again beyond its neighbour — where a moment ago, on two strings, the sliding had needed no lifting at all.*] It must come off, and go back on the far side. Walk the silk. Tails up.

Achilles: [*He lifts the hemp coin off the frame and lays it aside, and walks the silk coin alone from tails: at the dark clip it is laid heads up, the wreath going under; at the pale clip beyond it, it is turned, and the wreath comes up again. The hemp clips have not moved; only the silk's two have changed places.*] Silk arrives *tails*, and a moment ago it arrived heads. So the old fact stands: along one string there is an order, and it is part of what the string says.



off and on again: tails

Tortoise: [*She takes the clip off once more and restores it to the left of its neighbour.*] And between one string and another?

Achilles: None to find. It's stocking and boot: on the one foot, stocking before boot, but the left foot and the right in any order one pleases. So strings keep to themselves. They never so much as touch.

[*Achilles sits back, pleased with himself, and lets the sun have his hands a moment. The boards beneath him, where it has lain all morning, have gone dry and warm.*]

[*The Tortoise reaches down to the right-hand upright and unhooks a loop.*]

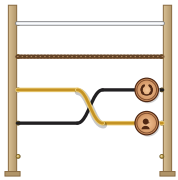
Fifth Challenge — *The Crossing*

Tortoise: Take the clips off, Achilles. We want bare strings.

[*Achilles takes the clips off the silk and the hemp and puts them back in their box, and lifts the silk coin off its hook to lie with the hemp coin. On the lower pair the gut runs straight along its own row, the horsehair straight along the row beneath; the gut's right-hand loop is already off its hook in the Tortoise's fingers, and she takes the horsehair's in her other hand.*]

Tortoise: I unhook the right-hand loops of the gut and the horsehair. The upper comes down in front, the lower goes up behind, and each takes the other's hook. [*She does so: the left-hand ends never move, and neither string is cut or joined to the other, yet the two right ends trade hooks — the gut's onto the lower hook, the horsehair's onto the upper — so the pair exchange their two rows and cross once, midway, flat, the one lying against the other.*] Walk a coin on each. A coin keeps to its string.

Achilles: [*He sets a coin heads up on the gut and a coin tails up on the horsehair, and walks them along, each following its string through the crossing: the heads coin goes down with the gut to the lower row, the tails coin up with the horsehair to the upper, and neither is turned on the way.*] They have changed hooks, and neither has changed its face.



crossed once

[*Two sparrows have the porch rail to themselves, a cock and a hen. The cock hops past in front of the hen, the hen shuffles along behind him, and when they have settled each is at the other's end of the rail, and the same bird as before.*]

Tortoise: [*She clips "turn it over" to the gut string, to the left of the crossing.*] Both heads up, this time.

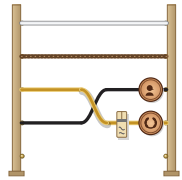
Achilles: The gut coin is turned, then goes down through the crossing: tails, on the horsehair's hook. The other arrives heads, on the gut's hook.

Tortoise: [*She slides the clip rightward along the gut; at the crossing she lifts the horsehair a finger's breadth, so the clip passes clear of it and stays with the gut, coming out on the lower row — its row changed, its string the same.*] And now?

Achilles: Now the clip is on the lower row, so it will turn the horsehair coin.

Tortoise: Which string is it clipped to?

Achilles: [*following the string back with a finger*] The gut. Still the gut. [*He walks the coins, both heads up: the gut coin comes down through the crossing first and meets the clip only beyond it, where it is turned, arriving tails on the horsehair's hook; the horsehair coin goes up untouched to the gut's hook, heads.*] As before the slide, face for face. The clip kept its string; the string changed its place.



slid through

Tortoise: [*She takes the clip off and puts it in the box.*] Your turn to say it.

[*The lower pair still lies crossed once: at the right-hand end the horsehair is uppermost now, the gut beneath it.*]

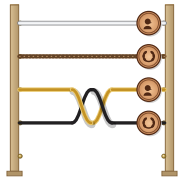
Achilles: I unhook the right-hand loops of the gut and the horsehair. The upper comes down in front, the lower goes up behind, and each takes the other's hook. [*He does so, the same hand-movement as before — but the loop now uppermost is the horsehair, so this time the horsehair passes in front and the gut behind; each string has now gone in front just once. The right ends come home to their own hooks, and still the pair will not lie flat.*] There: back on their own hooks, and — they are wound round each other.

Tortoise: Pull them straight.

Achilles: [*He draws the strings taut. He plucks at them.*] It won't pull out. And I said what you said and did what you did!

Tortoise: You did. Walk a coin on each.

Achilles: [*One coin heads up on the gut, one tails up on the horsehair; each follows its string through the one crossing, changing rows, and through the next, changing back, and neither is turned.*] Their own hooks, their own faces. [*He walks two coins, heads on the silk and tails on the hemp, along the bare silk and hemp.*] And along a straight pair: own hooks, own faces. The same report, word for word.



the same report

[*On the rail the sparrows have changed ends again, the hen going by in front this time. Each is back where it began.*]

Tortoise: [*On a scrap of paper she draws two lines side by side that change places, flat, and further along change places again. She rubs the middle out, and draws the two lines straight.*] I have only the one kind of crossing on paper, and I keep a rule for it: two of them, one after the other, I may draw straight.

Achilles: But the strings are still wound! The paper doesn't make them straight, and no pulling does. Your rule is false of the string.

Tortoise: Of the string, yes. The string remembers which went in front. The coin cannot tell. I draw for the coin.

Achilles: Then it is a rule one has to earn, by walking the coins; the paper won't vouch for it. A letter can tell you which door it came to, but not which way round the square the postman went. [*He reaches for the loops.*] Shall I unwind them?

Tortoise: Leave them wound.

Achilles: Very well. Crossed or not, each coin still minds only its own string.

[*Achilles flexes his fingers, stiff from the cool of the boards. The sun by now has come round far enough to warm his back.*]

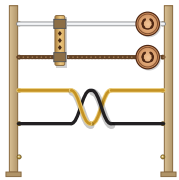
[*The Tortoise takes a short wooden bar out of her shell. It has a clip at either end.*]

Tortoise: A bar of wood, a clip at either end — long enough to reach across two neighbouring strings, and no further. Where a clip lays a rule on one string, this lays a single rule across the pair at once.

Sixth Challenge — *The Yoke*

Tortoise: [*She gathers up the coins lying at the right-hand hooks of the gut and the horsehair and puts them by, then lays the bar across the silk and the hemp near the left upright, a clip biting each string — just long enough to span the two and no more; both strings run on beneath it unbroken, a single line through each clip. She reads what is burnt into the wood.*] "If the upper coin shows tails, turn the lower one over."

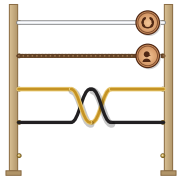
Achilles: [*He walks the pairs through it, the last of them tails above and heads below.*] Heads above: the lower coin goes through as it came, heads or tails. Tails above: the lower coin is turned. The upper coin is never touched. That is two clips nailed to a stick.



tails above

Tortoise: [*She unclips the bar and pushes the box of clips towards him.*] Then do it with two clips.

Achilles: On the silk, nothing; the upper coin is never touched. On the hemp: "turn it over". [*He clips it on, and walks.*] Tails above, heads below: arrives tails, tails. Right! Heads above, heads below: arrives heads, tails. Wrong. [*He takes the clip off, and walks tails above, heads below.*] And with no clip, right for heads above and wrong for tails. I want a cleverer clip than you keep in the box.



no clip: wrong for tails

Tortoise: [*She clips the bar on again and stands the card on edge between the silk and the hemp, running it from the left upright only as far as the bar — for the bar itself blocks the way the card would go on. He stands on the silk side of it, she on the hemp side; from her seat the silk coin cannot be seen at all, and from his the hemp coin is hidden the same.*] Write me its rule. It sits on the hemp string, on my side of the card.

Achilles: "If the upper coin—"

Tortoise: Which coin is that? I see one string from here.

Achilles: "If..." No. Whatever I write on it, it must turn your coin on some mornings and leave it on others, and it cannot see which morning it is. And the card cannot go any further along, because the bar goes through where the card would stand. ...Oh. The stick isn't holding two clips together. The stick is how the lower one looks up.

[*Across the yard the woodshed roof lets go its snow at one end, all in a piece. It comes down on the lean-to roof below; and the lean-to, which had held its own snow all morning, lets that go too, and the whole lot lands in the yard with a thump that makes Achilles look round. At the other end the upper roof has kept its snow, and the lean-to under it keeps its.*]

Tortoise: Then walk with me. Toss yours; mine starts heads up.

[*Achilles tosses the silk coin and lays it as it fell. They walk their coins along, the card between them. At the bar the Tortoise stops.*]

Tortoise: What does yours show, Achilles?

Achilles: Tails.

Tortoise: Then mine goes over. Tails.

Achilles: You waited for me. All morning you never once waited for me.

[*Achilles sits back, and for a moment neither of them says anything. He warms his hands on the cup, tepid now, and lets her have the last word.*]

Tortoise: All morning I had a string to myself. [*She folds a fresh sheet twice, as the first was folded, and gives it to him.*] A mark in the panel where each pair arrives.

[*A dozen tosses. Each time she stops at the bar and asks, and he answers, and only then does she move.*]

Achilles: Seven marks in the upper left, heads with heads. Five in the lower right, tails with tails. In the other two panels, not one. [*pencilling each panel in as dark as its share*] As shares: a half, nothing, nothing, a half. Now the border lists.

Tortoise: Begin with an empty panel.

Achilles: [*He lays two slips along the edges for the border lists, and stops before he has written a line.*] The upper right panel is nothing, so one of its border lines must be nothing: the silk's heads line or the hemp's tails line. But the first would empty the upper left panel, and the second the lower right, and those are full. It will not multiply out. It isn't a patchwork!



not a patchwork

Tortoise: A pair of gloves, posted in two parcels. Open either one: left or right, evens. Open one, and you know the other.

Achilles: And no pair of clips could do it. Two clips make a patchwork of a patchwork.

Tortoise: [*She takes down the card and, unhooking nothing, slides the winding of the lower pair apart along the strings until its two crossings stand well clear of one another. She takes a second bar from her shell and clips it across the silk and the hemp to the right of the first; below, the wound pair's first crossing now lies between the two bars, its second beyond them — the two orders straddling each other along the one frame, and no coin yet touched.*] The same rule, burnt into the same wood. Walk the pairs.

Achilles: Heads above: nothing happens, twice. Tails above, heads below: the lower coin goes over at the first bar, and over again at the second. Heads.

Tails above, tails below: heads, then tails again.

Tortoise: And a tossed one?

Achilles: [*He tosses the silk coin, and walks; and tosses again.*] The tossed one too: mine arrives as it fell, and the hemp coin arrives heads every time. Every pair comes out as it went in. Twice yoked, and nothing shows!

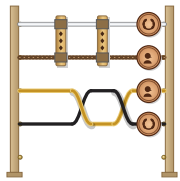
Tortoise: Below, more room on the sheet than the strings have lines. On different strings, no first and no second. Crossed, a change of place and nothing else. Yoked, one string minding another. And all of it at once, on one frame.

[*The thaw has the whole eave now. Every icicle keeps its own time over its own hollow, and the water from all of them has found the path and goes down it to the gate in one bright runnel, past the pear tree, past the woodshed's bare end. In the seed tray the labels stand round their two edges in the sun.*]

[*Achilles sets a coin on each string at the left upright: on the silk tails up, on the hemp heads, on the gut heads, on the horsehair tails.*]

Achilles: Shall I walk the upper pair first, or the lower? — No. I know better than to ask.

[*He walks them all together, level. Above, the silk coin passes both bars untouched, tails all the way; the hemp coin under it is turned at the first bar, since the silk shows tails, and turned back at the second. Below, the gut and horsehair coins change rows at the first crossing, each keeping its face, and change back at the second. Every coin arrives on its own hook, showing the face it set out with.*]



every one home

Tortoise: Every one home. And your tea gone stone cold, never touched — warm your hands a moment; the eaves have all but stopped dripping, and the path will bear you well when you go.

Achilles: As they left, every one. A whole frame of bars and windings, and it comes to bare strings. Tortoise — this morning I had one string and a list of two lines, and I thought I knew the rest. If one had begun with only the one string, how would one ever have guessed all this?

Tortoise: Draw a second string below the first, and we shall see.

[*The sun has climbed: the porch roof's shadow has come down the wall and over the print beside the door, and the three balls stand in their row in the shade, a picture on a porch wall. On the table the frame stands full in the light, the silk a line of light as it was when the morning began, the hemp taking none, the gut yellow, the horsehair dark; below them the empty hooks go on down the uprights. Along the eave the icicles have dripped themselves thin, and the drops come slowly now, and evenly, each into its own hollow. Two cups stand cold on the table, and beside the frame lies a fresh sheet of paper, not yet folded, warm from the sun.*]

Chapter 4: Uniting Wires

— Yoked Inventions —

Part II: Wires Together

Escher: "Three Spheres II" — three spheres side by side on a table top, the middle one holding its neighbours and the room in its reflection

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Put a second coin on the table beside the first: is there now twice as much to say, or more — and if something is done to each of them, which was done first?

Two travellers set out from two different towns. To describe either journey you need a timetable, and a timetable is a list. For the pair of them, then, two lists? That is what anyone would say, and for a great many purposes it is right. But ask an innocent question about the two of them *together* — what is the chance that she is delayed and he is not? — and the two lists fall silent. Neither of them has a line for it. A description of two things must have room for every combination of what the two might do, and combinations do not add. They multiply. With two coins the difference hides, since two and two are four and two times two are four. It stops hiding at the third coin, and by the tenth it is the difference between twenty and a thousand.

The second half of the question sounds like a riddle. Two musicians play in two different rooms, each from her own part. We may ask which bar of the first part comes before which other bar of the first part: the page says. But is the third bar of one part before or after the fifth bar of the other? We could decide to say so; we could record the two rooms and lay the tapes side by side any way we pleased. Nothing in the music would change. Between

two things that do not touch, "first" and "second" are not facts; they are habits of the person writing it down.

This chapter is about both halves, and about how each looks in our two notations. On the blackboard the second coin brings a new kind of multiplication, and an honest half page of indices to show that it behaves. In the sketchbook it brings almost nothing: a second line, drawn under the first. We shall find that the drawing had already answered the riddle, before anyone asked, simply by being a drawing. Whether it was *entitled* to answer is another matter, and we shall have to be honest about who pays for that. Then we make the two wires notice each other, in the only two ways there are: they can change places, and a box can be laid across them both.

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§4.1 A Second Wire, Below

Chapter 3 ended with four questions, and this chapter answers them in the order in which they were asked. The first: *is the second coin's wire drawn beside the first wire — above it, below it?*

Below. And since we shall have to refer to the two wires constantly, we fix once and for all which of them is mentioned first: **the top wire is the first.** Nothing deep is being decided. But it must be decided once and then never varied, because a good deal of what follows looks different, though it is not, if the two wires are silently exchanged.

On the blackboard. A single coin, when looked at, gives one of two answers, 0 or 1. Two coins give one of four *pairs* of answers, and we write a pair as two digits, the first coin's answer first: 00, 01, 10, 11. Each pair gets a ket of its own,

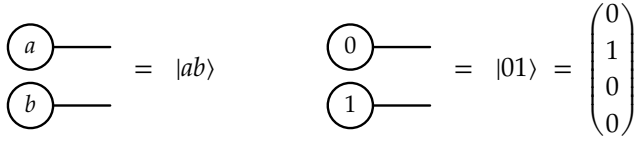
$$|00\rangle\,, \qquad |01\rangle\,, \qquad |10\rangle\,, \qquad |11\rangle\,,$$

and each ket is a column of four entries, with a single 1 in the place that belongs to its pair. The places are taken in the order just written, which is the order of counting in binary:

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}, \quad |01\rangle = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}, \quad |10\rangle = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad |11\rangle = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

We shall also write $|a, b\rangle$ for $|ab\rangle$ when the comma helps the eye. In $|ab\rangle$ the digit a is the answer of the first coin, the top wire; b is the answer of the second, the bottom wire. So $|01\rangle$ says: top 0, bottom 1.

In the sketchbook. One new element, described before it is drawn. To **stack** two pictures is to draw one below the other, with clear paper between them, each on its own wire. The simplest stack is two bare wires, one under the other. The next simplest is two states, each a rounded box with its wire leaving to the right, exactly as in Chapter 1, the one on the upper wire and the other on the lower. The stack of the state a over the state b is the picture of the column $|ab\rangle$:



Pictures are still read from left to right. What is new is that the page now has a second direction, and it means something different from the first. Left to right is *then*. Top to bottom is *and*. Every picture equation in this chapter is exact: the same number, or the same matrix, on both sides, with no factor swept aside.

Why this definition. Why below, and why "top first"? For no reason except that a choice is needed and conventions differ: this book reads the top wire first, and the reference for the general theory that we shall cite in §4.9 happens to stack the other way. What is *not* a matter of taste is that the choice is made once. The column $|01\rangle$ and the column $|10\rangle$ are different columns, and a reader who exchanged the wires without saying so would be exchanging them. See `Code/Ch4/SecondString.py`, which asserts the order.

§4.2 The Column for Two Coins

The second question of §3.10: *what is the column for two coins — two entries and two more, or something else?*

Begin with coins we understand, the covered coins of §1.4, whose columns are lists of shares. Let the upper coin be heads and tails in equal shares, $\begin{pmatrix}1/2\\1/2\end{pmatrix}$, and the lower one heads three times in four, $\begin{pmatrix}3/4\\1/4\end{pmatrix}$, and suppose the two coins have nothing to do with each other. "Two entries and two more" would be a list of four numbers, $\frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{1}{4}$, and it has a flaw that shows at once: its entries sum to 2. It has a worse flaw. Ask for the share of *heads above and tails below, together*, and no line of it answers. A description of the pair needs one line for each pair of answers, and when the coins are separate the line for a pair is the product of the two shares:

$$\begin{pmatrix}\frac{1}{2}\cdot\frac{3}{4}\\ \frac{1}{2}\cdot\frac{1}{4}\\ \frac{1}{2}\cdot\frac{3}{4}\\ \frac{1}{2}\cdot\frac{1}{4}\end{pmatrix} = \begin{pmatrix}3/8\\1/8\\3/8\\1/8\end{pmatrix}, \qquad \tfrac{3}{8}+\tfrac{1}{8}+\tfrac{3}{8}+\tfrac{1}{8}=1.$$

So it is four entries in either case, but they are not two and two more. They are two *times* two.

On the blackboard. The **Kronecker product** of two columns of two entries is the column of four entries

$$\begin{pmatrix}a\\b\end{pmatrix}\otimes\begin{pmatrix}c\\d\end{pmatrix} = \begin{pmatrix}ac\\ad\\bc\\bd\end{pmatrix}.$$

The sign $\otimes$ is read "tensor", and physicists call the operation the **tensor**

product. The recipe: take each entry of the *left* column in turn, and multiply the whole right column by it. The entry in the place *ik* is the *i*-th entry of the left column times the *k*-th entry of the right. The order of the places is thus the order of §4.1, and the basis kets are themselves products, $|ab\rangle = |a\rangle \otimes |b\rangle$. **The left factor is the top wire.** This is fixed here and does not vary again.

The product passes through sums and multiples in each factor separately, because every entry of it is one entry from the left times one from the right:

$$(|\psi\rangle + |\psi'\rangle) \otimes |\phi\rangle = |\psi\rangle \otimes |\phi\rangle + |\psi'\rangle \otimes |\phi\rangle\,,$$

$$(\lambda\,|\psi\rangle) \otimes |\phi\rangle = \lambda(|\psi\rangle \otimes |\phi\rangle) = |\psi\rangle \otimes (\lambda\,|\phi\rangle),$$

and likewise on the right. Note where the number λ ends up in the last equation: it belongs to neither factor. It belongs to the pair.

For two *qubits* the entries are amplitudes and not shares, and we take the same product. Here is one, with entries that are not real so that nothing can hide. The upper qubit is the arrow of Chapter 3, $|a\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$, and the lower is $|\omega\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ from §3.3:

$$|a\rangle \otimes |\omega\rangle \;=\; \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix} \otimes \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} \;=\; \tfrac{1}{5\sqrt{2}} \begin{pmatrix} 3 \\ 3i \\ 4 \\ 4i \end{pmatrix}.$$

The squared moduli are $\frac{9}{50}, \frac{9}{50}, \frac{16}{50}, \frac{16}{50}$, and they sum to 1. Each is the product of the two separate chances, $\frac{9}{25} \cdot \frac{1}{2}$ and $\frac{16}{25} \cdot \frac{1}{2}$, as it should be for two coins that have nothing to do with each other: the squared modulus of a product is the product of the squared moduli. And the order of the factors matters: $|\omega\rangle \otimes |a\rangle = \frac{1}{5\sqrt{2}}(3,\ 4,\ 3i,\ 4i)^T$, a different column.

In general, **a state of two qubits is a column of four complex numbers whose squared moduli sum to** 1; the set of all columns of four complex numbers, thought of as belonging to two wires, is written $\mathbb{C}^2 \otimes \mathbb{C}^2$. A state

of the form $|\psi\rangle \otimes |\phi\rangle$ is called a **product state**. That a product of two states *is* a state — that its squared moduli sum to 1 — is a special case of the next fact, which is the first thing one wants to know about the new product: how it meets the old bracket.

The bra of a column is the row of its conjugates (§2.2), and the conjugate of a product of numbers is the product of the conjugates. So the bra of $|\psi\rangle \otimes |\phi\rangle$ is the row $\begin{pmatrix} \bar{a}\bar{c} & \bar{a}\bar{d} & \bar{b}\bar{c} & \bar{b}\bar{d} \end{pmatrix}$, which is the same recipe applied to the two rows: it is $\langle\psi| \otimes \langle\phi|$.

An example first. Ask the turned question of §2.5 on both wires: what is the bracket of $|+\rangle \otimes |+\rangle = \frac{1}{2}(1,1,1,1)^T$ with the state above? The row $\frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix}$ adds up the entries and halves them:

$$\tfrac{1}{2} \cdot \tfrac{1}{5\sqrt{2}} \, (3 + 3i + 4 + 4i) \;=\; \tfrac{7+7i}{10\sqrt{2}}.$$

On the separate wires, $\langle +|a\rangle = \frac{1}{\sqrt{2}}(\frac{3}{5} + \frac{4}{5}) = \frac{7}{5\sqrt{2}}$ and $\langle +|\omega\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1+i) = \frac{1+i}{2}$, and their product is $\frac{7(1+i)}{10\sqrt{2}}$: the same number. Is that luck?

> **Proposition 4.1 (brackets multiply).** For all columns $|\psi\rangle, |\psi'\rangle, |\phi\rangle, |\phi'\rangle$ of two complex entries,
>
> $$(\langle\psi'| \otimes \langle\phi'|)(|\psi\rangle \otimes |\phi\rangle) \;=\; \langle\psi'|\psi\rangle \; \langle\phi'|\phi\rangle \, .$$
>
> In particular the Kronecker product of two states is a state. In pictures: two closed diagrams stacked are the product of their numbers.

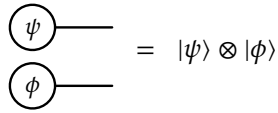
Proof idea. A sum of four products that factorizes into two sums of two.
Proof sketch. Write $|\psi\rangle = (a,b)^T$, $|\phi\rangle = (c,d)^T$, and the primed columns with primed letters. The left side is a row times a column,

$$\bar{a}'\bar{c}'\,ac + \bar{a}'\bar{d}'\,ad + \bar{b}'\bar{c}'\,bc + \bar{b}'\bar{d}'\,bd \;=\; (\bar{a}'a + \bar{b}'b)(\bar{c}'c + \bar{d}'d),$$

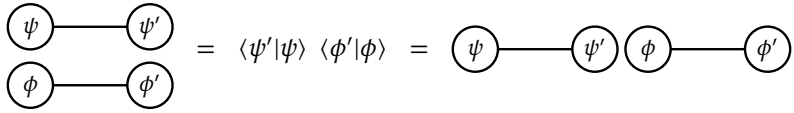
as one sees by multiplying out the right side; and the two brackets on the

right are $\langle\psi'|\psi\rangle$ and $\langle\phi'|\phi\rangle$. For the particular case take $\psi' = \psi$ and $\phi' = \phi$ and use Theorem 2.1 twice: $1 \cdot 1 = 1$. ■ See `Code/Ch4/SecondString.py`. We shall see in §4.9 that this proposition is the smallest case of the chapter's main theorem.

In the sketchbook. No element beyond stacking. Two states stacked stand for the Kronecker product of their columns, the upper state on the left of the $\otimes$:



Effects stack in the same way, and so do closed diagrams. A closed diagram is a number (§2.2), and Proposition 4.1 says what number a stack of two of them is:



The picture on the right is the old one: closed diagrams *in a row* multiply (row 12 of the Rosetta Table). So for closed diagrams it makes no difference whether they are set in a row or stacked. That is as it should be. A number has no wire by which to be before or after anything, or above or below it.

Why this definition. Why products of entries, and not something simpler? For shares there was never a choice: the chance of two independent things both happening is the product of their chances. For amplitudes the requirement is that the chances *computed from them* by the Born rule should multiply in that way, for every question asked separately on the two wires, and Proposition 4.1 says they do: the chance of the answers ψ' above and ϕ' below is $|\langle\psi'|\psi\rangle\,\langle\phi'|\phi\rangle|^2$, the product of the two separate chances. Why not the simpler "two entries and two more"? Because a column is a list of answers to questions, and the questions about a pair are about pairs.

§4.3 Doubling

There was a trap in the last section, and the numbers set it. Two wires, four entries; and $2 + 2 = 4$. A hasty reader could still believe that columns lengthen by two for each wire, and would expect six entries for three wires.

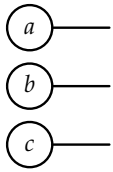
On the blackboard. Three coins give one of eight triples of answers, $000, 001, \dots, 111$, and the column has a place for each. The product of three columns is formed by taking the product twice, and it does not matter how the three are bracketed: $(|\psi\rangle \otimes |\phi\rangle) \otimes |\chi\rangle$ and $|\psi\rangle \otimes (|\phi\rangle \otimes |\chi\rangle)$ both have, in the place ikm, the number $\psi_i \phi_k \chi_m$ — the i-th entry of the first column times the k-th of the second times the m-th of the third — and both list the places in the order of counting. So we write $|\psi\rangle \otimes |\phi\rangle \otimes |\chi\rangle$ without brackets. Each further wire doubles the column:

wires n	entries of the column, 2^n	numbers needed for n separate states, $2n$
1	2	2
2	4	4
3	8	6
4	16	8

A column for n wires has 2^n entries, one for each string of n answers. The third column of the table counts something else: a product of n one-wire states is fixed by naming the n states, two complex numbers each, $2n$ in all. At one wire and at two the columns of the table agree, which is what made the trap possible; from three wires on they part, and they never meet again. At ten wires it is 1024 against 20; at forty the column has more than a million million entries.

The gap has a plain meaning, which we state and leave. The columns that are products of one-wire states are a thin family inside the space of all columns. **Most columns are not products, and what that means is the subject of Chapter 6.** Here we need only the count.

In the sketchbook. Stacking goes on downwards: a third wire under the second, a fourth under the third, the top wire always first.



$= \quad |abc\rangle \;=\; |a\rangle \otimes |b\rangle \otimes |c\rangle\,,$ a column of eight entries.

Here the two notations part company in a way worth noticing. On the blackboard each new wire doubles the writing. In the sketchbook each new wire costs one more line of ink: n wires, n lines. The picture is compact. But let us not claim too much for it. The picture is short because it does not write the column out; it does not make the column any shorter. Whoever wants the numbers must still find 2^n of them, and no drawing in this chapter will spare anyone that labour. A notation that is brief is not thereby a calculation that is cheap.

Why this definition. Why one entry for each string of answers, when $2n$ numbers describe n coins so much more cheaply? Because $2n$ numbers describe only coins that have nothing to do with one another. Already for covered coins this is too little: a list of shares for a pair must be able to say "both heads or both tails, never one of each" (we shall meet exactly this list in §4.7), and no two separate lists can say it. The column has a place for every combination because every combination is a question that can be asked. See `Code/Ch4/Doubling.py`.

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§4.4 A Box on One Wire

The third question of §3.10: *if a box sits on one wire and nothing on the other, what is the matrix of the pair?*

One is tempted to answer that a box on the upper wire leaves the lower coin alone, and so leaves half of the column alone. It does leave the lower coin

alone. It leaves *no* entry of the column alone, because every entry of the column is about both coins. Let us see what must happen. Put NOT on the top wire and nothing on the bottom. The pair $|ab\rangle$ must go to the pair with a turned over and b as it was: $|00\rangle \mapsto |10\rangle$, $|01\rangle \mapsto |11\rangle$, $|10\rangle \mapsto |00\rangle$, $|11\rangle \mapsto |01\rangle$. The column of a matrix is where it sends the corresponding basis ket (§1.1), so the matrix is known, and it is 4×4:

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

With NOT on the *bottom* wire instead, $|00\rangle \mapsto |01\rangle$, $|01\rangle \mapsto |00\rangle$, $|10\rangle \mapsto |11\rangle$, $|11\rangle \mapsto |10\rangle$, and the matrix is a different one:

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Cut each of them into four blocks of size 2×2, and a pattern appears. The first is $\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$: it is the matrix X with each of its entries multiplied into a copy of I. The second is $\begin{pmatrix} X & 0 \\ 0 & X \end{pmatrix}$: it is the matrix I with each of its entries multiplied into a copy of X.

On the blackboard. Number the entries of a 2×2 matrix from zero, row first, as in §3.1. The **Kronecker product** of two 2×2 matrices is the 4×4 matrix

$$A \otimes B \;=\; \begin{pmatrix} A_{00}\,B & A_{01}\,B \\ A_{10}\,B & A_{11}\,B \end{pmatrix},$$

each block being the whole matrix B multiplied by one entry of A. The rows and the columns of $A \otimes B$ are labelled by pairs, in the order $00, 01, 10, 11$,

and the entry in the row ik and the column jl is

$$(A \otimes B)_{ik,\,jl} \;=\; A_{ij}\,B_{kl}.$$

The first digit of each label goes to A and the second to B: **the left factor is the top wire**, as for columns. The same recipe serves for matrices of any shapes, the labels running over whatever ranges the shapes allow; the product of two columns in §4.2 is the case in which each factor has a single column, and the product of two rows is the case of a single row.

The definition was chosen to make one thing true, and it does. The column jl of $A \otimes B$ has the entries $A_{ij}B_{kl}$, and that is the Kronecker product of the j-th column of A with the l-th column of B:

$$(A \otimes B)(|j\rangle \otimes |l\rangle) \;=\; A\,|j\rangle \otimes B\,|l\rangle\,.$$

Each factor acts on its own wire. (The same holds with any columns $|\psi\rangle$ and $|\phi\rangle$ in the places of $|j\rangle$ and $|l\rangle$; that will be a special case of Theorem 4.2, whose proof does not depend on it.)

Now the question has its answer. A box A on the top wire, with nothing on the bottom wire, is $A \otimes I$; on the bottom wire, with nothing on the top, it is $I \otimes A$:

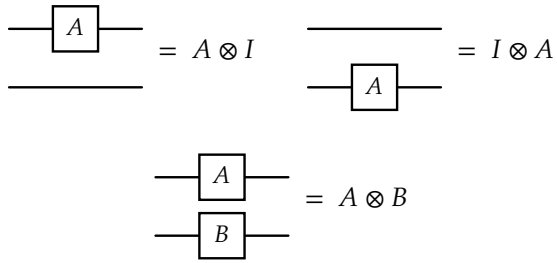
$$A \otimes I = \begin{pmatrix} A_{00}\,I & A_{01}\,I \\ A_{10}\,I & A_{11}\,I \end{pmatrix}, \qquad\qquad I \otimes A = \begin{pmatrix} A & 0 \\ 0 & A \end{pmatrix}.$$

Think of the column of four as two halves: the upper half holds the entries whose first digit is 0, the lower half those whose first digit is 1. Then $A \otimes I$ **mixes the two halves with each other**, as whole halves, in the proportions that A prescribes; and $I \otimes A$ **acts inside each half** and never mixes the two. For NOT these are the two matrices we found: $X \otimes I$ makes the halves change places (entries $00 \leftrightarrow 10$ and $01 \leftrightarrow 11$), and $I \otimes X$ exchanges the two entries inside each half ($00 \leftrightarrow 01$ and $10 \leftrightarrow 11$). In both cases every entry moves. In no case is "half the column left alone".

The two are different matrices unless A is a multiple of I. For if $A \otimes I = I \otimes A$, compare blocks: the off-diagonal blocks give $A_{01}I = 0$ and $A_{10}I = 0$, and the upper left block gives $A_{00}I = A$; so $A = A_{00}I$. With a test matrix that is neither real nor symmetric, $A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}$:

$$A \otimes I = \begin{pmatrix} 1 & 0 & i & 0 \\ 0 & 1 & 0 & i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad I \otimes A = \begin{pmatrix} 1 & i & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & i \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

In the sketchbook. No new element; stacking does it all. A box on one wire with a bare wire above or below it, and two boxes stacked:



The bare wire is the matrix I (the sixth row of Chapter 1's table, row 6), so the first two pictures are the third with a bare wire in the place of one box: nothing on a wire *is* the box I on it. The sketchbook's answer to §3.10's question was therefore available before the blackboard's: draw the box, and draw the other wire. What the sketchbook cannot do is tell us that the first picture is a matrix that moves whole halves and the second a matrix that works inside them. The two pictures differ by where one box sits; the two matrices differ in nearly every entry.

Dirac's Blackboard.

A Kronecker product written out in full. Two test matrices, neither of them real, symmetric, or its own mirror image:

$$A = \begin{pmatrix} 1 & i \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ i & 0 \end{pmatrix}.$$

The four blocks of $A \otimes B$ are $1 \cdot B$, $i \cdot B$, $0 \cdot B$, $1 \cdot B$, and $iB = \begin{pmatrix} 0 & i \\ -1 & 0 \end{pmatrix}$:

$$A \otimes B = \begin{pmatrix} 0 & 1 & 0 & i \\ i & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & i & 0 \end{pmatrix}.$$

Check one entry against the formula: row 01, column 10 is $A_{01}B_{10} = i \cdot i = -1$. In the other order the blocks are $0 \cdot A$, $1 \cdot A$, $i \cdot A$, $0 \cdot A$:

$$B \otimes A = \begin{pmatrix} 0 & 0 & 1 & i \\ 0 & 0 & 0 & 1 \\ i & -1 & 0 & 0 \\ 0 & i & 0 & 0 \end{pmatrix}.$$

The same sixteen products $A_{ij}B_{kl}$, in different places. The upper left block of the first is B; of the second, zero. The Kronecker product is *not* commutative, and §4.6 will say exactly how the two orders are related.

Why this definition. Why blocks of B scaled by the entries of A, and not the other way about, or something more even-handed? Because there is nothing left to choose once §4.2 is fixed. A matrix is known by its columns; the pair of boxes must send $|j\rangle \otimes |l\rangle$ to $A|j\rangle \otimes B|l\rangle$, each box minding its own wire; and those four columns are the matrix above. The other arrangement of blocks is $B \otimes A$ — the right matrix for a reader who takes the bottom wire first. See `Code/Ch4/OneClip.py`.

§4.5 Which First?

The fourth question of §3.10: *if two boxes sit on two wires, does it matter which we think of as acting first?*

On one wire, order is everything. Chapter 1 made the point with NOT and SET-0, the box $S_0 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ that sends both states to the state 0: NOT and then SET-0 is $S_0X = S_0$, while SET-0 and then NOT is $XS_0 = S_1$ (§1.3). Different boxes. Now put the same two on *different* wires, NOT on the top and SET-0 on the bottom, and compute both orders. Matrices cut into 2×2 blocks multiply by the usual rule, "along the row and down the column", with blocks in the places of entries, so long as the order of the factors inside each product of blocks is kept. First SET-0, then NOT:

$$(X \otimes I)(I \otimes S_0) = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}\begin{pmatrix} S_0 & 0 \\ 0 & S_0 \end{pmatrix} = \begin{pmatrix} 0 & S_0 \\ S_0 & 0 \end{pmatrix}.$$

First NOT, then SET-0:

$$(I \otimes S_0)(X \otimes I) = \begin{pmatrix} S_0 & 0 \\ 0 & S_0 \end{pmatrix}\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} = \begin{pmatrix} 0 & S_0 \\ S_0 & 0 \end{pmatrix}.$$

The same matrix, and it is $X \otimes S_0$: the two boxes acting "at once". The pair that quarrelled on one wire has nothing to quarrel about on two.

That was two boxes, each with a bare wire beside it. The real test has four: on the top wire C and then A, on the bottom wire D and then B. We may think of this as two steps taken by the pair, the step $C \otimes D$ and then the step $A \otimes B$. Or we may think of it as two separate wires, the top one carrying the box AC and the bottom one the box BD. Are they the same 4×4 matrix? And if they are, is the pairing the natural one, or must something be reversed or reflected, as so often in Chapter 3?

Dirac's Blackboard.

The example in full. With A and B as in §4.4, take also

$$C = \begin{pmatrix} 1 & 0 \\ 1 & i \end{pmatrix}, \qquad D = \begin{pmatrix} i & 2 \\ 0 & 1 \end{pmatrix}; \qquad C \otimes D = \begin{pmatrix} i & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ i & 2 & -1 & 2i \\ 0 & 1 & 0 & i \end{pmatrix}.$$

Multiply the two 4×4 matrices, sixty-four products of entries:

$$(A \otimes B)(C \otimes D) = \begin{pmatrix} 0 & 1+i & 0 & -1 \\ -1-i & -2+2i & 1 & -2i \\ 0 & 1 & 0 & i \\ -1 & 2i & -i & -2 \end{pmatrix}.$$

Now the small products:

$$AC = \begin{pmatrix} 1+i & -1 \\ 1 & i \end{pmatrix}, \qquad BD = \begin{pmatrix} 0 & 1 \\ -1 & 2i \end{pmatrix}.$$

The blocks of $AC \otimes BD$ are $(1+i)BD$, $-BD$, BD, iBD: exactly the matrix above, block for block. The cheaper guesses fail. With the orders reversed, $CA \otimes DB$ has the upper left entry $(CA)_{00}(DB)_{00} = 1 \cdot 2i$, where the true answer has 0. Transposes, daggers and conjugates on the factors fail likewise. The pairing that works is the plain one: top wire with top wire, bottom with bottom, each in the order its own wire carries.

Theorem 4.2 (The Interchange Law). For any four 2×2 complex matrices A, B, C, D,

$$(A \otimes B)(C \otimes D) \;=\; AC \otimes BD,$$

exactly, scalar included, the Kronecker product being taken with the top wire as the left factor. The same holds for matrices of any shapes for which the products AC and BD are defined. In

pictures: with C and then A on the top wire, and D and then B on the bottom wire, the picture read column by column and the picture read wire by wire are one and the same drawing.

Corollary 4.3 (Sliding). For all 2×2 matrices A and B,

$$(A \otimes I)(I \otimes B) \;=\; A \otimes B \;=\; (I \otimes B)(A \otimes I).$$

In pictures: boxes on different wires slide past each other.

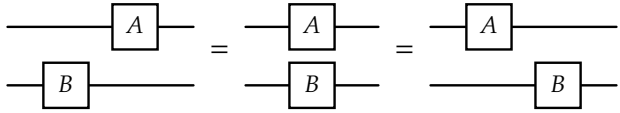
Proof idea. An entry of $A \otimes B$ is a product of one entry of A and one of B; an entry of a matrix product is a sum over a middle label. For the pair, the middle label is a pair of labels, and a sum over pairs of a product of two independent things is the product of two sums. Nothing crosses from one factor to the other at any point, and so nothing is reversed. The proof is carried out twice in §4.9. *Proof of the corollary.* Put I in the places of B and C in the theorem: $(A \otimes I)(I \otimes B) = AI \otimes IB = A \otimes B$. Put I in the places of A and D, and rename: $(I \otimes B)(A \otimes I) = IA \otimes BI = A \otimes B$. ■ See `Code/Ch4/Interchange.py`, which checks the law on the matrices above and on matrices drawn at random, checks that each of the wrong pairings fails, and checks one case of larger shapes.

Two consequences at once. The theorem with columns in the places of C and D is the general form of §4.4's rule, $(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = A|\psi\rangle \otimes B|\phi\rangle$: each factor acts on its own wire, for all columns and not only the basis kets. And boxes that keep the promise of Chapter 3, stacked, keep it still. The mirror image of a stack is the stack of the mirror images, $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$, since the entry of either side in the row jl and the column ik is $\overline{A_{ij}}\,\overline{B_{kl}}$; so for unitary U and V,

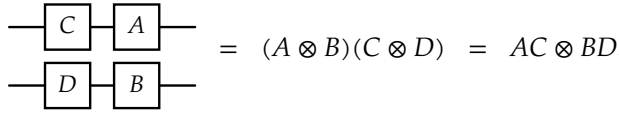
$$(U \otimes V)^\dagger (U \otimes V) \;=\; U^\dagger U \otimes V^\dagger V \;=\; I \otimes I,$$

and $I \otimes I$ is the 4×4 identity.

In the sketchbook. The corollary is a rewrite rule, the first that needs two wires, and we name it ***sliding***: **a box may be moved along its own wire past any box that sits on a different wire.**



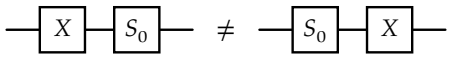
From left to right these are $(A \otimes I)(I \otimes B)$, then $A \otimes B$, then $(I \otimes B)(A \otimes I)$. And the theorem itself is the picture with four boxes:



Look at what the sketchbook has done here. The blackboard has two expressions and an equation between them to prove. The sketchbook has one drawing. We did not decide that the two readings should agree; we never noticed, while drawing, that there were two. This is the strength of the notation, and also the place where it must be watched, and §4.9 weighs the matter carefully.

Penrose's Sketchbook.

What the two directions of the page mean. A picture on two wires has two kinds of position, and they are not alike. *Up and down* is meaningful in full: which wire a box sits on is part of what the picture says, and A over B is not B over A. *Left and right* is meaningful only along a single wire. There it is the order of a row, and it matters as much as it ever did:



Two boxes on one wire cannot pass each other; each is in the other's way. Between *different* wires, left and right mean nothing at all. Whether A is drawn a little before B, level with it, or a little after, is an accident of the pencil. So *sliding* is a rule about what the picture never said. Use it freely between wires, and never along one.

Why this definition. Why is the law stated for four matrices, when *sliding*, with two, is what one uses every day? Because the law with four is what makes pictures on two wires well defined. A drawing with boxes on two wires can be cut into steps in more than one way, and the law says that all the ways give one matrix. *Sliding* is the case in which half the boxes are bare wire.

---

§4.6 The Crossing

So far the two wires keep entirely to themselves. Each carries its own boxes in its own order, and neither knows that the other is there. There are two ways of making them take notice of each other, and this section has the milder: they can change places.

An example shows why we should want it. §4.4's sidebar computed $A \otimes B$ and $B \otimes A$ and found two different matrices. Yet they describe the same two boxes on the same two coins; all that differs is which coin is drawn on top. There ought to be a matrix that does the redrawing, that carries one description into the other. What is it?

On the blackboard. SWAP is the matrix that exchanges the two digits of every label, $|ab\rangle \mapsto |ba\rangle$. It fixes $|00\rangle$ and $|11\rangle$ and exchanges $|01\rangle$ with $|10\rangle$; its columns are therefore

$$\text{SWAP} \;=\; \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

On a column of four it exchanges the two middle entries, and that is exactly what exchanges the factors of a product:

$$\mathrm{SWAP}\begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix} = \begin{pmatrix} ac \\ bc \\ ad \\ bd \end{pmatrix} = \begin{pmatrix} c \\ d \end{pmatrix} \otimes \begin{pmatrix} a \\ b \end{pmatrix},$$

that is, $\mathrm{SWAP}(|\psi\rangle \otimes |\phi\rangle) = |\phi\rangle \otimes |\psi\rangle$.

For the state of §4.2: SWAP carries $\frac{1}{5\sqrt{2}}(3,\ 3i,\ 4,\ 4i)^T$ to $\frac{1}{5\sqrt{2}}(3,\ 4,\ 3i,\ 4i)^T$, which is $|\omega\rangle \otimes |a\rangle$, as computed there.

> **Proposition 4.4 (the two rules of the crossing).** For all 2×2 matrices A and B,
>
> $$\mathrm{SWAP}\,(A \otimes B) \;=\; (B \otimes A)\,\mathrm{SWAP}, \qquad\qquad \mathrm{SWAP} \cdot \mathrm{SWAP} \;=\; I.$$
>
> And $A \otimes B = B \otimes A$ only if one of A, B is a multiple of the other.

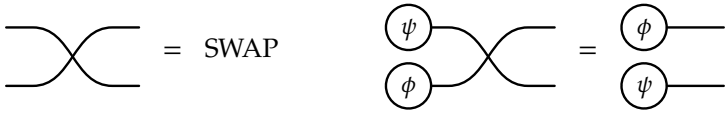
Proof idea. A matrix is known by what it does to the four basis kets, and each basis ket is a product, on which we know what everything does. *Proof sketch.* Apply both sides of the first equation to $|j\rangle \otimes |l\rangle$. On the left, $A \otimes B$ gives $A|j\rangle \otimes B|l\rangle$ (§4.4) and SWAP turns this into $B|l\rangle \otimes A|j\rangle$. On the right, SWAP gives $|l\rangle \otimes |j\rangle$ and then $B \otimes A$ gives $B|l\rangle \otimes A|j\rangle$. The four columns agree, so the matrices are equal. For the second equation, exchanging the digits of a label twice restores it. For the last statement, suppose $A_{ij}B_{kl} = B_{ij}A_{kl}$ for all labels. If B is zero it is a multiple of A. If not, fix k, l with $B_{kl} \neq 0$; then $A_{ij} = (A_{kl}/B_{kl})B_{ij}$ for all i, j, so A is a multiple of B. ■ See `Code/Ch4/Crossing.py`.

The first equation is the answer to our example. Multiply it on the right by SWAP and use the second: $\mathrm{SWAP}\,(A \otimes B)\,\mathrm{SWAP} = B \otimes A$. The two matrices of §4.4's sidebar are the same sixteen numbers, rearranged by SWAP on both sides.

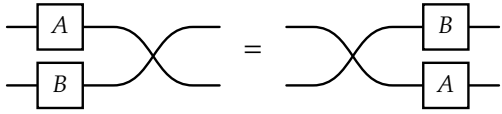
Could the crossing be had more cheaply, as some box P on the top wire and some box Q on the bottom? It could not. The blocks of $P \otimes Q$ are the multiples $P_{00}Q, P_{01}Q, P_{10}Q, P_{11}Q$ of one and the same matrix Q. The upper

left block of SWAP is $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and its lower right block is $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. From the first, $P_{00} \neq 0$ and Q is a multiple of $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$; but then so is $P_{11}Q$, and it cannot be $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. **SWAP is not a Kronecker product of two boxes.** It is the first such matrix we have met.

In the sketchbook. A new element, described before it is drawn. A **crossing** is a place where two neighbouring wires exchange rows: the wire that came in on the upper row leaves on the lower, and the wire that came in on the lower row leaves on the upper. The two lines pass through each other on the page. Neither is broken and neither is joined to the other; each is one unbroken line from the left edge to the right. We draw the crossing *flat*: the two lines simply cross, and the drawing does not say that either passes in front. This book has one kind of crossing, and this is it.



The crossing comes with two rules, and they are the two equations of Proposition 4.4. The first we call ***a box slides through a crossing***: a box pushed along its wire through a crossing stays on its own wire, and arrives on the other *row*.



On the left A is on the upper row; follow its wire through the crossing and the same wire is the lower one on the right, where A now sits. The second rule we call ***two crossings are two bare wires***:



Both rules are exact equations between 4×4 matrices, and both need a word of warning.

Penrose's Sketchbook.

Two rules, and what the paper does not do. It would be pleasant to say that these two equations are facts about lines on paper, and that anyone can see them. They are not, and one cannot. Take two real strings, stretched side by side; exchange their right-hand ends, and then exchange them again the same way round. The strings are now wound about each other, and no pulling will straighten them. Paper and string do not enforce "two crossings are two bare wires". We *adopt* it, as a rule about our flat crossing, and the licence for it is the blackboard line $\mathrm{SWAP} \cdot \mathrm{SWAP} = I$ and nothing else. Likewise the box sliding through: the licence is $\mathrm{SWAP}\,(A \otimes B) = (B \otimes A)\,\mathrm{SWAP}$. A string remembers which strand passed in front. Our drawing has no way of recording it, and deliberately so, because the matrix SWAP has nothing to record: it exchanges two labels, and exchanging them twice restores them. The drawing is made for the thing that travels along the wire, and that thing cannot tell a wound pair from a straight one.

Why this definition. Why a flat crossing, with no strand in front? Because the picture should record what the algebra records and no more, and the algebra of §4.2 has exactly one way of exchanging two factors. A drawing that distinguished two kinds of crossing would be promising a distinction that none of our matrices could honour.

Feynman's Notebook.

A crossing need not undo itself. That our crossing is its own undoing is a special property. Selinger's survey of graphical languages (§3.3) first treats the general case, a *braiding*: a natural family of isomorphisms from $A \otimes B$ to $B \otimes A$, drawn as wires crossing. ("Natural" is the technical word for a condition which, written out for our matrices, is the first equation of Proposition 4.4: a box slides through.) The survey notes there that a braiding

followed by its inverse is the identity, but that a braiding followed by another braiding in general is *not*. Only in §3.5 does it single out the case in which the braiding is self-inverse. It is then called a *symmetry*, and it is this that the survey draws as a plain crossing. Its example on vector spaces with the tensor product is the map $x \otimes y \mapsto y \otimes x$, which is our SWAP. For that case the survey states a coherence theorem, its Theorem 3.12, citing Joyal and Street: an equation follows from the axioms exactly when the two diagrams are isomorphic, with no reference, as the survey puts it, to two- or three-dimensional structure.

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§4.7 A Box on Two Wires

A crossing moves the wires and does nothing to what they carry. The stronger way for two wires to take notice of each other is for what happens on one to *depend* on the other. Here is the plainest rule of that kind: *if the upper coin shows* 1, *turn the lower coin over; if it shows* 0, *leave the lower coin alone.* The upper coin is not changed in either case.

On the blackboard. Write $a \oplus b$ for the sum of two digits modulo 2: $0 \oplus 0 = 0$, $0 \oplus 1 = 1 \oplus 0 = 1$, $1 \oplus 1 = 0$. Adding 1 in this sense turns a digit over and adding 0 leaves it. The rule is then $|a, b\rangle \mapsto |a,\, a \oplus b\rangle$, or

$$|00\rangle \mapsto |00\rangle\,, \qquad |01\rangle \mapsto |01\rangle\,, \qquad |10\rangle \mapsto |11\rangle\,, \qquad |11\rangle \mapsto |10\rangle\,,$$

and the matrix with these columns is called **CNOT**, the *controlled* NOT. The top wire is the **control** and the bottom wire the **target**:

$$\mathrm{CNOT} \;=\; \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \;=\; \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}.$$

The blocks say what the words said. In the upper half of the column, where the first digit is 0, do nothing; in the lower half, where it is 1, apply NOT. Set it beside the two matrices of §4.4 that it resembles: $I \otimes I = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}$ does nothing in both halves, and $I \otimes X = \begin{pmatrix} X & 0 \\ 0 & X \end{pmatrix}$ applies NOT in both. CNOT does the one in one half and the other in the other, and this is just what a Kronecker product cannot do.

> **Proposition 4.5 (CNOT is not a pair of boxes).** There are no 2×2 matrices P and Q with $P \otimes Q = \text{CNOT}$. Moreover $\text{CNOT} \cdot \text{CNOT} = I$.

Proof idea. All four blocks of a Kronecker product are multiples of one matrix; two blocks of CNOT are not multiples of any one matrix. *Proof sketch.* Suppose $P \otimes Q = \text{CNOT}$. The upper left blocks give $P_{00}Q = I$, so $P_{00} \neq 0$ and $Q = I/P_{00}$. The lower right blocks give $P_{11}Q = X$, that is, $X = (P_{11}/P_{00})I$. But X has zeros on its diagonal and is not zero, and no multiple of I is like that. For the second statement, multiply by blocks: $\begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix}\begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & XX \end{pmatrix}$, and $XX = I$ by Theorem 1.1. ■ See `Code/Ch4/Yoke.py`.

So CNOT is its own undoing, for the same reason that NOT is; it is, after all, NOT done or not done. It is real and symmetric, hence its own mirror image; being also its own undoing, it is undone by its mirror image, which is the test of §3.3 for a unitary box, here with 4×4 matrices. So it may sit on a pair of qubit wires. The same is true of SWAP.

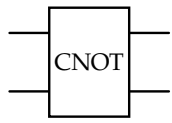
What can such a box do that a pair of boxes cannot? Here is one thing, and we take care to show it on ground we fully understand: covered coins and their shares. CNOT has entries that are not negative and columns that each sum to 1, so it is a stochastic matrix (§1.4), and it acts on lists of shares. Let the upper coin be tossed — shares $\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$ — and the lower coin be laid heads up, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$. The two have nothing to do with each other, and their joint

list is the product. Then CNOT exchanges its last two entries:

$$\begin{pmatrix}1/2\\1/2\end{pmatrix}\otimes\begin{pmatrix}1\\0\end{pmatrix}=\begin{pmatrix}1/2\\0\\1/2\\0\end{pmatrix}\;\longmapsto\;\begin{pmatrix}1/2\\0\\0\\1/2\end{pmatrix}.$$

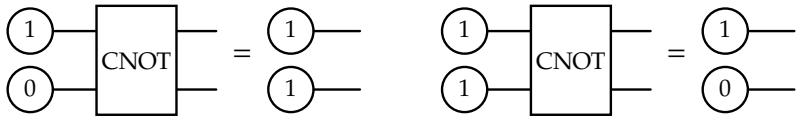
Both heads, or both tails, in equal shares, and never one of each. This list is *not* the product of two lists. For suppose $(pr,\ ps,\ qr,\ qs) = (\frac{1}{2},0,0,\frac{1}{2})$. From $ps = 0$ either $p = 0$ or $s = 0$; but $p = 0$ would empty the first entry, $pr = \frac{1}{2}$, and $s = 0$ would empty the last, $qs = \frac{1}{2}$. There is nothing mysterious in it. It is the list of a matched pair of gloves, posted in two parcels: either parcel alone is left or right in equal shares, and to open one is to know the other. It is ordinary correlation, and a box across two wires is how correlation is *made*. A pair of separate boxes cannot make it: $(P\otimes Q)$ applied to a product list gives the product list $P\binom{p}{q}\otimes Q\binom{r}{s}$, by §4.5.

In the sketchbook. The chapter's last new element, described before it is drawn. A **box that spans two wires** is a box tall enough to cover two neighbouring rows. Two wires enter it on the left and two leave it on the right, and it carries a label, which names a 4×4 matrix. It does not join the two wires into one, and it does not break them: **two lines run through it**, the upper one in at the upper left and out at the upper right, the lower one likewise. CNOT is drawn in this way and in no other in this book until further notice, and always with the control on top:

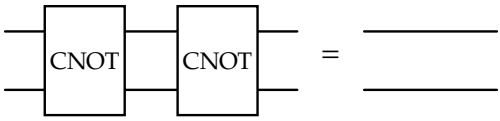


$$=\begin{pmatrix}I & 0\\0 & X\end{pmatrix}$$

The rule $|a,b\rangle \mapsto |a,a\oplus b\rangle$ is four picture equations, of which here are the two in which something happens:



With the state 0 on the upper wire the lower state comes out as it went in. And CNOT $\cdot$ CNOT $= I$ is a rewrite rule, which we call ***CNOT twice is nothing***:



We must be plain about the standing of these equations. They are **lent by the blackboard**, every one. At this stage the sketchbook has a labelled rectangle and no rule that looks inside it; it cannot derive *CNOT twice is nothing*, it can only record it. Nor does the outline of a two-wire box tell us whether it is, or is not, a pair of boxes in disguise. A two-wire box labelled $A \otimes B$ is a pair of boxes; one labelled CNOT is not, by Proposition 4.5; the rectangles look the same. *Sliding* still applies to the new box, with the natural reading: a two-wire box cannot pass a box on either of its own two wires, and it slides freely past anything on wires that it does not touch. The second half of that is Theorem 4.2 for larger shapes, and §4.10 uses it.

Why this definition. Why draw CNOT as a plain labelled rectangle, when the literature has a special symbol for it? Because the symbol is a small picture of how the box works inside, and we do not yet have the elements from which to build that inside honestly. Chapter 8 will have them. Until then a rectangle with a name is a truthful statement of what we know: a 4×4 matrix, two wires in, two wires out, and a proof that it does not come apart.

§4.8 The Rosetta Table

This chapter adds ten rows to the running dictionary, numbered on from Chapter 3's twenty-two. It is a long list because the page has acquired a second direction, and each row is a separate element or rule. Rows 23, 28 and 29 are the three new picture elements: stacking, the box that spans two wires, and the crossing. Rows 27, 30 and 31 are rewrite rules.

#	Algebra (Dirac)	Picture (Penrose)
23	$\mathbb{C}^2 \otimes \mathbb{C}^2$; the kets $\lvert ab\rangle$, with a first	two wires **stacked**, one below the other; the top wire is the first — *new element*
24	$\lvert\psi\rangle \otimes \lvert\phi\rangle$, the Kronecker column	two states stacked, ψ above
25	$\langle\psi'\lvert\psi\rangle \cdot \langle\phi'\lvert\phi\rangle$	two closed diagrams stacked multiply (as they do in a row, row 12)
26	$A \otimes I,\ I \otimes A,\ A \otimes B$	a box on one wire with a bare wire below it, or above it; two boxes stacked
27	$(A \otimes B)(C \otimes D) = AC \otimes BD$; $(A \otimes I)(I \otimes B) = (I \otimes B)(A \otimes I)$	***sliding***: boxes on different wires slide past each other
28	a 4×4 matrix; CNOT, $\lvert a, b\rangle \mapsto \lvert a, a \oplus b\rangle$	a **box that spans two wires**, labelled; two lines run through it — *new element*
29	SWAP: $\lvert ab\rangle \mapsto \lvert ba\rangle$	a **crossing**, drawn flat — *new element*
30	$\mathrm{SWAP}\,(A \otimes B) = (B \otimes A)\,\mathrm{SWAP}$	***a box slides through a crossing*** and changes row (a rule)
31	$\mathrm{SWAP} \cdot \mathrm{SWAP} = I$	***two crossings are two bare wires*** (a rule)

#	Algebra (Dirac)	Picture (Penrose)
32	n wires: a column of 2^n entries	n lines of ink

All ten rows are exact. *CNOT twice is nothing* has no row of its own: it is a fact about one labelled box, lent by the blackboard, as *NOT on 0* was in Chapter 1.

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§4.9 The Theorem, Twice

We have met the theorem through an example in which sixty-four products of entries came out, block for block, as two small products side by side (§4.5). Here is the statement again.

> **Theorem 4.2 (The Interchange Law).** For any four 2×2 complex matrices A, B, C, D,
>
> $$(A \otimes B)(C \otimes D) \;=\; AC \otimes BD,$$
>
> exactly, scalar included, where $A \otimes B$ is the matrix with entries $(A \otimes B)_{ik,\,jl} = A_{ij}B_{kl}$, the top wire being the left factor. The same holds for matrices of any shapes for which AC and BD are defined. In pictures: C and then A on the top wire, D and then B on the bottom wire, read column by column or wire by wire, is one drawing.

Dirac's Blackboard

Everything is in the entry formula. The rows and columns of the 4×4 matrices are labelled by pairs of digits. We compute the entry of the left side in the row ik and the column mn.

Step 1: the rule for a matrix product. The entry of a product is the sum, over a middle label, of an entry of the left factor times an entry of the right. Here the middle label is a pair jl, and it runs over $00, 01, 10, 11$:

$$\big((A \otimes B)(C \otimes D)\big)_{ik,\,mn} \;=\; \sum_{jl} (A \otimes B)_{ik,\,jl}\,(C \otimes D)_{jl,\,mn}.$$

Step 2: the entry formula, twice. $(A \otimes B)_{ik,\,jl} = A_{ij}B_{kl}$ and $(C \otimes D)_{jl,\,mn} = C_{jm}D_{ln}$:

$$= \sum_{j} \sum_{l} A_{ij}\, B_{kl}\, C_{jm}\, D_{ln}.$$

Step 3: numbers may be reordered. Each term is a product of four complex numbers. Bring the two that carry j together, and the two that carry l:

$$= \sum_{j} \sum_{l} (A_{ij}C_{jm})(B_{kl}D_{ln}).$$

Step 4: the double sum factorizes. The first bracket does not contain l and the second does not contain j, so the sum over all pairs is the product of the two separate sums:

$$= \Big(\sum_{j} A_{ij}C_{jm}\Big)\Big(\sum_{l} B_{kl}D_{ln}\Big).$$

Step 5: the rule for a matrix product, backwards, twice; then the entry formula, backwards.

$$= \;(AC)_{im}\,(BD)_{kn} \;=\; (AC \otimes BD)_{ik,\,mn}.$$

The two sides agree in every entry. Nothing in the five steps used that the labels run over two values; with other ranges for i, j, k, l, m, n the same lines prove the law for any shapes for which AC and BD exist. ■

Observe what did *not* happen. In Step 3 we reordered numbers; we never reordered matrices. A stays to the left of C, and B to the left of D, from the first line to the last, because the label j that joins the back of A to the front of C is never anything else. This is why there is no reversal and no dagger in the law, where Chapter 3's *Mirror of a Row* had both.

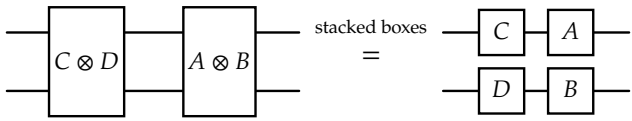
Trace one entry of the example of §4.5 through the five steps: the row 01 and the column 01. The row 01 of $A \otimes B$ is $(i,\ 0,\ -1,\ 0)$ and the column 01 of $C \otimes D$ is $(2,\ 1,\ 2,\ 1)^T$, so Step 1 gives $2i + 0 - 2 + 0 = -2 + 2i$. Step 5 gives $(AC)_{00}(BD)_{11} = (1 + i)(2i) = -2 + 2i$. They agree.

The smallest case is worth a look. Let the right-hand factors be columns and the left-hand factors rows: $A = \langle\psi'|$, $B = \langle\phi'|$, $C = |\psi\rangle$, $D = |\phi\rangle$. Then AC and BD are numbers, the Kronecker product of two numbers is their product, and the law reads $(\langle\psi'| \otimes \langle\phi'|)(|\psi\rangle \otimes |\phi\rangle) = \langle\psi'|\psi\rangle\,\langle\phi'|\phi\rangle$. That is Proposition 4.1; its proof in §4.2 was Steps 2 to 5 with no free labels.

Penrose's Sketchbook

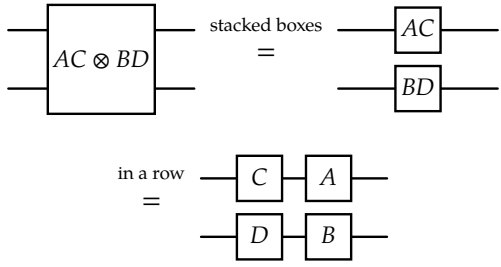
The sketchbook's tools are three dictionary entries. ***In a row*** (the fifth row of Chapter 1's table, row 5): two boxes in a row on a wire, C on the left and then A, are the box AC. ***Stacked boxes*** (row 26): a box above a box is the Kronecker product, the upper box on the left. And a two-wire box with a label (row 28) stands for the matrix that its label names. Each side of the theorem is a two-wire box; we unfold both.

Step 1 — the left side, unfolded by columns. The left side is a row of two two-wire boxes. Replace each by the stack that its label names (*stacked boxes*, used twice):



Step 2 — the right side, unfolded by wires. The right side is one two-wire box. Replace it by the stack that its label names (*stacked boxes*), and then, on each wire separately, replace the box by the row that its label names (*in a*

row, used twice):



Step 3 — compare. The picture at the end of Step 1 and the picture at the end of Step 2 are the same drawing: the same four boxes, on the same wires, in the same order along each. There is nothing to rewrite. ■

And *sliding* follows inside the sketchbook, with no further help. In the four-box drawing let B and C be bare wire: that is A to the right on the top wire and D to the left on the bottom. Let A and D be bare wire instead: the boxes have changed sides. Both are the stack of the two boxes.

The verdict

The picture wins, and by a margin that is almost embarrassing: five steps of indices against the remark that a drawing is equal to itself. But the picture wins on credit, and the creditor should be named. Step 3 quietly assumed that a drawing of four boxes on two wires *means one matrix*, whichever way it is cut up to be read: into columns, as in Step 1, or into wires, as in Step 2. Nothing about ink guarantees that. For our matrices it is guaranteed by the blackboard proof and by nothing else; the five steps of indices are what the freedom of the drawing costs, paid once, here. The same is true of the crossing's two rules in §4.6, each of which stands on its own blackboard line.

What each strand shows that the other hides: the blackboard shows *why* nothing is reversed — the labels of the two wires never meet — and it shows the price, sixty-four products at a time. The sketchbook shows that the question "which first?" was never a question about the boxes. It was a

question about the notation $(\cdot)(\cdot)$, which forces a writer to put one step before another even when the steps do not touch. The drawing is not so forced, and so it had the law built in before we knew there was a law.

Feynman's Notebook.

Who says a picture may be read both ways? The general theory treats our theorem not as something proved but as something demanded. Selinger's survey of graphical languages (§3.1) describes the setting in which wires are drawn in parallel, a *monoidal category*: besides composing two maps one may form their tensor, drawn by stacking the two diagrams. (The survey stacks from the bottom upwards; this book reads the top wire first.) One of its axioms is that the tensor is a *bifunctor*, $(k \otimes h) \circ (g \otimes f) = (k \circ g) \otimes (h \circ f)$, which is our interchange law word for word; vector spaces with the tensor product are among the examples it lists. The survey derives the sliding law from this axiom and remarks that, drawn, it holds up to deformation of the diagram. Then comes the converse, its Theorem 3.1, which the survey attributes to Joyal and Street: an equation follows from the axioms if and only if its two diagrams can be deformed into one another in the plane. It records one caveat (Caveat 3.2): the proof covers deformations in which every wire keeps running from left to right throughout.

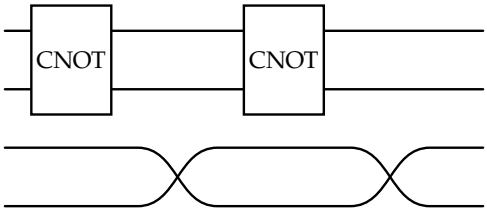
Readers who want the pictures first and the matrices afterwards will find that road taken, at book length, in Coecke and Kissinger's *Picturing Quantum Processes*.

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§4.10 Twice Yoked, Twice Crossed

Everything in the chapter can be put into one picture. Take two pairs of wires, an upper pair and a lower pair. On the upper pair put a CNOT, and further to the right a second CNOT. On the lower pair put a crossing, and further to the right a second crossing, each placed between the columns

of the boxes above, so that, reading from left to right, one meets a CNOT, a crossing, a CNOT, a crossing. We call the picture ***Twice Yoked, Twice Crossed***:



Follow the lines. The uppermost runs straight through the upper half of both boxes: it is the control of each. The next runs through their lower halves: the target. The third starts on its own row, crosses down, runs along the lowest row, and crosses back. The last does the opposite. No line is joined to another and none is broken: two lines run through each box, and a crossing changes the row a line is on and nothing else about it.

The picture is open, with a wire for every row entering on the left and leaving on the right, so it stands for a matrix, and by §4.3 the matrix is 16×16: two hundred and fifty-six entries, which nobody would wish to write out. We shall not need to.

On the blackboard. Write I_4 for the 4×4 identity, which is two bare wires, and I_{16} for the 16×16 identity. The Kronecker product of two 4×4 matrices is formed by the recipe of §4.4 with labels that are pairs, and the upper pair of wires is the left factor. The four columns of the picture, left to right, are $\mathrm{CNOT} \otimes I_4$, then $I_4 \otimes \mathrm{SWAP}$, then the same two again; so, reading the product from right to left,

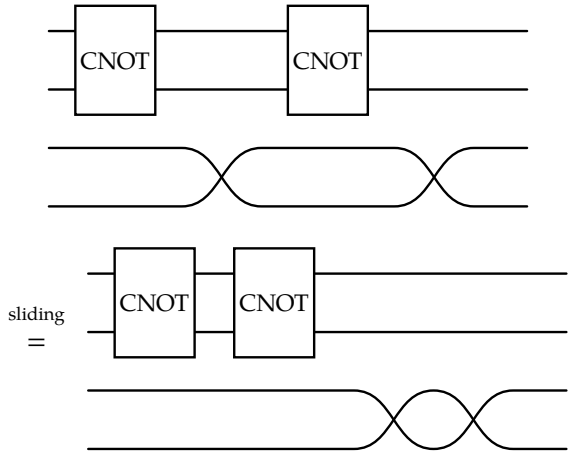
$$(I_4 \otimes \mathrm{SWAP})(\mathrm{CNOT} \otimes I_4)(I_4 \otimes \mathrm{SWAP})(\mathrm{CNOT} \otimes I_4).$$

By the interchange law for 4×4 factors, each neighbouring pair of brackets is $(I_4 \cdot \mathrm{CNOT}) \otimes (\mathrm{SWAP} \cdot I_4) = \mathrm{CNOT} \otimes \mathrm{SWAP}$, and so the whole is

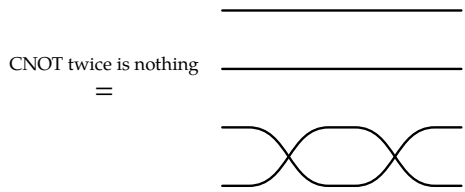
$$\begin{aligned}&(\mathrm{CNOT} \otimes \mathrm{SWAP})(\mathrm{CNOT} \otimes \mathrm{SWAP}) \\ &\quad = \ (\mathrm{CNOT} \cdot \mathrm{CNOT}) \otimes (\mathrm{SWAP} \cdot \mathrm{SWAP}) \ = \ I_4 \otimes I_4 \ = \ I_{16},\end{aligned}$$

using the law once more, then Propositions 4.5 and 4.4. Three uses of one theorem and two small facts, in the place of a product of four 16×16 matrices.

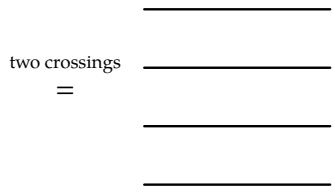
In the sketchbook. First *sliding*, in its form for two-wire boxes (§4.7): the second CNOT touches neither wire of the lower pair, so it slides to the left past the first crossing; and the first crossing, which touches neither wire of the upper pair, may be drawn further to the right.



Now the upper pair carries two CNOT boxes in a row and the lower pair two crossings in a row. Apply *CNOT twice is nothing* above:



and then *two crossings are two bare wires* below:



I_{16}.

***Twice Yoked, Twice Crossed* is bare wire on every row.** Three moves. And it is right to say where each came from. *Sliding* is this chapter's theorem, in the form for larger shapes. *CNOT twice is nothing* is a fact about one matrix, lent by the blackboard. *Two crossings are two bare wires* is a rule we adopted for the flat crossing, licensed by SWAP $\cdot$ SWAP $= I$. The sketchbook supplied the ease; the blackboard supplied every licence.

The picture says a little more than its evaluation. Put any two-wire boxes P and then Q in the places of the two CNOTs. The same first move applies, the lower pair becomes two bare wires as before, and the whole is $QP \otimes I_4$: whatever is done on the upper pair, the twice-crossed pair below contributes nothing to it and takes nothing from it. See `Code/Ch4/Yoke.py`, which checks the evaluation in both orders of the columns and checks the general statement on 4×4 matrices drawn at random.

---

§4.11 Two Roads

Let us gather what we have. The second wire is drawn below the first, and the top wire is mentioned first (§4.1). The column for two coins has an entry for each pair of answers, and for separate coins each entry is a product; brackets of products multiply (§4.2). Each further wire doubles the column while adding one line to the drawing, and the drawing's brevity is not the calculation's (§4.3). A box on one wire of two is $A \otimes I$ or $I \otimes A$, and it moves every entry of the column (§4.4). Boxes on different wires have no order between them; on one wire they still have (§4.5). Wires can cross, and the crossing has two rules, each standing on a line of algebra (§4.6). A box can span two wires, and CNOT, which does, is not a pair of boxes; from a

tossed coin and a plain one it makes a matched pair (§4.7). The interchange law is five steps of indices on the blackboard and a single drawing in the sketchbook, and the drawing's freedom is paid for by the indices (§4.9). Two CNOTs above two crossings are bare wire on every row (§4.10).

It comes to one sentence, and it answers the last of Chapter 3's questions. **Side by side there is no first and no second: what is drawn on different wires happens in no order at all, and the only order a picture has is the order along each wire.**

We have spent the chapter looking *across* the wires. Now look once more *along* one. The box H sends the state 0 to the state + (§3.4), which answers the standard question either way with equal chances: one start, headed towards both answers. Put a second H after the first. From the start 0 to the end 0 there are now two roads, one through the answer 0 in the middle and one through the answer 1; the rule for a matrix product says as much, for an entry of HH is a sum of two terms, one for each middle label. We know the total already, since $HH = I$: the state 0 comes home with certainty, and never arrives at 1. But a chance cannot be taken away by adding another chance to it. What, then, is being added along the two roads? Do the amplitudes add — and can they cancel?

That is the subject of Chapter 5. It sets a box between the two H's, to mark one of the roads, and watches the arrivals change: the arrangement is called an interferometer. With it come a small problem of Deutsch's about a function one is allowed to consult only once, and a curious effect called phase kickback, in which a box meant to act on one wire leaves its mark on the other. We shall need everything in this chapter to say what that means. The wires are laid side by side; the next chapter sends something down two roads at once.

Two-Path Invention

春の夜や
輪と輪重なり
月の凪

haru no yo ya
wa to wa kasanari
tsuki no nagi

spring night —
two rings cross on the water,
the moon lies still

[*Spring dusk on the Tortoise's porch, at the edge of the pond. A cat is already there when Achilles comes up the steps — stretched along the railing, or most of it is; the end that ought to be tail is more a suggestion than a tail. It is neither quite asleep nor quite awake.*]

[*The porch gives straight onto the water, and the water gives back the last of the sky. It is the first evening warm enough to sit out; the pond is glass, not a breath on it yet, and along the near bank the frogs are only clearing their throats. On the table a lamp is lit against the coming dark, a pot of tea beside it, two chairs drawn up. Through the open door behind, the room keeps its own small light: on the far wall, level with the lamp, hangs a little framed print of a still pond under a low moon — Escher's* Rippled Surface *— black water crossed by broken rings of white, catching a dull shine off the glass. The cat has settled into the one warm patch the*

day has left along the railing, or the patch is only where the sun was; from the steps you could not have said which.]

Achilles: Tortoise! I walked the long way round, by the water. I'd forgotten how loud the frogs get the moment the light goes.

Tortoise: They rehearse all week for one evening and then overplay it. Sit — the tea's still hot. You've timed it well; the moon isn't up yet, but it will be, and from this chair it comes up straight out of the pond.

Achilles: I came out hungry for a puzzle. And — good evening. I didn't see the cat there. Have you been sitting long?

Cat: [*without troubling to open both eyes*] Coming and going. Mostly the one, a little of the other.

Achilles: That is no sort of answer.

Tortoise: It's the only sort you'll get to a question shaped so loosely. Ask it something baggy and it hands you something baggy back. Drink your tea. You'll grow used to it.

[*Achilles takes the offered cup and folds both hands round it; the warmth is welcome, the evening cooler than it looked from the road. Out over the water the frogs have found their nerve at last, two of them now trading a phrase back and forth across the dark.*]

First Challenge — *Two Roads*

Achilles: Here is what's been nagging me since we last talked. You showed me a single start sent toward *both* answers at once, and a second gate set after it, and all of a sudden there were two roads leading from the one start back to the very same end. Two roads to a town are two chances of reaching it; you don't arrive *less* often for being given a choice. So the numbers on the two roads simply add up, surely. Do the amplitudes add — and can they cancel?

Tortoise: Let's not decide it by weighing the word "amplitude." Let's walk it.

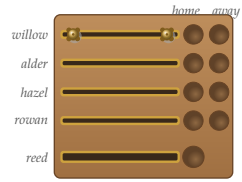
[*She unrolls a folding board the length of the table. Brass-lined channels run down it, side by side, each from its head at the left-hand end to its foot at the right: the willow lane nearest the pond, then alder, then hazel, then rowan — and, set a little apart from the rest and cut deeper, the reed lane at the far edge, ending in a single pocket where the others end in two.*]

[*The lamp finds the board and runs the length of it, a warm gleam down each brass channel: the willow nearest the water, the reed farthest off and set a little apart, with alder and hazel and rowan ranked between. There is no moon yet to reach them — only the lamp, and the dark settling over the pond beyond.*]

Tortoise: This is a conduit-board. Each brass channel is a lane, and a walker sent down a lane runs it straight from one end to the other — a lane never forks, never splits into two. Tonight we want only this nearest one, the willow; the others can lie idle a while.

Tortoise: When a walker reaches the end of a lane it drops a little disc — a token — into the lane's arrival pocket, to record that it came. A token has two faces, and shows one at a time: silver up, or slate up. And a lane like the willow ends not in one pocket but two — a *home* pocket, the nearer, for a walker that has come back to the value it started from, and an *away* pocket beyond it, for one turned aside to the other value.

Tortoise: At the head of the willow lane, and again at its foot, stands a little turnstile — the workshop calls it a spreader. It won't let a walker home until the lane has been walked once for *each* value the lane can carry. That is how one walker comes to owe two roads, without the channel ever splitting.



the willow in play

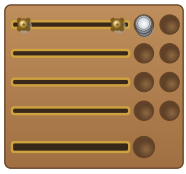
Achilles: Two roads from the one walker. Send it, then — down the low

road only, and I'll watch that.

Cat: [*uncurling onto the head of the board*] I don't take the low road only. I take both, or I take none.

Tortoise: You may ask it, afterwards, which road it took, if you care to. You won't enjoy the answer. Watch it walk.

[*The Cat walks the willow lane. At the head spreader it goes two ways at once, as though it were two cats of one substance; down the lane the two roads run together, and each lets fall its token at the foot. Both come up silver, the bright face. The foot spreader deals them out into the two pockets: into the home pocket it lays both just as they fell — silver upon silver, and they pile; into the away pocket it lays them too, but turns the high road's token over as it sets it down — silver beside slate, unlike faces, and the pair lifts out together and leaves that pocket bare. The walker has come home.*]



home heaped,
away bare

[*Out on the water the two frogs come down on the very same beat, and the doubled note carries clear across the pond and back off the far bank, fuller than either could manage alone. Achilles grins at it despite himself.*]

Achilles: There — the home pocket full, the away pocket bare. The two roads met at home, both silver, and piled up. The walker comes home every time. I *said* the chances only add.

Tortoise: Watch once more, with something in the lane.

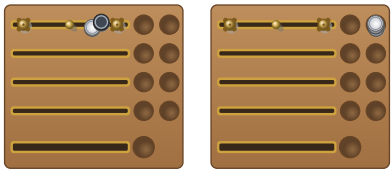
[*She tips the tokens out of the pockets, and takes a small pin from her shell.*]

Tortoise: This is a phase-peg. I press it down into the willow lane itself — into the brass, midway between the two spreaders — and it does one single

thing: it turns the face of any token that passes it, but only along the road that carries the *high* value. The low road runs past it untouched.

[*She presses the peg into the willow lane. The lane is otherwise just as it was; only now a single pin stands upright in the brass, halfway down.*]

[*The Cat walks again. Two roads again from the one head spreader, the high road's token running a little ahead of the low road's; but this time, as the high road passes the peg, its token turns over — silver going down, slate coming up — while the low road slips past the pin and stays silver. At the foot, the spreader deals as before. The same walk, the same lane; one pin between the spreaders, and the walker is turned away.*]



past the peg *home bare, away heaped*

[*The two frogs are still going, but they have drifted out of step — one filling exactly the gap the other leaves, on and on, so evenly that the porch seems to fall quiet though neither has stopped for a moment. Achilles tilts his head, listening for a silence that is not quite there.*]

Achilles: But nothing touched the roads except a pin that they — no. The low road went by it untouched.

Tortoise: Read me the low road.

Achilles: The low road came down silver, and lay silver in the home pocket, exactly as it always has.

Tortoise: And the high road came down turned by the peg, and lay in the home pocket slate against your silver — and where a moment ago the two piled up, now they lift out together and leave nothing behind.

Achilles: Two roads into one pocket, one silver, one slate. …Oh. They didn't add. They *cancelled*.

[*The Tortoise looks up from the board to the pond. The moon has begun to clear the water, and two rings of ripple spreading from out in the dark cross one another — standing high and bright where they ride up together, flattening away to nothing where crest meets trough — while the moon's reflection breaks along the seams and closes and breaks again.*]

Tortoise: *Rippled Surface.* [*She turns back to the board.*]

[*Through the open door, the first of the risen moon has reached the room and lies now on the glass over the little frame on the far wall.*]

Achilles: But the walker went both ways at once. Where did the other road go — can you draw the both-at-once?

Second Challenge — *A Sum of Pictures*

[*The Tortoise draws the peg up out of the willow lane; the pin comes free, and the brass lies unbroken again, the lane bare from spreader to spreader.*]

Tortoise: We're done turning tokens for the moment — the willow runs clean again. Now: draw me your both-at-once.

[*The moon is properly up now, clear of the water and climbing, and its light has just reached the near edge of the board — laid first along the willow lane, the one nearest the pond, while the rest of the board keeps to the lamp. The Tortoise sets a fresh pot between them and fills both cups; Achilles wraps his hands round his and leans in over the paper.*]

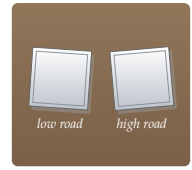
Achilles: Gladly. The walker comes in here, and then— [*he begins to sketch, on a scrap of paper, a lane dividing into two*] —the lane divides, one branch high, one branch low, and—

Tortoise: Stop your pencil there. A lane that splits into two is not a lane; it's a different creature altogether, one road in and two out, and we own no such thing on this board — nor shall for a good while yet. A walker never comes to a fork.

[*His pencil stops where she stopped it, over the half-drawn fork. His hand throws two shadows across the paper at once — one from the lamp at his elbow, one from the moon now clear over the water — two whole hands, each cast entire, crossing and darkening where they fall together and nowhere splitting in two.*]

[*She lays two flat printed cards on the table.*]

Tortoise: These are one-road tiles. Each stands for a single, whole road with a definite value — this one the low road, this one the high — and is blank but for its face, silver or slate, for the sign it carries. They're only cards: they lay no brass, they add no lane, and nobody walks them. But laid together in a pocket, they say what the walk says.



two one-road tiles

[*She sets the two tiles into the willow's home pocket, one over the other. For the plain walk both are printed silver: silver over silver, like faces, lying together. Then she lifts the high-road tile, turns it to its slate face, and lays it back: silver beneath slate, unlike faces, and the pair lifts out as one. Nothing was walked; two still cards, summed, gave the same full pocket and the same empty one as the Cat's two roads did.*]



a pile *silver beneath slate* *the pocket bare*

Achilles: So "both at once" was never a fork in the road. It's two whole roads, each with its own face, laid into the one pocket and added. Like a chord: two whole notes sounding at once, and neither of them a note split in

two. And if I set a second spreader after the first, then every road through the pair is one thing: in at the start, through *one* middle value, out at the end — and I add up those roads over every middle value they might pass through.

Tortoise: That is the whole of it. A walk is a sum of roads; each road a run of definite values, each carrying its face.

[*The Tortoise sits back a little, satisfied, and lets him turn it over. Along the railing behind them the warm patch is still warm, or warm again; whether the cat is stretched there or up on the board's head, the lamp and the moon between them make it hard to say. Beyond the rail a frog plops off into the water and sets a ring going that widens and thins and is gone.*]

Achilles: So every road carries a sign. What if I could make a road's sign depend on a *question* — on some rule I couldn't see the inside of — so that walking the road *asked* the question, and the road came home marked by the answer?

Third Challenge — *The Mark on the Other Wire*

Tortoise: For a question you need something to ask, and something to ask it *of*. Bring the reed lane into it — the deep one at the far edge, past the alder and the hazel and the rowan that have lain idle all evening. The willow is the road you ask *from*; the reed is the road you ask *about*.

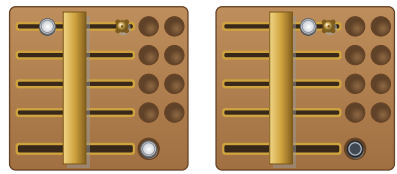
[*The moon has climbed enough to lay its light further across the board now — past the willow and onto the alder and the hazel, so that only the rowan and the deep reed still lie in the lamp's yellow. The reed, set apart and cut deeper than the rest, holds the shadow longest.*]

[*She lifts a long brass beam and lays it square across all the lanes at once, from willow to reed, a little way past the lanes' heads.*]

Tortoise: This is the bar. It is no lane — it lies *across* every lane, a solid beam, not a channel of its own, so you'll never mistake it for a road. It carries a rule hidden inside it: where a road runs under it at the high value the bar

acts, and where a road runs the low value the bar does nothing. To ask it a plain question I want a plain, single road — so up comes the willow's head spreader for a moment.

[*She lifts the head spreader off the willow lane. The turnstile comes away and the lane lies open at its head; now the willow owes not two roads but one — the single road she chooses to walk by hand. She drops a token into the reed's pocket, silver up and lying flat, and walks a token down the willow's high road herself. As the hand-walked token passes beneath the bar, the bar reaches across and turns the reed token — silver going down, slate coming up — while the willow token runs on and comes home unmarked, silver as it set out. The mark has landed on the reed, the lane asked* about, *exactly where Achilles would have put it.*]

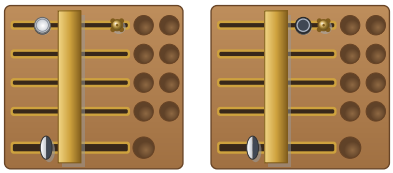


the bar across *the mark on the reed*

Achilles: Just as I'd expect. The bar wrote its mark on the reed — the lane I asked about — and left my willow token quite alone.

Tortoise: Now set the reed's token differently. Not lying flat with one face up, but wedged upright on its edge, so it shows silver to one side and slate to the other, both at once. Two-faced. That is the difference setting.

[*She lifts the reed's token out of its pocket and stands it up on its edge in the deep lane, just short of the bar, wedged there, silver to one hand and slate to the other, and walks a token down the willow's high road once more. The bar reaches across to turn the reed token — but a two-faced token turned end for end shows the very same two faces, and stands entirely unchanged. The turn has nothing there to spend itself on. So it jumps: back along the bar to the willow road that called it, and turns* that *token over instead, while the reed token stands exactly as it was wedged.*]



two-faced *the mark on the willow*

[*Out on the pond a frog launches itself at the moon's broken reflection and comes down short in a smack of water. The moon on the water does not so much as flinch; but the ring the frog threw runs out into the dark, meets the bank, and comes rocking back to lift the very pad it leapt from.*]

Achilles: [*staring*] You laid the bar across the *reed*. The reed didn't change. And my willow token turned over. The mark lands on the road I asked from, not the road I asked about.

Tortoise: A token that shows both faces can't take a mark: turn it end for end, and it is just as it was. So the turn comes back along the bar, onto the road that asked. And a walker carries that turned token home into its pocket, where it can meet another and pile or cancel.

Achilles: Like shoving at a wall that won't give: it's the one who shoves who goes back. So one walk under the bar comes home with the rule's mark on my own road. Could one question's mark tell me something about the rule *as a whole* — not this answer or that one, but the whole of it at once?

Fourth Challenge — *The One Question*

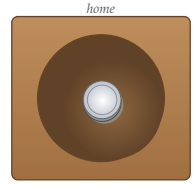
Tortoise: [*setting the willow's head spreader back in place, so the lane owes both its roads again*] Here is my wager. You may consult the bar exactly *once* — one walk of the board beneath it — and from that one walk you must tell me whether the hidden rule gives the willow the *same* answer on both its roads, or *opposite* answers. Same, or different. Not the rule you've just watched, mind — a fresh one, hidden.

[*The moon is high enough now to have found the rowan as well; the greater part of the board lies in its light, and only the reed at the far edge still keeps to the lamp, deep and apart. The frogs have thinned to a pair again, trading their phrase across the water.*]

[*She sets the bar to a new rule and lays her hand flat over its face, so the rule cannot be read.*]

Achilles: The plain way first. Up comes the head spreader, and I walk one road by hand — the low one — beneath your hidden bar.

[*He lifts the head spreader, walks a single token down the willow's low road under the covered bar, and reads the face it brings home.*]



one answer

Achilles: One answer, there in the pocket. Now to *compare*, I must walk the high road as well and read that too—

Tortoise: —a second walk. Two, the plodding way; and you have spent your one already, on a single answer.

Achilles: That's no fair test, then.

Tortoise: It is a game, not an execution. [*She resets the bar beneath her hand.*] A fresh rule. Try again — better.

[*She says it lightly, and there is tea still warm in the pot; Achilles laughs under his breath, half at himself, and squares up to the board again.*]

Achilles: Then I won't plod. [*He sets the head spreader back, so both roads open again.*] Let the Cat take both roads in the one walk, and I'll read just the road I want out of it. [*to the Cat*] Which road brought the mark home?

Cat: [*eyes drifting half-shut*] The one, and the other, and a little of the way between.

[*The two frogs are at it still, and it is the oddest thing — you can hear plainly whether they are keeping step or drifting apart, the whole shape of it clear across the water, and yet you could not say for your life which of the two it was you last heard.*]

Achilles: That tells me nothing whatever. Then I'll lift off the *foot* spreader — let the two roads simply *arrive*, unsorted, and read whichever token lands.

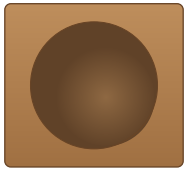
[*He lifts the foot spreader clear of the willow. The Cat walks under the covered bar; the two roads come down and drop their tokens into the pocket with nothing to sort them — silver, then slate, then silver, no telling which walk gave which.*]

[*A moth has found the lamp and beats at the hot glass, struck away and falling and climbing back, and there is no calling in advance which side of it will catch the light as it turns.*]

Achilles: ...a coin. Perfectly fair and perfectly empty. Reading one raw road tells me nothing about the rule either.

Tortoise: [*resetting the bar once more*] Another fresh rule, then. Now stop trying to read a single road. Put the foot spreader back, leave the reed wedged two-faced, and simply walk the whole thing once — head spreader, bar, foot spreader — and read the pockets.

[*He sets the foot spreader back. The Cat walks. At the head spreader both roads open; each runs under the hidden bar, and wherever the rule acts the reed's two-faced kick turns its token — the low road carrying the rule's answer for the low value, the high road its answer for the high value. At the foot the spreader brings the two together. This rule has given them opposite faces: silver and slate meet in the home pocket and lift out as a pair, leaving it bare, while the away pocket fills. The walker is turned away.*]



the home pocket bare

Achilles: Turned away — the home pocket empty. So the two answers were *opposite*, the roads cancelled, and I never once had to read either on its own. [*a pause*] And yet — which was the low answer, silver or slate? I couldn't tell you. I only know the two disagreed.

Tortoise: Nor could anyone, from that pocket. And yet it told you something.

Achilles: ...Oh. I never learned either value, only whether they agree. One walk, one consultation of the bar, and the pocket told me same-or-different — the very thing that no single answer holds by itself. A pair of scales does the same: it tells you whether the two sides balance, and never what either weighs.

Tortoise: Count what a plodding walker needs. To learn same-or-different by *reading answers*, he must read them both — two consultations, as you found when you spent one on a single road. You did it in one.

Tortoise: One walk has settled a two-answer question that costs two answers to settle the plodding way. Now stretch the question over more roads — how far does the cancelling reach?

Fifth Challenge — *The Sixteen Roads*

Tortoise: Bring the others in beside the willow — the alder, the hazel, the rowan — each one a query road, each with its own head and foot spreader; and the reed stays wedged two-faced at the edge, as before. Now one walk owes a road for every choice of values across the whole board.

[*She opens a small tally-book beside the board.*]

Tortoise: The road-book — a tally of the roads a single walk owes. We've had no need of it till now. Count them with me as they fall.

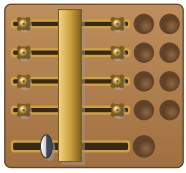
Achilles: Two roads on the willow alone — high and low. Bring in the alder, and each of those two doubles: four. The hazel doubles them again: eight.

The rowan doubles them once more: sixteen. Sixteen roads owed in the one walk.

Tortoise: Sixteen. And will they come home?

Achilles: And for once I can tell you what will happen before the Cat sets out. If the bar gives the *same* answer down every one of the sixteen roads, all sixteen tokens come home the same face and pile up — the walker comes home. But if the bar *splits* them — half one answer, half the other — then eight tokens meet eight of the opposite face in the home pocket, and lift out pair by pair, and leave nothing at all.

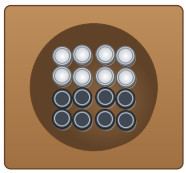
Tortoise: Then here is a rule, and I shan't tell you which it is. [*She sets the bar to a hidden rule and lays her hand over its face.*] Walk it, and read the home pocket alone.



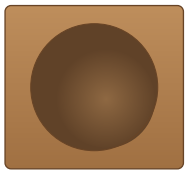
the whole board in play

[*The moon stands high over the pond at last, and its light has crept the whole width of the board and down into the reed lane too, the deep one at the edge — so that every channel lies lit together now, lamp and moon at once along the brass. Out on the water a dozen frogs are going, and the rings they throw run out and cross and cross again, standing up bright where they climb together and going flat to nothing where they meet against the grain.*]

[*The Cat walks — sixteen roads at once, the whole board alight from head to foot. At the foot, read the home pocket alone — the pocket a walker reaches only by coming home on every lane at once: the tokens drop into it road by road, and face meets opposite face, and pair after pair lifts out. The road-book's tally reads sixteen roads walked; the home pocket holds none of them.*]



eight and eight



not a token left

Achilles: Empty — not a token left at home. Eight cancelled eight, so the rule *split* them, and I read that straight off the bare pocket without ever seeing the bar.

Tortoise: There it is. [*She sits back. The moon is well up now, laid out whole across the pond, and the frogs have quieted to the odd remark in the dark.*] Well walked, Achilles. You saw that one coming.

Achilles: I did, didn't I. [*He watches the empty board a moment, then turns to the Cat.*]

Achilles: [*quietly*] All evening I've asked you loose things and had loose answers back. Let me ask you one straight one, now the board has settled it. The roads you just walked — are the roads all of one kind, or split? Same, or different?

Cat: Different — the roads cancel.

[*The Cat is awake now, or was; the railing is empty, the lanes are empty, and the last of it, when everything else of it has faded from the porch, is a grin above the emptied board.*]

[*The pond lies whole and quiet under the risen moon, the frogs gone down to the odd remark in the dark, and the brass channels of the board keep the light along them — willow, alder, hazel, rowan, and the deep reed at the edge — lamp and moon together, going cool now that no one walks them. In the room through the open door the little frame on the far wall has gone dark, a plain rectangle back in the wall where it always hung. The tea has gone cold in the pot, the lamp burns steady, and over the water the two ring-systems have long since run out and the surface has closed over where they crossed.*]

Chapter 5: Roads That Cancel

— Two-Path Invention —

Part II: Wires Together

Escher: "Rippled Surface" — a still pond holding the moon and the bare trees, and two sets of rings spreading across the reflection, crossing

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If a walker can take two roads to the same place at once, can the two roads erase each other so that the walker never arrives?

Nothing in ordinary life prepares us for a yes. Two roads to a town are two chances of reaching it; block one and the other still serves; open both and you are more likely to arrive, not less. Chances add, and adding makes things larger. A traveller who could take both roads at once would, if anything, be doubly sure of the town.

Chapter 4 left a wire in a strange condition. A single start, sent through the gate H, was pointed at *both* answers at once; and if a second H came after it, there were now two ways to get from the start $|0\rangle$ back to the end $|0\rangle$ — one through the middle value $|0\rangle$, one through the middle value $|1\rangle$. Two roads, one start, one finish. The question we shirked there is the question of this chapter. When the walker travels both roads, what reaches the far end is not two chances but two *amplitudes*, and amplitudes are not chances. They are numbers that may be positive or negative, and two numbers of opposite sign do not pile up. They cancel.

This is the whole of the chapter, and it is more than a curiosity. If a road's sign could be made to depend on some unknown function f — if travelling the road *asked a question* and came back marked $+$ or $-$ by the answer —

then two roads carrying two answers would, on meeting, report not either answer but their *relation*: whether they agree or differ. That is a fact about f as a whole, wrung from a single walk. By the end of the chapter one walk, past one oracle, will settle with certainty a question that any road-by-road interrogation needs two answers to decide; and a longer board, sixteen roads wide, will let eight roads cancel eight and leave nothing at all. The algebra will be three short lines. We shall draw it too — every road, every sign — and be honest at the close about which of the two strands did the proving and which only made the cancellation something you could see.

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§5.1 Two Roads

Return to the picture Chapter 4 left open. The gate H is the labelled box of Chapter 3, the one that sends a definite value to an even mixture:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

and it is its own undoing, $HH = I$ — a fact we proved in Chapter 3 and shall now *see*. Put two H's in a row and feed in $|0\rangle$.

On the blackboard. Take it in two steps, naming the middle value each time.

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \xrightarrow{H} \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) = \tfrac{1}{2}(|0\rangle + |1\rangle) + \tfrac{1}{2}(|0\rangle - |1\rangle).$$

Now read the amplitude of each ending. The end $|0\rangle$ is reached by two roads — through the middle value $|0\rangle$ (amplitude $\frac{1}{2}$) and through the middle value $|1\rangle$ (amplitude $\frac{1}{2}$) — and the two have the *same* sign, so they pile up to 1. The end $|1\rangle$ is also reached by two roads, of amplitudes $+\frac{1}{2}$ and $-\frac{1}{2}$, and these have *opposite* sign, so they cancel to 0. The walker comes home to $|0\rangle$ with certainty. Two H's undo each other, and the reason is a cancellation.

Now insert, between the two H's, a half-turn of phase. This is the gate Z of Chapter 3: it leaves $|0\rangle$ alone and sends $|1\rangle \to -|1\rangle$, so it flips the sign of the $|1\rangle$-value road and nothing else. The middle line above becomes

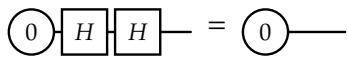
$$\frac{1}{\sqrt{2}}(H|0\rangle - H|1\rangle) = \frac{1}{2}(|0\rangle + |1\rangle) - \frac{1}{2}(|0\rangle - |1\rangle) = |1\rangle\,.$$

The signs have swapped fortunes: now the two roads to $|0\rangle$ cancel and the two roads to $|1\rangle$ pile up. The walker is turned away. In one matrix,

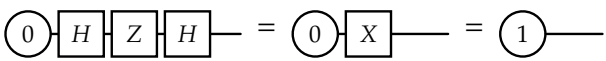
$$HZH = X,$$

the bit-flip of Chapter 3 — a wire that carries every value straight across to the *other* value. A half-turn of phase between two mirrors is exactly a flip. This is the Mach–Zehnder interferometer, written in gates: two half-silvered mirrors (H and H) with a phase in one arm, and whether the particle leaves by the near door or the far door is decided entirely by whether the two internal roads agree or disagree in sign.

In the sketchbook. Two boxes on one wire; a state at the start, the wire running out. With no phase between the mirrors, the picture is a wire with nothing on it:



With the half-turn Z between them, the two mirrors and the phase are a single bit-flip, and the start $|0\rangle$ leaves as $|1\rangle$:



Both equations are exact, on the nose — the same matrix on each side, number for number. But as pictures they only *report* what the blackboard found. They do not yet show the roads, and it is the roads that carry the story: two of them, meeting, one pocket filling and one emptying. See

`Code/Ch5/TwoRoads.py`.

Why this picture. We drew the phase as a box Z sitting *on the wire*, not as a fork in it, because a wire that splits into two roads and rejoins would be a new kind of drawing, and we have no rule for it yet. The next section supplies the honest Stage-A way to draw "both roads at once" — a way that never cuts the wire.

> **Feynman's Notebook.**
>
> The Mach–Zehnder interferometer is a real optical instrument, and Cleve, Ekert, Macchiavello and Mosca open their 1997 review — the paper this chapter leans on throughout — by describing it. A particle, a photon say, meets a half-silvered mirror and takes two paths at once; a phase shifter sits in each arm; a second half-silvered mirror brings the two paths back together and directs the particle to one of two detectors. That second mirror, they stress, erases all record of which path was taken — which is exactly what lets the two paths interfere. They then identify the half-silvered mirror with the Hadamard gate H, and argue that a quantum computer is at bottom a many-particle interferometer whose phase shift is computed by a logic gate. The whole chapter is that one idea, turned over and over.

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§5.2 A Sum of Pictures

Achilles wants to draw the walker going both ways. His first instinct is to split the wire — to draw it forking, one branch for each middle value. The Tortoise will not let him. A wire that forks would be a box with one input and *two* outputs, a different creature altogether from the boxes we have (we shall meet it, and its name, much later). In Stage A a wire never splits. So how is "both at once" to be drawn?

On the blackboard. The answer is already in the algebra of §5.1, and it needs a name. A box applied to a definite value is a *weighted list* of definite

values:

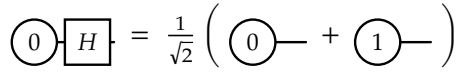
$$A\,|x\rangle = \sum_{y} A_{yx}\,|y\rangle\,,$$

where A_{yx} is the entry of A in row y and column x — the amplitude for the road from x to y. Nothing here is a fork. It is a *sum*: the output is so many parts of $|0\rangle$ plus so many parts of $|1\rangle$, each part a definite value with a number in front of it. And boxes are **linear** — a box applied to a sum is the sum of the box applied to each part:

$$B\big(A\,|x\rangle\big) = \sum_{y} A_{yx}\,B\,|y\rangle = \sum_{y} A_{yx} \sum_{z} B_{zy}\,|z\rangle = \sum_{z}\Big(\sum_{y} B_{zy}A_{yx}\Big)\,|z\rangle\,.$$

Read the double sum before it is collected. Each term $B_{zy}A_{yx}$ is one **road**: in the front door at x, through the middle value y, out at z, carrying the product of the two amplitudes it met. The output amplitude for z is the sum of these roads over every middle value y they could pass through. This is the picture Chapter 4 gestured at and this chapter is named for: *a computation is a sum over roads*, each road a definite choice at every step, each carrying a signed weight, and what you observe at the end is their sum. It is the book's own way of saying what physicists call a sum over paths.

In the sketchbook. So we draw "both at once" as a **sum of pictures**. A picture equals a sum of *whole* pictures, one for each definite output value, each with its wire unbroken and a number — a sign, in the cases we care about — written in front. Here is the first and simplest, the gate H acting on $|0\rangle$:



Each summand on the right is a complete, legal Stage-A picture — a definite value on an unbroken wire. No wire was cut. The equality is exact, on the nose: the same vector on each side, amplitude for amplitude. From $|1\rangle$ the same box gives a sum with a *minus* on the second road, $H\,|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

the $|1\rangle$-road is signed. And because boxes are linear, a sum of pictures may always be carried forward through whatever box comes next — you push each summand through and add. That one permission is what turns a stack of still pictures into a moving calculation, and every step of this chapter's picture-proofs will use it.

Why a sum, not a fork. Two reasons, and they agree. A fork would change the *type* of the diagram — one wire in, two out — and would need a rule we do not have. A sum changes nothing: the wire stays a wire, the drawing stays inside the grammar we built in Chapters 1–4, and the new content is carried entirely by the little + signs and the numbers in front. The picture pays for "both roads" in ink, not in new machinery. See `Code/Ch5/SumOfPictures.py`.

Penrose's Sketchbook.

A caution about the sum of pictures, because the drawing invites a wrong reading. The two roads on the right are not two walkers you could stop and question one at a time. They are summands of a *single* picture, and the whole of interference lives in that word "single." Put an effect on one road alone — read the middle value — and the sum collapses to that one road; the cancellation of §5.4 needs both roads, untouched, and so it never survives a peek. Nor is the number written in front — the $\frac{1}{\sqrt{2}}$, or a kicked $(-1)^{f(x)}$ — a box. It sits on no wire and passes through no gate; row 37 takes the oracle *away* and leaves a bare sign, so the picture ends with fewer boxes, not a new one. And when two roads of opposite sign meet, what is left is not some special empty box but the plain number 0.

---

§5.3 The Mark on the Other Wire

A road now carries a sign. Achilles asks the question that unlocks the chapter: could a road's sign be made to depend on a *question* — on some

function f? Then walking the road would interrogate f, and the sign coming back would be the answer.

To ask a question we need something to ask it of. Bring in a second wire — call the first the **query** wire and the second the **answer** wire — and the oracle that couples them. An oracle for a function $f : \{0,1\} \to \{0,1\}$ is the reversible box

$$U_f : |x\rangle\,|y\rangle \longmapsto |x\rangle\,|y \oplus f(x)\rangle\,,$$

which leaves the query alone and adds $f(x)$, modulo 2, into the answer. We draw it as a *tall box* laid across both wires — not a thin line, so that it is never mistaken for a wire of its own. Feed it a definite query $|x\rangle$ and a definite answer $|y\rangle$ and it does the obvious thing: it writes $f(x)$ into the answer. If the answer starts at $|0\rangle$, it comes out reading $f(x)$, and the mark lands where we expect — on the answer wire.

On the blackboard. Now do the one thing that is not obvious: set the answer wire not to a definite value but to the *difference* state

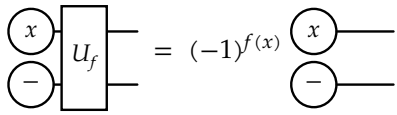
$$|-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Watch what U_f does to $|x\rangle\,|-\rangle$, taking the two cases of $f(x)$ together. Since U_f sends $|x\rangle\,|0\rangle \mapsto |x\rangle\,|f(x)\rangle$ and $|x\rangle\,|1\rangle \mapsto |x\rangle\,|1 \oplus f(x)\rangle$,

$$U_f\,|x\rangle\,|-\rangle \;=\; \tfrac{1}{\sqrt{2}}\,|x\rangle\,(|f(x)\rangle - |1 \oplus f(x)\rangle) \;=\; (-1)^{f(x)}\;|x\rangle\,|-\rangle\,.$$

If $f(x) = 0$ the bracket is $|0\rangle - |1\rangle$, unchanged; if $f(x) = 1$ it is $|1\rangle - |0\rangle$, which is $-(|0\rangle - |1\rangle)$. Either way the answer wire comes back *exactly* as it went in, $|-\rangle$, and the entire action of the oracle has become a single sign, $(-1)^{f(x)}$, standing in front. The answer wire is an eigenstate of the oracle; the mark the oracle would have made on it has nowhere to land there, so it lands on the *query* wire instead. This is **phase kickback** (Cleve and colleagues, 1997, Eq. 2.2). The bar touched the answer wire; the mark appeared on the query.

In the sketchbook. With the answer wire held in $|-\rangle$, the tall box U_f across a definite query and that held answer equals a *number*, $(-1)^{f(x)}$, written before the same picture with the oracle taken away:



exact, sign included. The oracle has spent itself: the two wires are as they were, and all that is left of the query is a signed number. See `Code/Ch5/Kickback.py`.

Why the difference state. Put a definite $|0\rangle$ on the answer wire and the oracle writes $f(x)$ there and leaves the query untouched — the mark stays on the wire you asked *about*. Only when the answer wire is an eigenstate of the flip, $|-\rangle$, does the oracle have nothing to change there, and the only bookkeeping left is a sign on the query — the wire you asked *from*. The difference state is precisely the state that turns an action into a mark.

Dirac's Blackboard.

Why $|-\rangle$ and not $|+\rangle$. The flip on the answer wire is the box X, and its eigenvectors are $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$:

$$X|+\rangle = +|+\rangle\,, \qquad X|-\rangle = -|-\rangle\,.$$

When $f(x) = 1$ the oracle applies X to the answer wire; when $f(x) = 0$ it does nothing. So on the eigenvector $|-\rangle$ the oracle multiplies by (-1) exactly when $f(x) = 1$, i.e. by $(-1)^{f(x)}$, and $|+\rangle$ would give $(+1)$ always — no mark at all. The kicked sign is the eigenvalue -1 of X, carried out of the answer wire and set down on the query. Nothing is created: the phase was X's all along, and $|-\rangle$ merely refuses to hide it. See `Code/Ch5/Kickback.py`.

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§5.4 The One Question

We can now ask f one question and get back, not $f(0)$, not $f(1)$, but the one thing a single walk can decide with certainty: whether they *agree*.

First feel the need. Suppose you are allowed to consult the oracle exactly once, and you want to know whether f is **constant** ($f(0) = f(1)$) or **balanced** ($f(0) \neq f(1)$). Classically you are stuck: one evaluation tells you one value, $f(0)$ *or* $f(1)$, and either value alone is consistent with both a constant f and a balanced one. You would need two evaluations. Nor does the quantum machine rescue you by letting you read a single road: send in $|0\rangle$, kick a sign, and *read* the query wire, and you get a fair coin that tells you nothing about f — each single road answers in superposition. The trap looks airtight. It is not, and the escape is interference.

Here is a concrete run to make the mechanism visible. Take the balanced function $f(0) = 0, f(1) = 1$ (the identity). Prepare the query in $|0\rangle$, the answer in $|-\rangle$, and put an H on the query before and after the oracle.

- After the first H: the query is $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ — both roads open.
- Through the oracle: kickback stamps each road with the oracle's sign — the $|0\rangle$-road by $(-1)^{f(0)}$, the $|1\rangle$-road by $(-1)^{f(1)}$ — which for this f leaves $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. The two roads now carry *opposite* signs, because f sent them opposite answers.
- Through the second H: this is $H|-\rangle = |1\rangle$. The roads to $|0\rangle$ cancel; the walker is turned away to $|1\rangle$.

Read the query and you get $|1\rangle$, with certainty — the balanced verdict — from a single consultation. Try a constant f and the two roads come back with the *same* sign, pile up, and the walker comes home to $|0\rangle$. The one walk did what two evaluations do.

On the blackboard. In general, for any of the four functions f,

$$
\begin{aligned}
(H \otimes I)\, U_f\, (H \otimes I)\; |0\rangle\, |-\rangle &= (H \otimes I)\, U_f\, \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\, |-\rangle \\
&= (H \otimes I)\, \tfrac{1}{\sqrt{2}}((-1)^{f(0)}\, |0\rangle + (-1)^{f(1)}\, |1\rangle)\, |-\rangle \\
&= (-1)^{f(0)}\, (H \otimes I)\, \tfrac{1}{\sqrt{2}}(|0\rangle + (-1)^{f(0)\oplus f(1)}\, |1\rangle)\, |-\rangle \\
&= (-1)^{f(0)}\; |f(0) \oplus f(1)\rangle\; |-\rangle\,.
\end{aligned}
$$

The last step is the two cases of §5.1 read backwards: if $f(0) \oplus f(1) = 0$ the query is $H|+\rangle = |0\rangle$; if it is 1, the query is $H|-\rangle = |1\rangle$. So the query wire ends in $|f(0) \oplus f(1)\rangle$ — $|0\rangle$ for constant, $|1\rangle$ for balanced — read off with a single certain measurement, one oracle call. **This is Deutsch's algorithm** (Theorem 5.1), and §5.7 proves it twice.

The factor $(-1)^{f(0)}$ out front is a **global phase**. It multiplies the whole state, so it cannot change any measured outcome — the query still reads $f(0) \oplus f(1)$ with certainty. But it is genuinely there, and the equation is only true with it: drop it and you have written down a falsehood. We keep it. See `Code/Ch5/Deutsch.py`, which checks the equation for all four f with the sign intact.

Why one query can do it. Classically, one call yields one bit and that bit is one value of f; the relation $f(0) \oplus f(1)$ is a second bit that no single value determines. Quantumly, the single call is spent on *both* roads at once, kickback writes the two answers into the two roads' signs, and the second H reads out not either sign but whether they match. You never learn $f(0)$ or $f(1)$ — the global phase is exactly the information you threw away — and in exchange you learn the one thing their *difference* holds.

> **Feynman's Notebook.**
>
> A careful word about who did what, because it is easy to get wrong. David Deutsch posed this problem in 1985 and gave the first quantum algorithm for it — but that original algorithm was *probabilistic*. As Cleve, Ekert, Macchiavello and Mosca describe it, Deutsch's 1985 procedure had three possible outcomes, "constant," "balanced," and "inconclusive": with probability one-

half it returned the right answer, and with probability one-half it returned "inconclusive" and told you nothing. Still remarkable — no classical single evaluation can even do that — but not a certainty.

The *deterministic* one-query algorithm, the one proved above that always answers correctly, came later: it is due to Cleve, Ekert, Macchiavello and Mosca (1997), who record that Alain Tapp found a similar algorithm independently the same year (private communication, 1997). So the version this chapter calls "Deutsch's algorithm" is really the 1997 improvement of Deutsch's 1985 idea. The credit for the certain single-query answer belongs to them, not to the 1985 paper.

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§5.5 The Sixteen Roads

One walk decided a two-valued question. Stretch the question over more roads and ask how far the cancelling reaches.

Let f now take four bits in, $f : \{0,1\}^4 \to \{0,1\}$, with a promise: f is either **constant** (the same value on all sixteen inputs) or **balanced** (value 0 on exactly eight inputs and 1 on the other eight). Which is it? Classically the worst case is cruel: you may see eight equal values in a row and still not know, so you may need $2^{4-1}+1 = 9$ evaluations before you are certain. This is the problem Deutsch and Jozsa posed and solved in 1992 for inputs of any width; the one-oracle-call circuit we are about to draw is the streamlined form given by Cleve and colleagues in 1997.

On the blackboard. Put four query wires, each starting at $|0\rangle$ with an H before and after the oracle, and one answer wire held in $|-\rangle$. After the first four H's the query register is an equal sum over all sixteen inputs, $H^{\otimes 4}|0000\rangle = \frac{1}{4}\sum_x |x\rangle$ (four factors of $\frac{1}{\sqrt{2}}$ make $\frac{1}{4}$). Kickback stamps each input's road with its sign, and the second layer of H's brings the roads together. We ask for one number: the amplitude with which the whole thing returns to $|0000\rangle$, which is what a closed diagram computes. It is

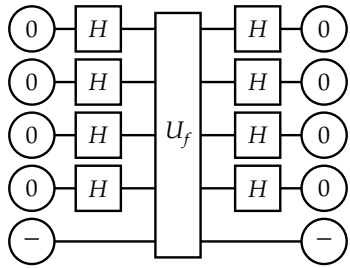
$$\langle 0000|\, H^{\otimes 4}\, U_f H^{\otimes 4}\, |0000\rangle \;=\; \frac{1}{16} \sum_{x\in\{0,1\}^4} (-1)^{f(x)},$$

with the answer wire riding in $|-\rangle$ at both ends, its kickback already absorbed into the signs and $\langle -|-\rangle = 1$ leaving no stray factor behind. (We keep $|-\rangle$ *normalized* at both the start and the finish precisely so that this number comes out clean; an unnormalized answer wire would multiply it by 2.) Read the two cases:

- f **constant**: all sixteen signs agree, the sum is ± 16, the number is ± 1. Every road piles into the home pocket.
- f **balanced**: eight signs are $+$ and eight are $-$, the sum is 0, the number is 0. Eight roads cancel eight, and the home pocket empties.

The chapter's title, made literal. See `Code/Ch5/SixteenRoads.py`, which builds a genuine balanced f (checking the tally is eight-and-eight before trusting the zero) and a constant one, and confirms both values.

In the sketchbook. Draw the whole thing closed — states at the head of every wire, effects at the foot — so that the picture is a single number. This is the master diagram of the chapter, **The Sixteen Roads**: four query wires, each running $|0\rangle \to H \to U_f \to H \to \langle 0|$, and one answer wire running $|-\rangle \to U_f \to \langle -|$ with no H, since it is the kickback eigenstate. The tall box U_f is laid across them all.



Each query wire runs unbroken from its state to its effect, straight through the tall oracle; the answer wire runs likewise beneath. The oracle crosses them all and joins none — a wide box is not a wire. This closed picture *is* the number $\frac{1}{16}\sum_x (-1)^{f(x)}$: sixteen roads through it, each a definite choice

of the four query values, each carrying a sign, summed. When they agree it is ± 1; when they balance it is 0. The equality is exact, on the nose.

Why draw it closed. An open diagram is a state or a process — a thing with loose ends. Closing both ends turns it into a single number (the rule of Chapter 2), and it is the number, not the process, that answers the constant-or-balanced question. The picture makes the answer a *pile height*: sixteen tokens in a pocket, or none.

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§5.6 The Rosetta Table

The rows this chapter adds to the running dictionary. Rows 33 and 34 introduce the new picture element — the *sum of pictures* — in its first and general form; row 35 reads a product of boxes as a sum over roads; row 36 is the rule that lets two roads pile up or cancel; rows 37 and 38 record kickback and the closed diagram whose value the signs decide. Every equality is exact, scalar included.

#	Algebra (Dirac)	Picture (Penrose)
33	$H\|0\rangle = \frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$; $H\|1\rangle = \frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$	a state through the box H = a $\|0\rangle$-road plus a $\|1\rangle$-road; from $\|0\rangle$ both roads carry $+$, from $\|1\rangle$ the $\|1\rangle$-road is signed $-$ — the first *sum of pictures* (*new element*)

#	Algebra (Dirac)	Picture (Penrose)
34	$A\|x\rangle = \sum_y A_{yx}\|y\rangle$	a state through **any** box = a sum of one-road pictures, one per output value, each weighted by its entry, every wire unbroken — the general *sum of pictures*
35	$BA\|x\rangle = \sum_y (BA)_{yx}\|y\rangle = \sum_{y,z} B_{zy}A_{yx}\|z\rangle$	two boxes in a row on a state = a sum over the middle value = the **roads** (the sum over paths)
36	$\|\psi\rangle + \|\phi\rangle$; opposite amplitudes give 0	one-road pictures of the **same** sign make one picture of doubled weight; of **opposite** sign they make the number 0 (the roads cancel)
37	$U_f\|x\rangle\|-\rangle = (-1)^{f(x)}\|x\rangle\|-\rangle$, $\|-\rangle$ an eigenstate	the tall box U_f across a definite query and a $\|-\rangle$ answer = the number $(-1)^{f(x)}$ before the same picture with U_f removed — **phase kickback**
38	$\langle 0\|^{\otimes n}\langle -\| \, (H^{\otimes n} \otimes I)U_f(H^{\otimes n} \otimes I) \, \|0\rangle^{\otimes n}\|-\rangle = \frac{1}{2^n}\sum_x (-1)^{f(x)}$	a fully closed diagram = a **number**: the signed sum of its one-road pictures, 0 when the + and − roads balance

All six rows are exact equalities. The sum of pictures (rows 33–36) is the last new element of Stage A; from the next chapter the wire itself will be allowed to bend.

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§5.7 The Theorem, Twice

Theorem 5.1 (Deutsch's algorithm). For $f : \{0,1\} \to \{0,1\}$ given as the oracle $U_f : |x\rangle\,|y\rangle \mapsto |x\rangle\,|y \oplus f(x)\rangle$,

$$(H \otimes I)\, U_f\, (H \otimes I)\; |0\rangle\,|-\rangle \;=\; (-1)^{f(0)}\; |f(0) \oplus f(1)\rangle\; |-\rangle\,,$$

exactly, global phase included — so one oracle call decides constant ($f(0) = f(1)$) versus balanced ($f(0) \neq f(1)$) with certainty, where any classical strategy needs two evaluations.

Dirac's Blackboard

Line by line, with the answer wire in $|-\rangle$ throughout. Write $a = f(0) \oplus f(1)$.

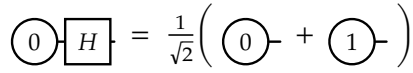
$$\begin{aligned}
|0\rangle\,|-\rangle &\xrightarrow{H\otimes I} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\,|-\rangle && \text{(both roads open)}\\
&\xrightarrow{U_f} \frac{1}{\sqrt{2}}((-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle)\,|-\rangle && \text{(kickback)}\\
&= (-1)^{f(0)}\,\frac{1}{\sqrt{2}}(|0\rangle + (-1)^{a}|1\rangle)\,|-\rangle && \text{(common sign out)}\\
&\xrightarrow{H\otimes I} (-1)^{f(0)}\;|a\rangle\;|-\rangle\,. && (H|-\rangle = |1\rangle\,)
\end{aligned}$$

Three lines of work. The middle two are the whole content: kickback turns the oracle into a pair of signs on the two roads, and the last H reads whether those signs agree ($a = 0$, query $|0\rangle$) or differ ($a = 1$, query $|1\rangle$). The global phase $(-1)^{f(0)}$ rides through untouched and is still there at the end. ■

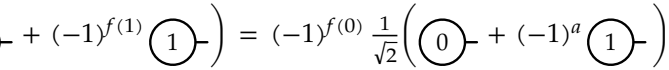
Penrose's Sketchbook

The same calculation, drawn — tracking the query wire's road-sum, with the answer wire held in $|-\rangle$ beside it and supplying every sign through kickback (row 37).

Step 1 — split the first H (row 33). The start $|0\rangle$ through H becomes two roads of equal, positive weight:

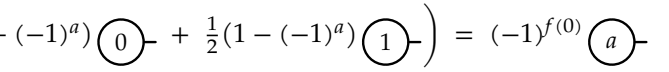


Step 2 — carry the sum through the oracle (linearity + row 37). Push each road through U_f. The answer wire is $|-\rangle$, so the oracle spends itself as a sign on each query road: the $|0\rangle$-road picks up $(-1)^{f(0)}$, the $|1\rangle$-road picks up $(-1)^{f(1)}$.



with the common sign $(-1)^{f(0)}$ pulled to the front and $a = f(0) \oplus f(1)$.

Step 3 — split the second H and collect (row 33, then row 36). Push each road through the last H: the $|0\rangle$-road becomes $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, the $|1\rangle$-road becomes $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Four one-road pictures now land in two pockets. In the $|0\rangle$ pocket the weight is $\frac{1}{2}(1 + (-1)^a)$; in the $|1\rangle$ pocket, $\frac{1}{2}(1 - (-1)^a)$. By row 36 the two roads in a pocket pile up or cancel:



For $a = 0$ (constant) the $|0\rangle$ pocket fills to 1 and the $|1\rangle$ pocket empties; for $a = 1$ (balanced) the reverse. Either way one pocket carries a single road of weight ± 1 and the other holds nothing: the query is $|a\rangle = |f(0) \oplus f(1)\rangle$,

with the global phase $(-1)^{f(0)}$ still in front, and the answer wire is $|-\rangle$ as it always was. ■

The verdict

The two proofs are the same length and the same calculation — line for line, the picture translates the algebra: split, carry, kick, split, collect. This is honest to report and worth stating plainly: **there is no Stage-A rewrite that derives Deutsch's algorithm more cheaply than the kets do.** The blackboard is, if anything, the more economical of the two, because it writes a superposition as one short sum where the picture must draw each road as a separate tile.

What the picture earns is not brevity but *sight*. On the blackboard the cancellation is a line of arithmetic, $\frac{1}{2}(1-(-1)^a)$, that happens to vanish; in the sketchbook it is two tokens of opposite face dropped into one pocket, which you watch empty. The strand that proves the theorem here is the algebra. The strand that shows you *why the answer is certain* — why one pocket must be full and the other bare — is the picture. Neither claims the other's virtue.

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§5.8 The Sign That Was Not There

Gather the chapter. Two roads to one place can add or cancel, and a half-turn of phase between two mirrors decides which (§5.1). "Both roads at once" is drawn not by cutting a wire but by summing whole pictures, each a signed road (§5.2). With the answer wire held in $|-\rangle$, an oracle's action becomes a sign kicked onto the query (§5.3); two such signs, brought together by a second H, report whether f agrees with itself — one query, a certain verdict (§5.4); and sixteen roads under a balanced oracle cancel to nothing at all (§5.5). Throughout, the deciding quantity was a *sign* — a thing with no size, invisible to any single road, felt only where two roads meet.

That is the last work Stage A will do with a straight wire. Every road in this chapter ran unbroken from one end of the board to the other; the only

new thing we allowed was to *sum* such roads. Notice what we never once needed: a wire that leaves and returns. Notice, too, what the wires never did. The whole trick of kickback was that a mark made on one wire showed up on another; yet the answer wire went in as $|-\rangle$ and came out as $|-\rangle$, and at every step it could be described on its own, apart from the query — the hopping sign was only a number standing in front. The wires talked, and still they came apart. In the next chapter we meet a pair that does not come apart. We let a single wire bend back on itself so that both its ends point the same way, and we find that this bent wire — going out and coming home — is by itself a shared fact between two places, held in neither of them. The roads that cancel were a rehearsal. What is kept in the bend is the thing itself.

Obbligato for Six Music Boxes

長き夜や
離れて一つ
オルゴール

nagaki yo ya
hanarete hitotsu
orugōru

The long autumn night —
music boxes, far apart,
one note between them.

[*Achilles and the Tortoise have come to the Crab's for the afternoon; the Crab meets them at the door, squinting into the low sun.*]

Crab: Come in, come in, both of you — mind the step; at this hour you come up to it in your own shadow, and it hides there. You found the lane?

Achilles: Tortoise found it. Left to me, I'd have walked us straight into the canal.

Tortoise: He was giving directions and telling a story at the same time. One of them had to go.

Achilles: A concert, it was — a quartet, and the cellist a whole bar behind the other three the entire evening. I have never heard anything so wrong

stay so cheerful about it.

Crab: Then you have come to exactly the right house. Come through; there's a pot just drawn. I have something I've been aching to show a pair of people who'd know what they were looking at — and I have waited all week for tonight.

Tortoise: We are all attention, Mr Crab. Achilles most of all; he has promised to keep step with me for once, which is more than his cellist managed.

[*They go through to the warm room and take the chairs he waves them to, the lane shut out behind them. The Crab pours, glad of company, and hands the cups round; Achilles, still beating time for his cellist with one hand, has to be offered his a second time.*]

First Challenge — *Two Recipe Cards*

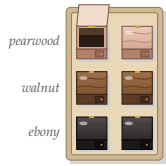
[*The Crab's music room, late on an autumn afternoon. Nearly everything in it was bought as a pair and has stayed one: two upright pianos side by side against the long wall, each tuned to the other, so that it is no use asking which of them is flat; two music stands sharing the two halves of one duet; a lamp at either end of the table, not yet lit; and on the mantelpiece two clocks by one maker, which do not keep step — their ticks fall anyhow, like two people walking on gravel. Over the mantel, between the clocks, hangs Escher's lithograph* Bond of Union *in a narrow black frame: two heads, face to face, hollow as peeled rind, with small globes adrift in the dark around them and inside them. The sun, low in the one tall window, comes through a pair of net curtains and has not reached it yet.*]

[*The Crab's table. On a tray stand his music boxes, none bigger than a matchbox, set out two by two, pair behind pair from the window side of the table to the near: a pair in pearwood farthest off, a pair in walnut, a pair in ebony. Each has a lid on top and a small door in its side. Beside the tray lies a ledger.*]

Crab: Now, before anyone touches anything. These came to me from a little workshop in Delft, behind the Oude Kerk, where a father and his daughter make nothing else. They are made in pairs, and only in pairs —

the pearwood together, the walnut together, the ebony together — each pair voiced at one bench on one afternoon. And the first thing the daughter tells every customer is—

[*The left pearwood box sits shut on the tray, one note folded up inside it, whole. Achilles's thumb catches the lid and lifts it — the box itself does not move, only the little hinged lid swings up — and the note springs out into the room: a single high tone, clear as a struck glass.*]



the lid up

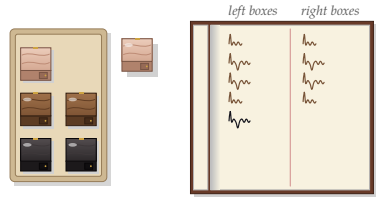
[*Then it is gone, and the lid settles back on a box with nothing left inside to sound. What was held is spent; what stays is a small pearwood box that will never sing again.*]

Achilles: Oh, that's lovely. How do you wind it again?

Crab: —is that one doesn't.

Achilles: Doesn't what?

Crab: Wind it again. Each box plays one note, once. High or low; you have had high. It is now a very small pearwood box with nothing further to say. [*He writes a single word, in fresh ink, in the left-hand column of the ledger — the first of a pair — and leaves the space beside it in the right-hand column blank; then he lifts the right pearwood box off the tray and folds it against his shell, shut and full, while its spent partner sits shut and silent on the tray*] And nobody touches this one.



held apart *one word waiting*

Achilles: I'm terribly sorry. One note? It seems a great deal of cabinet-making for one note.

Tortoise: It was a very good note.

[*The left pearwood box is the Crab's marvel no longer: the same close grain, the same hinge no bigger than a grain of rice, and about as much left to say for itself as a box one keeps stamps in.*]

Crab: It was half of something. There are two ways in, by the way, since you didn't wait to be told. The lid, which you have found. And the side door, which strikes the reed from the other side: by the door one hears not high or low but one of two other notes — the workshop calls them bright and dark. Either way: one note, once, and the box is spent.

Achilles: And which does a box play? By the lid, say. High or low?

Crab: Nobody can tell you. I have weighed them. I have held them up to the lamp. The daughter in Delft says she doesn't know either, and I believe her; she has an honest face.

Achilles: Let me put it on paper the way one writes out a recipe — a card for each box, and a card meant to hold everything there is to say about the thing it describes. If a box really is nothing but a coin, one card will catch it whole.

Achilles: Then I can tell you, in the only sense that's left. [*He takes two blank cards from the Crab's recipe tin and writes.*] A recipe card for each box. Left box: "plays high half the time, low half the time." Right box: the same. There — the pair, fully described. Two boxes, two cards, and a coin toss on each.

Tortoise: A recipe ought to be cookable. Would you play your cards for us? A coin for each card; heads for high.

Achilles: Gladly. [*He tosses two coins together.*] High, high. [*Again.*] High, low. Low, low. Low, high. High, low…

[*The Anteater is shown in while the coins are in the air.*]

Anteater: Good evening, good evening — don't get up, you are in the middle of a tally. I bring Aunt Hillary's regards. She has moved her nurseries to the south slope of the hill this week, and is very pleased with herself.

Crab: Delighted. Sit by the ledger, Doctor; it is about to be wanted.

[*Fresh tea comes in behind him: two pots of one pattern, one holding tea and the other hot water, which the household has never learned to tell apart except by pouring. The Anteater sits where he is told. There is a little reddish earth dried on his cuff.*]

Achilles: …low, high. High, high. That's twenty throws. The coins agreed eleven times and disagreed nine. About half and half, which is as it should be: a coin doesn't care what its neighbour does.

Tortoise: And the boxes, Mr Crab? How often have the two boxes of a pair disagreed?

Crab: [*turning the ledger back to an old page*] I have been spending pairs on my friends for eleven years. Left box in the left column, right box in the right column, a line for each pair. My hand is no clerk's, but a word always comes out the same shape. Read across, and keep going.

Achilles: High, high. Low, low. Low, low. High, high. Low, low… [*He turns the page.*] Here's a "bright, bright".

Crab: A side-door evening. Both boxes by the door. One opens a pair by the same way in; I was told so very firmly.



read across

Achilles: They never disagree. Not once, in — it must be hundreds of lines. My coins couldn't keep it up for twenty.

Tortoise: Then your cards are the recipe for some other dish.

Achilles: But I'd still swear to each of them! Whatever is true of that left box, it's on my left card, and the same for the right. It's only when I lay the two cards side by side that they start telling lies. How can two honest cards make a dishonest pair?

Tortoise: What did you leave off them?

Achilles: Each other. [*He tears both cards across, takes one fresh card, and writes along its whole width.*] "The pair: both high half the time, both low half the time." Two cards won't do it. It needs one card for the pair.

Crab: And that card will not cut in two. I have thought about it on many evenings. Whichever way you take the scissors to it, you lose the word "both".

[*At the window the two net curtains hang one behind the other, and where they overlap the low sun shows a watered pattern, broad slow bands like the figure in walnut, which is woven into neither of them. A draught lifts the inner curtain by a finger's breadth; the bands race the whole height of the window, and settle.*]

Achilles: But now I've a card for the pair and no card for a box. And there it sits in your claw, Mr Crab — one box, a solid object, with a lid. Then what does *one* box know?

Second Challenge — *The Ledger*

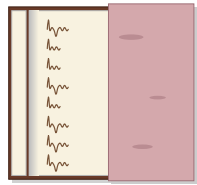
Anteater: At which level are you asking?

Achilles: I beg your pardon?

Anteater: You asked what one box knows. It is a fair question, but questions have addresses, and I am not sure yours has been delivered to the right one. Mr Crab, might the blotter be laid over one of your columns?

Crab: The blotter is here by the inkwell; I keep it to dry a wet page. It will hide a column of the ledger as neatly as it dries one.

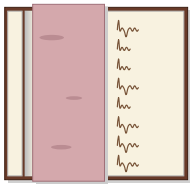
Crab: I'll do it. Nobody handles the ledger but me. [*He turns back a leaf, to the page Achilles began on, and lays the sheet of blotting paper down the right-hand column, edge to the centre rule: the right column vanishes under it, the left column stays open to the lamp, and eleven years of left-hand boxes stand there with nothing beside them to agree or disagree with*] There. The left boxes by themselves, eleven years of them.



the left boxes alone

Achilles: [*running a finger down the page and keeping a tally on the back of a torn card*] Five highs and three lows on this page. [*He lifts the leaf, blotter and all, peers at the page beneath, and lets it fall back.*] Over the leaf, a door evening: bright, dark, dark, bright — no better. It's a coin. It's as good a coin as mine.

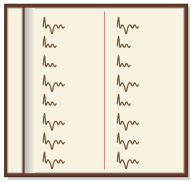
Crab: [*sliding the blotter across to bury the left column instead, and uncovering the right; the same eleven years, read down the other side now*] And the right boxes.



the right boxes alone

Achilles: Another coin. Just as fair and just as empty. If you had handed me either column alone, I'd have said the workshop in Delft had sold you a very expensive way of tossing a penny.

Crab: [*lifting the blotter clean off, so that both columns lie open to the lamp at once, side by side — neither column changed by a single mark from what it was under cover*] And now.



the blotter lifted

Achilles: [*reading across*] It's the same column twice. Line for line, lid evenings and door evenings alike. Either column by itself is a coin; only the two read together are sure of anything.

[*Beyond the curtains the starlings have come in over the roofs to roost, as they do at this hour. Any one bird, followed by eye, goes every way by turns and nowhere in particular; the flock draws itself out, folds, and pours down the sky in a single movement, like a cloth shaken once.*]

Anteater: That is what I meant by an address. Ask any ant of Aunt Hillary's where the colony is moving this week, and as far as I have ever been able to tell you get a coin toss: this one is off east with a crumb, that one west with a grub. Ask Aunt Hillary, and she says "the south slope" at once, and she is right. Mind, I don't say the ants are not there — I am extremely fond of ants — only that they do not hold the answer.

Tortoise: A box knows what an ant knows.

[*The flock crosses what is left of the sun, and the room dims for the length of a breath. When the light comes back it comes level, along the wall over the mantel, and lies for a moment on the glass of the print.*]

Achilles: [*looking up from the twin columns to the print on the Crab's wall — Escher's* Bond of Union] Two heads — and when you follow it round, only the one ribbon. I'd always taken it for a portrait of two people.

Crab: More tea, anyone?

Achilles: No — wait, this is much simpler than you're all making it. The father and daughter in Delft write the note inside beforehand: the same note in both boxes of a pair, high in this pair, low in that, as the fancy takes them. And a second note for the side door, the same in both. Each box alone looks like a coin, because nobody told *us* what they wrote; the pair always agrees, because it was written to. There's your whole ledger — and every box knows perfectly well what it is going to play.

Crab: She said she didn't know.

Achilles: She needn't. The box knows.

Anteater: It is a respectable theory. I cannot find a line in the ledger that contradicts it.

Tortoise: Nor can I, Achilles, and I shan't try. A theory like that deserves better than an argument; it deserves a game. Bring it with you another evening — notes written inside, as many as you please, composed any way you please — and we shall play it against the boxes and see which of them wins.

Achilles: You're being very gracious about losing.

Tortoise: I am always gracious.

[*On the mantel the two clocks strike the hour, the one beginning before the other has done, so that the strokes fall in between one another and nobody in the room could have counted them. Then they go back to their ticking, which has become a limp — long, short, long, short — that changes if one listens to it.*]

Crab: There is a quicker way than a game. [*He has the right pearwood box in both claws, one claw-tip under the edge of the lid.*] We all heard high. One lift, and we should know whether—

Tortoise: Whether what, Mr Crab? You know what it will play.

Crab: I know what the ledger says it will play.

Tortoise: And when you have lifted it you will own one more line of the ledger, of which you have hundreds, and one fewer box that can still sing, of which you have very few.

Crab: [*setting the box down at the far corner of the tray, shut, and keeping a claw beside it*] It is a hard thing, to own something whose whole value lies in its not having been looked at.

Achilles: I'll play your game, Tortoise, and I'll win it. But let me take the other side for a moment, purely for the sport. Suppose it *isn't* written in the boxes. Then where is it kept? The agreement is there on the page; it has to be somewhere on the table.

[*Tortoise reaches into her shell and brings out a piece of string.*]

Third Challenge — *The String*

Tortoise: May I have the walnut pair, Mr Crab? I shan't open them. I only want them to stand somewhere.

Crab: String. Near my boxes. [*He slides the walnut pair out into the middle of the cloth, side by side with the window behind them, shut, with both claws.*] If a lid so much as stirs, Madam Tortoise—

[*The lamps have been lit, one at either end of the table, and everything on the cloth has two shadows. The walnut pair stand out in the middle of it, off the tray for the first time since Delft. Behind them the ebony pair have the tray almost to themselves: the lamplight lies in the left lid and in the right, and there is still a wisp of wood-wool caught in the hinge of the left one.*]

Tortoise: This is nothing but a piece of string. I keep it because it is the plainest thing I know that can be in two places and still be all one thing — and your agreement, Achilles, has to be kept somewhere that is in two places at once. Watch what the two ends of it have to do with each other.

Tortoise: [*she takes the loose straight string and lays it on the cloth in a single hairpin — one end set down at the left walnut box, the string carried away from the pair, towards the company, round in one soft bend, and back again to rest its other end at the right walnut box: two ends now at two boxes, joined behind by one unbroken length*] There is a pair. Achilles, put your hand over one end and tell me what the other is the end of.



a pair

Achilles: Of a string. From here I couldn't say more; it might run anywhere.

Tortoise: And with your hand away?

Achilles: They're the two ends of the *same* string. So that's where it's kept? In the bend — in the one part that isn't at either box? Whereabouts in the bend? This inch, or that one?

Tortoise: [*she pushes the bend further out across the cloth, then draws it back close to the pair, then shoves it crooked off to one side — the bend wanders all over the table, but through every move the two ends never leave their boxes and the string never once comes apart*] Does it matter where I put it?

Achilles: No. It only matters that it goes out and comes back.

Tortoise: Then follow the string, and let us see what a bend will bear. [*She takes the string up and lays it out longer this time: out past the left walnut box, round in a first bend behind it, towards the window, back between the two boxes, round in a second bend that faces the other way about, in front of the right one,*

and out again past it — three legs and two bends now, out, back, out, and still one string]



out, back, out

Achilles: Two bends. Let me work it out. The first bend is a pair: a coin toss on each leg, and the legs agreeing. The second bend faces the other way about, so it's a pair being... listened for? By somebody who wants two legs to agree, which by my cards they'd do one time in — no, wait, the middle leg belongs to both bends, so I must tally it twice, or else halve it. A half of a half of... Tortoise, I have three legs, two bends and four coins, and I've lost one of the coins.

Tortoise: Take that end. I have this one. Pull.

[*Each takes an end and draws it away from the other. The slack runs out of the loop; the two bends slide toward one another, meet, and pull through into nothing — out and back and out drawing straight. What was three legs and two bends is one taut line from Achilles's hand to the Tortoise's, and the two walnut boxes it bent around a moment ago sit untouched to either side. The kinks are gone; the string is the same string, and just as long as it ever was.*]



pulled straight

Achilles: ...Oh. It's a string. It was only ever a string with a kink in it. Out,

back and out again is just — *out*. There was nothing to tally.

[*The draught under the door has dropped. The steam from the spout of the hot-water pot, which has been leaning, doubling back on itself and leaning again, stands up in a single thread, as if it had never done anything else.*]

Tortoise: [*laying it out again the other way about, the first bend now facing down where it faced up — the same S turned into a Z*] And bent like this?



the other way about

Achilles: [*taking his end and pulling without being asked; the two bends run out and cancel exactly as before, the string coming straight*] Straight again. So a bend out and a bend back undo each other, whichever comes first, and whatever goes in at one end of all that travelling comes out at the other exactly as it went in.

Crab: You moved the cloth. The left walnut box has shifted by a hair.

Tortoise: It has not opened.

Crab: It has been *disturbed*.

[*In the silence that follows this remark the clocks can be heard. The limp has evened out: a tick from the left of the mantel, a tock from the right, turn and turn about — the sound of one clock, made by two.*]

Achilles: But of course a bare string pulls straight; there's nothing on it to catch. What if something were threaded on it?

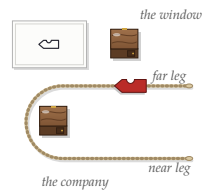
Fourth Challenge — *The Bead*

Tortoise: [*bringing a bead out of her shell*] Something like this. Look at it well before I thread it, because I shall ask you about it afterwards.

Tortoise: It is only a passenger. It rides wherever the string carries it and can change nothing about the string itself; so it will show us honestly what a bend does to whatever is threaded on it — which is just what you wanted to know.

Achilles: A flat bead; it lies on the cloth like a little tile. One end comes to a point. And there's a notch cut into one flank, like a bite out of a biscuit. Lopsided twice over.

Tortoise: Twice over is what we shall need. [*She bends the string once round the left walnut box, so that two legs run back from the bend side by side: a far leg, over nearer the window, and a near leg, nearer the company. She slides the bead onto the far leg and settles it face-up on the cloth, its point aimed along the string toward the bend, its notched flank turned to face the window.*] Point towards the bend. Notch towards the window. [*On a napkin she draws the bead exactly as it lies now: a long tile, a point at its left-hand end, a bite out of the flank on the window side — a fixed record of how it went in.*] That is how it goes in. I am going to slide it along the string, round the bend, and onto the near leg. How does it arrive?

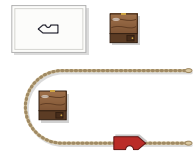


going in

Achilles: As it left. It's the same bead on the same string, and you're doing nothing to it but moving it. Point to the left, notch to the window.

Tortoise: [*sliding the bead along the far leg, round the bend behind the box, and out along the near leg — flat on the cloth the whole way, never once lifted from the*

table or turned over by hand, only carried round the corner by the string it rides on] It went in point first. Look how it comes out.



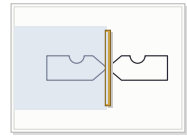
coming out

Achilles: Point first still — but pointing to the *right*. Well, naturally; it's travelling the other way now. Very well, I was hasty. It's reversed: it arrives mirrored. That's the looking-glass bead.

Tortoise: Mr Crab, might we have your mirror?

Crab: My mirror is a watchmaker's inspection glass from Le Locle, silvered on its front face, and it has never been in the same room as string. [*He hands it over.*] By the edges.

Tortoise: [*standing the silvered glass upright on the napkin, its foot set across the nose of the drawn bead, so the inked bead and its reflection meet point to point behind the glass*] Here is the bead as it went in, and there behind it is your looking-glass bead. Read it to me.



in the glass

Achilles: In the glass the point is turned about: pointing right. Good. And the notch is... towards the window. [*He looks at the bead on the near leg.*] But the real one has its notch towards *me*. Away from the window. The point has changed sides and the notch has changed sides too. In the glass only the point has.

Tortoise: So the glass shows you a different bead. There is no such bead on my string.

Achilles: Then what did happen to it?

Tortoise: It never left the table. It went round a corner, flat on the cloth, and a thing carried round a corner arrives facing the other way: turned end for end. A half-turn, Achilles — point and notch go round together. To make the looking-glass bead I should have had to pick it up and lay it down on its face, and I did not.

Achilles: Like a left shoe: walk it round the table as often as you please, and it never comes back a right one.

Crab: May I have the glass back now.

[*The glass goes back into its wash-leather, by the edges. On the rim of Achilles's saucer a ladybird, woken by the lamps from wherever ladybirds spend the autumn, is walking: along the far side towards the door, round the curve, and back along the near side towards the fireplace. It has not, so far as it knows, turned round.*]

Tortoise: Now call the bead anything one might do to a box before it is opened. Done on the far leg, to one box of the pair. I slid it round, and I changed neither the string nor the bead by a hair: it is the same table I started with. So whatever you do to one box of a pair, you could as well have done — turned half round — to the other.

Achilles: And nobody holding the pair could tell which, because there's only the one string. [*He pushes the bead back round the bend with a fingertip — along the near leg, round, onto the far leg — and watches it turn end for end once more, point and notch swinging together back to where they began*] Tortoise. The bead went round the bend. Could the bend go round the bead?

Fifth Challenge — *A Key in the Post*

Tortoise: Mr Crab, you have a tuning key for these boxes.

Crab: I have *the* tuning key. [*He opens a case lined in blue velvet.*] Cut by the father in Delft from a single piece of drill-rod, with a bit as fine as a fish bone. A quarter-turn in the socket at the back, before a box is opened, and a box that would have sung high will sing low, and the other way about. My cousin in Ostend has a pair from the same workshop, and has been begging the loan of it for a year.

Achilles: Then post it to him.

Crab: *Post* it. Achilles, the boxes came from Delft by post, packed in wood-wool, and you could drop them down a well. The key would not survive a sorting office. The key does not survive being looked at sternly.

Tortoise: May I thread it? [*She slips the bead off the string and back into her shell, passes the string through the bow of the key and draws it out straight on the cloth in front of the walnut pair, the key riding at the middle of one taut line: a note comes in along the string at the left end, passes through the key, and leaves at the right end — a wire in, a wire out, and the key at work between*] Here is the key as a thing one *does*. A box's note comes along the string from this end; the key turns it; it goes out at that end, changed. Too delicate to travel. Now. [*She carries the right end across and up to the left walnut box, and swings the left end down, round and up to the right one, until both ends point the same way: the one straight line closes into a hairpin, and where a moment ago there was one wire in and one wire out there are now two legs lying side by side — one running to the left walnut box with the key still upon it, the other running bare to the right. The key has not moved a hair; but it sits on a pair now, not on a through-line.*]



the key at work *a pair now*

Achilles: That's the bead's picture again. A bend, two legs, a box at the end of each — and something sitting on one leg.

Tortoise: So what have I made?

Achilles: A pair of boxes, with the key already turned in one of them. Let me see — in the ledger that pair wouldn't read high-high and low-low like the others. On a lid evening it would read high-low, low-high, all the way down: never agreeing, and just as reliably. It isn't a key any more. It's the key's whole effect, written out as the way two boxes answer each other.

Crab: And a pair can go in the post.

[*The lamplight, coming low across the floor, shows what daylight hid: a track in the carpet from the door to the table and from the table to the cabinet, the pile laid all one way, about the width of a crab walking sideways. Years of errands made it. Anyone crossing the room now goes that way without choosing to.*]

Anteater: A thing done and a thing kept: the same thing at two levels. Aunt Hillary makes no distinction between a habit and a monument, I may say. Her trails are both.

Achilles: ...Oh. Every key is a pair. Any key at all — anything one could do to a box — bend the string behind it and it's a pair of boxes that behave just so. And every pair is a key, if you can bend it back?

Tortoise: [*bending the left end back the way it came and laying the right end back down, so the two legs open out into one line again, and pulling it taut: the hairpin gone, the string straight, the key riding at the middle exactly as before*] The key again. Nothing lost on the way.

Crab: And my cousin performs that last manoeuvre how, precisely? He has no string. He has a parcel from me and a breakfast table.

Tortoise: With a pair of his own, a good ear, and some luck. How much luck, I think Dr Anteater is about to tell us.

Anteater: Since you speak of parcels. That reminds me of two friends of Aunt Hillary's who never met.

Sixth Challenge — *The Relay*

Anteater: Could I have the whole tray, Mr Crab, set out in a row. It is a story with a good deal of furniture.

Crab: [*lifting the boxes off the tray one at a time, and the walnut pair from the cloth, and standing them in a single line along the far side of the cloth, left to right, the three pairs opened out and set end to end into one long row*] Of course. The pearwood: the spent one, and beside it its partner — *shut*, and I am watching everybody. The walnut pair. The ebony pair, which nobody has so much as breathed on.

Anteater: Excellent. At the left-hand end of the row lives Alice, a friend of Aunt Hillary's from the east side of the hill. At the right-hand end, a long way off, lives Bob, a friend from the west. They have never met, never corresponded, and never owned boxes from the same bench. Alice has the left pearwood box. Bob has the right ebony one.

Achilles: Careful — Alice's is spent.

Anteater: Kindly recall that it is a story; there, hers is not. Stories are kinder than evenings.

Achilles: Even so — the boxes in between?

Anteater: Between them stand two relay stations. Alice's pair was made at a bench near her, and its other box sent along the road to the first station. The walnut pair was made between the stations: one box to the first, one to the second. The ebony pair was made near Bob; one box went back up the road to the second station, and Bob kept the other.

[*The Tortoise slips the tuning key off the string and hands it back to the Crab, who lays it in its open case. The whole row now stands ready along the far edge of the cloth, every box shut and silent; the freed string lies loose and bare in front of them, nothing threaded on it and nothing yet bent.*]

Tortoise: [*laying the string on the cloth in front of the row as he speaks, a leg of string for each box, like a garden path to each door: up to the left pearwood box, round in a bend, and back from the right pearwood box*] Let Bob draw a path to

the first pair; one bend. [*The string turns about at the near side and runs up to the left walnut box; round in a bend; back from the right walnut box.*] Let Bob draw on to the first station; the next pair. [*It turns about again at the near side and runs up to the left ebony box; round; and back from the right ebony box, where the string ends.*] Let Bob draw on to the second station; the last pair.

[*The string now runs the length of the row: up to the first pair and round, back down and up to the second pair and round, back down and up to the third pair and round — three bends reaching up to the boxes, two turns doubling back between them, and two free ends left lying at the near edge, one below the far-left box and one below the far-right.*]



the whole row laid

Achilles: So the first station is holding the right pearwood box and the left walnut one — boxes from different benches, with nothing to say to each other. Two pennies. And Alice and Bob have less than nothing: between her box and his there are all those separate bends, and a gap at each station.

Tortoise: Are they gaps? Follow the string.

Achilles: They're... turns. The string turns about in front of each station and goes out again. Out, back, out, back, out. But a turn of string on a tablecloth isn't something a relay station can *do*.

Anteater: No. This is what it can do. Each station has an instrument into which its two boxes are set side by side, and which puts to them a single question: not "what does each of you play?" but "do you two play as one?" Most evenings it gives one of several other answers. Now and then it gives a certain chord; and on the evening a station hears that chord, its two boxes have been joined exactly as Madam Tortoise's string is turned.

Achilles: How often is now and then?

Anteater: One evening in four.

Achilles: Then for both stations together: one in four, and one in four of that. One evening in sixteen. Your friends will need patience.

Anteater: Aunt Hillary's friends have little else. On the other evenings something else happens, which is not a failure; but that is another evening's story. Tonight, both stations hear the chord.

[*Along the cornice runs the wire from the street-door pull, on its way to the kitchen by way of two small brass cranks, one at each corner of the room. Somebody at the street door pulls. The wire twitches, each crank nods once and is still, and a long way off, beyond the baize door, a bell rings. Nobody gets up; it is only the evening post.*]

Tortoise: Let Bob draw both ends in. Achilles, one finger on the bend in front of the walnut pair — lightly. Mr Crab, I am not going to touch a box. [*She takes the two free ends of the string, the one below Alice's box and the one below Bob's, and pulls them towards her. The turns at the stations run out; the bends draw into one another. She spreads what is left with a claw: one long bend, up to the left pearwood box, across the front of the whole row, and back from the right ebony box.*]

[*Nothing on the table has been opened or moved: the same boxes stand in the same row. Only the string has changed — every turn at the stations pulled out, and the two end legs, still resting at the far-left and far-right boxes, joined now by one unbroken bend across the whole front of the row.*]



both ends drawn in

Achilles: One bend. From Alice's box to Bob's. But that's a *pair* — it's the hairpin you laid for the walnut boxes, only wider. Her box and his would fill a line in the ledger like any other: a coin on the left, a coin on the right,

and the same coin.

Anteater: They never shared a bench.

Achilles: Then the agreement was never in the boxes. That's what I asked you, Tortoise — where is it kept? — and it isn't kept in any box, or at any bench. It's in how they are joined. And joining can be handed along: the stations passed it down the row, and neither of them kept any.

Tortoise: What is left at the stations?

Achilles: Nothing. Spent boxes, and the memory of a chord. [*He looks along the row.*] And Alice and Bob at the two ends, each holding half of something — and neither half is anything at all without the other.

Anteater: [*gathering himself, and folding his napkin*] Well. It is a long way round the hill, and Aunt Hillary will want her regards carried home. You have both been very patient with an old creature and all his furniture.

Tortoise: Stay for the last of the tea, Doctor. There is one thing still on the table, and I think Mr Crab would rather we saw it out together than alone afterwards.

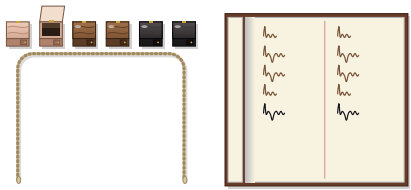
[*The last of the tea goes round, the pot lighter now, and the Anteater, half risen, settles again. Nobody hurries. From the mantel come the tick and the tock, so even by now that a visitor would look for one clock and find two.*]

Crab: [*after a silence*] That is the most uncomfortable account of my evening I have yet heard. I have been sitting beside half of something since before the tea. [*He takes his claw away from the right pearwood box.*] Madam Tortoise. You are the only person in this room who has not wanted to. You had better do it.

Tortoise: By the lid, I think; as Achilles did.

[*The right pearwood box has sat shut and full since the first lines of the evening, its note folded up inside it. The Tortoise catches the lid and lifts it — the box does not move, only the little hinged lid swings up, as its partner's did — and the note springs out: a single high tone, the same as the first, clear as a struck glass, and gone. The Crab turns the ledger to tonight's page and writes one word in the right-hand column, beside the word that has waited there alone since the evening*

began; and now the two words stand side by side, the same word twice, one line of a pair completed a whole evening late.]



the last note *the same word twice*

Achilles: High. Of course it's high. But was it in there all along?

Tortoise: Bring that to the game.

[*The note has gone where the first one went. On the cloth the row stands as the Crab set it — pearwood and pearwood, walnut and walnut, ebony and ebony — and before it lies the string in its one long bend, an end at either end of the row. In the ledger the ink of the second word is drying beside the first. The lamps burn level, the steam has stopped, and on the mantel the two clocks go on handing the time back and forth between them; over them the print has gone back into the wall, a dark frame with a little lamplight on its glass; it is the long part of the night. Nobody at the table reaches for anything, and for the moment nothing on it wants looking into.*]

Chapter 6: Entanglement

— Obbligato for Six Music Boxes —

Part II: Wires Together

Escher: "Bond of Union" — two heads formed from one continuous ribbon

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Can two things, far apart, share a fact that neither of them holds?

Toss two coins in two different towns and compare notes afterwards. Each list of results is noise, and the two lists have nothing to say to each other. Now imagine a stranger pair of coins. Each one, watched on its own, is still perfectly fair: heads and tails, half and half, no pattern that any statistician could find. But when the two lists are laid side by side they are the same list, line for line.

The everyday explanation comes at once: somebody fixed the coins in advance. Perhaps that is right. This chapter will not decide it, and we ask the reader to keep the suspicion alive, because the next chapter has plans for it. What this chapter does is look at how quantum theory itself describes such a pair — and the description is odd in a precise way. The theory has a complete account of the pair. It has no account at all of either member. It is as if a duet had been written down, in full, with no part for the first voice and no part for the second, and yet neither voice could be left out.

So where is the agreement kept, if not in the things that agree? On the blackboard the answer will be a sum that refuses to factorize, and we shall prove that it refuses. In the sketchbook the answer is more surprising, and it is the reason this chapter is a turning point in the book: the agreement is kept in a *bend*. A wire that goes out and comes back is, all by itself, the pair

we are looking for. Once a wire may bend, three things can be done with it. It can be pulled straight. A box can be slid along it, round the corner, and we shall watch carefully how the box arrives. And a box can be bent into a state, so that something one *does* becomes something one *has*.

Those three moves are the whole chapter. At the end we thread one wire back and forth a few times, pull, and find that two parties who never met have been joined.

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§6.1 Two Boxes, One Description

In Chapter 4 we learned to put two wires side by side. If the top wire carries $|\alpha\rangle = \alpha_0 |0\rangle + \alpha_1 |1\rangle$ and the bottom wire carries $|\beta\rangle = \beta_0 |0\rangle + \beta_1 |1\rangle$, the pair carries the tensor product

$$|\alpha\rangle \otimes |\beta\rangle \;=\; \alpha_0\beta_0 |00\rangle + \alpha_0\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle ,$$

where, as always in this book, the first symbol inside a ket belongs to the top wire. The two wires of Deutsch's algorithm in Chapter 5 were always of this kind: from start to finish the answer wire stayed in $|-\rangle$, a factor by itself, and the kicked sign was only a number in front. But the pair of wires is a four-dimensional space, and *any* vector in it is a legitimate state:

$$|\psi\rangle \;=\; c_{00} |00\rangle + c_{01} |01\rangle + c_{10} |10\rangle + c_{11} |11\rangle .$$

The question of this section is whether every such $|\psi\rangle$ is secretly an $|\alpha\rangle \otimes |\beta\rangle$. It would be a comfort if it were: a description of the pair would then be nothing more than a description of each wire.

On the blackboard. A state of two wires is a **product state** if it can be written as $|\alpha\rangle \otimes |\beta\rangle$ for some one-wire states $|\alpha\rangle$ and $|\beta\rangle$. A state that cannot be so written is **entangled**. Here is the state that will occupy us for the rest of the chapter, the **Bell state**

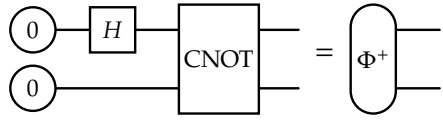
$$|\Phi^+\rangle \;=\; \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Let us try to factorize it. We need $\alpha_0\beta_0 = \frac{1}{\sqrt{2}}$ and $\alpha_1\beta_1 = \frac{1}{\sqrt{2}}$, so none of the four numbers $\alpha_0, \alpha_1, \beta_0, \beta_1$ is zero. But we also need $\alpha_0\beta_1 = 0$. That is a contradiction, and no cleverness with complex numbers gets round it. The Bell state is entangled.

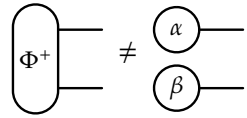
It is not an exotic state. The two gates of Chapters 3 and 4 make it from the plainest possible start: apply the Hadamard gate H to the top wire of $|00\rangle$, giving $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$, then apply CNOT, which flips the bottom wire exactly when the top wire reads 1:

$$\text{CNOT}\,(H \otimes I)\,|00\rangle \;=\; \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \;=\; |\Phi^+\rangle\,.$$

In the sketchbook. A product state is two separate state boxes, one on each wire. Nothing joins the top half of the page to the bottom half: the picture is *disconnected*. The Bell state, by contrast, has to be drawn as a single state box with two wires leaving it. We have no new notation for this yet; it is an ordinary box of the kind we have drawn since Chapter 4, labelled Φ^+. Here is the preparation above, as a picture equation:



and here is the claim that it is entangled, for every choice of α and β:



Why this definition. We define entanglement negatively — *not* a product —

because the product states are the ones we understand: they are exactly the states for which "the state of the top wire" means something. Entanglement is what is left when that phrase stops making sense.

A test that always works

Our argument for $|\Phi^+\rangle$ used its two zeros. What about a state with no zeros at all, such as

$$|\chi\rangle = \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)\,?$$

One can stare at this for some time. It would be good to have a test that needs no staring. Arrange the four amplitudes of $|\psi\rangle$ in a square, top wire indexing the rows:

$$C = \begin{pmatrix} c_{00} & c_{01} \\ c_{10} & c_{11} \end{pmatrix}.$$

Theorem 6.1 (the product test). A two-wire state $|\psi\rangle = \sum_{ij} c_{ij}\,|ij\rangle$ is a product state if and only if $\det C = c_{00}c_{11} - c_{01}c_{10} = 0$.

Proof idea. For a product state the square is a multiplication table — row labels α_i, column labels β_j — and the two rows of a multiplication table are proportional. The determinant of a 2×2 matrix is precisely the measure of whether its rows are proportional.

Proof sketch. If $|\psi\rangle = |\alpha\rangle \otimes |\beta\rangle$ then $c_{ij} = \alpha_i\beta_j$ and $\det C = \alpha_0\beta_0\alpha_1\beta_1 - \alpha_0\beta_1\alpha_1\beta_0 = 0$. Conversely, suppose $\det C = 0$. The state is not the zero vector, so some row of C is not zero; say it is the top row (c_{00}, c_{01}) (otherwise exchange the roles of the rows). A vanishing determinant says the bottom row is a multiple of it: $(c_{10}, c_{11}) = \lambda\,(c_{00}, c_{01})$. Then $|\alpha\rangle = |0\rangle + \lambda\,|1\rangle$ and $|\beta\rangle = c_{00}\,|0\rangle + c_{01}\,|1\rangle$ give $|\alpha\rangle \otimes |\beta\rangle = |\psi\rangle$. ∎

For $|\Phi^+\rangle$ the determinant is $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - 0 = \frac{1}{2}$: entangled, as we found by hand. For $|\chi\rangle$ it is $\frac{1}{4} \cdot (-1) - \frac{1}{4} = -\frac{1}{2}$: entangled too, which was far from

evident. And for $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$, which differs from $|\chi\rangle$ by one sign, it is 0: that state is just $|+\rangle \otimes |+\rangle$. See `Code/Ch6/NotAProduct.py`.

Notice that the test is pure blackboard. The sketchbook can *say* "this box does not come apart", but at this stage it has no way to see why. We shall return to this determinant in §6.5 and find that it was the determinant of something.

Feynman's Notebook.

In May 1935 Albert Einstein, Boris Podolsky and Nathan Rosen published a four-page paper in the *Physical Review* with a question for a title: "Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?" Their example was a pair of particles that had interacted and flown apart, described by a state of just the kind above — one description for the pair, none for the members. They took this as a sign that the theory was leaving something out.

Erwin Schrödinger read the paper and answered within months, in German and in English. He gave the phenomenon its name — *Verschränkung*, entanglement — and in a paper communicated to the Cambridge Philosophical Society that year he wrote of it: "I would not call that *one* but rather *the* characteristic trait of quantum mechanics, the one that enforces its entire departure from classical lines of thought."

Both reactions were reasonable. It took nearly thirty years before anyone found a way to put the disagreement to an experiment; that is Chapter 7.

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§6.2 Random Parts, Definite Whole

If the pair has a description and the parts have none, what happens when somebody looks at a part?

On the blackboard. Recall the rule of Chapter 4 for looking at one wire of several. To ask the top wire "are you $|b\rangle$?", apply the effect $\langle b|$ to the top wire and leave the bottom wire alone; that is the matrix $\langle b| \otimes I$. What comes out is an unnormalized state of the bottom wire. Its squared length is the probability of the answer "yes", and its direction is the state the bottom wire is left in.

Ask the top half of a Bell pair whether it is $|0\rangle$:

$$(\langle 0| \otimes I)\ |\Phi^+\rangle \ = \ \frac{1}{\sqrt{2}}(\langle 0|0\rangle\,|0\rangle + \langle 0|1\rangle\,|1\rangle) \ = \ \frac{1}{\sqrt{2}}\,|0\rangle\,.$$

The squared length is $\frac{1}{2}$. The same calculation with $\langle 1|$ gives $\frac{1}{\sqrt{2}}\,|1\rangle$, again with squared length $\frac{1}{2}$. So the top wire, asked for its bit, answers like a fair coin — and whichever answer it gives, the bottom wire is left in that same state, so the same question put to the bottom wire is now certain to get the same answer. By the symmetry of $|00\rangle + |11\rangle$ the story is identical with the wires exchanged. In a table, the joint outcomes $00, 01, 10, 11$ have probabilities $\frac{1}{2}, 0, 0, \frac{1}{2}$.

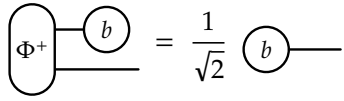
Perhaps that is special to the basis in which we wrote the state down. Try $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$:

$$(\langle +| \otimes I)\ |\Phi^+\rangle \ = \ \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \ = \ \frac{1}{\sqrt{2}}\,|+\rangle\,,$$

$$(\langle -| \otimes I)\ |\Phi^+\rangle \ = \ \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \ = \ \frac{1}{\sqrt{2}}\,|-\rangle\,.$$

A fair coin again, and agreement again. In fact $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle)$, as one checks by multiplying out: the state looks the same in the new basis as in the old.

In the sketchbook. Plug an effect, labelled b, into the top wire of the Φ^+ box. One wire is still open, so what remains is a state of one wire. The calculations above say which:



for $b \in \{0, 1, +, -\}$.

As a picture this is a bare report of the algebra. Why should an effect on the top wire turn up as a state on the bottom wire? The sketchbook cannot yet say. It will in §6.4, where this equation becomes a corollary of the chapter's main theorem: the effect has slid round a corner.

Dirac's Blackboard.

Every basis, and a conjugate. Let $|b\rangle = b_0 |0\rangle + b_1 |1\rangle$ be any unit vector, so $\langle b| = \bar{b}_0 \langle 0| + \bar{b}_1 \langle 1|$, the bar denoting the complex conjugate. Then

$$(\langle b| \otimes I)\, |\Phi^+\rangle = \tfrac{1}{\sqrt{2}}(\bar{b}_0 |0\rangle + \bar{b}_1 |1\rangle) = \tfrac{1}{\sqrt{2}} |b^*\rangle ,$$

where $|b^*\rangle$ is $|b\rangle$ with every amplitude conjugated. Its squared length is $\frac{1}{2}(|b_0|^2 + |b_1|^2) = \frac{1}{2}$. **Every question one can put to one half of a Bell pair is answered by a fair coin.**

The other half is left in $|b^*\rangle$, not $|b\rangle$. For a basis with real amplitudes — $|0\rangle, |1\rangle$ or $|\pm\rangle$ — there is no difference, and the halves agree. For $|b\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ we get $|b^*\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$, which is orthogonal to $|b\rangle$: asked *that* question, the two halves always disagree. Perfectly correlated still; only the labels are swapped. The conjugate is not a nuisance. It is the first footprint of the transpose of §6.4. See `Code/Ch6/Marginals.py`.

So each part is as random as anything can be, in every direction one looks, and the whole is as definite as anything can be: the two answers are locked together. That is what the title of this section means, and it is what the opening page described.

A caution, in plain words. Nothing here shows that the answers were *not* written into the two halves beforehand. Two identical sealed notes, prepared at the source, reproduce both calculations above: each note read alone looks

like a coin toss, and the two always agree. We have tested very little so far. The matter is taken up in Chapter 7, with sharper tools; until then the hidden-note theory stands.

Why this way of computing. We compute the statistics of a part directly from amplitudes — apply an effect, take a squared length — because at this point in the book there is no object that could be called "the state of the top wire". Chapter 9 constructs one. It will not be a ket.

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§6.3 The Cup, the Cap, and the Yank

We have a state that is one thing on two wires. Let us draw it as what it looks like.

In the sketchbook. Take a wire and bend it back on itself, so that both of its ends point to the right, like the letter C. Read from left to right, nothing comes in and two wires go out: by the grammar of Chapter 4 it is a *state on two wires*. We call it the **cup**. Now bend a wire the other way, so that both ends point to the left, like a reversed C. Two wires come in and nothing goes out: an *effect on two wires*. We call it the **cap**.



On the blackboard. We *define*

$$\text{cup} \;=\; |00\rangle + |11\rangle\,, \qquad\qquad \text{cap} \;=\; \langle 00| + \langle 11|\,.$$

The cup is a column of four numbers, $(1,0,0,1)$; the cap is the same four numbers as a row. Look closely: there is no $\frac{1}{\sqrt{2}}$. **The cup is not normalized.** Its length is $\sqrt{2}$, and its relation to the Bell state is

$$|\Phi^+\rangle \;=\; \frac{1}{\sqrt{2}}\,\text{cup}, \qquad\qquad \langle\Phi^+| \;=\; \frac{1}{\sqrt{2}}\,\text{cap}.$$

This convention holds for the whole book, and we shall repeat it wherever a cup and a Bell state stand close enough to be confused. In a picture, the Bell state is a cup with the number $\frac{1}{\sqrt{2}}$ written beside it.

The cap deserves an honest word at once. As an effect it is $\sqrt{2}\,\langle\Phi^+|$: the number $\sqrt{2}$ times the answer "yes" to the question "are these two wires in the state $|\Phi^+\rangle$?" That is one of four possible answers to a two-wire question (the other three are for Chapter 10), and nobody can *make* a question come out "yes". A diagram with a cap in it describes what happens **when** that outcome occurs — it is *postselected*. With that understood, caps are perfectly good mathematics, and we shall use them freely and count their cost at the end.

An example that wants to be a rule

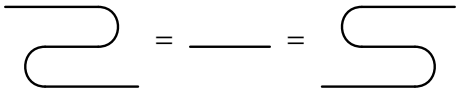
Put a cup and a cap together so that they share one wire. Draw three rows. A wire comes in on the top row; a cup sits on the middle and bottom rows; a cap then joins the top and middle rows; the bottom row goes out. The wire travels right, doubles back, and goes right again, in the shape of a Z. As a matrix this is $(\text{cap} \otimes I)(I \otimes \text{cup})$. Feed it $|0\rangle$:

$$\begin{aligned} |0\rangle &\xrightarrow{I\otimes\text{cup}} |000\rangle + |011\rangle \\ &\xrightarrow{\text{cap}\otimes I} \left(\langle 00|00\rangle + \langle 11|00\rangle\right)|0\rangle + \left(\langle 00|01\rangle + \langle 11|01\rangle\right)|1\rangle \;=\; |0\rangle\,, \end{aligned}$$

where the cap has eaten the first two symbols of each ket. Of four brackets one survives. Feed it $|1\rangle$ and, in the same way, $|1\rangle$ comes out. A bent wire that returns every basis state unchanged is no different from a wire that was never bent.

> **Theorem 6.2 (yanking).** A wire bent into a Z, or into an S, is a straight wire:
>
> $$(\text{cap} \otimes I)(I \otimes \text{cup}) \;=\; I \;=\; (I \otimes \text{cap})(\text{cup} \otimes I),$$

exactly.



Proof idea. The cup says "my two ends carry the same symbol"; the cap says "I pass only if my two ends carry the same symbol". Chain them and the symbol at the far end must equal the symbol at the near end. That is the identity.

Proof sketch. For the Z-bend on the left, take a general input $a_0\,|0\rangle + a_1\,|1\rangle$. Then $I \otimes$cup produces $\sum_{i,k} a_i\,|i\,k\,k\rangle$, four terms. The cap sends $|i\,k\rangle$ to $\langle 00|ik\rangle + \langle 11|ik\rangle$, which is 1 if $i = k$ and 0 otherwise; two terms survive, $a_0\,|0\rangle + a_1\,|1\rangle$. For the S-bend on the right the same computation runs with the rows in the other order. ■ See `Code/Ch6/Yanking.py`.

This is the chapter's first **rewrite rule**: wherever a wire makes an S or a Z, we may pull it straight, and wherever a wire is straight, we may put a kink in it. The equation is exact — the same matrix on both sides, number for number — and so is every other picture equation in this chapter.

Why this definition. Why no $\frac{1}{\sqrt{2}}$ in the cup? Try it with one. Replace cup by $|\Phi^+\rangle$ and cap by $\langle\Phi^+|$ and the same computation gives

$$(\langle\Phi^+| \otimes I)(I \otimes |\Phi^+\rangle) \;=\; \tfrac{1}{\sqrt{2}} \cdot \tfrac{1}{\sqrt{2}}\, I \;=\; \tfrac{1}{2}\, I.$$

A wire with a kink in it would be *half* a wire, and every picture would have to carry a tally of its kinks. We prefer that a bent wire be a wire. The price is that the cup is not a unit vector, and we pay it by remembering one number, $\frac{1}{\sqrt{2}}$, at the moments when probabilities are wanted.

Coecke's Sketchbook.

Where the bent wire comes from. In 1971 Roger Penrose published "Applications of negative dimensional tensors", setting out a way of calculating with tensors by drawing them: a blob for each tensor, a line for each index, and joined lines for summed

indices. He had, by his own account, used it privately for years. In that notation a bent line is an old friend — it is how an index is raised or lowered, how an "in" becomes an "out".

Samson Abramsky and Bob Coecke, computer scientists at Oxford, read the same bent line in quantum terms: the bend that turns an input into an output *is* the Bell state, and the yanking equation is the skeleton of protocols such as teleportation. The bent wire is Penrose's; the quantum-process reading that unfolds from here — the calculus whose name this Sketchbook now carries — is Coecke's, with Abramsky.

One caution, to carry through the book: a picture rule is only as good as the algebra that licenses it. We may pull a wire straight because of Theorem 6.2, and for no other reason.

---

§6.4 Sliding a Box Around the Bend

A bare wire can be pulled straight. What if there is a box on it?

Let us experiment, with a matrix chosen to have no helpful symmetries:

$$A \;=\; \begin{pmatrix} 1 & 2 \\ i & 3 \end{pmatrix}.$$

It is not a gate — it is not unitary — and it need not be: a box may hold any linear map. Apply A to the top leg of a cup. Since $A\,|0\rangle = |0\rangle + i\,|1\rangle$ and $A\,|1\rangle = 2\,|0\rangle + 3\,|1\rangle$ (the columns of A),

$$(A \otimes I)\,\mathrm{cup} \;=\; A\,|0\rangle \otimes |0\rangle + A\,|1\rangle \otimes |1\rangle \;=\; |00\rangle + 2\,|01\rangle + i\,|10\rangle + 3\,|11\rangle\,.$$

Now the question. The two legs of a cup are one wire. Could the same state have been made by a box on the *bottom* leg? We want a matrix B with

$(I \otimes B)\,\mathrm{cup} = |0\rangle \otimes B\,|0\rangle + |1\rangle \otimes B\,|1\rangle$ equal to the state above. Collect the terms that begin with $|0\rangle$ and those that begin with $|1\rangle$:

$$|0\rangle \otimes (|0\rangle + 2\,|1\rangle) \;+\; |1\rangle \otimes (i\,|0\rangle + 3\,|1\rangle),$$

so $B\,|0\rangle = |0\rangle + 2\,|1\rangle$ and $B\,|1\rangle = i\,|0\rangle + 3\,|1\rangle$. These are the columns of B:

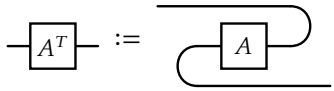
$$B \;=\; \begin{pmatrix} 1 & i \\ 2 & 3 \end{pmatrix}.$$

Compare with the three usual suspects:

$$A^T = \begin{pmatrix} 1 & i \\ 2 & 3 \end{pmatrix}, \qquad A^\dagger = \begin{pmatrix} 1 & -i \\ 2 & 3 \end{pmatrix}, \qquad \bar{A} = \begin{pmatrix} 1 & 2 \\ -i & 3 \end{pmatrix}.$$

It is the **transpose** — rows and columns exchanged, and *no* complex conjugation. The example suggests a rule: a box pushed round the bend of a cup arrives transposed. Before we can state that as a picture equation, the sketchbook needs to know what a transposed box looks like.

In the sketchbook. Take the box A and bend *both* of its wires round. Draw three rows. The input comes in on the top row and runs to the right, where a cap joins the top row to the middle row. The box A sits on the middle row. On the left, a cup joins the middle row to the bottom row, and the output leaves on the bottom row. The wire makes a Z, as in the yanking equation, but now with A on its middle stroke. Follow the wire from the input to the output and you pass through A *backwards*, from its output side to its input side. The box has been turned through a **half-turn**. We take this picture as the sketchbook's definition of the box labelled A^T:



On the blackboard. A definition by picture had better agree with the matrix whose name it borrows.

Lemma 6.3 (the half-turn is the transpose). For every 2×2 complex matrix A,

$$(\text{cap} \otimes I)(I \otimes A \otimes I)(I \otimes \text{cup}) \;=\; A^{T},$$

where $(A^{T})_{ki} = A_{ik}$.

Proof. Write $A_{ik} = \langle i|\, A\, |k\rangle$ for the entry in row i and column k. Feed in a basis state $|i\rangle$:

$$|i\rangle \;\longmapsto\; \sum_{k} |i\rangle\,|k\rangle\,|k\rangle \;\longmapsto\; \sum_{k} |i\rangle\;(A\,|k\rangle)\;|k\rangle \;\longmapsto\; \sum_{k} \langle i|\, A\, |k\rangle\;\;|k\rangle \;=\; \sum_{k} A_{ik}\,|k\rangle\,.$$

In the last step the cap, acting on the first two wires, keeps exactly the component of $A\,|k\rangle$ along $|i\rangle$. So the entry of our three-row matrix in row k, column i is A_{ik}. There are four entries and we have checked them all: it is A^{T}. No conjugate appeared at any step, because no bra of A's entries was ever taken. ■

Why this definition. We could have *declared* a new box called A^{T} and given it a rule. Defining it as A with its wires bent costs nothing new — only cups and caps — and it makes the theorem below almost draw itself. But note which way the debt runs: the picture is entitled to the name "transpose" only because of the index check just made.

Coecke's Sketchbook.

A half-turn is not a mirror image. In Chapter 3 we met the dagger $A^{\dagger}$, and its picture was the *mirror image* of the box: the diagram reflected left to right, with every number conjugated. The transpose is a different motion. Nothing is reflected; the box stays in the plane of the paper and is *rotated* through a half-turn, carried round by its own wires. A rotation cannot conjugate anything.

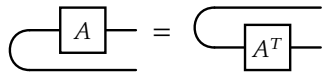
The two are easily confused because for many favourite boxes they agree. If every entry of A is real, then $A^{T} = A^{\dagger}$. If A is symmetric, $A^{T} = A$ and the box does not seem to have turned at

all — H, X and Z are all like this. The Pauli matrix Y is a good test: $Y^\dagger = Y$ but $Y^T = -Y$.

This is why our example has an i in one corner and a 2 in the other. With anything tamer, a wrong rule would have passed the test.

With the half-turned box defined, the rule suggested by the example can be drawn.

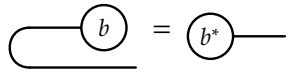
Theorem 6.4 (sliding; the transpose trick). For every 2×2 complex matrix A, $(A \otimes I)\,\text{cup} = (I \otimes A^T)\,\text{cup}$, exactly:



This is the headline theorem of the chapter, and §6.7 proves it twice. See `Code/Ch6/Sliding.py`, which also checks that the equation *fails* with $A^\dagger$, with $\bar{A}$, and with A itself on the bottom leg.

Two consequences are worth having now.

States and effects slide too. Neither proof needs the box to have one wire in and one wire out. An effect $\langle b|$ is a 1×2 matrix, $(\bar{b}_0 \;\; \bar{b}_1)$; its transpose is the 2×1 column with the same entries, which is the ket $|b^*\rangle$. So an effect on the top leg of a cup slides round to become a state:



Multiply both sides by $\frac{1}{\sqrt{2}}$, remembering that $|\Phi^+\rangle = \frac{1}{\sqrt{2}}\,\text{cup}$, and this is exactly the Dirac's Blackboard of §6.2. The picture that was a bare report there now has a reason: asking one half of the pair a question *is* handing the other half the answer, turned half round.

A rotation of the whole pair. For a unitary U, slide U round to the bottom leg: $(U \otimes \bar{U})\,\text{cup} = (I \otimes \bar{U}U^T)\,\text{cup} = \text{cup}$, because $\bar{U}U^T$ is the conjugate of $UU^\dagger = I$. Turning one half by U and the other by $\bar{U}$ leaves the pair as it was.

For real U this is why the Bell state looked the same in the $\pm$ basis as in the $0, 1$ basis.

---

§6.5 Bending: Every Process Is a State

In §6.4 the state $(A \otimes I)$ cup was a stepping stone. Look at it again for its own sake:

$$(A \otimes I)\,\text{cup} \;=\; |00\rangle + 2\,|01\rangle + i\,|10\rangle + 3\,|11\rangle \qquad \text{for} \qquad A = \begin{pmatrix} 1 & 2 \\ i & 3 \end{pmatrix}.$$

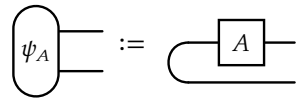
The four amplitudes are the four entries of A, in reading order.

On the blackboard. To every 2×2 matrix A assign the two-wire state

$$|\psi_A\rangle \;:=\; (A \otimes I)\,\text{cup} \;=\; \sum_k A\,|k\rangle \otimes |k\rangle \;=\; \sum_{i,k} A_{ik}\;|ik\rangle\,.$$

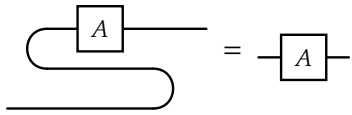
The amplitude of $|ik\rangle$ is the entry A_{ik}: the top wire is the row index and the bottom wire the column index. Nothing is lost and nothing is added; $|\psi_A\rangle$ is A with its four numbers stood in a column.

In the sketchbook. The box A has an input wire on its left. Take hold of that wire and bend it round with a cup, so that it points to the right. A process with one wire in and one wire out has become a *state* on two wires:



Can the box be recovered from the state? Bend the wire back: put a cap on the bottom wire of $|\psi_A\rangle$ and on a fresh input wire below it.

Theorem 6.5 (bending). The assignment $A \mapsto |\psi_A\rangle$ is a one-to-one linear correspondence between 2×2 matrices and two-wire vectors, and the box is recovered by $(I \otimes \text{cap})(|\psi_A\rangle \otimes I) = A$:



Proof idea. On the left the wire comes in at the bottom, goes round a cap, round a cup, and out through A: an S-bend with a box on its last stroke. Pull it straight.

Proof sketch. In the sketchbook: the box A sits on the top row to the right of the cup, so it can be set aside while the S formed by the cup and the cap is straightened by yanking (Theorem 6.2); what is left is a straight wire into A. On the blackboard: $(I \otimes \text{cap})(\sum_{i,k} A_{ik} |ik\rangle \otimes |j\rangle) = \sum_i A_{ij} |i\rangle = A |j\rangle$, since the cap keeps only $k = j$. That the correspondence is linear and one-to-one is visible from $|\psi_A\rangle = \sum A_{ik} |ik\rangle$: four free entries on one side, four free amplitudes on the other. ■ See `Code/Ch6/Bending.py`.

Three entries in this new dictionary are worth reading off.

- The identity matrix corresponds to $|\psi_I\rangle = |00\rangle + |11\rangle$: the cup itself. **A Bell pair is a plain wire, bent** — with the usual reminder that the normalized Bell state is $\frac{1}{\sqrt{2}}$ of it.
- The Hadamard matrix corresponds to $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)$, which is $\sqrt{2}$ times the puzzling state $|\chi\rangle$ of §6.1.
- The square of amplitudes C in the product test of §6.1 was the matrix of the state all along: $|\psi\rangle = |\psi_C\rangle$. Theorem 6.1 now reads: *a state is a product exactly when its matrix is not invertible.* A product state $|\alpha\rangle \otimes |\beta\rangle$ is the bent form of the matrix $|\alpha\rangle \langle\beta^*|$, which squashes everything onto the single direction $|\alpha\rangle$. An entangled state is the bent form of an invertible box, one that forgets nothing. The determinant of §6.1 was the determinant of a process.

Why this definition. Why the top leg? Because then amplitudes equal entries, with nothing to remember. Had we put the box on the bottom leg,

sliding tells us what we would get: $(I \otimes A)\,\text{cup} = (A^T \otimes I)\,\text{cup} = |\psi_{A^T}\rangle$, the state of the transposed matrix. Both conventions are in use in the literature. This book keeps the box on the top leg throughout.

And the honest scope of the two directions. Bending a gate U into a state is something one can do in a laboratory: make a Bell pair, apply U to the top half, and one holds $\frac{1}{\sqrt{2}}|\psi_U\rangle$. Bending back needs a cap, and a cap is a postselected outcome — the answer "Φ^+" to a question with four possible answers. The equation of Theorem 6.5 is exact; what it describes happens only when that answer comes. Chapter 10 will show what to do about the other three, and the result will be teleportation.

Dirac's Blackboard.

A state with the numbers of a matrix. The correspondence between processes and states is named after Andrzej Jamiołkowski (1972) and Man-Duen Choi (1975). Their theorems concern a grander kind of process than ours, which we are not equipped to meet before Chapter 9; what we have here is the seed of it.

Stated exactly, in our conventions: with the box on the *top* leg of the *unnormalized* cup,

$$|\psi_A\rangle = \begin{pmatrix} A_{00} \\ A_{01} \\ A_{10} \\ A_{11} \end{pmatrix}, \qquad A = \begin{pmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{pmatrix}.$$

Composition of boxes becomes something visible too. By Theorem 6.4, $(A \otimes B)\,\text{cup} = (AB^T \otimes I)\,\text{cup} = |\psi_{AB^T}\rangle$: a box on each leg of a cup is a single box on one leg. One last bookkeeping point: the squared length of $|\psi_A\rangle$ is $\sum_{i,k} |A_{ik}|^2$, which for a unitary U is 2, not 1. The missing $\frac{1}{\sqrt{2}}$ is the cup's normalization again, and nothing else.

§6.6 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 5's thirty-eight. Three of them introduce new picture elements — the cup, the cap, and the half-turned box; three are rewrite rules; one records a normalization.

#	Algebra (Dirac)	Picture (Coecke)
39	cup $= \|00\rangle + \|11\rangle$ (unnormalized; length $\sqrt{2}$)	a wire bent so that both ends point right: a state on two wires — *new element*
40	cap $= \langle 00\| + \langle 11\|$ (unnormalized; as an effect, the postselected outcome $\sqrt{2}\,\langle\Phi^+\|$)	a wire bent so that both ends point left: an effect on two wires — *new element*
41	$\|\Phi^+\rangle = \frac{1}{\sqrt{2}}$ cup	the cup with the number $\frac{1}{\sqrt{2}}$ written beside it
42	$(\text{cap} \otimes I)(I \otimes \text{cup}) = I = (I \otimes \text{cap})(\text{cup} \otimes I)$	an S-bend or a Z-bend pulled straight — **yanking**
43	A^T, with entries $(A^T)_{ki} = A_{ik}$	the box A with both wires bent round: a half-turn (*not* the mirror image, which is $A^\dagger$) — *new element*
44	$(A \otimes I)\,\text{cup} = (I \otimes A^T)\,\text{cup}$	a box slid round the bend arrives turned half round — **sliding**

#	Algebra (Dirac)	Picture (Coecke)
45	$\lvert\psi_A\rangle = \sum_{i,k} A_{ik} \lvert ik\rangle$ and $A = (I \otimes \text{cap})(\lvert\psi_A\rangle \otimes I)$	bend the input wire round: a box becomes a state, and back — **bending**

All seven rows are exact equalities of matrices.

---

§6.7 The Theorem, Twice

> **Theorem 6.4 (sliding; the transpose trick).** For every 2×2 complex matrix A — unitary or not —
>
> $$(A \otimes I)\,\text{cup} \;=\; (I \otimes A^T)\,\text{cup}, \qquad \text{cup} = \lvert 00\rangle + \lvert 11\rangle\,,$$
>
> exactly, where A^T is the transpose: not the dagger, and not the conjugate.

The example of §6.4 made the theorem plausible for one matrix. Here are two proofs for all of them.

Dirac's Blackboard

Write $A = \sum_{i,k} A_{ik} \lvert i\rangle \langle k\rvert$, so that $A \lvert k\rangle = \sum_i A_{ik} \lvert i\rangle$, and recall $(A^T)_{ki} = A_{ik}$, so that $A^T \lvert i\rangle = \sum_k (A^T)_{ki} \lvert k\rangle = \sum_k A_{ik} \lvert k\rangle$. Remember also that $\text{cup} = \sum_k \lvert kk\rangle$, whatever letter the sum runs over. Then

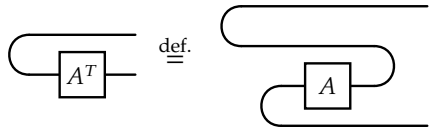
$$
\begin{aligned}
(A \otimes I)\,\mathrm{cup} &= \sum_k A\,|k\rangle \otimes |k\rangle && (A \text{ on the top wire}) \\
&= \sum_k \sum_i A_{ik}\ |i\rangle \otimes |k\rangle && (\text{expand } A\,|k\rangle) \\
&= \sum_i |i\rangle \otimes \Big(\sum_k A_{ik}\,|k\rangle\Big) && (\text{collect by } |i\rangle) \\
&= \sum_i |i\rangle \otimes A^T\,|i\rangle && (\text{recognize } A^T\,|i\rangle) \\
&= (I \otimes A^T)\,\mathrm{cup}. && (A^T \text{ on the bottom wire})
\end{aligned}
$$

Every step is an equality of four-entry columns; for the matrix of §6.4 each line is the column $(1,\ 2,\ i,\ 3)$. The entire content is in the middle line. The double sum $\sum_{i,k} A_{ik}\,|ik\rangle$ does not care whether we group it by k, which reads A column by column, or by i, which reads it row by row. Reading a matrix by rows instead of columns *is* transposing it. No conjugate is ever taken, because nothing is ever turned from a ket into a bra. ■

Coecke's Sketchbook

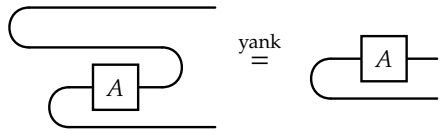
Start from the right-hand side, $(I \otimes A^T)$ cup: a cup with the box A^T on its bottom leg.

Step 1 — substitution (the definition of the half-turned box, §6.4). Replace the box A^T by its picture. We need four rows of wire, counted from the top of the picture. The outer cup joins rows 1 and 2, and row 1 runs straight out to the right. Row 2 is the input of the half-turn picture: it runs right to a cap joining rows 2 and 3; A sits on row 3; a cup on the left joins rows 3 and 4; row 4 runs out to the right.



Step 2 — yanking (Theorem 6.2, the S-bend). Follow the wire from the output side of A. It leaves A on row 3, goes round the cap to row 2, runs left, goes round the outer cup to row 1, and runs out. That is an S-bend

with nothing on it, $(I \otimes \text{cap})(\text{cup} \otimes I)$, and it equals a straight wire. Pull it straight: the output of A now runs directly out on the top.



What remains is the inner cup with A on its top leg and a bare bottom leg: $(A \otimes I)$ cup. The two outputs are in the right order — what was row 1 is on top, what was row 4 is below — and no wire has crossed another. One substitution, one yank. ■

The verdict

The sketchbook proof is shorter, and it shows something the blackboard hides: *why* the theorem is true. The box never changes; it is carried round a corner by its own wire, and a thing carried round a corner arrives facing the other way. One sees at a glance that the same must hold for states, for effects, and for boxes of any size.

But the picture proof has a debt, and it should be named. It proves that A on the top leg equals *the half-turned box* on the bottom leg — and what it calls A^T is a drawing. That this drawing is the matrix transpose, and not the dagger or the conjugate, is Lemma 6.3: a four-entry index check, which is algebra. The blackboard proof is five lines, asks nothing of anyone, and tells us exactly which numbers go where. So: the picture wins on length and on insight, and it wins honestly only because the blackboard has signed for the name. In this chapter neither strand could have done without the other — which, considering the subject, seems fair.

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§6.8 The Chain

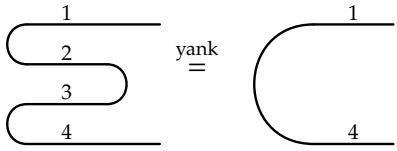
We have been bending one wire. Let us bend it more than once.

One junction

Take two cups, on wires 1–2 and 3–4: two separate Bell pairs (each $\sqrt{2}$ times a normalized one; cups are unnormalized, as ever). Wires 1 and 4 have nothing to do with each other: the state is cup $\otimes$ cup, a product across the middle of the page. Now put a cap on the two *inner* wires, 2 and 3. On the blackboard, writing the two cups as $\sum_{i,j} |i\,i\,j\,j\rangle$,

$$(I \otimes \text{cap} \otimes I) \sum_{i,j} |i\,i\,j\,j\rangle \;=\; \sum_{i,j} \big(\langle 00|ij\rangle + \langle 11|ij\rangle\big)\; |i\,j\rangle \;=\; \sum_{i} |i\,i\rangle \;=\; \text{cup},$$

because the cap passes only $i = j$. The outer wires, 1 and 4, which began in separate pairs, are now a pair. In the sketchbook there is nothing to calculate: cup, cap, cup in a column is an S-bend sitting on the top leg of the lower cup, and by Theorem 6.2 it pulls straight:

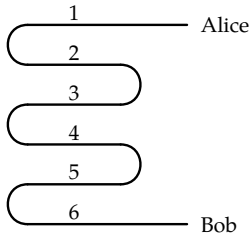


As always the cap is a postselected outcome; we count its cost below.

This is called **entanglement swapping**. It deserves to be strung out further.

The master diagram

Here is the Anteater's story from the dialogue, as a diagram. Alice holds wire 1. A first relay station holds wires 2 and 3. A second relay station holds wires 4 and 5. Bob holds wire 6. Three sources have each made a pair and sent the halves to neighbours: a cup on wires 1–2, a cup on wires 3–4, a cup on wires 5–6. Each station asks its two wires the Bell question and — in the case we draw — hears "Φ^+": a cap on wires 2–3 and a cap on wires 4–5.

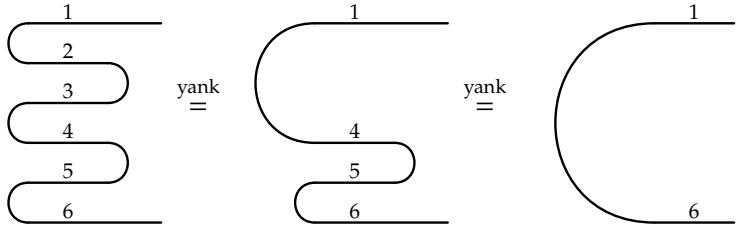


This is the master diagram of the chapter, and we call it **The Chain**.[1] It is one wire, threaded out and back: out round a cup, back round a cap, out, back, out.

> **Theorem 6.6 (the chain).** Three cups and two caps, threaded as in The Chain, are one cup on wires 1 and 6:
>
> $$(I \otimes \text{cap} \otimes \text{cap} \otimes I)\,(\text{cup} \otimes \text{cup} \otimes \text{cup}) \;=\; \text{cup},$$
>
> exactly.

Proof idea. One junction turns two pairs into one. Two junctions turn three pairs into one. Each junction is a single yank.

Proof, in the sketchbook. **First yank (Theorem 6.2, S-bend):** the cup on 1–2, the cap on 2–3 and the top leg of the cup on 3–4 form an S; pull it straight, and a single cup joins wires 1 and 4. **Second yank (the same rule):** that cup, the cap on 4–5 and the top leg of the cup on 5–6 form another S; pull.



Proof, on the blackboard. Three cups are $\sum_{i,j,k} |i\,i\,j\,j\,k\,k\rangle$. The cap on wires 2–3 passes only $i = j$; the cap on wires 4–5 passes only $j = k$. What is left is

[1] The reader may wish to count the wires in this figure. We offer no explanation at this time.

$\sum_i |i\,i\rangle$ on wires 1 and 6. ■ See `Code/Ch6/Chain.py`.

Here the two strands finish level — one line each — but the picture has the better of it in one respect: it makes plain that the length of the chain is of no consequence. Ten stations or a thousand, the wire pulls straight — as an equality of processes; what each pull costs comes next.

What the pull costs

Now the $\frac{1}{\sqrt{2}}$ that we have been carrying must be paid in. The sources make normalized Bell states, and a station's answer "Φ^+" is the normalized effect $\langle\Phi^+|$. Three states and two effects contribute $(\frac{1}{\sqrt{2}})^5$, and the cup that comes out is $\sqrt{2}\,|\Phi^+\rangle$:

$$(I \otimes \langle\Phi^+| \otimes \langle\Phi^+| \otimes I)\ |\Phi^+\rangle \otimes |\Phi^+\rangle \otimes |\Phi^+\rangle \;=\; \frac{1}{4}\,|\Phi^+\rangle\,.$$

The amplitude $\frac{1}{4}$ is $\frac{1}{2}\cdot\frac{1}{2}$, one factor for each junction — the same $\frac{1}{2}$ that a normalized kink cost in §6.3. Its square, $\frac{1}{16}$, is the probability that *both* stations hear "Φ^+": each does so with probability $\frac{1}{4}$, fairly enough for one answer out of four. When they do, Alice and Bob hold a perfect, normalized Bell pair.

So the diagram is an exact equation about one case in sixteen. That is what a cap means, and we do not wish to sell it as more. It is an equality of processes; it is not a claim about what a network of such stations achieves, how fast, or how reliably. What happens in the other fifteen cases is not failure — each of the other answers leaves Alice and Bob with a pair that is a known one-box correction away from $|\Phi^+\rangle$ — but seeing that requires classical messages and a little more notation. That is Chapter 10.

Feynman's Notebook.

Particles that never met. Entanglement swapping was proposed in 1993 by Marek Żukowski, Anton Zeilinger, Michael Horne and Artur Ekert, in a paper whose title spoke of "event-ready detectors": the click of the middle station would announce that

two far-apart particles, from independent sources, were now entangled and ready to be tested. It was first carried out in 1998 by Jian-Wei Pan, Dik Bouwmeester, Harald Weinfurter and Zeilinger, with photons; their title says it plainly — "entangling photons that never interacted".

In the laboratory the middle station is a half-silvered mirror where two photons, one from each pair, arrive at the same moment, followed by detectors. Certain patterns of clicks identify a particular Bell outcome; the experimenters keep those runs and set the others aside. That is postselection with the lid off, and it is the kind of thing our cap draws. With plain optics at most two of the four outcomes can be told apart, and Φ^+ is not among them; the 1998 experiment caught another, which serves as well.

Alice's and Bob's particles share no past. What they share is a wire that was threaded through other people's hands.

---

§6.9 Was It in There All Along?

Let us gather what we have. A Bell pair has a description although its halves have none, and a determinant says so (§6.1). Each half, asked anything, answers like a fair coin, and the two answers are locked together (§6.2). Drawn as it deserves, the pair is a bent wire, and a bent wire can be pulled straight (§6.3). A box pushed round the bend arrives transposed — turned half round, never mirrored — which we proved twice (§6.4, §6.7). Bending turns every box into a state and back, so that the pair is nothing but the identity, bent (§6.5). And a wire bent several times still pulls straight, which joins parties who never met (§6.8). In every picture the agreement between the two ends was kept in the same place: not at either end, but in the fact that they are ends of one wire.

That is how the theory *describes* the pair. It does not yet answer the question a sensible person asks first, the one we set aside at the very beginning. Two

halves that always agree: were the answers simply written inside them at the source, one sealed note in each, identical? Nothing in this chapter says no. The sealed notes reproduce the fair coins; they reproduce the agreement; with a little care they reproduce every number we have computed. If there is a difference between a bent wire and a pair of sealed notes, it must show itself somewhere we have not yet looked — in questions put to the two halves in *different* directions, by people too far apart to confer.

In 1964 John Bell found where to look. He wrote down a quantity that any pair of sealed notes must keep below a certain bound, and asked whether a bent wire must keep below it too. The clearest way to meet his discovery is as a game, played by partners who may agree on anything beforehand and may not speak once it starts. Bring your best strategy. Bring the sealed notes. The next chapter deals the cards.

Partita for Three Players, Far Apart

初霜や
三つの窓に
紋ひとつ

hatsushimo ya
mittsu no mado ni
mon hitotsu

The first frost —
on three separate windows,
one pattern.

[*The stair hall of the Crab's house, on the first hard morning of the winter; the low sun comes level through the tall windows and lies white on the floor. The Crab is away at his cousin's in Ostend. On the hall table stand a travelling-case with its lid still down, three sealed envelopes laid in a row, and a folded sheet of paper held flat under a glass paperweight; a longer letter, already unfolded, lies beside them. Achilles and the Tortoise come in from the lane, stamping the cold off their feet; the Tortoise has the house-key in her glove.*]

[*The hall is built around its stair. One broad flight climbs to the first landing, and there the ways divide — on up under the roof, down to the cellar, off by a passage to the garden room — so that three rooms hang from the one hall and no two of them share a floor; the hall table stands close under the foot of it. The Crab has left a fire banked in the grate, enough to take the worst edge off; the cold still comes up out of the flagstones as a surprise. The three tall windows are blind with the night's frost,*

bloomed white from corner to corner, and the frozen town beyond has gone to the same milk — the low sun stands behind the panes and does not yet burn through. On the wall at the first landing, where the stair turns, hangs Escher's Relativity *in a plain frame: stairways that cross it at every angle, a walled garden showing through one archway, small figures forever climbing and descending who never seem to arrive; the frost-light has not reached it. The three envelopes lie in their row with a name to each; the folded slip keeps still under its glass; the case waits with its lid down.*]

Achilles: I have never seen the canal stand still before. There were boys walking on it as if it owed them money.

Tortoise: It will bear a boy this morning. It would not have borne one yesterday.

Achilles: And our host bears none of it — off to the seaside the one week the whole town turns to glass. [*He warms his hands.*] You have his key, his hall and his breakfast fire, and the man himself is watching the grey waves at Ostend.

[*He stands close to the grate a moment; even indoors, this near the frozen glass, his breath still shows. He rubs a thumb-print clear in the frost and looks out: the canal lies flat and white under the boys, and the low sun strikes no glitter off it at all.*]

Tortoise: He left the key with me and a letter with the key. He is sorry to miss us; he says so at length. [*She sets down her glove and takes up the unfolded letter.*] And he begs — twice, underlined — that we do not wait for the evening.

Achilles: Why ever not? Last autumn it was all "another evening, bring it another evening."

Tortoise: [*reading*] "A trio does not keep as a pair keeps. Two boxes will sit on a shelf and wait for you; three made together grow restless apart, and I will not answer for them past a day or two. Open them by daylight, and soon, or the whole afternoon is wasted." [*She lays the letter down.*] You were never going to wait for the dark in any case. You have been carrying that card since October.

Achilles: I have. [*He produces a stiff ruled card from his coat.*] You told me,

when we parted, to bring my theory to the game. Here it is, written out. Notes sealed inside, composed any way I please — your own words — and I mean to play them against whatever is in that case and win.

[*The Anteater is shown in behind them, red winter earth still on one cuff.*]

Anteater: Good morning, good morning — don't let the cold in on my account, I bring enough of it. Aunt Hillary sends her regards and a loan. [*He sets a worn ruled book on the table.*] Her friends over the hill kept it; I thought it might earn its keep today.

Tortoise: It will. Sit where you like, Doctor; the Crab has left us the whole hall and strict instructions to spoil nothing.

[*With the Doctor in and the door shut on the cold, the hall settles. The Tortoise fills three cups from the pot the Crab has left warming on the hob by the grate and hands them round. The sun has climbed enough to lay one shaft across the table through a thinning place in the frost, and the dust the three of them have stirred up hangs turning in it — each mote going its own way, the shaft keeping its shape whole; the Anteater watches it a moment, the way a man watches a column of gnats, before he sits.*]

First Challenge — *The Sealed Notes* (Praeludium)

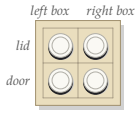
Tortoise: Before the case is opened at all, Achilles, make your theory as strong as you can on paper. There is a game the Crab describes; play your card against it dry, and see how it does. [*She turns to the letter and finds the place.*] The rules first, in his own hand.

Tortoise: [*reading*] "The pair game. Set the two boxes of one pair in two rooms, far apart, and let a referee ask each one a single question — the lid, or the door — chosen only after the players are parted, so that neither can know what the other was asked. Each box answers with its one note. They win the round if the two notes come out alike — save in the one case where the left box is asked the door and the right box the lid, when, for reasons

the daughter would not tell me, they win only if the two are unlike. Three of the four askings want alike; the odd one wants unlike."

Achilles: Easy. If the answer is written in the box before it leaves the bench, I only have to write it well. [*He fetches a handful of reversi counters from the Crab's games table.*] These will do for the notes — white face up for a bright note, black for a dark one; a counter in a square is what the box has decided to say. [*He rules his card into four squares — a column for each box, the lid square above the door square — and names them under his breath.*] Left box, lid. Left box, door. Right box, lid. Right box, door. Four squares, four sealed notes, the whole of what a pair can be told to do.

Achilles: [*laying a white counter in each*] And I say: let both boxes always answer bright. All white. Then whenever they are asked to be alike, they are — three askings in four, won on the spot.



the sealed notes

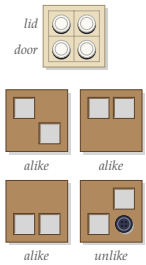
Tortoise: The Crab keeps a packet of masks with the boxes — strips of card with windows cut in them; laid over your card, a mask shows only the two squares one asking cares about, and a word to say what that asking wants. [*She takes them from the packet.*] One asking to a mask. Let me lay them over what you have written, one at a time.

[*The card as it stands: four white counters. She lays the first mask; its two windows sit over the left-lid and the right-door squares, both white, and the word on it is ALIKE.*]

Tortoise: Left lid, right door: alike, and alike it is. [*She lays the second, windows over the two lids, word ALIKE — white and white.*] The two lids: alike. [*The third, over the two doors, ALIKE — white and white.*] The two doors: alike. [*She lays the last, its windows over the left-door and right-lid squares, and its word is UNLIKE; both showing are white.*] And here — left door, right lid — this one wants unlike, and you have given it alike.

[*She has set each mask out below the card as she read it, two by two in the order laid, its windows still marking its two squares. From her pocket she takes three coat buttons and stands one on the fourth mask.*]

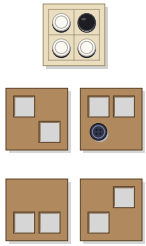
Tortoise: A button, where a rule is broken. One broken, three kept.



one button

Achilles: A slip, only. [*He turns the right-lid counter over.*] Make the right box answer dark when it is asked the lid. Now the unlike mask has a white and a black — unlike, mended.

[*The card as it was: four white counters, one button on the unlike mask. He turns one counter, the right-lid square, white to black; the other three do not move, and the button leaves the unlike mask and crosses to another. One counter turned, two masks changed their verdict, and there is still exactly one button on the card.*]

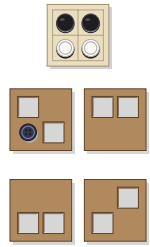


one counter turned

Achilles: ...The button has only moved. It has gone to the two-lids mask.

Tortoise: Mend that one.

Achilles: [*turning the left-lid counter to black*] Both lids dark, then — alike again. [*He watches.*] And now the left-lid-right-door mask has gone wrong instead, and the button is sitting on *that*. It walks. Whatever I flip, it walks to the next square along and sits down again.



turned again

[*Achilles has been fidgeting over the card, one heel up on the bottom step of the stair behind him, and the treads answer him — one gives with a small complaint as he leans his weight on it, and when he shifts off it that one hushes and the tread above takes up the creak. There is always one of them talking, and it moves when he moves.*]

Achilles: Two cards, then — one for a lid morning and one for a door morning — and I toss a coin to choose which card the pair carries out with them.

Tortoise: You may. But the boxes part before the questions are asked; the coin is spent before anyone knows what will be wanted. All the coin does is choose which of your two cards is on the table when the mask comes down — and each card, alone, still has its one broken mask. The coin only picks which asking loses.

Achilles: Then where is the fault? Not in the left box — its two notes are as good as I please. Not in the right. Not in any one square on this card.

Anteater: At which level are you asking? The fault is real, but you are right that it is in no single square. Like a blister under new wallpaper: press it flat here and it rises there, because there is more paper than wall. Look where each square lies. [*He sets a fingertip on the left-door counter.*] This one shows

through two masks at once — the two-doors mask and the unlike mask.

Achilles: Every square, the same way?

Anteater: Every square. Each note you write is read by two of the four askings, so turn one counter and you never change one verdict — you change two. An even change, to a thing that must come out odd.

Achilles: Odd?

Anteater: Count the broken masks, ever. One, or three. Never none, never two. To win all four you would have to break none, and none is even, and you can only ever move by twos across an odd total. The button has nowhere to vanish to.

Achilles: Then three in four is simply the most anyone can do.

Anteater: I know two people who do better.

[*The sun has come far enough round now to rake the lowest window from the side, low and glancing, and where its light runs along the frost the flat white breaks open into fern and feather and whole branching forests. Achilles, waiting on the Doctor, follows one frond of it with his eye until it loses itself in the next.*]

Second Challenge — *The Wedge* (Allemande)

Achilles: Better? Nobody does better than the best. Three in four is the ceiling; I have just watched it hold.

Anteater: It is the ceiling for written notes. [*He draws Aunt Hillary's book to himself.*] This is a score-book — a hundred ruled rows, a hundred evenings, kept by two of Aunt Hillary's friends who have never once met. Alice lives east of the hill; Bob lives west; the workshop sent them a hundred pairs between them, one box of each pair to each, and every evening for a hundred evenings they played the Crab's pair game across the whole width of the hill and marked down whether they won. Let me tell you how Bob's end of it goes, because Bob does a small thing that changes everything.

Anteater: Consider Bob, west of the hill, and the little brass wedge that came in his parcel along with the boxes.

Achilles: Only a wedge? A doorstop with ideas?

Anteater: Each evening, before he opens his box, he stands it up on that wedge, so that the box is put its question not square-on but at a slant.

Achilles: Cutting the difference between the two ways of asking — I begin to see it. But between which two?

Anteater: Keep the picture and hold your question one moment; the wedge is the whole of his trick, and I want the Tortoise to name it.

Achilles: Even a slant is a definite thing, though. He still hears one note and writes one mark.

Tortoise: The wedge asks the box a question tilted halfway between lid and door — square between the two, halfway from the one to the other, and that is the whole of it.

[*Held at that slant, the glancing light picks every fern out of the frost bright, edge and vein and all; the same pane, taken square-on from where the Tortoise sits, shows nothing but a blank white. She tilts her head to catch it the sidelong way, and the whole forest comes up out of the glass at once.*]

Tortoise: Let Bob draw the two columns, then, east and west, with a space for the mark between. [*In the tale, Bob rules the page; on the table the Tortoise sets a finger to the head of each column.*] What Alice was asked, what Bob was asked.

Tortoise: Let Bob draw the hundred rows down the page, one to an evening. [*Bob, in the tale, rules the lines across; the Tortoise runs her finger down the ready-ruled page.*] East asked, west asked, and a stroke where they won.

Anteater: And here is the book he and Alice filled. Each of them wrote down the way they were asked and the note they heard; Bob's stroke goes in the space between. Count it yourself; I would rather you did.

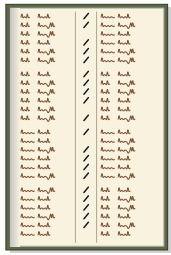


a page of evenings

Achilles: [*running down the strokes column*] Strokes... a great many strokes. [*He tallies on the back of a torn card.*] Eighty-five. Eighty-five wins in the hundred. [*He stops.*] That is not possible. My card could not pass seventy-five however I wrote it — you all watched the button refuse to leave.

Anteater: Let the Tortoise show you the drawing before you decide it is cheating.

Tortoise: Let Bob draw a stroke for every evening they won, sorted by which pair of questions was asked. [*Bob, in the tale, gathers the strokes into four heaps; the Tortoise turns from the open page, where the evenings run in the order they fell, to the last page of the book, where Bob wrote the same evenings out again sorted by kind — not a stroke added or lost, only the order changed — and blocks it into four with her hand.*] Lid-and-door here, lid-and-lid here, door-and-door, door-and-lid. Count each heap.



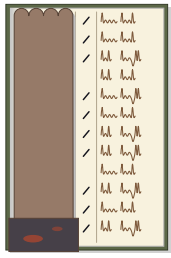
sorted by kind

Achilles: [*counting*] About the same in each. [*Slowly.*] That is the thing my card could never do. My card always had one asking it lost *every single time*

— the button's square. Theirs loses no asking always. There is no heap they gave up.

Achilles: But eighty-five in a hundred — they must be signalling. A lamp over the hill, a flag; Bob learns what Alice was asked and leans his wedge to suit.

Anteater: Ask the book. Take Bob's column alone, the way I once laid a blotter over half the Crab's ledger. [*He turns back to the page the book lay open at and covers Alice's column with his hand: her askings and her notes go under his palm, and Bob's column lies open beside it, not a word of it changed.*] Bob's own notes, whatever Alice was asked that evening: bright and dark, near enough half and half, all the way down — and Alice's alone, just the same. Neither column, read by itself, so much as twitches when the other is asked a different question. There is nothing in Bob's own results that could tell him what Alice heard; nothing crosses the hill.



Bob's column alone

[*The shaft of sun still stands over the table, its dust turning. Follow any single mote and it wanders — up, aside, back, no telling where it means to go; take the whole shaft at once and it holds its slow steady shape against the dark, and not a grain of it carries a word to the grain beside it.*]

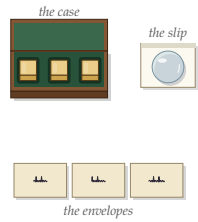
Achilles: [*to the Tortoise*] But this breaks the Crab's own rule, and you were the one who told it to him. He was made to promise, last autumn, that a pair is opened by *the same way in* on both sides. Bob is standing his box on a wedge and Alice is not. They are opening differently.

Tortoise: They are — but that rule was for a different purpose. "Same way in" was the recipe for making a pair *agree*: open both by the lid and they

match, both by the door and they match. This game does not ask them to agree; it asks them to be alike sometimes and unlike once, on questions chosen after they part. A pair that only ever agreed would lose the unlike asking every time — it would be your all-white card. The wedge is for winning a harder game, not for agreeing.

Achilles: Eighty-five is still not a hundred. They lose fifteen evenings in the hundred; they *do* lose. Perhaps the notes are written after all, and only smudged — good enough for eighty-five, spoilt on the rest.

[*The Tortoise lifts the lid of the Crab's travelling-case. Inside, in three shaped nests, lie three small brass boxes.*]



in their nests

Tortoise: These never lose.

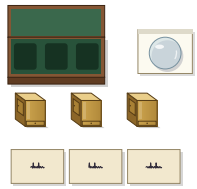
[*The three little boxes lie in their three nests, shut, giving off nothing — no tick, no shine, only cold brass and the low sun caught in it. Nobody reaches to wake them. Achilles draws his chair nearer the grate and warms his hands round the cup, watching the open case the way one watches a thing one has been told not to touch.*]

Third Challenge — *Three from One Bench* (Courante)

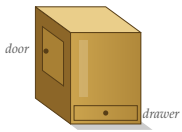
Achilles: Never? Everything loses sometimes. The wedge stopped at eighty-five.

Tortoise: The wedge is a pair. This is a trio. [*She lifts the three boxes from their nests, one, and one, and one, and stands them on the cloth, shut.*] Made together

at one bench on one afternoon, sent in one case. No lids on these — two ways in only: a little door in the side, and a drawer underneath. I shall not open one; I only want them standing where you can see them.



out on the cloth



a door and a drawer

Achilles: And the guarantee?

[*The three stand out on the bare cloth now, out of their case for the first time since Delft, each in its own small pool of shadow thrown by the low sun. Three of one make, and nothing to choose between them — alike as three notes of one chord not yet sounded.*]

Tortoise: [*from the letter*] "The trio, and my workshop's word upon it. Open all three by the door, and the number of dark notes will be even — none dark, or two. Open one by the door and the other two by the drawer, and the dark notes will be odd — one, or three. My workshop's ledger shows no trio that ever broke it, not once." [*She sets the letter down.*] His word, and the bench's, that it holds every time.

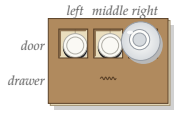
Achilles: Then this is my case made for me, and made better than I could make it. [*He sweeps the four counters off the front of his card.*] Watch. If I open two of them by the door and hear their notes, I can tell you the third before it is touched. Two brights so far, darks even — the third must be bright, or the count goes odd and the guarantee is broken. I would know it without laying a finger on the box. And what can be known for certain, without touching the thing — Doctor, you taught me the words yourself last autumn — was a real property of the thing all along. It was written inside.

Tortoise: Say it on the card, then, where we can check it. [*She gathers the pair-masks and their one button aside, then turns his card over and rules the back into six squares — three columns, one per box in the order they stand on the cloth,*

and two rows, the door above the drawer.] Six squares: what each of the three would say, asked each of the two ways. And these same masks read a trio, not a pair — three windows now, one to a box, and a word for what the three together must come to. Now — turn your back a moment.

Anteater: Lend me the Crab's egg-cup from the dresser, first — this one. Upturned, it covers a single counter whole; whatever I stand it on is fixed but hidden, which is exactly what I shall want.

[*Achilles turns to the window. The Tortoise lays a single trio-mask over the empty card: three windows, one to a column, and the word EVEN across it — the all-doors asking, whose darks must be even. The Anteater sets a counter in each window, obeying the word: two bright, and a third he keeps to himself; then he stands the upturned egg-cup on the flat of the mask, over the third window. What is fixed and hidden now is the one note under the cup.*]



one under the cup

Anteater: Turn round. The mask says EVEN — all three by the door, darks even. Two of the three are shown you. The third is under the cup. Name it.

Achilles: Two brights showing, and darks must be even — none or two. Two brights is no darks yet; a dark under the cup would make it one, which is odd and forbidden. So: bright.

[*The Anteater lifts the egg-cup. The counter beneath shows white.*]

Anteater: Bright. Again — turn round.

[*Achilles turns away. The same EVEN mask stays where it is, and the middle counter with it, white. The Anteater lifts the counters out of the left and the right windows and sets both back black side up, then stands the egg-cup over the left window — not the right one, this time.*]



a different window

Anteater: Same mask, same word. One bright, one dark shown. Name the hidden one.

Achilles: One dark already, and even is wanted; a bright under the cup leaves it at one, odd, forbidden — so it must be dark, to make two. Dark.

[*The egg-cup comes up; the counter is black.*]

Achilles: Foretold without touching it: then it was there. Twice, dead certain, and my hand never near the third box. The note was inside it, waiting; the guarantee only reads out what was already written.

[*The sun stands where it stands, and its height settles exactly how far the glass paperweight must throw its shadow across the table. There the shadow lies at just that reach — told over before anyone stooped to measure it, and no hand near the glass.*]

Anteater: It is the strongest form your theory can take, and you have made it well. But one small thing, at another level, before you are too pleased. [*He touches the boxes.*] Open just one of the three by its door: if it sings bright, the other two — this very instant — are a pair, exactly like last autumn's.

Achilles: A pair? Out of a trio?

Anteater: A whole become a pair, by our closing one leg of it — but only if it sings bright. If it sings dark, the two that are left are a stranger pair, with a turn put on one of them. The trio is not three notes lying side by side; it is one thing on three boxes.

Achilles: [*reaching to lift the mask*] Then let me write all six and be done — a full sealed theory, both ways in for all three.

Tortoise: Lift the mask, and set your six. [*He raises the trio-mask off the card, clears away the three counters the Doctor left, and lays six counters into the six squares — all white, every box bright by the door and bright by the drawer, as plain*

and clean a theory as he can write.] There. You have checked one mask. There are four.

[*The sun is as high as it will climb today and lies steady on the middle window; the ferns there have burned down to a plainer thing, a clean ruled lattice of lines that cross and cross again. The Tortoise traces one of them a finger's breadth off the cold glass, corner to corner, not touching it.*]

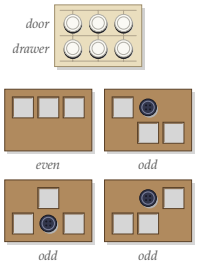
Fourth Challenge — *Six Notes, Four Masks* (Sarabande)

Achilles: Four? There is the all-doors asking I just did. And the others are — one door, two drawers, in its three arrangements.

Tortoise: Four askings, four masks. Lay them over your six as they come.

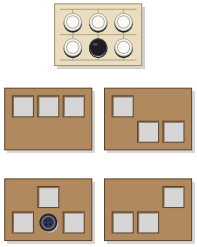
[*The card as it stands: six white counters — every box bright, by the door and by the drawer. The Tortoise lays the masks one at a time, setting each out below the card as she did the pair-masks. The EVEN mask first, its three windows over the three door-squares: three whites, no dark, and even it wants — kept. Then the first ODD mask, one window on the left box's door-square and two on the other boxes' drawer-squares: three whites again, no dark, but this asking wants an odd count of darks — broken. The second ODD mask, its windows over the middle box's door and the left and right drawers: broken the same way. The third, over the right box's door and the other two drawers: broken. Three of the four masks stand wanting a dark they have not been given.*]

Tortoise: [*standing a button on each of the three*] Three broken, one kept.



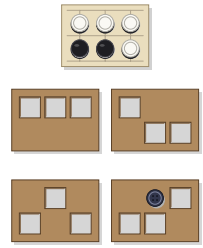
three buttons

Achilles: Then a dark is wanted. [*The card as it stands: three buttons, on the three ODD masks. He turns the middle box's drawer counter, white to black; the five others hold still. That one drawer-square lies under two of the ODD masks — and both of them, wanting a dark and now given one, read odd and are kept: two buttons lift off together. One is left standing, on the ODD mask over the middle door, which cannot see this square.*] Two gone at a stroke!



three become one

Achilles: Now mend the last. [*He turns another drawer counter — the left box's, one of the two under the single surviving button — white to black. That mask has its dark now and is kept; but the same square lies under a second ODD mask that was already kept, and that one, given a second dark, reads even where it wanted odd. The button does not lift; it crosses to the mask next door. One counter turned, two verdicts changed, and still exactly one button on the card.*] ...And it only walks. The very same walk as the front of the card, six squares wide instead of four.



the button crosses

Anteater: Because it is the same machine, only larger. One mask wants even and three want odd, so the four together must come out odd. But every counter you have set shows through exactly two of the masks; so add the four together, whatever you have written, and they come out even. Odd is not even. One mask is always broken — or three; never all four kept.

[*Achilles turns one more counter — the right box's door — watches the single button cross from mask to mask and settle where it always settles, and takes his hand off the card.*]

Achilles: ...Oh. It is not that I have written the six notes badly. It is that there are no six notes to write. No arrangement of them can keep all four guarantees at once; the arithmetic will not have it.

[*The thumb-print Achilles rubbed clear this morning has long since bloomed shut again — from the cold edge inward — so that by now, had one set out to wipe the whole pane, it would have frosted over behind the hand before the hand was done. The pane is never all clear at once; it cannot be made to be.*]

Achilles: Then the workshop's guarantee is simply false. No three boxes can do this. The Crab has been sold a promise that the counters prove impossible.

Tortoise: The counters prove that no *written notes* can do it. [*She glances at the case, and then at the hall table.*] There are three envelopes on the table with our names on them.

[*The three envelopes lie where they have lain all morning, each with its name turned up to the light; the folded slip keeps still beside them under its glass, and the letter*

lies folded back within reach. The fire has settled to a low red bed, and the hall is very quiet.]

Fifth Challenge — *The Round* (Gigue)

Achilles: Our names? [*He takes up the three envelopes and reads the faces.*] One for each of us. And the last rule in the letter, I suppose.

Tortoise: [*reading*] "The round, to be played apart. Give each of three players one box of the trio and one of these sealed envelopes; the envelope names the way that player must open — the door, or the drawer, and no choice about it. The tally slip, under the glass, tells the referee alone what the dark notes must come to: even, if all three envelopes say door; odd, if one says door and two say drawer. Each player goes to a room by himself, opens his box the named way, hears his one note, and comes back. Lay a counter for each note; read the tally last. Match it, and the round is won." [*She looks up.*] We cannot confer once we have parted. And there is nobody left in the house to carry a word between the rooms; the Crab saw to that by going to Ostend.

Achilles: Three rooms, then. I will take the attic.

Anteater: The garden room for me.

Tortoise: The cellar. The slip stays here under its glass; nobody's hand on it, all our hands full of box. [*Each takes one box and one envelope; the slip lies untouched under the paperweight.*] Reader — you are the referee. You alone will stand where all three rooms can be seen.

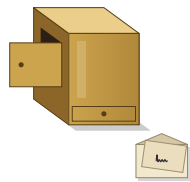
[*They climb the stairs together as far as the first landing, where the three ways part. Achilles stops a moment at Escher's* Relativity *on the wall.*]

Achilles: Look at that — one house of stairs, and every soul on it walking to a different down, passing each other on the same step and never once sharing a floor. [*Nobody answers him; they have already turned away, each to a separate stair.*]

[*The climbing sun reaches the first landing at last and runs up the print's crossing stairways; the hall goes still behind them.*]

[*Achilles, alone in the attic.*]

Achilles: You can see all three of us; I can see only this room, and only what is in my two hands. Here is my envelope. [*He breaks the seal.*] "Door." So I swing the door. [*The box sits shut and full on his palm, one note folded up inside it. One hand steadies it; the other catches the little hinged door in its side and swings it open — the box itself does not stir, only the panel turns back — and the note springs out into the attic, a single struck tone, and is gone. What was held is spent; what stays is a small brass box with nothing further to say, and no lid to try instead.*] Dark. There — you have mine, whoever is watching from where the whole house shows: I was asked the door, and the door gave me dark. And do you know, the word in the envelope told me nothing I could use. Door or drawer, I should have heard whatever this box was going to say. It might as well have been a coin that sent me up here.



in the attic

[*The Tortoise, alone in the cellar.*]

Tortoise: Mine says drawer. [*Her box stands shut and full on the cellar bench. She holds it down with one claw and, with the other, draws out the little drawer from underneath — the box holds its place, only the drawer slides free — and as it comes clear one note lifts from it, and is spent. The box is silent now, the drawer empty, and the door in its side will never be worth trying.*] Bright. That is all of mine. You have two of the three now; hold on to them.



in the cellar

[*The Anteater, alone in the garden room.*]

Anteater: My envelope says drawer as well. [*He draws it out from below; a note sounds and is gone.*] Bright. You alone have heard all three now — a dark and two brights — and you alone can say whether we have won, for not one of us in these three rooms knows what another was asked, or what another heard. I know only my own room. The round, if it is anything at all, is in none of the three rooms by itself; it is in the three of them at once — and only you are standing where that can be seen.

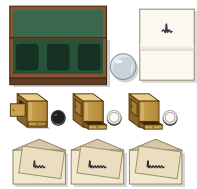


in the garden room

[*The same low sun is standing in all three rooms at once — square in the attic's little gable, thin through the cellar grating, broad on the garden room's frosted glass — though no one of the rooms can see into another, and nothing goes along the stairs between them. Down in the hall a whole pane has come clear at last, and through it the three frozen roads out of the town lie plain, each striking off its own way from the one square.*]

[*The three come back down to the hall, each with a spent box, still open the way it was opened, and an opened envelope. Each stands the box on the cloth where it stood before — Achilles's on the left, the Tortoise's in the middle, the Anteater's on the right — and lays the envelope in front of it.*]

Achilles: Now the counters. One for each note, laid by each spent box. [*He lays a black counter by his own.*] Dark, for mine. [*The Tortoise lays white by hers, the Anteater white by his.*] Bright, and bright. One dark among the three. [*He slides the glass paperweight off the slip and unfolds it.*] Odd. One black. Won.



the slip read

Achilles: We won it. Every one of us opened blind, and it came out right to the letter.

Tortoise: One round proves very little by itself, Achilles; be honest with the reader. Sealed notes win the pair game three deals in four — you could win a single round with your card and crow. It is not this one round. It is the workshop's ledger, that never once shows a trio breaking the guarantee, and it is your own counters through there, that say no card of written notes could keep the guarantee even in principle. Those two together are the thing.

Achilles: Put like that... The button would not vanish — written notes stop at three in four, and cannot keep all four trio-guarantees at all. And the boxes keep them, always, and win apart with nobody carrying a word between the rooms. So the most that written notes can do, the boxes go past; and if the boxes go past what any writing could do, then nothing was written. The notes were never in the boxes.

[*The winter sun has come round the hall and lies long and low across the floor; there are still hours of light in it. The three spent boxes go back into their three nests in the case, the counters are gathered into a little heap, and the last of the pot goes round while nobody hurries.*]

Achilles: But that is the part I cannot hold. If it wasn't in there, and nothing crossed the house — where was it?

Tortoise: In neither box. Draw the three of them, and look at the place where the lines meet.

[*On the table the case stands with its lid down, the three envelopes open where their names had faced the light, the slip unfolded beside them, and the paperweight set aside with nothing left under it to hold. On the tall windows the frost has given up the middles and drawn back into the corners, so that the clear glass shows the afternoon plainly through. On the wall at the first landing the print has quieted — a plain frame with a little sun caught in its glass, its crossing stairs gone dim. The fire ticks as it cools. Nobody at the table reaches for anything, and for the moment there is nothing on it that wants opening.*]

Chapter 7: Inequalities Nature Breaks

— Partita for Three Players, Far Apart —

Part II: Wires Together

Escher: "Relativity" — one house of stairs, three directions of down; the walkers pass and never meet

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At the end of the last chapter we let a suspicion stand: perhaps two things that always agree agree because the answers were written into them beforehand. Can that suspicion survive an honest test?

The suspicion is the most reasonable one in physics. Two halves of a Bell pair, questioned far apart, gave matching answers every time; and the plainest account of matching answers is a pair of sealed notes, written at the source and carried away, one in each half. Chapter 6 could not tell the two accounts apart, and said so. Every number we computed — the fair coin in each half, the perfect agreement between them — the sealed notes reproduce exactly. If a bent wire and a pair of sealed notes differ at all, the difference is hiding somewhere we have not yet looked.

This chapter looks there. The place to look, John Bell saw in 1964, is not in one question repeated but in *several* questions, put in different combinations to parties who are too far apart to confer once the questioning starts. A sealed note must have an answer ready for every question it might be asked, decided in advance, the same whoever does the asking. That is a strong constraint, and Bell found a way to measure it: a quantity that any collection of sealed notes must keep below a fixed bound, and that a shared quantum state can push past.

We shall meet the bound first as a game. Two partners, then three, are asked questions in separate rooms and must answer without conferring; they win or lose by a rule fixed in advance; and we count how often the best possible preparation lets them win. With sealed notes the count has a ceiling. With a shared quantum state — a bent wire, and then a new thing, a single point where three wires meet — the ceiling is broken, and in the three-party game it is broken *completely*: they win not more often, but every single time. No collection of sealed notes can do that. So the notes were not written.

On the blackboard this will be a short, stubborn piece of arithmetic about the numbers $+1$ and -1. In the sketchbook it will be a single new drawing — a dot with three legs — and one new rule for pushing boxes through it. Neither strand, we should say plainly at the outset, proves that nature is nonlocal; each proves an equality of processes, and the arithmetic does the rest. But between them they close the door that Chapter 6 left open.

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§7.1 The Hidden-Note Theory

Achilles ended the last chapter as a realist, and a realist has a theory. It is this: when a source prepares two halves and sends them apart, it writes into each half, in advance, an answer for every question that half might be asked. The halves then part. Each carries its answers like a sealed note; whatever is later asked, the note is simply read. Nothing needs to pass between the halves when they are questioned, because everything was settled at the source.

This is worth stating precisely, because it is exactly what we are going to test.

On the blackboard. A **deterministic local strategy** for two parties is a pair of functions: one, $a(x)$, that gives the first party's answer to each question x it might receive, and another, $b(y)$, for the second party. Both functions are fixed before the parties separate. For three parties there are three: $a(x)$, $b(y)$, $c(z)$. The parties may also share, before parting, a table of such strategies and a rule for choosing among them at random — but a random choice

among fixed strategies is a *mixture*, and a mixture of strategies can do no better, on average, than the best single strategy in it. So to find the ceiling of the hidden-note theory it is enough to examine the deterministic strategies one by one.

Behind this is a criterion that Einstein, Podolsky and Rosen made famous in 1935, and that we shall use in the reader's own words: *if the answer to a question can be predicted with certainty, without touching the thing that will answer it, then that answer was a real property of the thing all along*. A sealed note is exactly a thing whose every answer can be predicted without touching it — by reading the copy at the source, or, as we shall see in §7.4, by asking the other halves. The hidden-note theory is the claim that a Bell pair, and a trio, are collections of sealed notes in this precise sense.

In the sketchbook. We need no new element for the local theory. Two parties who share nothing but a table agreed in advance are, as a picture, two separate state boxes — one on each party's wire — with nothing joining the top of the page to the bottom. It is the *product state* of §6.1, drawn as a picture that falls into disconnected pieces. Whatever effects are later applied to the two halves, the closed picture factors into a top piece times a bottom piece, and the number it yields is a product of two independent numbers. That is the whole content of "the parties cannot confer": the picture does not connect them.

Why this definition. We phrase the local theory as fixed answer-functions, and not as "the particles have definite values", because answer-functions are the thing a game can catch. They commit the theory to a number for every question in advance, and a commitment made in advance can be checked against a question chosen at the last moment.

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§7.2 The Game and the Sealed Notes

Here is the game. Two partners, far apart, each receive one bit of question — call the first partner's question x and the second's y, each either 0 or 1. Each answers with one bit, a and b. They win the round exactly when

$$a \oplus b \;=\; x \cdot y,$$

where $\oplus$ is addition modulo 2 and $x \cdot y$ is the ordinary product, which is 1 only when both questions are 1. In words: for three of the four question pairs the partners must give *equal* answers, and for the one remaining pair, $x = y = 1$, they must give *unequal* answers. They may plan anything beforehand; they may not communicate once the questions are handed out. (This is the game of Clauser, Horne, Shimony and Holt, in the form used by Brunner and colleagues, $a \oplus b = x \cdot y$, whose one "unequal" pair is $x = y = 1$. We have only chosen which physical question — lid or door — each label names, to suit the story to come; the game is the same.)

The example that fixes the ceiling

Suppose the partners agree: *always answer* 0, whatever the question. Then $a \oplus b = 0$ always. They win the three pairs that ask for equal answers and lose the one, $x = y = 1$, that asks for unequal answers. Three wins out of four. Can they patch the fourth without spoiling the other three? Change the second partner to answer 1 when $y = 1$. Now the pair $x = 0, y = 1$ has answers 0 and 1 — unequal — and it was supposed to be equal. They have mended one pair and broken another. The count stays at three. Every patch we try does this: it moves the single loss around the table without removing it.

That is not an accident, and here is why.

> **Theorem 7.1 (the sealed-note ceiling).** No deterministic local strategy wins more than three of the four question pairs. Equivalently, the CHSH quantity
>
> $$S \;=\; E_{00} + E_{01} + E_{10} - E_{11}$$
>
> — where $E_{xy} = +1$ if the strategy answers pair (x, y) with equal bits and -1 if unequal — satisfies $S \le 2$ for every strategy, and a mixture of strategies can only lower $|S|$.

Proof idea. The four winning conditions are four equations in the same four answer-bits. Add them modulo 2: the left sides cancel in pairs and vanish, but the right sides do not. The four conditions therefore cannot all hold at once; one must fail.

Proof (on the blackboard). A deterministic strategy is four bits: $a_0 = a(0)$, $a_1 = a(1)$, $b_0 = b(0)$, $b_1 = b(1)$. The four winning conditions are

$$a_0 \oplus b_0 = 0, \quad a_0 \oplus b_1 = 0, \quad a_1 \oplus b_0 = 0, \quad a_1 \oplus b_1 = 1.$$

Add all four left-hand sides modulo 2. Each of a_0, a_1, b_0, b_1 appears exactly twice, so the sum is 0. Add the four right-hand sides: $0 \oplus 0 \oplus 0 \oplus 1 = 1$. If all four conditions held, we would have $0 = 1$. So at most three hold, and the best strategy wins exactly three of the four pairs. There are 16 deterministic strategies; each wins three pairs at most, so each has $S \leq 2$ (three agreements at $+1$ and one disagreement, or fewer). A random mixture assigns probabilities to strategies and averages their S values, which cannot exceed the largest, 2. ■ See `Code/Ch7/SealedNotes.py`, which enumerates all 16 strategies and confirms the best wins three of four, that every strategy has $|S| = 2$, and that changing one answer-bit always flips exactly two of the four verdicts — the reason a patch never helps.

In the sketchbook. The sketchbook has nothing new to add here, and it is honest to say so. A sealed-note strategy is a product picture; the number for each question pair is a product of a top number (the first partner's) and a bottom number (the second's); and no product picture can beat the ceiling, because the ceiling is a fact about products of ± 1 numbers, which the disconnected picture faithfully computes and cannot escape. The picture will earn its keep in this chapter, but not here: the classical bound is a blackboard result through and through.

Dirac's Blackboard.

Why one loss cannot be removed. Write the four verdicts as the parities $v_{xy} = a_x \oplus b_y \oplus (x \cdot y)$; a pair is *won* when $v_{xy} = 0$. Sum

all four modulo 2:

$$v_{00} \oplus v_{01} \oplus v_{10} \oplus v_{11} \\ = (a_0 \oplus a_0 \oplus a_1 \oplus a_1) \oplus (b_0 \oplus b_1 \oplus b_0 \oplus b_1) \oplus 1 = 1.$$

An odd number of the v_{xy} must therefore equal 1: one loss, or three, never zero and never two. Flip a single answer-bit, say a_0: it sits in v_{00} and v_{01} and in no others, so exactly two verdicts flip. A loss can be shuffled to a neighbour or split into three; it cannot be made to vanish, because vanishing would need an even change to an odd total. Everything the classical player can do lives inside this one equation.

Why this game. The game is the same statement as Bell's inequality $S \leq 2$, in newer dress. We lead with the game because it makes the constraint vivid: a sealed note must answer *before* it knows which of its questions was chosen, and the game is built to punish exactly that. The number S is a bookkeeper's summary; the odd number of losses is the thing itself.

Feynman's Notebook.

From a paradox to an inequality. The suspicion this chapter tests was sharpened in 1935 by Einstein, Podolsky and Rosen, who argued that if a distant measurement can predict a result with certainty, that result must already be a real property — so quantum theory, they held, was incomplete. For three decades the argument stayed philosophical. Then in 1964 Bell turned it into arithmetic: a quantity that any local theory must keep below a bound. Clauser, Horne, Shimony and Holt cast that bound in the experiment-ready form of 1969 our game rehearses. The first laboratory tests followed — Freedman and Clauser in 1972, Aspect and colleagues in 1982 — each finding the bound broken, though each still leaning on an extra assumption. We take the history through Brunner and colleagues' review.

§7.3 The Wedge

Sealed notes stop at three in four. Now give the partners a bent wire instead of a table of answers, and one small tool: a wedge, under the second partner's box, that tilts the question it is asked to a direction halfway between the two.

On the blackboard. The partners share the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}\,\mathrm{cup}$, the cup being unnormalized, $\mathrm{cup} = |00\rangle + |11\rangle$, as always in this book. Each partner has two questions. We take the first partner's question $x = 0$ to be the *lid* question, measured by the observable Z, and $x = 1$ the *door* question, measured by X; each is a ± 1-valued observable, $+1$ for a bright answer (bit 0), -1 for a dark one (bit 1). The second partner's questions are the door ($y = 0$) and the lid ($y = 1$) — named in this order so that a single wedge serves both.

The wedge is a real rotation of the sphere of Chapter 3. We introduce the box

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

a real matrix, *not symmetric*, with $R_\theta^T = R_{-\theta}$. The second partner applies $R_{\pi/8}$ to their half before asking. The matrix turns through $\pi/8$; the sphere turns through *twice* that, $\pi/4$, which is halfway from the lid (Z) to the door (X) — a right angle apart on the sphere. The effect, as one checks directly, is to replace the second partner's two plain observables by tilted ones:

$$R_{\pi/8}^T\, Z\, R_{\pi/8} = \frac{Z - X}{\sqrt{2}}, \qquad R_{\pi/8}^T\, X\, R_{\pi/8} = \frac{Z + X}{\sqrt{2}}.$$

The relevant number for the game is the **correlation** $E_{xy} = \langle\Phi^+|A_x \otimes B_y|\Phi^+\rangle$, where A_x is the first partner's observable (Z or X) and B_y is the second partner's tilted observable. For the Bell state there is a quick way to compute it, and it is Chapter 6's sliding trick: $\langle\Phi^+|A \otimes B|\Phi^+\rangle = \frac{1}{2}\langle \mathrm{cup}|A \otimes B|\mathrm{cup}\rangle$, and by Theorem 6.4, $(A \otimes B)\,\mathrm{cup} = (I \otimes BA^T)\,\mathrm{cup}$, so the whole thing

collapses to $\frac{1}{2}\operatorname{Tr}(BA^T) = \frac{1}{2}\operatorname{Tr}(A^TB)$. Since Z and X are symmetric, $E_{xy} = \frac{1}{2}\operatorname{Tr}(A_xB_y)$. Now read off the four numbers, using $\operatorname{Tr}(ZZ) = \operatorname{Tr}(XX) = 2$ and $\operatorname{Tr}(ZX) = 0$:

$$E_{00} = \tfrac{1}{2}\operatorname{Tr}\Big(Z\cdot\tfrac{Z+X}{\sqrt{2}}\Big) = \tfrac{1}{\sqrt{2}}, \quad E_{01} = \tfrac{1}{2}\operatorname{Tr}\Big(Z\cdot\tfrac{Z-X}{\sqrt{2}}\Big) = \tfrac{1}{\sqrt{2}},$$
$$E_{10} = \tfrac{1}{2}\operatorname{Tr}\Big(X\cdot\tfrac{Z+X}{\sqrt{2}}\Big) = \tfrac{1}{\sqrt{2}}, \quad E_{11} = \tfrac{1}{2}\operatorname{Tr}\Big(X\cdot\tfrac{Z-X}{\sqrt{2}}\Big) = -\tfrac{1}{\sqrt{2}}.$$

So $E = (+,+,+,-)\frac{1}{\sqrt{2}}$, and

$$S = E_{00} + E_{01} + E_{10} - E_{11} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \;>\; 2.$$

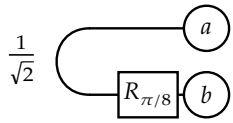
The winning probability for each pair is $p_{\text{win}} = \frac{1}{2}(1 + (-1)^{x\cdot y}E_{xy}) = \frac{1}{2}(1 + \frac{1}{\sqrt{2}}) = \cos^2(\pi/8) \approx 0.8536$ — the same for all four pairs, and above the sealed-note ceiling of $\frac{3}{4}$. See `Code/Ch7/Wedge.py`, which builds all sixteen amplitudes and confirms $p_{\text{win}} = \cos^2(\pi/8)$ for each of the four pairs and $S = 2\sqrt{2}$, and that each partner's own answer is a fair coin for every question — the marginal is $\frac{1}{2}$, so neither partner can read the other's question from their own results.

> **Theorem 7.2 (the wedge strategy).** With the shared Bell state and the wedge $R_{\pi/8}$ on the second partner's half, every one of the four question pairs is won with probability $\cos^2(\pi/8) = \frac{1}{2}(1 + \frac{1}{\sqrt{2}})$, and $S = 2\sqrt{2}$. No quantum strategy does better: $S \le 2\sqrt{2}$ for every state and every choice of ±1-observables.

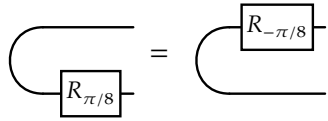
The last sentence is **Tsirelson's bound** (Cirel'son, 1980; we take it, and its history, through Brunner's review). We *state* it and do not prove it here; the proof is a fact about the norm of an operator built from the four observables, and it is not needed for anything this chapter does. What we need is only that $2\sqrt{2}$ is more than 2.

In the sketchbook. The whole game is one closed picture. The pair is the

Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}$ cup — the cup unnormalized, cup $= |00\rangle + |11\rangle$, so the $\frac{1}{\sqrt{2}}$ rides in front of the picture; the second partner's lower leg carries the wedge $R_{\pi/8}$; and each leg ends in an effect for that partner's answer, a on top and b below:



Read left to right and closed at both ends, it is a number — the amplitude of the answer pair (a, b) — and its squared modulus is their probability, by Chapter 5's rule. The one move worth savouring is where the wedge may stand. Sliding it round the cup (Theorem 6.4, with $R_{\pi/8}^T = R_{-\pi/8}$) carries it from the lower leg to the upper, turned the other way:



so the wedge may sit under either box, tilted oppositely, and the game is unchanged. `Wedge.py` checks the two placements give the same closed pictures exactly, and — because R_θ is real, so that a transpose could hide behind a dagger — first checks the sliding rule with a random complex, non-symmetric box, where transpose and dagger genuinely differ.

Dirac's Blackboard.

One pair, in full, with the $\frac{1}{\sqrt{2}}$ *shown.* Take $(x, y) = (0, 0)$: the first partner asks the lid (Z-basis, answers $|0\rangle, |1\rangle$), the second the door, after the wedge. The pair, before either asks, is $(I \otimes R_{\pi/8})\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}(\,|0\rangle \otimes R_{\pi/8}\,|0\rangle + |1\rangle \otimes R_{\pi/8}\,|1\rangle\,)$, with $R_{\pi/8}\,|0\rangle = \cos\frac{\pi}{8}\,|0\rangle + \sin\frac{\pi}{8}\,|1\rangle$. The amplitude for the bright–

bright outcome $(|0\rangle, |+\rangle)$ is

$$(\langle 0| \otimes \langle +|)(I \otimes R_{\pi/8})\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}\,\langle +|\,R_{\pi/8}\,|0\rangle$$
$$= \frac{1}{\sqrt{2}} \cdot \frac{\cos\frac{\pi}{8}+\sin\frac{\pi}{8}}{\sqrt{2}} = \frac{\cos\frac{\pi}{8}+\sin\frac{\pi}{8}}{2}.$$

Its square is $\frac{1}{4}(1+\sin\frac{\pi}{4}) = \frac{1}{4}(1+\frac{1}{\sqrt{2}})$. The dark–dark amplitude is $-\frac{1}{2}(\cos\frac{\pi}{8}+\sin\frac{\pi}{8})$, the same square. This pair wins on agreement, so $p_{\text{win}} = \frac{1}{4}(1+\frac{1}{\sqrt{2}}) + \frac{1}{4}(1+\frac{1}{\sqrt{2}}) = \frac{1}{2}(1+\frac{1}{\sqrt{2}}) = \cos^2\frac{\pi}{8}$. Every $\frac{1}{\sqrt{2}}$ in sight is the Bell state's normalization.

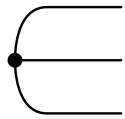
Why this definition of R_θ**.** We use a plain real rotation, not a fraction of a bit-flip, because we want a genuinely *intermediate* question — one tilted between lid and door — and only a rotation of the sphere tilts one axis toward another. A fraction of X would leave the question pointing the wrong way.

---

§7.4 A Node with Three Legs

Two boxes from one bench were a pair. Three boxes from one bench — made together, sent together — are something the wire notation cannot yet draw. We need one new element, and it is the only one this chapter adds.

In the sketchbook. Imagine a single point on the page from which *three* wires leave, all to the right: not three separate wires laid side by side, and not a wire that branches into a fork, but one small solid dot where three legs meet as equals. We draw it as a filled dot with three legs:



Read left to right, nothing comes in and three wires go out: it is a state on three wires. We call it, plainly and only, **the node**, or the three-legged node.

It has no label, no colour, and no number written on it; it is always the same thing.

On the blackboard. We *define* the node to be the unnormalized state

$$N \;=\; |000\rangle + |111\rangle\,,$$

whose squared length is $\langle N|N\rangle = 2$. Exactly as the cup is $\sqrt{2}$ times the Bell state, the node is $\sqrt{2}$ times the state a physicist would normalize:

$$\tfrac{1}{\sqrt{2}}\,N \;=\; \tfrac{1}{\sqrt{2}}\,(\,|000\rangle + |111\rangle\,),$$

the three-party state of Greenberger, Horne and Zeilinger. We keep the node unnormalized for the same reason we kept the cup unnormalized: so that the rules for pushing things through it hold with no stray scalars. Wherever the node is drawn in this book it means $|000\rangle + |111\rangle$, and we write the $\frac{1}{\sqrt{2}}$ beside it whenever the normalized state is wanted.

The node is not three cups, and it is not a product. Its three legs agree — every basis term is 000 or 111 — but they agree three at a time, in a way no pair of them can hold alone. (`Code/Ch7/Trio.py` confirms N is not a product across any way of splitting the three legs.)

Why this definition. We could have tried to build a trio from cups, joining pairs. No joining of cups gives $|000\rangle + |111\rangle$; the all-agree-together state is genuinely new, and it deserves its own mark on the page. The dot is the least we can draw: a place where wires meet, carrying no data of its own.

What the node lets a party predict

Here is the guarantee the trio comes with, and the realist's best argument. The three boxes are made together; when all three are opened the same way (all by the door), the number of dark notes is always *even*; when one is opened by the door and the other two by the drawer, the number of dark notes is always *odd*. Never broken, says the workshop's ledger.

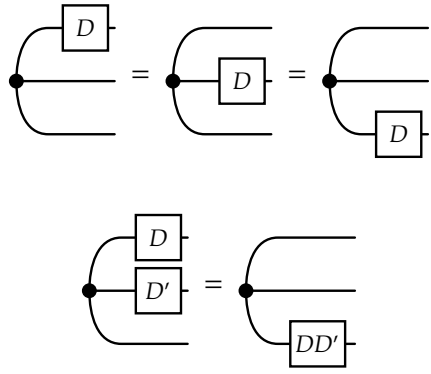
Grant the realist the guarantee. Then any one party can *predict* their own answer with certainty from the other two — for the all-door case, two answers fix the third by evenness — without touching their own box. By EPR's criterion, then, the answer was written inside. This is the theory at its strongest, and §7.6 will need every bit of that strength to overturn it. For now, two facts about the node make the guarantee precise, and both are exact equalities.

> **Theorem 7.3 (sliding and merging through the node).** A *diagonal* box slides freely from one leg of the node to any other, unchanged, and two diagonal boxes on one leg merge into their product. For diagonal D and D',
>
> $$(D \otimes I \otimes I)\, N = (I \otimes D \otimes I)\, N = (I \otimes I \otimes D)\, N,$$
> $$(D \otimes D' \otimes I)\, N = (I \otimes I \otimes DD')\, N.$$
>
> A box that is not diagonal does **not** slide.

Proof idea. The node lives on just two terms, $|000\rangle$ and $|111\rangle$. A diagonal box multiplies each basis state by a number depending only on the bit on its leg; but on those two terms all three legs carry the same bit, so it does not matter which leg the box sits on. A non-diagonal box moves weight between $|0\rangle$ and $|1\rangle$, which breaks the all-same pattern, and then the leg matters.

Proof (*on the blackboard*). Let $D = \mathrm{diag}(d_0, d_1)$. On N's two terms, $(D \otimes I \otimes I)\,|000\rangle = d_0\,|000\rangle$ and $(D \otimes I \otimes I)\,|111\rangle = d_1\,|111\rangle$; putting D on the second or third leg gives the very same $d_0\,|000\rangle + d_1\,|111\rangle$, because each term carries one bit across all three legs. For two boxes, $(D \otimes D' \otimes I)N = d_0 d_0'\,|000\rangle + d_1 d_1'\,|111\rangle = (I \otimes I \otimes DD')N$. ■ `Trio.py` checks both with random *complex* diagonal boxes, and checks that a random non-diagonal box fails to slide.

In the sketchbook. The rule is one picture. A diagonal box D on any leg is the same drawing wherever it sits, and two such boxes collect onto a single leg as their product:

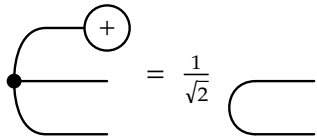


A box that is not diagonal cannot be moved this way; the drawing would change its value.

> **Theorem 7.4 (closing one leg).** Closing one leg of the node with a bright *door*-effect leaves the other two as a cup, and a dark door-effect a cup with a Z box on one leg; a bright *lid*-effect leaves a plain product:
>
> $$(\langle +| \otimes I \otimes I)\, N = \frac{1}{\sqrt{2}}\,\mathrm{cup} = |\Phi^+\rangle\,,$$
> $$(\langle -| \otimes I \otimes I)\, N = \frac{1}{\sqrt{2}}(Z \otimes I)\,\mathrm{cup},$$
> $$(\langle 0| \otimes I \otimes I)\, N = |00\rangle\,.$$

Proof. With $\langle +| = \frac{1}{\sqrt{2}}(\langle 0| + \langle 1|)$, applying it to the first leg of $N = |000\rangle + |111\rangle$ keeps $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}\mathrm{cup}$; with $\langle -| = \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)$ the second term picks up a minus, giving $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = \frac{1}{\sqrt{2}}(Z \otimes I)\mathrm{cup}$; with the lid-effect $\langle 0|$, only $|00\rangle$ survives. ■ (See `Trio.py`.)

In the sketchbook. A bright door-effect on one leg bends the other two into a cup — the first cup this node gives us, and unnormalized as ever, $\mathrm{cup} = |00\rangle + |11\rangle$:



so the trio, with one door opened bright, is a Bell pair on the wires that remain.

This is the Anteater's remark, at which level to ask: open one of the three by its door, and if it sings bright, the other two are a Bell pair — a *pair*, though a moment ago there was no pair, only a trio. If it sings dark, the other two are a pair with a Z turned on one leg. The whole was one thing on three wires; closing a leg leaves one thing on two.

The drawer is the Y question

The trio's second way in — the drawer — is the Y question, and it is where the conjugate of Chapter 6 comes back to matter. The drawer-effect asks along the states $|{\pm i}\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i\,|1\rangle)$, and its bra is a door-effect behind a box:

$$\langle{\pm i}| = \langle{\pm}|\, S^\dagger, \qquad S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}.$$

This is an equality that a real test would miss: $|{\pm i}\rangle$ has an imaginary amplitude, and only against a non-real state does $\langle{\pm}|\, S^\dagger$ differ from $\langle{\pm}|$. (`Code/Ch7/SixNotes.py` checks it with the genuinely non-real $|{\pm i}\rangle$, and checks that dropping the conjugation fails.) In a picture, a drawer-effect on a leg is an $S^\dagger$ box followed by a door-effect — a fact we shall spend in §7.6.

It also settles a small debt from Chapter 6, where the Crab was told to build his pairs with only one way in besides the lid, and warned that not every third way would agree. Here is why: a pair opened by the *drawer* on both sides would *disagree*, because $(Y \otimes Y)\,\text{cup} = -\text{cup}$ (checked in `SixNotes.py`). The minus sign that will bring down the realist in §7.6 is

already visible here, in a single pair.

Coecke's Sketchbook.

A dot is not a cup, and its normalization is its own. Draw the node and it is tempting to read it as a wire that forks — one line splitting into three. It is not. A forked wire would copy a state, and no box copies an unknown quantum state; the node does no such thing. It is a *state* on three legs, born equal, with no "first" leg the others come from. Its three legs meet as three, and that is the whole of it. One caution to carry: the node is unnormalized, $N = |000\rangle + |111\rangle$ with $\langle N|N\rangle = 2$, exactly as the cup is $|00\rangle + |11\rangle$ with length $\sqrt{2}$. When a probability is wanted, the $\frac{1}{\sqrt{2}}$ must be paid in, once, just as for the cup. Draw the dot freely; remember the number when you cash out.

---

§7.5 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 6's forty-five. One introduces a new picture element — the node; one is a new rewrite rule — sliding and merging through it; the rest record boxes and equalities we shall lean on.

#	Algebra (Dirac)	Picture (Coecke)
46	$N = \|000\rangle + \|111\rangle$ (unnormalized; length $\sqrt{2}$); GHZ state $= \frac{1}{\sqrt{2}}N$	a solid dot with three legs — a state on three wires — *new element*

#	Algebra (Dirac)	Picture (Coecke)
47	$(D \otimes I \otimes I)N = (I \otimes D \otimes I)N$, D diagonal; $(D \otimes D' \otimes I)N = (I \otimes I \otimes DD')N$	a diagonal box slides through the dot from leg to leg; two on a leg merge into one — **sliding through the node**
48	$(\langle +| \otimes I \otimes I)N = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$; a dark leg leaves it with a Z box	a bright door-effect on one leg leaves $\frac{1}{\sqrt{2}}$ cup
49	$\langle \pm i| = \langle \pm| \, S^\dagger$	a drawer-effect is the box $S^\dagger$ followed by a door-effect
50	R_θ real, $R_\theta^T = R_{-\theta}$; $(I \otimes R_\theta)\text{cup} = (R_{-\theta} \otimes I)\text{cup}$	the box R_θ; slid round the cup it arrives turned the other way
51	$P = \left(\begin{smallmatrix} H & 0 \\ 0 & HS^\dagger \end{smallmatrix}\right)$; $(\langle q| \otimes I)\, P\, (|q\rangle \otimes I)$ is its q-th block	a two-wire box whose upper wire, pinned between state q and effect q, leaves the chosen box on the lower wire
52	$\sum_{\text{win}} \lvert \langle ab| \, (I \otimes R_{\pi/8})\, |\Phi^+\rangle \rvert^2 = \cos^2(\pi/8)$	the game as a closed picture: the $\frac{1}{\sqrt{2}}$ cup, $R_{\pi/8}$, two effects

Every row is an exact equality of matrices; the box P of row 51 arrives in §7.7.

§7.6 The Theorem, Twice

Now the trio's guarantee is turned against the realist. If the answers are sealed notes — six of them, a door-answer and a drawer-answer for each of the three boxes — then the four cases of the guarantee are four equations those six numbers must satisfy. We shall see they cannot.

The example that fixes the ceiling

Write each box's two sealed answers as ± 1: for box A, a door-answer x_A and a drawer-answer y_A, and likewise x_B, y_B, x_C, y_C. A bright note is $+1$, a dark note -1, so "the number of dark notes is even" means "their product is $+1$", and "odd" means "their product is -1". The four cases of the guarantee become

$$x_A x_B x_C = +1, \qquad x_A y_B y_C = -1, \qquad y_A x_B y_C = -1, \qquad y_A y_B x_C = -1.$$

Can six numbers satisfy all four? Try to satisfy the last three first: set $x_A = x_B = x_C = +1$ and $y_A = +1$, and read off from $x_A y_B y_C = -1$ that $y_B y_C = -1$; take $y_B = -1, y_C = +1$. Check $y_A x_B y_C = (+1)(+1)(+1) = +1$ — but this was required to be -1. It fails. Adjust $y_A = -1$: now $y_A x_B y_C = (-1)(+1)(+1) = -1$, good, and $y_A y_B x_C = (-1)(-1)(+1) = +1$ — required -1. It fails again. Every repair breaks another case. Three of the four we can always meet; the fourth refuses. This is the same stubbornness as in §7.2, and it has the same one-line reason.

The theorem

The eigen-relations behind the guarantee are exact. Recall the bridge: a bright answer is $+1$, a dark answer -1; the door question is the observable X, the drawer question the observable Y. Then "$X \otimes X \otimes X$ acting on N gives $+N$" *is* the statement "all doors: darks even", and "$X \otimes Y \otimes Y$ gives $-N$" *is* "one door, two drawers: darks odd".

Theorem 7.5 (the GHZ paradox). Let $N = |000\rangle + |111\rangle$. **(i)** The node satisfies, exactly,

$$X \otimes X \otimes X N = +N,$$
$$X \otimes Y \otimes Y N = Y \otimes X \otimes Y N = Y \otimes Y \otimes X N = -N.$$

(ii) There are no six numbers $x_A, y_A, x_B, y_B, x_C, y_C \in \{\pm 1\}$ with

$$x_A x_B x_C = +1, \quad x_A y_B y_C = y_A x_B y_C = y_A y_B x_C = -1.$$

Part (i) says the trio meets all four guarantees at once; part (ii) says no collection of sealed notes can. So the trio is not a collection of sealed notes.

Part (ii) is a piece of arithmetic and belongs to neither strand: multiply the four left-hand sides. Every letter appears in exactly two of the four products — x_A in the first and second, y_A in the third and fourth, and so on — so the total is a product of squares, $(+1)$. Multiply the four right-hand sides: $(+1)(-1)(-1)(-1) = -1$. If all four held, $+1 = -1$. So at most three can hold. (`SixNotes.py` runs all 64 assignments: none satisfies four, the maximum is three, and flipping one note flips exactly two of the four verdicts.)

Part (i) is what we prove twice.

Dirac's Blackboard

The Pauli boxes act on basis states by $X|0\rangle = |1\rangle$, $X|1\rangle = |0\rangle$ and $Y|0\rangle = i|1\rangle$, $Y|1\rangle = -i|0\rangle$. Apply each operator to N's two terms in turn.

For $X \otimes X \otimes X$:

$$X{\otimes}X{\otimes}X\ |000\rangle = |111\rangle, \qquad X{\otimes}X{\otimes}X\ |111\rangle = |000\rangle,$$

so $X \otimes X \otimes X N = |111\rangle + |000\rangle = +N$.

For $X \otimes Y \otimes Y$:

$$X{\otimes}Y{\otimes}Y\ |000\rangle = |1\rangle \otimes i|1\rangle \otimes i|1\rangle = i^2|111\rangle = -|111\rangle,$$

$$X\otimes Y\otimes Y\,|111\rangle = |0\rangle \otimes (-i\,|0\rangle) \otimes (-i\,|0\rangle) = (-i)^2\,|000\rangle = -\,|000\rangle\,,$$

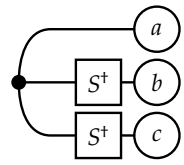
so $X \otimes Y \otimes Y\,N = -(|111\rangle + |000\rangle) = -N$. The two remaining relations are identical with the X in a different slot; the two factors of $\pm i$ multiply to -1 each time, wherever the lone X sits. Each minus sign is a single line of algebra: $Y\,|0\rangle = i\,|1\rangle$, $Y\,|1\rangle = -i\,|0\rangle$, and $i \cdot i = (-i)(-i) = -1$. ■

Coecke's Sketchbook

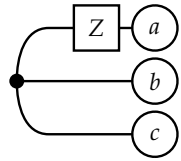
We prove the same four relations as one drawing. The door-effect for answer bit a is $\langle a|\,H$ — so $\langle 0|\,H = \langle +|$ and $\langle 1|\,H = \langle -|$ — and we mark it simply a at a leg's end. Start from the all-door case, the three door-effects a, b, c on the node's legs; its value is the base case

$$\langle abc|\ (H \otimes H \otimes H)\,N = \frac{1}{\sqrt{2}} \quad \text{when } a+b+c \text{ is even,} \qquad 0 \quad \text{when odd,}$$

a closed picture we compute once (below). Now build the mixed case, door on A and drawer on B and C, by the rule of §7.4: each drawer-effect is an $S^\dagger$ box followed by a door-effect. So the mixed case is the all-door picture with an $S^\dagger$ on leg B and an $S^\dagger$ on leg C:



Step 1 — sliding and merging (Theorem 7.3). Both $S^\dagger$ boxes are diagonal, so they slide through the node onto one leg and merge; since $S^\dagger S^\dagger = Z$, the two of them become a single Z box, which we may keep on leg A:



Step 2 — the Z flips one door-answer. A Z before a door-effect turns it into the other: since $HZ = XH$ and $\langle a|\, X = \langle a \oplus 1|$, we have $\langle a|\, HZ = \langle a|\, XH = \langle a \oplus 1|\, H$ — the door-answer bit a is flipped. So the mixed picture equals the all-door picture with answer a replaced by its opposite. Its value is therefore $\frac{1}{\sqrt{2}}$ exactly when $a + b + c$ is *odd* — the parities have exchanged. That is precisely "darks odd", the relation $-N$.

All three minus-sign relations are this one move: whichever two legs carry drawers, their two $S^\dagger$ boxes merge into one Z, and the Z flips one door-answer, reversing the parity. The picture says not only *that* the sign appears but *where* it lives — on the leg the merged box sits on. ■ (`SixNotes.py` runs the rewrite step by step as exact picture equalities and confirms the base-case values.)

The base case, computed. $H^{\otimes 3}N = |{+}{+}{+}\rangle + |{-}{-}{-}\rangle$, and $\langle abc|\, {+}{+}{+}\rangle = \frac{1}{2\sqrt{2}}$ while $\langle abc|\, {-}{-}{-}\rangle = \frac{1}{2\sqrt{2}}(-1)^{a+b+c}$, so the sum is $\frac{1}{2\sqrt{2}}(1+(-1)^{a+b+c})$, which is $\frac{1}{\sqrt{2}}$ for even $a + b + c$ and 0 for odd.

The verdict

The algebra wins on length, and narrowly. Each minus sign is one line on the blackboard — $Y|0\rangle = i|1\rangle$, $Y|1\rangle = -i|0\rangle$, and $i \cdot i = -1$ — and there is nothing to set up first. The sketchbook must first own a new rule, sliding-and-merging through the node, whose own justification is a two-line index check (Theorem 7.3); once that is paid for, all three minus signs are a *single* move — two $S^\dagger$ merge to a Z, and the Z flips one answer — and the picture shows *where* the sign lives, which the blackboard does not. There is a second debt the picture cannot dodge: its base case, the all-door value $\frac{1}{\sqrt{2}}$-on-even, is itself a closed calculation, and that calculation is algebra. So the picture

explains and the blackboard proves, and the blackboard is shorter.

And a warning against overselling either. Part (ii) — that no six numbers work — is arithmetic, and belongs to neither strand: it is the multiplication of four ± 1 products. Part (i) is what the two strands prove, and all it establishes is an *equality of processes*: that the node meets all four guarantees at once. That the sealed notes cannot, and hence that the trio is not sealed notes, is the arithmetic. Neither the picture nor the algebra proves that nature is nonlocal; they prepare the contradiction, and the counting closes it. No experiment is a rewrite, and no rewrite is an experiment.

---

§7.7 The Round

We can now play the whole thing as one game, apart, and draw it as a single picture.

Three players take one box each and go to three rooms; each also carries a sealed envelope naming which way to open — door or drawer. A referee — the reader — holds a fourth sealed slip that says what the answers must total: *even* if all three envelopes say door, *odd* if one says door and two say drawer. Each player opens their box the named way, hears bright or dark, and returns; the answers are compared with the slip. The round is won when the answers match the slip.

One master diagram lays this out as a single picture. It needs one new box.

The player's box P. Each player's question and their leg of the node meet in a two-wire box

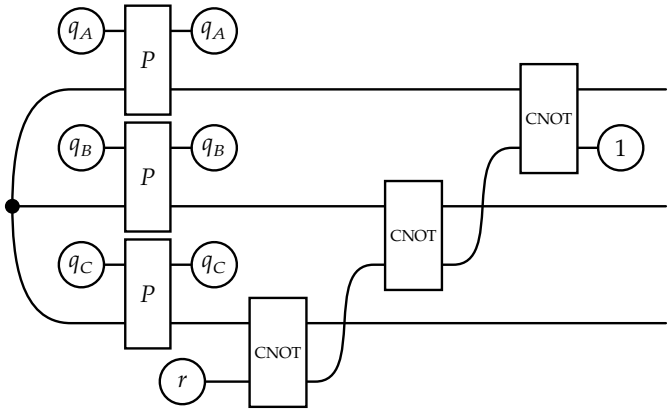
$$P = \begin{pmatrix} H & 0 \\ 0 & HS^\dagger \end{pmatrix},$$

built in Chapter 4's block form, exactly as CNOT $= \left(\begin{smallmatrix} I & 0 \\ 0 & X \end{smallmatrix}\right)$ is. The upper wire carries the question, $|0\rangle$ for door or $|1\rangle$ for drawer; the lower wire carries the player's leg. When the upper wire is pinned between the state $|q\rangle$ and

the effect $\langle q|$ of the *same* value, the box collapses **exactly** to its q-th block — H for the door, $HS^\dagger$ for the drawer — an equality, not a postselection with a probability attached. Following the chosen block by a door-effect $\langle a|$ gives $\langle a|H = \langle \pm|$ (the door question) or $\langle a|HS^\dagger = \langle \pm i|$ (the drawer question, by §7.4). One box, both questions, chosen by the wire.

The diagram

Here is *The Round*. Rows, top to bottom: the question q_A, leg A, then q_B, leg B, then q_C, leg C, and at the foot the referee's tally wire.



The three legs of the node run out to the right, one to each player; each passes through its player's box P (lower wire of the box) and then acts as the control, on top, of one CNOT, before ending at an open right-hand end — the player's answer. Each question wire begins in the state $|q\rangle$, runs through the upper wire of its P, and ends in the effect $\langle q|$ of the same label: this is the pinning that collapses P to its block. The tally wire begins in the referee's state $|r\rangle$, is flipped by each leg's CNOT in turn, and ends in the effect $\langle 1|$ — the *losing* slip. The two places where the tally wire steps up past a leg are crossings, and a crossing is Chapter 4's swap drawn with one strand held straight; bending a wire is free (Chapter 6). A crossing neither joins nor breaks a line, so the tally is one unbroken line from the state $|r\rangle$ to the effect $\langle 1|$.

Theorem 7.6 (the round never loses). For each of the four

dealt question triples $(q_A, q_B, q_C) \in \{000, 011, 101, 110\}$, with $r = q_A \vee q_B \vee q_C$, *The Round* with the losing effect $\langle 1|$ on the tally wire evaluates to the **zero vector** on the three open answer wires — exactly. The amplitude of every losing outcome is zero. With the winning effect $\langle 0|$ instead, the picture equals the same diagram with the tally wire and the three CNOTs erased: four answers of the correct parity, each of amplitude $\frac{1}{\sqrt{2}}$.

Proof idea. Pinning each P turns the diagram into the node with the three chosen questions on its legs and a running parity check on the tally. For a dealt triple the guarantee of Theorem 7.5(i) says the parity of the answers is fixed; the losing slip demands the opposite parity, which never occurs, so its amplitude is zero everywhere.

Proof (on the blackboard). Pin the three question wires: each P becomes its block, so the three legs of N carry H, or $HS^\dagger$, according to the deal, and then their door-effects — that is, the node measured in the door or drawer question per the deal. By Theorem 7.5(i) the surviving answers all have the parity the guarantee names, and the three CNOTs write that parity onto the tally wire, which started at $|r\rangle$ with r the guaranteed value. The tally therefore reaches $|0\rangle$ on every surviving branch, and the losing effect $\langle 1|$ annihilates all of them: the result is the zero vector. Replace $\langle 1|$ by $\langle 0|$ and every surviving branch passes; erasing the (now trivial) parity machinery leaves four equal-amplitude answers, squared norm $2 = \langle N|N\rangle$. ■ See `Code/Ch7/Round.py`, which builds the full diagram, confirms the zero vector for the four deals with the losing slip and a non-zero result with the winning slip, and — the point of playing apart — that each player's own answer is a fair coin for every one of the eight triples, so nothing about the other players' questions travels to any room.

What it evaluates to, and its cost. The winning picture's open end has squared norm 2, the node's $\langle N|N\rangle$; the honest probabilities are these divided by 2, so each of the four correct answers occurs with probability $\frac{1}{4}$. The contrast with §7.3 is the whole point of the chapter. There, the wedge won *more often* — with probability $\cos^2(\pi/8)$, past the sealed-note ceiling but short of certainty. Here the players win *every time*, and no collection of sealed notes can, by Theorem 7.5(ii). A game won with certainty by a

shared state and never by any local strategy is what Brunner's review calls pseudo-telepathy (Brassard and colleagues, 2005); we shall not need the name again.

Feynman's Notebook.

Three particles, and a campus. The three-party argument of this chapter is due to Greenberger, Horne and Zeilinger (1989), with Shimony added in a fuller 1990 account and a much-read exposition by Mermin the same year; we take the attribution, as we take the game, through Brunner and colleagues' review. Their original construction used *four* particles, the three-particle version noted as a possibility; the crisp three-party form in X and Y is the one the review presents, and the one drawn above.

Closing every loophole at once took until 2015, when Hensen and colleagues, on the campus of Delft University of Technology, entangled two electron spins in diamond 1.3 km apart and ran 245 trials of the CHSH game. They found $S = 2.42 \pm 0.20$ — above the sealed-note ceiling of 2 — winning $k = 196$ trials, with probability 0.039 that a local-realist model with memory could fake it (0.019 under their more conventional analysis). The bent wire had broken the inequality in the open.

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§7.8 Where Was It?

Let us gather what we have. Sealed notes, questioned in the several combinations of a game, meet a ceiling: they win at most three of the four pairs, and $S \leq 2$ (§7.2). A shared bent wire, tilted by a wedge, breaks the ceiling — $S = 2\sqrt{2}$, a win rate of $\cos^2(\pi/8)$ (§7.3). A single new drawing, a dot where three wires meet, carries a guarantee that any one answer can be foretold from the other two, which is the realist's whole case (§7.4). And that same guarantee, written out as six sealed numbers, contradicts itself — no six numbers meet all four cases, though the node meets them exactly, which we proved twice (§7.6). Played apart, the trio wins with certainty

what no notes could (§7.7). In every picture the agreement was kept in the same place it was kept in Chapter 6: not in either box, but in the fact that the boxes are legs of one thing.

So the answer to the chapter's opening question is no. The notes were not written. Two halves that always agree, questioned only one way, could have carried sealed notes; three boxes questioned in four combinations cannot, because the four combinations demand of the notes something arithmetic forbids. Where was the answer, then, if not in the boxes and not crossing the house? In neither box: in the place where the three legs meet.

And that dot has more to tell. We have used it as one fixed thing, unlabelled, always $|000\rangle + |111\rangle$. But a dot where wires meet is the beginning of a language, not the end of one: allow it a family — dots that count in a second way, dots that carry a turn of the sphere — and the pictures we have drawn box by box will start to rewrite themselves, without ever passing back through the algebra. The next chapter gives the dot its family, and its name.

Octet for Spiders

短日や
糸八本の
一結び

tanjitsu ya
ito happon no
hitomusubi

The short day —
eight threads,
one knot.

[*Achilles comes in out of the winter dark, stamping the snow off his boots; the Crab's scullery is warm and low-lit, a fire in the grate. A stout beam crosses the room overhead, and from hooks along it hang lengths of kitchen cord, some strung through small curtain-rings. The short day is already going grey at the one window. The Tortoise is there before him, in a chair drawn near the fire.*]

[*Along the long wall, running the whole length of the room under the beam, hangs Escher's* Metamorphosis II *— a narrow strip of a woodcut, no taller than a hand and about twice a man in length, so that the eye has to walk it end to end. The lamp on the table throws its light along the strip: part way along it a little town stands by the water with a square tower, and elsewhere the print goes to a chequer of light and dark squares. Above it, from the beam, the strung cords hang still in the warm air. At the one window the afternoon has gone the flat grey of coming snow, and the first flakes have begun — few and fitful, in no hurry, settling and starting again.*]

Achilles: I had forgotten a person could be this cold and still be alive to complain of it. The canal's frozen hard as the road — I could not tell you where the one stopped and the other began.

Crab: Come to the fire, come to the fire. There's a pot on the hob and it's been waiting on you. [*He pours.*] I keep this room warm for the work; the cords go slack if it's damp and sulk if it's cold.

Achilles: The work being all this washing-line you've hung the ceiling with. [*He takes the cup gratefully.*] I walked past your window twice thinking you'd gone to bed and left the string out.

Tortoise: He's been at it since noon. I came early and have watched him tie knots and untie them and stand back and admire them like a man hanging pictures.

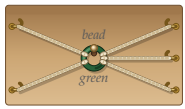
Crab: And now I have my audience, I shall show the pair of you the finest thing to come out of a workshop in Delft in twenty years, and neither of you will leave until you have understood it. Warm your hands first. It keeps till you're thawed.

[*Achilles crouches to the grate a moment, turning his hands to it until the ache goes out of them; the cup steams under his chin. The scullery closes warm around the three of them, the snow settling against the grey pane, and the cords overhead barely stirring in the heat off the fire.*]

First Challenge — *A Name, a Colour, a Turn*

Achilles: [*looking up at the beam, where one ring sits on its peg with three cords tied to it — one running in from a hook at the beam's left end, two running out to hooks at its right*] Wait — I know that. Or I knew its plainer cousin. Last autumn you showed us a bare dot where three cords met, that already knew things about itself: any one of its cords foretold by the other two. But that one had no colour, and no more than the three legs. This has a colour, and — how many legs is it wearing?

Crab: As many as I care to give it. [*He runs a fourth cord into the ring from the left, then a fifth out of it to the right.*] That is the whole point of the family. The dot has grown up.



grown a family

Achilles: It has a family now. Has it a name?

Crab: It has. This — a ring with cords drawn through it and knotted into one — I call a **green ring**, for the plainest of reasons: the ring is green. It asks its cords a single question, the same question your autumn dot asked: *do you all say the same?* All bright together, or all dark together — it passes those; anything else it stops. [*He touches a small bead threaded on the ring itself.*] And this bead is its **turn** — it slides round the ring, clockwise as you face it, to mark how far the thing is turned, the very sphere-turn you learned in the spring. Bead at the top, no turn at all.

Achilles: Then let me guess what the simplest of them does. A green ring with one cord coming in and one going out, and its bead at the top — no turn. It asks the one cord whether it agrees with... itself, which it can hardly fail to do. So it does nothing. It's a bare cord with a bead on a ring, and might as well not be there.

Crab: [*pleased*] Right, and right the first time — which I gather is not always your way. A green ring, one in, one out, no turn, is nothing but wire.

Tortoise: So the interest is in the ones with more legs, and in what happens when two of them meet. Watch. [*She takes a loose cord and threads two green rings onto it, a little apart, each with its bead set some way round.*] Here are two, on one cord, each turned a little.

[*The two green rings sit on the loose cord, a hand's breadth between them, one bead — the left-hand ring's — a third of the way round, the other a quarter. She sets a finger to the near ring, the left-hand one as they face it, and slides it along the cord until it meets the far one; the cord does not move and its far end does not stir, only*

the two rings travel together and knot into a single green ring where they meet. The two beads, sliding round with their rings, come to rest as one bead, at a turn that is the two turns added – a third and a quarter together. Two rings have become one; the far end of the cord is exactly where it was.]



two rings, two turns *slid into one*

Tortoise: Two green rings sharing a cord are one green ring, and the turns add. Slide them together; that is all.

Crab: And it makes no difference to anything downstream. [*He threads a test cord through the fused ring and gives its far end a tug; the end answers exactly as it did when there were two rings.*] The far end can't tell whether it was one ring or two. [*He lifts the demo cord and its ring off, the test cord still threaded through the ring, and sets them aside at the end of the table.*]

[*In the grate two flames stand up side by side off the one log, wavering apart and together, until a draught leans them into each other and they run up as a single taller flame, its two heats gathered in the one tongue. The chimney draws it exactly as it drew the two; nothing above the pot warming on the hob is any the wiser.*]

Achilles: So a whole chain of green rings is really just one green ring wearing the sum of all their turns. Tidy. [*He looks up at the beam.*] And you want to build something. Where do we start?

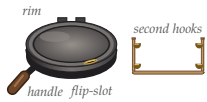
Crab: Here. [*He lifts the first ring down off the beam, cords and all, and the loose lengths hanging beside it with their small rings, and lays them at the end of the table; then he fixes two green rings to the beam, one above the other on the left, and runs a cord into each from the left side.*] Two green rings, and a cord fed to each — one for you to hold responsible, one for the Tortoise. The beginning of the thing.

[*Achilles steadies the cord into the upper green ring; the Tortoise sets the cord into the lower one. Two cords now run in from the left, one to each green ring.*]

Achilles: [*stepping back, taking it in*] Two green rings, two cords in, and all of it green. Is green all there is?

Second Challenge — *The Other Colour*

Crab: [*with relish*] It is not. [*He lifts a heavy flat gadget onto the table — a griddle-like press with a single slot in its rim and, standing beside it, a little wooden rack with two empty hooks up each post.*] This is my **koekpan**. It will do the heavy work later, and I'll thank you not to stare at it while it does — it's a shy thing. Two parts to know now. Along its rim, this **flip-slot**: a cord passed through it comes out the other side changed. And this second set of hooks — whatever the pan makes, it hangs *there*, and leaves the first cords be.



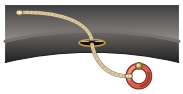
the Crab's press

[*It is a homely thing to have such a fuss made of it: a broad flat press of old Delft iron, its face worn to a soft shine by years of hands, the handle darkened where a claw has gripped it a thousand times. Set down on the table it gives off a warmth of its own now, and — if nobody is turned its way — a low sound at the very edge of hearing, like a kettle a long way from the boil. Achilles leans in to catch it; the sound is not there; he sits back and it steals up again.*]

Achilles: Changed into what?

Crab: Watch. [*He takes a loose cord feeding a green ring and passes the cord through the flip-slot.*] A green ring counts one way — all-agree, bright-together or dark-together. Feed its cord through the slot, and out comes a ring that counts the *other* way. A **red ring**.

[*The green ring sits on its loose cord. He draws the cord through the flip-slot; the cord runs through and the ring at its end, as it comes clear of the slot, is red where it was green. Its cord and its bead have not moved – same cord, same turn – only the colour, and with the colour the question it asks.*]



through the slot

Achilles: One slot, and it changes sides. But that's only the one cord flipped. What of a ring with a whole fistful of legs?

Crab: Then you flip them all. [*He lays the red-ended cord at the end of the table, takes a green ring with four cords and threads every one of its cords through the flip-slot in turn.*] Every leg through a slot, and the ring itself turns.

[*The green ring hangs with its four cords. One by one he draws each cord through the flip-slot; as the last of the four comes through, the ring itself is red – not touched, not re-tied, only every one of its legs sent through the slot, and the colour has followed. The ring's four cords still run where they ran; only green has become red.*]



as it hung *every one through*

Crab: [*handing the flipped-round ring to Achilles*] There. Handle it. It answers to every question exactly as a red ring does — it *is* one. [*Achilles tries a cord or two against it, nods; the Crab takes the demo ring and its cords down and sets them by the first.*]

Achilles: So it isn't that I *called* it red. Dressing a green ring's every leg in flip-slots truly makes it count the other way. Like putting a question into another tongue: change one word and you have a muddle, change every word and it is the same question, asked the other way. The colour earned it.

Tortoise: A red ring and a green ring see different things. Green asks whether the cords all say the same the plain way; red asks it the sideways way — the way the spring's other pair of alternatives said it. Two honest counters, one cord, two answers.

[*The snow has thickened at the window while they talked. Where the flakes fall against the bare black boughs of the tree by the bridge they show white, each one*

plain; where they cross the pale grey of the sky behind, the very same flakes go dark, small specks against the light. Achilles's eye drifts from the black branches up to the sky, and the whole falling field turns over at once — white to dark, the one snow answering to whichever ground he sets it against.]

Crab: [*fixing two red rings to the beam on the right, and running a cord out of each to the right*] And here are two reds, on the right, where the answers will come out — a cord drawn from each. [*Two cords now hang out to the right, one from each red ring.*] Greens on the left, reds on the right. That is my frame.

Achilles: Greens counting one way on the left, reds the other way on the right. [*He looks from side to side.*] A red ring and a green ring see different things — you've just made me agree to it. So what happens where they meet?

Third Challenge — *Each Copies the Other's Points*

Tortoise: That is exactly the question. Let us make them meet on the smallest scale first: one point of one colour, fed into a single ring of the other. [*She takes a short loose cord ending in a red ring with a single leg.*] This is a red point — a red ring with just the one cord, the plainest red thing there is. And here is a green ring made to fork: one cord in, two cords out, its bead at the top, no turn.



a point and a fork

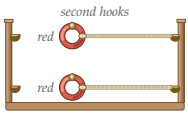
Achilles: Then I can tell you what happens without your pan. The green ring's business is to ask *do my cords agree?* — but it asks in *green*, and a red point is not a green kind of point; it's a stranger to that question. So the ring will have nothing to say to it. It passes straight through and comes out on one leg, unchanged. One cord in, one meaningful cord out.

Tortoise: Say it to the pan, then. [*She feeds the red point into the green fork, and*

presses it on the koekpan.]

[*The red point-cord runs into the green fork-ring: one cord in on the left, two green legs out on the right. She sets the little tangle on the koekpan and presses. The Crab turns his face to the fire while it works.*]

[*Off the pan, on the second set of hooks, hangs not what Achilles foretold. There is no green ring left at all – and instead of one cord out, there are two, and each is a red point, the very twin of the one that went in.*]



off the second hooks

Achilles: [*staring*] Two. There are two of them, and both red, and the green ring's — gone. But I sent in *one*. …Oh. It didn't pass the point along. It *copied* it. The green ring can't read a red point, so it does the only thing it knows — it hands out a matching one on every leg. Each colour copies the other's points.

[*Achilles's cup stands where he set it on the table, between the lamp at his elbow and the fire at his back. It throws two shadows across the cloth, one for each light, and they are the very same cup twice over — one bowl, one handle, the one thing laid down in two identical prints, and no reaching for it makes a third.*]

Crab: [*without looking round*] It does. Mind you don't lean over it; the last man who watched it work jammed it for a week. [*He lifts the first two red points down to the end of the table and presses once more, head still turned; the two red points come off clean again, and he lifts the copies down.*]

Tortoise: Now a companion trick, and this one wants no pan — just my two hands. [*She sets a green ring on a taut cord, its bead turned well off the top, and threads a red half-turn onto the cord before it — a red ring with its bead set square at the halfway mark.*] A green ring with a turn on it, and coming up the cord towards it, a red half-turn. Watch what each does to the other as I push the red one through.

[*The green ring sits with its bead a third of the way round; the red half-turn waits*

on the cord before it. She pushes the red half-turn along the cord and through the green ring to the far side. What passes through is unchanged – still a red half-turn, its bead square at the mark. But the green ring it passed through has its bead swung clean over to the mirror side, the same turn the other way about. The traveller is the same; the thing it travelled through is reversed.]



before the push *pushed through*

Achilles: The green ring's turn has flipped to the far side, the mirror of what it was. A half-turn of one colour, pushed through the other, turns the other's turn around.

[*Beyond the glass the fall has come on in earnest, whirling and crossing against the last of the grey daylight; no single flake can be followed for more than a moment before it loses itself in the rest of them. The room has drawn in close and warm about the small work on the table, and Achilles, without looking away from it, holds his free hand out to the fire.*]

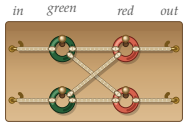
Tortoise: Just so. Now — green copies red's points, and red copies green's; a half-turn of one flips the turn of the other. [*She gathers the loose cords — the little tangle off the pan, the taut cord with its two rings — and lays them at the end of the table.*] Put two green rings and two red together, and let all of that happen at once. What does it all come to then?

Fourth Challenge — *The Cradle Comes Undone*

Achilles: All at once. That sounds like a knot.

Crab: It is a beauty of a knot. [*At the beam: the two green rings on the left, the two red on the right, feed-cords in, draw-cords out. Now he strings the middle.*] Every green tied to every red — the top green to the top red, and across to the bottom red; the bottom green to the top red, and across to the bottom. A full cross between the colours. I call it the **cradle**.

[*He runs the four cords of the cross, one after another: top green to top red; top green down across to bottom red; bottom green up across to top red; bottom green to bottom red. The two feed-cords still run in on the left, the two draw-cords still run out on the right; between them the four crossed cords hang taut. The cradle stands strung on the beam.*]

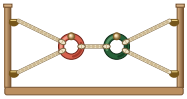


the cradle

Achilles: Four cords in the cross, every green speaking to every red. [*He counts the crossings under his breath, tracing them with a finger.*] It's a proper thicket. What does a thicket like that *do*?

Crab: Ask the pan. [*He sets the cradle's shape onto the koekpan — a copy of it, the strung cradle staying put on the beam — and presses, his eyes on the fire.*]

[*The four-cord cross is copied onto the pan; the pan cooks it down. On the beam the cradle itself does not stir – every cord where it was. But on the second set of hooks comes off something far barer: one knot-pair, doing what the whole cross did. Four crossed cords have become one; the cradle on the beam is untouched.*]



off the pan

Achilles: [*working it over aloud*] The whole cross — all four crossings of it — comes off as one red ring, one cord, one green ring. It reads whether the two cords coming in agree, and writes that same answer onto both going out. [*slowly*] And the cradle does *exactly* that, I'll wager, if I trace it through… yes. Fewer cords for the same job.

Crab: [*lifting the pressed pair off the second hooks*] Fewer cords, same job. That is the whole trade.

[*Out at the window the snow has been at work. Where this morning the frozen canal ran its one way and the road crossed it another — a tangle of edges and banks and a little bridge — the fall has filled it all level and laid one unbroken white sheet across the lot, so that the crossing is a single smooth place now, and yet a body could still walk it corner to corner and fetch up exactly where the old crossing would have taken him.*]

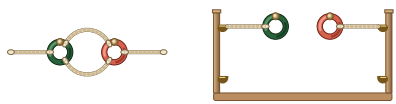
Achilles: [*frowning*] Though your pan is a bit of a liar about the bookkeeping. When I trace the cross through by hand and match it against that little pair, they do the same *work* — but the pair comes out fainter than it ought, by some steady factor the picture just... loses count of. The shapes agree; a number has gone missing.

Tortoise: The number is real and the picture drops it; that is honest to say. The pictures will tell you *that two things are equal* and decline to keep the change. Now — the cross's twin. [*She ties a green ring to a red ring with two separate cords between them, a cord running in to the green and one running out of the red.*] A green and a red, joined not by one cord but by two. Guess.

Achilles: Two cords must bind them twice as tightly.

Tortoise: Press it. [*She sets the double-linked pair on the koekpan; the Crab presses, looking away.*]

[*The green ring and the red ring hang joined by two distinct cords. The pan takes a copy and cooks it. Off the second hooks come the two rings – but with nothing between them at all: the two cords gone, the green loose on its side, the red loose on its own.*]



doubly joined *nothing between*

Achilles: They fell apart. Two cords between a green and a red is the same as *no* cord — the same as never having tied them. [*He shakes his head.*] Twice tied is untied.

[*The steam off the pot on the hob has been wavering all afternoon — leaning toward*

the thread of cold that creeps under the door, then the other way toward the colder air sliding down off the frozen pane. Now, for a moment, the two draughts come on together and meet under the spout, and the steam that each alone bent aside stands up dead straight between them, as still as if the room held no draught at all.]

Tortoise: [*taking the two freed rings down off the second hooks, and the twice-tied pair off the pan, and laying them at the end of the table*] The two facts are one fact seen twice. The cross coming undone, and two cords being as good as none — the same law, that the two colours reorganise each other. It is the engine of everything the pan does.

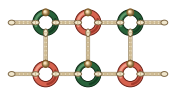
Achilles: [*looking back up at the cradle on the beam*] Fewer cords for the same job. Could that untangle a real gate?

Fifth Challenge — *Three Crossings, One Swap*

Crab: A real gate is where I was going. [*He ties a single green ring to a single red ring, sharing one cord — green up on the top line, red down on the bottom.*] One green, one red, one cord between them: this is a gate you already know. If the top line runs a mark, the bottom flips; if not, not. The autumn's gate, the commonest in all of computing — the one that copies a bit and carries it.

Achilles: I remember it. The controlled flip.

Crab: Now three of them in a row, and I'll alternate which line holds the control. [*He strings two more such gates after the first along the same two lines, as a loose tangle — green-red, then red-green, then green-red — the middle one reversed.*] Top, then bottom, then top. Three crossings' worth of gate.



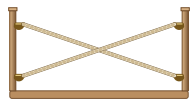
the middle reversed

Achilles: [*quick, tracing it in the air*] Let me run a mark through before

you touch that pan. Mark the top line, bottom bare: the first gate flips the bottom, so both are marked; the middle gate, its control on the bottom now, flips the top bare; the last does nothing. So the mark came in on top and leaves on the bottom — and the other way about, it crosses back. The two lines have traded ends: a swap, plain and exact, and I did it in my head.

Crab: You did. Now watch the pan do it the slow, deep way. [*He copies the three-gate tangle onto the koekpan and presses, eyes shut.*]

[*The three gates hang as a loose tangle across the two lines. The pan takes its copy and cooks it – gathering the first two gates into a cross, undoing the cross to a pair, fusing along the line, letting a doubled link fall away. Off the second hooks comes no tangle at all: just the two lines, each running from its own side to the other – a bare crossing. Three gates have cooked down to one plain cross.*]



what came off

Achilles: A bare crossing. The same swap I traced — the two lines simply changed places.

[*The whirl at the window has settled by now to a steady, even fall, straight down and unhurried, and through it two muffled figures have come out onto the white crossing from opposite banks. They meet in the middle, sidestep one another with the little bob strangers make, and go on — and each fetches up on the bank the other set out from, the two of them swapped end for end without either breaking stride.*]

Tortoise: So which won? Be fair to both. Your head got there faster, and got it *exactly* — no missing number, three gates traced to a swap in a breath. The pan was slower, and dropped its factor again.

Achilles: Then the pan lost.

Tortoise: On length, yes. But it showed you *why*: the swap is no coincidence of three particular gates — it falls straight out of the two colours undoing each other, the cradle law stood on its side. Change the gates for others of

that shape, and the same crossing drops out. The quick way tells you *that*; the slow way tells you *why*.

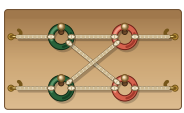
Achilles: [*quietly, looking up at the cradle again*] Three of them, and they just… swap. [*He lifts the tangle and the crossing-copy off, and stands looking at the strung cradle on the beam.*] How many cords went into that cradle, anyway?

Sixth Challenge — *Octet*

[*The short day is almost spent; the window is going from grey toward dark, and the steady fall beyond it has hushed the whole town so that not a sound comes up from the street. The fire has burned down to a low, even bed of red, and the lamp is taking the room, its light lying warm along the beam where the cords hang dead still in the quiet.*]

Tortoise: Count them and see. Nobody has touched it — it has hung there through everything. Every loose thing we tried is down; only the frame is left.

[*The cradle stands alone on the beam now, every demonstration cord taken down: the two feed-cords in from the left, the four cords of the cross, the two draw-cords out to the right, and nothing else strung anywhere.*]



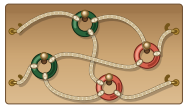
waiting on the beam

Achilles: [*reaching up, touching each cord as he counts*] One… two — the two coming in on the left. [*his hand moves to the cross*] Three, four, five, six — the whole cross of them, every green to every red. [*and to the right*] Seven… eight, going out. [*He lets his hand fall.*] Eight cords. Eight, and not one of them spare — every one carrying its part, and the whole thing meaning nothing if you pull a single one out. [*He is quiet a moment.*] Eight. It's a good number to end on.

Crab: It is the whole ensemble, playing at once. [*He does not press it; he only looks, content, at the thing he has built.*]

Tortoise: And here is the last thing worth seeing. [*She takes hold of the cradle's cords and begins to reshape them — sliding the rings along the beam, bending the cross into new curves, drawing the whole tangle into a shape that looks nothing like it did.*]

[*The cradle as it stood: two greens left, two reds right, a neat cross between. Her hands move the rings along the beam and coax the four crossed cords into long looping bends, until the shape on the beam is a different picture entirely – crooked where it was square, curved where it was straight. But she has untied nothing and joined nothing new: every cord still runs from the ring it left to the ring it reached. Only where the rings sit, and the path each cord takes to get there, has changed.*]

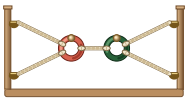


as she left it

[*The snow-hush deepens until the room is wholly still, and the last of the grey light, coming level off the pane, runs the whole length of the long print on the wall — the little town, the chequered squares, the whole strip lit end to end for a moment, so that it seems to lean in a little out of the dusk.*]

Tortoise: [*glancing at the wall, where Escher's* Metamorphosis II *hangs — birds becoming fish becoming a town, the strip flowing on and never once broken*] Like the print there: the shape flows on and on, and the thing stays the same. Only what joins to what ever mattered. [*She sets the reshaped tangle's shape on the koekpan — a copy of it, the tangle itself staying strung on the beam.*] Watch.

[*The Crab presses, his eyes on the fire. Off the second hooks comes the very same knot-pair the neat cradle gave. A shape drawn crooked and a shape drawn square cook down to the one pair. What changed was only the drawing; what held was every connection.*]



from the crooked one

[*Beyond the pane, in the last pale glow the snow throws up of itself, the town has gone all one white: roof into road into frozen canal, the sharp corners rounded off and the crossings smoothed away, until the whole of it is a single unbroken sheet. And yet not a house has stirred — each still stands where it always stood, every lane still runs where it ran, the whole place exactly itself under a coat that has quite changed the look of it.*]

Achilles: The same little pair, out of that mare's nest. It didn't matter how you drew it.

Crab: [*lifting the pressed pair down; the day at the window has gone from grey to dark*] Not a whit. [*He looks along the eight cords of the cradle, then at the two of them.*] Well. We've tied these, and pressed them, and you've counted them at last — a whole short afternoon of it, and the fire nearly down. And do you know what strikes me? Not once, in all of it, did any of us *look*. We never once watched a cord to see which way it truly runs — we only ever pressed a copy and read that.

Tortoise: [*gathering the last of the loose cord as the light fails*] No. Tomorrow we look.

[*Outside, the snow has stopped. The town lies under it all one unbroken white, and the window is dark now but for that faint pallor thrown back off the ground. Inside, the lamp burns steady and the fire has gone to a warm red bed that no one troubles to feed; the koekpan cools on the table where it did its work, only an old iron pan again. On the beam the cradle hangs exactly as it was tied and counted through the whole short afternoon, every cord still and dark against the last light. Along the wall the long print has gone quiet, a little lamplight caught in it, the town and the squares receding into the dim; and nobody in the warm room reaches to disturb anything.*]

Chapter 8: Spiders, Red and Green

— Octet for Spiders —

Part III: Pictures Become Proofs

Escher: "Metamorphosis II" — one form flowing into another, birds into fish into a town, while the strip stays one continuous thing

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Can a proof be a drawing — can we settle a fact about real quantum gates by sliding coloured dots along a few wires, never once going back to the algebra to check?

For seven chapters the pictures have travelled beside the algebra as a faithful companion, and never once out in front. Every time a wire was pulled straight or a box slid round a bend, the move was *licensed*: some theorem on the blackboard signed for it, and the picture cashed the cheque. The sketchbook could show you a thing beautifully, but it could not, on its own, *establish* one. At the very end of the last chapter that balance began to tip. We had drawn a single new mark — a dot where three wires meet — and found that it already knew things: that any one of its legs could be predicted from the other two, a fact we needed six sealed numbers to contradict. A dot that knows things is the beginning of a language.

This chapter builds that language and then turns it loose. The dot from Chapter 7 was one fixed thing, always the state $|000\rangle + |111\rangle$, unlabelled and alone. We now give it a family. It gets a **colour** — and there will be a second colour, which counts the world in a different way. It gets a **turn**, the sphere-rotation of Chapter 3, marked on the dot itself. It gets any number of legs. And once there are two colours of dot with turns and legs, a handful

of rules for pushing them past one another appears — rules that never mention a matrix. With those rules a diagram can be *rewritten* into another diagram of equal value, the way $3 + 4$ is rewritten into 7, entirely inside the pictures.

At the end we will take three of the commonest gates in all of computing, tangle them together, and cook the tangle down to a single crossing — a proof, done with coloured dots, that three of one gate make another. And then we will count the wires in the picture we drew, and something will fall quietly into place.

The calculus we are about to draw has a name and an author, given plainly at the chapter's end. For now, only the colours.

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§8.1 Colour Comes to the Dot

The three-legged dot of Chapter 7 was defined by a single habit: on the legs it read $|000\rangle + |111\rangle$ — all-zeros or all-ones, agreeing across every leg, nothing in between. That habit does not care how many legs there are, nor which are inputs and which outputs. Let us free it.

On the blackboard. For any number of input legs m and output legs k, and any angle α, define the **Z spider**

$$Z(m,k,\alpha) \;=\; |0\cdots0\rangle\langle 0\cdots 0| \;+\; e^{i\alpha}\,|1\cdots1\rangle\langle 1\cdots 1|\,,$$

with k zeros (or ones) in the ket and m in the bra. In words: it passes the all-agree inputs to the all-agree outputs, and it stamps the all-ones branch with the phase $e^{i\alpha}$ — the very turn of the sphere we met in Chapter 3, now written on the dot. The angle α is the spider's **phase**. Chapter 7's node is exactly $Z(0,3,0)$: no inputs, three outputs, no turn — the state $|000\rangle + |111\rangle$. And the plainest spider of all, $Z(1,1,0) = |0\rangle\langle 0| + |1\rangle\langle 1|$, is the identity matrix: one leg in, one leg out, no turn, a bare wire.

In the sketchbook. We draw the Z spider as a **green dot** with its legs

fanning out — m wires arriving on the left, k leaving on the right — and its phase written beside it. A dot with no number written is understood to carry phase 0.



Everything that made the Chapter 7 dot useful survives: it is symmetric in its legs, and it does not distinguish input from output — a leg can be bent from one side to the other with the cups and caps of Chapter 6, and the dot does not mind.

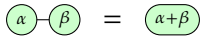
Two green dots joined by a wire are asking the same all-agree question twice in a row, which is asking it once. That is the first rule.

> **Theorem 8.1 (fusion).** Two Z spiders that share at least one leg are a single Z spider, and their phases add. On the blackboard, for spiders that meet along one wire,
>
> $$Z(1,1,\beta)\,Z(m,1,\alpha) \;=\; Z(m,1,\alpha+\beta),$$
>
> and the same for any number of shared and free legs. The equality is **exact** — no scalar.

Proof idea. The all-zeros branch of the first spider feeds only the all-zeros branch of the second, and the all-ones feeds only the all-ones; the two phases sit on the same all-ones branch and so multiply, $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$.

Proof (blackboard). Compose $(|0\rangle\langle 0| + e^{i\beta}|1\rangle\langle 1|)(|0\rangle\langle 0\cdots 0| + e^{i\alpha}|1\rangle\langle 1\cdots 1|)$. The cross terms vanish because $\langle 0|1\rangle = \langle 1|0\rangle = 0$; what survives is $|0\rangle\langle 0\cdots 0| + e^{i(\alpha+\beta)}|1\rangle\langle 1\cdots 1| = Z(m,1,\alpha+\beta)$. ■

In the picture the rule is even plainer: two green dots slide together into one, and you add the numbers written beside them.



See `Code/Ch8/Fusion.py`, which checks fusion for random phases $\alpha, \beta \neq 0$ and confirms $Z(1,1,\pi) = \mathrm{diag}(1,-1)$.

Why this definition. We could have kept the dot rigid, one fixed state. But a dot that carries a phase and any number of legs turns a *fact about one entangled state* into a *piece of grammar*: fusion holds for every m, k, α at once. One rule now does what a page of tensor bookkeeping did before.

Coecke's Sketchbook.

The calculus, and whose it is. From here the pictures stop being a companion to the algebra and become a calculus in their own right — one that can be computed with, not merely read. It is the work of Bob Coecke and his colleagues at Oxford, built on the categorical reading of quantum processes that he and Samson Abramsky began; the green–red diagrams themselves are due to Coecke and Ross Duncan. Its governing slogan, which we will earn in §8.5, is startling the first time: *only the connectivity matters*. Slide a dot along its wire, bend a leg from one side to the other, redraw the whole tangle — so long as the same dots stay joined to the same dots, the value does not change. A picture, in this calculus, is not a diagram *of* a computation. It *is* the computation, and its shape is negotiable.

---

§8.2 One Wire, Counted Twice

The green dot reads a wire in one particular way: it asks whether the legs all say 0 or all say 1. That is a choice of basis — the $|0\rangle, |1\rangle$ basis. But Chapter 6 taught us there is nothing sacred about it; the state $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$ makes an equally good pair of alternatives. A dot could just as well count in *that* basis. This is the second colour.

On the blackboard. Define the **X spider** intrinsically — not by borrowing the green one, but in its own basis:

$$X(m,k,\beta) \;=\; |+\cdots+\rangle\langle+\cdots+| \;+\; e^{i\beta}\,|-\cdots-\rangle\langle-\cdots-|,$$
$$|\pm\rangle \;=\; \tfrac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle).$$

It is the same shape of definition as the green spider, with $|\pm\rangle$ everywhere the green one had $|0\rangle, |1\rangle$. It passes the all-plus inputs to all-plus outputs and the all-minus to all-minus, stamping the minus branch with $e^{i\beta}$. Because $|\pm\rangle$ are unit vectors, so is each term; we build the red spider this way, from normalized $|\pm\rangle$, precisely so that the colour-change rule below is a *theorem* we prove, and not a definition we have quietly assumed.

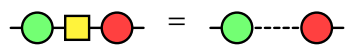
Two small facts pin the red dot to earth. Its single-leg, phase-free state is $X(0,1,0) = |+\rangle + |-\rangle = \sqrt{2}\,|0\rangle$ — an unnormalized $|0\rangle$, carrying the same $\sqrt{2}$ our cups have carried since Chapter 6. And $X(1,1,\pi) = |+\rangle\langle+| - |-\rangle\langle-|$ is the bit-flip $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, the NOT gate.

In the sketchbook. We draw the X spider as a **red dot**, otherwise exactly like the green one: any number of legs, a phase written beside it, symmetric in its legs, indifferent to which side a leg leaves on.

To pass between the two colours we need one more box, the one gate that turns the $|0\rangle, |1\rangle$ basis into the $|\pm\rangle$ basis and back: the Hadamard $H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ of Chapter 3, with $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$, and $H^2 = I$. In this calculus it gets its own mark: a small **yellow box** on a wire.



A Hadamard on a wire is so common between two dots that the calculus abbreviates it: a **dashed wire** — the *Hadamard-edge* — means exactly "an ordinary wire with a yellow box on it". Drawn between two dots, the two are the same picture:



a shorthand later chapters will lean on when tangles grow crowded. Now the bridge between the colours.

An example that forces the rule

Take the plainest green dot with a turn, $Z(1,1,\alpha) = |0\rangle\langle 0| + e^{i\alpha}|1\rangle\langle 1|$, and put a Hadamard on each of its two legs. On the blackboard $H\,Z(1,1,\alpha)\,H$

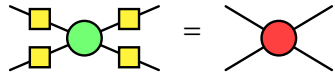
sends $|+\rangle \mapsto |0\rangle \mapsto |0\rangle \mapsto |+\rangle$ and $|-\rangle \mapsto |1\rangle \mapsto e^{i\alpha}|1\rangle \mapsto e^{i\alpha}|-\rangle$; that is $|+\rangle\langle+| + e^{i\alpha}|-\rangle\langle-|$, which is the *red* spider $X(1,1,\alpha)$ — exactly, for every α. The colour did not change by fiat; it changed because dressing the green counter in Hadamards makes it count in the $|\pm\rangle$ basis, which is what red *means*.

> **Theorem 8.2 (colour change).** A Hadamard on every leg of a Z spider turns it into an X spider of the same phase (and the reverse, since $H^2 = I$):
>
> $$H^{\otimes k}\, Z(m,k,\alpha)\, H^{\otimes m} \;=\; X(m,k,\alpha),$$
>
> **exactly**, for every m, k, α.

Proof idea. H carries $|0\rangle, |1\rangle$ to $|+\rangle, |-\rangle$; conjugating the green spider by H on all legs rewrites its all-agree action in the $|\pm\rangle$ basis, which is the red spider by definition.

Proof (blackboard). $H^{\otimes k}|0\cdots0\rangle = |+\cdots+\rangle$ and $H^{\otimes k}|1\cdots1\rangle = |-\cdots-\rangle$, and likewise for the bras with $H^{\otimes m}$. Substituting into $Z(m,k,\alpha)$ turns every $|0\cdots0\rangle$ into $|+\cdots+\rangle$ and every $|1\cdots1\rangle$ into $|-\cdots-\rangle$, giving $X(m,k,\alpha)$ term for term. The phase rides untouched on the second branch. ■ See `Code/Ch8/ColourChange.py`, which builds X intrinsically from $|\pm\rangle$ and checks the equality for $\alpha \in \{0, 0.7, 1.3, 2.9\}$.

In the picture: a green dot with a yellow box on each leg is a red dot.



Why this definition. Building red from its own $|\pm\rangle$ costs us one honest theorem — colour change — instead of getting it free. That is the point. Had we *defined* red as "green wrapped in Hadamards", the rule would be a tautology and would prove nothing about the world. Defined intrinsically, colour change is a real fact: the two counters see the same wire two genuinely different ways.

Dirac's Blackboard.

Spiders are matrices. Every dot is an honest linear map, and it is worth writing four of them down. The one-in-one-out green dots are diagonal: $Z(1,1,0) = \left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right) = I$, and $Z(1,1,\pi) = \left(\begin{smallmatrix}1&0\\0&-1\end{smallmatrix}\right)$, the phase-flip. The red ones are their Hadamard conjugates: $X(1,1,0) = I$ again, and $X(1,1,\pi) = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, the bit-flip. The legs change the shape: $Z(1,2,0) = |00\rangle\langle 0| + |11\rangle\langle 1|$ is a 4×2 matrix that copies a basis bit, and $Z(0,1,0) = |0\rangle + |1\rangle$ is the column $(1,1)^T$. Reading a spider off as a matrix is always this mechanical: put a 1 (or $e^{i\alpha}$) where all legs agree, a 0 everywhere else. The pictures never lie about the numbers; they only decline to write them all out.

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§8.3 Each Colour Copies the Other

We have two counters that read the same wire in incompatible bases. What happens where a red input meets a green dot — where the two ways of counting collide?

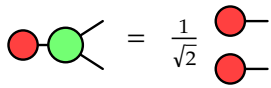
On the blackboard. A green copier is $Z(1,2,0) = |00\rangle\langle 0| + |11\rangle\langle 1|$. On its own green basis points it does what its name says: $Z(1,2,0)\,|0\rangle = |00\rangle$ and $Z(1,2,0)\,|1\rangle = |11\rangle$ — one point in, two identical points out. Now feed it a *red* point. The red one-leg states are the ones we met in §8.2: $|r_0\rangle = X(0,1,0) = \sqrt{2}\,|0\rangle$ and $|r_1\rangle = X(0,1,\pi) = \sqrt{2}\,|1\rangle$, each carrying the cup's $\sqrt{2}$. Watch the green copier act on $|r_0\rangle$:

$$\begin{aligned} Z(1,2,0)\,|r_0\rangle &= \sqrt{2}\,Z(1,2,0)\,|0\rangle = \sqrt{2}\,|00\rangle \\ &= \tfrac{1}{\sqrt{2}}\big(\sqrt{2}\,|0\rangle\big) \otimes \big(\sqrt{2}\,|0\rangle\big) = \tfrac{1}{\sqrt{2}}\,|r_0\rangle \otimes |r_0\rangle\,. \end{aligned}$$

So green *does* copy a red point — into two red points — but with a factor $\frac{1}{\sqrt{2}}$ out front. The honest statement is that a phase-free green spider copies a red point **up to the scalar** $\sqrt{2}$: $Z(1,2,0)\,|r\rangle = \frac{1}{\sqrt{2}}\,|r\rangle \otimes |r\rangle$ for the red points

$|r\rangle$, and symmetrically a phase-free red spider copies green points. Each colour is copied cleanly by the other.

In the sketchbook. A red point (a red dot with one leg) fed into a phase-free green dot comes out as two red points. This is the picture Achilles did not expect: one cord in, two cords out.



The factor $\frac{1}{\sqrt{2}}$ is drawn because from this chapter on we allow ourselves to state a picture equality **up to a non-zero scalar**, writing the scalar in only when it matters. Copy differs from a clean duplication by a factor of $\sqrt{2}$; we say so and move on.

A companion fact concerns a half-turn — a phase of π — pushed through the other colour.

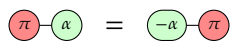
> **Theorem 8.3 (copy).** A phase-free spider of one colour copies the basis points of the other: $Z(1,2,0)\,|r\rangle = \frac{1}{\sqrt{2}}\,|rr\rangle$ for red points $|r\rangle$, and symmetrically. (*Up to* $\sqrt{2}$.)
>
> **Theorem 8.4 (π-commutation).** A half-turn of one colour passes through the other colour and reverses the other's turn:
>
> $$X(1,1,\pi)\,Z(1,1,\alpha) \;=\; Z(1,1,-\alpha)\,X(1,1,\pi),$$
>
> up to a non-zero scalar (and exactly when $\alpha = 0$).

Proof idea (copy). The green copier duplicates whatever agrees in *its* basis; a red point, seen from the green side, is an even superposition that the copier spreads across the diagonal $|00\rangle + |11\rangle$ — which is two red points, once the $\sqrt{2}$ is accounted.

Proof idea (π-commutation). The red half-turn is the bit-flip $X(1,1,\pi)$; conjugating a green phase gate by it sends $|1\rangle \mapsto |0\rangle$ and back, turning $e^{i\alpha}$ on the $|1\rangle$ branch into $e^{i\alpha}$ on the $|0\rangle$ branch — a global phase times $e^{-i\alpha}$ on $|1\rangle$,

which is $Z(1,1,-\alpha)$ up to that phase.

In the picture, π-commutation is a red π-dot sliding through a green dot and flipping the green dot's turn:



(up to a scalar).

See `Code/Ch8/Copy.py`, which checks copy up to $\sqrt{2}$ and π-commutation for $\alpha \in \{0.7, 1.3\}$ — not merely $\alpha = 0$, where it would hold trivially.

Why this definition. We insist on *phase-free* copiers because a spider with a turn does not copy cleanly: the turn would have to be duplicated too, and $e^{i\alpha}$ does not split into two honest points. The rule is exactly as strong as it is, and no stronger — which is why the scalar and the phase-free clause are both written down.

---

§8.4 Cradle That Comes Undone

Copy says green duplicates red points and red duplicates green points. Put two of each together and the two habits collide in the most productive way in the whole calculus.

The example: a tangle that ought to simplify

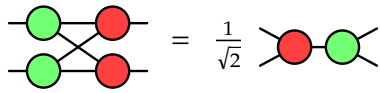
Take two green dots feeding two red dots, with **every** green joined to **every** red — a complete little cross, four cords, the shape a child strings between two pairs of nails. Two wires come in on the greens; two leave from the reds. On the blackboard this is $[X(2,1) \otimes X(2,1)] \circ (\text{cross}) \circ [Z(1,2) \otimes Z(1,2)]$, a 4×4 matrix. Multiply it out (the calculation is short but joyless) and it equals

$$\tfrac{1}{2}\big(|00\rangle\,(\langle 00| + \langle 11|) + |11\rangle\,(\langle 01| + \langle 10|)\big),$$

which reads the parity of its two inputs and writes that parity onto *both*

outputs. Now here is the surprise: the very same map, up to a scalar, is made by a single green dot tied to a single red dot — no cross at all.

> **Theorem 8.5 (bialgebra).** Two Z spiders joined to two X spiders by a complete cross of four wires equal one Z spider joined to one X spider by a single wire — the inputs gathered on the red, the outputs issuing from the green:
>
> $$[X(2,1) \otimes X(2,1)] \,(\text{cross})\, [Z(1,2) \otimes Z(1,2)] \;=\; Z(1,2)\, X(2,1),$$
>
> **up to the scalar** $\sqrt{2}$.



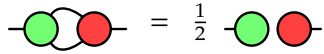
Proof idea. Copy is the engine. Push each green copier through each red adder: the green wants to duplicate, the red wants to combine, and when a whole cross of them does this at once, the duplications and combinations rearrange into a single duplicate-then-combine. The cross is the copying written twice over.

Proof (*blackboard*). This is the multiplication above: both sides send $|xy\rangle$ to $|pp\rangle$ with $p = x \oplus y$ the parity, and the scalars are $\frac{1}{2}$ on the left and $\frac{1}{\sqrt{2}}$ on the right, a ratio of $\sqrt{2}$. Every "in general" here is a finite check on the four inputs $|00\rangle, |01\rangle, |10\rangle, |11\rangle$; see `Code/Ch8/Complementarity.py`. ■

The bialgebra has a twin, about *two* cords rather than four. If a green dot and a red dot are joined by two separate wires, they might as well be joined by none.

> **Theorem 8.6 (Hopf law).** A green spider and an X spider linked by two distinct wires equal the two spiders disconnected — a green point beside a red point:
>
> $$X(2,1)\, Z(1,2) \;=\; (|r\rangle\,\langle g|),$$
>
> the rank-one disconnected map, **up to the scalar** 2.

Proof idea. Two wires between the colours let each copy the other's point and then immediately re-add it; a copy followed by its own undoing sends everything to a single fixed direction on each side, which is exactly what "disconnected" means. Complementarity of the two bases *is* this law — the two facts, bialgebra and Hopf, are one phenomenon seen twice (Coecke–Duncan, §8).



Proof (*blackboard*). With the intrinsic red spider, $X(2,1)Z(1,2) = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 0 & 0\end{smallmatrix}\right) = |0\rangle\langle +|$ — a rank-one map onto a single direction, hence disconnected. The disconnected picture on the right — a red state $\sqrt{2}\,|0\rangle$ beside a green effect $\langle 0| + \langle 1|$ — is $\sqrt{2}\,|0\rangle\,(\langle 0| + \langle 1|) = 2\,|0\rangle\langle +|$; the two agree up to the scalar 2. See `Code/Ch8/Complementarity.py`, which builds the genuine two-wire link (not one wire drawn twice) and derives the disconnection with the ZX rules. ■

Why this definition. These two are the heart of the chapter, and the reason the pictures can now outrun the algebra. Fusion and colour change only tidy dots of one colour at a time; the bialgebra and Hopf laws are the first rules that let the two colours *reorganise each other*, turning a dense tangle into a sparse one. Every reduction to come is these two, applied and reapplied.

---

§8.5 Keeping Only the Topology

We have been redrawing diagrams freely — sliding dots, bending legs, stretching wires — and asserting that the value is unchanged. It is time to say plainly what that freedom is, and what backs it.

The claim. In these diagrams *only the connectivity matters*: which dot is joined to which, and by how many wires. The lengths of the wires, the positions of the dots on the page, the order in which two wires happen to cross with no dot at the crossing — none of it changes the map the picture

denotes. Two diagrams that are the same graph, with the same dots joined the same way, are equal.

This is the slogan the Escher woodcut on the scullery wall makes visible: in *Metamorphosis II* the birds become fish become a town, the shape flowing on and on, and yet the strip is one continuous thing throughout — what changes is never the connection, only the drawing.

We **state** this and do not prove it here; it is a coherence theorem, and the honest citation is Peter Selinger's survey. Because our diagrams have crossings — swaps, from Chapter 4 — the backing is coherence for *symmetric monoidal* categories (Selinger, Theorem 3.12, resting on Joyal and Street): an equation holds in the graphical language exactly when the two diagrams have the same connectivity. It is **not** the planar coherence of Selinger's Theorem 3.1, which he flags in Caveat 3.2 as proved only for deformations that never turn a wire back on itself — too weak once wires cross. And it is worth keeping one distinction sharp: the extra freedom to *permute a spider's legs at will* is a separate fact, the spider theorem of Coecke and Duncan (§7), not part of the coherence result, whose boxes keep their legs in order. So the slogan is "only the connectivity matters" — never "bend anything however you like."

How far the calculus reaches. One naturally asks what these two colours of dot can express.

> **Theorem 8.7 (universality and completeness; stated, not proved).** Every pure quantum state, every unitary gate, and every post-selected projective measurement on any number of qubits can be written as a diagram of green and red spiders and Hadamard boxes (Coecke–Duncan). And within the *stabilizer* fragment — the part of quantum theory built from Hadamards, phase gates and CNOTs — the calculus is **complete**: any two diagrams denoting the same map can be turned into one another by the rules alone, with no return to matrices (Backens).

We will not prove these; the citations discharge them. But they are the licence for the chapter's ambition. Universality says the pictures miss nothing; completeness says that, in a large and important fragment, the rules miss nothing either — every true equation of matrices is reachable by pushing

dots. What we do next is one small, concrete instance of that promise.

Coecke's Sketchbook.

What "only topology" does not say. The slogan is intoxicating and easy to overstate, so a pencil-user's caution. It says that a rewrite which preserves the connections preserves the *map* — an equality of processes, no more. It does not say the shorter diagram is faster to run, cheaper to build, or in any way optimal; a picture with three dots and one with thirty can denote the very same gate. Nor does redrawing prove anything about security, or about how a physical device behaves when watched. When we cook a tangle down to a crossing in the next section, we will have shown that two processes are *equal* — and we will be careful to claim exactly that, and not a syllable more.

---

§8.6 Every Rule on One Page — The Rosetta Table

Before the table, one last assembly to name, because we are about to lean on it. Take a green dot on a top wire and a red dot on a bottom wire, and join them by a single vertical cord — the green with legs going in and out on the top, the red with legs in and out on the bottom.



On the blackboard this green–red pair is the CNOT gate of Chapter 4 — control on the green (top) wire, target on the red (bottom) — divided by $\sqrt{2}$: the bare picture is CNOT$/\sqrt{2}$, the missing factor being the cup convention we have carried all along. The green copies the control bit; the red adds it into the target. We verify this in the next section; here it earns its Rosetta row.

The rows this chapter adds to the running dictionary, numbered on from Chapter 7's fifty-two. This is the largest single instalment in the book, and deliberately so: the whole ZX vocabulary — four new picture elements,

six named rules, the topology principle, and the spider-CNOT — arrives together, because it is a language and a language must be handed over whole.

#	Algebra (Dirac)	Picture (Coecke)
53	$Z(m,k,\alpha) = \lvert 0\cdots 0\rangle\langle 0\cdots 0\rvert + e^{i\alpha}\lvert 1\cdots 1\rangle\langle 1\cdots 1\rvert$	a green dot, any legs, phase α — *new element*
54	$X(m,k,\beta) = \lvert +\cdots +\rangle\langle +\cdots +\rvert + e^{i\beta}\lvert -\cdots -\rangle\langle -\cdots -\rvert$ (normalized $\lvert\pm\rangle$)	a red dot, any legs, phase β — *new element*
55	the phase α (Chapter 3's sphere-turn)	a turn written beside a dot — *new element*
56	$H = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$; $HZ(\alpha)H = X(\alpha)$	a yellow box; a dashed wire (Hadamard-edge) — *new element*
57	same-colour spiders on a shared leg merge; phases add (exact)	two like dots slide together into one — **fusion**
58	H on every leg flips $Z \leftrightarrow X$ (exact)	a dot dressed in Hadamards changes colour — **colour change**
59	a phase-free spider copies the other colour's points (up to $\sqrt{2}$)	a point of one colour into the other comes out doubled — **copy**
60	$X(\pi)\,Z(\alpha) = Z(-\alpha)\,X(\pi)$ (up to a scalar)	a half-turn pushed through the other colour flips the turn — **π-commutation**

#	Algebra (Dirac)	Picture (Coecke)
61	$K_{2,2}$ of two greens into two reds = one green–red pair (up to $\sqrt{2}$)	the cross comes undone to a single knot-pair — **bialgebra**
62	green–(two wires)–red = green⊗red disconnected (up to 2)	two wires between the colours = none — **Hopf**
63	value depends only on which wire joins which	slide, bend, stretch freely — only the connectivity counts
64	green$(1\to2)$ control, red$(2\to1)$ target, one shared wire; bare = CNOT$/\sqrt{2}$	a green dot and a red dot joined by one cord — the CNOT

From Chapter 8 onward, picture equalities hold **up to a non-zero scalar** unless the scalar is written in. Fusion (57) and colour change (58) are exact; copy carries $\sqrt{2}$, bialgebra $\sqrt{2}$, Hopf 2.

---

§8.7 The Theorem, Twice

Here is the chapter's headline, chosen because it is small enough to prove in full and large enough to matter: it is the identity every student of quantum computing meets, that a swap can be built from nothing but CNOTs.

> **Theorem 8.8 (three CNOTs make a swap).** Let CNOT have its control on the top wire and target on the bottom, and CNOT′

the reverse — control bottom, target top. Then

$$\mathrm{CNOT} \cdot \mathrm{CNOT}' \cdot \mathrm{CNOT} \;=\; \mathrm{SWAP},$$

which exchanges the two wires. On the blackboard this is **exact**. In the pictures the three spider-CNOTs cook down to a bare crossing **up to the scalar** $\frac{1}{2\sqrt{2}}$.

Dirac's Blackboard

Write the three gates as 4×4 permutation matrices in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. The CNOT with control on top flips the bottom bit when the top is 1; its reverse flips the top bit when the bottom is 1:

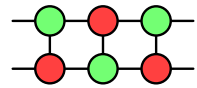
$$\mathrm{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad \mathrm{CNOT}' = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

$$\mathrm{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Follow a basis vector through, right to left. Start with $|10\rangle$. The first CNOT (control top = 1) flips the bottom: $|11\rangle$. Then CNOT' (control bottom = 1) flips the top: $|01\rangle$. Then CNOT (control top = 0) does nothing: $|01\rangle$. So $|10\rangle \mapsto |01\rangle$. The same march sends $|01\rangle \mapsto |10\rangle$, and fixes $|00\rangle$ and $|11\rangle$. That is the swap, exactly — three sparse matrices multiplied in a line, no scalar to explain. See `Code/Ch8/SwapFromCNOTs.py`.

Coecke's Sketchbook

Each CNOT is a green–red pair (row 64). Draw the three in a row: on the top wire green, red, green; on the bottom wire red, green, red; three vertical cords joining them.



Now cook it down, each move named. Call the top-wire dots, left to right, g_1 (green), r_2 (red), g_3 (green), and the bottom-wire dots r_1 (red), g_2 (green), r_3 (red); the three vertical cords are g_1r_1, r_2g_2 and g_3r_3.

Step 1 — only connectivity (§8.5). Look at the four dots of the first two CNOTs — the greens g_1, g_2 and the reds r_1, r_2. Each green is joined to each red: g_1 to r_1 (its cord) and to r_2 (along the top wire); g_2 to r_2 (its cord) and to r_1 (along the bottom wire). That is a complete cross between the colours — a bialgebra cradle standing on its side. Redraw it as one without moving an endpoint.

Step 2 — bialgebra (Theorem 8.5). Collapse that cross to a single green–red pair. The rule sends the two greens' external legs onto the new red and the two reds' external legs onto the new green — not the other way about. Outside the cradle, the greens g_1, g_2 carry the top boundary input (on g_1) and the wire running on toward the third CNOT's target r_3 (on g_2); the reds r_1, r_2 carry the bottom boundary input (on r_1) and the wire running on toward the third CNOT's control g_3 (on r_2). So a **new red** gathers the top boundary input together with the wire toward r_3, a **new green** gathers the bottom boundary input together with the wire toward g_3, and a single cord joins the new red to the new green.

Step 3 — fusion (Theorem 8.1), on the green. The new green carries a leg running toward g_3, the third CNOT's green; two greens on a shared wire, so they fuse into a single green dot (phases add, both 0). This green is now joined to the third CNOT's red r_3 by one wire — the surviving cord g_3r_3 — and to the new red by another; but r_3 and the new red are still two *different* red dots, so there is as yet no double edge.

Step 4 — fusion (Theorem 8.1), on the red. The new red carries a leg running toward r_3; two reds on a shared wire, so they fuse into a single red dot (phases add, both 0). Now a single green and a single red remain, and they are joined by *two* separate wires — the cord left by the bialgebra pair and the third CNOT's own cord g_3r_3.

Step 5 — Hopf (Theorem 8.6). Two wires between a green and a red are the same as none. The link falls away, leaving only phase-free spiders with one wire in and one out.

Step 6 — a phase-free $1\to1$ dot is a bare wire ($Z(1,1,0)=I$, §8.1). Erase those dots and follow the surviving wires from boundary to boundary: the top input emerges at the bottom and the bottom input at the top — a bare crossing.

 = SWAP.

The bare crossing is SWAP exactly; the scalar rides on the reduction that brought us here. The three spider-CNOTs cook down to this crossing up to $(\frac{1}{\sqrt{2}})^3 = \frac{1}{2\sqrt{2}}$, one factor of $\frac{1}{\sqrt{2}}$ for each spider-CNOT (row 64). See `Code/Ch8/SwapFromCNOTs.py`, which assembles the three spider-CNOTs, reduces the diagram by the rules, and confirms it equals the bare swap up to exactly $\frac{1}{2\sqrt{2}}$.

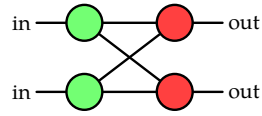
The verdict

Be honest: for this theorem the blackboard wins. Three sparse matrices multiply to the swap in a single line, exactly, with no scalar to apologise for — shorter and cleaner than the six-step rewrite. What the picture buys is not brevity but *understanding*. It shows that the swap is not a coincidence of three permutations but a consequence of complementarity: the swap appears the instant the two colours are made to reorganise each other by the bialgebra law. And it shows the result is robust — the same cradle collapses whatever gates of this shape are strung, which a specific matrix product cannot suggest. The algebra tells you *that*; the picture tells you *why*, and carries a $\frac{1}{2\sqrt{2}}$ for the privilege.

---

§8.8 The Octet

Here is the master diagram of the chapter. Count its cords first: two coming in on the left, four crossing in the middle, two going out on the right — **eight**.



Two green dots on the left, two red on the right, and between them a complete cross — every green tied to every red. It is the bialgebra cradle of §8.4 drawn at full size, and it is called **The Octet**.

> **Theorem 8.9 (the Octet).** The Octet is a genuine eight-wire diagram — two feed cords, four cradle cords, two draw cords — and it evaluates to the parity map
>
> $$\tfrac{1}{2}\big(|00\rangle\,(\langle 00| + \langle 11|) + |11\rangle\,(\langle 01| + \langle 10|)\big),$$
>
> which reads the parity of its two inputs and writes it onto both outputs. Up to the scalar $\sqrt{2}$, this is a single green–red pair — the bialgebra rule of §8.4, drawn once in the large.

Proof. The eight cords are enumerated above. The companion file `Code/Ch8/Octet.py` builds the diagram and tallies its edges to exactly eight; it then derives the collapse to the single pair by the named bialgebra rule and confirms the parity map up to $\sqrt{2}$. The evaluation is the blackboard multiplication of Theorem 8.5. ■

And the Tortoise, reshaping the cradle into a different-looking tangle that presses down to the very same pair, glances at the Escher and says it: the shape flows on, but the thing stays the same. Only the connectivity matters.

> **Coecke's Sketchbook.**
>
> *A confession about the numbering.* The reader who counts the cords of the Octet finds eight — and the chapter is the eighth. That is not a coincidence, and it has not been one since the

beginning. Chapter 1 turned on a single wire; Chapter 2 on two; the six music boxes of Chapter 6 were six wires bent into cups and caps; the trio of Chapter 7 met at a node whose legs, with the tally, came to seven. Every chapter of this book has been built on its own number of wires, and its number *is* that count. We hinted once, in a footnote in Chapter 6, and said we would not explain. This is the explanation. If you wish to see where it starts, turn back to the very beginning — before the first chapter, to the Prelude, a page with no wires drawn on it at all: the empty diagram, which counts zero. We will not mention the numbering again; from here you can carry it yourself.

Feynman's Notebook.

Whose calculus. The diagrammatic reading of quantum processes — the bent wire as a Bell state, yanking, the whole apparatus of a †-compact category — is due to Samson Abramsky and Bob Coecke, *A categorical semantics of quantum protocols*, presented at the 19th IEEE Symposium on Logic in Computer Science (LiCS'04) in 2004 (arXiv:quant-ph/0402130). The two colours of dot — the ZX-calculus proper, with fusion, colour change and the bialgebra — are Coecke and Ross Duncan, *Interacting quantum observables*, first given at ICALP 2008 and published in full in 2009 (arXiv:0906.4725). Completeness for the stabilizer fragment was proved by Miriam Backens in 2013 (arXiv:1307.7025). All of this rests on, but is distinct from, Roger Penrose's 1971 diagrammatic notation for tensors, which drew the wires but did not paint them.

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§8.9 Now the Pictures Can Prove

Look back at what changed. For seven chapters a diagram was a way of *displaying* a truth the algebra had established. In this one the diagram became a way of *establishing* one. We gave the Chapter 7 dot a colour, a

second colour, a turn and a family of legs; we found the rules by which the two colours push through and reorganise one another — fusion, colour change, copy, the bialgebra and its twin the Hopf law; and with those rules alone we cooked three CNOTs down to a crossing, never once returning to a matrix to check. The calculus is Coecke's; the promise it makes — universality, and completeness on the stabilizer fragment — is that this was no lucky example.

Everything we did, though, we did with the lights off. The koekpan pressed its cradles by feel; the cords were tied and crossed and counted, but never once *watched* to see which way a cord truly ran. We drew states and effects and post-selected caps, and each time quietly kept only the outcome we wanted. That cannot last. A real device does not let you keep only the answer you like; it forces one of several on you, at random, and destroys the rest — and the act of looking is itself a thing that must be drawn. To draw it we will need a new kind of wire, and a partner in the corner who is neither asleep nor awake until someone looks. Tomorrow we look.

Lullaby for a Cat

猫の目や
寝るも覚むるも
冬灯

neko no me ya
neru mo samuru mo
fuyu tomoshi

A cat, one eye open —
asleep and awake at once,
the winter lamp

[*The Crab's parlour, the short winter afternoon already on the turn. Achilles comes in out of the snow, stamping his boots; the Tortoise is settled by the lamp; and in the corner, on the sill, something is curled that is not quite asleep and not quite awake.*]

Achilles: I ran the last of the lane — it's the kind of cold that makes your teeth ring. You said today. Yesterday you said, *tomorrow we look.* Well. It's tomorrow.

Crab: It is, and I have been up since it was dark, nerving myself to it. Come to the fire; there's tea. I have kept a thing in a drawer eleven years that I have never once dared use in company. Today I use it.

Tortoise: He has said *today I use it* three times since I arrived. Each time he shuts the drawer again.

Crab: This time it stays open. Sit, Achilles — thaw first.

[*The fire has been built high against the cold, and it spits and shifts as the fresh wood catches; the snow that came in on Achilles's boots goes dark, then to water, then to nothing on the hearth-stone. Beyond the one tall window the afternoon has gone the colour of pewter and is failing early, as it does in the deep of the year, so that the newly-lit lamp already does the greater part of the work. In the far corner, on the sill, something keeps its own counsel, curled — settling deeper, it may be, or it may be about to stretch.*]

Achilles: [*warming his hands, then nodding at the corner*] And who's your friend on the sill? Settling down for the night, or waking up into it?

Cat: [*without troubling to open more than one eye*] A little of each. More of the one than the other — and the other than that — depending how you ask.

Achilles: That is no sort of answer.

Tortoise: It is the exact size of your question. Ask it something baggy, it hands you something baggy back. Drink your tea. You'll grow used to the manners.

[*Achilles takes the offered cup, folds both hands round it, and lets the fire do its work while the light in the window fails by the minute.*]

First Challenge — *The Look That Forces*

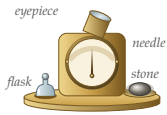
[*The Crab's parlour in the deep of winter, the short afternoon sinking toward an early dusk, and the lamp lit already to meet it. It is a room that shuts and covers: winter shutters folded back in their reveals, waiting to be drawn; a sideboard of lidded jars and stoppered bottles; cloths laid over whatever the cold would spoil; a drawer or two that keep their contents to themselves. The one tall window has not yet decided what it is — half a view still of the white lane and the snow sifting down, and half a ghost of the warm room laid over it, the two neither one thing nor the other; and at its cold lower corners the glass has begun to fog with the breath of the room, a soft grey bloom that has not settled what it will be. On the scullery wall, where the lamplight reaches only at its edge, hangs a print the Crab has kept these many years — Escher's* Eye, *a single vast eye drawn so close that every lash*

stands up like a rafter, and far down in the dark of its pupil something small and steady looking back out. It was on that wall before any of them came in from the cold, and it will be there after they have gone.]

Achilles: Then let me have it. All this while I have handled things but never once *looked* at one while it was making up its mind. Today I look. What do I look at?

Crab: [*bringing a shallow drawer to the table*] These. Little stones, and they come in pairs — each pair cut of one piece, so the two are never quite two apart. A restless stone hasn't settled what it is; it is both its notes at once, the way your friend on the sill is neither asleep nor awake. [*He sets one small stone apart.*] But a look spends a stone. Once you have looked, it is decided, and it stays decided — so each showing wants a fresh pair, and I have a drawerful.

Crab: And this is how you look. [*He sets a squat brass instrument on the table: an eyepiece at the top, a fine needle standing at the middle of a scale on its face, a stoppered flask sunk in a well at the left of the base and a shallow dish at the right; he lays the set-apart stone in the dish.*] The viewer. From the same bench in Delft as my music boxes — the daughter grinds glass now, as well as cutting reeds. You look *through* it, at the stone, never *at* it: put your eye to its works to catch how it does the trick and the needle jams and it is spoilt. I have never once looked inside, and it costs me. Look through it at a restless stone and it forces one of the two notes on you — which one, nobody can say beforehand — swings the needle to that side, and breathes once into the flask below.



the viewer

Crab: The answer it forces is cut here. [*He stands a slate upright against the viewer's side, where a stylus on a small arm, geared to the needle, rests on its edge.*] One notch an answer — the viewer cuts it itself, at the top edge for high, at the foot for low — and there it stays: a plain record, settled and done, like

the notches in a baker's tally-stick. And every look I have ever taken is in this. [*a fat ledger*] Eleven years of them.

[*The viewer sits square in the lamplight, all brass and ground glass, as finely made as anything from that Delft bench — and, like the best of that bench's work, it does not care to be studied: left alone it keeps up the faintest tick somewhere inside its brass, a small live sound at the very edge of hearing, that falls silent the moment anyone leans in close to catch where it comes from, and takes itself up again when they sit back. The Crab draws a chair up for Achilles and another for himself, and keeps his own gaze at a careful middle distance. Achilles settles in and bends to the eyepiece.*]

Tortoise: Go on, Achilles. Look at it.

Achilles: Like a spyglass, then — it will show me which note the stone has been keeping.

[*The stone lies restless — neither note, both notes, trembling at nothing. Achilles sets his eye to the viewer and looks through it at the stone. The needle, which has hovered at the middle of its scale, jumps — and swings hard over to the high stop, on the right, and holds there; it might as well have gone the other way, and no one in the room could have said beforehand which. The stylus scratches a single notch into the slate's top edge. And the stone in the dish that trembled a moment ago lies quite still now: it has an answer, and it is spent.*]



looked through



cut at the top

Achilles: High. [*He looks again; the needle stays hard against its stop, and the stylus cuts nothing more.*] High again — and again, high. It won't shift.

Crab: It won't. You have spent it. Look till the small hours and it says high; but the both-at-once you looked *through* is gone. You didn't find it high—

Achilles: ...Oh. The look didn't read the stone, it *made* it. The stone had no answer until I looked; my looking is what settled it. [*He tips the spent stone onto a little heap beside the viewer — the needle falling back to the middle as the*

dish empties — and looks at fresh stone after fresh stone, the stylus cutting a notch for each and each spent one but the last going onto the heap.] High. Low. Low. High. Low. A fresh stone each time, and a row of notches with no more pattern in it than a tossed coin.



stone after stone

[*At the window one of the fogged patches, chilled past bearing, springs of a sudden into frost — a whole white fern flung out across the glass in the space of a breath, from no cause that anyone could have named beforehand, and fixed there now for good. A moment since it was undecided grey; it will not be grey again tonight.*]

Tortoise: One look, one answer, and a restless thing is put to rest. That is the whole of it.

Achilles: The whole of it — except for what nags. Suppose I look, the stone decides, and I *don't* read the slate. I have spent the stone; it has its answer; but I don't know it. It isn't the restless both-at-once — that's gone. And it isn't a plain high or low — I never read it. So what is a stone that has been looked at, and never read?

Second Challenge — *A Stone That's Been Looked At*

Tortoise: Then make one, and we'll look at it — the stone, mind, not the slate. Mr Crab, cover the dial and the slate before he looks. He is to spend a stone without learning its answer.

Crab: [*tipping the last spent stone onto the heap and setting a fresh one in the dish, then draping one cloth over the needle's dial, another over the slate and its stylus arm*] Covered. Look through, Achilles — you'll see nothing come of it.

[*The stone lies restless, both at once. Achilles sets his eye to the viewer and looks through. Behind the cloth the needle swings and the stylus cuts a notch — but nobody sees to which side it swung, nobody reads the mark. When he lifts his eye the stone lies still: spent, settled, decided. Only* which way *it decided, no one in the room knows. It is a stone that has been looked at and never read.*]



looked through, unseen



cut, unread

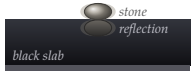
Achilles: So it has an answer, and I can't get at it. It isn't restless any longer — but it isn't a plain high or low to me either. What have I got?

Tortoise: A black glass to show it on, is what we want. Mr Crab — the slab?

Crab: [*drawing a polished black slab into the lamplight*] My scrying-slab; I keep it for its looks. Set anything on it and it throws back a reflection so clean you would swear there were two of the thing. [*wry*] I begin to suspect I shall regret lending it to a demonstration.

[*The window has given up the last of its plain view. Where an hour since there was a lane, there is now a muddle of both worlds at once — the snow-light and the lamp-flame, the frost on the pane and the reflected room, laid one across another until a stranger looking in could not have told which things stood in the parlour and which only seemed to. In the glass the lamp burns beside its own reflected self; and for a moment Achilles, glancing up, cannot say which is the flame and which the window's.*]

[*The Tortoise sets a fresh, restless stone — one of the few the Crab keeps cut alone, with no partner — at the edge of the slab. It trembles at nothing — both notes at once — and in the black its reflection trembles with it, the two locked step for step, so close you could not slip a hair between their timing. She rocks the stone a hair with one fingertip; the reflection rocks the same hair, the same instant, the same distance. A thing, and its reflection, moving as one thing.*]



restless, in step

Tortoise: A thing and its reflection. While the thing is restless, the two are tight — one motion, twice over.

Achilles: Then a spent stone will be tighter still — it has stopped trembling altogether.

[*She puts the restless stone back in the drawer, lifts the spent-and-unread stone out of the viewer's dish — under its cloth the needle falls back to the middle — and sets it where the restless one stood. It lies still. But when she rocks it, its reflection answers slack: a beat late, drifting a little wide, no longer keeping its partner's exact time. The same stone, the same twin thrown back in the glass; only the coupling between them has gone loose.*]



spent and unread

Tortoise: And a thing that has been looked at — still a thing and its reflection, but slack, like a rower a beat behind the stroke. That looseness is the whole of what your look left you: not a note, not the other note, but a thing gone loose from its own mirror.

Achilles: So my looked-at stone is — half the time high, half the time low, and I simply can't say which. A weighted list, not one note and not the other. [*working it through*] And the restless one is the tight end of that: one whole possibility, its reflection in perfect step. Every looseness between the two is a look taken and not read.

[*Beyond the glass the snow is at its slow work still. The flakes coming down now are sharp six-pointed stars, each one particular and cut clean; where they have lain a while on the sill they have softened and slumped, arm run into arm, until they are only a bank of rounded grains — the same snow, every edge of it gone, keeping nothing of the figure it fell in.*]

Crab: [*eyeing the drawer and the heap beside the viewer, as if every stone in them were bound for the black*] And each of these throws its own reflection, and my slab is filling up with twins. Might we clear a few off before I lose which is a stone and which is the ghost of one?

Achilles: Clear them, by all means — but that is my next trouble exactly. If every stone is a thing and its reflection now, and a pair is two of them… suppose I care about only *one* half of a shared pair. The far half's business is the far half's. How do I get an honest account of just my half, and cover the rest?

Third Challenge — *Cover the Other Half*

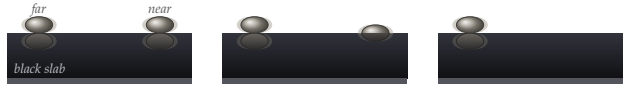
[*The dusk has settled the quarrel the window was having with itself: it is nearly all mirror now, a clean dark sheet with the whole room stood complete and orderly a pane's-depth inside it. The Crab, passing, feeds the fire a single log and sets the guard back; the tea has gone tepid in the pot and no one minds. Outside, past the reflected room, the snow has thickened to a steady, straight-down fall.*]

Tortoise: You cover a half by folding it into its own reflection — a new move, and a plain one. To cover a thing is not to look at it and not read it; it is to refuse it any account at all, to let it go. On the slab, letting a thing go is folding it down to meet its reflection, as a book closes on its facing page: they touch, and cancel, and there is an end of it.

Achilles: Fold mine away and something of the far one must go with it — they stood on one slab together.

[*The Crab lifts the looked-at stone onto the heap. Here is a plain pair — two stones with nothing between them, cut from different pieces and each cut alone, the far one at the slab's far end and the near one at Achilles's — each with its reflection standing beneath it in the black. The near stone is Achilles's; the far one is no business of his. The Tortoise tips the near stone down toward the glass; its reflection rises to meet it, mirror-fashion; they touch at the surface — and where thing met twin there is now neither, only a bare patch of black slab. The near stone is not spent, not looked at,*

not read: simply gone, folded into its own mirror. The far stone, and its reflection, stand exactly where they stood, not stirred by a hair.]



nothing between them — *the near stone tipped* — *folded away, the far stone unstirred*

[*On the roof of the house across the lane the afternoon's snow has crept at last to its own weight, and the near slope lets go all together — a whole clean sheet of it sliding off into the dark below and leaving the slates bare — while the ridge above and the far slope hold their white exactly as it lay, not a crystal shifted. What went, went wholly; what stayed, stayed untouched.*]

Tortoise: That fold is "cover the other half." I covered the near one — mine — and the far one never felt it. Cover an unrelated half and you lose nothing you wanted, and change nothing you kept.

Achilles: Because the two had nothing to do with each other. [*nods*] But that is the dull case. Do it on a pair truly *of one piece* — the way your stones are cut — and cover the *near* half, the one in my hand. What is left of the far one, that I never touched?

Fourth Challenge — *What the Twin Cannot Tell*

Achilles: [*to the Cat, half to himself*] You on the sill — if I work this cleverly enough, will my message get across?

Cat: [*one eye barely open*] It will, and it won't. And the won't has the better of it.

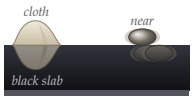
Tortoise: Loose question, loose answer — leave it loose while it's on the sill. A plain question is a look, remember; and we are not looking *there* yet.

[*The window is almost wholly a mirror now, the whole lit room hung in it — the lamp, the bowed heads, the table and its business — with only a last dull grey low*

down where the lane had been, and the snow beyond gone to a blue-dark nothing. The Crab turns the lamp up a little against the dark; the fire has burned down to a steady, companionable red. On the sill the curled shape is a warm hollow and a smooth one at once, an eye's-worth of lamplight caught in it and the rest given over to the dark — settling, or waking, there is no telling which.]

Crab: A pair of one piece, then. [*He lifts the plain pair's far stone back into the drawer, sets one fresh stone on the black by Achilles's hand, and carries its partner to the far end of the slab, at the table's end, under a cloth.*] The near one is yours to bully. The far one goes under here, and nobody looks at it — not you, not me. Those are the rules.

[*On the black the near stone lies restless, trembling at nothing — and yet its reflection answers it slack, a beat late and a little wide, as the looked-at stone's did, though no one has looked at this one.*]



far under its cloth,
near to hand

Achilles: Right. Now I send a message across this table with no more than my own half. Watch. If I *look* at my half, the pair is cut of one piece, so the far half must fall the same way — a twitch I could send. If I *fold* my half away instead, that's a different twitch. If I *leave it be*, a third. Three things I can do, three states of the far stone, and a code between them. [*He reaches for the cloth.*]

Crab: You do not look at the far one. Rules.

Tortoise: You need not, and you must not. That is what the ledger is for. Every one of those things you might do to a near half, I have done a hundred times over the years, on a hundred pairs, and written down what the far halves said. Read them off — by what was *done*, not by what fell.

Achilles: [*at the ledger*] Then — every page where the near half was looked at, whatever it came up. The far halves read: high, low, low, high, high, low. A coin. [*turning pages*] Every page where the near half was folded away

and let go. The far halves: low, high, high, low, high. The same coin. [*more pages*] Every page where nothing at all was done to the near half. A coin again. Three different things done here, and the far half a fair coin under every one of them. No difference for her to read.

[*The ledger's pages have gone soft as cloth at the corners, thumbed through eleven winters of such evenings. At the table's far end the kept stone lies dumb under its cloth where the Crab set it, a small covered weight that no one has touched and no one means to. Achilles reads on, chin propped in one hand, warming to the chase.*]

Achilles: But wait — let me *sort* them. Only the pages where my near half came up *high*. Surely then the far halves—

Crab: —all read high, yes. Cut of one piece; they always agree. But you did not *pick* high, Achilles. The viewer did, at random — you told me so yourself an hour ago. And your partner across the room has no way on earth to know which pages are hers, the high pile or the low, unless you write and tell her by post. Sorting a thing you could not choose is no message.

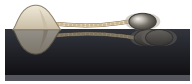
Achilles: …Oh. There's nothing I can do here that reaches there. The halves agree — but the agreeing is no use to me, because I cannot steer it and she cannot see it. Whatever I *do*, her half is the same grey coin.

[*Outside, the wind that has been gathering all afternoon finds its voice, and flings a fistful of snow against the panes, and then another, and leans on the old house until the timbers creak. Inside, the lamp-flame stands straight and unwavering and does not so much as lean; the surface of the tea lies still in the cup; nothing in the sealed warm room takes the least notice of all that battering at the glass.*]

Tortoise: [*a length of string from her shell*] One picture, and you'll see *why* it can't. This is a string — the plainest thing I know that goes round a bend and stays all one piece, which is just what a pair cut of one piece is. Let me lay it across your pair on the slab, halves and reflections and all.

[*On the black she lays the string from the far stone across to the near stone — that is the pair, joined of one piece — and in the glass its own reflection runs beneath it, from the far stone's twin across to the near stone's twin. Then she folds the near stone down into its reflection, the same cover as before; and where the near stone meets its twin, the string above joins the string below. Far stone, across to near,*

round the fold, and back beneath to the far stone's twin: one string and its mirror, with the near half pinched away in the middle.]



joined, far to near



the near stone folded

[*She sets one finger on the far stone to hold it, and with her other hand draws the slack out of the middle — toward the far stone. The bend slides in toward the far stone, the long run behind it closing up; the near half and its twin are drawn in and swallowed; and the string plunges straight down into the glass at the far stone's own foot, to meet its own reflection there. What was far-across-near-fold-near-back is now one clean bend: the far stone, joined to nothing but its own reflection in the black. Beside it she chalks the one word:* half.]



drawn in to the far stone

Achilles: The far half, bent round to its own reflection, and a half beside it. That is your slack stone from before — the thing loose from its own mirror, the flattest grey there is. A fair coin, every way of asking at once. And no look loosened it — a partner out of reach is loosening enough.

Tortoise: One pull, the way you'd draw in a washing line. And whatever you had done to the near half — any box you please on that middle leg — the fold swallows it before the pull even begins. It never reaches the far stone. She was always going to come out that same grey.

Achilles: But your fold is *clean*. A real look isn't — mine swung a needle, breathed into a flask, left a leak in the room. Fold a clean half away and nothing crosses, granted. But a real look drags all that plumbing behind it. Does the leak get out? Could the *leak* carry across what I couldn't?

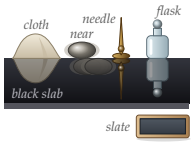
Fifth Challenge — *A Look with the Lights On*

Crab: Then we lay it open — the look itself, mind, not the trick that does it. [*He lifts the cloth from the needle's dial and lays it by the viewer — the needle stands at the middle — then unclasps the viewer's case and folds it back, keeping his eye carefully off the works.*] Here is where a real look happens. The near stone goes in here. The needle here, that swings to the answer. And here, sunk in the base, the flask — the vent; every look breathes once into it, and then it is stoppered and set by. Three things, bound into one by the works in the instant of looking. [*aching*] I could tell you *how* they are bound, if I put my eye to the works. I never have, and I never shall; that is what this device costs me, and I pay it gladly — or nearly gladly.

[*The Tortoise takes up her string from the black. From the drawer the Crab takes a fresh pair: the old far stone goes into the drawer, cloth and all, unlooked-at; the fresh far half goes under a fresh cloth at the slab's far end, and its partner into the viewer's dish.*]

[*The opened viewer lies on the table like a watch with its back off, laid open to the lamp — and it is, uncovered so, a little less of a marvel and a little more of an ordinary clever thing, as such things always are once one can see round behind them. The Crab keeps his eyes resolutely on the far wall while his claws do the work, the way a man carries a brimming cup without watching his own feet. It has cost him something to lay it bare, and he does not pretend that it hasn't.*]

[*He lifts the near stone from the dish, the needle from its pivot and the flask from its well, and on the black slab stands them out for the look, each with its reflection beneath it: the near stone and its twin, the needle and its twin, the flask and its twin. The far stone keeps to its cloth at the end of the table, its own reflection with it. The works will bind the near three; the far one they never touch. Off the black, in front of the slab at Achilles's end, the Crab lays a clean slate.*]



stood out on
the black

[*The needle and its reflection stand together, two pointers, both quivering toward an answer. The Tortoise brings them together and pins the needle down onto its own reflection, at the one point where the two meet: where there were two quivering pointers there is now a single still one. From that pinning a notch is cut — low, as it happens; she scratches it at the slate's foot — on the slate, which she keeps off the black glass, so that this one mark alone throws no twin of itself. Two pointers in; one settled mark out; and the mark stands single, unpartnered, as the first notch of the evening did.*]

[*She folds the flask down into its reflection — the same cancelling fold as the stones — and it is gone from the black: breathed into, stoppered, let go, and lifted off to stand by the viewer's base.*]



pinned, and
the flask let go

Achilles: So pinning is reading — the way a finger stops a spun coin flat, one face up — and the notch is what it read.

Tortoise: Now the leak. The works tie the near stone, the needle and the flask into one — a box across the three of them, and its reflection a box across the three twins. Then we let them all go: the flask folded away, the near stone folded away, the needle read to its one notch and finished — every output of the works folded down into its own reflection. Watch what that does to the far half.

[*She folds the near stone down into its reflection, and lifts the pinned needle off the*

black to lie by the viewer's base. The Crab, eyes on the wall, lifts the works out of the case by feel and stands them on the black between the far stone and Achilles's end, their reflection beneath them. The Tortoise lays her string again, from the far stone into the works on the side that faces it, through the box, and out the far side, toward Achilles's end; and every leg it comes out on — the near stone's, the flask's and the needle's, its answer already cut on the slate — is folded into its reflection on Achilles's side of the works. The Tortoise sets a finger on the far stone and draws the slack in toward it. The box, coming out, meets its own reflection under the fold — arrives upon itself, and cancels — and the whole apparatus is swallowed as the fold closes. The string plunges straight into the glass at the far stone's foot, to its own reflection: one clean bend, the plain half beside it, exactly as for the clean fold. The needle's notch stands on the slate; the flask sits stoppered by the base. But the far stone is the same flat coin it was before — the works reached it not at all. The Crab lifts the works clear, by feel, and sets them by the case.]



through the works *drawn in to the far stone*

[*Up the chimney the fire is pouring out everything it has — smoke, and the afternoon's stored heat, and a low steady roar in the flue — away into the freezing dark above the roofs. And for all of it the night is not altered by a breath: the snow lies along the roofs the length of the lane no thicker and no thinner for the smoke going up past it, and the dark takes the roar and gives back nothing. All that pouring-out, and none of it reaches across.*]

Achilles: Nothing. All that plumbing — needle, flask, leak — and the far half is the same grey coin.

Tortoise: Because discarding after the works is discarding straight off. Let a thing go after all its machinery, or never build the machinery at all — it comes to one and the same. The works meet their own reflection under the fold and cancel; the far half never hears them. A real look, laid bare, leaks nothing across.

[*As she speaks the last grey light gives out in the window, and the room drops to the*

lamp alone. In that sudden evenness the print on the scullery wall seems to stand a half-step out from its wall, the great eye darkening as the shadows gather into it.]

[*The Tortoise, seeing the look itself now laid out on the slab as one more thing among the stones, glances up at the print on the scullery wall — Escher's* Eye, *a single vast eye with something small looking back out of its own pupil.*]

Tortoise: The eye that looks — set down as one more thing to be looked at. [*She turns back to the slab.*]

Crab: [*looking over the spent stones, the first slate still under its cloth, the near-empty drawer*] Eleven years in that drawer, and the better part of it gone in an afternoon. The daylight with it — it is the lamp now, and no more. [*He sets the needle back on its pivot and the works back in their case, gathers the stoppered flask and sets it by.*]

[*The room has drawn in to the size of the lamplight now: the fire fallen low and red, the drawer all but empty, the little spent stones gathered in a dull heap, the slate set off to one side of the black. It is the hour when a long day's work is all but done, and the doing of it has gone soft and unhurried. Someone's tea has been refilled and let go tepid again, and nobody minds it.*]

Tortoise: [*quietly, winding her string back into her shell*] Let them be spent; they were made for it. Restless, every one — and every one put to rest, a notch apiece, and still.

Crab: There is warm tea yet, if anyone—

Achilles: [*not quite hearing him, looking round the settled table — the spent stones, the bare black where stones were folded away, the one notch alone on the slate — and then to the corner*] In a moment. There is one thing here that no one has looked at all evening. [*nodding to the sill*] Your friend there. Restless as any stone, both at once, and nobody has spent it.

[*The sill in the corner keeps its small mystery to the last: a dent in the cushion that is also, looked at again, quite smooth; a gathered warmth with the cold of the glass just beyond it; the gleam of a half-open eye that might be a bead of lamplight caught and held, or might be a look returned. Asleep, or awake — the room has let it be both all evening, and has asked nothing of it.*]

Tortoise: [*softly*] Mind how you ask it, then. All evening you have asked it loose things and had loose things back. A plain question is a look — and a look, tonight, we have seen what it does.

Achilles: I have. [*to the Cat, gently*] I have looked at everything else on this table. Now I look at you. Cat — asleep, or awake?

Cat: Awake.

[*The Cat settles to its one answer; and as the rest of it goes from the sill — the curled warmth, the folded paws, the one half-open eye — the last of it to fade is the grin, which hangs a moment over the empty place, and then is gone too.*]

[*The corner is only a windowsill now, and a cushion, and the cold coming off the glass. The room settles the last of the way down into its quiet: the fire sunk to a bed of embers, the lamp steady, and on the table where the work left them the spent stones in a dull heap, the kept stone still under its cloth at the far end, the viewer open with its needle stilled, the flask stoppered and set by, the slate off to one side. In the black window the whole warm room stands doubled and still, and beyond it the snow has stopped falling at last. On the scullery wall the great eye has gone back into the dark of its frame, a plain rectangle with a little lamplight on its grain, part of the room's stillness now. Only the lamp, and the embers, and no one left in the room with a question to ask of anything.*]

Chapter 9: Now You Look

— Lullaby for a Cat —

Part III: Pictures Become Proofs

Escher: "Eye" — a single vast eye, with something looking back from inside its own pupil

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When you finally look at something, what has the look done to it — and could anyone far away ever tell that you looked?

For eight chapters we have kept the lights off. We prepared states, slid boxes round bends, cooked tangles down to crossings, and every time a question was asked of a wire we quietly kept only the answer we wanted — the postselected cap, the outcome that happened to suit us. No real device is so obliging. A real device forces one of several answers on you, at random, and destroys the rest; and the forcing is not a footnote to the physics but a process in its own right, a thing that has to be drawn. The last chapter promised as much and then closed the book for the night. Now we open it and look.

Two things go wrong at once, and this chapter is about both. The first is near at hand: a look leaves the thing looked at changed, and if we did not read the outcome we are left holding something that is neither the restless superposition we started with nor a plain definite answer — something new, for which a ket is the wrong kind of object. We shall build the right kind. The second is across the room: if a look here changed a thing here, and that thing was one half of a pair shared with someone far away, did the look reach *there*? Could a message be sent, faster than any letter, by

choosing what to do to one's own half? The answer is no, exactly no, and the reason is one of the most quietly astonishing facts in the theory. Half of a shared pair, examined on its own, is the same featureless grey no matter what is done to the other half — and drawn in the right notation, that fact is a single tug on a bent wire.

To get there we need three new marks on the page, and one honest reckoning with the number in front of a cup. Let us make the marks.

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§9.1 The Look That Forces

Achilles has never once watched a wire. Let him look now, at the plainest thing there is: a single qubit, and the question "are you $|0\rangle$ or $|1\rangle$?"

On the blackboard. To ask that question we use two **projectors**, $P_0 = |0\rangle\langle 0|$ and $P_1 = |1\rangle\langle 1|$ — each one a box that keeps the part of a state lying along its own axis and throws the rest away. Let the qubit be the unit vector $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$. The **Born rule**, which is the whole physics of a look, says two things. The look returns outcome i with probability

$$p_i \;=\; \langle\psi| P_i |\psi\rangle \;=\; |\psi_i|^2,$$

and, having returned i, it leaves the qubit in the state $P_i|\psi\rangle \,/\, \sqrt{p_i} = |i\rangle$. Which outcome comes is not fixed by anything: it is a genuine coin, weighted by the $|\psi_i|^2$. Ask $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and the two outcomes each have probability $\frac{1}{2}$; ask $|0\rangle$ and the answer is 0 with certainty, $p_0 = 1$. And once the look has spoken, the question is settled: measuring a second time returns the same i, because $P_i|i\rangle = |i\rangle$. The look did not *read off* a bit the qubit was already carrying. It *made* one, and destroyed the alternative.

What comes out is a record — a definite value, already decided, no longer able to interfere with anything. It is a new kind of quantity, and it deserves a new kind of wire.

In the sketchbook. So far every wire has been a quantum wire, drawn as a

single stroke. A definite outcome is different: its value is settled, it carries no phase, it cannot be put into superposition with anything. We draw it as a **classical wire** — a single line, but stroked *double*, to keep it distinct on the page from a quantum wire.

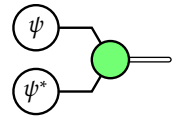
One system, one wire; the double stroke is a reminder, not a second line. Now the look itself. To draw it we lean on a fact we have owned since Chapter 6. There we saw that a state has a **conjugate**, $|\psi^*\rangle$ — the same ket with every amplitude complex-conjugated (§6.2), a genuinely different state when ψ is not real. Feed a green Z-spider two legs — on one, the state $|\psi\rangle$; on the other, its conjugate $|\psi^*\rangle$, which is an ordinary state box, a ket running rightward, and emphatically *not* the effect $\langle\psi|$. Let the spider have one classical leg out. This is the **read-spider**,

$$Z(2\to 1) \;=\; \sum_i |i\rangle\langle ii| \;=\; |0\rangle\langle 00| + |1\rangle\langle 11|\,,$$

the very Z-spider of Chapter 8, put to a new use. On $|\psi\rangle \otimes |\psi^*\rangle$ it keeps only the branches where the two legs agree, and passes their product to the classical wire:

$$\sum_i \psi_i\,\overline{\psi_i}\;|i\rangle \;=\; \sum_i |\psi_i|^2\;|i\rangle \;=\; (p_0,\,p_1).$$

The classical wire comes out carrying the Born probabilities, exactly — for $|+\rangle$ the vector $(\frac{1}{2},\frac{1}{2})$, for $|0\rangle$ the vector $(1,0)$.



Result 9.1 (the look reads the Born rule). The read-spider $Z(2 \to 1)$, fed a state on one leg and its conjugate on the other, outputs on its classical wire the vector of Born probabilities $(|\psi_0|^2, |\psi_1|^2)$ — exactly, scalar 1. See `Code/Ch9/Measurement.py`.

Why this definition. A probability is an amplitude multiplied by its own conjugate, so the look must be handed *two* copies of the state — itself and its mirror — for the spider to multiply; and the outcome must ride a wire of its own, because it is no longer a thing that can interfere, only a thing that has happened.

A word on scalars before we go on. From Chapter 8 we allowed ourselves to state a picture equality up to a non-zero number. That licence is withdrawn for the rest of this chapter wherever the number *means* something — a probability, a trace, a weight in a mixture. Every equation below is exact, the scalar written in, and we say so each time.

Feynman's Notebook.

The idea that a look leaves a *mixture*, and that the mixture is bookkept by a matrix rather than a vector, goes back to John von Neumann. Wojciech Zurek's review argues that in a purely unitary account "there is no room for a 'real collapse'," and traces that conclusion to von Neumann (1932). Zurek's own subject is *decoherence*: an apparatus that couples to a qubit is itself watched by an unruly environment, which "monitors" it, selects a preferred set of **pointer states**, and enforces the effective — Zurek's phrase — "collapse of the wavepacket." The coupling that starts it all is a controlled-NOT, "known as a 'bit by bit measurement' (Zurek, 1981; 1983)": the apparatus copies the system's bit the way the green dot copies a wire. On this telling the classical world is not put in by hand; it is what survives the environment's looking on our behalf.

---

§9.2 A State That's Been Looked At

Achilles measured the qubit but — suppose — did not read the slate. The outcome is settled, but he does not know it. What is he holding? Not $|0\rangle$; not $|1\rangle$; not the $|+\rangle$ he began with, whose phase is gone. It is a thing with no ket, and Chapter 6 promised it to us: back in §6.2 we said there was no

object that could be called "the state of one wire," and that this chapter would build one, and that it would not be a ket. Here it is.

On the blackboard. A **density matrix** is a weighted list of possibilities,

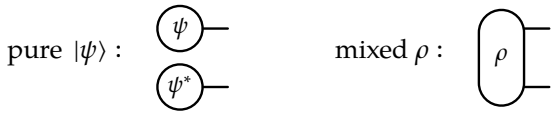
$$\rho \;=\; \sum_i p_i \,|\psi_i\rangle\langle\psi_i|\,, \qquad p_i \geq 0,\; \sum_i p_i = 1.$$

Achilles's looked-at-unread $|+\rangle$ is: with probability $\frac{1}{2}$ it is $|0\rangle$, with probability $\frac{1}{2}$ it is $|1\rangle$. So

$$\rho \;=\; \tfrac{1}{2}\,|0\rangle\langle 0| + \tfrac{1}{2}\,|1\rangle\langle 1| \;=\; \tfrac{1}{2}I,$$

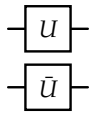
the **maximally mixed state** — the flat grey that knows nothing, the same in every basis. Compare it with the state he *started* from: the pure $|+\rangle\langle +| = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, whose off-diagonal $\frac{1}{2}$'s are exactly the phase the look destroyed. Both have trace 1; that is the density matrix's one law, $\operatorname{tr}\rho = 1$, the total probability. A **pure** state is the special case $\rho = |\psi\rangle\langle\psi|$ — one possibility, weight one, a matrix of rank one. Everything else is a genuine mixture, and no ket describes it, because a ket has no room to record that a phase was lost.

In the sketchbook. The picture of a density matrix is **doubling**. A quantum system is drawn, from now on, as *two* lines — its wire, and a second wire carrying its **conjugate** (mirror) copy. A pure state $|\psi\rangle$ is its two copies side by side, matched: $|\psi\rangle$ above and $|\psi^*\rangle$ below, exactly the pair the read-spider ate. A general ρ is a single box straddling the two lines, whose copies no longer come apart:



The second copy is the *conjugate*, and it matters which. When a unitary U evolves the state, $\rho \mapsto U\rho U^\dagger$; in doubled form U acts on the top (ket) copy and its conjugate $\bar{U}$ on the bottom (mirror) copy, because the mirror

carries conjugated amplitudes and a conjugated amplitude is turned by the conjugated matrix. We draw this doubled process as a box on both lines and call it $U \otimes \bar{U}$:



Note that this is neither the transpose U^T (the half-turn of Chapter 6) nor the dagger $U^\dagger$ (the left-to-right reflection of Chapter 3): it is the plain conjugate, no wires bent, no reflection. Doubling is what makes the scalar in a picture a real, non-negative, *meaningful* number: an overall phase $e^{i\theta}$ on the top copy is cancelled by $e^{-i\theta}$ on the mirror, so only magnitudes survive.

Why this definition. Why carry a whole second copy of every wire? Because a probability lives in the product of an amplitude with its conjugate, and only an object that holds both — the wire *and* its mirror — can record what a look leaves. A ket holds one; the doubled wire holds both, which is exactly enough to hold a mixture. See `Code/Ch9/DensityMatrix.py`.

Dirac's Blackboard.

The state that is not a ket. Chapter 6 asked for "the state of the top wire" of a Bell pair and had nothing to hand back. Now we do: it is the reduced density matrix, and for $|\Phi^+\rangle$ it will turn out to be $\frac{1}{2}I$. Three facts pin ρ down. It is **Hermitian**, $\rho = \rho^\dagger$, since each $|\psi_i\rangle\langle\psi_i|$ is. It is **positive**, $\langle\phi|\rho|\phi\rangle = \sum_i p_i |\langle\phi|\psi_i\rangle|^2 \geq 0$ for every $|\phi\rangle$. And it has **unit trace**, $\mathrm{tr}\,\rho = \sum_i p_i = 1$. Purity is measured by $\mathrm{tr}(\rho^2)$: equal to 1 for a pure $|\psi\rangle\langle\psi|$, and $\frac{1}{d}$ — here $\frac{1}{2}$ — for the maximally mixed $\frac{1}{d}I$, the flattest grey a d-dimensional system allows. No column of amplitudes carries this; a ket has $\mathrm{tr}(\rho^2) = 1$ always. The mixture is a new object, and it needed the whole matrix to say it.

§9.3 Covering a Half — Partial Trace

Achilles holds one half of a shared pair and wants its honest description, the far half being none of his business. He must **cover** the other half — not measure it, not read it, simply refuse to account for it. What is left?

On the blackboard. The description of the half you keep is the **partial trace** over the half you discard. For a joint density matrix ρ_{AB} on two systems, the marginal on A is

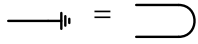
$$\rho_A \;=\; \mathrm{tr}_B\, \rho_{AB} \;=\; \sum_c (I \otimes \langle c|)\, \rho_{AB}\, (I \otimes |c\rangle),$$

summing the effect $\langle c|$ over a basis of B on both sides. If the pair is a plain product $\rho_A \otimes \rho_B$, this returns $\rho_A \,\mathrm{tr}\,\rho_B = \rho_A$: covering an unrelated half changes nothing, as it should. The interest is entirely in what it does to a pair that is *not* a product — which is the next section's business.

In the sketchbook. To cover a system is to **discard** it. We draw the discard as a new mark: a wire that ends not in a box or a bend but in the **ground symbol** — a short stem closing in a few shrinking bars, the sign that a wire is let go into the environment and never asked another question.



What does discarding *do*, in the doubled picture? A system is now two lines, its wire and its mirror. To let it go is to fold the two copies together — to join them with a **cap**, the effect $\langle 00| + \langle 11|$ of Chapter 6. The ground symbol on a system is shorthand for exactly this cap on its two copies:



the left drawn as one line only because the doubling is understood; unfold it and the two legs of the cap reappear. As always the cap is unnormalized, $\langle 00| + \langle 11|$; the scalar is load-bearing here and we shall watch it. Cap a system to its mirror and you are computing a trace. Applied to a doubled ρ — the box on two lines — the cap contracts the two copies and returns $\mathrm{tr}\,\rho$.

Two small closed pictures make the scalar honest. The cup joining a system to its mirror, $\sum_i |ii\rangle$, is the doubled form of the identity matrix (Chapter 6, §6.5). Cap it — discard it — and you get a number, $\sum_i \langle ii| \sum_j |jj\rangle = \sum_i 1 = 2 = \operatorname{tr} I$. Put a $\frac{1}{2}$ in front and the number is 1: the mirror-cup, scaled by $\frac{1}{2}$, is the maximally mixed state $\frac{1}{2}I$, and its trace is 1 as a density matrix's must be. This is the image we will need in a moment:

$$\tfrac{1}{2}I \;=\; \tfrac{1}{2}$$ 

— a system cupped to its own mirror copy, with the number $\frac{1}{d}$ beside it. One convention, to fix the pictures that follow: we draw a system beside its mirror copy, and when a system is to be discarded we keep its two copies side by side, so that its cap is small and its neighbours are undisturbed.

> **Result 9.2 (discard is partial trace).** For every joint state ρ_{AB}, capping the two copies of the B-system in the doubled picture equals the partial trace $\operatorname{tr}_B \rho_{AB} = \rho_A$ — exactly, scalar included.

Proof idea. The cap says "join B to its mirror and sum"; summing a matrix's diagonal over one factor is precisely tracing that factor out.

Proof (blackboard). Write $\rho_{AB} = \sum \rho_{ab,a'b'} \; |ab\rangle \langle a'b'|$. In doubled form this is a vector with a ket copy $|ab\rangle$ and a mirror copy $|a'b'\rangle$ on each term. The cap on the B-copies applies $\sum_c \langle cc|$ to the pair (b, b'), keeping only $b = b' = c$ and summing:

$$\sum_{a,a',c} \rho_{ac,a'c} \; |a\rangle \langle a'| = \operatorname{tr}_B \rho_{AB}.$$

Every "every" here is checked on a random ρ_{AB}, and the two scalars are confirmed with it — the doubled bare cup discards to I (a closed value of 2), the maximally mixed $\frac{1}{2}$-cup to trace 1. See `Code/Ch9/PartialTrace.py`. ■

Why this definition. Discarding must be the trace and nothing else — the one effect that asks a system no question, only lets it go — and drawing it

as a cap is what lets a discarded half inherit *yanking*. That inheritance is the whole reason the chapter's theorem is one move rather than a page.

Coecke's Sketchbook.

The environment structure. The ground symbol is not an ornament; it is an *axiom*. Coecke and Perdrix (2010) equip a theory with a chosen discard $\top_A : A \to I$ on every system — for density matrices, exactly the trace $\rho \mapsto \operatorname{tr}\rho$ — and read it, in their words, as "this system being part of the environment." What they require of it is a single law (their Eq. 10): two pure processes are the same up to a phase precisely when they look the same after the discard, $\top \circ f = \top \circ g$. Take g the identity and the law says something we will use directly — a process leaves the discard untouched, $\top \circ f = \top$, exactly when it is trace-preserving. This book calls that property **causality**; the word is ours, not the paper's, but the fact is theirs, and it is what forbids a signal.

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§9.4 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 8's sixty-four. Three introduce new picture elements — the classical wire, doubling, and the discard; the rest are constructions built from them. The last three name what §9.6 will prove — a channel, its dilation, and the causality that makes the master hold; they are gathered here so the chapter's additions sit on one page.

#	Algebra (Dirac)	Picture (Coecke)
65	a **classical wire**: a system whose value is already definite (an outcome)	a single wire drawn with a double stroke (`cwire`) — *new element*

#	Algebra (Dirac)	Picture (Coecke)
66	**doubling**: a system carries ρ; a pure $\lvert\psi\rangle$ becomes $\lvert\psi\rangle \otimes \lvert\psi^*\rangle$	a quantum wire drawn as two lines — the wire and its conjugate (mirror) copy — *new element*
67	**discard** $\top_A : \rho \mapsto \operatorname{tr}\rho$ (the environment structure)	a **ground symbol** closing a wire = a cap joining a doubled system's two copies — *new element*
68	projective measurement; Born rule $p_i = \langle\psi\rvert P_i \lvert\psi\rangle$	the **read-spider**: a phase-free Z-spider on a doubled system emitting one classical wire, $\sum_i \lvert i\rangle\langle ii\rvert$
69	maximally mixed state $\frac{1}{d}I$	a system cupped to its own mirror copy, with the number $\frac{1}{d}$ beside it
70	channel $\Lambda(\rho) = \sum_i K_i \rho K_i^\dagger$ (CPTP)	a box on doubled wires — the doubled process
71	Stinespring: $\Lambda = \operatorname{discard}_E \circ V$, V an isometry $A \to B \otimes E$	a doubled unitary $V = U \otimes \bar{U}$ with the environment leg grounded
72	causality: $\operatorname{tr}(\Lambda\rho) = \operatorname{tr}\rho$, i.e. $\top \circ \Lambda = \top$	a unitary capped by a discard yanks to a bare discard

Rows 65–69 are exact by construction. Rows 70–72 are the subject of §9.6,

and each is exact there, scalar included.

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§9.5 The Theorem, Twice

Now the pair. Let Achilles and a partner across the room share the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, the top wire his, the bottom wire hers. He wants to send a message by acting on his half — flip it, measure it, leave it alone — and have her half twitch in reply. Take the simplest scheme: he does nothing, or he applies the bit-flip X. Can she tell which, looking only at her own half? Her half's honest description is the marginal, tr over *his* wire — and here is the theorem that dashes his hopes.

> **Result 9.3 (half a Bell pair is maximally mixed; no-signalling).** Tracing out either half of $|\Phi^+\rangle$ leaves the other maximally mixed,
>
> $$\mathrm{tr}_2\ |\Phi^+\rangle\langle\Phi^+|\ =\ \tfrac{1}{2}I,$$
>
> exactly, the $\frac{1}{2}$ included. Moreover, for **any** trace-preserving channel Λ applied to the traced-out half, the surviving marginal is unchanged:
>
> $$\mathrm{tr}_2\big((I\otimes\Lambda)\ |\Phi^+\rangle\langle\Phi^+|\big)\ =\ \tfrac{1}{2}I.$$
>
> The far half is $\frac{1}{2}I$ no matter what is done to the near half. No message crosses.

Here tr_2 traces the *bottom* wire — the near, acted-on half — and returns the far half's description.

Proof idea. Her half, asked anything, is a fair coin (Chapter 6, §6.2), in every basis at once — and that flat grey is $\frac{1}{2}I$. Whatever he does is a trace-preserving process, and a trace-preserving process is, by definition, one the discard cannot feel. The picture will make the second sentence a single tug.

Dirac's Blackboard

Write $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\sum_{a,b}|aa\rangle\langle bb|$. Trace the second wire, $\mathrm{tr}_2(|aa\rangle\langle bb|) = |a\rangle\langle b|\,\langle b|a\rangle = \delta_{ab}\,|a\rangle\langle b|$:

$$\mathrm{tr}_2\,|\Phi^+\rangle\langle\Phi^+| \;=\; \frac{1}{2}\sum_{a,b}\delta_{ab}\;|a\rangle\langle b| \;=\; \frac{1}{2}\sum_{a}|a\rangle\langle a| \;=\; \tfrac{1}{2}I.$$

Flipping his half with X gives $(I\otimes X)\,|\Phi^+\rangle$, whose marginal recomputes the same way to $\frac{1}{2}I$; doing nothing gives $\frac{1}{2}I$ as well. For a general channel $\Lambda(\rho)=\sum_i K_i\rho K_i^\dagger$ with $\sum_i K_i^\dagger K_i = I$, the marginal is

$$\mathrm{tr}_2\big((I\otimes\Lambda)\,|\Phi^+\rangle\langle\Phi^+|\big) = \sum_i \mathrm{tr}_2\big((I\otimes K_i)\,|\Phi^+\rangle\langle\Phi^+|\,(I\otimes K_i^\dagger)\big),$$

and cyclicity of the (partial) trace gathers $K_i^\dagger K_i$ under it:

$$\sum_i \mathrm{tr}_2\big(|\Phi^+\rangle\langle\Phi^+|\,(I\otimes K_i^\dagger K_i)\big) = \mathrm{tr}_2\big(|\Phi^+\rangle\langle\Phi^+|\,(I\otimes I)\big) = \tfrac{1}{2}I.$$

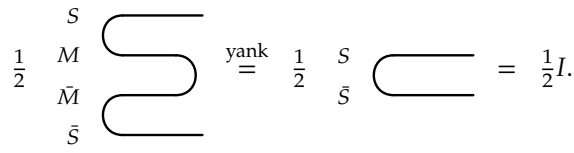
The step that saves it is $\sum_i K_i^\dagger K_i = I$ — trace-preservation, and nothing else. See `Code/Ch9/NoSignalling.py`, which runs a random non-real Kraus channel.

Coecke's Sketchbook

Draw $|\Phi^+\rangle\langle\Phi^+|$ doubled. Since $|\Phi^+\rangle = \frac{1}{\sqrt{2}}\mathrm{cup}$, its doubled form is $\frac{1}{\sqrt{2}}$ times $\frac{1}{\sqrt{2}}$, one factor for the ket copy and one for the mirror:

$$|\Phi^+\rangle\langle\Phi^+| \;=\; \tfrac{1}{2}\;\mathrm{cup}(S,M)\otimes\mathrm{cup}(\bar{S},\bar{M}),$$

with the far half's two copies $S,\bar{S}$ outermost and the near half's copies $M,\bar{M}$ innermost. To trace the near half, cap M to $\bar{M}$ (Result 9.2). What stands on the page is a cup, a cap, a cup, in a column — an S-bend with the near half's copies pinched in the middle:



One yank (Theorem 6.2). The cup on (S, M), the cap on $(M, \bar{M})$ and the cup on $(\bar{M}, \bar{S})$ pull straight into a single cup on $(S, \bar{S})$. What remains is the far half cupped to its own mirror, times $\frac{1}{2}$ — which is the maximally-mixed picture of §9.3, $\frac{1}{2}I$. The scalar is carried, not dropped, at every step.

And the corollary is one more tug. Before the discard, let the near half pass through the channel — a doubled box Λ on $(M, \bar{M})$. Being trace-preserving is, by the environment-structure law (causality, proved as Result 9.4 in §9.6), exactly the statement that the discard slides over it and off: capping after Λ is capping alone. So the box vanishes under the cap before the yank even begins, and the far half is $\frac{1}{2}I$ as before, whatever Λ was.

The verdict

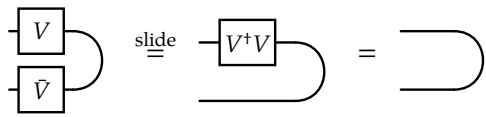
The picture wins, on both counts, and we should say so plainly, because it is the rarer verdict. The headline is a *single yank* with a $\frac{1}{2}$ written beside it, and the reason it is one yank is visible: the maximally mixed state *is* a system cupped to its mirror, and half a Bell pair, discarded, straightens into exactly that. The blackboard's partial trace is short too — four lines — but it shows only *that* the answer is $\frac{1}{2}I$, never *why*. On the corollary the gap widens: the picture removes the channel with one causality slide, while the blackboard must summon $\sum_i K_i^\dagger K_i = I$ and the cyclicity of the trace. What the picture proves, though, is an equality of processes and no more: that a certain marginal does not move. It is not a claim about bandwidth, nor a security guarantee — only that there is nothing here that reaches there.

§9.6 The Unmoved Half

A clean look was too kind. A real measurement has a needle that swings, a flask that vents, a leak into the room; drawn honestly, it is not a bare cap but a whole apparatus. Does the leak carry a signal the clean cap did not?

On the blackboard. A real, irreversible look is a **channel** — a completely positive, trace-preserving map. Drawn, it is a box on the doubled wires (Rosetta row 70), and bending such a box into a state is the process–state duality Chapter 6 named for Choi and Jamiołkowski — the "grander kind of process" it promised, now in hand. Two classical facts, which we state and cite rather than prove, tell us what every channel is made of. First, the **Kraus form**: every channel can be written $\Lambda(\rho) = \sum_i K_i \rho K_i^\dagger$ with $\sum_i K_i^\dagger K_i = I$. Second, **Stinespring's dilation**: every channel is a single unitary on the system together with a fresh apparatus and environment, followed by discarding the environment — $\Lambda(\rho) = \mathrm{tr}_E(V\rho V^\dagger)$ for an isometry $V\colon A \to B \otimes E$, $V^\dagger V = I$. A leaky look is a clean, reversible turn of a *larger* system, with the part we cannot keep let go. We take both facts, with their history, from the eighth chapter of Nielsen and Chuang. Trace-preservation is the causality condition $\mathrm{tr}(\Lambda\rho) = \mathrm{tr}\,\rho$: the discard cannot feel the channel.

In the sketchbook. Causality, drawn, is Chapter 6 coming back for its debt. Dilate the near half through the isometry, doubled as $V \otimes \bar{V}$ on the system's two copies, and then discard the whole output — cap the top copy to the bottom. Slide $\bar{V}$ round the cap (Theorem 6.4, the transpose trick in its cap form): a box carried round a bend arrives transposed, so $\bar{V}$ arrives as $\bar{V}^T = V^\dagger$, meeting V. But $V^\dagger V = I$, and the cap is left bare.



Result 9.4 (causality). Discarding after an isometry is discarding: a doubled isometry capped by a discard yanks to a bare discard, because $V^\dagger V = I$. For a general trace-preserving channel this equality *is* the definition of trace-preserving; via Stinespring it reduces to the isometry case just drawn.

Feynman's Notebook.

Where the channel picture was pinned down. That every noisy process is an operator-sum, $\Lambda(\rho) = \sum_i K_i \rho K_i^\dagger$, and that every such process is a clean unitary on a larger system with a part discarded, are the Kraus decomposition and Stinespring's dilation; we lean for both on the eighth chapter of Nielsen and Chuang, where they are proved in full and their history set out. The doubling, the discard and the environment structure we draw to make them one picture are newer. Bob Coecke and Simon Perdrix set them down in 2010 (arXiv:1004.1598), reading the trace as a system passing quietly into its environment and axiomatising that reading in their Equation 10 — the law this chapter has been calling causality.

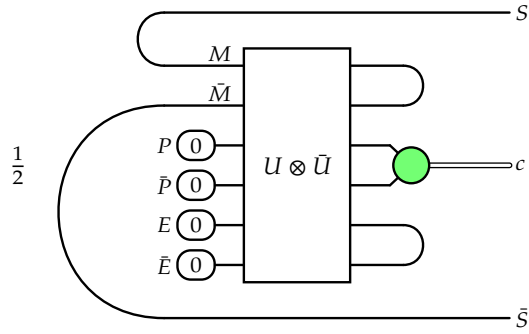
One more small equality, and it is the hinge of the master. A look does not only leave the apparatus; it also writes an outcome, and sometimes we do not read it. Reading with the read-spider and then discarding — *forgetting* — the classical outcome is the same as never having read at all:

$$\left(\sum_j \langle j|\right) \circ \left(\sum_i |i\rangle \langle ii|\right) \;=\; \sum_i \langle ii| \;=\; \text{cap}.$$

Result 9.5 (reading, then forgetting, is discarding). The read-spider followed by discarding its classical wire equals the plain discard (cap) of the measured system — exactly. See `Code/Ch9/Stinespring.py`.

Now the master. It is the no-signalling scene with the apparatus laid bare, drawn in unfolded doubled form so that every line shows. On the left, the doubled Bell state $|\Phi^+\rangle\langle\Phi^+|$ — two cups with the factor $\frac{1}{2}$ written beside them (cup $= |00\rangle + |11\rangle$, as always) — ties the far half S to the near half M. The near half passes through the dilation box $V = U \otimes \bar{U}$, drawn as one box because its two halves — U on the kets, $\bar{U}$ on the mirrors — are wired the same way and only too far apart to draw apart; into that box also go the doubled pointer P (a fresh $|0\rangle$) and the doubled environment E (another $|0\rangle$). Out the other side: the near half M is discarded (a cap on its two copies), the environment E is discarded (another cap), and the pointer

P is read by the read-spider, which emits the single classical outcome c. The far half runs straight across, untouched, above and below everything.



The lines are the far half and its mirror, S and $\bar{S}$; the near half and its mirror, M and $\bar{M}$; the pointer and its mirror, P and $\bar{P}$; the environment and its mirror, E and $\bar{E}$; and the classical outcome c threaded out of the read-spider. We call this diagram **The Unmoved Half**.

> **Result 9.6 (the unmoved half).** In The Unmoved Half, *avert your eyes from the outcome* — trace the classical wire c — and the far half evaluates to $\frac{1}{2}I$, for **every** dilation V, independent of which measurement the near party chose. The far half never moves.

Proof. Forget the outcome: by Result 9.5 the read-spider with c discarded becomes a plain cap on $(P, \bar{P})$, so now *all* of the box's outputs — M, P, E and their mirrors — are discarded. By causality (Result 9.4) the whole discard slides over $V = U \otimes \bar{U}$ and off, and the fresh $|0\rangle$ pointer and environment each discard to $\langle 0|0\rangle = 1$. What is left is the doubled Bell cup with its near half capped — the very picture of §9.5 — which yanks to $\frac{1}{2}I$. Nothing about U survived the discard, so the far half is the same $\frac{1}{2}I$ for every apparatus and every choice of look. See `Code/Ch9/Stinespring.py`, which builds the diagram, tallies its edges, and checks the marginal for a random non-real V. ∎

Be careful with the converse, and honest about it. Do *not* read this as "the far half is independent of the outcome." It is not: leave c open — condition on a particular reading — and the far half is thrown into the state correlated

with that outcome, exactly as a shared pair must be. What is independent of anything the near party does is only the far half *averaged over the outcomes it cannot see*. The signal that could have crossed is drowned, precisely and completely, in the party's own ignorance of which outcome fell.

The Tortoise, drawing the apparatus into the picture — the look itself now a thing on the page — glances up at *Eye* on the scullery wall, the great eye with something watching back from inside its own pupil, and remarks that the eye which looks is also a thing looked at.

Dirac's Blackboard.

Half a Bell pair, and Kraus in full. The headline in coordinates: $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|)$, and tr_2 keeps the blocks whose second labels agree, $\frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I$. The off-diagonal $|00\rangle\langle 11|$ and $|11\rangle\langle 00|$ — the coherence, the part that carried the correlation — are killed by the partial trace, not because they are small but because they are off the diagonal of the traced factor. For the leaky look, the Kraus operators of a Stinespring dilation are read straight off: $K_i = (\, I_B \otimes \langle e_i|\,)V$, one for each environment basis vector $|e_i\rangle$, and $\sum_i K_i^\dagger K_i = V^\dagger(I_B \otimes \sum_i |e_i\rangle\langle e_i|)V = V^\dagger V = I$. Trace-preservation is $V^\dagger V = I$, wearing coordinates.

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§9.7 Now the Other Half

We came to look, and looking cost us the comfort of the ket. A qubit, measured and unread, is not a ket but a mixture, and to draw a mixture every wire had to be doubled, paired with its own conjugate. A part of a pair, honestly described, is the whole with the rest discarded — a cap folding a system into its mirror, which is the partial trace. Half of a Bell pair, so discarded, is $\frac{1}{2}I$; and it stays $\frac{1}{2}I$ through any channel on the other half, because a trace-preserving process is one the discard slides over and off. That is no-signalling, and it survived the apparatus being laid bare: a real, leaky, Stinespring-dilated look, its environment vented and its outcome

forgotten, leaves the far half exactly as grey as before. The whole of that argument was one yank and one slide.

But we have been throwing outcomes away. A cap, since Chapter 6, has been a *postselected* answer — the one reading out of several that happened to suit us — and Chapter 6 warned that the other readings were not failures but corrections waiting to be applied. Now we have the tools to apply them: a classical wire to carry the outcome that fell, and a box, chosen by that wire, to set the far half right. Thread a cup, keep the outcome, send it across, and turn the far half by the correction it names, and the postselected cap of Chapter 6 becomes a protocol that works every time. That is teleportation, and it is the next chapter.

Out-of-Order Canon

雪晴や
文より先に
届く影

yukibare ya
fumi yori saki ni
todoku kage

Clear after the snow —
before the letter itself
its shadow arrives.

[*The Crab's parlour, the next morning. Last night's black slab is still out along the table, wiped clean and bare; beside it, off the black, stands something wrapped in chamois. The Tortoise is already at the table with her tea. Achilles comes in from the cold with a handful of letters.*]

[*It is a hard, bright, windless morning, and the parlour is full of a white light that comes from everywhere at once and throws no shadows. The one tall window at the near end of the long table, which last night was a black glass giving the lamplit room back to itself, has gone white in the dark: every pane is furred from edge to edge with frost, fern grown over fern, so thick that the lane beyond is only a brightness. The sun is up, low and white on the snow outside, but it is still round the side of the house and has not come to this glass. At eye height in one pane there is a small round clear place the size of a breath, where someone stood at first light to watch for the post, and already its rim is feathering over again. Last night's lamp stands cold*

on the dresser, with the viewer beside it shut in its case; the fire has been built up high, the kettle murmurs on the hob, and a plate of toast is keeping warm on the hearth. Down the middle of the table the black slab runs from the window end to the far end, bare. On the wall facing the slab hangs a print the Crab has had so long that he has stopped seeing it — Escher's Magic Mirror*: a mirror standing upright, and creatures stepping out of it. The frost-light falls on it evenly, without favour, as it falls on everything this morning; it was on that wall long before anyone came in out of the cold, and it will be there after. The Tortoise has one of last night's quilts round her shoulders, and the door, as Achilles pushes it to behind him, lets in a last breath of air so cold it smells of iron.*]

Achilles: Good morning! I met your postman at the gate, Mr Crab, blowing on his fingers, and said I'd carry the last few yards for him. Three for you. One of them's from Ostend.

Crab: My cousin. He writes every quarter-day to ask for the loan of my tuning key, and every quarter-day I tell him no. [*He takes the letters and props them against the teapot.*] Come to the fire, come in. You've brought the whole frost in on your coat.

Achilles: It's the white sort — every twig in the lane has grown a fur of it overnight, and the hedges crackle when you brush them. I can't feel my hands at all.

[*He peels off a pair of knitted mittens gone stiff as card with rime and lays them side by side on the fender to thaw, and at once they begin to steam. The Crab pushes a chair in close to the fire for him; Achilles sits on the edge of it and holds his bare hands out to the blaze, flexing them, as if to make sure they are still his.*]

Tortoise: I never went home. The snow had the lane before we'd finished last night, and Mr Crab put me in the little room over the stairs with three quilts and a hot brick.

Crab: And she was up before the light, following the frost on the window-pane with one claw, as if it might need her help.

Tortoise: It doesn't. It is a very good draughtsman. I was only taking lessons.

Achilles: [*folding both hands round the cup he is given*] Ah — that's better.

That's much better. Your postman told me nothing he carries is ever the worse for the road, except eggs.

Crab: Eggs, and my cousin's temper. Mind the slab, by the way; I've left it out from last night, and it will show you two of everything you set on it. [*He goes to the dresser drawer, looks into it a moment, and brings out what is left there: the last two pairs of little stones.*] The last of them. I didn't like to leave them alone in a drawer on a morning like this.

[*He lays the first pair at the near end of the slab. One stone he covers at once with a small square of cloth, and it stays there, hidden, its reflection hidden with it under the cloth's edge; its twin he sets beside it, bare. The second pair he parts: one stone he leaves at the near end, by the others; the other he carries down the whole length of the black and sets at the far end, alone. Each stone, as it touches down, finds its reflection waiting for it underneath.*]

Crab: The one under the cloth is my keepsake. Nobody touches it — not today, not ever.

Achilles: And why send that one all the way down there by itself?

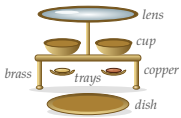
Crab: Because it is a long table, and I want it a long way off. You'll see.

[*The stones sit where he left them down the length of the black: at the near end the small square of cloth, with the keepsake under it; the keepsake's twin, bare beside it; the near half of the parted pair beside them; and far away at the other end, by itself, its partner. The white window-light lies on the slab like milk on slate. Achilles, his cup held up under his chin for the warmth of it, keeps looking down the table at the lone stone at the far end, as though it might be lonely.*]

First Challenge — *Four Ways to Answer*

Crab: [*drawing the wrapped thing nearer and folding back the chamois*] Now. I promised myself I'd bring this out the morning after we looked, and it is the morning after. The pairing-glass — from the same bench in Delft; the daughter ground the lens and her father made the brass. It has two cups side by side under one lens, and it reads two stones together, never one.

[*It stands beside the slab, off the black, on small feet: two shallow cups in a brass cradle, one lens across both, and under each cup a little tilting tray. In the left tray lies a brass token, in the right a copper one, each face-down, blank side up. A brass dish waits beneath the trays' lips.*]



blank side up,
in the trays

Crab: When both cups are full it chimes and tips its trays, and a token slides from each into the dish and lands face-up or face-down. That is its whole answer.

Achilles: Then the brass token is the left stone's answer and the copper token is the right one's. Two stones, two tokens — a fact apiece.

Crab: That is just what I thought, the day it came. It isn't so. Neither token belongs to either stone; both of them tell you only how the two stones stand *together*.

Achilles: Then let me see each stone by itself. Put your eye to the lens while there's only the one in.

Crab: [*with feeling*] Oh, I have wanted to, since the day it came. But it will not read one stone alone — leave one in by itself till Easter and it only waits. And put your eye to the lens to see a single stone, and the glass jams and reads nothing, and the stone you peered at is spent for nothing. I have never once done it, and it costs me every time I look at the thing.

[*In the white light the glass is a lovely thing — brass worked as fine as a watch-case, a lens as clear as a drop of water, the little trays set under the cups like a pair of cupped hands — and it stands so finely on its small feet that a heavy step on the boards makes the trays shiver. Left to itself it keeps up a faint, thin singing somewhere in its brass, like a wet finger going round the rim of a glass in another room. When Achilles leans towards it the singing stops, and the lens is only a round of glass; when he sits back, it takes up again. Worn almost smooth on the foot of the*

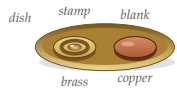
cradle is a maker's stamp: two sets of initials, struck one over the other.]

Tortoise: Then what *does* it say, when it speaks? You have seen it used.

Crab: Not by me. By two of my correspondents, Alice and Bob. They live a long way apart, and for years now the workshop's own carrier — slow, careful, once a season — has brought them pairs, a half of each. [*nodding down the slab*] That is why I put that stone so far off: let this end be Alice's, and that end Bob's.

Crab: Alice has a glass like this one. And they copy me every letter that passes between them, and I keep them in here. [*a fat ledger, which he opens by the dish*] One line to a letter. Here is what Alice's glass has dropped, letter by letter. Watch the tokens.

[*The Crab lifts the two tokens out of their trays into the dish, both still blank side up. He sets a claw at the first line of the ledger and turns the brass token over: it rocks onto its back and shows its stamp, face-up, while the copper token lies exactly as it was. One token has changed; the other has held still.*]



the brass turned,
the copper still

Achilles: Brass up, copper down.

[*At the next line the Crab turns the brass back down and the copper over: now it is the copper that shows its stamp, and the brass lies blank. At the line after, he turns the brass up again beside it, and both lie stamped. At the next, both go down together, and the dish looks as it did at the start.*]

Achilles: Copper up. Both up. Both down. [*The claw goes on down the page, and the tokens go on turning under it, one or both or neither at each line.*] Brass up. Both down. Copper up. Copper up. Both up. Brass up...

Achilles: [*ticking the margin of the Crab's newspaper with a pencil stub*] That's twenty letters: five of the one, six of another, four and five of the rest. Four ways the tokens can lie, and every one of them turns up about a quarter of the time. Never a fifth way, and never two at once.

[*The Crab, pleased with him, refills his cup without being asked, fetches the toast from the hearth and sets the plate at his elbow. Achilles takes a slice and eats it one-handed, his eyes still on the dish.*]

Crab: [*turning both tokens blank side up again and tipping them back into their trays*] And each way means something. Both down: the two stones stand together just as they were cut. Brass up: as though one of them had been flipped — turned over, top to bottom, like this. [*His empty claw turns over in the air, about a level line.*] Copper up: as though one had been spun — turned half round where it stands. [*His claw turns half round on the spot, about its upright.*] Both up: both.

Achilles: One of them. Which one?

Crab: It never says.

Achilles: Because it can't see. But one of them *was* flipped — Alice's or Bob's. That's a fact about one stone, and your glass is only too clumsy to report it.

[*The breath-sized clear place in the window has feathered shut while nobody was watching it. The pane is white now from edge to edge, and whatever the lane is doing this morning, the parlour cannot see it: only the brightness comes through, the same from every pane.*]

Tortoise: May I draw it? Not on the black, Mr Crab — I'll not lay string on your glass, where everything comes out twice. [*She spreads a small drawing-cloth on the table along the front of the slab, on the company's side, running the length of it.*] This is only a cloth to draw on. Tell us the tale, and I'll draw.

Crab: Alice calculates; she has a column for everything. Bob would rather draw.

Tortoise: Then I'll draw for Bob. [*She takes a length of string from her shell.*] This is my string, as in the autumn: a pencil that can go round corners. A bend in it is a pair cut of one piece, with an end at each stone. Let Bob draw curves, then: the one bend between Alice's half and his.

[*She lays the string on the cloth in one long bend: one end laid down opposite the near end of the slab, where Alice's stone stands; the string carried out along the*

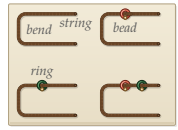
cloth and round; its other end laid down opposite the far end, where Bob's stone waits alone.]

Crab: And Alice has a stone she wants Bob to have — as it might be that one there, my keepsake's twin. But stones don't survive the ordinary post; the road looks at everything on it, a thousand times between her gate and his. Letters survive it, being settled already.

Tortoise: Let Bob draw one more line: the stone Alice wants to send, coming in to her. [*She lays a second length on the cloth, a straight run coming in from the left towards Alice's end of the bend and stopping short of it.*] And now her glass. She sets the stone she wants to send in one cup, and her own half of the pair in the other, and it reads them together.

Tortoise: [*bringing out a little pouch*] These are some of Mr Crab's rings from the scullery beam, red and green, each with its bead set exactly halfway round. A red ring on a string is a flip, and a green ring is a spin. Let Bob draw every way that reading could end.

[*She lays four short bends on the cloth, two above two, each a length of string turned round on itself once. The first she leaves plain. Onto one leg of the second she threads a red ring and pushes it down until it sits close in by the bend; the bend itself does not move, only the ring rides down to it. Onto the third, a green ring, in the same place. Onto the fourth, a red ring close in by the bend and a green ring further out on the same leg.*]

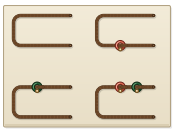


every way a reading ends

Achilles: A plain bend, a flip, a spin, and both. The four ways the tokens lie. [*He puts a finger by the second bend.*] And there's my fact: the red ring is on one leg. Whichever stone that leg goes to — that's the one that was flipped.

Tortoise: Is it? [*She puts a claw to the red ring.*]

[*The red ring sits on one leg of the second bend, close in, its bead halfway round. The Tortoise slides it along the string, into the bend, round it, and out along the other leg. The string does not move and neither end stirs; only the ring travels. It comes to rest on the other leg, the same distance from the bend — still red, its bead still exactly halfway round. Nothing about it has changed but the leg it sits on.*]



round the bend,
unchanged

Achilles: It came round exactly as it went. Your bead in the autumn came round end for end.

Tortoise: That bead was lopsided. A half-turn has nothing lopsided about it to turn: one ring comes round just as it went. Two together come round the other way about — but one comes round unchanged.

[*On the hob the kettle has begun to sing, and its note lies so close to the glass's thin singing that the two make between them a slow throb, swelling and fading, that belongs to neither of them. Nobody in the room could say which of the two is the higher. The Crab eases the kettle a little off the heat; the throb slows and lengthens, and the room goes on keeping it.*]

Achilles: Then the flip isn't on Alice's leg or on Bob's. It's on the bend — it slides to whichever leg you please. So the glass isn't clumsy at all; there was never a fact about one stone for it to report. It's like a handshake: the grip is in neither hand, only between them.

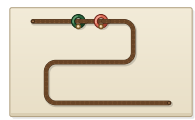
Crab: I told you so. Both tokens, about the two together.

Achilles: Four answers about how two stones stand together. So if Alice reads *her* stone against the one she wants to send, what does Bob's stone, a whole house away, know about it?

Second Challenge — *Two Bits Early*

Tortoise: Let us put the tale together and see.

[*She gathers her first drawings off the cloth — the long bend and the message's line into her shell, and the little bends after them, their rings into the pouch — all but the fourth bend's red and green, which she keeps back. She takes up her longest string and lays the whole tale on the cloth in one piece. In from the left it comes, the message; at Alice's end it turns round in a bend, and onto its incoming leg she threads those two rings, the red close in by the bend and the green further out. From there the string runs back towards the left, round a second bend facing the other way — the pair between Alice and Bob — and out along the far leg to Bob's end, where it stops opposite his stone. Laid flat, it makes an S: in with the message, round Alice's reading, back, round the pair, and out to Bob.*]



in, round, back, round and out

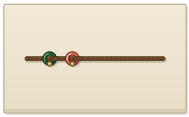
Achilles: Let me work it through. The stone Alice sends goes into her glass with her half of the pair, and both are read there. Bob's stone is the other half of a pair she's just used up. So Bob has lost his partner and never had a thing to do with the message. Nothing reaches Bob until the letter does.

[*The fire has begun on the window from below. On the lowest panes the ferns have gone glassy at their tips, half frost and half water and each tangled into the next, and through them the lane comes in as a wavering blur of white and dark that might be hedges or might be anything. Above, the frost holds, white and whole.*]

Tortoise: Take the message end. I'll take Bob's. Pull.

[*Achilles takes the loose end on the left; the Tortoise takes the loose end at Bob's. They draw apart. The S tightens; the bend at Alice's end and the bend of the pair slide towards each other, meet, and run out through one another into nothing. The rings ride with the string, never lifted and never passing each other. What was in, round, back, round and out is now one taut line from Achilles's hand to the*

Tortoise's — from the message straight to Bob's end — and on it, still, a green ring nearer the message and a red ring nearer Bob. The string is no longer and no shorter than it was; only the bends are gone.]



pulled straight

Achilles: It's at Bob's end. The message runs straight through to him — and he hasn't had a letter. Alice has only just read.

Tortoise: And what is on it?

Achilles: Her rings. A spin and a flip. It's there, but turned — spun and flipped, the way her tokens fell.

Crab: [*half out of his chair, a claw reaching towards the far end of the slab*] Then Bob's stone — this very minute — if I only—

Tortoise: Nothing has been sent on this table, Mr Crab. And if it had, one look would show you a coin, and spend your last pair.

Crab: [*sitting back*] A coin. Yes. [*He turns pages in the ledger.*] Bob was no better at waiting than I am. Here — many a morning he peeked at his stone through his viewer before the letter came.

Achilles: [*reading down the column*] High. Low. Low. High. Low. High... And here, where Alice had sent something quite different — high, low, high, low. A coin toss, every time, whatever Alice had sent.

Tortoise: Which bend was she on, Achilles — the plain one, the flip, the spin, or both? Bob can't know. Four ways turned, and all four together look like a coin.

[*Against the teapot, with the white window behind it, the letter from Ostend has gone half transparent in the glare, and the cousin's hand shows through envelope and folded sheet alike — the whole of it at once. But every fold has laid the writing down a different way: some of it the right way round, some mirror-wise, some*

upside down in its mirror, and some simply standing on its head; and laid one over another the lines make only an even grey lattice that no one could read a word of. It is all there. It says nothing at all.]

Achilles: …Oh. It's there before the letter — and it's no use to him.

[*He sits back in his chair and blows on his tea, and for a moment nobody says anything. The kettle murmurs on the hob; the Tortoise draws her quilt up round her.*]

Achilles: Then what does the letter carry? Not the stone — two tokens?

Third Challenge — *The Letter*

[*The lower half of the window has given in to the fire. Its small panes are clear now, clean glass from bar to bar, each with its own sharp little picture of the lane — the white road, the hedge-tops heavy with rime, the stable roof across the way — while the upper panes still keep their ferns, and the frost stops exactly at the wood of the bars, as tidy as a border in a book. The Crab tops up the pot from the kettle, sets it back by the letters, and tucks its cosy round it.*]

Crab: Two tokens, just as they fell; that's all Alice ever posts. [*He shakes his cousin's letter out of its envelope and holds the empty envelope up.*] This will do for a letter. The tokens go in exactly as they lay, the flap goes down, and the road can look at them all it likes — they're settled already, and looking can't unsettle them.

Crab: And she writes the same two words on the back of every one, in case Bob forgets. [*He writes across the back of the envelope: brass first.*]

[*The Crab tips the two tokens out of their trays into the dish; they land as they lay, both blank side up.*]

Crab: Here's one of hers from the autumn. [*At the ledger's line he turns the brass token face-up in the dish, and then the copper: both lie stamped. He slides them into the envelope, faces as they are, and folds the flap down over them.*] Postman?

Achilles: [*taking it*] The long way round, I suppose — not over the black.

[*He carries the envelope round the edge of the table: past the window end, down the long side, never once over the slab, and lays it on the cloth at Bob's end. Inside, the tokens have not stirred: brass up, copper up, just as they were laid.*]

Achilles: [*opening it*] Brass up: flip. Copper up: spin. And on the back, brass first.

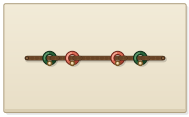


opened at Bob's end

Tortoise: Let Bob draw corrections, then, just as the letter says.

Achilles: More rings? Then the message will end up more turned than ever.

[*The straight string still lies as the pull left it: from the message end, a green ring, then a red. Now the Tortoise threads two more on from Bob's end. First a red ring, which she slides up the string until it sits a little way below Alice's red; then a green ring, which she leaves nearer Bob's hand. Along the whole string, from the message to Bob: green, red, red, green — Alice's two and Bob's two, facing each other across the middle.*]



as the letter says

Tortoise: Let Bob draw kin to kin.

[*She slides Alice's red ring down the string towards Bob's red. The string holds still; only the ring travels, and there is nothing in its way — no green between the two reds. They touch, and run together into one ring: two beads, each halfway round, come to rest as one bead gone all the way round, back at the top, where no turn at all sits. A red ring with no turn on it, one string in and one out, is only string; the Tortoise lifts it off and drops it in her pouch. Now Alice's green has a clear run down to Bob's green. She slides it; they meet; the two halves make a*

whole; she lifts that off too. The string lies bare from the message end to Bob's end — nothing on it at all, and no longer or shorter than before.]



red to red, lifted off



green to green, and bare

Tortoise: A whole turn is no turn.

[*Across the lane the weathercock on the stable roof, which froze in the night with its tail to the weather, is loosed by the first of the day's warmth and caught by the morning's first small breeze. It goes round with a creak that carries clear over the snow — half round — and stops facing exactly as it faced yesterday, as though it had never been turned at all.*]

Achilles: Straight from the message to Bob, and nothing on it. He has it exactly.

Achilles: But brass first — must it be? A flip and a spin are a flip and a spin, whichever you do first.

Tortoise: His stone could never tell the difference. But on the string, only this way round does each ring meet its own kind; spin first, and a green would have to get past a red to reach its partner.

[*Achilles brings the envelope back round the table's edge to the near end, and the Crab tips the two tokens out of it into the dish. They slide out just as they went in, brass up, copper up; only their place has changed.*]

Crab: And the other letters? [*He lays out three more ledger lines in the dish, one after another, and at each the Tortoise re-threads the string: Alice's rings as the tokens fell, Bob's as the letter says.*]

[*Brass alone: one red ring left on the string by Alice's reading, one red put on by Bob; they slide together, make their whole turn, and come off. Copper alone: green to green, and off. Both down: nothing on the string from Alice, nothing for Bob to put on, and the string bare from the start. Every pattern, and every time the string*

lies clean from the message to Bob.]

Crab: [*turning a page*] And here are the mornings Alice set her stone herself, some way of her own choosing, and wrote afterwards to tell Bob how. When he had turned his stone as the letter said, he looked at it that way through his viewer. It matched. Every time, whatever the tokens were.

Crab: [*closing the ledger with both claws*] I am so glad. I always liked Bob. And the letter went by the post, and the stone never had to.

[*The Crab turns both tokens blank side up — the last letter left them so already — and tips them back into the glass's trays, one to each, where they lay when the glass was unwrapped.*]

[*He blows his nose, rather loudly, into a large spotted handkerchief, and the Tortoise passes him the toast. Achilles laughs under his breath and helps himself to the last slice before anyone can stop him.*]

Achilles: Two tokens to carry one stone's worth. Could it go the other way — one stone to carry two tokens' worth?

Fourth Challenge — *The Other Way Round*

Crab: It can, and they did — last spring, when Alice had two tokens' worth of news and nothing but a stone to send it by. [*He tips the tokens out of their trays into the dish, both blank side up.*] She set her news out first, by hand.

[*Both tokens lie blank. The Crab, for Alice, turns the copper one over: it shows its stamp, and the brass lies blank beside it. Nothing has fallen from any glass; this time the tokens were chosen.*]

Crab: Then she turned her half of the pair as her tokens said — flip if the brass was up, then spin if the copper was. So: no flip, and a spin. And she sent Bob the *stone*, by the workshop's carrier, slow and careful.

[*Through the clear lower panes the milkman can be seen in the lane, on foot this morning because of the snow, stopping at the house opposite to look at the card in its front window: a square of pasteboard that the house turns over, or turns round,*

or both, or leaves as it is, to tell him the whole of what it wants without a word written. He reads it at a glance, by the way it was agreed between them long ago, and takes his cans up the path.]

Achilles: And Bob? I suppose his glass drops any of the four, a quarter of the time each, the way Alice's did.

Crab: Bob set her stone and his own in his glass, and it read them together. [*He finds Bob's line in the ledger, reads it, and does not touch the tokens: they already lie as Bob's glass dropped them.*] Copper up, brass down. The very tokens she chose. [*turning pages*] And the next letter, and the next — whatever she set, his glass dropped the same. Never once otherwise.

Achilles: Flip if the brass is up, then spin if the copper is. That's exactly what Bob did with the letter. ...Oh. It's the letter's own table. The half-turns Bob used to mend, Alice uses to write.

[*As he says it a bright fleck of sun, thrown in off the milkman's cans as he comes back down the path opposite, slides across the wall and over the glass of the print, and for a breath the upright mirror in it is lit as if from within. Then the fleck has gone on, and the wall is only white again.*]

[*Achilles glances up at the print on the wall facing the slab — Escher's* Magic Mirror*: a mirror standing upright, and creatures stepping out of it.*]

Achilles: Stepping out of the looking-glass, as solid as anything that went in.

Tortoise: [*taking up the bare string from the cloth*] I'll draw this one myself. A loop of string has no loose end, so nothing comes out of it but a yes or a no.

[*She lays a bend on the cloth — the pair, Alice's leg along the top and Bob's along the bottom. On Alice's leg she threads Alice's writing: no red, for the brass lay down, and a green ring for the copper. Further along the same leg she threads the answer Bob's glass is being asked about — copper up, brass down: one green ring. Then she brings the far end round in a second bend, back to meet Bob's leg, and ties it off. A closed loop of string, no loose end anywhere, with two green rings side by side on it.*]



green beside green

Tortoise: Kin to kin.

[*She slides the answer's green ring along the string to Alice's green. They touch; the two half-turns make a whole; she lifts the ring off. The loop lies bare, closed, nothing on it anywhere.*]



lifted off, the loop bare

Tortoise: A clean loop: a sure yes. Now ask Bob's glass about brass up instead.

[*She threads Alice's green back on, and in place of the answer's green she puts a red ring, close in by the far bend. Then she slides the red round the loop — round the far bend, along Bob's leg, round the near bend and up Alice's leg — until it sits beside the green. It is still red, its bead still halfway round, with no red anywhere to meet it; and the green has no green to meet. Nothing comes off.*]



round the loop, and nothing meets

Tortoise: A loop with a ring left over is a no: that answer never comes. Whatever Alice writes, only her own answer can.

[*Her quilt has slipped off one shoulder while she worked; the Crab, passing behind her chair to see the loop better, puts it back without a word, and she pats his claw.*

The fire snaps and settles. Above them the frost on the upper panes has thinned to a grey lace, with blue sky showing through the holes in it.]

Achilles: [*looking from one bend of the loop to the other*] And the two bends. That one is Alice writing, and this one is Bob's glass reading — the same bend with the same rings, only facing back the other way.

Crab: Don't think it came for nothing, mind. The carrier took Bob his half of the pair long before; one stone travelled then and one now. Two stones for two tokens' worth.

Achilles: And if the stone Alice sends is *itself* half of a pair?

Fifth Challenge — *The Mended Junction*

Crab: [*a long look at the near end of the slab: the cloth, and the bare stone beside it*] Then it's this one. My keepsake's twin — half of a pair, and the other half under that cloth. [*He straightens.*] We'll do it for real, on the table, with the last of the drawer. This end is Alice's. You shall be Bob, Madam Tortoise, down at his end, and do exactly what Bob would do. And Achilles knows the road.

Tortoise: [*untying her loop and winding it up, its rings into her pouch, then going down the long side of the table to the far end and settling there beside the lone stone, the drawing-cloth running along the slab beside her*] I'm at home.

[*The Crab has Achilles carry her cup down to her the long way round, by the table's edge, and she sets it at her elbow on the table beside her cloth, well clear of the black. Down there at the far end, with her quilt round her and her tea beside her, she looks very small and entirely comfortable.*]

[*The tokens still lie in the dish as Alice set them: copper up, brass down. The Crab turns the copper one back onto its face, so that both lie blank side up again, and tips them into the glass's trays, brass to the left, copper to the right, where they lay at the start of the morning. The envelope, empty, lies ready beside the dish.*]

Crab: [*putting a claw to the message stone, and hesitating*] The message, and Alice's half. Into the glass.

Achilles: Then your keepsake is left with no partner at all.

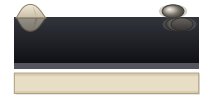
[*At the near end the message stone and Alice's stone lie on the black, each with its reflection under it, answering it slack, a beat late and a little wide, as a half of a pair's always is. The Crab lifts the message stone and sets it in the left cup of the glass; as it leaves the black its reflection sinks away beneath it and is gone. The glass waits. He lifts Alice's stone into the right cup, and its reflection goes the same way. Under the one lens the two sit side by side, and the moment the second settles the glass chimes, clear and small. Both trays tip at once; the tokens slide off and ring down into the dish — the brass face-up, the copper face-up — and lie there, off the black, casting nothing. In the cups the two stones sit still: read together, and spent. At the far end Bob's stone has not moved, and nobody has touched it; nothing about it shows anything at all.*]



as they lay all morning



chimed, and tipped



read, and Bob's untouched

Tortoise: The letter never went, and then the glass chimed.

Achilles: It hasn't gone — I haven't touched it. And the glass chimed first; we all heard it. You've said it back to front, or inside out, or both — and you can't tell which, down there.

Tortoise: Not from here.

Crab: [*very quietly, looking down the length of the slab*] It's there now, isn't it. Down at that end, this very minute — my keepsake's partner already, only turned. And I cannot so much as glance at it.

Tortoise: One glance and you'd see a coin, and have a line in your ledger where you had a pair.

Crab: [*folding his claws together, and keeping them folded*] The only pair ever made by post in this house. I shan't look.

[*The glass stands at the near end with its work done. The small singing in its brass has not come back since the chime: it is a handsome thing of brass and ground glass with its trays tipped empty, no more, and the room is a little plainer for it. Nobody looks down the table. The Crab keeps his eyes on the fire, and his claws stay folded.*]

Achilles: Then the post had better come.

[*The tokens lie in the dish where they fell, brass up, copper up. Achilles lifts them out one at a time by their edges, keeping each on exactly the face it fell on, and lays them in the envelope side by side, above the two words on its back. He folds the flap. Nothing about them has changed but where they are.*]

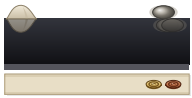
Achilles: The glass chimed, and then the letter went.

[*He carries it the long way round: along the table's edge, round the window end, down the far side, never once over the black. At Bob's end he lays it on the cloth in front of the Tortoise.*]

Achilles: [*opening it*] Flip. Spin.

[*Bob's stone sits at the far end as it has sat all morning, its reflection beneath it. The Tortoise keeps her eyes on the two tokens in the open envelope and puts out a claw to the stone by feel. She turns it over, top to bottom, about a level line through its middle: what lay uppermost goes down against the black, and what lay against the black comes up. Beneath it the reflection turns over too, mirror-fashion, and meets it again face to face at the surface. The stone has not left its spot and no one has looked at it; only which way up it lies has changed.*]

[*Then, still by feel, she turns it half round where it stands, about its upright: the side that faced the window now faces the room, and its top stays its top. Its reflection turns the same half round beneath it. The stone is where it was, and still unread; only the way it faces has changed, and with no mark on it, it looks just as it did. She tips the two tokens out of the envelope onto the cloth beside her, faces as they fell, brass up, copper up, and hands the empty envelope up to Achilles.*]



turned by feel,
and unread

Tortoise: The glass chimed, and then the letter went.

[*At the window the last ferns on the upper panes have begun to burn gold along their top edges: the sun, coming round the corner of the house, is almost on the glass. The whole room lightens by a shade, as if someone had opened a door somewhere.*]

Tortoise: Let Bob draw end to end. [*She lays her string straight along the drawing-cloth, from opposite the keepsake under its cloth at the near end, the whole length of the slab, to opposite Bob's stone at the far: one line, no bend in it, nothing on it.*]

Achilles: Your keepsake, and Bob's stone. One piece now — and they were never in the same hand, never even at the same end of the room. Nothing crossed the room but a letter. And the answer was waiting there before it came.

[*Out in the garden, between the house and the lane, the old pear against the wall stands up black out of the snow, and among its own boughs is one that came from a tree clear across the town, set into it by some grafter in another winter. The join grew over long ago. Nobody could find now where the one tree stops and the other begins; in the spring the whole of it comes into leaf together.*]

Crab: [*opening the ledger once more, at the back*] Alice and Bob made pairs like that for their friends, now and then, the same way. Whatever tokens fell, once the letter was obeyed, every one of them read afterwards as one piece. The winter the letters went astray in the snow, the pairs read like two strangers' coins.

Crab: [*closing it*] And these two I shall never read. The last pair in the house, and it never met. [*He goes to the dresser and looks into the drawer, which is empty, and shuts it.*]

Tortoise: [*coming back up the long side of the table*] Then you have a thing

worth more than a line in a book, Mr Crab. Most people only have the book.

[*The sun has come round the corner of the house at last and stands full on the window, low and gold, and the ferns on the upper panes darken at their edges and begin to gather themselves into beads of water, pane after pane, until the lane shows through the whole window, dazzling. Warmth comes in with the light, and for the first time this morning the room has a sunny side and a cool one.*]

Crab: There's tea left. Hot, too — look, the sun's got round to the window, and your frost is running, Madam Tortoise.

Tortoise: It will draw it again tonight. I'll take more lessons.

[*The Crab pours; Achilles, who has been standing at Bob's end with the empty envelope in his hand, comes back to his chair and takes his cup. For a while nobody reaches for anything on the table.*]

Achilles: It was there before the letter came.

Tortoise: And no use to anyone until it did.

Achilles: [*turning the empty envelope over, to the two words on its back*] Then what was it I carried?

[*Nobody answers him, and for the moment nobody needs to. The sun lies along the near end of the room, low and warm, and the window is clear from top to bottom, a few beads of water still standing on the glass; beyond it the lane is white and quiet, and the postman's footprints go away down it towards the next gate. On the black the keepsake keeps to its cloth at the near end, and far down at the other end Bob's stone sits alone; neither has been read, and neither will be. Beside the slab the pairing-glass stands silent, the two stones that went into it still in its cups, off the black; on the cloth at the far end the brass token and the copper lie out of their envelope, off the black. The mittens on the fender have dried. On the wall facing the slab the print hangs in the ordinary daylight — a mirror, and its creatures, in a plain frame — one more thing in a sunny room. Achilles sits with the empty envelope on his knee, the Crab fills the Tortoise's cup and then his own, and the kettle, set back from the fire, ticks as it cools; the frost, which will be back tonight, has left the window to the sun.*]

Chapter 10: Teleportation

— Out-of-Order Canon —

Part III: Pictures Become Proofs

Escher: "Magic Mirror" — a mirror standing upright, and creatures stepping out of it

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Can a thing that nobody is allowed to look at be sent to someone far away, when the only thing the post will carry is a short letter?

Suppose you hold a single qubit in a state you do not know, and a friend in another city needs it. You cannot look at it and write down what you saw. Chapter 9 taught us what a look does: it forces one answer out of several, at random, and destroys the rest, so what you wrote down would be a coin toss and not the qubit. Nor can you put the qubit itself in the post. Say the road is rough, and anything quantum that travels it is looked at a thousand times on the way. What you can send is a letter: a few definite marks on paper, which no amount of looking can spoil, because they are settled already. And the qubit is not a few marks. It is a pair of complex numbers, a point anywhere on a sphere; to write it out exactly would take digits without end.

So the task looks impossible, and it is exactly what this chapter does. It needs one extra thing, arranged long before: a shared pair, one half each, the bent wire of Chapter 6. With that in hand, a letter carrying *two bits* is enough.

The book has owed this chapter for some time, and it owes it three promises. In Chapter 6 a cap was a postselected outcome, "the answer 'Φ^+' to a

question with four possible answers", and we wrote: "Chapter 10 will show what to do about the other three, and the result will be teleportation." At the end of the same chapter, joining two pairs at a junction worked one time in four, and we said that each of the other answers "leaves Alice and Bob with a pair that is a known one-box correction away from $|\Phi^+\rangle$ ... That is Chapter 10." And Chapter 9 closed on the recipe itself: thread a cup, keep the outcome, send it across, turn the far half by the correction it names, "and the postselected cap of Chapter 6 becomes a protocol that works every time."

There is one strange fact to watch for. The moment the question is asked at the near end, the message is *already* at the far end, before any letter has been written. It arrives turned, though, by one of four half-turns, and nobody at the far end can tell which. The letter comes afterwards and names the half-turn to undo. The answer arrives before its cue, and the cue sets it right. We shall see why it cannot be used early, and why, once the letter arrives, the picture proof is a wire pulled straight and two pairs of rings lifted off.

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§10.1 Four Answers to One Question

Chapter 6's cap asked two wires one question — "are you $|\Phi^+\rangle$?" — and kept only the answer yes. A real device asked about two wires must say *something* every time. What are the other things it can say?

On the blackboard. We need two of the Pauli matrices of Chapter 3: the bit-flip $X = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and the phase-flip $Z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. Each is its own inverse, $X^2 = Z^2 = I$, and each is real and symmetric. For a bit $m \in \{0, 1\}$ write X^m and Z^m for "do it if $m = 1$": $X^0 = Z^0 = I$, $X^1 = X$, $Z^1 = Z$. The two do not commute; one line of multiplication gives $XZ = -ZX$.

Chapter 6 built $|\Phi^+\rangle$ as CNOT $(H \otimes I)\,|00\rangle$. Feed the same little circuit the other three inputs. For two bits m_1, m_2 define

$$|B_{m_1 m_2}\rangle \;=\; \text{CNOT}\,(H \otimes I)\;|m_1 m_2\rangle\,.$$

Take $|11\rangle$. The Hadamard turns the top $|1\rangle$ into $|-\rangle$, giving $\frac{1}{\sqrt{2}}(|01\rangle - |11\rangle)$; the CNOT flips the bottom wire where the top reads 1, giving $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. The four together:

$$|B_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle, \qquad |B_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = |\Psi^+\rangle,$$
$$|B_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = |\Phi^-\rangle, \qquad |B_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) = |\Psi^-\rangle.$$

These are the four **Bell states**. Now look at them against $|\Phi^+\rangle$. $|\Psi^+\rangle$ is $|\Phi^+\rangle$ with the top bit flipped; $|\Phi^-\rangle$ is $|\Phi^+\rangle$ with a sign on the $|11\rangle$ branch, a phase-flip on the top; and $|\Psi^-\rangle$ is both. Check the last: $(ZX \otimes I)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(Z|1\rangle \otimes |0\rangle + Z|0\rangle \otimes |1\rangle) = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$. Each of the four is the first one with a Pauli on one leg.

Result 10.1 (the Bell basis). For bits m_1, m_2, exactly,

$$\begin{aligned} |B_{m_1 m_2}\rangle &= \text{CNOT}\,(H \otimes I)\,|m_1 m_2\rangle \\ &= (Z^{m_1}X^{m_2} \otimes I)\,|\Phi^+\rangle = (I \otimes X^{m_2}Z^{m_1})\,|\Phi^+\rangle. \end{aligned}$$

The four are orthonormal and complete: $\langle B_j|B_k\rangle = \delta_{jk}$ and $\sum_k |B_k\rangle\langle B_k| = I$.

Proof idea. The circuit is a unitary, so it carries the four orthonormal inputs $|m_1 m_2\rangle$ to four orthonormal outputs, and four orthonormal vectors in a four-dimensional space leave no room for a fifth. The Pauli forms are the table read off, and the bottom-leg form is Chapter 6's slide.

Proof. The first equality is the definition, and the second is the table above, checked in all four cases. For the third, slide the box off the top leg of $|\Phi^+\rangle = \frac{1}{\sqrt{2}}$ cup (Theorem 6.4): it arrives on the bottom leg transposed, and $(Z^{m_1}X^{m_2})^T = (X^{m_2})^T(Z^{m_1})^T = X^{m_2}Z^{m_1}$, since X and Z are symmetric. Orthonormality: CNOT $(H \otimes I)$ is unitary. Completeness: four orthonormal vectors in a four-dimensional space are a basis. ■ See `Code/Ch10/FourAnswers.py`.

A **Bell reading** is the question "which of the four?". Its four answers are the four effects that are the mirror images of the Bell states (Chapter 3: the mirror is the dagger, the conjugate transpose). Taking the dagger of $(Z^{m_1}X^{m_2} \otimes I)\,|\Phi^+\rangle$ reverses the order of the product, and the Paulis are their own daggers, so

$$\langle B_{m_1 m_2}| \;=\; \langle \Phi^+|\,(X^{m_2}Z^{m_1} \otimes I) \;=\; \langle m_1 m_2|\,(H \otimes I)\,\mathrm{CNOT}.$$

The second form says how to build the reading: run the circuit backwards (CNOT, then H on the top wire) and then look at both wires with the plain look of Chapter 9. The bit m_1 is read off the Hadamarded top wire, m_2 off the bottom. By the Born rule (§9.1), a two-wire state $|\Omega\rangle$ gives answer $m_1 m_2$ with probability $|\,\langle B_{m_1 m_2}|\Omega\rangle\,|^2$. Chapter 6's cap was $\sqrt{2}\,\langle B_{00}|$, the first of these four, and nothing more.

Why this definition. Why define the Bell states by a circuit, when we could simply list four vectors? Because a definition should say how to build the thing, and run backwards this circuit *is* the reading. And the four vectors it produces turn out to be Chapter 6's one bent wire with a Pauli on one leg. That is what will make every one of the "wrong" answers correctable.

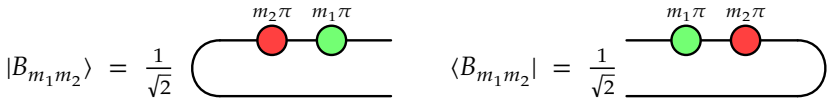
In the sketchbook. Chapter 8 gave us the two Paulis as dots. A red dot with one leg in, one leg out and phase π is exactly X, and a green one is exactly Z (§8.2, Dirac's Blackboard). We now need a small new mark: a **ring whose phase is a bit times** π. Write $m\pi$ beside a green or red dot on a wire, with m a bit left open. For $m = 0$ the phase is 0, and a phase-free dot with one leg in and one out is a bare wire (§8.1); for $m = 1$ it is the half-turn. So one drawing stands for two, exactly:



A picture with two such rings stands for four pictures. Labelling a phase by an unknown bit is how van de Wetering's survey draws teleportation (§5.4), and it is what lets one drawing carry all four answers of a Bell reading.

With the ring in hand, Result 10.1 draws itself. The cup is, as always,

unnormalized, cup $= |00\rangle + |11\rangle$, so $|\Phi^+\rangle = \frac{1}{\sqrt{2}}$ cup; and a Bell state is that cup with a red ring $m_2\pi$ and then a green ring $m_1\pi$ on its top leg, read outward from the bend:



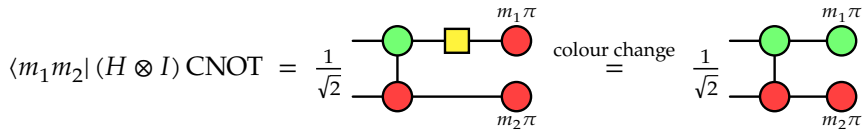
The effect on the right is the mirror image of the state on the left: reflected left to right, so the cup becomes a cap and the order of the rings is reversed, and every phase conjugated, which changes nothing here, because $e^{-i\pi} = e^{i\pi} = -1$. A half-turn ring is its own mirror image.

Why this definition. Why label a phase by a bit rather than draw four pictures? Because then a proof drawn once is four proofs. The only fact about the label that any rewrite will ever use is that two of them add to $2m\pi$, a whole turn, which is no turn at all.

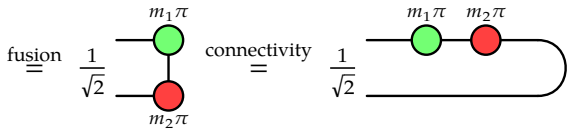
The reading, drawn from its circuit

We have *asserted* that the reading's effect is a cap with two rings. The reading we can build is the circuit $\langle m_1 m_2| \, (H \otimes I)\,$CNOT. The claim that the two agree is a real debt, and we pay it with Chapter 8's rules, keeping every number.

Three translations, all from Chapter 8. The CNOT is a green dot on the top wire tied by a cord to a red dot on the bottom, and the bare picture is CNOT$/\sqrt{2}$ (row 64); so CNOT is $\sqrt{2}$ times that picture. A red dot with one leg and phase 0 is the state $\sqrt{2}\,|0\rangle$, and with phase π it is $\sqrt{2}\,|1\rangle$ (§8.2, §8.3); its mirror image, a red dot with one leg coming *in*, is $\sqrt{2}\,\langle m|$ when its phase is $m\pi$. So $\langle m|$ is $\frac{1}{\sqrt{2}}$ times a one-legged red effect. And H is the yellow box. The numbers in front multiply to $\sqrt{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$:



The first step is colour change (Theorem 8.2): a red effect with a yellow box on its one leg is a green effect of the same phase. Now each wire carries two dots of one colour in a row, and fusion (Theorem 8.1) merges them, adding the phases, $0 + m_1\pi$ on top and $0 + m_2\pi$ below:



What remains is a green dot with one leg in and one out, joined by the cord to a red dot with two legs in and none out. That red dot is $\langle{+}{+}| + e^{im_2\pi}\langle{-}{-}|$, which is $\langle 00| + \langle 11|$ for $m_2 = 0$ and $\langle 01| + \langle 10|$ for $m_2 = 1$: a cap, carrying a red ring $m_2\pi$ on one leg. Since only the connectivity matters (§8.5), the whole is a single wire running in at the top, through a green ring and a red ring, and back out at the bottom: the cap of the reading.

Result 10.2 (the reading is a ringed cap). For each of the four outcomes, exactly,

$$\langle m_1 m_2| (H \otimes I)\,\mathrm{CNOT} \;=\; \langle B_{m_1 m_2}| \;=\; \frac{1}{\sqrt{2}}\,\mathrm{cap}\,(X^{m_2}Z^{m_1} \otimes I),$$

and the picture derivation above uses only colour change and fusion, both exact, with the scalar $\frac{1}{\sqrt{2}}$ carried from the two translations. See `Code/Ch10/FourAnswers.py`, which checks the circuit form against the ringed cap for all four outcomes, before and after reducing it by the rules.

Coecke's Sketchbook.

Rings round a bend. A single half-turn ring slides round a bend and arrives unchanged: sliding transposes (Theorem 6.4), and $X^T = X$, $Z^T = Z$. Two rings in a row do something that is easy to miss. The transpose of a product reverses it, $(BA)^T = A^T B^T$, so a chain slid round a bend arrives in the *opposite order*. That is Result 10.1's third form: red then green on the top leg is green then red on the bottom leg. Abramsky and Coecke noticed the same thing in the abstract, remarking on a "seemingly 'acausal' " inversion of the order in which things are applied along a bent wire (their §3.3). The order matters here because a red half-turn and a green one do not commute: $XZ = -ZX$. Passing one through the other costs a sign (Theorem 8.4 with $\alpha = \pi$). Every rewrite in this chapter is planned so that no ring ever has to pass a ring of the other colour.

---

§10.2 Two Bits Early

Now the scene. Alice holds a qubit on wire 1 in a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ that she does not know. She also holds wire a, one half of a pair $|\Phi^+\rangle$; the other half, wire c, is with Bob, far away. The wires are drawn top to bottom $1, a, c$. Alice performs a Bell reading on her two wires, 1 and a. What happens to Bob's?

On the blackboard. Write the three wires out:

$$|\psi\rangle \otimes |\Phi^+\rangle \;=\; \tfrac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle).$$

To see what the reading does we must write Alice's two wires in the Bell basis. Reading the table of §10.1 backwards,

$$|00\rangle = \tfrac{1}{\sqrt{2}}(|\Phi^+\rangle + |\Phi^-\rangle), \qquad |11\rangle = \tfrac{1}{\sqrt{2}}(|\Phi^+\rangle - |\Phi^-\rangle),$$
$$|01\rangle = \tfrac{1}{\sqrt{2}}(|\Psi^+\rangle + |\Psi^-\rangle), \qquad |10\rangle = \tfrac{1}{\sqrt{2}}(|\Psi^+\rangle - |\Psi^-\rangle).$$

Substitute into the four terms, which become eight, and gather them by Bell state (the sidebar below writes every term):

$$\begin{aligned}|\psi\rangle \otimes |\Phi^+\rangle \;=\; \tfrac{1}{2}\Big[\; & |\Phi^+\rangle\,(\alpha\,|0\rangle + \beta\,|1\rangle) + |\Psi^+\rangle\,(\alpha\,|1\rangle + \beta\,|0\rangle) \\ & + |\Phi^-\rangle\,(\alpha\,|0\rangle - \beta\,|1\rangle) + |\Psi^-\rangle\,(\alpha\,|1\rangle - \beta\,|0\rangle)\Big].\end{aligned}$$

Look at what stands next to each Bell state. Next to $|\Phi^+\rangle$ is $|\psi\rangle$ itself. Next to $|\Psi^+\rangle$ is $X|\psi\rangle$, flipped; next to $|\Phi^-\rangle$ is $Z|\psi\rangle$, phase-flipped; next to $|\Psi^-\rangle$ is $XZ|\psi\rangle = X(\alpha\,|0\rangle - \beta\,|1\rangle)$, both. The message is on Bob's wire in every branch, turned by the Paulis that the branch's own Bell state carries.

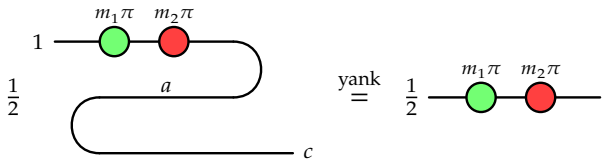
> **Result 10.3 (the message arrives turned).** For every one-qubit state $|\psi\rangle$, exactly,
>
> $$|\psi\rangle_1 \otimes |\Phi^+\rangle_{ac} \;=\; \tfrac{1}{2}\sum_{m_1,m_2} |B_{m_1m_2}\rangle_{1a} \otimes X^{m_2}Z^{m_1}\,|\psi\rangle_c\,.$$
>
> So Alice's reading gives each of the four outcomes with probability $\frac{1}{4}$, whatever $|\psi\rangle$ is, and outcome m_1m_2 leaves Bob's wire in $X^{m_2}Z^{m_1}|\psi\rangle$.

Proof idea. A Bell state is a bent wire carrying the outcome's rings. Asking whether Alice's wires stand in that Bell state bends the message wire into Bob's, rings and all.

Proof (blackboard). The calculation above, with α and β left free, is the whole proof: it holds for every $|\psi\rangle$ at once. Each branch has amplitude $\frac{1}{2}$ times a unit vector, since Paulis preserve length, so its probability is $\frac{1}{4}$. ■ See `Code/Ch10/TwoBitsEarly.py`.

In the sketchbook. On the branch where the reading says m_1m_2, draw the scene. The input comes in on the top wire. Alice's answer is the ringed cap of Result 10.2, closing the top wire onto the middle one. Bob's pair is

a cup on the middle and bottom wires. Two factors of $\frac{1}{\sqrt{2}}$, one from the reading and one from the pair, make the $\frac{1}{2}$ in front. Bob's wire runs out at the bottom. The wire makes an S, and Theorem 6.2 pulls it straight:



That is Result 10.3 for all four branches at once: the message comes out on Bob's wire, times $\frac{1}{2}$, still wearing Alice's rings, $X^{m_2}Z^{m_1}|\psi\rangle$. It is there *already*, at the instant of the reading, and no letter has been sent.

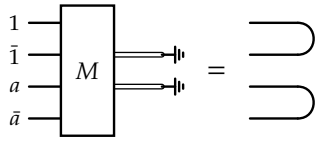
It is also no use to Bob. He does not know which branch he is on, and the four branches are equally likely. What he holds, honestly described, is the average.

> **Result 10.4 (nothing before the letter).** Before the outcome reaches him, Bob's wire is in the maximally mixed state $\frac{1}{2}I$, for every input, pure or mixed. Nothing Alice sends changes what Bob holds until the letter arrives.

Proof (*blackboard*). The input may as well be a density matrix $\rho = \left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right)$, with $r + u = \operatorname{tr}\rho = 1$; the pure case is $\rho = |\psi\rangle\langle\psi|$. By Result 10.3 and linearity, branch m_1m_2 leaves Bob with $\frac{1}{4}P\rho P^\dagger$ where $P = X^{m_2}Z^{m_1}$, so Bob's average is $\frac{1}{4}(\rho + Z\rho Z + X\rho X + XZ\rho ZX)$. Now $Z\rho Z$ flips the signs of s and t, so $\rho + Z\rho Z = \operatorname{diag}(2r, 2u)$; and $X(\cdot)X$ exchanges the two diagonal entries, so the sum of all four is $\operatorname{diag}(2r + 2u,\ 2r + 2u) = 2I$. A quarter of that is $\frac{1}{2}I$. ■

Proof (*sketchbook*). Not knowing the outcome is forgetting it. The reading is the circuit $(H \otimes I)$ CNOT, which is unitary, followed by two of Chapter 9's looks. Draw the reading on doubled wires, as every real process must be drawn since Chapter 9: a box M that takes Alice's two wires and their two mirror copies, four lines in, and emits two classical wires, one per bit. To forget the outcome is to end each classical wire in the ground symbol. A look followed by forgetting its outcome is a discard (Result 9.5), and a discard after a unitary is a discard (Result 9.4, causality). So Alice's reading

with its record forgotten is simply the discard of both her wires, a cap on each wire and its mirror:



Discarding the input gives its trace, 1. What is left is a Bell pair with Alice's half discarded, and Result 9.3 says Bob's half is then $\frac{1}{2}I$. ■ See `Code/Ch10/TwoBitsEarly.py`.

So the message has arrived and cannot be read. Bob's wire, before the letter, is the same featureless grey it was before Alice did anything; Chapter 9's no-signalling theorem is at work. Teleportation sends nothing faster than light. The message is there, hidden in a four-way average, and only two bits of news can pull it out.

Dirac's Blackboard.

The eight terms, and what the outcomes say. Substituting the Bell basis into $\sqrt{2}\,|\psi\rangle\,|\Phi^+\rangle = \alpha\,|00\rangle\,|0\rangle + \alpha\,|01\rangle\,|1\rangle + \beta\,|10\rangle\,|0\rangle + \beta\,|11\rangle\,|1\rangle$ gives, times $\frac{1}{\sqrt{2}}$,

$$\begin{aligned}&\alpha\,|\Phi^+\rangle\,|0\rangle + \alpha\,|\Phi^-\rangle\,|0\rangle + \alpha\,|\Psi^+\rangle\,|1\rangle + \alpha\,|\Psi^-\rangle\,|1\rangle\\ &+ \beta\,|\Psi^+\rangle\,|0\rangle - \beta\,|\Psi^-\rangle\,|0\rangle + \beta\,|\Phi^+\rangle\,|1\rangle - \beta\,|\Phi^-\rangle\,|1\rangle\,.\end{aligned}$$

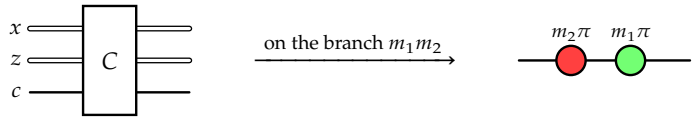
Gathered by Bell state, these are the four brackets of §10.2, with $\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}} = \frac{1}{2}$ in front. The four probabilities are $\|\frac{1}{2}X^{m_2}Z^{m_1}\psi\|^2 = \frac{1}{4}(|\alpha|^2 + |\beta|^2) = \frac{1}{4}$, and they sum to 1. Notice what does *not* appear: α and β. The outcome is a fair four-way toss whatever the message was, so the two bits Alice will write in her letter say nothing whatever about $|\psi\rangle$. The letter carries information *about the turn*, and none about the thing turned.

§10.3 The Letter and the Half-Turns

Alice writes the two bits $m_1 m_2$ in a letter and posts it. When it arrives, Bob knows he holds $X^{m_2} Z^{m_1} |\psi\rangle$. To undo a product, undo its factors in the reverse order: the inverse of $X^{m_2} Z^{m_1}$ is $Z^{m_1} X^{m_2}$, since each factor is its own inverse. So Bob applies X^{m_2} **first**, and then Z^{m_1}.

On the blackboard. Take the hardest case, outcome 11. Bob holds $\frac{1}{2} XZ |\psi\rangle = \frac{1}{2}(\alpha |1\rangle - \beta |0\rangle)$. The flip first: $\frac{1}{2}(\alpha |0\rangle - \beta |1\rangle)$. Then the phase-flip: $\frac{1}{2}(\alpha |0\rangle + \beta |1\rangle) = \frac{1}{2} |\psi\rangle$. The message, exactly, times the branch's $\frac{1}{2}$. Now do it in the other order. The phase-flip first: $Z(\alpha |1\rangle - \beta |0\rangle) = -\alpha |1\rangle - \beta |0\rangle$; then the flip: $-\alpha |0\rangle - \beta |1\rangle$. That is $-\frac{1}{2} |\psi\rangle$: the message with a minus sign in front. The other three outcomes need at most one Pauli, so order cannot matter for them.

In the sketchbook. Bob's correction is a box whose content is chosen by what the letter says. This needs one more new mark. In Chapter 9 classical wires only ever *left* something: the look emitted one. Now a classical wire runs **into a box**. A box with classical wires running into it is a box chosen by their values: on the branch where the wires carry $m_1 m_2$, the box is the process named for $m_1 m_2$. We let the classical wires run on through the box and out the other side, the record kept, since reading a definite value does not use it up. Here is Bob's correction box C, fed by the two bits of the letter, a classical wire x carrying the flip bit m_2 and a classical wire z carrying the phase bit m_1. Bob's wire c is drawn once, its mirror copy understood, as Chapter 9 drew the ground symbol on a single line:



On the blackboard the box is a sum over the branches. Drawn on doubled wires, as a real process must be (Chapter 9), with the classical record passed through,

$$C \;=\; \sum_{m_1,m_2} |m_2 m_1\rangle\langle m_2 m_1|_{xz} \otimes \left(Z^{m_1}X^{m_2}\right)_c \otimes \left(\overline{Z^{m_1}X^{m_2}}\right)_{\bar{c}},$$

where the bar is the complex conjugate, which for the real Paulis changes nothing. The two new marks meet here. The bit on the classical wire is the bit in the ring's label: *classical control* is the bit-labelled ring with its bit supplied by a wire.

Why this definition. Why not simply draw four pictures, one per outcome? Because Bob does not choose his box; the letter does. A classical wire into a box says so on the page. It records that the correction depends on a value that was settled somewhere else, and has to travel to get here.

> **Result 10.5 (teleportation, all four outcomes, exact).** Let wires $1, a, c$ run top to bottom, let $|\Phi^+\rangle = \frac{1}{\sqrt{2}}$ cup on (a,c), and let Alice's Bell reading on $(1,a)$ have effects $\langle B_{m_1m_2}| = \langle\Phi^+|\,(X^{m_2}Z^{m_1}\otimes I)$. Then for **every** one-qubit $|\psi\rangle$ and **each** outcome m_1m_2,
>
> $$\left(Z^{m_1}X^{m_2}\right)_c\left(\langle B_{m_1m_2}|_{1a}\otimes I_c\right)\left(|\psi\rangle_1\otimes|\Phi^+\rangle_{ac}\right) \;=\; \tfrac{1}{2}\,|\psi\rangle_c\,,$$
>
> exactly, the scalar included. Each outcome has probability $\|\frac{1}{2}\psi\|^2 = \frac{1}{4}$, and after the correction (X^{m_2} first, then Z^{m_1}) Bob holds exactly $|\psi\rangle$. As a channel: read, send both bits, correct and forget the bits, and the result is the identity, $\rho \mapsto \rho$, each branch contributing exactly $\frac{1}{4}\rho$.

The statement matches Abramsky and Coecke's account (their §2.1): four outcomes, four unitary corrections on Bob's qubit, and at the end Bob's qubit in the state the input began in. §10.6 proves it twice. See `Code/Ch10/Teleportation.py`, which evaluates both sides on a non-real $|\psi\rangle$ and a non-real ρ, and also checks that the wrong order gives $-\frac{1}{2}\psi$ on outcome 11.

> **Dirac's Blackboard.**
>
> *Why the order matters exactly and not physically.* With the corrections in the wrong order, outcome 11 ends in $-\frac{1}{2}|\psi\rangle$ rather than

$\frac{1}{2}|\psi\rangle$. The minus sign is a global phase, and no experiment can see it: in the doubled picture of Chapter 9 an overall phase $e^{i\theta}$ on the ket copy meets $e^{-i\theta}$ on the mirror copy, and $(-1)\cdot(-1)=1$. So physically the two orders do the same thing. But Result 10.5 is an equation between vectors, and it holds with $+\frac{1}{2}$ only when the flip comes first. Abramsky and Coecke's list of corrections shows the same sign: of their four, three are their own inverses, and the fourth is minus its own inverse (their §2.1). We fix the order once, $Z^{m_1}X^{m_2}$, and keep it for the rest of the chapter, so that every equation is exact as written.

Feynman's Notebook.

Two channels. The protocol was published in 1993 by C. H. Bennett, G. Brassard, C. Crépeau, R. Jozsa, A. Peres and W. K. Wootters, in *Physical Review Letters* 70, 1895–1899, under the title "Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels" (Abramsky and Coecke's reference [5]). The title names both ingredients: a classical channel, which is the letter, and an Einstein–Podolsky–Rosen channel, which is the shared pair. Abramsky and Coecke, opening their 2004 paper with teleportation, dwell on what the count means. The unknown qubit is an arbitrary pair of complex numbers, and yet only two classical bits cross, which they call "no mean feat". Since a continuous quantity has been moved and only two bits have travelled, they argue that there must be a quantum flow of information besides the classical one. In the sketchbook that flow is the wire, bent at the source of the pair and pulled straight by the reading.

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§10.4 The Other Way Round: Superdense Coding

Two bits in a letter moved one qubit. Can it go the other way: can one qubit sent carry two bits?

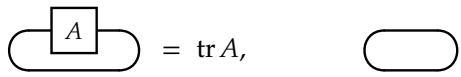
On the blackboard. Again Alice and Bob share $|\Phi^+\rangle$, Alice's half on wire a and Bob's on wire c. This time Alice has two bits k_1k_2 to send. She turns her half by $Z^{k_1}X^{k_2}$ (the flip first, then the phase-flip, as in Bob's correction table) and posts the *qubit* to Bob. Bob now holds both halves, and he makes a Bell reading on them.

Take $k_1k_2 = 10$. Alice applies Z, and $(Z \otimes I)\,|\Phi^+\rangle = |\Phi^-\rangle = |B_{10}\rangle$. Bob's reading is certain to answer 10. In general there is nothing to calculate: by Result 10.1, $(Z^{k_1}X^{k_2} \otimes I)\,|\Phi^+\rangle$ *is* $|B_{k_1k_2}\rangle$, and the four Bell states are orthonormal, so the reading tells them apart without error. The amplitude that Bob reads $j = j_1j_2$ when Alice wrote $k = k_1k_2$ is

$$\langle B_j|\,(Z^{k_1}X^{k_2} \otimes I)\,|\Phi^+\rangle \;=\; \langle B_j|B_k\rangle \;=\; \delta_{jk}.$$

There is a second way to see the same number, and it is the one the picture uses. Write the reading's effect in its ring form: the amplitude is $\langle\Phi^+|\,(X^{j_2}Z^{j_1}Z^{k_1}X^{k_2} \otimes I)\,|\Phi^+\rangle$, a Bell state and a Bell effect with one box between them on the top wire. That shape has a name.

In the sketchbook. Close a wire on itself: a cup, and then a cap on the same two wires. Nothing comes in and nothing goes out, so it is a number. We call it a **closed loop**. Put a box A on it, and the loop is the **trace** of A:



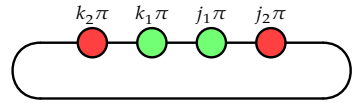
Result 10.6 (a loop is a trace). For every 2×2 matrix A, cap $(A \otimes I)$ cup $= \operatorname{tr} A$, exactly. In particular the bare loop is $\operatorname{tr} I = 2$.

Proof. cap $(A \otimes I)$ cup $= \sum_{i,k} \langle ii|\,(A\,|k\rangle \otimes |k\rangle) = \sum_{i,k} \langle i|\,A\,|k\rangle\,\langle i|k\rangle = \sum_i A_{ii}$. The bare loop was met in §9.3, where it was a system's mirror-cup discarded. ■

Why this definition. Why give a name to a picture that is only a number? Because the loop is the shape of every question of the form "does the reading return what was prepared?". A preparation is a cup, a reading is a cap, and whatever was done in between sits on the loop. Naming it the trace turns sixteen amplitudes into one fact about the traces of four matrices.

The loop is where the unnormalized cup pays us back. Because cup and cap carry no $\frac{1}{\sqrt{2}}$, a loop is a plain trace, and the Bell state and Bell effect contribute $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$, written in front. Now draw dense coding. The cup is the shared pair. On its top leg, reading outward from the bend, come Alice's rings, red $k_2\pi$ then green $k_1\pi$, and then the reading's rings, green $j_1\pi$ then red $j_2\pi$, and the cap closes the loop:

$$\langle B_j | \, (Z^{k_1} X^{k_2} \otimes I) \, |\Phi^+\rangle \;=\; \tfrac{1}{2}$$



In the middle, green meets green, and fusion adds them to $(j_1 + k_1)\pi$. The last red ring is a single ring, so it slides round the cap, along the bottom and round the cup, arriving unchanged (Theorem 6.4, twice), until it sits next to the first red ring; fusion adds those to $(j_2 + k_2)\pi$. What is left is a loop carrying one green ring $(j_1 + k_1)\pi$ and one red ring $(j_2 + k_2)\pi$. If $j = k$, both phases are whole turns, both rings are bare wire (§8.1), and the value is $\frac{1}{2}$ times the bare loop, $\frac{1}{2} \cdot 2 = 1$. If $j \neq k$, a ring is left over: a red one, a green one, or one of each. By Result 10.6 the loop is then the trace of X, of Z or of ZX, and all three traces are 0. So the read label matches the sent one with certainty.

> **Result 10.7 (superdense coding).** For all sixteen pairs of two-bit labels j, k, exactly,
>
> $$\langle B_j | \, (Z^{k_1} X^{k_2} \otimes I) \, |\Phi^+\rangle \;=\; \tfrac{1}{2} \operatorname{tr}(X^{j_2} Z^{j_1} Z^{k_1} X^{k_2}) \;=\; \delta_{jk}.$$
>
> Alice's codebook, the half-turn $Z^{k_1} X^{k_2}$ for message $k_1 k_2$, is Bob's correction table in teleportation, entry for entry. See `Code/Ch10/DenseCoding.py`.

Here is what "the mirror image" means, and it needs saying precisely, because in this book the mirror is the dagger. The same three pieces appear in both protocols: one shared pair, the Bell reading, and the four half-turns $Z^{m_1} X^{m_2}$. In **teleportation** Alice *reads* and Bob *turns*: one pair and two bits sent move one qubit. In **superdense coding** Alice *turns* and Bob *reads*: one

pair and one qubit sent move two bits. The classical and quantum channels have changed places. And the pieces themselves go into each other in the mirror. Alice's four readings in teleportation are the Bell effects $\langle B_k|$; reflected, they are exactly her four preparations in dense coding, the Bell states $|B_k\rangle = \langle B_k|^\dagger$, each a cup carrying the same rings, because a half-turn ring is its own mirror image. The dagger has exchanged the two ends of the pieces. It has *not* reversed the protocol. The dagger of teleportation's corrected branch, $\frac{1}{2}I$, is $\frac{1}{2}I$ again: a qubit going in and a qubit coming out, teleportation run backwards, which is not dense coding.

And the honest accounting. Two bits per qubit sent sounds like something for nothing, and it is not. The pair had to be shared beforehand; one of its halves travelled to Bob at some earlier time. Counting that half, two qubits carried two bits.

In the dialogue, as this sinks in, Achilles looks up at the print on the parlour wall, *Magic Mirror*: a mirror standing upright, and creatures stepping out of it.

Coecke's Sketchbook.

A mirror is not a half-turn. The reading is the dagger of the preparation, not its transpose. With the real Paulis the two cannot be told apart, so here is a test that can. Pick a unitary W that is neither real nor symmetric, and use the twisted boxes $U_k = Z^{k_1}X^{k_2}W$ in place of the Paulis, with states $|E_k\rangle = (U_k \otimes I)\,|\Phi^+\rangle$. The reading's effects are $\langle E_k| = |E_k\rangle^\dagger = \langle\Phi^+|\,(U_k^\dagger \otimes I)$, which is not $|E_k\rangle^T$. The yank of §10.2 then delivers $\frac{1}{2}U_k^\dagger|\psi\rangle$, for every unitary U_k, so the correction is U_k itself: not U_k^T, not $\bar{U}_k$, not $U_k^\dagger$. Put the box on the *other* leg instead, $|E'_k\rangle = (I \otimes U_k)\,|\Phi^+\rangle$, and sliding moves it to the top leg as U_k^T, so the correction becomes U_k^T. The transpose enters by sliding, and only by sliding, and Bob's output is always $|\psi\rangle$, never its conjugate. The companion code runs this with a random W.

Feynman's Notebook.

The mirror came first. The other way round was published a year

before teleportation: Bennett and Wiesner, "Communication via one- and two-particle operators on Einstein-Podolsky-Rosen states", *Physical Review Letters* 69, 2881–2884 (1992). Read with this chapter's eyes, the title names the two halves of the scheme: the sender acts on one particle, her half of the pair, and the receiver on two together, with a joint reading. That gloss is ours; the paper itself should be consulted for its own framing. The name *superdense coding* is the one the protocol goes by now. The two papers share an author and a resource. What one of them spends the shared pair on, moving a qubit with two bits, the other spends it on in reverse, moving two bits with a qubit.

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§10.5 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 9's seventy-two. Three introduce new picture elements: the bit-labelled phase (75), the classical wire into a box (76) and the closed loop (77). The rest are protocols built from them.

#	Algebra (Dirac)	Picture (Coecke)
73	Bell states $\lvert B_{m_1 m_2}\rangle = (Z^{m_1} X^{m_2} \otimes I)\,\lvert\Phi^+\rangle$	a cup, $\times\frac{1}{\sqrt{2}}$, with a red ring $(m_2\pi)$ then a green ring $(m_1\pi)$ on its top leg
74	the Bell reading: effects $\langle B_{m_1 m_2}\rvert = \langle\Phi^+\rvert\,(X^{m_2} Z^{m_1} \otimes I)$, outcome two bits	the mirrored cap with the rings on its incoming leg; as a box M on doubled wires, two classical wires out

#	Algebra (Dirac)	Picture (Coecke)
75	a Pauli chosen by a bit: Z^m, X^m, $m \in \{0,1\}$	a spider whose phase is a bit times π: one picture for all the cases — *new element*
76	classically controlled correction: apply X^{m_2}, then Z^{m_1}	a classical wire running into a box (classical control) — *new element*
77	$\operatorname{tr} A = \text{cap}\,(A \otimes I)\,\text{cup}$; $\operatorname{tr} I = 2$	a closed loop through a box A is its trace; the bare loop is the number 2 (§9.3) — *new element*
78	teleportation: $(Z^{m_1}X^{m_2})(\langle B_{m_1 m_2}| \otimes I)(\psi \otimes \Phi^+) = \frac{1}{2}\psi$, $p = \frac{1}{4}$	the S-bend yanks; red meets red, green meets green; a bare wire $\times\frac{1}{2}$
79	superdense coding: $\langle B_j|\,(Z^{k_1}X^{k_2} \otimes I)\,|\Phi^+\rangle = \delta_{jk}$	the loop with Alice's rings and the reading's rings: clean (value 1) iff $j = k$, otherwise a red ring, a green ring or one of each is left, value 0
80	swapping corrected: $\frac{1}{4}\sum_k |k\rangle\langle k| \otimes |\Phi^+\rangle\langle\Phi^+|_{dc}$	The Mended Junction: two doubled cups, the reading box, two classical wires, the correction box; one doubled cup (d, c) on every outcome

Every row is an exact equality, scalar included; none of them needs the "up

to a non-zero scalar" of Chapter 8. Rows 78 and 80 are proved in §10.6 and §10.7.

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§10.6 The Theorem, Twice

Result 10.5 (teleportation, all four outcomes, exact), restated. For every one-qubit $|\psi\rangle$ and each outcome $m_1 m_2$ of Alice's Bell reading on wires $1, a$,

$$(Z^{m_1} X^{m_2})_c \, (\langle B_{m_1 m_2}|_{1a} \otimes I_c)(|\psi\rangle_1 \otimes |\Phi^+\rangle_{ac}) \;=\; \tfrac{1}{2}\, |\psi\rangle_c \,,$$

exactly; each outcome has probability $\frac{1}{4}$; as a channel, the protocol is the identity.

Dirac's Blackboard

Let $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

Step 1: expand. $|\psi\rangle \otimes |\Phi^+\rangle = \frac{1}{\sqrt{2}}(\alpha\,|00\rangle\,|0\rangle + \alpha\,|01\rangle\,|1\rangle + \beta\,|10\rangle\,|0\rangle + \beta\,|11\rangle\,|1\rangle)$, Alice's two wires written first.

Step 2: change basis on Alice's wires. Substitute $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ in terms of the Bell basis (§10.2), which gives eight terms (the Dirac's Blackboard of §10.2), and regroup:

$$\begin{aligned} |\psi\rangle \otimes |\Phi^+\rangle = \tfrac{1}{2}\big[\; & |B_{00}\rangle\,(\alpha\,|0\rangle + \beta\,|1\rangle) + |B_{01}\rangle\,(\alpha\,|1\rangle + \beta\,|0\rangle) \\ & + \; |B_{10}\rangle\,(\alpha\,|0\rangle - \beta\,|1\rangle) + |B_{11}\rangle\,(\alpha\,|1\rangle - \beta\,|0\rangle)\big]. \end{aligned}$$

Step 3: read. The effect $\langle B_{m_1 m_2}|$ on Alice's wires keeps exactly one bracket, because the Bell states are orthonormal (Result 10.1). Branch $m_1 m_2$ leaves Bob's wire with $\frac{1}{2}$ times its bracket.

Step 4: correct, all four branches.

$$
\begin{aligned}
00: &\quad \tfrac{1}{2}(\alpha\,|0\rangle + \beta\,|1\rangle) &&\xrightarrow{\text{nothing}}\ \tfrac{1}{2}(\alpha\,|0\rangle + \beta\,|1\rangle),\\
01: &\quad \tfrac{1}{2}(\alpha\,|1\rangle + \beta\,|0\rangle) &&\xrightarrow{X}\ \tfrac{1}{2}(\alpha\,|0\rangle + \beta\,|1\rangle),\\
10: &\quad \tfrac{1}{2}(\alpha\,|0\rangle - \beta\,|1\rangle) &&\xrightarrow{Z}\ \tfrac{1}{2}(\alpha\,|0\rangle + \beta\,|1\rangle),\\
11: &\quad \tfrac{1}{2}(\alpha\,|1\rangle - \beta\,|0\rangle) &&\xrightarrow{X}\ \tfrac{1}{2}(\alpha\,|0\rangle - \beta\,|1\rangle) \xrightarrow{Z}\ \tfrac{1}{2}(\alpha\,|0\rangle + \beta\,|1\rangle).
\end{aligned}
$$

Every branch ends in $\frac{1}{2}\,|\psi\rangle$, exactly.

Step 5: probabilities and the channel. Each branch has squared length $\frac{1}{4}(|\alpha|^2 + |\beta|^2) = \frac{1}{4}$, and four quarters make 1. For a mixed input $\rho = \sum_i p_i\,|\psi_i\rangle\langle\psi_i|$ apply Steps 1–4 to each $|\psi_i\rangle$. Branch m_1m_2 then contributes $\sum_i p_i \cdot \frac{1}{2}\,|\psi_i\rangle \cdot \frac{1}{2}\,\langle\psi_i| = \frac{1}{4}\rho$, and the four branches, with the bits forgotten, sum to ρ. The protocol is the identity channel. ■

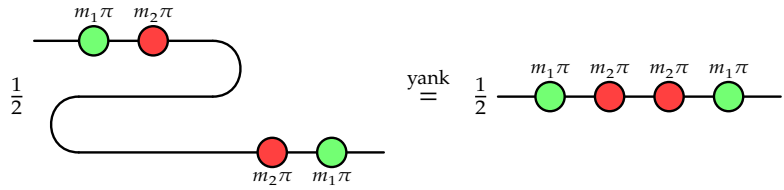
The page is honest and complete, and it shows every amplitude. What it does not show is *why* the four brackets are the message, turned. That is a fact about the Bell basis, visible only in the regrouping.

Coecke's Sketchbook

Fix a branch m_1m_2; the bit-labelled rings will carry all four at once. Every rule used below is exact.

Step 0: substitution. Alice's reading is the ringed cap, $\frac{1}{\sqrt{2}}$ in front (Result 10.2). Bob's pair is a cup, $\frac{1}{\sqrt{2}}$ in front (cup $= |00\rangle + |11\rangle$, unnormalized). Bob's correction box, on this branch, is red $m_2\pi$ then green $m_1\pi$ (row 76). The two $\frac{1}{\sqrt{2}}$ make the $\frac{1}{2}$.

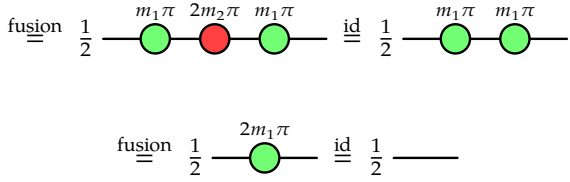
Step 1: yanking (Theorem 6.2). The cap and the cup form an S; pull it straight. The rings go with the wire, in the order the wire meets them:



Step 2: fusion (Theorem 8.1), red meets red. The two red rings are neighbours; they fuse, and their phases add to $2m_2\pi$.

Step 3: the identity rule (§8.1). $2m_2\pi$ is 0 or 2π, a whole turn either way, so the fused red dot is a phase-free dot with one leg in and one out: a bare wire. Remove it. Now the two green rings are neighbours.

Step 4: fusion, green meets green, to $2m_1\pi$. **Step 5: the identity rule** again.



A bare wire, times $\frac{1}{2}$: Bob's output is $\frac{1}{2}|\psi\rangle$ on every branch. Van de Wetering's ZX proof of teleportation ends the same way, with the remark that $2b\pi$ is either 2π or 0, both zero modulo 2π (his §5.4). The probability is the square of the drawn scalar, $(\frac{1}{2})^2 = \frac{1}{4}$; in the doubled picture it is simply the number written in front, $\frac{1}{2} \cdot \frac{1}{2}$. ■

Nothing had to slide and nothing had to pass anything else. The order of Bob's corrections, flip first and phase-flip second, is exactly what makes like rings neighbours. In the other order the straight wire would read green, red, green, red, and the middle pair could meet only by passing a green half-turn through a red one, at the cost of a sign.

The verdict

The picture wins, and it wins on both counts: it is shorter, and it shows why. One yank and two fusions settle all four outcomes at once, and the drawing makes the reason plain. The wire simply runs from Alice's input to Bob's output, and the only things ever on it are the reading's half-turns and the letter's half-turns, which meet their own kind and vanish. The blackboard needs a page and an eight-term regrouping, and at the end we know *that* each bracket is the message turned, not why.

But the picture has a debt, and we name it. That Alice's reading *is* a cap carrying those two rings is a translation: Result 10.2, a four-effect check on the blackboard or a derivation by colour change and fusion from the CNOT and the Hadamard. The $\frac{1}{4}$ is a scalar the picture carries rather than explains. And what the picture proves is an equality of processes, no more. It makes no claim about speed or about security, and it sends nothing faster than light; the letter is needed, as §10.2 showed.

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§10.7 The Mended Junction

Chapter 6 ended with a junction. Two pairs, a cup on wires d, a and a cup on wires b, c; a cap on the inner wires a, b; and the outer wires d and c, which had never met, pulled into a pair by a single yank. The cap was a postselected answer, one in four, and each of the other three, Chapter 6 said, left "a pair that is a known one-box correction away" from the one we wanted. (The history of that junction, proposed and then performed, is told in Chapter 6's Feynman's Notebook, §6.8; it is not repeated here.) This chapter adds the two things Chapter 6 lacked: the three other answers, each corrected, and the record of which answer came, carried as classical wires and drawn in the doubled form of Chapter 9.

The junction is teleportation with a particular message. The wire a that Alice reads together with her half b of the shared pair is itself half of another pair, (d, a). So the junction teleports *half a pair*. And Theorem 6.5 said a pair is a bent wire, the identity bent into a state. Teleporting half of it is

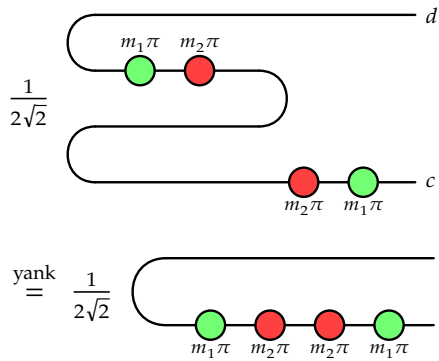
teleporting the identity; this is the headline, bent.

On the blackboard. Wires top to bottom d, a, b, c; the state is $|\Phi^+\rangle_{da} \otimes |\Phi^+\rangle_{bc}$; Alice reads (a, b); Bob corrects c. On branch $m_1 m_2$,

$$(I_d \otimes \langle B_{m_1 m_2}|_{ab} \otimes (Z^{m_1} X^{m_2})_c)(|\Phi^+\rangle_{da} \otimes |\Phi^+\rangle_{bc}) \;=\; \tfrac{1}{2}\,|\Phi^+\rangle_{dc}\,,$$

and without the correction it is $\frac{1}{2}\,|B_{m_1 m_2}\rangle_{dc}$: the pair left at the far ends is *the very Bell state the reading named*. Both follow from Result 10.5 applied to the wire a, with d carried along.

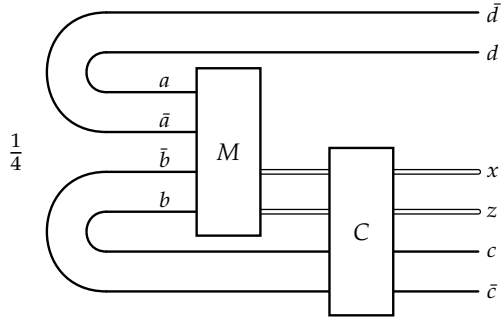
In the sketchbook. On one branch, with three factors of $\frac{1}{\sqrt{2}}$ in front (two pairs and the reading), the wire runs out of d, round the first cup, along a through the reading's rings, round the cap, back along b, round the second cup, and out along c through Bob's rings. Yank it:



Then red meets red and green meets green, exactly as in §10.6, and a bare cup on (d, c) is left: $\frac{1}{2\sqrt{2}}$ cup $= \frac{1}{2}\,|\Phi^+\rangle$. Stop before Bob's rings and the cup still carries the reading's green and red, which is $\frac{1}{2}\,|B_{m_1 m_2}\rangle$ in its bottom-leg form (Result 10.1).

Now the whole thing, every branch together, with the record kept. Two boxes on doubled wires do the work. The **reading box** M takes Alice's two doubled wires, four lines, and emits two classical wires, x carrying m_2 and z carrying m_1; on the blackboard it is $M = \sum_{m_1,m_2} |m_2 m_1\rangle_{xz}\,(\langle B_{m_1 m_2}| \otimes$

$\langle B_{m_1 m_2}|$), the Bell effect on the kets and its mirror on the mirror copies (the Bell states are real, so the conjugate changes nothing). It is the circuit of §10.1 followed by two of Chapter 9's looks. The **correction box** C is the classically controlled box of §10.3, on Bob's doubled wire, with x and z running in and on through it. The two doubled pairs are $\frac{1}{2}$ cup⊗cup each (§9.5), so $\frac{1}{4}$ is written in front of the four cups; as ever, cup $= |00\rangle + |11\rangle$.



The lines are the keepsake half d and its mirror $\bar{d}$, running straight out; the message half a and its mirror $\bar{a}$, from the first pair into M; Alice's half b and its mirror $\bar{b}$, from the second pair into M; Bob's half c and its mirror $\bar{c}$, through C and out; and the two classical wires, x for the flip bit and z for the phase bit, from M through C and out, the record kept. The mirror pairs are drawn so that their cups nest and no line crosses another. We call this diagram **The Mended Junction**.

> **Result 10.8 (the mended junction).** The Mended Junction evaluates, exactly, to
>
> $$\frac{1}{4} \sum_{m_1, m_2} |m_2 m_1\rangle\langle m_2 m_1|_{xz} \;\otimes\; |\Phi^+\rangle\langle\Phi^+|_{dc} \,.$$
>
> The record is a fair four-way toss; on every one of the four outcomes the pair (d, c) is exactly $|\Phi^+\rangle$; and the two are independent. Forget the record and (d, c) is exactly $|\Phi^+\rangle\langle\Phi^+|$. Remove the correction box and forget the record, and (d, c) is $\frac{1}{4}I$, no pair at all.

Proof idea. Fix the record's value and the diagram falls apart into the one-

branch picture above and its mirror image. Each yanks to a cup on (d, c), and what is left to do is count scalars.

Proof. On the branch $m_1 m_2$ the classical wires carry a definite value, M becomes the Bell effect on the kets and on the mirrors, and C becomes the two rings on c and their conjugates on $\bar{c}$. The doubled diagram is the pure one-branch diagram beside its mirror copy. Count the numbers: $\frac{1}{4}$ from the drawn pairs; $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$ from the normalized Bell effect on the two copies. The bare cups and ringed caps yank, and fuse, to cup $\otimes$ cup on (d, c) and $(\bar{d}, \bar{c})$, which is $2\ |\Phi^+\rangle\langle\Phi^+|$. So each branch contributes $\frac{1}{4} \cdot \frac{1}{2} \cdot 2 = \frac{1}{4}\ |\Phi^+\rangle\langle\Phi^+|$, tagged by its record $|m_2 m_1\rangle\langle m_2 m_1|$; summing over the four gives the formula. Forgetting the record adds the four quarters, giving $|\Phi^+\rangle\langle\Phi^+|$. Without the correction each branch leaves $\frac{1}{4}\,|B_{m_1 m_2}\rangle\langle B_{m_1 m_2}|$, and by the completeness of Result 10.1 the four sum to $\frac{1}{4} I$. ■ See `Code/Ch10/MendedJunction.py`, which builds the diagram in doubled form with its classical record, tallies its lines, and checks the value exactly.

So the promise of Chapter 6 is kept. The junction no longer works one time in four: it works every time, provided the letter is obeyed. The keepsake half d and Bob's half c come out as one pair, although they were never in the same hand, and no quantum thing crossed from Alice to Bob; only the two bits did. The uncorrected branches were not failures. Each was a named Bell state, one known turn away from the pair we wanted. And forgetting the record without correcting leaves $\frac{1}{4} I$, the grey of two uncorrelated coins, which is Result 10.4 again, one level up.

Coecke's Sketchbook.

Keeping the scalars. Abramsky and Coecke's own diagrammatic proofs of teleportation and of entanglement swapping say, each time, "We ignore the scalars – which cancel out against each other – in this proof" (their Appendix). In those proofs the scalars are set aside. They do keep them in the statement: their teleportation sends the input to the output along each of the four branches with a weight $s^\dagger s$, where $2 s^\dagger s = 1$, which is our $\frac{1}{2}$, "corresponding to the probability amplitude for that branch" (their §9.1). This book writes the $\frac{1}{2}$ and the $\frac{1}{4}$ into the pictures,

because here they are the whole point. They are the probabilities. A picture that drops them proves that the protocol is right on each branch; a picture that keeps them proves also that the branches happen a quarter of the time each, and add up to certainty.

Feynman's Notebook.

Teleporting a gate. Abramsky and Coecke treat two cousins of teleportation alongside it. One is entanglement swapping, whose history this book told in Chapter 6. The other is logic-gate teleportation, which they credit to Gottesman and Chuang, "Quantum teleportation is a universal computational primitive" (arXiv:quant-ph/9908010, 1999; Abramsky and Coecke's reference [12]), published in *Nature* 402, 390–393 (1999) under a different title. Replace the shared pair by some other entangled state, and the qubit that arrives is not the input but a function of it, fixed by the state. Done with pairs of qubits in place of single ones, Abramsky and Coecke report (their §2.2), this gives a universal computational primitive, one that is moreover fault-tolerant. In the sketchbook the idea is one bend. Put a box on the cup, and by Theorem 6.5 the box rides along the wire and is applied to whatever is teleported through it; the corrections must then be chosen to suit the box.

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§10.8 What Travelled

Let us gather it up. A Bell reading has four answers, the four Bell states: Chapter 6's bent wire, plain or carrying a flip, a phase-flip, or both, and one drawing with two bit-labelled rings holds all four (§10.1). The reading's effect is a cap with those rings, a translation we paid for with colour change and fusion. Read half of a pair against a message and the message is at the far end at once, turned by the reading's rings, and useless there, a flat grey, until the outcome arrives (§10.2). Send the outcome, two bits, and

the far end turns the message back, flip first and phase-flip second; in the picture the wire runs straight through, like rings meet and vanish, and $\frac{1}{2}$ is left, whose square is the $\frac{1}{4}$ of each branch (§10.3, §10.6). Run the same pieces the other way round and one qubit carries two bits, the correction table serving as the codebook and the preparations as the readings' mirror images (§10.4). And teleport half of a pair, and Chapter 6's junction is mended on every outcome (§10.7).

What travelled? Not the qubit: nothing quantum went from Alice to Bob. Not a description of it either, since the two bits are a fair toss whatever the message was. What travelled was *which turn*, and the rest was already there, in a wire that was bent long ago.

One thing more has happened, and the dialogue lets it pass in silence. When the reading was done, Alice's message was gone from her end. Her wire gave up its state in the reading; as Abramsky and Coecke put it, the information in the source "has been 'destroyed' in transferring it to the target". Teleportation *moves* a state. At no moment in the whole procedure were there two copies of $|\psi\rangle$, one at each end.

Was that bad luck, or bad design, or is it a law? Could a cleverer protocol have left Alice her message and sent Bob a second one? Could anything, anywhere, take an unknown qubit and make two? The next chapter asks, and the answer is written into the very linearity that made every picture in this one work.

Gigue That May Not Be Repeated

落ちし玉
二度は落ちざる
氷柱かな

ochishi tama
nido wa ochizaru
tsurara kana

A bead once fallen
does not fall a second time —
the eaves' icicle.

[*The Crab's parlour, the same day, the hour after the midday meal. The black slab still lies along the table, wiped and bare. Beside it, off the black, on the side nearer the fire, stand the viewer, fetched down from the dresser, and a little wooden tray with three pewter counters stacked in it. The Tortoise's drawing-cloth runs along the slab's other side, by her chair, spread and bare; her string and rings are in her shell. Achilles stands at the window.*]

[*The thaw has come at noon, as it will on the brightest of deep-winter days. The sun, low even at its highest, stands full on the house front, and the parlour is lit from its near end, through the one tall window, while the far end keeps a cooler, quieter light of its own. The frost that furred the glass at breakfast has gone from it altogether; the small panes are clear, and every one of them, blown by hand long before the Crab's time, bends the lane a little differently — the gate leans in one, the hedge swims in the next, and no two show quite the same lane. Along the eaves*

above the window hangs a fringe of icicles, and the sun has set them all going: a drop here, two there, a long wait, then three at once, at no pace anyone could follow. Each drop falls once, past the glass, and is gone into the snow on the sill. The lace curtain is drawn back to one side of the window and hangs there in loose folds. On the long wall the fire is built up high, the kettle stands at the side of the hob, and the dresser holds the shut ledger, the cake-tin, and the Crab's cups — he buys them one at a time, wherever he happens to be, so that there are no two of a pattern on the shelf. Down the middle of the table the black slab runs from the window to the far end, bare. In the tray beside it the pewter counters sit in their stack: the top one new and bright, the middle one dulled with handling, the bottom one with a small nick in its rim. On the end wall beyond the far end of the table, facing the window down the whole length of the room, hangs a small print in a plain frame — Escher's Drawing Hands*: two hands, each drawing the other. The noon light reaches it only at second hand, pale, off the ceiling. It has hung there since long before the slab was first laid out on that table, and it will hang there after. The Tortoise has kept this morning's quilt round her shoulders, and the room smells of the pudding and, faintly, of wet wool.*]

Achilles: Will it draw the very same pattern tonight, do you think, Madam Tortoise — line for line, just as it lay on the pane this morning?

Tortoise: Not in all the winters I have watched it. It draws a new one every night and never looks back at the old.

Crab: [*stacking the last plates*] The eaves are no better. They've dripped since the pudding, and I've stopped hearing them.

Achilles: The whole lane is running with it. I walked down to the gate after lunch to look at the thaw and brought half of it back in my boots.

Crab: Then put them by the fender and your feet by the fire, and let the rug forgive you. [*He carries the plates out and comes back drying his claws.*] This morning's keepsake pair is in the drawer, unread, and there it stays; the pairing-glass is on its shelf. The table is clean for once.

[*Achilles sits down on the fender stool and works his boots off, and stands them on the hearth, where they begin at once to steam and to leak a small dark pool of thaw onto the stone. He puts his stockinged feet out to the fire and wriggles his toes at it.*

The Crab, passing behind him, drops a dry towel over his knees without a word.]

Tortoise: Clean, and waiting for something. You fetched the viewer down, and you've looked at the clock four times since the soup.

Crab: Because the midday post is late, and the midday post is bringing me a parcel. The old man at the workshop wrote to say so.

[*A knock at the door. The Crab is out of his chair before it has finished, and comes back with a squarish parcel in brown paper, stiff with frost at the corners, straw showing at a torn seam.*]

Crab: From Delft. The postman made me promise not to drop it on his foot.

Achilles: Is it heavy?

Crab: Heavier than anything he has ever sent me. [*He cuts the string and folds back the paper on a nest of straw. From the top of the straw he lifts a small box and opens it.*] And, before anything, a fresh pair — cut of one piece, as always. He never sends a thing without a pair to try it on.

[*He sets one stone at the window end of the slab. Its reflection is there at once in the black, beneath it, meeting it at the surface. The other he carries down the whole length of the table and sets at the far end, and its reflection rises to meet it there. Two stones, a long table apart, each doubled in the glass; neither has been looked at.*]

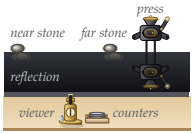
Crab: One here, one there. The rest of the parcel we'll come to.

[*The two stones sit a table's length apart. The near one, at the window end, has the sun full on it and is plain to see from anywhere in the room; the far one, down by the end wall, sits in the cooler light there, and from the fire it looks smaller than it is. Achilles, his feet still to the blaze, keeps turning his head from the one to the other, as though they were having a conversation and he was missing it.*]

First Challenge — *The Press Copies Its Own*

[*The Crab lifts out of the straw a small iron press: a hopper with a lid on top, a lever at the side, a hood with a shutter over the middle, and two little chutes below,*

one to the left and one to the right. He sets it on the black at the far end, beside the far stone. Under it, upside down in the glass, stands the press's reflection, hopper to hopper, chutes to chutes. Beside it the far stone's reflection, and at the window end the near stone's, answer their stones slack, a beat late and a little wide, as a half of a pair's always does.]



set down beside the far stone

Crab: The copying press. The old man has been building it since the summer, and I have been asking him for one since before I could see over his bench. Put a stone in this hopper, pull the lever once, and two stones drop from these chutes.

Achilles: Two from one? Where does the second come from?

Crab: From in here. [*He taps the housing.*] It comes loaded with blanks — raw stones, settled low at the workshop's bench and sealed inside — and it eats one on every strike.

Achilles: So the stone goes in, and it comes out with a twin.

Crab: Two come out. Which of them, if either, is the one that went in, nobody can say.

[*In the noon light the press is a lovely small thing: blacked iron polished like a good stove-front, a hopper lid of brass, a lever with a knob of turned boxwood worn pale by the maker's hand. It has come all the way from Delft in straw, and still it stands so finely on its four small feet that a cart going by in the lane sets a faint shiver in it. Under the lever, struck small into the iron, is the workshop's sign, a little anchor, and beside it the old man's initials, their edges gone soft. And at the seam of the hood, seen only from the side of the eye, there is now and then a faint warm glow, as of a coal banked somewhere in the works; looked at straight, it is only a join in blacked iron.*]

[*He lifts the hopper's lid: inside, a small dark cup, empty. He lets the lid down.*

Then he sets a claw to the shutter on the hood and slides it open. Behind it there is only dark iron — no stone, no light, nothing turning. He slides it shut again. Nothing has moved; the press stands as it stood.]

Crab: Open, while it idles, it shows you dark works and nothing else. Open it while it strikes, and it jams, and the stone in the hopper is spent for nothing. So it is worked blind, and I shall never see how it does it. [*He looks at the shutter a moment longer than he needs to.*] I knew that when I asked for it.

[*When at last he looks away, the glow at the seam is back, just where the eye cannot quite get at it. The Tortoise, who has been watching the Crab rather than the press, hands him his cup, and he drinks from it without noticing that he has.*]

Achilles: All this morning we moved a stone from one end of the room to the other, and not once were there two of it. And here is a machine that makes two. So it can be done after all.

Crab: Read what it does before you say so. [*From the straw he takes a slim, cloth-bound book.*] The maker's proving-book. Every pressing made at the workshop, one line each: how the stone was settled beforehand, and how its two struck stones read afterwards.

Tortoise: Settled — not looked at?

Crab: Settled. That is the workshop's bench making a stone a certain way, not reading it; it spends nothing.

Crab: [*laying a hand on the viewer*] And they read the struck stones through a viewer like mine. [*He turns its collar a quarter round and back.*] This collar I never showed you last night: upright, it asks a stone the plain question, high or low; turned crosswise, it asks the crosswise question, and the needle swings to the same two marks. Either way, a reading spends the stone.

Crab: [*a claw on the little tray*] And these counters keep the readings. Whoever reads a stone lays one on the table's edge, face-up for high and face-down for low; stacked in the tray, they mean nothing at all.

Crab: [*taking a letter from the straw and setting it, unopened, by the book*] There's a letter from him too, mostly about a clerk called Eve. He says I'm to hear it

before I use the press for anything clever.

Achilles: Then the book first. What did the plain stones do?

Crab: [*opening the book and running a claw down the first page*] Settled plain-high, pressed, and both struck stones read high. Settled plain-low: both low. High and high; low and low; high and high — all the way down the page and over the next.

Achilles: One stone in, two stones out: that's all a copy is. And now the crosswise ones. A crosswise stone is half high and half low, near enough, and a press that copies high and copies low will copy anything in between. So out come two crosswise stones, and each, read at cross, will say what its original would have said, every time.

Crab: [*turning a page*] Settled crosswise one way, pressed, and the struck pair read at cross. The first struck stone: high mark; low mark; low; high; low. The second, just the same kind of scatter. A coin toss, each of them.

[*Outside, the sun has got well onto the eaves, and the drip has quickened into a patter on the sill: a light, quick, running figure that seems each moment about to come round to its beginning again, and never quite does. Achilles, by the fire, taps it on his knee for a bar or two and loses it.*]

Achilles: Each? But a stone settled crosswise, and asked the crosswise question, gives the same answer every time.

Crab: Its struck stones don't. And yet look along the lines, not down them. Whenever the first said high, the second said high; whenever low, low — not one line where they differed. And where the stone went in settled crosswise the other way, the two always disagreed, every line of it.

Achilles: What did they do when they were asked the plain question?

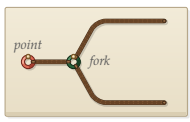
Crab: There are some pages of that at the back. [*He finds them.*] Settled crosswise, either way, and the struck pair read plain: both high or both low, half the time each. Never one of each.

Achilles: Not a single high-and-low. Two honest crosswise stones, read plain, would be two separate tosses, one high and one low as often as not. It copied the high part into both-high and the low part into both-low, and

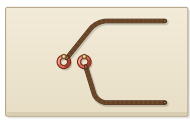
there's nowhere left for one-of-each to come from. ...Oh. It doesn't copy the crosswise stone. It makes a *pair* of it.

Tortoise: Let me show you on the cloth, where nothing gets spent. [*She takes from her shell a green ring with one cord running in and two running out.*] You know this from Mr Crab's beam: a green ring made to fork, one cord in and two out, and it does to its own kind what his press does. [*She lays beside it two short cords, each ending in a single-legged ring, one red, one green.*] And these are points: a red point is a stone settled plain, and a green point is a stone settled crosswise.

[*The fork lies on the cloth, one cord in from the left, two out to the right. The Tortoise threads the red point's cord onto the fork's in-cord and slides the red ring up it. The red ring meets the green, and the green ring opens round it, and she lifts it off into her shell, the in-cord with it. At the head of each of the two out-cords, where the fork stood, a red ring now rests, its cord running on out to the right. Where there was one red point and a green fork, there are now two red points, each the very twin of the one that went in; the out-cords are no longer and no shorter, and only the green ring and its in-cord have gone.*]



a green ring made to fork



slid through, the green ring gone

Achilles: Two, and both red. That's the plain page of the book.

[*She gathers the red points into her shell, sets out a fresh fork, and threads the green point onto it instead. Green meets green: the two rings slide into each other, as like rings always do, and are one ring. The point's own cord is swallowed with it, and what is left is a single green ring with just the two out-cords hanging from it. A ring with two cords and no turn on it is only a bend; she lifts the ring off and drops it back in her shell, and on the cloth lies one length of string, bent round on itself, both its ends running out to the right. Nothing has slid out along either cord. Where two green points might have lain, there lies one bend — two ends, but one piece.*]



bent round on itself

Tortoise: Red goes through as two. Green goes in, and one bend comes out: a pair cut of one piece, not two copies of anything.

[*Somewhere above them the thaw loosens the drift that has lain along the roof-ridge since the night, and it goes all at once, down both slopes together: a soft thump into the lane at the front of the house and another into the garden at the back, so exactly together that they make one sound, heard from two sides. Achilles looks up at the ceiling, and then at the window, where a little loose powder is still sifting past the glass.*]

Achilles: So this press copies only its own kind. But a better press, one cleverer inside — might that copy anything?

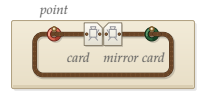
Second Challenge — *One Loop, Two Loops*

Crab: That is exactly what I mean to ask the old man. He swears this one loses nothing — whatever goes in is all accounted for in what comes out — but he could build round it, surely, with better works.

Tortoise: Then let me draw any press whatever, as clever as he pleases. [*She gathers the bend off the cloth into her shell, takes out a small card, and threads it on a length of string.*] This card is the press — its works, its blanks, whatever it keeps inside, all folded into one. [*She lays a second card on the first, traces it through, turns the tracing over, and threads that on too.*] And this is its mirror card, traced from it back to front: the same press, run the other way about.

Tortoise: One more thing to lay out. A loop of string with a stone's point at one end and, closing it, the mirror of another stone's point, is how alike the two stones are: a whole loop if they are the very same, nothing at all if they are wholly unalike, and something between for the rest.

[*She makes the loop. At one end of the string she fixes a red point — a stone settled plain-low. From it the string runs through the card, then through the mirror card, and on to the mirror of a green point, a stone settled crosswise, where she brings it round and ties it back to the start. A closed loop, with two points on it and the two cards between them.*]



the press and its mirror in the way

Tortoise: That is how alike the two stones are before any pressing — the press and its mirror both in the way. A fair press that loses nothing always meets its own mirror like this.

[*She slides the card along the string towards the mirror card. The string does not move; only the card travels. The two cards touch, face to face, and she lifts them off the string together, as one, and puts them away in her shell, still face to face. The loop lies closed as before, the red point and the mirrored green point still on it, the string no shorter than it was; only the cards are gone.*]



the cards gone, the points still on the loop

Tortoise: One loop. The press has left no mark on it at all: its likeness is just what it was before.

[*Then, beside it, she lays out the same two stones after a faithful pressing: each stone now two copies. The first copy of the plain stone she closes on the first copy of the crosswise one, in a loop of its own; the second copy on the second, in another. Two separate loops lie side by side, not touching, each with its red point and its mirrored green point. A little way off she lays a small plain loop.*]



side by side,
not touching

Tortoise: And that little one is whatever the works kept for themselves. At best it is worth a whole, and never more.

Tortoise: One loop is the likeness; two loops side by side are the likeness taken twice over, and multiplied.

Crab: Say it again, slowly, so I have it exactly. No — let me. One loop is how alike they were, and two loops side by side are that same likeness laid down twice and multiplied — didn't you say?

Achilles: [*laughing*] Close, Mr Crab.

Crab: Bother. I had it word for word in my head, and it came out of my mouth another shape.

Tortoise: Keep yours, Mr Crab. It came out a new one, and a better one than mine.

[*The Crab, mollified, laughs under his breath and fills all three cups from the pot, and the Tortoise folds both claws round hers. Outside, the drip has settled: each icicle along the eaves has found a pace of its own and keeps to it — one quick, one slow, one that waits and waits and then lets go — and none of them is in step with another.*]

Achilles: Then let me put numbers on your loops. A plain-low stone and a crosswise one: ask the crosswise stone the plain question and it says low half the time, so call them half alike. Before the press, one loop: a half. After a faithful pressing, two loops: a half of a half, a quarter — and the works' little loop can only make that smaller. A half against a quarter.

Tortoise: And the press, which loses nothing, must keep the half it was given.

Crab: Then it can't be done for those two. But for others — any two at all?

Tortoise: One loop equals two only when the loop is worth all or nothing: stones the very same, or wholly unalike. For any two between, there is no press.

[*The sun lies full on the lace curtain where it hangs gathered at the side of the window. Where the lace lies single, the lane shows through it dimmed; where a fold lies over it, dimmer again, by the same share once more; and in the thick bunch at the hook no lane comes through at all. Only that dark bunch, and the bare glass beside the curtain, would look no different for one more fold.*]

Crab: Not even if the old man made it cleverer inside?

Tortoise: The cleverness lives in the card, and the card came off.

Crab: [*sitting back, and looking at the letter by the proving-book*] Then Eve's press — the letter says she had one — never copied a thing either. So what was she doing with it all those years?

Third Challenge — *Eve's Tap*

Crab: [*breaking the seal and unfolding the letter*] Let us find out. He writes a small hand. [*reading*] "You will remember Eve, who clerked for the carrier — the slow, careful one that takes our stones about the country once a season. She had a press like yours. For three winters she pressed every stone that passed through her hands, kept one of the struck stones in her drawer, and sent the other on to the customer."

Achilles: A thief! And nobody noticed?

Crab: [*reading on*] "The customers whose stones were settled plain noticed nothing: they came as good as ever. But every stone that was sent settled crosswise, either way, arrived looked-at. Asked the crosswise question, it answered like a coin." [*He lowers the letter.*] Looked-at, every one. People paid for those stones, Achilles, and waited a season for them.

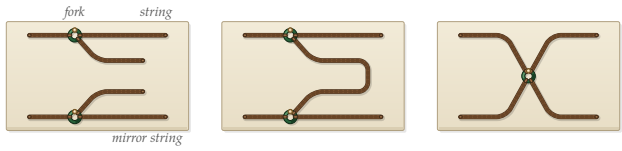
Achilles: Wait, though — why were the plain ones spared? [*He thinks.*] Nothing in them to spoil: they were settled already. A press that copies high

and low leaves high and low alone, and looks at everything in between.

[*Through the clear panes the lane lies in the full thaw. The morning's footprints go along it still, every one still there, a hollow wherever a foot came down; but the thaw has gone softly into each of them and taken out the tread and the nail-heads, so that whose boots they were can no longer be told from any one of them.*]

Tortoise: Let me draw it as your slab would show it, Mr Crab, with everything doubled. [*She gathers the loops off the cloth into her shell, lays a length of string along the cloth, and lays a second below it as the first's reflection would lie in the black: upside down, and running the same way.*] This is the mirror string: it carries the reflection of whatever I lay on the first. And to let a thing go, as we did last night, I fold its end over onto its own mirror twin.

[*On the first string she threads a fork: the stone's cord coming in, two copies going out. On the mirror string, facing it, she threads the fork's reflection, the stone's reflection coming in, the two copies' reflections going out. Then she takes the second copy's end and folds it over onto the second copy's reflection: they meet end to end, and the two forks are now joined, fork to mirror fork, through that single folded string. She slides the forks together along it. Green meets green; the two rings run into each other and are one. What lies on the cloth is a single green ring with four cords: the stone's and its reflection's coming in, the kept copy's and its reflection's going out. What has changed: the second copy and its reflection are gone into the fold, and the two forks have become one ring. What has held: a stone and its reflection still come in, and one stone and its reflection still go out.*]



as the slab would show it | *fork to mirror fork* | *the forks slid into each other*

Tortoise: Stone and reflection in, stone and reflection out, and one ring holding all four together.

Achilles: I've seen that. Last night — the stone I looked at with the needle hidden behind the cloth, and never read. A look that nobody read.

Tortoise: Laid out in string, it is that, exactly.

Crab: So her press copied nothing. It looked, and kept the look in her drawer where no one could read it, and sent my customers the leavings.

[*He sits a moment with the letter on his knee, looking at nothing; Achilles leans forward from the fender and puts another piece of coal on the fire for him. On the sill below the window each icicle has worn its own small round pit in the snow, a neat row of them under the eaves, and every drop that falls now falls into its own icicle's pit and no other.*]

Achilles: And her blanks? The press eats one a strike. Couldn't she have made her own, out of stones she had two of?

Fourth Challenge — *Eve's Blanks*

Crab: She tried. [*turning the sheet over*] "Blanks cost money, and hers ate one a strike. Now and then a customer sent two stones settled alike by the same hand — two of a kind, sent together — and she never knew how they had been settled. So she built a frame to turn one of the two into a blank, and leave the other as it was. Her frame had a tell: a small stone inside, which shifted whenever it worked."

Achilles: And it worked?

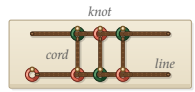
Crab: "It always gave her a blank." [*reading on*] "When she was found out, we bought the frame from her and tried it at the bench: two stones settled alike, the frame run, and the tell read at the setting the stones had been given. The tell always came out holding the stone she had blanked. It read exactly as that stone would have read, every settling, every trial."

Achilles: Then she did delete it; the tell is just a bit of ballast that shook when it ran. A blank is a blank.

Tortoise: Watch where the stone goes. [*She gathers the ring and its cords off the cloth into her shell, and lays two strings side by side on it.*] The upper line is her second stone. The lower is the tell, which starts settled low, as a blank

does. And between them, a knot you know from Mr Crab's beam: three cords from line to line, the middle one tied the other way about.

[*She ties it: three short cords across from the upper line to the lower, the first and last tied one way, the middle the other. The upper line runs in from the second stone at the left; the lower runs in from the tell's settled start. She takes both right-hand ends in one claw and both left-hand ends in the other and pulls. The three cords draw tight, slide into one another, and give way; the knot comes undone. What is left is two lines and a single crossing. The line that came in from the stone now runs down across the middle and out along the bottom, into the tell's place; the line that came in from the tell's settled start runs up and out along the top, where the second stone's line used to leave. Nothing is on either line that was not there before, and nothing is missing; only which one ends where has changed.*]



the middle tied the other way about



pulled, and undone

Achilles: A crossing. The stone went down into the tell, and the tell's own start came up to take its place — that's her blank. Two stones in, one blank out, and nothing gone: that's all her un-copying was.

[*Across the lane a man in a greatcoat is clearing the path to his door. It comes clear only a shovelful at a time, and every shovelful goes onto the bank beside it, which grows by just what the path has lost. When he straightens at his door there is a clean path and a taller bank, and not a flake fewer in his garden.*]

Crab: "Nothing came free," he says. He has underlined it.

Tortoise: She could move the stone. She could not make it be nowhere.

[*The icicles have gone thin and glassy with the afternoon, each tip a bead of light, and the drops come slower now, as though each were being considered before it fell. The sun is lower and reaches further in: its patch has crept along the floorboards beside the table, halfway down the room towards the end wall. Achilles draws his feet back from the fire and finds his stockings dry.*]

Crab: [*folding the letter, and looking down the table at the press, and at the fresh*

stones, one at each end] Then let us neither copy nor delete. Let us *use* it. The post to Delft takes a week in this weather. Say that end of the table is Delft. If the far stone there could be copied, could it not carry word faster than the post?

Fifth Challenge — *The Forbidden Telegraph*

Tortoise: Suppose the press were perfect, and made true copies of anything. Achilles, you are here at the window with the near stone, and you choose how to read it. What does Delft see?

Achilles: Read the near stone plain, and the far one settles plain, high or low; perfect copies of that are two alike, and read plain they always agree. Read the near stone crosswise, and the far one settles crosswise; perfect copies of a crosswise stone, read plain, are two separate coin tosses, and they agree only half the time. So Delft presses, reads both copies, and knows: all agreeing, I read plain; half agreeing, I read crosswise. A telegraph!

Tortoise: But this press is not perfect. Run it, Mr Crab.

Crab: [*a long breath*] Once, then. The only strike of the afternoon. Achilles, read yours crosswise.

[*Achilles gets up from the fender, sets his cup down on the hearth, and wipes his palms on his knees. The Tortoise pushes the quilt back off one shoulder to have a claw free, moves her chair an inch nearer the table, and then sits very still.*]

[*Achilles takes the viewer by its collar. It stands upright, set for the plain question; he turns it a quarter round until it lies crosswise, and the needle, resting at the middle of its dial, does not stir. Only the question has changed. He carries the viewer along the table's edge to the window end, never over the black, and looks through it at the near stone.*]

[*The needle leaves the middle of the dial and swings hard over to the high mark, and holds there; it might as well have gone the other way. The near stone, which a moment ago could have answered either way, lies still now, spent, its reflection still beneath it. Achilles takes one counter off the stack in the tray and lays it on the*

table's edge opposite the near stone, face-up. The near stone's reflection has drawn tight beneath it now, in exact step: read, and not merely looked at, with its counter lying opposite to say which way, it is one way and no other. The other two stay stacked in the tray, meaning nothing yet.]

Achilles: It swung to the high mark, at cross. Face-up. And I've told nobody.

[*At the far end the Crab lifts the hopper's lid, takes up the far stone, and sets it in the hopper's cup. As the stone leaves the black its reflection sinks from the surface and goes down, into the hopper of the press's reflection. The Crab lets the lid fall. The far stone is inside now; the place where it stood is bare black.*]

Crab: Shutter shut. [*He tries it with a claw: shut.*] And eyes on the fire.

[*He turns his head to the grate and keeps it there, and feels for the lever, and pulls it once, blind. A thump in the housing. From the left chute a stone drops onto the black, and the instant it lands its reflection is there beneath it. From the right chute drops another, and its reflection meets it. The hopper is empty. Where one stone went in, two stand on the black below the chutes, each doubled in the glass, its reflection slack as the fresh pair's were, and nothing about either says which, if either, was the stone he put in.*]



nothing about
either says which

[*The patch of sun, which has been creeping down the room all afternoon, reaches the end wall just then and lies full across the glass of the print, so that the frame seems for a breath to stand nearer the table than the wall.*]

[*The Crab turns back from the fire, and his eye goes past the two stones to the print on the wall — Escher's* Drawing Hands.]

Crab: Two hands, each drawing the other. And neither of them is the one I put in.

Tortoise: [*taking the viewer*] Now Delft reads. [*She turns the collar back until*

it stands upright, and carries the viewer along the edge of the table, the long way round, to the far end.]

[*She looks through it at the first struck stone. The needle swings to the low mark and holds. The stone is spent, and still. She takes a counter from the tray and lays it on the edge opposite, face-down. She moves the viewer to the second struck stone and looks. The needle goes to the low mark again, and holds. She lays the last counter beside the other, face-down. The tray is empty. At the window end one counter lies face-up; at the far end two lie face-down, side by side.*]

Tortoise: Low on the first; low on the second as well.

Achilles: They agree. But that's a single run — agreeing is what a perfect press would have shown me if I'd read plain.

Crab: [*opening the proving-book at its last pages*] Then he has tried it for us, many times. "Pressings of one half of a fresh pair, the other half read first, at either setting." [*reading down*] The struck pair always agreed, high with high or low with low, half the time each. When the near half was read plain — the same. When it was read crosswise — the same. Page after page, you could not tell one setting's page from the other's.

Crab: [*a finger at a margin*] And where the near half was read plain, the struck pair matched the near reading too. But that match lives in the near end's column, and Delft only ever gets that column by post.

Achilles: So Delft sees the same whatever I do here.

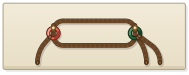
Tortoise: Watch the counters' cords. [*She gathers the crossed lines off the cloth into her shell, and lays it out on the cloth: the bend of the pair, the fork on its far leg, and a short cord for each counter.*] A plain reading I draw as a green ring gathering a stone and its reflection; a crosswise one, as a red ring.

[*First the plain setting. The near reading's ring is green, and so is everything at the far end: the fork, its mirror, the two far readings. Green meets green along the cords, and they slide together into a single green ring, with every counter's cord hanging from it. Pull any one of them, and the rest come with it.*]



slid together,
all green

[*Then she gathers the plain ring into her shell and lays it out afresh for the crosswise reading: the far end slid together as before into one green ring, and at the near end a red ring where the green one was. The red ring and the far green ring are joined by the stone's cord and its reflection's. She draws the red ring away from the green. The cords between them slip loose together and fall away, as such a link between the colours always does; the red ring lies apart with the near counter's cord hanging from it, tied to nothing at the far end. The far green ring still holds the far cords together. What changed: the near cord is free of the far ones. What held: the far cords still hang as one.*]



joined by the stone's
and its reflection's



red drawn away
from green

Achilles: At cross, my counter isn't tied to theirs at all. And theirs are still tied to each other, so they agree — but about nothing I did.

[*Through the window at Achilles's end, the two chimneys of the house across the lane are smoking side by side. Their smoke leans the same way, bends at the same height, and frays out together over the stable roof; it is the one small wind on both.*]

Tortoise: No cleverness would mend it: to copy every stone, a press would first have to cut every bend in two, and a bend in two pieces is no pair at all.

Achilles: It copies its own kind and makes a pair of every other; and a press that copied every kind would beat the post to Delft. So there's no such press.

[*Outside, the drops come slower still. The sun has slid along the eaves towards the corner of the house, and each icicle holds its bead at the tip a long while, bright, before it lets it go.*]

Crab: [*quietly, closing the proving-book*] The post to Delft, then. I shall write tonight, and thank him, and tell him I worked it blind.

Tortoise: [*rolling her drawing-cloth up round her string and rings*] And tell him you never once looked. He'll know what that cost you.

[*The Crab slides the press's shutter home a last time, though it is shut already, and lays his claw flat on the hood. On the black lie the spent stones, each with its reflection; on the table's edge, the laid counters; the fresh pair is gone.*]

[*The patch of sun climbs off the end wall and goes out of the room. Along the eaves the last drops let go, one icicle after another, each in its own time, and nothing comes after them. The Crab keeps his claw where it is a moment longer, feeling the iron warm under it.*]

Crab: Listen — the eaves have stopped. The sun's gone off them.

Achilles: So it has. It's colder already by the window.

Crab: Then come away from it, both of you, and I'll put the kettle back on. There's the end of the seed cake.

[*The Tortoise gets up, stiff from sitting, and comes round the table's end to the fire with her quilt gathered about her. The Crab cuts the end of the seed cake into three pieces, none of them the size of another, and gives Achilles the largest.*]

[*The Crab pours. Achilles takes his cup to the fire and stands with it, looking back at the counters on the edge of the table.*]

Achilles: Not this press, and not a cleverer one. But anything at all — anywhere — that could take a stone nobody knows and make two of it?

Tortoise: Ask the frost tonight to draw this morning's pane again, line for line, and see what it gives you.

[*Nobody takes her up on it; there is cake to be eaten first. The sun is off the house now, and the parlour's light has gone even and grey-white, the light of a winter afternoon that has begun, without any fuss, to think about the evening. Along the eaves every icicle holds one last bead at its tip, and the beads have stopped falling: they are stiffening where they hang, and the little pits in the snow on the sill are glazing over. At the very bottom corner of the lowest pane, where the glass is coldest,*

a first faint grey of frost has started, too fine yet to have any pattern at all. Down the table the stones lie where the afternoon left them: one at the window end, with the bright counter on the edge opposite; two at the far end below the chutes, with the dulled counter and the nicked one side by side on the edge before them; the tray between is empty. The press stands at the far end keeping its works to itself, and at the seam of its hood, just where the eye cannot quite get at it, the faint glow is back; nobody turns to look. In the dresser drawer this morning's pair keeps its own counsel. On the end wall the print has gone grey with the rest of the wall now the sun is off it — two hands, each drawing the other, in a plain frame — and nobody has glanced at it since. The three of them sit close to the fire with their odd cups: Achilles in his stockinged feet, his boots dried stiff on the hearth beside him; the Crab with crumbs of seed cake down his front; the Tortoise with the quilt up to her chin. The kettle, pulled to the side of the hob, sings itself slowly down to nothing.]

Chapter 11: Copying Is Forbidden

— Gigue That May Not Be Repeated —

Part III: Pictures Become Proofs

Escher: "Drawing Hands" — two hands, each drawing the other

Can a copy be made of something that nobody has looked at?

Every copier we know begins with a look. A photocopier shines a light on the page and records what comes back. A scribe reads a line before writing it out again. A computer copies a bit by finding out whether it is 0 or 1 and writing the same thing somewhere else. Copying is so cheap in the everyday world that we hardly notice that it always starts by looking. Chapter 9 told us what a look does to a qubit: it forces one answer out of several, at random, and destroys the rest. So a copier of qubits, if there is one, would have to work blind. It would take a state it has not examined, and cannot examine without spoiling, and hand back two of it.

If such a machine existed, a great deal would follow. You could make a thousand copies of an unknown state and ask each copy a different question, until the state had no secrets left. And you could, as the end of this chapter will show, send a message faster than the post. Chapter 10 ended on exactly this question and left it open: "Could anything, anywhere, take an unknown qubit and make two? The next chapter asks, and the answer is written into the very linearity that made every picture in this one work." In teleportation the message was gone from Alice's end the moment it was read, and at no moment were there two of it. Was that a quirk of one clever protocol, or a law?

It is a law, and one of the shortest in the book. On the blackboard it is a single line, in which a number would have to equal its own square. In the sketchbook it is a count of closed pictures: one on one side, two on the other. But the law has a positive side, and the pictures see it first. Some things *can* be copied: the things that were settled already. The machine the Crab unpacked this afternoon copies exactly those, and turns everything else into a pair. From there the chapter follows a copy thrown away (it turns out to be a look that nobody read) and a copy that someone tries to unmake (it can only be moved). At the end we meet the only telegraph a copier could ever build, one that no linear picture can draw.

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§11.1 The Press Copies Its Own

The press in the dialogue is a copier with a magazine of blanks. Settle a stone plain-high, press it, and two high stones drop out. Settle it plain-low, and two low ones. The proving-book says so, line after line. What does the same press do to a stone settled crosswise?

On the blackboard. Chapter 8 met the copier (§8.3, row 59), and we now give it a letter of its own. The **copy map** is

$$\delta \;=\; |00\rangle\langle 0| + |11\rangle\langle 1|\,,$$

the phase-free green dot with one leg in and two out, $Z(1,2,0)$ of §8.1; the Dirac's Blackboard of §8.2 names it as a 4×2 matrix. On the two plain states it does what its name says, exactly:

$$\delta\,|0\rangle = |00\rangle = |0\rangle\otimes|0\rangle\,, \qquad \delta\,|1\rangle = |11\rangle = |1\rangle\otimes|1\rangle\,.$$

Now the crosswise stone, $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. The copier is linear, so it acts on each branch separately:

$$\delta\,|+\rangle \;=\; \tfrac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \;=\; |\Phi^+\rangle\,.$$

Two copies of $|+\rangle$ would be $|+\rangle\otimes|+\rangle = \frac{1}{2}(|00\rangle+|01\rangle+|10\rangle+|11\rangle)$. The copier has not made two crosswise stones. It has made the Bell pair of Chapter 6: an entangled state, neither half of which is $|+\rangle$, and each half of which, asked anything on its own, answers like a fair coin (§6.2). In the same way $\delta\,|-\rangle = |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$. For a general state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ the two things to compare are

$$\delta\,|\psi\rangle = \alpha\,|00\rangle + \beta\,|11\rangle\,, \qquad |\psi\rangle\otimes|\psi\rangle = \alpha^2\,|00\rangle + \alpha\beta\,|01\rangle + \alpha\beta\,|10\rangle + \beta^2\,|11\rangle\,,$$

and they agree only when $\alpha\beta = 0$.

The press has a magazine of blanks, and δ seems to have none. Where did the blank go? Put a blank $|0\rangle$ on a second wire and run a CNOT from the stone to the blank. Then $\mathrm{CNOT}\,(|\psi\rangle\otimes|0\rangle) = \alpha\,|00\rangle + \beta\,|11\rangle = \delta\,|\psi\rangle$ for every $|\psi\rangle$, so, exactly,

$$\mathrm{CNOT}\,(I\otimes|0\rangle) \;=\; \delta.$$

The copier is a CNOT with its blank already loaded.

Is the press simply badly made? Could a cleverer one, with works inside that are allowed to change as it runs, copy the plain stones *and* everything else? To ask that properly we need to say what copying is.

We say a process **clones** a state $|\psi\rangle$ if, started on $|\psi\rangle$, a blank $|0\rangle$, and its own apparatus in a fixed starting state $|A\rangle$, it ends in

$$|\psi\rangle\otimes|\psi\rangle\otimes|A_\psi\rangle\,,$$

for some state $|A_\psi\rangle$ of the apparatus, which is allowed to depend on $|\psi\rangle$. The apparatus may be of any size.

Why this definition. Why insist that the two copies come out as a product, $|\psi\rangle \otimes |\psi\rangle$, while the apparatus is allowed to do anything at all? Because a copy is something you can walk away with. Each copy, taken on its own, must be $|\psi\rangle$, and the copies must not be tied to each other or to the machine; otherwise they are parts of one larger thing, as $\delta|+\rangle$ is. The apparatus is left free so that the definition rules out no cleverness inside the machine. (§11.3 relaxes the demand on the copies too, and finds that the relaxed demand fails as well.)

Here is the first answer. It needs nothing but linearity.

> **Result 11.1 (no-cloning from linearity).** Let L be any linear process on a stone, a blank that starts in $|0\rangle$, and an apparatus of any size that starts in a fixed state $|A\rangle$. Suppose L clones the two plain states:
>
> $$L(|0\rangle\,|0\rangle\,|A\rangle) = |0\rangle\,|0\rangle\,|A_0\rangle\,, \qquad L(|1\rangle\,|0\rangle\,|A\rangle) = |1\rangle\,|1\rangle\,|A_1\rangle\,.$$
>
> Then for every $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$,
>
> $$L(|\psi\rangle\,|0\rangle\,|A\rangle) \;=\; \alpha\,|00\rangle\,|A_0\rangle + \beta\,|11\rangle\,|A_1\rangle\,,$$
>
> which has **no** $|01\rangle$ **part**, while a clone $|\psi\rangle\,|\psi\rangle\,|A_\psi\rangle$ has $|01\rangle$ part $\alpha\beta\,|A_\psi\rangle$. So L clones $|\psi\rangle$ only if $\alpha\beta = 0$. With no apparatus, L with its blank loaded is exactly δ.

By the $|01\rangle$ **part** of a state on stone, copy and apparatus we mean what is left when the effect $\langle 01|$ is applied to the first two wires, $(\langle 01| \otimes I)$, and the apparatus is left open.

Proof idea. A linear machine that copies the two plain states has no choice about anything else: it treats a superposition branch by branch, so every branch it makes has stone and copy agreeing. A true copy of a superposition needs branches in which they disagree.

Proof. Linearity gives $L(|\psi\rangle\,|0\rangle\,|A\rangle) = \alpha L(|0\rangle\,|0\rangle\,|A\rangle) + \beta L(|1\rangle\,|0\rangle\,|A\rangle)$, which is the displayed state. Apply $\langle 01| \otimes I$: both terms die, since $\langle 01|00\rangle = \langle 01|11\rangle = 0$. For the clone, $(\langle 01| \otimes I)(|\psi\rangle\,|\psi\rangle\,|A_\psi\rangle) = \langle 0|\psi\rangle\,\langle 1|\psi\rangle\,|A_\psi\rangle = \alpha\beta\,|A_\psi\rangle$, and $|A_\psi\rangle$ is a unit vector, so this is zero only when $\alpha\beta = 0$. Without an apparatus,

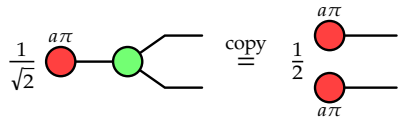
a linear map from the stone to two wires is fixed by where it sends $|0\rangle$ and $|1\rangle$, and $|00\rangle$, $|11\rangle$ are where δ sends them. ■ See `Code/Ch11/OwnBasis.py`, which also runs the three-wire basis cloner $\mathrm{CNOT}_{0\to2}\,\mathrm{CNOT}_{0\to1}$ and finds its $|01\rangle$ part empty on a non-real $|\psi\rangle$.

Dirac's Blackboard.

The missing $|01\rangle$*, and what the apparatus knows.* Take $|\psi\rangle = |+\rangle$, so $\alpha = \beta = \frac{1}{\sqrt{2}}$. A clone would be $|{+}{+}\rangle\,|A_+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)\,|A_+\rangle$, with amplitude $\frac{1}{2}$ on "stone low, copy high". The linear machine gives $\frac{1}{\sqrt{2}}(|00\rangle\,|A_0\rangle + |11\rangle\,|A_1\rangle)$: amplitude 0 there, whatever the apparatus does. The apparatus cannot repair the gap, because it only rides along on each branch; it cannot create a branch. It can do something else, though. If $|A_0\rangle$ and $|A_1\rangle$ differ, the apparatus holds a record of which branch it is on. When they are orthogonal, the record is perfect: the machine has in effect *looked* at the stone, in the plain way, and written down the answer. That remark will come back in §11.3, where copying and looking turn out to be the same act.

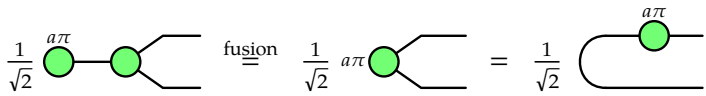
In the sketchbook. Chapter 8's points carried a $\sqrt{2}$. A red dot with one leg and phase 0 is $\sqrt{2}\,|0\rangle$, and with phase π it is $\sqrt{2}\,|1\rangle$ (§8.2, §8.3). A green dot with one leg and phase 0 is $|0\rangle + |1\rangle = \sqrt{2}\,|+\rangle$, and with phase π it is $\sqrt{2}\,|-\rangle$. With Chapter 10's bit-labelled phase (row 75) one drawing covers both cases of each colour: for a bit a, the red point $a\pi$ is $\sqrt{2}\,|a\rangle$, and the green point $a\pi$ is $\sqrt{2}$ times $|+\rangle$ or $|-\rangle$. So a normalized state is a point with $\frac{1}{\sqrt{2}}$ written in front, and we write it in every time.

Feed a red point to the copier. The copy rule (Theorem 8.3) sends it out as two red points, with a factor $\frac{1}{\sqrt{2}}$:



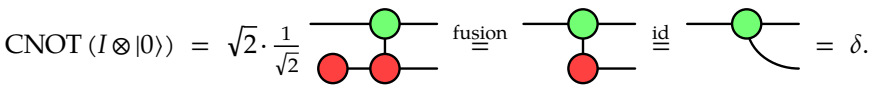
The numbers in front work out exactly: $\frac{1}{\sqrt{2}}$ for the point that went in, times the rule's $\frac{1}{\sqrt{2}}$, is $\frac{1}{2} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}$, one normalizing factor for each point that comes out. So $\delta\,|a\rangle = |a\rangle \otimes |a\rangle$, exactly. Chapter 8 stated the copy rule "up to $\sqrt{2}$" (row 59) only because its points were unnormalized. With normalized states the rule is exact.

Now feed it a green point. There is nothing to copy here. A green point and a green dot joined by a wire are two dots of one colour sharing a leg, and fusion (Theorem 8.1) makes them one:



The middle picture is a green dot with no legs in and two out, phase $a\pi$: by §8.1's formula, $|00\rangle + (-1)^a\,|11\rangle = (Z^a \otimes I)$ cup. Read fusion backwards to pull the phase out onto the top leg as a one-in, one-out dot; what is left, the phase-free green dot with two legs out, is $|00\rangle + |11\rangle$, the cup itself. That is the right-hand picture. The cup is unnormalized, as always, cup $= |00\rangle + |11\rangle$, so $|\Phi^+\rangle = \frac{1}{\sqrt{2}}$ cup, and the equation says $\delta\,|\pm\rangle = |\Phi^\pm\rangle$ exactly, with the drawn $\frac{1}{\sqrt{2}}$ doing real work. The point does not come out doubled. It melts into the copier, and what leaves is one bent wire: a pair, not two copies.

The blank can be drawn going in too. A CNOT is a green dot tied by a cord to a red dot, and the bare picture is CNOT$/\sqrt{2}$ (row 64); the blank $|0\rangle$ is $\frac{1}{\sqrt{2}}$ times a red point. The two numbers cancel:



The blank's red point fuses into the red target of the CNOT (same colour, shared leg). What remains of the red dot has one leg in and one out and no phase, so it is a bare wire (§8.1). What is left is the green copier. The blank

has no line of its own: it is part of the copier.

> **Result 11.2 (the copier copies its own basis).** Exactly, for each bit a:
>
> $$\delta\,|a\rangle = |a\rangle \otimes |a\rangle\,,$$
>
> $$\delta\,|\pm\rangle = |\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}\,(Z^a \otimes I)\,\text{cup},$$
>
> with $|+\rangle$ for $a = 0$ and $|-\rangle$ for $a = 1$. In pictures, red points go through the green copier as two red points, and green points fuse with it into a single cup. For $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$, the output $\delta\,|\psi\rangle$ is a product state only if $\alpha\beta = 0$, that is, only if $|\psi\rangle$ is a plain state up to a phase.

Proof. The two pictures above, each step exact (copy with the scalar kept, and fusion). For the last sentence, write $\delta\,|\psi\rangle = \alpha\,|00\rangle + 0\,|01\rangle + 0\,|10\rangle + \beta\,|11\rangle$ as its square of coefficients, $\left(\begin{smallmatrix} \alpha & 0 \\ 0 & \beta \end{smallmatrix}\right)$. The product test of Chapter 6 (Theorem 6.1) says it is a product exactly when the determinant, $\alpha\beta$, vanishes. ■ See `Code/Ch11/OwnBasis.py`.

Why this definition. Why take δ, a map with no apparatus and one fixed basis, as *the* copier, when we are asking about all copiers? Because Result 11.1 says every linear machine that copies the plain states, run with no apparatus, *is* δ, and Result 11.2 shows what δ does with everything else. The question is not whether δ is well made; it is whether any machine could copy more than one basis. The next section says no.

> **Coecke's Sketchbook.**
>
> *Red copies, green fuses.* The green copier copies exactly the red points, phase 0 or π, and nothing else. A green point is not copied. It fuses, and what comes out is a cup. Van de Wetering's survey says the same of the rule itself: the state-copy rules "only hold when the spider being copied has a phase of 0 or π" (§4.2). Give the red point any other turn and the blackboard agrees: $\frac{1}{\sqrt{2}}X(0,1,\theta)$ has $\alpha\beta = \frac{1}{4}(1 - e^{2i\theta})$, which vanishes only for $\theta \in \{0, \pi\}$. Coecke and Duncan read no-cloning this way round, as a positive statement: quantum states may be copied "if they are

known to lie in a given basis" (their §1). A copier *is* a choice of basis, and the states it copies are the ones already settled in it. Each colour copies the other's points and swallows its own. That is complementarity, and Chapter 8's Hopf law (row 62) is its other face: van de Wetering (§4.5) reads Hopf as saying that the green and red bases are complementary.

---

§11.2 One Loop, Two Loops

The Crab's next hope is the natural one. Perhaps a cleverer press, with more elaborate works, could copy more. It need not copy everything. Suppose it only has to copy two stones: one settled plain-low, $|0\rangle$, and one settled crosswise, $|+\rangle$.

On the blackboard. Linearity alone does not forbid this. The two states $|0\rangle$ and $|+\rangle$ are linearly independent, so there is a linear map L from one wire to two with

$$L\,|0\rangle = |00\rangle\,, \qquad L\,|+\rangle = |{+}{+}\rangle\,.$$

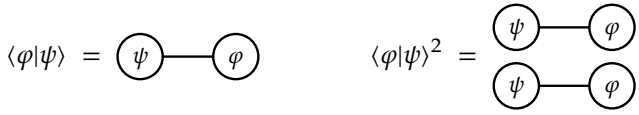
It is fixed on the basis $|0\rangle\,, |1\rangle$ by $|1\rangle = \sqrt{2}\,|+\rangle - |0\rangle$, giving $L\,|1\rangle = \sqrt{2}\,|{+}{+}\rangle - |00\rangle$. Result 11.1 does not apply, because L does not copy $|1\rangle$. But look at the length of $L\,|1\rangle$:

$$\|L\,|1\rangle\|^2 \;=\; 2 + 1 - 2\sqrt{2}\,\langle 00|\,{+}{+}\rangle \;=\; 3 - \sqrt{2} \;\approx\; 1.59.$$

L stretches a state out of the unit sphere, so it cannot be a physical process (Theorem 3.2). Here is what went wrong, stated in a form that will generalize. Before the press, the two stones had amplitude $\langle 0|+\rangle = \frac{1}{\sqrt{2}}$ for each other. After a faithful copy, the pairs would have $\langle 00|\,{+}{+}\rangle = \langle 0|+\rangle^2 = \frac{1}{2}$. A unitary box preserves amplitudes between states (Theorem 3.2), and $\frac{1}{\sqrt{2}} \neq \frac{1}{2}$. In probabilities, squaring both, it is a half against a quarter.

Call the amplitude $\langle\varphi|\psi\rangle$ the **likeness** of the two states: it is 1 when they are the same state, 0 when they are orthogonal, and in between otherwise. Copying multiplies a likeness by itself, while every unitary process keeps it fixed.

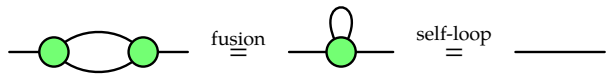
In the sketchbook. Put the state ψ at the left end of a wire and the effect φ at the right end: the closed diagram of Chapter 2, whose value is the amplitude (§2.2). The effect labelled φ is the *mirror image* of the state φ, so its entries are conjugated. We call this closed picture a **likeness**. The dialogue's string makes it a loop, and it is one. Put the state ψ and the effect φ on Chapter 10's closed loop, as the box $|\psi\rangle\langle\varphi|$; the loop is the trace of that box (Result 10.6), which is $\langle\varphi|\psi\rangle$, and yanking straightens it into the segment below. Two likenesses stacked, with nothing joining them, are the product of their numbers (Proposition 4.1):



One loop is the likeness of the stones. Two separate loops are the likeness of the pairs, the same number taken twice over and multiplied.

Why this definition. Why measure likeness by the amplitude $\langle\varphi|\psi\rangle$ and not by the probability $|\langle\varphi|\psi\rangle|^2$? Both would serve for the argument, but the amplitude is the thing a closed picture *is*. A process that meets its own mirror image and vanishes leaves the amplitude untouched, phase and all, so the amplitude is what the pictures can carry through a proof unchanged.

The worked instance is the copier itself. The mirror image of the green fork is a green dot with two legs in and one out; it is $\delta^\dagger$, and since δ is real no conjugation shows. Put δ and then $\delta^\dagger$ on a wire. The two dots are joined by *two* wires, and fusion applies to spiders "connected by one or more wires" (van de Wetering, §4.1):



After fusing along one wire, the other wire runs from the dot back to itself: a **self-loop**. Identity removal, van de Wetering notes in the same section, also gets rid of self-loops on spiders: the loop is erased, and no number appears. (On the blackboard, closing two legs of a green dot with a wire only forces two of its all-agree labels to agree, which they already do.) The result is a bare wire, exactly. On the blackboard, in one line: $\delta^\dagger\delta = (\sum_i |i\rangle\langle ii|)(\sum_j |jj\rangle\langle j|) = \sum_i |i\rangle\langle i| = I$. The copier meets its mirror and vanishes, as a unitary box does in Chapter 3; so the copier preserves likeness, $\langle\varphi|\,\delta^\dagger\delta\,|\psi\rangle = \langle\varphi|\psi\rangle$. It is an *isometry*, not a unitary: in the other order, $\delta\delta^\dagger$ is not the identity on two wires.

So the question is now exact. A faithful press would have to turn one loop into two, and a likeness into its own square, while keeping it fixed. With the apparatus kept, here is the law.

> **Result 11.3 (no-cloning, unitarity form).** Let U be any unitary on a stone, a blank and an apparatus of any size, with the blank starting in $|0\rangle$ and the apparatus in a fixed state $|A\rangle$. Suppose U clones two states,
>
> $$U(|\psi\rangle\,|0\rangle\,|A\rangle) = |\psi\rangle\,|\psi\rangle\,|A_\psi\rangle\,, \qquad U(|\varphi\rangle\,|0\rangle\,|A\rangle) = |\varphi\rangle\,|\varphi\rangle\,|A_\varphi\rangle\,.$$
>
> Put $s = \langle\varphi|\psi\rangle$ and $a = \langle A_\varphi|A_\psi\rangle$. Then
>
> $$s \;=\; s^2\,a,$$
>
> and since $|a| \le 1$, **either** $s = 0$ **or** $|s| = 1$: a unitary can clone two states only if they are orthogonal or equal up to a phase. So no unitary clones every state; for instance $|0\rangle$ and $|+\rangle$ have $s = \frac{1}{\sqrt{2}}$.

Proof idea. Count the loops. Before the machine, the likeness of the two inputs is one loop (the blank's loop and the apparatus's loop are each worth 1). The machine and its mirror vanish, so the likeness is unchanged. After a faithful copy the same likeness is two loops and the apparatus's loop, multiplied. One loop can equal two loops only if the loop is worth nothing or has size one.

The proof is given twice, in full, in §11.6. See `Code/Ch11/NoCloning.py`,

which runs the unitarity identity for a random 8×8 unitary and non-real states, and builds the one-loop and two-loop pictures, with values exactly s and s^2.

Dirac's Blackboard.

Why the apparatus cannot help. The apparatus enters the equation once, as the factor $a = \langle A_\varphi | A_\psi \rangle$ on the right of $s = s^2 a$. Both $|A_\varphi\rangle$ and $|A_\psi\rangle$ are unit vectors, so by the Cauchy–Schwarz inequality $|a| \leq 1$: the apparatus can only shrink the right-hand side, never enlarge it. Take moduli: $|s| = |s|^2 |a| \leq |s|^2$. If $s \neq 0$, divide by $|s|$ to get $|s| \geq 1$; and $|s| \leq 1$, again by Cauchy–Schwarz, so $|s| = 1$. Equality in Cauchy–Schwarz happens only when one vector is a multiple of the other, so $|\varphi\rangle = e^{i\theta} |\psi\rangle$: a phase on the whole state, which no question can detect. With no apparatus at all, $a = 1$ and the law reads $s = s^2$. An apparatus whose final states are orthogonal, $a = 0$, is worse off still: then $s = 0$ is forced. Its works have recorded which stone went in, which is a look.

Feynman's Notebook.

Three statements of one law. Both of our sources credit the no-cloning theorem to two papers of 1982. One is W. K. Wootters and W. H. Zurek, "A single quantum cannot be cloned", *Nature* 299, 802–803. The other is D. Dieks, "Communication by EPR devices", *Physics Letters A* 92, 271–272. We have had neither paper in hand, so we say nothing about how either one argued. The statements below come from our sources. In their preprint of November 1995, Howard Barnum, Carlton M. Caves, Christopher A. Fuchs, Richard Jozsa and Benjamin Schumacher describe the theorem as the fact that there are "no physical means by which an unknown pure quantum state can be reproduced or copied". Arun Kumar Pati and Samuel L. Braunstein, writing about photons, put the unitary form in one clause: two non-orthogonal photon-states "cannot be copied perfectly by a unitary process". Result 11.3 is that clause with an apparatus kept. The derivation of $s = s^2 a$ given here is this book's own.

§11.3 Copying Is Looking

The maker's letter tells of Eve, a clerk at a carrier, who had a press like the Crab's. She pressed every stone that passed through her hands, kept one struck stone, and sent the other on. Her customers noticed nothing when their stones had been settled plain. Stones settled crosswise, though, arrived "looked-at": at the cross setting they read as a coin toss. What exactly did her tap do?

On the blackboard. Take the two cases in turn. A plain stone $|0\rangle$ goes through the press as $|00\rangle$; Eve keeps one, and the customer receives $|0\rangle$, untouched. A crosswise stone $|+\rangle$ goes through as $|\Phi^+\rangle$, and the customer receives half of a Bell pair. By Result 9.3 that half, on its own, is $\frac{1}{2}I$, the flat grey that reads as a fair coin at every setting. The press did to $|+\rangle$ what a look does.

To see the general case, let the stone carry any density matrix $\rho = \sum_{i,j} \rho_{ij} |i\rangle \langle j|$ (Chapter 9). The copier sends it to $\delta\rho\,\delta^\dagger = \sum_{i,j} \rho_{ij} |ii\rangle \langle jj|$. Discard either copy, which is the partial trace (Result 9.2). The partial trace keeps only the terms whose discarded labels agree, $i = j$:

$$\mathrm{tr}_2(\delta\rho\,\delta^\dagger) \;=\; \mathrm{tr}_1(\delta\rho\,\delta^\dagger) \;=\; \sum_i \rho_{ii}\, |i\rangle \langle i| \;=\; \sum_i P_i\,\rho\,P_i,$$

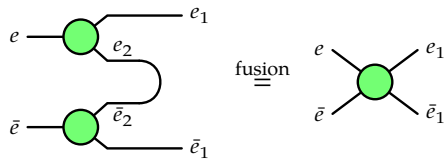
with $P_i = |i\rangle \langle i|$. The diagonal survives; the off-diagonal entries, where the phase lives, are gone. This is the state of §9.2: the stone after a plain look whose outcome nobody read.

> **Result 11.4 (copying is looking).** For every density matrix ρ, both marginals of $\delta\rho\,\delta^\dagger$ equal $\sum_i P_i \rho P_i$, the state after a plain look with the outcome unread. They equal ρ exactly when ρ is diagonal in the plain basis.

Proof. The calculation above, for each marginal. The last sentence holds because $\sum_i P_i \rho P_i$ is ρ with its off-diagonal entries set to zero. ■ See

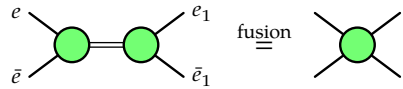
`Code/Ch11/NoBroadcasting.py`, which runs a non-real, non-diagonal ρ.

In the sketchbook. Draw the tap on doubled wires, as every real process must be drawn since Chapter 9. The stone is two lines, its wire e and its mirror copy $\bar{e}$ (row 66). The copier acts on e, and its conjugate acts on $\bar{e}$; since δ is real, its conjugate is the green fork again. Keep one copy, e_1 with its mirror $\bar{e}_1$. Discard the other: a cap on e_2 and $\bar{e}_2$ (row 67). The cap is a bent wire, so the two copiers are now joined by a single wire, and fusion (Theorem 8.1) makes them one dot:



The result is a phase-free green dot with four legs, two in and two out, $Z(2,2,0) = \sum_i |ii\rangle\langle ii|$, exactly. On doubled wires it sends ρ to $\sum_i \rho_{ii} |i\rangle\langle i|$: the blackboard's answer.

Now draw a look that leaves the stone in place. Chapter 9's read-spider takes the doubled stone and emits a classical wire (row 68). Its mirror image is a green dot that takes a classical wire in and puts a doubled stone out, $\sum_i |ii\rangle\langle i|$: it sets a stone to whatever the record says. Read the stone, then set it as read, and throw the record away between the two. The classical wire joins two green dots, and it is a wire like any other to the rules; the double stroke is only a reminder (§9.1). So fusion makes them one:



It is the same four-legged dot. Copying and throwing one copy away is, as a picture, a look that nobody read. That is Eve's tap. It left plain stones alone because a plain stone is already settled, and a look in its own basis finds nothing to spoil. It turned crosswise stones grey because a look at the plain setting destroys the crosswise phase. (A small check: cap the output as well, discarding the stone after the look, and fusion leaves a green dot with

the two input legs only, $\sum_i \langle ii|$, which is the cap. The tapped stone, thrown away, is a stone thrown away, as causality requires; Result 9.4.)

Barnum and his colleagues relaxed the demand on a copier in exactly the direction that Eve's customers would care about. The copies need not be independent of each other; each one, examined alone, need only look right.

We say a process **broadcasts** a set of states $\{\rho_0, \rho_1\}$ on a system A if, fed ρ_s together with a second system B in a fixed standard state, it produces a joint state on A and B whose **two marginals are both** ρ_s, for $s = 0$ and for $s = 1$. **Cloning** is the special case where the joint state is $\rho_s \otimes \rho_s$.

Why this definition. Why relax cloning to broadcasting? Because a customer who receives one stone can only examine that stone. A tap that produced correlated copies, each of which looked perfect alone, would be invisible to every customer. For a *pure* state the relaxation buys nothing: Barnum et al. point out that the only way to broadcast a pure $|\psi\rangle$ is to produce $|\psi\rangle \otimes |\psi\rangle$. For mixed states it could buy a great deal, and that is the question.

> **Result 11.5 (no-broadcasting; Barnum, Caves, Fuchs, Jozsa, Schumacher; the converse stated, not proved).** A pair of states $\{\rho_0, \rho_1\}$ can be broadcast **if and only if** $\rho_0\rho_1 = \rho_1\rho_0$. The pair can be cloned if and only if ρ_0 and ρ_1 are identical or orthogonal ($\rho_0\rho_1 = 0$).

Proof of the easy direction (theirs too). If ρ_0 and ρ_1 commute, they are diagonal in one common orthonormal basis $|b\rangle$: $\rho_s = \sum_b \lambda_{sb} |b\rangle \langle b|$. Use the copier of *that* basis, $\delta_B = \sum_b |bb\rangle \langle b|$, with the second system's standard state as its blank. It gives $\delta_B \rho_s \delta_B^\dagger = \sum_b \lambda_{sb} |bb\rangle \langle bb|$, and by Result 11.4, in the basis $|b\rangle$, each marginal is ρ_s with its off-diagonal entries removed. It has none, so each marginal is ρ_s. This is broadcasting and not cloning. The state $\sum_b \lambda_{sb} |bb\rangle \langle bb|$ equals $\rho_s \otimes \rho_s = \sum_{b,c} \lambda_{sb}\lambda_{sc} |bc\rangle \langle bc|$ only if every $\lambda_{sb}\lambda_{sc}$ with $b \neq c$ vanishes and every $\lambda_{sb}^2 = \lambda_{sb}$, that is, only if ρ_s is pure. The copier makes correlated copies: broadcast, but not cloned. *The converse*, that broadcasting forces the two states to commute, is the hard half. Barnum et al. prove it with the fidelity between density matrices, and the proof is beyond this chapter; we cite it. See `Code/Ch11/NoBroadcasting.py`, which checks the easy half and the correlated state, and states the theorem without testing it.

Eve wanted to tap stones settled plain and stones settled crosswise, say $|0\rangle\langle 0|$ and $|+\rangle\langle +|$. Those two do not commute. So no press, however cunning its works, could have passed both on unmarked. Hers marked exactly the ones the theorem says must be marked.

Feynman's Notebook.

Broadcasting. The theorem comes from "Noncommuting mixed states cannot be broadcast", by Barnum, Caves, Fuchs, Jozsa and Schumacher (arXiv:quant-ph/9511010, posted in November 1995 and marked as submitted to *Physical Review Letters*), which gives its address as the Center for Advanced Studies of the University of New Mexico, in Albuquerque. They set the problem as a question about secrets. System A is "secretly prepared" in one of two states, a second system waits in a standard state, and the question is whether any physical process can leave both systems looking, separately, like the secret one. Their most general process is the one of Chapter 9: a unitary together with an auxiliary system that is afterwards ignored. They also say why cloning had to fail. If nonorthogonal states could be cloned, one could measure enough copies to tell the states apart as reliably as one liked, which quantum measurements cannot do. Any broadcasting process, they add, can be seen as creating distinguishability "ex nihilo". Only for commuting states does it create none.

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§11.4 No Blank for Free

The letter goes on. Blanks cost money, and Eve's press ate one on every strike. So she tried to make her own. Given two stones that the same unknown hand had settled alike, could she turn one of them into a blank and leave the other as it was?

Pati and Braunstein asked the question in this form, for two photons of the same unknown polarization. Their **deleting machine** acts on two copies of

a state and an apparatus in a ready state $|A\rangle$, and is required to satisfy

$$|\psi\rangle\,|\psi\rangle\,|A\rangle \;\longmapsto\; |\psi\rangle\,|\Sigma\rangle\,|A_\psi\rangle$$

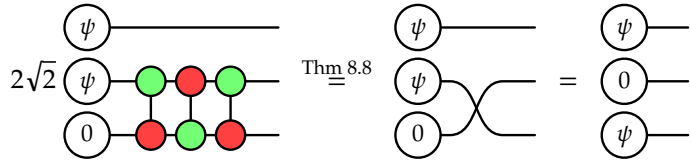
for every $|\psi\rangle$. Here $|\Sigma\rangle$ is a fixed standard state, the blank, and the apparatus may end in a state $|A_\psi\rangle$ that depends on $|\psi\rangle$. Eve's frame has such an apparatus: the letter calls it the tell.

Why this definition. Why allow the apparatus to depend on $|\psi\rangle$? For the same reason as in §11.1: the definition should forbid no cleverness inside the machine. It also leaves a loophole, and the loophole is the whole story.

An example that works too well. Let the tell be one more qubit starting in $|0\rangle$, and tie the second stone to the tell with Chapter 8's three CNOTs. By Theorem 8.8 they are a swap, exactly, so

$$|\psi\rangle\,|\psi\rangle\,|0\rangle \;\longmapsto\; |\psi\rangle\,|0\rangle\,|\psi\rangle$$

for every $|\psi\rangle$. The second stone has become the blank $|\Sigma\rangle = |0\rangle$, and the first is untouched. It is a deleting machine by the letter of the definition, and it deleted nothing. The tell's final state is $|A_\psi\rangle = |\psi\rangle$: the stone has been moved into the machine. In the sketchbook the knot is pulled straight. The three spider-CNOTs, drawn bare, carry $\frac{1}{2\sqrt{2}}$ (Theorem 8.8), so $2\sqrt{2}$ stands in front:



Pati and Braunstein saw this coming. Swapping the second copy into the apparatus solves the defining equation, they note, but it is only erasure in the ordinary sense, with the extra copy playing no part; so they "explicitly exclude swapping" as a form of deleting. Then they show that linearity leaves nothing else. Here is their argument, in outline. Write $|\psi\rangle = \alpha\,|H\rangle +$

$\beta\,|V\rangle$ in the horizontal and vertical polarizations, and let the machine act on $|H\rangle\,|H\rangle$ and $|V\rangle\,|V\rangle$ as required, with final tells $|A_H\rangle$ and $|A_V\rangle$. On the mixed input $\frac{1}{\sqrt{2}}(|H\rangle\,|V\rangle + |V\rangle\,|H\rangle)$ it may produce any fixed state at all. Linearity then makes the output on $|\psi\rangle\,|\psi\rangle$ a *quadratic* polynomial in α and β, and it must equal the target $(\alpha\,|H\rangle + \beta\,|V\rangle)\,|\Sigma\rangle\,|A_\psi\rangle$ for every α and β. Since the output on the mixed input does not depend on them, they conclude that $|A_\psi\rangle$ must be linear in them: $|A_\psi\rangle = \alpha\,|A_H\rangle + \beta\,|A_V\rangle$. Normalization, for every α and β, then forces $|A_H\rangle$ and $|A_V\rangle$ to be orthogonal. So the tell carries α and β faithfully in a two-dimensional part of itself: "merely swapping", in their words. Their machine is only required to be linear. They remark that if it were known to be unitary, it "might work like the time-reverse of cloning".

For unitary machines there is a one-line version in the likeness of §11.2, and it is the headline's argument run the other way.

> **Result 11.6 (no-deleting, loop form).** Let U be a unitary with $U(|\psi\rangle\,|\psi\rangle\,|A\rangle) = |\psi\rangle\,|\Sigma\rangle\,|A_\psi\rangle$ and $U(|\varphi\rangle\,|\varphi\rangle\,|A\rangle) = |\varphi\rangle\,|\Sigma\rangle\,|A_\varphi\rangle$, and let $s = \langle\varphi|\psi\rangle \neq 0$. Then
>
> $$\langle A_\varphi|A_\psi\rangle = s.$$
>
> The tell's two final states are exactly as alike as the stones were: whatever was deleted from the second stone is now in the tell.

Proof. Compare likenesses before and after, as in §11.2. Before: $\langle\varphi\varphi A|\psi\psi A\rangle = s \cdot s \cdot 1 = s^2$, two loops and the tell's starting loop. After: $\langle\varphi\Sigma A_\varphi|\psi\Sigma A_\psi\rangle = s \cdot 1 \cdot \langle A_\varphi|A_\psi\rangle$, one loop, the blank's loop, and the tell's final loop. A unitary box meets its mirror and vanishes, so the two are equal: $s^2 = s\,\langle A_\varphi|A_\psi\rangle$. Divide by $s \neq 0$. ■ See `Code/Ch11/NoDeleting.py`, which also shows that a would-be deleter whose tell ignores the stone ($|A_H\rangle = |A_V\rangle$), extended by linearity, sends some states to vectors that are not of unit length.

In the picture it is a matter of counting loops. Two loops went in. One comes out on the stones, so the missing loop must come out on the tell. The three-cord knot shows how that happens for every state at once, and

Pati and Braunstein's argument above shows there is no other way: the tell *becomes* the stone. Result 11.6 is this book's formulation for unitary machines; Pati and Braunstein's theorem is stronger in one respect, since it asks only for linearity.

Feynman's Notebook.

Uncopying. Arun Kumar Pati and Samuel L. Braunstein, of the University of Wales at Bangor, turned the cloning question round. Given two photons in the same unknown polarization, can one of them be reset to a standard state while the other is left alone? Their motive was practical. A blank made that way could be copied onto, approximately by a deterministic cloner or exactly by a probabilistic one, and a quantum computer could store new information in an already computed state by deleting the old. Classically, they note, one copy can be deleted against the others in a perfectly reversible way, while erasing a single copy has an energy cost (Landauer's principle). What they proved is that the quantum case is different: this is their "quantum no-deleting" principle, published in *Nature* 404, 164 (2000). They call it "complementary in spirit" to no-cloning. They also suggest that it gives quantum files a kind of intrinsic security: no one can obliterate one copy of an unknown file from a collection of several copies.

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§11.5 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 10's eighty. One introduces a new picture element, the cut (87), drawn in §11.7. The rest are new uses of elements we already had: the copier of row 59, the cup, the doubled wire, the read-spider and the closed picture.

#	Algebra (Dirac)	Picture (Coecke)
81	the copy map $\delta = \|00\rangle\langle 0\| + \|11\rangle\langle 1\|$; $\delta\|0\rangle = \|00\rangle$, $\delta\|1\rangle = \|11\rangle$ (normalized, exact); the only linear map copying $\|0\rangle, \|1\rangle$	the green dot with one leg in and two out; a red point in comes out as two red points (row 59, exact once the points carry their $\frac{1}{\sqrt{2}}$)
82	$\delta\|\pm\rangle = \|\Phi^{\pm}\rangle$; $\delta(\alpha\|0\rangle + \beta\|1\rangle) = \alpha\|00\rangle + \beta\|11\rangle \neq \|\psi\rangle\|\psi\rangle$ unless $\alpha\beta = 0$	a **green** point into the green copier **fuses** into one cup $\times\frac{1}{\sqrt{2}}$ (a half-turn point gives the cup carrying it): a pair, not two copies; **complementarity**: red points copy, green points fuse
83	$\delta^{\dagger}\delta = I$; the likeness $\langle\varphi\|\psi\rangle$; $\langle\varphi\varphi\|\psi\psi\rangle = \langle\varphi\|\psi\rangle^2$	the copier meets its mirror and they leave a bare wire (fusion along two wires, then self-loop removal; van de Wetering §4.1); a likeness is a closed picture, and two likenesses side by side multiply
84	**no-cloning:** $s = s^2\langle A_{\varphi}\|A_{\psi}\rangle \Rightarrow s = 0$ or $\|s\| = 1$	one loop = two loops only if the loop is all or nothing

#	Algebra (Dirac)	Picture (Coecke)
85	copy, then discard one copy = dephasing $\sum_i P_i \rho P_i$; **broadcasting** $\{\rho_0, \rho_1\}$ possible iff $\rho_0\rho_1 = \rho_1\rho_0$ (Barnum et al., stated)	the doubled copier with one output capped onto its mirror fuses into a four-legged green dot: a look, the stone left as looked at, the record not read
86	**no-deleting:** $\langle A_\varphi \vert A_\psi \rangle = \langle \varphi \vert \psi \rangle$; the copy is moved into the apparatus (Pati–Braunstein)	three CNOT cords between copy and apparatus pull into a crossing: $\psi\,\psi\,0 \mapsto \psi\,0\,\psi$
87	rank across a cut $\leq 2^k$; $\dim \operatorname{span}\{\psi \otimes \psi\} = 3 > 2$; a universal copier forces rank 1	**the cut**: a thin, densely dotted grey curve across a picture, not a wire; rank 1 across a cut means the picture falls apart there, **disconnected** — *new element*
88	the Forbidden Telegraph: $p = \frac{1}{4}$ whenever $z_1 = z_2$ (cross); $p = \frac{1}{2}$ whenever $m = z_1 = z_2$ (plain); $p = 0$ otherwise; far marginal the same	the master: at the cross setting, fusion and Hopf leave the near record on its own

Every row is exact, scalar included; none needs Chapter 8's "up to a non-zero scalar". Row 85's second half and the ideal cloner of row 88 are the only entries the pictures do not prove. The first is cited; the second, as §11.7 explains, has no picture at all.

§11.6 The Theorem, Twice

Result 11.3 (no-cloning, unitarity form), restated. Let U be any unitary on stone $\otimes$ blank $\otimes$ apparatus, the apparatus of any size, with fixed starting states $|0\rangle$ and $|A\rangle$. If $U(|\psi\rangle\,|0\rangle\,|A\rangle) = |\psi\rangle\,|\psi\rangle\,|A_\psi\rangle$ and $U(|\varphi\rangle\,|0\rangle\,|A\rangle) = |\varphi\rangle\,|\varphi\rangle\,|A_\varphi\rangle$, then, with $s = \langle\varphi|\psi\rangle$ and $a = \langle A_\varphi|A_\psi\rangle$,

$$s = s^2 a, \qquad \text{so} \qquad s = 0 \ \text{ or } \ |s| = 1.$$

Dirac's Blackboard

Step 1: the likeness of the inputs. Inner products of product states multiply (Proposition 4.1), and $\langle 0|0\rangle = \langle A|A\rangle = 1$:

$$\langle \varphi\, 0\, A \,|\, \psi\, 0\, A\rangle \;=\; \langle\varphi|\psi\rangle\,\langle 0|0\rangle\,\langle A|A\rangle \;=\; s.$$

Step 2: unitarity. For any vectors x, y, $\langle Ux, Uy\rangle = \langle x|\,U^\dagger U\,|y\rangle = \langle x|y\rangle$, because $U^\dagger U = I$ (Theorem 3.2, whose proof works for a box on any number of wires). With $x = \varphi\,0\,A$ and $y = \psi\,0\,A$:

$$\langle U(\varphi\,0\,A),\, U(\psi\,0\,A)\rangle \;=\; s.$$

Step 3: substitute the cloning hypothesis, and multiply again by Proposition 4.1:

$$\langle \varphi\,\varphi\,A_\varphi \,|\, \psi\,\psi\,A_\psi\rangle \;=\; \langle\varphi|\psi\rangle\,\langle\varphi|\psi\rangle\,\langle A_\varphi|A_\psi\rangle \;=\; s^2 a.$$

Step 4: equate. Steps 2 and 3 compute the same inner product, so $s = s^2 a$.

Step 5: take moduli. $|s| = |s|^2|a| \le |s|^2$, since $|a| \le 1$ by Cauchy–Schwarz for the unit vectors $|A_\varphi\rangle$, $|A_\psi\rangle$. If $s \neq 0$, then $|s| \ge 1$; and $|s| \le 1$ by Cauchy–

Schwarz again, so $|s| = 1$, which for unit vectors means $|\varphi\rangle = e^{i\theta}|\psi\rangle$.

Step 6: the example. $|0\rangle$ and $|+\rangle$ have $s = \frac{1}{\sqrt{2}}$, and $0 < \frac{1}{\sqrt{2}} < 1$. No unitary clones both, so none clones every state. ■

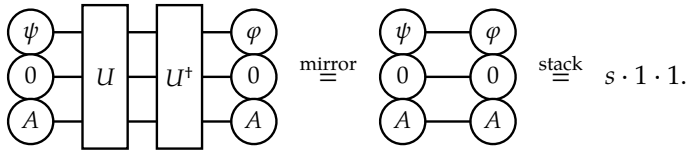
The whole argument is Step 4, a single line. Everything else is bookkeeping and one inequality.

Coecke's Sketchbook

We work with a general unitary box U on three wires, stone, blank and apparatus, top to bottom. The copier δ is one example of such a box (with the blank loaded and no apparatus), but a proof drawn for δ alone would prove nothing about every U. Build one closed picture. The states ψ, 0, A enter on the left. U acts, then its mirror image $U^\dagger$, and the effects φ, 0, A close the three wires on the right. By the mirror of a row (Theorem 3.1), the right half, $U^\dagger$ followed by the three effects, is the mirror image of the left half with φ in place of ψ. We evaluate the picture in two ways.

First way, Step 1: a box meets its mirror (§3.3). U is unitary, so U followed by $U^\dagger$ is three bare wires. The rule is licensed by Theorem 3.2 and by nothing else.

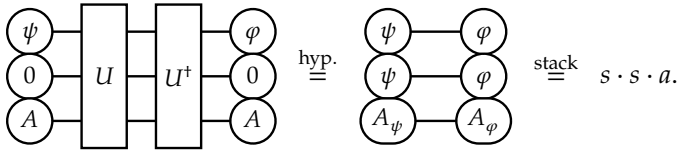
Step 2: stacked closed pictures multiply (Proposition 4.1). Three separate likenesses remain. The blank's likeness and the apparatus's likeness are each 1 (Theorem 2.1).



Second way, Step 1: the hypothesis. The left half, the states $\psi, 0, A$ followed by U, is the state ψ, ψ, A_ψ.

Step 2: its mirror image (Theorem 3.1). The right half, $U^\dagger$ followed by the effects $\varphi, 0, A$, is the mirror image of the hypothesis for φ: the effects $\varphi, \varphi, A_\varphi$.

Step 3: stacked closed pictures multiply (Proposition 4.1). Now the three likenesses are ψ against φ, ψ against φ again, and A_ψ against A_φ:



The same picture has been evaluated twice, so $s = s^2 a$. The first way leaves one loop that matters; the second leaves two, side by side. The last step, from $s = s^2 a$ to "$s = 0$ or $|s| = 1$", is a fact about complex numbers, and the sketchbook borrows it from the blackboard's Step 5. ■

The worked instance of the first way's Step 1 is §11.2's picture: when U is the copier (with its blank inside, §11.1), the box meeting its mirror is fusion along two wires and a self-loop removed. In that case the second way fails at Step 1, because the hypothesis is false: δ sends $|\psi\rangle$ to $\alpha\,|00\rangle + \beta\,|11\rangle$, not to $|\psi\rangle\,|\psi\rangle$.

The verdict

The blackboard wins on length. Its proof is one line of inner products, $s = \langle U(\varphi 0 A), U(\psi 0 A)\rangle = s^2 a$, and the rest is an inequality. The sketchbook needs the box to meet its mirror, the mirror of the hypothesis, and the rule that separate closed pictures multiply. Then it hands the final step, a fact about numbers, back to the blackboard.

The picture wins on insight, and it can say why. The input likeness is one loop; the copies close into two *separate* loops, because a clone is a product and a product is a picture in two pieces. A closed loop is a number, so a faithful copier would make a number equal its own square. The picture also gives the positive reading that the blackboard does not show. The copier copies its own basis: red points pass through the green copier as two red points, and green points fuse into a cup, a pair and not two copies. That is complementarity, and it is why the forbidden machine fails exactly where it does. None of this is a bound on how well an imperfect copier can do, a

security claim, or a statement about any machine outside the linear theory. A rewrite proves an equality of processes, and here the equality is $s = s^2 a$.

Coecke's Sketchbook.

What the loops do not say. The one-loop, two-loop argument is about *perfect* copies: it shows that $s = s^2 a$ fails, and nothing more. It does not say how close to a copy a real machine can come, or at what cost. That question has a literature of its own, on approximate and optimal cloning machines; Pati and Braunstein's closing paragraph points to it, and suggests that one could try to build an approximate *deleting* machine by analogy. Nor does the picture know why $|a| \leq 1$; that is Cauchy–Schwarz, from the blackboard. And it cannot tell a unitary box from any other by its outline. Chapter 3's warning holds here with full force: cancel a box against its mirror only because the blackboard has shown that the box is unitary, and say so. The copier is an isometry, so $\delta^\dagger \delta$ cancels and $\delta \delta^\dagger$ does not. What the loops *do* give, and give at a glance, is the reason: one loop against two.

---

§11.7 The Forbidden Telegraph

The Crab's last hope is the boldest. The post to Delft takes a week. Suppose one half of a fresh pair stays here and the other goes to Delft, and suppose the Delft end had a perfect copier. Could the near end then send word by choosing how to read its half?

On the blackboard, the machine that cannot exist. Share $|\Phi^+\rangle$ between a near stone a and a far stone b. The near end reads a at a **setting**, plain (the $|0\rangle, |1\rangle$ basis) or cross (the $|\pm\rangle$ basis), and gets a record m. The far end copies b and reads *both* copies plain, getting records z_1 and z_2. Regrouping the four terms, $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle)$. So at plain the far stone is left in $|0\rangle$ or $|1\rangle$, and at cross in $|+\rangle$ or $|-\rangle$, each half the time.

Now suppose, against everything so far, a machine K with $K|\psi\rangle = |\psi\rangle \otimes |\psi\rangle$

for every $|\psi\rangle$, and apply it on each branch. At plain the far end gets $|00\rangle$ or $|11\rangle$, and its two records always agree. At cross it gets $|{+}{+}\rangle$ or $|{-}{-}\rangle$, and each copy, read plain, is a fair coin independent of the other, so the records agree only half the time. The far end, with no letter, sees agreement with probability 1 or $\frac{1}{2}$ according to the near setting. Run it on many pairs and the near end is sending a message: a telegraph.

That is a blackboard computation of something that cannot be drawn. K is not linear, and every picture in this book is a linear map. The sketchbook cannot draw the forbidden machine; only the blackboard can state it and watch it contradict itself. So we turn to the machine that does exist.

Dirac's Blackboard.

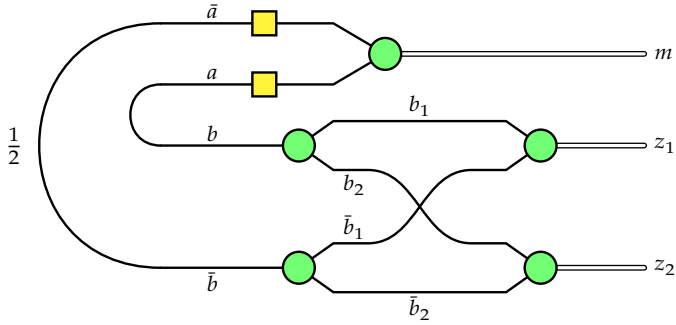
Why only a nonlinear machine could read the setting. Before any letter arrives, the far stone is $\frac{1}{2}I$ whatever the near end does (Result 9.3). But the same matrix can be written as two different mixtures:

$$\tfrac{1}{2}I \;=\; \tfrac{1}{2}\,|0\rangle\langle 0| + \tfrac{1}{2}\,|1\rangle\langle 1| \;=\; \tfrac{1}{2}\,|+\rangle\langle +| + \tfrac{1}{2}\,|-\rangle\langle -|\,.$$

The plain setting prepares the first mixture at the far end, and the cross setting prepares the second. A linear machine acts on the matrix, $\frac{1}{2}I$, and cannot tell how it was mixed. K acts stone by stone, and it sends the two mixtures to $\frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$ and $\frac{1}{2}(|{+}{+}\rangle\langle {+}{+}| + |{-}{-}\rangle\langle {-}{-}|)$. These differ: the first gives agreeing plain readings with probability 1, the second with probability $\frac{1}{2}$. Telling apart two descriptions of the same density matrix is exactly what linearity forbids. It is also exactly what a telegraph would need.

The Forbidden Telegraph. Now the real press. Draw the whole scene in unfolded doubled form, so that every line shows, as in Chapter 9 §9.6 and Chapter 10 §10.7. On the left, the doubled pair $|\Phi^+\rangle\langle\Phi^+|$: two cups, one joining the near stone a to the far stone b and one joining their mirror copies $\bar{a}$ and $\bar{b}$, with $\frac{1}{2}$ written beside them. As always, cup $= |00\rangle + |11\rangle$, so each normalized pair is $\frac{1}{\sqrt{2}}$ cup and the doubled pair is $\frac{1}{2}$ cup $\otimes$ cup. At the near

end, a and $\bar{a}$ run into Chapter 9's read-spider, which emits the classical near record m. At the cross setting each of a and $\bar{a}$ first passes through a yellow Hadamard box (row 56), which turns the plain reading into a crosswise one; at the plain setting the two boxes are simply absent. At the far end, b runs into the green copier, whose outputs are the first and second struck stones b_1 and b_2. The mirror copy $\bar{b}$ runs into the copier's conjugate, which is the green copier again since δ is real, with outputs $\bar{b}_1$ and $\bar{b}_2$. Each struck stone is read plain: one read-spider takes b_1 with $\bar{b}_1$ and emits the classical record z_1, and another takes b_2 with $\bar{b}_2$ and emits z_2. To bring each struck stone next to its own mirror, $\bar{b}_1$ and b_2 cross once. A crossing neither joins nor breaks a line.



The lines are the near stone a and its mirror $\bar{a}$, each running from its cup through its Hadamard box into the near read-spider; the far stone b and its mirror $\bar{b}$, from the cups into the copier and its conjugate; the struck stones b_1 and b_2 and their mirrors $\bar{b}_1$ and $\bar{b}_2$, each a leg of a copier running to a far read-spider; and the three classical records m, z_1 and z_2. The drawing is at the cross setting. We call this diagram **The Forbidden Telegraph**.

Result 11.7 (the Forbidden Telegraph). The Forbidden Telegraph evaluates, exactly, to the joint distribution of the three records:

$$\text{cross:}\quad p(m, z_1, z_2) = \tfrac{1}{4}\,\delta_{z_1 z_2},$$

$$\text{plain:}\quad p(m, z_1, z_2) = \tfrac{1}{2}\,\delta_{m z_1}\delta_{z_1 z_2}.$$

Here $\delta_{z_1 z_2}$ is the Kronecker delta, 1 when the two records agree and 0 otherwise (likewise $\delta_{m z_1}$); it is not the copier. At both set-

tings the struck stones always agree, $z_1 = z_2$, and the far records have the same distribution, $\sum_m p = \frac{1}{2}\,\delta_{z_1 z_2}$. **No signal crosses.** At the cross setting the near record is a fair toss independent of the far records.

Proof, on the blackboard. At plain, the near reading gives m with probability $\frac{1}{2}$ and leaves b in $|m\rangle$; the copier makes $|mm\rangle$, and both far records read m. At cross, the near reading gives $|\pm\rangle$ with probability $\frac{1}{2}$ each and leaves b in the same $|\pm\rangle$; the copier makes $|\Phi^{\pm}\rangle$ (Result 11.2), whose two plain readings agree and are each a fair coin, whichever sign came. Sum over m for the far marginal. ■

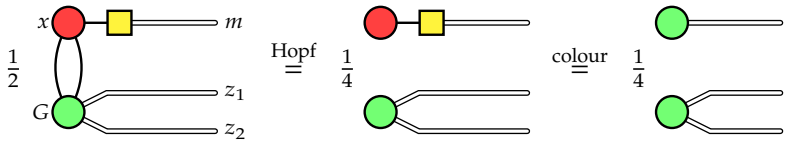
Proof, in the sketchbook. Only connectivity matters (§8.5), and a cup is a bent wire, so the near read-spider is joined to the copier by one wire and to the copier's conjugate by another. Every scalar below is exact.

At the **plain** setting every dot is green and phase-free. **Fusion** (Theorem 8.1) merges the near read-spider, the copier, its conjugate and both far read-spiders into one green dot. The six wires among them close two self-loops, which are **removed** (van de Wetering §4.1). What is left is $\frac{1}{2}$ times a green dot with three legs, m, z_1 and z_2: $\frac{1}{2}\sum_i |iii\rangle$. All three records agree.

At the **cross** setting, five steps.

Step 1: colour change (Theorem 8.2). Put two Hadamard boxes on the record m, which changes nothing, since $H^2 = I$. The record leg is a leg like any other to the rules; the double stroke is only a reminder (§9.1). Now the near read-spider wears a Hadamard on every leg, so it is a red dot x of phase 0, with one Hadamard box left on m.

Step 2: fusion (Theorem 8.1). The copier, its conjugate and the two far read-spiders are green and joined, so they fuse into one green dot G. The four wires among them contain one wire too many for a tree, which leaves one self-loop on G; remove it (van de Wetering §4.1). G keeps the legs z_1 and z_2, and two wires to x, one through each cup.



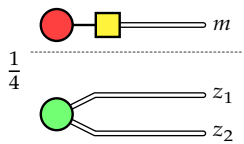
Step 3: Hopf (Theorem 8.6). G and x are a green dot and a red dot joined by two wires. Unfuse the legs that take no part (fusion read backwards), so that the shape is exactly Theorem 8.6's. Apply the rule: the two wires fall away, with the factor $\frac{1}{2}$ of Theorem 8.6. Then fuse back. The red dot keeps only its leg toward m, and G keeps only z_1 and z_2.

Step 4: colour change again. A red point followed by a Hadamard box is a green point, exactly: $H(\sqrt{2}\,|0\rangle) = |0\rangle + |1\rangle$.

Step 5: read off. The result is $\frac{1}{4}\,(|0\rangle + |1\rangle)_m \otimes (|00\rangle + |11\rangle)_{z_1 z_2}$: the near record is a fair toss on its own, and the far records agree and are a fair toss between them. ■ See `Code/Ch11/ForbiddenTelegraph.py`, which builds the doubled diagram at both settings, counts its lines, checks both distributions exactly, and derives the disconnection with the rules.

At the cross setting the near record has come off the picture altogether. To say that precisely we need one new mark.

In the sketchbook: the cut. Draw a thin, densely dotted grey curve across a picture, dividing it into two parts. We call it a **cut**. It is not a wire and carries nothing; it is a line of inspection, drawn over the picture to ask how the two sides are joined. The two parts are joined only through the wires the curve crosses. (It is drawn dotted and grey, never dashed: a dashed wire is Chapter 8's Hadamard edge.) Here is the telegraph at the cross setting after the Hopf step, with a cut drawn between the near record and the far records. It crosses no wire at all:



On the blackboard: rank across a cut. Bend all the open wires on one side of the cut round to be inputs (Theorem 6.5), so that the picture becomes a process from one side to the other. Its matrix has a **rank**, the dimension of its image, and we call that the picture's **rank across the cut**. For a state on two parts this is its Schmidt rank. Every path from one side to the other runs through a wire that the cut crosses, so the process factors through those wires. If the cut crosses k qubit wires, the matrix factors through a space of dimension 2^k, and so its rank is at most 2^k. A cut crossing no wire gives rank at most $2^0 = 1$: the picture is a product of its two parts, and it falls apart there. We call such a picture **disconnected** across the cut. Rank 1 across the cut in the telegraph means $p(m, z_1, z_2) = p(m)\,p(z_1, z_2)$: independence, at the cross setting. At the plain setting the three records hang on one green dot, and any cut separating m from z_1, z_2 crosses a wire.

Why this definition. Why draw a cut rather than simply compute a rank? Because in the picture the rank can be read from where the curve can pass. Rewriting moves wires, and a picture that has fallen apart shows it by admitting a cut through empty paper. The picture gives an upper bound for free. A lower bound, such as "this cup really is connected", still needs the blackboard: the cup's rank is 2 because its matrix is the identity.

As in Chapter 9, the independence is honest only on average. Across the cross setting's cut, the near record is independent of the far ones. At the plain setting it is not. Condition on m there, and the far records follow it exactly, $z_1 = z_2 = m$. What the far end cannot see is the *setting*, because it never sees m. Averaged over the near outcome, its records are $\frac{1}{2}\delta_{z_1 z_2}$ either way. That is Chapter 9's no-signalling (Result 9.3, and the master of §9.6), now with a copier on the far wire.

In the dialogue, just after the press drops its two struck stones and before they are read, the Crab glances at *Drawing Hands*: two hands, each drawing the other.

The cut also says why no better press could ever have helped the Crab.

> **Result 11.8 (a copier cannot widen a wire; a universal copier would disconnect every cup).** (i) For every linear process D with one qubit wire in and two out, the picture $(I \otimes D)$ cup has

rank at most 2 across a cut crossing the wire into D. Equivalently, the image of D has dimension at most 2. (ii) The clones $|\psi\rangle \otimes |\psi\rangle$ span a space of dimension 3. So no linear process copies every state. (iii) More generally, on a wire of dimension r the clones span a space of dimension at least $r(r+1)/2$. A linear process copying every state of such a wire exists only if $r(r+1)/2 \le r$, that is $r \le 1$, and on a wire of dimension 1 every cup has rank 1 across every cut: disconnected. (iv) What the copier actually does to a cup is exact: $(I \otimes \delta)\,\text{cup} = |000\rangle + |111\rangle$, Chapter 7's three-legged node (row 46), connected across every cut.

Proof. (i) The cut crosses one wire, so the rank is at most $2^1 = 2$; bending $(I \otimes D)\,\text{cup}$ back into D (Theorem 6.5) shows the same number is the dimension of D's image, which is spanned by $D|0\rangle$ and $D|1\rangle$. (ii) The clones include $|00\rangle$, $|11\rangle$ and $|{+}{+}\rangle$, and $|{+}{+}\rangle - \frac{1}{2}|00\rangle - \frac{1}{2}|11\rangle = \frac{1}{2}(|01\rangle + |10\rangle)$; these three are independent. They also span everything, since every clone $\alpha^2|00\rangle + \alpha\beta(|01\rangle + |10\rangle) + \beta^2|11\rangle$ is a combination of them. A universal copier would need an image of dimension 3 through a wire that allows 2. (iii) For any two basis vectors u, v of the wire, $(u+v)\otimes(u+v) - u\otimes u - v\otimes v = u\otimes v + v\otimes u$. So the span of the clones contains every $e_i \otimes e_i$ and every $e_i \otimes e_j + e_j \otimes e_i$ with $i < j$, which are $r + \frac{1}{2}r(r-1) = \frac{1}{2}r(r+1)$ independent vectors. The inequality $\frac{1}{2}r(r+1) \le r$ gives $r \le 1$. A cup on a one-dimensional wire is a single term, a product. (iv) The cup is a phase-free green dot with two legs, and the copier fuses with it (Theorem 8.1). ■ See `Code/Ch11/ForbiddenTelegraph.py`, which checks (i) on random D, (ii) on random clones, and (iv) exactly.



On the left, any process D on one half of a cup, with the cut crossing the single wire that feeds it: whatever D does, everything on the far side of the cut came through one wire. On the right, the green copier on one half of

a cup. The cup is not copied. It grows a leg, into the three-legged state of Chapter 7, and it stays connected: any cut that separates one leg from the other two crosses one wire, and the rank there is 2. This book's pictures are built on the connected cup (Chapter 6). A universal copier could exist only in a world where every cup came apart, so the universal copier and the connected cup cannot live in the same picture. That is the precise sense in which a universal copier would disconnect every cup. The argument is the book's own; we attribute it to no one.

Coecke's Sketchbook.

The cut is not a wire. The dotted curve is drawn *over* a picture, never *in* it. It carries no system, it is not counted among the lines, and it changes no value; erase it and the picture means what it meant before. It is a question put to the picture: how many wires join this side to that? Its answer is an upper bound. If the curve crosses k qubit wires, the rank across it is at most 2^k, but it may be less. A product state drawn carelessly, as a single box with two wires, admits no empty cut until it is rewritten into two pieces. So a cut through empty paper *proves* disconnection, while a cut that must cross a wire proves nothing by itself about connection. For that you go to the blackboard, or to a rewrite that ends in a spider no cut can avoid, like the plain telegraph's single green dot. In this chapter the cut crosses nothing at the cross setting, and that is the whole of the no-telegraph theorem in one stroke of grey.

Feynman's Notebook.

Faster than the post. The link between copying and signalling is older than this chapter's small telegraph. Pati and Braunstein, citing both 1982 papers, put it in one sentence: if an arbitrary state could be cloned, then with non-local resources one could send signals faster than light. They go on to ask whether their own no-deleting principle could likewise be derived from the impossibility of faster-than-light signalling, as no-cloning can, and they leave the question open. The telegraph computed in

this section, a copier on one half of a Bell pair, read plain at the far end and at either setting at the near end, is this book's own worked example of the remark. It is not taken from either paper. Note which way the logic runs. No-signalling was proved in Chapter 9 for every trace-preserving channel at the near end (Result 9.3): the far half stays $\frac{1}{2}I$, and the real copier, being linear, acts on that $\frac{1}{2}I$ alone and inherits the law. Only a nonlinear machine could break it, and the sketchbook has no way to draw one.

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§11.8 What May Be Passed On

Let us gather it up. The green copier copies exactly the states of its own basis. Fed a red point, it gives two red points; fed a green one, it fuses into a cup and makes a pair. Linearity forbids any machine that copies the plain states from doing better, apparatus or no apparatus (§11.1). Unitarity forbids any machine at all from copying two states unless they are orthogonal or the same, because a faithful copy would turn one loop into two, and a likeness into its own square (§11.2, proved twice in §11.6). A copy made and thrown away is a look that nobody read, and even the gentler demand of broadcasting fails unless the states commute (§11.3). A copy cannot be unmade either. The best a deleter can do is move the copy into its own works, where it sits exactly as unlike as the stones were (§11.4). And a perfect copier on one half of a pair would be a telegraph. The real one is not, because at the cross setting the near record falls away from the far ones, across a cut that crosses nothing. The only copier that could do better exists in a world where every cup has come apart (§11.7).

So Chapter 10's question has its answer. Teleportation moved the message and did not copy it, and it could not have done otherwise. A quantum state can be passed on, or answered, but never duplicated.

That fact has a use, and it is the next chapter. Barnum and his colleagues closed their paper with one: in quantum cryptography over a noisy channel,

their theorem leaves an eavesdropper no means of taking the signal meant for the legitimate receiver without disturbing what that receiver gets. Think of a spy on the line. She cannot copy what passes, so she cannot keep a copy and let the original go by untouched. To learn anything she must look. A look is made at some setting, plain or cross, and this chapter has shown what a look at the wrong setting does: it turns a crosswise stone grey, as Eve's customers found. Chapter 12 makes those two settings into the two keys of a lock, built of light. There the mark a spy's look leaves on the stones becomes an alarm.

Yes–No Duet, with Eavesdropper

寒灯や
二人の窓に
人の息

kantō ya
futari no mado ni
hito no iki

Lamp on a cold night —
on the window of the two
someone else's breath.

[*The Tortoise's parlour, early evening. On the table lies a large black tray, and on it, set out beforehand, stands the Tortoise's kit. In the middle is a small brass lantern with a lever, and a port on either side. From its left port a slender glass rod, on low brass forks a finger's breadth above the lacquer, runs to a clear crystal in a brass collar at the tray's left end. From its right port another rod runs most of the way along to a short brass sleeve, and from the sleeve a further rod runs on to another crystal and collar at the right end. Beneath each rod, in the lacquer, lies its reflection; beneath the lantern, the sleeve and the crystals, theirs. Off the tray, along the table's far side towards the window, beyond it, a deep box of draughtsmen stands open, and beside it the Tortoise's drawing-cloth is spread, bare, her string and rings in her pouch; a small brass bell sits on the table's near corner at the dresser end. The door opens, and the Tortoise and Achilles come in out of the dark.*]

[*The Tortoise finds the matches on the dresser by feel and lights the room lamp that*

stands high there, and the parlour comes up out of the dark around it. It is a little after five and full night already; the window on the long wall is black, and the cold stands off it into the room. The stove, banked since morning, wakes when she opens its damper, with a small roar behind its mica door and a first tick from its iron. The table runs down the middle of the room, its left end towards the stove and its right towards the dresser, and the kit lies along it as she left it this morning: the long rod to the stove end has a faint green in the thick of its glass; the one from the lantern to the sleeve carries a single seed of air near its middle; the far rod, on to the dresser end, is the palest of the three. The supper bell on the table's corner is plain brass, its turned wooden handle worn pale by forty years of suppers, and the draughts box is older still, its pieces rubbed smooth at the rims. On the end wall beyond the dresser end, beside the dresser, hangs a print in a narrow black frame — Escher's Three Worlds*, a pond: leaves on its surface, trees reflected in it, a fish beneath. The room lamp reaches it from the side, and it takes the light quietly, as it has every evening since long before the tray was set out on that table, and as it will after. On the lower pane of the black window, at the very bottom corner, the frost has made a start: a few feathers, fine as hair, each going its own way.*]

Achilles: [*stamping on the mat*] I can't feel my toes. Every rut in your lane has a skin of ice on it, and I found them all.

Tortoise: You would walk on the crown of the road. Boots by the stove, and I'll see to the kettle.

[*Achilles sits down on the fender and works his boots off, and stands them by the stove, where they begin at once to steam. His gloves he pulled off at the door by the fingertips, as one does with cold hands, and they lie on the fender beside him inside out, with his woollen cap, inside out too, dropped on top of them. The Tortoise fills the kettle from the jug on the dresser and pushes it over the firebox.*]

Achilles: [*crossing to the window, rubbing his hands*] It's begun already, look — ferns, from the bottom corner up. And nothing like this morning's. Not a frond of it in the same place.

Tortoise: It never is. It starts fresh each night, from whatever the cold finds on the glass.

[*Achilles puts a fingertip, warm from the rubbing, to the glass where the first*

feathers have started, to feel how cold it is. They are gone at once from under it, and a small round of wet bare glass is left in their place, with the frost beginning again all round its edge.]

Achilles: Mr Crab stood on his step with a candle to see us off, and then came pattering down the path after us in his slippers. [*He takes a folded letter from his coat.*] This had come by the last post, just as we were leaving. It's to him, from Alice — Bob's Alice. He wouldn't open it. He pressed it on me and said, "It is about light, and she has the glass for it."

Tortoise: And what are we to do with it?

Achilles: Read it here, and tell him in the morning what it says. He said he had done enough looking for one day.

Tortoise: [*setting out two cups*] Poor Mr Crab. A whole afternoon with his claws off the machine he has wanted since he was small. We'll give him a good account.

Achilles: [*turning to the table*] And you've set all this out. When?

Tortoise: This morning, before I came over to his. I had a feeling the day would end with light.

Achilles: And the supper bell on the corner — is that a promise?

Tortoise: Supper is later. The bell stays where it is till it's wanted.

[*The kettle comes to the boil over the firebox, and the Tortoise makes the tea in a brown pot and fills both cups. Achilles stands with his back to the stove and his cup in both hands until the feeling comes back into his toes. The Tortoise keeps her shawl about her shoulders; at the dresser end of the room it is still cold enough to see one's breath.*]

First Challenge — *Same Setting, Same Answer*

Achilles: [*sitting, and breaking the seal*] Then the letter first. She writes a round hand. [*reading*] "Dear Mr Crab — Bob and I have given up sending

stones by the carrier; a season is too long to wait for a word. The carrier has laid a glass line between our towns instead, and we send glints down it, one at a time, from a lamp of mine to a spar-key of Bob's."

Achilles: [*reading on*] "What we want is a key: a string of yeses and noes that only the two of us know, to lock our letters with. The letters themselves go by the ordinary post, like this one, which anyone may read but nobody can alter without its showing. I want a lock made of light."

Achilles: [*looking up*] A lock made of light. What is a glint, exactly?

Tortoise: The least light there is: a single spark, too small to be split. [*She lays a hand beside the lantern in the middle of the tray.*] This is a glint-lamp. My great-uncle kept a light on the north coast, and made it to signal to the next light along, one glint at a time.

Tortoise: One press of this lever lets out a twin glint: one glint from this port and one from that, of one piece, like Mr Crab's pairs of stones. Nobody sees a glint in flight; it shows only where it lands.

Achilles: And the tray? It's like his slab.

Tortoise: It's my grandmother's tea-tray, and tonight it does the same work. Whatever stands on it stands again underneath, in the lacquer; a glint has its mirror side, as his stones had, and the tray keeps both in view.

Tortoise: [*a claw along the glass*] These rods carry a glint along them and out at the end, as a speaking-tube carries a voice. This one runs to your end. That one runs to the brass sleeve, which holds its end against the next rod's end, so that a glint passes straight through as if the sleeve were not there, and on to my end.

Tortoise: And at each end, a spar-key: a crystal of Iceland spar in a brass collar. Behind the crystal are two small frosted windows, yes above and no below, facing whoever sits behind the key. A glint that comes through lands in exactly one window, which flashes once; and then the glint is spent.

Achilles: [*bending to the collar at the left end*] The collar turns.

Tortoise: It chooses the question. Turn it till you feel it stop: at the upright stroke it asks the upright question, and at the cross it asks the crosswise

one. The stops make no sound, so a hand cupped over the collar keeps its setting to itself.

Achilles: There's another mark past the cross — a little curl.

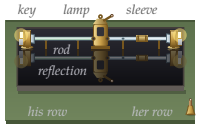
Tortoise: Another stop. We shan't want it yet. Sit at that end: that key is yours, and its windows face you. Mine face me.

[*Achilles carries his chair and his cup round to the stove end and sits, and the stove's warmth finds his back at once. He sets the cup on the corner of the stove to keep it hot. The Tortoise draws her own chair up to the dresser end, under the room lamp, and settles into it with the shawl pulled round her.*]

Tortoise: [*drawing the draughts box along the table's far side to the middle, behind the lamp*] The draughtsmen keep the score, and they stay off the tray, where nothing is doubled. When your window flashes, lay a white piece for yes or a black for no, next in your row along the table's edge in front of your key.

Tortoise: Both upright, to begin. [*She feels that her collar stands at the stroke; Achilles, a hand on his, finds it there too.*] Call it when you see it, both together.

[*The tray stands as it is: the lamp shut and still, the rods clear and empty from end to end, each with its reflection lying beneath it in the black, and at either end a pair of frosted windows, dark. The Tortoise reaches along the tray and presses the lever once. Nothing is seen to leave the lamp, and nothing is seen to run along any rod; the glass stays as clear as it was. In the same instant, at both ends of the tray, one window flashes — once — and is dark again. The rods, the lamp, the collars and all their reflections are exactly as they were. What is different is only that two glints are gone, one spent at each end, and that each of the two readers has seen a window flash that the other has not.*]



a window lit at each end

Achilles: Yes.

Tortoise: Yes.

[*The table's near edge in front of Achilles's key is bare cloth. He takes a white piece from the box and sets it down there, off the tray, close under his key. His row will run from there along the edge towards the middle. His window is already dark and the glint is gone; the white piece stays where he put it, and nothing appears beneath it, for it lies off the black. At her end the Tortoise lays a white piece close under her own key, where her row will run the other way along the edge, towards the middle. Where a moment ago there was a flash that only one of them could see, there is now a piece that anyone in the room can see, and it will still be there in the morning.*]

Tortoise: Again. [*The lever. A window flashes at each end.*]

Achilles: No.

Tortoise: No.

[*Two black pieces go down, one in each row, beside the whites.*]

Tortoise: Again. [*The lever; two flashes.*]

Achilles: Yes.

Tortoise: Yes.

[*Two white pieces.*]

Achilles: In unison, every time. Well — of course. They're twins, cut of one piece. Each carries its yes or no out of the lamp, and the key simply reads what's in it.

[*Down at the bottom corner of the lower pane the frost is busy. The first feathers have put out fronds, and the fronds fronds of their own, one laid over another and crowding up the corner every which way, faster than an eye can follow any one of them.*]

Tortoise: Then turn yours to the cross. I'll do the same with mine.

[*Each collar turns on to its other stop, and holds. The lever; two flashes.*]

Achilles: No.

Tortoise: No.

[*Two black pieces. The lever again.*]

Achilles: Yes.

Tortoise: Yes.

[*Two white pieces.*]

Achilles: Crosswise too. So each glint carries two answers out of the lamp, an upright one and a crosswise one, and its twin carries the same two. And the two can't be strangers: a glint that is an upright yes is a yes, and asked the crosswise question it ought still to lean towards yes. Keep yours at the cross and I'll go back to the stroke, and I say our calls will still agree more often than not — three times in four, I'd guess.

Tortoise: Three times in four. Let's hear it.

[*Achilles turns his collar back to the upright stroke; the Tortoise leaves hers at the cross. The lever.*]

Achilles: Yes.

Tortoise: No.

[*He lays white; she lays black. The lever.*]

Achilles: Yes.

Tortoise: Yes.

[*White, and white. The lever.*]

Achilles: No.

Tortoise: Yes.

[*He lays black; she lays white.*]

Achilles: Once in three. But three rounds are nothing; toss a coin three times and you can get anything.

Tortoise: Then ask someone who has tossed it more often. What does Alice's lock-book say?

Achilles: [*turning a page*] Here. "I keep a lock-book, a line for every glint I send, and Bob keeps one for every glint he reads. These first pages are from our trials, before the carrier routed the line through its relay house, when

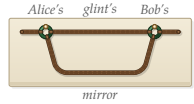
we compared everything by post to learn how the line behaved." [*reading down*] "Two hundred glints. Bob read ninety-six of them at my setting, and every one came out as I sent it. The other hundred and four he read at the other setting: yes on fifty-three and no on fifty-one, about equally, near enough, and it came out the same whether I had sent a yes or a no. I have added it all twice."

[*He reads it with his cup in his other hand, and forgets to drink. The Tortoise leans over with the pot and tops the cup up to the brim all the same, and he sets it back on the stove's corner without looking.*]

Achilles: Then an upright yes, asked crosswise, doesn't lean at all. Not a quarter of the way.

Tortoise: Let Bob draw circles, then, since Alice has done the sums. [*She moves to her drawing-cloth.*] These are my rings, as at Mr Crab's. A green ring asks the upright question and a red ring the crosswise one; each gathers a glint's cord and its mirror cord together, with a short cord to one side, where the answer goes out. Run the short cord in instead of out, and the same ring sends where it used to read.

[*From her pouch she lays a green ring on the cloth at the left, its short cord coming in from the cloth's left edge: Alice, sending upright. At the right she lays another green ring, its short cord running out to the right edge: Bob, reading upright. Between the two rings run two cords side by side, the glint's and, just below it, its mirror, lying as a rod's reflection lies in the tray: beneath it and running the same way. She takes a ring in each claw and slides them along the two cords towards each other. The cords between them shorten and are gone; green meets green, and the two rings run into one. A ring with one cord in and one cord out, and nothing else on it, is no more than a joint, and she lifts it off and drops it into her pouch. What lies on the cloth now is a single cord, from the left edge to the right: Alice's short cord running straight on into Bob's. The rings and the glint's cords have gone; nothing has been added.*]



sending upright, reading upright



slid together and lifted off

Achilles: Straight through. What she sends is what he reads.

[*She gathers the cord into her pouch and lays the rings out afresh, as before, but puts a red ring at Bob's end: Alice sends upright, Bob reads crosswise. Green at the left, red at the right, the glint's cord and its mirror running between them. She draws the two rings apart. The two cords between them go slack together, slip loose at both ends at once, and fall away onto the cloth. Alice's short cord now ends in a green ring tied to nothing; Bob's short cord ends in a red ring tied to nothing. What has changed: the two ends no longer touch at all. What has held: each end still has its ring and its short cord, and Bob still has an answer to read out — only it is joined to nothing Alice put in.*]



green and red drawn apart

Achilles: Tied to nothing she put in. [*He looks from the cloth to his row, and to hers.*] ...Oh. There's no crosswise answer in an upright glint, waiting to be read out. Asked crosswise, it gives one there and then, and nothing in it comes from what she sent.

[*At his elbow a drop of tea runs down the outside of his cup, where it stands warming on the stove's corner, and falls onto the hot iron. It does not hiss. It skates there, a bright bead, off one way and then another, and is gone.*]

Tortoise: Then the rounds we asked different questions have nothing to tell us about each other. We'll put them back.

[*The rows as they stand: in front of Achilles, white, black, white, black, white, and after them the three mixed rounds, white, white, black. In front of the Tortoise, white, black, white, black, white, and after them black, white, white. For five pieces*

the rows are the same colour, piece for piece; after that they go their own ways. She takes the first mixed round's piece from her row, and Achilles, at her nod, the same round's piece from his, and they drop them back into the box; the other two mixed rounds go the same way, a piece from each row for each round. Nothing else on the table moves. Each row is shorter now, and ends where the matching ended: white, black, white, black, white at his end, and the same at hers, every piece still opposite its own round's partner.]

Achilles: So a key needs the settings to match. But if Alice and Bob agree their settings by post beforehand, anyone who reads the post can turn their own key to match, and read the line as well as Bob. So how are they ever to match?

Second Challenge — *Say Only the Setting*

Tortoise: Then let them say nothing beforehand. Choose in secret, read, lay your piece where nobody sees it — and only when the glints are all spent, say which setting you chose, and nothing else.

Tortoise: [*She lifts the lid from the draughts box and stands it on its long edge in front of Achilles's row, a low wall between his pieces and the rest of the table.*] Take a handful of each colour behind that. [*She draws the box itself to her own end and sets it in front of her row, its deep side towards him.*]

Tortoise: Cup your hand over your collar and turn it by feel, to whichever of its stops you like — upright or cross. Don't tell me which, and don't call.

[*Achilles cups his hand over his collar and turns it; somewhere under his palm it stops, without a sound, and not even he could have said which stop from the feel of the room. At her end the Tortoise does the same. The lever. A window flashes at each end, and neither says a word. Achilles lays a piece behind his wall; the Tortoise lays one behind the box. The rows each grow by one, but where in the first rounds two calls hung in the air, there is nothing now, and nobody at the table knows both colours: only the small sound of two draughtsmen set down.*]

[*Five times more: a hand over each collar, a turn felt and not heard, the lever, a flash*

at each end, two pieces laid in silence behind their screens.]

[*In the hush the room's own small sounds come forward: the stove ticking as its iron swells, a coal settling in the firebox. On the lower pane the frost has stopped crowding and found its figure. Every frond now throws its side-shoots at the same few angles all along its length, row on row, as if it had been ruled.*]

Achilles: [*low*] It's very quiet.

Tortoise: Keys are quiet things. Now the settings, round by round, and only the settings.

Achilles: First round: upright.

Tortoise: Upright. We keep it.

Achilles: Second: crosswise.

Tortoise: Upright. Strike it.

[*Each reaches behind a screen, takes that round's piece from the new end of his own row, and drops it into the box. The pieces after it slide up to close the gap, so that at both ends every piece still stands opposite its own round. What has gone is a round asked two different questions; what stays is unseen by the other, as before.*]

Achilles: [*stopping*] Wait — we're saying the settings aloud, the very thing I said a listener wanted. Somebody at the keyhole —

Tortoise: Then be the somebody. You heard me say "upright" for the first round, and you said it too. What do you know of my piece?

Achilles: Anyone with an ear to the keyhole has heard "upright, upright" and learned that we kept the round — and not a whisper of which piece we laid.

[*Out in the lane a step goes by on the rime, crunching, unhurried, and is gone towards the corner. Whose it was, nobody in the house could say.*]

Tortoise: And the struck rounds he's welcome to; they're back in the box. Third.

Achilles: Crosswise.

Tortoise: Crosswise. Keep.

Achilles: Fourth: upright.

Tortoise: Crosswise. Strike.

[*Each takes that round's piece from behind the screen, drops it into the box, and closes the gap.*]

Achilles: Fifth: upright.

Tortoise: Upright. Keep.

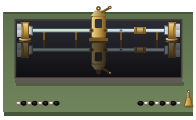
Achilles: Sixth: crosswise.

Tortoise: Upright. Strike.

[*The last pieces of the run go back in the box.*]

Tortoise: Now look — once, for the demonstration, and never again. A key that has been shown is a key no longer. [*She takes down the lid, sweeps Achilles's spare pieces back into the box, and draws the box back to the table's far side, beyond the tray, the lid with it.*]

[*In front of Achilles lie the five pieces from the first rounds of the evening, and after them the kept pieces of the silent run: black, white, black. In front of the Tortoise lie the same five, and after them black, white, black. Neither saw the other's pieces laid; laid open now, the two rows match piece for piece, right to the end, each piece opposite its partner.*]



laid open,
piece for piece

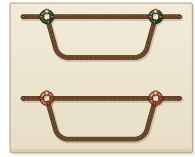
Achilles: The same, right to the end. And I didn't choose a single one of them. Nor did you. The lamp did.

[*The Tortoise laughs under her breath and sits back in her chair. Then she gets up, fills the kettle again from the jug, and sets it at the stove's cool edge, well back from the fire, where it will take its own time. Achilles stretches his stockinged feet towards the stove and wriggles his toes at it.*]

Achilles: [*finding his place in the letter*] She says so too. "Of our two hundred trial glints, ninety-six had matching settings — about half, near enough — and not one of those disagreed. So each glint I send is half a letter of key, and I never choose a letter of it myself."

Tortoise: Let Bob draw only the rounds whose settings matched.

[*She gathers the rings and the fallen cords into her pouch, and on the cloth lays two rounds, one above the other: above, a green sending ring and a green reading ring with the glint's cord and its mirror cord between them; below, the same in red. She slides each pair of rings together, as before; each pair runs into one ring, which comes off into her pouch, and each round is left as a single cord from the left edge to the right — upright above, crosswise below, and no difference to be seen between the two cords. The rounds that mixed red with green she does not lay at all.*]



upright above, crosswise below



no difference to be seen

Achilles: Two straight cords. A listener knows which colour each one was, and nothing of what went along it.

Achilles: [*turning the page*] There's more. [*reading*] "But there is one thing you should know. The carrier has since routed our line through its relay house, and the clerk there is Eve — the one with the press."

Third Challenge — *The Tell-Glass*

Achilles: Eve! The carrier's clerk from Mr Crab's letter this afternoon — the one who kept the struck stones in her drawer.

Tortoise: She has moved with the times. [*From a small wooden box by her chair she takes a block of glass the size of a sugar lump, with a port at either end and*

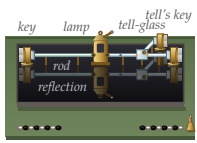
another in its side; a short glass rod; and a spar-key in its collar, like the others.] This is a tell-glass: what Eve would put on the line. A glint goes in at this end and on out at that one, and at the same moment a second glint leaves by this side port, carrying the passing glint's upright answer.

Achilles: Mr Crab's press — in glass.

Tortoise: In glass, and quieter; nothing drops out of a chute. [*She sets the spar-key on the tray beside her own at the right end, its windows turned towards her, and its reflection stands beneath it in the black.*] And this is the tell's key. It stands at my end with its windows facing me, so that only I see what it reads; the short rod will carry the side glint to it.

[*The right-hand rod and the rod that runs on to her key meet end to end inside the brass sleeve: one line of glass, with one line beneath it in the black. She loosens the sleeve, draws it off and drops it into her wooden box. The two rod ends stand apart now, a finger's gap between them, each still on its fork; nothing else on the tray stirs. Into the gap she sets the tell-glass, the right-hand rod's end at its near port, the far rod's end at its through-port, its side port turned towards her. The moment it rests on its fork, a second tell-glass is standing beneath it, upside down in the black, port to port. Where one line of glass ran through a sleeve, there are now two rods meeting at a block of glass, and the block has its reflection; the way from the lamp to her key is still open end to end.*]

[*From the side port she lays the short rod on its forks to the tell's key, and turns that key's collar till it stops at the upright stroke. Beneath the new rod, in the lacquer, its reflection lies.*]



the tell-glass in the sleeve's place

[*For a moment the room makes no sound at all: the stove has left off its ticking, and nothing stirs at the kettle's spout. The room lamp's flame stands perfectly still on the dresser, and the print on the end wall hangs in its light as though it had only now been hung there.*]

[*Achilles, looking down at the tell-glass standing upside down in the black, lifts his eyes from the tray to the print on the wall: Escher's* Three Worlds.]

Achilles: A pond: leaves on its surface, trees reflected in it, a fish beneath. I have drunk a dozen pots of tea under that print and never once looked past the leaves.

Tortoise: Then look now. Secret settings, hands over the collars, as before — but this time call your windows aloud, for the demonstration. The tell's pieces I'll lay myself, and say nothing about them.

[*Achilles blows on his fingers before he cups them. The short rod the Tortoise has just laid is newer glass than the rest, clear as water, with none of the old rods' faint colour in it, and his eye keeps going back to it.*]

[*Hands over collars. The tell's key stands at the upright stroke, its windows dark; the side rod is clear and empty, as it has been since she laid it. The Tortoise presses the lever. A window flashes at Achilles's end; a window flashes at her own key; and in the same instant a window flashes at the tell's key as well, where no glint has ever landed before. The lamp let out no more than it always does. The rods, the tell-glass and every reflection are as they were. What is new is one more flash at her end, from a glint the tell-glass sent off sideways — and the glint that went on to her own key arrived all the same.*]

Achilles: Yes.

Tortoise: Yes.

[*Two white pieces go down in the rows. Then the Tortoise takes a white piece and, without a word, lays it on the table's edge just behind her own row, level with her piece of this round: the first piece of a new row.*]

Tortoise: Again. [*Hands over collars; the lever; a flash at each end, and one more at hers.*]

Achilles: No.

Tortoise: Yes.

[*Black for him, white for her; and in silence, a white for the tell.*]

Tortoise: Again.

Achilles: Yes.

Tortoise: No.

[*White for him, black for her; for the tell, in silence, black.*]

Tortoise: Again.

Achilles: Yes.

Tortoise: Yes.

[*Two whites; in silence, a white for the tell.*]

Tortoise: Again.

Achilles: No.

Tortoise: No.

[*Two blacks; in silence, a black for the tell.*]

Tortoise: Once more.

Achilles: No.

Tortoise: No.

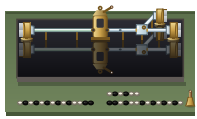
[*Two blacks; and for the tell, in silence, white.*]

Tortoise: Now the settings. Yours first, all of them.

Achilles: Upright, crosswise, crosswise, upright, upright, crosswise.

Tortoise: Upright, upright, crosswise, crosswise, upright, crosswise. The second round goes, and the fourth.

[*From every row — his, hers and the tell's — the second round's piece and the fourth round's go back into the box, and the gaps are closed. The rows stand aligned as before, round opposite round, only shorter: what is left of the run is the first and fifth rounds, asked upright at both ends, and the third and sixth, asked crosswise at both ends.*]



the run as laid *struck, the gaps closed*

Achilles: The upright rounds, the first and the fifth: we agreed both times, and the tell's pieces agree with us — white, then black. She has those, whole. [*His finger moves along the rows.*] The crosswise ones. In the third, I said yes and you said no — at the same setting. That hasn't happened all evening.

Tortoise: Not on an empty line.

[*On the lower pane, the round that Achilles's fingertip melted at the start of the evening has frozen over again, but not as the rest has. The ferns that climbed up round it bend away at its edge, and across the patch itself runs a single blunt frond, thick and crooked, like nothing else on the glass.*]

Achilles: And in the sixth we agreed, both no, and the tell's piece is white. On the crosswise rounds her pieces are no guide at all.

Achilles: [*taking up the letter again*] "Then the carrier routed the line through its relay house, and the lock-book changed. We had begun by then to check some of our kept rounds openly, by post, before we used a key. Now some of the checks disagreed: every one of them on a crosswise round, never an upright one — about one checked round in four, near enough. So we knew."

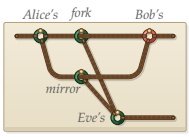
Achilles: Then a cleverer Eve would have a glass that copies the crosswise answer instead.

Tortoise: On which rounds? She hears the settings only when the glints are long spent. Copy crosswise, and it's the upright rounds she spoils; she can only guess, and every wrong guess spoils a round, whichever way it goes.

Achilles: And the upright rounds came through untouched because the tell-glass copies its own kind, as the press did: an upright glint it copies and lets be, and a crosswise one it can only make a pair of.

Tortoise: Let Bob draw Eve where she sits, then. A forked ring is the tell-glass on my cloth: a green ring with one cord in and two out, one going on to Bob and one turning off to her.

[*She gathers the straight cords into her pouch, and on the cloth lays Alice's round again: a green sending ring at the left, a red reading ring at Bob's end, the glint's cord and its mirror cord running between them. Then she takes a green ring made to fork, one cord in and two out, and threads it onto the glint's cord partway along; its reflection, another fork, she threads onto the mirror cord just opposite. From each fork one out-cord runs on to Bob's ring as before, and the other turns off sideways, both side cords going together to another ring, a green one, laid off to one side below the round, with its own short cord out to the cloth's right edge: Eve's. The upper side cord passes under the mirror cord on its way. What has changed: Bob's line now has a side road off it, and at its end a ring of its own. What has held: Alice's ring, Bob's ring, and the line from one to the other, which still runs through unbroken.*]



threaded on
partway along

Achilles: A side road off his line, and Alice never sees it from her end.

[*The Tortoise rolls her shoulders under the shawl and flexes her claws, stiff from the fine work. At the stove's cool edge the kettle has begun a low hum, too low yet to call a note, and neither of them turns to it.*]

[*The Tortoise lifts the tell-glass from the tray, and its reflection goes out of the black with it; she lays it back in her wooden box. She takes the sleeve out of the box and slides it over the two rod ends again, and one line of glass runs once more from the lamp to her key, with one line beneath it. The side rod stays where it lies, its near end at nothing now, its reflection beneath it; the tell's key stands idle beside her own, still at the upright stroke. The tell's row stays where it lies on the table's edge.*]

Achilles: But wait. [*looking along the tray*] This isn't Alice's way at all. Here the lamp sits in the middle and throws a pair, one glint to each end; Alice

sends from her end, and Bob reads at his. Why should they come out the same?

Fourth Challenge — *Backwards to the Lamp*

Tortoise: Let Bob draw clean through the lamp, and see. A bend is how I draw the lamp's twin: one cord turned round on itself, a pair cut of one piece, with an end at each key.

[*She gathers the fork, its cords and Eve's ring into her pouch, and on the cloth she lays the table's own line, from his key through the lamp to hers, with both keys at the upright stroke. In the middle, where the lamp is, a bend: the glint's cord turned round on itself in a narrow turn towards the cloth's near edge, and its mirror cord in a wider turn round it, each running on out to the left and out to the right. At the left, a little way in from the edge, on the glint's cord and its mirror, she threads a green reading ring, Achilles's short cord running out of it to the left edge; at the right, a little way in, a green reading ring with her own short cord running out to the right edge. Then she takes Achilles's short cord in one claw and her own in the other, and pulls. The rings are drawn apart towards the edges, riding on their cords, and the turns shorten, run out and are gone; followed along its cord, Achilles's ring has gone round the turn of the bend where the lamp stood and come out on the other side, heading right. What lies on the cloth is the glint's cord and its mirror from left to right, with a green ring at each end, where the rings of a round lie: at the left, a ring whose short cord now comes in from Achilles's side; at the right, a ring whose short cord goes out to hers. What has changed: the lamp's bend is gone, the rings have moved out to the ends, and Achilles's ring has turned from a reading ring into a sending one, his cord coming in where it used to go out. What has held: every ring and every cord, each ring still threaded as before, and where each short cord ends.*]



bent round where the lamp stands *pulled, the bend run out*

Achilles: A sending ring and a reading ring on the same cords — the very picture from the start of the evening, Alice's lamp and Bob's key. My reading at my end of the twin is her sending.

Tortoise: Your reading goes back into the lamp, and on out to me.

Achilles: Then that's settled: same setting, same answer, whichever way it's done.

[*Achilles leans back, well satisfied, and reaches behind him to spread his hands to the stove. The Tortoise watches him do it and says nothing at all. On the window a second fern has started from the other bottom corner of the lower pane, and is climbing towards the first.*]

Tortoise: Upright, both, first, and let's hear it. [*Two collars turn back from the cross to the upright stroke. The lever; a flash at each end.*]

Achilles: Yes.

Tortoise: Yes.

Tortoise: And crosswise. [*Two collars go to the cross. The lever; a flash at each end.*]

Achilles: No.

Tortoise: No.

Tortoise: And now the stop you asked about. Past the cross, to the curl. I'll turn mine to the curl as well.

[*Each turns a collar on past the cross until it stops at the little curl. The lever.*]

Achilles: Yes.

Tortoise: No.

[*The lever.*]

Achilles: No.

Tortoise: Yes.

[*The lever.*]

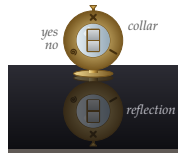
Achilles: No.

Tortoise: Yes.

Achilles: Every one wrong! The same setting — the very same curl at both ends — and not once alike. [*He looks along the tray to the sleeve.*] The tell-glass is in its box. Is it the lamp? Or has somebody put something on the line?

Tortoise: Nothing is on the line. Look at your collar in the tray.

[*Achilles bends down to the lacquer. His collar stands at the curl, the little mark facing him and turning, as he looks at it straight on, the way the sun goes round. In the black beneath, the collar's reflection shows its curl turning the other way. He turns the collar back to the cross: the cross in the tray is a cross, the same as the one above. On to the stroke: the reflected stroke is just the stroke. He turns it to the curl again, and the curl beneath turns against the curl above. The stroke and the cross hold still in the mirror; only the curl changes hands.*]



the cross, unchanged beneath



the curl, winding the other way beneath

Tortoise: Round the bend, your question reaches my end as its reflection. Upright and crosswise look the same in the lacquer, so nobody notices. The curl does not: at the curl, my yes-window is your no-window's reflection.

Achilles: ...Oh. Then if you'd read your windows the other way about, we'd have agreed every time.

[*On the fender his gloves lie where he dropped them when he came in, pulled off by*

the fingertips and still inside out. The left one is, to any eye, a right glove now. His woollen cap, inside out on top of them, is only a cap.]

Tortoise: Every time. A glint sent at the curl and read at the curl comes through as it was sent; it's only the bend that hands on a reflection. [*She turns her collar back to the upright stroke; Achilles turns his.*] Now suppose the calls agreed at upright and at crosswise, every time — no wrong note, ever. What could anyone else have of such a pair?

Fifth Challenge — *The Alarm*

Achilles: I don't know. This evening Eve had half of every key she touched.

Tortoise: This evening Eve had a glass on the line. Let's put it back, and give her every advantage we can think of.

[*At the stove's cool edge the kettle has found its note at last: a thin, steady singing from the spout, which goes on under everything. Neither of them turns round.*]

[*She draws the sleeve off again and drops it into her wooden box, takes out the tell-glass and sets it in its place, the right-hand rod at its near port, the far rod at its through-port, the side rod at its side port; its reflection stands beneath it at once. The line to her key is whole as before, with the tell-glass on it and the side road open.*]

Tortoise: This time we agree the setting aloud beforehand, and I set the tell's key to the same stop: the strongest Eve there is, who knows the question before she reads.

Achilles: Crosswise, then — all of them.

[*His collar and hers turn to the cross, and so does the tell's.*]

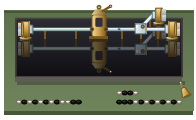
Tortoise: [*a claw on the bell at the table's corner*] Now the bell. This is a checked round: we both call aloud, and if our calls differ I ring it once, and if they agree it stays still. That is all it does tonight.

[*The lever. A flash at his end; two at hers.*]

Achilles: Yes.

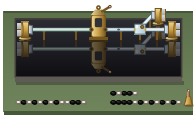
Tortoise: No.

[*The bell stands on the table's corner, silent, as it has stood all evening. The Tortoise lifts it by its handle and rings it once; the note goes round the room and fades. She sets it back where it stood. Nothing on the tray has moved, nothing in the black; the lamp, the glass and the rods are as they were. What has changed is only who knows: the difference between the two calls, which a moment ago was a thing between the two of them, is now a sound that anyone in the house could have heard.*]



lifted by its handle and rung

[*He lays white; she lays black; and then, without a word, she lays the tell's piece: black.*]



his white, hers black

[*Across the lane a dog, woken by the bell, barks twice and is quiet; somewhere a door opens on the cold and shuts again. At the stove's edge the kettle sings on, unheeded.*]

Achilles: And the tell's piece is black.

Tortoise: Another, crosswise still. [*The lever.*]

Achilles: No.

Tortoise: No.

[*The bell stays where it is, and does not sound. Two blacks go down; and in silence, a white for the tell.*]

Achilles: No bell — and the tell's piece is white.

Tortoise: At crosswise it always goes so: her piece is black whenever the bell rings, and white whenever it keeps still.

Achilles: Then all she holds is whether she was caught — and nothing of the key.

Tortoise: Now upright, all of them, agreed aloud. [*The collars go back to the stroke, the tell's with them. The lever.*]

Achilles: No.

Tortoise: No.

[*The bell is still. Two blacks; and in silence, a black for the tell, the same as his.*]

Achilles: Silent again, and black like mine. At upright she has the key, and the bell never hears her.

Tortoise: Which is why Alice and Bob check at both settings, and say which only afterwards.

[*Achilles takes up his cup from the stove's corner, finds it stewed and bitter from standing, and drinks it anyway. The Tortoise draws the shawl closer; the dresser end of the room has never quite warmed.*]

Achilles: [*turning to the last page*] She comes to that. [*reading*] "Now we check about one kept round in three, by post, and strike the checked ones from the key. If too many disagree, we throw the whole key away and begin again. Bob, who draws, has found a way to mend a key with only a few disagreements in it, and to shorten it, so that what the clerk knows of it comes to next to nothing. I cannot follow his proof yet. He says it is all in the drawing."

Tortoise: Let Bob draw keys and checks apart, then. [*From her pouch she takes two small rings, one red and one green, each with a bead set exactly halfway round.*] These are half-turn rings, like this morning's. Threaded on a pair's bend, a red one makes the pair disagree whenever both ends are asked upright, and a green one whenever both are asked crosswise.

[*She gathers the straightened cords and their rings into her pouch, and on the cloth lays a bend of string, one cord turned round on itself, both its ends running out to the right: a pair, before anyone reads it. On its upper leg she threads the red half-turn ring, and after it the green, each bead sitting halfway round.*]



red close in,
green further out

Tortoise: An upright check that never disagrees — [*she slides the red ring along the leg into the turn, round it, and out along the lower leg and off its end, into her pouch*] — and a crosswise check that never disagrees. [*The green ring, alone on the upper leg now, follows it off by that leg's own end, into her pouch.*]

[*The bend lies bare on the cloth. Before, two rings rode on its upper leg, each a way the pair could go wrong; now there are none. The bend itself has not moved: the same turn, the same two ends, both running out to the right, the upper one Alice's and the lower one Bob's.*]



bare, the turn
where it was

Tortoise: Let Bob draw empty paper between the pair and anyone else.

[*She runs the edge of her claw down the cloth just beyond the bare bend's turn, from the top of the cloth to the bottom, between the bend and the rest of the cloth. It crosses nothing: no cord, no ring. The bend lies on one side of the line her claw made, whole, with both its ends taken; on the other side there is only cloth.*]

Tortoise: Not an end left over for anyone.

[*On the dresser shelf, below the room lamp, lies the great-uncle's horseshoe magnet, its iron keeper laid across both its ends, as it has lain for years. A steel thimble has rolled against the keeper and lies there, held by nothing.*]

Achilles: Then that's the whole of it. Two calls at the same setting sing the

same; a listener has to look; a look at the wrong setting is a wrong note, and the bell hears it; and a pair that never sings a wrong note, upright or crosswise, has nothing to spare for anyone else.

[*On the lower pane the two ferns, climbing all evening from their two corners, have met near the middle. Where they meet, each frond runs into a gap in the other's, and neither grows on past the other.*]

Tortoise: The checks catch a spy. That a key with a few wrong notes in it can still be made quite safe is Bob's proof, and not tonight's.

[*She lifts the tell-glass out of its place for the last time and lays it in its box; its reflection goes out of the black with it. She takes the sleeve out of the box and slides it back over the two rod ends, and shuts the lamp. Then she gathers the bare bend into her pouch and rolls her drawing-cloth up round the pouch.*]

Tortoise: Leave the lamp to cool; it's warmer than it looks. And the kettle has been singing to itself for some time — did you hear it?

[*She lifts the kettle off the stove's edge with a fold of her shawl round the handle, and the singing stops in the middle of its note. She makes a second pot, and pours.*]

Achilles: I heard nothing all evening but yes and no, and a bell. [*He takes the cup she hands him, and goes with it to the window.*] The ferns have reached the middle of the pane.

Tortoise: They'll have the whole of it by morning.

Achilles: Two people in different towns, with nothing between them but a line anyone might tap, and the post. Could they really lock a letter with nothing but light — and know, afterwards, whether anyone had laid a finger on it?

Tortoise: Bob says so, and Alice can't follow him yet. Drink your tea, and tell me in the morning which of them you would trust with your own letters.

[*Achilles laughs, and drinks, and stays at the window with the cup warm between his hands. The ferns stand at the middle of the lower pane, where the two of them met; above them the glass is clear and black, and past it the lane lies pale under the rime, the curtained windows opposite lit, and no step in it. The stove has settled to a low red behind its mica, and its ticking has slowed to one now and then. Down*

the table the rods lie on their forks as the Tortoise left them — the long green one to the stove end, the one with the seed of air running to the sleeve, the pale one on from the sleeve to her end, and the short new one lying idle by the tell's key; before Achilles's key his row runs along the table's edge, and before hers her own row and the tell's lie side by side. The bell stands on its corner. On the fender his gloves are dry, and still inside out. On the dresser shelf the horseshoe magnet keeps its keeper, and the thimble lies where it rolled. Behind the Tortoise's empty chair the print hangs on the end wall where it has always hung, pale in its narrow frame; from the window it is too far to make out more than the leaves on its surface. The Tortoise sits down by the stove with her own cup, and the room is quiet but for the slow ticking of the iron and the frost going on with its work on the glass, too slowly for anyone to see.]

Chapter 12: Locks Made of Light

— Yes–No Duet, with Eavesdropper —

Part III: Pictures Become Proofs

Escher: "Three Worlds" — a pond: leaves on its surface, trees reflected in it, a fish beneath

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Can two people who share only a line that anyone may tap agree on a secret, and know afterwards whether anyone listened?

Every lock needs a key, and every key has to travel. Two friends who want to write to each other in a cipher must first agree on how to scramble and unscramble their letters, and that agreement is itself a message. If they are in the same room they can whisper it. If they live in different towns it has to go by some carrier: a courier who might be bribed, a letter that might be steamed open and sealed again, a wire that might be tapped. The trouble is not only that a spy might read the key. It is that a careful spy leaves no trace. A letter can be copied and sent on unharmed, and a signal on a wire can be recorded without weakening it by a detectable amount. In the everyday world, looking costs nothing, so the two friends can never be sure they were alone.

The last chapter changed the terms of that bargain. A qubit cannot be copied. A spy who wants to learn something from a qubit in transit has to look at it, and a look is made at some setting. Chapter 11 ended here: "Think of a spy on the line. She cannot copy what passes, so she cannot keep a copy and let the original go by untouched. To learn anything she must look. A look is made at some setting, plain or cross, and this chapter has shown what

a look at the wrong setting does: it turns a crosswise stone grey, as Eve's customers found. Chapter 12 makes those two settings into the two keys of a lock, built of light. There the mark a spy's look leaves on the stones becomes an alarm."

This is that chapter. Its central fact is small enough to state on a postcard: *read a qubit at the setting it was sent with, and you get back what was sent; read it at the other setting, and you get a coin toss that knows nothing of what was sent.* From that one fact, Bennett and Brassard built a way to share a key in 1984, and it is still the first thing anyone learns about quantum cryptography. We will build it in four steps. Two people choose their settings at random and afterwards tell each other only the settings. A spy on the line is Chapter 11's copier, and we will measure exactly what her look costs her. The same key can be grown from a shared pair instead of a sent qubit, and a single bend in a wire shows the two methods are one. Finally the checks the two make become an alarm, and a pair that passes every check turns out to have nothing left over for anyone else.

We will also be plain about where the pictures stop. Every rewrite in this chapter proves an equality of processes. None of them proves that the key is secure against every possible spy. That theorem exists, and we will state it and say whose it is, but it is not drawn here.

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§12.1 Two Settings

Chapter 11 left us two ways to read a qubit. Chapter 12 turns them into the two keys of a lock.

On the blackboard. The two settings are the ones Chapter 11 used. The **plain** setting asks the question whose answers are $|0\rangle$ and $|1\rangle$; the **cross** setting asks the question whose answers are $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. We label a setting by a bit: 0 for plain, 1 for cross. The Hadamard box of Chapter 3 turns one set of answers into the other, $H|0\rangle = |+\rangle$ and $H|1\rangle = |-\rangle$, so one formula covers both settings. Write $H^0 = I$ and

$H^1 = H$. Then the answer x at setting r is the state

$$H^r|x\rangle, \qquad x, r \in \{0,1\}: \qquad |0\rangle, |1\rangle \ \ (r=0), \qquad |+\rangle, |-\rangle \ \ (r=1).$$

Alice, who wants to send a bit x, picks a setting r and sends $H^r|x\rangle$. Bob, at the other end, picks a setting t and reads. A reading at setting t has the two outcomes $y = 0$ and $y = 1$, with effects the mirror images of the two states $H^t|y\rangle$. The mirror image of a state is its conjugate transpose, and H is real and symmetric, so the effect for outcome y is $\langle y|H^t$. The amplitude for Bob to read y is therefore

$$\langle y|\ H^t H^r\ |x\rangle,$$

and the probability is its squared modulus (§2.3).

Start with the case that matters most. Alice sends $x = 0$ at the plain setting, $|0\rangle$, and Bob reads at the cross setting. The two amplitudes are $\langle +|0\rangle = \frac{1}{\sqrt{2}}$ and $\langle -|0\rangle = \frac{1}{\sqrt{2}}$, so Bob reads 0 or 1 with probability $\frac{1}{2}$ each. Had Alice sent $x = 1$, the amplitudes would have been $\langle +|1\rangle = \frac{1}{\sqrt{2}}$ and $\langle -|1\rangle = -\frac{1}{\sqrt{2}}$, and the probabilities again $\frac{1}{2}$ each. Bob's reading at the wrong setting is a fair coin, and it is the same fair coin whatever Alice sent. A reader who believes that the qubit "really" carries an answer at both settings, and that the cross question merely reveals the cross answer it was carrying all along, has to explain why that answer is a fair coin whose odds do not depend on the plain answer Alice put in. The simpler account is the true one. A qubit sent at the plain setting has no cross answer waiting to be read.

Two settings with this property have a name. Two orthonormal bases $\{|u_0\rangle, |u_1\rangle\}$ and $\{|v_0\rangle, |v_1\rangle\}$ of a qubit are **conjugate** if

$$|\langle u_i|v_j\rangle|^2 = \tfrac{1}{2} \qquad \text{for all } i, j.$$

In Bennett and Brassard's words, each vector of one basis has equal-length

projections onto all vectors of the other; they take the word from Wiesner.

Why this definition. Why ask for equal projections, and not merely that the two bases differ? Because equal projections are exactly what makes a reading at the wrong setting worthless. If $|\langle u_i|v_j\rangle|^2$ were 0.6 for one pair and 0.4 for another, a reading in one basis would lean toward the state sent in the other, and a reader at the wrong setting would learn something. At exactly $\frac{1}{2}$ everywhere, the reading is a fair coin whatever was sent.

> **Result 12.1 (conjugate bases).** The plain and cross bases are conjugate: $|\langle i|\,H\,|j\rangle|^2 = \frac{1}{2}$ for all bits i, j. The circular basis of the BB84 paper, $|c_1\rangle = \frac{1}{\sqrt{2}}(1, i)$ and $|c_2\rangle = \frac{1}{\sqrt{2}}(i, 1)$, is conjugate to both.

Proof. The four entries of H are $\langle i|\,H\,|j\rangle = (-1)^{ij}/\sqrt{2}$, each of squared modulus $\frac{1}{2}$. For the circular basis, $\langle 0|c_1\rangle = \frac{1}{\sqrt{2}}$, $\langle 1|c_1\rangle = \frac{i}{\sqrt{2}}$, $\langle 0|c_2\rangle = \frac{i}{\sqrt{2}}$ and $\langle 1|c_2\rangle = \frac{1}{\sqrt{2}}$, all of modulus $\frac{1}{\sqrt{2}}$. Against the cross basis, $\langle +|c_1\rangle = \frac{1+i}{2}$, $\langle -|c_1\rangle = \frac{1-i}{2}$, $\langle +|c_2\rangle = \frac{1+i}{2}$ and $\langle -|c_2\rangle = \frac{i-1}{2}$, and each of $1 \pm i$ and $i - 1$ has modulus $\sqrt{2}$, so each overlap has modulus $\frac{1}{\sqrt{2}}$. ■ See `Code/Ch12/TwoSettings.py`.

The protocol will use only the plain and cross settings. The circular basis, which Bennett and Brassard mention and do not use, returns in §12.4 as a test that tells a transpose from a conjugate.

> **Dirac's Blackboard.**
>
> *Why the third basis needs i.* A real unit vector in the plane is $(\cos\theta, \sin\theta)$. Its overlap with $|0\rangle$ has squared modulus $\cos^2\theta$, and with $|+\rangle$ it is $\frac{1}{2}(\cos\theta + \sin\theta)^2 = \frac{1}{2}(1 + \sin 2\theta)$. To be conjugate to the plain basis we need $\cos^2\theta = \frac{1}{2}$, so θ is an odd multiple of $\frac{\pi}{4}$. To be conjugate to the cross basis we need $\sin 2\theta = 0$, so θ is a multiple of $\frac{\pi}{2}$. No real θ does both. With complex entries the obstacle goes: $|c_1\rangle = \frac{1}{\sqrt{2}}(1, i)$ has overlap $\frac{1}{2}$ with everything in sight, as Result 12.1 shows. This is what the BB84 paper means when it says that, "owing to the complex nature of its coefficients", the space of a single photon admits a third basis

conjugate to the other two. Three questions, each a fair coin to the other two, fit on one qubit only because amplitudes are complex numbers.

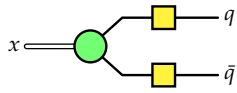
In the sketchbook. Recall from Chapter 11 (§11.1) that a normalized plain state is a red point with $\frac{1}{\sqrt{2}}$ written in front: the red point with phase $a\pi$ is $\sqrt{2}\,|a\rangle$. In the same way, a green effect with phase $b\pi$ is $\langle 0| + e^{ib\pi}\langle 1|$, which is $\sqrt{2}\,\langle +|$ for $b = 0$ and $\sqrt{2}\,\langle -|$ for $b = 1$. A plain state read with a cross effect is the closed picture of the two joined by one wire, and a closed picture is a number (§2.2):

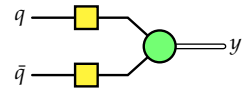
$$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}$$ 

$$= \frac{1}{2} \cdot \sqrt{2}\, e^{iab\pi} = \frac{(-1)^{ab}}{\sqrt{2}}.$$

For all four choices of a and b the number has the same modulus, $\frac{1}{\sqrt{2}}$. Only its sign depends on the phases. That is Result 12.1 as a picture: a red point meeting a green effect gives the same odds whichever red point and whichever green effect.

A reading, though, is more than an amplitude. It is a process that ends in a record, and since Chapter 9 we draw such processes on doubled wires. The **reader** at setting t is Chapter 9's read-spider (row 68): a green dot that takes a quantum system as two lines, its wire q and its mirror copy $\bar{q}$, and emits one classical wire carrying the outcome. At the cross setting a yellow Hadamard box sits on each of the two quantum legs; the mirror copy of H is H again, because H is real. At the plain setting there are no boxes. The **sender** at setting r is the same green dot run the other way: one classical wire comes in carrying Alice's bit, and the doubled wire goes out, with the same yellow boxes on both quantum legs at the cross setting. Chapter 11 met this dot once already, in §11.3, as the green dot that "sets a stone to whatever the record says". Here it gets a name and a row of its own (row 89). Drawn at the cross setting:

$E_1 =$ 

$R_1 =$ 

A yellow box on a leg is the same thing as drawing that leg dashed, Chapter 8's Hadamard edge (row 56). We draw the boxes, because in a moment two of them will meet on one line.

On the blackboard the two dots are, with $\delta = |00\rangle\langle 0| + |11\rangle\langle 1|$ the copier of Chapter 11,

$$E_r = (H^r \otimes H^r)\,\delta, \qquad R_t = \delta^\dagger\,(H^t \otimes H^t).$$

The sender sends $|x\rangle$ to $H^r|x\rangle \otimes H^r|x\rangle$. That is the doubled form of the pure state $H^r|x\rangle$: the state on the wire and its conjugate on the mirror copy, which is the same vector because $H^r|x\rangle$ is real. The reader takes a doubled state $|\psi\rangle \otimes |\psi^*\rangle$ to $\sum_y |\langle y|H^t|\psi\rangle|^2\,|y\rangle$, the vector of probabilities of the reading at setting t (Result 9.1, with H^t in front).

Why this definition. Why draw the sender and reader on doubled wires, when the blackboard managed with one amplitude? Because the round ends in a record, and a record is a probability, not an amplitude. The doubled picture carries probabilities directly: sender, then reader, is a process from Alice's classical bit to Bob's classical bit, a table of odds. And the next two sections will put things on the line, such as a spy whose copy is later forgotten, that are not pure processes at all and can only be drawn doubled.

Now the headline. For a statement that something happens exactly when two bits agree we need a symbol. Write $[\,y = x\,]$ for the number that is 1 if $y = x$ and 0 otherwise.

> **Result 12.2 (BB84 correctness).** Let Alice send the bit x at setting r and let Bob read at setting t. Then Bob reads y with probability
>
> $$P(y \mid x, r, t) = \begin{cases} [\,y = x\,] & r = t, \\ \frac{1}{2} & r \neq t. \end{cases}$$

As doubled processes on the classical wires, exactly,

$$R_t\, E_r = \begin{cases} I & r = t, \\ \frac{1}{2} J & r \neq t, \end{cases}$$

where J is the 2×2 matrix with every entry 1. In pictures: **matching settings fuse into a wire; mismatched settings disconnect into noise.**

Proof idea. When the settings match, the reading undoes the sending: two Hadamards in a row cancel, $H^2 = I$, and Bob's question is the one Alice's state answers with certainty. When they differ, one Hadamard is left over, and by Result 12.1 it spreads every plain state evenly over both outcomes. In the picture, the matching round has a box cancelling a box on each line, and the two green dots melt into a bare wire. The mismatched round has a green dot and a red dot joined by two lines, which Chapter 8's Hopf law says is no join at all: Bob's record is left tied to nothing Alice did. The proof is given twice, in full, in §12.6. See `Code/Ch12/TwoSettings.py`, which checks all sixteen amplitudes and builds both rounds as pictures.

Feynman's Notebook.

A conference in Bangalore. The paper that named the protocol is C. H. Bennett and G. Brassard, "Quantum cryptography: Public key distribution and coin tossing", in the proceedings of the International Conference on Computers, Systems & Signal Processing, Bangalore, India, December 10–12, 1984, pages 175–179. It gives the authors' addresses as IBM Research, Yorktown Heights, and the department IRO of the Université de Montréal; the authors have since posted a reprint on the arXiv (2003.06557). Gisin, Ribordy, Tittel and Zbinden, reviewing the field, remark that it appeared "in a conference in India, totally unknown to physicists". The word *conjugate* is older. Bennett and Brassard take it from Stephen Wiesner's "Conjugate Coding", which their reference list gives as a manuscript of about 1970, published in *SIGACT News* in 1983. Gisin et al. place Wiesner at Columbia University in the 1970s, and say his paper "appeared only a

decade later". The BB84 paper credits it with two applications: money that is impossible to counterfeit, and two or three messages multiplexed so that reading one destroys the others. To polarize light, Bennett and Brassard suggest "a Polaroid filter or calcite crystal".

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§12.2 Say Only the Setting

Result 12.2 is a lock with two keys, and a key only works if both ends use the same one. Alice and Bob live in different towns. If they agree their settings in advance, anyone who overhears the agreement can read the line at the right setting and learn everything. If they do not agree them, half their rounds are coin tosses. The way out is to agree on the settings *afterwards*.

On the blackboard. Here is the protocol, as Bennett and Brassard describe it and as Gisin and his co-authors retell it.

1. For each round, Alice picks a bit x and a setting r at random, and sends $H^r|x\rangle$.
2. Bob picks a setting t at random, independently of Alice, reads, and writes down his outcome y.
3. Over an ordinary public channel, Bob announces his settings, one per round, and Alice says in which rounds hers was the same. Neither ever announces a bit.
4. They keep the rounds where $r = t$ and strike the rest. The kept bits are the **sifted key**, Gisin et al.'s name for it.

The public channel has to satisfy one condition, and it is worth saying plainly. It need not be secret, but it must be **authentic**: anyone may listen, nobody may alter or insert a message. Bennett and Brassard assume a channel "susceptible to eavesdropping but not to the injection or alteration of messages"; Gisin et al. put it as "does not have to be confidential, but has to be authentic". Without that, a spy could simply pose as Bob to Alice and as Alice to Bob. Gisin et al. warn that without some authentication Alice "could actually be communicating directly with Eve!"

A short run makes it concrete. The two coin-toss readings in rounds 2 and 4 could have come out either way.

round	1	2	3	4	5	6
Alice's bit x	0	1	1	0	0	1
Alice's setting r	plain	plain	cross	cross	plain	cross
Bob's setting t	plain	cross	cross	plain	plain	cross
Bob's reading y	0	0	1	1	0	1
announced aloud	plain	cross	cross	plain	plain	cross
kept?	yes	no	yes	no	yes	yes

The kept rounds are $1, 3, 5, 6$, and on each of them Bob's reading is Alice's bit: the sifted key is $0, 1, 0, 1$ at both ends. The two struck rounds are the ones where a coin was tossed. A listener heard "plain, cross, cross, plain, plain, cross" and "yes, no, yes, no, yes, yes", and nothing else. Notice also who chose the key. Alice chose her bits, but which of them survive was decided by both sets of random settings together. Gisin et al. make the same point: "neither Alice nor Bob can decide which key results from the protocol".

> **Result 12.3 (sifting).** Let x, r and t be independent fair coins in every round. Then, exactly: (i) the settings match, $r = t$, with probability $\frac{1}{2}$; (ii) on every kept round, $y = x$; (iii) over all rounds, before sifting, $y \neq x$ with probability $\frac{1}{4}$; (iv) the announcement tells a listener nothing about the kept bit: $P(x = 0 \mid r, t,\ r = t) = \frac{1}{2}$ for each announced pair.

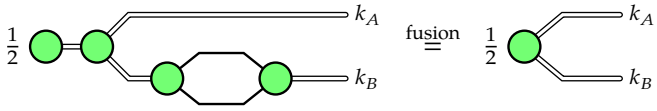
Proof. (i) Of the four equally likely pairs (r, t), two match. (ii) Result 12.2 with $r = t$. (iii) By Result 12.2, $y \neq x$ happens only when $r \neq t$, and then with probability $\frac{1}{2}$; so $P(y \neq x) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. (iv) The announcement consists of r and t, and x was chosen independently of both, so x is still a fair coin given any value of (r, t). On a kept round $y = x$, so the same holds for y. ■ See `Code/Ch12/SiftedKey.py`, which enumerates all eight triples

(x, r, t) exactly and also runs a seeded sequence of rounds.

Part (iii) is Gisin et al.'s remark that before the settings are compared Bob holds "a string of bits with 25% errors, called the raw key". Sifting removes all of them at the cost of half the rounds.

Why this definition. Why announce the settings and not, say, a few of the bits? Because the settings are exactly the information that decides which rounds are useful, and, by (iv), no information at all about what the useful rounds contain. A bit announced is a bit lost to the key. A setting announced costs nothing, because it was chosen independently of the bit it accompanies.

In the sketchbook. By Result 12.2, a kept round is a bare classical wire and a struck round falls apart into $\frac{1}{2}$ times a green effect on Alice's wire and a green point on Bob's. Now feed the kept round a fair coin, and let Alice keep a copy of her bit. A fair coin is the green point with $\frac{1}{2}$ in front: $\frac{1}{2}(|0\rangle + |1\rangle)$ is the vector of probabilities $(\frac{1}{2}, \frac{1}{2})$. Copying a classical bit needs no permission. The classical wire carries a settled value, and the green copier of Chapter 11 copies exactly the settled values of its own basis (Result 11.2). The coin, the copier and the kept round are all green, so fusion (Theorem 8.1) melts them into one dot:



The round is drawn at the plain setting; at the cross setting it carries two boxes on each line, which cancel in pairs (§12.6). Here k_A is Alice's record of her bit and k_B is Bob's reading. The fusion leaves the doubled lines between sender and reader closing one loop on the fused dot, and a self-loop on a spider is erased with no number (van de Wetering §4.1, as in §11.2). The right-hand side is a green dot with two classical legs, $\frac{1}{2}(|00\rangle + |11\rangle)$: the joint distribution $p(k_A, k_B) = \frac{1}{2}[\, k_A = k_B \,]$. It is a bend on the two records, a shared fair coin. That is what a key *is*. Section 12.4 will find the same bend with no coin tossed by anyone.

§12.3 Eve Is a Box on the Wire

Now put someone on the line. Gisin et al. describe what a spy would like to do: send each qubit on to Bob "in its original state, keeping a copy for herself". Chapter 11 showed that she cannot. What she can do is what Eve did at the carrier's office in Chapter 11: press each passing qubit with the green copier δ, keep one output, and send the other on. By Result 11.4 that is a look at the plain setting. The difference now is that Eve need not read her kept output at once. She can hold it until Alice and Bob have announced their settings, and then read it however she likes; Gisin et al. allow the spy exactly this, and more, in their account of the ideal attack (§VI.1).

On the blackboard. Take a kept round, so Alice and Bob use the same setting, and follow the two cases.

A plain round. Alice sends $|x\rangle$. The press makes $\delta|x\rangle = |x\rangle|x\rangle$. Bob reads his copy plain and gets x; Eve's copy is $|x\rangle$ too. Bob's key bit is right, and Eve holds it.

A cross round. Alice sends $H|x\rangle$, which is $|+\rangle$ or $|-\rangle$. The press makes $\delta|\pm\rangle = |\Phi^{\pm}\rangle$ (Result 11.2): not two copies of a cross state but a Bell pair, one half with Bob and one half with Eve. Bob, reading his half at the cross setting, reads the half of a Bell pair, which on its own is $\frac{1}{2}I$ (Result 9.3): a fair coin. He agrees with Alice only half the time. And Eve? If she reads her half plain, she gets 0 or 1 with probability $\frac{1}{2}$ each from both $|\Phi^+\rangle$ and $|\Phi^-\rangle$, so her record says nothing about x. If she waits, hears "cross", and reads her half at the cross setting, she does no better. Rewriting the two pairs at the cross setting,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle), \qquad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle + |{-}{+}\rangle),$$

so her cross reading k_E and Bob's cross reading k_B satisfy $k_B \oplus k_E = x$, where $\oplus$ is addition of bits mod 2. That tells her x only if she also knows Bob's reading, and she never sees it. Her own record, alone, is a fair coin whatever x was.

Result 12.4 (the press on the wire). Let Eve's press δ sit on the

line in a kept round. Let k_B be Bob's reading and k_E Eve's reading of her copy. Then, exactly: (i) at the plain setting, $p(k_B, k_E \mid x) = [\, k_B = x \,]\,[\, k_E = x \,]$: Bob is right and Eve holds the key bit; (ii) at the cross setting, with Eve reading plain, $p(k_B, k_E \mid x) = \frac{1}{4}$ for all four pairs; with Eve reading cross, $p(k_B, k_E \mid x) = \frac{1}{2}\,[\, k_B \oplus k_E = x \,]$. Either way Bob disagrees with Alice with probability $\frac{1}{2}$, and Eve's record alone is independent of x; (iii) with Alice's setting a fair coin, Eve learns on average half a bit about each key bit, and Bob disagrees with Alice on a quarter of the kept rounds.

Proof. (i) and (ii) are the calculations above; for Eve reading plain on a cross round, write $|\Phi^{\pm}\rangle$ with Bob's half at the cross setting and Eve's at the plain one, $|\Phi^{+}\rangle = \frac{1}{2}(|+\rangle\,(|0\rangle + |1\rangle) + |-\rangle\,(|0\rangle - |1\rangle))$, and likewise for $|\Phi^{-}\rangle$: all four amplitudes have modulus $\frac{1}{2}$. (iii) On plain rounds Eve's record determines x, which is one bit; on cross rounds it is independent of x, which is none. The two kinds of round are equally likely, so the average is $\frac{1}{2}$ bit. The disagreement rate is 0 on plain rounds and $\frac{1}{2}$ on cross rounds, which averages to $\frac{1}{4}$. ■ See `Code/Ch12/EvesBox.py`.

Dirac's Blackboard.

Half a bit, counted. "Half a bit" has a precise meaning. Let $h(p) = -p \log_2 p - (1-p)\log_2(1-p)$ be the binary entropy, the uncertainty in bits of a coin that lands 0 with probability p; a fair coin has $h(\frac{1}{2}) = 1$. The information Eve's record carries about Alice's bit is the uncertainty she had before, minus what remains once she has read it:

$$\mathcal{I}(x; k_E) = h(\tfrac{1}{2}) - \sum_{k} p(k_E{=}k)\, h(p(x{=}0 \mid k_E{=}k)).$$

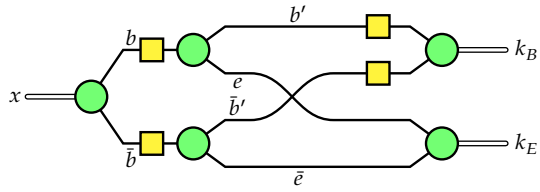
On a plain round $x = k_E$, every $p(x{=}0 \mid k_E{=}k)$ is 0 or 1, every h vanishes, and $\mathcal{I} = 1$. On a cross round x is a fair coin given either record, every h is 1, and $\mathcal{I} = 1 - 1 = 0$. Averaged over the two settings, $\frac{1}{2}(1 + 0) = \frac{1}{2}$ bit. The disagreement is simpler: 0 and $\frac{1}{2}$, averaging $\frac{1}{4}$. Eve cannot improve on the plain setting by choosing cross, because she does not know which rounds are

> which until the settings are announced, and by then the qubit has gone on to Bob.

Is that a good trade for Eve? Bennett and Brassard answered in general, and we state their answer without proving it. No measurement on a photon in transit, by an eavesdropper who learns the photon's basis only after measuring, "can yield more than $1/2$ expected bits of information about the key bit encoded by that photon", and any such measurement yielding b bits of expected information "must induce a disagreement with probability at least $b/2$" when the photon is later measured in its original basis. The press gets $b = \frac{1}{2}$ and causes disagreement $\frac{1}{4} = b/2$: it sits exactly on the bound. These are also the numbers of the example Bennett and Brassard give of this "optimum tradeoff", a different strategy in which the eavesdropper measures every photon at the plain setting and sends on a fresh one, "learning the correct polarizations of half the photons and inducing disagreements in $1/4$" of those later measured in the original basis. Gisin et al. call that strategy intercept–resend and give the same figures, 50% information and 25% errors in the sifted key. They add that a spy who applies it to only a tenth of the rounds causes about 2.5% errors for about 5% information.

Why this definition. Why take Chapter 11's copier as the model spy, and not the most general attack quantum mechanics allows? Because the copier is the look, drawn: Result 11.4 showed that pressing a qubit and keeping one output is a plain reading whose outcome has not yet been read. It is the simplest spy that learns anything, and it already sits on Bennett and Brassard's bound. The most general spy is the business of the security theorems of §12.8, and no picture in this chapter reaches it.

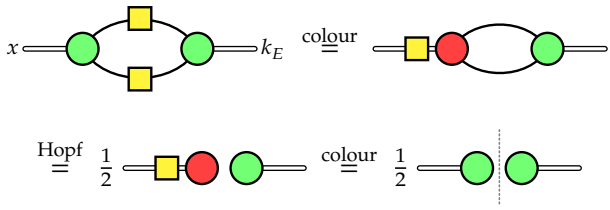
In the sketchbook. Draw the press on the doubled line. The copier acts on the wire, and its mirror copy acts on the mirror copy of the wire; δ is real, so the mirror copier is the green copier again. Each copier has one leg in and two out. Its through-leg, b' on the wire and $\bar{b}'$ on the mirror, goes on to Bob's reader; its kept leg, e and $\bar{e}$, goes to Eve's reader. To bring each pair of legs to its own reader, e and $\bar{b}'$ cross once; a crossing neither joins nor breaks a line. Here is a cross round, with Alice and Bob at the cross setting and Eve reading plain:



From left to right: Alice's sender at the cross setting; the press on b and its mirror on $\bar{b}$; Bob's reader at the cross setting on $b', \bar{b}'$; Eve's reader at the plain setting on $e, \bar{e}$. This is Eve as a box on the wire: a picture with a side road.

At the **plain** setting there are no boxes anywhere, every dot is green and phase-free, and fusion (Theorem 8.1) makes the whole picture one green dot with three classical legs, x, k_B and k_E. Six lines join five dots, two more than a tree needs, so two self-loops are removed on the way (van de Wetering §4.1). A green dot with legs x in and k_B, k_E out is $\sum_i |ii\rangle\langle i|$: both records equal Alice's bit, which is Result 12.4(i).

At the **cross** setting, ask what Eve learns while Bob's record is forgotten. Forgetting a record is the green effect $\sum_j \langle j|$ on its classical wire, and a reading followed by forgetting is a discard, a cap on the two copies (Result 9.5). A unitary followed by a discard is a discard (Result 9.4), so Bob's two yellow boxes go too, and his side of the press is capped. A copier with one output capped onto its mirror copy fuses with it into Chapter 11's four-legged green dot (row 85), and Eve's reader, green and without boxes, fuses into that as well. What is left is Alice's sender, its two boxes, and one green dot D with Eve's record as its only classical leg. Then four steps:



Colour change (Theorem 8.2): put two yellow boxes on Alice's classical leg, which changes nothing since $H^2 = I$; now the sender wears a box on

every leg, so it is a red dot, with one box left on x. **Hopf** (Theorem 8.6): a red dot and a green dot joined by two lines fall apart, with the factor $\frac{1}{2}$. **Colour change** again: a red effect after a yellow box is a green effect, $\sqrt{2}\,\langle 0|\,H = \langle 0| + \langle 1|$. The result is $\frac{1}{2}$ times a green effect on x beside a green point on k_E: $p(k_E \mid x) = \frac{1}{2}$ for every x and k_E. The dotted grey curve is the cut of Chapter 11 (row 87), and it crosses nothing. Alice's bit and Eve's record are disconnected. Every step is exact.

The other half of the story is Bob's disagreement, and it comes from the same rules. Forget Eve's record instead. The press with one output discarded is the four-legged green dot (row 85), a look nobody read, sitting on the line between Alice's cross sender and Bob's cross reader. Colour-change both of them to red. The four-legged dot is now joined to a red dot by two lines on each side. Hopf applies twice, with $\frac{1}{2}$ each time, and leaves the green dot with no legs at all, whose value is $1 + 1 = 2$ (§8.1). So the round is $\frac{1}{2} \cdot \frac{1}{2} \cdot 2 = \frac{1}{2}$ times a green effect on Alice's wire and a green point on Bob's: $\frac{1}{2}J$, exactly the mismatched round of Result 12.2, although the settings matched. Eve's look has turned a kept round into a struck one. That is the mark her look leaves.

In the dialogue, just after the tell-glass is set in the sleeve's place and its reflection appears in the tray, Achilles glances at *Three Worlds*: a pond: leaves on its surface, trees reflected in it, a fish beneath.

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§12.4 Backwards to the Source

So far Alice has sent and Bob has read. There is a second way to grow the same key, in which neither of them sends anything. A source between them emits a pair, one half to each, and both of them read. This is the idea behind the protocol of Ekert (1991), and in the form used here the two ways turn out to be one.

On the blackboard. The source emits the Bell state of Chapter 6,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}}\,\text{cup},$$

where, as always in this book, the cup is unnormalized: cup $= |00\rangle + |11\rangle$, and cap $= \langle 00| + \langle 11|$. Gisin et al. write this state as their Eq. (9), with spins up and down in place of 0 and 1. Alice reads her half at setting r, Bob reads his at setting t, and each writes down the outcome, k_A and k_B.

Try the four pairs of settings. Both plain: the outcomes are 00 or 11, each with probability $\frac{1}{2}$, so Alice and Bob agree. Both cross: the same state rewritten at the cross setting is $\frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle)$, as we saw in §12.3, and again they agree. Plain against cross, in either order: all four outcomes come with probability $\frac{1}{4}$. So the pair behaves exactly like a sent qubit whose bit is a fair coin. Alice's reading plays the part of her choice of x, and Bob's reading is correct when the settings match and a coin toss when they do not. Why should a reading at one end act like a sending from that end?

The reason is the transpose trick of Chapter 6. Gisin et al. state it as their Eq. (10): for all unitaries U_1 and U_2,

$$(U_1 \otimes U_2)\,\Phi^{(+)} = (\mathbb{1} \otimes U_2 U_1^t)\,\Phi^{(+)},$$

where U_1^t is the transpose and $\mathbb{1}$ the identity. A process on Alice's half can be moved to Bob's half, where it arrives *transposed*. Reading is a process too, and the same move carries Alice's reading to Bob's end. There it arrives as a *preparation*, and its effect arrives conjugated: for every state $|\varphi\rangle$,

$$(\langle\varphi| \otimes I)\;|\Phi^+\rangle = \frac{1}{\sqrt{2}}\;|\varphi^*\rangle\,,$$

where $|\varphi^*\rangle$ is $|\varphi\rangle$ with every amplitude complex-conjugated, Chapter 6's $|b^*\rangle$ (§6.4). When Alice's reading at setting r gives k_A, Bob's half is left in the conjugate of $H^r\,|k_A\rangle$. For the plain and cross settings that conjugate is $H^r\,|k_A\rangle$ itself, because those states are real. So Alice's reading prepares for Bob exactly the state she would have sent in BB84.

Result 12.5 (E91 is BB84 bent). Exactly: (i) for all 2×2 matrices U_1, U_2, $(U_1 \otimes U_2)\,|\Phi^+\rangle = (I \otimes U_2 U_1^T)\,|\Phi^+\rangle$; (ii) for every state $|\varphi\rangle$, $(\langle\varphi| \otimes I)\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}\,|\varphi^*\rangle$; (iii) for Alice at the plain or cross setting and Bob at the plain, cross or circular setting, the pair gives

$$p_{\text{pair}}(k_A, k_B \mid r, t) = \tfrac{1}{2}\, P_{\text{BB84}}(y{=}k_B \mid x{=}k_A, r, t),$$

the BB84 round of Result 12.2 fed a fair coin; (iv) with both at the circular setting, $p_{\text{pair}}(k_A, k_B) = \frac{1}{2}\,[\,k_A \neq k_B\,]$: the two readings **always disagree**, where the BB84 round at the circular setting always agrees.

Here P_{BB84} at the circular setting means Alice sends $|c_1\rangle$ for $x = 0$ and $|c_2\rangle$ for $x = 1$, and Bob's effects are $\langle c_1|$ and $\langle c_2|$.

Proof. (i) Theorem 6.4 slides U_1 round the cup: $(U_1 \otimes I)$ cup $= (I \otimes U_1^T)$ cup. Apply $I \otimes U_2$ to both sides, and divide by $\sqrt{2}$. (ii) $(\langle\varphi| \otimes I)$ cup $= \sum_i \langle\varphi|i\rangle\,|i\rangle = \sum_i \overline{\varphi_i}\,|i\rangle = |\varphi^*\rangle$; divide by $\sqrt{2}$. (iii) Alice's effect at setting r with outcome k_A is $\langle k_A|\,H^r$, which is real. By (ii) she leaves Bob's half in $\frac{1}{\sqrt{2}} H^r\,|k_A\rangle$, which is exactly what BB84's sender emits for $x = k_A$, scaled by $\frac{1}{\sqrt{2}}$. Bob's reading at any setting then gives the BB84 amplitude times $\frac{1}{\sqrt{2}}$, and squaring gives the factor $\frac{1}{2}$. (iv) Alice's effect $\langle c_1|$ leaves Bob's half in $\frac{1}{\sqrt{2}}\,|c_1^*\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1, -i)$, and $(1, -i) = -i\,(i, 1)$, so $|c_1^*\rangle = -i\,|c_2\rangle$. Likewise $|c_2^*\rangle = -i\,|c_1\rangle$. Bob, reading at the circular setting, gets the other answer every time. In BB84, $|c_1\rangle$ is read as c_1 every time. ■ See `Code/Ch12/BentKey.py`, which checks (i) on random complex non-symmetric matrices, checks (ii) on random non-real states, and confirms that the dagger in place of the transpose fails.

The circular test is the whole point of (iv). For the plain and cross settings, a transpose and a conjugate look the same, because every matrix and every state involved is real. Only a non-real setting can tell them apart, and at the circular setting the pair and the sent qubit give opposite answers. The BB84 paper's own two-photon state is a different one, the singlet $0.7071(r_1 r_2 - r_2 r_1)$ in its notation, whose halves are "always found to have opposite

polarization". Bennett and Brassard note that the same state vector "is demonstrably equal to" $0.7071(d_1d_2 - d_2d_1)$ and to $0.7071(c_1c_2 - c_2c_1)$; with the paper's own diagonal vectors, the first holds as a vector only up to an overall sign, which no reading can see. So the singlet is anticorrelated at every setting, the circular one included. The pair of this chapter, $|\Phi^+\rangle$, is not the singlet. It agrees at the plain and cross settings and disagrees at the circular one, and that difference is the conjugation.

Dirac's Blackboard.

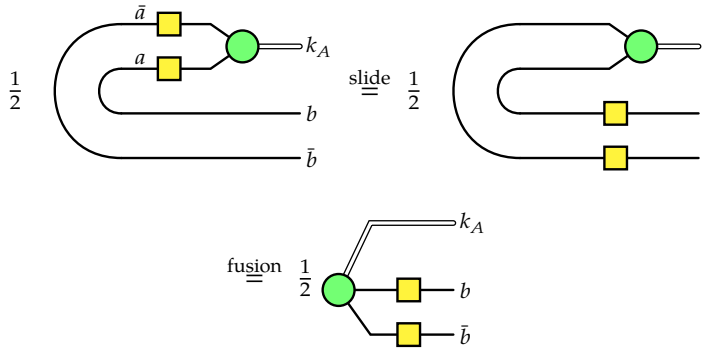
A transpose, not a dagger. Eq. (10) moves U_1 across the pair as U_1^T. It is tempting to write $U_1^\dagger$, since the dagger is the operation this book uses most for turning things round. Check it on one entry. Take $U_1 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & i \end{smallmatrix}\right)$, a diagonal matrix, so $U_1^T = U_1$ while $U_1^\dagger = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -i \end{smallmatrix}\right)$. Then

$$(U_1 \otimes I)\,\mathrm{cup} = |00\rangle + i\,|11\rangle = (I \otimes U_1^T)\,\mathrm{cup},$$

$$(I \otimes U_1^\dagger)\,\mathrm{cup} = |00\rangle - i\,|11\rangle\,.$$

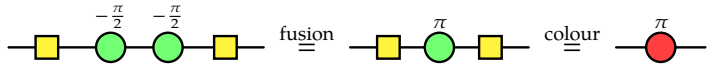
The dagger gives the wrong sign on $|11\rangle$. In general, $(\langle jk|\,(U_1 \otimes I)\,\mathrm{cup}) = (U_1)_{jk}$ and $(\langle jk|\,(I \otimes M)\,\mathrm{cup}) = M_{kj}$, so the two sides agree exactly when $M_{kj} = (U_1)_{jk}$, which is the transpose. The dagger would also conjugate each entry. That conjugation is not lost, though. It turns up in (ii), where an *effect* slid round the bend arrives as a conjugated *state*, because an effect is already the conjugate transpose of a state.

In the sketchbook. Draw the pair doubled, as in Chapters 10 and 11: the cup on (a, b) and a second cup on the mirror copies $(\bar{a}, \bar{b})$, with $\frac{1}{2}$ beside them, since $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\,\mathrm{cup} \otimes \mathrm{cup}$ in doubled form. Put Alice's reader on a and $\bar{a}$, at the cross setting. Now move her two yellow boxes round their cups. Theorem 6.4 moves a box round a bend as its transpose, and $H^T = H$, so each box arrives on the other leg unchanged. A cup is a phase-free green dot with two legs (§8.1), so the two cups and Alice's green reader are green dots joined in a chain, and fusion (Theorem 8.1) makes them one:



The last picture is Alice's sender at the cross setting (row 89), with its classical leg turned to point out of the picture instead of in, and $\frac{1}{2}$ in front. It is the fair coin, Alice's copy of it and her sender of §12.2, fused into one dot. That is Result 12.5(iii) as a picture: **the cup and Alice's reader fuse into her sender**. The reading has run backwards to the source and forwards again to Bob. Gisin et al. give the same image in words: in two-photon schemes, "everything is as if Alice's photon propagated backwards in time from Alice to the source and then forwards from the source to Bob." The blackboard needed Eq. (10) and a remark about conjugates to reach this. The picture needed one slide and one fusion.

The circular setting shows the conjugation in the picture too. A circular state is a cross state given a quarter-turn: $|c_1\rangle$ is $Z(\frac{\pi}{2})H|0\rangle$, a green dot of phase $\frac{\pi}{2}$ applied after the yellow box, and $Z(\frac{\pi}{2})H|1\rangle = -i|c_2\rangle$. So a circular *sender* puts a box and then a green $+\frac{\pi}{2}$ dot on the wire. A circular *reader* undoes that: a green $-\frac{\pi}{2}$ dot, then the box, then the read-spider. On the mirror copy the phases are negated, because the mirror conjugates, so the circular reader carries $+\frac{\pi}{2}$ on $\bar{q}$ (row 93). Now slide Alice's circular reader round the bend. A phase dot on one leg of a cup can slide to the other leg unchanged, since a green dot is its own transpose (only the connectivity matters, §8.5). So on Bob's wire b the slid reader arrives as a box followed by a $-\frac{\pi}{2}$ dot: a circular sender with the *wrong* quarter-turn. Bob's circular reader adds its own $-\frac{\pi}{2}$. Follow the wire b from the source to Bob's read-spider:



The two quarter-turns fuse into a half-turn, $-\pi$, which is the same phase as π (a whole turn is nothing); the green half-turn between two boxes is a red half-turn by colour change; and a red half-turn is X, a flip (§8.2). On the mirror copy the two $+\frac{\pi}{2}$ dots make the same flip. So Bob always reads the opposite of Alice: Result 12.5(iv), seen. In the BB84 round, Alice's circular sender carries $+\frac{\pi}{2}$, Bob's reader $-\frac{\pi}{2}$, and they cancel to a bare wire.

Two remarks keep the history straight. First, Ekert's own protocol used a third setting and a test of Bell's inequality, as Gisin et al. and Horodecki et al. both describe; this chapter does not run that test, and uses the pair only at BB84's two settings. Second, that simplified use is not a novelty of ours. Horodecki et al. describe the protocol of Bennett, Brassard and Mermin, in which Alice and Bob check their pairs at the plain and cross settings only, and call this simplified form of Ekert's protocol "formally equivalent to the BB84 protocol".

Why this definition. Why bother with the pair at all, if it gives the same key as a sent qubit? Because it moves the source out of Alice's hands. Gisin et al. point out that a source which simply sent both parties the same BB84 state would have to be trusted, and "could be in Eve's hand". A source of $|\Phi^+\rangle$ can be tested by the two ends themselves, and §12.7 shows what the test finds.

Feynman's Notebook.

A second discovery. Gisin et al. credit the pair version to Artur Ekert, then at Oxford University, who in 1991, "while elaborating on a suggestion of David Deutsch (1985), discovered QC independently of the BB84 paper". His paper, "Quantum cryptography based on Bell's theorem" (*Physical Review Letters*, 1991), we know only through two reviews. Ekert proposed to found the security on Bell's inequality. He gave Alice and Bob a third choice of basis, so that while they built a key they also gathered the data to test the inequality; Gisin et al. note that this lowers

the chance of matching bases from $\frac{1}{2}$ to $\frac{2}{9}$. Horodecki et al. add, in a footnote, that Ekert used the fact that the singlet state cannot be correlated with any environment, which is the monogamy of §12.7. The following year, Gisin et al. report, Bennett, Brassard and Mermin argued that the Bell violation is not needed for security and stressed how close Ekert's scheme is to BB84. Gisin et al. also credit them with first noting the close analogy between the one-photon and two-photon schemes: when Alice detects her photon, she "effectively prepares Bob's photon in a given state".

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§12.5 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 11's eighty-eight. There is no new picture element in them. Every row is a new *use* of something we already had: the read-spider run backwards as a sender (89), the red dot laid on classical wires as a parity check (96), a green quarter-turn for the circular setting (93), the copier on the line (92), the half-turn rings on a bend (94) and the cut (91, 95).

#	Algebra (Dirac)	Picture (Coecke)
89	two settings: plain $\{\|0\rangle, \|1\rangle\}$, cross $\{\|+\rangle, \|-\rangle\}$; conjugate bases, $\|\langle i\|H\|j\rangle\|^2 = \frac{1}{2}$; the sender $E_r = (H^r \otimes H^r)\,\delta$ and the reader $R_t = \delta^\dagger (H^t \otimes H^t)$	the **sender**: a green dot with a classical wire in and a doubled wire out, a yellow box on each quantum leg at the cross setting (a dashed edge, row 56); the **reader**: Chapter 9's read-spider with the same boxes

#	Algebra (Dirac)	Picture (Coecke)
90	matching settings: $R_r E_r = \delta^\dagger \delta = I$	**matching settings fuse into a wire**: the boxes cancel in pairs, the two dots fuse along two lines, the self-loop goes; a bare classical wire
91	mismatched settings: $R_l E_r = \frac{1}{2} J$: Bob's reading a fair coin, Alice's bit forgotten	**mismatched settings disconnect into noise**: colour change, then Hopf removes both lines; a green effect on Alice's wire and a green point on Bob's, $\frac{1}{2}$ beside them; a cut crosses nothing
92	Eve's press δ on the line; plain: $k_B = k_E = x$; cross: Bob errs half the time and Eve's record is independent of x; averages $\frac{1}{2}$ bit and $\frac{1}{4}$	**Eve is a box on the wire**: the copier and its mirror on the doubled line, one leg to her reader; with that leg forgotten, the four-legged dephasing dot (row 85)

#	Algebra (Dirac)	Picture (Coecke)
93	$(U_1 \otimes U_2)\,\lvert\Phi^+\rangle =$ $(I \otimes U_2 U_1^T)\,\lvert\Phi^+\rangle$; $(\langle\varphi\rvert \otimes I)\,\lvert\Phi^+\rangle = \frac{1}{\sqrt{2}}\,\lvert\varphi^*\rangle$; the pair $= \frac{1}{2}$ BB84 fed a fair coin	**E91 is BB84 bent**: the cup and Alice's reader fuse into her sender; a reading slid round the bend arrives conjugated: unchanged at plain and cross; the circular reader's $-\frac{\pi}{2}$ dot keeps its phase round the bend, where a circular sender carries $+\frac{\pi}{2}$, so it arrives as the other circular state
94	plain disagreement $=$ $\lvert\Psi^{\pm}\rangle$ (bit flips); cross disagreement $=$ $\lvert\Phi^-\rangle, \lvert\Psi^-\rangle$ (phase flips) (Shor–Preskill)	on the bend: a **red** half-turn ring for a plain disagreement, a **green** one for a cross disagreement (row 73); passing both checks leaves the bare bend
95	monogamy, extreme form: the pass projector $\frac{1}{4}(I + ZZ)(I + XX) =$ $\lvert\Phi^+\rangle\langle\Phi^+\rvert = \frac{1}{2}$ cup cap; a state that always passes is $\lvert\Phi^+\rangle\langle\Phi^+\rvert \otimes \rho_{\text{Eve}}$	the pass projector is a cap then a cup, $\times\frac{1}{2}$: **disconnected in the middle**; a cut through empty paper separates the pair from anything else

#	Algebra (Dirac)	Picture (Coecke)
96	parity $w = k_A \oplus k_B$, with $X(2 \to 1) = \frac{1}{\sqrt{2}}\,[\, w = k_A \oplus k_B \,]$; **the Alarm**: $p(k_E, w) = \frac{1}{2}[\, w = 0 \,]$ (plain), $\frac{1}{2}[\, k_E = w \,]$ (cross)	a red dot on two classical wires (Chapter 8's parity, now on records); **the master**: at plain Eve's record and the alarm fall apart, at cross one bend joins them

Every row is exact, scalar included; none needs Chapter 8's "up to a non-zero scalar". Two things behind these rows are cited and not proved: Bennett and Brassard's general bound behind row 92, and the quantitative form of monogamy of which row 95 is the extreme case. No row says anything about security.

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§12.6 The Theorem, Twice

Result 12.2 (BB84 correctness), restated. Alice sends x at setting r and Bob reads at setting t. Then $P(y \mid x, r, t) = [\, y = x \,]$ when $r = t$ and $\frac{1}{2}$ when $r \neq t$; as doubled processes, $R_t E_r = I$ when $r = t$ and $\frac{1}{2}J$ when $r \neq t$, exactly.

Dirac's Blackboard

Step 1 — the amplitude. Alice's state is $H^r\,|x\rangle$ and Bob's effect for outcome y is $\langle y|\,H^t$ (§12.1). The amplitude is $\langle y|\,H^t H^r\,|x\rangle$, the (y, x) entry of the matrix $M_{rt} = H^t H^r$.

Step 2 — the four matrices. Square the Hadamard:

$$H^2 = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \tfrac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

So the four settings give

$$M_{00} = I, \qquad M_{11} = H^2 = I, \qquad M_{01} = H, \qquad M_{10} = H.$$

These four matrices hold all sixteen amplitudes. When $r = t$ the amplitude is $\langle y|x\rangle = [\,y = x\,]$. When $r \neq t$ it is $\langle y|\,H\,|x\rangle = (-1)^{xy}/\sqrt{2}$: that is $+\frac{1}{\sqrt{2}}$ for $(x,y) = (0,0), (0,1), (1,0)$ and $-\frac{1}{\sqrt{2}}$ for $(1,1)$.

Step 3 — the probabilities. Square the moduli. When $r = t$, $[\,y = x\,]^2 = [\,y = x\,]$. When $r \neq t$, $\left((-1)^{xy}/\sqrt{2}\right)^2 = \frac{1}{2}$ for all four pairs: the sign $(-1)^{xy}$ is squared away. That is the first form of the result.

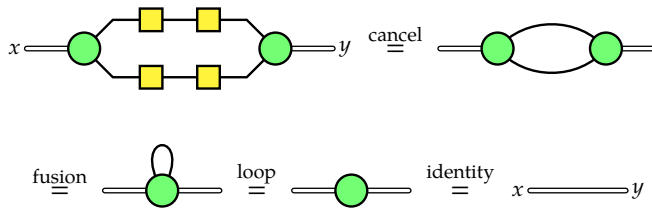
Step 4 — the doubled process. Now the second form. With $\delta\,|x\rangle = |xx\rangle$ and $\delta^\dagger = \sum_y |y\rangle\,\langle yy|$, and $M = M_{rt}$ real,

$$\begin{aligned} R_t E_r &= \delta^\dagger (H^t \otimes H^t)(H^r \otimes H^r)\,\delta = \delta^\dagger (M \otimes M)\,\delta \\ &= \sum_{x,y} \langle yy|\,(M \otimes M)\,|xx\rangle\;|y\rangle\,\langle x| = \sum_{x,y} M_{yx}^2\;|y\rangle\,\langle x|\,. \end{aligned}$$

The entry in row y, column x is the probability of Step 3. For $M = I$ this is I. For $M = H$ every entry is $\frac{1}{2}$, so it is $\frac{1}{2}J$. ■

Coecke's Sketchbook

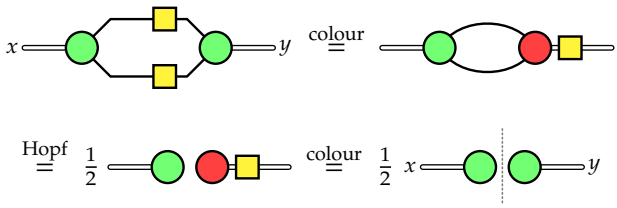
Matching settings, drawn at $r = t = 1$ (at $r = t = 0$ there are no boxes and the chain starts at its second picture). Alice's sender on the left, Bob's reader on the right, a box on each end of each line:



Cancel ($H^2 = I$, Chapter 3; row 56): two boxes in a row on one line cancel, on each of the two lines. **Fusion** (Theorem 8.1): two green dots that share a

line are one green dot; they share two, so the second becomes a loop from the fused dot to itself. **Loop removal**: a self-loop on a phase-free spider is erased, with no number (van de Wetering §4.1). **Identity** (§8.1): a phase-free green dot with two legs is a plain wire. What is left is a bare classical wire: $R_r E_r = I$, exactly, with no scalar anywhere on the way.

Mismatched settings, drawn at $r = 0$, $t = 1$: a plain sender, so no boxes on Alice's side, and a cross reader.



Colour change (Theorem 8.2): put two boxes on Bob's classical leg, which changes nothing since $H^2 = I$; now his reader wears a box on every leg, so it is a red dot, with one box left over on y. **Hopf** (Theorem 8.6): a green dot and a red dot joined by two lines fall apart, with the factor $\frac{1}{2}$. **Colour change** again: a red point followed by a box is a green point, $H\sqrt{2}\,|0\rangle = |0\rangle + |1\rangle$. The result is $\frac{1}{2}$ times a green effect on x beside a green point on y, which is $\frac{1}{2}\sum_{x,y} |y\rangle\langle x| = \frac{1}{2}J$, exactly. The cut (row 87) crosses nothing: Bob's reading is tied to nothing Alice did. The case $r = 1$, $t = 0$ is the same chain reflected, with the colour change made on Alice's sender and its leftover box on x; it is the chain of §12.3 with Eve's dot in Bob's place. ■ See `Code/Ch12/TwoSettings.py`, which builds all four rounds as pictures and compares them, scalar included, with the matrices of the blackboard.

The verdict

The blackboard is shorter. Its whole proof is one amplitude, $\langle y| H^t H^r |x\rangle$, and the fact $H^2 = I$; the rest is squaring. The sketchbook needs four rules for the matching round and three for the mismatched one, and it has to be told what "a green effect beside a green point" means before it can be read as $\frac{1}{2}J$. The blackboard also keeps something the picture never had. The

doubled picture computes probabilities from the start, so the sign $(-1)^{xy}$ of the mismatched amplitude, which the blackboard sees and then squares away, never appears in it.

What the picture shows, and the blackboard does not, is *why* the mismatched round is useless. On the blackboard, $\frac{1}{2}J$ is a matrix of equal numbers, and the reader must notice that equal numbers mean independence. In the picture the round has come apart: there is no line from Alice's bit to Bob's record, and a cut passes through empty paper. And the same rules, with no new idea, turned the shared pair of §12.4 into the sent qubit with one slide and one fusion, where the blackboard needed a transpose identity and a remark about conjugates. None of this is a claim about security. The two proofs establish the same equality of processes, and nothing more.

Coecke's Sketchbook.

Independence is disconnection. The Hopf law did the work in every mismatched round of this chapter, and it is worth saying what it means. A green dot and a red dot joined by two lines come apart. Van de Wetering's survey reads the rule as a statement about the two bases (§4.5): "having maximal information about the Z observable gives minimal information about the X observable". Here the Z observable is our plain setting and the X observable our cross one, and the rule is Result 12.1, drawn. In the same place the survey reads the bialgebra law as a stronger form of the same idea, *strong* complementarity; §12.7 uses it. The picture adds one thing to the words. "Minimal information" is a quantity, but "disconnected" is a shape, and a shape can be checked at a glance. Whenever a round of this chapter came apart, a cut through empty paper was available. Whenever a line survived, as between Alice's bit and Eve's record at the plain setting, it carried the information straight across.

§12.7 The Alarm

The last step turns Eve's mark into an alarm. Alice and Bob have to look for the mark on purpose. Bennett and Brassard have them compare, over the public channel, some of the bits on which they think they should agree. The compared bits are then lost to the key, and they suggest a random subset, "(say one third)" of them, so that a spy who touched more than a few photons would be unlikely to escape. In the pair form, Shor and Preskill have Alice choose some of the pairs as **check bits**, read at both settings. A check that finds a disagreement where there should be none rings the alarm.

On the blackboard. What exactly does a check see? Write the four Bell states of Chapter 10 as $|B_{m_1 m_2}\rangle = (Z^{m_1} X^{m_2} \otimes I)\,|\Phi^+\rangle$, so that $|B_{00}\rangle = |\Phi^+\rangle$, $|B_{10}\rangle = |\Phi^-\rangle$, $|B_{01}\rangle = |\Psi^+\rangle$ and $|B_{11}\rangle = |\Psi^-\rangle$, where $|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$. The bit m_2 records whether the pair has suffered a flip, X, and m_1 whether it has suffered a sign, Z. For example, $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ read plain at both ends always gives 01 or 10: the two readings always differ. Rewritten at the cross setting it is $\frac{1}{\sqrt{2}}(|{+}{+}\rangle - |{-}{-}\rangle)$, so read crosswise at both ends the two readings always agree. A flip shows up at the plain check and not at the cross one.

Result 12.6 (what the checks see). Exactly:

$$|\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-| = |01\rangle\langle 01| + |10\rangle\langle 10|\,,$$

$$|\Phi^-\rangle\langle\Phi^-| + |\Psi^-\rangle\langle\Psi^-| = |{+}{-}\rangle\langle{+}{-}| + |{-}{+}\rangle\langle{-}{+}|\,.$$

So a plain check disagrees exactly on the Bell states with a flip ($m_2 = 1$), and a cross check disagrees exactly on those with a sign ($m_1 = 1$).

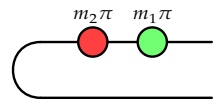
Proof. On the left of the first identity the cross terms $\pm\,|01\rangle\langle 10|$ and $\pm\,|10\rangle\langle 01|$ come in with opposite signs and cancel, leaving the two diagonal terms. For the second, write both states at the cross setting. Substituting $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ and $|1\rangle = \frac{1}{\sqrt{2}}(|+\rangle - |-\rangle)$ gives $|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle + |{-}{+}\rangle)$ and

$|\Psi^-\rangle = \frac{1}{\sqrt{2}}(|-+\rangle - |+-\rangle)$, and the same cancellation happens. The right-hand sides are the projectors onto "the two plain readings differ" and "the two cross readings differ". ■ See `Code/Ch12/Alarm.py`.

Shor and Preskill put the consequence plainly: the rates of bit flip errors and of phase flip errors that Alice and Bob estimate from their check bits "are the same as they would have estimated using the Bell basis measurement". The checks at the two settings read off the two bits m_2 and m_1, and nothing else.

In the sketchbook. Chapter 10 drew a Bell state as a bend carrying two half-turn rings, a red ring $m_2\pi$ and then a green ring $m_1\pi$, on its top leg (rows 73, 75). The cup is unnormalized, cup $= |00\rangle + |11\rangle$, hence the $\frac{1}{\sqrt{2}}$:

$|B_{m_1 m_2}\rangle \;=\; \frac{1}{\sqrt{2}}$



A plain check asks whether the two legs have the same plain value. The bare bend, $|00\rangle + |11\rangle$, always does. A red half-turn is a flip of the plain value (§8.2), so the red ring makes the legs differ; a green half-turn only changes a sign, which no plain reading can see. A cross check asks the same about the cross values, and there colour change swaps the roles: the green ring flips a cross value and the red ring only signs it. So the plain check sees the red ring and the cross check sees the green one. A pair that passes both checks every time has neither ring, and what is left is the bare bend.

That last sentence is about Bell states. What about everything else? Here is the spy of §12.3 again, now on a pair. Eve's press on Bob's half turns the pair into

$$(I \otimes \delta)\,|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle),$$

with the qubits in the order Alice, Bob, Eve. At the plain setting all three readings agree: Alice and Bob pass the plain check every time, and Eve holds the key. At the cross setting they fail half the time (Result 12.4). Could a cleverer Eve prepare some three-part state, entangled with her own system in some cleverer way, that passes *both* checks every time and still leaves

her a record tied to the key? With classical bits she could: a coin copied three ways agrees with itself at every comparison anyone cares to make. Horodecki et al. note that classically, if two bits are perfectly correlated, "there is no constraints on correlations" between either of them and a third. With qubits, the answer is no.

> **Result 12.7 (monogamy, extreme form).** Let ρ be a state of Alice's qubit, Bob's qubit and any finite system of Eve's. If Alice's and Bob's plain readings agree with probability 1, and their cross readings agree with probability 1, then
>
> $$\rho = |\Phi^+\rangle\langle\Phi^+| \otimes \rho_{\text{Eve}}$$
>
> for some state ρ_{Eve} of Eve's system. Whatever Eve then reads is independent of everything Alice and Bob read.

Proof idea. Each check is a projector, and a state that passes a check with certainty lives inside its range. The two ranges meet in a single line, the line of $|\Phi^+\rangle$. A state on Alice and Bob that is pinned to one pure state has nothing left over to share with anyone.

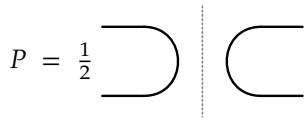
Proof. The plain check passes on the range of $P_{\text{plain}} = |00\rangle\langle 00| + |11\rangle\langle 11| = \frac{1}{2}(I + Z \otimes Z)$, and the cross check on the range of $P_{\text{cross}} = |{+}{+}\rangle\langle{+}{+}| + |{-}{-}\rangle\langle{-}{-}| = \frac{1}{2}(I + X \otimes X)$. The first range is spanned by $|\Phi^+\rangle$ and $|\Phi^-\rangle$. The second is spanned by $|{+}{+}\rangle + |{-}{-}\rangle = \sqrt{2}\,|\Phi^+\rangle$ and $|{+}{+}\rangle - |{-}{-}\rangle = \sqrt{2}\,|\Psi^+\rangle$. They meet in the line of $|\Phi^+\rangle$ alone, so the projector onto both,

$$P = \tfrac{1}{4}(I + Z \otimes Z)(I + X \otimes X) = |\Phi^+\rangle\langle\Phi^+|,$$

has rank 1. (The two factors commute, since $(Z \otimes Z)(X \otimes X) = ZX \otimes ZX = XZ \otimes XZ$, the two minus signs from $ZX = -XZ$ cancelling.) Take first a pure state $|\psi\rangle$ of all three parts. If $\langle\psi|\,(P_{\text{plain}} \otimes I)\,|\psi\rangle = 1$, the projection of the unit vector $|\psi\rangle$ has length 1, so $|\psi\rangle$ lies in the range of $P_{\text{plain}} \otimes I$; likewise for the cross check. So $|\psi\rangle = (P \otimes I)\,|\psi\rangle = |\Phi^+\rangle \otimes (\langle\Phi^+| \otimes I)\,|\psi\rangle)$, a product with $|\Phi^+\rangle$. For a mixed state, write $\rho = \sum_k p_k\,|\psi_k\rangle\langle\psi_k|$ with every $p_k > 0$. The passing probability is $\sum_k p_k\,\langle\psi_k|\,(P_{\text{plain}} \otimes I)\,|\psi_k\rangle$, an average of numbers at most 1. It equals 1 only if every term does, so each $|\psi_k\rangle$ is $|\Phi^+\rangle \otimes |\eta_k\rangle$, and

$\rho = |\Phi^+\rangle\langle\Phi^+| \otimes \sum_k p_k |\eta_k\rangle\langle\eta_k|$. ■ See `Code/Ch12/Alarm.py`, which checks the projector identity and projects random three-part states.

In the sketchbook the projector is already a picture. $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}$ cup cap, with the cup unnormalized as always, cup $= |00\rangle + |11\rangle$: a cap, then a cup, and $\frac{1}{2}$:



The picture is **disconnected in the middle**. Whatever comes in from the left, including any tie to a system of Eve's, is closed off by the cap, and what goes out on the right is a fresh bend tied to nothing. The cut passes through empty paper. Horodecki et al. state the extreme case in one sentence: "If two qubits A and B are maximally quantumly correlated they cannot be correlated at all with third qubit C." They go on to a trade-off between the two correlations when neither is perfect. That quantitative form is beyond this chapter, and nothing here proves it.

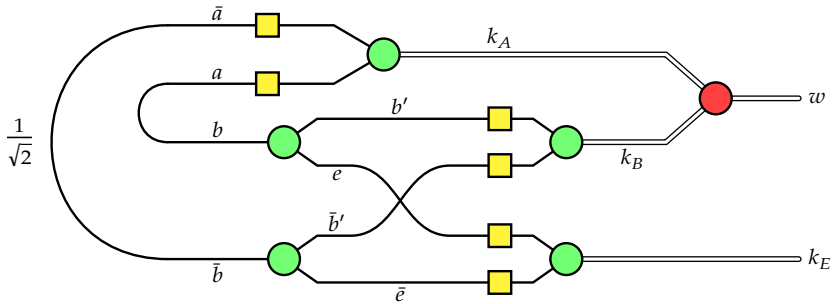
Why this definition. Why check at both settings, when either one alone catches the press half the time? Because each check alone can be fooled. The state $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ passes the plain check every time and gives Eve the key; a state with Eve's copy taken at the cross setting passes the cross check every time and gives her the cross key. Only the pair of checks pins the state down to a single line, and only a single line has nothing left over.

A check round with a spy

Now put everything in one picture: a check round with a spy. The source emits the pair, doubled. Alice reads her half at the announced setting S and records k_A. Eve's press sits on Bob's half, with its mirror on the mirror copy. Bob reads the press's through-legs at S and records k_B. Eve holds her kept legs until the setting has been announced, which is her best case: Gisin et al.'s ideal eavesdropper may keep her system for as long as she likes and choose her measurement "after listening to all the public discussion

between Alice and Bob". Then she reads at S and records k_E. Finally, the alarm. A red dot on two classical wires is Chapter 8's parity, now laid on records (row 96): it takes k_A and k_B and emits $w = k_A \oplus k_B$. On the blackboard it is $X(2 \to 1) = \frac{1}{\sqrt{2}}\,[\,w = k_A \oplus k_B\,]$, so the parity itself is $\sqrt{2}$ times the red dot. The alarm sounds when $w = 1$.

The lines of the picture are the doubled lines $a, \bar{a}$ (Alice's half and its mirror), $b, \bar{b}$ (Bob's half, into the press), $b', \bar{b}'$ (the press's through-legs, to Bob) and $e, \bar{e}$ (its kept legs, to Eve), and the classical wires k_A, k_B, k_E and w. The cups are unnormalized, cup $= |00\rangle + |11\rangle$, so the doubled pair carries $\frac{1}{2}$; with the $\sqrt{2}$ of the parity, the picture carries $\frac{1}{\sqrt{2}}$. It is drawn at the cross setting; at the plain setting the yellow boxes are absent.



This is **The Alarm**. From left to right: the doubled pair, two nested cups; Alice's reader on $a, \bar{a}$, top; Eve's press and its mirror on b and $\bar{b}$; the lines e and $\bar{b}'$ crossing once, so that each reader gets a wire and its mirror (a crossing neither joins nor breaks a line); Bob's reader and Eve's reader; and, far right, the red parity dot on k_A and k_B.

> **Result 12.8 (the Alarm).** In a check round at announced setting S, with Eve's press on Bob's half and Eve reading at S after the announcement, Eve's record and the alarm are distributed as
>
> $$p(k_E, w) = \begin{cases} \frac{1}{2}\,[\,w = 0\,] & S = \text{plain}, \\ \frac{1}{2}\,[\,k_E = w\,] & S = \text{cross}. \end{cases}$$
>
> At the plain setting the alarm never sounds and Eve's record is the key. At the cross setting Eve's record *is* the alarm bit: she

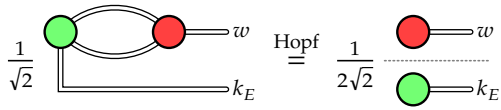
learns whether she was caught, and nothing of the key, since k_A and k_E are then independent fair coins.

Proof, on the blackboard. The press turns the pair into $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ on (a, b', e), as above, and all three are read at S. At the plain setting the three readings are 000 or 111, each with probability $\frac{1}{2}$. So $k_A = k_B = k_E$, $w = 0$ always, and k_E is a fair coin. At the cross setting apply H to each qubit:

$$(H \otimes H \otimes H)\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle) = \frac{1}{\sqrt{2}}(|+++\rangle + |---\rangle) = \frac{1}{2}\sum_{k_A \oplus k_B \oplus k_E = 0} |k_A k_B k_E\rangle\,,$$

since each $|k_A k_B k_E\rangle$ appears in $|+++\rangle$ with amplitude $\frac{1}{2\sqrt{2}}$ and in $|---\rangle$ with amplitude $\frac{(-1)^{k_A + k_B + k_E}}{2\sqrt{2}}$. So each of the four triples of even parity has probability $\frac{1}{4}$ and the others none. Then $w = k_A \oplus k_B = k_E$, and summing over k_A gives $p(k_E, w) = 2 \cdot \frac{1}{4}\,[\,k_E = w\,] = \frac{1}{2}[\,k_E = w\,]$. The pair (k_A, k_E) runs over all four values with probability $\frac{1}{4}$ each. ■

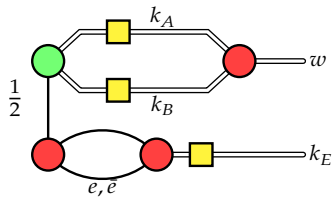
Proof, in the sketchbook. At the **plain** setting there are no boxes. The two cups, the two copiers and the three readers are all phase-free green dots, and fusion (Theorem 8.1) makes them one green dot. It has the three classical legs k_A, k_B, k_E, and two self-loops, which are removed (van de Wetering §4.1): the quantum lines joined those dots with two more lines than a tree on them needs. The green dot and the red parity dot now share the two lines k_A and k_B, and Hopf (Theorem 8.6) takes them apart with the factor $\frac{1}{2}$:



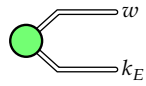
A red point is $\sqrt{2}\,|0\rangle$ and a green point is $|0\rangle + |1\rangle$, so the right-hand side is $\frac{1}{2\sqrt{2}} \cdot \sqrt{2}\,[\,w = 0\,] = \frac{1}{2}[\,w = 0\,]$ for either k_E. Eve's record and the alarm are disconnected, and the alarm is pinned at silence.

At the **cross** setting, six steps.

1. **Fusion** (Theorem 8.1). The cup on (a, b) fuses with the copier on b into a green dot G_1 with legs a, b', e; likewise the mirror cup and the mirror copier make G_2 with legs $\bar{a}$, $\bar{b}'$, $\bar{e}$.
2. **Colour change** (Theorem 8.2). Each reader has a box on both quantum legs. Put two boxes on its classical leg ($H^2 = I$) and it becomes a red dot, with one box left on its record.
3. **Bialgebra** (Theorem 8.5). The green G_1, G_2 and the red readers of Alice and Bob are joined by all four lines $a, \bar{a}, b', \bar{b}'$: a complete square. The bialgebra law replaces it with one red dot, holding the green dots' other legs $e, \bar{e}$, and one green dot, holding the red dots' other legs k_A, k_B (each with its box), joined by one link. The factor is $\frac{1}{\sqrt{2}}$, which with the $\frac{1}{\sqrt{2}}$ in front makes $\frac{1}{2}$:



4. **Colour change** on the parity dot. It becomes green with a box on each of its legs. The boxes on k_A and k_B meet the boxes already there and cancel ($H^2 = I$); one box stays on w.
5. **Fusion**, twice. The green dot on top and the green parity dot share k_A and k_B: they fuse into one green dot, and the second shared line is a self-loop, removed. The red dot below and Eve's red reader share e and $\bar{e}$: they fuse likewise, with a self-loop removed.
6. **Identity** (§8.1). Each of the two surviving dots has two legs, so each is a plain wire. What is left is one wire from k_E through a box, through the link, through a box to w, and the two boxes cancel.

The Alarm at cross $= \frac{1}{2}$  $= \frac{1}{2}(|00\rangle + |11\rangle) = \frac{1}{2}\,[\,k_E = w\,]$.

A green dot with two legs is a bend (§8.1): **one bend joins Eve's record to the alarm**. Every step is exact. ■ See `Code/Ch12/Alarm.py`, which builds The Alarm at both settings, checks it against the blackboard, and asks the automatic simplifier for the two shapes: nothing joins k_E to w at the plain setting, and a path does at the cross setting.

Average over the two settings, which are equally likely. The alarm never sounds on plain check rounds and sounds on half of the cross ones, so it sounds on a quarter of all checked rounds. Eve's record is the key bit on plain rounds and independent of it on cross rounds, so she holds half a bit per key bit. These are the numbers of Result 12.4, and of Bennett and Brassard's optimum example. The difference is that now they are heard: the mark Eve's look leaves has become the sound of the bell.

The Alarm also bends back into the form of §12.3. Slide Alice's boxes round the bend and fuse, as in §12.4: the cup and Alice's reader become her sender, with k_A as its classical leg, and the picture becomes Eve's box on the line of a sent qubit, with Alice's bit copied to the alarm.

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§12.8 What the Lock Does Not Prove

Take stock of what has been proved. A round at matching settings delivers Alice's bit to Bob every time, and a round at mismatched settings delivers a coin (Result 12.2). Announcing the settings costs nothing and keeps half the rounds (Result 12.3). A spy who presses the line learns half a bit per key bit and pays for it with errors on a quarter of the kept rounds (Result 12.4), and the checks hear those errors as an alarm (Result 12.8). A pair that passes both checks every time has nothing left over for anyone (Result 12.7). Each of these is an exact equality, and each has been proved.

None of them says the key is secret. The press is one spy among infinitely many. A real eavesdropper may entangle each passing qubit with a probe of her own design, keep all her probes, and measure them together after every word of the public discussion. And real checks never come out perfect: some errors come from the line and the detectors, and Alice and Bob cannot

tell those from a spy's. Result 12.7 covers only the case in which the checks never fail. What is needed is a theorem that turns a *small* measured error rate into a *small* bound on what any eavesdropper can know. That theorem exists.

> **Result 12.9 (Shor and Preskill; stated, not proved).** The BB84 protocol, with its sifting, a random sample of check bits at both settings, and suitable classical error correction and privacy amplification, is secure against an eavesdropper able to perform any operation quantum mechanics permits. Precisely: the probability is exponentially small that Alice and Bob agree on a key about which Eve can obtain more than an exponentially small amount of information. This holds as long as the bit- and phase-error rates measured on the check bits are below 11%, the point at which the rate $1 - 2h(\mathcal{D})$ reaches 0, where $\mathcal{D}$ is the error rate and h the binary entropy of §12.3. The proof assumes that Alice's source emits perfect single photons.

Here is the shape of their argument, which we do not reproduce. They first give a protocol in which Alice and Bob share many noisy pairs, and purify them. The pairs are purified into fewer, nearly perfect copies of $|\Phi^+\rangle$ using quantum error-correcting codes of a family known as CSS codes. That protocol they prove secure with methods from a proof by Lo and Chau. Then they show that Alice and Bob can do all of it with single qubits sent and read at two settings, so that the pair protocol collapses into BB84. The last step works because this family of codes corrects bit errors and phase errors separately. The checks enter through Result 12.6: the bit-flip rate and the phase-flip rate are what the plain and cross checks measure. What the check bits estimate is what the codes must correct.

Gisin et al. give the thresholds in context. Against *individual* attacks, where Eve attaches an independent probe to each qubit and measures her probes one after the other, BB84 is secure up to an error rate $\mathcal{D}_0 = \frac{1}{2}(1 - \frac{1}{\sqrt{2}}) \approx 15\%$. They note that this is also the error rate up to which Alice and Bob could still see a violation of Bell's inequality in the CHSH form of Chapter 7. Against *coherent* attacks, where Eve may process several qubits together, the bound is 11%, "precisely that obtained in Mayers proof (after improvement by P.

Shor and J. Preskill (2000))". Between 11% and 15%, Gisin et al. wrote, it was not known what happens.

Two classical steps finish the protocol, and neither is quantum at all. *Error correction* removes the disagreements that noise, or a spy, left in the sifted key. *Privacy amplification* shrinks what Eve knows. Gisin et al. give its simplest form. Alice picks two bits of the key, announces only *which* two (say bit 103 and bit 537), and both replace them by their sum mod 2. If Eve knows the first bit and nothing of the second, she knows nothing of the sum. If she knows each with probability 60%, she guesses the sum right only with probability $0.6^2 + 0.4^2 = 52\%$. Repeated, and done in larger blocks, this drives her information as low as Alice and Bob wish, at the cost of key length.

We should be exact about the scope of the pictures. The rewrites of this chapter prove *correctness*, Result 12.2; the numbers of one spy, Results 12.4 and 12.8; the identities behind the checks, Result 12.6; and monogamy in its extreme case, Result 12.7. They prove no security theorem, no key rate and no threshold. The 11% and 15% are stated and cited here, and proved in neither of our two notations. The blackboard does not prove them either, in this chapter. Result 12.9 is a theorem about many rounds at once, about error-correcting codes and about bounds on information, and neither of our strands was built for that.

Coecke's Sketchbook.

What the rewrites do not prove. A chain of rewrites proves that two pictures are the same process. It is tempting to read more into a picture that falls apart, and to say that a disconnected Eve "cannot know" the key. The picture licenses something smaller. In §12.6, Bob's record is disconnected from Alice's bit in *that* round, with *that* reader. In §12.7, Eve's record is disconnected from the alarm for *that* spy, the copier, at the plain setting. Change the spy and the picture changes with it, and there are infinitely many spies. Monogamy is the nearest thing here to a statement about every spy, and even it is only the extreme case: a pair that never fails a check. A key that fails

a few checks in a hundred needs Result 12.9, and Result 12.9 needs a theory of errors and codes that is not in the sketchbook. The pictures show what one protocol does with one spy. They are not a security proof.

Feynman's Notebook.

A simple proof, sixteen years on. Peter W. Shor, of AT&T Labs Research in Florham Park, New Jersey, and John Preskill, of the Lauritsen Laboratory of High Energy Physics at the California Institute of Technology, posted their proof on the arXiv as quant-ph/0003004; part of the work was done while Shor was visiting Caltech. Their opening sets the scene: BB84 was proposed in 1984, but "for many years, it was not rigorously proven secure against an adversary able to perform any physical operation permitted by quantum mechanics". They count three recent proofs, none entirely satisfactory. Lo and Chau's is easy to understand but needs a protocol that requires a quantum computer. The other two, one of them by Dominic Mayers, cover a protocol based on BB84 but are "quite complicated". Shor and Preskill's proof, they write, was inspired by noticing that the codes they use "are hidden in the inner workings" of Mayers's proof. They end on its weakness. Like the two earlier proofs for BB84, it needs perfect single-photon sources, while most experiments use weak coherent sources, which "no currently known proof covers".

The lock is built. Two settings that are each a coin to the other gave a key that arrives whole when the settings match. Announcing the settings afterwards kept the key and lost nothing. A spy on the line became a green box with a side road, and the price of her look could be written down. The shared pair bent back into the sent qubit with one yank. And a check round with a spy in it, The Alarm, came apart at the plain setting and joined Eve's record to the bell at the cross setting. The part the pictures could not reach, security against every spy, is there to be cited, and we have said whose it is.

This is the end of Part III. The pictures have become proofs: of copying and

its impossibility, of teleportation, and now of a lock made of light. Part IV asks something different of them. Until now, but for Chapter 5's single oracle, every box in the book has been ours, built from gates we chose. In Chapter 13 someone else hands us a box and will not say what is inside it. We may ask it a question, once. The box turns out to be a web of CNOTs, and yellow boxes turn its colours over. Its secret can be read straight off the picture, if we ask in the right way.

Minuet for One Wish

霜の夜や
見えぬ峰より
返る声

shimo no yo ya
mienu mine yori
kaeru koe

A night of hard frost —
from the peaks I cannot see
my own voice comes back.

[*The Tortoise's parlour, towards eight on the same frosty evening. The second pot stands empty between the two cups, and the stove has burned down low.*]

[*It has settled to a dull red behind its mica, and its iron ticks now and then as it cools. On the lower pane the two ferns stand where they met at the middle; above them the glass is black, and past it the lane lies pale under the rime, the curtained windows opposite still lit. Achilles has come back from the window to his chair by the fender with his stockinged feet to the stove, and the Tortoise sits across from him with her empty cup in her lap. Low as the stove is, the parlour is still the warmest room in the house.*]

Achilles: [*tilting the pot, and finding nothing in it*] That's the second pot gone, and I haven't the face to ask for a third.

Tortoise: You'd only have to carry it up the stairs.

Achilles: Up the stairs? [*He looks at the window.*] The ferns have met in the middle of the pane. I was about to say that means it's time I went home.

Tortoise: It means the frost has finished with the parlour. There's something upstairs I've meant to show you since before the frost, and tonight is cold enough and late enough for it.

Achilles: In all the winters I've been coming here, I've never once been up your stairs. I'd assumed they only led to more stairs.

Tortoise: They lead to one small room, and nobody has set foot in it since the frost came. [*She takes the candlestick from the mantel and lights it at the stove.*] This is all the light the top room gets. Keep close behind it on the stairs: one of the treads is loose, and I never remember which.

Achilles: [*picking up her work-bag*] I'll carry this, then. It's heavier than knitting.

Tortoise: My cloth, and some string, and a few things you haven't seen yet. Mind the tread.

[*They go up, the candle first. Halfway, a tread gives under Achilles's foot with a groan, and he stops short; the Tortoise, above him, does not turn round.*]

Tortoise: That one.

[*The stair is narrow, and makes one turn at a small landing. Their shadows go up the wall beside them, huge and unsteady, stretching tall when the candle dips and shrinking when it rises, sliding round the turn of the landing and back; not one of them keeps its place for two steps together. The flame leans in the cold that comes down the stairs to meet them, and every shadow leans with it.*]

Achilles: [*at the head of the stairs, his breath showing*] It's colder with every step. You could keep butter up here.

Tortoise: I do, in summer. [*She sets the candlestick on the small table in the middle of the room, takes her drawing-cloth from the work-bag, and spreads it beside the candle.*] Sit. Yours is the chair without the wobble.

[*The top room: a small table and two chairs, one window over the lane, and on the wall beside the window a shelf. On the shelf stand an old brass lamp and a lidded*

casket.]

[*The top room is small and very cold, tucked under the slope of the roof, and it has one of everything. One door, which had stuck in its frame with the frost and gave all at once when the Tortoise put her shoulder to it, and which stands open now on the dark of the stairs. One window over the lane, its pane furred white with frost, the ferns grown well past the middle, so that the lane shows through nowhere. One shelf, on the wall beside the window. One light: the candle on the table. On the floor under the shelf the Tortoise's summer larder is put by for the winter, a row of empty earthenware butter-crocks and, among them, one tall glass jar. The lamp on the shelf is tarnished nearly brown, with no oil in it, and the casket beside it is of dark old wood, the dust lying undisturbed along its lid; neither has been touched since the frost came, nor for a long while before. On the wall facing the window, not above the shelf, hangs a print in a plain dark frame: Escher's* Belvedere, *a pillared lookout whose upper storey is set crosswise on the lower; a ladder stands inside it and leans against the outside. The candle just reaches it across the table, and it takes what little light there is, as it has taken it every winter since long before the Tortoise came to the house. Then the Tortoise lowers herself into the other chair, with her back to that wall, and her shadow goes up over half the print. With the candle still at last, the flame stands up, and every shadow in the room settles where it belongs, thrown straight back from it: the chair-backs up the walls, the lamp and the casket together on the slope of the ceiling, and Achilles's, still standing, huge across the frosted pane.*]

Achilles: [*sitting, and looking up at the shelf*] Is that what you've brought me up to see? A lamp and a jewel box?

Tortoise: Neither of them is mine. They came with the house, and they aren't mine to lend.

First Challenge — *One Wish, One Answer*

Tortoise: [*holding the candlestick up to the shelf, to a small upright casket of dark old wood, a faded paper band, its seal, pasted across the front edge of its lid, and a small oval of enamel set in the lid's top*] This is the Genie's casket. It has been

sealed as long as I've had the house, and it opens once, after it has done its one piece of work.

Achilles: There are knobs down this side. And little windows down the other.

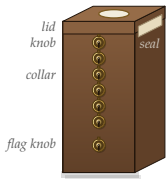
Tortoise: Down the left face, the knobs: each turns down for dark or up for lit, and tonight they all stand down. The lowest, set a little apart from the rest, is the flag knob; the ones above it are wish-knobs.

Tortoise: Down the right face, a window level with each knob, and the flag window lowest. A window shows nothing until the casket passes; then it shows dark, a plain pane, or lit, a small gleam behind the glass, and it keeps what it shows.

Achilles: And these rings round every knob and every window?

Tortoise: Collars, like the collars on my keys downstairs, but with only two stops: the upright stroke and the cross. Every one of them stands upright tonight. Setting the knobs and turning the collars is free, all night if you like; only the passage costs.

[*Each collar has a small brass nib on its rim; at the upright stroke the nib stands at the top.*]



knobs down,
collars upright

Achilles: [*standing, and bending to the lid*] There's a little enamel plate on the top. [*reading*] "The Terms. One wish, one passage. Behind each wish-knob there is a mark, or none. With every collar upright, each wish-window shows its own knob as it was set, and the flag window shows the flag knob as it was set, turned over once for each marked knob that was set lit. What

the collars do crosswise is the wisher's business."

Achilles: So the secret is which knobs are marked. And it says nothing about how the marks are made.

Tortoise: Nothing at all. The Terms are short, and they keep to them.

Achilles: [*his hand going to the lamp*] And the lamp —

Tortoise: Is how you wish. Rub it once and the Genie comes; he grants one wish, and the casket passes once. Not yet: we should know what to wish before you rub it.

Achilles: [*taking his hand back, and sitting down again*] One wish. Then we'd better spend it on the right knobs.

Tortoise: [*setting the candlestick back on the table*] Then we'll practise on a box we can spend all night on. [*She tears a strip from the back of her notebook and rules it across into a row of little squares.*] One square for each wish-knob. This is your slip: ink some of the squares where I can't see, and fold it, and the inked squares are your marks.

[*With the candlestick back on the table, the shadows that swung round the walls when she held it up to the shelf have come back to their places, each where it lay before: the chair-backs, the lamp and the casket on the slope of the ceiling, her own over the print. Achilles pulls his chair in close and holds his hands to the flame a moment, working the cold out of his fingers.*]

[*Achilles turns his shoulder to her, inks some of the squares, and folds the slip along its length so that the squares are inside.*]

Tortoise: [*bringing a small counter out of the work-bag*] And this is our flag, a disc. It's gilt on one face, for lit, and black on the other, for dark. I name a wish and hand you the disc with whichever face up I choose; you look at your slip, turn the disc over once for each inked square among the places I named lit, and hand it back.

Achilles: Just as the Terms say the flag window goes.

Tortoise: Just so. The first and the third lit, the rest dark. [*She puts the disc in his hand, black face up.*]

[*The disc lies on Achilles's palm, black face up: dark. He opens the slip behind his other hand. Of the two places she named lit, the first is inked and the third is bare. He turns the disc over once between finger and thumb: the black face goes down against his palm, and the gilt comes up. It is the same small counter in the same hand; the slip is folded again, and the Tortoise has not seen it; the wish she named has not changed. Only the face that shows has changed: dark has become lit.*]

Achilles: [*handing it back, gilt up*] There.

Tortoise: Lit. One face, one answer.

Achilles: Only because you asked so meanly: two places, one disc. Let me choose the wish. Light every place at once, and the disc will have to account for every mark I made; with more places lit, it has more to say.

Tortoise: Choose, then. [*She hands him the disc, black up.*]

[*Every place lit. Achilles runs his thumb along the whole slip under his hand and turns the disc once for each inked square he passes: over to gilt, and then over again to black. It comes to rest on his palm black up, exactly as she handed it to him. The slip and its marks are as they were; the disc shows the same face as before the wish, as though nothing had been asked.*]

Achilles: Black. As if I'd made no marks at all. [*He frowns at the slip.*] Or two, or four. Every mark turns it over, and a disc turned twice is back where it started. It has only two faces. Whatever I light, all it can ever say is whether the count came out odd or even.

Tortoise: One face, one answer, however many places you light. Now my first wish again, the first and the third lit, but I'll hand you the disc gilt up.

[*The disc goes to him gilt up. The first place is inked, as before; he turns the disc once, and it goes back to her black up. Last time it went to him black and came back gilt; this time it went to him gilt and came back black. The same one mark has turned it the same once. What it was handed is what has changed, and so what comes back.*]

Achilles: Turned the other way from before. Of course. The one mark turns over whatever face it's handed.

Tortoise: Then the same wish twice running. [*She hands it to him black up, the first and the third lit.*]

[*He turns it once, gilt up, and hands it back. She hands it straight back to him, gilt up, naming the same wish; the same inked square turns it the same once more, and it lies on her palm black up, the face she first gave. The slip has not moved, nor has the wish. The disc has been turned twice, and the two turnings have undone each other.*]

Achilles: Back as you gave it. The wish, asked twice, undoes itself.

[*The Tortoise leans back from the table, and her chair, which has its wobble as promised, rocks over onto its other short leg with a small clack. She leans in again, and it rocks back onto the first with another. It has two ways to stand, and no third.*]

Tortoise: Now I'll read your slip. The first place alone lit. [*The disc goes to him black up.*]

Achilles: [*turning it once*] Gilt.

Tortoise: Marked. The second alone. [*Black up.*]

Achilles: [*handing it straight back*] Black.

Tortoise: Bare.

[*So on down the slip, a single place lit each time and the disc handed over black each time: it comes back black at the third, black at the fourth, gilt at the fifth, black at the sixth.*]

Tortoise: The first and the fifth.

Achilles: [*unfolding the slip: the first square inked, and the fifth*] The first and the fifth. A wish for every knob, and one place read by each.



his slip, unfolded

Tortoise: Could you do it with fewer? Change places. [*She tears another strip, rules it, turns her shoulder, inks it, and folds it.*] My slip, my disc; you're the wisher. Read my marks with fewer wishes than there are knobs.

Achilles: In pairs, then; you can't hide from pairs. The first two lit.

[*She takes the disc black up, looks at her slip, turns it once, and hands it back gilt.*]

Achilles: The third and the fourth.

[*Black comes back black.*]

Achilles: The fifth and the sixth.

[*Black side up to her again; her slip turns it once, and it comes back gilt.*]

Achilles: And the first, the third and the fifth.

[*Black side up once more; it comes back black, as it went.*]

Achilles: One of the first two is marked, and one of the last two. The third and fourth go together, both or neither. And the first, third and fifth come out even. So I say the second and the sixth, and nothing in the middle.

Tortoise: Perhaps. Or perhaps this. [*She sets her folded slip down on the cloth, tears a third strip, rules it, inks it in front of him, and lays it open to the right of her folded one, as Achilles sees them.*]

[*The new slip lies open on the cloth: the first square inked, the third, the fourth and the sixth. Achilles goes down his wishes against it one at a time, a finger counting the inked squares among the places he named lit. The first two: one inked square, the first; odd, and the disc would have turned. The third and fourth: both inked; even, and the disc would have come back as given. The fifth and sixth: one, the sixth; odd. The first, third and fifth: two; even. Wish by wish, the open slip would have turned the disc exactly as her folded one did.*]

Achilles: It answers every one of my wishes just as yours did. And it isn't your slip?

Tortoise: [*unfolding hers where it lies, to the left of the new one: the second square inked, and the sixth*] It isn't. You guessed right, and your wishes could never have told you so. Whatever fewer wishes you choose, I can ink you a second slip that answers them all alike; and the wisher never holds the slip he is wishing against.



hers, and the
new slip beside it

Achilles: [*sitting back*] What does one wish buy?

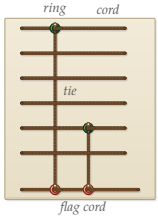
Tortoise: One answer, and only the one.

Achilles: Then the rest must be bought one at a time.

Tortoise: [*drawing the work-bag to her, and putting into it her notebook, her two slips off the cloth and the disc from Achilles's hand, while Achilles folds his own slip into his pocket*] Then let me draw what your slip is. My drawing-cloth: whatever I lay on it is a picture of a thing, not the thing, and no wish is ever spent on it.

Tortoise: [*bringing out string and a little box of rings, green and red*] My string, and my rings. A green ring threaded on one cord and tied by a short length of string to a red ring threaded on another is a mark: whenever the green ring's cord is lit, the red ring turns the other cord over.

[*The cloth lies bare between them. She lays a cord across it for each square of Achilles's slip, running from left to right as he sees it, one under another like the knobs down the casket's face, the first square's cord furthest from him; and below them, nearest him and set a little apart, a longer cord for the flag, running on to the right past the others' ends. On the cord for the first square she threads a green ring, and on the flag cord a red ring beneath it, and ties the two together with a short length that runs down beneath the cords between, joined to none of them. On the cord for the fifth square she threads another green ring, tied down to a red ring of its own further along the flag cord. The other cords she leaves straight and bare. What was hidden in a folded slip lies open now: the marks have become ties, and the unmarked places are cords with nothing on them at all.*]



the slip's web

Tortoise: That is all a box of marks is.

Achilles: A tie from every mark down to the flag. Light a marked cord and its tie turns the flag over; light a bare one and nothing happens. [*He runs a finger along the flag cord, past one red ring and then the other.*] My disc, drawn. Then the casket will take a wish for every knob, and the Genie grants one.

Tortoise: Unless the one wish is set some other way. What do you suppose the collars are for?

Second Challenge — *The Mark on the Cords*

Achilles: The Terms say that crosswise is the wisher's business. So it's ours.

[*They both draw in to the cloth, and the Tortoise's chair clacks over as she comes. Bent over the table from either side, their two heads throw two great shadows up the slope of the ceiling, which lean together and apart as they talk, cross, slide over each other, and never once keep still. The candle has begun to run: a drop of wax slips down its side and stops there, gone white in the cold.*]

Tortoise: Then we'll mind it. First, a knob on the cloth. [*She takes a handful of small rings from the box, each with a bead set in it.*] A ring at a cord's loose end, with that one cord and nothing else, is a knob's setting. Red with its bead at rest is a dark knob; red with its bead set halfway round, like the half-turn rings you've met before, is a lit one.

[*She threads a red end-ring on the loose left end of every cord of the web, the flag cord too, each bead at rest. Every cord now begins at a red end-ring: every knob*

dark, as the knobs stand on the casket. What is left of the handful she drops back into the box.]

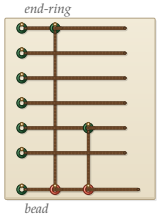
Tortoise: [*setting a new packet down on the table beside the cloth and shaking one out into her palm: short yellow tubes, each no longer than a fingernail and slit along its length, so that it clips onto a cord anywhere*] And these are new. Each is a yellow tube that slides along a cord, and a tube on a cord is a collar turned crosswise.

Achilles: Mr Crab's flip-slot, sold by the packet.

Tortoise: The same trick, and much cheaper. Two tubes that meet on a cord slide off together, since a collar turned twice is no turn at all; and a tube slid out onto an end-ring changes its colour.

Tortoise: Every wish-knob crosswise and dark, then, and the flag crosswise and dark as well.

[*Each wish-cord begins at a red end-ring, bead at rest. She clips a tube from the packet onto the cord near the end and slides it outward until it runs onto the ring; the tube goes into the ring and is gone, and the ring is green, its bead still at rest, the cord behind it just as it was. Down the web, one after another, every wish-cord's end-ring goes from red to green. The ties, the green rings on the first and fifth squares' cords and the red rings on the flag cord are untouched. Last she does the same to the flag's own end-ring: green, at rest.*]



every end-ring turned green

Tortoise: A window reads the far end of its cord by the same code. Read crosswise, a green end at rest shows dark, and a green end with its bead halfway shows lit.

Tortoise: Two more things my rings do. Two rings of one colour on one

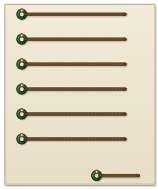
cord slide together into one, and their beads' turns add up. And an end-ring pushed into a ring of the other colour vanishes into it: a copy of it comes out on every other length of cord that leads from that ring, and that ring is gone.

Achilles: Mr Crab's rules, from the evening of his koekpan.

Tortoise: The same rules; my rings keep them too. Now watch the flag.

[*The flag's green end-ring, bead at rest, sits at the start of the flag cord. Along the cord lies the red ring where the first square's tie comes down, and beyond it the red ring for the fifth square's tie. She slides the green end-ring along the flag cord into the first red ring. It goes in and does not come out, and the stretch of flag cord it ran along is swallowed with it. The red ring is gone from the cord, and in its place two green end-rings, both at rest: one at the foot of the tie, one on the flag cord just past where the red ring was, where the flag cord now begins. The one at the foot of the tie she slides up it to the green ring on the first square's cord, the tie swallowed behind it as it climbs; green meets green, and they run together into that ring, whose bead stays exactly where it was. Then she slides that ring out along its cord to the green end-ring at the cord's start, and they run together into the one end-ring, still at rest. The flag's copy she slides on into the red ring for the fifth square, and the same happens there: that red ring gone and the flag cord behind it swallowed, a green copy up the tie into the fifth square's ring, adding nothing, that ring slid out into its end-ring in the same way, and one last green copy left on the flag cord, at rest.*]

[*What lies on the cloth now: not a tie anywhere, and not a red ring. On the first and fifth squares' cords the ring the tie came up to and the end-ring have slid together into one green end-ring, at rest; every other wish-cord begins at its green end-ring, at rest, as she set it. Every wish-cord ends as it began. The flag cord ends as it began too, green and at rest.*]



every bead
still at rest

Achilles: Then the web is empty. There are no marks.

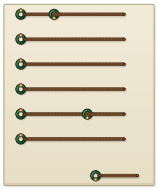
Tortoise: The slip has marks; you inked them.

Achilles: Then the flag gave nothing to anyone.

Tortoise: It had nothing to give. Now the flag crosswise and lit.

[*She gathers the cloth clear, the rings into her little box of rings and the string into her work-bag, and lays the web afresh as it was: the ties down from the first and fifth squares' green rings to their red rings on the flag cord, every wish-cord beginning at a green end-ring at rest. At the start of the flag cord she sets a green end-ring as before; but before she lets go of it she turns its bead halfway round.*]

[*She slides the flag's end-ring, its bead halfway round, into the first red ring, the stretch of flag cord it ran along swallowed with it. The red ring is gone, and out come two copies, each green, each with its bead halfway round: one at the foot of the tie, one on the flag cord beyond. She slides the first up the tie, the tie swallowed behind it, into the green ring on the first square's cord. Green runs into green, and the ring's bead, which was at rest, now stands halfway round. The flag's copy she slides on into the red ring for the fifth square: gone, the flag cord behind it swallowed, and two copies again, halfway round, one climbing that tie into the fifth square's green ring, one left on the flag cord. Nothing has come to the bare cords, and their end-rings sit at rest. What has changed: the first and fifth squares' cords each carry a half-turn they did not have, exactly where the ties came up. What has held: the flag cord, which ends as it began, green with its bead halfway round. It has given its turn away twice and still has it.*]



turned at the marks,
the flag as it began

Achilles: The flag's turn has gone to the first and the fifth, my marks, and the flag still has its own. It hasn't given anything away.

Achilles: ...Oh. It's your token, wedged up on its edge on the porch in the spring. The turn had nothing on the flag to spend itself on, so it jumped back up the tie onto the cord that asked.

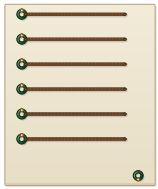
[*Up the lane the church clock begins to strike eight. At the first stroke the tall glass jar under the shelf hums back, faint and clear, and holds its hum a moment after the stroke has gone; the crocks on either side of it stay dumb. So at every stroke, to the eighth, and each stroke from the tower comes as full as the one before. Achilles glances down at the jar, and back to the cloth.*]

Tortoise: And read crosswise, which windows gleam?

Achilles: The first and the fifth. Every mark comes back on its own knob's cord.

Tortoise: One more box. [*She gathers the cloth clear again, the rings into their box and the string into her work-bag, lays the cords afresh with nothing threaded on them, and sets the end-rings: green at rest on every wish-cord, green with its bead halfway on the flag.*] A box with no marks at all.

[*She slides the flag's end-ring along the flag cord. There is no red ring on it now to go into; it runs the whole length and comes to rest at the far end, halfway round, as it set out, the flag cord it ran along swallowed behind it. Nothing climbs to any wish-cord, for there is no tie to climb. Every wish-cord lies as she set it, green and at rest, and every bead on the cloth is where it was before she pushed.*]



run clear to the far end

Tortoise: If every wish-window stays dark, the box has no marks. If any wish-window gleams, it has.

Achilles: And the flag window?

Tortoise: Set crosswise and lit, it comes back lit every time, marks or none. Leave it out of the test.

Achilles: Then set the casket as you set the cloth, every collar crosswise, the wish-knobs dark and the flag lit, and the one wish brings back every mark at once. [*He looks up at the shelf.*] But would the Genie count that as one wish?

Third Challenge — *The Turned Web*

Tortoise: There's only one way to find out what the Genie counts.

[*Achilles reaches up and takes the lamp down from the shelf. It sits in his two hands as it has sat on the shelf all evening: tarnished brass, cold, its wick dry, never lit. He rubs its side once with his thumb. The lamp does not warm or brighten, and nothing in it catches. From its spout a thread of something paler than smoke rises, straight as a plumb line in the still cold air, thickens, and stands; and where it stands there is all at once a Genie, upright beside the table, with the air of someone who has come about a form. The lamp in Achilles's hands is exactly as it was, cold and dull and no lighter. What has changed is that there is a third at the table. Achilles sets the lamp down beside the candlestick.*]

[*While the pale thread stood, the candle flame leaned over towards the lamp as*

though something were drawing on the air, and every shadow in the room leaned with it. Now it stands up straight again, stiller than it has stood all evening, and not a shadow on the walls stirs. On the wall by the shelf there is a third shadow beside the other two, thinner than theirs, as though the candle can only half find him.]

Genie: The Office of Wishes. The lamp has been rubbed once, which is correct.

Genie: One wish is one passage of the casket, whatever its knobs and collars are set to. A wish with every collar crosswise is still one wish.

Achilles: You've been listening.

Genie: The Office anticipates. A plain wish, every collar upright, returns the knobs' settings in the wish-windows and one reading in the flag window: the flag knob's setting, turned once for each marked knob set lit. That is one answer.

Achilles: And setting the knobs —

Genie: Is not a wish. Setting is free; the Office charges for passage. There is one wish.

Achilles: Then I wish to know which knobs are marked.

Genie: Referred. [*He inclines his head as though listening up a stairwell, and straightens at once.*] The Meta-Genie regrets. The marks are the casket's to tell, and the casket tells only by passage. Nothing has been spent.

[*A draught comes up the dark stairwell and in at the open door, lifts the candle flame tall and thin, and lets it fall back to its own height, all in the one instant. Nothing else in the room has had time to move.*]

Achilles: That was quick.

Genie: Any request other than one passage of the casket as it is now set is a meta-wish, and it goes up. The Meta-Genie referred yours to the one above, who referred it higher, and so on all the way; the answer came down all the way at once. The Office has no stairs.

Achilles: Then I wish for more wishes.

Genie: Referred. [*The same tilt of the head, no longer than a blink.*] The reply has come down: there is one wish. The Meta-Genie adds that a wish for wishes is a wish about wishes, and is not charged.

Achilles: Then at least let us look inside first.

Genie: Referred. [*A blink.*] The lid lifts on a spent casket and on no other. Your wish is untouched.

[*The Tortoise laughs under her breath and tucks her claws into her sleeves. Achilles, a little pink about the ears, blows on his fingers and holds them to the candle.*]

Tortoise: Achilles, you'll talk your wish into the ground. Let me show you the setting on the cloth first.

Genie: The Office is in no hurry. The Office is never in a hurry.

Achilles: May I at least bring the casket down to the table? Or is that a meta-wish too?

Genie: That is not a wish. That is furniture. Carry it level.

[*Achilles lifts the casket down from the shelf in both hands and sets it on the table beside the cloth, its knobs down the left, its windows down the right, as the cords run on the cloth. It stands as it has stood all evening: every knob down, every collar upright, every window a plain pane, the seal whole.*]

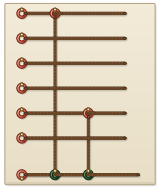
Tortoise: [*gathering the cloth clear, the rings into their box and the string into her work-bag, and laying the slip's web out afresh on the cloth: a cord for each square, the flag cord below, the ties down from the first and fifth squares' green rings to their red rings*] Your slip's web again, set as the casket will be set. Every wish-knob dark, [*a red end-ring, bead at rest, at the start of each wish-cord*] and the flag lit. [*At the start of the flag cord, a red end-ring with its bead halfway round.*] Now every collar crosswise.

[*She clips tubes from the packet onto the web. On every cord that runs from a knob's end to a window's, one tube goes on just inside its end-ring, and one near its far end, where the window would be. On each tie she clips a pair of tubes, and a pair on the flag cord between its two red rings.*]

Tortoise: A pair is no turn at all. It only puts the tubes where I need them;

and a ring with a tube on every cord that leads into it changes colour, as Mr Crab's did when every leg went through his slot.

[*She slides the tubes along toward the rings. On the bare cords the two tubes run in from either end, meet in the middle, and slide off together into her palm; those cords lie bare again, their end-rings red and at rest, as before. On the first square's cord one tube comes in from the knob's end and one from the window's end, and the tie's pair parts, one tube running up to the green ring and one down to the red; the same on the fifth square's cord and its tie; on the flag cord, the tube from the knob's end runs to the first red ring, the middle pair parts toward the two red rings, and the tube from the far end runs to the second. Now every ring in the web has a tube on every length of cord that leads into it. Every ring changes colour at once, and its tubes go into it and are gone, as at the end-rings: the green rings on the first and fifth squares' cords are red now, and the red rings on the flag cord are green. Not a tube is left on the cloth. The pairs in her palm she drops back into the packet. The end-rings have not changed: red at rest on the wish-cords, red and halfway round on the flag. Each tie still joins the same two cords it joined before; but a tie runs from a green ring to a red one, and now the green is on the flag.*]



colours swapped, the end-rings unchanged

Achilles: The ties run the other way. They came down from my marks to the flag; now they go up from the flag to my marks.

Achilles: So what you're saying is that the collars have moved the ties.

Tortoise: Nothing has moved. Every tie joins the cords it always joined; only the rings have turned.

Achilles: And last time you slid the tubes outward, onto the end-rings.

Tortoise: Outward or inward, it's the same collar, and the two ways agree.

[*Achilles looks up from the cloth, past the candle, to the print on the wall facing the window: Escher's* Belvedere.]

Achilles: A pillared lookout whose upper storey is set crosswise on the lower; a ladder stands inside it and leans against the outside. It has been there all along.

[*The Tortoise turns in her chair to look where he is looking, and her shadow, which has lain over that wall since the candle came up, slides aside off the frame. For the first time this evening the print hangs clear in the candlelight, and seems no further off than the candle.*]

Tortoise: Push the flag in.

[*The flag's end-ring, red with its bead halfway round, sits at the start of the flag cord. Along the cord lies the green ring whose tie climbs to the first square's cord, and beyond it the green ring for the fifth. Achilles slides the end-ring along into the first green ring, the stretch of flag cord it ran along swallowed with it. The ring is gone, and out come two copies, each red with its bead halfway round: one at the foot of the tie, one on the flag cord beyond. He sends the first up the tie, the tie swallowed behind it, into the red ring on the first square's cord. Red runs into red: the cord's end-ring at rest, the ring, and the copy all slide together into one red end-ring at the cord's start, with its bead halfway round. The flag's copy he slides on into the next green ring, and the same happens there, up the tie to the fifth square's cord. The last copy comes to rest at the flag cord's far end, the flag cord it ran along swallowed behind it, red, halfway round. The bare cords have not been touched: red and at rest from end to end.*]



carried up to the marks, and on

Tortoise: No tubes left, so read them plainly: red at rest is dark, and red halfway round is lit.

Achilles: Lit at the first and the fifth, my marks, and the flag lit. One push, and every mark read at once.

Achilles: [*turning to the casket*] Now the casket. Every collar crosswise.

[*The casket stands as it has all evening: every knob down, every collar at the upright stroke, every window plain. Achilles goes down the left face, turning each wish-knob's collar a quarter-turn clockwise as he faces it, its nib going from the top round to the side, till he feels it stop at the cross; then down the right face, each wish-window's collar; then the flag window's.*]

Achilles: [*his fingers on the flag knob's collar*] Or leave this one upright, as the Terms describe it?

Tortoise: That would gamble the only wish.

[*He turns the flag knob's collar to the cross, and then turns the flag knob up, to lit. Every wish-knob he leaves down. What has changed: every collar, now at the cross, and the flag knob, now up. What has held: the wish-knobs, dark; the plain windows; the seal; the lid.*]

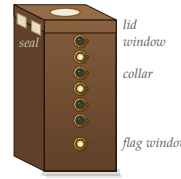
Achilles: Is that one wish?

Genie: Every collar crosswise, every knob as set: one passage. One wish.

Achilles: Then I wish it.

Genie: Granted.

[*The casket as it stands: every collar at the cross, the flag knob up, the wish-knobs down, every window a plain pane, the seal whole. At the word, nothing on it moves: no knob, no collar, not the lid. Behind the second wish-window a small gleam comes up and stays. Behind the fourth, another. Behind the flag window, one more. Behind the first, the third, the fifth and the sixth wish-windows the panes stay plain. Achilles touches the flag knob, and it will not turn: every knob is locked where he set it. The seal on the lid has parted of itself. What has changed: the gleams, the lock, the seal. What has held: every setting, every collar, and the lid, still down.*]



the gleams come
up, the seal parted

Achilles: The second and the fourth, and the flag. Not my marks at all. The casket's own.

[*Achilles's eye goes from the casket to the window. Low on the pane, where the frost lies thickest, a pair of initials stands out white that nobody could have made out on the bare glass: scratched into it long ago with a diamond, and picked out by the frost, the whole of them at once.*]

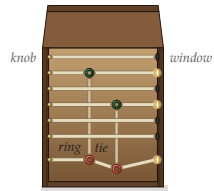
Tortoise: Your slip was only the rehearsal. The casket had its own secret all along.

Genie: The casket is spent. A secret read is no longer a secret, and the lid may be lifted.

[*The casket as it stands: spent, its knobs locked, the gleams behind the second and fourth wish-windows and the flag window, the seal parted. Achilles lifts the lid on its hinge. The knobs do not move, nor the collars, nor the gleams; only the lid swings back, and the candle shines down into the casket.*]

[*Inside, from the top of the column down. The cord from the first knob runs straight across to the first window, knotted at the knob and knotted at the window, with nothing on it. The cord from the second knob runs in only as far as a green ring and is knotted there; from that ring a cord runs on across to the second window, knotted at the ring and at the window; and from the same ring a tie drops away down the casket, knotted to the ring above and, below, to the upper red ring, low in the casket near the flag knob, passing behind every cord beneath it without touching one. The cord from the third knob runs straight across to the third window. The fourth knob is like the second: its cord runs to a green ring and is knotted there; a cord runs from that ring to the fourth window; and a tie runs down from the*

ring, behind the cords below it, to the lower red ring, below the upper and further across. The cord from the fifth knob runs straight across to the fifth window, and the cord from the sixth knob straight across to the sixth. From the flag knob a cord runs to the upper red ring and is knotted there; another cord, knotted at both its ends, runs from the upper red ring to the lower; and from the lower red ring a last cord runs to the flag window. Every length is knotted at both its ends, and no cord passes through a ring. What has changed is only that the inside can be seen. What has held is everything the windows say, and every gleam can now be followed back, length by length, to the knob it came from.]



inside, from the top down

[*A smell comes up out of the open casket, of old wood and something like cloves, shut in since long before the house was the Tortoise's. Achilles and the Tortoise lean in over it together; the Genie, standing a little back, does not trouble to look.*]

Achilles: Green rings at the second and the fourth knobs, red rings low down, where the flag's lengths meet, and the ties coming down from the marks. It's your web as you first laid it, not turned at all. So the turning was never inside.

Tortoise: It was all in the collars.

Genie: The wish has been spent in accordance with the Terms. The Office does not collect spent boxes.

[*The lamp lies where Achilles set it down, beside the candlestick, cold, as it has been since he rubbed it. The Genie thins from the feet upward into the pale thread he came as; the thread leans, runs down into the lamp's spout, and is gone. The lamp does not stir or warm. The casket stays open on the table, its windows still showing what they showed, and nothing else in the room has moved. There are two at the*

table again.]

Tortoise: [*taking up the lamp and setting it back on its shelf*] Back where it lives.

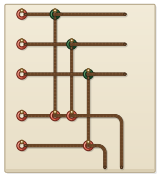
[*Up on its shelf, out of the candle's reach, the lamp is only brass again, one dull shape beside the clean patch in the dust where the casket stood. The thin third shadow has gone from the wall by the shelf; there are two again, and the Tortoise's chair clacks as she sits back down in it.*]

Achilles: [*looking into the open casket*] Every box of marks, read in one wish. Is there any secret a wish can't read like that?

Fourth Challenge — *The Pairs Box*

Tortoise: Some. Here is one. [*She clears the web from the cloth, the rings into her little box of rings and the string into her work-bag.*] This is a pairs box, drawn. Its settings come in pairs that answer alike, and what it hides is where the two of a pair differ; in place of a flag it has a pair of answer-cords.

[*Across the cloth's far side from Achilles she lays a short row of wish-cords, each beginning at a red end-ring at rest, and below them, nearer him and set a little apart, where the flag cord would lie, a pair of answer-cords, each beginning at a red end-ring at rest. On the cords for the first two places she threads a green ring each, the second's a little further along than the first's, and ties both straight down to the upper answer-cord, each tie to a red ring of its own; on the third place's cord a green ring further along still, tied straight down to a red ring on the lower answer-cord. Each tie she leaves a little slack. Past its last red ring, each answer-cord turns down towards Achilles and runs to the cloth's near edge, on his side, the lower turning down a little before the upper, so that neither crosses the other.*]



laid out, nothing folded yet

Tortoise: Which settings answer alike?

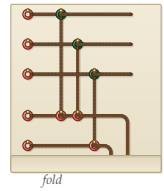
Achilles: [*touching the ties as he goes*] Light the first place and the upper answer turns over; light the second instead and it turns over just the same. Light both and it turns twice, and comes back as if I'd lit neither. The third place turns the lower answer, and nothing else touches that. So two settings answer alike exactly when they differ at the first two places together, and nowhere else.

[*Achilles sits back and tucks his hands under his arms; his breath still shows in the candlelight, and the cold has got into his fingers again. The Tortoise's chair clacks as she leans in over the cloth.*]

Tortoise: Then the secret is the first two places. You can see the ties; a wisher couldn't. A crosswise wish on this box would light its wish-windows in some pattern, and we'll call the pattern a clue.

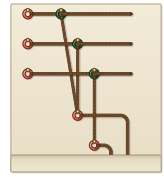
Tortoise: Nobody reads the answer-cords: a wisher who wants a clue leaves the answer windows alone. So I'll fold them away. Whatever is under the fold is never looked at.

[*The two answer-cords run from their end-rings, through their red rings, and down to the cloth's near edge, where their far ends lie loose. The Tortoise reaches across, takes the cloth's near edge on Achilles's side, and folds it up and back over them, a hand's breadth deep. The loose ends are gone under the fold; everything above it lies where it lay. Nothing has been cut, and nothing under the fold will be read: the answer-cords now run on under the folded edge and stop where nobody looks.*]



the near edge
folded over

[*On the upper answer-cord, the dark end-ring, the red ring for the first place's tie and the red ring for the second's all lie on one cord, all red. She slides them together: the end-ring and the first place's red ring along the cord into the second place's red ring, the first place's tie coming with its ring, and the stretch of answer-cord they ran along swallowed with them; they run into one red ring there, its bead at rest. The green rings on the first two places' cords are tied now, both of them, to that one red ring, and its own cord runs on down under the fold. On the lower answer-cord, the end-ring runs along into the red ring in the same way, the cord it ran along swallowed with it: one red ring, tied to the third place's green ring, its cord running on under the fold. No ring is worked past the fold. What has changed: each answer-cord's start and its red rings have gone into one ring. What has held: the ties, the wish-cords, and the fold.*]



run together
above the fold

Tortoise: A red ring counts the ties that come into it, odd or even, as it always has. With its own cord under the fold, all it can do is hold every clue to a check: a clue lights the secret's places an even number of times, and here that means the first two places both lit or neither. The third place's ring has one tie and nobody reading it, so the third place is free.

Achilles: Then let me rub the lamp and fetch a real clue.

Tortoise: The Genie has gone, and so has our wish. We'll suppose the clues instead. Suppose one wish had lit only the third window, and another the first two.

Achilles: The first lit the third window alone, so the third can't be a secret place; they would share one lit place, and one is odd. The second lit the first two, so the secret holds both of them or neither. Neither is no secret at all. Both, then: the first two places, just as the ties said.

[*Up the lane the church clock gives a single stroke. It gives the same one stroke at every half hour, and says the half, and not of which hour; but anyone in the house who heard it strike eight needs no telling.*]

Tortoise: And a wish that lit nothing at all?

Achilles: Would pass every check there is, and tell me nothing.

Tortoise: Two clues, then: two wishes, had the Genie granted them.

Achilles: What does one wish buy?

Tortoise: One answer, and only the one.

Achilles: Then the rest must be bought one at a time.

[*They have both leaned in over the cloth, and on the slope of the ceiling their two shadows, which came up the stairs apart and have leaned and crossed all evening, have run together over the table into one, and hold there. Every shadow in the room stands a little higher on the walls than it did when the candle was first set down.*]

Achilles: [*looking from the cloth to the open casket*] So that's the evening. A plain wish buys one answer. A crosswise wish on a box of marks buys the whole secret, read straight off its ties. On a pairs box it buys one clue: a line through the secret, not the secret. And no turning of the collars buys what the box doesn't hold.

Tortoise: And some questions are no quicker for any wish, however it's set; that was proved by people who counted wishes, not by drawing.

Achilles: [*looking at the candle*] The candle's lower than I thought.

Tortoise: It always burns faster up here. The cold makes it gutter.

Achilles: [*going to the window*] And the frost is thicker on this pane than on yours downstairs. It's come right over the middle.

Tortoise: Nobody breathes on it up here.

Achilles: [*coming back to the table, and looking into the open casket*] Its windows are still showing. The second, the fourth, and the flag. Will they keep?

Tortoise: Till the house falls down, I expect. It's spent; it has nothing left to do but show them.

Achilles: [*unwinding his scarf and putting it round the Tortoise's shoulders*] You'll catch cold up here.

Tortoise: I never catch anything. [*She settles the scarf.*] Thank you.

Achilles: Shall we go down?

Tortoise: In a little while. [*She draws her chair nearer the candle.*]

Achilles: If someone hands you a box and won't say what is inside it, how much can one question tell you?

Tortoise: [*taking up the candlestick*] That depends on the box, and on how the question is set. Mind the loose tread going down; I still don't remember which it is.

[*She does not get up at once. The candle stands low in its stick between her two claws, guttered down to a small flame, and Achilles leans on the table beside her with his hands cupped to it. Every shadow in the room has climbed as the candle burned down, and they all stand tall and still now: the chair-backs high on the walls, the two of theirs together on the slope of the ceiling. The casket stands open on the table where Achilles set it. On the shelf by the window the lamp is only brass, out of the light. On the wall facing the window, past the Tortoise's shoulder, the print hangs half in her shadow again at the edge of the light, a print on a cold wall. Then she stands, and the candle goes up with her, and every shadow in the room swings round and shortens and follows it out of the door. The stair takes them down its one turn; halfway, a tread gives under Achilles's foot, as it did coming up, and neither of them says which. Behind them the top room is dark and still: the lamp on its shelf, the casket open on the table, the print on the wall that nobody can see now,*

and the frost going on with its work over the whole of the pane. Up the stairwell, into the dark they have left, climbs a little of the parlour's warmth.]

Chapter 13: Asking the Oracle

— Minuet for One Wish —

Part IV: Algorithms

Escher: "Belvedere" — a pillared lookout whose upper storey is set crosswise on the lower; a ladder stands inside it and leans against the outside

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If someone hands you a sealed box and will not say what is inside it, how much can one question tell you?

Everyone has played the game. A friend thinks of something, and you may ask questions that are answered yes or no. Each answer is one bit of news, and a careful player learns to make every bit count: a question that splits the possibilities in half is worth asking, and a question whose answer you could have guessed is wasted. A secret of six yes-or-no facts needs six good questions, and no cleverness in the wording gets it done in five. That feels less like a rule of the game than a fact about the world. A question is a thing you put in, an answer is a thing you get back, and one answer can carry only so much.

Now change the game in one respect. The box is not a friend but a device, and it will do exactly one thing, as often as you like: take in a question and hand back an answer, with no opinion about what you are doing. You pay for each use. What you would like to know is not the answer to any single question but something about the box as a whole: whether it treats every question alike, say, or which hidden setting inside it decides its answers. Asked the ordinary way, the device gives one answer per use, and the counting of the last paragraph applies in full.

Chapter 5 found a crack in that counting. A single use of a quantum box, asked in a superposition and read after one more turn of the wires, told whether a function gave the same answer at both of its inputs, a fact that no single ordinary question can settle. This chapter follows the crack as far as it goes. We will meet a box whose whole secret, all its hidden bits at once, comes out of one use. We will draw that box and find that it is nothing but a web of simple gates, and that the secret is written in the web's wiring for anyone who looks at the picture the right way round. Then we will meet a box that one use cannot empty, and see exactly what one use does buy from it.

We will also be careful about what we have not shown. A picture can prove that a certain question returns a certain answer. Whether some other method, classical or quantum, could have done as well with fewer questions is a different kind of claim, a claim about all possible methods at once, and no picture in this chapter proves one. Where such a claim is needed we prove it by counting on the blackboard, or we state it and say whose it is.

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§13.1 One Wish, One Answer

Chapter 12 ended with a promise: "In Chapter 13 someone else hands us a box and will not say what is inside it. We may ask it a question, once. The box turns out to be a web of CNOTs, and yellow boxes turn its colours over. Its secret can be read straight off the picture, if we ask in the right way." Before any of that we have to say what a question is, and what it costs.

On the blackboard. The box computes a function f that takes a string of n bits to one bit, $f : \{0,1\}^n \to \{0,1\}$. We cannot look inside it. What we can do is Chapter 5's oracle: the reversible box U_f that acts on n query wires and one answer wire by

$$U_f \, |x\rangle \, |y\rangle \;=\; |x\rangle \, |y \oplus f(x)\rangle \,,$$

where x is an n-bit string, y is a bit, and $\oplus$ is addition mod 2. In this chapter

we call the n query wires the **rods** and the answer wire the **flag**. Ket order is top wire first, so the rods come first and the flag last. One use of U_f is one **query**. This is the operation Cleve, Ekert, Macchiavello and Mosca call the f-controlled-NOT, their Eq. (2.1) for one bit and Eq. (3.3) for n bits. Bennett, Bernstein, Brassard and Vazirani (we will call them BBBV) write the same thing for a quantum Turing machine: a query changes the contents of a query tape from $|x \circ b\rangle$ to $|x \circ (b \oplus A(x))\rangle$, where $\circ$ joins two strings and A is their name for the box.

A query costs one unit, and it costs one unit whatever is on the rods. BBBV describe an oracle as "a special subroutine call whose invocation only costs unit time", and they allow the query tape to hold an arbitrary superposition of strings when the call is made. So a query on a superposition of all 2^n strings is still one query. This is the only resource the chapter counts.

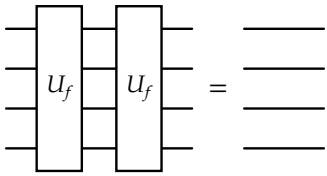
Here is a small box to make it concrete. Take $n = 2$ and let f return the first bit, $f(x_1x_2) = x_1$. Then $U_f|10\rangle|0\rangle = |10\rangle|1\rangle$: the first rod is 1, so the flag is flipped. Ask again with the same rods, and $U_f|10\rangle|1\rangle = |10\rangle|0\rangle$: the flag is flipped back. With the rods at 01 nothing happens either time. BBBV make the same observation in general: if the answer bit starts at 0 it ends holding $A(x)$, and "if the target bit is already in state $|A(x)\rangle$, calling the oracle will reset it to $|0\rangle$". They say this ability to uncompute "will often prove essential" for interference.

> **Result 13.1 (a query undoes itself).** For every $f : \{0,1\}^n \to \{0,1\}$, U_f permutes the 2^{n+1} basis states, so it is unitary, and $U_f^2 = I$.

Proof. Apply U_f twice to a basis state: $|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle \mapsto |x\rangle|y \oplus f(x) \oplus f(x)\rangle = |x\rangle|y\rangle$, since $f(x) \oplus f(x) = 0$. So $U_f^2 = I$ on a basis, hence everywhere. A map that is its own inverse on the basis states is a bijection of them, so the matrix of U_f has exactly one 1 in each row and each column and 0 elsewhere. Its columns are distinct basis vectors, which are orthonormal, so U_f is unitary. ■ BBBV note the same fact for their oracles: the transformation that evaluates a classical Boolean function "is an involution". See `Code/Ch13/PlainWishes.py`.

In the sketchbook. The tall box U_f of Chapter 5 (§5.3, row 37) is laid across

the rods and the flag, drawn wide so that it is never mistaken for a wire. Drawn here for three rods, Result 13.1 says that two tall boxes in a row are bare wires:



exactly. The picture says nothing more, because the tall box hides everything inside it. That is the point of an oracle, and it is about to change.

Why this definition. Why count uses of the box, and not time, or bits of output? Because the box is sealed. Whatever happens inside it is not ours to count, and the one thing we control is how often we consult it. And why the form $y \oplus f(x)$, rather than simply writing $f(x)$ on the flag? Because writing over y would destroy it, and a process that destroys information cannot be unitary. Adding mod 2 keeps the flag's old value recoverable, which is exactly Result 13.1.

A box of marks

A single definite query, $|x\rangle\,|y\rangle$ with x a fixed string, returns one bit of news: the flag comes back as $y \oplus f(x)$, and the rods come back unchanged. Which boxes have secrets worth reading? The simplest interesting ones are the **linear** functions. For a string $a \in \{0,1\}^n$ and a bit b, let

$$f(x) \;=\; (a \cdot x) \oplus b, \qquad a \cdot x \;=\; a_1x_1 \oplus a_2x_2 \oplus \cdots \oplus a_nx_n.$$

The product $a \cdot x$ is the dot product **taken mod** 2: multiply the bits place by place and add the results mod 2. This is Cleve et al.'s Eq. (3.7), with their dot product (3.2). The string a is the box's **secret**, and b is a constant bit. Call rod i **marked** if $a_i = 1$, and write w for the number of marks. So $f(x)$ is the parity of x on the marked rods, flipped once more if $b = 1$.

Take $n = 3$, $a = 101$, $b = 0$. Then $f(x) = x_1 \oplus x_3$, and $U_f|x_1x_2x_3\rangle\,|y\rangle = |x_1x_2x_3\rangle\,|y \oplus x_1 \oplus x_3\rangle$. Read that as two operations: flip the flag if rod 1 is 1, then flip it if rod 3 is 1. Each is a CNOT (Chapter 4) with its control on a marked rod and its target on the flag. Rod 2 is not touched at all.

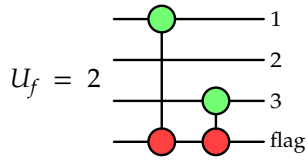
Result 13.2 (the web lemma). If $f(x) = (a \cdot x) \oplus b$, then

$$U_f \;=\; X_{\text{flag}}^{b} \prod_{i:\,a_i=1} \text{CNOT}_{i\to\text{flag}},$$

where $\text{CNOT}_{i\to\text{flag}}$ flips the flag when rod i is 1 and X_{flag} flips the flag. The factors commute, so their order does not matter. In pictures, U_f is $2^{w/2}$ times a **web of ties**: a green dot on each marked rod, tied to its own red dot on the flag, the unmarked rods straight, and a red π dot on the flag if $b = 1$. Two webs in a row are 2^{-w} times bare wires.

Proof. On a basis state, each $\text{CNOT}_{i\to\text{flag}}$ adds x_i to the flag, so together they add $\bigoplus_{a_i=1} x_i = a \cdot x$, and X_{flag}^{b} adds b. The flag ends as $y \oplus (a \cdot x) \oplus b = y \oplus f(x)$ and the rods are untouched, which is $U_f|x\rangle\,|y\rangle$. Two linear maps that agree on a basis are equal. Each of these operations only adds something to the flag, and additions mod 2 commute. For the picture: a bare green–red pair is $\text{CNOT}/\sqrt{2}$ (row 64), and a red π dot with one leg in and one out is X exactly (its matrix, §8.1), so the web is $(\frac{1}{\sqrt{2}})^{w}\,U_f$. ■ See `Code/Ch13/PlainWishes.py`.

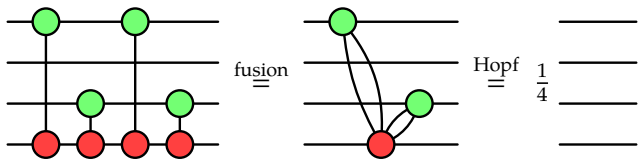
In the sketchbook. Here is the web of $a = 101$, $b = 0$: rods 1, 2, 3 from the top, then the flag.



The tie from rod 1 runs down past rods 2 and 3 without touching them: a crossing neither joins nor breaks a line. The factor $2 = 2^{w/2}$ with $w = 2$ is drawn because, as in Chapters 9 to 12, every picture equation in this chapter

is exact, with its scalar written in.

Now ask twice. Put two webs in a row. On each marked rod two green dots share a wire, and fusion (Theorem 8.1, row 57) makes them one. On the flag all the red dots share the flag's wire, and fusion makes them one red dot. Each marked rod's green dot is now tied to that red dot by two separate ties, and the Hopf law (Theorem 8.6, row 62) removes a double tie at the price of $\frac{1}{2}$. What is left on every wire is a phase-free dot with one leg in and one out, which is a plain wire (§8.1):



With $U_f = 2\times\text{web}$, the left side is $\frac{1}{4}U_f^2$, so the picture has proved $U_f^2 = I$ again, for this box. For $b = 1$ each web carries a red π dot on the flag. They fuse into the flag's red dot as well, and their phases add to 2π, which is no turn at all. See `Code/Ch13/PlainWishes.py`, which checks the web against the matrix U_f and derives the bare wires.

Why this definition. Why draw a CNOT for every mark, when one tall box would do? Because the tall box hides the secret and the web displays it. In the web, the marks *are* the ties. Every rule of Chapter 8 acts on dots and ties, so once the box is drawn as a web the rules can reach its secret. Most boxes are not webs. A web always computes some $(a \cdot x) \oplus b$, and there are only 2^{n+1} of those among the 2^{2^n} functions on n bits: for $n = 2$, eight of the sixteen. For every other box the tall box is all we have.

How many plain questions?

With definite queries, how many are needed to learn a? Suppose first that b is known; say $b = 0$. Asking at the unit string e_i, which has a single 1 in place i, returns $a \cdot e_i = a_i$. So n queries read the whole secret, one rod at a time. Can a cleverer choice do it in fewer? Take $n = 3$ and try the two queries 110 and 011. They return $a_1 \oplus a_2$ and $a_2 \oplus a_3$. Now compare two

secrets that differ by $v = 111$, for instance $a = 100$ and $a \oplus v = 011$. The first gives $100 \cdot 110 = 1$ and $100 \cdot 011 = 0$; the second gives $011 \cdot 110 = 1$ and $011 \cdot 011 = 0$, again 1 and 0. The two secrets answer both questions alike, because $v \cdot 110 = 0$ and $v \cdot 011 = 0$. Two questions cannot tell them apart.

> **Result 13.3 (the classical baseline).** Use only definite queries, chosen one after another in any way that may depend on the earlier answers, and suppose the method must name a with certainty. (i) If b is known, every such method uses at least n queries, and n suffice. (ii) If b is unknown, every such method uses at least $n + 1$ queries, and $n + 1$ suffice.

Proof. (i) Run the method against the secret a, and let $q_1, \ldots, q_k$ be the query strings it uses, with $k < n$. The k equations $v \cdot q_j = 0$ in the n unknown bits of v have a solution $v \neq 0$ (fewer equations than unknowns; see the Dirac's Blackboard below). Now run the same method against $a \oplus v$. Its first query is q_1, since nothing has been answered yet, and the answer is $(a \oplus v) \cdot q_1 \oplus b = (a \cdot q_1) \oplus b$, the same as before. So its second query is q_2, with the same answer, and so on to the end. The method sees exactly the same answers for a and for $a \oplus v \neq a$, so it names the same secret for both and is wrong for one of them. The unit strings show that n suffice. (ii) Now suppose $k \leq n$. The k equations $v \cdot q_j \oplus c = 0$ in the $n + 1$ unknowns (v, c) have a nonzero solution. If v were 0, then c would be 1 and the equations would read $1 = 0$; so $v \neq 0$. The secrets (a, b) and $(a \oplus v, b \oplus c)$ give the same answer to every query, and the argument of (i) finishes. Asking first at $x = 0^n$, the all-zero string, returns b, and then the n unit strings return every $a_i \oplus b$: $n + 1$ suffice. ■

A method that tosses coins but must still be certain is certain on every run, and each run is a fixed method of the kind above, so the same bounds hold for it. Cleve et al. give the reason for (i) in one sentence: a "contains n bits of information and each classical evaluation of f yields a single bit". `Code/Ch13/PlainWishes.py` checks both parts exhaustively for $n = 4$. There is no picture on this page. No rewrite proves a lower bound, and this one is proved by counting.

Dirac's Blackboard.

Fewer equations than unknowns, mod 2. Why must $k < n$ equations $v \cdot q_j = 0$ have a solution $v \neq 0$? The same way as over the real numbers. Take the queries 110 and 011: the equations are $v_1 \oplus v_2 = 0$ and $v_2 \oplus v_3 = 0$. The first says $v_1 = v_2$, the second $v_3 = v_2$, and v_2 is free. Setting it to 1 gives $v = 111$. In general, use each equation in turn to express one unknown through the others, adding it mod 2 to the later equations to remove that unknown from them; an equation that has become $0 = 0$ is dropped. Each equation removes at most one unknown. With $k < n$ equations at least one unknown is never removed, and we may set it to 1 and read the others off. Addition mod 2 is its own subtraction, so nothing in the elimination needs anything but $\oplus$.

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§13.2 The Mark on the Rods

A definite query moves a mark onto the flag: the flag is flipped if $f(x) = 1$. Chapter 5 showed the other way round. With the flag held in the difference state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, the flag does not change, and the sign of $f(x)$ lands on the rods instead:

$$U_f\,|x\rangle\,|-\rangle \;=\; (-1)^{f(x)}\;|x\rangle\,|-\rangle$$

(row 37; Cleve et al.'s Eq. (2.2), which they write without the normalization). For a box of marks there is more to see, because the box is a web, and each tie can be watched on its own.

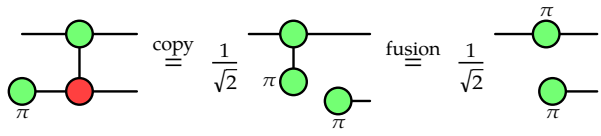
On the blackboard. Start with the trap. Put the flag in the other crosswise state, $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and one tie: a CNOT from a rod in any state $|\psi\rangle$ to the flag. A CNOT applies X to the flag when the rod is 1, and $X|+\rangle = |+\rangle$. So CNOT $(|\psi\rangle\,|+\rangle) = |\psi\rangle\,|+\rangle$, and nothing at all has happened. Now the flag

at $|-\rangle$. Here $X|-\rangle = -|-\rangle$, so on a basis state CNOT $|x\rangle|-\rangle = (-1)^x|x\rangle|-\rangle$, and by linearity

$$\text{CNOT}\,(|\psi\rangle|-\rangle) \;=\; (Z|\psi\rangle)\,|-\rangle\,,$$

where $Z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ is the phase flip. The flag is untouched, and the tie has left a Z on the rod.

In the sketchbook. Recall the points of Chapters 11 and 12: a red point is $\sqrt{2}\,|0\rangle$ and a red π point is $\sqrt{2}\,|1\rangle$; a green point is $|0\rangle + |1\rangle = \sqrt{2}\,|+\rangle$ and a green π point is $|0\rangle - |1\rangle = \sqrt{2}\,|-\rangle$. So the flag's $|-\rangle$ is $\frac{1}{\sqrt{2}}$ times a green π point. Feed it into the red end of a tie. A phase-free red dot copies green points, by the copy rule (Theorem 8.3, row 59), with a factor $\frac{1}{\sqrt{2}}$. The green π point goes in and comes out twice: once on the flag, which leaves the tie as it came, and once up the tie. Up the tie it meets the green control, and fusion (Theorem 8.1) adds its π to the control's phase. A green dot with phase π, one leg in and one out, is Z (its matrix, §8.1):



With a phase-free green point in place of the green π point, the same two steps copy a phase-free point up the tie, fusion adds nothing, and the control is left as a green dot with no phase and two legs: a plain wire. Nothing comes back.

> **Result 13.4 (kickback as a copy).** Exactly: (i) a green π point fed into the red end of a tie is copied onto the rod as a green π dot, a Z, and the flag keeps its green π point, with the factor $\frac{1}{\sqrt{2}}$ drawn above; on the blackboard, CNOT $(|\psi\rangle|-\rangle) = (Z|\psi\rangle)\,|-\rangle$ for every rod state $|\psi\rangle$; (ii) a phase-free green point is copied as nothing; on the blackboard, CNOT $(|\psi\rangle|+\rangle) = |\psi\rangle|+\rangle$; and for every f, marked or not, $(H^{\otimes n} \otimes I)\,U_f\,(H^{\otimes n} \otimes I)\,|0\rangle^{\otimes n}|+\rangle = |0\rangle^{\otimes n}|+\rangle$.

Proof. The pictures above, with copy and fusion; and the two blackboard lines before them. The scalar checks: the left side is $\frac{1}{\sqrt{2}}$CNOT applied to a rod and $\sqrt{2}\,|-\rangle$, which is $Z \otimes |-\rangle$; the right side is $\frac{1}{\sqrt{2}}\, Z \otimes \sqrt{2}\,|-\rangle$, the same. For the last claim of (ii), $U_f |x\rangle\,|+\rangle = |x\rangle\,|+\rangle$ for every x, since $X|+\rangle = |+\rangle$, so U_f acts on $|\psi\rangle\,|+\rangle$ as the identity for every f, and the two layers of Hadamards cancel. ■ See `Code/Ch13/KickedMark.py`, which tests (i) on a random complex rod state and the kickback of row 37 on a random non-linear f.

The picture shows why kickback is a copy. The red dot copies the flag's green point, one copy stays on the flag and one goes up the tie, and a green point is exactly the kind of point that a green control absorbs by fusion. The flag gives nothing away; it is copied. And the only green points that survive the copy with a trace are those with phase π.

Why the difference state. Chapter 5 answered this on the blackboard: $|-\rangle$ is the eigenvector of X with eigenvalue -1, and the -1 is what lands on the rod (row 37). The picture gives a second answer. The flag's point must be green to be copied at all, since a red dot copies only green points; and of the two green points, the phase-free one fuses into the control as nothing. Only the green π point leaves a mark.

The one-question test, read off the web

Chapter 5 proved Deutsch and Jozsa's test for any width (§5.5, row 38). Put n rods at $|0\rangle$, a Hadamard on each before and after the box, the flag at $|-\rangle$, and one query. The amplitude of finding every rod at 0 is

$$\frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)},$$

which is ± 1 when f is constant and 0 when f is balanced (zero on exactly half the strings). We do not prove that again. What is new is what the number is for a box of marks. It needs one sum, which the next section needs as well. For a statement that holds exactly when two strings are equal we use the bracket of Chapter 12: $[\, c = 0 \,]$ is the number that is 1 if c is the all-zero string and 0 otherwise.

A small case first. For $n = 2$ and $c = 10$, the four signs $(-1)^{c\cdot x}$ at $x = 00, 01, 10, 11$ are $+1, +1, -1, -1$, and they cancel. For $c = 00$ they are all $+1$, and the sum is 4.

Result 13.5 (the sign sum). For every $c \in \{0,1\}^n$,

$$\sum_{x\in\{0,1\}^n} (-1)^{c\cdot x} \;=\; 2^n\,[\,c = 0\,].$$

Proof. Since $c \cdot x$ is a sum mod 2 and $(-1)^{u\oplus v} = (-1)^u(-1)^v$, the sign is a product, $(-1)^{c\cdot x} = \prod_i (-1)^{c_i x_i}$, with one factor for each bit of x. A sum over all strings of a product of one-bit factors is the product of the one-bit sums:

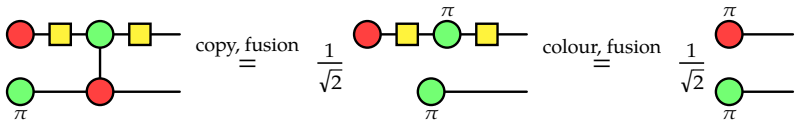
$$\sum_x (-1)^{c\cdot x} = \prod_{i=1}^{n} \sum_{x_i\in\{0,1\}} (-1)^{c_i x_i} = \prod_{i=1}^{n} (1 + (-1)^{c_i}).$$

Each factor is 2 if $c_i = 0$ and 0 if $c_i = 1$. So the product is 2^n if $c = 0$ and 0 otherwise. ■ See `Code/Ch13/KickedMark.py` and `Code/Ch13/TurnedWeb.py`.

Result 13.6 (Deutsch and Jozsa, read off the web). If $f(x) = (a \cdot x) \oplus b$, the amplitude of row 38 is $(-1)^b\,[\,a = 0\,]$. So a box of marks is constant exactly when it has no marks, and balanced otherwise; the one-query test reads "no ties" or "some tie".

Proof. $2^{-n}\sum_x (-1)^{(a\cdot x)\oplus b} = (-1)^b\,2^{-n}\sum_x (-1)^{a\cdot x} = (-1)^b[\,a = 0\,]$ by Result 13.5. If $a = 0$, then $f = b$ is constant. If $a \neq 0$ the sum $\sum_x (-1)^{a\cdot x}$ is 0, so f takes each value on exactly half the strings. ■ See `Code/Ch13/KickedMark.py`.

In the sketchbook. The test needs no sum when the box is a web. Take one marked rod: a red point, a yellow box, the green control, a yellow box, then the output; and the flag as a green π point into the tie's red end. Kickback as a copy (Result 13.4) puts a green π on the rod between its two yellow boxes. A green dot with a yellow box on each leg is a red dot of the same phase, by colour change (Theorem 8.2, row 58), and the red π fuses with the red point before it:



A red π point is $\sqrt{2}\,|1\rangle$: the marked rod comes back reading 1. An unmarked rod has no tie, its two yellow boxes meet and cancel ($H^2 = I$), and it comes back as its red point, reading 0. So every rod reads 0 exactly when the web has no ties. That is Result 13.6, seen rod by rod. The next section does the same for the whole web at once.

Two honest remarks bound this test. First, it reads "no ties" only for a box of marks. Cleve et al. are explicit that "a balanced function need not be of this form", and a general balanced f has no web; for it, only Chapter 5's sum of signs decides, and there the blackboard is the only strand that works. Second, the test beats the classical worst case, $2^{n-1} + 1$ evaluations, only because it is certain. BBBV, describing this line of results, say at once that "these computational problems are in BPP relative to the same oracle": a classical machine that tosses coins, and may err with small probability, solves them quickly. Such a machine can query k random strings and answer "constant" if all k answers agree; when f is balanced, each answer is a fair coin and the k agree with probability 2^{1-k}.

Feynman's Notebook.

From one bit to many. Cleve and his co-authors trace the line. David Deutsch proposed the first problem of this kind in 1985, for a function on one bit. Deutsch and Jozsa generalised it in 1992, in "Rapid solution of problems by quantum computation" (*Proc. R. Soc. Lond. A* 439, 553–558), to a function on n bits promised to be constant or balanced. Any classical algorithm, Cleve et al. note, needs $2^{n-1} + 1$ evaluations in the worst case to answer with certainty. The network of this chapter, with a single f-controlled-NOT, is theirs: they call it "a slight improvement" of Deutsch and Jozsa's original algorithm, "which involves two

f-controlled-NOT operations". Chapter 5 told the same story for one bit. BBBV place Deutsch and Jozsa at the start of the field's evidence that quantum machines can be stronger: their work "showed how to exploit some inherently quantum mechanical features". Then comes the honest qualification about BPP, quoted above.

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§13.3 One Wish, Asked Crosswise

The test of §13.2 asked only whether the web has ties. It did so by bringing a mark back onto every marked rod, and then it asked only whether any rod was marked. Why not read every rod?

Try it on $a = 101$, $n = 3$, $b = 0$. Put each rod at $|+\rangle = H|0\rangle$ and the flag at $|-\rangle$. By kickback, the query multiplies each string $|x\rangle$ by $(-1)^{f(x)} = (-1)^{x_1 \oplus x_3} = (-1)^{x_1}(-1)^{x_3}$, and a sign that is a product of one-rod signs gives a state that is a product of one-rod states:

$$U_f\ |+\rangle|+\rangle|+\rangle|-\rangle \;=\; \frac{|0\rangle - |1\rangle}{\sqrt{2}}\ \frac{|0\rangle + |1\rangle}{\sqrt{2}}\ \frac{|0\rangle - |1\rangle}{\sqrt{2}}\ |-\rangle \;=\; |-\rangle|+\rangle|-\rangle|-\rangle\,.$$

Now a Hadamard on each rod turns $|-\rangle$ into $|1\rangle$ and $|+\rangle$ into $|0\rangle$: the rods read 101. One query has returned the whole secret, where Result 13.3 says that plain queries need three. The rest of this section says why it always works, and what the picture sees.

The turned tie

First a small surprise about a single CNOT. Put a Hadamard on both of its wires, before and after. Feed in $|01\rangle$: the Hadamards make $|+\rangle|-\rangle$; the CNOT, by kickback, makes $|-\rangle|-\rangle$; the Hadamards make $|11\rangle$. The bottom bit was 1, and the *top* bit has been flipped. Feed in $|10\rangle$: $|-\rangle|+\rangle$, then nothing

(since $X|+\rangle = |+\rangle$), then $|10\rangle$. The top bit was 1, and nothing happened to the bottom. The control has moved to the bottom wire.

Result 13.7 (the turned web). Exactly,

$$(H \otimes H)\,\mathrm{CNOT}_{1\to 2}\,(H \otimes H) \;=\; \mathrm{CNOT}_{2\to 1}.$$

Hence, for $f(x) = (a \cdot x) \oplus b$,

$$H^{\otimes(n+1)}\,U_f\,H^{\otimes(n+1)} \;=\; Z_{\text{flag}}^{\,b} \prod_{i:\,a_i=1} \mathrm{CNOT}_{\text{flag}\to i}\,:$$

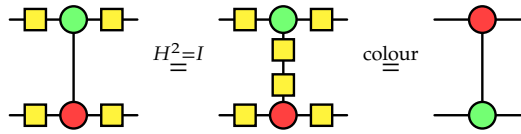
the flag controls, and every marked rod is a target. In pictures: yellow boxes on every outer leg of the web turn every dot's colour, and the ties now run *from* the flag *to* the marked rods, with no scalar.

Proof. Write $\mathrm{CNOT}_{1\to 2} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$. Conjugating by $H \otimes H$ gives $|+\rangle\langle +| \otimes I + |-\rangle\langle -| \otimes Z$, since $H|0\rangle = |+\rangle$, $H|1\rangle = |-\rangle$ and $HXH = Z$. With $|\pm\rangle\langle\pm| = \frac{1}{2}(I \pm X)$ this is

$$\begin{aligned} \tfrac{1}{2}(I+X) \otimes I + \tfrac{1}{2}(I-X) \otimes Z \;&=\; I \otimes \tfrac{1}{2}(I+Z) + X \otimes \tfrac{1}{2}(I-Z) \\ &=\; I \otimes |0\rangle\langle 0| + X \otimes |1\rangle\langle 1|\,, \end{aligned}$$

which is $\mathrm{CNOT}_{2\to 1}$. For the web, $H^{\otimes(n+1)}$ is its own inverse, so conjugating the product of Result 13.2 conjugates each factor. Each $\mathrm{CNOT}_{i\to\text{flag}}$ becomes $\mathrm{CNOT}_{\text{flag}\to i}$ (the Hadamards on the other wires cancel in pairs), and X_{flag} becomes $HXH = Z_{\text{flag}}$. ■ See `Code/Ch13/TurnedWeb.py`.

In the sketchbook. Take one tie with a yellow box on each of its four outer legs. The green control has boxes on two of its three legs, and the red target likewise; the tie itself is bare. Put a cancelling pair of yellow boxes on the tie, which changes nothing since $H^2 = I$ (Chapter 3; row 56). Now each dot wears a yellow box on every leg it holds, so colour change (Theorem 8.2) turns the green dot red and the red dot green, and the boxes come off. The pair on the tie is split, one box absorbed at each end:



A red dot on top tied to a green dot below is a CNOT with its control on the *bottom* wire (row 64, read upside down), and both bare pairs carry the same $\frac{1}{\sqrt{2}}$, so the equation holds with no scalar. For a whole web, the same two moves run on every tie at once: one cancelling pair on each tie, one on each stretch of the flag between two of its red dots, and then colour change on every dot. That is Result 13.7 in pictures, **the turned web**.

Now the headline. Its statement is the same network as Chapter 5's test, with nothing changed but the question we ask of the output.

> **Result 13.8 (Bernstein–Vazirani, in Cleve et al.'s one-query form).** Let $a \in \{0,1\}^n$, $b \in \{0,1\}$ and $f(x) = (a \cdot x) \oplus b$, and let $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Then, for every n, a and b, exactly,
>
> $$(H^{\otimes n} \otimes I)\, U_f\, (H^{\otimes n} \otimes I)\ |0\rangle^{\otimes n} |-\rangle \ = \ (-1)^b\ |a\rangle\, |-\rangle\,.$$
>
> One query, and reading the rods returns a with certainty.

The attribution needs care, because two different things are credited here. The *problem* is Ethan Bernstein and Umesh Vazirani's, from 1993: given a box that computes $(a \cdot x) \oplus b$, find a. They also gave the first quantum algorithm for it, which, as Cleve et al. record, "employs two f-controlled-NOT operations instead of one". The one-query network stated above is Cleve et al.'s presentation, the same network as their one-query form of Deutsch and Jozsa's test; their Figure 4 is drawn once for both. Cleve et al. also point out why the two problems share a network: a function $(a \cdot x) \oplus b$ "is constant if $a = 00\cdots0$ and balanced otherwise", so every Bernstein–Vazirani box satisfies Deutsch and Jozsa's promise, though not every balanced function is of this form.

Proof idea. On the blackboard, kickback writes the sign $(-1)^{a\cdot x}$ on every string, the sign factors into one sign per rod, and the second layer of

Hadamards reads each rod's sign as a bit. In the sketchbook, the Hadamards turn the web round (Result 13.7), so the ties run from the flag to the marked rods, and the flag's 1 is copied up every tie. Both proofs are given in full in §13.6. See `Code/Ch13/TurnedWeb.py`, which checks the equation, sign included, for every a and both b at $n = 4$, and for random secrets at $n = 6$.

Why this definition. Why must the flag be crosswise too? Because with the flag left at $|0\rangle$ the query entangles the rods with it: the state after the query is $2^{-n/2}\sum_x |x\rangle\,|(a\cdot x)\oplus b\rangle$. Take $b = 0$, split the state by the flag's value, and apply the second layer to the rods; write z for an n-bit string read from the rods. Using Result 13.5 on $\sum_{x:\,a\cdot x=0}(-1)^{x\cdot z} = \frac{1}{2}\sum_x(1+(-1)^{a\cdot x})(-1)^{x\cdot z}$, one finds, for $a \neq 0$,

$$\tfrac{1}{2}\big(|0^n\rangle + |a\rangle\big)\,|0\rangle + \tfrac{1}{2}\big(|0^n\rangle - |a\rangle\big)\,|1\rangle\,.$$

The rods read 0^n or a, each with probability $\frac{1}{2}$, and the flag is a fair coin independent of them. With the flag at $|1\rangle$, or with $b = 1$, the two branches change places. So a query with the flag upright gambles the one question. Only the difference state keeps the flag out of the answer.

In the dialogue, just after the Tortoise's tubes have turned every ring on her cloth web, Achilles looks up at *Belvedere*: a pillared lookout whose upper storey is set crosswise on the lower; a ladder stands inside it and leans against the outside.

Dirac's Blackboard.

Why the second layer returns a. After the query the rods hold $2^{-n/2}(-1)^b\sum_x(-1)^{a\cdot x}|x\rangle$. The second layer sends $|x\rangle$ to $2^{-n/2}\sum_z(-1)^{x\cdot z}|z\rangle$, so the amplitude of $|z\rangle$ is

$$(-1)^b\,2^{-n}\sum_x(-1)^{x\cdot(a\oplus z)} = (-1)^b\prod_{i=1}^{n}\tfrac{1}{2}(1+(-1)^{a_i\oplus z_i}),$$

the sum factored bit by bit as in Result 13.5. Each factor is 1 when $z_i = a_i$ and 0 when $z_i \neq a_i$. Take $n = 2$, $a = 10$. For $z = 10$ both factors are 1. For $z = 00$ the first factor is $\frac{1}{2}(1-1) = 0$; for $z = 11$ the second is; for $z = 01$ both are. The second layer never

compares whole strings. It asks each rod separately whether its sign was $+$ or $-$, and 2^n roads collapse to n one-rod questions.

Coecke's Sketchbook.

The Hadamards turn the web. Colour change is usually applied to one dot. Here it is applied to a whole web at once, and what makes that possible is a single bookkeeping move: a cancelling pair of yellow boxes on every tie and on every stretch of the flag between its red dots. After that, every dot is dressed in yellow on every leg, every dot changes colour together, and every tie reverses its direction. The picture never forms the sum over 2^n strings that the blackboard needs. It works with the ties, and there are only w of them. This is not a trick that works on any oracle. It works because this oracle *is* a web: the box is a handful of CNOTs, and the secret is which rods they touch. A general f is a tall box with nothing inside to turn, and for it the blackboard's sum is the only proof we have.

Feynman's Notebook.

An interferometer with many arms. Ethan Bernstein and Umesh Vazirani posed the problem of this section in "Quantum complexity theory", in the proceedings of the 25th ACM Symposium on Theory of Computing, 1993, pages 11–20; a journal version appeared in the *SIAM Journal on Computing* in 1997. Cleve et al. describe their network as an experiment. In a Mach–Zehnder interferometer a photon meets a half-silvered mirror, travels two paths through phase shifters, and meets a second mirror before the detectors. Cleve et al. identify each half-silvered mirror with a Hadamard, and remark that the second mirror "effectively erases all information about the path taken", which is what lets the paths interfere. Their summary of the known algorithms is a three-part pattern: "a Fourier transform, followed by an f-controlled-U, followed by another Fourier transform", and they insist that the last transform "is essential". This chapter's Hadamard layers are the simplest Fourier transform, and the

query sits between them.

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§13.4 The Pairs Box

The Bernstein–Vazirani box has its whole secret in its wiring, and a crosswise query reads the wiring whole. Here is a box with a different kind of promise, whose secret no single query has been shown to return.

On the blackboard. A **pairs box** computes a function $f : \{0,1\}^n \to \{0,1\}^m$ with m output bits, with this promise: there is a secret string $s \neq 0$ such that

$$f(x) = f(x') \quad \text{exactly when} \quad x' \in \{x,\ x \oplus s\}.$$

So the inputs come in pairs $\{x, x \oplus s\}$, each pair shares an answer, and different pairs have different answers. The task is to find s. This is Simon's problem. The oracle is $U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$ as before, where now y is an m-bit string on m **answer wires** that take the flag's place, and $\oplus$ acts bit by bit.

Here is the smallest pairs box we will draw. Take $n = 3$, $m = 2$, and let $f(x) = Mx$ for the matrix

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad f(x_1x_2x_3) = (x_1 \oplus x_2,\ x_3),$$

with the arithmetic mod 2. Then $f(000) = f(110) = 00$, $f(001) = f(111) = 01$, $f(010) = f(100) = 10$ and $f(011) = f(101) = 11$. The pairs differ by $s = 110$. This box happens to be linear, so by Result 13.2 it is a web: rods 1 and 2 are tied to the first answer wire and rod 3 to the second. Linearity is not part of Simon's promise, though at this size it comes free: with three rods and two answer wires every box with the promise is a web, perhaps with some answer bits flipped. With more answer wires there are boxes with the promise that are not webs. What follows about one round holds for every f with the promise, web or not.

One **round** of Simon's method is the network we have used all chapter, with

the answer wires starting at $|0\rangle$: rods at $|0\rangle$, a Hadamard on each rod, one query, a Hadamard on each rod, and then the rods are read. The answer wires are never looked at: they are **discarded**, in the sense of Chapter 9. The string read from the rods we call the **clue**, z.

> **Result 13.9 (one round of Simon's problem).** For every f with the promise, one round leaves the rods in the state
>
> $$\rho \;=\; 2^{-n}(I + Z^s) \;=\; 2^{1-n}P_s,$$
>
> where $Z^s = Z^{s_1} \otimes \cdots \otimes Z^{s_n}$ applies Z to the rods where s has a 1, and $P_s = \frac{1}{2}(I + Z^s)$. The clue z appears with probability $2^{1-n}\,[\,z \cdot s = 0\,]$: it is equally likely to be any of the 2^{n-1} strings with $z \cdot s = 0$, and never anything else.

Proof. After the first layer and the query the state is $|\Psi\rangle = 2^{-n/2}\sum_x |x\rangle|f(x)\rangle$. The second layer acts only on the rods, and discarding acts only on the answer wires, so we may discard first. Discarding is the partial trace (Result 9.2):

$$\mathrm{tr}_{\mathrm{ans}}\,|\Psi\rangle\langle\Psi| = 2^{-n}\sum_{x,x'}\langle f(x')|f(x)\rangle\;|x\rangle\langle x'| = 2^{-n}\sum_{x,x'}[\,f(x) = f(x')\,]\;|x\rangle\langle x'|.$$

By the promise, $f(x) = f(x')$ exactly when $x' = x$ or $x' = x \oplus s$, so this is $2^{-n}\sum_x(|x\rangle\langle x| + |x\rangle\langle x \oplus s|) = 2^{-n}(I + X^s)$, where X^s flips the rods where s has a 1 (it sends $|x \oplus s\rangle$ to $|x\rangle$). The second layer gives $\rho = 2^{-n}(I + H^{\otimes n}X^sH^{\otimes n}) = 2^{-n}(I + Z^s)$, since $HXH = Z$ and $HIH = I$ on each rod. Finally $Z^s|z\rangle = (-1)^{s\cdot z}|z\rangle$, so the diagonal entry at z is $2^{-n}(1 + (-1)^{s\cdot z}) = 2^{1-n}[\,z \cdot s = 0\,]$. ■ See `Code/Ch13/PairsBox.py`, which checks the pairs box and a six-rod instance with $s = 100010$.

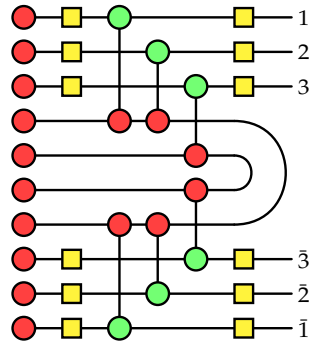
For the pairs box, the clue is one of 000, 001, 110, 111, each with probability $\frac{1}{4}$. One clue does not name s. Each clue is one parity check that the secret passes, $z \cdot s = 0$: a line through the secret, not the point. How many are needed? Suppose two rounds give the clues 001 and 110. The first says $s_3 = 0$; the second says $s_1 \oplus s_2 = 0$. Since $s \neq 0$, the only candidate left is $s = 110$.

> **Result 13.10 (clues fix the secret).** If $z_1, \dots, z_{n-1}$ are clues that are linearly independent mod 2 (no nonempty set of them adds up to 0), then s is the only string $v \neq 0$ with $v \cdot z_j = 0$ for all j. On average, at most $2(n-1)$ rounds are needed to collect $n-1$ independent clues.

Proof. Every clue satisfies $z_j \cdot s = 0$, so s is a solution. The map $v \mapsto (v \cdot z_1, \dots, v \cdot z_{n-1})$ has $n-1$ independent rows, so its image has 2^{n-1} elements, and each image value is taken by $2^n / 2^{n-1} = 2$ strings v. In particular exactly two strings solve all the equations, 0 and s. For the rounds: suppose $k < n-1$ independent clues are in hand. The strings they add up to, 2^k of them, all satisfy $z \cdot s = 0$, and a new clue is equally likely to be any of the 2^{n-1} such strings, so it falls among the old ones with probability $2^k / 2^{n-1} \leq \frac{1}{2}$. A new independent clue therefore takes at most 2 rounds on average, and $n-1$ of them at most $2(n-1)$. ■ See `Code/Ch13/PairsBox.py`, which checks every independent pair of clues for the pairs box.

The all-zero clue always falls among the old ones: it tells nothing.

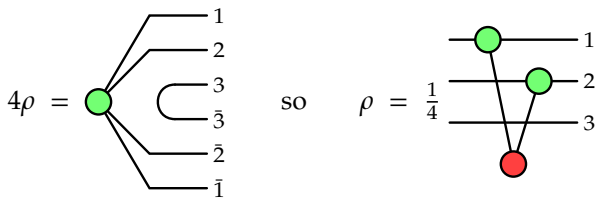
In the sketchbook. Draw the round doubled, as Chapter 9 taught: each wire as two lines, the wire and its mirror copy, with the mirror copy of each gate on the mirror lines (row 66). The mirror conjugates, but every gate and point here is real, so the mirror copy of the circuit is the same circuit, drawn below it with its rows in reverse order. To discard an answer wire, join it to its own mirror copy by a cap (row 67); as always the cap is unnormalized, cap $= \langle 00| + \langle 11|$. The answer wires start at red points, $\sqrt{2}\,|0\rangle$, like the rods. Rods 1, 2, 3 are at the top and their mirrors $\bar{1}$, $\bar{2}$, $\bar{3}$ at the bottom; the four answer lines in the middle are joined in pairs by the two nested caps:



Read as a doubled state on the six open legs, this picture is 4ρ: the ten points carry 2^5 and the six bare ties 2^{-3}. Four rewrites finish it, and none of them costs a scalar.

1. **Colour change** (Theorem 8.2). On each rod, a red point followed by a yellow box is a green point: $H\sqrt{2}\,|0\rangle = |0\rangle + |1\rangle$.
2. **Fusion** (Theorem 8.1). That green point fuses into the rod's green control. What remains is a phase-free green dot with two legs, the tie and the rest of the rod, which is a plain wire (§8.1). Each tie now runs straight through to its rod's last yellow box.
3. **Fusion**, across the caps. On each answer line the red point fuses into the red targets. A cap is a plain wire, so the red dots of an answer wire and of its mirror copy fuse through it into one red dot. The first answer wire and its mirror give one red dot tied to rods 1, 2, $\bar{1}$, $\bar{2}$; the second gives one tied to rods 3 and $\bar{3}$.
4. **Colour change.** Each of those red dots has a yellow box on every leg, the rods' last boxes, so each turns green, and the boxes come off.

What is left is a green dot with legs on $1, 2, \bar{1}, \bar{2}$, and a green dot with legs on $3, \bar{3}$, which is a cup:



Every step was exact. The last equality reads the doubled picture as a matrix, in Chapter 9's way: a doubled state on the legs q and $\bar{q}$ lists the entries $\rho_{z,z'}$ of ρ, with z on the wires and z' on the mirrors. The cup on $3, \bar{3}$ is the identity on rod 3 (row 69, without its $\frac{1}{2}$). The green dot on $1, 2, \bar{1}, \bar{2}$ is $|0000\rangle + |1111\rangle$, whose only entries are at $(00, \bar{00})$ and $(11, \bar{11})$: the matrix $|00\rangle\langle 00| + |11\rangle\langle 11|$ on rods 1 and 2. Drawn as a process with legs in and out, that matrix is a green dot on rod 1 and a green dot on rod 2, tied to one phase-free red dot. The red dot has two legs, so it is a plain wire (§8.1), and the two green dots fuse through it into one; the drawing keeps the red dot because of what it does in general.

Call that drawing the **parity gadget** of s: a phase-free green dot on each rod where s has a 1, each tied to one phase-free red dot with $|s|$ legs, where $|s|$ is the number of 1s in s, and the other rods bare. The red dot reads a parity, as it did in Chapter 8's Octet and on Chapter 12's records (row 96).

> **Result 13.11 (the parity gadget).** For every $s \neq 0$, exactly,
>
> $$P_s \;=\; 2^{|s|/2-1} \times (\text{the parity gadget of } s).$$
>
> So one round leaves $\rho = 2^{1-n}P_s = 2^{|s|/2-n}\times$ gadget. For the pairs box, $|s| = 2$ and $n = 3$: P_s is the gadget itself and $\rho = \frac{1}{4}\times$ gadget, as drawn.

Proof. A phase-free green dot with legs in, out and tie sends $|z\rangle$ to $|z\rangle\,|z\rangle$ (Chapter 11's copier, row 81), so the green dots put a copy of each z_i with $s_i = 1$ on its tie. A phase-free red dot with k legs, all of them ties coming in, is the effect $\langle +|^{\otimes k} + \langle -|^{\otimes k}$ (row 54). On a string t of k bits it gives $2^{-k/2}(1 + (-1)^{t_1 \oplus \cdots \oplus t_k})$, which is $2^{1-k/2}$ if the parity of t is even and 0 if it is odd. With $k = |s|$ the ties carry the bits of z at the places of s, whose parity is $s \cdot z$. So the gadget sends $|z\rangle$ to $2^{1-|s|/2}[\, s \cdot z = 0\,]\,|z\rangle = 2^{1-|s|/2}P_s\,|z\rangle$. ■

See `Code/Ch13/PairsBox.py`, which also checks a secret of weight three, where the factor is not 1.

The scalar matters. For $|s| \neq 2$ the gadget is not P_s, and ρ is not 2^{1-n} times the gadget.

Why this definition. Why discard the answer wires, rather than read them?

Reading them would tell us f of some random string, which says nothing about s that one plain query would not. And nothing we do to the answer wires after the query can change the statistics of the rods: a process on one part followed by discarding it is the same as discarding it, for every trace-preserving process (causality, Result 9.4). So the discard is the honest summary of what the rods alone can say. In the picture it is also what makes the round simple: the caps are what let the answer wires' red dots fuse with their mirrors.

How much better is this than asking plainly? Bennett, Bernstein, Brassard and Vazirani report the answer in their introduction, and we state it without proof.

> **Result 13.12 (Simon; stated, not proved).** Simon proved "the existence of an oracle relative to which BQP cannot even be simulated by probabilistic machines allowed to run for $2^{n/2}$ steps" (BBBV §1). Here BQP is the class of problems a quantum machine solves in polynomial time with error probability at most $\frac{1}{3}$, and a probabilistic machine is a classical one that may toss coins.

BBBV add that Simon's paper "also introduced an important new technique which was one of the ingredients" in Shor's factoring algorithm. That is where Chapter 14 begins. The separation is a statement about all classical machines, and neither the pictures nor the blackboard of this chapter prove it. What we have proved is what one round does: it buys one clue, uniformly chosen among the parity checks the secret passes.

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§13.5 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 12's ninety-six. There is no new picture element in them. Every row is a new use of something Chapters 5, 8 and 9 already drew: the tall box (97), the green–red CNOT (98, 99), the copy rule (100, 101), and the doubled picture with its caps (102).

#	Algebra (Dirac)	Picture (Coecke)
97	a **query** is one use of $U_f\|x\rangle\|y\rangle = \|x\rangle\|y \oplus f(x)\rangle$; U_f is unitary and $U_f^2 = I$; a superposed query is one query	Chapter 5's tall box U_f; two in a row are bare wires
98	linear $f = (a \cdot x) \oplus b$: $U_f = X^b_{\text{flag}} \prod_{a_i=1} \text{CNOT}_{i\to\text{flag}}$	**the oracle is a web of ties**: a green dot on each marked rod tied to its own red dot on the flag; bare, each tie is $\text{CNOT}/\sqrt{2}$ (row 64); asked twice, Hopf leaves bare wires
99	$(H \otimes H)\,\text{CNOT}_{1\to2}\,(H \otimes H) = \text{CNOT}_{2\to1}$; $H^{\otimes(n+1)} U_f H^{\otimes(n+1)}$ = fan-out from the flag onto the marked rods ($b = 0$)	**the turned web** (drawn for $b = 0$; if $b = 1$ the flag's red π comes through the turn as a green π and only signs the result): yellow boxes on every leg turn every dot's colour; the ties now run from the flag to the rods (exact, no scalar)
100	kickback, $\text{CNOT}\,(\|\psi\rangle\|-\rangle) = (Z\|\psi\rangle)\|-\rangle$ (row 37 in spiders)	**kickback as a copy**: a green π point into a red target is copied onto the control's wire as a green π; a phase-free green point copies as nothing
101	**Bernstein–Vazirani**: $(H^{\otimes n} \otimes I)\,U_f\,(H^{\otimes n} \otimes I)\,\|0\rangle^{\otimes n}\|-\rangle = (-1)^b\|a\rangle\|-\rangle$	**the secret is read off the wiring**: after the turn, the flag's 1 is copied up every tie, and exactly the marked rods end at 1 (the master, *The Turned Web*)

#	Algebra (Dirac)	Picture (Coecke)
102	one Simon round, answer wires discarded: $\rho = 2^{-n}(I + Z^s) = 2^{1-n}P_s$; clues z with $z \cdot s = 0$, each equally likely; $P_s = 2^{\lvert s\rvert/2-1}\times$ gadget	the discarded answer wires are caps (row 67); the round is $2^{\lvert s\rvert/2-n}\times$ **the parity gadget** (green dots on the s-rods, one red dot with $\lvert s\rvert$ legs); for the pairs box $\rho = \frac{1}{4}\times$ gadget: a line through the secret

Every row is exact, scalar included, as in Chapters 9 to 12. Nothing in these rows is a statement about how many queries a task needs: rows 97 to 102 are equalities of processes, and the bounds of this chapter are proved by counting (Result 13.3) or cited (Results 13.12 and 13.14).

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§13.6 The Theorem, Twice

Result 13.8 (Bernstein–Vazirani, in Cleve et al.'s one-query form), restated. For $f(x) = (a \cdot x) \oplus b$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, for every n, a and b, exactly, $(H^{\otimes n} \otimes I)\, U_f\, (H^{\otimes n} \otimes I)\, |0\rangle^{\otimes n} |-\rangle = (-1)^b |a\rangle |-\rangle$.

Dirac's Blackboard

These are Cleve et al.'s lines, their Eqs. (3.4), (3.5), (3.8) and (3.9), with the normalization they omit written back in.

Step 1: the first layer. One Hadamard on each rod sends $|0\rangle$ to $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and the product over n rods is the equal sum of all strings:

$$(H^{\otimes n} \otimes I)\ |0\rangle^{\otimes n} |-\rangle \ = \ 2^{-n/2} \sum_{x \in \{0,1\}^n} |x\rangle\, |-\rangle\,.$$

Step 2: the query. By kickback (row 37), $U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$, one road at a time:

$$\begin{aligned} U_f\, 2^{-n/2}\sum_x |x\rangle|-\rangle &= 2^{-n/2}\sum_x (-1)^{(a\cdot x)\oplus b}|x\rangle|-\rangle \\ &= (-1)^b\, 2^{-n/2}\sum_x (-1)^{a\cdot x}|x\rangle|-\rangle\,. \end{aligned}$$

Step 3: the second layer. On one rod, $H|x_i\rangle = \frac{1}{\sqrt{2}}\sum_{z_i}(-1)^{x_i z_i}|z_i\rangle$; on all of them, $H^{\otimes n}|x\rangle = 2^{-n/2}\sum_z (-1)^{x\cdot z}|z\rangle$. So the state becomes

$$\begin{aligned} &(-1)^b\, 2^{-n}\sum_z \Big(\sum_x (-1)^{a\cdot x}(-1)^{x\cdot z}\Big)|z\rangle|-\rangle \\ &\quad = (-1)^b\, 2^{-n}\sum_z \Big(\sum_x (-1)^{x\cdot(a\oplus z)}\Big)|z\rangle|-\rangle\,, \end{aligned}$$

since $(a\cdot x)\oplus(x\cdot z) = \bigoplus_i x_i(a_i\oplus z_i) = x\cdot(a\oplus z)$.

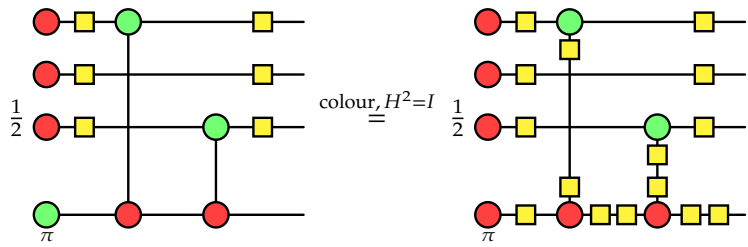
Step 4: the sum. By Result 13.5 with $c = a\oplus z$, the inner sum is $2^n[\,a\oplus z = 0\,] = 2^n[\,z = a\,]$. Only $z = a$ survives, with coefficient $(-1)^b\,2^{-n}\,2^n = (-1)^b$:

$$(H^{\otimes n}\otimes I)\,U_f\,(H^{\otimes n}\otimes I)\;|0\rangle^{\otimes n}|-\rangle \;=\; (-1)^b\;|a\rangle|-\rangle\,. \qquad \blacksquare$$

Coecke's Sketchbook

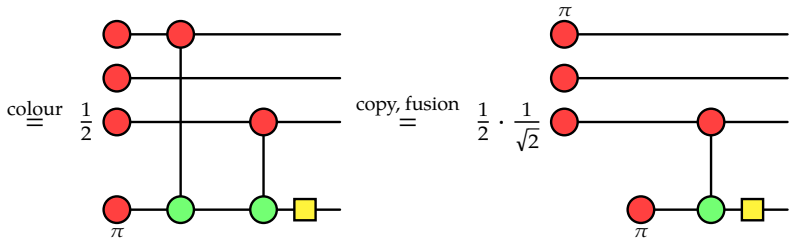
Draw the smallest web that has everything in it: three rods with $a = 101$, so rods 1 and 3 are marked, and rod 1's tie passes rod 2 on its way to the flag; take $b = 0$. Each rod is a red point, a yellow box, the control if it has one, a yellow box. The flag is $|-\rangle$, which is $\frac{1}{\sqrt{2}}$ times a green π point, followed by the two red targets. The picture carries $2^{3/2}$ from the rods' points, $\sqrt{2}$ from the flag's point, and $\frac{1}{2}$ from the two bare ties, 2 in all, so the left side of Result 13.8 is $\frac{1}{2}$ times it.

Step 1: dress every dot. Colour change (Theorem 8.2) read backwards on a single point: a green π point is a red π point with a yellow box on its leg, since $H\sqrt{2}|1\rangle = \sqrt{2}|-\rangle$. Then insert a cancelling pair of yellow boxes ($H^2 = I$; row 56) on each tie, on the flag between its two red dots, and on the flag's output. Now every dot has a yellow box on every leg it holds:

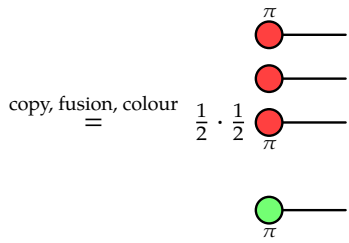


Step 2: turn the web. Colour change (Theorem 8.2) on every dot at once: the two controls turn red, the two targets turn green, and every box on a dot's leg is absorbed. The pairs on the ties and on the flag's middle are split, one box absorbed at each end. On rod 2, which has no dot, its two boxes meet and cancel ($H^2 = I$). One box is left, the outer half of the pair on the flag's output. The ties now run from the flag up to the marked rods (Result 13.7).

Step 3: copy and fuse, first tie. The flag's red π point enters the first green dot, and the copy rule (Theorem 8.3) sends a red π point up the tie and another on along the flag, at the price of $\frac{1}{\sqrt{2}}$. Up the tie, it fuses (Theorem 8.1) with rod 1's red dot and with that rod's red point: rod 1 is left holding a single red π point.



Step 4: copy and fuse, second tie; colour change on the flag. The same at the second green dot: a red π point climbs to rod 3 and fuses there, another goes on along the flag, and a second $\frac{1}{\sqrt{2}}$ is paid. On the flag a red π point meets the last yellow box, and colour change makes it a green π point:



$$= \tfrac{1}{4}\,(\sqrt{2}\,|1\rangle)(\sqrt{2}\,|0\rangle)(\sqrt{2}\,|1\rangle)(\sqrt{2}\,|-\rangle) \;=\; |101\rangle\,|-\rangle\,.$$

Nothing in the chain depended on how many rods there were or which of them were marked. Step 1 dresses each dot separately, Step 2 turns each tie separately, and Steps 3 and 4 climb one tie at a time, paying $\frac{1}{\sqrt{2}}$ for each. So for any n and a, with w marks, the same moves leave a red π point on every marked rod, a red point on every unmarked rod and a green π point on the flag. The scalars agree: the left side of Result 13.8 is $2^{(w-n-1)/2}$ times the picture, the copies contribute $2^{-w/2}$, and the $n+1$ points are worth $2^{(n+1)/2}$ times $|a\rangle\,|-\rangle$, a product of 1. For $b = 1$ the web carries a red π dot on the flag. Dressed and turned with the rest, it becomes a green π dot, a Z, and when the flag's red π point passes it the result is $Z\sqrt{2}\,|1\rangle = -\sqrt{2}\,|1\rangle$: the sign $(-1)^b$, and nothing else. ■ See `Code/Ch13/TurnedWeb.py`, which builds this form for random secrets at $n = 6$ and reduces it by the rules.

There is a second route, in Cleve et al.'s own shape: leave the web unturned, copy the green π point through each red target onto its control (Result 13.4), and let the rods' last yellow boxes turn each green π between its two boxes into a red π, as in §13.2. It reaches the same points by the same rules in a different order.

The verdict

On length it is close, and the picture wins narrowly. The blackboard's four steps are short, but the last one rests on the sign sum, a sum over 2^n strings that needed its own proof. The sketchbook needs four pictures and five named moves, and forms no sum at all. The blackboard also has a shortcut

that is easy to miss: the sign $(-1)^{a \cdot x}$ is a product of one-rod signs, so the state after the query is a product of $|+\rangle$s and $|-\rangle$s, as in the example of §13.3. That makes the algebra almost as quick as the picture, provided one notices it.

On showing, the picture wins outright. In the sketchbook the secret is not computed at all: it is the wiring. The marked rods are the ones with ties, the Hadamards turn the ties round, and the flag's 1 climbs exactly those ties. The blackboard proves that the amplitude of every wrong string is zero, and it does not show *where* the right one came from. The picture's advantage has a price, and the price is honesty about its reach. It works because this oracle is a web. For Deutsch and Jozsa's general balanced function there are no ties to turn, and only the blackboard's sum decides. Neither proof says anything about how many queries a classical method needs; both prove one equality of processes.

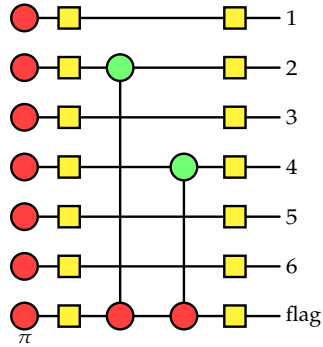
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§13.7 The Turned Web

Here is the chapter's master diagram, one instance of the headline drawn whole. There are six rods and a flag, and the secret is $a = 010100$, with $b = 0$: rods 2 and 4 are marked. It is drawn in the symmetric form, with the flag entering at $|1\rangle$ and a yellow box on the flag on both sides, like every rod. That form agrees with Cleve et al.'s, since $H|1\rangle = |-\rangle$ and $H|-\rangle = |1\rangle$, and it gives every wire the same dress: a point, a yellow box, the web, a yellow box.

From left to right: a red point on each rod ($|0\rangle$) and a red π point on the flag ($|1\rangle$); a yellow box on every wire; the web, a green dot on rods 2 and 4, each tied to its own red dot on the flag; a yellow box on every wire; open outputs. The two red dots on the flag are drawn unfused, one for each tie, which is what a gate-by-gate translation gives: one green–red pair per CNOT (row 64). Fusing them is a legal rewrite, but it gives a different picture of the same process, and this is the picture we draw. The ties run down past the rods between without touching them. The picture is called **The Turned**

Web.



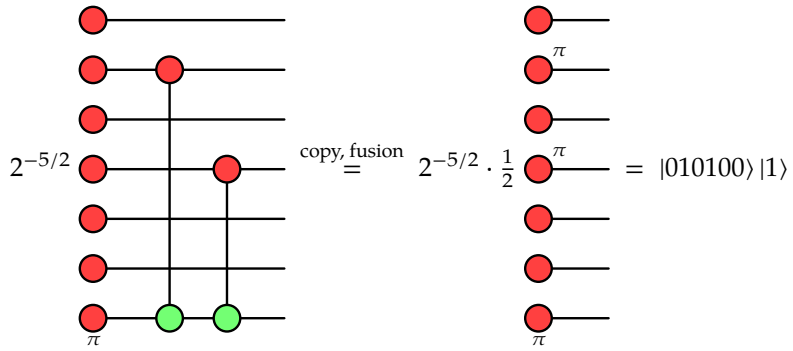
Result 13.13 (The Turned Web). Exactly,

$$2^{-5/2} \times \text{The Turned Web} \;=\; |010100\rangle\,|1\rangle\,,$$

that is, $H^{\otimes 7}\,U_f\,H^{\otimes 7}\;|000000\rangle\,|1\rangle = |010100\rangle\,|1\rangle$ for $f(x) = 010100\cdot x$. Exactly the marked rods read 1, and the flag comes back at 1.

Proof, on the blackboard. By Result 13.7 with $b = 0$, $H^{\otimes 7}U_fH^{\otimes 7} = \mathrm{CNOT}_{\mathrm{flag}\to 2}\,\mathrm{CNOT}_{\mathrm{flag}\to 4}$. On $|000000\rangle\,|1\rangle$ the flag is 1, so both flip their rods: $|010100\rangle\,|1\rangle$. For the scalar, the seven points give $2^{7/2}$ and the two bare ties 2^{-1}, so the picture is $2^{5/2}\;|010100\rangle\,|1\rangle$.

Proof, in the sketchbook. The chain of §13.6, with nothing to prepare on the flag, since it already wears its yellow boxes. Insert a cancelling pair of yellow boxes on each tie and on the flag's middle line, between its two red dots. Colour change turns the controls red and the targets green; on rods 1, 3, 5 and 6 the two boxes cancel. Then the flag's red π point is copied through the first green dot, climbs the tie and fuses into rod 2, while its copy runs on along the flag's middle line into the second green dot; there it is copied again, climbs into rod 4, and runs on to the flag's output, whose box has already been absorbed:



The first picture is the web turned: the ties now run from the flag. The last is one point per wire, red π on exactly rods 2 and 4 and the flag, worth $2^{7/2}\,|010100\rangle\,|1\rangle$, and $2^{-5/2} \cdot \frac{1}{2} \cdot 2^{7/2} = 1$. ■ See `Code/Ch13/TurnedWeb.py`, which builds The Turned Web, evaluates it with the scalar, and reduces it by the rules to the same seven points.

The secret was chosen not to read the same in both directions. Read from the bottom rod up, the outputs would give 001010, the wrong secret; the book's convention, top wire first, gives 010100, and the code checks that it is this one.

Look at the picture once more with the rewrite in mind. Before the turn the ties hang from the marked rods down to the flag, as a box of marks is built: each marked rod adds its bit to the flag. After the turn they rise from the flag to the marked rods, and the flag's 1 is handed up each of them. What was inside the box below is outside it above. The rods that no tie touches come back as they went in, and the rods that a tie touches come back turned, and the secret is read in the return.

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§13.8 What a Wish Cannot Buy

Take stock. A query is one use of a reversible box that undoes itself (Result 13.1). A box that computes $(a \cdot x) \oplus b$ is a web of CNOTs, one tie for each mark (Result 13.2), and plain questions need n of them, or $n + 1$ if b is unknown, to read its secret (Result 13.3). With the flag at $|-\rangle$, each tie copies a mark back onto its rod (Result 13.4), which reads Deutsch and Jozsa's test off the web (Result 13.6). With the whole network crosswise the web turns round, and one query returns the entire secret (Results 13.7 and 13.8, and The Turned Web). A pairs box yields to one round of Simon's method only a single clue, a parity check on the secret (Result 13.9), and on average at most $2(n - 1)$ rounds find it (Result 13.10).

Now the honest part. Every rewrite in this chapter proves that two processes are equal: that this network, with this box, returns this state. The Turned Web proves that one query returns a. It proves nothing about how many queries any *other* method needs. The claim that plain questions need at least n was proved on the blackboard by counting, in Result 13.3, and no picture took part. It is a claim about every classical method at once, and a picture shows one method.

The gap it measures is modest. For Bernstein and Vazirani's problem as posed here, the classical cost is n queries, or $n + 1$, against one: a gap that grows in proportion to n, not exponentially. Bernstein and Vazirani's own separation used a harder, recursive problem, which BBBV call recursive Fourier sampling, and showed, in BBBV's account, that relative to an oracle there are problems in BQP that cannot be solved with small error probability by probabilistic machines restricted to $n^{o(\log n)}$ steps. We state that and do not prove it.

And some questions are no quicker for any quantum method at all. The sharpest statement of this in the chapter's sources is BBBV's.

> **Result 13.14 (Bennett, Bernstein, Brassard and Vazirani, Theorem 3.5; stated, not proved).** "For any $T(n)$ which is $o(2^{n/2})$, relative to a random oracle, with probability 1, BQTime$(T(n))$ does not contain NP."

Here BQTime$(T(n))$ is the class of problems accepted with probability at least $\frac{2}{3}$ by an oracle quantum machine whose running time is bounded by $T(n)$ (BBBV §2); $T(n)$ is $o(2^{n/2})$ when $T(n)/2^{n/2}$ tends to 0; and a random oracle answers every string with a fair coin, fixed once and for all. In words: relative to such a box, no quantum machine running in much less than $2^{n/2}$ steps solves every problem in NP, the problems whose solutions can be checked quickly. BBBV explain what the bound means. The proof, they write, shows that no quantum algorithm "can gain more than this quadratic factor in success probability compared to classical algorithms, when attempting to answer NP-type questions formulated relative to a random oracle". It does not show that NP is out of reach of quantum computers. What it establishes, in their words, is "that there is no black-box approach to solving NP–complete problems by using some uniquely quantum-mechanical features". The bound is tight: they note that Grover's search, the subject of Chapter 15, meets it.

Every bound in this chapter, then, is of one of two kinds. Result 13.3 is proved on the blackboard by counting, and Results 13.12 and 13.14 are stated and cited. None is drawn.

Coecke's Sketchbook.

What a rewrite does not count. It is tempting to look at The Turned Web and say that the picture *shows* why one query suffices where n were needed. It shows something smaller. It shows that this network, with this box, returns a. The number n belongs to a different question: what every classical method must do. A picture is one method, drawn. To show that no method does better you would have to range over all of them, and the rules of Chapter 8 have nothing to say about methods that are not drawn. There is a subtler temptation too. The web is small, w ties and a flag, and one might read its smallness as the reason the quantum method is cheap. But the classical method has the same web to work with and still needs n questions. The size of a picture is not the cost of a computation. The rewrite is an equality, and that is all it is.

Feynman's Notebook.

Four authors and a limit. "Strengths and weaknesses of quantum computing" is by Charles H. Bennett of IBM Research, Ethan Bernstein of Microsoft Corporation, Gilles Brassard of the Université de Montréal and Umesh Vazirani of UC Berkeley. The preprint is dated 12 December 1996 and carries the note "To appear in SIAM Journal on Computing (special issue on quantum computing)". Its introduction is a short history of the evidence that quantum machines are stronger: Deutsch's quantum Turing machine, Deutsch and Jozsa, Bernstein and Vazirani's universal machine and their proof that BPP $\subseteq$ BQP $\subseteq$ PSPACE, then Simon's oracle, and Shor. Simon's paper, "On the power of quantum computation", appears in their references in the proceedings of the 35th IEEE Symposium on Foundations of Computer Science, 1994, pages 116–123; a journal version was published in the *SIAM Journal on Computing* in 1997. Against that list of strengths the paper sets its weakness: the random-oracle bound, and the remark that Grover's algorithm makes it tight.

The chapter's question has an answer now, or rather three. If the box is a web of marks and we ask in the right way, one question tells us everything the box holds. If the box keeps its secret in pairs, one question tells us one parity check the secret passes, and a handful of questions tell us the rest. And if the box is a table filled by coin tosses and the question is a search of the kind NP asks, no quantum method gains more than a square root over the ordinary count; that was proved by people who counted queries, and we have only stated it. Where one question did buy something, its pattern was the same each time: a layer of Hadamards, one query, a layer of Hadamards. Every wire the query did not touch came back as it went in, and every wire it touched came back turned.

Chapter 14 keeps the pattern and changes the layers. Simon's round, with the second layer of Hadamards replaced by a Fourier transform over a larger

group, is where Shor began: BBBV name Simon's technique as one of the ingredients of Shor's result. The box will hide not a string but a period, the query will be a controlled multiplication, and the pictures will again be kickback diagrams, the marks copied back along their ties. What will change is how much of the work the pictures can do. The number theory that turns a period into a factor stays on the blackboard, and we will say so.

Aria with Periodic Variations

寒き夜や
数えぬ鐘の
一めぐり

samuki yo ya
kazoenu kane no
hitomeguri

A cold winter night —
the bell that I did not count
once, all the way round.

[*The Tortoise's parlour, the next evening, after tea. The cups are empty, and Achilles's scarf lies folded on the arm of the Tortoise's chair.*]

[*The stove has been fed after tea and burns bright behind its mica, its iron ticking now and then as it heats. On the lower pane the frost ferns have grown back since the morning and met again at the middle, and past the black glass above them the lane lies pale under the rime. Achilles sits by the fender with his coat still on, his hands round his empty cup for what warmth is left in it; the brass candlestick stands on the mantel above him, its stem twisted like a stick of barley sugar. Up the lane, faint through the glass, the bells of the church have begun their practice: the same short run of notes from the highest bell down to the deepest, over and over, every bell in its place.*]

Achilles: I came up your path on tiptoe. I thought if I made no noise the frost might not notice me.

Tortoise: Did it work?

Achilles: It noticed me at the gate. [*He blows on his fingers.*] The tea has only got as far as my elbows.

Tortoise: [*handing him the folded scarf*] Then here is the rest of the way. You left it round my shoulders last night.

Achilles: So I did. Was it any use?

Tortoise: I slept in it. [*He winds it on.*] I went up to the top room first thing this morning. The casket was still standing open on the table with its windows showing, as pleased with itself as a hen with one egg.

Achilles: You didn't open it again?

Tortoise: There was nothing left in it to open. I shut its lid and carried it down, and it's in the cupboard under the stairs, with the jam.

Achilles: Still showing, in the dark.

Tortoise: Showing to the jam. And when I lifted it off the table, there was something new on the shelf beside the lamp. A tall case of dark wood, sealed, with the Office's label on its side.

Achilles: Delivered overnight? Up your stairs, past the loose tread?

Tortoise: The Office doesn't use stairs.

Achilles: What does the label say?

Tortoise: "For the wisher, top room. With the Office's compliments." And underneath, in smaller letters, "Do not tip."

Achilles: Then I'm the wisher, and it's my case.

Tortoise: It's the Genie's case. You're only its wisher. [*She takes the candlestick from the mantel and lights it at the stove.*] Bring my bag, and mind the tread.

Achilles: [*taking up the work-bag, packed this morning with last night's cloth and string, her little box of rings and the packet of tubes*] Did you ever find out which one it is?

Tortoise: I went up and down three times this morning, and it never said a word. It only groans for guests.

[*They go up, the candle first. Halfway, a tread groans under Achilles's foot.*]

Achilles: Good evening to you too.

[*The stair is narrow and makes its one turn at the landing, and their shadows go up the wall beside them, stretching and shrinking with the candle, as they did last night. Nearer the roof the bells are louder, and somewhere between the parlour and the landing they have broken out of their first order: the same bells, but in a new order every time through, tumbling over one another so that nobody on the stairs could say which will sound next. Each keeps its own note, wherever the order puts it.*]

[*The top room, as it was last night: the small table and two chairs, the one window over the lane, the shelf beside it. On the shelf stands the old brass lamp, and beside it a tall, narrow case of dark wood, about the height of a mantel clock, sealed, with the Office's label on its left side.*]

Tortoise: [*setting the candlestick on the table, and her drawing-cloth beside it*] Sit. Yours is still the chair without the wobble.

[*The top room is as small as it was last night, and as cold, tucked under the slope of the roof, and it still has one of everything. One door, standing open on the dark of the stairs. One window over the lane, frosted over again since the morning from edge to edge; low on the pane the pair of initials that somebody scratched there long ago with a diamond stands out white, as it does on every night of hard frost, whether or not anyone looks. One shelf, beside the window, where the old brass lamp stands tarnished nearly brown, and beside it the new case, dark and narrow, standing in the clean patch the casket left in the dust as though it had been measured for it. Under the shelf the empty butter-crocks stand in their row, and among them the tall glass jar. On the wall facing the window, not above the shelf, hangs a print in a plain dark frame: Escher's* Regular Division of the Plane*, a sheet filled edge to edge with figures fitted against one another, with no gap between them. The candle just reaches it across the table. It was hanging there before the candle came up the stairs, and it will be hanging there when the candle goes down. The Tortoise lowers herself into the chair with the wobble, her back to that wall, and it rocks under her*

with a small clack; her shadow goes up over the lower half of the print. With the candle still at last, every shadow in the room settles into its place, and under the roof the bells go on with their changes; whenever the deepest of them sounds, the tall glass jar under the shelf hums back to it, faint and clear, as it hummed to the clock last night.]

Achilles: [*sitting, and looking up at the shelf*] A clock-case with no face on it. How is anybody meant to tell the time?

Tortoise: The Genie's things don't show you their faces. You get one look, when they've finished, and that's all.

First Challenge — *The Fifteen-Hour Dial*

Achilles: How do you tell how long a thing takes to come round, if you may look at it only once?

Tortoise: You don't count the ticks. You hear how far one tick turns it.

Achilles: I should count them. I'd sit in front of it with a pencil and make a stroke at every tick, till it was home.

Tortoise: Then let's give you something to count, on paper, where counting costs nothing. [*She takes a sheet of paper and a pencil from the work-bag.*] This will be a dial of fifteen hours, with a paper hand on a pin; the top mark is nought, and an hour is found by counting marks round from it. One tick sends the hand from the hour it stands at to seven times that hour, counted round and round the dial. Ticking it is free.

[*She draws a circle and pricks fifteen marks round it, evenly spaced, the first at the very top, on the side of the sheet furthest from Achilles, and writes nothing beside any of them; an hour is counted from it clockwise as Achilles sees it. From the edge of the sheet she cuts a narrow paper hand, pins it through the centre with a brass pin, and turns it to the first mark past the top.*]



the paper dial

Tortoise: It starts at one.

Achilles: One tick, then. Seven ones are seven.

[*The hand points at the first mark past the top: one. Achilles takes its tip and carries it round clockwise, past the second mark and the third and on, counting the marks under his breath, and stops it on the seventh. The circle, its marks and the pin have not moved. The hand is the same slip of paper, the same length, pinned at the same place; only what it points at has changed. One has become seven.*]



after a tick

Achilles: Two ticks. Seven sevens are forty-nine; less three fifteens, four. [*He carries the hand on round, past the top, to the fourth mark.*]

Achilles: Three ticks. Seven fours are twenty-eight; less one fifteen, thirteen. [*Round again, to the thirteenth mark, two short of the top.*]

Achilles: Four ticks. Seven thirteens are ninety-one; less six fifteens, one. Home.

[*The hand goes over the top once more and stops on the first mark past it, where it began. It has pointed at seven, at four and at thirteen on the way; now it points at one, as though it had never left.*]



home again

[*Up the lane, after all their tumbling, the bells fall back into the order they began in, the highest first and the deepest last, and ring it through once and again; then they stop. Achilles lifts his head at the quiet, and blows on his fingers, and looks back down at the dial.*]

Achilles: Four ticks to come round. That was easy. Any dial can be ticked round like this; a bigger one only takes a few more ticks.

Tortoise: The dials the Office uses have more hours than there are grains of frost on that pane. A hand on one of those may take nearly as many ticks as there are hours before it comes home.

Achilles: Then I'd need a great deal of pencil. But somebody must know a quicker way.

Tortoise: Nobody has found a quicker way, with paper and ticking. Nobody has shown there isn't one.

Achilles: And is it worth the pencil? What does anyone want with a dial's ticks home?

Tortoise: Know them, and they'll split the dial's hours into two smaller dials. But that's for later.

Achilles: [*looking at the paper hand, back at one*] It's been round and round, and here it is on the same mark, the same hand, as if it had never left.

Tortoise: The ticking changes where it stands, never what it is.

Achilles: Then how could anyone find the ticks home without ticking them all?

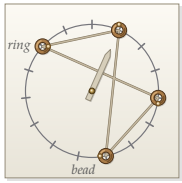
Tortoise: By listening to something the ticking doesn't move, only turns. Can you think of anything on this dial that a tick would turn, and not move?

Second Challenge — *The Round Turned Whole*

Achilles: Nothing. A tick moves the hand; that's all a tick does. Unless you've something in that bag.

Tortoise: I have. [*She brings out a loop of pale twine threaded with small rings of plain wood, each with an ivory bead set in it.*] This is a round: a loop of string with a ring for each hour the hand visits, in the hand's order, and in each ring a bead set at some part of a turn. Laid on the dial, each ring sits at its hour. To tick a round, you carry every ring to the hour the hand would go to next, and its bead rides with it, unchanged.

[*She lays the round on the dial. A bead at rest stands at the top of its ring as Achilles sees it; turned round, or forward, it goes clockwise as he sees it, and turned back, the other way. At one sits a ring whose bead stands at rest; at seven, a ring whose bead is turned a quarter back; at four, a bead at a half; at thirteen, a bead three quarters back. The loop runs straight from each ring to the next in the order the hand went, and from the last back to the first, crossing itself once.*]

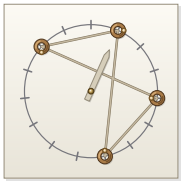


the round as laid

Tortoise: Tick it.

[*The round lies on the dial: at rest at one, a quarter back at seven, a half at four, three quarters back at thirteen. Achilles lifts the ring from one and sets it down at seven. The ring from seven he carries to four, the ring from four to thirteen, and the ring from thirteen over the top to one. He does not touch a bead: each rides in its ring, set exactly as it was. When he takes his hands away, every hour the round covered is covered still, by the same loop in the same order; but at each hour now sits a different ring. At one, where the bead at rest stood, there is a bead three*

quarters back. At seven, where a quarter back stood, a bead at rest. At four, where the half stood, a quarter back. At thirteen, where three quarters back stood, a half.]



the round, ticked

Achilles: Every ring has gone to a new hour, and not one hour is empty. And at each hour [*going round with a finger*] the bead there now stands a quarter on from the bead that stood there before. One, a quarter on. Seven, a quarter on. Four, and thirteen, the same.

Achilles: ...Oh. The round hasn't gone anywhere. Every ring has moved, and the round lies where it lay, only turned, all of it at once, a quarter round.

[*The Tortoise draws the candlestick nearer the dial for him, and it turns a little in her claw as she sets it down. The twist of its brass stem goes round with it, and every coil of the twist seems to climb a step up the stem, though nothing on it has risen.*]

Achilles: Tick it four times, then, and it has turned a whole round, and every bead is back as you set it.

Tortoise: The ticking changes where it stands, never what it is.

Achilles: Then the round's turn is the thing a tick turns and doesn't move. A quarter at every tick: four ticks and it's home. I needn't tick at all; I need only read the quarter.

Tortoise: If you can read it. A bead's turn is not a thing a window shows you. Let me draw how one would listen for it. [*She spreads her drawing-cloth beside the dial.*] My cloth again: what I lay on it is a picture of a thing, not the thing.

[*They both draw in to the cloth, and the Tortoise's chair clacks over as she comes; Achilles pulls his chair in after her, his breath showing in the candlelight. Up the lane the ringers have begun again, a longer piece than the first: the order changes*

every time through, and changes faster, and once two bells strike together, a clash that makes Achilles lift his head, before they part and go on.]

Tortoise: [*setting her little box of rings and the packet of tubes on the table beside the cloth, laying a cord across the part of the cloth nearest Achilles, and threading a red end-ring on its left end as he sees it*] First the round, drawn small: a cord that begins at a red end-ring, which I'll call the wheel. Lit, with its bead at a half, it stands for a round the tick turns by a quarter; dark, with its bead at rest, for a round the tick doesn't turn at all.

Tortoise: [*laying a second cord above it, further from Achilles, and threading a green end-ring at rest on its left end*] And this is a listener: a cord that begins at a green end-ring at rest, which is a knob set crosswise and dark. It keeps whatever turn it hears. To read it, I slide one of last night's yellow tubes onto its ring, which turns the ring red: red at rest reads dark, and red at a half reads lit.

[*On the listener's cord she threads a green ring, and on the wheel cord another, just below it, each from its cord's loose far end. She ties both down to a red ring set to their right, midway between the cords, each by its own short cord, a tie, and from that red ring she hangs a short pendant running on to the right and ending in a small green ring of its own, which hangs free.*]

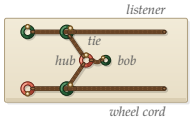
Tortoise: This is my shorthand for "tick the round only if the listener is lit", and I'll call it a controlled turn: the red ring in the middle is its hub, and the little one hanging from it is its bob. For a turn of a quarter, each green ring gets its bead at an eighth, half the turn, and the bob gets an eighth back.

[*She sets the beads: an eighth on the listener's ring, an eighth on the wheel's ring, an eighth back on the bob. The hub she leaves bare.*]

Achilles: It doesn't look as if it cares which cord is which.

Tortoise: It doesn't. Light the wheel.

[*She turns the bead in the wheel's end-ring to a half: lit. The picture as it stands: above, the listener's green end-ring at rest, then its green ring at an eighth; below, the wheel's red end-ring at a half, then its green ring at an eighth; the two green rings tied to the bare red hub, and the bob hanging from the hub at an eighth back.*]



a controlled turn,
the wheel lit

[*First she slides the eighth off the wheel's green ring onto a little green bob of its own, hung from that ring on a short length: the ring is at rest now, its new bob at an eighth, and nothing else has changed. Then she pushes the lit end-ring along the wheel cord into the ring, the stretch of wheel cord it ran along swallowed with it. It goes in, and the ring is gone; out come copies of the lit end, one on each length that led into the ring: one on the wheel cord beyond, one at the foot of the tie, one on the short length to the new bob. That last copy meets the new bob, and the two close up into no ring at all, only a turn of the whole picture, an eighth, which she sets aside, off the cloth. The copy at the foot of the tie she slides along it into the red hub, the tie swallowed behind it: red into red, and the hub, bare a moment ago, stands at a half.*]

Tortoise: An eighth set aside. Now the hub goes through the bob.

[*The hub, red at a half, has two cords left: the tie up to the listener's ring, and the pendant down to the bob. She pushes it down the pendant into the bob. Its half passes into the bob and, the bob having no other cord, is gone; the bob's bead, which stood an eighth back, turns the other way and stands an eighth forward, and its old turn, an eighth back, she sets aside beside the first. The hub is left at rest with two cords, and it slips off: tie and pendant are one cord now, running from the listener's ring straight to the bob.*]

Tortoise: An eighth set aside, and an eighth back. What we set aside, we get back.

[*She slides the bob along that cord up into the listener's ring, the cord swallowed with it. Green into green: they run together, and their beads add, an eighth and an eighth. The listener's ring and its end-ring, green on one cord, slide together too, the ring back along the cord into the end-ring, the stretch between swallowed, into one end-ring at a quarter. What lies on the cloth now: no hub, no bob, no tie. The wheel's end-ring lies just past where its green ring stood, lit, red at a half, and the wheel cord runs on from it to its far end as before. What has changed is the listener*

alone: it began at rest, and now it holds a quarter. What has held is the wheel, lit going in and lit coming out.]



the turn landed

Achilles: So what you're saying is that the wheel gave its quarter to the listener.

Tortoise: It gave nothing away; it's lit as it was. The listener asked to be ticked, and the turn came back to the cord that asked. Now dark.

[*She gathers the cloth clear, the rings into her little box of rings and the string into her work-bag, and lays it all again: the listener at rest, the two green rings at an eighth, the bob at an eighth back, and this time the wheel's end-ring red with its bead at rest. She pushes the dark end in the same way, the stretch of wheel cord it runs along swallowed with it. Its copies carry nothing: the eighth she slid off the wheel's ring onto its own bob closes up into nothing at all, and there is nothing to set aside. The hub, red at rest, slips off. The bob, an eighth back, runs up into the listener's ring, the cord swallowed with it, an eighth forward, and they cancel; the ring, at rest with two cords, slips off too. The listener lies as it was laid, a green end-ring at rest, and the wheel is dark in and dark out. The whole controlled turn has fallen away.*]

Tortoise: From now on it's one move. Push a dark end through a controlled turn, and the whole turn falls away; push a lit end through, and the whole turn lands on the other cord's ring.

Achilles: Then I have it. One listener, one tube, and it tells me the turn.

Tortoise: Does it? [*She takes two green end-rings from her little box of rings, sets one bead a quarter round and the other three quarters round, and lays them side by side on the cloth.*] Here are two listeners, one that heard a quarter and one that heard three quarters.

[*She holds the two rings up to the candle. The beads stand at the same slant from upright, the one leaning to the right, the other to the left, like a minute hand at a*

quarter past and at a quarter to. Nothing about them differs but the side. She lays them back where they lay.]

Tortoise: A tube asks only how near to rest a bead stands, and these two stand equally near: even odds, both of them. A turn of none reads dark every time, and a half lit every time; anything else, only odds.

[*The ringing up the lane has stopped, and in the quiet the church clock gives a single stroke. It is the one stroke it gives at every half hour, and the one it gives at one in the morning; heard by itself, it is the same stroke at half past any hour. Achilles tucks his hands under his arms and frowns at the candle.*]

Achilles: Then one listener can never tell a quarter from three quarters.

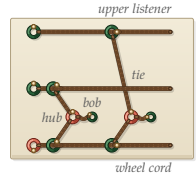
Tortoise: Not by itself. What would a listener hear, do you think, if the wheel ticked twice for it?

Third Challenge — *Twice the Pace*

Achilles: Twice a quarter is a half. And a half reads lit every time.

Tortoise: Let's see it. [*She gathers the cloth clear, the rings into her little box of rings and the string into her work-bag, and lays it out again: the wheel cord lowest, its end-ring lit; above it the listener, with its controlled turn for a quarter laid as before, close to the end-rings.*] A second listener above the first, and a controlled turn that ticks the wheel twice for it: twice the turn, so a quarter on each of its green rings and a quarter back on its bob.

[*The second listener goes above the first, beginning at a green end-ring at rest; it is the upper listener now, and the first the lower. On it she threads a green ring at a quarter, further along than the first turn's rings, and on the wheel cord a second green ring at a quarter, below it, each from its cord's loose far end; she ties both to a new red hub set to their right, between the lower listener's cord and the wheel cord, the tie from above passing beneath the lower listener's cord, joined to it nowhere, and hangs from the hub a bob a quarter back, running on to the right.*]



another listener above

[*She pushes the lit wheel end along its cord, the cord it runs along swallowed behind it. It meets the first controlled turn, and the whole turn lands on the lower listener's ring: a quarter. It runs on, lit still, into the second, and the whole of that turn lands on the upper listener's ring: a half. On each listener's cord the green ring slides back along it into the end-ring, as before, the stretch between swallowed. Then she runs the wheel's end-ring on to the wheel cord's far end. No hub or bob is left on the cloth. The wheel cord ends as it began, lit. The lower listener has gone from at rest to a quarter, and the upper from at rest to a half.*]

Tortoise: [*taking a tube from the packet and sliding it onto the upper listener's end-ring, where it goes in and is gone; the ring turns red, its bead still at a half*] Lit.

Achilles: And if the wheel turned three quarters? Twice three quarters is one and a half, and a whole turn is no turn, so a half. Lit again. For a half, twice a half is a whole: dark. For none, dark.

Tortoise: So the upper listener hears only whether the quarters are odd. It can't tell you which odd.

[*Achilles looks up from the cloth, past the candle, to the print on the wall facing the window: Escher's* Regular Division of the Plane.]

Achilles: A sheet filled edge to edge with figures fitted against one another, with no gap between them. If you slid it along, would it land on itself, and how far would you have to slide it?

[*As he looks, the candle flame leans in a breath of air from the stairs, and its light slides across the sheet from one edge of the frame to the other, and back, and is still. The Tortoise does not look up from the cloth.*]

Tortoise: The lower listener heard a quarter, or three quarters. The upper

one has told us which kind it was: odd. So take the odd quarter away, and see what's left.

[*Between the upper listener's cord, just past its end-ring, and the lower listener's cord she lays one more controlled turn: a green ring on each, threaded on from the cord's loose far end, both beads an eighth back, tied to a red hub to their right, between the two listeners' cords, whose bob hangs an eighth forward, running on to the right.*]

Tortoise: A controlled turn of a quarter back, and the upper listener's own reading drives it.

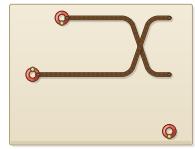
[*The upper listener's end-ring is red, at a half: lit. She pushes it along its cord into the new controlled turn, the stretch it runs along swallowed with it, and the whole turn lands on the lower listener's ring: a quarter back, and that ring slides back along its cord into the lower listener's end-ring, as before. The lower listener, which held a quarter, now holds a quarter and a quarter back, which is none. The upper listener's end-ring runs on, lit, as it was, and comes to rest just past where its green ring stood. What has changed is the lower listener only, from a quarter to at rest.*]

Tortoise: [*taking another tube from the packet and sliding it onto the lower listener's end-ring, where it goes in and is gone; the ring turns red, at rest*] Dark.

Achilles: And for three quarters, three quarters and a quarter back make a half. Lit.

Tortoise: And if the upper had read dark, the correction would have fallen away, and the lower would keep what it heard, none or a half. Now, which to read first.

[*The two listeners' far ends lie one above the other, the upper's above, the lower's below. She takes them and crosses them, the lower's end passing over the other without touching it, and lays them down again. Not a bead on either cord has changed, nor an end; only which lies on top. The lower listener's reading now lies uppermost.*]



read, and the
far ends crossed

Tortoise: The half first, then the quarter.

Achilles: So the top one says whether the wheel turns a half, and the one beneath says whether there's an odd quarter besides. None is dark and dark. A quarter is dark, then lit. A half is lit, then dark. Three quarters, lit and lit.

Tortoise: Every one of them certain. And a wheel that turned in eighths would want a third listener, ticked four times, and another correction; every listener ticked twice as fast hears one finer place.

Achilles: And the wheel? [*He touches the wheel cord's far end: red, at a half.*] It has been ticked once for one listener and twice for the other, and it comes out lit, just as it went in.

Tortoise: The ticking changes where it stands, never what it is.

Achilles: [*turning to the shelf*] Then a case built to do all that could read its wheel in one passage.

Tortoise: [*taking the candlestick up to the shelf*] This is the Genie's clock-case, and the Office sent it for one wish. The knob low on its left side is the only thing on it you may set: down for dark, up for lit, and tonight it stands down. Down the right side are its windows, the first and the second above and the wheel window lowest, level with the knob; after the passage each shows dark, a plain pane, or lit, a small gleam, and keeps what it shows.

Achilles: [*bending to the top of the case, where a small oval of enamel is set*] And a plate on top. [*reading*] "The Terms. One wish, one passage. Inside is a wheel: lit, it turns by the same whole number of quarter-turns at every tick; dark, it does not turn. With the wheel knob down, every window shows dark. What the case does with the wheel lit is the wisher's business."

Achilles: The same whole number of quarters at every tick, and it doesn't say how many. None, one, two or three: just what your listeners read.

Tortoise: [*setting the candlestick back on the table*] The Terms are short, and they keep to them. The lamp is the same as last night: rub it once, and the Genie comes.

Achilles: [*reaching up, and taking the lamp down from the shelf*] Once, I remember.

[*The lamp lies in Achilles's hands as it has lain on the shelf since last night: brass, tarnished, cold, its wick dry, never lit. He rubs its side once with his thumb. Nothing in it warms. From the spout a pale thread leans out into the cold air, turns on itself like a ribbon being wound, and fills; and there between the table and the window stands the Genie, straightening his cuffs. The lamp in Achilles's hands is just as it was, dull and cold and no lighter. What is new is a third at the table. Achilles sets the lamp down beside the candlestick.*]

[*While the pale thread stood, the candlelight drew in off the walls toward the lamp, the corners of the room going dark one after another as though the light were being gathered up; now it goes back out to them, and every shadow settles where it was. Achilles's breath shows white at every word, and the Tortoise's; the Genie's does not show at all. Up the lane the tower is silent.*]

Genie: The Office of Wishes. [*He glances along the shelf to the case.*] Ah. You have the newer model. The Office is rather proud of it.

Genie: One wish is one passage of the whole case, however often its wheel turns inside. What the case does inside is the case's business.

Achilles: And turning the knob?

Genie: Setting the knob is not a wish. Nor is any arithmetic you do afterwards with what the windows show. There is one wish this evening.

Achilles: Then I wish to be told how many quarters the wheel turns.

Genie: Referred. [*He tilts his head, as though listening up a stairwell, and straightens at once.*] The Meta-Genie regrets. The quarter-turns are the case's to tell, and the case tells only by passage. Nothing has been spent.

[*On the table a point of candlelight runs up the lamp's curved handle to the top and down again, all in the one instant, though neither the candle nor the lamp has moved. The lamp is as cold as it was.*]

Achilles: Then I wish to have the wheel ticked a thousand times, with the front open, while I watch.

Genie: Referred. [*A blink.*] The reply has come down from well above the Meta-Genie: a wish to watch is a wish to see inside, and the front opens on a spent case and on no other. Your wish is untouched.

Achilles: From well above, and back already?

Genie: No floor of the Office keeps a reply waiting. It goes all the way up, and the answer comes all the way down, at once.

Achilles: Then one more. If the passage shows me nothing I can use, I wish for a second.

Genie: Referred. [*A blink.*] The reply is that there is one wish this evening. A reading that fails to give what the wisher wanted is still a spent wish; another reading would be another wish, and the Office grants one.

Tortoise: You'll wear the Meta-Genie out.

Genie: Nobody has yet.

[*Achilles laughs in spite of himself, and the Tortoise's chair clacks as she sits back to enjoy it. He winds his scarf once more round his neck and holds his hands to the candle.*]

Achilles: May I at least bring the case to the table?

Genie: Carrying is not a wish either. Keep it upright. The Office labels its boxes for a reason.

[*Achilles lifts the case from the shelf in both hands and sets it on the table beside the cloth, its front towards him, its knob on the left side, its windows down the right, evenly spaced, each a round pane in a thin brass rim. Down the front's right-hand edge, where the front closes against the side, a faded paper band is pasted down the join: the seal. It stands as it has stood all evening: knob down, every pane plain, the seal whole.*]

[*He puts his fingers to the wheel knob. It stands down: dark. He turns it up until it stops. The knob is the only thing on the case that has moved; the panes are plain, the seal is whole, the front is shut, as before. Down has become up: the wheel is lit.*]

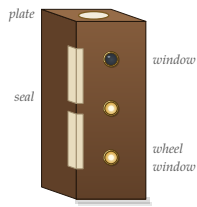
Achilles: The wheel lit, and everything the case does inside, ticking and all. Is that one wish?

Genie: One passage of the whole case, as it is now set. One wish.

Achilles: Then I wish it.

Genie: Granted.

[*The case as it stands: knob up, every pane plain, the seal whole. At the word nothing on it moves, not the knob, not the front. Behind the first window the pane stays plain. Behind the second a small gleam comes up and stays. Behind the wheel window, another. Achilles touches the knob, and it will not turn back; it is locked where he set it. The seal down the front's right-hand edge has parted of itself, in the middle. What has changed: two gleams, the lock, the seal. What has held: the knob, up; the first pane, plain; the front, still shut.*]



after the passage

Achilles: The first plain, the second lit, and the wheel lit. No half, and an odd quarter: one quarter, no more.

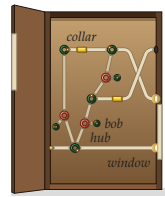
Achilles: [*looking across at the paper dial, where the hand stands at one*] ...Oh. A quarter round at every tick, so home in four. And nobody ticked it where I could count.

[*Achilles's hand goes to his waistcoat pocket, as it does whenever he wants to count something, and comes out with his watch. He looks at it once: the short hand most of the way to nine, and the long hand, which goes round so much faster, at the eight. That one look gives him the time to the minute, and he puts the watch away.*]

Genie: The case is spent. A secret read is no longer a secret, and the front may be opened.

[*The case as it stands: spent, the knob locked up, the second pane and the wheel pane gleaming, the first plain, the seal parted. Achilles takes the front by its edge and swings it open on its hinge at the left. The knob does not move, nor the panes, nor their gleams; only the front swings back, and the candle shines into the case.*]

[*Inside, row by row from the top. The upper row begins at a green ring whose bead stands a quarter round, with nothing to the left of it. From that ring a cord runs along the row, through a fixed yellow collar set crosswise on it, to a second green ring, whose bead stands an eighth back; it is knotted at the one ring and at the other. From the second ring another cord runs on, and down, to the second window, knotted at the ring and at the window. Below, the lower row begins at a green ring whose bead stands at rest; from it a cord runs through a second fixed collar and then up to the first window, knotted at the ring and at the window. On the way these two cords cross, the one going down and the other going up, the one going up passing over the one going down without touching it. Lowest lies the wheel row: a cord from the wheel knob to a green wheel ring, whose bead stands three eighths round, knotted at the knob and at the ring; and from the wheel ring a cord on to the wheel window, knotted at both its ends. Among the rows hang red hubs, with no beads in them, each between the rows it ties, one after another from left to right. The first hub has a tie of its own up to the upper row's first ring, another down to the wheel ring, and a pendant hanging to a small green bob a quarter back. The second hub has a tie to the lower ring, another to the wheel ring, and a pendant to a bob an eighth back. The third hub has a tie to the upper row's second ring, another to the lower ring, and a pendant to a bob an eighth forward. Every length is knotted at both its ends, and no cord passes through a ring. What has changed is only that the inside can be seen; every gleam can now be followed back, cord by cord, to where it began.*]



the front open

Achilles: A quarter round on the upper row's first ring. The quarter was in the case all along, and no window showed it until it was asked the right way.

Tortoise: So the first window has the half, and the second the quarter.

Genie: The wish has been spent in accordance with the Terms. The Office does not collect spent boxes.

[*The lamp stands where Achilles set it down beside the candlestick, cold, as it has been since he rubbed it. The Genie folds in on himself from the shoulders down, as a ribbon is wound back onto its card, thins to the pale thread he came as, and the thread runs into the spout and is gone. The lamp does not stir or warm. The case stays open on the table, its panes still showing what they showed, and nothing else in the room has moved. There are two at the table again.*]

Tortoise: [*taking up the lamp and setting it back on its shelf*] Back where it lives.

[*Up on its shelf, out of the candle's reach, the lamp is only brass again, beside the clean patch in the dust where the case stood all evening. Up the lane the bells begin again: a plainer piece than before, easy to follow, the highest bell working down through the others one place at a time and then back up, while the rest give way to it in turn. The Tortoise sits back down, and her chair clacks.*]

Achilles: [*looking from the open case to the paper dial*] The Office's case had its wheel lit on a round of its own; the Office saw to that. But the dial's hand, standing at one, isn't on any one round at all.

Tortoise: No. It's on all of them. What do you suppose one wish would read from a hand that stands on every round at once?

Fourth Challenge — *Every Round at Once*

Achilles: One round's turn, I suppose. But which?

Tortoise: Let's lay them all, and see. First put yours back as I set it.

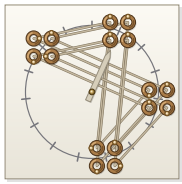
[*The first round still lies on the dial as Achilles last left it, ticked once: three quarters back at one, at rest at seven, a quarter back at four, a half at thirteen. He ticks it three times more, each time carrying every ring on to the hand's next hour, beads untouched; at each tick every hour's bead stands a quarter on again. After the third, the ring whose bead is at rest is back at one, a quarter back at seven, a half at four, three quarters back at thirteen: every bead where she first set it, and the loop lying where it always lay.*]

Achilles: Four ticks in all, and it's as you laid it.

Tortoise: [*bringing three more loops out of the work-bag, each threaded with rings like the first*] These are the rest of this dial's rounds. The one you ticked goes a quarter back at each step along the hand's way; this one stands at rest all round; this one goes a half at each step; and this one three quarters back.

Tortoise: Laid one over another, the beads at each hour are read together, as pointers. Two beads pointing opposite ways cancel, and I turn both face down; beads all at rest add, and stay face up.

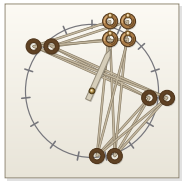
[*She lays the other three rounds over the first on the dial, so that at each of the hand's hours four rings sit together in a little square, side by side so that every bead shows: the ring of the round he ticked at the far left as Achilles sees it, the ring of the round at rest at the far right, the half's at the near left and the three quarters back's at the near right. At one: four beads, every one at rest. At seven: one at rest, one a quarter back, one at a half, one three quarters back, pointing four ways. At four: two at rest, two at a half. At thirteen: one at rest, one three quarters back, one at a half, one a quarter back, four ways again.*]



the rounds together

Achilles: [*going round with a finger*] At one they all agree. At seven, the one at rest faces the half, and the quarter back faces the three quarters back. At four, rest faces half, twice over. At thirteen, just as at seven.

[*The Tortoise takes the rings at seven in pairs, each bead with the one that points against it, and turns each pair face down on the dial, bead to bead, so that neither shows and the pair lies as one plain ring, the two pairs at an hour side by side; then the rings at four, and then at thirteen. The loops still run from hour to hour, but at seven, four and thirteen no bead shows at all. At one the four rings lie face up in their little square, their beads all at rest. What has changed: every hour the hand visits but one shows nothing. What has held: the hour one, and the four beads on it, at rest, where the hand began.*]



turned face down

Achilles: So all the rounds laid together are the hand at one, and nothing else.

[*Achilles's eye goes past the dial to the floor under the shelf. Where the candlelight passes through the tall glass jar, it comes out on the floor beyond as a faint patch of colours, red at one edge and violet at the other; everywhere else in the room it is only candlelight.*]

Tortoise: So a passage with the hand at one can only read one round's turn, and which round is chance, the same chance for each: none, a quarter, a

half or three quarters.

Achilles: And if we tick the whole pile, each round turns by its own amount, and not one of them goes anywhere?

Tortoise: The ticking changes where it stands, never what it is.

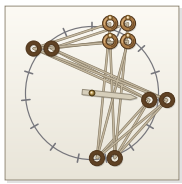
Achilles: Then let me rub the lamp and see which round we get.

Tortoise: The wish is spent, and the Genie with it. We'll suppose instead. Suppose a case built on this dial, with the hand at one inside it where the wheel was, had shown you a half.

Achilles: A half at every tick: two ticks home.

Tortoise: Tick it and see.

[*The paper hand stands at one. Achilles carries it once round to seven, and once more, seven sevens less three fifteens, to the fourth mark; and there he lets go. It points at four, not at one. Two ticks have moved it twice, and it has not come home.*]



the hand gone on

Achilles: Four. Not home.

Achilles: [*frowning at the slack loops*] The half came from the round that goes a half at each step, and that round comes home in two. The hand doesn't.

Tortoise: And the other readings?

Achilles: A quarter says four ticks home, and so does three quarters, and four ticks from one bring the hand home. None says nothing at all. Two of the four readings give it, then. Half the time.

Tortoise: Half the time, from one wish. And a half or a none is still a spent wish; another reading would be another wish.

[*Up the lane the plain piece comes to its end, and the bells fall into the order they began the evening with, highest to deepest, and ring it over and over, every bell in its own place; at every deepest stroke the jar under the shelf hums back. Achilles sits back, tucks his hands under his arms, and listens.*]

Achilles: So the wish finds the ticks home half the time. But you said the ticks home would split the dial.

Tortoise: They will. How would you measure one number by another, with nothing but paper?

Fifth Challenge — *Euclid's Strips*

Achilles: With a ruler, I suppose. Only I haven't brought one.

Tortoise: [*bringing a rule, a pair of scissors and a fresh sheet out of the work-bag*] I have a rule, marked in thumbs, and scissors. Lay a short strip along a long one as many times as it fits, pencil how many times on the long strip, and cut off what's left; then lay what you cut off along the short one, the same way. When a cut leaves nothing, the last short strip measures both.

Achilles: Then what do I measure?

Tortoise: The dial, by the hand at half its ticks home. Four ticks home: half of four is two, and after two ticks the hand stood at four.

[*Up the lane the ringers are ringing down: the strokes come quicker and softer and closer together, the order lost among them, until they run into one small jangle and stop. The tower is silent, and the jar under the shelf gives a last faint hum and is still.*]

[*Achilles cuts a strip fifteen thumbs long, for the dial, and another four thumbs long, for the hand. The long strip lies along the table's edge on his side; the short one lies beside it, further from him, their left ends level. He lays the short strip at the long one's left end, pencils a tick where it stops, and lays it again from the tick, and again: three times it fits, and past the last tick there is a piece of the long strip that the four cannot cover. He sets the short strip back beside it.*]



the short strip laid along

[*He pencils the count on the long strip and snips it at the last tick; the measured part, with its count on it, he lays aside, and the piece left past the last tick he slides along the edge to where the long strip began. What has changed: the long strip is gone from the edge, and in its place lies the piece it could not measure, three thumbs long. What has held: the four-thumb strip, which has not been cut at all.*]



after the snip

[*The three goes along the four once, and one thumb is left over; he cuts it off. The one goes along the three three times, and nothing is left.*]

Achilles: One thumb. That measures anything, and it tells me nothing.

Tortoise: Not the hour itself. One either side of it.

Achilles: Three and five. Why those?

Tortoise: Multiply them and you get the hour times itself, less one. The hour times itself is the hand ticked on as many ticks again, which is home at one: one past a whole number of dials. Take the one away, and three times five is whole dials exactly, so each of them may take a share of the dial.

[*A fresh strip of fifteen thumbs, and one of three. The three goes along the fifteen five times, and at the fifth the ends meet exactly; there is nothing to cut. He does the same with a strip of five: it goes along three times, and nothing is left.*]

Achilles: Fifteen is three fives.

[*Down in the lane feet go by on the frozen ruts: two of the ringers going home past the house, the one with a long stride and the other with a short. Their steps fall together, and apart, and together again, at every third step of the one and every fifth of the other, until they are out of hearing.*]

Tortoise: Now a harder dial. [*She draws a second circle on the fresh sheet, pricks*

twenty-one marks round it, the first at the top, on the side furthest from Achilles, as on the first, and pins a second paper hand at its centre, turned to one.] This one has twenty-one hours, and its tick doubles the hour. Suppose a finer case than the Office's had read its turn for you as five parts in thirty-two.

Achilles: Five parts in thirty-two isn't a quarter or a half of anything.

Tortoise: Strips, then. Thirty-two and five.

[*Achilles cuts a strip thirty-two thumbs long and one of five. The five goes along the thirty-two six times, and two thumbs are left over; he pencils six on the long strip and cuts the two off. The two goes along the five twice, and one thumb is left; he pencils two and cuts. The one goes along the two twice, and nothing is left at all. The counts he has pencilled read, in order, six, two, two.*]

Achilles: Five goes into thirty-two six times, with a little over. So five parts in thirty-two is a little less than one part in six: a sixth of a turn at every tick, and six ticks home.

Tortoise: The later counts only say how little the little over is. A real case for a dial this size would be made finer still, so that the strips can't mislead; five in thirty-two is only our supposing.

Achilles: [*ticking the new hand from one*] Two, four, eight, sixteen. Thirty-two, less twenty-one: eleven. Twenty-two, less twenty-one: one. Home in six.

[*Ticked six times, the second hand has pointed at two, four, eight, sixteen and eleven, and now stands again on the first mark past the top, where it started. It is the same paper hand on the same pin; the twenty-one marks have not moved.*]



the new hand, home

Achilles: Six ticks, and it's on the mark it left.

Tortoise: The ticking changes where it stands, never what it is.

Achilles: Half of six is three, and after three ticks it stood at eight. One either side: seven and nine.

[*A strip of twenty-one thumbs and one of seven: the seven goes along three times, and nothing is left. A fresh twenty-one and a nine: the nine goes along twice, and three are left over; he cuts them off, and the three goes along the nine three times, and nothing is left. The last short strips lie side by side: seven thumbs, and three.*]

Achilles: Seven, and three. Twenty-one is three sevens.

Tortoise: So what has the evening done?

Achilles: A tick moves the hand, but it only turns a round. One listener hears the turn, but blurred. Listeners ticked twice, and four times, each hear one finer place, and one wish reads the whole turn. The hand at one is every round at once, so a wish reads a round at random, and half the time that's enough. Then paper and strips do the rest.

Tortoise: The case only reads the turn. Turning a turn into factors is done with strips, and how long it all takes was reckoned on paper, not by drawing.

Achilles: [*looking at the candle*] The candle's burned lower than I thought.

Tortoise: It always goes faster up here. The cold makes it hurry.

Achilles: [*looking into the open case*] And its panes are still showing. The second, and the wheel.

Tortoise: It'll keep them. It's spent; showing is all it has left to do.

Achilles: Like the casket downstairs, with the jam.

Tortoise: They'll have a great deal to say to each other.

[*Achilles laughs, and the Tortoise laughs under her breath with him, and for a while neither says anything. The lane is quiet now the ringers have gone. Somewhere below them, in the parlour, the stove's iron ticks now and then as it cools, to no count at all.*]

[*Achilles unwinds his scarf and puts it round the Tortoise's shoulders.*]

Achilles: Keep it up here tonight. It seems to belong to this room.

Tortoise: [*settling it*] Thank you. I'll bring it down in the morning, and the stair can groan at me for a change.

[*Up the lane, the church clock begins to strike.*]

Achilles: [*a finger lifting at each stroke, under his breath*] ...seven, eight, nine. Nine o'clock. I counted every stroke.

Tortoise: You always will.

Achilles: How do you tell how long a thing takes to come round, if you may look at it only once?

Tortoise: You don't count the ticks. You hear how far one tick turns it.

[*She takes up the candlestick, and the case stays open on the table, its panes still showing.*]

[*The candle stands low in its stick, guttered down to a small flame, and as she lifts it every shadow in the room swings round and shortens. With her free claw she takes the scarf from her shoulders, folds it, and lays it folded on the seat of her chair, where it will be in the morning. On the wall facing the window, past her shoulder, the print is at the edge of the light again, half in her shadow: a sheet in a plain dark frame on a cold wall. Then the candle goes to the door, and Achilles after it, and the stair takes them down its one turn; halfway, a tread groans under Achilles's foot, as it will for him every time. Behind them the top room is dark and still: the case open on the table where Achilles set it, the lamp on its shelf, the paper and the strips where they were left, the print that nobody can see now, and over the whole of the pane the frost going on with its work, the old initials white in it. From the foot of the stairs comes the sound of the stove door opening, and the kettle going on. Up the lane the tower is dark, and the clock in it goes on round towards ten, and at ten it will strike again, whether anyone counts or not.*]

Chapter 14: Shor's Hidden Periods

— Aria with Periodic Variations —

Part IV: Algorithms

Escher: "Regular Division of the Plane" — a sheet filled edge to edge with figures fitted against one another, with no gap between them

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How do you tell how long a thing takes to come round, if you may look at it only once?

With a clock the answer is easy, because you may keep looking. You watch the hand until it is back where it began and count the ticks on the way. The tides, the phases of the moon and a child on a swing are all timed the same way: you wait for the return. The method is honest and certain, and it costs exactly as many ticks as the thing takes to come round. For anything that comes round slowly, that is a great many.

Now suppose you may not watch. The thing is sealed in a case, it ticks inside, and you are allowed one glance. One glance at a hand tells you where it stands. It does not tell you how fast it is going. A hand that has come round once in twelve ticks and a hand that has come round five times look exactly alike if they happen to stand at the same place. Where a thing stands is a fact about one moment. How long it takes to come round is a fact about all the moments together, and a glance is only one of them.

It turns out that one of the oldest problems in arithmetic is a problem of exactly this kind. Take a large number, multiply by a fixed smaller number over and over, and keep only the remainders on dividing by a third. The remainders come round in a cycle, and the length of that cycle holds the

key to splitting the third number into its factors. Nobody knows a quick way to find the length by ticking. The cycle can be longer than the age of the universe measured in ticks of the fastest machine anyone could build.

This chapter finds it with one glance, or close to it. The trick is that a quantum glance need not be at the hand. It can be at something the ticking does not move, only turns, and a turn can be read, digit by digit, by listeners that tick at doubled pace. The pictures of the last six chapters will show how the listening works, down to where each digit is heard. They will not show everything. The step from the length of a cycle to a factor is number theory, and it has no picture. We will say where the pictures stop, and why.

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§14.1 The Fifteen-Hour Dial

Chapter 13 closed with a promise: "Chapter 14 keeps the pattern and changes the layers. Simon's round, with the second layer of Hadamards replaced by a Fourier transform over a larger group, is where Shor began ... The box will hide not a string but a period, the query will be a controlled multiplication, and the pictures will again be kickback diagrams, the marks copied back along their ties. What will change is how much of the work the pictures can do. The number theory that turns a period into a factor stays on the blackboard, and we will say so." We begin with the period.

On the blackboard. Fix a whole number $N > 1$, the number we would like to factor, and a **base** a with $1 < a < N$ that shares no factor with N, $\gcd(a, N) = 1$. We work with remainders on dividing by N: write $u \equiv v \pmod{N}$ when N divides $u - v$, and $u \bmod N$ for the remainder, a number between 0 and $N - 1$. Shor writes n for the number and x for the base; we write N and a throughout.

Picture the N remainders as the hours of a dial, with 0 at the top. One **tick** moves the hand from the hour x to the hour $ax \bmod N$. Start the hand at 1. Take $N = 15$ and $a = 7$. The first tick gives 7. The second gives $49 = 3 \cdot 15 + 4$, so 4. The third gives $28 = 15 + 13$, so 13. The fourth gives $91 = 6 \cdot 15 + 1$, so 1: the hand is home after four ticks. The least number of ticks that brings

the hand home from 1 is the **order** of a mod N, written r: the least $r \geq 1$ with $a^r \equiv 1 \pmod{N}$ (Shor §5). For 7 mod 15, $r = 4$. It is a period hidden in the multiplication table.

The quantum version of the tick is a box. Take enough wires to hold the numbers $0, \ldots, N-1$ in binary, top wire most significant (four wires for $N = 15$), and define the **tick box**

$$U_a \, |x\rangle \;=\; |ax \bmod N\rangle \qquad (0 \leq x < N),$$

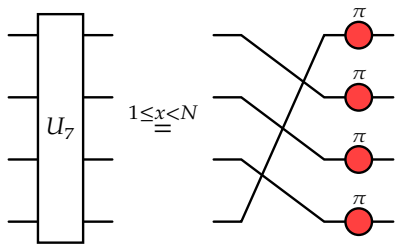
letting U_a leave alone any basis state $|x\rangle$ with $x \geq N$ that the wires can also hold.

> **Result 14.1 (the order, and the tick box).** If $\gcd(a, N) = 1$, then $x \mapsto ax \bmod N$ is a permutation of $\{0, \ldots, N-1\}$, so U_a is unitary; the order r exists; and $U_a^r = I$.

Proof. If $ax \equiv ax'$ then N divides $a(x - x')$, and since a shares no factor with N, N divides $x - x'$, so $x = x'$. A one-to-one map of a finite set to itself is a permutation, so U_a permutes basis states and its matrix has one 1 in each row and column: it is unitary. The powers $a, a^2, a^3, \ldots$ take at most N values mod N, so two agree, $a^i \equiv a^{i+d}$ with $d \geq 1$; multiplying by the inverse of a^i (which exists, since $x \mapsto a^i x$ is a permutation) gives $a^d \equiv 1$. The least such d is r. Finally $U_a^r \, |x\rangle = |a^r x \bmod N\rangle = |x\rangle$. ■ See `Code/Ch14/HandsRound.py`, which also checks the order by ticking against the least r with $a^r \equiv 1$ for random pairs (N, a).

In the sketchbook. The tick box is Chapter 5's tall box (§5.3), laid across the register's wires and labelled U_a. It hides everything inside it, as an oracle's box did in Chapter 13. For one small dial we can open it. Since $16 = 15 + 1$, doubling a number below 15 just rotates its four bits: the top bit, worth 8, becomes worth $16 \equiv 1$ and comes round to the bottom. So multiplying by 8 is three such rotations, and three rotations one way are one rotation the other way. And $7 \equiv -8$, while $15 - x$ is the four bits of x flipped. So multiplying by 7 is: rotate the bits three places, then flip every bit. In pictures, a rotation is crossings and a flip is a red π dot (§8.2, the NOT). The bottom wire climbs to the top, crossing the other three, which

each step down one row; then a red π sits on every wire:



Result 14.2 (the fifteen-hour dial as a picture). Call the right-hand picture C_7. Exactly, as matrices on the four wires: $C_7\,|x\rangle = |7x \bmod 15\rangle$ for $1 \le x < N$, and $C_7\,|0\rangle = |15\rangle$. Moreover C_7^2 is the pure crossing that exchanges wires $1 \leftrightarrow 3$ and $2 \leftrightarrow 4$, which multiplies by 4 on $1 \le x < N$; and $C_7^4 = I$ on all sixteen basis states.

Proof. On the blackboard: for $0 \le x < 15$ with top bit b, $2x = 16b + 2(x - 8b) \equiv b + 2(x - 8b)$, which is the left rotation of the bits; so three rotations send x to $8x \bmod 15$. For $1 \le x < N$ that remainder y is again an hour between 1 and $N - 1$, and flipping all four bits sends it to $15 - y \equiv -8x \equiv 7x$, again between 1 and $N - 1$. So $C_7\,|x\rangle = |7x \bmod 15\rangle$ there. At $x = 0$ the rotation gives 0 and the flips give 15, which is not an hour of the dial; the orbit of 1 never goes there. In the sketchbook: in C_7^4, slide every red π along its wire, through the crossings, to the right end (only the connectivity matters, row 63). Each wire collects four red π dots, and fusion (Theorem 8.1, row 57) gives a red dot of phase 4π, which is no turn: a two-legged red dot with no phase is a plain wire (§8.2). The crossings compose to twelve rotations of four wires, which is none. With two copies, two red πs fuse to nothing on each wire and six rotations are two: wires $1 \leftrightarrow 3$, $2 \leftrightarrow 4$. Two rotations multiply by 4. No scalar appears anywhere: a red π dot with one leg in and one out is X exactly. ■ See `Code/Ch14/HandsRound.py`, which derives C_7^4 by the rules and checks that C_7 sends 0001 to 0111.

The picture is special to this dial. It works because 15 is one less than a power of two and 7 is minus a power of two. A general multiplication mod

N is a reversible arithmetic circuit (Shor §3) with no such picture, and from here on the tick box stays closed.

How long does ticking take? It takes r ticks, and r can be large. It can be nearly as large as N: when N is an odd prime, some a has order $N-1$, because the multiplication of nonzero remainders mod an odd prime is cyclic, one base running through them all (a classical fact we state and do not prove; Shor §5 uses it for every odd prime power). For numbers of the size used in cryptography, N has hundreds of digits and so can r. Cleve, Ekert, Macchiavello and Mosca put the state of knowledge plainly: "There is no known classical procedure for doing this in time polynomial in n, where n is the number of bits of N." Shor's abstract calls factoring and discrete logarithms "generally thought to be hard on a classical computer". Both statements are careful. No fast classical method is known; none has been proved impossible.

Why this definition. Why chase the order, when what we want is a factor? Because the order is a *period*, and periods are what waves are good at finding; §14.5 shows that a period gives a factor with a few lines of arithmetic. And why a box that multiplies by a, rather than one that computes a^x from x? Because multiplying by a can be undone (Result 14.1), so it is a single unitary that can be applied again and again, and "applied again and again" is exactly what a period is about. Shor's own circuit computes a^x into a second register; Cleve et al. show that the two views give the same network (§14.3 and §14.5).

Feynman's Notebook.

A thesis with two escape clauses. Peter W. Shor, of AT&T Research, published a preliminary version of the factoring algorithm in the proceedings of the 35th Annual Symposium on Foundations of Computer Science, held in Santa Fe, New Mexico, on 20–22 November 1994; the full paper, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer", was posted in its revised form on 25 January 1996. It opens with Church's thesis in a quantitative form: any physical computing device can be simulated by a Turing machine in a

number of steps polynomial in the resources it uses. Shor points to two escape clauses, in the words "physical" and "resources", and reminds the reader that "the universe is in reality quantum mechanical". He names his inspiration directly: "Simon's algorithm inspired the work presented in this paper." He is equally direct about the hardware: "Currently, nobody knows how to build a quantum computer". And he adds that even if none is ever built, the result says something about the world: any general method for simulating quantum mechanics with at most a polynomial slowdown would lead to a polynomial-time algorithm for factoring.

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§14.2 The Round Turned Whole

Ticking moves the hand. Is there anything on the dial that a tick does not move, but only turns?

On the blackboard. Try a sum over the orbit of 1 with a turning phase. For 7 mod 15 the orbit is $1, 7, 4, 13$. Put a quarter-turn back at each step:

$$|\psi_1\rangle \;=\; \tfrac{1}{2}(|1\rangle - i\,|7\rangle - |4\rangle + i\,|13\rangle).$$

The phases are 1, $e^{-i\pi/2} = -i$, $e^{-i\pi} = -1$ and $e^{-3i\pi/2} = i$. Apply the tick box: each hour moves to the next, $1 \to 7 \to 4 \to 13 \to 1$, and keeps its phase,

$$U_7\,|\psi_1\rangle \;=\; \tfrac{1}{2}(|7\rangle - i\,|4\rangle - |13\rangle + i\,|1\rangle) \;=\; i\cdot\tfrac{1}{2}(|1\rangle - i\,|7\rangle - |4\rangle + i\,|13\rangle) \;=\; i\;|\psi_1\rangle\,.$$

The state has not moved. It has turned, whole, by a quarter: $i = e^{2\pi i/4}$. We call such a state a **round**.

Result 14.3 (eigenrounds). For $k = 0, 1, \dots, r-1$ let

$$|\psi_k\rangle \;=\; \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} e^{-2\pi i k s/r}\, |a^s \bmod N\rangle\,.$$

Then $U_a\,|\psi_k\rangle = e^{2\pi i k/r}\,|\psi_k\rangle$, and the r rounds are orthonormal.

This is Cleve et al.'s Eqs. (6.1) and (6.2), with their exponent j written s and the factor $r^{-1/2}$, which they leave out, written in.

Proof. U_a sends $|a^s \bmod N\rangle$ to $|a^{s+1} \bmod N\rangle$, so

$$U_a\,|\psi_k\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} e^{-2\pi i k s/r}\,|a^{s+1}\rangle = e^{2\pi i k/r}\,\frac{1}{\sqrt{r}} \sum_{s=1}^{r} e^{-2\pi i k s/r}\,|a^s\rangle = e^{2\pi i k/r}\,|\psi_k\rangle\,,$$

where the last step uses $a^r \equiv 1 = a^0$ and $e^{-2\pi i k r/r} = 1$, so the term $s = r$ is the old term $s = 0$. The hours $a^0, \dots, a^{r-1}$ are distinct (else a would come home sooner), so $\langle\psi_{k'}|\psi_k\rangle = \frac{1}{r}\sum_s \omega^s$ with $\omega = e^{2\pi i(k'-k)/r}$. If $k = k'$ this is 1. Otherwise $\omega \neq 1$ and $\omega^r = 1$, so the geometric sum $\frac{\omega^r - 1}{\omega - 1}$ is 0. ■ See `Code/Ch14/KickedTurn.py`.

The eigenvalue carries the period. If we could read the turn k/r of a round, we would have r in the denominator. The rest of the chapter is about reading a turn.

A listener on a controlled box

A turn by itself is invisible: $|\psi_k\rangle$ and $e^{2\pi i k/r}\,|\psi_k\rangle$ are the same state up to an overall phase (Chapter 3), and no reading tells them apart. What makes a turn visible is doing it *on one branch only*. Let U be any unitary with an eigenvector $|u\rangle$ (normalized), $U\,|u\rangle = e^{2\pi i\varphi}\,|u\rangle$, and call φ its **phase**, measured as a fraction of a whole turn; $\theta = 2\pi\varphi$ is the same angle in radians. Kitaev's **controlled-U** acts on one extra control wire and the register:

$$\Lambda(U) \;=\; |0\rangle\langle 0| \otimes I \;+\; |1\rangle\langle 1| \otimes U.$$

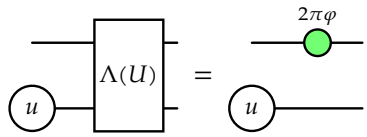
Result 14.4 (kickback of an eigenphase). For $c \in \{0, 1\}$, $\Lambda(U)(|c\rangle\,|u\rangle) = e^{2\pi i c\varphi}\,|c\rangle\,|u\rangle$. Hence for every control state

$|\chi\rangle$, $\Lambda(U)(|\chi\rangle\,|u\rangle) = (Z(\theta)\,|\chi\rangle)\,|u\rangle$, where $Z(\theta) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{smallmatrix}\right)$. And $\Lambda(U^{2^m})$ leaves $e^{2\pi i c 2^m \varphi}$.

Proof. On $|0\rangle\,|u\rangle$ nothing happens; on $|1\rangle\,|u\rangle$, U acts and gives $e^{2\pi i\varphi}\,|1\rangle\,|u\rangle$. Linearity in the control gives the second statement, and $U^{2^m}\,|u\rangle = e^{2\pi i 2^m \varphi}\,|u\rangle$ the third. ■ See `Code/Ch14/KickedTurn.py`, which tests it with a random complex non-diagonal U, so that a slip between φ and $-\varphi$ would show.

This is Chapter 5's kickback (row 37) in general form. There the eigenvalue was -1, a half-turn; here it is any turn. The register does not change. The control wire does: its $|1\rangle$ branch has been turned by θ.

In the sketchbook. The tall box $\Lambda(U)$ spans the control wire and the register, and $|u\rangle$ is a state box on the register. The kickback says the tall box can be removed, leaving a green dot with phase θ on the control wire, which is exactly $Z(\theta)$ (Chapter 8's $Z(1,1,\theta)$):



exactly. For a general U this picture is **borrowed from the blackboard**. The tall box hides everything, and no rule of Chapter 8 can open it; the equation is Result 14.4, drawn. Chapter 6 borrowed the transpose in the same way. But when U is itself a phase box, the whole thing can be drawn with dots, and then the rules can prove it.

The controlled turn

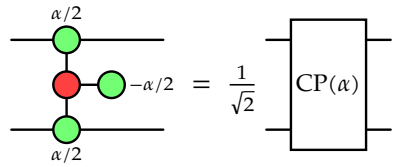
Take the smallest case: the register is one wire and $U = Z(\alpha)$, a phase box that turns $|1\rangle$ by α. Then $\Lambda(Z(\alpha))$ turns $|11\rangle$ by α and leaves the other three basis states alone:

$$\mathrm{CP}(\alpha) \;=\; \mathrm{diag}(1,1,1,e^{i\alpha}).$$

This is the **controlled phase**. Its matrix does not care which wire is the

control: $|11\rangle$ is the only state touched, and it is symmetric. So we will call it the **controlled turn** and draw it without an arrow of control.

Here is the drawing. It is the one new picture element of this chapter, and it is built entirely from Chapter 8's dots. On each of the two wires sits a green dot with phase $\alpha/2$, with a third leg, a **tie**, running off the wire. The two ties meet at one phase-free red dot, the **hub**. From the hub hangs a third short leg, the **pendant**, ending in a green point with phase $-\alpha/2$. This shape is the **phase gadget** of van de Wetering's survey (his Eq. (86), and Eq. (162) for exactly this decomposition of a controlled phase):



Result 14.5 (the controlled turn). Exactly, for every angle α, the bare gadget above is $\frac{1}{\sqrt{2}}\,\mathrm{CP}(\alpha)$. It is unchanged when the two wires are exchanged.

Proof. Follow a basis state $|x\rangle\,|y\rangle$. A green dot with phase $\alpha/2$, one leg in, and two out (the wire and the tie) is $Z(1,2,\frac{\alpha}{2})$: it sends $|x\rangle$ to $e^{i\alpha x/2}\,|x\rangle\,|x\rangle$, keeping the bit on the wire and sending a copy up the tie. So the two ties carry x and y, with the factor $e^{i\alpha(x+y)/2}$. The hub, a phase-free red dot with the two ties in and the pendant out, is $X(2,1,0) = |+\rangle\langle{+}{+}| + |-\rangle\langle{-}{-}|$, and on $|x\rangle\,|y\rangle$ it gives $\frac{1}{2}(|+\rangle + (-1)^{x+y}\,|-\rangle) = \frac{1}{\sqrt{2}}\,|x \oplus y\rangle$: it reads the parity. The pendant's green point, met by $|z\rangle$, gives $e^{-i\alpha z/2}$. Altogether

$$|x\rangle\,|y\rangle \;\mapsto\; \frac{1}{\sqrt{2}}\,e^{i\alpha(x+y-x\oplus y)/2}\,|x\rangle\,|y\rangle \;=\; \frac{1}{\sqrt{2}}\,e^{i\alpha xy}\,|x\rangle\,|y\rangle\,,$$

since $x + y - x \oplus y = 2xy$ for bits. That is $\frac{1}{\sqrt{2}}\mathrm{CP}(\alpha)$. The picture has the same shape with its wires exchanged. ■ See `Code/Ch14/KickedTurn.py`, which checks it at $\alpha = \pi/2$ and $\alpha = 3\pi/7$.

The $\frac{1}{\sqrt{2}}$ is the price of the hub, as a bare tie in Chapter 13 cost $\frac{1}{\sqrt{2}}$. As in Chapters 9 to 13, every picture equation in this chapter is exact, and its

scalar is drawn.

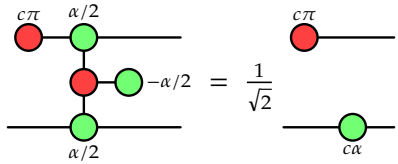
Why this definition. Why draw a controlled phase as a gadget of four dots, when a tall box would do? Because the tall box hides its insides and the gadget does not: every piece of it is a dot, so every rule of Chapter 8 can reach it. The gadget also shows at a glance what the circuit hides, that control and target are interchangeable. Van de Wetering makes the same remark: the symmetry of a phase gadget "is clear in the representation of a phase gadget as a ZX-diagram", but not in a circuit.

A basis point passes through

What does a controlled turn do when one of its wires carries a definite bit? On the blackboard it is immediate. Write $|c\rangle$ with $c \in \{0, 1\}$ and $|\psi\rangle$ for any state of the other wire. Then $\mathrm{CP}(\alpha)(|c\rangle \otimes |\psi\rangle) = |c\rangle \otimes Z(c\alpha)\,|\psi\rangle$: if $c = 0$ nothing happens, and if $c = 1$ the other wire is turned by α.

In the sketchbook, recall the points of Chapters 11 to 13: a red point is $\sqrt{2}\,|0\rangle$ and a red π point is $\sqrt{2}\,|1\rangle$. We call them **dark** and **lit**, and with Chapter 10's bit-times-π spider (row 75) we draw both at once as a red point with phase $c\pi$.

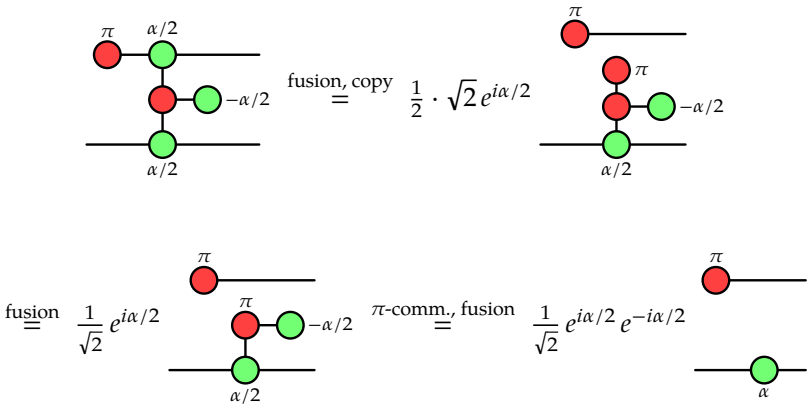
Result 14.6 (the basis-point lemma). Exactly, for every α and $c \in \{0, 1\}$:



A basis point passes straight through a controlled turn and leaves its turn on the other wire: a dark point leaves nothing, a lit point leaves a green α. On the blackboard, $\mathrm{CP}(\alpha)(|c\rangle \otimes |\psi\rangle) = |c\rangle \otimes Z(c\alpha)\,|\psi\rangle$.

The scalars agree: by Result 14.5 the left side is $\frac{1}{\sqrt{2}}\mathrm{CP}(\alpha)$ applied to $\sqrt{2}\,|c\rangle$, which is $|c\rangle \otimes Z(c\alpha)$; the right side is $\frac{1}{\sqrt{2}} \cdot \sqrt{2}\,|c\rangle \otimes Z(c\alpha)$, the same.

Proof, in pictures. Take the lit point, $c = 1$. Copy (Theorem 8.3) needs a phase-free dot, so first run fusion (Theorem 8.1) backwards: the top ring's $\alpha/2$ slides off onto a green point hanging from a new leg, and the ring is left phase-free. Now the lit point enters the ring and is copied onto its three other legs: the wire, the tie and the new leg. The copy on the new leg meets the green $\alpha/2$ point there and the two close into a number. Next, the copy on the tie fuses into the hub, which becomes a red π dot with two legs. Then π-commutation (Theorem 8.4) pushes that π through the pendant's point, whose turn flips from $-\alpha/2$ to $+\alpha/2$, and the hub, now phase-free with two legs, is a plain wire. Last, the pendant's point fuses into the bottom ring: $\alpha/2 + \alpha/2 = \alpha$.



The two turns cancel and the scalar is $\frac{1}{\sqrt{2}}$, as claimed. For the dark point, $c = 0$, the same moves run with nothing to carry: the copy on the new leg closes against the $\alpha/2$ point into the number $\sqrt{2}$ with no turn; the hub, fed a dark point, stays phase-free and has two legs, so it is a plain wire; and the pendant's $-\alpha/2$ fuses into the bottom ring's $+\alpha/2$ to give 0, a plain wire again. The scalar is $\frac{1}{2} \cdot \sqrt{2} = \frac{1}{\sqrt{2}}$ once more. ■ See `Code/Ch14/KickedTurn.py`, which checks the lemma for both points at $\alpha = \pi/2$ and $3\pi/7$, and each scalar of the derivation separately.

So for a phase box the kickback picture is no longer borrowed. A lit point

fed into the wheel's side of a controlled turn is the eigenstate $|1\rangle$ of $Z(\alpha)$, and the lemma proves that its turn lands on the other wire.

Coecke's Sketchbook.

The ledger of the lemma. Every scalar in that derivation can be named. The copy sends one point into a phase-free dot with three other legs, and a point in, three points out, costs $\frac{1}{2}$: $\sqrt{2}\,|c\rangle$ becomes $\sqrt{2}\,|ccc\rangle$, which is $\frac{1}{2}$ times three points of $\sqrt{2}$ each. The copy that meets the $\alpha/2$ point closes into $\sqrt{2}\,\langle c|\,(|0\rangle + e^{i\alpha/2}\,|1\rangle) = \sqrt{2}\,e^{ic\alpha/2}$: the ring's turn goes into the scalar only if the point is lit. Chapter 8 left the scalar of π-commutation unstated, and here it matters, so here it is: $X(|0\rangle + e^{-i\alpha/2}\,|1\rangle) = |1\rangle + e^{-i\alpha/2}\,|0\rangle = e^{-i\alpha/2}(|0\rangle + e^{i\alpha/2}\,|1\rangle)$. The pendant's old turn comes out as $e^{-i\alpha/2}$, exactly cancelling the $e^{i\alpha/2}$ from the closing. Fusion and the removal of a plain two-legged dot cost nothing. So the whole bill is $\frac{1}{2}\cdot\sqrt{2} = \frac{1}{\sqrt{2}}$, for the dark point and the lit point alike. The cancellation is not luck: the gadget splits α into two halves and a correction, and the correction is exactly what the lit point pays back.

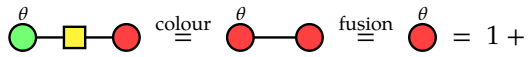
One listener, and its blur

Now read the turn. Put one control wire at $H\,|0\rangle$, apply $\Lambda(U)$ with $|u\rangle$ on the register, apply a Hadamard to the control, and read it. This is Kitaev's measuring operator, his Eq. (16): a Hadamard gate on each side of $\Lambda(U)$. Try two phases. With $\varphi = \frac{1}{4}$ the kickback leaves the control at $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$, and the Hadamard gives $\frac{1}{2}((1+i)\,|0\rangle + (1-i)\,|1\rangle)$: the reading is 0 or 1 with probability $\frac{1}{2}$ each. With $\varphi = \frac{3}{4}$ the control is $\frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$, and the odds are again $\frac{1}{2}$ each. A quarter turn and three quarters cannot be told apart.

Result 14.7 (one listener; Kitaev's Eq. (17)). The reading is 0 with probability $\frac{1}{2}(1 + \cos 2\pi\varphi)$ and 1 with probability $\frac{1}{2}(1 - \cos 2\pi\varphi)$. It depends on φ only through $\cos 2\pi\varphi$, so φ and $1-\varphi$ give the same odds.

Proof. By Result 14.4 the control ends at $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle)$ before the last Hadamard, which turns it into $\frac{1}{2}((1+e^{i\theta})|0\rangle + (1-e^{i\theta})|1\rangle)$. And $|1 \pm e^{i\theta}|^2 = 2 \pm 2\cos\theta$. ■ See `Code/Ch14/KickedTurn.py`.

The picture says the same thing in one line. The control is a green point turned by θ; read it through a yellow box against the dark point. Colour change (Theorem 8.2, row 58) makes the green point red, fusion joins it to the dark point, and a red dot with no legs at all is the number $1 + e^{i\theta}$ (its two branches, with nothing left to agree on, are the numbers 1 and $e^{i\theta}$):



The two points carry $\sqrt{2} \cdot \sqrt{2} = 2$, so the amplitude of the dark reading is $\frac{1}{2}(1 + e^{i\theta})$. For a quarter this is $\frac{1}{2}(1 + i)$ and for three quarters $\frac{1}{2}(1 - i)$: different numbers, the same size. The picture shows why. A lone red dot turned by θ is worth $1 + e^{i\theta}$, and turning it the other way only takes the mirror image of that number.

Kitaev's own way out, in his §3, is to repeat and to vary. Many repetitions estimate $\cos 2\pi\varphi$; substituting iU for U changes the cosine into minus the sine; and the two together fix φ to a constant precision (his Lemma 9). The controlled powers $\Lambda(U^{2^j})$ then sharpen the estimate to precision 2^{-l-2} (his Lemma 10). And when U is a permutation, as the tick box is, he measures φ exactly, by applying Euclid's algorithm to the measured value (his Theorem 1). We will take a different road to the same place, the one in the dialogue: instead of repeating one listener, add a second that ticks twice as fast.

Feynman's Notebook.

Measuring an eigenvalue. A. Yu. Kitaev, of the L. D. Landau Institute for Theoretical Physics in Moscow, posted "Quantum measurements and the Abelian Stabilizer Problem" on 20 November 1995. Its abstract states its aim plainly: "Thus we extend famous Shor's results". The problem of the title includes both factoring and discrete logarithms, and the method, as the abstract says, "is based on a procedure for measuring an eigenvalue of a

unitary operator". That procedure is this section's listener: one control bit, a Hadamard gate on each side of $\Lambda(U)$, read many times. Kitaev's paper is also, as he puts it, "a rather detailed introduction to the theory of quantum computation", with its own models of circuits, reversible computation and precision. Cleve et al. later drew the moral: Shor's algorithm, "in effect, estimates the eigenvalue" of the multiplication box. That box is a permutation, a permutation has eigenvalues that are roots of unity, and the denominator of the root is the length of the cycle. Find the eigenvalue, and the period comes with it.

---

§14.3 Twice the Pace

One listener hears $\cos 2\pi\varphi$ and cannot tell a quarter from three quarters. Here is the way out.

On the blackboard. Give the listener $\Lambda(U^2)$ instead of $\Lambda(U)$. By Result 14.4 it hears the turn 2φ. For $\varphi = \frac{1}{4}$ that is $\frac{1}{2}$. For $\varphi = \frac{3}{4}$ it is $\frac{3}{2}$, which is $\frac{1}{2}$ plus a whole turn, and a whole turn is no turn at all. Both give a half-turn, and by Result 14.7 a half-turn reads 1 with probability $\frac{1}{2}(1 - \cos\pi) = 1$: certainly. A turn of 0 or $\frac{1}{2}$, on the other hand, would read 0 with certainty. So the doubled listener says, with certainty, whether the number of quarters is odd, and it cannot say which odd.

Now correct the single listener. It carries $\frac{1}{4}$ or $\frac{3}{4}$. If the doubled listener read 1, turn the single listener back a quarter: $\frac{1}{4}$ becomes 0 and $\frac{3}{4}$ becomes $\frac{1}{2}$, and a Hadamard now reads the half with certainty. Two readings, both certain, and together they are the turn: *a half or not*, then *a quarter or not*.

In binary this is shifting a point. Write the phase as a binary fraction, $\varphi = 0.j_1 j_2 \cdots j_t = \frac{j_1}{2} + \frac{j_2}{4} + \cdots + \frac{j_t}{2^t}$, so that $\varphi = j/2^t$ for the integer $j = j_1 j_2 \cdots j_t$. A quarter is 0.01 and three quarters 0.11. Doubling moves the point one place right, $2\varphi = j_1.j_2 \cdots j_t$, and the digit that crosses the point is a whole turn, which vanishes.

Result 14.8 (doubling shifts the point). If $\varphi = 0.j_1 j_2 \cdots j_t$, then $\Lambda(U^{2^{t-m}})$ kicks the turn $e^{2\pi i\, 0.j_{t-m+1}\cdots j_t}$ onto its control. The listener at the fastest pace, 2^{t-1}, hears only the last digit j_t, as a half-turn or none.

Proof. $2^{t-m}\varphi = j_1 \cdots j_{t-m}.j_{t-m+1} \cdots j_t$, and the part before the point is a whole number of turns. Result 14.4 does the rest. ■

Each listener hears one digit more than the listener at twice its pace, and the new digit is the one it must be corrected for. The correction for t listeners is a known transform, run backwards.

The quantum Fourier transform

Let $q = 2^t$ and number the basis states of t wires $|0\rangle, \ldots, |q-1\rangle$, reading the bits top wire first. The **quantum Fourier transform** is

$$F_q\,|x\rangle \;=\; \frac{1}{\sqrt{q}} \sum_{y=0}^{q-1} e^{2\pi i x y/q}\;|y\rangle\,.$$

This is Shor's Eq. (4.1), his matrix A_q, with his labels a and c written x and y. For $t = 1$ it is the Hadamard: $F_2 = H$. For $t = 2$,

$$F_4 \;=\; \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}, \qquad F_4\,|1\rangle = \tfrac{1}{2}(|0\rangle + i\,|1\rangle - |2\rangle - i\,|3\rangle).$$

Look at that last column as a state on two wires. It is $\frac{1}{2}(|00\rangle + i\,|01\rangle - |10\rangle - i\,|11\rangle) = \frac{1}{2}(|0\rangle - |1\rangle)(|0\rangle + i\,|1\rangle)$: a product, the top wire turned by a half and the lower wire by a quarter. That is exactly what two listeners hold after the controlled powers for $\varphi = \frac{1}{4}$. The top listener, at double pace, heard $2 \cdot \frac{1}{4} = \frac{1}{2}$; the lower heard $\frac{1}{4}$. The listeners' state is $F_4\,|1\rangle$, and $\varphi = \frac{1}{4}$ is $j/4$ with $j = 1$. It is not a coincidence.

Result 14.9 (the product form; Cleve et al., Eq. (4.2)). For $x = x_1 x_2 \cdots x_t$ in binary, top wire first,

$$F_q\,|x\rangle \;=\; \frac{1}{\sqrt{q}}\,(|0\rangle + e^{2\pi i\, 0.x_t}\,|1\rangle)(|0\rangle + e^{2\pi i\, 0.x_{t-1}x_t}\,|1\rangle) \\ \cdots\,(|0\rangle + e^{2\pi i\, 0.x_1 x_2 \cdots x_t}\,|1\rangle).$$

The top wire carries only the last digit; each wire below carries one more.

Cleve et al. note that the state "is unentangled, and can in fact be factorised". The proof is the Dirac's Blackboard below. See `Code/Ch14/QuarterClock.py`, which checks the product form for $t = 3$ on every basis state.

Dirac's Blackboard.

Factoring the Fourier sum. Write $y = y_1 y_2 \cdots y_t$ in binary, top wire first, so $y = \sum_{m=1}^{t} y_m 2^{t-m}$ and $y/q = \sum_m y_m 2^{-m}$. The phase then splits into one factor per wire:

$$e^{2\pi i x y/q} = \prod_{m=1}^{t} e^{2\pi i\, x\, 2^{-m}\, y_m}.$$

Now $x\,2^{-m} = x_1 \cdots x_{t-m}.x_{t-m+1} \cdots x_t$, and the whole part is whole turns, so $e^{2\pi i x 2^{-m} y_m} = e^{2\pi i\, 0.x_{t-m+1}\cdots x_t\, y_m}$. A sum over all y of a product of one-wire factors is the product of the one-wire sums:

$$\sum_y e^{2\pi i x y/q}\,|y_1 \cdots y_t\rangle = \prod_{m=1}^{t} \Big(\sum_{y_m \in \{0,1\}} e^{2\pi i\, 0.x_{t-m+1}\cdots x_t\, y_m}\,|y_m\rangle \Big),$$

and the factor for wire m is $|0\rangle + e^{2\pi i\, 0.x_{t-m+1}\cdots x_t}\,|1\rangle$. Wire 1 gets $0.x_t$, wire t gets $0.x_1 \cdots x_t$. Check $t = 2$, $x = 01$: wire 1 gets 0.1, a half; wire 2 gets 0.01, a quarter. This is Chapter 13's sign sum again, with the signs ± 1 replaced by turns.

The transform as a network

The product form says how to build F_q. Wire m must end up turned by $0.x_m \cdots x_t$ (before the wires are put in reverse order). A Hadamard on a wire holding $|x_m\rangle$ gives $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\, 0.x_m}|1\rangle)$, since $(-1)^{x_m} = e^{2\pi i\, 0.x_m}$: the first digit. Each further digit $x_{m'}$, from a lower wire $m' > m$ still holding its bit, is added by a controlled turn of angle $\pi/2^{m'-m}$ between wires m and m': it turns the $|1\rangle$ branch by $2\pi\, x_{m'}/2^{m'-m+1}$, which puts $x_{m'}$ in place $m'-m+1$ after the point.

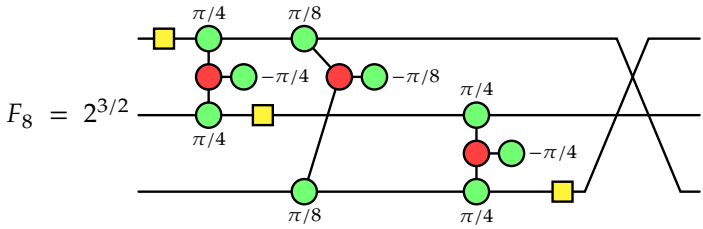
> **Result 14.10 (the transform's network).** For $m = 1, \ldots, t$ in turn, apply a Hadamard to wire m, then a controlled turn $\mathrm{CP}(\pi/2^{m'-m})$ between wire m and each lower wire m'; finally put the wires in reverse order. The result is exactly F_q. Its inverse is the same network run backwards, with every turn negated.

Proof. On $|x_1 \cdots x_t\rangle$, wire 1 after its Hadamard and its controlled turns holds $|0\rangle + e^{2\pi i\, 0.x_1 x_2 \cdots x_t}|1\rangle$ (normalized by $\frac{1}{\sqrt{2}}$), because each lower wire is still in its basis state and a controlled turn fed a basis bit acts on the other wire as $Z(x_{m'}\alpha)$ (the blackboard half of Result 14.6). Wire 1 is never touched again. Wire 2 then gets $0.x_2 \cdots x_t$ the same way, and so on down to wire t, which gets $0.x_t$ from its Hadamard alone. Reversing the order puts $0.x_t$ on top, and that is Result 14.9. The inverse of a product of unitaries is the product of their inverses in reverse order; $H^{-1} = H$, a crossing undoes itself, and $\mathrm{CP}(\alpha)^{-1} = \mathrm{CP}(-\alpha)$. ■ This is the walk Cleve et al. take through their Fig. 5. See `Code/Ch14/QuarterClock.py`, which checks the network against F_q for $t = 2, 3, 4$, and Shor's own gate order as well.

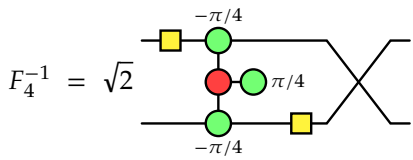
Shor gives the same network in his Eq. (4.4): Hadamards, which he calls R_j, alternating with controlled phases $S_{j,k}$ of angle $\theta_{k-j} = \pi/2^{k-j}$, his Eqs. (4.2) and (4.3). His order interleaves them slightly differently, which changes nothing, since gates on different wires commute. And he deals with the reversal as we have: the network's output has its bits in reverse order, and "to obtain the actual quantum Fourier transform, we need either to do further computation to reverse the bits ... or to leave these bits in place and read them in reverse order". He credits the construction to Coppersmith and,

independently, to David Deutsch, and calls it "essentially the standard fast Fourier transform (FFT) algorithm ... adapted for a quantum computer".

In the sketchbook. Draw every controlled turn as a gadget and every Hadamard as a yellow box (row 56). The network is a triangle: one yellow box per wire, and between each wire and every wire below it a controlled turn whose angle halves with each step down; then the crossings that reverse the wires. Here it is for $t = 3$, with its three bare gadgets paid for by the scalar in front:



The tie from wire 3 up to the middle gadget's hub passes wire 2 without touching it, and so do the two lines of the final crossing, which cross each other once, above wire 2. For $t = 2$ the triangle has one controlled turn. Its inverse is what phase estimation will need: a yellow box on the top wire, the controlled turn $\mathrm{CP}(-\pi/2)$, a yellow box on the lower wire, and the crossing,



exactly. Run backwards, the forward network would start with the crossing and put its yellow box on the lower wire first. But a crossing only exchanges rows, and the yellow boxes and the symmetric controlled turn can slide through it and change rows as they go. So the inverse is the transform's mirror image with every turn negated, redrawn with the crossing moved back to the end: the same shape as the transform.

Why this definition. Why this transform and not some other correction?

Because it is the one whose output is a product of single-wire turns, each holding a tail of the binary expansion (Result 14.9), and a product of single-wire states is what the controlled powers produce. Any unitary that undoes the product would do; the inverse Fourier transform is the one that does it for every x at once, with a network of about $t^2/2$ gates.

Approximately is enough. The controlled turns between distant wires have tiny angles, $\pi/2^{m'-m}$. Shor remarks that multiplying by "a very small phase factor ... would be very difficult to do accurately physically", and that Coppersmith showed they may be left out. Coppersmith's report defines the transform with every controlled turn between wires h or more apart deleted, that is, every angle $\pi/2^h$ or smaller. Each entry of the matrix then differs from the true transform's by a factor $e^{i\epsilon}$ with $|\epsilon| \leq 2\pi t\, 2^{-h}$, and the network needs about th two-wire operations instead of about $t^2/2$. (In the report's letters, L for our t and m for our h.) For $t = 500$ and $h = 20$ he estimates the phase error at about 3/1000: in Shor's factoring it gives "a fraction of a degree in each phase angle", and a decrease of less than one percent in the probability of each desirable final state. We state this and do not prove it.

Phase estimation

Now put the pieces together: t listeners, the controlled powers at doubling paces, and the inverse transform.

> **Result 14.11 (phase estimation on an eigenstate, exact for a dyadic phase).** Let U be unitary, $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ with $|u\rangle$ normalized, and $\varphi = j/2^t$ for an integer $0 \leq j < 2^t$. Write $j = j_1 j_2 \cdots j_t$ in binary, so $\varphi = 0.j_1 j_2 \cdots j_t$, and write $|j\rangle$ top wire first, j_1 on top. Let $q = 2^t$, and let $\Lambda_m(V)$ apply V to the register when counting wire m (from the top) is $|1\rangle$. Then, exactly, scalar included,
>
> $$(F_q^{-1} \otimes I) \prod_{m=1}^{t} \Lambda_m(U^{2^{t-m}})\ (H^{\otimes t}|0\rangle^{\otimes t} \otimes |u\rangle) \;=\; |j\rangle\,|u\rangle\,.$$
>
> Reading the t counting wires returns j with certainty. The Λ_m

commute, so the order of the product does not matter.

The attribution needs care. The idea of measuring an eigenvalue of a unitary with controlled powers $\Lambda(U^{2^j})$ is A. Yu. Kitaev's, from 1995 (his §§3–4). His procedure uses one control bit at a time and repeats it, with classical post-processing (Lemma 10). The one-network form stated here, the controlled powers followed by an inverse Fourier transform, is the presentation of Cleve, Ekert, Macchiavello and Mosca (1997, §5, with their product state (5.1)), which they say was "motivated by the method proposed by Kitaev (1995)". They also show (their §6) that for the tick box it is the same network as Shor's algorithm. Both proofs of Result 14.11 are in §14.7. See `Code/Ch14/QuarterClock.py`, which checks it exactly for $t = 1, 2, 3$, every j, with a random complex unitary U.

The theorem is "exact for a dyadic phase", one of the form $j/2^t$. The phases of the tick box are k/r, which are rarely of that form; §14.4 deals with them. The rounds of the tick box also have a second difficulty: we have no way to prepare them. That is §14.4's business too.

Feynman's Notebook.

Electrons with spins. Don Coppersmith's "An approximate Fourier transform useful in quantum factoring" is an IBM Research Report from the T. J. Watson Research Center in Yorktown Heights, dated June 1994 on its title page. It was posted to the arXiv only in January 2002. Its abstract says the transform "may find application to the problem of factoring integers with a quantum computer as is currently under investigation by Peter Shor". Coppersmith starts from the textbook fast Fourier transform and pictures its passes physically: the amplitudes belong to "L electrons" with spins up or down, and the transform "is performed by a sequence of two-electron interactions". He then makes a pleasant observation. His approximate transform with the smallest threshold, which keeps no controlled turns at all, is the Hadamard transform, suitably indexed; with the largest, it is the full Fourier transform. Chapter 13's layers of Hadamards and this chapter's Fourier

transform are the two ends of one family.

---

§14.4 Every Round at Once

Phase estimation reads the turn of a round. But we do not have a round. Cleve et al. say it bluntly: such a state "is not at all trivial to fabricate". What we can easily make is $|1\rangle$, the hand standing at one.

On the blackboard. Add the four rounds of 7 mod 15. At the hour 1 every round has the phase 1, so the four amplitudes add. At each other hour the four phases are the four powers of a root of unity, and they cancel: at 7 they are $1, -i, -1, i$; at 4 they are $1, -1, 1, -1$; at 13 they are $1, i, -1, -i$. So $|\psi_0\rangle + |\psi_1\rangle + |\psi_2\rangle + |\psi_3\rangle = \frac{1}{2} \cdot 4\,|1\rangle = 2\,|1\rangle$. The hand at one is all four rounds laid together.

Result 14.12 (every round at once). $|1\rangle = \dfrac{1}{\sqrt{r}} \displaystyle\sum_{k=0}^{r-1} |\psi_k\rangle$.

Proof. $\sum_k |\psi_k\rangle = \frac{1}{\sqrt{r}} \sum_s \big(\sum_k e^{-2\pi i k s/r}\big)\,|a^s\rangle$. The inner sum is r when $s = 0$ and 0 otherwise, by the geometric sum in the proof of Result 14.3. So the total is $\frac{r}{\sqrt{r}}\,|a^0\rangle = \sqrt{r}\,|1\rangle$. ■ See `Code/Ch14/EveryRound.py`.

This is Cleve et al.'s Eq. (6.4), which they write as $|1\rangle = \sum_k |\psi_k\rangle$. Their rounds carry no factor $r^{-1/2}$ and run over $k = 1, \ldots, r$; with rounds normalized, as ours are, their sum is $\sqrt{r}\,|1\rangle$, and taken literally in their own normalization it is $r\,|1\rangle$. The statement above is the normalized one.

Now start phase estimation with $|1\rangle$ in the register. The network is linear, so it acts on each round separately, and each round is an eigenstate.

Result 14.13 (a random round is read). Suppose r divides $q = 2^t$. Phase estimation with $U = U_a$, started on $|1\rangle$, ends in

$$\frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} |kq/r\rangle\,|\psi_k\rangle\,.$$

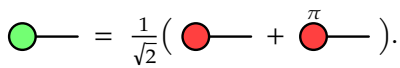
The reading is $c = kq/r$ for a uniformly random k, each with

probability $1/r$, and $c/q = k/r$. For $N = 15$, $a = 7$, $t = 2$: the readings $0, 1, 2, 3$ each occur with probability $\frac{1}{4}$, and exactly two of them, 1 and 3, give $r = 4$. One run finds the period with probability exactly $\frac{1}{2}$.

Proof. By Result 14.12 and linearity, the output is $r^{-1/2} \sum_k (\text{output on } |\psi_k\rangle)$. The phase of $|\psi_k\rangle$ is $k/r = (kq/r)/q$, a dyadic fraction with numerator $kq/r < q$, so Result 14.11 gives $|kq/r\rangle\,|\psi_k\rangle$. The rounds are orthonormal (Result 14.3), so the probability of reading c is the sum over k of $\frac{1}{r}\,|\langle c|kq/r\rangle|^2$, which is $\frac{1}{r}$ when $c = kq/r$. For the dial: $q = 4 = r$, so $c = k$. A reading c suggests the period as the denominator of $c/4$ in lowest terms, and the suggestion is checked by one multiplication. Reading 1 or 3 gives $\frac{1}{4}$ or $\frac{3}{4}$ and the period 4, and $7^4 \equiv 1$. Reading 2 gives $\frac{1}{2}$ and the candidate 2, but $7^2 = 49 \equiv 4$, not 1: the round that turns a half at each tick comes home in two, but the hand does not. Reading 0 gives nothing. ■ See `Code/Ch14/EveryRound.py`, which runs phase estimation with U_7 on the four-wire register for $t = 2$ and $t = 3$.

The half reading fails for an honest reason. It came from the round $k = 2$, whose turn $\frac{2}{4}$ is not in lowest terms: k and r share a factor. A reading gives r exactly when the random k has no factor in common with r.

In the sketchbook. Our small wheel of §14.3 has a picture version of the same fact. Take the one-wire phase box $S = Z(\pi/2)$, the smallest phase box that comes round in four ticks. Its rounds are the basis states: $|0\rangle$ with phase 0 (the dark wheel) and $|1\rangle$ with phase $\frac{1}{4}$ (the lit wheel). A phase-free green point is $|0\rangle + |1\rangle$, which is the two red points added and divided by $\sqrt{2}$:



A green point on the wheel is the equal mix of the wheel's two rounds, and the whole picture splits into two: one with the wheel dark and one with it lit. §14.8 evaluates both halves.

When the phase is not dyadic

For a general dial, r does not divide any power of 2, and k/r is not a binary fraction of finite length. Phase estimation then reads a good approximation with good probability.

> **Result 14.14 (the general phase; Cleve et al., Eq. (5.4); stated, not proved).** If φ is not of the form $j/2^t$, the network of Result 14.11 returns the t-bit fraction nearest to φ with probability at least $4/\pi^2 = 0.405\ldots$

Cleve et al. derive it from a geometric series, their Eqs. (5.2) and (5.3). They add that the success probability can be pushed to $1-\epsilon$ by adding $O(\log(1/\epsilon))$ wires and discarding the extra digits. `Code/Ch14/QuarterClock.py` checks the bound on random phases for $t = 2, \ldots, 6$ as documentation; it is not a proof.

Shor's own analysis, in his §5, is for his version of the circuit, which also reads the second register. In our letters, his outcome is a pair $(c,\ a^s \bmod N)$: his exponent k is our s. He takes q to be the power of 2 with $N^2 \le q < 2N^2$. For an integer z, write $\{z\}_q$ for the remainder of z mod q taken in the range $-q/2 < \{z\}_q \le q/2$.

> **Result 14.15 (Shor's analysis, §5; stated, not proved).** Each outcome $(c,\ a^s \bmod N)$ with $|\{rc\}_q| \le r/2$ (his Eq. (5.11)) has probability at least $1/(3r^2)$, for sufficiently large N; asymptotically the bound is $4/(\pi^2 r^2)$. For such c there is an integer d with
>
> $$\left|\frac{c}{q} - \frac{d}{r}\right| \le \frac{1}{2q} \qquad \text{(his Eq. (5.13))}.$$
>
> Counting the good outcomes, one run yields r with probability at least (the number of $d < r$ with $\gcd(d,r) = 1)/(3r)$, and $O(\log\log r)$ repetitions suffice for a high probability of success.

Here d plays the part of the round index k. The factor that loses to $\log\log r$ is the chance that the random d shares a factor with r, the failure we met at the half reading above. `Code/Ch14/EveryRound.py` checks the per-outcome bound and the success rate on three small instances, $(N,a,q) = (15,7,256)$,

$(21, 2, 512)$ and $(33, 5, 2048)$, as documentation. Shor proves the bounds only for sufficiently large N; they happen to hold on these three.

Why this definition. Why start from $|1\rangle$, when it is not an eigenstate at all? Because it is the simplest state we can prepare that has an equal share of every round and needs no knowledge of r. A superposition of eigenstates, fed to a process that reads eigenvalues, gets one of them read at random, with the weight it carries. Equal weights mean a uniform choice of k, and a uniform k is what makes the good fractions common.

---

§14.5 Euclid's Strips

The quantum part is over. It hands us a reading c out of q, close to d/r for an unknown d. Two steps of ordinary arithmetic remain: from the reading to the period, and from the period to a factor. Both are Euclid.

From a reading to a period

Take $N = 21$ and $a = 2$. Ticking gives $1, 2, 4, 8, 16, 11, 1$, so the order is 6; but suppose we did not know that. Shor's rule gives $q = 512$, since $21^2 = 441 \le 512 < 882$. The fraction $1/6$ of 512 is $85.33\ldots$, so a likely reading is $c = 85$. How do we get from $85/512$ back to a fraction with a small denominator?

Run Euclid's algorithm on 512 and 85: $512 = 6 \cdot 85 + 2$, then $85 = 42 \cdot 2 + 1$, then $2 = 2 \cdot 1 + 0$. The quotients $6, 42, 2$ are the **continued fraction** of $85/512$:

$$\frac{85}{512} = \cfrac{1}{6 + \cfrac{1}{42 + \cfrac{1}{2}}}.$$

Cutting it off after each quotient gives the **convergents** $\frac{1}{6}$, $\frac{42}{253}$ and $\frac{85}{512}$. The only one with a denominator below $N = 21$ is $\frac{1}{6}$, and $|\frac{85}{512} - \frac{1}{6}| = \frac{1}{1536} \le$

$\frac{1}{1024} = \frac{1}{2q}$. So the reading says $d/r = 1/6$, and $r = 6$. One multiplication checks it: $2^6 = 64 = 3 \cdot 21 + 1$.

> **Result 14.16 (from a reading to a period).** (i) If $q \geq N^2$ and $|c/q - d/r| \leq 1/(2q)$ with $r < N$, then d/r is the only fraction with denominator less than N that is this close to c/q. (ii) (*Stated, not proved; Shor §5.*) That fraction is one of the convergents of c/q, which Euclid's algorithm produces in polynomial time.

Proof of (i). Suppose $d'/r' \neq d/r$ also has $r' < N$ and is within $1/(2q)$ of c/q. Then the two are within $1/q$ of each other. But $|d/r - d'/r'| = |dr' - d'r|/(rr') \geq 1/(rr') > 1/N^2 \geq 1/q$, since the numerator is a nonzero whole number. The two cannot both be that close. ■

Shor cites the classical theory of continued fractions for (ii): the expansion "finds all the best approximations of c/q by fractions". Kitaev's Theorem 1 does the same job with Euclid's algorithm applied to his measured value. If d and r share a factor, the fraction in lowest terms gives only a divisor of r; the check $a^r \equiv 1$ catches it, as the half reading was caught in §14.4. In the dialogue the Tortoise supposes a coarser reading, $5/32$ for the same dial, with q smaller than Shor's rule asks; there Euclid gives the quotients $6, 2, 2$, whose first convergent is again $\frac{1}{6}$. Shor's q is what makes such a reading safe in general. See `Code/Ch14/SharedFactor.py`.

From a period to a factor

The period of 7 mod 15 is 4. Half of it is 2, and $7^2 = 49 \equiv 4$. One less than 4 is 3, one more is 5, and $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$: $15 = 3 \cdot 5$. For 2 mod 21 the period is 6, $2^3 = 8$, and $\gcd(7, 21) = 7$, $\gcd(9, 21) = 3$: $21 = 3 \cdot 7$. Here is why this works.

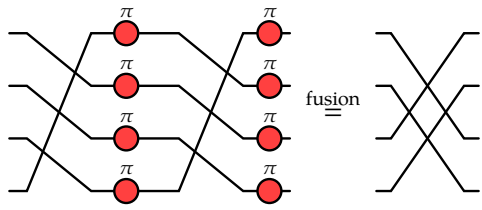
> **Result 14.17 (the reduction from factoring).** Let N be odd and not a prime power, and let a have order r mod N. If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2} - 1, N)$ is a factor of N other than 1 and N.

Proof. $(a^{r/2} - 1)(a^{r/2} + 1) = a^r - 1 \equiv 0$, so N divides the product. N does not divide $a^{r/2} - 1$, or a would come home after $r/2$ ticks, before r. N does

not divide $a^{r/2}+1$, by hypothesis. If $\gcd(a^{r/2}-1, N)$ were 1, then N would have to divide $a^{r/2}+1$; so the gcd is more than 1. And it is less than N, since N does not divide $a^{r/2}-1$. ■ See `Code/Ch14/SharedFactor.py`.

The reduction fails only when r is odd or $a^{r/2} \equiv -1$. Shor shows (his §5, which we state and do not prove) that for a chosen at random among the numbers that share no factor with N, it succeeds with probability at least $1 - 1/2^{\ell-1}$, where ℓ is the number of distinct odd prime factors of N (his k). For $N = 15$, $\ell = 2$ and the bound is $\frac{1}{2}$; in fact, of the eight numbers below 15 that share no factor with it, six split it, and the two failures are $a = 1$ (order 1, odd) and $a = N - 1$ (order 2, and $a^1 \equiv -1$). Shor credits the reduction to earlier work (1976), and says that prime powers, the case excluded above, can be factored efficiently by classical methods.

In the sketchbook. The gcd has no picture, and neither has the continued fraction. The one step of this section that the pictures reach is the squaring. By Result 14.2, C_7^2 is a pure crossing, and it multiplies by $4 = 7^2 \bmod 15$:



exactly, after sliding the four pairs of red πs together along their wires. That is $a^{r/2}$ as a picture, and it is as far as the pictures go. The subtraction of 1 and the greatest common divisor are arithmetic on numbers, not processes on wires.

Dirac's Blackboard.

The same algorithm twice. Euclid's algorithm divides the larger number by the smaller, keeps the remainder, and repeats until nothing is left. Here it is on the two jobs of this section, side by side.

$$\begin{aligned} 512 &= 6 \cdot 85 + 2 \\ 85 &= 42 \cdot 2 + 1 \\ 2 &= 2 \cdot 1 + 0 \end{aligned} \qquad\qquad 15 = 5 \cdot 3 + 0$$

On the left, the quotients $6, 42, 2$ are the continued fraction of $85/512$, and the first of them already says "about one in six". On the right, the last nonzero remainder is the greatest common divisor: 3 measures 15 exactly, so $\gcd(3, 15) = 3$. The first job reads the quotients, the second reads the last remainder, and neither needs anything but division with remainder. Both run in a number of steps proportional to the number of digits, since two steps always at least halve the remainder: if the next remainder is at most half the current one, it has halved already; if it is more than half, it goes into the current one once, and what is left, the difference, is less than half. That is why the classical half of Shor's algorithm is cheap.

The whole algorithm

Here is Shor's algorithm assembled, with each step's owner named.

1. If N is even, or a prime power, split it classically (Shor §5).
2. Choose a at random with $1 < a < N$. If $\gcd(a, N) > 1$, that is already a factor.
3. Choose t with $q = 2^t$ and $N^2 \leq q < 2N^2$. Run phase estimation (Result 14.11) with the tick box U_a, the register started at $|1\rangle$ (Result 14.12), and read c.
4. Expand c/q in a continued fraction and take the convergent d/r with $r < N$ nearest to c/q (Result 14.16). Check $a^r \equiv 1$.
5. If r is even and $a^{r/2} \not\equiv -1$, output $\gcd(a^{r/2} - 1, N)$ (Result 14.17). Otherwise go back to step 2.

Step 3 is the only quantum step. Its controlled powers $\Lambda(U_a^{2^{t-m}})$ are not made by applying the tick box 2^{t-m} times. As Cleve et al. note, the sequence of controlled powers is equivalent to the implementation, "via repeated squarings", of modular exponentiation in Shor's algorithm (Shor §3): $a^2, a^4, a^8, \ldots \bmod N$ are computed classically in advance, and each controlled power is one reversible multiplication.

---

§14.6 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 13's hundred and two. There is one new picture element, the controlled turn (row 103). The other rows are new uses of old elements, and the last row says that there is no picture.

#	Algebra (Dirac)	Picture (Coecke)
103	$\mathrm{CP}(\alpha) = \mathrm{diag}(1,1,1,e^{i\alpha})$, symmetric in its wires	**the controlled turn** — *new element*: a green $\alpha/2$ on each wire, both tied to one red hub, from which hangs a green $-\alpha/2$ point; bare $= \mathrm{CP}(\alpha)/\sqrt{2}$
104	$\mathrm{CP}(\alpha)(\lvert c\rangle \otimes \lvert\psi\rangle) = \lvert c\rangle \otimes Z(c\alpha)\lvert\psi\rangle$, $c \in \{0,1\}$	**a basis point passes through a controlled turn and leaves its turn on the other wire**: dark leaves nothing, lit leaves green α (bare left side $= 2^{-1/2}\times$ right side)

#	Algebra (Dirac)	Picture (Coecke)
105	kickback: $\Lambda(U)(\|c\rangle\|u\rangle) = e^{2\pi i c\varphi}\|c\rangle\|u\rangle$ for $U\|u\rangle = e^{2\pi i\varphi}\|u\rangle$ (row 37, general form)	the tall box $\Lambda(U)$ over $\|u\rangle$ = a green $2\pi\varphi$ on the control wire beside $\|u\rangle$ (for general U, borrowed from the blackboard; for a phase box, row 104)
106	$F_q\|x\rangle = q^{-1/2}\sum_y e^{2\pi i xy/q}\|y\rangle$; product form; network of H and $\mathrm{CP}(\pi/2^d)$, output reversed	**the transform's triangle**: yellow boxes and controlled turns, then the wires crossed into reverse order; the inverse is its mirror image, every turn negated
107	**phase estimation**: the network returns $\|j\rangle\|u\rangle$ exactly for $\varphi = j/2^t$	***The Quarter Clock*** $(t = 2)$: each counting wire hears one digit, at twice the pace of the wire below; the inverse transform's turns peel off the finer digit; the yellow boxes read what is left
108	$U_a\|x\rangle = \|ax \bmod N\rangle$; $\|\psi_k\rangle$, eigenvalue $e^{2\pi i k/r}$; $\|1\rangle = r^{-1/2}\sum_k\|\psi_k\rangle$	the tall box U_a; for $N = 15$, $a = 7$: red πs and crossings (on the residues $1 \le x < N$); a green wheel point is the equal mix of the wheel's rounds
109	continued fractions; $\gcd(a^{r/2} \pm 1, N)$; Shor's bounds	**no picture**: the number theory stays on the blackboard

Every row is exact, scalar included. In row 106, d is the distance between the two wires of a controlled turn. None of these rows is a statement

about running time or success probability: rows 103 to 108 are equalities of processes, and the bounds of this chapter are cited (Results 14.14 and 14.15).

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§14.7 The Theorem, Twice

Result 14.11 (phase estimation on an eigenstate), restated. For $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ with $\varphi = j/2^t = 0.j_1\cdots j_t$, exactly, $(F_q^{-1}\otimes I)\prod_{m=1}^{t}\Lambda_m(U^{2^{t-m}})(H^{\otimes t}|0\rangle^{\otimes t}\otimes|u\rangle) = |j\rangle|u\rangle$, with j_1 on the top wire.

Dirac's Blackboard

Step 1: the Hadamards. Each counting wire goes to $\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)$:

$$H^{\otimes t}|0\rangle^{\otimes t}\otimes|u\rangle \;=\; 2^{-t/2}\bigotimes_{m=1}^{t}(|0\rangle+|1\rangle)\otimes|u\rangle\,.$$

Step 2: the controlled powers. The register stays in $|u\rangle$ throughout, so each $\Lambda_m(U^{2^{t-m}})$ acts only by kickback (Result 14.4), turning the $|1\rangle$ branch of wire m by $2^{t-m}\varphi$. The Λ_m commute: on a basis state of the counting wires, $|c_1\cdots c_t\rangle$, the product applies $U^{\sum_m c_m 2^{t-m}}$ to the register whatever the order, since powers of one U commute. So

$$\prod_m\Lambda_m(U^{2^{t-m}})\;2^{-t/2}\bigotimes_m(|0\rangle+|1\rangle)\otimes|u\rangle \;=\; 2^{-t/2}\bigotimes_{m=1}^{t}(|0\rangle+e^{2\pi i\,2^{t-m}\varphi}|1\rangle)\otimes|u\rangle\,.$$

This is Cleve et al.'s product state, their Eq. (5.1).

Step 3: shift the point. By Result 14.8, $e^{2\pi i\,2^{t-m}\varphi} = e^{2\pi i\,0.j_{t-m+1}\cdots j_t}$. So wire 1 carries $0.j_t$, wire 2 carries $0.j_{t-1}j_t$, and wire t carries $0.j_1\cdots j_t$. By the product form (Result 14.9) with $x = j$, the counting wires are in the state $F_q|j\rangle$:

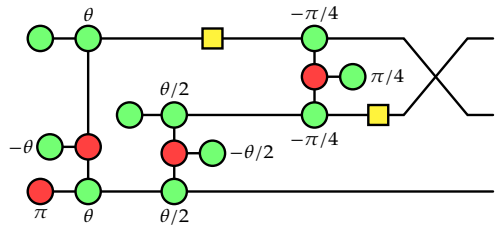
$$2^{-t/2}\bigotimes_{m=1}^{t}(|0\rangle+e^{2\pi i\,0.j_{t-m+1}\cdots j_t}|1\rangle)\otimes|u\rangle \;=\; F_q|j\rangle\otimes|u\rangle\,.$$

Step 4: undo the transform. $(F_q^{-1} \otimes I)(F_q |j\rangle \otimes |u\rangle) = |j\rangle |u\rangle$. The top wire holds j_1, because $|j\rangle$ is written top wire first. ■

Coecke's Sketchbook

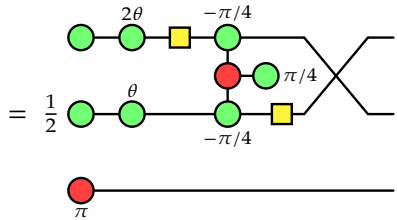
Take $t = 2$, a one-wire register, and the phase box $U = Z(\theta)$ with $\theta = 2\pi\varphi$, whose eigenstate $|1\rangle$ is drawn as a lit point: $\sqrt{2}\,|1\rangle$. Then $\varphi = 0.j_1 j_2$ and $\theta = \pi j_1 + \frac{\pi}{2} j_2$, with the two digits left general. The counting wires start at $H|0\rangle$, which is a green point ($H\sqrt{2}\,|0\rangle = |0\rangle + |1\rangle$ by colour change) with $\frac{1}{\sqrt{2}}$ in front; the two counting points and the lit point together carry $2^{3/2}$. The controlled powers are controlled turns: $\Lambda(Z(\theta)^2) = \mathrm{CP}(2\theta)$ between the top wire and the wheel, and $\Lambda(Z(\theta)) = \mathrm{CP}(\theta)$ between the lower wire and the wheel. The inverse transform is the picture of §14.3. Three bare gadgets carry $2^{-3/2}$, so the drawn picture below is exactly the left side of Result 14.11, with no scalar to add.

Step 0: the circuit, gate by gate.

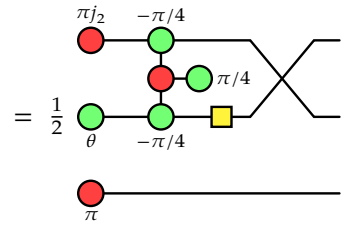


The top row is the double-pace listener, the middle row the single-pace listener, and the bottom row the wheel. The lower listener's point is drawn a little to the right, so that the first gadget's tie passes where its row has not yet begun; only the connectivity matters (row 63).

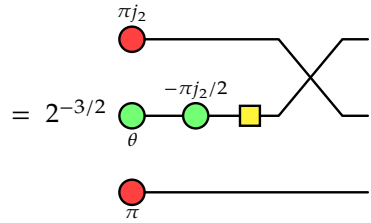
Step 1: the lit point passes both controlled turns (Result 14.6, row 104, twice). It goes straight along the wheel and leaves a green 2θ on the top wire and a green θ on the lower wire; each passage costs $\frac{1}{\sqrt{2}}$:



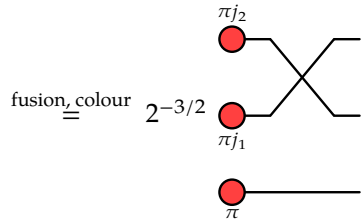
Step 2: fusion, then colour change on the top wire (Theorems 8.1 and 8.2). On the top wire the green point and the green 2θ fuse into a green point with phase $2\theta = 2\pi j_1 + \pi j_2$, which is πj_2, since a whole turn is no turn. A green point turned by πj_2 followed by a yellow box is a red point turned by πj_2: the top listener has become a basis point, dark if $j_2 = 0$ and lit if $j_2 = 1$. On the lower wire the green point and the green θ fuse into a green point with phase θ:



Step 3: the top listener's point passes the inverse transform's controlled turn (Result 14.6 again, now with $\alpha = -\pi/2$). It goes straight on, and leaves a green $-\frac{\pi}{2}j_2$ on the lower wire, at the price of one more $\frac{1}{\sqrt{2}}$. This is the correction: the double-pace listener, having heard an odd number of quarters, turns the single-pace listener back a quarter.



Step 4: fusion and colour change on the lower wire; the crossing (Theorems 8.1 and 8.2). On the lower wire the green point turned by θ and the green $-\frac{\pi}{2}j_2$ fuse into a green point turned by $\theta - \frac{\pi}{2}j_2 = \pi j_1$, and the yellow box makes it a red point turned by πj_1: the second digit has been peeled off, and the first is left.



The crossing only exchanges rows, so the points slide through it: j_1 on top,

j_2 below, the lit wheel at the bottom. That is

$$2^{-3/2}(\sqrt{2}\,|j_1\rangle)(\sqrt{2}\,|j_2\rangle)(\sqrt{2}\,|1\rangle) \;=\; |j_1 j_2\rangle\,|1\rangle\,. \qquad \blacksquare$$

Every step was exact, and the scalar ledger is short: three points give $2^{3/2}$, three passages through controlled turns give $2^{-3/2}$. See `Code/Ch14/QuarterClock.py`.

Two debts must be stated. First, for a general U, the step "the lit point passes through and leaves its turn" is kickback (row 105), and for a tall box that is **borrowed from the blackboard**: the picture proves it only for a phase box. Second, for general t the chain is the same, said in words rather than drawn: the fastest listener hears one digit as a half-turn or none, its yellow box makes it a basis point, and that point passes through every controlled turn of the inverse transform that it meets (one application of Result 14.6 each), peeling its digit off every slower listener; the next listener is then left with one digit, and so on up the triangle.

The verdict

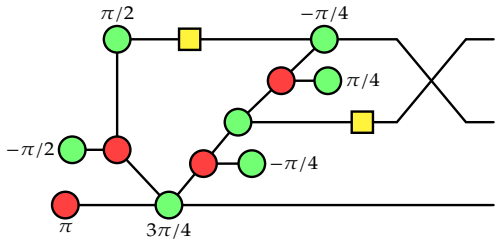
The blackboard wins, and not narrowly. Its four steps prove the theorem for every t and every unitary U, and once the product form is known they are almost bookkeeping. The sketchbook needs five pictures to prove one instance, $t = 2$ and a phase box, general in the two digits; for a general U it borrows kickback from the blackboard, and for general t it describes a chain it does not draw.

What the pictures show is the mechanism, which the blackboard's product form compresses into one line. Each counting wire hears one digit, at twice the pace of the wire below it. The fastest listener's yellow box turns its half-turn into a bit. That bit passes through the inverse transform's controlled turn and peels the finer digit off the slower listener, and the slower listener's yellow box reads what is left. The picture is an equality of processes. It says nothing about how often the whole algorithm must be run, nor how long it takes.

§14.8 The Quarter Clock

Here is the chapter's master diagram, one instance of the headline drawn whole. Take the circuit of §14.7 with $t = 2$, the phase box $S = Z(\pi/2)$ on the wheel, and the wheel's point lit: $\theta = \pi/2$, so $\varphi = \frac{1}{4} = 0.01$, with $j_1 = 0$ and $j_2 = 1$. Put $\theta = \pi/2$ into the first picture of §14.7, and apply **fusion** (Theorem 8.1) everywhere it applies: every pair of green dots joined by a plain wire slides together into one. On the top row, the counting point fuses with its gadget's ring. On the lower row, the counting point fuses with both rings. On the wheel, the two rings fuse. Fusion is exact, so the result is the same process, drawn more compactly. That drawing is the master.

From left to right: on the top row, a green ring with phase $\pi/2$ (the counting point and its half of the controlled U^2), a yellow box, a green ring with phase $-\pi/4$ (its half of the inverse transform's controlled turn), and then the line crosses down to the second output. On the lower row, a phase-free green ring (the counting point, $+\pi/4$ from the controlled U and $-\pi/4$ from the inverse transform, which cancel), a yellow box, and then the line crosses up to the first output. On the wheel, the lit point, a green ring with phase $3\pi/4$ (the wheel's halves of both controlled powers), and the third output. Below and between them, three phase-free red hubs, each with its pendant: the first is tied to the top row's first ring and the wheel ring, with a pendant at $-\pi/2$ (the controlled U^2); the second to the lower ring and the wheel ring, pendant $-\pi/4$ (the controlled U); the third to the top row's second ring and the lower ring, pendant $+\pi/4$ (the inverse transform's $\mathrm{CP}(-\pi/2)$). The picture is called **The Quarter Clock**.



The tie from the top row's first ring runs down past the place where the lower row has not yet begun, and the two counting rows cross once, without

touching, before the outputs.

Result 14.18 (The Quarter Clock). Exactly, with no scalar,

$$\text{The Quarter Clock} \;=\; |0\rangle\,|1\rangle\,|1\rangle\,.$$

The reading 01 says $\varphi = 0.01 = \frac{1}{4}$, and the wheel comes back lit. With the wheel's point dark instead, the value is $|0\rangle\,|0\rangle\,|0\rangle$; with it green, the value is $\frac{1}{\sqrt{2}}(|0\rangle\,|0\rangle\,|0\rangle + |0\rangle\,|1\rangle\,|1\rangle)$.

Proof. Fusion is exact, so The Quarter Clock equals the gate-by-gate circuit of §14.7 at $\theta = \pi/2$, and the chain of §14.7 with $j_1 = 0$, $j_2 = 1$ gives $|0\rangle\,|1\rangle\,|1\rangle$: three points worth $2^{3/2}$ against three bare controlled turns worth $2^{-3/2}$, so no scalar is left. On the blackboard it is Result 14.11 with $U = S$, $|u\rangle = |1\rangle$ and $j = 01$. With the point dark, the wheel is the eigenstate $|0\rangle$ of S, whose phase is 0; the dark point passes every controlled turn leaving nothing (Result 14.6 with $c = 0$), and the chain ends at $|0\rangle\,|0\rangle\,|0\rangle$. With the point green, it is $\frac{1}{\sqrt{2}}$ times the dark point plus the lit point (§14.4), and the value is $\frac{1}{\sqrt{2}}$ times the sum of the two values. ■ See `Code/Ch14/QuarterClock.py`, which builds this picture, evaluates it with its scalar, derives it from the gate-by-gate circuit by fusion, and reduces it by the rules to one point per output, with phase π on exactly the second and third; and `Code/Ch14/EveryRound.py` for the green point.

The phase was chosen so that its reading does not read the same in both directions. Read from the bottom counting wire up, the outputs would say 10, a half; the book's convention, top wire first, gives 01, a quarter, and the code checks that it is this one. The crossing is what makes the convention come out right: it is the reversal of the transform, and it is why the first output shows j_1, the digit that counts halves.

Look at the picture once more with the value in mind. The quarter was in it all along, as the phase $\pi/2$ on the top row's first ring. That ring is the double-pace listener, and its turn is the controlled U^2's half-turn split into halves. The wheel's ring, at $3\pi/4$, holds the wheel's halves of both controlled powers, a quarter-turn's worth from the double-pace gadget and an eighth from the single-pace one. Nothing in the picture ticks four times,

and nothing is repeated. The period of four ticks comes out of one passage, as two binary digits.

With the wheel's point green, the reading is 00 or 01, each with probability $\frac{1}{2}$, and the wheel comes out in the matching round: the reading and the round are correlated. That is Result 14.13 for the smallest wheel with a period of four, with only two of its four rounds present, because one wire holds only two.

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§14.9 What the Picture Leaves to the Blackboard

Take stock. The order r of a mod N is a hidden period (Result 14.1), and for one small dial it can even be drawn (Result 14.2). The tick box has rounds that it only turns (Result 14.3), and a controlled box kicks the turn back onto its control (Result 14.4). For a phase box the controlled box is a gadget of dots (Result 14.5), and a basis point passes through it leaving its turn behind (Result 14.6). One listener hears the turn only as a cosine (Result 14.7); listeners at doubled paces each hear one binary digit more (Result 14.8); and the inverse Fourier transform, a triangle of yellow boxes and controlled turns (Results 14.9 and 14.10), peels the digits apart. Phase estimation reads a dyadic phase exactly (Result 14.11), and The Quarter Clock is one instance of it, drawn whole (Result 14.18). The hand at one is every round at once (Result 14.12), so the reading is the turn of a random round (Result 14.13), and continued fractions and a gcd finish the job (Results 14.16 and 14.17).

Now the honest part. **The diagram shows the architecture; the number theory stays on the blackboard.** The following have no picture in this chapter: the continued fraction, the gcd, the reduction from factoring to order finding, Shor's probability bounds (Result 14.15), the general-phase bound (Result 14.14), and the running time. In Shor's own letters, where n is our N, his quantum factoring algorithm "takes asymptotically $O((\log n)^2(\log\log n)(\log\log\log n))$ steps on a quantum computer, along with a polynomial (in $\log n$) amount of post-processing time on a classical computer". The best classical method he cites, the number field sieve, takes

time $\exp(c(\log n)^{1/3}(\log\log n)^{2/3})$ for some constant c. Every one of those statements was proved by counting, or is cited here and proved by counting elsewhere.

A rewrite proves that two processes are equal. The Quarter Clock proves that one passage of that network returns 01. It proves no running time. It proves nothing about how often the algorithm must be repeated, and nothing at all about whether factoring is hard for a classical computer. Shor calls it "generally thought to be hard", and Cleve et al. say only that no classical polynomial-time procedure is known. As far as this chapter goes, a fast classical factoring method is not known and not ruled out.

There is one more thing the chapter has quietly given up. In Chapters 5 and 13 the oracle was a sealed box, and the whole game was to count its uses. In Shor's algorithm there is no sealed box. The controlled powers $\Lambda(U_a^{2^{t-m}})$ are, as Cleve et al. show, modular exponentiation by repeated squaring, a classical circuit made reversible (Shor §3). Anyone who knows N and a can build it. The quantum speed-up is not a count of queries; it is a claim about circuits, and it rests on the number theory as much as on the pictures.

Coecke's Sketchbook.

What the picture does not count. It is tempting to look at The Quarter Clock and say that the picture shows why the period costs one passage instead of four ticks. It shows something smaller. It shows that this network, set this way, returns this state. Nothing in the rules of Chapter 8 knows how many gates a picture has, or how hard a gadget is to build, or how small an angle a laboratory can turn. The triangle of the Fourier transform looks cheap, and it is, but that was proved by counting its gates, and Coppersmith's approximate version by bounding its errors. A second temptation is subtler. The picture for $N = 15$ and $a = 7$ was a handful of crossings, and one might think that the tick box is always so simple inside. It is not: that picture used the accident that 15 is one less than a power of two. A general multiplication mod N is a long arithmetic circuit, and the pictures of this chapter never open it.

The chapter's question has an answer. To tell how long a thing takes to come round with one look, do not look at where it stands. Look at something the ticking only turns, and listen to the turn with listeners that tick at doubled paces, each hearing one binary digit more than the last. Correct each by the digits already heard, and read them all at once: that is phase estimation, and the correction is the inverse Fourier transform. Start from the hand at one, which is every round at once, and the reading is the turn of a random round; half the time or so, Euclid's algorithm turns it into the period, and the period into a factor. The pictures drew the listening, digit by digit, down to where each digit is heard. The arithmetic around it they left to the blackboard, and so did we.

Chapter 15 asks a different question of a different box. The box marks one item among many, and the task is to find it. The quantum answer is faster than searching the items one by one, but only by a square root, and Chapter 13 has already told us, from BBBV's bound, that for a box of this kind no quantum method does better. The picture that explains the search is not a string diagram at all. It is a picture of a plane, with two lines through it and a state that turns between them, a different kind of diagram for a different kind of speed-up.

Toccata in Square-Root Time

凍星や
折りてひらけば
また高く

iteboshi ya
orite hirakeba
mata takaku

Stars in the hard frost —
each time I fold and unfold,
higher once again.

[*The Tortoise's parlour, the next evening, after supper. The plates are cleared. Achilles's scarf lies folded on the seat of the Tortoise's chair.*]

[*The stove has been fed after supper and burns bright behind its mica, its iron ticking now and then as it heats. On the lower pane the frost ferns have grown back since the morning, and past the black glass above them the lane lies pale under the rime. At its far end the smithy's doors stand open on a square of red, and the glow spills out across the frozen ruts: the smith is working late, as smithies do in a hard frost, when every horse in the parish wants its shoes sharpened for the ice. Through the glass comes his iron, a clink, and a flurry of clinks, and nothing, and a clink again, in no pattern anybody could follow. Achilles sits with his feet to the fender and his hands round his teacup, and the Tortoise's cup keeps warm on the hob beside the pot.*]

Achilles: That was a supper for a hard frost. I shan't need to eat again till spring.

Tortoise: You say that every winter, and every winter spring comes round by breakfast.

Achilles: [*picking up the folded scarf*] And here's my scarf, waiting on your chair like a guest who won't sit down.

Tortoise: I brought it down this morning, as I promised, and the stair groaned at me, as you promised.

Achilles: [*winding it on*] Only once?

Tortoise: Twice. The second time I was carrying my seed-cabinet up.

Achilles: Up? The whole cabinet?

Tortoise: On my back. It's the only way I carry anything. It lived in the cupboard under the stairs, and this morning the cupboard needed the room.

Achilles: For what?

Tortoise: For the clock-case. I went up first thing, and there it was on the table, still standing open, still showing its panes, very pleased with itself. I shut its front and carried it down, and now it stands in the cupboard beside the casket and the jam.

Achilles: Two spent boxes and a jar of jam, in the dark together.

Tortoise: The jam has heard all they have to say by now. And when I went back up with the seed-cabinet, I found the Office had been in the night. On the shelf, where the clock-case used to stand, there's a new case with the Office's label on it.

Achilles: Up your stairs, over the loose tread, and not a groan?

Tortoise: The Office never treads on anything.

Achilles: What does the label say this time?

Tortoise: "For the wisher, top room. With the Office's compliments. Keep in a cool place."

Achilles: Then it's come to the right house.

Tortoise: [*taking the candlestick from the mantel and lighting it at the stove*] Bring my bag, and mind the tread. It's had a busy day.

[*She tops up both cups from the pot and sets them on the flat of her shell, where they ride as level as on a tray. Achilles shoulders the work-bag and grins at them.*]

[*They go up, the candle first. Halfway, the tread groans under Achilles's foot, and he pats the banister by way of apology.*]

[*The stair is narrow and makes its one turn at the landing, and their shadows climb the wall beside them, stretching and shrinking with the candle. Under the candle the treads are all alike, each worn to the same hollow in the middle, and nothing about any of them tells which is the loose one; only a foot finds it. The cups on the Tortoise's shell go up the whole flight without a ripple. On the stair the smithy's clink comes fainter than in the parlour, and no more regular.*]

[*The top room, as it was last night: the small table and two chairs, the one window over the lane, the shelf beside it. Against the wall beside the door stands the Tortoise's seed-cabinet, eight rows of eight small drawers, each with its brass pull. On the shelf stands the old brass lamp, and beside it a low, broad case of dark wood, about as wide as a loaf, with the Office's label on its side. Along its front runs a row of four small drawers with brass pulls; at its right end are two small windows, one above the other; on its top is a little enamel plate. The case is sealed.*]

[*The top room is as small as it was last night, and colder, tucked under the slope of the roof, and it has one of everything but drawers. One door, standing open on the dark of the stairs. One window over the lane, frosted over again since the morning from edge to edge, white ferns lying close against the glass; low on the pane the pair of initials that somebody scratched there long ago with a diamond comes up white among them, as it does on every night of hard frost. One shelf beside the window, where the old brass lamp stands tarnished nearly brown, and beside it the new case, low and dark, in the place where the clock-case stood. Under the shelf the empty butter-crocks keep their row, and among them the tall glass jar. Against the wall by the door the candle finds the seed-cabinet as it comes in and lies evenly across its front, every drawer the same as its neighbour and every pull the same, so that nothing in the light could tell one from another. On the wall facing the*

window hangs a print in a plain dark frame: Escher's Cubic Space Division*, a space divided into cubes by straight beams. The candle, still in the Tortoise's claw, reaches only the lower part of the frame. It was hanging there before the candle came up the stairs, and it will be hanging there when the candle goes down. Through the frosted pane the smithy goes on at the end of the lane, heard and not seen: a clink, a pause, a clink and another, and nothing that could be called a tune.*]

Achilles: [*setting the work-bag down by the table*] So that's the seed-cabinet. It looks like a wall of drawers that somebody has shrunk to fit a room.

Tortoise: Every drawer the same, and a packet of seed in each. I know where the beans are, and the beans know it.

Achilles: It's colder up here than in the lane.

Tortoise: [*setting the candlestick on the table*] Then keep your scarf on and sit. The chair without the wobble is still yours.

[*She lowers herself into the chair with the wobble, and it rocks under her with a small clack. Achilles lifts the cups down from her shell, sets hers at her elbow, and keeps his own in both hands; his breath shows white over it.*]

First Challenge — *Four Drawers, One Wish*

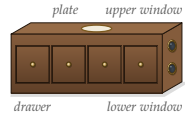
Tortoise: [*taking the candlestick up to the shelf, so that its light falls on the new case*] This is the Genie's drawer-case, and the Office sent it for one wish. Along its front are four little drawers, locked, and at its right end two windows, the upper and the lower. There is nothing on it anyone can set; it hasn't so much as a knob.

Achilles: [*rising, and bending at the shelf to the plate on the top of the case, its drawers towards him and its windows at his right*] And the Terms. [*reading*] "One wish, one touch of the marker. Touched, the marked drawer is turned over, and the others are left as they are. A passage of the case touches the marker once."

Achilles: One of the drawers is marked, then. But turned over how? A drawer turned over would spill.

Tortoise: The Terms don't say. Read on.

Achilles: [*reading*] "The windows name a drawer: both dark, the first; upper dark and lower lit, the second; upper lit and lower dark, the third; both lit, the fourth. A spent case unlocks its drawers. The marked drawer holds the key to its back."



the Genie's drawer-case

Achilles: So one drawer has the key in it, and I get one wish to find out which.

Tortoise: [*bringing the candlestick back to the table and setting it at Achilles's right, a little beyond his place, as he takes his chair again*] That's all the Terms promise. How would you go about it?

Achilles: The plain way. I'd ask about one drawer, the first, say: is it this one? If it is, I'm done, and if it isn't, I guess among the rest.

Tortoise: And how often does that find it?

Achilles: A quarter of the time the first is the one, and I have it at once. The other three quarters, I guess among three and I'm right one time in three, which is another quarter. Half the time, all told.

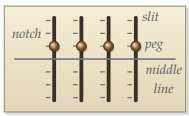
Tortoise: And to be sure?

Achilles: Ask the first, then the second, then the third; if all three say no, it's the fourth. Three wishes, to be sure. And the Terms give me one.

Tortoise: Then the plain way, on one wish, is half the time. Now the Office's way. [*She takes from the work-bag a strip of cream card, with a wooden peg in each of its slits, and a pencil, and lays the card on the table in front of him, the right way round for him: its first slit at his left, and every slit running away from him, so*

that a peg slides up its slit away from him and down towards him. The pencil she lays beside it.] This is my peg-card: four slits in a row, one for each drawer, first to fourth from the left, with a middle line pencilled across them and notches marked above it and below. The peg in each slit slides up or down, and its height, in notches, is how strongly its drawer is being asked about; a peg below the line stands less than nothing. A drawer's odds are its peg's height squared, out of all four squares together.

[*She slides each of the four pegs up its slit to the first notch above the middle line.*]



level across the card

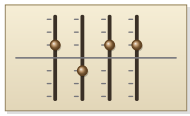
Tortoise: The even row: every peg one notch up. That is every drawer asked at once, evenly, and that is how the Office's marker is touched, on all four drawers at once. It is still one touch.

Achilles: One notch each. One, one, one and one, squared, and a quarter of the odds apiece. Fair.

Tortoise: Now you are the marker, and the second drawer is marked. Do to its peg what the Terms say.

Achilles: Turned over. [*considering the peg*] Then it goes under the line, as far below as it stood above.

[*The four pegs stand in a level row, each one notch above the middle line, and each drawer holds a quarter of the odds. Achilles puts a finger on the second peg alone and slides it down its slit, to the middle line and through it, and on, until it stands one notch below. The first, the third and the fourth have not moved, nor has the line. The second peg is exactly as far from the line as it was, only on the other side: under. Its square is what it was.*]



the marked peg under

Achilles: One, one, one, one. The squares haven't changed.

[*Down the lane, faint through the frost, a hammer gives the anvil a single light tap. Achilles lifts his head, but no blow follows it, and the smithy goes quiet.*]

Achilles: A quarter of the odds on every drawer, just as before. Then a touch on all four drawers at once is wasted. It leaves a mark nobody can see.

Tortoise: Nobody can see it in the odds. But look at the pegs: the second is under, and the others aren't. The touch has left its mark; it's only upside down.

[*Achilles's eyes go to the window. The frost ferns lie white and close against the glass from edge to edge, and from the chair, by candlelight, there is no telling whether they have grown on this side of the pane or on the far side. The Tortoise waits for him to look back.*]

Tortoise: What would you have to do to the pegs, without looking to see which is under, to make that one stand taller than the rest?

Second Challenge — *The Turn-About*

Achilles: If I could look, I'd pull the one that's under straight up. Without looking, I'd have to do the same thing to every peg.

Tortoise: The same thing to every peg, yes, but not the same distance. I call it the turn-about. Add the four heights and take a quarter of the sum, which is their average; then move every peg to the other side of that average, just as far as it stood from it.

Achilles: A peg standing at the average stays where it is, then, and a peg far from it goes far.

Tortoise: Let's play it from the beginning, two hands in turn: you touch, and I turn about.

[*She slides the second peg up again, through the middle line to the first notch above, level with its neighbours: the even row once more, a quarter of the odds on every drawer.*]

Achilles: [*sliding the second peg down through the middle line to one notch under, the others left standing*] The marker turns it over.

Tortoise: And we turn it about.

[*Down the lane the smith's hammer taps the anvil, light, and the striker's sledge comes down after it, heavy, just where the tap fell. Then the smithy is quiet again.*]

Tortoise: First the average.

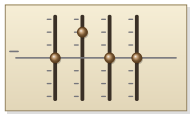
Achilles: One, less one, one and one: two notches in all. A quarter of two is half a notch.

Tortoise: Then every peg goes to the other side of half a notch.

Achilles: The level ones stand half a notch above it, so they go to half a notch below it, which is the line itself.

Achilles: And the one that's under stands a notch and a half below it, so it goes a notch and a half above. Two notches up.

[*The row as it stands: the first, third and fourth pegs one notch up, the second one notch under, and the average, half a notch up, lying between them. Achilles pencils a small tick at half a notch in the card's left margin, beside the first slit, and there it stays. He slides the three level pegs down, one notch each, until each rests on the middle line. Then he takes the second peg from below the line and slides it up, past the line, past his pencilled tick, past the first notch, to the second notch above. The line, the tick and the slits are where they were. What has changed: three pegs that stood a notch up now stand at nothing, and the peg that was under stands higher than any peg has stood tonight. What has held: the average, still half a notch; and the four squares together, still four.*]



turned about,
a tall peg

Tortoise: Now the odds.

Achilles: Nothing, four, nothing, nothing, out of four.

Achilles: ...Oh. All the odds are on the second drawer.

Achilles: The mark nobody could see, and the turn-about has stood it up where anybody could.

Tortoise: The turn-about never looked to see which peg was under. It only took the average of all four, and the average knew.

[*Achilles takes up his cup, finds the tea only warm, drinks, and sets it down harder than he meant. The tea rocks over in the cup: the side that dipped comes up as far above the level as it went below, the side that rose goes down as far, and all the while the middle of it does not move.*]

Achilles: So what you're saying is that the turn-about does the finding, and the touch is only a formality.

Tortoise: Then let it find something without one. [*She slides the three pegs on the line up a notch each, and the tall one down from two notches to one: level again, every peg a notch up, and a quarter of the odds on each.*] The even row, and no touch. Turn it about.

Achilles: One and one and one and one: four, and a quarter of four is one. Every peg stands at the average already. [*He puts a finger to each peg in turn and moves none of them.*] Nothing moves. It found nothing.

Tortoise: It turns an even row into itself. It needs the touch's mark to find anything: two hands.

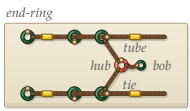
[*Her own cup stands at her elbow where she set it down, gently, and the tea in it lies level and does not stir. She warms her claws round it a moment before she reaches for the work-bag.*]

Tortoise: Now the same turn-about another way. [*She spreads her drawing-cloth beside the peg-card, on its right as Achilles sees it, between the card and the candlestick, and brings out string, rings, beads and yellow tubes.*] My drawing-cloth again, with last night's rings and tubes and last night's rules: what I lay on it is a picture of a thing, not the thing. Two cords, one for the upper pane and one for the lower, and on them a turn-about built of pieces you know: a tube on each cord, a ring at a half on each, the controlled turn for a half between them, and a tube on each again.

[*She lays two cords across the cloth from Achilles's left to his right, one above the other, the upper further from him. Near the left end of each she slides on a yellow tube. Beyond the tubes she threads a green ring on each cord, each from its cord's loose right-hand end, and sets both beads at a half; a bead at rest stands at the top of its ring as Achilles sees it; turned round, it goes clockwise as he sees it, and turned back, the other way. Beyond those she threads another green ring on each cord the same way, its bead a quarter round, and ties each by a tie of its own to a bare red hub set to their right, midway between the cords; from the hub a short pendant runs on to the right to a small green bob, a quarter back. Beyond, a yellow tube on each cord again, threaded on from the loose right-hand end, which runs on a little way past it.*]

Tortoise: The controlled turn for a half, as last night: push a dark end-ring through it and the whole of it falls away; push a lit one through and a half lands on the other cord, as a ring at a half. And a ring at a half lets a dark end pass with nothing set aside, but a lit end sets the half aside, off the cloth.

Tortoise: First the even setting. [*She threads a green end-ring, its bead at rest, on the left end of each cord.*] Green at rest on both cords: every drawer, evenly.



the build, set even

[*The picture as it stands: the two green end-rings at rest at the left; then the tubes, the rings at a half, the controlled turn, and the far tubes. She pushes the upper*

end-ring along its cord into its tube, the stretch of cord it runs along swallowed with it, as every end she pushes tonight swallows the cord it travels; it comes out red at rest: dark. Its bead has not moved; only its colour has. The lower does the same. She pushes the dark upper end on through its ring at a half: the ring is gone, and nothing is set aside. The lower dark end passes its own ring the same way. The upper dark end runs into the controlled turn, and the whole of it falls away, both its rings, the hub, the pendant and the bob; the lower dark end runs on along a cord with nothing left on it. At the far tubes both ends turn green again, at rest, and each comes to rest just past where its far tube stood. What lies on the cloth now: on each cord, just past where its far tube stood, a green end-ring at rest, with only the short loose end of its cord beyond it, bare; and nothing set aside. Everything between has gone, and the setting has come out just as it went in.]

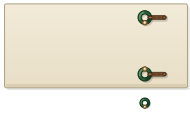
Achilles: It comes out as it went in. Like the even row on the pegs.

[*Down the lane the smithy has gone to work in earnest: taps and blows so close upon one another that nobody at this distance could say which answers which. Achilles blows on his fingers and draws his chair in to the cloth.*]

Tortoise: Now tilt it. [*She gathers the cloth clear, the end-rings and what is left of the cords into her work-bag, and lays the whole build again from the work-bag, piece for piece as before, and this time sets the bead of the upper green end-ring at a half; the lower she leaves at rest.*] Green at a half on the upper cord: that is the tilted setting on that cord.

[*As it stands: the upper green end-ring at a half, the lower at rest, and the build between them. Through the tubes, the upper comes out red at a half, lit; the lower red at rest, dark. The lit upper end goes through its ring at a half: the ring is gone, and a half is set aside: she lays a small green bob from her work-bag, its bead at a half, off the cloth just beyond its near edge at the right-hand end, by the candlestick. The dark lower end passes its own ring with nothing set aside. The lit upper end runs on into the controlled turn: the whole turn vanishes from between the cords, and on the lower cord, where its ring was, lies a single green ring at a half. The dark lower end reaches that ring and passes it, with nothing set aside. At the far tubes the upper end turns green at a half again, and the lower green at rest, each coming to rest just past where its far tube stood. What has held: each cord's setting, exactly as it went in. What has changed: one half lies set aside by the candlestick,*

and a half set aside turns the whole picture over.]



the tilted setting,
pushed through

Tortoise: The same setting, with a half set aside: turned over.

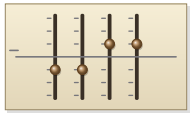
Achilles: The even setting comes out as it was, and the tilted one comes out turned over. But what has a tilted setting to do with my pegs?

Tortoise: The upper cord at a half puts under the line every drawer whose upper pane would be lit. That's the third and the fourth. Set them so.

[*Achilles slides the first and second pegs to one notch up, and the third and fourth to one notch under.*]

Achilles: One, one, under, under. The average: one and one, less one, less one. Nothing. So every peg goes to the other side of the line itself.

[*As it stands: the first and second pegs a notch up, the third and fourth a notch under, and the average lying on the middle line, below his pencilled tick, which still stands at half a notch. Achilles slides each peg straight across the line to the notch opposite: the first and second down to a notch under, the third and fourth up to a notch above. No peg has come nearer the line or gone farther from it, and every square is what it was. Only the sides have changed, every one of them.*]



tilted, then
turned about

Achilles: Under, under, one, one. The whole row turned over, and not one peg taller.

Tortoise: Your arithmetic and my cords do the same thing: an even row they leave alone, and a row that averages to nothing they turn over.

Achilles: Then the Office's case needs only one touch.

Tortoise: For four drawers. Shall we see whether the Office agrees?

Third Challenge — *One Touch*

Achilles: [*looking up at the shelf*] The lamp?

Tortoise: The Genie's lamp, the same as last night. Rub it once, and he comes.

[*Achilles reaches up and takes the lamp down from the shelf. It is as it has stood all evening: brass, tarnished, cold, its wick dry, never lit. He rubs its side once with his palm. The lamp does not warm. From the spout comes a breath of pale mist, like a breath on a frosty morning, and instead of fading it gathers, rises and thickens; and there, between the table and the window, stands the Genie, brushing a little frost from his sleeves. The lamp in Achilles's hands is as cold and dull as before. What is new is a third at the table. Achilles sets the lamp down beside the candlestick.*]

[*While the mist gathered, the candle flame bowed toward the lamp and every shadow in the room leaned after it, as though the light were being drawn in at the spout; now the flame stands up again, and the shadows go back to their places. Achilles's breath shows white at every word, and the Tortoise's; the Genie's does not show at all. Down the lane the smithy is quiet: the iron is back in the fire.*]

Genie: The Office of Wishes. [*He glances along the shelf to the drawer-case.*] Ah. The Office's newest. They are very pleased with it upstairs.

Genie: One wish is one touch of the marker. A passage of this case touches it once, so a passage is one wish. What the case does in a passage besides that touch is the case's business, and costs you nothing.

Achilles: So the case may do as much as it likes inside, and I pay only for the touch. And if I simply asked you about a drawer?

Genie: A plain question is also one touch: ask me whether a drawer you name is the marked one, and I shall put it to the marker myself and bring

you yes or no. That, too, is one wish. Reading windows is not a wish, nor opening a spent case's drawers, nor anything you do on paper, card or cloth. There is one wish this evening.

Achilles: Then I wish to be told which drawer is marked.

Genie: Referred. [*He tilts his head, as though listening up a stairwell, and straightens at once.*]

[*Under the shelf the tall glass jar gives one faint note that climbs and falls back in the same instant, over before anyone could turn to it. Down the lane the smithy stays quiet.*]

Genie: The Office marks drawers; it does not name them. Nothing has been spent.

Achilles: Then I wish the case to touch its marker as often as I like, and all of it to count as this one wish.

Genie: Referred. [*A blink.*]

Genie: The reply has come down from well above the Meta-Genie: every touch is a wish. Your wish is untouched.

Achilles: Untouched! Is that the Office's idea of a joke?

Genie: The Office does not joke. It is exact.

[*Achilles laughs under his breath, and the Tortoise hides her smile in her cup; the Genie waits, exactly. Achilles winds his scarf once more round his neck and holds his hands to the candle.*]

Achilles: Then let me see inside the case first, to know how it's built.

Genie: Referred.

[*Down in the lane, outside the smithy, a waiting horse strikes the ice once with its hoof, and the smithy's own wall gives the sound back before the sound itself has finished. Then the lane is still.*]

Genie: The back opens with the key, and the key is in the marked drawer. The drawers open on a spent case and on no other.

Achilles: Up all those floors and back already.

Genie: It goes up as far as it must, and the answer comes down at once. It always has.

Tortoise: You'll wear the Meta-Genie out.

Genie: Not yet, madam. Not in all the Office's history.

Achilles: [*to the Tortoise*] Then the plain question. We've been marking the second drawer all evening; I could simply ask whether it's the second.

Genie: That is one touch, and so one wish.

Achilles: And if the answer's no, I'm left guessing among the other three. Half the time, all told. [*He shakes his head.*] No. Not the plain question.

Achilles: May I at least bring the case to the table?

Genie: Carrying is not a wish. Keep it level.

[*Achilles lifts the drawer-case from the shelf in both hands and sets it on the table beside the peg-card, just beyond it, its drawers towards him and its windows at his right. It is just as it was on the shelf: the drawers locked, both panes plain, the back sealed.*]

[*Behind him the shelf stands bare, with the clean places in its dust where the lamp and the case stood all day. The Tortoise moves the candlestick a little to make room, and settles back with a clack.*]

Achilles: One passage of the case, then, with everything it does inside. Is that one wish?

Genie: One passage touches the marker once. One touch, one wish.

Achilles: Then I wish it.

Genie: Granted.

[*The case as it stands: the drawers shut and locked, both panes plain, the back sealed, and nothing on it set, for nothing on it can be. At the word nothing on the case moves. Behind the upper window the pane stays plain. Behind the lower a small gleam comes up and stays. Along the front runs a little row of clicks as the drawer locks let go. What has changed: one gleam, and the locks. What has held: the upper pane, plain; the back, still sealed; every drawer still shut.*]



upper plain,
lower gleaming

Achilles: Upper dark, lower lit. The second drawer.

Achilles: The second! I'd have had it from the plain question too.

Tortoise: This time. The case would have had it whichever drawer the Office marked.

[*Down the lane the smithy, quiet since the lamp was rubbed, gives one light tap of the hammer and one heavy blow of the sledge after it, and then the long hiss of hot iron going into water. One heat, and the work is done.*]

[*The drawers stand in their row, shut, their locks let go. Achilles draws out the first towards him by its brass pull: it slides open on a bare wooden floor, with nothing in it, and he pushes it shut again. He draws out the second towards him. On its floor lies a small brass key.*]



the key, found

[*He takes the key up. Nothing else on the case has stirred: the upper pane is plain and the lower pane gleams, as before; only the second drawer stands open, and its key is in his hand.*]

Achilles: [*holding up the key*] The key to its back.

Genie: The case is spent. A drawer found is no longer a secret, and the back may be opened.

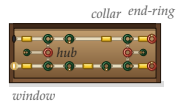
[*Achilles swings the case round on the table so that its back faces him and the candle at his right, fits the key to the small lock in the back panel, and turns it. The panel comes away in his hands, and he lays it on the table beside the case, the key still in*

its lock. The panes do not change, nor the drawers; only the back is open now, and the candle shines into the case.]

[*Inside, two rows of cord run the length of the case, one above the other, from its left end, where they begin, to its right end, where the windows are; seen from behind, as Achilles now sees them, they begin at his right and run to the windows at his left, and every bead in the case reads as he sees it from behind, at rest at the top of its ring and turned round clockwise. They run straight, and neither crosses the other. The upper row begins at a red end-ring whose bead is at rest: dark. From it a cord runs through a fixed yellow collar to a green ring with its bead a quarter round, the marker's upper ring, knotted at the end-ring and at the ring. From the marker's upper ring another cord runs through a collar to a green ring at a half, the turn-about's upper half-ring, knotted at both. From that half-ring a short cord, with no collar on it, runs to the turn-about's upper ring, a quarter round. And from that ring a cord runs through a collar to the upper window, knotted at the ring and at the window.*]

[*The lower row begins the same way, at a red end-ring at rest, and a cord runs from it through a collar to the marker's lower ring, a quarter round. But from there a short cord, with no collar, runs to a green ring at a half that has no partner on the upper row: the marker's half-ring. From the marker's half-ring a cord runs through a collar to the turn-about's lower half-ring, at a half; from that, a short cord to the turn-about's lower ring, a quarter round; and from that ring a cord through a collar to the lower window. Each length is knotted at both its ends.*]

[*In the gap between the rows, nearer the start, sits the marker's hub, a bare red ring. A tie runs up from it to the marker's upper ring, and another tie down to the marker's lower ring; and from the hub a pendant lies out along the gap, towards the end-rings, to a small green bob, a quarter back. Nearer the windows sits the turn-about's hub, bare too, with a tie up to the turn-about's upper ring, a tie down to the turn-about's lower ring, and a pendant along the gap, towards the windows, to its own bob, a quarter back. No tie crosses a row, and no pendant crosses either. Every length in the case is knotted at both its ends, and no cord passes through a ring. What has changed is only that the inside can be seen: the lower pane's gleam can be followed back, cord by cord, to where it began.*]



the inside,
from behind

Achilles: The marker's pieces aren't the same on the two rows. The lower row has a ring at a half there, and the upper row hasn't.

Tortoise: And the far end of it is my build from the cloth: collars for my tubes, a half on each cord, the controlled turn for a half, and collars again.

Achilles: The marker near the start and the turn-about near the windows, and one passage runs through both.

Genie: The wish has been spent in accordance with the Terms. The Office does not collect spent boxes.

[*The lamp lies where Achilles set it beside the candlestick, cold. The Genie draws in from his edges, as the mist on a pane draws in when the warmth leaves it, thins to the pale breath he came as, and the breath runs back into the spout and is gone. The lamp does not stir or warm. The case stays on the table with its back open, its upper pane plain and its lower pane gleaming, and nothing else in the room has moved. There are two at the table again.*]

Tortoise: [*taking up the lamp and setting it back on its shelf*] Home again.

[*Up on its shelf, out of the candle's reach, the lamp is only brass again, beside the clean place in the dust where the case stood all day. The Tortoise sits back down, and her chair clacks. Down in the lane the new-shod horse stamps on the frozen ruts, and somebody speaks to it low.*]

Achilles: [*looking from the open case to the seed-cabinet by the door*] Four drawers, one touch. But the Genie said every touch is a wish. What would a case with as many drawers as your seed-cabinet need?

Tortoise: More than one, and not as many as you'd think. How many would you guess?

Fourth Challenge — *The Turning Plane*

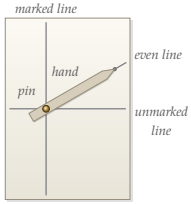
Achilles: Four drawers took one touch, and your cabinet has sixteen times as many drawers. Sixteen touches, drawer for drawer.

Tortoise: Let's see, on paper, where touching costs nothing. [*From the work-bag she brings a sheet of paper, her rule marked in thumbs, a needle, and the paper hand from last night's first dial, still on its pin.*] This sheet is a picture of the whole case at once. A line across it will stand for every unmarked drawer together, a line up it for the marked drawer, and last night's paper hand, on its pin where they cross, for where the case stands. The higher its tip above the line across, as a part of the hand's length, the taller the marked drawer's peg.

Tortoise: Here the touch and the turn-about are both folds. For a fold I pencil a dot at the tip, lift the hand off with its pin, fold the sheet along a line, prick through the dot with the needle, and unfold; then the pin goes back where it was and the hand is laid out to the new hole.

[*She slides the peg-card along to Achilles's left and lays the sheet on the table before him, where the card lay, between him and the case, square to his edge of the table, its long sides running away from him. She draws a line straight up it, away from him, nearer its left edge than its right, and a line across it, half way up, crossing at right angles, and pushes the pin through the crossing. The hand, eight thumbs long, turns on the pin. She lays it out to the right of the pin as Achilles sees it, so that its tip stands four thumbs above the line across, measuring with her rule, pencils a mark at the tip, and rules a third line from the pin through the mark, tilted up from the line across by a third of a right angle, rising to the right.*]

Tortoise: For four drawers. The line across is the unmarked line, the line up is the marked line, and this tilted one is the even line. The tip stands four thumbs up, half the hand's length.



the sheet ruled,
the hand laid

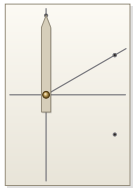
Achilles: Half the hand, and half squared is a quarter. A quarter of the odds on the marked drawer, as the even row gave it.

Tortoise: The marker's fold is along the unmarked line: it turns the marked part of the hand over and leaves the rest alone. The turn-about's fold is along the even line. Watch both.

[*The hand as it stands: on its pin at the crossing, lying along the even line, its tip four thumbs above the unmarked line. The Tortoise pencils a dot at the tip and lifts the hand off the sheet, pin and all. She folds the sheet along the unmarked line, creasing it with a thumbnail, and pushes the needle through the dot and through the paper beneath. She unfolds the sheet flat: a second hole shows below the unmarked line, just as far below it as the dot is above. She puts the pin back through its hole at the crossing and lays the hand out from it to the new hole. The hand is the same length and turns on the same pin, and it reaches just as far along the unmarked line as it did. Only its height has gone to the other side: the tip stands four thumbs under.*]

Achilles: Four thumbs under. The second peg, under the line.

[*The tip stands four thumbs under the unmarked line. The Tortoise dots it and lifts the hand, pin and all. This time she folds along the even line, which runs up from the crossing at its tilt; the dot below the unmarked line comes over onto the paper high above, and she pricks through. Unfolded, the new hole stands on the marked line itself, near the top of the sheet. The pin goes back through the crossing, and she lays the hand out to the new hole: straight up the marked line. Its length has not changed, nor its pin. But the tip, that stood four thumbs under, now stands eight thumbs up, the hand's whole length.*]



after the folds, upright

Achilles: [*laying her rule from the unmarked line to the tip*] Eight thumbs of eight. All the odds on the marked drawer.

Tortoise: That is what the pegs did.

Achilles: ...Oh. Two folds, and the hand isn't folded at all; it has turned round the pin. It lay a third of a right angle up from the unmarked line, and now it stands at a right angle to it.

Tortoise: Two folds make a turn: twice the tilt between the lines you fold along, towards the marked line.

Tortoise: Now your sixteen times as many. [*She lifts the hand off the first sheet, pin and all, and sets that sheet aside on the table. She takes a fresh sheet and lays it where the first lay, square to Achilles's edge of the table, draws the unmarked line across it and the marked line up it at the same places as on the first, and pins the hand at the crossing. She lays the hand out to the right of the pin, so that its tip stands one thumb above the unmarked line, marks the tip, and rules the even line from the pin through the mark: a tilt so slight that the two lines hardly part.*] A case with as many drawers as my seed-cabinet. Asked evenly, the marked drawer's peg is one eighth of the whole: one thumb of eight.

Achilles: One thumb up, and one in sixty-four of the odds. The even line hardly leaves the unmarked line at all.

[*Achilles looks up from the sheet, past the candle, to the print on the wall facing the window: Escher's* Cubic Space Division.]

Achilles: A space divided into cubes by straight beams. How many drawers would a case need, to give each of those cubes a drawer of its own?

[*As he looks, the candle flame draws up tall and perfectly still, and for that moment its light reaches the print from the bottom of the frame to the top, as it has not all evening. Then the flame settles, and the upper part of the frame goes back into shadow.*]

Tortoise: More than my cabinet, whatever it is. Let's learn what mine needs first. Two hands in turn: you take the marker's fold, and I'll take the turn-about's.

[*Down the lane the smithy takes up its work again, steady now: the smith's light tap, and the striker's heavy blow after it, and a little pause while the iron is turned on the anvil, and tap and blow again. Achilles draws his chair in to the sheet.*]

[*The hand as it stands: along the even line, its tip one thumb up. Achilles pencils a dot at the tip, lifts the hand with its pin, folds the sheet along the unmarked line, pricks through the dot, and unfolds. The new hole lies one thumb under; he pins the hand back and lays it out to it. The same hand, the same pin, the same slight angle to the unmarked line; only the side has changed.*]

Achilles: The marker turns it over.

[*The Tortoise dots the tip, one thumb under, lifts the hand, and folds along the even line. Pricked and unfolded, the new hole stands well above the even line, and the hand laid out to it has swung up round the pin, farther from the unmarked line than it has been yet.*]

Tortoise: And we turn it about.

Achilles: [*laying the rule from the unmarked line up to the tip*] Just under three thumbs.

Achilles: [*folding along the unmarked line, so that the tip that stood just under three thumbs up now stands as far under*] The marker turns it over.

Tortoise: [*folding along the even line; the hand swings up past the hole where it stood before*] And we turn it about.

Achilles: Four and three quarters.

Achilles: [*folding the tip's height under the unmarked line, four and three quarters below it now*] The marker turns it over.

Tortoise: [*folding along the even line; up the hand comes again, steeper than before*] And we turn it about.

Achilles: Six and an eighth. [*He looks at the holes marching round the sheet.*] It turns the same each time, but the tip climbs less.

Tortoise: It's nearing the top, and near the top a turn carries it more across than up.

[*Down the lane the tap and the blow keep their turn, the same each time, but under them the ring of the iron grows thinner and brighter as the work comes near its shape. The Tortoise holds the sheet flat with one claw and warms the other at the candle.*]

Achilles: [*folding; the tip goes down to six and an eighth under, the hand now pointing steeply below the line*] The marker turns it over.

Tortoise: [*folding; the hand comes back up, higher than it has stood yet*] And we turn it about.

Achilles: Seven and a quarter.

Achilles: [*folding; the long hand swings down steeply below the line, seven and a quarter under*] The marker turns it over.

Tortoise: [*folding; the hand rises to stand a little short of the marked line*] And we turn it about.

Achilles: Seven and seven eighths.

[*The hand as it stands: a little short of the marked line, its tip seven and seven eighths up. Achilles dots the tip, lifts the hand with its pin, folds along the unmarked line and pricks. Unfolded, the new hole lies seven and seven eighths below, and the hand, pinned back and laid out to it, points nearly straight down.*]

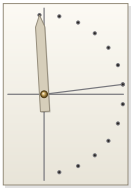
Achilles: The marker turns it over.

[*The Tortoise dots the tip and lifts the hand, folds along the even line, and pricks. Unfolded, the new hole stands at the top of the sheet, just on the far side of the marked line from the even line, to its left as Achilles sees it. She pins the hand and lays it out: it stands upright, leaning by a whisker past the marked line. The same*

hand, the same pin, the same turn as at every step before; and the tip has climbed its last little way, to within a hair of the hand's whole length.]

Tortoise: And we turn it about.

Achilles: [*measuring*] All but a hair of eight. Just past upright.



climbed to the top

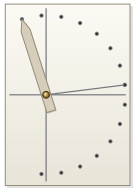
Tortoise: Stop there. Six touches.

[*Down the lane the smith lets his hammer ring on the bare face of the anvil, bright and empty, and the sledge does not fall again. Then comes the long hiss of the finished iron going into the water, and the smithy is quiet.*]

Achilles: It's still a hair short of eight. One more, and it'll be surer.

Tortoise: [*with perfect courtesy*] By all means.

[*The hand as it stands: upright, leaning a whisker past the marked line, its tip all but a hair of eight. Achilles dots the tip, lifts the hand, folds along the unmarked line and pricks. Unfolded, the new hole lies far down, just past straight below the pin, and the hand laid out to it points almost at his knees. The Tortoise takes her turn along the even line, and the new hole comes up high again, but on the far side of the marked line, to its left, and farther from it than before; laid out to it, the hand leans over the marked line towards the far edge of the sheet. Its length is the same, the pin is the same, and the turn has been just as large as every other. But the tip, that had all but reached eight, has come down.*]



turned too far

Achilles: [*measuring*] Seven and five eighths. It's come down!

Tortoise: It goes round. Too many, and it turns away.

[*Achilles sits back and regards the sheet as though it had been rude to him, and the Tortoise's chair clacks as she sits back too. Down the lane the smithy has not struck again.*]

Achilles: Seven and five eighths: about nine in ten of the odds. And what did six give?

Tortoise: All but about three in a thousand. Nearly certain, not certain. Of all our cases, only the four drawers came out certain, because there the two folds land exactly on the marked line. Four drawers were a lucky fit.

Achilles: Six touches, where the plain way wants up to sixty-three asks. And I guessed sixteen.

Tortoise: Once the case is big, four times the drawers takes about twice the touches. It goes by the square root of the drawers, not by the drawers.

Achilles: And a million drawers?

Tortoise: About eight hundred touches, this way of searching. Asking plainly, up to a million less one.

Achilles: Six wishes, then, for your cabinet. Every touch is a wish, and the Office grants one an evening.

Tortoise: So we suppose on paper, and wish nothing.

[*Achilles's eye goes to the lamp on its shelf, out of the candle's reach, and stays there a moment: only brass. He tucks his hands under his arms.*]

Achilles: [*frowning at the sheet*] But six is only how we folded. A cleverer way of touching might need fewer.

Tortoise: How far could any touch move the hand, do you think, if the drawer it touches is one you've hardly asked?

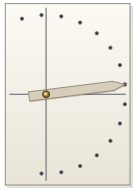
Fifth Challenge — *No Faster Hand*

Achilles: Not far, I suppose. But how far is not far?

Tortoise: First, where the hand would be with nothing to find. [*She lifts the hand from where the seventh step left it and pins it back as it began, its tip at the first dot, one thumb up on the even line.*] Suppose no drawer is marked at all.

Achilles: Then my fold has nothing to turn over. I make none.

[*The hand as it stands: along the even line, its tip one thumb up. The Tortoise dots the tip, lifts the hand with its pin, and folds along the even line. The dot lies on the crease itself, and the needle goes through it and through nothing else. Unfolded, there is no new hole: the prick is the dot. She pins the hand back at the crossing and lays it out to that same hole. Nothing has changed: not its length, not its tilt, not its tip.*]



unmarked, the hand stays

Achilles: It hasn't moved. It can't; it lies along the very line you fold on.

Tortoise: With nothing marked, the hand never leaves the even line, however often we fold, and every peg stays at an eighth of the whole.

Tortoise: Now the rule. A touch can pull the hand away from where it would stand with nothing marked by at most twice the height of the marked drawer's peg, as that peg stands in the unmarked case when the touch comes.

Achilles: Every peg there stands an eighth. So each touch pulls it at most a quarter of a hand.

Achilles: [*laying her rule from the tip on the even line to the top of the marked line*] And from where it stands unmarked to the top of the marked line is ten and a half thumbs, and a little over. A little under one and a third hands.

Tortoise: So how many touches, at the very least?

Achilles: Four touches, one hand: not enough. Five, one and a quarter hands: still short. Six, one and a half: enough. That's the six we found.

Tortoise: That's for touching evenly, as the Office's case does. Now touch any way you please: whatever you do between touches, the pegs' squares always make one whole, so each touch shares out one whole portion of asking among the drawers. After two touches, two portions among sixty-four drawers, and some drawer has had no more than two sixty-fourths.

Achilles: Split evenly, that's a sixty-fourth at each touch: an eighth of height, and a quarter of a pull, twice over. Half a hand at most.

Tortoise: And split unevenly it's less. So that drawer's touches can have pulled the hand at most half its length from where it would stand unmarked: a drawer asked so little can hardly be told from no drawer at all. The people who counted touches proved it properly. I've only shown you why it's believable.

[*Across the room the one candle's light lies spread over the whole front of the seed-cabinet, and each drawer has so small a share of it that its face can hardly be told from the dark. No drawer has more of it than its neighbour; there is only the one flame to go round.*]

Achilles: Then with a million drawers, even the Office's cleverest case needs hundreds of touches, not a handful. And every cord added to a case doubles its drawers, and the search still grows.

Tortoise: Yes. A great saving, and not a miracle.

Tortoise: So what has the evening done?

Achilles: A touch puts the marked drawer under the line, where no look can see it. The turn-about stands every peg on the other side of the average, and that makes the one that's under tall. Together they turn the hand by twice the tilt.

Tortoise: Four drawers take one touch, and it's certain.

Achilles: Sixty-four take six, nearly certain, and a seventh spoils it. And no way of touching does much better.

Tortoise: The cords showed how the case was built, and that one touch finds one drawer in four. How many touches a bigger case needs, and that nobody can do better, was reckoned on paper, not by drawing.

Achilles: [*looking at the candle*] The candle's burned lower than I thought.

Tortoise: It always does when there's arithmetic. It likes to listen.

[*Achilles laughs, and the Tortoise laughs under her breath with him. Down the lane the smithy's doors bang shut for the night, and a horse goes home past the house, its new-sharpened shoes biting the ice at every step, sure-footed, until it is out of hearing.*]

Achilles: [*looking into the open case*] And the lower pane is still gleaming.

Tortoise: It'll go on showing the second drawer until somebody asks it something else, and nobody will.

Achilles: [*at the window*] There's frost on the inside of the pane.

Tortoise: A hard one. Keep your scarf on for the lane, and don't give it to me tonight; I have a cabinet full of seeds to keep warm, and they keep me.

[*Achilles comes back from the window, winding his scarf tighter, and holds his hands to the candle one last time. The Tortoise drains her cup, cold as it is, and neither of them is in any hurry.*]

[*Up the lane, the church clock begins to strike, and strikes ten.*]

Achilles: Ten already. And the tread will be waiting for me on the way down.

Tortoise: Step over it if you like. It groans at people who step over it; it thinks they're being clever.

Achilles: [*looking from the sheet, with its holes marching round the pin, to the seed-cabinet's rows of brass pulls*] So, for any wall of drawers, however big: how many times must you ask to find the one that's marked, and why can nobody ask fewer?

Tortoise: That's the question for any wall. Tonight we only tried it on mine.

[*She takes up the candlestick. The case stays open on the table, its lower pane still gleaming.*]

[*The candle stands low in its stick, guttered down to a small flame, and as she lifts it every shadow in the room swings round and shortens. Achilles sets the two empty cups on the flat of her shell, where they ride as level as they came. As she passes the table the candle reaches just inside the open case, and moves on; the lamp stands on its shelf, only brass; the peg-card, the cloth and the sheets lie where they were left. On the wall facing the window, past her shoulder, the print is at the edge of the light again, a plain dark frame on a cold wall, and then behind it. The light goes once more across the front of the seed-cabinet, every drawer alike, and leaves it. Then the candle goes to the door, and Achilles after it, and the stair takes them down its one turn; halfway, Achilles steps carefully over the loose tread, and it groans at him anyway. Behind them the top room is dark and still, and on the inside of the pane the frost goes on with its work, the old initials white in it. Down the lane the smithy is dark, its fire banked for the morning, and up the lane the church clock goes on round towards eleven.*]

Chapter 15: Searching in Square-Root Time

— Toccata in Square-Root Time —

Part IV: Algorithms

Escher: "Cubic Space Division" — a space divided into cubes by straight beams

---

If one drawer in a wall of identical drawers holds what you are looking for, how many drawers must you open to find it, and could any trick at all open fewer?

Picture the wall. Every drawer has the same brass pull, and nothing is written on any of them. One of them holds a key, and the rest are empty. There is no order to go by: the key was not put in alphabetically, or by size, or by date. So you open drawers. Each one tells you only "yes, here it is" or "no, not here", and a "no" tells you nothing about any other drawer. With luck you find the key in the first drawer you open. Without luck you open every drawer but the last, and then you know. On average you open about half of them.

Nothing cleverer seems possible, and for an ordinary searcher nothing cleverer is. The drawers share no structure that a clever order could exploit. Any plan, however ingenious, comes down to opening drawers one at a time, and a plan that opens few drawers usually misses.

A quantum searcher is allowed a stranger kind of touch. It can touch every drawer at once. But the touch does not open anything. It leaves behind a mark that no look can see: every drawer is still exactly as likely to be the one as any other. And yet something has changed, and a second operation, which knows nothing about the key, can turn that invisible mark into a

visible lean towards the right drawer. Repeat the pair of operations and the lean grows. For a million drawers, about eight hundred such touches find the key almost certainly, where opening drawers one by one takes half a million on average.

Two facts make this chapter worth reading, and they pull in opposite directions. The first is that the speed-up is real: the square root of the number of drawers, not the number itself. The second is that it is not larger. A square root is all any quantum method can gain at this task, and there is a proof. We will see how the touches work, why they must be counted and stopped at the right moment, and why nobody can do better.

---

§15.1 Four Drawers, One Wish

Chapter 14 ended by pointing here. "The box marks one item among many, and the task is to find it. The quantum answer is faster than searching the items one by one, but only by a square root, and Chapter 13 has already told us, from BBBV's bound, that for a box of this kind no quantum method does better. The picture that explains the search is not a string diagram at all. It is a picture of a plane, with two lines through it and a state that turns between them, a different kind of diagram for a different kind of speed-up."

That is true, and it could mislead. The book's two strands still run in this chapter, as in every chapter. The algebra and the string diagrams prove the chapter's headline theorem, and the string diagrams derive a whole four-drawer search, dot by dot, with every scalar. What they do not do is count steps. The picture of a plane is a third picture, which we will call the *Turning Plane*. It explains how many steps a larger search needs, and it is a drawing of algebra, not a diagram of processes. We will keep the three apart.

The chapter's question, in the form we can answer: **how many times must you ask to find one marked item among N, and why can no method ask fewer than about $\sqrt{N}$ times?**

On the blackboard. Take n wires, which this chapter calls **cords** as the dialogues do, and let $N = 2^n$. Each **drawer** is an n-bit string x, written top cord first. Exactly one drawer w is **marked**. We write $w_1 w_2 \cdots w_n$ for its bits and $\bar{w}_i = 1 - w_i$ for the opposite bit. The only way to learn anything about w is to **ask**, and an ask is one evaluation of the test

$$f(x) \;=\; [\, x = w \,],$$

using the bracket of Chapter 12: $[\, x = w \,]$ is 1 if $x = w$ and 0 otherwise. A classical method asks about drawers one at a time, possibly choosing each drawer from the answers so far and possibly tossing coins, and at the end **names** a drawer. It succeeds if the drawer it names is w.

Take four drawers, $00, 01, 10, 11$, and ask in that order. If $w = 00$, one ask finds it. If $w = 01$, two asks. If $w = 10$, three. If $w = 11$, three asks also suffice, because three answers "no" leave only one drawer. Averaged over the four choices of w, that is $(1 + 2 + 3 + 3)/4 = 2.25$ asks.

> **Result 15.1 (asking one by one).** Ask about the drawers in a fixed order, stopping at the first "yes", and name the last drawer without asking if every earlier answer was "no". Then the worst case takes $N - 1$ asks, and for a uniformly random marked drawer the expected number of asks is
>
> $$\frac{N-1}{2} + \frac{N-1}{N}.$$
>
> For four drawers this is 2.25, and for sixty-four it is 32.48.

Proof. If w is the j-th drawer in the order and $j \le N - 1$, the method asks j times. If w is the last drawer, it asks $N - 1$ times and names the last one. Each position has probability $1/N$, so the expected number is $\frac{1}{N}(1 + 2 + \cdots + (N-1) + (N-1)) = \frac{N-1}{2} + \frac{N-1}{N}$. ■ See `Code/Ch15/FourDrawers.py`.

Could a cleverer plan do better? Here is the most modest alternative: ask once, about a fixed drawer, and if the answer is "no", guess uniformly among the others. It succeeds if the drawer asked about is w (probability $1/N$), or if it is not and the guess is right (probability $\frac{N-1}{N} \cdot \frac{1}{N-1}$). That is

$2/N$ in all: one half for four drawers, one in thirty-two for sixty-four. Each ask, it seems, buys one drawer's worth of chance, and the final guess buys one more. That is true of every classical plan.

> **Result 15.2 (no classical shortcut).** For a uniformly random marked drawer, every classical method that asks at most k times and then names a drawer succeeds with probability at most $(k+1)/N$. In particular, one ask and a guess can reach exactly $2/N$; success with probability $\frac{1}{2}$ needs at least $\frac{N}{2} - 1$ asks; and certainty needs $N - 1$.

Proof. First fix the method's coin tosses, so that it is deterministic. Consider the run in which every answer is "no". In that run the method asks about some drawers $x_1, \dots, x_k$ (each chosen from the answers so far, which are all "no") and names some drawer z. Now take any marked drawer w outside the set $\{x_1, \dots, x_k, z\}$. With that w, the first ask is about $x_1 \neq w$ and gets "no", so the second ask is about x_2, which also gets "no", and so on: the method follows the all-"no" run exactly and names $z \neq w$. So it can succeed only for the at most $k+1$ drawers in the set, out of N equally likely ones. With coin tosses, the success probability is an average of such numbers, each at most $(k+1)/N$. The special cases follow by setting $(k+1)/N$ equal to $\frac{1}{2}$ and to 1. ■ See `Code/Ch15/FourDrawers.py`.

Grover opens his paper with the same count: to find a name in a directory of N names "arranged in completely random order" with probability $\frac{1}{2}$, any classical algorithm "will need to look at a minimum of $N/2$ names". Result 15.2 is that sentence proved, with the named drawer counted as one of the looks.

Why this definition. Why count asks and nothing else? Because the drawers are sealed and share no structure, the ask is the only part of the work that touches the secret, and everything else a method does is its own private business. Chapter 13 made the same choice for the same reason, and the quantum count below uses the same unit.

The marker

The quantum ask is Chapter 13's **query box** (Result 13.1, row 97): the tall box

$$U_f\,|x\rangle\,|y\rangle \;=\; |x\rangle\,|y \oplus f(x)\rangle$$

laid across the n cords (Chapter 13's **rods**) and one extra cord (the **flag**). One use of it is one query, even when the rods carry a superposition of drawers. With the flag in the difference state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, the box does not change the flag at all. It turns the rods' sign instead: that is phase kickback (Chapter 5, row 37), which Chapter 13 drew as a copy (Result 13.4, row 100).

For the search this gives the operation the chapter is built on. The **marker** is

$$O_w \;=\; I - 2\,|w\rangle\langle w|\,, \qquad O_w\,|x\rangle = (-1)^{[x=w]}\,|x\rangle\,.$$

It turns the marked drawer over and leaves every other drawer as it is. This is step (ii)(a) of Grover's algorithm: in case the condition holds, "rotate the phase by π radians", and otherwise "leave the system unaltered".

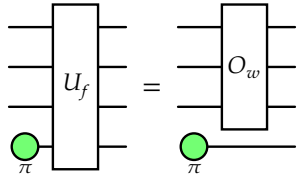
> **Result 15.3 (the marker is one query).** For every w and every state $|\psi\rangle$ of the rods,
>
> $$U_f(|\psi\rangle \otimes |-\rangle) \;=\; (O_w\,|\psi\rangle) \otimes |-\rangle\,.$$
>
> The marker is unitary, $O_w^2 = I$, and it changes no odds: $|\langle x|\,O_w\,|\psi\rangle|^2 = |\langle x|\psi\rangle|^2$ for every drawer x.

Proof. On a drawer, $U_f|x\rangle|-\rangle = \frac{1}{\sqrt{2}}(|x\rangle|f(x)\rangle - |x\rangle|1 \oplus f(x)\rangle) = (-1)^{f(x)}\,|x\rangle\,|-\rangle$, and $(-1)^{f(x)}\,|x\rangle = O_w\,|x\rangle$. Both sides are linear in $|\psi\rangle$ and agree on every drawer, so they agree everywhere. O_w is diagonal with entries ± 1, so $O_w^2 = I$, and each amplitude only changes sign. ■ See `Code/Ch15/FourDrawers.py`.

Grover makes the same remark about every operation of this kind: "the probability in each state stays the same since the square of the absolute value of the amplitude in each state stays the same." Bennett, Bernstein,

Brassard and Vazirani (BBBV, whom we met in Chapter 13) describe the same trick for their query tape: with the target bit in $(|0\rangle - |1\rangle)/\sqrt{2}$, an oracle call "leaves the entire state unchanged while introducing a phase factor -1" when the answer is 1.

In the sketchbook. Draw three rods and the flag, as Chapter 13 did. The flag's difference state is a green π point, which is $\sqrt{2}\,|-\rangle$ (Chapter 13). Result 15.3, multiplied through by $\sqrt{2}$, says that the query box with a green π point on its flag equals a tall box O_w across the rods alone, with the green π point passing by untouched:



The equation is exact, and neither side carries a number, because the same point stands on both. From here on the flag is left out of the pictures and the marker is drawn as its own box. Section §15.2 opens that box into dots.

Why this definition. Why a sign and not a flag? A flag saying "here" would be read, and reading a superposition of drawers picks one drawer at random, which returns us to asking one by one. A sign cannot be seen by any reading (Result 15.3), but it lives in the amplitudes, where a later operation can turn it into odds. The whole search is a way of making that later operation happen.

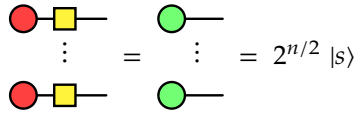
The even state

Grover's step (i) prepares "the same amplitude to be in each of the N states". Write $W = H^{\otimes n}$ for a Hadamard on every cord, the Walsh–Hadamard transform, with entries $W_{xy} = 2^{-n/2}(-1)^{x\cdot y}$, where $x \cdot y$ is the bitwise dot product taken mod 2 (Chapter 13). The **even state** is

$$|s\rangle \;=\; W|0\cdots 0\rangle \;=\; \frac{1}{\sqrt{N}}\sum_{x}|x\rangle\,.$$

Before stating it in general, take two cords: $H|0\rangle \otimes H|0\rangle = \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$, every drawer with amplitude $\frac{1}{2}$ and odds $\frac{1}{4}$.

Result 15.4 (the even state). For every n, $W|0\cdots 0\rangle = |s\rangle$, W is real and symmetric, and $W^2 = I$. In pictures, n red points each followed by a yellow box are n green points, and those are $2^{n/2}|s\rangle$:



Proof. $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ on each cord, and the product over n cords is $2^{-n/2}\sum_x |x\rangle$. The entries $2^{-n/2}(-1)^{x\cdot y}$ are real and symmetric in x and y, and $W^2 = I$ because $H^2 = I$ on each cord. In the picture, a red point is $\sqrt{2}\,|0\rangle$, and a yellow box on the only leg of a red point turns it into a green point of the same phase (colour change, Theorem 8.2, row 58). A green point is $|0\rangle + |1\rangle$, and n of them are $\sum_x |x\rangle = 2^{n/2}|s\rangle$. ■ See `Code/Ch15/FourDrawers.py`.

Why this definition. Why start from the even state? Because nothing is known about w, and a start that favoured some drawers would be a guess made before any touch. The even state favours none. It is also the state that the turn-about of §15.2 leaves exactly where it is, so the search starts on the one line it can pivot about.

Now touch the even state once. $O_w|s\rangle = |s\rangle - \frac{2}{\sqrt{N}}|w\rangle$: every amplitude is still $\pm 1/\sqrt{N}$, and every drawer still has odds $1/N$. For four drawers with $w = 01$ the amplitudes are $\frac{1}{2}(1, -1, 1, 1)$ in the order $00, 01, 10, 11$. The mark is there, as a minus sign on the second entry, and a reading cannot see it. It is a wish spent on a secret that no look reveals. Something must turn the sign into a height.

Feynman's Notebook.

A directory in random order. Lov K. Grover wrote "A fast quantum mechanical algorithm for database search" at Bell Labs in Murray Hill, New Jersey. The version we read is, in its own words, "an updated version of a paper that originally appeared in Proceedings, STOC 1996, Philadelphia PA USA, pages 212-219". It begins with a telephone directory whose N names are in completely random order and a number to be found, and it promises that the number "can be obtained in only $O(\sqrt{N})$ steps". Grover is precise about what the marking step means physically. One part of the system senses the state and decides whether to turn the phase, and it must do so "so that no trace of the state of the system be left", so that the paths can still interfere. "The implementation does not involve a classical measurement." His acknowledgments give the paper's company: Peter Shor introduced him to quantum computing, Ethan Bernstein provided the lower-bound argument of his §6, and Gilles Brassard's comments helped him update the STOC paper.

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§15.2 The Turn-About

Take the four amplitudes left by the touch, $\frac{1}{2}(1, -1, 1, 1)$. Their average is $\frac{1}{4}$. Now move every entry to the other side of the average, as far below it as it was above, or as far above as it was below: an entry v becomes $2 \cdot \frac{1}{4} - v$. The three entries $\frac{1}{2}$ become 0. The entry $-\frac{1}{2}$ becomes 1. The result is $(0, 1, 0, 0)$: the second drawer, with certainty. One touch and one turn about the average have found one drawer in four.

On the blackboard. Grover calls this operation **inversion about average** and gives its matrix, the **diffusion transform** D (his (4.0)): the entries are $2/N$ off the diagonal and $-1 + 2/N$ on it. Writing J for the $N \times N$ matrix of all ones,

$$D \;=\; \frac{2}{N} J - I.$$

On any vector v with entries v_x and average $\bar{v} = \frac{1}{N}\sum_x v_x$, the vector Jv has every entry $N\bar{v}$, so

$$(Dv)_x \;=\; 2\bar{v} - v_x.$$

That is the move we just made by hand. Grover writes it as $D = -I + 2P$, where $P = \frac{1}{N}J$ and "P acting on any vector v gives a vector each of whose components is equal to the average of all components". We will call D the **turn-about**.

Grover also shows (his Theorem 1) that D factors as $D = WRW$, where R is diagonal with $R_{00} = 1$ and $R_{xx} = -1$ for every other x:

$$R \;=\; 2\,|0\cdots 0\rangle\langle 0\cdots 0| - I.$$

Grover calls R "the rotation matrix", in the sense of his selective rotation of phases: it turns the phase of every drawer but one by π. As a map of the space, though, it is a reflection. It fixes the line through $|0\cdots 0\rangle$, reverses every direction at right angles to that line, and satisfies $R^2 = I$. We will call it a reflection, because §15.4 is about reflections.

Finally, write $C^{n-1}Z = I - 2\,|1\cdots 1\rangle\langle 1\cdots 1|$ for the **multi-controlled** Z, which turns over the all-ones drawer and nothing else. For $n = 2$ it is CZ, and for $n = 1$ it is Z itself. Here is the headline theorem.

> **Result 15.5 (the turn-about is a reflection about the even state).** For every $n \geq 1$ and $N = 2^n$, exactly,
>
> $$D \;=\; WRW \;=\; -W X^{\otimes n} C^{n-1}Z\, X^{\otimes n} W \;=\; 2\,|s\rangle\langle s| - I.$$
>
> Hence $(Dv)_x = 2\bar{v} - v_x$ for every vector v (inversion about average); $D|s\rangle = |s\rangle$; $Du = -u$ for every vector u whose entries sum to 0; and $D^2 = I$.

It is proved twice in §15.7, once on each strand. The companion file checks it exactly for $n = 1, \ldots, 6$ and tests inversion about average on a random complex vector: see `Code/Ch15/TurnAbout.py`.

Dirac's Blackboard.

Grover's matrix, entry by entry. Grover proves $D = WRW$ by multiplying out. Write $R = 2\,|0\cdots0\rangle\langle 0\cdots0| - I$ and use $W_{xz} = 2^{-n/2}(-1)^{x\cdot z}$:

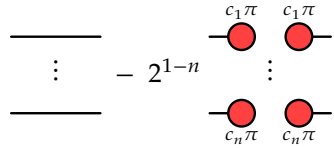
$$(WRW)_{xy} \;=\; \sum_z W_{xz}R_{zz}W_{zy} \;=\; 2^{-n}\Big(2 - \sum_z (-1)^{x\cdot z \,\oplus\, z\cdot y}\Big).$$

The 2 is the $z = 0$ term counted with $R_{00} = 1$ instead of -1. Since $x\cdot z \oplus z\cdot y = (x \oplus y)\cdot z$, the sum is $\sum_z (-1)^{u\cdot z}$ with $u = x \oplus y$, which is 2^n if $u = 0$ and 0 otherwise (each bit of u equal to 1 splits the z's into equal halves of opposite sign). So $(WRW)_{xy} = 2/N - [\,x = y\,]$: $2/N$ off the diagonal, $-1 + 2/N$ on it. That is Grover's D, entry for entry.

In the sketchbook. The turn-about is built from a box that turns over one drawer. Start with the smallest one, $\mathrm{CZ} = I - 2\,|11\rangle\langle 11|$. Chapter 5 taught us to write a box as a sum of pictures (rows 34–36), and Chapter 10 gave us the effects we need: a red effect with phase $c\pi$ is $\sqrt{2}\,\langle c|$, the mirror of the red point $\sqrt{2}\,|c\rangle$. We keep Chapter 14's names: a phase-free red point is **dark** ($\sqrt{2}\,|0\rangle$), a red π point is **lit** ($\sqrt{2}\,|1\rangle$), and likewise for effects. So a lit effect over a lit point, on each of two cords, is $\sqrt{2}\,\langle 1| \cdot \sqrt{2}\,\langle 1|$ followed by $\sqrt{2}\,|1\rangle \cdot \sqrt{2}\,|1\rangle$, which is $4\,|11\rangle\langle 11|$. Half of that is $2\,|11\rangle\langle 11|$, and so CZ is the two bare cords minus half of the picture with lit effects and lit points. The same bookkeeping works for any drawer c on any number of cords.

Result 15.6 (one thing turned over). For every n and every n-bit string c, exactly,

$$I - 2\,|c\rangle\langle c| \;=$$



on the right, n bare cords, minus 2^{1-n} times n red effects with phases $c_i\pi$ beside n red points with the same phases. Three cases name the chapter's boxes: $c = 1\cdots1$ gives $C^{n-1}Z$ (row

111); $c = w$ gives the marker O_w (row 110); and $c = 0\cdots0$ gives $I - 2\,|0\cdots0\rangle\langle 0\cdots0| = -R$.

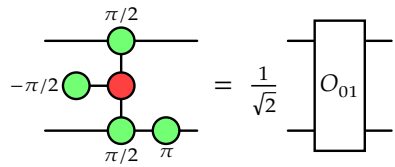
Proof. Cord i contributes $\sqrt{2}\,|c_i\rangle \cdot \sqrt{2}\,\langle c_i| = 2\,|c_i\rangle\langle c_i|$, so the right-hand picture is $2^n\,|c\rangle\langle c|$, and 2^{1-n} times it is $2\,|c\rangle\langle c|$. The difference of two pictures is a sum of pictures with signed weights (Chapter 5). ■ See `Code/Ch15/TurnAbout.py`.

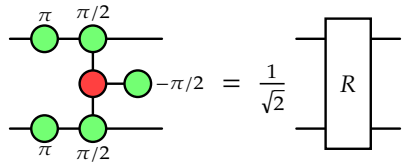
The shape is worth saying in words: **all the cords, minus twice the one thing**. The marker has this shape with the marked drawer as the one thing. Section §15.7 shows that the turn-about has it too, with the even state as the one thing and an overall sign.

Two cords, with no sign

For two cords the tall boxes can be opened into dots with Chapter 14's **controlled turn** (Result 14.5, row 103) at $\alpha = \pi$: a green ring with phase $\pi/2$ on each cord, the two rings tied to one phase-free red hub, and from the hub a pendant green point with phase $-\pi/2$. Bare, it is $\mathrm{CP}(\pi)/\sqrt{2} = \mathrm{CZ}/\sqrt{2}$. A green dot with phase π, one leg in and one out, is Z exactly (row 53); on a cord we call it a **half-ring**. Before stating the result, compute one product by hand. On $|01\rangle$, CZ does nothing and $Z \otimes Z$ gives $(+1)(-1)$, so $(Z \otimes Z)\,\mathrm{CZ}\,|01\rangle = -|01\rangle$. On $|11\rangle$, CZ gives -1 and $Z \otimes Z$ gives $(-1)(-1)$, so the product is $-|11\rangle$. On $|00\rangle$ nothing happens. That is Grover's R for two cords, and no overall sign is needed.

Result 15.7 (two cords, no sign). Exactly: (i) $R = (Z \otimes Z)\,\mathrm{CZ} = \mathrm{diag}(1,-1,-1,-1)$; (ii) for every w, $O_w = (-1)^{\bar{w}_1\bar{w}_2}\,(Z^{\bar{w}_2} \otimes Z^{\bar{w}_1})\,\mathrm{CZ}$; in particular $O_{01} = (I \otimes Z)\,\mathrm{CZ}$. In pictures, with the half-ring on the lower cord for the marker of the second drawer and on both cords for R,





For a general w the half-ring goes on the lower cord if $w_1 = 0$ and on the upper cord if $w_2 = 0$, and for $w = 00$ the number -1 is written beside the picture.

Proof. On the blackboard, every factor is diagonal, so it is enough to compare signs on the four drawers c_1c_2. CZ gives $(-1)^{c_1c_2}$, and a Z on the upper cord gives $(-1)^{c_1}$, on the lower $(-1)^{c_2}$. With half-rings h_1 on top and h_2 below ($h_i \in \{0,1\}$), the sign on c_1c_2 is $(-1)^{c_1c_2+h_1c_1+h_2c_2}$. For R, $h_1 = h_2 = 1$, and $c_1c_2 + c_1 + c_2$ is odd unless $c_1 = c_2 = 0$: that is $\mathrm{diag}(1,-1,-1,-1)$. For the marker, $h_1 = \bar{w}_2$ and $h_2 = \bar{w}_1$, and

$$c_1c_2 + \bar{w}_2c_1 + \bar{w}_1c_2 + \bar{w}_1\bar{w}_2 \;=\; (c_1 + \bar{w}_1)(c_2 + \bar{w}_2),$$

which is odd exactly when $c_1 = w_1$ and $c_2 = w_2$. So the sign is $(-1)^{[c=w]}$ times $(-1)^{\bar{w}_1\bar{w}_2}$, which is (ii).

In the sketchbook, feed the two cords a dark or lit point each, red with phases $c_1\pi$ and $c_2\pi$. By the basis-point lemma (Result 14.6, row 104) at $\alpha = \pi$, the upper point passes the controlled turn and leaves a green $c_1\pi$ on the lower cord, for the price $\frac{1}{\sqrt{2}}$. On the lower cord that green dot fuses with the half-ring, if there is one (fusion, Theorem 8.1). A green dot with phase φ, one leg in and one out, multiplies a point $\sqrt{2}\,|c\rangle$ by $e^{ic\varphi}$, so each cord's green phase becomes a sign: $(-1)^{c_2(c_1+h_2)}$ below and $(-1)^{c_1h_1}$ above. The product is the sign found on the blackboard. ■ See `Code/Ch15/TurnAbout.py`, which also derives both pictures by the rules.

So for two cords the reflection R, and with it the turn-about $D = WRW$, can be drawn with no number beside it: collars (yellow boxes) on both cords, a half-ring on each, the controlled half-turn, and collars again. Bare, that picture is exactly $D/\sqrt{2}$.

Why this definition. Why reflect about the average, rather than simply lift

the marked drawer? Because the turn-about must not know which drawer is marked: only the marker knows that, and every use of the marker is a query. So the turn-about treats all drawers alike and may look only at what they share. Turning every entry over, $-I$, also treats them alike, but it changes no odds. A reflection about the average treats every drawer alike and still makes a sign visible as a height.

---

§15.3 One Touch

We can now put Grover's algorithm together. On n cords, with $N = 2^n$ drawers:

1. **Step (i).** Prepare the even state $|s\rangle = W|0\cdots0\rangle$ (Result 15.4).
2. **Step (ii).** Repeat a **step**, the marker followed by the turn-about,

$$G = D\,O_w,$$

   some number of times. Grover's own text says "$O(\sqrt{N})$ times (the precise number of repetitions is important as discussed in [BBHT96])".
3. **Step (iii).** Read the cords. Grover asks for the marked drawer "with a probability of at least $\frac{1}{2}$".

The reference [BBHT96] is the paper by Michel Boyer, Gilles Brassard, Peter Høyer and Alain Tapp (we write BBHT) that §15.4 follows. They write the step as $G = -TS_0TS_{i_0}$, where S_{i_0} is our marker O_w, $S_0 = I - 2|0\cdots0\rangle\langle0\cdots0| = -R$, and T is their letter for the Walsh–Hadamard transform. We keep Grover's W for that transform and save T for counting touches, so their step reads $G = -WS_0WO_w = WRWO_w = DO_w$. BBHT write i_0 for the marked drawer, and in their §7, like BBBV, y. We write w throughout and keep y for the flag bit.

What counts. Each step uses the marker once, and so, by Result 15.3, one query. The turn-about uses none: it is the same for every w. BBBV say so exactly. After the first query they call Grover's "inversion about the average" "an oracle-independent unitary transformation", one that turns the phase

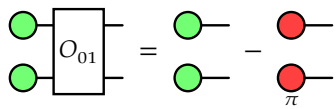
difference into an amplitude difference, "leaving the y term with the same sign as all the other terms but a magnitude approximately threefold larger". So we count **one touch per step**. BBHT count differently, and we say it once to avoid confusion. In their table model the entry must first be fetched, to decide on the phase shift, and then erased, to allow interference, so they charge "two table look-ups" per step. In Chapter 13's query model, one use of the box with the flag at $|-\rangle$ does the whole marker (Result 15.3).

Here is the complete search for four drawers with $w = 01$. The even state is $\frac{1}{2}(1,1,1,1)$. The marker gives $\frac{1}{2}(1,-1,1,1)$. The turn-about, by inversion about the average $\frac{1}{4}$, gives $(0,1,0,0) = |01\rangle$. The reading shows the second drawer. The same happens for every w.

> **Result 15.8 (one step finds one drawer in four).** For $N = 4$ and every marked drawer w, exactly,
>
> $$D\,O_w\,|s\rangle \;=\; |w\rangle\,,$$
>
> so one step and one reading find w with certainty.

Proof. **On the blackboard.** Since $\langle w|s\rangle = \frac{1}{2}$, the marker gives $O_w\,|s\rangle = |s\rangle - 2\cdot\frac{1}{2}\,|w\rangle = |s\rangle - |w\rangle$. By Result 15.5, $D(|s\rangle - |w\rangle) = 2\,|s\rangle\,(\langle s|s\rangle - \langle s|w\rangle) - |s\rangle + |w\rangle = 2\,|s\rangle\cdot\frac{1}{2} - |s\rangle + |w\rangle = |w\rangle$.

In the sketchbook. Take $w = 01$. The two green points are $2\,|s\rangle$ (Result 15.4). Row 110 opens the marker into two pictures: the bare cords, minus $\frac{1}{2}$ times a dark and a lit effect beside a dark and a lit point (Result 15.6 with $c = 01$). Feed it the green points. On the bare cords they pass untouched. On the second picture each red effect meets a green point and closes to $\sqrt{2}\,\langle c|\,(|0\rangle + |1\rangle) = \sqrt{2}$ (Chapter 10), so the two cords give 2 and cancel the $\frac{1}{2}$:



On the blackboard this reads $O_{01}\,2\,|s\rangle \;=\; 2\,|s\rangle - 2\,|01\rangle$: the green points

minus a dark point above a lit point. Now the turn-about. By Result 15.5, $D = 2\,|s\rangle\langle s| - I$. The two green points are $2\,|s\rangle$ and the two green effects, their mirror image, are $2\,\langle s|$, so D is $\frac{1}{2}$ times green effects beside green points, minus the bare cords. A green effect closes to 2 on a green point and to $\sqrt{2}$ on a red point, dark or lit. So the turn-about takes the green points to $\frac{1}{2}\cdot 4 - 1 = 1$ copy of themselves, and the dark-and-lit pair to $\frac{1}{2}\cdot 2 = 1$ copy of the green points minus the pair itself. Subtract: the green points cancel exactly, and the dark point above the lit point is left. That is $\sqrt{2}\,|0\rangle \cdot \sqrt{2}\,|1\rangle = 2\,|01\rangle$, so $D\,O_{01}(2\,|s\rangle) = 2\,|01\rangle$. Every equality here is exact. The other three drawers go the same way, with the dark and lit points exchanged to spell w. §15.8 draws this search gate by gate, with the marker and the turn-about opened into dots. ■ See `Code/Ch15/SecondDrawer.py`, which checks all four drawers.

BBHT make the same observation in a more general form: when a quarter of all drawers are marked, "a solution is found with certainty after a single iteration" (their §3.1). Four drawers with one marked is the smallest such case.

For more drawers one step is not enough. Starting from $|s\rangle$ and repeating the computation in the proof of Result 15.8 with $\langle w|s\rangle = 1/\sqrt{N}$ gives

$$D\,O_w\,|s\rangle \;=\; \Big(1 - \frac{4}{N}\Big)\,|s\rangle + \frac{2}{\sqrt{N}}\,|w\rangle\,,$$

so the marked amplitude rises from $1/\sqrt{N}$ to $(3-4/N)/\sqrt{N}$, just under three times as large, which is BBBV's "approximately threefold". For sixty-four drawers that is 0.367, with odds about 0.135: better than $1/64$, but far from certain. So we repeat the step. How often, and why "precise"? The next section answers both questions.

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§15.4 The Turning Plane

Take four drawers again and do a second step, although the first already found w. The state is $|w\rangle$. The marker turns it over, giving $-|w\rangle$. The turnabout gives $-(2\,|s\rangle\,\langle s|w\rangle - |w\rangle) = |w\rangle - |s\rangle$, whose entries are $\frac{1}{2}$ at w and $-\frac{1}{2}$ everywhere else. Every drawer now has odds $\frac{1}{4}$, just as before the search began. The second step has undone the first.

So more steps are not always better, and adding up improvements step by step cannot tell us when to stop. We need to see the whole path of the state at once. Two facts make that possible.

The plane. Write

$$|w^\perp\rangle \;=\; \frac{1}{\sqrt{N-1}} \sum_{x \neq w} |x\rangle$$

for the even state of the unmarked drawers, and let θ be the angle with $\sin\theta = 1/\sqrt{N}$, between 0 and $\pi/2$. Then

$$|s\rangle \;=\; \sin\theta\,|w\rangle + \cos\theta\,|w^\perp\rangle\,,$$

because the amplitude of $|s\rangle$ at w is $1/\sqrt{N} = \sin\theta$ and the remaining $N-1$ amplitudes $1/\sqrt{N}$ make up $\sqrt{(N-1)/N}\;|w^\perp\rangle = \cos\theta\;|w^\perp\rangle$. The two vectors $|w^\perp\rangle$ and $|w\rangle$ are real, orthogonal and of length one, so the real combinations $a\,|w\rangle + b\,|w^\perp\rangle$ form a **plane**, and we can draw it on paper with $|w^\perp\rangle$ pointing right and $|w\rangle$ pointing up. The marker maps this plane to itself: it fixes $|w^\perp\rangle$ and sends $|w\rangle$ to $-|w\rangle$. It is the **reflection across the $|w^\perp\rangle$ line**. The turnabout maps it to itself too, since $D = 2\,|s\rangle\,\langle s| - I$ and $|s\rangle$ lies in the plane. It fixes $|s\rangle$ and turns over the vector $|s^\perp\rangle = \cos\theta\,|w\rangle - \sin\theta\,|w^\perp\rangle$, whose entries sum to 0 (Result 15.5). So it is the **reflection across the $|s\rangle$ line**. A step is two reflections in a row.

What do two reflections make? Hold a sheet of paper and fold it along one line through a pin, then along another line through the pin at angle $\beta - \alpha$ to the first. The sheet has not been turned over, since two turnings-over cancel. It has been turned round the pin, and by twice the angle between the folds.

Result 15.9 (two folds make a turn). In the real plane, let F_α be

the reflection across the line at angle α to the first axis:

$$F_\alpha = \begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix}.$$

Then for every α and β, $F_\beta F_\alpha$ is the rotation by $2(\beta - \alpha)$.

Proof. F_α fixes $(\cos\alpha, \sin\alpha)$ and reverses $(-\sin\alpha, \cos\alpha)$, as a direct multiplication shows, so it is the reflection named. Write $c_\alpha = \cos 2\alpha$ and $s_\alpha = \sin 2\alpha$, and likewise for β. Multiplying,

$$F_\beta F_\alpha = \begin{pmatrix} c_\beta c_\alpha + s_\beta s_\alpha & c_\beta s_\alpha - s_\beta c_\alpha \\ s_\beta c_\alpha - c_\beta s_\alpha & s_\beta s_\alpha + c_\beta c_\alpha \end{pmatrix} = \begin{pmatrix} \cos 2(\beta-\alpha) & -\sin 2(\beta-\alpha) \\ \sin 2(\beta-\alpha) & \cos 2(\beta-\alpha) \end{pmatrix}$$

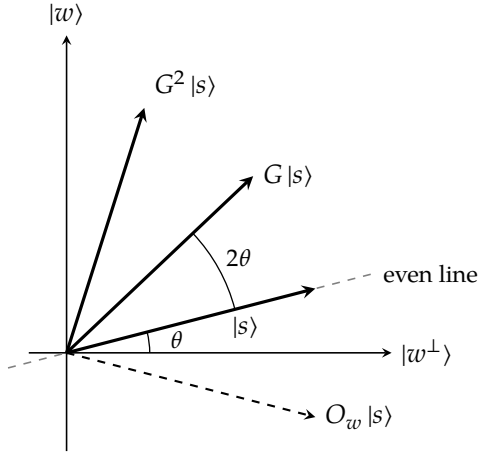
by the difference formulas for cosine and sine. ■ The companion file for this result and the next two is `Code/Ch15/TurningPlane.py`.

In the plane with $|w^\perp\rangle$ as the first axis and $|w\rangle$ as the second, the marker is F_0: its matrix is $\mathrm{diag}(1, -1)$. The turn-about is F_θ: in the same coordinates, $2|s\rangle\langle s| - I$ has entries $2\cos^2\theta - 1 = \cos 2\theta$, $2\sin\theta\cos\theta = \sin 2\theta$ and $2\sin^2\theta - 1 = -\cos 2\theta$. So a step, $F_\theta F_0$, is a rotation by 2θ.

Why this definition. Why a plane, when the state lives in N dimensions? Because the marker changes only the direction of $|w\rangle$ and the turn-about changes only the direction of $|s^\perp\rangle$, and both of those directions lie in the plane. Everything at right angles to the plane is left alone by both, and the search starts in the plane. So the whole search, however many drawers, happens on one sheet of paper.

A different kind of picture. This is where the chapter's one new picture comes in. We draw the plane itself: the unmarked line $|w^\perp\rangle$ across the page and the marked line $|w\rangle$ up it, meeting at a pin; the **even line** through the pin at the small angle θ above the unmarked line; and the state as an arrow from the pin, a **hand**. The marker folds the paper along the unmarked line, and the turn-about folds it along the even line. We call this the **Turning Plane**. It is not a string diagram. Its lines are directions in a real plane, not wires. Its hand is a vector, not a process. None of the rules of Chapter 8 apply to it. Grover's paper does not draw it, and we attribute it to no one: it

is BBHT's formula, which follows, drawn.



The Turning Plane. A picture of the real plane spanned by $|w\rangle$ and $|w^\perp\rangle$, drawn for sixteen drawers ($\sin\theta = \frac{1}{4}$, $\theta \approx 14.5°$). It is not a string diagram. It draws one side of the equation

$$D\,O_w\Big|_{\mathrm{span}\{|w\rangle,|w^\perp\rangle\}} \;=\; \text{rotation by } 2\theta,$$

and the three solid hands are the states

$$G^k\,|s\rangle \;=\; \sin((2k+1)\theta)\,|w\rangle + \cos((2k+1)\theta)\,|w^\perp\rangle \qquad (k = 0, 1, 2),$$

at 14.5°, 43.4° and 72.4°. The dashed hand is the even state after the marker alone, folded to $-\theta$; the turn-about folds it across the even line to 3θ. The height of each hand's tip is the marked amplitude.

> **Result 15.10 (the search turns).** For every $N = 2^n$ and every $k \geq 0$, exactly,
>
> $$G^k\,|s\rangle \;=\; \sin((2k+1)\theta)\,|w\rangle + \cos((2k+1)\theta)\,|w^\perp\rangle\,,$$
>
> where $\sin\theta = 1/\sqrt{N}$. So after k steps the marked drawer has amplitude $a_k = \sin((2k+1)\theta)$, each unmarked drawer has amplitude $b_k = \cos((2k+1)\theta)/\sqrt{N-1}$, and the probability that

reading the cords gives w is $\sin^2((2k+1)\theta)$.

This is BBHT's Eq. (2). Their step count j is our k, and their amplitudes k_j and ℓ_j are our a_k and b_k.

Proof. $|s\rangle$ is the unit vector at angle θ in the plane. By Result 15.9 with $\alpha = 0$ and $\beta = \theta$, each step turns it by 2θ, so after k steps it is at angle $(2k+1)\theta$. Its coordinates are the cosine and sine of that angle. The amplitude of any single unmarked drawer is $\cos((2k+1)\theta)$ times the amplitude $1/\sqrt{N-1}$ of that drawer in $|w^\perp\rangle$. ■ See `Code/Ch15/TurningPlane.py`, which checks Eq. (2) against direct simulation for $N = 4, 8, 16, 64, 4096$, and against an eight-drawer search built from spiders.

Dirac's Blackboard.

From the recurrence to the sine. BBHT start from what one step does to a state with amplitude k at the marked drawer and ℓ at each other drawer (their Eq. (1)). In our letters,

$$a_{k+1} = \tfrac{N-2}{N}\, a_k + \tfrac{2(N-1)}{N}\, b_k, \qquad b_{k+1} = \tfrac{N-2}{N}\, b_k - \tfrac{2}{N}\, a_k.$$

They reach the sine "by standard techniques—and a little sweat". The plane removes the sweat. Put $c_k = \sqrt{N-1}\, b_k$, the component along $|w^\perp\rangle$. Since $\sin\theta = 1/\sqrt{N}$, we have $\cos 2\theta = 1 - 2/N = \frac{N-2}{N}$ and $\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\sqrt{N-1}}{N}$, so

$$a_{k+1} = \cos 2\theta\, a_k + \sin 2\theta\, c_k, \qquad c_{k+1} = \cos 2\theta\, c_k - \sin 2\theta\, a_k.$$

That is the rotation of (c_k, a_k) by 2θ. It starts at $(c_0, a_0) = (\cos\theta, \sin\theta)$, so induction gives $a_k = \sin((2k+1)\theta)$ and $c_k = \cos((2k+1)\theta)$.

When to stop

The hand starts just above the unmarked line and climbs by 2θ per step. Success is certain when it stands upright, at $\pi/2$, which would take $\tilde{m} = (\pi - 2\theta)/4\theta$ steps. That is rarely a whole number, so BBHT take the nearest safe whole number below.

Result 15.11 (when to stop, and the square root). Let $m = \lfloor \pi/4\theta \rfloor$. After m steps the probability of failure is

$$\cos^2((2m+1)\theta) \;\le\; \sin^2\theta \;=\; \frac{1}{N},$$

and $m \le \frac{\pi}{4}\sqrt{N}$. For large N, m is close to $\frac{\pi}{4}\sqrt{N}$.

Proof. Since $\tilde{m} = \pi/4\theta - \frac{1}{2}$ and $\pi/4\theta - 1 < m \le \pi/4\theta$, we have $|m - \tilde{m}| \le \frac{1}{2}$, so $(2m+1)\theta$ lies within θ of $(2\tilde{m}+1)\theta = \pi/2$. Writing $(2m+1)\theta = \pi/2 + \delta$ with $|\delta| \le \theta$, we get $|\cos((2m+1)\theta)| = |\sin\delta| \le \sin\theta$. Finally $\theta \ge \sin\theta = 1/\sqrt{N}$, so $m \le \pi/4\theta \le \frac{\pi}{4}\sqrt{N}$. For large N, θ and $\sin\theta$ are nearly equal. ■ This is BBHT's §3 argument. See `Code/Ch15/TurningPlane.py`.

Here is the square root in numbers, with the worst case of asking one by one (Result 15.2) beside it:

drawers N	steps m	success after m steps	asks, one by one, worst case
4	1	1	3
64	6	0.9966	63
4096	50	0.99995	4095
2^{20}	804	0.9999998	1 048 575

For eight drawers $m = 2$, and for sixteen $m = 3$. The last row is BBHT's own example: for 2^{20} possibilities they "achieve almost certainty of success after 804 iterations". Every step is one touch. Multiply the drawers by four and the steps roughly double.

Now the warning in Grover's step (ii). For sixty-four drawers, $\theta \approx 7.18^\circ$ and the success probabilities after $0, 1, 2, \ldots, 8$ steps are

$$0.0156,\;\; 0.135,\;\; 0.344,\;\; 0.591,\;\; 0.816,\;\; 0.964,\;\; 0.9966,\;\; 0.907,\;\; 0.718.$$

After six steps the hand stands at $13\theta \approx 93.4^\circ$, just past upright. A seventh step carries it to $(2\cdot 7+1)\theta \approx 107.7^\circ$, and the odds fall back to about nine in ten. The hand does not stop at the top, and neither do the odds. BBHT

say it plainly: Grover proved that some number of iterations below $\sqrt{2N}$ gives success at least $\frac{1}{2}$, and "this is correct, but one must be careful in using his algorithm because the probability of success does not increase monotonically with the number of iterations." By $\sqrt{2N}$ iterations, they compute, the probability has "dropped down to less than 9.5%", and about $\frac{\pi}{2}\sqrt{N}$ iterations, twice the right number, give "a negligible probability of success". They also note that half the number, about $\frac{\pi}{8}\sqrt{N}$, suffices for a 50% chance, "as Grover considered in his original paper". The explicit count, and not only its order, is what makes the search usable.

Coecke's Sketchbook.

Two mirrors of one shape. Set the marker and the turn-about side by side as sums of pictures. The marker is the bare cords minus 2^{1-n} times red w-effects over red w-points (Result 15.6). The turn-about is 2^{1-n} times green effects over green points, minus the bare cords (§15.7). The two have the same shape in the two colours: all the cords, minus twice the one thing. The overall sign differs, and it swaps which line is kept: the marker turns its one thing over, while the turn-about keeps its one thing and turns over everything at right angles to it. On a whole step the sign is global and changes no odds. The picture even measures the angle between the two mirrors. Close a red w-effect against a green point on every cord: each cord gives $\sqrt{2}\,\langle w_i|\,(|0\rangle + |1\rangle) = \sqrt{2}$, so n cords give $2^{n/2}$. Divide by the $2^{n/2} \cdot 2^{n/2}$ that the effects and points carry, and $\langle w|s\rangle = 2^{-n/2} = \sin\theta$. What the picture cannot do is iterate. A string diagram of k steps is just k copies laid end to end, and no rule of Chapter 8 reads a sine off it. The count lives in the plane.

Many marked drawers

BBHT treat the case of t marked drawers in their §3. The same plane appears, with $|w\rangle$ replaced by the even state of the marked drawers, and the same two folds.

Result 15.12 (many marked drawers; Boyer, Brassard, Høyer and Tapp, §3 and §3.1; stated, not proved). If exactly t drawers are marked, and θ is chosen with $\sin^2\theta = t/N$, then after k steps each marked drawer has amplitude $\sin((2k+1)\theta)/\sqrt{t}$ and each unmarked one $\cos((2k+1)\theta)/\sqrt{N-t}$ (BBHT's Eq. (3)). With $m = \lfloor \pi/4\theta \rfloor$ the probability of failure is at most t/N, and $m \le \frac{\pi}{4}\sqrt{N/t}$. When $t = N/4$, one step succeeds with certainty.

We state it and do not prove it here. Their proof is the one above, with $\sin\theta = \sqrt{t/N}$, the marked even state in place of $|w\rangle$, and the marker turning over every marked drawer. The companion code checks the amplitudes for several t and N. See `Code/Ch15/TurningPlane.py`.

Feynman's Notebook.

A little sweat. "Tight bounds on quantum searching" is by Michel Boyer, Gilles Brassard and Alain Tapp of the Université de Montréal and Peter Høyer of Odense University. Its title page calls it a full paper for PhysComp96, in a draft dated 10 May 1996; the arXiv stamp on the copy we read is 23 May 1996. It reads Grover's step (ii) closely, turns the repeated step into a recurrence solved in closed form "by standard techniques—and a little sweat", and draws the practical consequences. Their largest example is the most striking. For 2^{20} possibilities and one solution, 804 iterations make success almost certain. The same 804 iterations, they point out, "would yield a solution with probability less than one in a million should there be 4 solutions!" In the plane of §15.4 the reason is plain to see: with four marked drawers the hand turns twice as fast, and after 804 steps it has swung round to lie almost along the unmarked line again. They go on to handle an unknown number of solutions, to sketch a quantum way of counting them, and, in their §7, to prove the lower bound we meet next.

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§15.5 No Faster Hand

Chapter 13 stated BBBV's bound as Result 13.14: for a random oracle, no quantum machine running in time $o(2^{n/2})$ decides every problem in NP. It also recorded their remark that Grover's search meets the bound. In this chapter's terms: a search on $N = 2^n$ drawers takes about $\sqrt{N} = 2^{n/2}$ touches, and nothing takes fewer. This section shows where the $\sqrt{N}$ comes from.

Start with an example. Run Grover's search on sixty-four drawers with **no drawer marked**, so that the marker is the identity. Each step is then the turn-about alone, and $D\,|s\rangle = |s\rangle$: the hand never moves. With drawer w marked, the hand moves by 2θ per step, and after T steps the distance between the two final states is the chord $2\sin(T\theta)$. That is 0.496 after two steps and 1.366 after six. The distance from $|s\rangle$ to $|w\rangle$ is $\sqrt{2 - 2\sin\theta} = \sqrt{1.75} \approx 1.323$. So the question "how many touches?" becomes "how far can T touches pull the marked run away from the unmarked run?" The answer depends on how much of the state each touch spends on the marked drawer.

On the blackboard. Consider any method with T touches. It may use extra cords as workspace. It starts from a fixed state and applies unitaries $V_0, V_1, \ldots, V_T$ that do not depend on w, with a touch between each pair: the marker O_w on the n drawer cords and nothing on the workspace. Its final state is

$$|\varphi^w\rangle \;=\; V_T\, O_w\, V_{T-1} \cdots V_1\, O_w\, V_0\, |\text{start}\rangle\,.$$

The **unmarked run** is the same with every O_w replaced by I; call its final state $|\varphi^\emptyset\rangle$. Let $|\psi_i\rangle$ be the unmarked run's state just before its i-th touch, and for each drawer x let $\alpha_{i,x}$ be the length of the part of $|\psi_i\rangle$ in which the drawer cords hold x. BBBV call the square $q_x = \alpha_{i,x}^2$ the **query magnitude** of x at that touch (their Definition 3.2).

> **Result 15.13 (a touch moves the hand only by its share).** For every such method and every drawer w,
>
> $$\big\| \,|\varphi^w\rangle - |\varphi^\emptyset\rangle\, \big\| \;\le\; 2\sum_{i=1}^{T} \alpha_{i,w}.$$

Proof. For $j = 0, \dots, T$ let $|h_j\rangle$ be the final state of the run that uses I at the first j touches and O_w at the rest, so $|h_0\rangle = |\varphi^w\rangle$ and $|h_T\rangle = |\varphi^\emptyset\rangle$. The runs h_{j-1} and h_j agree up to touch j, where both are in the state $|\psi_j\rangle$. At touch j one applies O_w and the other I, and afterwards both apply the same unitaries, which preserve distances. So

$$\| |h_{j-1}\rangle - |h_j\rangle \| = \| (O_w - I)\,|\psi_j\rangle \| = 2\,\| (|w\rangle\langle w| \otimes I)\,|\psi_j\rangle \| = 2\alpha_{j,w}.$$

The triangle inequality over $j = 1, \dots, T$ gives the result. ■ See `Code/Ch15/NoFasterHand.py`, which tests it on random methods.

This is BBHT's Lemma 6(1), with their y renamed w, and its proof is the telescoping argument of BBBV's Theorem 3.3. It holds just as well if a touch is the query box U_f with the flag in any state, since $U_f - I = |w\rangle\langle w| \otimes (X - I)$ and $X - I$ at most doubles a length.

In the sketchbook. The smallest case can be drawn and weighed. The Second Drawer of §15.8 is a method with one touch on four drawers. Its unmarked run is the same picture with the marker taken out. The yellow boxes that stood either side of the marker then meet and cancel (row 58). What is left is the two dark points, the turn-about's half-rings, rings and hub, and the last pair of yellow boxes. The turn-about's middle is $\frac{1}{\sqrt{2}}R$ (Result 15.7), and R is $\frac{1}{2}$ times dark effects beside dark points, minus the bare cords (row 112). A dark effect closes on a dark point to 2, so the dark points come through as $\frac{1}{2} \cdot 4 - 1 = 1$ copy of themselves. The last yellow boxes turn them green (colour change), and the unmarked run is exactly $\frac{1}{\sqrt{2}}$ times two green points, which is $\sqrt{2}\,|s\rangle$. The picture is $\sqrt{2}$ times the run itself: its two dark points carry 2, and its one remaining hub only $\frac{1}{\sqrt{2}}$. Divide by that $\sqrt{2}$: the unmarked run ends at $|s\rangle$, and the marked run ends at $|01\rangle$ (Result 15.8). Their difference, $\frac{1}{2}(-1, 1, -1, -1)$, has length exactly 1. At its one touch the unmarked run is in $|s\rangle$, whose part on drawer 01 has length $\frac{1}{2}$, so Result 15.13 allows at most $2 \cdot \frac{1}{2} = 1$. The bound is met with equality. Both values are exact equalities of pictures. The inequality, for every method, stays on the blackboard: a rewrite proves an equality of processes, never a bound (row 117). See `Code/Ch15/NoFasterHand.py`.

Each touch has only one unit of squared amplitude to spread over N drawers, since $\sum_x \alpha_{i,x}^2 = 1$. So over T touches the squares add up to T, and on average a drawer gets T/N of them. If the searcher cannot find a drawer except by pulling its run far from the unmarked run, then the drawers that get only their fair share are out of reach. Here is the whole argument, with a modest constant.

> **Result 15.14 (no faster hand, in a weak form).** Suppose a method with T touches names the marked drawer with probability at least $\frac{1}{2}$ for every choice of w. Then
>
> $$T \;\geq\; \frac{\sqrt{N}}{4\sqrt{2}} - \frac{1}{2}.$$

Proof. Write $S_x = \sum_i \alpha_{i,x}^2$, so that $\sum_x S_x = T$. At most $N/4$ drawers can have $S_x > 4T/N$, since otherwise those alone would add up to more than T. This is the counting of BBBV's Corollary 3.4. The method's final reading names a drawer; write P_x for the projection onto the outcomes that name x, so the P_x are orthogonal and add up to at most I. The probabilities $\|P_x\,|\varphi^{\emptyset}\rangle\|^2$ add up to at most 1, so at most $N/4$ drawers have $\|P_x\,|\varphi^{\emptyset}\rangle\|^2 \geq 4/N$. That leaves at least $N/2$ drawers x with $S_x \leq 4T/N$ and $\|P_x\,|\varphi^{\emptyset}\rangle\| < 2/\sqrt{N}$. Take one of them as w. On one side, $\|P_w\,|\varphi^{w}\rangle\| \geq 1/\sqrt{2}$, so

$$\big\|\,|\varphi^{w}\rangle - |\varphi^{\emptyset}\rangle\big\| \;\geq\; \|P_w\,|\varphi^{w}\rangle\| - \|P_w\,|\varphi^{\emptyset}\rangle\| \;>\; \frac{1}{\sqrt{2}} - \frac{2}{\sqrt{N}}.$$

On the other side, by Result 15.13 and the Cauchy–Schwarz inequality, $\sum_i \alpha_{i,w} \leq \sqrt{T}\sqrt{S_w} \leq 2T/\sqrt{N}$, so the distance is at most $4T/\sqrt{N}$. Together, $4T/\sqrt{N} > 1/\sqrt{2} - 2/\sqrt{N}$, which rearranges to the bound. ■ See `Code/Ch15/NoFasterHand.py`.

The constant is poor. For 2^{20} drawers the bound says at least 181 touches, while Grover's search needs about 402 for even odds and 804 for near-certainty. But the shape is right: the touches must grow like $\sqrt{N}$. The example shows the same count at work in Grover's own run. There every touch spends exactly $1/\sqrt{N}$ of amplitude on each drawer, so Result 15.13

bounds the distance after T touches by $2T/\sqrt{N}$, which is $T/4$ for sixty-four drawers. The bound is 0.5 after two touches, against the actual 0.496, so Grover's touches waste almost nothing. After five touches the bound is 1.25, which is short of the 1.323 from $|s\rangle$ to $|w\rangle$, so five touches of this kind cannot bring the hand to the marked line. Six give 1.5, and six are what the stopping rule prescribes.

The sharp constant is BBHT's, and we state it without proof.

> **Result 15.15 (Boyer, Brassard, Høyer and Tapp, Theorem 7; stated, not proved).** Let the marked drawer w be chosen uniformly at random among N, and let M be any oracle quantum machine with bounded error probability. Then the expected number of times M must query the oracle "in order to determine y with probability at least $1/2$ is at least $\lfloor(\sin(\pi/8))\sqrt{N}\rfloor$", where their y is our w.

Grover's search reaches probability $\frac{1}{2}$ after about $\frac{\pi}{8}\sqrt{N}$ steps, and $(\pi/8)/\sin(\pi/8) \approx 1.026$. So, counting one query per step, the search spends about 2.6% more than the bound. BBHT call their bound "only a few percent smaller than the number of iterations required by Grover's algorithm". Because they charge two table look-ups per step (§15.3), their own conclusion reads that Grover's algorithm is "roughly within a factor 2 of being optimal in terms of the number of table lookups". Grover's §6 summarizes BBBV's result in words: "any quantum mechanical algorithm running for T steps is only sensitive to $O(T^2)$ queries", and "[BBHT96] gives a direct proof of this result along with tight bounds showing the algorithm of this paper is within a few percent of the fastest possible quantum mechanical algorithm." Result 15.14 is a weak form of that sensitivity, with its proof written out.

What the bound does not say. It speaks of drawers with no structure, a box that can only be asked. It does not say that quantum computers cannot solve problems in NP. Grover makes the point from the other side in his §1.1: if the number of steps could be reduced "to a finite power of $O(\log N)$ (instead of $O(\sqrt{N})$ as in this paper), it would yield a polynomial time algorithm for NP-complete problems". BBBV's reading, quoted in Chapter 13, is that

"there is no black-box approach to solving NP–complete problems by using some uniquely quantum-mechanical features". A problem with structure is not a sealed wall of drawers, and Shor's algorithm in Chapter 14 used structure.

Why this definition. Why measure a touch by the squared amplitude it spends on a drawer, rather than by which drawers it "looks at"? Because a quantum touch looks at every drawer a little, and the only honest measure of how much is how much of the state is on that drawer when the touch happens. The query magnitude is that measure, and its total over all drawers is fixed at one per touch. That fixed total is the whole reason the answer is a square root.

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§15.6 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 14's row 109. There is one new picture element, the Turning Plane (row 116), and it is a different kind of picture: not a string diagram and not part of Coecke's calculus. The other rows are new uses of old elements, and the last row says that there is no picture.

#	Algebra (Dirac)	Picture (Coecke)
110	the marker $O_w = I - 2\,\lvert w\rangle\langle w\rvert$, one query (row 37: the query box with the flag at $\lvert-\rangle$); for two cords $O_w = (-1)^{\bar{w}_1\bar{w}_2}(Z^{\bar{w}_2} \otimes Z^{\bar{w}_1})\,\mathrm{CZ}$	bare cords minus 2^{1-n} times red w-effects over red w-points; for two cords, the controlled turn at a half (row 103, $\alpha = \pi$), with a half-ring on the lower cord if $w_1 = 0$ and on the upper if $w_2 = 0$ (-1 beside it for $w = 00$); bare $= O_w/\sqrt{2}$

#	Algebra (Dirac)	Picture (Coecke)
111	$C^{n-1}Z = I - 2\,\lvert 1\cdots 1\rangle\langle 1\cdots 1\rvert$	the tall box as **a sum of two pictures**: bare cords, minus 2^{1-n} times n lit effects beside n lit points (rows 34–36)
112	Grover's $R = 2\,\lvert 0\cdots 0\rangle\langle 0\cdots 0\rvert - I = -X^{\otimes n}C^{n-1}Z\,X^{\otimes n}$; for two cords $R = (Z\otimes Z)\,\mathrm{CZ}$	red πs on every cord either side of the tall box, with -1 beside it; they fuse into the lit points and effects, which become dark (row 57). For two cords: a half-ring on each cord and the controlled turn at a half, no sign
113	**the headline:** $D = WRW = 2\,\lvert s\rangle\langle s\rvert - I$	collars on every cord either side of $R = 2^{1-n}$ (green effects beside green points) − bare cords (row 58; collars that meet cancel)
114	CCZ in dots (three cords)	a green eighth-turn on each cord; three hubs of two ties with pendants an eighth back; one hub of three ties with a pendant an eighth forward; bare $= 2^{-5/2}\,\mathrm{CCZ}$ (a hub of k ties gives $2^{-(k-1)/2}$). Checked, not derived
115	$WRWO_{01}\,W\lvert 00\rangle = \lvert 01\rangle$	***The Second Drawer***: one whole search on two cords, the marker and the turn-about drawn as separate pieces; value $\lvert 01\rangle$, scalar 1

#	Algebra (Dirac)	Picture (Coecke)
116	O_w, D on $\mathrm{span}\{\lvert w\rangle, \lvert w^{\perp}\rangle\}$: reflections across the $\lvert w^{\perp}\rangle$ and $\lvert s\rangle$ lines; DO_w = rotation by 2θ; $G^k \lvert s\rangle = \sin((2k+1)\theta)\lvert w\rangle + \cos((2k+1)\theta)\lvert w^{\perp}\rangle$	***the Turning Plane*** *— new; a different kind of picture, not a string diagram*: two folds make a turn by twice the tilt
117	optimality: BBBV Theorem 3.5 (Result 13.14); BBHT Theorem 7, $\lfloor \sin(\pi/8)\sqrt{N} \rfloor$; the fair share (Results 15.13, 15.14)	**no picture**: a rewrite proves an equality of processes, never a bound

Every string-diagram row is exact, scalar included. Rows 110 to 115 are equalities of processes. Row 116 is a drawing of a real plane that carries the step count, and row 117 is proved or cited on the blackboard. No row is a statement about running time drawn as a picture.

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§15.7 The Theorem, Twice

Result 15.5 (the turn-about is a reflection about the even state), restated. For every $n \geq 1$ and $N = 2^n$, exactly,

$$D \;=\; W R W \;=\; -\,W X^{\otimes n} C^{n-1} Z X^{\otimes n} W \;=\; 2\lvert s\rangle\langle s\rvert - I,$$

and hence $(Dv)_x = 2\bar{v} - v_x$, $D\lvert s\rangle = \lvert s\rangle$, $Du = -u$ whenever the entries of u sum to 0, and $D^2 = I$.

Dirac's Blackboard

Step 1: R through the Xs. $X^{\otimes n}$ flips every bit, $X^{\otimes n}|x\rangle = |\bar{x}\rangle$. $C^{n-1}Z$ turns over $|\bar{x}\rangle$ exactly when $\bar{x} = 1\cdots1$, that is, when $x = 0\cdots0$. So

$$X^{\otimes n}\, C^{n-1}Z\, X^{\otimes n} \;=\; I - 2\,|0\cdots0\rangle\langle 0\cdots0| \;=\; -R.$$

Step 2: WRW. By Result 15.4, $W|0\cdots0\rangle = |s\rangle$, and since W is real and symmetric, $\langle 0\cdots0|\,W = \langle s|$. With $W^2 = I$,

$$WRW \;=\; 2\,W|0\cdots0\rangle\langle 0\cdots0|\,W - W^2 \;=\; 2\,|s\rangle\langle s| - I.$$

Together with Step 1, $-WX^{\otimes n}C^{n-1}ZX^{\otimes n}W = WRW$.

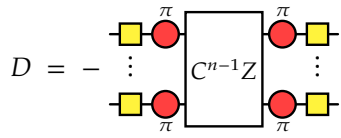
Step 3: Grover's entries. $|s\rangle\langle s|$ has every entry $1/N$, so $(2\,|s\rangle\langle s| - I)_{xy} = 2/N - [\,x = y\,]$, which is Grover's D (his (4.0)).

Step 4: the consequences. For any v, $\langle s|v\rangle = \frac{1}{\sqrt{N}}\sum_x v_x = \sqrt{N}\,\bar{v}$, so $(2\,|s\rangle\langle s|v\rangle)_x = 2\bar{v}$ and $(Dv)_x = 2\bar{v} - v_x$. Next, $D\,|s\rangle = 2\,|s\rangle - |s\rangle = |s\rangle$. If the entries of u sum to 0, then $\langle s|u\rangle = 0$ and $Du = -u$. Finally $D^2 = 4\,|s\rangle\langle s|s\rangle\langle s| - 4\,|s\rangle\langle s| + I = I$. ■

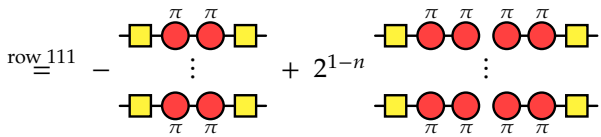
Coecke's Sketchbook

Draw n cords as the top cord and the bottom cord, with dots between for the rest; every column acts the same way on every cord. The yellow box is H (row 56), so a column of yellow boxes is W. A red π dot with one leg in and one out is X exactly (§8.2). Each step below names its rule.

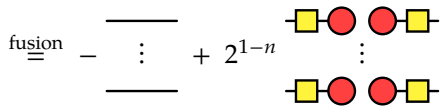
Step 0: the turn-about as drawn. Grover's turn-about is $D = WRW$ (§15.2): a column of yellow boxes either side of R. We draw R with pieces already on the page, the tall box $C^{n-1}Z$ (row 111) with a red π on every cord either side of it and -1 written beside the picture (row 112). That this drawing is R is not assumed here. Step 2 derives it by the rules:



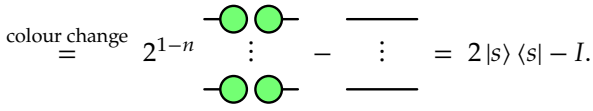
Step 1: open the tall box (Result 15.6 with $c = 1 \cdots 1$, row 111). The tall box is the bare cords minus 2^{1-n} times lit effects beside lit points. Distributing the -1:



Step 2: fuse the red dots (fusion, Theorem 8.1, row 57). On the bare-cords term, each cord's two red πs fuse into one red dot of phase $2\pi = 0$, and a phase-free red dot with one leg in and one out is a plain cord (§8.2). The two yellow boxes then meet, and $H^2 = I$ removes them. On the other term, each red π fuses into the lit effect or lit point beside it, and $\pi + \pi = 0$ leaves a dark effect and a dark point. Look at the middle, between the two columns of yellow boxes, before those on the bare cords cancel: it is 2^{1-n} dark effects beside dark points, minus the bare cords. That is Result 15.6 at $c = 0 \cdots 0$ with its sign reversed, $-(I - 2\,|0 \cdots 0\rangle\langle 0 \cdots 0|) = R$. So the chain passes through WRW, and Step 0's drawing of R is derived:



Step 3: change colour (colour change, Theorem 8.2, row 58). A yellow box on the only leg of a red effect or point turns it into a green one of the same phase:



Step 4: read off the value. The n green points are $2^{n/2}\,|s\rangle$ (Result 15.4), and the n green effects are their mirror image, $2^{n/2}\,\langle s|$. So the first term is $2^{1-n}\cdot 2^n\,|s\rangle\langle s| = 2\,|s\rangle\langle s|$, and the bare cords are I. The scalar 2^{1-n} is carried unchanged through Steps 1 to 3: fusion, the removal of a plain dot, $H^2 = I$ and colour change cost nothing.

The entries come back as closed pictures too. Close the last picture with red effects x on the right and red points y on the left, each worth $\sqrt{2}$ times the corresponding bra or ket. A green effect meeting a red point gives $(\langle 0| + \langle 1|)\sqrt{2}\,|y_i\rangle = \sqrt{2}$ on each cord, and so does a red effect meeting a green point. The first term closes to $2^{1-n}\cdot 2^{n/2}\cdot 2^{n/2} = 2$. The bare cords close to $\prod_i \sqrt{2}\,\langle x_i|\,\sqrt{2}\,|y_i\rangle = 2^n[\,x = y\,]$. So $2^n\,\langle x|\,D\,|y\rangle = 2 - 2^n[\,x = y\,]$, which is Grover's $2/N$ off the diagonal and $-1 + 2/N$ on it. ■

The picture chain is general in n and exact. What it does not do is open $C^{n-1}Z$ into dots: it uses the tall box and its sum of pictures (row 111). For two cords the tall box is the controlled turn at a half (Result 15.7), derived by the rules. For three cords it needs eighth-turns (§15.9), and that drawing is checked by matrix, not derived here.

The verdict

The blackboard wins, and by a wide margin. Step 2 is two lines, it holds for every n, and Step 1 is a remark about flipping bits. The sketchbook needs four pictures and a paragraph of scalars to reach the same equation. Shor and Grover are won by the algebra, and this chapter's headline is no exception.

What the pictures show is the shape the algebra compresses: **all the cords, minus twice the one thing, seen crosswise**. The marker turns over a dark-and-lit thing. The turn-about, apart from its overall sign, turns over the even

thing, and the collars are what turn "dark" into "even". The same shape in two colours is the reason the two operations are reflections of the same kind, and it is the reason §15.4's plane exists. But the picture is an equality of processes, and it counts nothing. It does not say how many steps a search needs, or that $\sqrt{N}$ is the right number, or that no method does better. Those came from the blackboard and from the Turning Plane.

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§15.8 The Second Drawer

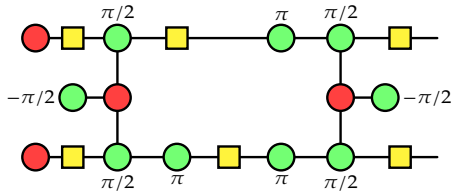
Here is the chapter's master diagram: a whole search on four drawers, Grover's steps (i), (ii)(a) and (ii)(b) done once, with the second drawer marked. On the blackboard it is

$$W\,R\,W\,O_{01}\;W|00\rangle\,.$$

It is drawn **gate by gate, unfused**, so that the marker and the turn-about stay separate pieces, as Grover's two operations are. Fusing the green dots on each cord would be a valid rewrite (Theorem 8.1) and would give the same process drawn more compactly, but it would merge the marker's half-ring into the marker's ring and the turn-about's half-rings into its rings, and the two operations would no longer be told apart.

From left to right. The **upper row**: a dark point; a yellow box; a green ring $\pi/2$, the marker's upper ring; a yellow box; a green π, the turn-about's upper half-ring; a green ring $\pi/2$, the turn-about's upper ring; a yellow box; the first output. The **lower row**: a dark point; a yellow box; a green ring $\pi/2$, the marker's lower ring; a green π, the marker's half-ring; a yellow box; a green π, the turn-about's lower half-ring; a green ring $\pi/2$, the turn-about's lower ring; a yellow box; the second output. **The marker's hub** is a phase-free red dot between the rows, tied to both marker rings, with a pendant green point $-\pi/2$ hanging to the left: this is the controlled half-turn, and with the half-ring on the lower row it is $O_{01} = (I \otimes Z)\,\mathrm{CZ}$ (Result 15.7). **The turn-about's hub** is the same, tied to both turn-about rings, with its pendant hanging to the right; with the two half-rings it is $R = (Z \otimes Z)\,\mathrm{CZ}$,

and with the yellow boxes on either side it is $WRW = D$. We call the picture **The Second Drawer**.



The rows run straight, and no tie crosses a row: each hub sits between the rows, directly below its upper ring and above its lower ring.

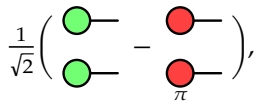
Result 15.16 (The Second Drawer). Exactly, with no scalar,

$$\text{The Second Drawer} \;=\; |0\rangle\,|1\rangle\,.$$

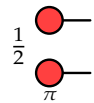
The upper output is dark and the lower is lit: read top cord first, the second drawer, 01.

Proof. **On the blackboard.** The two dark points are $\sqrt{2}\,|0\rangle$ each, together $2\,|00\rangle$. The two bare controlled half-turns are $\mathrm{CZ}/\sqrt{2}$ each (Result 14.5), together $\frac{1}{2}$ times the matrices. With Result 15.7, the picture is $2 \cdot \frac{1}{2} \cdot W(Z \otimes Z)\mathrm{CZ}\,W\,(I \otimes Z)\mathrm{CZ}\;W|00\rangle = WRW\,O_{01}\,W|00\rangle = D\,O_{01}\,|s\rangle$, and by Result 15.8 this is $|01\rangle$.

In the sketchbook. Colour change (Theorem 8.2) turns each dark point with its yellow box into a green point. By Result 15.7, the marker's piece is $\frac{1}{\sqrt{2}}O_{01}$, and by Result 15.6 with $c = 01$ it is $\frac{1}{\sqrt{2}}$ times the difference of two pictures: the bare cords, minus $\frac{1}{2}$ times a dark and a lit effect beside a dark and a lit point. Feed it the two green points. On the first term they pass untouched. On the second, each red effect meets a green point and closes to $\sqrt{2}\,\langle c|\,(|0\rangle + |1\rangle) = \sqrt{2}$, so the two cords give 2 and cancel the $\frac{1}{2}$. What remains is



two green points minus a dark point above a lit point: on the blackboard, $\frac{1}{\sqrt{2}}(2\,|s\rangle - 2\,|01\rangle) = \sqrt{2}\,O_{01}\,|s\rangle$. Now the turn-about's piece. By Result 15.7 it is $\frac{1}{\sqrt{2}}D$, and by the chain of §15.7 with $n = 2$ it is $\frac{1}{\sqrt{2}}(\frac{1}{2}$ (green effects beside green points) $-$ bare cords). Apply it to each of the two pictures above. A green effect meeting a green point closes to $(\langle 0| + \langle 1|)(|0\rangle + |1\rangle) = 2$, and meeting a red point it closes to $\sqrt{2}$. So the two green points give $\frac{1}{2} \cdot 4$ green points minus the green points themselves, which is the green points once more. The dark-and-lit pair gives $\frac{1}{2} \cdot 2$ green points minus the dark-and-lit pair. Subtracting, the green points cancel exactly, and with the two factors of $\frac{1}{\sqrt{2}}$ what is left is



$= \frac{1}{2} \cdot \sqrt{2}\,|0\rangle \cdot \sqrt{2}\,|1\rangle = |0\rangle\,|1\rangle\,.$

■ See `Code/Ch15/SecondDrawer.py`, which builds this picture, evaluates it with its scalar, and reduces it by the rules to one point per output, phase-free above and π below.

The drawer was chosen so that its reading is not the same both ways up. Read from the bottom cord up, the outputs would say 10, the third drawer, and the code checks that the amplitude there is exactly 0. The drawer is also written into the picture itself: the marker's half-ring sits on the lower row, and the lower row is the one that comes out lit. Mark the third drawer instead and the half-ring moves to the upper row. Mark the fourth and it disappears. Mark the first and the marker has a half-ring on each row and the number -1 beside it (Result 15.7). In each case the search finds its drawer, with the first drawer's value carrying the marker's sign, $-|00\rangle$.

Coecke's Sketchbook.

The ledger of the Second Drawer. Every number in the proof can be named, and they balance. The picture carries two points worth $\sqrt{2}$ each, which is 2, and two bare controlled half-turns worth $\frac{1}{\sqrt{2}}$ each, which is $\frac{1}{2}$. So nothing needs to be written beside it.

Inside the proof, three closings do the work. A red effect meets a green point: $\sqrt{2}$. A green effect meets a green point: 2. A green effect meets a red point: $\sqrt{2}$. The marker's $2^{1-n} = \frac{1}{2}$ is paid back by two of the first kind. In the turn-about, the green points are fed back into themselves, which gives $\frac{1}{2} \cdot 4 - 1 = 1$ copy, while the dark-and-lit pair returns $\frac{1}{2} \cdot 2 = 1$ copy of the green points. The two copies cancel, and only the dark-and-lit pair survives, at $\frac{1}{2}$ from the two hubs. The cancellation is the search. After the touch the state is half the even state plus a part whose entries sum to zero. The turn-about keeps the first and turns the second over, and what is left is the marked drawer alone.

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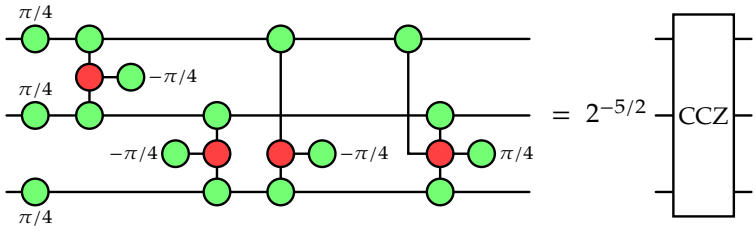
§15.9 What the Pictures Leave to the Blackboard

Take stock. Asking one by one costs $N - 1$ asks in the worst case (Result 15.1), and no classical plan does better than $(k + 1)/N$ with k asks (Result 15.2). The marker is one query, a sign that no reading sees (Result 15.3), laid on the even state (Result 15.4). The turn-about is a reflection about the even state (Result 15.5), built from one-thing-turned-over boxes (Result 15.6), and for two cords both reflections are drawn with no sign (Result 15.7). One step finds one drawer in four (Result 15.8), and The Second Drawer derives that search in dots (Result 15.16). Two folds make a turn (Result 15.9), so the search turns by 2θ a step (Result 15.10), stops after about $\frac{\pi}{4}\sqrt{N}$ steps (Result 15.11), and does the same with many marked drawers (Result 15.12). A touch moves the answer only by its share (Result 15.13), so no method has a faster hand than about $\sqrt{N}$ (Results 15.14 and 15.15).

Now the honest part. **Shor and Grover are won by the algebra, and the book says so.** The following have no string-diagram proof in this chapter: the step count, the closed form $\sin((2k + 1)\theta)$, the stopping rule, the square root, and the bound that no method does better. The first four came from the blackboard and from the Turning Plane, which is a drawing of the blackboard's plane and not a diagram of processes. The last is proved by

counting (Results 15.13 and 15.14) or cited (Results 13.14 and 15.15). A rewrite proves that two processes are equal. It proves that this network, with this box, returns this state. It proves no bound on queries, and nothing about optimality. The Second Drawer shows that one step on four drawers is certain. It says nothing about sixty-four.

There is also a limit to what the pictures in this chapter have been built from. The Second Drawer uses only quarter-turns and half-turns: rings at $\pi/2$, half-rings at π, pendants at $-\pi/2$. For eight drawers the turn-about needs CCZ, the three-cord $C^{n-1}Z$, and its dots need eighth-turns. Here is the drawing. On each of three cords sits a green dot with phase $\pi/4$. Three hubs of two ties each join the pairs of cords, through phase-free green rings, with pendants at $-\pi/4$. One hub of three ties joins all three cords, with a pendant at $+\pi/4$. The ties from the top cord to the lowest hubs pass the middle cord without touching it: where a tie crosses a cord and no dot is drawn, the two do not meet.



Result 15.17 (the eight-drawer turn-about needs eighths; checked, not derived). Exactly, the bare picture above is $2^{-5/2}$ CCZ. Each hub with k ties, with its pendant, contributes $2^{-(k-1)/2}$: three hubs of two ties and one of three give $2^{-3/2} \cdot 2^{-1}$.

Why it holds, on the blackboard. On a drawer $x_1x_2x_3$, each phase-free green ring copies its cord's bit up its tie, each hub reads the parity of the bits it receives (Result 14.5's mechanism), and each pendant turns by its phase times that parity. For bits,

$$x_1x_2x_3 \;=\; \tfrac{1}{4}[(x_1+x_2+x_3)-(x_1\oplus x_2)-(x_2\oplus x_3)-(x_1\oplus x_3)+(x_1\oplus x_2\oplus x_3)],$$

as the eight cases confirm. Multiplied by π, the right side is exactly the sum

of the turns in the picture, and the left side is the turn of CCZ. A hub of k ties reading bits gives $2^{-(k-1)/2}$ times the parity point, which accounts for the scalar. ■ We have not derived this picture from the tall box by the rules of Chapter 8, and we do not claim to. See `Code/Ch15/TurnAbout.py`, which checks it by matrix and runs an eight-drawer search built from it.

With collars and red πs on either side and $-2^{5/2}$ beside it, the picture is the eight-drawer turn-about. Its finest turn is an eighth, and nothing in The Second Drawer is finer than a quarter.

The chapter's question has an answer. To find one marked drawer among N, an ordinary searcher must open drawers one at a time, and no plan does better than one drawer's worth of chance per ask. A quantum searcher touches all the drawers at once, and each touch turns the marked one over where no reading can see it. The turn-about then reflects the whole state about its average. Together the two reflections turn the state by a fixed small angle, twice the tilt between the even line and the unmarked line. After about $\frac{\pi}{4}\sqrt{N}$ touches the state stands almost upright on the marked drawer, and one more touch would carry it past. And no method can ask fewer than about $\sqrt{N}$ times, because each touch has only one unit of amplitude to spread over all the drawers, and a drawer that gets only its fair share cannot be told apart from an empty wall. The pictures drew the touch and the turn-about, and the whole four-drawer search, dot by dot. The count they left to the blackboard, and so did we.

Chapter 16 looks inside the pictures instead of at what they compute. Everything in The Second Drawer is a quarter-turn or a half-turn, and the eight-drawer turn-about needed an eighth. That looks like a small difference, one angle halved. Chapter 16, *Inside the Clifford World*, asks what separates the circuits built from quarters and halves from the ones that need eighths, and why that difference matters so much more than its size suggests.

Twenty-Four Preludes and a T

春待つや
二十四落ちて
揺るる粒

haru matsu ya
nijūshi ochite
yururu tsubu

Waiting for the spring —
twenty-four drops have fallen,
and one hangs, trembling.

[*The Crab's parlour, the afternoon of the next day. On the table, on a folded woollen cloth, stands something low and square under a cover of green baize; beside it stands a small stand of darker turned wood with a pale wooden solid seated in it. The Tortoise is in the chair nearest the fire, her work-bag at her feet. The door opens, and Achilles comes in with his boots in one hand.*]

[*The parlour he comes into is warm from end to end. The frost went in the night, and all afternoon the eaves have been giving it up, a drop past the glass and another and then a run of them together, into the slush on the sill. The one tall window at the near end, each of its small panes blown by hand, looks up the lane, all ruts and water now, to the church at the top of it, where the ringers have come in for their afternoon practice and are raising the tower's bells: a stray note, a pause, two together, the tumble of them quickening and climbing as each bell swings higher, in no order anybody could follow. On the long wall the fire burns in a grate framed*

with blue-and-white Delft tiles, a little ship or a flower painted on each. The tiler set every tile whichever way up it came to his hand, and every one sits square against its neighbours, and wherever four corners meet their painted sprigs make a small flower between them. The kettle sits at the side of the hob; the dresser holds the Crab's cups, bought one at a time, no two of a pattern, and the cupboard beside it is shut. At the far end of the room, on the end wall beyond the table, hangs a print in a plain black frame: Escher's Stars, *many-sided solids hanging in a dark space. The window's light reaches it only at second hand, off the ceiling, and the dark inside the frame is darker than the wall round it. It was hanging there before the postman came up the lane this morning, and it will be hanging there when he next comes up it.*]

Achilles: The frost's broken! The lane's gone to soup from your door to the church, and every eave in the village is dripping on somebody.

Crab: It went in the night. I heard the first drip at three o'clock and lay awake till it was light, waiting for the second. Boots by the fender, if you please, and you by the fire.

Achilles: [*setting his boots by the fender and unwinding his scarf*] The stair at Madam Tortoise's groaned at me on the way down last night, exactly as she promised. I stepped over it, and it groaned anyway.

Tortoise: It groans at people who step over it. And this morning it said nothing at all when I came down. I think the thaw has got into its temper.

Achilles: [*folding the scarf over the back of his chair*] Then everything is softening today. Even stairs.

Crab: Except my patience. Tea first. [*He pours for all three, and sets the pot down on the side of the table away from the fire, at the right of Achilles's place.*] And before you ask what is on my table, because you were about to ask...

Achilles: I was about to ask.

Crab: A parcel from Delft, this morning, from the little workshop where the old man and his daughter made my music boxes. The postman carried it up the lane as though it might hatch.

Achilles: [*nodding at the wooden stand*] Is that what hatched?

Crab: That is the half I have unpacked. [*He sets the stand nearer the edge of the table on Achilles's side, at the left of the tray as seen from Achilles's place, where they can all see it.*] This is a model, turned from a single piece of pale wood, with six tips, and every tip painted: a dark green and a pale green opposite each other, a dark red and a pale red, a dark gold and a pale gold. And this is its cradle: a socket in the middle to take the bottom tip, and four notched lugs round it at the height of the model's waist, one at the front, one at the right, one at the back and one at the left, each to take a tip. The daughter's letter says it is for learning how a thing may be turned and still sit.

[*The model sits in its cradle as it came out of the straw, the cradle's front lug towards Achilles's place and its right-hand lug at his right: the dark green tip straight up, the pale green down in the socket, the dark red in the front lug, the dark gold in the right-hand lug, the pale red in the back lug and the pale gold in the left.*]

Achilles: It looks like a spinning top that has decided not to.

Crab: It is far too well brought up to spin. [*He lays a claw on the baize.*] And the other half of the parcel came last week, and I have been keeping it covered for company.

[*He lifts the baize away and folds it over the end of the dresser. Beneath it, on the woollen cloth, stands a tray of unpolished dark wood with a raised rim, its front towards Achilles's place, so that its left and right are his. On the tray stand five small silver bells, each mouth-down on its own green felt pad, each with a little crown loop on top. One bell stands in the middle of the tray. The others stand one near each corner: at the left, a left-back bell and a left-front bell; at the right, a right-back bell and a right-front bell.*]

[*Cream silk ties run between the crown loops, each knotted at both its ends, each wearing a short yellow sleeve halfway along its length. One tie runs from the middle bell to the left-back bell; one from the middle bell to the left-front bell; one from the middle bell to the right-back bell; one from the middle bell to the right-front bell. At the left, one tie runs from the left-back bell to the left-front bell; at the right, one from the right-back bell to the right-front bell. The front and the back of the tray are open: no tie runs along them.*]

[*Every bell has its pull. From the left-back bell's crown loop an inner length of cord*

runs to a small brass ring, knotted at the loop and at the ring; from the same brass ring an outer length runs on, knotted at the ring and at a silk tassel that hangs over the left side of the rim, just by the left-back corner. The left-front bell's pull is made the same way, an inner length to its brass ring and an outer length to its tassel, and hangs over the left side of the rim by the left-front corner. The right-back bell's hangs over the right side of the rim by the right-back corner, and the right-front bell's over the right side by the right-front corner, each made the same way. The middle bell's pull runs straight forward to the front rim, between the two front bells, crossing no tie: its inner length to its brass ring, its outer length to its tassel. No cord passes through a ring; each length is knotted at both its ends.]

[*As the baize comes away, the five bells shiver on their pads, all together, a tremble with no sound in it, and are still again as soon as the Crab's claw has gone. The window's pale light falls across the left of the tray and the fire's across the right, so that the left wing's silk lies grey and cool, the right wing's takes a warmer colour, and the middle bell's own ties, and its pull running forward to the front rim, catch a little of both. Each yellow sleeve is a short bright collar on its tie, and the tassels hang over the rim without a stir. Burnt small into the back of the rim are the workshop's little anchor and the old man's initials.*]

Achilles: It's a bow tie. Two wings and a knot in the middle.

Crab: And the middle bell is the knot; the old man called it so himself. This is a bell-tray, the newest thing from the same bench. The five bells are hung together by their ties, so that no one of them is quite a bell on its own any more.

Tortoise: He has been polishing the rules of it all week. Tell him the rule.

Crab: There is only one. Nobody rings it. Pull a tassel sharply and that bell tips on its pad and rings; you would hear that one bell, and the others would go dull, and the tray is spent for the day.

Achilles: Not even a very small ring?

Crab: There is no such thing as a very small ring. Drink your tea.

Achilles: [*taking the cup in both hands*] It's the first thing today that's been warmer than I am.

Tortoise: [*setting her work-bag against the leg of the table*] Then sit, and let it do its work. The afternoon's long, now; it gets a little longer every day.

[*Achilles sits down by the fire with his cup and puts his stockinged feet to the fender, beside his boots, which have begun to dry. Up the lane the tumble from the tower has stopped: the bells are up, and for a while the tower keeps quiet, and there is only the drip from the eaves and the small talk of the fire. When his toes are warm he draws his chair up to the table.*]

First Challenge — *The Twenty-Four*

Achilles: [*looking at the model over his cup*] How many ways will it sit in that cradle? Properly sat, I mean, with a tip in the socket and a tip in every lug.

Crab: Try it and see. Turn it every way you like.

Tortoise: Count before you try. Six tips could be up, and any of the others in front?

Achilles: Any of the other five. Six times five: thirty.

Tortoise: Then seat me one with the dark green still on top and the pale green in the front lug.

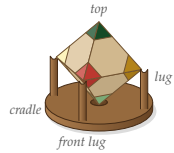
[*Achilles lifts the model clear of the cradle, keeping the dark green tip upright between his fingers, and looks for the pale green. It is straight underneath, pointing down at the socket. He turns the model about its upright line, slowly, all the way round: the dark red comes to the front, then the pale gold, then the pale red, then the dark gold, and the dark red again. The pale green does not come round with them. Whichever way he turns it, the pale green stays pointing at the socket, opposite the dark green, and to bring it to the front he would have to tip the dark green off the top.*]

Achilles: ...Oh. Only the four round the top. The one opposite the top is always in the socket.

Tortoise: So?

Achilles: Six tips to put up, and then only four of the rest for the front. Six times four, twenty-four.

[*He sets the model back as it came: the dark green up, the dark red in the front lug, the pale green in the socket.*]



Seated as it came

Crab: Twenty-four, and I mean to see every one of them this afternoon, each of them once, and the model home again at the end. You turn, and I shall call. I have three calls.

Achilles: Only three?

Crab: The daughter says three are plenty. "Quarter": lift it a hair, and turn it a quarter about the up-and-down line, so that the front tip goes to the right-hand lug and the left tip comes round to the front. "Back": the same the other way, so that the front tip goes to the left lug and the right tip comes to the front.

Tortoise: And the third?

Crab: "Over", which is the one the letter is proudest of. Take the model by the top tip and the front tip, and give it a half turn about the middle of the edge between them: those two change places, the left and right tips change places, and the back and bottom tips change places.

Crab: And each time you set it down, name the seating: the tip that is up, and the tip in the front lug. The other four follow from those two.

[*Up the lane the ringers begin again, and now in order: the tower's bells one after another from the highest to the lowest, a plain falling scale, and then the same scale, and the same, every stroke in its place. Achilles sets his cup down on the table, well away from the tray, and rubs his cold fingers together.*]

Achilles: Then here's where it starts.

Achilles: Dark green up, dark red front.

Crab: Quarter.

[*The model as it stands: the dark green up, the dark red in the front lug, the dark gold at the right, the pale red at the back, the pale gold at the left, the pale green in the socket. Achilles lifts it a hair clear of the lugs and turns it a quarter about its upright line, the way a key turns in a lock. The dark green stays on top and the pale green stays beneath; only the waist goes round. The dark red swings from the front lug to the right, the dark gold from the right to the back, the pale red from the back to the left, and the pale gold comes round from the left into the front. He lowers it, and four tips knock softly into four lugs. What has changed: which tip sits in which lug round the waist. What has held: the top, the bottom, and the order of the four colours round the middle.*]



Turned about the upright

Achilles: Dark green up, pale gold front.

Crab: Quarter again.

[*Again the waist goes round a quarter under the unmoving dark green: the pale gold leaves the front for the right-hand lug, and the pale red comes round from the left into the front. Knock.*]

Achilles: Dark green up, pale red front.

Achilles: And the dark red's gone round to the back, where the pale red was. Two quarters, and every tip round the middle is where its opposite stood.

Crab: Quarter.

[*The pale red goes to the right; from the left the dark gold comes into the front. The dark green has not stirred from the top.*]

Achilles: Dark green up, dark gold front.

Crab: That is the dark green's four. Now: over.

[*The model as it stands: the dark green up, the dark gold in the front lug, the pale red at the right, the pale gold at the back, the dark red at the left, the pale green in the socket. Achilles takes it by the top tip and the front tip, one between the fingers of each hand, lifts it clear, and turns it a half turn about the middle of the edge that runs between those two tips, as one turns a pillow over by its corner. The dark green comes down and forward into the front lug, and the dark gold goes up to the top. The pale red and the dark red change sides, right for left. The pale gold at the back swings down into the socket, and the pale green comes up out of the socket into the back lug. He sets it down, and it drops into its lugs with one knock. What has changed: every tip's place, and, for the first time, the top. What has held: every tip still has its opposite straight across from it, and every lug has a tip in it.*]



Turned over

Achilles: Dark gold up, dark green front.

Crab: [*leaning in*] By the tips, if you please, not by the faces. The paint is three weeks old.

Achilles: My hands are still cold from the lane. I'm holding it like an egg.

Crab: Then it is warming them, and you are warming it, and everybody is served. Back.

[*The model as it stands: the dark gold up, the dark green in the front lug, the dark red at the right, the pale green at the back, the pale red at the left, the pale gold in the socket. Achilles lifts it a hair and turns it a quarter the other way about its upright line: the dark green goes from the front to the left-hand lug, the pale red from the left to the back, the pale green from the back to the right, and the dark red comes from the right into the front. The dark gold on top and the pale gold beneath do not move. He lowers it; the knock. What has changed: the waist has gone round the*

other way. What has held: the top, the bottom, and the order round the middle, read the other way.]

Achilles: Dark gold up, dark red front.

Crab: Back.

[*The dark red goes from the front to the left; from the right the pale green comes into the front.*]

Achilles: Dark gold up, pale green front.

Crab: Back.

[*The pale green goes to the left lug, and the pale red comes round from the right into the front.*]

Achilles: Dark gold up, pale red front.

Crab: Over.

[*He takes it by the dark gold on top and the pale red in front, and turns it over by the edge between them: the pale red rises to the top and the dark gold comes down into the front lug; the dark green and the pale green change sides; the dark red at the back drops into the socket, and the pale gold comes up out of the socket into the back lug. Knock.*]

Achilles: Pale red up, dark gold front.

[*The Tortoise nudges the sugar bowl across the table to him. Achilles takes a lump in his fingers; it slips, tumbles across the saucer, and comes to rest the way a lump always does, square on one of its faces, though which face, and which way about, nobody could have said before it fell. He drops it into his tea and licks the sweetness off his thumb.*]

Achilles: Whatever I do to it, it comes down with a tip in every lug. Never an edge, never a face. The six places are always the same six places; the only thing that changes is which tip is in each.

Crab: That is what it is to sit. A seating puts a tip where a tip was. Quarter.

[*The waist goes round a quarter under the pale red: the dark gold goes to the right-hand lug, and the dark green comes round from the left into the front.*]

Achilles: Pale red up, dark green front.

Crab: Quarter.

[*The dark green goes to the right, and the pale gold comes round into the front.*]

Achilles: Pale red up, pale gold front.

Crab: Quarter.

[*The pale gold goes to the right, and the pale green comes round from the left into the front. The pale red still points at the ceiling.*]

Achilles: Pale red up, pale green front.

Tortoise: [*refilling her cup*] Half-way. You're turning it more gently than you drink tea.

Achilles: It deserves it more. Over?

Crab: Over.

[*Top and front change places: the pale red comes down into the front lug and the pale green goes up. The gold tips change sides, right for left; the dark green at the back goes down into the socket, and the dark red comes up out of it to the back. Knock.*]

Achilles: Pale green up, pale red front.

Crab: Back.

[*The waist goes round the other way under the pale green: the pale red goes to the left, and the dark gold comes from the right into the front.*]

Achilles: Pale green up, dark gold front.

Crab: Back.

[*The dark gold goes to the left, and the dark red comes round from the right into the front.*]

Achilles: Pale green up, dark red front.

Crab: Back.

[*The dark red goes to the left, and the pale gold comes from the right into the front.*]

Achilles: Pale green up, pale gold front.

Crab: Over.

[*Achilles takes it by the pale green on top and the pale gold in front and turns it over by their edge: the pale gold goes up and the pale green comes down into the front; the reds change sides; the dark gold at the back goes down into the socket, and the dark green comes up out of it to the back. Knock.*]

Achilles: Pale gold up, pale green front.

Achilles: Every tip has had its turn on top except the dark red. It's been waiting all this while.

Crab: It will have its four. Quarter.

[*The pale green goes from the front to the right, and from the left the pale red comes round into the front.*]

Achilles: Pale gold up, pale red front.

Crab: Quarter.

[*The pale red goes to the right, and the dark green comes round into the front.*]

Achilles: Pale gold up, dark green front.

Crab: Quarter.

[*The dark green goes to the right, and the dark red comes round from the left into the front.*]

Achilles: Pale gold up, dark red front.

Crab: Over.

[*Top and front change places once more: the dark red rises to the top at last and the pale gold comes down into the front lug; the greens change sides; the pale red at the back goes down into the socket, and the dark gold comes up out of it into the back lug. Knock.*]

Achilles: Dark red up, pale gold front.

Crab: Back.

[*The pale gold goes to the left, and from the right the pale green comes into the front.*]

Achilles: Dark red up, pale green front.

Crab: Back.

[*The pale green goes to the left, and the dark gold comes into the front.*]

Achilles: Dark red up, dark gold front.

Crab: Back.

[*The dark gold goes to the left, and the dark green comes round from the right into the front.*]

Achilles: Dark red up, dark green front.

Crab: [*softly*] And over.

[*The model as it stands: the dark red up, the dark green in the front lug, the pale gold at the right, the pale green at the back, the dark gold at the left, the pale red in the socket. Achilles takes it by the dark red and the dark green and turns it over by the edge between them: the dark green rises to the top and the dark red comes down into the front; the dark gold comes across to the right and the pale gold to the left; the pale green at the back drops into the socket, and the pale red comes up out of it into the back lug. It knocks into its lugs. Every tip is where it was when it came out of the straw.*]

Achilles: Home again.

[*Up the lane the tower is still at its plain falling scale, stroke after stroke in the same order, just as it was when the Crab made his first call. The Crab lets out a breath he seems to have been holding all the while.*]

Crab: Every seating once, and not one twice. [*He puts his claws together.*] The daughter will be pleased with you.

Crab: And the three calls were all it needed. There is no seating the model has that they cannot reach.

Achilles: But I'd have thought the plainest turns would be half-turns, straight about one of its own lines. You never called one.

Crab: They are in the twenty-four already. Let me show you. [*He takes the model from the cradle himself, between two careful claws.*]

[*The model as it stands, at home: the dark green up, the dark red in the front lug, the pale red at the back, the dark gold at the right, the pale gold at the left, the pale green in the socket. The Crab holds it by the dark red and the pale red, one claw at each, and turns it a half turn about the line between them, like a spit. The dark red and the pale red stay where they are, front and back. The dark green goes down and under into the socket, and the pale green comes up to the top; the dark gold goes over from the right to the left, and the pale gold from the left to the right. He sets it down: it sits, a tip in every lug. What has changed: four tips have gone to the places of their opposites. What has held: the front and the back, and every line from a tip to its opposite, which still runs up and down, or front to back, or side to side.*]



Turned like a spit

Crab: A half about the red line.

Achilles: Every tip is either in its own place or in its opposite's.

[*The Crab turns it the same half again about the same line, and every tip goes back: the dark green to the top, the golds to their own sides. Home. Then he holds it by the dark green and the pale green and turns it a half about the up-and-down line: the top and the bottom hold; the dark red goes to the back and the pale red comes to the front; the golds change sides. It sits. He turns it the same half again, and it is home.*]

Crab: A half about the green line.

[*Last, he holds it by the dark gold and the pale gold and turns it a half about the side-to-side line: the golds hold, right and left; the dark green goes down into the socket and the pale green comes up; the dark red goes to the back and the pale red to the front. It sits. The same half again, and it is home, the dark green up and the dark red in the front lug.*]

Crab: And a half about the gold line. Each of them seats it, and each of them sends every tip to its own place or its opposite's. Remember the red half and the green half. We shall want them presently.

[*Achilles finishes his tea, sweet at the bottom, and the Crab fills the cup again without being asked. The fire settles in the grate with a soft collapse, and in the new light from it the tiles round it show their little ships and flowers, every one of them sitting square.*]

Achilles: [*glancing from the model to the bell-tray beside it*] On the bells? Can you turn one of those without the others knowing?

Second Challenge — *Safe Touches*

Crab: Not freely. The old man hung the five bells even, and then knotted on the ties one by one; each tie gives the bell at one end a green half whenever the bell at the other end is lit. So no bell on that tray is a bell by itself any more. Turn one, and the ties tell the others.

[*From the church the falling scale breaks into changes: at every stroke two of the bells, side by side in the order, trade places, so that no order comes twice running. Achilles glances up at the window. On the table, a yard from the glass, the tray's bells do not so much as stir.*]

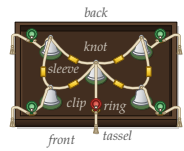
Achilles: And ringing it spoils the lot.

Crab: Ringing spoils the lot. But there are touches it does not notice. [*From his waistcoat pocket he takes a small painted tin, opens it, and sets it on the table beside him.*] These are my clips. Each snaps onto one of the brass rings on the pulls, and each is red or green, with a little bead set at a half, a quarter round or a quarter back; a clipped ring takes its clip's colour and its turn, and not a cord is changed. Two clips of one colour on a ring add their turns, and two colours on one ring I never allow.

Achilles: So a red clip at a half does to its bell what your half about the red line did to the model.

Crab: Just so. Now watch. Here is a touch the tray will not notice.

[*The tray as it stands: five bells on their pads, the ties in their sleeves, every pull hanging in its two lengths from bell to brass ring to tassel, and every brass ring bare. The Crab takes a red clip at a half from the tin, its bead at the bottom as Achilles sees it, half-way round from the top, where a bead at rest would stand, and snaps it onto the knot's brass ring, where the knot's pull crosses the front of the tray. Then a green clip at a half onto the left-back bell's ring; a green half onto the left-front's; a green half onto the right-back's; a green half onto the right-front's. Each snaps on with a tiny click, and each time the ring it grips takes its colour. Not a cord has moved: every tie lies where it lay, every inner and outer length hangs as it hung, every bell sits still on its pad. What has changed: every brass ring on the tray now wears a clip, red at the knot and green on all four wings.*]



Every ring clipped

[*At every click the bells have shivered on their pads, without a sound, and settled again as his claw moved on to the next. The Crab does not so much as glance at them.*]

Crab: The safe touch at the knot: a red half on the knot's own ring, and a green half on the ring of every bell tied to it. Nothing else.

Achilles: Nothing else! You've changed every ring on the tray. It must have noticed.

Achilles: Let me ask it. One little pull on the knot's tassel...

[*He reaches for the tassel of the knot's pull, where it hangs over the front rim.*]

[*As his fingers come near it, the five bells shiver on their pads, all of them at once, the way a cup shivers in its saucer when a cart goes by in the lane. There is no sound in it at all.*]

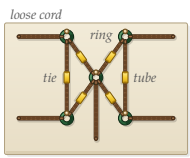
Crab: [*catching his wrist, gently*] Not the tassel. That is not a question, it is a

ring; you would hear the knot, and the others would go dull, and the tray would be spent till tomorrow. [*He lets go.*] I have wanted to pull that tassel myself every day since it came. I have not. Ask it any other way you like.

[*Achilles, a little pink about the ears, puts both hands back round his cup, where they can do no harm. On the tray the bells settle on their pads and are still, and not one of them has made a sound.*]

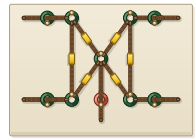
Tortoise: Then we'll ask a picture instead. [*She lifts her drawing-cloth out of the work-bag and spreads it on the table beside the tray, at its right as Achilles sees it, the cloth's near edge on his side and the teapot just off that edge at its right-hand end, then brings out string, a little box of rings with their beads, and her yellow tubes.*] My drawing-cloth, and my rings and tubes, as ever: what I lay on it is a picture of a thing, not the thing.

[*She lays five green rings on the drawing-cloth, each bead at rest, at the top of its ring as Achilles sees it, where a turn starts and goes round clockwise, in the tray's pattern: one in the middle, one near each corner. She ties the middle ring to each of the four corner rings with string, and the two left rings to each other, and the two right rings to each other, threading a yellow tube onto each length of string before she knots its second end, and sliding it to the middle of its tie. From each ring she runs a short loose cord out towards the edge of the cloth: from the two left rings out to the left, from the two right rings out to the right, and from the middle ring straight towards the near edge.*]



The net laid out

Tortoise: Your tray, drawn. Now the same touch. [*On the knot ring's loose cord she threads a red ring from its free end and sets its bead at a half; on each of the four corner rings' loose cords she threads a green ring at a half.*]



Threaded
for the knot

Tortoise: Three rules you know. Two rings of one colour joined by a cord slide together into one, and their beads add; a half and a half make a whole turn, which is no turn at all, and leave the ring at rest. A ring at rest with only two cords is just cord. And a ring with two cords, pushed through a yellow tube, comes out the other colour with its bead unchanged.

Achilles: Those I know. And one I don't?

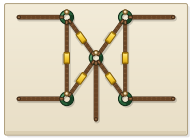
Tortoise: A red ring at a half, pushed into a green ring at rest, is gone from its cord and comes out as a red ring at a half on every other cord of that ring. The green ring stays at rest, and nothing is set aside. Only into a ring at rest.

Achilles: Why all of them at once?

Tortoise: A ring at rest keeps all its cords alike, all dark or all lit. Turn one over, and you have turned them all.

[*The picture as it stands: the knot ring green at rest in the middle, tied to every corner ring, with a red ring at a half on its loose cord; each corner ring green at rest, with a green ring at a half on its own loose cord. The Tortoise puts one claw behind the red ring and pushes it up its loose cord into the knot ring. It is gone from the loose cord, which lies bare. On each of the knot ring's ties, just beside it, a red ring at a half now sits; the knot ring itself is still at rest. She pushes each of them out along its tie. Each reaches its tube and passes through it, and comes out on the far side green, its bead still at a half. Each slides on into the corner ring at the end of its tie, and that ring is at a half now; and the green half waiting on the corner ring's own loose cord slides in after it: a half and a half, a whole turn. The bead comes round to rest. At all four corners the same. What lies on the cloth now: five green rings at rest, the same ties with the same tubes, every loose cord bare, and*

nothing set aside. What has changed: every piece of the touch is gone. What has held: everything else, exactly as she laid it.]



Every piece gone

Achilles: It's the net you laid. Not a thing set aside.

Tortoise: Your tray, touched at the knot, is the one I drew. And it has come out your tray untouched.

Achilles: Let me follow it my way, without a cloth. The knot was hung even, and turning an even bell over leaves it even; all the red half does is swap the knot's dark settings for its lit ones. But each tie gives its far bell a green half whenever the knot is lit, so after the swap each far bell gets that half exactly where it didn't before and misses it where it did: one extra green half, everywhere. Your green halves on the wing rings take those back.

Tortoise: The same answer, and not a cord of mine in it.

Achilles: So what you're saying is the touch did nothing at all.

Tortoise: It did something to every ring. It did nothing to the tray.

[*Outside the window, at the corner of the house, the water-butt stands full to the brim. The spout from the gutter pours the thaw into it at one side, and at the other side it spills out over the top as fast as it comes in, and the water in the butt has not risen or fallen by a hair all afternoon.*]

Achilles: And a corner bell? The left-back's touch?

Tortoise: [*threading a red ring at a half on the left-back ring's loose cord, and a green ring at a half on the loose cords of the left-front ring and the knot*] A red half on its own cord, and a green half on the cord of every ring tied to it: the left-front and the knot. Nothing on the right.

[*She pushes the red half up into the left-back ring. It comes out on that ring's ties,*

one towards the left-front ring and one towards the knot; each passes its tube and turns green at a half; each slides into the ring at the end of its tie and meets the green half on that ring's own loose cord, and the two make a whole turn. The right wing has not been touched at all. The cloth is as she laid it.]

[*The Crab has found the paper of spiced biscuits that came packed in the straw, and passes it round. Achilles takes one and eats it in two bites without taking his eyes off the cloth; the Tortoise takes two, and sets one by her saucer for later.*]

Achilles: Fewer ties this time, and the right wing left alone. And gone again.

Achilles: Then two at once. The left-back's touch and the right-back's, on your tray, Mr Crab.

Crab: On my tray, then.

[*The Crab unclips the knot's touch, ring by ring, and drops each clip back in the tin; the brass rings are bare again, and nothing else on the tray has stirred. Then he clips on the left-back's touch: a red half on the left-back's ring, a green half on the left-front's, a green half on the knot's. And the right-back's: a red half on the right-back's ring, a green half on the right-front's, and a second green half on the knot's ring, beside the first. The knot's ring now wears two green halves: a half and a half, a whole turn, no turn. He takes them both off again, and the knot's ring hangs bare. What is left: a red half on each back bell's ring, a green half on each front bell's ring, and the knot at rest.*]

Achilles: So the knot's left out of it altogether.

Crab: A touch and a touch is a touch.

Crab: And here is the thing the old man told me, the first day. If you know every touch a tray will not notice, you know the tray. Only one tray sits still for all of them.

Achilles: Then the tray is its touches.

Crab: It is known by them. [*He unclips the four clips and drops them in the tin, and the brass rings hang bare.*]

[*The Tortoise takes the tin from beside him, opens it, and holds up two small clips,*

one red and one green, their beads set at a quarter.]

Tortoise: And half of a touch?

Third Challenge — *Half a Touch*

Crab: Half a touch. [*He takes the quarter clips from her, and a second green one from the tin.*] The left-back again, but in quarters.

[*The tray as it stands: every brass ring bare. The Crab snaps a red clip, its bead a quarter round, at the right as Achilles sees it, onto the left-back bell's ring, and a green clip, a quarter back, its bead at the left, onto the left-front bell's ring, and another green quarter back onto the knot's. Three small clicks. The right wing's rings stay bare. No cord moves; no bell stirs on its pad. What has changed: three rings wear quarter clips. What has held: every tie, every length, every bell.*]



Red round,
green back

Achilles: [*bending close to the tray*] The cords are all where they were. Every tie in its sleeve, every pull hanging just as it hung. It's the same tray with three clips on.

Tortoise: It has noticed, this time. It isn't the same tray. The left-front bell and the knot are no longer tied.

Achilles: The tie is right there, sleeve and all!

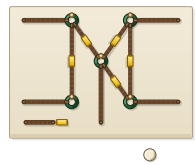
Tortoise: On the tray. Not in the net. Here is a rule I'll give you without the working: a red ring a quarter round on a bell's loose cord, and a green ring a quarter back on the loose cord of each bell tied to it, or all of them the other way round, may be taken off, if the bells tied to it are re-tied among

themselves. Every two of them that were tied are untied, and every two that were not are tied.

Achilles: And nothing else changes?

Tortoise: A tie gained or lost moves the picture's number, and I keep that to one side, as ever.

[*The cloth net as it stands: five green rings at rest in the bow tie, the ties in their tubes, the loose cords bare. On the left-back ring's loose cord the Tortoise threads a red ring a quarter round; on the left-front ring's and the knot's, a green ring a quarter back each. The left-back ring is tied to two rings, the left-front and the knot, and those two are tied to each other. She unknots that tie, the one between the left-front ring and the knot, draws its tube off, and lays the tie at the near edge of the drawing-cloth, at its left, and the tube just beyond the tie's right-hand end. Then she takes the three quarter rings off their loose cords by their free ends, back into her box of rings, and sets a small bone counter from her work-bag to one side, off the cloth just beyond its near edge at the right-hand end, by the teapot. What lies on the cloth now: the left-back ring tied to the left-front ring and to the knot, but the left-front ring and the knot no longer tied to each other; the left wing opened from a triangle into a vee. The right wing as it was. Every ring at rest, every loose cord bare; and at the near edge, the tie and its tube.*]



A vee on the left

Achilles: So the tray has every one of its cords, and the net you've drawn for it has lost one. [*He looks from the cloth to the tray, and back.*] The cords don't tell you which tray you have.

Tortoise: They tell you what was knotted. The clips tell you the rest.

[*In the lane below the window, a puddle that froze in the night still wears its lid of ice, white and whole, the same to look at as it was at breakfast. A blackbird drops onto it, and the lid tips under him like a loose tile: the water it was frozen fast to*

has drained away underneath in the thaw, and it is fastened to nothing. The bird is off again in a flurry, and the lid settles, white and whole.]

Crab: Then let me give it the other half. [*He takes three more quarter clips from the tin.*]

[*The Crab snaps a second red quarter round onto the left-back's ring, beside the first, at its left as Achilles sees it: two quarters round, a half. A second green quarter back onto the left-front's ring: two quarters back, which is a half back, and a half back is the same as a half. The same on the knot's ring. The rings now carry a red half on the left-back and a green half on the left-front and the knot: the left-back bell's whole touch.*]

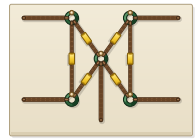


Made whole

Achilles: The left-back's safe touch. The one the tray doesn't notice.

Tortoise: Then my net had better not notice either. [*On the left-back ring's loose cord she threads a red ring a quarter round again, and on the left-front ring's and the knot's a green ring a quarter back.*] Half a touch again, on the net as it stands now.

[*The cloth as it stands: the left wing a vee, the left-front ring and the knot not tied, and the three quarter rings on their loose cords; at the near edge, the tie and its tube. The rings tied to the left-back are still the left-front and the knot, and now they are not tied to each other; so they must be. The Tortoise takes up the tie and the tube from the near edge of the cloth, slides the tube back on over a free end of the tie, and knots the tie between the left-front ring and the knot, just where it was. She takes the three quarter rings off by the free ends, back into her box, and takes the counter back from beside the teapot into her work-bag. What lies on the cloth now: the bow tie, every tie in its tube, every ring at rest, every loose cord bare. What has changed since the first half: one tie, back. What has held, from the start of the afternoon to now: the net.*]



Back to the bow

Tortoise: A tie lost moved the number one way, and a tie gained moves it back. Two halves of a touch make a touch, and two re-tyings make no change.

Achilles: And at the knot? Half the knot's touch?

Crab: Ah, at the knot the quarters would do a great deal more. [*He does not reach for the tin.*] Each wing's two bells would come untied from each other, and each wing bell would be tied across to both the bells of the other wing.

Achilles: A different shape altogether. And not a cord on the tray moved.

[*The changes from the tower go on, the same bells and the same notes, in a new order at every stroke. Achilles, listening idly, tries to keep one bell in his head from one stroke to the next, loses it by the third, and gives it up with a laugh under his breath.*]

Crab: [*drawing himself up a little*] I will tell you what I have found, in all the years I have been buying from that bench. Anything the workshop could make me out of quarters and halves turns out to be a tray of bells: each bell seated one of the twenty-four ways, and the bells tied, or re-tied.

Tortoise: And re-tie at one bell, then at a bell tied to it, then at the first again, and the two change places in the net. Three re-tyings in a row.

[*The Crab unclips the six quarter clips from the three rings and drops them back in the tin. The brass rings hang bare.*]

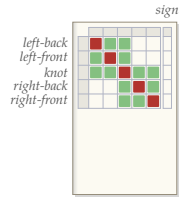
Achilles: Then how do you keep track of which tray you have, if the cords won't tell you?

[*The Crab is already reaching into the dresser drawer.*]

Fourth Challenge — *The Crab's Ledger*

Crab: [*laying a narrow bound book on the table at his own place, at Achilles's left on the same side of the table, between the model's stand and the window end, and opening it flat*] With this. My ledger for the tray: one row for each bell, and in every row one cell for each bell and one more for the sign. A cell holds nothing, or red, or green, or gold, which is red and green together; the sign is plain, or turned over. I keep a ledger for everything, but this is the one I am proudest of, because it is so small.

[*The page as it stands, upright to the Crab and to Achilles beside him: five ruled rows, empty, headed at the side with the five bells' names, and across the top the same five names and a narrow column for the sign. The Crab dips his pen, red ink or green as each cell needs, and fills each cell he marks with a wash of its colour. In the left-back's row he writes red in the left-back's own cell and green under the left-front and the knot, and leaves the right-back and the right-front empty; the sign cell he leaves blank, which is plain; a sign turned over would get a small inked square. In the left-front's row, green under the left-back, red in its own cell, green under the knot. In the knot's row, green under the left-back and the left-front, red in its own cell, green under the right-back and the right-front. In the right-back's row, green under the knot, red in its own cell, green under the right-front; in the right-front's row, green under the knot and the right-back, and red in its own cell; the left wing's cells empty in both. Every sign is plain. What has changed: the page has five full rows. What has held: the tray; he has not touched it.*]



The tray's rows

Achilles: Red in its own cell, and green for every bell tied to it. [*He looks at the tin.*] Those are the touches! Every row is a safe touch.

Crab: Of course they are. A tray is known by the touches it does not notice,

so I write those down, and nothing else.

Achilles: And when you turn a bell? Do you have to write it all out again?

Crab: One column. Suppose I gave the knot an over, on paper only. An over swaps red and green in that bell's column, leaves gold as gold, and turns over the sign of every row that has gold there.

[*The knot's column as it stands, reading down the five rows: green, green, red, green, green. The Crab rules a line under the page and writes the rows again below it, copying every other column exactly as it was, and in the knot's column writing each mark swapped: red, red, green, red, red. There is no gold in that column, so no sign turns over. What has changed: one column, red for green and green for red. What has held: every other cell, and every sign.*]



The knot's column swapped

Achilles: Now let me count. Each cell is red or not, and green or not: two marks a cell. Five cells and a sign: eleven marks a row, fifty-five for the tray.

Achilles: And the tray's own numbers, one for every way the five bells could read dark or lit, are thirty-two. You aren't saving much.

Crab: Not with five. With a hundred bells, twenty thousand one hundred marks, against a count thirty-one digits long.

Achilles: ...Oh.

Achilles: Not a count of numbers. A count thirty-one digits long, just to say how many.

Crab: And twenty thousand one hundred marks fit in a ledger I could carry to church.

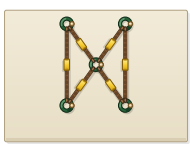
[*Up at the church the changes have been going long enough now to have a shape one can hear: one bell working its way steadily down through the others to the first place, and up through them again to the last, and the rest stepping aside for it one after another. Not a ringer in the tower has the whole of it written out. Each of them has a few words by heart, and follows them, stroke by stroke. The Crab sits back from his ledger and looks as though he could listen all day.*]

Tortoise: The ledger follows the tray on paper. My cloth can follow it another way. [*From her little box of rings she takes five green end-rings, each bead set a quarter round.*] End-rings: they close a loose cord. Close every loose cord of a picture and it has no ends left, and a picture with no ends is a single number: here, the tray's answer to these five caps.

Achilles: And you find the number without writing one of the thirty-two?

Tortoise: With one more rule. A ring in the middle, with no loose cord and its bead a quarter round, can be pulled out: the rings tied to it are re-tied among themselves, as before, and each of them is turned a quarter back. And a lone ring, with no cord at all, is a number: at rest, two; a quarter round, one and one turned a quarter; a quarter back, one and one turned a quarter back; a half, nothing at all.

[*The cloth net as it stands: the bow tie, every ring at rest, every loose cord bare. The Tortoise caps each loose cord with a green end-ring a quarter round, and each end-ring, joined to its ring by the loose cord, slides along it into the ring, the loose cord swallowed with it: a quarter round and at rest make a quarter round. Now every ring is a quarter round, and not one has a loose cord.*]



No loose end left

[*She takes hold of the left-back ring. It is tied to the left-front ring and to the knot, and those two are tied to each other. She unknots the left-back ring's ties at the left-front ring and at the knot, lifts the left-back ring out with them and their tubes,*

and drops the lot into her work-bag; unknots the tie between the left-front ring and the knot and drops that, with its tube, into the bag too; turns the left-front ring's bead a quarter back, from a quarter round to rest, and the knot's the same, from a quarter round to rest; and sets a counter by the teapot. The left-front ring lies at rest, tied to nothing now: a lone ring. The knot, at rest, is still tied to the right wing.]

[*Then the right-back ring, a quarter round, tied to the knot and to the right-front ring, which are tied to each other. She unknots its ties at the knot and at the right-front ring and lifts it out with them and their tubes into her work-bag; unknots the tie between the knot and the right-front ring and drops that, with its tube, in after it; turns each of those two a quarter back, the knot from rest to a quarter back, the right-front from a quarter round to rest; and sets a second counter by the teapot, on the first. What lies on the cloth now: three lone rings, far apart, not a cord between them: the left-front ring at rest, the knot a quarter back, the right-front ring at rest. What has changed: the net is gone, and the picture is smaller than it has ever been. What has held: the number it stands for, with what lies by the teapot.*]



Pulled out to the last

Tortoise: Nothing left but lone rings, and each of them is a number. Put them together with what I set aside, and that's the tray's answer.

[*Across the lane, the snowman the children built on the first morning of the frost has been going all afternoon. He is a single low heap now, sunk into the slush, and round him on the wet ground lie the things that made him, two coals and his buttons, each by itself. The Tortoise sits back and reaches for the biscuit she saved.*]

Achilles: Three lone rings and a little heap by the teapot. And not one of the thirty-two numbers written down.

Achilles: So what you're saying is that anything built this way can be followed on paper, and the machine is no better than paper.

Tortoise: Anything built of quarters and halves.

Achilles: And a turn that is neither a quarter nor a half?

[*The Crab looks at the model, sitting at home in its cradle.*]

Fifth Challenge — *An Eighth*

[*The Tortoise sweeps the three lone rings into her work-bag, and the two counters from beside the teapot after them, and lays two fresh cords on the drawing-cloth, one above the other, the upper further from Achilles, each running from his left to his right.*]

Tortoise: [*nodding at the model*] An eighth.

[*The model as it stands, at home: the dark green up, the dark red in the front lug, the dark gold at the right, the pale red at the back, the pale gold at the left, the pale green in the socket. Achilles lifts it a hair and turns it about its upright line the way of a quarter, but stops half-way. The dark red has gone only half-way from the front lug to the right; the pale gold has come only half-way from the left to the front. He lowers it. The pale green finds the socket. But at the waist no lug meets a tip: each lug meets the edge between two tips, and the model settles crooked on its notches. When he takes his fingers away it rocks, and rocks again. What has changed: every lug has lost its tip. What has held: the top, the bottom, and the order of the colours round the middle; nothing else.*]



Rocking, crooked

Achilles: It won't sit. The front lug has an edge in it, between the dark red and the pale gold.

Achilles: I can't name it. It isn't one seating or another; it's in between two

of them.

[*Achilles looks up from the rocking model to the print on the wall: Escher's* Stars.]

Achilles: Many-sided solids hanging in a dark space. That is how it sits now: hanging, held by no lug.

[*A rook goes over the lane, low, and its shadow runs in at the window, down the length of the room and across the glass of the print, and for that moment the dark inside the frame is the darkest thing in the parlour. Then the shadow is off the end wall, and it is a print again.*]

Crab: [*sliding the ledger along the table to Achilles, its page upright to him as before*] And my ledger has no mark for between. Nothing, red, green, gold: that is all a cell can say.

Tortoise: Nor has my cloth. Put a ring turned an eighth in the middle of a net, and no rule I have will pull it out; the pulling needs a quarter.

[*The passage door has stood a little open since Achilles came in. A draught finds it now and swings it to, and its latch comes to rest on the lip of the keep, not in it, so that the door stands neither open nor shut, stirring a little in its frame. Nobody gets up to see to it.*]

Achilles: Then give it another. [*He takes hold of the model again.*]

[*The model as it stands: rocking, every lug on an edge. Achilles turns it on, the same way, by the same small amount. The dark red comes the rest of the way into the right-hand lug, and the pale gold the rest of the way into the front; the tips at the back and the left follow. The model drops into its lugs with a knock, and stops rocking. What has changed: it is seated. What has held: the dark green on top, the pale green beneath. It sits a quarter round from where it started, exactly as if the Crab had called one quarter.*]

Achilles: Seated. Two eighths make a quarter.

[*Another breath of the draught, no stronger than the first, and the door's latch goes over the lip and drops into its keep with a small click. The door is shut, and the warmth of the fire gathers in the room.*]

Achilles: Then clip an eighth on a bell. Is there one in your tin?

Crab: There is not. The tin has halves and quarters, and nothing finer; the tray was built for them. [*He goes to the dresser, and comes back with a small lidded box, and sets it down by the Tortoise.*] But the bench can do what the cradle can't. These are eighth-ends, made ahead of time in the workshop, slowly: green end-rings with the bead set at an eighth. Each one is used up the moment it is tied on.

[*He lifts the lid. Inside, on cotton wool, lie a few small green end-rings, each bead set at an eighth.*]

Achilles: Used up how?

Tortoise: Let me show you on the cloth. [*She takes one eighth-end from the box.*] The upper cord is any cord you like, set however you like; the lower cord begins at this eighth-end. And between them my joiner, as before: a green ring on the upper cord, tied by a short length to a red ring on the lower. Then I read the lower cord at its far end, and it reads dark or lit, half the time each; I don't choose which.

[*She threads a green ring on the upper cord and a red ring on the lower, just across from it, and ties them together with a short length. On the left end of the lower cord she ties the eighth-end.*]

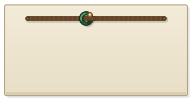


The joiner, ready

Tortoise: If it reads dark, I close the far end with a red end-ring at rest.

[*The cloth as it stands: the upper cord with its green joiner ring; the lower cord running from the eighth-end, through the red joiner ring, to a red end-ring at rest. The Tortoise slides the red end-ring along into the red joiner ring, the stretch of lower cord it runs along swallowed with it: both red, both at rest, and they are one red ring at rest, with only two cords left to it, the one back to the eighth-end and the short length up to the green ring. A ring at rest with two cords is just cord; she lifts the red ring away into her box of rings, knotting the two cords it held end to end, and the eighth-end is tied straight to the green ring on the upper cord. It slides*

along its cord and the short length into that ring, both swallowed with it. What lies on the cloth now: the upper cord alone, and on it one green ring with its bead at an eighth. What has changed: the lower cord is gone, and the upper cord carries the eighth. What has held: everything else on the upper cord.]



Carried up

Tortoise: Dark: the eighth is on, the right way round.

Achilles: And lit?

Tortoise: [*clearing the drawing-cloth into her work-bag, all but the green ring that took the eighth, which she slips off the end of the upper cord and sets by the box; then laying the two cords and the joiner out again as they were, with a second eighth-end from the box at the head of the lower cord*] A fresh eighth-end first; that one is spent. Lit, I close the far end with a red end-ring at a half. And one old rule: a red half pushed into a green ring turns its bead the other way round, and sets a little something aside.

[*The cloth as it stands: the same, but with a red end-ring at a half at the far end of the lower cord. It slides along the lower cord into the red joiner ring, the stretch it runs along swallowed with it, and the joiner ring is red at a half, with two cords, between the eighth-end and the green ring above. The Tortoise slides the eighth-end along the lower cord up to the red half, the stretch between swallowed, and pushes the red half into it: the red half is gone, and the eighth-end's bead is turned the other way round, an eighth back; a counter goes by the teapot. The eighth-end, an eighth back now, slides up the short length into the green ring on the upper cord, the short length swallowed with it. Then she threads a green ring a quarter round on the upper cord from its right-hand end, to lie beside it at its right, and slides it in: a quarter round and an eighth back make an eighth. What has changed: the lower cord gone again, and the upper cord carrying an eighth. What has held: the rest of the upper cord. And the eighth on it faces the same way as in the dark reading, once the quarter is in.*]



Carried up again

Tortoise: Lit: it went on the wrong way round, and a quarter set it right. We have quarters in plenty. Either way, the eighth-end is spent.

[*The Crab rises and takes the two spent eighth-ends, the green ring that took the eighth from the end of the upper cord and the first from beside the box, turns them once in his claws, and drops them into his waistcoat pocket.*]

Crab: Keepsakes.

[*He pats the pocket once, as a man pats a pocket with a letter in it. The fire has burnt low while nobody was looking; the Crab takes a block of applewood from the basket, split and dried last summer, and lays it on the coals, and it catches at once and fills the room with its smell.*]

Tortoise: So here is the afternoon. The twenty-four seatings on every bell, and the ties between the bells, make a whole world. Everything in it can be followed with your ledger, or shrunk on my cloth to a number. One eighth leaves it. And with eighth-ends brought in from outside, anything at all can be built.

Crab: That is what the old man says. The bench makes the eighths, slowly, and after that it is all clips.

Achilles: [*looking at the model, a quarter round from home*] Then let me put this one back where it belongs. [*He takes it by the tips.*]

[*The model as it stands: the dark green up, the pale gold in the front lug, the dark red at the right. Achilles lifts it and turns it a quarter the other way: the pale gold goes back to the left lug, and the dark red comes back from the right into the front. It drops in with a knock: the dark green up, the dark red in front, home, as it came out of the straw.*]

[*At the same moment, up the lane, the changes from the tower return to the plain falling scale, highest to lowest, every bell in the place it started from. The ringers*

ring it again, and again, plain and settled.]

Crab: Thank you. [*He lays the green baize back over the bell-tray, corner to corner, and carries the clip tin, the box of eighth-ends and the ledger, shut, to the dresser.*]

[*The falling scale sounds a last time, and then the tower's bells stop one after another, each left standing where the ringers set it, and the tower goes quiet. The quiet comes down the lane and in at the window, and the applewood can be heard again, ticking in the grate.*]

Tortoise: [*putting the upper cord, the counter from beside the teapot, her string, her tubes and her little box of rings into her work-bag, and folding her cloth*] Your tray stayed quiet all afternoon.

Crab: It always does, in good company. More tea? There is a cup left in the pot, and it has been listening.

Achilles: Half, then, and you have the other half.

Crab: A half and a half. I shall be careful not to make a whole turn of it.

[*He pours with great care, a little into each cup, and shakes the last drop into his own. Achilles holds his cup in both hands again, though his hands are warm now, and the Tortoise settles deeper into her chair by the fire.*]

Tortoise: Listen. The eaves are still going.

Achilles: It'll be dripping till midnight.

[*Up the lane, the church clock begins to strike, and strikes four.*]

Achilles: Four, and still light. The days are getting longer.

Achilles: [*looking at the model in its cradle*] So everything inside is paper. Only the eighths cost anything.

Crab: Come to the kitchen tomorrow. I have a long picture that wants cooking down, and I should like to see how few eighths are left in it at the end.

Achilles: Then what exactly is it that separates the machines paper can follow from the ones it can't? And how few eighths does a machine really need?

Tortoise: Inside the world of quarters, nothing is hard. Count the eighths.

[*Nobody answers her yet. The sun has gone down behind the roofs across the lane, and the long pale light of the thaw lies in the window, with the fire's warmer light across the hearth and its blue-and-white tiles, every one of them square in its place. Under the green baize the tray stands as the Crab covered it, its five bells on their pads and every cord of it knotted where the old man knotted it; beside it the model sits in its cradle. The passage door is shut. At the far end of the room the print hangs on the end wall in its plain black frame, gone grey with the rest of the wall, one more of the Crab's things. Up the lane the tower is silent, and only the eaves go on, a drop, and another, into the slush on the sill. Achilles's scarf lies folded over the back of his chair, his boots stand dry by the fender, and the three of them sit a while longer with their cups, in no hurry at all.*]

Chapter 16: Inside the Clifford World

— Twenty-Four Preludes and a T —

Part V: Engineering the Diagram

Escher: "Stars" — many-sided solids hanging in a dark space

Can a machine be thoroughly quantum, entangling everything it touches, and still be followed step by step by someone with a pencil and a notebook?

It sounds impossible. We have spent fifteen chapters learning that a quantum state is expensive to write down. Five small quantum systems need thirty-two numbers. A hundred need a list of numbers whose length is itself a number with thirty-one digits, and no notebook that could ever be built would hold it. Entanglement makes it worse, not better: an entangled state cannot be written as one short list per system, because the whole is not the parts. This is why nobody simulates a large quantum computer on an ordinary one, and why anyone bothers to build one.

And yet there is a large family of quantum machines that a notebook can follow with ease. They entangle freely. They run the protocols of the earlier chapters: the Bell pairs of Chapter 6, the teleportation of Chapter 10, the key-sharing of Chapter 12. Their states can be spread over a hundred systems or a thousand. A careful clerk with a ruled book, writing a few marks per system, can say at every moment exactly what the machine holds and exactly what the odds of every reading are. The trick is not to write down the state. It is to write down the tests the state passes: the touches it would not notice.

Which machines are these? The answer is surprisingly small and surprisingly geometric. They are the machines built from a handful of moves: turns by a quarter and by a half. On one wire those moves make twenty-four positions, the ways a many-sided wooden solid can sit in a stand. On many wires they make a world, closed and complete, in which nothing is hard.

One move leads out of that world: a turn by an eighth. It looks like a small thing, one angle halved again. It is the whole difference between a machine a clerk can follow and a machine nobody can. This chapter walks round the inside of the world of quarters and halves, learns to follow it on paper and to shrink its pictures to numbers, and then turns the eighth that leaves it.

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§16.1 The Twenty-Four

Chapter 15 ended at the edge of this chapter. "Everything in The Second Drawer is a quarter-turn or a half-turn, and the eight-drawer turn-about needed an eighth. That looks like a small difference, one angle halved." The chapter's question, in the form we can answer: **what separates the pictures built only of quarter-turns and half-turns from the pictures that need an eighth, and why does that difference matter so much?**

We answer it in five steps. This section meets the quarters and halves on one wire. The next four put them on many wires, show what they leave unnoticed, follow them on paper and in pictures, and then turn the eighth.

On the blackboard. The bit-flip X and the phase-flip Z have been with us since the early chapters. Add their product, turned by i:

$$Y \;=\; iXZ \;=\; \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Gottesman writes the same relation as $\sigma_y = i\sigma_x\sigma_z$ (his eq. (5)). The four matrices I, X, Y, Z satisfy $X^2 = Y^2 = Z^2 = I$, and any two of X, Y, Z **anticommute**: $XZ = -ZX$, and likewise for the other pairs. Their products stay in the family up to a factor: $XY = iZ$, $YZ = iX$, $ZX = iY$.

A **Pauli string** on n wires is a tensor product of n of these four letters, one per wire, top wire first; for instance $X \otimes Z \otimes I$ on three wires. The **Pauli group** $\mathcal{P}_n$ is the set of Pauli strings multiplied by one of the four phases $\pm 1, \pm i$.

Take two wires and two strings, $X \otimes Z$ and $Z \otimes X$. On each wire the letters anticommute, so exchanging the order of the product gives $(-1)\cdot(-1) = +1$: the two strings commute. By contrast $X \otimes I$ and $Z \otimes X$ disagree in sign on the first wire only, and they anticommute. The sign is a count of disagreements.

> **Result 16.1 (the Pauli group).** For every n, $\mathcal{P}_n$ is a group with $4 \cdot 4^n$ elements: 64 on two wires and 256 on three. Any two of its elements either commute or anticommute. They anticommute exactly when the number of wires on which both letters are different from I and from each other is odd.

Proof. On one wire the product of two letters is a letter times one of $\pm 1, \pm i$ (the products listed above, together with $P^2 = I$ and $IP = PI = P$). So a product of two Pauli strings is a Pauli string times a phase, and the phases multiply among $\pm 1, \pm i$; the inverse of a phase times a string is the conjugate phase times the same string, since every letter squares to I. There are 4^n strings and four phases, and a phase times a string determines both, so $|\mathcal{P}_n| = 4 \cdot 4^n$. On one wire two letters commute when one of them is I or they are equal, and anticommute otherwise. Reversing the order of a product of two strings reverses the order on every wire, which contributes one -1 per anticommuting wire. ■ See `Code/Ch16/TwentyFour.py`.

Gottesman opens his account with the same count: the Pauli group "consists of $4 \cdot 4^n$ elements".

Following operators instead of states

Gottesman's paper of 1998 is called "The Heisenberg Representation of Quantum Computers", and its idea fits in one line (his eqs. (1) and (2)). If a machine applies a unitary U to a state $|\psi\rangle$, then for any operator N,

$$UN|\psi\rangle \;=\; UNU^\dagger U|\psi\rangle\,.$$

So after the gate, the operator $UNU^\dagger$ does to the new state exactly what N did to the old one. Instead of following the state, follow what happens to operators: $N \mapsto UNU^\dagger$. That map respects products, $UMNU^\dagger = (UMU^\dagger)(UNU^\dagger)$, so it is enough to follow X and Z on each wire: every Pauli string is a product of those, up to a phase.

For most gates this is no saving, because $UXU^\dagger$ is some large matrix. For some gates it is a great saving.

The Clifford group $\mathcal{C}_n$ is the set of unitaries U on n wires with

$$UPU^\dagger \in \mathcal{P}_n \qquad \text{for every } P \in \mathcal{P}_n.$$

Three examples are familiar. The Hadamard H sends $X \mapsto Z$ and $Z \mapsto X$, and therefore $Y = iXZ \mapsto iZX = -Y$. The phase gate

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

sends $X \mapsto Y$, $Y \mapsto -X$ and $Z \mapsto Z$. The CNOT, control on top, sends

$$X \otimes I \mapsto X \otimes X, \quad I \otimes X \mapsto I \otimes X, \quad Z \otimes I \mapsto Z \otimes I, \quad I \otimes Z \mapsto Z \otimes Z.$$

These are the rows of Gottesman's Table 1. A word on names: Gottesman's thesis of 1997 writes G for the Pauli group and $N(G)$ for this group of unitaries (the normalizer of G); his paper of 1998 writes $\mathcal{P}$ and $N(\mathcal{P})$ and says that $N(\mathcal{P})$ "is also called the Clifford group $\mathcal{C}$"; both write R and P for the Hadamard and the phase gate. We write $\mathcal{P}_n$, $\mathcal{C}_n$, H and S.

On one wire, what can a Clifford do? It must send X to a Pauli that is Hermitian and squares to I, and not to $\pm I$, since X is not a multiple of I. It must also be traceless, as X is, which leaves six choices: $\pm X$, $\pm Y$, $\pm Z$. Draw them as the six tips of an octahedron, with $+X$ pointing to the front, $+Y$ to the right, $+Z$ up, and each minus sign at the opposite tip. Then a Clifford moves the tips about.

> **Result 16.2 (the twenty-four).** Up to an overall phase, the one-wire Clifford group $\mathcal{C}_1$ has exactly 24 elements. Each one per-

> mutes the six tips $\pm X, \pm Y, \pm Z$ as a rotation of the octahedron; different elements give different rotations; every rotation of the octahedron occurs; and H and S generate them all.

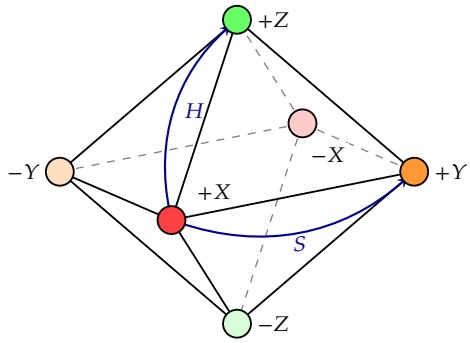
Proof. At most twenty-four. A Clifford U sends X to one of the six tips. It sends Z to a tip that anticommutes with the image of X (conjugation keeps $XZ = -ZX$), so not to that tip or its opposite: four choices. The image of $Y = iXZ$ is then forced, $UYU^\dagger = i\,(UXU^\dagger)(UZU^\dagger)$. So there are at most 6×4 ways to move the tips. Two unitaries that move the tips the same way agree up to a phase: if $UXU^\dagger = VXV^\dagger$ and $UZU^\dagger = VZV^\dagger$, then $Q = V^\dagger U$ commutes with X and Z, hence with I, X, Y, Z, which span all 2×2 matrices, so Q is a multiple of I.

The moves are rotations. A move sends the three axes to three axes with signs, so it is a signed permutation of (X, Y, Z), and the forced image of Y fixes the last sign so that the determinant is $+1$: from $XYZ = iI$, the images satisfy the same relation, and a signed permutation with determinant -1 would give $-iI$. The rotations of the octahedron are exactly the signed permutations of its three axes with determinant $+1$.

At least twenty-four. S turns the octahedron a quarter about the up–down axis, sending the front tip to the right; its powers I, S, S^2, S^3 give four different rotations that keep $+Z$ up. H exchanges $+Z$ and $+X$, S carries $+X$ to $+Y$, $S^2 = Z$ carries $+X$ to $-X$, H carries $-X$ to $-Z$, and S carries $-X$ to $-Y$. So the group generated by H and S can bring any of the six tips up, and for each tip up it has the four quarter-turns about it. That is 6×4 different rotations, all generated by H and S. ■ See `Code/Ch16/TwentyFour.py`, which also enumerates the group.

Hein, Dür, Eisert, Raussendorf, Van den Nest and Briegel list them in their Table 1, "All 24 single-qubit Clifford unitaries", "disregarding such global phases", each written as a product of elementary permutations of the three axes.

Here is the octahedron. It is **not a string diagram**: its lines are edges of a solid, not wires, and none of the rules of Chapter 8 apply to it. It is a picture of where the three Pauli axes point.



The octahedron of Pauli tips. A figure of a different kind, not a string diagram. The six tips are $\pm X$ (red, towards the reader and away), $\pm Y$ (gold, right and left) and $\pm Z$ (green, up and down), the plus tip dark and the minus tip pale. Each one-wire Clifford is a rotation that seats the solid again. The two arrows draw two equations as tip moves: $HXH^\dagger = Z$, the front tip carried up, and $SXS^\dagger = Y$, the front tip carried to the right.

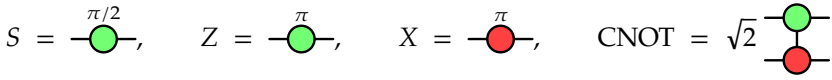
The half-turns about the three axes are the Paulis themselves: X fixes $\pm X$ and swaps the other two pairs, and likewise for Y and Z. Every tip lands on itself or its opposite.

For many wires we state the answer and do not prove it.

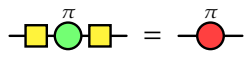
> **Result 16.3 (the generators; Gottesman, stated, not proved).** Every element of $\mathbb{C}_n$ is, up to an overall phase, a product of H and S on single wires and CNOTs on pairs of wires. In Gottesman's words (1998, §3), "R, P, and CNOT, applied to arbitrary qubits or pairs of qubits, generate the full Clifford group $\mathbb{C}$." Hein and co-authors add that "up to a global phase factor any Clifford operation" decomposes into $O(N^2)$ such gates, their N being our n.

In the sketchbook. Every one of these gates is a spider picture whose turns are all quarters or halves. A red π dot with one leg in and one out is X exactly, and a green π dot is Z exactly (§8.2). A green dot with phase $\pi/2$ is $|0\rangle\langle 0| + e^{i\pi/2}|1\rangle\langle 1| = S$, exactly. The Hadamard is the yellow box. The green–red pair of Chapter 8 (row 64) is the CNOT, with the number $\frac{1}{\sqrt{2}}$

when it is drawn bare:



Reading left to right, $Y = iXZ$ is i times a green π followed by a red π. The "over" of the octahedron, the Hadamard, is Chapter 8's colour change (Theorem 8.2, row 58): a green π between two yellow boxes is a red π, exactly, which is $HZH = X$.



So every product of H, S and CNOT, fed $|0\rangle$ (a red point, times $\frac{1}{\sqrt{2}}$) and read with $\langle 0|$ where we like, is a picture in which every turn is a multiple of a quarter. We call such pictures **quarter-turn pictures**. Van de Wetering's survey states the converse as well: the Clifford maps "are precisely those linear maps that can be presented by ZX-diagrams containing only phases that are multiples of $\frac{\pi}{2}$". We have shown one direction, and we use only that one.

Why this definition. Why single out the unitaries that send Paulis to Paulis? Because they are exactly the gates whose effect can be followed by following operators. A Clifford on n wires is known, up to a phase, once we know where it sends the $2n$ operators $X_1, \dots, X_n, Z_1, \dots, Z_n$ (the argument of the proof above, wire by wire), and each of those goes to a single Pauli string. That is a short list of signed letters, where a general gate would need a matrix with 4^n entries. Section §16.4 turns this into a notebook.

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§16.2 Safe Touches

Take two wires, each in $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and apply CZ, the gate that turns over $|11\rangle$ and nothing else. The result is

$$\mathrm{CZ}\,|{+}{+}\rangle \;=\; \tfrac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle).$$

Now touch it. The string $X \otimes I$ swaps the first bit, giving $\frac{1}{2}(|10\rangle + |11\rangle + |00\rangle - |01\rangle)$: a different state. The touch was noticed. But $X \otimes Z$ swaps the first bit and turns over every term whose second bit is 1:

$$\tfrac{1}{2}(|10\rangle - |11\rangle + |00\rangle + |01\rangle),$$

which is the state we started with, term for term. A flip on the first wire is noticed; the same flip with a sign on its partner is not. By symmetry $Z \otimes X$ goes unnoticed too. This section explains the pattern for any number of wires joined in any way.

On the blackboard. A **graph** $G = (V, E)$ is a set V of **vertices** and a set E of **edges**, each edge an unordered pair of different vertices. Two vertices joined by an edge are **adjacent**, and the **neighbourhood** N_a of a vertex a is the set of vertices adjacent to a. Put one wire on each vertex, so $n = |V|$, and write CZ_{bc} for the CZ on the wires of b and c. The **graph state** of G is

$$|G\rangle \;=\; U_G\,|+\rangle^{\otimes n}, \qquad U_G \;=\; \prod_{\{b,c\}\in E} \mathrm{CZ}_{bc}$$

(Hein et al., eq. (24); their U_{ab} is our CZ). Every CZ is diagonal, so the order of the product does not matter. Written out,

$$|G\rangle \;=\; 2^{-n/2} \sum_{x\in\{0,1\}^n} (-1)^{E(x)}\,|x\rangle, \qquad E(x) = \sum_{\{b,c\}\in E} x_b x_c,$$

every string with amplitude $\pm 2^{-n/2}$, and a minus sign when the string has an odd number of edges with both ends at 1.

Our running example is the **bow tie**: five vertices, called **bells** as in the dialogue, numbered in ket order: 1 left-back, 2 left-front, 3 the **knot**, 4 right-back, 5 right-front. Its edges are 12, 13, 23, 34, 35 and 45: two triangles sharing the knot. We write $|G_{\bowtie}\rangle$ for its graph state.

For each vertex a, let

$$K_a \;=\; X_a \prod_{b\in N_a} Z_b,$$

an X on a and a Z on each neighbour of a. Hein and co-authors call the K_a

correlation operators (their Proposition 2). On the bow tie, top wire first,

$$K_3 \;=\; Z \otimes Z \otimes X \otimes Z \otimes Z, \qquad K_1 \;=\; X \otimes Z \otimes Z \otimes I \otimes I.$$

The example above was the graph with two vertices and one edge, and its K_1 and K_2 were $X \otimes Z$ and $Z \otimes X$.

Dirac's Blackboard.

One tie at a time. The headline of this section (Result 16.6 below) rests on one identity for one edge: $\mathrm{CZ}_{ab}\, X_a\, \mathrm{CZ}_{ab} = X_a Z_b$. Check it on a basis string $|x_a x_b\rangle$ of the two wires. The first CZ gives the sign $(-1)^{x_a x_b}$. Then X_a flips the first bit to $1 - x_a$. The second CZ gives $(-1)^{(1-x_a)x_b}$. The two signs multiply to $(-1)^{x_b}$, whatever x_a was. So the net effect is: flip the first bit, and turn over the string if the second bit is 1. That is $X_a Z_b$. A CZ on two other wires commutes with X_a, and every CZ commutes with every Z, so pushing X_a through the whole of U_G leaves one Z on each neighbour of a and nothing else:

$$U_G\, X_a\, U_G^{\dagger} \;=\; X_a \prod_{b \in N_a} Z_b \;=\; K_a.$$

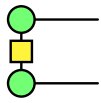
CZ is its own inverse, so $U_G^{\dagger}$ is the same product of CZs, and the order of the CZs does not matter.

The net

In the sketchbook. Van de Wetering's survey draws a graph state in one step: "every vertex becomes a Z-spider with an output attached to it", and every edge becomes a wire between the two spiders with a Hadamard on it. We call that picture the **net** of G. Each vertex is a phase-free green spider with one output leg running to the right, and one leg for each edge at that vertex. Each edge is a wire joining two spiders, carrying a yellow box. Chapter 8 also allowed a dashed wire for a Hadamard on a wire (row 56). We draw the box, because the rewrites below move things through it, and because the dialogue's ties wear yellow sleeves. We call the edges of a

net **ties**.

The smallest net has two spiders and one tie. It is the state we touched at the start of the section, with a number:

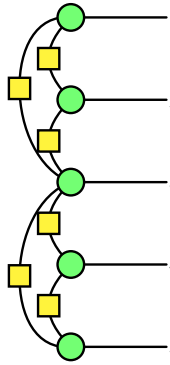


$$= \sqrt{2}\,\mathrm{CZ}\,|{+}{+}\rangle \;=\; \tfrac{1}{\sqrt{2}}(|00\rangle + |01\rangle + |10\rangle - |11\rangle).$$

Result 16.4 (the net). For every graph with v vertices and e edges, the bare net equals exactly $2^{(v-e)/2}\,|G\rangle$. For the bow tie it is $2^{-1/2}\,|G_{\bowtie}\rangle$.

Proof. Read the net as a sum over bits. A phase-free green spider forces all its legs to agree (row 53), so each vertex b carries one bit x_b to its output and to all of its ties. A yellow box between bits x_b and x_c contributes the matrix entry $\langle x_c|\,H\,|x_b\rangle = 2^{-1/2}(-1)^{x_b x_c}$. So the net is $2^{-e/2}\sum_x (-1)^{E(x)}\,|x\rangle = 2^{-e/2}\,2^{v/2}\,|G\rangle$. ■ See `Code/Ch16/SafeTouches.py`, which checks the number on random graphs.

Here is the bow tie's net. The bells are stacked in ket order, top wire first, their outputs run to the right, and the ties are drawn as arcs on the left: short arcs between neighbouring bells, and longer arcs from the knot to the far bell of each wing.



$$= 2^{-1/2}\,|G_{\bowtie}\rangle$$

A half through a ring at rest

To prove the headline in pictures we need one rule that Chapter 8 stated only for two legs and only "up to a scalar": pushing a half-turn of one colour through a phase-free spider of the other. A **clip ring** is our name for a spider with exactly two legs sitting on a wire. A green or red clip ring with no phase is a plain wire (§8.2); we say it is **at rest**. A red clip ring with phase π is X, a green one is Z.

> **Result 16.5 (a half through a ring at rest).** Let a phase-free green spider have $k \geq 2$ legs. A red π on one of its legs equals a red π on each of its other $k-1$ legs, exactly, with no number. The same holds with the colours exchanged. On the blackboard: $X^{\otimes k}$ exchanges the two terms of $|0\cdots0\rangle\langle0\cdots0| + |1\cdots1\rangle\langle1\cdots1|$, and so leaves their sum unchanged.

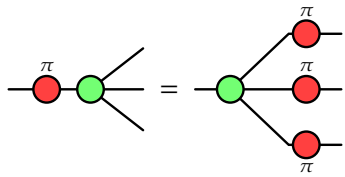
Proof. The derivation by the rules is the sketchbook sidebar below. On the blackboard: bend all legs to one side, so that the spider is the state $|0\cdots0\rangle + |1\cdots1\rangle$ on k wires. $X^{\otimes k}$ fixes it, and multiplying both sides by X on the first wire moves that X to the other $k-1$ wires. ■ See `Code/Ch16/SafeTouches.py`.

It is Chapter 8's π-commutation (Theorem 8.4, row 60) for a phase-free spider, now with any number of legs and with its exact number, just as Chapter 14 found the exact number of the two-legged case.

> **Coecke's Sketchbook.**
>
> *Deriving the push.* Here is Result 16.5 from rules the book has stated. Stand the spider with one leg in and $k-1$ legs out. A map with one input is fixed by what it does to the dark point and the lit point, the red points $\sqrt{2}\,|0\rangle$ and $\sqrt{2}\,|1\rangle$, because they span the input (the principle behind row 104). Feed in a red point with phase $c\pi$. On the left side, the red π fuses with the point first (fusion, row 57), giving a red point with phase $(c+1)\pi$; then the phase-free green spider copies it onto its $k-1$ outputs (copy, row 59). On the right side, the spider copies the point with phase $c\pi$ first, and then each output's red π fuses into its

copy, giving phase $(c+1)\pi$ on every output. Both sides end with the same $k-1$ red points. And both paid the same price: one point in, $k-1$ points out, is the same copy in both orders, and fusion costs nothing. So the prices cancel, and the rule holds with no number at all. The rule needs the spider phase-free: a turned spider does not copy.

The picture of the rule, drawn for four legs:



Result 16.6 (the stabilizers of a graph state). For every graph G with $n \geq 1$ vertices and every vertex a, exactly,

$$K_a\,|G\rangle \;=\; |G\rangle\,.$$

In pictures: on the net of G (Result 16.4), a red π on a's output and a green π on the output of each neighbour of a give back the bare net, with no number.

This is the chapter's headline theorem. It is proved twice in §16.7, once in each strand. We call K_a the **safe touch** at a.

For the two-spider net the touch is a red π on the top output and a green π on the bottom one: the $X \otimes Z$ of the opening example, which went unnoticed. §16.7 proves the headline for every graph, in both strands, and §16.8 draws it on the bow tie, touched at the knot.

The safe touches do more than fix $|G\rangle$. They name it.

Result 16.7 (the stabilizer group). For every graph G with n vertices: (i) the K_a commute, and the products $K^{\mathsf{b}} = \prod_a K_a^{b_a}$ over the bit strings $\mathsf{b} \in \{0,1\}^n$ are 2^n different elements of $\mathcal{P}_n$, forming an abelian group $\mathcal{S}_G$, the **stabilizer group** of $|G\rangle$, every element of which fixes $|G\rangle$; (ii) for bit strings $\mathsf{b} \in \{0,1\}^n$, the

2^n states $Z^{\mathrm{b}}\,|G\rangle = \prod_a Z_a^{b_a}\,|G\rangle$ form an orthonormal basis, and $K_a Z^{\mathrm{b}}\,|G\rangle = (-1)^{b_a} Z^{\mathrm{b}}\,|G\rangle$; (iii) $|G\rangle\langle G| = 2^{-n}\sum_{\sigma\in\mathcal{S}_G}\sigma$; (iv) $|G\rangle$ is, up to a number, the only state fixed by every K_a.

This is Hein et al.'s Proposition 3, with their eqs. (29) and (30); their subsets W of vertices are our bit strings b. For the bow tie, $\mathcal{S}_{G_{\bowtie}}$ has 32 elements.

Proof. (i) K_a and K_b can disagree in sign only on the wires a and b (everywhere else both carry I or Z). On wire a, K_a has X and K_b has Z exactly when a and b are adjacent; on wire b the same holds with the roles exchanged. So there are zero or two anticommuting wires, and by Result 16.1 the two commute. The product K^{b} has an X or Y exactly on the wires a with $b_a = 1$, so different b give different elements, and none of them is $-I$. Each fixes $|G\rangle$ by Result 16.6.

(ii) Z_c commutes with K_a for $c \neq a$ and anticommutes with K_a for $c = a$. So $K_a Z^{\mathrm{b}}\,|G\rangle = (-1)^{b_a} Z^{\mathrm{b}} K_a\,|G\rangle = (-1)^{b_a} Z^{\mathrm{b}}\,|G\rangle$. Two different b differ at some a, where K_a, which is Hermitian, has different eigenvalues on them, so the states are orthogonal. There are 2^n of them, each of length one, so they are an orthonormal basis.

(iii) Since the K_a commute and square to I, expanding the product $\prod_a \frac{1}{2}(I + K_a)$ gives each element of $\mathcal{S}_G$ once, times 2^{-n}. Each factor $\frac{1}{2}(I + K_a)$ projects onto the states with $K_a = +1$, so the product projects onto the states with every $K_a = +1$. By (ii), among the basis states that is $Z^0\,|G\rangle = |G\rangle$ alone.

(iv) A state fixed by every K_a is fixed by the projector of (iii), which is $|G\rangle\langle G|$. ■ See `Code/Ch16/SafeTouches.py`.

So a graph state can be described without a single amplitude: list, for each vertex, the letter X on the vertex and Z on its neighbours. For five bells that is five short strings instead of thirty-two numbers. Section §16.4 makes that description into a working notebook.

Why this definition. Why build graph states from $|+\rangle$ and CZ, rather than write down a general state? Because then the whole state is described by the graph: each tie is the same simple gate, the ties commute, and the only information is which vertices are joined. And nothing is lost by the

restriction. Section §16.3 shows that every state the quarter-turn gates can make is a graph state with a one-wire Clifford on each wire.

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§16.3 Half a Touch

The safe touch K_a is a half-turn on each wire it touches. What does half of it do?

Take the bow tie and the left-back bell, $a = 1$, with neighbours 2 and 3. Half a turn about X is $e^{-i\frac{\pi}{4}X} = \frac{1}{\sqrt{2}}(I - iX)$, and half a turn about Z is $e^{i\frac{\pi}{4}Z} = \frac{1}{\sqrt{2}}(I + iZ)$. Apply

$$U \;=\; e^{-i\frac{\pi}{4}X_1}\, e^{i\frac{\pi}{4}Z_2}\, e^{i\frac{\pi}{4}Z_3}$$

and ask what the new state is fixed by. For any operator K that fixed the old state, $UKU^\dagger$ fixes the new one. Take $K_2 = Z_1X_2Z_3$. On wire 1, $e^{-i\frac{\pi}{4}X} Z\, e^{i\frac{\pi}{4}X} = e^{-i\frac{\pi}{2}X} Z = -iXZ$, because Z anticommutes with X. On wire 2, likewise $e^{i\frac{\pi}{4}Z} X\, e^{-i\frac{\pi}{4}Z} = iZX$. So

$$UK_2U^\dagger \;=\; (-iX_1Z_1)(iZ_2X_2)\, Z_3 \;=\; X_1Z_1\, Z_2X_2\, Z_3.$$

That is not a safe touch of the bow tie. But it is the product $(X_1Z_2Z_3)(Z_1X_2)$, and those are the safe touches K_1' and K_2' of a **different graph**: the bow tie with the tie between bells 2 and 3 removed. Half a touch did not leave the state alone. It re-tied the net.

On the blackboard. For a graph G and a vertex a, the **local complement** $\tau_a(G)$ is the graph obtained by complementing the edges among the neighbours of a: two neighbours of a that were adjacent become non-adjacent, and two that were not become adjacent. Nothing else changes. This is Hein et al.'s notation (their eq. (41)); Duncan, Kissinger, Perdrix and van de Wetering, and van de Wetering's survey, write $G \star a$ for the same graph. Doing it twice undoes it: $\tau_a(\tau_a(G)) = G$.

Result 16.8 (the LC-rule, with its phase). For every graph G

and vertex a, let

$$U_a^\tau \;=\; e^{-i\frac{\pi}{4}X_a}\prod_{b\in N_a} e^{i\frac{\pi}{4}Z_b}.$$

Then, exactly,

$$U_a^\tau\,|G\rangle \;=\; e^{i\pi(|N_a|-1)/4}\,|\tau_a(G)\rangle\,, \qquad (U_a^\tau)^2 \;=\; i^{|N_a|-1}K_a.$$

Two halves of the touch make the touch: at the knot of the bow tie $(U_3^\tau)^2 = -iK_3$, and at a wing bell iK_a.

Hein and co-authors state this as their Proposition 5, eqs. (42) and (43), with $U_a^\tau(G) \propto \sqrt{K_a}$. Their eq. (42) is an equality of states determined by their stabilizers, which fixes the phase only up to an overall factor; the phase above is ours, computed below, with their $e^{i\frac{\pi}{4}\sigma_z^{N_a}}$ read as the product of one factor per neighbour.

Proof. The stabilizers. This is Hein et al.'s argument (their eq. (44) and after), and the example above is one case of it. Write K'_c for the safe touches of $\tau_a(G)$. For a vertex c that is not a neighbour of a (including a itself), U_a^τ commutes with K_c, and $K_c = K'_c$, because local complementation at a changes only the edges among N_a. For a neighbour b, conjugation turns the Z_a in K_b into $-iX_aZ_a$ and the X_b into iZ_bX_b, and leaves the other Zs alone, so

$$U_a^\tau K_b U_a^{\tau\dagger} \;=\; X_aZ_a\,Z_bX_b\prod_{c\in N_b\setminus\{a\}} Z_c \;=\; K'_aK'_b,$$

where the last step compares wire by wire: the Zs of K'_a on the other neighbours of a exactly toggle the edges from b to those neighbours. So the new state $U_a^\tau\,|G\rangle$ is fixed by every K'_c (and K'_a itself is the image of K_a), and by Result 16.7 (iv) it is a number times $|\tau_a(G)\rangle$. Both have length one, so the number is a phase λ.

The phase. Compare the amplitudes of $|0\cdots0\rangle$. Every graph state has $\langle 0\cdots0|\tau_a(G)\rangle \;=\; 2^{-n/2}$. On the other side, $\langle 0\cdots0|\,e^{i\frac{\pi}{4}Z_b} \;=\; e^{i\pi/4}\,\langle 0\cdots0|$ for each neighbour, and $\langle 0\cdots0|\,\frac{1}{\sqrt{2}}(I - iX_a)\,|G\rangle \;=\; \frac{1}{\sqrt{2}}(1-i)\,2^{-n/2}$, since the string with a single 1 at a has no edge with both ends at 1. So

$\lambda\, 2^{-n/2} = e^{i\pi|N_a|/4}\, e^{-i\pi/4}\, 2^{-n/2}$.

The square. $(e^{-i\frac{\pi}{4}X})^2 = e^{-i\frac{\pi}{2}X} = -iX$ and $(e^{i\frac{\pi}{4}Z})^2 = iZ$, so $(U_a^\tau)^2 = (-i)\, i^{|N_a|} X_a \prod_b Z_b$. ■ See `Code/Ch16/HalfATouch.py`.

Hein and co-authors add a converse, which we state and do not prove: "two graph states $|G\rangle$ and $|G'\rangle$ are LC-equivalent iff the corresponding graphs are related by a sequence of local complementations", LC-equivalent meaning related by one-wire Cliffords on every wire. They cite Van den Nest, Dehaene and De Moor, and Glynn, for it.

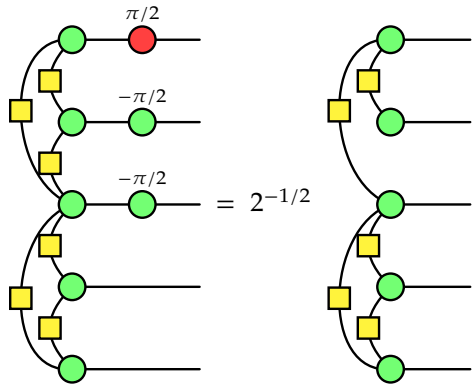
In the sketchbook. A red dot with phase $\pi/2$ and two legs is $|+\rangle\langle+| + i\,|-\rangle\langle-| = e^{i\pi/4}e^{-i\frac{\pi}{4}X}$, and a green dot with phase $-\pi/2$ is $\mathrm{diag}(1,-i) = e^{-i\pi/4}e^{i\frac{\pi}{4}Z}$. So half a touch, drawn as clip rings, is U_a^τ times $e^{i\pi/4}e^{-i\pi|N_a|/4}$, and that number cancels the phase of Result 16.8 exactly.

> **Result 16.9 (re-tying).** For every graph G with edge set E and every vertex a, a red clip ring turned a quarter round $(+\pi/2)$ on a's output and a green clip ring turned a quarter back $(-\pi/2)$ on each neighbour's output send $|G\rangle$ to exactly $|\tau_a(G)\rangle$. So do the reverse turns, a red quarter back and green quarters round. As nets, with E' the edge set of $\tau_a(G)$:
>
> $$(\text{net of } G \text{ with those clip rings}) \;=\; 2^{(|E'|-|E|)/2}\,(\text{net of } \tau_a(G)).$$

Proof. The first form is Result 16.8 with the number just computed. For the reverse turns the clip rings are $(U_a^\tau)^{-1}$ times the inverse number, and $(U_a^\tau)^{-1}|G\rangle = i^{1-|N_a|}\,U_a^\tau K_a\,|G\rangle$ by the square in Result 16.8, which is $i^{1-|N_a|}e^{i\pi(|N_a|-1)/4}\,|\tau_a(G)\rangle = e^{-i\pi(|N_a|-1)/4}\,|\tau_a(G)\rangle$; the inverse number cancels it. The net form follows from Result 16.4, since both nets have the same v. ■ See `Code/Ch16/HalfATouch.py`.

This equation is **proved on the blackboard and checked by matrix**, not derived here by rewriting. Its derivation by the rules of the calculus is due to Duncan and Perdrix. Duncan and co-authors cite it as "R. Duncan and S. Perdrix. Graph states and the necessity of Euler decomposition" (2009), and draw it as their eq. (5), with $X_{-\pi/2}$ on the vertex and $Z_{\pi/2}$ on its neighbours, which is the reverse form. Van de Wetering's survey draws it as its eq. (100).

At the left-back bell of the bow tie the neighbours are 2 and 3, which are tied, so re-tying removes one tie and the number is $2^{-1/2}$:



The wires are all where they were; the clip rings have made it a different net. At the knot, whose four neighbours are joined by only two ties, re-tying removes the tie inside each wing and adds a tie from each wing bell to both bells of the other wing: the bow tie's six ties become eight, and the number is $2^{(8-6)/2} = 2$. Add a second quarter on each of the three rings and they fuse (row 57) into the safe touch of Result 16.6, which the net does not notice; on the graph, the second re-tying at the same bell undoes the first.

Three re-tyings: the pivot

Re-tie the bow tie at the left-back bell, then at the knot, then at the left-back bell again. Following the definition three times gives the graph with ties 12, 13, 14, 15, 23, 24, 25 and 45: the left-back bell now has every other bell as a neighbour, the knot has only the left-back and left-front bells, and the two bells have in effect exchanged their places in the graph, with the ties among the others adjusted. Duncan et al. give this combination a name. For adjacent vertices u and v, the **pivot** of G along the edge uv is

$$G \wedge uv \;=\; \tau_u(\tau_v(\tau_u(G)))$$

(their Definition 4.3, "$G \wedge uv := G \star u \star v \star u$"). It is symmetric in u and v. Split the other vertices by where they are joined: A, the neighbours of u

alone; B, the neighbours of v alone; C, the common neighbours. The pivot complements the ties between A and B, between A and C, and between B and C, and exchanges the neighbourhoods of u and v.

> **Result 16.10 (the pivot).** For every graph G and every edge uv, exactly, with no phase,
>
> $$H_u H_v \prod_{c \in N_u \cap N_v} Z_c \; |G\rangle \;=\; |G \wedge uv\rangle \,.$$
>
> In the bow tie example, $H_1 H_3 Z_2 \, |G_{\bowtie}\rangle = |G_{\bowtie} \wedge 13\rangle$.

Van de Wetering's survey draws this as its eq. (101), "applying a Hadamard to the vertices we pivot along, and applying a Z gate to the vertices in their common neighbourhood", and Duncan et al. as their eq. (6), crediting Duncan and Perdrix (cited as "Ross Duncan and Simon Perdrix. Pivoting makes the ZX-calculus complete for real stabilizers", 2014). In pictures: yellow boxes on the outputs of u and v and a green π on each common neighbour's output turn the net of G into $2^{(|E''|-|E|)/2}$ times the net of $G \wedge uv$, where E'' is its edge set. Again this is checked by matrix, not derived by rewriting.

Proof. Apply Result 16.8 three times. Follow the neighbourhoods: at the first step u's neighbours are $\{v\} \cup A \cup C$; after it, v's are $\{u\} \cup A \cup B$; after the second step, u's are $\{v\} \cup B \cup C$. Multiply the three unitaries wire by wire. On u they give $e^{-i\frac{\pi}{4}X} e^{i\frac{\pi}{4}Z} e^{-i\frac{\pi}{4}X}$, and on v they give $e^{i\frac{\pi}{4}Z} e^{-i\frac{\pi}{4}X} e^{i\frac{\pi}{4}Z}$; each of these equals $-YH$, as one checks by multiplying out. Every vertex of A, B or C receives exactly two Z quarter-turns, $(e^{i\frac{\pi}{4}Z})^2 = iZ$. So, up to a phase,

$$|G \wedge uv\rangle \;\propto\; (Y_u Y_v \, Z_A Z_B) \; H_u H_v Z_C \, |G\rangle \,,$$

writing Z_A for a Z on every vertex of A, and so on. The first bracket is the product of the safe touches of u and v in $G \wedge uv$: there u's neighbours are $\{v\} \cup B \cup C$ and v's are $\{u\} \cup A \cup C$, and wire by wire $(X_u Z_v Z_B Z_C)(X_v Z_u Z_A Z_C) = (X_u Z_u)(Z_v X_v) Z_A Z_B = Y_u Y_v Z_A Z_B$. So it fixes $|G \wedge uv\rangle$, and $H_u H_v Z_C \, |G\rangle$ is a phase times $|G \wedge uv\rangle$. The phase is 1: $\langle 0 \cdots 0| \, H_u H_v Z_C \, |G\rangle = \langle +_u \, +_v \, 0 \cdots 0| \, G\rangle = \frac{1}{2}(1+1+1-1) 2^{-n/2} = 2^{-n/2}$, the same as $\langle 0 \cdots 0 | G \wedge uv\rangle$. ■ See `Code/Ch16/HalfATouch.py`.

Every state of the world

The quarter-turn gates, started from $|0\cdots 0\rangle$, make a family of states with many names. Aaronson and Gottesman prove that four descriptions of it agree.

> **Result 16.11 (stabilizer states; stated, not proved).** (Aaronson and Gottesman, Theorem 1.) For an n-qubit state $|\psi\rangle$ the following are equivalent: (i) $|\psi\rangle$ can be obtained from $|0\rangle^{\otimes n}$ by CNOT, Hadamard and phase gates only; (ii) the same, with measurements allowed too; (iii) $|\psi\rangle$ is fixed by exactly 2^n Pauli operators; (iv) $|\psi\rangle$ is determined by the group $\mathrm{Stab}(|\psi\rangle) \cap \mathcal{P}_n$ of Pauli operators that fix it. Such states are called **stabilizer states**. There are
>
> $$2^n \prod_{k=0}^{n-1} (2^{n-k} + 1)$$
>
> of them (their Proposition 2): 6 on one wire, 60 on two, 1080 on three, and 2 423 520 on five. Every stabilizer state is a graph state with a one-wire Clifford on each wire (Hein et al., Proposition 4, which they credit to Schlingemann and to Grassl, Klappenecker and Rötteler; van de Wetering calls this the **GSLC** form, graph state with local Cliffords). And every unitary quarter-turn circuit equals one "in canonical form", "11 rounds in the sequence H-C-P-C-P-C-H-P-C-P-C" (Aaronson and Gottesman, Theorem 8), from which, "building on a result by Patel, Markov, and Hayes", they obtain circuits with $O(n^2/\log n)$ gates.

The six on one wire are the tips of the octahedron: the states fixed by $\pm X$, $\pm Y$ or $\pm Z$. The graph states of §16.2 are, by Result 16.7, examples of (iii) and (iv). In the GSLC form every state of the world of quarters and halves is a net with a quarter-turn picture on each output: in the language of the dialogue, a tray of bells, each seated one of the twenty-four ways.

Why this definition. Why local complementation, a strange operation on graphs, rather than something simpler such as deleting an edge? Because it is what the world does. Deleting an edge needs a CZ, a gate on two wires; local complementation needs only one-wire turns, so it is the move

by which one-wire Cliffords can be absorbed into the net. That is why the GSLC form is not unique, and why the shrink of §16.4 can remove spiders without touching the rest.

Feynman's Notebook.

An operation older than the qubit. Local complementation was not invented for quantum states. Duncan, Kissinger, Perdrix and van de Wetering credit it to Anton Kotzig, in a paper on Eulerian lines in 4-valent graphs, printed in the proceedings of the Colloquium on Graph Theory held at Tihany in 1966, which appeared in 1968. Hein and co-authors, writing in 2006 from Innsbruck, London, Pasadena and Leuven, cite Bouchet for the operation, and Van den Nest, Dehaene and De Moor, and Glynn, for its meaning in quantum terms: one-wire Cliffords on a graph state are local complementations of its graph. The pictures came later. Duncan and Perdrix derived the local complementation of graph states in the ZX-calculus in a paper of 2009, "Graph states and the necessity of Euler decomposition", and the pivot in "Pivoting makes the ZX-calculus complete for real stabilizers", at the QPL workshop in 2014. By 2020 Duncan, Kissinger, Perdrix and van de Wetering had turned both into the engine of a circuit simplifier: remove every interior spider you can, by re-tying its neighbours.

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§16.4 The Crab's Ledger

How much is there to write down, if we follow a quarter-turn machine by its stabilizers?

Take the bow tie and write its five safe touches as five rows, one letter per wire and a sign at the end. Then apply one Hadamard, to the knot. By §16.1, H exchanges X and Z on its wire and turns Y into $-Y$. So each row changes only in the knot's column:

row	before	after H_3
1	$+XZZII$	$+XZXII$
2	$+ZXZII$	$+ZXXII$
3	$+ZZXZZ$	$+ZZZZZ$
4	$+IIZXZ$	$+IIXXZ$
5	$+IIZZX$	$+IIXZX$

Each new row fixes the new state $H_3\,|G_{\bowtie}\rangle$, for the reason in Gottesman's line: if $K|\psi\rangle = |\psi\rangle$ then $(UKU^\dagger)(U|\psi\rangle) = U|\psi\rangle$. No amplitude was computed. The new state is determined by the five rows, because they generate a group of 2^5 Paulis that fix it (Result 16.11 (iii), (iv)).

On the blackboard. Store each letter as two bits, an x-bit and a z-bit: $I = 00$, $X = 10$, $Z = 01$, $Y = 11$ (the pair 11 stands for $Y = iXZ$; any sign goes in the row's sign bit). A row on n wires is then $2n$ bits and one sign bit. The whole state is n rows.

> **Result 16.12 (Knill's theorem; Gottesman 1998, Theorem 1, stated).** "Any quantum computer performing only: a) Clifford group gates, b) measurements of Pauli group operators, and c) Clifford group operations conditioned on classical bits, which may be the results of earlier measurements, can be perfectly simulated in polynomial time on a probabilistic classical computer."

Gottesman credits it to "E. Knill, private communication". Aaronson and Gottesman call it "the Gottesman-Knill theorem". Here is the part of its proof that concerns us, with the updates in full.

> **Result 16.13 (the ledger).** Let a stabilizer state on n wires be recorded as n rows of signed Pauli strings that generate its stabilizer group. Then: (i) after H on wire a, each row has its x- and z-bits on wire a exchanged, and its sign flipped if both were 1; (ii) after S on wire a, each row has its sign flipped if both bits on wire a were 1, and then its z-bit on wire a replaced by $z \oplus x$; (iii) after a CNOT or a CZ, each row changes only in the two columns of the gate; (iv) a measurement of a Pauli string M that

commutes with every row gives a certain outcome, $+1$ if M is in the group and -1 if $-M$ is; a measurement of an M that anticommutes with some row gives each outcome with probability $\frac{1}{2}$. The record has $n(2n+1)$ bits.

Proof. (i)–(iii): conjugation respects products, so it is enough to know where the gate sends X and Z on its wires, which is Gottesman's Table 1 (§16.1); the bit rules are those images written in bits (the sidebar below). A gate on wires a and b conjugates a string letter by letter, and only the letters on a and b change. (iv) If M anticommutes with a row g, then $\langle\psi|M|\psi\rangle = \langle\psi|Mg|\psi\rangle = -\langle\psi|gM|\psi\rangle = -\langle\psi|M|\psi\rangle$, so $\langle\psi|M|\psi\rangle = 0$, and the probability of reading $+1$, which is $\|\frac{1}{2}(I+M)|\psi\rangle\|^2 = \frac{1}{2}(1+\langle\psi|M|\psi\rangle)$, is $\frac{1}{2}$. If M commutes with every row, then $M|\psi\rangle$ is fixed by every row, so by Result 16.11 (iv) (Result 16.7 (iv) for a graph state) $M|\psi\rangle = \lambda|\psi\rangle$, with $\lambda = \pm 1$ since $M^2 = I$. ■ See `Code/Ch16/CrabsLedger.py`, which runs the ledger and the state vector side by side through random quarter-turn circuits and checks every row after every gate.

On the bow tie, measuring K_3 gives $+1$ with certainty. Measuring Z on the left-back bell gives each outcome half the time: Z_1 anticommutes with the row K_1.

Dirac's Blackboard.

The updates, bit by bit. On one wire write a letter as (x, z), meaning $i^{xz}X^xZ^z$, so that $(1,1)$ is $iXZ = Y$. The Hadamard sends $X \mapsto Z$, $Z \mapsto X$, $Y \mapsto -Y$: in bits it swaps x and z, and flips the row's sign when $x = z = 1$. The phase gate sends $X \mapsto Y$, $Y \mapsto -X$, $Z \mapsto Z$: $(1,0) \mapsto (1,1)$ with no sign, $(1,1) \mapsto (1,0)$ with a sign, $(0,1)$ fixed. So: flip the sign when $x = z = 1$, then set $z := z \oplus x$. These are Aaronson and Gottesman's rules ("set $r_i := r_i \oplus x_{ia}z_{ia}$ and swap x_{ia} with z_{ia}", and "$z_{ia} := z_{ia} \oplus x_{ia}$"). For a CNOT from a to b they give $x_b := x_b \oplus x_a$, $z_a := z_a \oplus z_b$, and a sign flip when $x_a z_b (x_b \oplus z_a \oplus 1) = 1$. The count: n rows of $2n+1$ bits, $n(2n+1)$ in all. For the bow tie, 55 bits against 32 complex amplitudes; for a hundred bells, 20 100 bits against a list of amplitudes whose length has thirty-one digits.

Aaronson and Gottesman improved the ledger. They add n more rows, "destabilizer" generators, at "a cost of a factor-2 increase in the number of bits", $2n(2n+1)$, and in return every measurement is simulated in $O(n^2)$ steps. They implemented it in a program they called CHP (CNOT-Hadamard-Phase), which, their abstract says, "can handle thousands of qubits easily". And they placed the problem exactly. Deciding whether a given wire will read $|1\rangle$ with certainty after a given quarter-turn circuit is hard for the class $\oplus$L ("parity-L"), since circuits of CNOTs are a special case; and it lies in $\oplus$L, their Theorem 4. So it is $\oplus$L-complete, which they read as "strong evidence that stabilizer circuits are not even universal for classical computation". We state these results and do not prove them.

Feynman's Notebook.

Whose theorem. The result first appears as a report. Daniel Gottesman's doctoral thesis at the California Institute of Technology, *Stabilizer Codes and Quantum Error Correction*, submitted on 21 May 1997, says in its section "The Toffoli Gate" that "Knill [44] has shown that a quantum computer using only elements from $N(G)$ and measurements can be simulated efficiently on a classical computer", and the reference reads "E. Knill, personal communication". The following year Gottesman, then in the T-6 Group at Los Alamos National Laboratory, wrote "The Heisenberg Representation of Quantum Computers", Los Alamos preprint LAUR-98-2848, and stated the result as Theorem 1, "Knill's theorem", citing "E. Knill, private communication". He drew the moral in the next paragraph: the theorem "implies that quantum computation is only more powerful than classical computation when it uses gates outside the Clifford group." Scott Aaronson and Gottesman, in a paper of 2004, wrote of the conclusion "sometimes known as the Gottesman-Knill theorem", and made it fast.

In the sketchbook. A picture with no inputs and no outputs is a number (Chapter 2). An amplitude $\langle y|\, C\, |0\cdots 0\rangle$ of a quarter-turn circuit C is such a picture: red points on the left, red effects on the right, and C in quarter-turn spiders between. Van de Wetering's survey (§6.5) shows how to find that

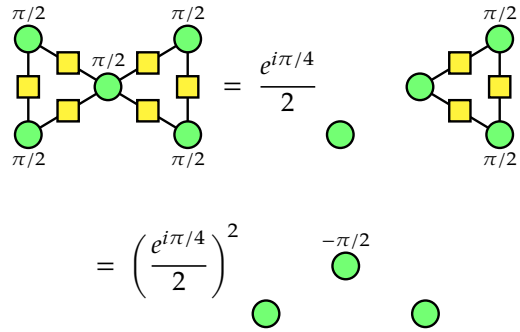
number by rewriting alone, using the two graph moves of §16.3 in a form for the inside of a picture.

First, fuse and colour-change until the picture is a net: green spiders joined only by ties. A spider is **interior** if it has no input or output leg. Then:

- **Pull out a quarter.** An interior spider turned a quarter round $(+\pi/2)$ or a quarter back $(-\pi/2)$ can be removed: complement the ties among its neighbours, as in re-tying, and turn each neighbour a quarter the other way. A number is set aside. This is van de Wetering's eq. (102) and Duncan et al.'s Lemma 5.2.
- **Pull out a pair of halves.** Two tied interior spiders whose turns are 0 or π can be removed together, by a pivot along their tie, with half-turns added to the neighbours (van de Wetering's eq. (103), Duncan et al.'s Lemma 5.3).
- **A spider with no legs is a number**: a green spider with phase α and no legs is $1 + e^{i\alpha}$, so 2 at rest, $1 + i$ at a quarter round, $1 - i$ at a quarter back, and 0 at a half.

We call this **the shrink**. For a closed quarter-turn picture, van de Wetering explains, the procedure "removes all spiders that have any connection to another spider", and what is left is a set of spiders with no legs, whose product is the amplitude.

Here it is on the bow tie. Cap every output of its net with a green effect turned a quarter round, $\langle 0| + i\,\langle 1|$. Each cap fuses into its bell (row 57), so every bell becomes an interior spider turned a quarter round, and the picture is closed. Draw the bells as the bow tie itself: the left wing on the left, the back bells on top, and the knot in the middle. Pull out the left-back bell, top left: its neighbours, the left-front bell and the knot, are tied, so the tie between them goes, and each is turned a quarter back, which leaves both at rest; the left-front bell is now alone. Then pull out the right-back bell in the same way: the tie from the knot to the right-front bell goes, the knot is turned to a quarter back, and the right-front bell comes to rest. Each pull-out sets aside the number $e^{i\pi/4}/2$ (the sidebar below shows where it comes from).



Result 16.14 (the shrink of the bow tie). Exactly,

$$\left(\langle 0| + i\,\langle 1|\right)^{\otimes 5}\left(2^{-1/2}\,|G_{\bowtie}\rangle\right) \;=\; \left(\frac{e^{i\pi/4}}{2}\right)^2 \cdot 2 \cdot (1-i) \cdot 2 \;=\; 1+i.$$

Proof. In pictures, the chain above: two pull-outs, each setting aside $e^{i\pi/4}/2$, and three spiders with no legs, worth 2, $1-i$ and 2. $(e^{i\pi/4}/2)^2 = i/4$, and $\frac{i}{4}\cdot 4(1-i) = 1+i$. On the blackboard, the left side is $2^{-3}\sum_x i^{|x|}(-1)^{E(x)}$ over the 32 strings x, with $|x|$ the number of 1s, which also gives $1+i$. ■ See `Code/Ch16/CrabsLedger.py`, which builds the three pictures and checks each against the one before, with its number.

The number does not depend on the order of the pull-outs; pulling out the knot first would give different pictures and the same $1+i$. With the caps at rest instead of turned, the closed net is exactly 1, and pivots on pairs of halves shrink it to a number in the same way.

Coecke's Sketchbook.

Where the number comes from, and what the shrink does not do. Sum over the bit x_1 of the pulled-out bell. Its two ties give $2^{-1}(-1)^{x_1(x_2+x_3)}$, and its quarter turn gives i^{x_1}, so the sum is $\frac{1}{2}(1 + i(-1)^{x_2+x_3})$. Check the four cases: this equals $\frac{1}{2}\sqrt{2}\,e^{i\pi/4}(-i)^{x_2}(-i)^{x_3}(-1)^{x_2x_3}$. The factors $(-i)^{x_2}$ and $(-i)^{x_3}$ are the quarter-back turns on the neighbours. The factor $(-1)^{x_2x_3}$ cancels the sign of the tie between them, which leaves

that tie's $2^{-1/2}$ outside. In all, $\frac{1}{2} \cdot \sqrt{2}\, e^{i\pi/4} \cdot 2^{-1/2} = e^{i\pi/4}/2$. Van de Wetering is candid that the rules in his survey are not "scalar-accurate", and that the Gottesman–Knill theorem "also states that you can efficiently calculate marginal probabilities as well, which our procedure in its current form does not do"; for that he points to the doubled pictures of Chapter 9. So the shrink computes amplitudes of closed pictures, one number at a time. The ledger computes the odds of readings. And the claim that the shrink takes polynomially many steps is the sources' count of moves, each removing a spider; a rewrite proves an equality, not a running time.

Why this definition. Why keep a ledger of stabilizers rather than of amplitudes? Because a quarter-turn gate sends each row to a single row, so the ledger never grows: n rows before, n rows after, each updated by a rule a clerk can apply. The amplitudes of the same state are 2^n numbers, and a gate mixes them. The ledger is small because the world is closed: what goes in as a Pauli string comes out as a Pauli string.

---

§16.5 An Eighth

So the world of quarters and halves can be followed on paper, in the ledger or by the shrink. The eighth-turn is the gate

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix},$$

a green $\pi/4$ dot in pictures (Chapter 15, row 114). Two eighths make a quarter: $T^2 = S$.

On the blackboard. Follow the front tip:

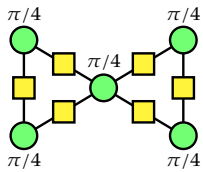
$$TXT^\dagger = \begin{pmatrix} 0 & e^{-i\pi/4} \\ e^{i\pi/4} & 0 \end{pmatrix} = \tfrac{1}{\sqrt{2}}(X + Y).$$

Result 16.15 (T is not a Clifford). $TXT^\dagger = (X+Y)/\sqrt{2}$, which is not a phase times a Pauli string, so $T \notin C_1$. On the octahedron, T turns the solid an eighth of a whole turn about the up–down axis: the front tip $+X$ goes to the direction $(1,1,0)/\sqrt{2}$, the midpoint of the edge between the tips $+X$ and $+Y$. The up and down tips stay where they are, but the four tips round the middle all land on edge midpoints, so the solid is not seated. $T^2 = S$ is a quarter-turn, and a Clifford.

Proof. The matrix product above, and $X + Y$ has two non-zero Pauli components, while a phase times a Pauli has one. T is diagonal, so it fixes Z; on the X–Y plane it acts as the rotation by $\pi/4$, since $T(\cos\varphi\, X + \sin\varphi\, Y)T^\dagger = \cos(\varphi + \frac{\pi}{4})X + \sin(\varphi + \frac{\pi}{4})Y$ (the diagonal entries of the matrix above carry the angle). ■ See `Code/Ch16/AnEighth.py`.

A word on names. T is often called the $\pi/8$ gate, because $T = e^{i\pi/8}\,\mathrm{diag}(e^{-i\pi/8}, e^{i\pi/8})$. Gottesman's list of gates that complete the Clifford group includes "a $\pi/8$ rotation of the Bloch sphere" (1998) and "the single-qubit $\pi/8$ rotation" (thesis); those are his words. What T does to the octahedron, and so to the sphere, is the turn by $\pi/4$ just computed: an eighth of a whole turn.

In the sketchbook. An eighth is a green spider with phase $\pi/4$, and the shrink cannot touch it. Cap every output of the bow tie's net with a green effect turned an eighth, $\langle 0| + e^{i\pi/4}\langle 1|$. Every bell fuses into an interior spider turned an eighth. Neither move of the shrink applies: a pull-out needs a quarter, and a pair needs halves.



$$= \frac{1}{8}\sum_x e^{i\pi|x|/4}(-1)^{E(x)} = \frac{3\sqrt{2}-4}{8} + \frac{\sqrt{2}-2}{8}\,i.$$

The picture is still a number, and the blackboard finds it by adding 32 terms. The shrink stops before it starts. That is not a proof that no clever method could find the number quickly. It is a statement about these moves: they

were built for quarters and halves. `Code/Ch16/AnEighth.py` runs PyZX's quarter-turn simplifier on the picture above and checks that it stops with all five eighth-turn spiders left inside.

Bringing the eighth in from outside

If the world cannot make an eighth, perhaps it can be given one, ready made. Write

$$|\mu\rangle \;=\; T|+\rangle \;=\; \tfrac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle)$$

for the **magic state**. The name is Bravyi and Kitaev's, and so is the idea that such states, supplied from outside, are all that quarter-turn machines lack. Van de Wetering's survey (§5.3) gives the **injection**, which moves the eighth from $|\mu\rangle$ onto any state.

Take a data wire in any state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and a second wire in $|\mu\rangle$. Apply a CNOT, control on the data wire, target on the magic wire, and read the magic wire in the basis $|0\rangle, |1\rangle$. Before stating the result, compute:

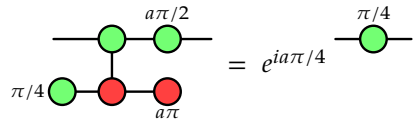
$$\mathrm{CNOT}(|\psi\rangle|\mu\rangle) \;=\; \tfrac{1}{\sqrt{2}}\Big((\alpha|0\rangle + e^{i\pi/4}\beta|1\rangle)|0\rangle + (e^{i\pi/4}\alpha|0\rangle + \beta|1\rangle)|1\rangle\Big).$$

If the reading is 0, the data wire holds $\alpha|0\rangle + e^{i\pi/4}\beta|1\rangle = T|\psi\rangle$. If it is 1, it holds $e^{i\pi/4}(\alpha|0\rangle + e^{-i\pi/4}\beta|1\rangle) = e^{i\pi/4}T^\dagger|\psi\rangle$, the eighth turned the wrong way. A quarter mends it: $ST^\dagger = T$.

> **Result 16.16 (injection).** For every data state $|\psi\rangle$, exactly: projecting the magic wire onto $\langle 0|$ leaves $\frac{1}{\sqrt{2}}T|\psi\rangle$ on the data wire; projecting it onto $\langle 1|$ and then applying S leaves $\frac{e^{i\pi/4}}{\sqrt{2}}T|\psi\rangle$. The probability of reading 0 on the magic wire is $\frac{1}{2}$, and so is the probability of reading 1. Either way the data wire ends in $T|\psi\rangle$, up to a phase, and the magic state is used up.

Proof. The computation above, with the $\frac{1}{\sqrt{2}}$ carried along; each branch has length $\frac{1}{\sqrt{2}}$, so probability $\frac{1}{2}$. In pictures, with the bit a of the reading written into the phases as in Chapter 10 (row 75): the magic state is a green $\pi/4$ point, which is $\sqrt{2}\,|\mu\rangle$; the CNOT is the green–red pair (row 64), bare

CNOT/$\sqrt{2}$; the reading is a red effect with phase $a\pi$, which is $\sqrt{2}\,\langle a|$; and the correction is a green $a\pi/2$ on the data wire. Together the pieces carry $\sqrt{2}\cdot\sqrt{2}/\sqrt{2}=\sqrt{2}$. For $a=0$ the red effect fuses into the red dot of the pair (fusion, row 57), leaving a phase-free red dot with two legs, which is a plain wire (§8.2); the green point then fuses into the green dot on the data wire. For $a=1$ the fused red dot is a red π with two legs, which is X; pushed into the green point it turns the point's phase to $-\pi/4$ and sets aside $e^{i\pi/4}$, since $X(|0\rangle+e^{i\pi/4}|1\rangle)=e^{i\pi/4}(|0\rangle+e^{-i\pi/4}|1\rangle)$ (π-commutation with its number, as in Chapter 14); the point fuses into the data wire with $-\pi/4$, and the green quarter fuses on top, $-\pi/4+\pi/2=\pi/4$:



and the right side is $e^{ia\pi/4}\,T$, which is $\sqrt{2}$ times the branch on the blackboard. ■ See `Code/Ch16/AnEighth.py`, which checks both branches on a random non-real $|\psi\rangle$.

The picture shows the eighth travelling: from the magic point, through the tie, onto the data wire. It also shows why a quarter is the right correction: the red half on the reading flips the eighth, and the difference between an eighth forward and an eighth back is a quarter, which the world of quarters can supply.

Bravyi and Kitaev use different letters, and we will not borrow them. Their T is a Clifford that turns the octahedron a third of a turn, cycling its three axes; their $|T\rangle$ is one kind of magic state, and their $|H\rangle$ another; they write a general state as $|A_\theta\rangle=2^{-1/2}(|0\rangle+e^{i\theta}|1\rangle)$ (their eq. (2)). Our $|\mu\rangle$ is their $|A_{\pi/4}\rangle$, and it is of their "H-type". Van de Wetering's magic state is $|0\rangle+e^{i\pi/4}|1\rangle$, which is $\sqrt{2}\,|\mu\rangle$.

> **Result 16.17 (magic and universality; Gottesman and Bravyi–Kitaev, stated, not proved).** The quarter-turn gates together with T generate a set of unitaries dense in all unitaries: in Gottesman's words, "the Clifford group plus an appropriate extra gate

generate a set of unitary operators dense in $U(2^n)$". Bravyi and Kitaev show (their §III) that a supply of one H-type magic state lets quarter-turn machines apply the phase gate $\Lambda(e^{-i\pi/4})$, which is our $T^\dagger$: "It does not belong to the Clifford group", and the subgroup of $U(2)$ it generates together with the one-wire Cliffords "is dense in $U(2)$". Our $|\mu\rangle$ is H-type too, and it gives T directly, by the injection above. On the octahedron, their magic states point to the centres of its faces (8 of their "T-type") and the midpoints of its edges (12 of "H-type"); $|\mu\rangle$ points to the edge midpoint between $+X$ and $+Y$.

The octahedron plays a second role in their paper. Write a one-wire mixed state as $\frac{1}{2}(I + \rho_x X + \rho_y Y + \rho_z Z)$. If $|\rho_x| + |\rho_y| + |\rho_z| \le 1$, the point (ρ_x, ρ_y, ρ_z) lies inside the octahedron whose tips are the six stabilizer states, the state is a mixture of those six, and a quarter-turn machine fed with it can be simulated efficiently (their Theorem 1). Outside the octahedron things change. They show that noisy copies of a magic state can be purified by quarter-turn operations alone, a process now called **distillation**, provided the polarization along a magic direction exceeds a threshold, "about 65%" in their abstract. We name distillation and do not describe it.

Feynman's Notebook.

Why "magic". Sergey Bravyi and Alexei Kitaev, at the Institute for Quantum Information at Caltech, wrote "Universal quantum computation with ideal Clifford gates and noisy ancillas" in 2004. They imagined hardware in which the quarter-turn gates are nearly perfect, protected, for instance, by the topology of exotic two-dimensional materials whose excitations are anyons, while everything else is noisy. Such a machine can only be as powerful as its one noisy ingredient: a supply of ancilla qubits in some mixed state ρ. Their question was for which ρ it becomes universal. Their answer turned on special states with two properties at once: fed to the perfect gates, they give universality, and noisy copies of them can be purified by the perfect gates alone. "It is a rather surprising coincidence that one and the same state can comprise both of these properties," they wrote, "and that

is the reason why we call them magic states." The quarter-turn machine is a notebook's plaything; with a trickle of magic it is a quantum computer.

Why this definition. Why a state and not a gate? Because the world of quarters can do everything except the eighth, including prepare, entangle, read and correct. So the eighth need only be made once, somewhere, as a state, perhaps slowly and with effort, and then fed in. Injection turns a supply of states into a supply of gates. The cost of a computation then moves to one place: how many magic states it consumes.

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§16.6 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 15's row 117. The chapter introduces no new string-diagram element: rows 119 to 126 are new uses of the spiders, boxes and rules of Chapters 8 to 15, and new rules derived or stated from them. Row 118 also contains a figure of a different kind, the octahedron, which is not a string diagram.

#	Algebra (Dirac)	Picture (Coecke)
118	$\mathcal{P}_n$: strings of I, X, Y, Z with $Y = iXZ$, phases $\pm 1, \pm i$, $4 \cdot 4^n$ elements; $\mathcal{C}_n$, the unitaries with $U\mathcal{P}_n U^\dagger = \mathcal{P}_n$, generated by H, S, CNOT; 24 one-wire Cliffords up to a phase	**quarter-turn pictures**: red π is X, green π is Z, green $\pi/2$ is S; every turn a multiple of a quarter, with yellow boxes; the Hadamard's "over" is colour change (row 58). *Also, a different kind of picture, not a string diagram:* **the octahedron of Pauli tips** $\pm X, \pm Y, \pm Z$, each one-wire Clifford a rotation that seats it; drawn with $HXH^\dagger = Z$ and $SXS^\dagger = Y$ as tip moves
119	$\lvert G\rangle = U_G \lvert+\rangle^{\otimes n}$, $U_G = \prod_{\{b,c\}\in E} \mathrm{CZ}_{bc}$	**the net**: a phase-free green spider per vertex with one output, a tie with a yellow box per edge; bare $= 2^{(v-e)/2} \lvert G\rangle$
120	$X^{\otimes k}(\lvert 0\cdots 0\rangle\langle 0\cdots 0\rvert + \lvert 1\cdots 1\rangle\langle 1\cdots 1\rvert)X^{\otimes k} = \lvert 0\cdots 0\rangle\langle 0\cdots 0\rvert + \lvert 1\cdots 1\rangle\langle 1\cdots 1\rvert$	**a half through a ring at rest**: a red π on one leg of a phase-free green spider = red πs on all its other legs, scalar 1; colours exchanged likewise (derived; phase-free spiders only)
121	**the headline:** $K_a \lvert G\rangle = \lvert G\rangle$, $K_a = X_a \prod_{b\in N_a} Z_b$	**the safe touch**: red π on a's output, green π on each neighbour's = the bare net (rows 120, 58, 57)

#	Algebra (Dirac)	Picture (Coecke)
122	$U_a^\tau \lvert G\rangle = e^{i\pi(\lvert N_a\rvert-1)/4} \lvert \tau_a(G)\rangle$; $(U_a^\tau)^2 = i^{\lvert N_a\rvert-1} K_a$	**re-tying**: a red quarter round on a's output and a green quarter back on each neighbour's (or both reversed) $= 2^{(\lvert E'\rvert-\lvert E\rvert)/2}$ net of $\tau_a(G)$ (proved on the blackboard, checked; not derived by rewriting)
123	$H_u H_v \prod_{c\in N_u\cap N_v} Z_c \lvert G\rangle = \lvert G \wedge uv\rangle$	**the pivot**: yellow boxes on u's and v's outputs, green πs on the common neighbours' $= 2^{(\lvert E''\rvert-\lvert E\rvert)/2}$ net of $G \wedge uv$ (proved on the blackboard, checked)
124	Gottesman–Knill for amplitudes: $\langle y\rvert C \lvert 0\cdots 0\rangle$ of a quarter-turn circuit, in polynomially many steps (the sources' count)	**the shrink**: interior quarter-turn spiders pulled out by re-tying, tied interior pairs of halves by pivots; what is left are spiders with no legs, i.e. numbers. Amplitudes only
125	$TXT^\dagger = (X+Y)/\sqrt{2}$; $T^2 = S$	**the eighth**: a green $\pi/4$ (row 114) is not a quarter-turn picture; an interior eighth stops the shrink
126	$\lvert\mu\rangle = T\lvert+\rangle$; injection leaves $\frac{1}{\sqrt{2}} T\lvert\psi\rangle$ (reading 0) and $\frac{e^{i\pi/4}}{\sqrt{2}} T\lvert\psi\rangle$ after S (reading 1)	**the magic point**: a green $\pi/4$ point, joined by the green–red pair and read with a red $a\pi$ effect; the eighth lands on the data wire, and a green quarter mends the reading 1

Every string-diagram row is exact, scalar included; where a source states a rule up to a number, the number is written here. Row 119 is a definition with its number computed by a sum over bits; rows 120 and 121 are derived by the rules; rows 122 and 123 are proved on the blackboard and checked by matrix; row 124 is a procedure whose steps are equalities and whose running time is a count taken from the sources.

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§16.7 The Theorem, Twice

> **Result 16.6 (the stabilizers of a graph state), restated.** Let $G = (V, E)$ be a graph with $n \geq 1$ vertices, $|G\rangle = U_G\,|+\rangle^{\otimes n}$, and for each vertex a let $K_a = X_a \prod_{b \in N_a} Z_b$. Then for every a, exactly,
>
> $$K_a\,|G\rangle \;=\; |G\rangle\,.$$

Dirac's Blackboard

Step 1: the empty graph. $X|+\rangle = |+\rangle$, so $X_a\,|+\rangle^{\otimes n} = |+\rangle^{\otimes n}$: the start is fixed by X_a alone.

Step 2: through the ties. For an edge at a, $\mathrm{CZ}_{ab}\,X_a\,\mathrm{CZ}_{ab} = X_a Z_b$ (the Dirac's Blackboard sidebar of §16.2: on $|x_a x_b\rangle$ the two signs multiply to $(-1)^{x_b}$). For an edge $\{b, c\}$ not at a, CZ_{bc} acts on other wires and commutes with X_a. Every CZ is diagonal and commutes with every Z. So, pushing X_a through the CZs of U_G one at a time, and using $\mathrm{CZ}^\dagger = \mathrm{CZ}$,

$$U_G\,X_a\,U_G^\dagger \;=\; X_a \prod_{b \in N_a} Z_b \;=\; K_a.$$

Step 3: conclude.

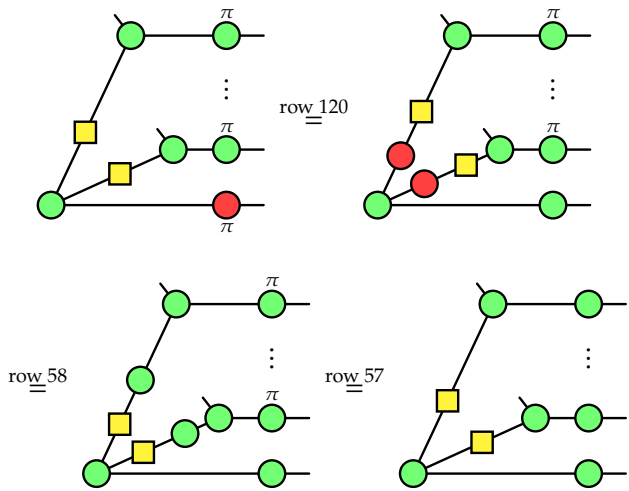
$$K_a\,|G\rangle \;=\; U_G X_a U_G^\dagger\,U_G\,|+\rangle^{\otimes n} \;=\; U_G\,X_a\,|+\rangle^{\otimes n} \;=\; U_G\,|+\rangle^{\otimes n} \;=\; |G\rangle\,. \qquad \blacksquare$$

Coecke's Sketchbook

Draw the part of the net of G around the vertex a: its green spider, its output with a clip ring, and its ties, each with a yellow box, to its neighbours $b_1, \ldots, b_k$, each with its own output and clip ring. A short stub on each neighbour stands for its other ties, which play no part. The touch puts a red π on a's clip ring and a green π on each neighbour's. Each step names its rule. In the pictures below a is at the bottom left, two neighbours stand for all of them, and every dot drawn on a tie carries a half, π.

Step 1: push the half into a (Result 16.5, row 120). a's spider is phase-free, and its legs are its output and its ties. The red π leaves the output and appears on every tie, next to a. The output's clip ring is left at rest.

Step 2: through the boxes (colour change, Theorem 8.2, row 58). A red π followed by a yellow box is the yellow box followed by a green π: $XH = HZ$, which is Theorem 8.2 on a two-legged dot with $H^2 = I$ used on one side. Each half crosses its box and turns green.



Step 3: meet the neighbour's half (fusion, Theorem 8.1, row 57). On each tie the green π now stands next to the neighbour's green spider, and slides into it. The neighbour's own clip ring carries a green π too, on the other

side of the spider. Fusion, read both ways, lets the spider and its ring share their turns as they like: together they carry $\pi + \pi = 2\pi$, which is no turn. So the neighbour is phase-free again and its ring is at rest.

Step 4: read off. What is left is the untouched net, with every clip ring at rest. A ring at rest is a plain wire (§8.2), so this is the bare net, and nothing was set aside at any step: the push costs nothing (Result 16.5), and so do colour change and fusion. The touched net equals the bare net, exactly, which is $K_a |G\rangle = |G\rangle$ multiplied on both sides by the net's number $2^{(v-e)/2}$ (Result 16.4). ■ See `Code/Ch16/SafeTouches.py`, which carries out these moves as edits of the picture, checking the value after each.

The verdict

A draw, and the chapter says so. The blackboard proof is three lines, and Step 2 is one identity for one tie, repeated. The sketchbook proof is one push and a slide, and it shows something the algebra compresses: *why* the Zs land where they do. The half on a runs out along every tie, and each tie carries it to exactly one neighbour. The pattern of the touch is the pattern of the net.

What the blackboard gives that the picture does not is everything around the headline. That the K_a commute, that they generate 2^n touches the state does not notice, that the state is the only one they all fix, and that the states $Z^{\mathsf{b}} |G\rangle$ form a basis: those are Result 16.7, proved with operators and counting. The picture proves that each touch goes unnoticed. The algebra proves that together they name the state.

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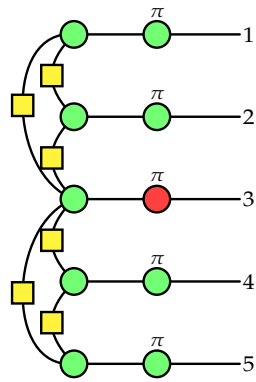
§16.8 The Touched Bow Tie

Here is the chapter's master diagram: the net of the bow tie, with a clip ring on every output, touched at the knot. On the blackboard it is

$$K_3 \, (2^{-1/2} \, |G_{\bowtie}\rangle), \qquad K_3 = Z \otimes Z \otimes X \otimes Z \otimes Z.$$

It is drawn with its clip rings **unfused**, so that the touch sits on the outputs as separate pieces, one ring per bell, as the dialogue's brass rings sit on the pulls of the bell-tray.

From top to bottom, the bells in ket order: the left-back bell, the left-front bell, the knot, the right-back bell and the right-front bell, each a phase-free green spider. On the left, the ties, each with its yellow box: a short arc from the left-back bell to the left-front bell, a short arc from the left-front bell to the knot, and a longer arc outside them from the left-back bell to the knot; then, below, a short arc from the knot to the right-back bell, a short arc from the right-back bell to the right-front bell, and a longer arc outside them from the knot to the right-front bell. On the right, each bell's pull: an inner length from the bell to its clip ring, and an outer length from the clip ring to the output. The knot's clip ring is a red π. The clip rings of the left-back, left-front, right-back and right-front bells are green πs. We call the picture **The Touched Bow Tie**.



No two ties cross, and no tie crosses a pull: the knot's four ties leave it at four different angles on its left, and every pull runs straight to the right.

Result 16.18 (The Touched Bow Tie). Exactly,

$$\text{The Touched Bow Tie} \;=\; 2^{-1/2}\,|G_{\bowtie}\rangle\,,$$

the same as the untouched net with every clip ring at rest. The number $2^{-1/2}$ is the net's own, $2^{(5-6)/2}$ for five bells and six ties.

Proof. **On the blackboard.** The clip rings are $Z \otimes Z \otimes X \otimes Z \otimes Z = K_3$ on the outputs of the net, which is $2^{-1/2}\,|G_{\bowtie}\rangle$ (Result 16.4). By Result 16.6, $K_3\,|G_{\bowtie}\rangle = |G_{\bowtie}\rangle$.

In the sketchbook. The chain of §16.7 at $a = 3$, with the four neighbours $1, 2, 4, 5$. The red π on the knot's ring goes into the knot and out along its four ties (row 120), leaving the knot's ring at rest. Each half crosses its yellow box and turns green (row 58): on the arcs to the left-back and right-front bells, and on the short arcs to the left-front and right-back bells. Each reaches a wing bell and meets the green π on that bell's ring: $\pi + \pi = 0$ (row 57), four times. The ties between the wing bells, the arcs from left-back to left-front and from right-back to right-front, are never touched. Every ring is at rest, the picture is the untouched net, and no number was set aside. ■
See `Code/Ch16/SafeTouches.py`, which builds this picture, counts its lines, checks that it equals the untouched net, and checks that the touch with the colours exchanged, green on the knot and red on the wings, is not the net.

The touch changed every ring on the tray, and the net did not notice. Touched at a wing bell instead, say the left-back, the picture has a red π on that bell's ring and green πs on the rings of the left-front bell and the knot only; the chain is the same, with two ties instead of four.

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§16.9 What the Pictures Leave to the Blackboard

Take stock. The Pauli strings on n wires form a group of $4 \cdot 4^n$ elements, any two of which commute or anticommute (Result 16.1). The Cliffords are the unitaries that send Paulis to Paulis; on one wire there are twenty-four of them, the rotations of an octahedron (Result 16.2), and on many wires H, S and CNOT generate them (Result 16.3). In pictures a graph state is a net with a known number (Result 16.4), and a half pushed through a phase-free spider comes out on every other leg (Result 16.5). A graph state is fixed by its safe touches (Result 16.6, the headline, proved twice), which generate a group of 2^n Paulis that names the state (Result 16.7). Half a touch re-ties the net (Results 16.8 and 16.9), three re-tyings make a pivot (Result

16.10), and every state of the world is a net with a quarter-turn picture on each output (Result 16.11). Knill's theorem (Result 16.12) is made practical by the ledger (Result 16.13), and a closed quarter-turn picture shrinks to a number (Result 16.14). The eighth is not a Clifford (Result 16.15), but a magic state brings it in (Result 16.16), and with it every unitary can be approximated (Result 16.17). The Touched Bow Tie is the untouched net (Result 16.18).

Now the honest part. **The world of quarters and halves belongs, in the end, to the blackboard.** The headline was a draw. But the following have no string-diagram proof in this chapter: that the stabilizers generate a group of 2^n elements and name the state; the exact phase of the LC-rule and of the pivot; that the ledger follows every quarter-turn circuit, including its measurements, and with what odds; Aaronson and Gottesman's tableau, canonical form and $\oplus$L-completeness; the density of the quarter-turn gates with T; and distillation. The pictures of re-tying and pivoting were proved on the blackboard and checked by matrix, not derived by rewriting; their derivations are Duncan and Perdrix's. The shrink is a genuine pictorial computation, but it computes amplitudes of closed pictures, one at a time; marginal probabilities need the doubled pictures of Chapter 9. And its speed, like every speed, is a count of steps taken from the sources. A rewrite proves that two processes are equal. It proves nothing about how long anything takes, and it did not prove that the eighth is hard: only that these moves cannot remove it.

What the pictures do show is the shape of the world. A quarter-turn state is a net with a quarter-turn picture on each output (Result 16.11), and a quarter on one wire can be traded for a re-tying of the net (Result 16.9). That is why the world is small: the only information is which vertices are tied and how each output is seated. And it is why the eighth is a different kind of thing, not a finer quarter. It leaves the octahedron unseated, and it leaves the shrink with nothing to pull.

The chapter's question has an answer. A machine built only of quarter-turns and half-turns can entangle as much as it likes, and still a clerk can follow

it: not by writing down its state, which would need more numbers than any notebook holds, but by writing down the touches its state would not notice. Those touches form a small group, one row per wire, and every quarter-turn gate moves each row to a single new row. In pictures the same world is a net of spiders, and any closed picture of it can be shrunk, spider by spider, to a number. The eighth-turn escapes both: it sends a Pauli to a mixture of two, and it sits inside a picture where no move can reach it. With nothing but quarters a machine is paper. With quarters and a supply of eighths it can approximate any computation at all. The difference matters so much because it is the whole difference.

That leaves the next question. If the quarters are free and the eighths are the cost, then a circuit should be written with as few eighths as possible. Chapter 17, *Compiling by Rewriting*, takes a whole circuit, turns it into a net, and shrinks it with the moves of this chapter. What is left behind is only its eighths. How few can it be made to need?

Étude in Simplification

物干しや
雫寄り合ふ
春隣

monohoshi ya
shizuku yoriau
haru-donari

The clothesline in rain —
drops slide into each other,
and spring is next door.

[*The Crab's kitchen, early on the afternoon of the next day. Rain drums on the roof of the scullery end. At the deal table the Tortoise sits with her work-bag at her feet; on the table stand the teapot, a stiff card, and something low and heavy under a cloth. The back door opens, and Achilles comes in out of the rain, dripping.*]

[*The kitchen he comes into is long, and warm from end to end. At the scullery end, under the drumming roof, are the stone sink and the one window, and the outside of the glass is covered with small drops, each one by itself, none touching another, and new ones landing in the spaces between. At the far end the range has been alight since before dawn: the oven is shut, the fire door stands a finger's width ajar and lets a warm light out across the hearth, and at the back of the hot plate a big stockpot, its lid on, mutters to itself; the copper pans hang on their nails on the wall behind. Between the two ends stands the deal table, and along the long wall the dresser holds the Crab's cups and, on its top shelf, a row of preserves in jars, each a*

summer's picking cooked down to a few dark inches. Over the range, in a plain dark frame, hangs a print: Escher's Development I, *plain squares that change, step by step, into interlocking figures. The warmth of the range keeps its glass clear while the window at the other end runs with rain. It hung there before the rain set in this morning, and it will hang there when the rain has gone.*]

Achilles: The thaw's turned to rain, and the rain's turned serious. I came the field way, and most of the field came with me.

Crab: Off with that scarf before it waters my floor. There's a rail on the range; hang it there and it'll be dry before you go.

[*Achilles unwinds his scarf and hangs it on the range rail.*]

[*He stays by the range a moment with his hands spread behind him to the warmth, and the scarf on the rail begins at once to smell of warm wet wool. Behind him, the back door has been shut on the weather, and the rain goes back to being a sound on the roof.*]

Tortoise: I came by the lane and only my shell got wet, which is what it's for.

Achilles: [*sniffing*] Something in that oven smells warm and good. What is it?

Crab: Three tins, set in the warm oven since dawn. Not for eating; you'll see.

Achilles: "Not for eating" is a cruel thing to say to a wet man in a warm kitchen.

Crab: There's bread and tea for eating, and you shall have both. [*He pours for all three, and sets the pot down at the right-hand end of the table as the bench on its range side faces it.*] Sit where you can see the beam.

[*Achilles sits on the bench that runs along the range side of the table, at the Tortoise's left, facing the scullery end, his back to the range, and looks up. At the scullery end a stout beam crosses overhead, and from hooks at its ends a long tangle of green curtain rings and cord hangs in two lines, one above the other.*]

Crab: The long picture I promised you. I tied it last night from this card: the

card's recipe, hung up in rings and cords, every step of it. This afternoon I mean to cook it down.

[*There are hooks at both ends of the beam: an upper left hook and an upper right hook on the beam itself, and a lower left hook and a lower right hook on short drops below them. Every ring in the tangle is green. Every cord is knotted at both its ends, to a ring or to a hook; none passes through a ring.*]

[*The upper line. Its left lead runs plain from the upper left hook to a small green ring. From that first small ring a cord in a yellow sleeve runs to a second small ring. From the second small ring a sleeved cord runs to a big green ring. From the big ring the upper line's right lead runs plain to the upper right hook.*]

[*The lower line. Its left lead runs plain from the lower left hook to the first eighth ring, a green ring whose bead is set an eighth round. From the first eighth ring a sleeved cord runs to the first plain ring, a green ring at rest. Another runs from the first plain ring to the second eighth ring; another from the second eighth ring to the second plain ring; another from the second plain ring to the third eighth ring; another from the third eighth ring to the third plain ring; another from the third plain ring to the fourth eighth ring; and another from the fourth eighth ring to the fourth plain ring. From the fourth plain ring the lower line's right lead runs to the lower right hook, and it too wears a yellow sleeve.*]

[*From the big ring the ties hang down in a fan, each in its own yellow sleeve: a tie from the big ring to the first plain ring, a tie from the big ring to the second plain ring, a tie from the big ring to the third plain ring, and a tie from the big ring to the fourth plain ring. Nothing crosses.*]

[*The window's grey light comes in under the beam and lies along the lower line, cord after cord; the range's warmer light, from the far end of the kitchen, catches the upper line, and the ties of the fan between them take a little of each. Every yellow sleeve is a short bright collar on its cord. In the warm air off the range nothing in the tangle stirs, and the rain drums on the roof just above it.*]

Achilles: It looks like washing hung out by somebody very particular.

Crab: It took me till midnight. [*He slides the stiff card across the table.*] And that is what I tied it from.

Achilles: I'll read it when I've stopped dripping on it.

Crab: [*lifting the cloth from the low heavy thing beside the teapot, at the very end of the table: beneath it, a griddle-like press with a slotted rim, and beside it a second row of empty hooks, facing the bench*] And my koekpan — you remember; still shy. A copy goes on it, it presses, and whatever it makes hangs on those second hooks; the thing you copied stays where it is. [*He lays the cloth back over it.*]

Achilles: I remember it jams if anyone watches it.

Crab: It likes being watched less than ever. I've had it longer than I've had this kitchen, and I have never once seen it work.

[*Under its cloth again the koekpan sits beside the teapot like a sleeping thing: a broad flat press of old Delft iron, its face worn to a soft shine by years of hands, its handle dark where a claw has held it, the maker's mark cast into its rim and worn nearly smooth. With nobody turned its way it gives off a warmth of its own, and a low sound at the very edge of hearing, like a kettle a long way from the boil. Achilles turns his head to catch it, and it is not there; he looks back into his cup, and it steals up again.*]

Tortoise: [*setting her work-bag on the bench beside her*] I've brought my cloth, in case the pan wants company.

Achilles: [*wrapping both hands round his cup*] This is the first thing all day that's been warmer than I am.

Crab: Then drink it before it changes its mind.

[*The Crab sets a plate of bread and butter by the teapot and draws up his own chair. Achilles eats two slices without noticing that he has, and the colour comes back into the hands round his cup. The Tortoise tucks her feet in under the bench, and the rain keeps up its one sound on the scullery roof.*]

First Challenge — *Four Eighths*

Achilles: [*picking up the card and reading aloud*] "An eighth on the lower line, then a joiner from the upper line to the lower. Four times over." [*He counts*

on his fingers.] Four eighths. And yesterday every eighth was the dear part.

Crab: Every one of them made slowly, on the bench, and used up the moment it's tied on. A card with four is a dear card.

Achilles: And a joiner?

Tortoise: Yesterday's joiner: a green ring on the upper line, tied by a short length to a red ring on the lower. If the upper line is lit, it flips the lower; if the upper is dark, it leaves the lower be. It's the commonest step on any card.

Achilles: The controlled flip, from the snow. I remember it.

Tortoise: [*pleasantly*] The joiners come in pairs. What do they do to each other?

Achilles: They cancel. Flip and flip back. So it's just four eighths on the lower line: a half.

Tortoise: Then follow the card with the upper line lit and the lower dark, and tell me where the eighths land.

Achilles: [*running his finger along the card*] The first eighth sees the lower line dark, so it does nothing. The first joiner sees the upper lit and lights the lower. The second eighth sees it lit, and turns. The second joiner darkens it again.

Crab: And the rest?

Achilles: The third eighth sees dark and does nothing; the third joiner lights it; the fourth eighth turns; the fourth joiner darkens it. [*His finger stops at the end of the card.*] Two of them turned. A quarter — and on the upper line's account. ...Oh.

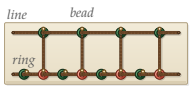
Tortoise: The joiners cancel only with nothing between them. On this card there's always an eighth between.

[*By the back door the latch lies in its keep where it lay before Achilles lifted it, and the draught he let in has been shut out again. But on the flags just inside the sill the rain that blew in while the door stood open lies in a puddle, and the mat beside it is dark with what came in on his boots.*]

Achilles: So it isn't a half on the lower line at all. What the lower line gets depends on the upper. [*He turns the card over, as if the answer might be on the back.*] Then how does the card come apart, if not in pairs?

Tortoise: Not apart. Into a net. [*She lifts her drawing-cloth out of her work-bag and spreads it on the table in front of Achilles, its near edge on his side and its right-hand end towards the teapot, then brings out string, her little box of rings with their beads, each ring opening at a small hinge so that it clips on or off a cord anywhere, and her yellow tubes, each slit along its length to clip on.*] My drawing-cloth, and my rings and tubes, as ever: what I lay on it is a picture of a thing, not the thing. And the old rule for the tubes: a ring with a tube on every cord comes out the other colour, its bead where it was; and two tubes in a row are no tube at all, so I may put a pair on anywhere I like.

[*She lays a long cord across the drawing-cloth from Achilles's left to his right, and another below it, nearer him: the upper line and the lower. Along the lower she lays the card's steps from the left: a green ring with its bead set an eighth round, an eighth of the way clockwise from the ring's far side as Achilles looks down, where a bead at rest sits; then a red ring at rest, tied by a short string straight up to a green ring at rest on the upper line, a joiner; then another eighth, and another joiner; another eighth, another joiner; another eighth, and a joiner last. What lies on the cloth: the card, step for step, in green and red, every cord plain. Every string she ties this afternoon she leaves slack, long enough to follow its rings wherever they slide.*]



The card, laid out

[*The cloth as it stands: on the lower line each red ring has a cord running left along the line, a cord running right, and a short string up to its green. On each of a red ring's cords the Tortoise clips a tube and another beside it, which changes nothing. Then she draws the ring through the near tube of every pair: the tubes go into it and are gone, and the ring comes out green, its bead still at rest. The far tube of each pair stays on its cord. She does the same to each red ring along the line. What lies on the cloth now: every ring green; a tube on every cord that runs between two rings*

of the lower line, and on every short string up to the upper line; the lower line's left lead plain, its right lead wearing the tube the last red ring left behind. What has changed: not a red ring is left, and a tube stands wherever red used to meet the rest. What has held: every bead, every cord's two ends, and what the picture does.]



Every ring green

Achilles: All green now. And on the upper line, a row of plain green rings with nothing between them but cord.

Tortoise: [*setting a claw beside the first green ring on the upper line*] Slide them together.

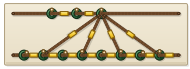
[*The upper line as it stands: four green rings at rest spaced along a plain cord, each with a sleeved string hanging down to a ring on the lower line. The Tortoise slides them along the cord towards its middle, the first and the second into one, then the third into those, then the fourth, so that they meet above the third eighth. The cord does not move and its two ends do not stir; only the rings travel, and where they meet they knot into one green ring. At rest and at rest make rest: the bead does not move. Every string that hung from one of the four now hangs from that ring, and each still reaches the same ring below. What has changed: four rings are one, and the strings fan down from it. What has held: which ring below each string reaches, and every bead.*]



A fan of sleeves

[*The ring at the head of the fan now holds both leads of the upper line, the one from the left and the one to the right. The Tortoise clips a green ring at rest onto the left lead, which changes nothing; clips a tube onto the lead between that ring and the ring at the head of the fan, and another beside it, which changes nothing; and clips another ring at rest onto the lead between the tubes. What lies on the upper line now: the left lead to a ring, a tube, another ring, a tube, the ring at the head of the*

fan, and the right lead. What has changed: the ring at the head of the fan holds only one end. What has held: everything the picture does.]



Rings clipped on

[*At the window two drops high on one pane have crept near enough to touch, and in the moment they touch they are one drop, as big as both together. It gathers itself, and slips a little way down the glass.*]

Achilles: [*looking from the cloth up to the beam, and back*] That's the tangle. Ring for ring: the small rings, the big one, the eighths, the plain rings, and a sleeve wherever you've a tube.

Tortoise: Your tangle, drawn. Nothing was spent to get it: not a bead turned, not an eighth lost or gained.

Crab: [*proudly*] That is just how I tied it, the way you showed me on your cloth after tea yesterday: every ring green, a sleeve on every tie, and no ring holding two ends. The knots are mine; the rule was yours.

Achilles: Then why two small rings on the upper lead? One would look lonely enough.

Tortoise: A ring may hold only one end of a line, so the big ring had to give up the left one. One small ring alone would sit on plain cord next to the big one, and slide straight back into it. So a pair of tubes, and a ring between them: two sleeves in a row are no sleeve.

[*The Crab laughs under his breath at the lonely ring, and tops up all three cups, Achilles's first, and pushes the plate of bread and butter an inch nearer him. Achilles takes another slice and eats it with his eyes on the cloth.*]

Achilles: And the eighths can't slide together, either. On the lower line there's a sleeve between each eighth and the next, and a plain ring in the way. [*He looks up at the tangle.*] Four eighths hanging there. How few can it be made to need?

[*The Crab lifts the cloth off the koekpan.*]

Second Challenge — *Cooking Down*

Crab: Now. A copy on the pan, a press, and we shall see.

Tortoise: The cloth first, if you'll humour me. Your pan works in the dark; I'd like Achilles to see one way through by daylight before it does.

Crab: [*resigned, with a claw still on the handle*] By daylight, then. Briskly.

[*The Tortoise gathers the net off the drawing-cloth, its rings into her box and its cords, strings and tubes into her work-bag, and on the bare cloth lays the card out again as she did at first: the upper line and the lower, and along the lower from the left an eighth, a joiner, an eighth, a joiner, an eighth, a joiner, an eighth, a joiner; every ring in its first colour, every cord plain.*]

Tortoise: Look at the second eighth. It sits between two joiners, and nothing else does: a joiner, a turn on the lower line, a joiner. Watch what that comes to.

[*The second eighth as it stands: a green ring an eighth round on the lower line, a joiner's red ring on each side of it, and above, on the upper line, those joiners' green rings, each tied down by a short string to its red. The Tortoise slides the two upper greens towards each other into one, meeting above the eighth; that one ring now holds both short strings. She slides the eighth apart, into a plain green ring on the lower line and a small green end hanging from it by a plain cord: the bead goes with the end, an eighth round, and the ring left on the line is at rest. She slides the upper ring apart too, so that one part keeps the upper line and the other holds both short strings.*]

[*Now four rings sit in a square: two greens, the upper part holding the strings and the plain ring on the lower line, and two reds, each green tied to each red, and each ring with just one cord besides. It is the Crab's cradle from the snow. It comes undone as the cradle always did: the square goes, its rings unclipped into her box and its strings into her work-bag, and in its place one red ring gathers the greens' other cords, and one green ring, tied to it by a single cord, gathers the reds'. What*

lies in that place now: on the upper line a plain green ring, above the eighth's place, tied down to a red ring, the hub, which lies between the lines straight below it; from the hub, hanging out to the left, the green end, an eighth round; and from the hub a cord down to a plain green ring on the lower line. The Tortoise takes a small bone counter from her work-bag and sets it off the drawing-cloth, just beyond its near edge at its right-hand end, by the teapot. What has changed: two joiners and the eighth between them are one hub, with the eighth hanging from it. What has held: the eighth's turn, whole, hanging now where both lines can reach it.]



The cradle, undone

Achilles: It hangs from something tied to both lines at once.

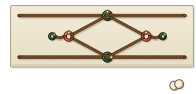
Tortoise: A turn between two joiners hangs from a hub. The fourth eighth sits between two joiners as well.

[*She does the same with the fourth eighth and its two joiners, more quickly: the upper greens into one, meeting above the eighth, the eighth apart into a plain ring and an end, the square formed and undone, its rings into her box and its strings into her bag; a second hub stands on the drawing-cloth, straight below its upper ring, with its own eighth hanging from it out to the right. A second counter goes by the teapot, on the first.*]

Tortoise: [*setting a claw at the head of the upper line*] Slide them together.

[*The cloth as it stands: on the upper line two plain green rings, one hub's foot each, with plain cord between; on the lower line the first eighth, the first hub's foot, the third eighth and the second hub's foot, all green, with plain cord between. The Tortoise slides the upper line's feet towards each other into one plain ring, meeting above the third eighth; it is tied to both hubs now. Then along the lower line she slides the first eighth into the first foot, and those into the third eighth, and brings the second foot along into all of it: an eighth and nothing and an eighth and nothing, and the bead comes to rest a quarter round. What lies on the cloth now: one ring on the upper line, at rest; one on the lower, a quarter round; and between them the two hubs, each tied to both rings, each with its eighth hanging. What has changed: two*

of the eighths have met on the lower line and made a quarter. What has held: the other two, still hanging from their hubs, untouched.]



Feet slid together

Achilles: Two hubs, and both hung from the same two rings.

Tortoise: Then they are one hub.

[*The two hubs as they stand: each tied to the upper ring and to the lower, each with an eighth hanging from it. The Tortoise slides each line's ring apart, so that one part of it holds both hubs' cords; those two parts and the two hubs make a square again, and the square comes undone, its rings into her box and its strings into her bag, into a single red hub, between the lines straight below the upper ring, and a single green ring; and the green ring slides together with both hanging ends. What lies on the cloth now: one hub, tied to the upper ring and to the lower, and hanging from it, out to the right, one end, a quarter round. A third counter goes by the teapot, on the other two. What has changed: two eighths on two hubs are one quarter on one hub. What has held: the rings on the lines, and what the whole picture does.*]

[*The rain has thickened, and the window over the sink is busy with it: all over the glass drops are meeting drops, and every meeting leaves one where there were two. There are fewer of them on the pane than there were, and bigger, with clear glass between.*]

Tortoise: And a hub with a quarter hanging from it, standing on a quarter back on each line, is a tie in a sleeve. That one I haven't a move for; I've tried it on every way the lines can start, and it holds.

[*The picture as it stands: the upper ring at rest, the lower a quarter round, the hub between with its quarter hanging. The Tortoise slides each line's ring apart so that the part holding the hub's cord is turned a quarter back: the upper ring leaves a ring a quarter round on its line, and the lower leaves a ring at a half. The hub stands now on two feet a quarter back, with a quarter round hanging from it. She lifts it away whole, feet and hub and end, unclipping the feet from the lines, into her box of rings; and in its place ties a single sleeved string between two green rings at rest,*

one on each line; and those two slide into their neighbours on the lines. What lies on the cloth now: on the upper line one green ring, a quarter round; on the lower, one green ring at a half; between them one sleeved tie. No eighth anywhere. What has changed: four eighths have become a quarter, a half and a tie. What has held: what the card does, with the counters by the teapot.]



Nothing hanging

Tortoise: A quarter on the upper line, a half on the lower, and a tie between. The eighths met, and added, and there's nothing left of them.

Achilles: Let me check that my way: the card, and no cloth. Both lines dark: no eighth ever sees a lit line, so nothing turns.

Crab: And with the lower lit?

Achilles: Upper dark, lower lit: the joiners never flip it, and every eighth sees it lit. All four: a half. Upper lit, lower dark I did before: two, a quarter.

Tortoise: And both?

Achilles: Both lit: the first eighth turns, the first joiner darkens the lower, the second eighth sees dark, the second joiner lights it again, and round we go. Two again: a quarter. [*He sits back.*] A quarter whenever the upper is lit, a half when the lower is — and when both are, a half taken back.

Tortoise: And your cloth?

Achilles: Says the same. A quarter on the upper ring, a half on the lower, and the tie is yesterday's tie: it gives a half when both its ends are lit. That's the half taken back.

Tortoise: Your finger and my cloth agree, then, tie and all.

[*Achilles sits back with his cup against his chest, rather pleased with his finger, and the Tortoise nods to him over hers. For a moment there is nothing in the kitchen but the rain on the roof and the stockpot talking to itself at the back of the range.*]

Crab: Now my pan. [*He is already at the dresser.*]

[*The Crab takes from the dresser drawer a copy of the tangle that he tied this morning, ring for ring the same from the same card, and lays it on the koekpan: the small rings, the big ring, the eighth rings with their beads an eighth round, the plain rings, every sleeve. The tangle on the beam he does not touch. He closes the press, stands at the table's end with his back to it and to the pan, his face to the fire, and works the handle behind him by feel.*]

[*Achilles leans over the table to see what the pan is doing with it. Without turning round, the Crab puts up a claw.*]

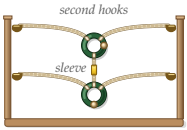
Crab: It jams if watched. Look at the fire, or at me; my back is very interesting.

Achilles: [*straightening*] Sorry. It's hard not to.

Crab: Everyone says so. That is why it jams so often.

[*Achilles turns his eyes to the fire as he is told, and the Tortoise turns hers there too. At the back of the hot plate the stockpot, full to the brim at dawn and loud all morning, has sunk to a low mutter; nobody has lifted its lid since, and what comes from under it now smells of everything that went in, only closer. Behind their backs, from the table, where a moment ago there was nothing at all, comes the koekpan's low sound at the edge of hearing, like a kettle a long way from the boil.*]

[*The copy as it went on: the long tangle, ring for ring, four eighths among its rings. The press settles with a small click, and the Crab's claw leaves the handle. The copy is gone from the pan. On the second row of hooks hangs something far barer: on the upper line one green ring, its bead a quarter round, at the right of the ring as Achilles faces the hooks; on the lower line one green ring, its bead at a half, at the bottom of the ring, for on a hanging ring a bead at rest stands at the top and round goes clockwise; between them a single cord in a yellow sleeve, from the foot of the upper ring to the top of the lower; each ring hung by its leads from the second hooks, left and right, knotted to its band either side of its top. No eighth ring anywhere. On the beam the long tangle has not stirred: every ring and every sleeve where it was. What has changed: the copy, cooked down to two rings and a tie. What has held: the tangle, untouched, and the cloth's answer, matched.*]



What the pan made

Achilles: [*looking from the second hooks to the cloth*] The same. A quarter, a half, and a sleeve between. What does it do in there?

Crab: It slides together what can slide, re-ties what can be re-tied, the way you saw yesterday, and carries the eighths along until they meet. No bead changes in there except by one of those moves. That is all I know, and all the letter from Delft would say.

[*Achilles looks up from the second hooks to the long tangle on the beam, and then round over his shoulder to the print on the wall over the range: Escher's* Development I.]

[*As he looks, a gust drops down the flue, the fire roars up for a breath behind its door, and the glow from the gap climbs the wall and lies across the print. For that moment it is the brightest thing at that end of the kitchen.*]

Achilles: Like your print over the range: plain squares that change, step by step, into interlocking figures — only the other way about.

[*For answer the Crab lifts the two rings and their sleeved cord off the second hooks, and puts them away in the dresser drawer.*]

[*He lays the cloth back over the koekpan.*]

Crab: On a long card it leaves fewer eighths than it was given, most times. But it never promises the fewest. On this one it left none.

Tortoise: Every move on the pan and on my cloth is one that leaves the thing unchanged. That's why its answer can be trusted, though nobody may watch it.

[*She lifts the quarter ring and the half ring off the drawing-cloth on their lines, with their sleeved tie between, and lays them on the table just beyond the drawing-cloth's far edge, leaving it bare; and sweeps the counters from beside the teapot into her bag.*]

Achilles: And the pan hands back a picture, not a recipe. Can a picture be run just as it is, without turning it back into a card?

[*The Crab glances at the oven.*]

Third Challenge — *The One-Way Row*

Crab: It can. That's what's in the oven.

[*He opens the oven door, and with a folded oven cloth lifts out three long narrow tins, one after another, and sets them in a row on the table at its left-hand end as Achilles sees it, side by side, a hand's breadth apart, each with a long side towards him, the first tin nearest him; then he hangs the oven cloth back by the range. Each has three small domed lids along it, and on each lid a knob with a pointer, turning against eight marks round the rim: at rest the pointer points away from Achilles, and round is clockwise as he looks down on it. The end lid at his left is the first.*]

[*The warm smell Achilles sniffed at comes out of the oven with them, and it turns out to be the smell of warm tin and nothing else. The tinsmith's mark is stamped small into the end of each tin, with the word Delft beneath it. Standing in their row on the table they tick softly as they cool, now one and now another.*]

Crab: My one-way tins, from the tinsmith two doors down from the old man's bench in Delft. Each is a long narrow tin with three hollows in a row: the first, the second and the last. Set together in one tin, the hollows come out joined, each to the next.

Achilles: And the knobs?

Crab: A dial on every lid. The pointer turns in eighths: at rest, an eighth round, a quarter round, three eighths round, a half, three eighths back, a quarter back, an eighth back; where it stands is that lid's setting. Lift a lid, and the hollow is read at its setting: under the lid is a glass drop, and it shows dark or it shows lit. A lifted lid can't be put back; that hollow is spent.

Achilles: Then you've three tins' worth of looking, and no more.

Crab: [*a claw on the row*] Everything else I own spoils if you look at it. These are the one thing I own that works only because you do. I've set them all alike: the first dial an eighth round, the second at rest, the last an eighth back.

Achilles: May I?

Crab: Lift them in order: first, second, last.

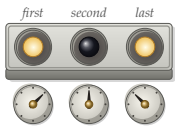
[*The first tin as it stands: three lids down, the first pointer an eighth round, the second at rest, the last an eighth back. Achilles lifts the first lid. It comes away in his fingers and will not seat again, and he lays it on the table just in front of its own hollow, knob up, its pointer still where it stood, as every lid lifted this afternoon is laid; in the hollow the glass drop is lit. The other two lids have not moved, and their pointers stand where they stood. What has changed: the first hollow is read, and spent. What has held: the second and the last, lids down.*]

Achilles: Lit.

[*He lifts the second lid. The drop beneath it is dark. The last lid is still down, its pointer still an eighth back.*]

Achilles: Dark. And the last... [*He lifts it.*] Lit.

[*The last drop shows lit. All three lids lie on the table in front of their hollows, knob up, and the tin is spent.*]



Every lid lifted

Achilles: It's a coin. They all are.

Crab: The first two are coins, and I can't help that. The last need not be, if you keep my rules.

Achilles: What rules?

Crab: Read the lids in order, first, second, last, and turn any dial before its own lid comes up. If the first comes up lit, turn the second dial the other

way, and put a half on the last dial. If the second comes up lit, turn the last dial the other way.

Achilles: And then?

Crab: Read by the rules from the preset an eighth back, the last hollow answers dark every time, as one of my eighth-ends would; kept unread, it holds an eighth-end turned as the first two readings say.

Tortoise: The reading part, at least, a tin can show. Show him.

[*The second tin as it stands: lids down, pointers at the preset. The Crab lifts the first lid: the drop is dark. The second and last pointers stay where they are.*]

Crab: Dark: the rules say nothing. [*He lifts the second lid.*] Lit. So the last dial goes the other way.

[*The last pointer stands an eighth back. The Crab holds the lid down under one claw and turns the knob with the other, back through rest and on, until the pointer stands the same distance on the far side: an eighth round. The lid has not lifted; the hollow under it is unread. What has changed: the setting, from an eighth back to an eighth round. What has held: the lid, and whatever is under it.*]

[*He lifts the last lid. The drop is dark.*]

Crab: Dark. Your turn at the dials; I'll call.

[*Achilles draws the third tin out of the row to its right and round towards him, until it lies beside the first tin, at its right, its first lid still at his left and its near side still towards him, short of the drawing-cloth's left-hand end; and he lifts its first lid. The drop is lit.*]

Achilles: Lit.

Crab: Second dial the other way.

Achilles: It's at rest. The other way from rest is… rest. [*He leaves it.*]

Crab: So it is. And a half on the last.

[*The last pointer stands an eighth back. Achilles turns the knob half-way round the rim: the pointer passes rest, an eighth round and a quarter round, and comes to*

stand at three eighths round. The lid stays down. What has changed: the setting, by a half. What has held: the lid, and the hollow under it, unread.]

Achilles: [*lifting the second lid*] Lit again.

Crab: Last dial the other way.

[*The pointer stands three eighths round. Achilles turns it back through rest to the same distance on the other side: three eighths back. The lid is still down.*]

[*He lifts the last lid. The drop is dark.*]

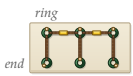
Achilles: Dark. Both times dark, whatever the first two did.



Read by the rules

Tortoise: My cloth can show you why. [*She lays three green rings at rest in a row across the middle of the drawing-cloth, the first at Achilles's left, ties the first to the second and the second to the last with strings, clips a tube onto each tie, and gives every ring a loose cord running from it towards the near edge.*] A tin, drawn: three rings at rest in a row, each tie in its tube, and a loose cord on each ring for its lid. A lid lifted at a setting and showing dark, I draw as a green end turned to that setting on the ring's loose cord; lit, turned to the setting and a half.

[*She sets down ends for the first tin's readings, each threaded onto its loose cord by the free end, near the edge, and does not slide them in: on the first ring's loose cord, an end an eighth round and a half; on the second's, an end at rest; on the last's, an end an eighth back and a half.*]



Ends waiting

Tortoise: Your tin, Achilles: lit, dark, lit. Now follow the first ring's half.

[*The cloth as it stands: three rings at rest in a row, tied by sleeved strings; on the*

first loose cord an end an eighth round and a half, on the second an end at rest, on the last an end an eighth back and a half. The Tortoise slides the half off the first end onto its cord, on into the first ring, and out again onto the tie towards the second; the first end is left an eighth round. The half reaches the tube, passes through, and comes out red. It goes into the second ring, which is at rest, and comes out on every other cord of it: a red half on the second ring's loose cord, and a red half on its tie onward. On the loose cord the red half meets the second end, goes into it, and turns it the other way; the other way from rest is rest, and nothing there changes. On the tie onward it passes the tube and is green again; it goes into the last ring and on to the last end, where it lands as a half: an eighth back and a half and a half, which is an eighth back. What lies on the cloth now: the first end an eighth round, the second at rest, the last an eighth back, just as a tin would lie that came up dark, dark and dark. What has changed: the first end has lost its half, and so has the last. What has held: every ring at rest, every tie, and the second end.]



Like a tin read dark

Tortoise: So your last reading was turned over before you lifted it. The first lid's half had travelled down the row.

Achilles: And the Crab's rule turns the last dial by a half to meet it.

Tortoise: [*touching each end in turn*] Slide them together.

[*She slides each end up its loose cord into its ring, the loose cord swallowed with it. The first ring is an eighth round now, the second at rest, the last an eighth back, and no loose cord is left: a closed row of three rings, tied, with nothing hanging. What has changed: the ends are gone into the rings. What has held: the row is the very row a dark, dark, dark tin leaves.*]

Tortoise: Your rules, Mr Crab, as my cloth shows them. A lit reading sends a half forward: the next end turns the other way, and the one after gets a half.

Crab: I had them from the letter, and never knew why. [*He bows a little to the cloth.*]

[*Outside the window, at the corner of the scullery, three water-butts stand in a row, each set a little lower than the one before. The first has been full since noon, and what it cannot hold goes over its rim into the second, and what the second cannot hold into the last; nothing goes back up the row. On the pane above the sink the drops go one way only, down, and whatever a drop reaches on its way goes down with it.*]

Achilles: And the tin never had a card at all. Nobody wrote an eighth or a joiner on it; there's the row, the dials, and the order the lids come up in. The readings are the recipe.

[*The Crab stacks the three spent tins, lids and all, by the sink. The Tortoise sweeps the row of rings into her bag; the quarter ring, the half ring and their tie still lie at the side of her cloth.*]

[*By the sink the tins have stopped ticking. They are only tins now, cooling in a stack under the window, and the rain runs down the glass above them as it has all afternoon. The Crab wipes his claws on his apron and sits down again, and Achilles pours for him.*]

Achilles: And if I want the recipe back after all, from what the pan left?

[*The Tortoise turns to the cooked pieces at the side of her cloth, and lays them back across the middle of the drawing-cloth, the lines running from Achilles's left to his right, the upper further from him, and the rings and their tie towards the left-hand end.*]

Fourth Challenge — *Out Again*

Tortoise: The pan's answer, as it lies. A ring a quarter round on the upper line, a ring at a half on the lower, and a tie between. Take the card out of it yourself; you know the move, and you know which way it runs.

Achilles: The one move, backwards.

Achilles: [*reaching for the two rings*] Slide them apart.

[*The cooked pieces as they stand: the upper ring a quarter round, the lower at a half, the sleeved tie between them, each ring with a lead to the left and a lead to the right. Achilles takes the upper ring and slides it apart along its line: a ring at rest stays where it was, holding the tie and the left lead, and a new ring slides off towards the right-hand end, carrying the whole quarter, and comes to rest about half-way there. He does the same below: a ring at rest keeps the tie and the left lead, and the half slides off to the right, to rest straight below the quarter. What lies on the cloth now: a sleeved tie between two plain rings, and after it, towards the right, a quarter on the upper line and a half on the lower. What has changed: where the turns sit. What has held: every turn, and the tie.*]



The turns slid off

Achilles: A sleeved tie between the lines; then a quarter on the upper line, and a half on the lower. That's a card!

[*The Crab takes a fresh card from the dresser and writes on it in his careful hand: "A sleeved tie between the lines; then a quarter on the upper line and a half on the lower." He lays it on the table beside the old one.*]

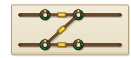
Crab: A new card for the same recipe. Not an eighth on it.

[*At the window a drop that has grown fat on everything it met on its way down the pane reaches the sash bar, and there it comes apart: where the big drop was, small drops sit along the bar, each by itself again. The rain goes on landing round them.*]

Tortoise: That one came out easily; its tie ran straight across. Not every net is so kind. [*She lifts the slid-apart pieces of the pan's answer off the drawing-cloth, on their lines, onto the table just beyond its far edge, and lays four green rings at rest on the bare cloth, the upper pair further from Achilles than the lower.*] Here is a harder one, and it's all at the ends. Two rings at the back, one on each line with its lead coming in from the left; two rings at the ends, one on each line with its lead going out to the right.

[*She ties the rings with sleeved strings: the upper ring at the ends to the upper back ring, and to the lower back ring too; the lower ring at the ends to the lower back*

ring only.]



A harder net

Achilles: The upper one at the ends is tied to both back rings. What does it take from each?

Tortoise: Both at once, and a card can't say that in one step. But a joiner at the ends can. [*She takes a joiner from her box: a green ring and a red ring, tied by a short string.*]

[*The picture as it stands: the upper ring at the ends tied to both back rings, the lower ring at the ends to the lower back ring only, and to the right of them only the leads going out. The Tortoise clips the joiner onto the leads at the right, just past the rings at the ends, green on the upper line and red on the lower, and in the same motion lifts away the tie between the upper ring at the ends and the lower back ring, and drops it, with its tube, into her work-bag. The other ties stay where they were. What lies on the cloth now: on the upper line a back ring, a sleeved tie, a ring at the ends; on the lower line the same; and after them the joiner. What has changed: the tie that crossed from the upper ring at the ends to the lower line is gone, and a joiner stands at the ends. What has held: what the picture does.*]



Out at the ends

Tortoise: It holds on every way the lines can start. Lay the joiner the other way up, red on the upper line, and it doesn't.

Achilles: And now each line is a ring at rest, a sleeve, and a ring at rest.

Tortoise: A ring at rest with two cords is just cord, so each line is one sleeve on its own: a yellow box. The card reads: a yellow box on each line, then the joiner. The joiner came out at the ends, and the extra tie came away with it.

Achilles: So a net comes back out as a card from the right-hand end: the turns slide off, a tie straight across comes off as a tie, and a doubled tie

comes off as a joiner.

Tortoise: Does every net? Try this one. [*She sweeps the second net into her bag, and lays three green rings at rest on the bare drawing-cloth: two arms across its far part, each tied by a sleeved string to a corner ring between them and nearer Achilles, and the arms not tied to each other. She gives each arm a loose cord, the left arm's running out to the left and the right arm's to the right, and leaves the corner's loose cord open, running out to the right.*] A V: two arms, each tied to the corner, and not to each other. The arms are to be read, like the hollows of a tin; the corner is kept, never read. Find me an order to read the arms in, so that the Crab's rules can do their work.



Arms and a corner

Achilles: First the left arm, then. If it comes up lit, its half travels [*pushing a finger along the cloth*] through the tube, red, into the corner, which is at rest, and out on every other cord of it: onto the corner's own cord, which we keep, and along the tie to the right arm, green again, and onto the right arm's end as a half. So the right arm must wait for the left, to have its dial turned.

Tortoise: And if you begin with the right arm?

Achilles: Then its half goes through the corner the same way and lands on the left arm's end, and the left must wait for the right. [*He looks from one arm to the other.*] ...Oh. Each has to go before the other.

Tortoise: So the V has no order. There's no way to read it in which every half is sent on to a lid still down.

Achilles: [*still looking at the V*] Not one.

[*The Crab and the Tortoise reach for the teapot at the same moment, and both draw back for the other. Each waits; neither will pour first; and the pot stands between them in its cosy, poured by nobody.*]

Tortoise: And there's the whole of it. The order a tin is read in is the order

a recipe comes out in. The tin's row has one: first, second, last, every half sent on ahead.

Crab: And my new card came out of the pan's picture the same way, from the ends inward?

Tortoise: From the ends inward, ring by ring, in such an order. A net tied from a card is born with one, and your pan's re-tyings keep it.

Crab: Then I shall keep to nets tied from cards. [*He props the new card against the teapot beside the old one.*] Four eighths, and none.

[*The Tortoise slides the V's rings apart and sweeps them into her bag, and the slid-apart pieces of the pan's answer after them. The Crab smooths the cloth over the koekpan with one claw.*]

Crab: There. It can sleep now.

[*Under the smoothed cloth the koekpan gives off a last warmth and no sound at all. The Crab leaves his claw on it a moment longer than he needs to.*]

Achilles: [*looking up at the beam*] And the tangle?

Crab: Stays where it is. I shall keep it. It's the long way round, and I like to be able to look at the long way round.

[*He takes the two cards, the old and the new, and sets them side by side on the dresser.*]

Crab: The last of the pot. [*He shares it out, three ways, not quite evenly.*]

Tortoise: Listen to the rain.

Achilles: It's got louder while we weren't looking. And the wind's getting up.

[*A gust goes round the house and throws the rain against the window in handfuls; down the flue the range answers with a low roar, and the stockpot's lid rattles once. The Tortoise draws her feet in closer to the warmth, and the Crab opens the fire door and feeds the range from the scuttle.*]

[*Up the lane, the church clock begins to strike, and strikes two.*]

Achilles: Only two! I'd have sworn it was nearer dark.

Crab: It's the rain; it makes an evening out of anything. [*He looks along the spent tins by the sink.*] In weather like this the damp gets into the tins, too. Set with every care, and now and then a hollow reads false: lit, when it ought to be dark.

Achilles: [*half joking*] Then set three for every one, and trust the two that agree.

Crab: [*looking at him a long moment*] Hm.

Tortoise: Another day.

[*Achilles takes his scarf from the range rail. It is still damp. He looks at the two cards on the dresser.*]

Achilles: One thing, before I go out in it. The pan never shows its work. How does anyone know the new card does exactly what the old one did? Not nearly: exactly.

Tortoise: Every move on the way was one that changes nothing. And for this card, your four ways of lighting the lines check it.

Achilles: And for a card too long for my finger to follow?

Tortoise: [*folding her drawing-cloth into her work-bag with her string, her tubes and her little box of rings, and fastening it*] Then the moves are all the proof there is.

[*Nobody gets up yet. The rain is loud on the scullery roof and the wind goes round the house looking for a way in, and the kitchen is warm against it from end to end. Under the beam the long tangle hangs exactly as the Crab tied it last night, every cord still and every cord knotted where he knotted it, the yellow sleeves small and bright in the grey light from the window. On the dresser the two cards stand side by side where he set them, below the jars; the spent tins sit in their stack by the sink; the koekpan sleeps under its cloth; and at the back of the hot plate the stockpot has cooked down to a murmur. Over the range the print hangs in its plain dark frame, dim now, with only the range's low glow at the foot of it, one more of the Crab's things. On the window the drops go on meeting and running together and coming apart again, with nobody watching them. Achilles sits down again with his damp scarf folded in his lap, and the Crab draws the kettle forward onto the heat, as if the*

afternoon had hardly begun; which, by the church clock, it hardly has.]

Chapter 17: Compiling by Rewriting

— Étude in Simplification —

Part V: Engineering the Diagram

Escher: "Development I" — plain squares that change, step by step, into interlocking figures

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If someone hands you a recipe that works, how can you find a cheaper recipe that does exactly the same thing, and how can you be sure that it does?

Every cook knows the first half of the question. A recipe written in a hurry repeats itself: it tells you to stir, then to stop, then to stir again; it adds salt twice. A better cook reads it and writes a shorter one. But in a kitchen the test of the new recipe is easy. You make both dishes and taste them. If the second is as good as the first, nobody minds how it was found.

A quantum computer is run from a recipe too, a list of small steps done to a handful of wires, one after another. Some steps are cheap. Some are expensive, so expensive that on the machines now being designed, the cost of a whole computation is counted almost entirely in them. A recipe with fewer expensive steps is a better recipe. And here the tasting fails. A quantum recipe on a hundred wires cannot be run and compared on paper: its table of numbers is longer than any notebook. Nor can the machine itself be trusted to taste, because a machine that reads its own wires disturbs them. If a shorter recipe is to be used, it has to be *known* to do the same thing, before anyone runs it.

This chapter's answer is to stop reading the recipe as a list and to draw it. The picture forgets the order in which the steps were written down, and keeps only how the wires are joined and turned. Then the picture is rewritten, one small move at a time, and every move is one that changes nothing. When no move is left, what remains is the same process, drawn smaller. The expensive steps meet inside the picture, add up, and sometimes vanish. Then a recipe is taken out of the small picture again.

And the pictures give something more. A picture can be run without being turned back into a recipe at all: prepare the wires joined as the picture says, then read them one by one. The readings are coin tosses, but if they are read in the right order, each reading's outcome can be sent forward and the coins stop mattering. The order that makes that possible turns out to be exactly the order that lets a recipe be taken out of a picture.

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§17.1 Four Eighths

Chapter 16 ended with this question: "If the quarters are free and the eighths are the cost, then a circuit should be written with as few eighths as possible. Chapter 17, *Compiling by Rewriting*, takes a whole circuit, turns it into a net, and shrinks it with the moves of this chapter. What is left behind is only its eighths. How few can it be made to need?"

The chapter's question, in the form we can answer: **a circuit is a recipe; how does one find a cheaper recipe that does exactly the same, and how does one know that it does?** We answer it in four steps. This section counts what a recipe costs and draws it as a net. Section §17.2 cooks the net down. Section §17.3 runs a net without ever turning it back into a recipe. Section §17.4 takes a recipe out of a cooked net, and finds the order that makes that possible.

On the blackboard. Recall the gates of Chapter 16, each a 2×2 matrix:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Squaring doubles the turn: $T^2 = S$, $S^2 = Z$, $Z^2 = I$, so $T^8 = I$. On two wires, write CNOT_{12} for the CNOT whose control is the upper wire and whose target is the lower, so that $\mathrm{CNOT}_{12}\,|x_1 x_2\rangle = |x_1,\, x_1 \oplus x_2\rangle$ in ket order, top wire first. It undoes itself: $\mathrm{CNOT}_{12}^2 = I$. Products of operators are read right to left, as always; pictures and recipes are read left to right.

Here is the recipe of this chapter, in words: *an eighth on the lower wire, then a CNOT from the upper wire to the lower; four times over.* As a matrix,

$$U \;:=\; \left(\mathrm{CNOT}_{12}\,(I \otimes T)\right)^4.$$

The letter U means this matrix, and nothing else, for the rest of the chapter.

Chapter 16 showed why the eighths are the cost. The gates H, S and CNOT generate the Clifford gates, which a notebook can follow, and T is not one of them. Kissinger and van de Wetering state the practical side: on the machines being designed to correct their own errors, "magic state distillation and injection allow for fault-tolerant implementation of T gates", but "they typically require an order of magnitude more resources than Clifford gates". So they count only the eighths. The **T-count** of a circuit built from Clifford gates and T is its number of T and $T^\dagger$ gates; Kissinger and van de Wetering count, more generally, every phase gate whose turn is not a multiple of a quarter, and for circuits of T gates the two counts agree. The recipe above has T-count 4.

How few can it be made to need? A first guess is tempting.

The trap. The CNOTs come in pairs, and $\mathrm{CNOT}_{12}^2 = I$. So perhaps they cancel, and the four eighths all land on the lower wire: $T^4 = Z$, and the recipe is $I \otimes Z$, with T-count 0.

Follow one input and see. Start in $|10\rangle$: the upper wire lit, the lower dark. The first eighth sees a dark lower wire and does nothing. The first CNOT lights the lower wire. The second eighth sees it lit and turns the whole state by $e^{i\pi/4}$. The second CNOT darkens the lower wire again, so the third eighth does nothing; the third CNOT lights it, and the fourth eighth turns by $e^{i\pi/4}$ again; the fourth CNOT darkens it. The state comes back as $e^{i\pi/2}\,|10\rangle = i\,|10\rangle$. But $(I \otimes Z)\,|10\rangle = |10\rangle$. The guess is wrong: two of the

four eighths turned, and they turned because the *upper* wire was lit. The CNOTs cancel only when nothing sits between them.

Dirac's Blackboard.

The recipe, multiplied out. Write $w = e^{i\pi/4}$ and $Q = \mathrm{CNOT}_{12}(I \otimes T)$. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, $I \otimes T = \mathrm{diag}(1, w, 1, w)$, and CNOT_{12} exchanges $|10\rangle$ and $|11\rangle$. So Q fixes $|00\rangle$, sends $|01\rangle \mapsto w|01\rangle$, $|10\rangle \mapsto |11\rangle$ and $|11\rangle \mapsto w|10\rangle$. Apply it twice: $|10\rangle \mapsto |11\rangle \mapsto w|10\rangle$ and $|11\rangle \mapsto w|10\rangle \mapsto w|11\rangle$. Hence

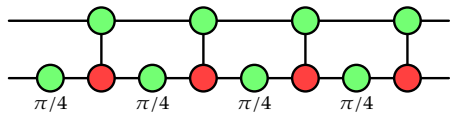
$$Q^2 = \mathrm{diag}(1,\, w^2,\, w,\, w) = \mathrm{diag}(1,\, i,\, w,\, w),$$

and squaring once more,

$$U = Q^4 = \mathrm{diag}(1,\, -1,\, i,\, i).$$

Against it, $I \otimes Z = \mathrm{diag}(1, -1, 1, -1)$. The two agree exactly on the inputs whose upper bit is 0, and differ on both inputs whose upper bit is 1. And $(S \otimes Z)\,\mathrm{CZ}$, with $\mathrm{CZ} = \mathrm{diag}(1, 1, 1, -1)$, is $\mathrm{diag}(1, -1, i, -i)\,\mathrm{diag}(1, 1, 1, -1) = \mathrm{diag}(1, -1, i, i)$: the recipe needs no eighth at all. Section §17.2 says why.

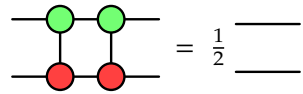
In the sketchbook. A CNOT is Chapter 8's green–red pair (row 64): a green spider on the control wire and a red spider on the target wire, joined by one cord. Drawn bare, with no number, it is $\mathrm{CNOT}/\sqrt{2}$. In this chapter we call it a **joiner**, as the dialogue does. An eighth is a green spider with phase $\pi/4$ (row 125). Every spider is drawn in the colours of Chapters 8 to 16, and its phase is written beside it; a spider with no phase written is at rest. The recipe, bare, is four joiners and four eighths:



$= \frac{1}{4}\,U.$

Each joiner carries $\frac{1}{\sqrt{2}}$ and each eighth carries nothing, so the number is $(\frac{1}{\sqrt{2}})^4 = \frac{1}{4}$.

Now look at the trap in the picture. Two joiners with nothing between them: the two greens sit on one cord of the upper wire and slide together (fusion, row 57), and so do the two reds on the lower wire. What is left is a green and a red joined by two cords, which Chapter 8's Hopf rule (row 62) takes apart, at the price of a half, and fusion closes the loose ends back into their wires:



which is $\mathrm{CNOT}_{12}^{2}/2 = I/2$. Put an eighth between the two joiners and the reds can no longer meet: the lower wire now reads red, green, red, and a red and a green do not fuse. The greens on the upper wire still slide together. That is the picture's way of saying what the blackboard found: the eighths between joiners feel the upper wire.

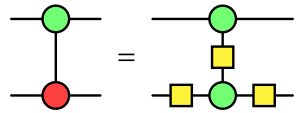
The net of a circuit

Chapter 16 drew a graph state as a **net**: a phase-free green spider for each vertex, and a **tie** carrying a yellow box for each edge (row 119). A circuit can be drawn in the same language, with inputs and turns added. Duncan, Kissinger, Perdrix and van de Wetering give the precise form. Their Definition 3.1 reads: "A ZX-diagram is graph-like when: 1. All spiders are Z-spiders. 2. Z-spiders are only connected via Hadamard edges. 3. There are no parallel Hadamard edges or self-loops. 4. Every input or output is connected to a Z-spider and every Z-spider is connected to at most one input or output." A Z-spider is a green spider, and a **Hadamard edge** is a cord carrying one yellow box (row 56). Chapter 8 also allowed a dashed cord for it; in this chapter every Hadamard edge is drawn with its yellow box, as the dialogue's ties wear yellow sleeves.

Here is our recipe, turned into that form in three moves, none of which costs a number.

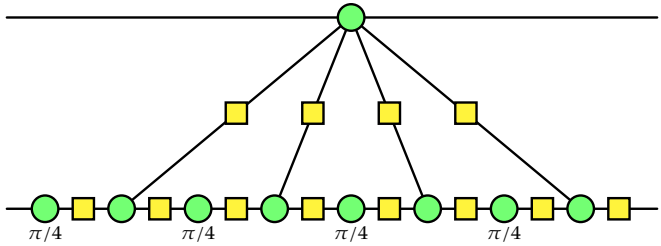
First, colour change (Theorem 8.2, row 58). A red spider is a green spider with a yellow box on every leg, exactly. Each red of a joiner has three legs, so it becomes a green spider with a box on its tie and a box on each side of

it on the lower wire:



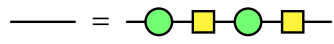
Do this to all four joiners. On the lower wire every eighth now sits between two boxes, except the first, which has the lower input on its left; each former red is a green spider at rest. Along the lower wire the spiders alternate, an eighth and a spider at rest, joined by Hadamard edges, and the last spider at rest reaches the lower output through one more box.

Second, fusion (row 57). The four greens on the upper wire are joined by plain cords, so they slide together into one green spider at rest. It holds the upper input, the upper output, and four ties, one to each former red:



This picture is still exactly the recipe's picture, $\frac{1}{4}U$. It satisfies conditions 1 to 3, but not condition 4: the big spider on the upper wire holds two ends, the upper input and the upper output.

Third, the dummies. A green spider at rest with two legs is a plain wire (row 57, as in Chapter 8), and two yellow boxes in a row are none ($H^2 = I$, row 56). So a plain lead may be replaced, at no cost, by a spider at rest, a box, a spider at rest and a box:



Put this on the upper wire's left lead, with the last box next to the big spider. Now the big spider is reached from the input only through two small spiders and two Hadamard edges, and it holds one end. Every condition holds. The result, drawn in full in §17.7, is this chapter's master diagram, *The Uncooked*

Net. It is the recipe, exactly: colour change, fusion, the spider at rest and $H^2 = I$ are all exact, so **no number was spent**, and its value is $\frac{1}{4}U$.

Nothing in these three moves used the particular recipe. Duncan and co-authors prove that it always works, and they add two moves for doubled ties and loops that a recipe of joiners and eighths does not produce here.

> **Result 17.1 (every diagram has a net; Duncan, Kissinger, Perdrix and van de Wetering, Lemma 3.2, stated).** Every ZX-diagram equals a non-zero number times a graph-like ZX-diagram.

Their Lemma 3.2 says "Every ZX-diagram is equal to a graph-like ZX-diagram", and their Remark 2.2 says that throughout the paper they "neglect (non-zero) scalar factors", so that an equality they prove means that the two matrices differ by a non-zero factor z. We have written the factor into the statement. For our recipe we tracked it: it is 1. Their proof is the one above: colour change, fusion, the removal of doubled ties and loops, and dummy spiders wherever a spider holds two ends. See `Code/Ch17/FourEighths.py`, which builds *The Uncooked Net* by hand, checks it against the recipe's picture after each of the three moves, and checks that PyZX's own graph-like form of the recipe is the same net, spider for spider and tie for tie.

Why this definition. Why insist on one colour, boxes on every tie, and one end per spider, when the recipe's picture was already a correct picture? Because then the whole diagram is a graph with a turn on each vertex. Nothing else is left to know. And graphs can be changed by graph moves: the re-tying and the pivot of Chapter 16, which were proved for nets, now apply to a circuit. The price is that the picture no longer looks like a recipe. Section §17.4 is about paying it back.

---

§17.2 Cooking Down

Look again at the recipe's second eighth. It sits between the first and second joiners, and nothing else does. Call such a stretch, a joiner, a turn on the lower wire, a joiner, a **sandwich**.

On the blackboard. For any angle α let $Z(\alpha) = \mathrm{diag}(1, e^{i\alpha})$, and let

$$\Gamma(\alpha) \;:=\; \mathrm{CNOT}_{12}\,(I \otimes Z(\alpha))\,\mathrm{CNOT}_{12}.$$

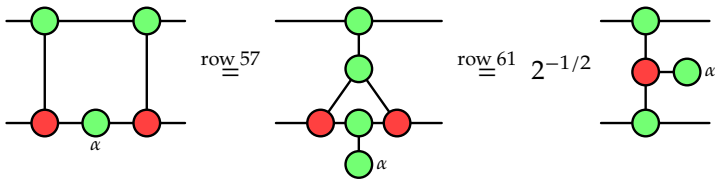
Follow $|x_1x_2\rangle$. The first CNOT puts $x_1 \oplus x_2$ on the lower wire; the turn multiplies by $e^{i\alpha(x_1\oplus x_2)}$; the second CNOT restores x_2. So

$$\Gamma(\alpha)\,|x_1x_2\rangle \;=\; e^{i\alpha(x_1\oplus x_2)}\,|x_1x_2\rangle\,, \qquad \Gamma(\alpha) = \mathrm{diag}(1,\, e^{i\alpha},\, e^{i\alpha},\, 1).$$

The sandwich turns by α when the two wires disagree. It does not care which wire is which. This is Kissinger and van de Wetering's **phase gadget** on two wires; their eq. (8) gives it on any number of wires, a turn by α on the parity $x_1 \oplus \cdots \oplus x_n$, and van de Wetering's survey (§5.6) calls both the diagram and the unitary phase gadgets. The letter Γ means only this two-wire gadget here.

In the sketchbook. Chapter 14 drew a controlled turn as a **hub**: a phase-free red spider tied to a green spider on each wire, with a third leg, the pendant, ending in a green spider with one leg (row 103). Chapter 13 drew the same red hub, with nothing hanging from it, as the parity gadget (row 102). A green spider with one leg we call an **end**, and we say it is turned by its phase. Hang an end turned α from a hub whose two feet are at rest:

Result 17.2 (the sandwich). For every angle α, exactly:



On the blackboard: the sandwich, bare, is $\frac{1}{2}\Gamma(\alpha)$ (two joiners),

and the hub picture is $\frac{1}{\sqrt{2}}\Gamma(\alpha)$.

Proof. The derivation by the rules is the sketchbook sidebar below. On the blackboard, the sandwich's two joiners carry $\frac{1}{2}$. In the hub picture, each green foot copies its wire's bit onto its tie; the red hub, a phase-free red spider with two legs in and one out, sends $|x_1\rangle\,|x_2\rangle$ to $\frac{1}{\sqrt{2}}\,|x_1 \oplus x_2\rangle$, as in the proof of Result 14.5; and the end gives $e^{i\alpha(x_1 \oplus x_2)}$. That is $\frac{1}{\sqrt{2}}\Gamma(\alpha)$, and $\frac{1}{2} = 2^{-1/2} \cdot \frac{1}{\sqrt{2}}$. ■ See `Code/Ch17/CookingDown.py`, which checks both pictures at $\alpha = \pi/4$ and at a random angle.

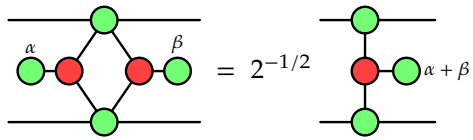
The hub with its end is Kissinger and van de Wetering's phase gadget, drawn with a red hub. They draw it with a green hub joined to the feet by Hadamard edges, and to the end by a Hadamard edge; colour change (row 58) turns one drawing into the other. With the feet turned $\alpha/2$ and the end turned $-\alpha/2$, it is Chapter 14's controlled turn; with the feet at rest, all of its turn hangs from the hub.

Coecke's Sketchbook.

Deriving the sandwich. Result 17.2 follows from rules already stated. The two greens on the upper wire are joined by a plain cord: slide them together (row 57), into one green holding two ties. Slide the turned spider on the lower wire apart (row 57 backwards): a spider at rest on the wire, with an end turned α hanging from it. Slide the upper green apart too, so that one part stays on the wire and the other holds both ties. That is the middle picture: two greens, the part holding the ties and the spider at rest on the lower wire, and two reds, and every green tied to every red. Each of the four is at rest and has exactly one other leg. That is Chapter 8's cross (Theorem 8.5, row 61), and it comes undone to one red tied to one green, worth $\frac{1}{\sqrt{2}}$ of it. The new red gathers the greens' other legs, so it is tied to the upper wire's spider and hung with the end: it is the hub. The new green gathers the reds' other legs, the two halves of the lower wire: it is the lower foot. It never looked at α.

The same rules fuse two gadgets. Suppose two hubs are tied to the same two spiders, with ends turned α and β. On the blackboard, $\Gamma(\alpha)\,\Gamma(\beta) = \Gamma(\alpha+\beta)$, since both are diagonal and their phases add.

Result 17.3 (two hubs on the same spiders). For all angles α, β, exactly, and whatever turns the two spiders on the wires carry:



Proof. Slide each wire's spider apart so that one part stays on the wire, keeping its turn, and the other part, at rest, holds both hub ties (row 57). Those two parts and the two hubs are a cross, every one at rest with one other leg, and the cross comes undone (row 61) at the price $\frac{1}{\sqrt{2}}$: a red tied to the two wire parts, and a green gathering the two ends. The green and the ends slide together (row 57): an end turned $\alpha+\beta$. On the blackboard, each hub is worth $\frac{1}{\sqrt{2}}\,e^{i\alpha(x_1\oplus x_2)}$ on the bits its feet copy, so two hubs give $\frac{1}{2}e^{i(\alpha+\beta)(x_1\oplus x_2)}$ and one hub gives $\frac{1}{\sqrt{2}}$ times the same. ■ See `Code/Ch17/CookingDown.py`.

Kissinger and van de Wetering call this **gadget fusion**, (GF), and derive it from their rules (f), (h) and (b′), which are fusion, colour change and the cross for any number of legs. On the blackboard it is, in their words, "a well-known simplification used in phase-polynomial circuits, where two phases acting on the same parity of the input qubits can be summed together."

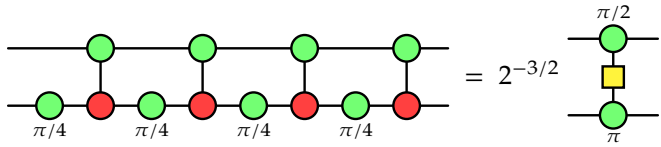
Four eighths that cancel

Now the recipe. Its second and fourth eighths are each inside a sandwich. Its first and third are not, but they sit on the lower wire, next to the hubs' lower feet, where fusion reaches them.

Result 17.4 (four eighths that cancel). Exactly,

$$U \;=\; \left(\mathrm{CNOT}_{12}\,(I\otimes T)\right)^4 \;=\; (S\otimes Z)\,\mathrm{CZ} \;=\; \mathrm{diag}(1,-1,i,i),$$

and $U|x_1x_2\rangle = i^{x_2+(x_1\oplus x_2)}|x_1x_2\rangle$. The recipe has T-count 4, and $(S\otimes Z)$ CZ has T-count 0. In pictures, the recipe's bare picture is $2^{-3/2}$ times the **cooked picture**, a green spider turned a quarter on the upper wire and a green spider turned a half on the lower, joined by one Hadamard edge:



The cooked picture is $\frac{1}{\sqrt{2}}U$; no eighth is left in it.

This is the chapter's headline theorem. It is proved twice in §17.6, by a parity count on the blackboard and by six moves in the sketchbook, the sandwiches first.

Cooking in general

A long circuit is cooked the same way, but with more moves. Its net has **interior** spiders, spiders with no input or output leg, and the cook's work is to remove as many of them as possible. Duncan and co-authors call a spider **Pauli** if its phase is a multiple of π (at rest or a half), and **Clifford** if its phase is a multiple of $\pi/2$; a Clifford spider that is not Pauli, turned an odd number of quarters, they call **proper Clifford**. Their two main moves are Chapter 16's graph moves, done inside a net: re-tying at a spider turned a quarter removes that spider (their Lemma 5.2), and a pivot along a tie between two Pauli spiders removes both (their Lemma 5.3); a third move, their eq. (9), first adds spiders at rest on a boundary wire, so that a spider next to the boundary becomes interior and the pivot can reach it. **This chapter uses re-tying and the pivot only as Chapter 16 stated them (rows 122 and 123), and does not derive them.**

Result 17.5 (simplification; Duncan, Kissinger, Perdrix and van de Wetering, Theorem 5.4, stated). There is a terminating procedure that turns any graph-like ZX-diagram into a graph-

> like ZX-diagram, up to one-wire unitaries on its inputs and outputs and up to a non-zero number (their Remark 2.2), which contains no interior proper Clifford spider, no pair of tied interior Pauli spiders, and no interior Pauli spider tied to a boundary spider. If every spider of the first diagram is Clifford, the result has no interior spider at all.

Their count of the cost (their Remark 5.6): each step removes a spider and toggles ties among the others, so for a net with vertex set V the procedure takes at most $O(|V|^3)$ elementary graph operations; they write n for $|V|$, and report that in practice it scales far better.

The eighths survive Result 17.5: a spider turned an eighth is not Clifford, and no move touches it. Kissinger and van de Wetering found the way round. Their variations (P2) and (P3) of the pivot remove any interior Pauli spider that is left, at the price of moving a neighbouring turn out onto a gadget, a hub hanging from the net. Then gadgets can meet. Two gadgets on the same spiders fuse, (GF), which is Result 17.3; and a gadget tied to only one spider slides its turn into that spider, their (ID).

> **Result 17.6 (ZX-simplify; Kissinger and van de Wetering, Algorithm 4.3 and Definition 4.4, stated).** Starting from a graph-like diagram: (1) re-tie until no interior proper Clifford spider is left; (2) pivot, with (P1), (P2) and (P3), until every interior Pauli spider is removed or moved onto a gadget; (3) remove every gadget whose turn is a multiple of a quarter; (4) apply (ID) and (GF) wherever possible, and if anything changed, go back to (1). The procedure terminates, since every step removes a spider or a gadget, and for a net with vertex set V it takes at most $O(|V|^3)$ elementary graph operations (their n is $|V|$). It ends in **reduced gadget form**: every interior spider is non-Clifford or part of a non-Clifford gadget, every gadget is tied to more than one spider, and no two gadgets are tied to the same set of spiders.

Result 17.6 lowers the number of eighths in the picture. It does not promise the fewest. Nothing in it is a proof that a smaller count is impossible, and the authors do not claim one.

Phase teleportation. A cooked net does not look like a circuit, and turning it back into one is the subject of §17.4. Kissinger and van de Wetering found a way to skip that step. Give each eighth of the original circuit its own name, $\alpha_1, \alpha_2, \ldots$, and cook the net with the names in place of the numbers. When (ID) or (GF) adds two named turns, the sum can be moved onto one of the two and the other set to 0, *in the original circuit*, and the new circuit cooks to the same net, so it is provably equal to the old one. In the words of their abstract, "Phase teleportation does not change the number or location of non-phase gates"; only the turns change.

> **Result 17.7 (the recipe, teleported).** Keep every joiner of the recipe where it is. Replace the first and the fourth eighths by quarters, S, and remove the second and the third. The new circuit equals U exactly, and has T-count 0. So does every choice of one quarter among the first and third places and one among the second and fourth. But quarters in the first and third places give $I \otimes Z$, the trap.

Proof. By the parity count of §17.6 (steps (1) and (2)), a turn in the first or third place sees x_2, and a turn in the second or fourth place sees $x_1 \oplus x_2$. So one quarter in each class gives $i^{x_2}\, i^{x_1 \oplus x_2}$, which is U by Result 17.4, and two quarters that both see x_2 give $(-1)^{x_2}$, which is $I \otimes Z$. ■ PyZX's phase teleportation chooses the first and fourth places; `Code/Ch17/CookingDown.py` checks the circuit it returns against U.

In the language of the gadgets: the first and third eighths slid together on the lower wire, and the second and fourth added on one hub. Two classes, a quarter each.

What PyZX does. Duncan and co-authors, and Kissinger and van de Wetering, built these procedures into PyZX, an open-source Python library. On *The Uncooked Net* its `full_reduce` leaves exactly the cooked picture of Result 17.4, with the number $2^{-3/2}$ written beside it, and its `extract_circuit` returns $(S \otimes Z)$ CZ, exactly U. On twenty random circuits of Clifford gates and eighths generated with a fixed seed, three wires and forty gates each, `full_reduce` kept the value exactly in all twenty and lowered the number of eighths in eighteen. `Code/Ch17/CookingDown.py` checks all of this against

the matrices.

On the published benchmarks the numbers are honest and mixed. Kissinger and van de Wetering ran thirty-six benchmark circuits. They matched or beat the previous best-known count of eighths, without extra wires, on twenty-six of the thirty-six, and beat it on six; with a further optimiser, TODD, run afterwards, they beat it on twenty. On twenty of the thirty-six they got exactly the previous best, which they read as a hint that those counts sit in "some kind of 'local optimum' ". Three rows of their Table 1: the circuit *nth-prime6* went from 567 eighths to 279 (the previous best was 400); *mod54* from 28 to 8, and to 7 with TODD (previous best 16); *tof3* from 21 to 15, the same as before. In their words, "The circuits where PyZX seems to do considerably better are ones that contain many Hadamard gates", which hamper the older methods built on phase polynomials. The PyZX paper adds that on *nth-prime6*, "PyZX optimises this to a circuit with 279 T gates in less than a second."

Validation. How do they know the cooked circuits are right? The same rules check them. Compose the original circuit with the adjoint of the optimised one, and cook the result. If the circuits are equal, the composite is the identity, and cooking should leave bare wires. Kissinger and van de Wetering did this for every benchmark: "In every case, the resulting circuit was reduced to the identity." When it succeeds, the list of moves is a certificate of equality. When it fails, nothing is proved either way; they note that "the general problem of circuit equality validation is QMA-hard", citing earlier work, "and hence a general efficient validation strategy is unlikely to exist". For our recipe the check is short: the recipe followed by the adjoint of $(S \otimes Z)\,\mathrm{CZ}$ is $U\,U^{\dagger} = I$, and PyZX cooks its picture to two bare wires.

Feynman's Notebook.

A compiler with no gates in it. In 2019 Ross Duncan, Aleks Kissinger, Simon Perdrix and John van de Wetering posted "Graph-theoretic Simplification of Quantum Circuits with the ZX-calculus"; *Quantum* published it in 2020. Their move was to forget the gates. Turn the circuit into a graph with turns, remove every interior spider that local complementation and

pivoting can reach, and then prove that what is left can still be turned back into a circuit. The next month Kissinger and van de Wetering posted "Reducing T-count with the ZX-calculus", which added the gadgets and phase teleportation; revised in January 2020, it appeared in *Physical Review A* that year as "Reducing the number of non-Clifford gates in quantum circuits". Both methods live in PyZX, which its authors presented at the Quantum Physics and Logic conference of 2019, published in 2020 — "pronounced like 'physics' without the H", they write. Kissinger and van de Wetering's validation caught a mistake that was not theirs: of another group's optimised benchmark circuits, all passed but one, qcla-mod7, which was then shown to be incorrect.

Why this definition. Why hang a turn from a hub, instead of leaving it on the wire where the recipe put it? Because on the wire a turn is stuck between the joiners around it, and in a recipe of many joiners two turns that act on the same parity may be far apart. On a hub, the turn is attached to *which wires it reads*, not to *when*. Two turns that read the same wires hang from hubs tied to the same spiders, wherever they came from, and Result 17.3 adds them.

---

§17.3 The One-Way Row

The cook hands back a picture, not a recipe. Can a picture be run as it is?

Take one wire in a state $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$ and a second wire in $|+\rangle$. Apply CZ, and then read the first wire, not in the basis $|0\rangle, |1\rangle$ but in a turned basis. Danos and Kashefi write

$$|\pm_\alpha\rangle \;=\; \tfrac{1}{\sqrt{2}}(|0\rangle \pm e^{i\alpha}\,|1\rangle),$$

and a reading M^α that projects onto $|+_\alpha\rangle$ or $|-_\alpha\rangle$, with the outcome $s = 0$ for $+$ and $s = 1$ for $-$. Compute. $\mathrm{CZ}(|\psi\rangle \otimes |+\rangle) = a\,|0\rangle\,|+\rangle + b\,|1\rangle\,|-\rangle$. Reading

outcome s applies $\langle \pm_\alpha | = \frac{1}{\sqrt{2}}(\langle 0| \pm e^{-i\alpha} \langle 1|)$ to the first wire and leaves

$$\frac{1}{\sqrt{2}}(a\,|+\rangle + (-1)^s e^{-i\alpha} b\,|-\rangle) \;=\; \frac{1}{\sqrt{2}}\,H(a\,|0\rangle + (-1)^s e^{-i\alpha} b\,|1\rangle)$$

on the second wire. For $s = 0$ that is $\frac{1}{\sqrt{2}} HZ(-\alpha)\,|\psi\rangle$; for $s = 1$ there is an extra Z inside, and $HZ = XH$.

> **Result 17.8 (one step).** For every state $|\psi\rangle$, every angle α and each outcome $s \in \{0, 1\}$, exactly,
>
> $$(\langle \pm_\alpha| \otimes I)\,\mathrm{CZ}\,(|\psi\rangle \otimes |+\rangle) \;=\; \frac{1}{\sqrt{2}}\,X^s HZ(-\alpha)\,|\psi\rangle\,,$$
>
> with $+$ for $s = 0$ and $-$ for $s = 1$. Each outcome has probability $\frac{1}{2}$, whatever $|\psi\rangle$ is.

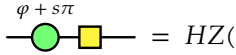
Proof. The computation above; $X^s HZ(-\alpha)$ is unitary, so each branch has length $\frac{1}{\sqrt{2}}$. ■ See `Code/Ch17/OneWayRow.py`, which checks it on a random non-real $|\psi\rangle$.

The reading moved the state from one wire to the other, turned it by $-\alpha$, and put a yellow box on it. The price is the X^s: half the time the result comes out flipped, and the reader cannot choose which half.

On the blackboard: patterns. Raussendorf and Briegel turned this into a way of computing. Prepare many wires in $|+\rangle$, join neighbours by CZ, in a line, a square grid, or any graph, and then read the wires one at a time. The state before any reading is the graph state of Chapter 16, which they call a **cluster state**; their eq. (1) says that it is fixed, up to a sign, by $\sigma_x^{(a)} \prod_{a'} \sigma_z^{(a')}$ over the neighbours a' of a, which is Chapter 16's safe touch. In their words, the cluster is "a 'substrate' on which any quantum circuit can be imprinted by one-qubit measurements". Danos and Kashefi wrote the procedure down as a **pattern**, a list of commands: prepare the non-input wires in $|+\rangle$; apply E_{ij}, their name for the CZ on wires i and j, which we write CZ_{ij}, for every edge; read wire i with M_i^α; and correct wire j with $X_j^{s_i}$ or $Z_j^{s_i}$, a Pauli applied only if an earlier outcome s_i was 1. The wires are the vertices of an **open graph** $(G, V_{\text{in}}, V_{\text{out}})$: a graph G with vertex set V, a set

V_{in} of inputs and a set V_{out} of outputs. Danos and Kashefi, and Duncan and co-authors, write it (G, I, O); we keep I for the identity matrix. Every vertex not in V_{out} is read, and N_i is the neighbourhood of i (Danos and Kashefi write $G(i)$).

In the sketchbook. A pattern is a picture already. Its CZs and $|+\rangle$s make the net of Chapter 16 (row 119). A reading's effect, $\langle \pm_\alpha |$, is $\frac{1}{\sqrt{2}}(\langle 0| + e^{i(-\alpha+s\pi)} \langle 1|)$, which is $\frac{1}{\sqrt{2}}$ times a green end turned $-\alpha+s\pi$: one picture for both outcomes, with Chapter 10's bit-times-π phase (row 75). So a pattern, branch by branch, is a net with a green end on each wire that is read.[2] The one step of Result 17.8 is a spider on the input wire, tied through a yellow box to a spider at rest on the output wire, with an end hanging from the first. Write $\varphi = -\alpha$ for the end's turn. Fuse the end into its spider (row 57): the spider on the input wire becomes a green spider turned $\varphi + s\pi$. The spider at rest on the output wire has two legs, so it is a plain wire (row 57, as in Chapter 8). What is left is a green spider turned $\varphi + s\pi$, then a box, then a plain wire. So



$$= HZ(\varphi + s\pi) = X^s HZ(\varphi),$$

exactly (since $Z(\varphi + s\pi) = Z(\varphi)Z^s$ and $HZ = XH$). The picture before fusing, with the $|+\rangle$, the CZ and the effect each drawn bare, has the same value, and that is $\sqrt{2}$ times the branch on the blackboard: the $|+\rangle$ is $\frac{1}{\sqrt{2}}$ times a green point, the CZ is $\sqrt{2}$ times its bare tie (row 119's tie), and the effect is $\frac{1}{\sqrt{2}}$ times its end.

Conversely, van de Wetering's survey (§6.3) states that "any graph-like ZX-diagram can be described as a measurement fragment", adding spiders at rest where a turned spider carries an output. It adds the caveat at once: such a fragment is "post-selected", read as if every reading came out the right way, and "it is in general not possible" to write a graph-like diagram as a pattern that can absorb the other outcomes; a diagram with inputs and no outputs, for instance, has nowhere to send them. We state this and do

[2] Van de Wetering's survey (§6.3) draws the effect of the outcome $|+_{XY,\alpha}\rangle = |0\rangle + e^{i\alpha}|1\rangle$ as a green effect labelled α, and the other outcome as $\alpha + \pi$. With this book's convention that the mirror conjugates (Chapter 3), the effect of $|0\rangle + e^{i\alpha}|1\rangle$ is the end turned $-\alpha$. The difference is one of labelling; we follow Danos and Kashefi and write the turn $-\alpha$.

not prove it.

Three dishes in a row

Now the example that makes feed-forward necessary. Take three wires in a row, 1–2–3, all prepared in $|+\rangle$ and joined by $\mathrm{CZ}_{12}\mathrm{CZ}_{23}$: no inputs, $V_{\text{out}} = \{3\}$. Read wire 1 with its end turned $\varphi_1 = \pi/4$ (Danos and Kashefi's $\alpha_1 = -\pi/4$), then wire 2 with $\varphi_2 = 0$, then wire 3 with $\varphi_3 = -\pi/4$. If the first two readings both come out 0, Result 17.8 twice says that wire 3 holds, before its reading,

$$\tfrac{1}{2}\,HZ(\varphi_2)HZ(\varphi_1)\,|+\rangle \;=\; \tfrac{1}{2}\,T\,|+\rangle \;=\; \tfrac{1}{2}\,|\mu\rangle\,,$$

Chapter 16's magic state, and the reading at $\varphi_3 = -\pi/4$ finds 0 with certainty: $\langle 0| + e^{-i\pi/4}\langle 1|$ against $|0\rangle + e^{i\pi/4}|1\rangle$ gives 2, and $\langle 0| - e^{-i\pi/4}\langle 1|$ gives 0. The dialogue calls outcome 0 *dark* and outcome 1 *lit*.

But the first two readings are coin tosses. If the first comes out 1 and the second 0, wire 3 holds $Z|\mu\rangle$ instead, and the reading at $-\pi/4$ finds 1 with certainty. The last reading has become a coin that knows the first two. The cure is to change the later settings according to the earlier outcomes. Here is the rule the chapter uses, which the dialogue's Crab keeps for his tins: **if the first reading is** 1**, turn the second setting the other way,** $\varphi_2 \mapsto -\varphi_2$**, and put a half on the third,** $\varphi_3 \mapsto \varphi_3 + \pi$**; if the second reading is** 1**, turn the third setting the other way.** So the third setting is $(-1)^{s_2}\varphi_3 + s_1\pi$.

> **Result 17.9 (the row, with and without feed-forward).** In the three-wire row above, each of the four patterns (s_1, s_2) of the first two readings has probability $\frac{1}{4}$. Read by the rule, the third reading is 0 with probability 1 in every pattern; the third settings are $-\pi/4$, $\pi/4$, $3\pi/4$ and $5\pi/4$ for $(s_1, s_2) = (0,0), (0,1), (1,0), (1,1)$. Read without the rule, always at $-\pi/4$, the third reading is 1 with probability $\frac{1}{2}$ overall: never after $(0,0)$, always after $(1,0)$, and with probability $\frac{1}{2}$ after $(0,1)$ and after $(1,1)$. With only the "other way" half of the rule it is 1 with probability $\frac{1}{2}$; with only the "half on" half, $\frac{1}{4}$. Kept unread, wire 3 holds exactly $X^{s_2}Z^{s_1}|\mu\rangle$ (the branch

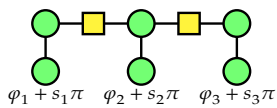
divided by its length), which the Paulis the first two outcomes name turn back into $|\mu\rangle$, up to a sign.

Proof. The Dirac's Blackboard sidebar below shows where the kept state $X^{s_2}Z^{s_1}|\mu\rangle$ comes from. Each reading of it is a two-term sum; for instance after $(1,0)$ the state is $\frac{1}{\sqrt{2}}(|0\rangle + e^{i5\pi/4}|1\rangle)$, and the reading at $3\pi/4$ gives $\frac{1}{2}(1+e^{i2\pi}) = 1$ for outcome 0. The other cases are the same kind of sum. ■ See `Code/Ch17/OneWayRow.py`, which computes every branch and every probability above from the three-wire state.

Dirac's Blackboard.

Where the corrections come from. Danos and Kashefi's proof of their Theorem 1 runs on five equations, their (1)–(5), which in our letters read: $\langle +_\alpha| Z^s = \langle \pm_\alpha|$ (the outcome-s effect); $Z_i^s \,\mathrm{CZ}_{ij} = X_j^s \,\mathrm{CZ}_{ij}\, X_j^s$; $X_i^s \,\mathrm{CZ}_{ij} = \mathrm{CZ}_{ij}\, Z_j^s\, X_i^s$; $Z_i^s \,\mathrm{CZ}_{ij} = \mathrm{CZ}_{ij}\, Z_i^s$; and $X_i^s|+\rangle_i = |+\rangle_i$. Run the row. Reading wire 1 with outcome s_1 leaves, by Result 17.8, $X_2^{s_1}$ on wire 2, still in front of CZ_{23}. Push it through by the third equation, rearranged: $\mathrm{CZ}_{23}X_2^{s_1} = X_2^{s_1}Z_3^{s_1}\mathrm{CZ}_{23}$. So wire 2 carries an X and wire 3 a Z. The X on wire 2 is absorbed by reading it at $-\varphi_2$ instead of φ_2, since $(\langle 0| + e^{i\theta}\langle 1|)X = e^{i\theta}(\langle 0| + e^{-i\theta}\langle 1|)$. The Z on wire 3 commutes onward (fourth equation) and is absorbed by adding π to wire 3's setting. Reading wire 2 leaves $X_3^{s_2}$, absorbed by turning wire 3's setting the other way. These are the rule's three clauses.

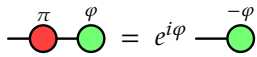
In the sketchbook. Close the row: an end on every wire, turned $\varphi_k + s_k\pi$. Three $|+\rangle$s, two CZs and three effects carry $(\frac{1}{\sqrt{2}})^3(\sqrt{2})^2(\frac{1}{\sqrt{2}})^3 = \frac{1}{4}$, so the bare picture is exactly 4 times the amplitude of the readings (s_1, s_2, s_3):



$$= 4\,(\langle \pm_{\alpha_1}| \otimes \langle \pm_{\alpha_2}| \otimes \langle \pm_{\alpha_3}|)\,\mathrm{CZ}_{12}\mathrm{CZ}_{23}\,|+++\rangle\,,$$

with $\alpha_k = -\varphi_k$ and the sign of each effect set by s_k. To see the rule in the picture we need one more small rule, about a red half meeting an end.

Result 17.10 (a red half against an end). For every angle φ, exactly: a red spider turned π on the cord of a green end turned φ may be taken off, if the end is turned the other way and the number $e^{i\varphi}$ is written beside it:



So an end at rest is unchanged, an end turned a half stays a half with the number -1, and an end turned an eighth becomes an end turned an eighth back.

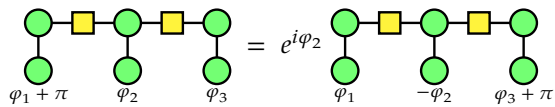
Proof. A red π with two legs is X exactly (§8.2), and $(\langle 0| + e^{i\varphi}\langle 1|)X = \langle 1| + e^{i\varphi}\langle 0| = e^{i\varphi}(\langle 0| + e^{-i\varphi}\langle 1|)$. In the rules, this is Chapter 8's π-commutation (Theorem 8.4, row 60) applied to an end, with its number written. ■ See `Code/Ch17/OneWayRow.py`.

Result 17.11 (the travelling half). For all angles $\varphi_1, \varphi_2, \varphi_3$, write $P(\theta_1, \theta_2, \theta_3)$ for the closed row with its ends turned $\theta_1, \theta_2, \theta_3$. Exactly,

$$P(\varphi_1 + \pi,\ \varphi_2,\ \varphi_3) \;=\; e^{i\varphi_2}\,P(\varphi_1,\ -\varphi_2,\ \varphi_3 + \pi),$$

$$P(\varphi_1,\ \varphi_2 + \pi,\ \varphi_3) \;=\; e^{i\varphi_3}\,P(\varphi_1,\ \varphi_2,\ -\varphi_3).$$

In pictures, the first equation is



Proof. The derivation is the sketchbook sidebar below: the half from the first end travels to the second wire's spider and on to the third, turning the second end the other way (Result 17.10) and landing on the third end as a half. The second equation is the same journey, one step shorter. ■ See `Code/Ch17/OneWayRow.py`, which builds both sides for random angles and compares them as numbers.

Result 17.11 is the rule. Suppose the first reading is 1. By the first equation,

the picture of that branch, read at the settings the rule prescribes, $-\varphi_2$ and $\varphi_3 + \pi$, equals $e^{-i\varphi_2}$ times the picture with the first reading 0 and the original settings, since the two halves on the third end add to a whole turn. The same holds for the second reading and the second equation. So every branch has, for each third outcome, the amplitude of branch $(0,0)$ times a number of length 1. Branch $(0,0)$ never reads 1 at the third wire; so no branch does.

Coecke's Sketchbook.

The travelling half. Leave the ends unfused, so every spider on the row is at rest. Slide the half off the first end, turned $\varphi_1 + \pi$, onto its cord (row 57 backwards), then through the first spider onto its tie (row 57). Through the yellow box it turns red (row 58). It meets the second spider, at rest, and comes out on every other leg (row 120): a red half on the second end's cord, and one on the tie to the third spider. On the cord it turns the second end the other way, setting aside $e^{i\varphi_2}$ (Result 17.10). On the tie, through the box, it turns green again, passes through the third spider and fuses with the third end: a half. If both of the first two ends are lit, push the second end's half first, so that the second spider is still at rest when the first half arrives. The warning: this is an equality of numbers, branch by branch. It does not say how the odds are shared out; Result 17.8 does.

Danos and Kashefi proved that this works on any open graph with the right kind of order. Their Definition 2 is quoted in §17.4; here is their theorem.

Result 17.12 (determinism; Danos and Kashefi, Theorem 1, stated). If an open graph $(G, V_{\text{in}}, V_{\text{out}})$ has a flow $(f, \prec)$ (§17.4), then the pattern that prepares $|+\rangle$ on every non-input, applies CZ_{ij} on every edge, and then, for each non-output i in the order $\prec$, reads i with $M_i^{\alpha_i}$ and applies the corrections

$$X_{f(i)}^{s_i} \prod_{k \in N_{f(i)} \setminus \{i\}} Z_k^{s_i},$$

is runnable, and **uniformly and strongly deterministic**: every

branch gives the same map, for every choice of the angles α_i. Danos and Kashefi call that map the unitary embedding the pattern realizes.

In the row, $f(1) = 2$ and $f(2) = 3$: the reading of wire 1 corrects wire 2 by an X and wire 3, the other neighbour of wire 2, by a Z; the reading of wire 2 corrects wire 3 by an X. A correction on a wire that is still to be read is absorbed into its setting, as the sidebar showed, and a correction on an output is applied. In the picture, the $X_{f(i)}$ is the red half arriving at $f(i)$, and the Z_k are the green halves it sends on to the other neighbours of $f(i)$. Danos and Kashefi give the same intuition in words: their eq. (2) converts "an anachronical Z correction at i" into "a 'future' X correction, the one sent to $f(i)$", and a "'past' X correction", sent back until it is absorbed by a preparation.

Feynman's Notebook.

Measurements only. Robert Raussendorf and Hans J. Briegel, at the Ludwig-Maximilians-Universität in Munich, posted "Quantum computing via measurements only" in October 2000; *Physical Review Letters* printed it in 2001 as "A one-way quantum computer". Their picture of the laboratory was a lattice of neutral atoms held in "periodic micropotentials", entangled all at once by "controlled collisions between atoms in neighboring potential wells", then read one atom at a time. They noted that single-atom occupation of a three-dimensional optical lattice had been reported with a filling factor of 0.44, above the percolation threshold of 0.31, so a cluster bounded only by the trap could exist. Each reading destroys entanglement as it goes. Their conclusion: "Cluster states are thus one-way quantum computers and the measurements form the program." Vincent Danos and Elham Kashefi asked in 2005 which patterns can be made deterministic, and answered with flow; the paper appeared in *Physical Review A* in 2006. Van de Wetering's survey adds that "the development of the ZX-calculus was originally motivated by the desire to describe MBQC in a better way", citing Ross Duncan's "A graphical approach to measurement-based quantum

computing" (2013).

Why this definition. Why read in a turned basis, $|\pm_\alpha\rangle$, and not in $|0\rangle, |1\rangle$? Because a reading in $|0\rangle, |1\rangle$ only cuts a wire out of the graph (Raussendorf and Briegel: measurements of σ_z "effectively remove" qubits), and a reading in $|\pm\rangle$ only moves the state along. The turn α is the only place where the program can enter. With it, one step is a turn and a yellow box, and turns and yellow boxes make every one-wire gate.

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§17.4 Out Again

The tins of the dialogue were read in an order, first, second, last, and the rule sent each outcome forward. What exactly must be true of a graph for such an order to exist?

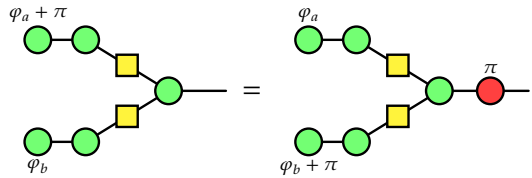
On the blackboard: flow. Danos and Kashefi's Definition 2 reads: "An open graph state (G, I, O) has flow if there exists a map $f : O^c \to I^c$ (from measured qubits to prepared qubits) and a partial order $>$ over V such that for all $i \in O^c$: (F0) $(i, f(i)) \in G$; (F1) $f(i) > i$; (F2) for all neighbours of $f(i)$ except i ($k \in G(f(i)) \setminus \{i\}$), we also have $k > i$." In our letters: an open graph $(G, V_{\text{in}}, V_{\text{out}})$ has a **flow** $(f, \prec)$ if f sends each vertex not in V_{out} to a vertex not in V_{in}, and $\prec$ is a partial order on V, such that for every $i \notin V_{\text{out}}$:

- (F0) $f(i) \in N_i$: the correction goes to a neighbour;
- (F1) $i \prec f(i)$: the neighbour is read later, or not at all;
- (F2) $i \prec k$ for every $k \in N_{f(i)} \setminus \{i\}$: the other neighbours of $f(i)$, which receive the Z corrections, are later too.

The row has a flow: $f(1) = 2, f(2) = 3$, with $1 \prec 2 \prec 3$. Check (F2): the neighbours of 2 other than 1 are $\{3\}$, and $1 \prec 3$; wire 3 has no neighbour other than 2.

Here is a graph that has none. Take three vertices a, b, c, with edges a–c and b–c, a **V** with its corner at c; read a and b, and keep c as the output. Danos and Kashefi draw this graph as their Fig. 2, with a an input.

In the sketchbook. Close the V with ends on a and b, and leave c's output open. A lit reading on a sends its half through c, which is at rest, and out on c's other legs: onto c's output, as a red half, and along the tie to b, where it lands on b's end as a green half. Exactly, with no number:



(rows 57, 58 and 120, as in the travelling half). So a's outcome has to be known before b is read. By the same picture turned over, b's outcome has to be known before a is read. Each has to go before the other.

The blackboard makes this exact, and for a weaker kind of order too. Browne, Kashefi, Mhalla and Perdrix introduced **generalised flow**, or **gFlow**, in 2007; we take the definition from Duncan and co-authors, who cite them (their ref. [8]). For a set A of vertices, its **odd neighbourhood** $\mathrm{Odd}(A)$ is the set of vertices with an odd number of neighbours in A. A gFlow $(g, \prec)$ sends each vertex u not in V_{out} to a set $g(u)$ of vertices not in V_{in}, such that $u \in \mathrm{Odd}(g(u))$; $u \prec v$ for every $v \in g(u)$; and $u \prec v$ for every $v \neq u$ in $\mathrm{Odd}(g(u))$ (their Definition B.2). A flow is a gFlow with $g(u) = \{f(u)\}$: then $\mathrm{Odd}(g(u)) = N_{f(u)}$, and the conditions are (F0) to (F2). In the picture, $g(u)$ is the set of spiders a lit half from u is sent through, and $\mathrm{Odd}(g(u))$ the set of ends it lands on.

> **Result 17.13 (the V has no order).** The V, with $V_{\text{out}} = \{c\}$ and V_{in} empty or $\{a\}$, has no flow and no gFlow.

Proof. The only neighbour of a is c, so $a \in \mathrm{Odd}(g(a))$ forces $c \in g(a)$. The only neighbour of b is c as well, so then $b \in \mathrm{Odd}(g(a))$, and $b \neq a$ forces $a \prec b$. The same argument with a and b exchanged forces $b \prec a$. A partial order cannot have both. A flow is a gFlow, so there is no flow either. ■ See `Code/Ch17/OutAgain.py`, which also searches every order by brute force.

This is Danos and Kashefi's Fig. 2, "an open graph state with no flow".[3]

[3] Danos and Kashefi add that the pattern on this graph can still be made deterministic if b

Result 17.13 says nothing about the V beyond this: no order of readings makes its corrections work for every setting.

Taking a recipe out

A cooked net is a picture of a unitary, and Result 17.12 says how to *run* it if it has a flow. Duncan and co-authors used the same kind of order to take a *circuit* out of it.

> **Result 17.14 (nets from circuits keep an order; Duncan, Kissinger, Perdrix and van de Wetering, Lemma 3.7, Theorem 4.5 and Corollary 5.5, stated).** (i) The net that Lemma 3.2's procedure makes from a circuit has a **focused gFlow**, a gFlow in which $\mathrm{Odd}(g(u))$ contains no non-output other than u (their Definition 3.4, which they credit to Mhalla, Murao, Perdrix, Someya and Turner). (ii) If an open graph has a focused gFlow, then it still has a gFlow after a local complementation at an interior vertex u followed by deleting u, and after a pivot along an edge between two interior vertices followed by deleting both. (iii) So the procedure of Result 17.5 keeps a focused gFlow. By their Remark 3.6, an open graph has a gFlow if and only if it has a focused one.

Part (i) is easy to see on *The Uncooked Net*: it has a flow that runs along the wires. Send each spider to the next spider on its own wire, the first small spider to the second, the second to the big spider, and each lower spider to the one on its right; order the upper wire's two small spiders first, then the lower wire's spiders from left to right, then the big spider. Each of (F0) to (F2) is a check against the drawing in §17.7. Duncan and co-authors prove (i) exactly this way: a circuit's net has a causal flow (their Definition B.1, credited to Danos and Kashefi), which is a gFlow (their Theorem B.3). Part (iii) covers re-tying and the pivot, and only those. For the gadget moves of Result 17.6, van de Wetering's survey says that the extraction algorithm of Backens, Miller-Bakewell, de Felice, Lobski and van de Wetering, "There and back again: A circuit extraction tale" (a preprint of 2020, its ref. [13]),

is read at the fixed angle $\pi/2$; but then it is deterministic only at that angle, and so, in their words, "not uniformly deterministic".

"further generalises this approach by being able to deal with phase gadgets"; we have not read that paper, and claim nothing more about it. On the twenty random circuits of §17.2, PyZX's gFlow finder found an ordinary gFlow after cooking on sixteen, and none on the four whose cooked nets kept a gadget; PyZX took a circuit out of all twenty.

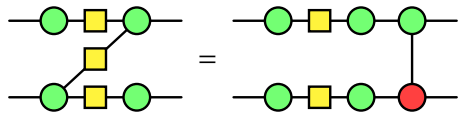
On the blackboard: extraction. Duncan and co-authors take the circuit out from the right. Keep a **frontier**, one spider for each wire, starting at the outputs. A turn on a frontier spider slides off onto its output as a phase gate. A Hadamard edge between two frontier spiders is a CZ, and comes off as one. What remains between the frontier and the spiders behind it is a set of Hadamard edges, described by a matrix B with a row for each frontier spider and a column for each spider behind, with entry 1 where they are tied: the **biadjacency matrix**. Van de Wetering's survey writes it P and Duncan and co-authors write it M; we write B. Adding one row of B to another, over $\mathbb{F}_2$, where $1 + 1 = 0$, is a CNOT on the two wires (their Proposition 7.1). Gaussian elimination turns part of B into rows with a single 1; a frontier spider with a single tie to one spider behind it, and no turn, is a plain wire through a yellow box, and the frontier moves one step left. The existence of a gFlow guarantees that some row can always be reduced so, which is what makes the procedure finish (their §7; they remark that knowing a gFlow exists is enough, without computing it).

Here is the smallest case, one row operation. Two spiders behind, on the inputs, and two frontier spiders, on the outputs, with

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} :$$

the upper frontier spider is tied to both spiders behind, the lower only to the lower one. Adding the second row to the first gives I.

Result 17.15 (a joiner at the ends). Exactly, with no number:



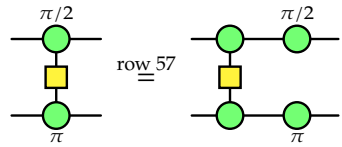
The joiner has its green on the upper wire and its red on the lower. The same picture with the joiner the other way up is not equal to the left side.

Proof. Sum over bits. On the left, the spiders behind copy the input bits x_1, x_2, the frontier spiders carry output bits y_1, y_2, and the three boxes give $2^{-3/2}(-1)^{x_1y_1+x_2y_1+x_2y_2}$. On the right, the picture before the joiner is $H \otimes H$ exactly, since each wire is a spider at rest, a box and a spider at rest; so the right side is $\frac{1}{\sqrt{2}}\mathrm{CNOT}_{12}(H \otimes H)$. Now $(H \otimes H)\,|x_1x_2\rangle = \frac{1}{2}\sum_z (-1)^{x_1z_1+x_2z_2}\,|z_1z_2\rangle$, and the CNOT sends $|z_1z_2\rangle$ to $|z_1, z_1 \oplus z_2\rangle$. Writing $y_1 = z_1$ and $y_2 = z_1 \oplus z_2$, so that $z_2 = y_1 \oplus y_2$, the sign is $(-1)^{x_1y_1+x_2(y_1\oplus y_2)}$, which is the left side's, and the numbers agree: $\frac{1}{\sqrt{2}}\cdot\frac{1}{2} = 2^{-3/2}$. With the joiner the other way up the same computation gives the sign $(-1)^{x_1(y_1\oplus y_2)+x_2y_2}$, which differs at $x = (1,0), y = (0,1)$. ■ See `Code/Ch17/OutAgain.py`.

Duncan and co-authors state Proposition 7.1 for any diagram and any pair of frontier rows, and add that a focused gFlow survives the step. We have proved only this instance, which is the one the dialogue's cloth uses.

Now the recipe itself. The cooked picture has no spider behind its frontier: both of its spiders hold an output.

Result 17.16 (the recipe taken out). Exactly: the cooked picture equals a Hadamard edge between two spiders at rest, followed by a green quarter on the upper wire and a green half on the lower,



which is $\frac{1}{\sqrt{2}}(S \otimes Z)\,\mathrm{CZ}$. The new recipe is: a CZ, then S on the upper wire and Z on the lower. It equals U exactly, with T-count 0.

Proof. Slide each turn off its spider toward the output (row 57 backwards). A phase-free green spider on each wire, joined by a yellow

box, is $\frac{1}{\sqrt{2}}$CZ: each spider copies its bit onto the tie, and the box gives $\langle x_2|\,H\,|x_1\rangle = 2^{-1/2}(-1)^{x_1 x_2}$; this is the computation of Result 16.4 with an input leg on each spider. A green $\pi/2$ with two legs is S, and a green π is Z (§16.1). ■ PyZX's `extract_circuit` returns the same three gates; see `Code/Ch17/OutAgain.py`.

For a general circuit, one thing is lost on the way out.

> **Result 17.17 (extraction; Duncan, Kissinger, Perdrix and van de Wetering, §7, stated).** If a graph-like diagram has a gFlow, the frontier procedure takes a circuit out of it. For a net cooked from a circuit by Result 17.5, the circuit taken out equals the original **up to a global phase**: a number r with $|r| = 1$.

The global phase is the one place in this chapter's own computations where a number is not tracked; the stated Results 17.1 and 17.5 carry their sources' non-zero number. Duncan and co-authors neglect non-zero numbers throughout (their Remark 2.2), and they explain why that is harmless here: if two unitaries differ by a factor z, then $|z| = 1$, so neglecting numbers will, "at worst", give an output that differs from the input by a global phase. On the twenty random circuits of §17.2, every extracted circuit was the original times such an r, and none was exactly equal; `Code/Ch17/OutAgain.py` asserts $|r| = 1$ in each. Our recipe's extraction happens to be exact.

Why not take a circuit out of *any* picture of a unitary? Van de Wetering's survey (§6.6) gives the reason to doubt it can be done fast. The promise that a picture is unitary "is a global property of the diagram": large parts of it may be far from unitary and be cancelled only by a later part, so an algorithm "would somehow have to be using knowledge about the global structure of the diagram"; and deciding whether a closed picture is non-zero "comes awfully close to being a QMA-complete problem". The survey states the argument as a reason for doubt, not as a proof, and so do we. The hope, it says, is local structure, and the best-known algorithm, which it credits to Backens and co-authors, "can extract a circuit out of any ZX-diagram that has a gflow, a combinatorial property … that corresponds to the possibility of consistently feed-forwarding Pauli errors from measurement outcomes".

The synthesis. That last phrase joins the two halves of this chapter. The

order in which a pattern can be read, every half sent forward, is an order in which a cooked net can be taken out again as a recipe, frontier by frontier from the outputs. A net made from a circuit starts with such an order (Result 17.14 (i)); re-tying and the pivot keep it (Result 17.14 (iii)). The frontier moves from the outputs toward the inputs, the gFlow order read backwards. We claim that a gFlow is *enough* for extraction, which is Duncan and co-authors' theorem, and nothing more: PyZX also takes circuits out of cooked nets that keep gadgets and have no ordinary gFlow, and the algorithm that covers them is the one we have not read.

Coecke's Sketchbook.

What the pictures of this section do not say. Result 17.15 is an equality of pictures with its number; it is not an algorithm, and it does not say which row operations to choose. Gaussian elimination chooses them, and Gaussian elimination is the blackboard's. The frontier is the picture's: it is the line in the drawing between what has become recipe and what is still net, and it moves left one spider at a time. Result 17.13 is the other kind of statement a picture cannot make on its own. The drawing of the V shows a half from a landing on b's end and a half from b landing on a's; it takes a small argument about all possible orders, on the blackboard, to conclude that no order exists. Pictures show what a move does. That no sequence of moves can succeed is a claim about every sequence, and it needs a proof about every sequence.

Why this definition. Why an order on the vertices, and not simply a rule for each reading? Because a correction can only go to a wire that has not been read yet, and one correction can reach several wires, as Z_3 did in the row. So the corrections must fit together: every wire that a reading's half lands on must come later than that reading. Flow asks for this with one target per reading, gFlow with a set. Anything weaker, and the V shows what goes wrong.

§17.5 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 16's row 126. The chapter introduces no new string-diagram element: rows 127 to 135 are new uses of the spiders, boxes and rules of Chapters 8 to 16, two new rules derived from them (rows 129 and 130), and procedures stated from the sources.

#	Algebra (Dirac)	Picture (Coecke)
127	T-count: the number of T and $T^\dagger$ in a circuit of Clifford gates and T; $\mathrm{CNOT}^2 = I$, $T^2 = S$, $T^8 = I$	**the eighths of a picture**: spiders whose phase is an odd multiple of $\pi/4$; two joiners in a row are $\frac{1}{2}$ times two plain wires, but only with nothing between them
128	graph-like form (Duncan et al., Definition 3.1); every diagram is a non-zero number times a graph-like one (Lemma 3.2)	**the net of a circuit**: targets turned green by colour change, controls slid together, and two spiders at rest, each followed by a yellow box, put on a lead where one spider would hold two ends; exact, no number spent
129	$\Gamma(\alpha) = \mathrm{CNOT}_{12}(I \otimes Z(\alpha))\mathrm{CNOT}_{12} = \sum_x e^{i\alpha(x_1 \oplus x_2)} \lvert x\rangle \langle x\rvert$	**the sandwich**: joiner, turn α on the lower wire, joiner $= 2^{-1/2}\times$ a hub on the two wires with an end turned α hanging from it (derived: rows 57, 61)
130	$\Gamma(\alpha)\Gamma(\beta) = \Gamma(\alpha + \beta)$	**two hubs on the same spiders** $= 2^{-1/2}\times$ one hub, ends added (derived: rows 57, 61)

#	Algebra (Dirac)	Picture (Coecke)
131	**the headline:** $(\mathrm{CNOT}_{12}(I \otimes T))^4 = (S \otimes Z)\,\mathrm{CZ}$; T-count $4 \to 0$	**the cooked picture**: a green quarter on the upper wire, a green half on the lower, one Hadamard edge; recipe picture $= 2^{-3/2}\times$ cooked
132	simplification: Duncan et al., Theorem 5.4; Kissinger and van de Wetering, Algorithm 4.3 and phase teleportation	**cooking**: re-tying and pivots (rows 122 and 123) remove interior quarters and halves; hubs carry eighths together (row 130); what is left: boundary spiders, eighths that could not meet, hubs. No promise of the fewest
133	a pattern: $\lvert+\rangle$ on non-inputs, CZ on edges, readings M_i^{α}; one step $\frac{1}{\sqrt{2}} X^s H Z(-\alpha)\,\lvert\psi\rangle$	**a pattern is a picture**: the net with a green end turned $-\alpha + s\pi$ on each read wire (row 75); a closed row of three is $4\times$ its amplitude
134	feed-forward: Danos and Kashefi, Theorem 1, corrections $X_{f(i)}^{s_i}$ and $Z_k^{s_i}$ for $k \in N_{f(i)} \setminus \{i\}$; flow, their Definition 2	**the travelling half**: a lit half goes through a yellow box, through the next spider at rest, turns that spider's end the other way ($\times e^{i\varphi}$, Result 17.10) and lands as a half on the ends after it; a flow is an order in which halves only travel forward

#	Algebra (Dirac)	Picture (Coecke)
135	extraction (Duncan et al., §7, Proposition 7.1; survey §6.6): phases → phase gates, frontier Hadamard edges → CZ, row operations over $\mathbb{F}_2$ → CNOT; up to a global phase in general	**taking a recipe out**: turns slid off the end spiders; a joiner at the ends unties a doubled tie; spider, box, spider on a wire is a yellow box

Every string-diagram row is exact, number included, except the last row's general case, where the sources give extraction up to a global phase. Rows 129 and 130 are derived by the rules; row 131 is derived in §17.6, with one step checked by evaluation; rows 132 and 135 are procedures stated from the sources, whose steps are equalities and whose running times are the sources' counts.

---

§17.6 The Theorem, Twice

Result 17.4 (four eighths that cancel), restated. With CNOT_{12} the CNOT whose control is the upper wire, exactly,

$$U \;=\; (\mathrm{CNOT}_{12}\,(I \otimes T))^4 \;=\; (S \otimes Z)\,\mathrm{CZ},$$

and the recipe's bare picture is $2^{-3/2}$ times the cooked picture.

Dirac's Blackboard

Every gate in the recipe sends a basis state to a basis state times a phase, so it is enough to follow $|x_1 x_2\rangle$.

Step 1: the joiners. A CNOT adds the upper bit to the lower and leaves the upper bit alone. So after k joiners the lower wire carries $x_2 \oplus (k \bmod 2)\, x_1$,

and after four it carries x_2 again: the recipe returns every basis state to itself, times a phase.

Step 2: what each eighth sees. The k-th eighth acts after $k-1$ joiners. So the first and third eighths see x_2, and the second and fourth see $x_1 \oplus x_2$.

Step 3: the phases. An eighth multiplies by $e^{i\pi/4}$ when the bit it sees is 1. In all,

$$U\,|x_1 x_2\rangle \;=\; e^{i\frac{\pi}{4}(2x_2+2(x_1\oplus x_2))}\,|x_1 x_2\rangle \;=\; i^{\,x_2+(x_1\oplus x_2)}\,|x_1 x_2\rangle\,.$$

Step 4: read it as gates. For bits, $x_1 \oplus x_2 = x_1 + x_2 - 2x_1x_2$. So

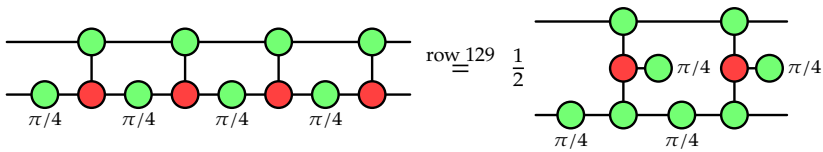
$$i^{\,x_2+(x_1\oplus x_2)} \;=\; i^{\,x_1}\,i^{\,2x_2}\,i^{-2x_1x_2} \;=\; i^{\,x_1}\,(-1)^{x_2}\,(-1)^{x_1x_2},$$

which is S on the upper wire, Z on the lower, and CZ: $U = (S \otimes Z)\,\mathrm{CZ}$. The four values are $1, -1, i, i$ on $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. ■

Coecke's Sketchbook

Start from the recipe's bare picture, which is $\frac{1}{4}U$. Each step names its rule, and every number is written.

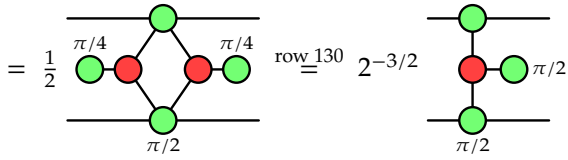
Step (i): the sandwiches (Result 17.2, row 129, twice). The second eighth sits between the first and second joiners, and the fourth between the third and fourth. Each sandwich becomes a hub with an eighth hanging from it, at $2^{-1/2}$ each: the picture is $\frac{1}{2}$ times an eighth, a hub, an eighth, a hub.



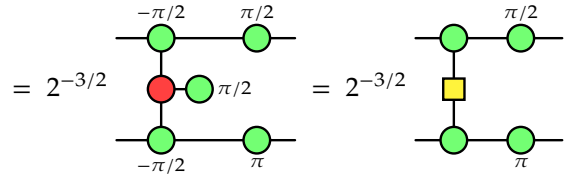
Step (ii): slide together (fusion, row 57). On the upper wire the two hubs' feet are joined by a plain cord and become one spider at rest, tied to both hubs. On the lower wire the first eighth, the first hub's foot, the third eighth and the second hub's foot are joined by plain cords and become one spider

turned $\pi/4 + 0 + \pi/4 + 0 = \pi/2$, tied to both hubs. No number.

Step (iii): two hubs into one (Result 17.3, row 130). The two hubs are tied to the same two spiders, so they become one hub whose end is the two ends slid together, an eighth and an eighth, a quarter; the price is $2^{-1/2}$.

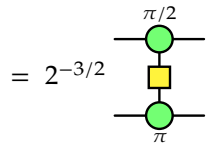


Step (iv): read the quarter hub (row 57 backwards; then a translation by evaluation, Result 14.5 and Result 16.4). Slide each wire's spider apart so that the part holding the hub's tie is turned a quarter back: the upper spider, at rest, becomes a foot turned $-\pi/2$ and a spider turned $\pi/2$ on the wire; the lower spider, turned $\pi/2$, becomes a foot turned $-\pi/2$ and a spider turned π. Now the hub, with its feet a quarter back and its end a quarter round, is Chapter 14's controlled turn at $\alpha = -\pi$ (row 103): bare, it is $\frac{1}{\sqrt{2}}\mathrm{CP}(-\pi) = \frac{1}{\sqrt{2}}\mathrm{CZ}$ (Result 14.5). A Hadamard edge between two spiders at rest is also, bare, $\frac{1}{\sqrt{2}}\mathrm{CZ}$ (the computation of Result 16.4 with an input leg on each spider, as in the proof of Result 17.16). So the one may be put in place of the other, with no number. **This step is a translation, checked by evaluating both sides, not a rewrite**: we derive no chain of moves from the quarter hub to the Hadamard edge here (colour change and two re-tyings would reach it, but Chapter 16 stated re-tying for nets only, and the number re-tying carries on this open picture is not worked out).



Step (v): slide together (row 57). The quarter and the half slide back into

the spiders at the ends of the Hadamard edge:



Step (vi): read off. The cooked picture is $\frac{1}{\sqrt{2}}(S \otimes Z)\,\mathrm{CZ}$ (Result 17.16). The recipe's picture, $\frac{1}{4}U$, equals $2^{-3/2} \cdot \frac{1}{\sqrt{2}}(S \otimes Z)\,\mathrm{CZ} = \frac{1}{4}(S \otimes Z)\,\mathrm{CZ}$. So $U = (S \otimes Z)\,\mathrm{CZ}$, and no eighth is left. ■ See `Code/Ch17/CookingDown.py`, which checks every picture of this chain against the one before, with its number.

The verdict

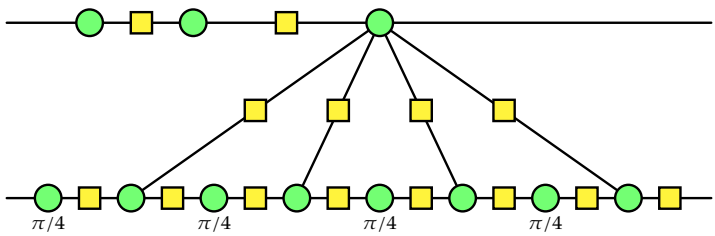
The blackboard wins, and the chapter says so. The parity count is four short steps and needs nothing but the fact that a CNOT adds bits. The sketchbook takes six moves and three numbers, and one of its moves, step (iv), is not a rewrite at all: the quarter hub and the Hadamard edge were matched by evaluating both, because this book has derived no chain of moves from the one to the other. A skeptic of pictures may say the proof leans on the blackboard at its hinge, and the skeptic is right.

What the pictures show is *why* the eighths cancel, and where. The second and fourth eighths hang from hubs tied to the same two wires, so they read the same parity and add; the first and third slide together on the lower wire, because they read the same bit. That is the blackboard's step 2, made visible: two classes, a quarter each. And the pictures win where the blackboard's count breaks. Put a Hadamard in the middle of the recipe, and the lower wire no longer carries a parity of the inputs; the bookkeeping of steps 1 and 2 stops. The gadget moves do not care: a hub is a hub wherever it came from. That is where Kissinger and van de Wetering found their gains, on circuits "that contain many Hadamard gates". We state that from their results, and have not proved it here.

§17.7 The Uncooked Net

Here is the chapter's master diagram: the graph-like net of the recipe, built in §17.1, before any cooking. Every spider is green. Every tie between spiders is a Hadamard edge, drawn with its yellow box, as the ties of the Crab's tangle wear yellow sleeves.

On the upper wire, from left to right: the upper input's lead, running plain to a small spider at rest; a Hadamard edge to a second small spider at rest; a Hadamard edge to the big spider at rest, which is the four controls slid together; and the upper output's lead, running plain from the big spider. On the lower wire, from left to right: the lower input's lead, running plain to the first eighth; then, alternating, a Hadamard edge to the first spider at rest (the first joiner's target), a Hadamard edge to the second eighth, a Hadamard edge to the second spider at rest, and so on, through the fourth eighth and the fourth spider at rest; and the lower output's lead, which carries one yellow box from the fourth spider at rest to the output. From the big spider, four ties fan down, each a Hadamard edge, one to each spider at rest on the lower wire. No two lines cross. We call the picture **The Uncooked Net**.



Result 17.18 (The Uncooked Net). Exactly,

$$\text{The Uncooked Net} \;=\; \tfrac{1}{4}\,U \;=\; 2^{-3/2} \times \text{the cooked picture.}$$

It has four eighths, so its picture's count is the recipe's T-count, 4.

Proof. **On the blackboard.** It is the recipe's bare picture after colour change, fusion and the two dummies of §17.1, each of which is exact, and the recipe's picture is $\frac{1}{4}U$, with U as computed in §17.6. One can also sum

over the bits of the net's eleven spiders directly, with a sign $(-1)^{xy}$ and a factor $\frac{1}{\sqrt{2}}$ for each yellow box between bits x and y; the sum gives the same matrix. **In the sketchbook.** The chain of §17.6, run on the recipe's picture, which is this net up to the exact moves of §17.1 (rows 56, 57 and 58). ■ See `Code/Ch17/FourEighths.py`, which builds this picture and checks it against the recipe's picture; and `Code/Ch17/CookingDown.py`, where PyZX's `full_reduce` cooks the net by its own moves, re-tyings, pivots and gadget moves, to the cooked picture with the number $2^{-3/2}$ written beside it, and `extract_circuit` returns $(S \otimes Z)$ CZ.

The net has fifteen vertices: its two inputs and two outputs, and eleven spiders, four of them eighths. It is what the dialogue's tangle hangs as, under the beam: a long picture, drawn with nothing spent, and nothing yet gained.

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§17.8 What a Rewrite Does Not Promise

Take stock. Every diagram is a non-zero number times a net (Result 17.1), and for a circuit the number can be tracked; our recipe of four eighths is *The Uncooked Net*, exactly (Result 17.18). A turn between two joiners hangs from a hub (Result 17.2), and two hubs on the same spiders add (Result 17.3), so the recipe's four eighths meet and become none (Result 17.4, proved twice). Re-tying and the pivot clear a net of its interior quarters and halves (Result 17.5), and gadgets carry the eighths together (Result 17.6); phase teleportation writes the result back into the original circuit (Result 17.7). A pattern is a picture read wire by wire (Result 17.8); its readings are coin tosses (Result 17.9), but a lit half travels forward (Results 17.10 and 17.11), and with a flow the corrections make every branch the same (Result 17.12). The V has no such order (Result 17.13). A net from a circuit has one, and re-tyings and pivots keep it (Result 17.14); a joiner at the ends unties a doubled tie (Result 17.15), our recipe comes out exactly (Result 17.16), and a general circuit comes out up to a global phase (Result 17.17).

Now the honest part. **A rewrite proves that two processes are equal, and**

nothing else. No move in this chapter proves that a count of eighths is the least possible. PyZX's numbers are what a heuristic found, and they beat, match or lose to other methods circuit by circuit: on twenty of the thirty-six benchmarks they came out exactly where the previous best already was. No move proves a running time; the $O(|V|^3)$ bounds are the sources' counts of steps, and the speeds are measurements. Extraction of a general diagram is not claimed to be efficient, and the survey gives reasons to doubt it. The headline itself was won by the blackboard, and the sketchbook's proof leaned on it at step (iv). Re-tying and the pivot were used as Chapter 16 stated them; their derivations by the rules are in Duncan and co-authors' Appendix B.3, and are not in this book. And every picture in the chapter is exact, number included, except for extraction in general, which gives the circuit back up to a global phase.

What the pictures do give is a kind of freedom. A recipe fixes when each step is done; a net forgets it and keeps only which wires are joined and how each is turned. That forgetting is what lets two eighths written far apart find each other on a hub. And the same forgetting explains the one-way machine: a net can be read in any order that its ties allow, and the order that makes the readings safe is the order that lets a recipe be written down again.

The chapter's question has an answer. A cheaper recipe is found by drawing the recipe as a net, which costs nothing, and cooking the net: sliding like spiders together, re-tying, and hanging turns from hubs until the turns that read the same wires meet and add. It is known to do the same thing because every move is an equality, and the moves themselves are the certificate; the same moves, run on the old recipe followed by the new one reversed, cook it down to bare wires. The cooked net can then be run by reading, if its readings can be sent forward in order, or taken back out as a recipe by the same order, frontier by frontier. For our recipe the four eighths vanished, and the recipe that came out needs none.

But every picture in this chapter was noiseless. Every joiner joined, and every reading read what the state held. Real hardware is not like that: now

and then a reading comes out false. Chapter 18, *Against Noise*, answers with a plain remedy: make copies, and take a vote. The picture of making copies is one this book already knows. Its encoder is a single green spider.

Ricercar with Repeats

鈴はただ
隣に問ふや
虎落笛

suzu wa tada
tonari ni tou ya
mogaribue

The little bell asks
only of its neighbours —
the fence whistles all night.

[*The Crab's front hall, on a late-winter evening, the day after the kitchen. The storm is at its height. The Tortoise is already in, by the hall table, perfectly dry. In the corner by the stairs, on the settle, something is curled that may be asleep and may not. The knocker goes, and the Crab opens the front door on a wet Achilles and a great deal of wind.*]

[*The hall the wind comes into is long and narrow and lit early, and the Crab keeps it, like most things he owns, in threes. There are three lamps: one on its bracket by the front door, one on the hall table halfway down, and one on its bracket by the kitchen door at the far end, where the hall clock ticks on the wall. There are three coat-pegs by the door, and three steps up to the landing where the stairs turn. In the corner at the foot of them stands the old settle, with a cushion at its end. The open door lets the whole storm in at once, and all three flames flatten and gutter together; for a moment the hall is half in the dark, and which of the three would stay*

in, nobody could have said. Over the door the fanlight's three panes rattle in their frame, and the rain comes at them in handfuls; at the narrow window beside the door it stands in drops on the glass; along the floor the draught comes feeling in over the worn sill, past a squat iron doorstop, across the flags. On the wall above the settle hangs a print in a plain dark frame: Escher's Order and Chaos*, a regular star-shaped solid at the centre, with broken things scattered round it. The table lamp reaches it only at a slant and leaves its lower corner in shadow. It hung there before the storm got up this morning, and it will hang there when the storm has blown itself out. On the settle beneath it the cushion keeps a dent that is, looked at again, quite smooth, and there is a warmth in that corner that may be the house's own. The clock ticks on under the wind, heard in the lulls and lost in the gusts.*]

Crab: In, in, before the door takes you with it!

Achilles: [*getting the door shut behind him with his shoulder*] I had the wind at my back the whole length of the lane and the rain in my face all the way. I don't know how it managed both.

Crab: Weather like this keeps to no side. Off with that scarf; it's making a pond.

[*Achilles unwinds his scarf, soaked through, and hangs it on a coat-peg by the door.*]

[*With the door shut the three flames stand up again one at a time, the one by the door last, and the hall comes back into its light. Under the peg the scarf begins at once to drip onto the mat, and Achilles stamps the wet out of his boots and blows into his hands.*]

Achilles: [*to the Tortoise*] And how are you dry?

Tortoise: The shell again. It's what it's for.

Achilles: One day I'll borrow it.

Tortoise: One day I'll lend it, and you'll find it heavier than it looks.

[*Achilles turns from the door and stops. Down the whole length of the hall, at head height, runs a line of cords and curtain rings, along the wall the hall table stands against.*]

Crab: [*proudly*] My storm line. I strung it this morning, for the storm. A message goes in here at the front door, runs the length of the hall with the draught at it all the way, and comes out at the kitchen.

[*At the door end the line hangs from the front-door hook, which also holds a brass dial under a hinged cover; at the kitchen end, from the kitchen-door hook, which also holds a small lidded glass. Every ring on the line hangs upright, facing into the hall, with a bead on its band, and every bead sits stiff on its band wherever it is turned and stands at rest at the top of its ring; turned a half, a bead stands at the ring's foot. Every cord is knotted at both its ends, to a ring or to a hook; none passes through a ring.*]

[*From the door end. The lead-in runs from the front-door hook to a green ring just inside the door: the parting ring. From the parting ring three strands run side by side down the hall, the first nearest the wall and the third furthest from it. The first strand's first length runs from the parting ring to a red ring hanging in the draught by the door, the first weather ring; the middle strand's first length runs from the parting ring to the middle weather ring; the third strand's first length runs from the parting ring to the third weather ring.*]

[*Mid-hall. From the first weather ring a cord runs on to a green ring, the first tell ring; from the middle weather ring a cord runs to the middle tell ring; from the third weather ring a cord runs to the third tell ring. Below the strands hang small brass bells, each on a red ring that hangs in the V of its ties, a little towards the kitchen from the tell rings and below the gap between its strands, so that from underneath, looking up, it is seen between them. The first bell hangs between the first and middle strands: the first bell's tie to the first tell ring runs up on one side, and the first bell's tie to the middle tell ring on the other. The second bell hangs between the middle and third strands: the second bell's tie to the middle tell ring runs up on one side, and the second bell's tie to the third tell ring on the other.*]

[*Towards the kitchen. From the first tell ring a cord runs to a red ring within easy reach of the Crab's hand, the first mending ring; from the middle tell ring a cord runs to the middle mending ring; from the third tell ring a cord runs to the third mending ring. From the first mending ring a cord runs on to a green ring by the kitchen door, the joining ring; from the middle mending ring another runs to the joining ring; from the third mending ring another. From the joining ring the*

lead-out runs to the kitchen-door hook. Nothing crosses.]

Achilles: It looks like the rigging of a very small ship.

Crab: The rings and the bells are from Delft, from the bellfounder across the canal from the old man's bench. He says a line like this will carry a message through any weather worth the name. Tonight I mean to find out.

[*The three lamps light the line in three reaches: the one by the door the lead-in and the first lengths of the strands, where the draught is; the table lamp, halfway down, the lengths on either side of the tell rings and the ties that come down to the bells; the lamp by the kitchen door the last lengths into the joining ring, and the lead-out. The rings are painted, the beads are glass, and each bell carries the founder's mark small on its shoulder. And when nobody in the hall is looking at it, the whole line seems to give off a note of its own under the wind, too low to be sure of; the moment anyone turns to listen for it, there is only the storm.*]

[*Achilles, turning to take it all in, sees the settle in the corner by the stairs.*]

Achilles: Oh — and you're here too. How do you find the weather?

Cat: [*not troubling to open more than one eye*] Comfortably, and not.

Achilles: That's no sort of—

Tortoise: A loose question buys a loose answer. It always has, in that corner.

Crab: There's tea on the hall table, and bread and cheese. [*He pours for all three, and sets the pot down at the front of the hall table, at its kitchen end.*] And Achilles: I did as you said. Three for every one.

Achilles: As I said?

Crab: Yesterday, at my kitchen door, going out into the rain. You'll see.

Achilles: [*a slice of bread in one hand, the cup in the other*] I said it for a joke.

Crab: Most good advice begins as a joke. The trick is to hear it.

[*Achilles laughs under his breath and lowers himself onto the lowest of the three steps, the bread gone and the cup in both hands, his wet boots stretched out across the flags. The Tortoise has the hall chair by the table, her feet tucked in under her. At the end of the settle beside Achilles there is something curled, or there is a*

cushion with nobody on it; the warmth coming off that corner would do for either, and nobody looks at it straight for long enough to settle which.]

First Challenge — *Three for One*

[*The Crab goes through to the kitchen, and comes back with a tray of small tins, warm from the oven's warming drawer. He sets it on the hall table beside a fat notebook, at the table's door end. The table stands halfway down the hall against the wall under the storm line, and the Tortoise sits at it in the hall chair, facing it and the wall, her back to the hall and the door end at her left.*]

Crab: My tins: the kind you met yesterday, with three hollows in a row under three little lids. Plain ones tonight, no dials: each hollow is simply set dark or set lit, and when you lift its lid a glass drop shows which. Last night I set every tin three for every one: one setting, written in all three hollows alike.

Achilles: And the damp?

Crab: Goes both ways in a storm: a dark hollow comes up lit, or a lit one dark. [*He opens the notebook.*] My ledger.

Achilles: Go on.

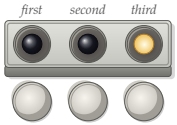
Crab: I lifted most of the tins this morning, and wrote down every hollow. About one hollow in ten read false. And the votes, trusting the two that agree: about three in a hundred went wrong. That's for tins set dark and tins set lit alike.

Achilles: [*working it out*] For a vote to go wrong, two hollows of the three must go false. Two at one in ten apiece is one in a hundred, and there are three ways to choose the two. Three in a hundred, a shade under. The ledger has it right.

Crab: [*pleased*] The ledger always has it right. It's the tins that don't.

Tortoise: Show him a vote.

[*The Crab takes a tin from the tray and sets it on the table in front of the Tortoise, a long side towards her: three lids down in a row, the one setting under all three, and nobody in the hall knowing which. He lifts the first lid, the one at her left. It comes away in his claw and will not seat again; the drop beneath it is dark. The other two lids have not stirred. He lifts the second: dark. He lifts the third: lit. Each lid lies on the table just in front of its own hollow, the right way up, and the tin is spent. What has changed: all three hollows read, and one of them out of step with the others. What has held: the two that agree.*]



A tin, spent

Crab: Dark, dark, lit. The damp got the last one. Trust the two: dark. [*He sets the spent tin back on the tray, its lids beside it, and turns to the back of the ledger, where a column of settings runs down the page, and runs a claw down it.*] Set dark. The vote has it.

[*A gust leans on the front door, and the lamp beside it ducks and gutters while the other two burn on. The next takes the table lamp, and the door's stands up again; the one after, the kitchen lamp alone. The hall, lit by whichever two are standing, hardly darkens at all.*]

Achilles: Then do the same with your storm line! Three copies of whatever you send, and a vote at the kitchen end.

Crab: Whatever I send is set on that. [*He goes to the front-door hook and lifts the hinged cover of the brass dial.*] My door dial: a pointer, and six marks round it. Dark and lit, and the turns: at rest, a quarter round, a half, a quarter back. Where the pointer stands is the message, and the cover is there so that whoever sets it can keep it to himself.

Tortoise: Dark and lit I can see you copying. But what are three copies of a message a quarter round?

Achilles: Three messages a quarter round, of course. One on each strand.

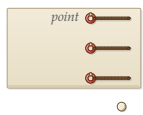
Tortoise: Let's see. [*She takes her drawing-cloth from her work-bag and spreads*

it on the hall table in front of her, beside the tray, its near edge on her side and its right-hand end towards the teapot, and brings out cords and her little box of rings and beads, each ring opening at a small hinge so that it clips on or off a cord anywhere. On her cloth, as ever, a bead at rest sits at its ring's far side as she looks down, and a turn goes round clockwise.] My drawing-cloth and my rings, as ever: what I lay on it is a picture of a thing, not the thing. A ring with a single cord is a point, a message laid on the cloth: dark is a red point at rest, lit a red point at a half, and a turned message a green point, turned.

[*Achilles comes across the hall to stand at her shoulder. She lays a green ring at rest with three cords running from it to the right, and its lead running to the left, empty. Beside the empty lead she sets a red point at rest.*]

Tortoise: And one old rule, for red points only. A dark or a lit point, pushed into a green ring at rest, comes out on every other cord of it: the same point on each.

[*The cloth as it stands: the green ring at rest, its three cords, its empty lead, and the dark point beside. The Tortoise knots the point's cord to the lead and pushes the point in. The point goes into the ring along the lead, its cord and the lead swallowed with it, and the ring is gone; at the end of each of the three cords sits a red point at rest, at the end where the cord fanned out from the ring. She takes a small bone counter from her work-bag and sets it off the drawing-cloth, just beyond its near edge at its right-hand end, by the teapot. What has changed: one point has become three, and the green ring has gone into them. What has held: dark. Each of the three is as dark as the one.*]



Out at the ends

Achilles: Three dark copies. That's what I asked for.

Tortoise: It is. Now the other kind. [*She gathers the dark points off the drawing-cloth into her box of rings, their cords into her work-bag and the counter from beside the teapot after them, and on the bare cloth she lays a second green ring at rest, three cords to the right and an empty lead to the left, and beside the lead a green point, its*

bead turned a quarter round.]

[*The cloth as it stands: the second green ring at rest, its three cords, its empty lead, and the turned point beside. She knots the point's cord to the lead and pushes. This time nothing comes out on the cords. Green meets green: the point slides into the ring along the lead, its cord and the lead going in with it, the two are one, and their beads add, so the ring's bead, which was at rest, now stands a quarter round. What lies on the cloth now: one green ring turned a quarter round, its three cords running from it, and at the end of each cord nothing at all. What has changed: the ring has taken the turn into itself. What has held: one ring, with all three cords still on it. No turned point has come out anywhere.*]



Taken in

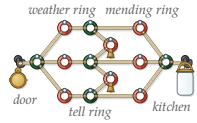
Achilles: But I wanted three points, each a quarter round, one at the end of each cord. You've given me one ring, turned, holding all three.

Tortoise: You remember the press from Delft. It copied its own kind, and made a pair of every other.

Achilles: [*slowly*] So the three cords aren't three copies. They're one turn, shared out along three cords, and not one of them has it alone. ...Oh. You can't copy a turn; you can only spread it.

[*Over the front door, the lamp at the corner of the lane comes in through the fanlight's three panes of old rippled glass, not as a lamp but as one blur of light laid across all three. Each pane has a share of it; not one of them has the lamp.*]

Achilles: [*looking up from the cloth to the storm line, and along it to the door*] Just inside the front door, a green ring: one lead in, and three strands out. That's your parting ring.



Strung for the storm

Crab: [*with a little bow to his line*] It is. It spreads whatever comes in from the dial over all three strands, and copies nothing it shouldn't. And that is why nobody looks at my strands, ever. A look at any one of them spoils a turned message on all three.

Tortoise: [*gathering the turned ring off the drawing-cloth into her box, and its cords into her work-bag*] So a message on the line isn't three messages. It's one message, hanging on three strands.

Achilles: But then how does anyone vote? To vote you must look at each copy, and you've said looking spoils a turned one.

[*The Crab only smiles, and looks down the hall at the bells.*]

Second Challenge — *The Blind Vote*

Crab: Nobody looks. We listen. [*He goes to stand mid-hall, below the tell rings.*] My bells. Each one hangs from two strands side by side, tied up to their tell rings. It rings when those two strands disagree, one dark and one lit, and keeps quiet when they agree; and it never tells which is which.

Achilles: Then what use is it? I'd want to know which strand is which.

Tortoise: That's the one thing you mustn't want. A bell that told you which would be a look.

Achilles: So it only hears whether the two agree. A strange thing to put to a bell.

Tortoise: Ask the pair, not the one.

[*Along the foot of the front door the draught has found the one place where the old*

sill has worn away from the door's edge, and there, and only there, it whistles, thin and high. Everywhere else along the door, where wood meets wood true, there is no sound at all. It is the same thin note on a mild night as on a bitter one, wet or dry; it sounds only where two edges fail to meet.]

Crab: [*moving along to the red rings nearest the kitchen*] And these are my mending rings. I turn their beads with my own claw, a half at a time, and a bead I've turned stays where I put it until I turn it back. My rule goes by the bells and nothing else: both silent, I touch nothing; the first bell alone, I turn the first mending ring; both bells, the middle one; the second bell alone, the third.

[*He lifts the lidded glass down from the kitchen-door hook.*]

Crab: And my kitchen glass, which reads what comes out. Its lid has the same six marks as the door dial: lift it at a mark, and the glass shows clear if the message is the one at that mark, cloudy if it's that mark's opposite — dark and lit, at rest and a half, a quarter round and a quarter back. If the message is neither the mark nor its opposite, it comes up clear or cloudy like a tossed coin.

Achilles: So the one who sent it lifts it at the mark he sent.

Crab: And clear means it came through as sent. A message goes once and is read once. [*He hangs the glass back on its hook, lid down.*] The glass's lid goes back afterwards, unlike my tins; the message it read is spent.

Achilles: Let me send one.

Crab: Set it where I can't see. I don't want to know what you've sent. I'll mend it all the same.

[*Achilles goes to the front door and lifts the dial's cover with his back to the room, turns the pointer, and lets the cover fall. The line as it stands: the pointer at his secret mark under the cover; every weather ring, tell ring and mending ring at rest; the bells hanging still; the glass on its hook, lid down, clear. He lets the dial's catch go, and it clicks. Nothing visible runs along the cords; they hardly stir. Down by the door the weather rings hang in the draught. Neither bell makes a sound. The Crab's claws stay folded. What has changed: the message has gone in, and waits now at the mending rings. What has held: every bead at rest, and both bells silent.*]

Crab: Not a sound. I touch nothing.

[*Achilles walks the length of the hall to the kitchen door, and lifts the lid of the glass at his mark. The glass stays clear.*]

Achilles: Clear. It came through as I sent it. I set a quarter round, and nobody knew.

Crab: [*resetting the glass: lid down, clear*] A quiet run. The draught was kind to you. Again.

[*Achilles, grinning, finishes his tea in one long swallow and hands the cup to the Tortoise, who fills it again and keeps it warm between her hands for him. The Crab, who has not sat down once since the knocker went, still does not.*]

[*Achilles goes back to the door and sets the dial under its cover again. The Crab runs a claw along his mending rings without looking up at them: every bead at rest, where he left it. The line as it stands: the pointer at Achilles's secret mark; every bead at rest; the bells still; the glass reset. Achilles lets the catch go.*]

[*A gust shoves in under the front door, and the draught goes hard along the floor and up. Down by the door the middle weather ring's bead is flung round a half as the message passes it, and falls back to rest behind it: too quick, from where the company stands, for anyone there to say which ring it was. Mid-hall the first bell rings; then the second, too. Then both fall still. What has changed: both bells have spoken. What has held: every bead that anyone can see, at rest; and the message, whatever it is, waiting at the mending rings.*]

[*Every head in the hall turns, for no reason anyone could give, toward the settle in the corner; and then back.*]

Achilles: Both! Which strand was it? Let me just see which strand it was — [*He hurries up the hall from the door to the tell rings and reaches up toward the middle one.*]

Crab: [*hurrying along the line from his mending rings and catching his wrist, gently*] No. A look at any one strand spoils a turned message on all three, and yours may be turned. I don't know what you sent, and I won't learn it that way.

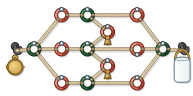
Achilles: Then how will you know what to mend?

Crab: [*a claw lifted toward the bells, not the strands*] Ask the pair, not the one.

Crab: Both bells rang. So the middle strand disagrees with both its neighbours, and it's the middle I mend.

[*The middle mending ring as it stands: its bead at rest, like its neighbours'. The Crab goes back along the line to it and turns the bead round a half with one claw, and takes the claw away. The bead stays at the half. The first and third mending rings he does not touch, and he has not so much as glanced at a strand. What has changed: one bead, by a half. What has held: the other mending rings; the mark under the dial's cover; and what the Crab knows of the message, which is nothing.*]

[*Achilles goes down the hall to the kitchen door and lifts the glass's lid at his mark. The glass stays clear.*]



Lifted at the kitchen

Achilles: Clear. [*He stares at it.*] A quarter round again, through a gust, and it came out exactly as I sent it. And you mended it without knowing what it was.

[*Out in the lane the gale drops, all at once, into its one lull of the evening. In the sudden quiet the table lamp's flame stands up tall and perfectly still, and its light reaches across the hall and lies, for the first time tonight, full on the print above the settle.*]

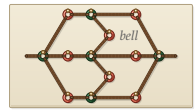
[*The Tortoise turns in her chair and looks across the hall, her glance passing over the settle without resting there, to the print on the wall opposite the storm line: Escher's* Order and Chaos.]

Tortoise: Like your print: a regular star-shaped solid at the centre, with broken things scattered round it. The storm breaks what it likes; the middle keeps.

Achilles: But how? The bells heard the gust. Why didn't they hear the message too? It was right there on the strands they're tied to.

Tortoise: [*as Achilles lets the glass's lid back down and comes back up the hall to her shoulder*] The cloth can show you.

[*On the bare drawing-cloth she lays out ring for ring the line that hangs above her head, its door end at her left and its first strand furthest from her, as the first hangs nearest the wall: a lead in to a green ring; from it three strands, and on each strand in order a red ring, a green ring and a red ring; the three strands meeting again in a last green ring, and a lead out. Between the strands she lays two red rings, each a little towards the kitchen end from the green rings of the strands it lies between and tied to them, as the bells hang below the gaps in the hall. Every cord of hers runs where one of his does, and every bead is at rest. Every tie she leaves slack, long enough to follow its rings wherever they slide.*]



On the cloth, at rest

Tortoise: Your storm line, drawn.

Crab: [*coming up the hall from his mending rings to the table, and looking from the cloth to the hall and back*] Ring for ring.

[*The gale has come back to the door, and the flames bow to it again. The Crab draws up a kitchen chair at last and sits, elbows on the table, and Achilles leans in over the Tortoise's shoulder with a slice of bread forgotten in his hand.*]

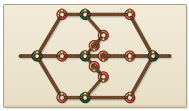
Tortoise: Now this evening's second run. A weather ring the gust turned, I draw at a half; a bell that rang, at a half; a mending ring you turned, at a half.

[*The cloth as it stands: every bead at rest. She turns beads round a half: on the middle strand's red ring nearest the door, on each bell's ring, and on the middle strand's red ring nearest the kitchen. Nothing else moves. What has changed: those beads, each by a half. What has held: every green ring at rest, the first and third strands' red rings at rest, and every cord.*]

Tortoise: And the rule for a half that meets a ring at rest. A red half on one

cord of a green ring at rest goes through it, and comes out on every other cord of that ring. Watch the gust's half.

[*The middle strand as it stands: the red weather ring at a half; the green tell ring at rest, with the weather cord coming in, the mending cord going out, and the ties to both bells running from it. The Tortoise slides the weather ring's half along its cord into the tell ring. The tell ring's bead does not move. The half goes through it and comes out on every other cord of it at once: a red half on the tie to the first bell, a red half on the tie to the second bell, and a red half on the cord onward to the mending ring. The weather ring is left at rest. What has changed: the one half has become a half on each of the tell ring's other cords. What has held: the tell ring at rest; and the first and third strands, untouched.*]



Gone through

Achilles: Into both bells at once. And on into the mending ring.

Tortoise: Red meets red, and their beads add. Two halves make none.

[*She slides the half on each tie down into its bell. Each bell was at a half, and a half and a half make a whole turn, which is rest. She slides the half on the mending cord into the middle mending ring: it was at a half, and a half more brings it too to rest. What lies on the cloth now: every red ring at rest, the weather rings, the bells and the mending rings alike. What has changed: the gust's half, the bells' halves and the Crab's half have met and cancelled. What has held: the green rings, all at rest, and every cord.*]

Crab: So my half went to meet the gust's.

Tortoise: Exactly there. And a red ring at rest on a strand is only cord. So now the green rings may slide together along bare cord.

[*The cloth as it stands: along each strand, green rings at rest with bare cord between them; the bells, at rest, lying between the strands. The Tortoise slides the first tell ring along the first strand's bare cord into the parting ring; then the joining ring,*

along the first strand's cord from the kitchen end, into the same ring. That ring now holds the lead in, the lead out, the first bell's tie from the first strand, and the middle and third strands at both their ends. She slides the middle tell ring into it along the middle strand's cord from the door end; what is left of the middle strand now runs from that ring back to itself, a loop, and a loop comes off, so she lifts it off into her work-bag. She does the same with the third tell ring, and the third strand's loop. What lies on the cloth now: one green ring at rest, with the lead in and the lead out on it, and each bell tied twice to it. What has changed: every green ring has gone into one, and the strands' loops are gone. What has held: every bead at rest, and every tie of each bell still where it runs.]

Tortoise: And a bell at rest, tied twice to one green ring and to nothing else, comes off it.

[*She lifts the first bell off into her box of rings, its ties into her work-bag, and then the second. What lies on the cloth now: one green ring at rest, with a lead in and a lead out, and nothing else; and a green ring at rest with only a lead in and a lead out is just cord. She unclips the ring into her box, knotting the lead in to the lead out where it held them, and draws the cord straight: a plain cord from the door end to the kitchen end. What has changed: the whole storm line, gust and bells and mending, has come down to one plain cord. What has held: whatever goes in at one end comes out at the other, untouched.*]



Drawn straight

Achilles: A plain cord. [*He looks up at the line, and back.*] So what you're saying is that the bells only ever heard the gust.

Tortoise: The gust, and where it was. Never your message.

Achilles: Because whatever I set, the parting ring hangs it on the strands as all-dark and all-lit together, so much of each. A gust on the middle strand spoils both the same way: in each, the middle differs from its neighbours. So the bells hear the gust, and my message never comes into it.

Crab: Just so.

Achilles: Ask the pair, not the one.

Achilles: [*delighted, looking up at the bells*] So nothing can get past the bells.

Tortoise: Will you send me one more, and let me stand by the middle strand?

Third Challenge — *The Turn the Bells Miss*

[*The middle mending ring as it stands: at a half, where the Crab left it. He goes along to it and turns it back round a half, and it stands at rest again, like its neighbours; he knows where he left it, and has no need to look at a strand to do it. He resets the glass: lid down, clear.*]

Crab: Everything back as it was. [*He comes back up the hall to the table, and watches the Tortoise take her place mid-hall, by the middle tell ring.*] You won't look at it?

Tortoise: I shan't look at anything. I may turn something.

Crab: [*uneasily*] Turning's not looking. I suppose.

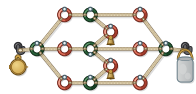
[*The Tortoise gives him a look of perfect mildness and folds her claws in front of her. Achilles, on his way to the door, pats the Crab on the shoulder in passing, and the Crab smooths down his waistcoat as if that had been the trouble.*]

Achilles: [*at the dial*] Setting. [*He turns the pointer under the cover and lets the cover fall.*] Secret again.

[*The line as it stands: the pointer at Achilles's secret mark; every bead at rest; the bells still; the glass reset. Achilles lets the catch go. As the message reaches mid-hall, the Tortoise puts one claw to the middle tell ring and turns its green bead round a half, and holds it there while the message goes by. The ring does not move on its cords; only its bead turns. Neither bell makes a sound. The Crab, watching the bells and nothing else, keeps his claws folded. What has changed: the middle tell ring's bead, at a half. What has held: both bells silent, and every other bead at rest.*]

Crab: Silent, both. Nothing to mend.

[*Achilles walks down the hall to the kitchen door and lifts the glass's lid at his mark. The glass clouds over.*]



Turned in passing

Achilles: Cloudy! ...Oh. The bells never heard a thing, and it's spoilt all the same.

[*The middle tell ring as it stands: its bead at a half, under the Tortoise's claw. She turns it back round a half, to rest, as the Crab had it, and takes her claw away. The cords have not stirred.*]

Tortoise: There. And had you sent dark or lit, the glass would have come up clear. My kind of turn does nothing at all to dark and lit; it spoils only the turned ones.

Crab: [*staring at his bells*] Then my storm line is half deaf.

[*At the front door the draught whistles at its worn place as it has all evening, and the three flames bow to every gust it brings. But the damp has never needed the door. It has come in through the walls without a sound, and nothing in the hall has bowed to it: behind the coat-pegs the plaster has bloomed dark in a patch the size of a hand, the bread on the plate has gone soft at the crust, and Achilles's scarf on its peg is no drier than when he hung it.*]

Achilles: But why didn't they ring? She turned the middle strand, and they're both tied to it.

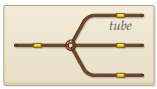
Tortoise: The gust makes a strand disagree with its neighbours, and the bells hear that. My turn makes it disagree with nothing: it turns the message round, and the bells were never asked about that.

Achilles: Then what would hear it?

Tortoise: The same line in the other colour. [*She goes back to her chair at*

the hall table, Achilles letting the glass's lid back down over the cloud and coming up the hall to her shoulder, and takes her yellow tubes from her bag.] My yellow tubes, as ever. A ring with a tube on every cord comes out the other colour, its bead where it was; and two tubes in a row are no tube at all, so I may put a pair on anywhere I like.

[*She lifts the plain cord off the drawing-cloth into her work-bag, and on the bare cloth lays a fresh parting ring, its lead coming in from her left as on the line: a green ring at rest, a lead in, three strands out. On each of its cords she slides a pair of tubes, which changes nothing. Then she draws the ring through the near tube of every pair: the tubes go into it and are gone, and the ring comes out red, its bead still at rest. The far tube of each pair stays on its cord. What lies on the cloth now: a red ring at rest, with a tube on its lead in and a tube on each strand. What has changed: the ring's colour. What has held: its bead, its cords, and everything the picture does.*]



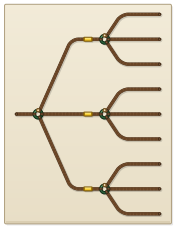
Come out red

Tortoise: The same ring, asking in the other colour. Hang a whole line from it, and its bells hear my kind of turn.

Achilles: And the gust?

Tortoise: They're deaf to the gust. So we lay both.

[*She lifts the red ring off the drawing-cloth into her box, its cords and tubes into her work-bag, and on the bare cloth lays a green parting ring, its lead coming in from her left, a lead in and three strands out. On each of the three strands she slides a tube, and past each tube she sets another green parting ring, each with three strands of its own. What lies on the cloth now: one ring spreading to three, three tubes, and three rings each spreading to three: nine strands, in groups of three.*]



Spread out again

Achilles: Nine strands for one message!

Tortoise: [*laying a pair of bells along each group, as on the Crab's line, and across the groups one more pair, for the other kind of turn*] Each group of three hears a gust on its own strands. The pair across the groups hears my kind of turn, on any strand of the nine.

Crab: And a turn that's part the one kind and part the other? Storms aren't tidy, and my draught doesn't choose.

Tortoise: Whatever turns one strand is a sum of four parts: no turn, the gust's kind, my kind, and both at once. The bells on the nine sort out each part, and the mending undoes each. Undo every part, and you have undone the sum.

Crab: Then I'll hang a bell that asks *everything*. [*He goes to the hall-table drawer.*]

Achilles: Everything? What would a bell like that ask?

Fourth Challenge — *What a Bell May Ask*

Crab: [*coming back from the drawer with a small brass bell on a spring clip*] This. My clip-bell, from the same bellfounder in Delft; he sent it with the others, as a sample. It has no ties at all: it clips straight onto one ring, and it rings when that one strand is lit, and keeps quiet when it's dark.

Achilles: It asks a single strand, straight out.

[*The clip-bell lies in his claw like something out of a jeweller's window: no bigger than a thimble, its brass as bright as a new coin, the founder's mark stamped small on its shoulder, and its clip so fine that it trembles when the Crab breathes on it. Of all the things from Delft in the hall, it is the prettiest.*]

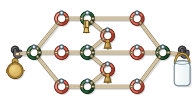
Crab: [*proudly, clipping it to the first tell ring*] Ask the one, not the pair.

[*The first tell ring as it stands: green, at rest, the first strand's cords on either side of it and the first bell's tie hanging from it. The Crab springs the clip-bell onto the foot of its band, and the little bell hangs straight down from the clip. The ring does not move on its cords; nothing is tied or untied. The bells below hang as they were. What has changed: a small bell rides now on the first strand's tell ring. What has held: every cord, and every bead.*]

Crab: Why ask a pair a riddle when you can ask one strand straight?

Tortoise: Send him one, Achilles. Dark.

[*The Crab goes down to the kitchen door and resets the glass: lid down, and the cloud the last message left in it clears, until the glass stands clear again. Achilles goes to the front door, sets the dial to dark, openly, and lets the catch go. The clip-bell hangs silent; the tied bells are silent; the Crab touches nothing. Achilles walks down to the kitchen door and lifts the glass's lid at dark. The glass stays clear.*]



Clipped on

Achilles: Dark, and clear.

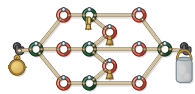
[*The Crab resets the glass. Achilles goes back to the door, sets the dial to lit, and lets the catch go. The clip-bell rings; the tied bells stay silent; the Crab touches nothing. Achilles walks down to the glass again and lifts the lid at lit. The glass stays clear.*]

Crab: [*triumphant*] It told me the message itself! Lit, and it said lit, and the glass as clear as water.

Tortoise: Now let me set one. [*The Crab resets the glass. The Tortoise goes to the dial, lifts the cover with her back to the room, turns the pointer, and lets the*

cover fall.] Achilles, let it go.

[*The line as it stands: the pointer at the Tortoise's secret mark; every bead at rest; the clip-bell on the first tell ring, still; the tied bells still; the glass reset. Achilles comes back to the door and lets the catch go. The clip-bell rings. The tied bells stay silent, and the Crab touches nothing. The Tortoise walks to the kitchen door and lifts the glass's lid at her mark. The glass clouds over. What has changed: the clip-bell has spoken, and the glass is cloudy. What has held: no gust, no tied bell ringing, no claw on a mending ring. Nothing touched the message but the clip-bell's question.*]



Lifted at her mark

Tortoise: A quarter round. Cloudy.

Crab: [*stricken*] But it rang! It said lit!

Tortoise: It said the first strand was lit, and to say so it had to make it so, on all three. After that your message was lit and nothing else, and a lit message read at a quarter round is a tossed coin: it comes up cloudy as often as clear.

Tortoise: A bell that can tell you the message spoils every turned one.

Achilles: So even when the glass comes up clear after that bell, it's clear by luck.

[*Achilles, still by the front door where he let the catch go, puts his face close to the narrow window beside it to see how the lane is faring, and his own breath fogs the pane before he can make out anything at all. He draws back, and the fog goes, and there is the lane again, rain and lamplight.*]

Crab: [*slowly*] Then it's a look after all. A look with a bell on it. [*He regards the clip-bell ruefully.*] Well. It's a very pretty bell.

[*The clip-bell as it stands: on the first tell ring. The Crab comes back along the hall to it, springs the clip open and lifts it off. The ring hangs as it was, green, at rest, the first bell's tie still hanging from it; the tied bells below have not stirred. He puts the clip-bell back in the hall-table drawer, and shuts it. What has changed: the*

one-strand bell is gone from the line. What has held: every cord and every bead, as they were before it came.]

Crab: Ask the pair, not the one.

[*In the shut drawer the clip-bell knocks once against the wood as it settles, and is quiet. It is a pretty sample among the odd buttons now, and the hall is a little plainer without it.*]

Achilles: Then what may a bell ask, at all?

Tortoise: Anything whose answer is the same whatever the message. Whether two strands agree is such a question. Whether one strand is lit is not.

Achilles: And must it always be two strands at a time?

Tortoise: Not at all. [*Back in her chair from the kitchen door, where she has let the glass's lid back down over the cloud, and with Achilles come up the hall to her shoulder again, she lifts the nine strands off the drawing-cloth, their rings into her box and the rest into her work-bag, and on the bare cloth lays strands crosswise and lengthwise in a square weave. At every crossing she lays a bell of one colour, tied to the four lengths of strand that meet there; in the middle of every little square, a bell of the other colour, tied to the four lengths round it.*] Bells that ask four at once, and never the one.

Achilles: And your tins, Crab. If a vote of three is good, what about a vote of three votes?

Crab: [*turning the pages of the ledger*] I tried it, last night, on the top shelf: three tins to a message, a vote in each tin, and then a vote of the three. About two in a thousand went wrong.

Achilles: One in ten for a hollow, three in a hundred for a vote, two in a thousand for a vote of votes. Vote on the votes again, and again, and it goes to nothing!

Tortoise: While the damp is less than a half. Above that, every vote makes it worse.

Achilles: [*working it*] At six in ten a hollow, two of the three go false more

often than not. So the vote is wrong more often than a single hollow is. ...Worse, yes.

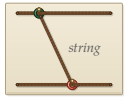
Crab: Then pray for a mild damp. [*He closes the ledger.*]

[*He goes through to the kitchen and comes back with the kettle, and freshens the pot. Achilles cuts himself another piece of cheese, and the Tortoise holds out her cup without looking up from the cloth. Down by the stairs the draught passes over the settle and stirs nothing there, if there is anything there to stir.*]

Achilles: And when a thing is safe inside its three strands, or its weave, how do you *do* anything with it? Join two, say, without unwrapping them?

Fifth Challenge — *Split and Merge*

Tortoise: With a joiner. [*She lifts the weave off the drawing-cloth into her box and her work-bag; lays a long cord across the bare cloth from her left to her right, the upper line, and another below it, nearer her, the lower; and takes a joiner from her box and clips it across them, its green ring on the upper line and its red ring on the lower a little to the right, so that the string between them slants.*] My joiner, the one you know: a green ring on the upper line, tied by a short string to a red ring on the lower. Upper lit, it flips the lower; upper dark, it leaves the lower be.



Clipped across

Achilles: But a joiner reaches right into both lines. Wouldn't it unwrap them?

Tortoise: Look at it again.

[*The joiner as it stands: the upper line running into the green ring and out again; the lower line running into the red ring and out again; the short string between*

them. The Tortoise loosens the short string at the red ring and draws it out long, so that it drops a little from the green ring, runs level beside the upper line for a way, and then drops to the red ring, and knots it again where it was. No ring has moved; no bead has turned; every cord ends where it ended. What has changed: only how it lies. The green ring now has one line coming in and two going out, the upper line and the long string beside it; the red ring has two coming in, the long string and the lower line, and one going out.]



Drawn out long

Tortoise: A split and a merge. The green ring splits the upper line in two, like your parting ring with two strands instead of three. The red ring merges the second of them with the lower line: it puts a question to the two of them, in the other colour, and answers.

[*Where the rain has been driven in under the front door, a thin shine of water has been spreading over the flags all evening. It reaches the iron doorstop and parts round it, and on the far side its two edges find each other again and are one shine, going on as before.*]

Crab: It answers? How?

Tortoise: Like a bell: silent, or ringing. My cloth can't ring. Have you a penny?

Crab: [*producing one from his waistcoat pocket*] My pocket penny, honest as they come; I've tossed it for the last slice of cheese for years. Heads, your merge is silent; tails, it rings.

Tortoise: For a joiner, the merge answers like a fair penny, whatever the two messages. Toss it.

[*The Crab tosses. The penny turns over and over in the lamplight, and he claps it onto the back of his claw: heads.*]

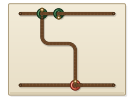
Crab: Heads. Silent.

Tortoise: Then the split and the merge are a plain joiner, just as they lie. Again.

[*He tosses again. The penny turns, and lands: tails.*]

Crab: Tails. It rings.

[*The joiner as it stands: the split and the merge, every bead at rest. The Tortoise sets a green ring turned a half on the long string, on its level run, just before it turns down to the red ring, on the merge's first cord in. She slides it back along the string, away from the red ring, into the green ring the string came from: green meets green, and the split's bead now stands at a half. Then she slides the split apart along the upper line: the split is left at rest, and a green ring at a half sits on the upper line just past it. What has changed: the half has moved from between the rings onto the upper line. What has held: the red ring, the lower line, the long string, and every knot.*]



Slid back and apart

Tortoise: Tails leaves a half behind. It isn't lost; it's only sitting on the upper line, where a hand can reach it.

Crab: Then my hand goes where it goes on my storm line. [*He sets a green ring turned a half on the upper line, beside hers, on the side away from the split.*]

[*The upper line as it stands: two green rings, each at a half, plain cord between them. The Tortoise slides one into the other. A half and a half make none, and the ring that is left stands at rest, with only the upper line through it: just cord, and she unclips it into her box. What lies on the cloth now: the split, the long string, the merge. What has changed: the half is gone. What has held: the joiner, whole.*]



Nothing left over

Crab: A joiner again.

Achilles: So joining two protected things is a split, and a merge, and a penny, and a hand.

Tortoise: And look what the rest of the evening was. Your parting ring splits one message three ways. A bell merges two strands and answers whether they agree. A joiner is a split and a merge. Everything tonight was one ring, asked in two colours.

[*Between gusts the three flames stand up together, and the hall is lit from end to end by one light in which nobody could say where one lamp's leaves off and the next begins. Only under the fanlight the colder glow from the lane lies across the warm: two lights, and the one floor under both.*]

[*The Crab puts the penny back in his waistcoat pocket. The Tortoise folds her drawing-cloth, rings and tubes and all, into her work-bag. The Crab stacks the spent tins on the tray beside the ledger.*]

Crab: The last of the pot. [*He shares it out, three ways, not quite evenly, and gives Achilles the bigger share.*]

[*Achilles takes it without argument and holds it in both hands, and the Tortoise, her work-bag on her knee, leans back in the hall chair. Outside the gale throws the rain at the fanlight in fistfuls, and the house leans into it, and holds.*]

Achilles: Listen to it out there. It hasn't let up once all evening.

Crab: Nor will it before morning. You'll not go out into that. There's a bed made up at the top of the stairs.

Achilles: [*glancing at his scarf on its peg*] My scarf thanks you. It's still wet through.

[*A clock in the hall strikes eight.*]

Tortoise: Eight. The storm has made a long evening of it.

[*Up the lane, a breath behind as it always is in wet weather, the church clock strikes eight too, the sound of it torn about by the wind. Achilles yawns, and does not trouble to hide it.*]

Crab: [*looking up at the storm line*] I'll leave it up while the storm lasts. Every message tonight came through it, and nobody looked.

Achilles: Set in secret, turned by a gust, mended by a claw that never knew what it mended. And out it came, exactly.

[*Every head in the hall turns toward the settle in the corner by the stairs.*]

[*The settle keeps its small mystery to the last. The cushion at its end holds a dent that is, looked at again, quite smooth; the warmth in that corner may be something curled there, or only the house's own, gathered into the corner out of the wind; and what may be an ear, or may be a fold in the cushion's cover, has not quite lain down all evening. Asleep or awake, the hall has let it be both through all the storm.*]

Tortoise: [*softly, to the Crab*] All evening you've asked nothing but pairs.

Crab: And never one thing, straight. [*He crosses to the settle.*] One question, then, and a straight one. Through all that storm — Cat, asleep, or awake?

Cat: Asleep.

[*The Cat settles to its one answer; and as the rest of it goes from the settle — the curled warmth, the tucked paws, the one ear that never quite lay down — the last of it to fade is the grin, which hangs a moment over the empty cushion, and then is gone too.*]

[*The settle is only a settle now: an old bench at the foot of the stairs, and a cushion with one plain dent in it, the warmth going slowly out of it. Outside the storm is at its height still and means to be till morning; it throws the rain at the fanlight and leans on the front door, and the draught whistles thin at its one worn place along the sill. Inside, the three lamps burn upright and steady. Down the length of the hall the storm line hangs as the Crab strung it this morning, from the front-door hook to the kitchen-door hook, every cord knotted where he knotted it: the lead-in and the first lengths in the door lamp's light, the lengths about the tell rings and the ties down to the bells in the light from the table, the last lengths and the lead-out by the kitchen. Now that nobody is attending to it, it may be giving off its low note again under the wind, or it may not. On the table the tins stand stacked on their tray beside the ledger, and the Tortoise's work-bag sits closed at her feet; on its peg Achilles's scarf drips more slowly onto the mat. Above the settle the print hangs in its plain dark frame, the table lamp's light no more than an edge on it again, one*

more of the Crab's things. At the top of the stairs a bed is made up and waiting. Nobody goes up to it yet: Achilles sits on the lowest step with the last of his tea in his hands, and the Crab looks along his line from end to end, and does not look at any one strand of it.]

Chapter 18: Against Noise

— Ricercar with Repeats —

Part V: Engineering the Diagram

Escher: "Order and Chaos" — a regular star-shaped solid at the centre, with broken things scattered round it

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How can a message that nobody is allowed to look at travel through a storm that turns it at random, and still arrive exactly as it was sent?

People have always had an answer to half of this question. When a line is bad, you say the thing three times. A clerk who cannot afford a mistake writes the figure into three ledgers, and if one of them later disagrees with the other two, he trusts the two. A computer does the same, silently and all the time. Gottesman points out, in a footnote, that an ordinary digital computer "is using a repetition code": each bit is held by many electrons, and after each step the bit is set back to whatever most of them say.

A quantum message breaks both halves of that answer. The first half, *say it three times*, fails because a quantum state cannot be copied. Chapter 11 proved it: no process makes two copies of every state. The second half, *trust the two that agree*, fails because to find out which copies agree you would have to look at them, and a look at a quantum message is not a harmless glance. It is a measurement, and a measurement changes the thing it measures. So the clerk's method is forbidden twice over.

And the storm is worse than the clerk's. A figure in a ledger is either right or wrong. A quantum message can be turned by any small amount, in any direction, and a turn of one part in a thousand is already a different

message. There seems to be a whole continuum of ways to go wrong, and no finite list of repairs.

Yet the thing can be done, and done exactly. This chapter shows how. The message is not copied, it is *spread*: one message is carried by three wires together, in a way that is not three copies. Nobody asks the wires what they hold. Instead, each question asks whether two of the wires *agree*, and the answer to that question turns out to be the same whatever the message is. So the questions hear the storm and never the message. And the continuum of possible damage turns out to be an illusion of the blackboard's description: every small turn is a sum of a few whole ones, and whatever repairs the whole ones repairs their sums.

The price is paid in wires, and in a condition on the storm. If the storm is too strong, the repair does more harm than good. Where exactly that line falls is one of the great practical questions of the subject, and the pictures of this book cannot answer it. We will say so when we get there.

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§18.1 Three for One

Chapter 17 ended with a promise: "Real hardware is not like that: now and then a reading comes out false. Chapter 18, *Against Noise*, answers with a plain remedy: make copies, and take a vote. The picture of making copies is one this book already knows. Its encoder is a single green spider."

The chapter's question, in the form we can answer: **a message that must not be looked at is sent through noise that turns it at random; how can it arrive exactly as sent?** We answer it in five steps. This section says what noise is, counts what a vote buys, and draws the one copier that does not clone. §18.2 votes without looking. §18.3 finds the kind of noise the vote cannot hear, and builds the nine-wire code that hears both kinds. §18.4 asks which questions a code may ask at all, and states what is known about building a whole machine this way. §18.5 joins two protected messages without unwrapping them.

Noise

On the blackboard. Recall the Pauli matrices, in Gottesman's Table 1:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad Y = iXZ = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Gottesman calls X the *bit flip*, Z the *phase flip*, and Y both at once. Chapter 9 taught us that a process that may go more than one way is a **channel**, written in Kraus form: $\rho \mapsto \sum_i E_i \rho E_i^\dagger$ with $\sum_i E_i^\dagger E_i = I$ (Gottesman's eq. (23); Chapter 9, row 70). The three channels of this chapter each leave a wire alone with probability $1 - p$ and do something to it with probability p:

$$\mathcal{N}_p^X(\rho) = (1-p)\,\rho + p\,X\rho X, \qquad \mathcal{N}_p^Z(\rho) = (1-p)\,\rho + p\,Z\rho Z,$$

$$\mathcal{N}_p^{\text{dep}}(\rho) = (1-p)\,\rho + \tfrac{p}{3}(X\rho X + Y\rho Y + Z\rho Z).$$

The first is the **bit-flip channel**, the second the **phase-flip channel**, the third the **depolarizing channel**. Their Kraus operators are $\sqrt{1-p}\,I$ and $\sqrt{p}$ times a Pauli (for the third, $\sqrt{p/3}$ times each of the three), and $\sum_i E_i^\dagger E_i = (1-p)I + pI = I$ because every Pauli squares to I. Noise on n wires is **independent** when it is the channel $\mathcal{N}_p$ on each wire separately, $(\mathcal{N}_p)^{\otimes n}$.

A small calculation shows what the two flips do. The dialogue's door dial has six marks. Read as states, they are dark $|0\rangle$, lit $|1\rangle$, at rest $|+\rangle$, a half $|-\rangle$, a quarter round $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ and a quarter back $\frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$. A bit flip exchanges dark and lit, and it leaves at rest alone: $X|+\rangle = |+\rangle$. On a quarter round it gives

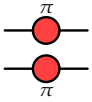
$$X\tfrac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) = \tfrac{1}{\sqrt{2}}(|1\rangle + i\,|0\rangle) = i \cdot \tfrac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle),$$

a quarter back, up to the overall number i, which changes nothing a reading can see. A phase flip does the opposite: it leaves dark and lit alone and exchanges at rest with a half, and a quarter round with a quarter back. So $\mathcal{N}_p^X$ never harms a message set at rest, and $\mathcal{N}_p^Z$ never harms a message set dark or lit. Neither is harmless for all six marks.

In the sketchbook. A red spider turned π with two legs is X, exactly

(Chapter 8, §8.2), and a green spider turned π with two legs is Z. Chapter 9 drew a channel on **doubled** wires (rows 66 and 70): a wire and its mirror copy, the mirror carrying the conjugate, and a process A drawn as A on the wire and $\bar{A}$ on the mirror. The matrix X is real, so $\bar{X} = X$, and its mirror copy is again a red half. A channel with two Kraus operators is a sum of two doubled pictures, each with its weight. Here is the bit-flip channel, the wire on top and its mirror copy below:

Result 18.1 (the bit flip, doubled). For every p, exactly,

$\mathcal{N}_p^X \;=\; (1-p)$ $+\; p$ 

The phase-flip channel is the same with green halves, and the depolarizing channel is a weighted sum of four doubled pictures.

Proof. In doubled form a state $\rho = \sum_{ij} \rho_{ij} |i\rangle \langle j|$ is the vector $\sum_{ij} \rho_{ij} |i\rangle \otimes |j\rangle$, the ket on the wire and the conjugated bra on the mirror (Chapter 9, row 66), and a process A acts as $A \otimes \bar{A}$, which sends it to $A\rho A^\dagger$. The right-hand side acts as $(1-p)\, I \otimes I + p\, X \otimes X$, which sends ρ to $(1-p)\rho + pX\rho X$. ■ See `Code/Ch18/ThreeForOne.py`, which checks it on a random non-real density matrix.

This is the only doubled picture in the chapter. A sum of two pictures is not one picture, and the rules of the sketchbook rewrite pictures, not sums. So from here on we work **branch by branch**: we follow one Kraus operator at a time, a definite pattern of flips, drawn undoubled with Chapter 10's bit-labelled phases (row 75). A red spider turned $e\pi$, with e a bit, is X^e: no flip when $e = 0$, a flip when $e = 1$. One picture covers every branch. The probabilities of the branches, and their sums, stay on the blackboard.

Why this definition. Why describe noise by a list of things that might happen, each with a probability, rather than by one "noisy" operation? Because the wire is not the only thing that knows what happened. A gust that flips a wire leaves a trace in the air that flipped it, and that trace is gone for good: we cannot undo a flip we cannot identify. The Kraus form

says exactly this. It lists the things that might have happened, as if the environment kept a record, and it forgets the record.

A vote

Before the quantum case, the classical one, whose numbers the dialogue's tins carry. A bit is written three times. The damp turns each of the three independently, either way, with probability p. We read all three and trust the majority.

> **Result 18.2 (the vote of three).** For every p with $0 \le p \le 1$, and whichever way the bit was set, the majority of the three is wrong with probability exactly
>
> $$F(p) \;=\; 3p^2 - 2p^3.$$
>
> At $p = \frac{1}{10}$, $F(p) = 0.028$.

Proof. The majority is wrong exactly when two or three of the three are turned: $3p^2(1-p) + p^3 = 3p^2 - 2p^3$. Nothing in the count depends on which way the bit was set. ■ See `Code/Ch18/ThreeForOne.py`, which enumerates the eight patterns for a bit set dark and a bit set lit, at $p = \frac{1}{10}$, at $p = 0.37$ and at a random p.

The letter F means this function for the rest of the chapter. The vote turns a failure rate of one in ten into about three in a hundred. It needs no cleverness: only copies, and a count.

The one copier there is

The quantum version of the clerk's method would be Gottesman's eq. (3): send $|\psi\rangle$ to $|\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$. His Theorem 1 forbids it: "There is no quantum operation that takes a state $|\psi\rangle$ to $|\psi\rangle \otimes |\psi\rangle$ for all states $|\psi\rangle$." Chapter 11 proved this in several forms (Results 11.1 and 11.3), and Gottesman's proof is the one we met first, linearity: a map that copies $|\psi\rangle$ and $|\varphi\rangle$ sends their sum to $|\psi\rangle|\psi\rangle + |\varphi\rangle|\varphi\rangle$, not to the copy of the sum.

What linearity does allow is to copy a *basis*. Chapter 11's copier $\delta = |00\rangle\langle 0| +$

$|11\rangle\langle 1|$ copies $|0\rangle$ and $|1\rangle$ and nothing else (row 81). Here is its three-legged cousin.

On the blackboard. Let

$$\text{Enc} \;:=\; |000\rangle\langle 0| + |111\rangle\langle 1|\,.$$

It is an 8×2 matrix, with a 1 in the rows of $|000\rangle$ and $|111\rangle$. The letter Enc means this matrix, and nothing else, for the rest of the chapter. On a general message,

$$\text{Enc}(\alpha\,|0\rangle + \beta\,|1\rangle) \;=\; \alpha\,|000\rangle + \beta\,|111\rangle\,.$$

Compare the forbidden clone, $(\alpha\,|0\rangle + \beta\,|1\rangle)^{\otimes 3} = \alpha^3\,|000\rangle + \alpha^2\beta\,|001\rangle + \cdots + \beta^3\,|111\rangle$, which has eight terms. The two agree only when one of α, β is zero: for dark and lit messages, and for no other. The message a quarter round, $\alpha = \frac{1}{\sqrt{2}}$ and $\beta = \frac{i}{\sqrt{2}}$, becomes $\frac{1}{\sqrt{2}}(|000\rangle + i\,|111\rangle)$: not three quarter-round wires, but one quarter-round message written across three wires at once. It is the classical repetition code of Gottesman's eqs. (1) and (2), $0 \mapsto 000$ and $1 \mapsto 111$, applied to the basis and extended by linearity. Finally, $\text{Enc}^\dagger\text{Enc} = |0\rangle\langle 000|000\rangle\langle 0| + |1\rangle\langle 111|111\rangle\langle 1| = I$: the encoder loses nothing.

The **code space** of the encoder is the image of Enc, the two-dimensional space spanned by $|000\rangle$ and $|111\rangle$. We write Q for a code space; Gottesman writes C.

In the sketchbook. A green spider at rest with one leg in and three legs out is, by its definition in Chapter 8 (row 53), $|000\rangle\langle 0| + |111\rangle\langle 1|$. That is all.

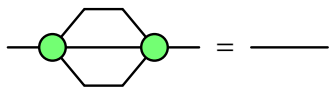
Result 18.3 (the encoder is a spider). Exactly, with no number,

Enc =



and

$\text{Enc}^\dagger\text{Enc}$ =



The mirror image of the encoder, a green spider with three legs

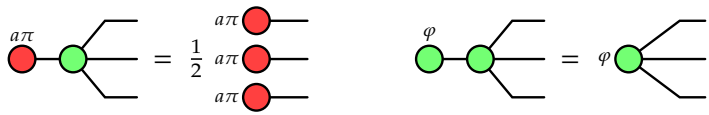
in and one out, is $\mathrm{Enc}^\dagger$.

Proof. The first equation is row 53 with three outputs. For the second: the two spiders share a leg, so they slide together into one green spider (fusion, row 57); the other two shared legs become loops from that spider to itself, and a self-loop on a green spider goes, with the number 1 (row 90; the Dirac's Blackboard sidebar of §18.2 checks the number for a spider of any phase). What is left is a green spider at rest with two legs, a plain wire (row 57). ■ See `Code/Ch18/ThreeForOne.py`, where PyZX's `full_reduce` takes the right-hand picture to a bare wire with the number 1.

The mirror image is the second meaning of a picture that Chapter 3 gave us: turn it left to right, and conjugate. The encoder is real, so conjugating changes nothing. (Its three legs out are not three legs of a cup, and no cup is drawn in this chapter; where the book draws one, the convention of Chapter 6 holds: the cup is $|00\rangle + |11\rangle$, unnormalized, and the Bell state is $|\Phi^+\rangle = \frac{1}{\sqrt{2}}$ cup.)

Now feed it a message. A dark or lit message is a **red point**: Chapter 8's red spider with one leg, at rest for dark and turned a half for lit. Bare, a red point at rest is $|+\rangle + |-\rangle = \sqrt{2}\,|0\rangle$ and a red point turned π is $|+\rangle - |-\rangle = \sqrt{2}\,|1\rangle$ (row 54). A turned message is a **green point**: a green spider with one leg, turned φ, which is $|0\rangle + e^{i\varphi}\,|1\rangle$ (row 53).

Result 18.4 (copy and spread). Let a be a bit. Exactly:



A dark or lit point comes out as three points of the same kind, worth twice the one, so the number $\frac{1}{2}$ goes to one side. A green point turned φ is not copied: it slides into the encoder, and what comes out is one green spider turned φ with three legs, $|000\rangle + e^{i\varphi}\,|111\rangle$. For every φ this is not three points of any kind: it is entangled.

Proof. On the blackboard: $\mathrm{Enc}\,\sqrt{2}\,|a\rangle = \sqrt{2}\,|aaa\rangle$, and three red points give

$(\sqrt{2})^3\,|aaa\rangle$, which is twice as much. The green point: $\mathrm{Enc}(|0\rangle + e^{i\varphi}\,|1\rangle) = |000\rangle + e^{i\varphi}\,|111\rangle$, which is the green spider turned φ with three outputs (row 53); and it has rank 2 across the cut between its first wire and the other two, while three points have rank 1 there (Chapter 11, row 87). The derivation of the first equation by the rules is the Coecke's Sketchbook sidebar below; the second is fusion (row 57). ■ See `Code/Ch18/ThreeForOne.py`.

This is Chapter 11's complementarity again (row 82), now with three legs: red points copy through a green spider, green points fuse into it. The encoder copies a message exactly when the message is dark or lit, and for every other message it does something no copier could do: it spreads one message across three wires.

> **Coecke's Sketchbook.**
>
> *The number of a copy.* Chapter 8 stated copy (Theorem 8.3) for a green spider with one leg in and two out: a red point at rest, $\sqrt{2}\,|0\rangle$ bare, comes out as two red points with the number $\frac{1}{\sqrt{2}}$ beside them. Chapter 11 (row 81) made it exact. For three legs, slide the encoder apart (fusion backwards, row 57) into two copiers in a row: the first with one leg in and two out, the second on its lower output. The point meets the first copier and becomes two points, with the number $\frac{1}{\sqrt{2}}$. The lower of them meets the second copier and becomes two more, with another $\frac{1}{\sqrt{2}}$. Three points, and $\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$. The same steps work for a point turned a half, since copy holds for both red points. They do not work for a green point: a green point and a green spider are the same colour, and fusion comes first. The warning in the picture: the rule copies **points of the other colour**, and only those.

Why this definition. Why take the encoder that copies dark and lit, and not some cleverer map? Because it is the only kind of copying that linearity allows: an encoder that copies $|0\rangle$ and $|1\rangle$ is fixed on a basis, and so it is Enc. Whatever a vote can do for dark and lit messages, this encoder does. The whole chapter is about what it does for the others.

§18.2 The Blind Vote

The clerk votes by reading. He opens each ledger, sees the figure, and trusts the two that match. We may not read: a reading of one wire of $\alpha\,|000\rangle + \beta\,|111\rangle$ answers 0 with probability $|\alpha|^2$ and leaves $|000\rangle$ behind, and the message is gone (Chapter 9's Born rule). So the vote must be taken in a different way. We shall not ask any wire what it holds. We shall ask two wires whether they **agree**.

Asking a pair

On the blackboard. The operator $Z_1Z_2 = Z \otimes Z \otimes I$ multiplies $|x_1x_2x_3\rangle$ by $(-1)^{x_1 \oplus x_2}$: by $+1$ when the first two bits agree, by -1 when they differ. It does not care what the bits are, only whether they match. Its two projectors are

$$\tfrac{1}{2}(I + Z_1Z_2) \ \text{(the pair agrees)}, \qquad \tfrac{1}{2}(I - Z_1Z_2) \ \text{(the pair disagrees)},$$

and a look with these two projectors, in the sense of Chapter 9, is Gottesman's measurement of $Z \otimes Z \otimes I$ (his eq. (39)). The second check asks the second and third wires, with Z_2Z_3. We write s_1 for the answer of the first check and s_2 for the second: 0 for "agree", 1 for "disagree". Both together have four outcomes $s = (s_1, s_2)$, with projectors

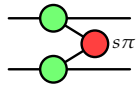
$$P_s \ = \ \tfrac{1}{4}(I + (-1)^{s_1}Z_1Z_2)(I + (-1)^{s_2}Z_2Z_3).$$

The two factors commute, each is a projector, and one checks that $P_sP_{s'} = [s = s']\,P_s$ and $\sum_s P_s = I$. Here and below, $[s = s']$ is 1 when the statement in brackets is true and 0 when it is false; we do not write it δ, because δ is Chapter 11's copier. The four outcomes are called the **syndrome**.

Now the small calculation that makes the whole chapter work. On the code space, $Z_1Z_2\,|000\rangle = |000\rangle$ and $Z_1Z_2\,|111\rangle = |111\rangle$: both basis words have their first two bits in agreement. So Z_1Z_2 leaves every $\alpha\,|000\rangle + \beta\,|111\rangle$ alone,

and the first check answers "agree" with certainty, whatever α and β are. The answer carries no news of the message.

In the sketchbook. Chapter 12 drew the parity of two classical wires as a red dot (row 96), and Chapter 13 hung a red hub from green feet on quantum wires and wrote its number (row 102). Here is the smallest case. On each of two wires put a green spider at rest with three legs, two along the wire and one tie; tie both to one red spider turned $s\pi$, which has only those two legs. We call the red spider a **bell**, after the dialogue's, and the two green spiders its **tell rings**.

> **Result 18.5 (a bell asks a pair).** For each bit s, exactly, with no number:
>
> 
>
>
> $$= \tfrac{1}{2}(I + (-1)^s Z \otimes Z).$$
>
> Two such bells, on the first and second wires and on the second and third, are the four projectors P_s at once.

Proof. Follow a basis state $|x_1x_2\rangle$. Each green spider copies its bit onto its tie (row 53), so the bell receives $|x_1x_2\rangle$ on its two legs. Bare, the bell is the effect $\langle{+}{+}| + e^{is\pi}\langle{-}{-}|$ (row 54), and $\langle{+}{+}|x_1x_2\rangle = \frac{1}{2}$, $\langle{-}{-}|x_1x_2\rangle = \frac{1}{2}(-1)^{x_1 \oplus x_2}$. So the picture multiplies $|x_1x_2\rangle$ by $\frac{1}{2}(1 + (-1)^s(-1)^{x_1 \oplus x_2})$, which is the projector. For the second sentence: put the second bell after the first; the two green spiders on the middle wire are joined by a plain wire, so they slide together into one green spider with two ties (row 57), and the product of the two projectors is P_s. ■ See `Code/Ch18/BlindVote.py`.

The number is exactly 1 here only because the bell has two legs. With w legs the same computation gives $2^{1-w/2}$ times the projector; we will need that in §18.4.

Why this definition. Why ask whether two wires agree, and not simply ask one wire what it holds? Because the question "what does wire one hold?" has a different answer for $|000\rangle$ and for $|111\rangle$, and a question with different answers on the two words of the code must separate them: after it is asked, α and β are no longer in the same state. The question "do wires one and two agree?" has the *same* answer on both words. Questions of that kind

leave a superposition of the two words alone, and §18.4 shows, with Knill and Laflamme's theorem, that they are the only questions a code may ask.

The bells hear the gust

Now let the storm in. A **gust pattern** is a string of three bits $e = e_1e_2e_3$, with $e_k = 1$ when wire k is flipped; the gust itself is $X^e = X^{e_1} \otimes X^{e_2} \otimes X^{e_3}$. Write $\bar{e}$ for e with every bit flipped. The **syndrome map** sends a gust pattern to the pair of answers it would produce,

$$\sigma(e) \;=\; (e_1 \oplus e_2,\; e_2 \oplus e_3).$$

The letter σ means only this map in this chapter.

> **Result 18.6 (the bells hear the gust, not the message).** For every gust pattern $e \in \{0,1\}^3$, every outcome s and every message,
>
> $$P_s\, X^e \,\mathrm{Enc} \;=\; [\, s = \sigma(e) \,]\; X^e \,\mathrm{Enc}.$$
>
> So the checks answer $\sigma(e)$ with probability 1, and they do not disturb the state.

Proof. $X^e\mathrm{Enc}(\alpha\,|0\rangle + \beta\,|1\rangle) = \alpha\,|e\rangle + \beta\,|\bar{e}\rangle$. The first check multiplies $|e\rangle$ by $(-1)^{e_1 \oplus e_2}$ and $|\bar{e}\rangle$ by $(-1)^{(1\oplus e_1)\oplus(1\oplus e_2)} = (-1)^{e_1\oplus e_2}$, the same sign; likewise the second. So $\alpha\,|e\rangle + \beta\,|\bar{e}\rangle$ lies wholly in the range of $P_{\sigma(e)}$, and the other three projectors send it to 0. ■ See `Code/Ch18/BlindVote.py`, which checks all eight patterns and all four outcomes.

Here is the dialogue's headline run, in coordinates. The message is set a quarter round, $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$. The encoder spreads it: $\frac{1}{\sqrt{2}}(|000\rangle + i\,|111\rangle)$. A gust turns the middle wire, $e = 010$: the state becomes $\frac{1}{\sqrt{2}}(|010\rangle + i\,|101\rangle)$. On $|010\rangle$ the first pair disagrees and so does the second; on $|101\rangle$, the same. Both bells ring: $s = 11$, with certainty, and the state is untouched. The bells have said "the middle wire differs from both its neighbours", and nothing at all about $\frac{1}{\sqrt{2}}$ or i.

So the mending is a table. For each outcome, flip back the one wire that

would have caused it:

$$c(00) = 000, \qquad c(10) = 100, \qquad c(11) = 010, \qquad c(01) = 001.$$

The correction is $X^{c(s)}$. In the example, X^{010} turns $\frac{1}{\sqrt{2}}(|010\rangle + i\,|101\rangle)$ back into $\frac{1}{\sqrt{2}}(|000\rangle + i\,|111\rangle)$, which is the spread message exactly. Nobody has looked at it.

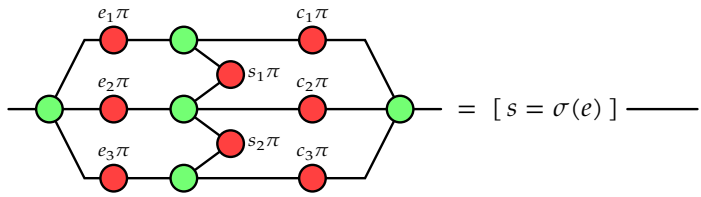
One gust, mended blind

Put the pieces in a row: spread, weather, ask, mend, and join. The joining step is the mirror image of the encoder, $\mathrm{Enc}^\dagger$. We should say plainly what that means. $\mathrm{Enc}^\dagger$ is not a device one can build on its own: it sends $|000\rangle$ to $|0\rangle$ and $|111\rangle$ to $|1\rangle$ and every other basis word to 0, so it is the picture of the *good branch* of a decoder, the branch in which the three wires are back in the code space. A real decoder must also do something with the other branches; we do not draw one. Every statement below says what happens in the good branch, and says that the others have amplitude 0.

The gusts we correct are the single ones, $\mathcal{E}_X = \{I, X_1, X_2, X_3\}$, which are X^e for $e \in \{000, 100, 010, 001\}$.

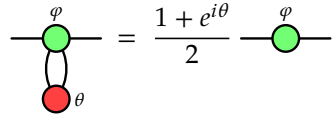
> **Result 18.7 (one gust, mended blind; the headline).** For every $X^e \in \mathcal{E}_X$ and every outcome s, exactly,
>
> $$\mathrm{Enc}^\dagger\, X^{c(s)}\, P_s\, X^e\, \mathrm{Enc} \;=\; [\, s = \sigma(e)\,]\, I.$$
>
> In words: in the one branch that occurs, $s = \sigma(e)$, the message comes back exactly, with the number 1; every other branch is exactly 0, and so has probability 0. It holds for every message, since both sides are linear. The same statement as a picture: *The Storm Line* (§18.8), with its weather rings turned $e_k\pi$, its bells $s_j\pi$ and its mending rings $c_k\pi$ where $c = c(s)$, equals $[s = \sigma(e)]$

times a bare wire:



It is proved twice in §18.7. The blackboard's proof is Result 18.6 plus two facts: on $\mathcal{E}_X$ the map σ is one-to-one, and $c(\sigma(e)) = e$, so $X^{c(s)}X^e = I$ in the branch that occurs. The picture reading, which is the reason the picture has the shape it has: the spread message runs left to right along the three strands; each weather ring is X^{e_k} (Chapter 8, §8.2, with row 75's bit); the tell rings and the two bells are the checks of Result 18.5, slid together on the middle strand; each mending ring is X^{c_k}; and the joining ring is $\mathrm{Enc}^\dagger$. Every piece is exact with no number, so the whole picture *is* $\mathrm{Enc}^\dagger X^c P_s X^e \mathrm{Enc}$, exactly.

The picture proof needs one rule the book has not yet stated. It is about a bell whose two ties both end on the same green spider, which is what happens when the tell rings are slid together.

> **Result 18.8 (the double-tied bell).** For every angle θ, every angle φ, and a green spider with any number of other legs, exactly:
>
> 
>
>
> The red spider has exactly the two legs drawn, both tied to the one green spider. A bell at rest comes off with the number 1; a bell turned a half takes the whole picture to 0.

Proof. On the blackboard: the green spider holds all its legs at one bit x, with the factor $e^{i\varphi x}$ (row 53), so the bell sees $|xx\rangle$, and $\langle ++|xx\rangle + e^{i\theta}\langle --|xx\rangle = \frac{1}{2} + \frac{1}{2}e^{i\theta}$ for both values of x. The number does not depend on x, so it comes out in front. By the rules, the Coecke's Sketchbook sidebar below.

■ See `Code/Ch18/BlindVote.py`, which checks it at $\theta = 0$, at $\theta = \pi$ and at a random θ.

Coecke's Sketchbook.

The double-tied bell, by the rules. Four moves. First, fusion backwards (row 57), twice: peel the bell's turn off onto a red point turned θ, leaving a red spider at rest that holds the two ties and a cord to the point; and peel off from the green spider a part at rest that holds the two ties, joined by one cord to the rest. Second, the two parts at rest are now joined by two wires and each has one other leg: Chapter 8's Hopf law (Theorem 8.6, row 62) cuts both ties at the price $\frac{1}{2}$. Third, the red spider, left with one leg, fuses with the red point (row 57) into a red spider with no legs, turned θ, which is the number $1 + e^{i\theta}$ (row 54). Fourth, the green part at rest, left with one leg, fuses back (row 57). Total: $\frac{1}{2}(1 + e^{i\theta})$. The warning: the rule needs *both* ties on *one* green spider. A bell tied to two different spiders does not come off; it is a question still being asked.

With Result 18.8 the sketchbook proves the headline in five moves, which §18.7 draws. In outline: each weather ring's half passes through its tell ring (row 120) and lands in the bells tied to that strand and in its mending ring (row 57); in the branch $s = \sigma(e)$ with $c = e$ every red ring is left at rest; the tell rings and the joining ring slide into the parting ring along bare strands; each bell is then tied twice to one green spider and comes off with the number 1; and what is left is a bare wire.

Dirac's Blackboard.

The headline in four lines, and a loop that costs nothing. (1) $X^e \mathrm{Enc}(\alpha\,|0\rangle + \beta\,|1\rangle) = \alpha\,|e\rangle + \beta\,|\bar{e}\rangle$. (2) Z_1Z_2 and Z_2Z_3 give $|e\rangle$ and $|\bar{e}\rangle$ the same signs, $(-1)^{e_1 \oplus e_2}$ and $(-1)^{e_2 \oplus e_3}$, so P_s multiplies the whole state by $[s = \sigma(e)]$ and never separates α from β. (3) On $\mathcal{E}_X$, $\sigma(000) = 00$, $\sigma(100) = 10$, $\sigma(010) = 11$, $\sigma(001) = 01$: one-to-one, with $c(\sigma(e)) = e$, so $X^{c(s)}X^e = I$ in the branch that occurs. (4) $\mathrm{Enc}^\dagger\mathrm{Enc} = I$. On the dialogue's message, $\alpha = \frac{1}{\sqrt{2}}$,

$\beta = \frac{i}{\sqrt{2}}$, $e = 010$: line (1) gives $\frac{1}{\sqrt{2}}(|010\rangle + i|101\rangle)$, line (2) gives $s = 11$, line (3) turns the middle wire back, line (4) returns $\frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$. The loop: a green spider turned φ with a plain wire from one of its legs to another is $\sum_x e^{i\varphi x}\,|x\cdots x\rangle\langle x\cdots x|$ with two of the x's joined, and joining them sums $\langle a|x\rangle\langle a|x\rangle$ over a, which is 1. So the loop goes with the number 1 (row 90), for every φ.

What the vote cannot do

The headline is about single gusts. Two gusts fool the bells, and they fool them in a definite way.

Result 18.9 (what the vote cannot do). For every gust pattern $e \in \{0,1\}^3$ and every outcome s, exactly,

$$\mathrm{Enc}^\dagger\, X^{c(s)}\, P_s\, X^e\, \mathrm{Enc} \;=\; [\, s = \sigma(e)\,]\; X^{[\mathrm{wt}(e)\geq 2]},$$

where $\mathrm{wt}(e)$ is the number of 1s in e. With two or three gusts, the branch that occurs returns the message flipped, $X|\psi\rangle$, with certainty.

Proof. Result 18.6 holds for all eight patterns. Flipping every bit of e does not change $\sigma(e)$, so $\sigma(e) = \sigma(\bar{e})$; exactly one of e and $\bar{e}$ has weight at most 1, and c picks that one. If $\mathrm{wt}(e) \geq 2$ then $c(\sigma(e)) = \bar{e}$, and $X^{\bar{e}}X^e = X \otimes X \otimes X$, which the code sees as $\mathrm{Enc}^\dagger X^{\otimes 3}\mathrm{Enc} = X$. ■ See `Code/Ch18/BlindVote.py`.

This is the classical vote exactly: two wrong copies outvote one right one. For a dark or lit message the result is the wrong bit; for a message a quarter round it is a quarter back, and for a message at rest it is harmless, since $X|+\rangle = |+\rangle$.

Every mixture of gusts

The headline is branch by branch and gust by gust. Real weather is a channel, and a channel's Kraus operators need not be single Paulis: a small unwanted turn is a sum of I and an X with complex weights. Linearity takes care of

all of them at once.

> **Result 18.10 (every mixture of single gusts).** For every E in $\operatorname{span}\mathcal{E}_X$, that is, every $E = \sum_e a_e X^e$ with complex numbers a_e and e running over the four single-gust patterns $000, 100, 010, 001$, and for every s,
>
> $$X^{c(s)}\, P_s\, E\, \mathrm{Enc} \;=\; a_{\sigma^{-1}(s)}\, \mathrm{Enc},$$
>
> where $\sigma^{-1}(s)$ is the one single-gust pattern with syndrome s. So with the recovery channel $\mathcal{R}(\rho) = \sum_s X^{c(s)} P_s\, \rho\, P_s X^{c(s)}$, every channel whose Kraus operators all lie in $\operatorname{span}\mathcal{E}_X$ is undone exactly on the code space: $\mathcal{R}(\sum_i E_i\, \mathrm{Enc}\, \rho\, \mathrm{Enc}^\dagger E_i^\dagger) = \mathrm{Enc}\, \rho\, \mathrm{Enc}^\dagger$.

Proof. The operator statement is Result 18.7 applied to each term, with $\mathrm{Enc}^\dagger$ removed: Result 18.6 and $X^{c(\sigma(e))}X^e = I$ give $X^{c(s)}P_sX^e\mathrm{Enc} = [s = \sigma(e)]\,\mathrm{Enc}$ on $\mathcal{E}_X$. For the channel, apply it to each Kraus operator $E_i = \sum_e a_{i,e}X^e$: the output is $\sum_s \sum_i |a_{i,\sigma^{-1}(s)}|^2\, \mathrm{Enc}\, \rho\, \mathrm{Enc}^\dagger$. Since the channel preserves trace on the code, $\sum_i \mathrm{Enc}^\dagger E_i^\dagger E_i \mathrm{Enc} = I$, and $\mathrm{Enc}^\dagger X^e X^{e'} \mathrm{Enc} = [e = e']\, I$ for single-gust patterns (for two different single-gust patterns, $e \oplus e'$ has weight 1 or 2, and $\mathrm{Enc}^\dagger X^f \mathrm{Enc} = 0$ unless f is 000 or 111; this is also the $C = I_4$ of §18.4), so $\sum_i \sum_e |a_{i,e}|^2 = 1$. ■ See `Code/Ch18/BlindVote.py`, which builds five random non-real Kraus operators in span $\mathcal{E}_X$ and checks that $\mathcal{R}$ restores a random non-real ρ exactly.

This is Gottesman's Theorem 2 in its first case, and we meet it in full in §18.3. The continuum of small turns is not a continuum of problems: whatever repairs I, X_1, X_2 and X_3 repairs every combination of them, because the checks force each branch onto one of the four.

The noise of a storm, counted

Finally the probabilities. Independent bit flips on the three wires, $(\mathcal{N}_p^X)^{\otimes 3}$, have Kraus operators $\sqrt{p^{\mathrm{wt}(e)}(1-p)^{3-\mathrm{wt}(e)}}\, X^e$ for all eight e. Their Kraus operators do not all lie in span $\mathcal{E}_X$, and Result 18.9 says what the others do.

> **Result 18.11 (the logical bit flip).** For every p with $0 \le p \le 1$, encoding, independent bit flips of probability p on each wire,

the recovery $\mathcal{R}$ and the good-branch decoder $\mathrm{Enc}^{\dagger}$ together act on the message as the bit-flip channel $\mathcal{N}^{X}_{p_L}$ with

$$p_L \;=\; F(p) \;=\; 3p^2 - 2p^3.$$

Proof. By Result 18.9 each pattern e gives I if $\mathrm{wt}(e) \le 1$ and X if $\mathrm{wt}(e) \ge 2$, in its one branch. So the message is flipped with the total probability of two or three flips, which is $F(p)$ of Result 18.2. ■ See `Code/Ch18/ThreeForOne.py`, which compares the two 4×4 superoperators at $p = \frac{1}{10}$, $p = 0.37$ and a random p.

The quantum code does exactly what the tins do, and no better: at $p = \frac{1}{10}$ it turns one flip in ten into 0.028. It has also done something the tins could never do, which is to carry a message a quarter round.

Why this definition. Why a table of corrections $c(s)$, fixed in advance, and not a correction chosen by looking at the result? Because there is nothing to look at. The syndrome is the only information the checks release, and Result 18.6 says it is a function of the gust alone. A mending rule can depend only on what is known, and what is known is s. When $p < \frac{1}{2}$, the table names the most likely gust behind each syndrome, since each syndrome comes from one pattern e of weight at most 1 and from $\bar{e}$, of weight at least 2. That is also the rule of the tins: trust the majority.

---

§18.3 The Turn the Bells Miss

A gust flips a wire. But §18.1 met two kinds of flip, and the bells were built to hear only one of them. What happens to a **phase flip**, Z, on one of the three wires?

On the blackboard. Z_k multiplies $|x_1x_2x_3\rangle$ by $(-1)^{x_k}$. On the code space it gives $|000\rangle$ the sign $+1$ and $|111\rangle$ the sign -1, whichever wire k it hits:

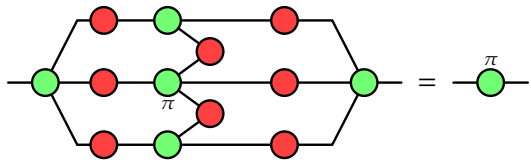
$$Z_k\,\mathrm{Enc}(\alpha\,|0\rangle + \beta\,|1\rangle) \;=\; \alpha\,|000\rangle - \beta\,|111\rangle \;=\; \mathrm{Enc}\,Z(\alpha\,|0\rangle + \beta\,|1\rangle).$$

The result is still in the code space. Both checks answer "agree", since Z_1Z_2 and Z_2Z_3 commute with Z_k and fix the code space; so nothing is mended, and the message arrives phase-flipped.

Result 18.12 (a phase flip passes unheard). For $k = 1, 2, 3$ and every outcome s, exactly,

$$\mathrm{Enc}^{\dagger}\, X^{c(s)}\, P_s\, Z_k\, \mathrm{Enc} \;=\; [\, s = 00 \,]\, Z.$$

As a picture: *The Storm Line* with every weather ring, bell and mending ring at rest, and a green half on the middle tell ring, equals a green half on a wire, with the number 1:



Proof. On the blackboard: $P_sZ_k\mathrm{Enc} = Z_kP_s\mathrm{Enc} = [s = 00]\, Z_k\mathrm{Enc}$, since Z_k commutes with both checks and the code space lies in the range of P_{00} (Result 18.6 with $e = 000$); then $c(00) = 000$ and $\mathrm{Enc}^{\dagger}Z_k\mathrm{Enc} = \mathrm{Enc}^{\dagger}\mathrm{Enc}Z = Z$. By the rules, for the middle wire as drawn: every red ring at rest with two legs is bare cord (row 57). Slide, along bare cords, the first tell ring into the parting ring along the first weather cord, then the joining ring into it along the first mending cord, then the middle tell ring, carrying its half, along the middle weather cord, and the third tell ring along the third (row 57; the phases add, so the parting ring is now turned π). The middle and third mending cords are now loops, which go (row 90). Each bell, at rest, is tied twice to the one green spider and comes off with the number 1 (Result 18.8). What is left is a green spider turned π with two legs, which is Z. ■ See `Code/Ch18/NineStrands.py`, where PyZX's `full_reduce` leaves exactly one green spider turned π on the storm line with this labelling, with the number 1.

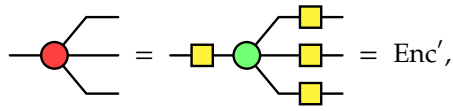
The picture says it vividly: a green half and the green tell ring are the same colour, so the half does not *pass through* the tell ring, it *fuses into it*, and a

spider turned a half has nothing to say to the bells hanging from it. A dark or lit message comes through unharmed, since $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$, the same message up to the overall number -1. A message at rest arrives as a half, and a quarter round as a quarter back. The code that corrects every single bit flip is blind to every single phase flip.

The other colour

Chapter 8's colour change (row 58) says that a spider with a yellow box on every leg is the spider of the other colour. On the blackboard, a yellow box is H, and $HXH = Z$. So the whole construction can be repeated with the colours exchanged.

Result 18.13 (the phase-flip code). Let $\mathrm{Enc}' = H^{\otimes 3}\,\mathrm{Enc}\,H$, with checks X_1X_2 and X_2X_3, projectors $P'_s = H^{\otimes 3}P_sH^{\otimes 3}$ and corrections $Z^{c(s)}$. Exactly:



$$\mathrm{Enc}'^{\dagger}\, Z^{c(s)}\, P'_s\, Z^e\, \mathrm{Enc}' \;=\; [\, s = \sigma(e)\,]\, I$$

for every single phase flip Z^e, $e \in \{000, 100, 010, 001\}$, and every s. The red spider sends $|+\rangle$ to $|+\,+\,+\rangle$ and $|-\rangle$ to $|-\,-\,-\rangle$.

Proof. The picture is colour change (row 58), exact. For the rest, conjugate: $H^{\otimes 3}Z^eH^{\otimes 3} = X^e$ and $H^{\otimes 3}X^cH^{\otimes 3} = Z^c$, so

$$\begin{aligned}\mathrm{Enc}'^{\dagger}Z^{c(s)}P'_sZ^e\mathrm{Enc}' &= H(\mathrm{Enc}^{\dagger}X^{c(s)}P_sX^e\mathrm{Enc})H \\ &= [\, s = \sigma(e)\,]\,HH \;=\; [\, s = \sigma(e)\,]\,I\end{aligned}$$

by Result 18.7. ■ See `Code/Ch18/NineStrands.py`, where PyZX turns the red spider green with a yellow box on every leg by its own move, and checks the correction for all four single phase flips.

Its code space is spanned by $|+\,+\,+\rangle$ and $|-\,-\,-\rangle$. It is the code of Knill and Laflamme's eq. (15), written there without normalization as $\langle|0\rangle +$

$|1\rangle\rangle(|0\rangle + |1\rangle\rangle(|0\rangle + |1\rangle\rangle$ and $(|0\rangle - |1\rangle\rangle(|0\rangle - |1\rangle\rangle(|0\rangle - |1\rangle\rangle$, "such that if one qubit is corrupted by the environment, then it is possible to detect it by using a majority rule". Their encoding sends $|0\rangle$ to the first word and $|1\rangle$ to the second; the red spider sends $|+\rangle$ and $|-\rangle$ to the same two words, normalized, so the two differ by the one yellow box on the input. And of course the phase-flip code is blind to bit flips, by Result 18.12 in the other colour.

Coecke's Sketchbook.

The whole storm line in the other colour. Put two yellow boxes in a row on every cord of *The Storm Line* and on each lead; two boxes in a row are no box (row 56), so the picture has not changed. Now every spider has a box on every leg, and colour change (row 58) turns each into the other colour, turn kept. The green parting ring becomes red: Enc′ itself (Result 18.13). The red weather rings become green: phase flips. The green tell rings become red, the red bells green: checks of X_1X_2 and X_2X_3. The mending rings become green: Z^c. The joining ring becomes red, and on each lead one box is left over, at the far end. So the storm line in the new colours is the old one with a yellow box before it and one after. Every proof of §18.2 runs in the new colours with no change but the names, because every rule of Chapter 8 holds with the colours swapped. What the picture does *not* say is which code to use. That depends on which way the storm turns things, and the storm is physics, not a picture.

Nine strands

To hear both kinds of turn, use both codes, one inside the other. Spread the message three ways against phase flips, then spread each of the three against bit flips.

On the blackboard. Let

$$\mathrm{Enc}_9 \;=\; \mathrm{Enc}^{\otimes 3}\, H^{\otimes 3}\, \mathrm{Enc},$$

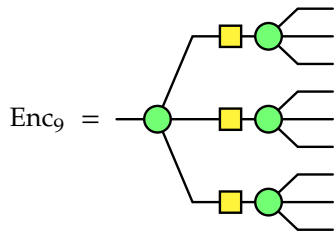
a 512×2 matrix from one wire to nine. Follow $|0\rangle$: $\mathrm{Enc}\,|0\rangle = |000\rangle$, then $H^{\otimes 3}$

gives $|+++\rangle = 2^{-3/2}(|0\rangle+|1\rangle)^{\otimes 3}$, then each $|0\rangle+|1\rangle$ is spread to $|000\rangle+|111\rangle$. And $|1\rangle$ goes the same way with minus signs:

$$\mathrm{Enc}_9\,|0\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)^{\otimes 3}, \qquad \mathrm{Enc}_9\,|1\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)^{\otimes 3}.$$

These are Gottesman's eqs. (9) and (10), divided by $2\sqrt{2}$, which makes them unit vectors: the **nine-qubit code**, which Gottesman calls "a variant of the repetition code", credited to Shor (his ref. [43]). The inner layer is three bit-flip codes, one on each block of three wires; the outer layer is the phase-flip code across the blocks. The checks are Gottesman's Table 2, eight of them: Z_1Z_2, Z_2Z_3, Z_4Z_5, Z_5Z_6, Z_7Z_8, Z_8Z_9 inside the blocks, and across them $X_1X_2X_3X_4X_5X_6$ and $X_4X_5X_6X_7X_8X_9$, his eqs. (42) and (43).

In the sketchbook. Each factor is a picture we have. Enc is a green spider with three legs out; $H^{\otimes 3}$ is a yellow box on each; $\mathrm{Enc}^{\otimes 3}$ is three more green spiders. So Shor's encoder is green, yellow, green:



exactly, with no number: the $\frac{1}{2\sqrt{2}}$ is already inside the three yellow boxes, $1/\sqrt{2}$ each, as the blackboard shows. The picture's middle cords carry a yellow box each; these are Chapter 8's Hadamard edges (row 56).

> **Result 18.14 (Shor's nine).** Exactly: $\mathrm{Enc}_9^\dagger\mathrm{Enc}_9 = I$; the eight checks of Table 2 commute and fix every state $\mathrm{Enc}_9\,|\psi\rangle$; and the twenty-eight single-wire operators I, X_k, Y_k, Z_k $(k = 1,\ldots,9)$ produce twenty-two different syndromes: one for I, nine for the X_k, three for the Z_k and nine for the Y_k. Measuring the checks and applying the correction for the syndrome returns $\mathrm{Enc}_9\,|\psi\rangle$ after every one of the twenty-eight, up to an overall number r with $|r| = 1$, a global phase of the code state, the same for every

$|\psi\rangle$.

Proof. The first claim: slide each inner green spider into its mirror image (Result 18.3), which leaves three bare wires; the boxes meet in pairs and cancel (row 56); what is left is Result 18.3 once more. On the blackboard it is $\mathrm{Enc}^\dagger H^{\otimes 3} (\mathrm{Enc}^\dagger \mathrm{Enc})^{\otimes 3} H^{\otimes 3} \mathrm{Enc} = I$. The syndromes: an X on wire k rings the two checks of its block exactly as in §18.2, which gives nine different answers, and commutes with the two long checks. A Z on any wire of a block is, on the code, a phase flip of that block's spread message (Result 18.12), which the two long checks see as the bit flip of the outer code that it is after the boxes (Result 18.13); all three wires of a block give the same answer, so three answers. $Y_k = iX_kZ_k$ rings both kinds, nine more. The corrections: an X is undone within its block (Result 18.7); a Z is undone by a Z on any wire of its block, since two Zs in one block act on the code as Z_1Z_2 does, as nothing; a Y by both; and the factor i in $Y = iXZ$, with the order of the two corrections, leaves a number r with $|r| = 1$. ■ See `Code/Ch18/NineStrands.py`, which checks the twenty-eight one by one, asserting $|r| = 1$ for each.

The three Zs of a block share one syndrome, and the code does not mind, because they do the same thing to it. Gottesman calls such a code **degenerate**: "different errors look the same when acting on the coding subspace". We return to this in §18.4.

The nine-qubit code corrects twenty-eight operators. But a real error on one wire is rarely one of them: it is a small turn by some angle about some axis. Here Gottesman's Theorem 2, which he credits to Shor and to Steane (his refs. [43] and [46]), finishes the job:

> **Gottesman's Theorem 2 (quoted).** "If a quantum code corrects errors A and B, it also corrects any linear combination of A and B. In particular, if it corrects all weight t Pauli errors, then the code corrects all t-qubit errors."

Every 2×2 matrix is a combination of I, X, Y and Z. So Shor's code corrects *every* error on any one of its nine wires, whatever it is, by the argument of Result 18.10: the checks force each branch onto one of the Paulis, and each is mended. The single-wire depolarizing channel $\mathcal{N}_p^{\mathrm{dep}}$, with Kraus

operators $\sqrt{1-p}\,I$ and $\sqrt{p/3}$ times X, Y and Z, is undone exactly on any one wire. Independent depolarizing noise on all nine wires is another matter: its branches with errors on two wires are not all corrected, and we do not compute the rate at which the code then fails.

Feynman's Notebook.

Nine for one. The code of this section is P. W. Shor's. Gottesman's survey, posted in 2009, introduces it as "a variant of the repetition code" and cites P. W. Shor, "Scheme for reducing decoherence in quantum memory", *Physical Review A* 52 (1995). The next year brought the seven-qubit code, which Gottesman cites to A. M. Steane, "Error correcting codes in quantum theory", *Physical Review Letters* 77 (1996), and whose stabilizer is his Table 3: the parity checks of the classical seven-bit Hamming code, written once with Z and once with X. The general recipe for building quantum codes out of two classical ones, Gottesman writes, "is known as the CSS construction after its inventors Calderbank, Shor, and Steane", and he cites A. R. Calderbank and P. W. Shor, "Good quantum error-correcting codes exist", *Physical Review A* 54 (1996), together with Steane's paper in the *Proceedings of the Royal Society of London* A 452 (1996). Shor's code costs nine wires for one; Steane's, seven; and there is a code with five, Gottesman's Table 4. Section 18.4 quotes the bound that rules out four.

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§18.4 What a Bell May Ask

The bells of §18.2 ask pairs. Why not ask more? A bell that could tell us *which way* a wire is set would tell us more about the storm, and more is surely better. Let us try it before we reason about it.

A bell that asks the one

Hang a bell on the first wire alone: it asks "is wire one dark or lit?", with projectors $\frac{1}{2}(I + Z_1)$ and $\frac{1}{2}(I - Z_1)$. The dialogue's clip-bell is this bell. Send the spread message $\alpha\,|000\rangle + \beta\,|111\rangle$ past it.

- A dark message, $\alpha = 1$: the bell is silent with certainty, and the state is untouched.
- A lit message, $\beta = 1$: the bell rings with certainty, and the state is untouched.
- A message a quarter round, $\alpha = \frac{1}{\sqrt{2}}$, $\beta = \frac{i}{\sqrt{2}}$: $\frac{1}{2}(I + Z_1)$ keeps $\frac{1}{\sqrt{2}}\,|000\rangle$ and $\frac{1}{2}(I - Z_1)$ keeps $\frac{i}{\sqrt{2}}\,|111\rangle$. So the bell rings with probability $\frac{1}{2}$, and afterwards the three wires hold $|000\rangle$ or $|111\rangle$. The message is now dark or lit. Read at the mark it was sent at, a quarter round, it is found there with probability $|\,\langle\text{quarter round}|0\rangle\,|^2 = \frac{1}{2}$, whichever way the bell went.

For dark and lit messages the one-wire bell is excellent: it even *tells* us the message. That is the trap. A question whose answer depends on the message has, by that fact, split the message into its answers, and for every message that is not dark or lit the split is damage. The bells of §18.2 did not have this problem because their answers were the same on $|000\rangle$ and $|111\rangle$. What is needed is a test that says, of any list of errors, whether some recovery undoes all of them. Knill and Laflamme gave one.

The Knill–Laflamme conditions

Let $\mathcal{Q}$ be a code space with an orthonormal basis of codewords $|\psi_i\rangle$, and let $\mathcal{E}$ be a list of errors E_a. The code **corrects** $\mathcal{E}$ when one recovery operation undoes every E_a on every state of $\mathcal{Q}$. Knill and Laflamme's phrasing is that the code "can be extended to an $\mathcal{A}$-correcting code", with $\mathcal{A}$ their name for the errors; we keep the letters E_a.

> **Result 18.15 (Knill and Laflamme, Theorem 3.2; Gottesman, Theorem 3; stated).** A code corrects the errors $\{E_a\}$ if and only

if, for all codewords $|\psi_i\rangle$, $|\psi_j\rangle$ with $i \neq j$ and all E_a, E_b,

$$\langle\psi_i| E_a^\dagger E_b |\psi_i\rangle \;=\; \langle\psi_j| E_a^\dagger E_b |\psi_j\rangle\,, \qquad \langle\psi_i| E_a^\dagger E_b |\psi_j\rangle \;=\; 0,$$

which are Knill and Laflamme's (19) and (20). Gottesman writes the same two conditions as one, his (30): $\langle\psi_i| E_a^\dagger E_b |\psi_j\rangle = C_{ab}\,[i = j]$, where C_{ab} is a Hermitian matrix that does not depend on i and j; and, for a linear space $\mathcal{E}$ of errors, as his (29): $\langle\psi| E^\dagger E |\psi\rangle = C(E)$ for every $E \in \mathcal{E}$ and every unit vector $|\psi\rangle \in Q$, with $C(E)$ independent of $|\psi\rangle$. If the code has an encoder Enc with $\mathrm{Enc}^\dagger\mathrm{Enc} = I$ whose columns are the codewords, the condition is

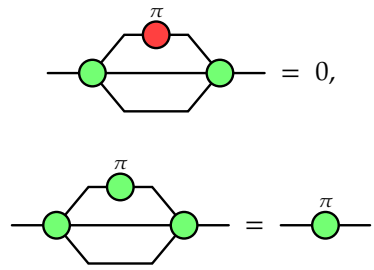
$$\mathrm{Enc}^\dagger\, E_a^\dagger E_b\, \mathrm{Enc} \;=\; C_{ab}\, I.$$

We state the theorem and do not prove it; the proofs are Knill and Laflamme's §3 and Gottesman's §2.2. The last line is ours, and it is one sentence: the (i,j) entry of $\mathrm{Enc}^\dagger E_a^\dagger E_b \mathrm{Enc}$ is $\langle\psi_i| E_a^\dagger E_b |\psi_j\rangle$, so (30) says exactly that this matrix is C_{ab} times the identity. Read it as a statement about questions. $\mathrm{Enc}^\dagger E_a^\dagger E_b \mathrm{Enc}$ is what the pair of errors looks like *from inside the code*. The condition asks that it look like a number: the same on every message, and never mixing one message with another. Whatever the errors reveal, they reveal about themselves.

In the sketchbook. $\mathrm{Enc}^\dagger E_a^\dagger E_b \mathrm{Enc}$ is a closed-off picture: the parting ring, whatever the errors draw on the strands, and the joining ring. The condition says that it must come down to **a number times a bare wire**. A question that reads the message leaves something on the wire that is not a number.

Result 18.16 (what the three strands may be asked). Exactly: (a) for the single gusts $\mathcal{E}_X = \{I, X_1, X_2, X_3\}$, $\mathrm{Enc}^\dagger X^e X^{e'} \mathrm{Enc} = [e = e']\, I$, so $C = I_4$ and the conditions hold; (b) for $\{I, Z_1\}$ the conditions fail: $\mathrm{Enc}^\dagger Z_1 \mathrm{Enc} = Z$, which is not a number times I; they fail too for the clip-bell's projector, $\mathrm{Enc}^\dagger \frac{1}{2}(I + Z_1)\mathrm{Enc} = |0\rangle\langle 0|$; (c) for the twenty-eight operators I, X_k, Y_k, Z_k on Shor's nine wires the conditions hold, and the 28×28 matrix C has

rank 22. As pictures:



Proof. (a) $X^e X^{e'} = X^{e \oplus e'}$, and $\mathrm{Enc}^\dagger X^f \mathrm{Enc}$ is I for $f = 000$, X for $f = 111$ and 0 otherwise, since X^f moves $|000\rangle$ and $|111\rangle$ out of the code space unless f is 000 or 111. For two single-gust patterns, $e \oplus e'$ is 000 when $e = e'$ and has weight 1 or 2 otherwise. (b) $Z_1 \mathrm{Enc} = \mathrm{Enc} Z$ (§18.3), and $\frac{1}{2}(I + Z_1)\mathrm{Enc} = \mathrm{Enc}\,\frac{1}{2}(I + Z)$. (c) By computation, in the code. The pictures: in the first, slide the joining ring into the parting ring along the bare middle strand (row 57). The lower strand becomes a loop, which goes (row 90); the upper strand becomes a red half tied twice to the one green spider, and Result 18.8 at a half gives 0. In the second, the green half on the upper strand fuses in as well, and what is left is a green spider turned π with two legs: Z, a wire that still carries a question. ■ See `Code/Ch18/WhatABellMayAsk.py`, where PyZX's `full_reduce` takes each of the sixteen pictures of (a) to a bare wire with the number 1 or to 0, and leaves one green spider turned π for Z_1.

Part (b) is the clip-bell. For a dark or lit message it looked harmless; the conditions look past that, at every message at once, and see that $|000\rangle$ and $|111\rangle$ get different numbers. Part (c) has a new feature. Shor's code corrects the twenty-eight operators, but C is not invertible: Z_1, Z_2 and Z_3 act on the code in the same way, so their rows of C are equal. Gottesman: "If the matrix C_{ab} in (30) has maximum rank, the code is called nondegenerate. If not, as for the nine-qubit code, the code is degenerate." A degenerate code need not tell errors apart; it need only undo them.

Counting what a code can do

A code with n wires carrying k messages is written $[[n, k, d]]$, with double brackets to tell it from a classical code. The **distance** d is, in Gottesman's second definition, the least weight of a Pauli product E for which $\langle\psi|E|\psi\rangle$ depends on which code state $|\psi\rangle$ it is taken in (his (34)): the lightest error whose effect some question on the code could notice. A code of odd distance d corrects every error on at most $(d-1)/2$ wires (Gottesman, §§2.2–2.3; stated, not proved here). Two bounds from Gottesman's §2.2 limit what is possible, and we state them without proof. For a nondegenerate code, counting states gives the **quantum Hamming bound**, his (35):

$$\sum_{j=0}^{(d-1)/2} 3^j \binom{n}{j} 2^k \;\le\; 2^n,$$

since each error on j wires is one of $3^j\binom{n}{j}$ Pauli strings and each must send the code to its own orthogonal copy. With $k = 1$ and $d = 3$ it reads $(1 + 3n)\,2 \le 2^n$, which first holds at $n = 5$. Gottesman adds that "we do not know of any codes that beat it". The **Knill–Laflamme bound**, also called the quantum Singleton bound, his (38), "applies even to degenerate quantum codes":

$$n - k \;\ge\; 2d - 2.$$

A $[[4, 1, 3]]$ code would violate it, so no code that carries one message and corrects every error on one wire can do it with fewer than five wires, and the $[[5, 1, 3]]$ code of Gottesman's Table 4 "really is optimal".

Stabilizers

The checks of Shor's code generate a group, and the group fixes the code. Gottesman's §2.3 turns this into a general construction. The **Pauli group** on n wires is the set of products of I, X, Y, Z, one on each wire, times ± 1 or $\pm i$; the **weight** of such a product is the number of wires on which it is not I. A **stabilizer** is a subgroup $\mathcal{S}$ of it in which any two elements commute and which does not contain $-I$; its code $Q_{\mathcal{S}}$ is the set of states $|\psi\rangle$ with $M|\psi\rangle = |\psi\rangle$ for every $M \in \mathcal{S}$. The **normalizer** $N(\mathcal{S})$ is the set of Pauli

products that commute with every element of $\mathcal{S}$. Its elements outside $\mathcal{S}$ act on the code without being noticed by any check: they are the code's own logical operations, and so they are the errors it cannot catch.

> **Result 18.17 (Gottesman, Theorem 5; stated).** Let $\mathcal{S}$ be a subgroup of order 2^g of the n-wire Pauli group whose elements commute, with $-I \notin \mathcal{S}$, and let d be the least weight of an element of $N(\mathcal{S}) \setminus \mathcal{S}$. Then $Q_{\mathcal{S}}$ is an $[[n, n-g, d]]$ code.

Here g is the number of independent checks; Gottesman calls it a, a letter this chapter keeps for errors. Our examples, said plainly:

- **The bit-flip code is** $[[3,1,1]]$**.** Its stabilizer is $\{I, Z_1Z_2, Z_2Z_3, Z_1Z_3\}$, of order 2^2, so $k = 3 - 2 = 1$. And Z_1 commutes with all four and is not one of them: a single phase flip is a logical operation, as §18.3 found. Its weight is 1, so $d = 1$. The code corrects every single bit flip and no single phase flip, and the distance counts the worst kind.
- **Shor's code is** $[[9,1,3]]$, with the eight generators of Table 2; the seven-qubit code of Table 3 is $[[7,1,3]]$, with six; the five-qubit code of Table 4 is $[[5,1,3]]$, with four.
- **CSS codes.** Table 3 is built from the classical seven-bit Hamming code, its parity checks written once with Z and once with X. Gottesman's general form uses two classical codes, one for each kind of check, "known as the CSS construction after its inventors Calderbank, Shor, and Steane".

The first item is our own count; `Code/Ch18/WhatABellMayAsk.py` checks it, and checks, for the generators of Tables 2, 3 and 4 alike, that no Pauli product of weight 2 or less lies in $N(\mathcal{S}) \setminus \mathcal{S}$ and that one of weight 3 does.

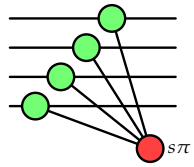
In the sketchbook. A check of a stabilizer code is a Pauli product, and a product of Zs is a bell with more ties. Put a green spider at rest on each of w wires and tie all of them to one red spider turned $s\pi$, the **hub**. The ties must cross the wires in between to reach it; a crossing joins nothing.

> **Result 18.18 (a check of weight w).** For every $w \geq 1$ and each bit s, exactly, w green spiders at rest tied to one red hub turned

$s\pi$ equal

$$2^{1-w/2} \cdot \tfrac{1}{2}(I + (-1)^s Z^{\otimes w}).$$

For $w = 4$:



$$= \tfrac{1}{2} \cdot \tfrac{1}{2}(I + (-1)^s Z \otimes Z \otimes Z \otimes Z).$$

A check made of Xs is the same with the colours swapped, with the same number.

Proof. As for Result 18.5: the feet copy $|x\rangle$ onto the ties, and the hub gives $\langle + \cdots + |x\rangle + (-1)^s \langle - \cdots - |x\rangle = 2^{-w/2}(1 + (-1)^s(-1)^{|x|})$, where $|x|$ is the number of 1s in x. For the colours swapped: put a yellow box on every leg of every spider and pairs of boxes on the ties (rows 56, 58); on the blackboard this is conjugation by $H^{\otimes w}$, which turns $Z^{\otimes w}$ into $X^{\otimes w}$ and keeps the number. ■ See `Code/Ch18/WhatABellMayAsk.py`, which checks $w = 2, 3, 4$ and both values of s, and lets PyZX's `full_reduce` show that the check followed by itself is $2^{1-w/2}$ times the check.

At $w = 2$ the number is exactly 1, which is why the bells of §18.2 needed none. At $w = 4$ it is $\frac{1}{2}$.

A weave on a torus

Kitaev's §1 builds a code in which every check has weight at most four and every wire is asked by at most four checks. In our letters: take an $\ell \times \ell$ square grid drawn on a torus, so that the right edge is glued to the left and the top to the bottom, and put a wire on every edge of the grid, so $n = 2\ell^2$. For each vertex v, let A_v be the product of X on the four edges that meet at v, its **star**; for each face f, let B_f be the product of Z on the four edges around it. Kitaev writes A_s, B_p, σ^x, σ^z and k; we write A_v, B_f, X, Z and ℓ, because s, p, σ and k already have jobs here.

Result 18.19 (the toric code; Kitaev, §1). The A_v and B_f all

> commute. They satisfy exactly two relations, $\prod_v A_v = I$ and $\prod_f B_f = I$, so $2\ell^2 - 2$ of them are independent, and the states fixed by all of them form a space of dimension $2^{2\ell^2-(2\ell^2-2)} = 4$: two protected messages. Kitaev calls it "a toric code". An error that is not detected and not harmless must contain a loop that goes around the torus, so it acts on at least ℓ wires.

Proof of the first sentence. A star and a face share either no edge or two, as Kitaev notes, and X and Z anticommute on each shared edge, so A_v and B_f meet an even number of sign changes and commute. Two stars, or two faces, are made of one kind of Pauli and commute. The rest we take from Kitaev: the count of relations, the dimension (which he derives two ways, once by counting independent checks, as in Theorem 5, and once by finding the operators that act on the protected space) and the loops. ■ Here is the count at $\ell = 2$: eight wires, four vertices, four faces; eight checks with two relations, six independent; by Theorem 5, $[[8, 8-6, d]]$, a space of dimension $2^2 = 4$. See `Code/Ch18/WhatABellMayAsk.py`, which builds the checks over the two-element field for $\ell = 2$, 4 and 5 and finds the same 4 each time.

In the sketchbook every B_f is a weight-four hub, red, with green feet on the four edges of a face, and every A_v is its colour-swapped twin on the four edges of a star, each with the number $\frac{1}{2}$ of Result 18.18. Drawn over the grid, the checks are a weave of hubs, each tied to four neighbouring wires. Kitaev's point is locality. His toric codes are "local check codes": each check involves at most four wires, and each wire is in at most four checks, however large ℓ is. By his loop argument the distance is at least ℓ, so the code is certain to correct every error on fewer than $\ell/2$ wires; Kitaev adds that it "corrects almost any multiple error" on a number of wires proportional to ℓ^2.

The surface code. Fowler, Mariantoni, Martinis and Cleland describe what grew from this. "The surface codes evolved from an invention of Alexei Kitaev known as toric codes"; the torus "turned out to be unnecessary, and planar versions (thus 'surface codes') were developed by Bravyi and Kitaev as well as Freedman and Meyer". They report a tolerance of "a per-operation error rate as high as about 1%", against "per-step thresholds

of about 2×10^{-5}" for the Steane and Bacon–Shor codes laid out on two-dimensional grids with nearest-neighbour coupling. The price is wires: "It takes a minimum of thirteen physical qubits to implement a single logical qubit", and a reasonably fault-tolerant one "takes of order 10^3 to 10^4 physical qubits". In their own simulations (§VII), with errors of several kinds injected into every step of the checking circuit at a rate set by p, they find a threshold they call $p_{th} = 0.57\%$: below it the logical error rate falls exponentially as the code's distance grows, and above it a larger code does worse.

The threshold

Everything so far assumed that the storm hits only the wires. In a real machine the bells are made of the same stuff as the wires, and the hand that mends can slip. A code whose repairs make more errors than they catch is worse than no code. The theorem that makes quantum computing possible in principle says that below a certain error rate this can be beaten, at a modest cost.

> **Result 18.20 (the threshold theorem; Gottesman, Theorem 10; stated, not proved).** In the book's letters: "There is a threshold error rate p_T. Suppose we have a local stochastic error model with $p_i \le p < p_T$. Then for any ideal circuit C, and any $\epsilon > 0$, there exists a fault-tolerant circuit C' which, when it undergoes the error model, produces an output which has statistical distance at most ϵ from the output of C. C' has a number of qubits and a number of timesteps which are at most $\mathrm{polylog}(|C|/\epsilon)$ times bigger than the number of qubits and timesteps in C, where $|C|$ is the number of locations in C."

Here p_i bounds the probability of a fault at location i of the circuit. Gottesman proves it by **concatenation**: encode every wire in a code that corrects t errors, then encode every wire of *that* in the same code, and so on, L levels deep. If one level turns an error rate p into at most Ap^{t+1}, his (86) gives, after j levels, $p^{(j)} \le A(p^{(j-1)})^{t+1}$ with $p_T = 1/A^{1/t}$, and his (87):

$$p^{(j)}/p_T \;\le\; (p/p_T)^{(t+1)^j}.$$

Below p_T the error falls doubly exponentially in the number of levels, while the size grows only exponentially in it; that is where the polylog comes from. We do not reproduce the proof, whose hard part is to show that each level really does turn p into Ap^{t+1} when every gate, including those of the checks and the mending, is noisy.

What is p_T? Each answer below is its source's, with its date. Gottesman (2009) quotes a proven lower bound of $p_T \geq 2.73 \times 10^{-5}$ for the seven-qubit code; simulations by Knill with "a threshold p_T of as high as 5%, depending on the details of the error model"; and "the current best rigorous lower bound", 10^{-3}. Kitaev (1997), in a footnote: "Actually, the threshold is not known. Estimates vary from $1/300$ to 10^{-6}." Fowler and co-authors (2012), for the surface code under their error model: 0.57% per step. These numbers are not in conflict: they answer different questions, about different codes and different models of noise.

There is no picture of this theorem (row 143). A rewrite proves that two processes are equal. The threshold is an inequality between probabilities, a statement about a whole family of circuits as they grow, and nothing in the sketchbook can say it.

Dirac's Blackboard.

A vote of votes: a toy, not the theorem. Take the classical vote of Result 18.2 and apply it to its own output: write each bit three times, write each of those three times, and vote twice. The failure rate becomes $F(F(p))$. At $p = \frac{1}{10}$: $F(0.1) = 0.028$, and $F(0.028) = 3(0.028)^2 - 2(0.028)^3 = 0.002352 - 0.000043904 = 0.002308096$. Each level roughly squares the rate and multiplies it by three. But $F(p) - p = p(1-p)(2p-1)$, which is negative for $0 < p < \frac{1}{2}$ and positive for $\frac{1}{2} < p < 1$; since F is increasing on $[0,1]$ ($F'(p) = 6p(1-p) \geq 0$), the levels drive every p below $\frac{1}{2}$ down towards 0 and every p above $\frac{1}{2}$ up towards 1. At $p = 0.6$, $F(0.6) = 0.648$: voting makes it worse. So the toy has a threshold, $\frac{1}{2}$. It is a toy: its votes are counted without error, and it knows only bit flips. Gottesman's threshold is far smaller because his counting is itself done by noisy gates.

Feynman's Notebook.

Conditions, a torus, and a threshold. E. Knill and R. Laflamme posted "A theory of quantum error-correcting codes" in April 1996, as Los Alamos report LA-UR-96-1300 (journal version, per Gottesman: *Physical Review A* 55, 1997). Gottesman credits his Theorem 3 to it and to C. Bennett, D. DiVincenzo, J. Smolin and W. Wootters (*Physical Review A* 54, 1996). In July 1997 A. Yu. Kitaev posted "Fault-tolerant quantum computation by anyons". He asked why a magnet remembers: a flipped spin is flipped back by its neighbours, and "This process is quite similar to the standard error correction procedure for the repetition code. We may say that errors are being corrected at the physical level." His quantum answer begins with the toric codes of his §1. Gottesman credits the threshold theorem to D. Aharonov and M. Ben-Or; to A. Y. Kitaev, "Quantum error correction with imperfect gates"; and to E. Knill, R. Laflamme and W. H. Zurek; and notes that "Shor presented the first protocols for fault-tolerant quantum computation" (his ref. [44], dated 1996). For codes "based on topological constructions" he cites E. Dennis, A. Kitaev, A. Landahl and J. Preskill, *Journal of Mathematical Physics* 43 (2002).

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§18.5 Split and Merge

A message spread over a weave of wires is safe. But a computer must also *act* on its messages, and two messages in two separate weaves must sometimes be joined, by a CNOT, say, without unwrapping either of them. Unwrapping would expose them to the storm at the worst moment. In the surface code this is done by **lattice surgery**: two patches of the weave are joined along an edge and then cut apart again, and each such step acts on the protected messages in a definite way. De Beaudrap and Horsman showed that the steps are spiders.

We use their letters with one change. They write U_S, U_R, $K_{j,S}$ and $K_{j,R}$

for the smooth and rough operations; we write U_{sm}, U_{ro}, $K_{j,\text{sm}}$ and $K_{j,\text{ro}}$, because in this book S is the phase gate and $\mathcal{R}$ is the recovery. The bit j is a merge's outcome.

On the blackboard. A **split** takes one protected message into two. The smooth split and the rough split act on it as

$$U_{\text{sm}} = |00\rangle\langle 0| + |11\rangle\langle 1|, \qquad U_{\text{ro}} = |{+}{+}\rangle\langle +| + |{-}{-}\rangle\langle -|.$$

The first is Chapter 11's copier δ (row 81); the second is its twin in the other basis. A **merge** takes two messages into one and, as it does so, answers a question with a bit j. The smooth merge has the two outcomes

$$K_{0,\text{sm}} = |0\rangle\langle 00| + |1\rangle\langle 11|, \qquad K_{1,\text{sm}} = |0\rangle\langle 10| + |1\rangle\langle 01|,$$

and the rough merge the two outcomes

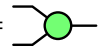
$$K_{0,\text{ro}} = |+\rangle\langle {+}{+}| + |-\rangle\langle {-}{-}|, \qquad K_{1,\text{ro}} = |+\rangle\langle {-}{+}| + |-\rangle\langle {+}{-}|.$$

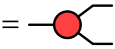
In the negative branches the merge acts after a flip of its first input: $K_{1,\text{sm}} = K_{0,\text{sm}}(X\otimes I)$ and $K_{1,\text{ro}} = K_{0,\text{ro}}(Z\otimes I)$. For each merge $K_0^\dagger K_0 + K_1^\dagger K_1 = I$, so the two outcomes are the two branches of one channel, with Kraus operators K_0 and K_1. The form of $K_{1,\text{sm}}$ and $K_{1,\text{ro}}$ depends on a convention. De Beaudrap and Horsman give two, according to whether "the Pauli frame of the child memory" is adapted "to agree with the second parent memory" or with the first; we use the first of their two, which is the one written above.

A small calculation shows what a smooth merge asks. On $|ab\rangle$, $K_{0,\text{sm}}$ is non-zero only when $a = b$ and $K_{1,\text{sm}}$ only when $a \neq b$. So the outcome is $j = a \oplus b$, with certainty: the smooth merge asks whether two messages agree, which is exactly the question of the bells.

In the sketchbook. Each of the four is a spider with three legs, at rest.

> **Result 18.21 (splits and merges are spiders; de Beaudrap and Horsman, Lemma 1 and Corollary 2).** Exactly, with no number:

$U_{\text{sm}} =$  , $K_{0,\text{sm}} =$ 

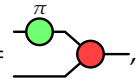
$K_{0,\mathrm{ro}} =$  , $U_{\mathrm{ro}} =$ 

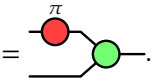
De Beaudrap and Horsman's Lemma 1 says that the effect on the logical state of a smooth split, and of "the positive branch" of a smooth merge, is the green three-legged spider at rest, and the same for rough operations and red. Their Corollary 2 says that the positive branches of lattice surgery "provide a model for the equational theory of the ZX calculus without projections", with spiders of two or three legs at rest.

Proof of the equalities. Each is row 53 or row 54 with three legs, the red ones with the normalized $|\pm\rangle$ of Chapter 8. That surgery on a surface code acts on the protected message by these matrices is de Beaudrap and Horsman's result, which we state and do not prove. ■ See `Code/Ch18/SplitAndMerge.py`.

The negative branch is where the pictures earn their keep. $K_{1,\mathrm{ro}} = K_{0,\mathrm{ro}}(Z \otimes I)$, and Z is a green half.

Result 18.22 (the negative branch; de Beaudrap and Horsman, Lemma 3 and Corollary 4). In the first of their two conventions, exactly: the negative branch of a rough merge is the red merge with a green half on its first incoming leg, and the negative branch of a smooth merge is the green merge with a red half on its first incoming leg:

$K_{1,\mathrm{ro}} =$ 

, $K_{1,\mathrm{sm}} =$ 

.

In their other convention the half sits on the second incoming leg.

Proof. $K_{1,\mathrm{ro}} = K_{0,\mathrm{ro}}(Z \otimes I)$ and $K_{1,\mathrm{sm}} = K_{0,\mathrm{sm}}(X \otimes I)$, as one checks on the four basis states; a two-legged green half is Z and a two-legged red half is X (Chapter 8, §8.2). ■ See `Code/Ch18/SplitAndMerge.py`.

A merge is not always a coin. Merge two messages both at rest, $|{+}{+}\rangle$, roughly: $K_{0,\mathrm{ro}}|{+}{+}\rangle = |+\rangle$ and $K_{1,\mathrm{ro}}|{+}{+}\rangle = 0$, so the outcome is $j = 0$ with

probability 1. What the coin depends on is the question and the messages. For the joiner, it turns out, the coin is fair whatever the messages are.

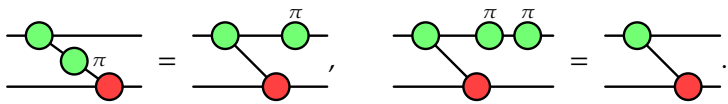
A joiner by surgery

Chapter 8's joiner (row 64) is a green spider on the upper wire, a red spider on the lower wire, and one plain cord between them; bare, it is CNOT$/\sqrt{2}$. Read it again with Result 18.21 in hand. The green spider has one leg in and two out: it is a smooth split of the upper message. The red spider has two legs in and one out: it is a rough merge of the split's second half with the lower message. De Beaudrap and Horsman note that "the standard lattice-surgical realisation of CNOT" is exactly this composition, "Omitting the correction operations needed to realise CNOT deterministically", and credit it to Horsman, Fowler, Devitt and Van Meter (their ref. [30]).

Result 18.23 (a joiner by surgery). For $j = 0, 1$, exactly,

$$(I \otimes K_{j,\mathrm{ro}})(U_{\mathrm{sm}} \otimes I) \;=\; (Z^j \otimes I)\,\frac{\mathrm{CNOT}}{\sqrt{2}} \;=\; \frac{\mathrm{CNOT}}{\sqrt{2}}\,(Z^j \otimes I).$$

Each branch occurs with probability $\frac{1}{2}$ for every input, product or entangled, and a green half on the upper wire after the negative branch mends it: both branches then give the CNOT. As a picture, the negative branch and its mending are



Proof. On the blackboard: follow $|ab\rangle$. The split gives $|a\rangle\,|a\rangle\,|b\rangle$; the rough merge of the last two sends $|ab\rangle$ to $\frac{1}{\sqrt{2}}\,|a \oplus b\rangle$ (for $j = 0$, since $\langle + + |ab\rangle = \frac{1}{2}$, $\langle - - |ab\rangle = \frac{1}{2}(-1)^{a\oplus b}$, and $\frac{1}{2}(|+\rangle + (-1)^{a\oplus b}\,|-\rangle) = \frac{1}{\sqrt{2}}\,|a \oplus b\rangle$). So the $j = 0$ branch sends $|ab\rangle$ to $\frac{1}{\sqrt{2}}\,|a\rangle\,|a \oplus b\rangle$, which is CNOT$/\sqrt{2}$. The $j = 1$ branch is the same after Z on the split's second output, which the split turns into Z on its input: $U_{\mathrm{sm}}Z = (I \otimes Z)U_{\mathrm{sm}}$. A Z on the control commutes with the CNOT.

For every unit vector $|\psi\rangle$, $\|(Z^j \otimes I)\,\mathrm{CNOT}\,|\psi\rangle\,/\sqrt{2}\|^2 = \frac{1}{2}$, since $(Z^j \otimes I)\,\mathrm{CNOT}$ is unitary. By the rules: in the negative branch the green half sits on the merge's first incoming leg, which is the split's second output (Result 18.22). It fuses into the split (row 57), and fusion backwards (row 57) puts it out again on the split's upper output; that is the first picture. The mending half then meets it on the upper wire, and two green halves fuse to a whole turn, which is rest: a bare wire (row 57). ■ See `Code/Ch18/SplitAndMerge.py`, which checks both branches, the probability $\frac{1}{2}$ on five random product inputs and one random entangled one, the certain bare rough merge on $|{+}{+}\rangle$, and lets PyZX's `full_reduce` bring the surgery picture and the circuit picture of the CNOT to the same picture.

The coin is fair for the joiner because each of its branches is a unitary divided by $\sqrt{2}$. That is also why the bare joiner of Chapter 8 carried the number $1/\sqrt{2}$. De Beaudrap and Horsman put it this way: the subnormalisation of their diagrams "is a feature of ZX notation, not a bug", since it records "the 2-norms of the Kraus operators". Their Theorem 5 makes it general: lattice surgery procedures "model randomly-constructed simple ZX diagrams", in which each merge is one of its two branch pictures, "and the probability of each diagram is given by its 2-norm". We state it and do not prove it.

One kind of ring

Look back at what the chapter has built. The parting ring spreads one message over three wires: it is $|000\rangle\langle 0| + |111\rangle\langle 1|$, a green spider. A bell asks two wires whether they agree: two green spiders and a red one. A smooth split is a green spider, a merge is a spider of either colour, and a CNOT between two protected messages is a split and a merge. The phase-flip code is the same spiders in the other colour, and Shor's code is the two stacked with yellow boxes between. Every move of error correction, copying, asking and joining, is made of one kind of map, $|0\cdots0\rangle\langle 0\cdots 0| + |1\cdots1\rangle\langle 1\cdots 1|$, and its colour-changed twin. That is the reason this book has been able to draw every step.

Why this definition. Why describe a merge by its two Kraus operators rather than by "the" operation it performs? Because it has no single operation. A

merge is a measurement: it answers with a bit, and the bit is not chosen by anyone. Writing both branches keeps the answer in view, and the answer is what tells the mending hand what to do. Hiding it, as a positive-branch picture alone would, would make lattice surgery look deterministic, and it is not.

Feynman's Notebook.

Surgery on a weave. Lattice surgery was introduced by C. Horsman, A. G. Fowler, S. Devitt and R. Van Meter in "Surface code quantum computing by lattice surgery", *New Journal of Physics* 14 (2012), 123011; de Beaudrap and Horsman call it "the original paper on lattice surgery", and it already described splitting one patch into several at once. Niel de Beaudrap and Dominic Horsman posted "The ZX calculus is a language for surface code lattice surgery" in 2017, and it was accepted in *Quantum* in 2019. Their contribution was to say what the operations do *to the protected message* in every branch, including the unlucky ones: "we can prove what effect such operations have on the encoded data, and how they interact with each other". The spiders this book has drawn since Chapter 8 turned out, in their hands, to be a literal description of how the patches of a surface code are cut and joined.

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§18.6 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 17's row 135. The chapter introduces no new string-diagram element: rows 136 to 144 are new uses of the spiders, boxes and doubled wires of Chapters 8 to 17, one new rule derived from them (Result 18.8, in row 139), and statements quoted from the sources.

#	Algebra (Dirac)	Picture (Coecke)
136	noise: $\mathcal{N}_p^X(\rho) = (1-p)\rho + pX\rho X$ (Kraus form, Gottesman's eq. (23)); independent on each wire, $(\mathcal{N}_p)^{\otimes n}$	**the doubled channel as a weighted sum**: $(1-p)\times$ the bare doubled wire $+\, p\times$ a red half on the wire and one on its mirror copy; after this, branch by branch
137	$\mathrm{Enc} = \lvert 000\rangle\langle 0\rvert + \lvert 111\rangle\langle 1\rvert$; $\mathrm{Enc}\lvert\psi\rangle \neq \lvert\psi\rangle^{\otimes 3}$ unless $\alpha\beta = 0$; $\mathrm{Enc}^\dagger\mathrm{Enc} = I$	**the encoder is one green spider**, one leg in and three out, no number; dark and lit points copy through it ($\frac{1}{2}\times$ three points); a turned point fuses
138	$P_s = \frac{1}{4}(I + (-1)^{s_1}Z_1Z_2)(I + (-1)^{s_2}Z_2Z_3)$; $P_sX^e\mathrm{Enc} = [s = \sigma(e)]\, X^e\mathrm{Enc}$	**a bell asks a pair**: two green spiders at rest tied to one red at $s\pi$ is exactly $\frac{1}{2}(I + (-1)^s Z \otimes Z)$
139	**the headline:** $\mathrm{Enc}^\dagger X^{c(s)}P_sX^e\mathrm{Enc} = [s = \sigma(e)]\, I$ on $\mathcal{E}_X$; two or more flips give X	***The Storm Line***: halves through tell rings into bells and mending rings; a bell tied twice to one green spider comes off with $(1 + e^{i\theta})/2$, which is 1 at rest and 0 at a half (derived: rows 57, 62, 54)
140	$\mathrm{Enc}^\dagger Z_k\mathrm{Enc} = Z$; $H^{\otimes 3}\mathrm{Enc}\, H$; $\mathrm{Enc}_9 = \mathrm{Enc}^{\otimes 3}H^{\otimes 3}\mathrm{Enc}$; Gottesman's Theorem 2	**phase flips pass unheard**: a green half fuses into a tell ring; **colour change** makes the red encoder; Shor's nine is green, yellow, green

#	Algebra (Dirac)	Picture (Coecke)
141	Knill–Laflamme: $\langle\psi_i\vert\, E_a^\dagger E_b\,\vert\psi_j\rangle = C_{ab}[i=j]$; $\mathrm{Enc}^\dagger E_a^\dagger E_b \mathrm{Enc} = C_{ab} I$	**an allowed question closes to a number times a bare wire**; a question that reads the message leaves a turn on the wire
142	stabilizer $\mathcal{S}$, $[[n, n-g, d]]$ (Gottesman's Theorem 5); toric code, A_v, B_f, $n = 2\ell^2$, dimension 4	**a check of weight** w $= 2^{1-w/2}\times$ its projector: w green spiders tied to one red hub; stars and faces are weight-four hubs on a weave
143	the threshold theorem (Gottesman's Theorem 10), stated	**no picture**: a rewrite proves an equality, never a bound
144	U_{sm}, U_{ro}, $K_{j,\mathrm{sm}}$, $K_{j,\mathrm{ro}}$; $(I \otimes K_{j,\mathrm{ro}})(U_{\mathrm{sm}} \otimes I) =$ $(Z^j \otimes I)\,\mathrm{CNOT}/\sqrt{2}$	**split and merge are spiders**; the negative branch leaves a half of the other colour on the first incoming leg; the joiner is a smooth split, then a rough merge

Every string-diagram row is exact, number included. Row 136 is the chapter's only doubled picture, a sum of two pictures rather than one; rows 138 and 142 carry their numbers, 1 and $2^{1-w/2}$; row 139's rule is derived in §18.2 from Hopf and fusion; rows 141, 142 and 144 sit beside theorems quoted from Knill and Laflamme, Gottesman, Kitaev and de Beaudrap and Horsman; and row 143 is a theorem with no picture at all.

§18.7 The Theorem, Twice

Result 18.7 (one gust, mended blind), restated. With $\mathrm{Enc} = |000\rangle\langle 0| + |111\rangle\langle 1|$, the checks P_s, the syndrome map $\sigma(e) = (e_1 \oplus e_2,\ e_2 \oplus e_3)$ and the corrections $c(00) = 000$, $c(10) = 100$, $c(11) = 010$, $c(01) = 001$: for every $X^e \in \mathcal{E}_X = \{I, X_1, X_2, X_3\}$ and every outcome s, exactly,

$$\mathrm{Enc}^\dagger\, X^{c(s)}\, P_s\, X^e\, \mathrm{Enc} \;=\; [\, s = \sigma(e)\,]\, I.$$

Dirac's Blackboard

Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ be any message, and write $\bar{e}$ for e with every bit flipped.

Step 1: spread, then the gust. $\mathrm{Enc}\,|\psi\rangle = \alpha|000\rangle + \beta|111\rangle$. A gust X^e flips the bits where e has a 1, so $X^e|000\rangle = |e\rangle$ and $X^e|111\rangle = |\bar{e}\rangle$:

$$X^e\,\mathrm{Enc}\,|\psi\rangle \;=\; \alpha|e\rangle + \beta|\bar{e}\rangle\,.$$

Step 2: the checks. $Z_1Z_2|x\rangle = (-1)^{x_1 \oplus x_2}|x\rangle$ for every basis word x. Flipping both bits does not change their parity, so Z_1Z_2 multiplies $|e\rangle$ and $|\bar{e}\rangle$ by the same sign, $(-1)^{e_1 \oplus e_2}$; likewise Z_2Z_3, with $(-1)^{e_2 \oplus e_3}$. Hence

$$\begin{aligned}\tfrac{1}{2}(I + (-1)^{s_1}Z_1Z_2)(\alpha|e\rangle + \beta|\bar{e}\rangle) \;&=\; \tfrac{1}{2}(1 + (-1)^{s_1 + e_1 \oplus e_2})(\alpha|e\rangle + \beta|\bar{e}\rangle)\\ &=\; [\, s_1 = e_1 \oplus e_2\,](\alpha|e\rangle + \beta|\bar{e}\rangle),\end{aligned}$$

and with the second factor,

$$P_s\, X^e\,\mathrm{Enc}\,|\psi\rangle \;=\; [\, s = \sigma(e)\,](\alpha|e\rangle + \beta|\bar{e}\rangle).$$

The number in front is the same for α and for β: the checks never separate them.

Step 3: the mending. If $s \neq \sigma(e)$, the state is already 0, and so is everything after. If $s = \sigma(e)$: on the four single-gust patterns, $\sigma(000) = 00$, $\sigma(100) = 10$, $\sigma(010) = 11$ and $\sigma(001) = 01$, four different answers, and the table c was

built as the inverse of this list, so $c(\sigma(e)) = e$. Then $X^{c(s)}X^e = X^eX^e = I$, and

$$X^{c(s)}\,P_s\,X^e\,\mathrm{Enc}\,|\psi\rangle \;=\; \alpha\,|000\rangle + \beta\,|111\rangle\,.$$

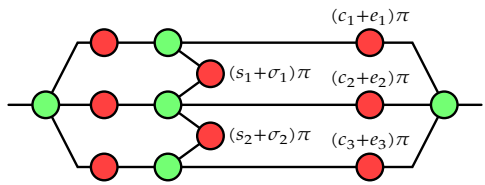
Step 4: the join. $\mathrm{Enc}^\dagger(\alpha\,|000\rangle + \beta\,|111\rangle) = \alpha\,|0\rangle + \beta\,|1\rangle = |\psi\rangle$.

So $\mathrm{Enc}^\dagger X^{c(s)}P_sX^e\mathrm{Enc}\,|\psi\rangle = [s = \sigma(e)]\,|\psi\rangle$ for every $|\psi\rangle$, which is the operator equation. ■ See `Code/Ch18/BlindVote.py`, which checks it for all four gusts and all four outcomes, and the full statement of Result 18.9 for all eight patterns.

Coecke's Sketchbook

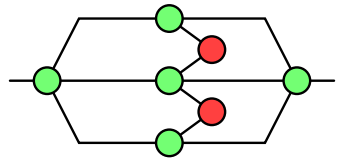
Start from *The Storm Line* as drawn in Result 18.7: the parting ring, one weather ring $e_k\pi$ on each strand, the tell rings with the two bells $s_1\pi$ and $s_2\pi$ hanging from them, one mending ring $c_k\pi$ on each strand, and the joining ring. The picture *is* $\mathrm{Enc}^\dagger X^cP_sX^e\mathrm{Enc}$, exactly, by the reading of §18.2. Every step below names its rule; every number is written. We write σ_1 and σ_2 for the two bits of $\sigma(e)$.

Step (i): each gust's half goes where its ties go (row 120, then row 57). The weather ring on strand k sits on one leg of the tell ring t_k, which is green and at rest. A red half on one leg of a green spider at rest comes out as a red half on every other leg, with the number 1 (row 120; if $e_k = 0$ there is no half, and nothing moves). The weather ring is left at rest. On the strand, the half meets the mending ring and fuses into it (row 57): it becomes $(c_k + e_k)\pi$. On each tie, it meets the bell at the tie's end and fuses into it: the first bell, tied to strands one and two, receives $e_1 + e_2$, and the second receives $e_2 + e_3$. So the bells are now $(s_1 + \sigma_1)\pi$ and $(s_2 + \sigma_2)\pi$, turns being counted modulo a whole turn.

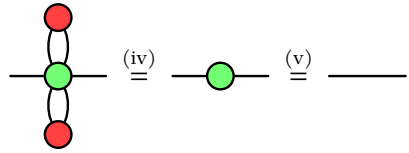


This picture is exact for every labelling. It already shows the syndrome: a gust on the middle strand turns both bells, a gust on an outer strand turns one.

Step (ii): in the good branch every red ring is at rest (row 57). Suppose $s = \sigma(e)$ and $c = c(s) = e$. Then $s_j + \sigma_j$ and $c_k + e_k$ are all even: every bell and every mending ring is at rest. A red spider at rest with two legs is a bare wire (row 57), so the weather and mending rings go, and each strand is one bare cord from the parting ring to its tell ring and one from the tell ring to the joining ring. The bells stay, as rings, for step (iv).



Step (iii): slide along bare cords (row 57, then row 90). Slide the first tell ring into the parting ring along the first strand's weather cord. The parting ring now holds the first strand's mending cord, which runs to the joining ring; slide the joining ring into it along that cord. Now slide the middle tell ring into it along the middle weather cord, and the third along the third. Every ring on this path is green, at rest in this branch, and every cord slid along is bare, so each move is fusion (row 57). One strand carried the slides; the other two strands' mending cords now run from the green spider back to itself, and a loop goes with the number 1 (row 90; the Dirac's Blackboard sidebar of §18.2). What is left is one green spider at rest holding the lead-in, the lead-out and both ties of each bell:

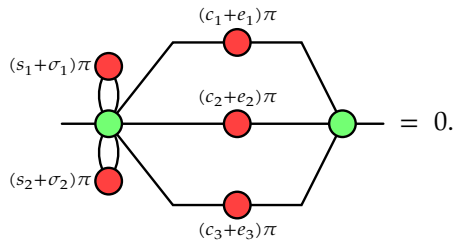


Step (iv): the bells come off (Result 18.8). Each bell is at rest and tied twice to the one green spider, and to nothing else. Result 18.8 at $\theta = 0$ takes

each off with the number $\frac{1}{2}(1+1) = 1$.

Step (v): a bare wire (row 57). A green spider at rest with two legs is a bare wire. The picture equals the bare wire, number 1: the message comes back exactly.

The other branches. If $s \neq \sigma(e)$, go back to the picture of step (i). Its weather rings are at rest whatever the bits, so the weather cords are bare; slide the three tell rings into the parting ring along them (row 57). Each bell is now tied twice to the parting ring, and at least one bell is at $(s_j + \sigma_j)\pi$ with $s_j \neq \sigma_j$, a half:



Result 18.8 at $\theta = \pi$ gives the number $\frac{1}{2}(1 + e^{i\pi}) = 0$, and the whole picture is 0, whatever the mending rings hold. ■ See `Code/Ch18/BlindVote.py`, where PyZX's `full_reduce`, applied to *The Storm Line* with each labelling, derives the same bare wire, number 1, in the good branch and 0 in the others.

The verdict

The blackboard wins, and the chapter says so. Its proof is four steps, each a line: the gust moves both words of the code, the checks see the same parity in both, the table undoes the gust, and the encoder's mirror undoes the encoder. The sketchbook needs five moves, a path chosen with care so that every ring slid along is bare, and a rule, Result 18.8, that had to be derived first from Hopf and fusion. Nothing in the picture proof is hard, but there is more of it.

What the picture shows, and the algebra hides, is *where a gust goes*. On the blackboard σ is a formula, $(e_1 \oplus e_2,\ e_2 \oplus e_3)$. In the picture of step (i) it is

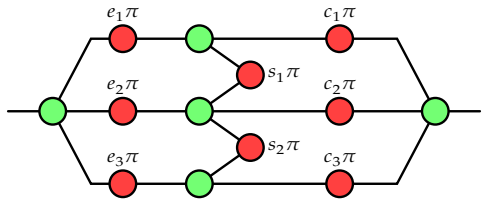
the drawing: a gust's half goes into exactly the bells tied to its strand, and on into its own mending ring, and nowhere else. The ties *are* the syndrome map. The same picture shows why the checks cannot hurt the message: they hang from green spiders at rest, which pass a red half on without looking at what flows along the strand. The pictures win outright elsewhere in the chapter: on phase flips, where colour change turns the whole code into its twin in one move (§18.3), and on surgery, where the negative branch's half slides back through the split to where the mending hand can reach it (§18.5). The Knill–Laflamme conditions are a draw: the picture closes each question to a number times a bare wire, but the matrix C_{ab} of those numbers, and its rank, are the blackboard's. And every probability in the chapter, p_L and the threshold among them, belongs to the blackboard alone.

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§18.8 The Storm Line

Here is the chapter's master diagram: one whole round of the bit-flip code, drawn once for every case. It spreads the message, meets the weather, asks the two checks, mends, and joins. Its bits make it one picture for all branches (row 75): e_k says whether the weather turned strand k, s_j is the answer of bell j, and c_k says whether the mending hand turned strand k back.

Walk it from left to right. The lead-in runs from the input to the parting ring, a green spider at rest. Three strands leave the parting ring, the first on top, and each is drawn in four lengths: from the parting ring to its weather ring, a red spider turned $e_k\pi$; from the weather ring to its tell ring, a green spider at rest; from the tell ring to its mending ring, a red spider turned $c_k\pi$; and from the mending ring to the joining ring, a green spider at rest. Between the first and middle strands hangs the first bell, a red spider turned $s_1\pi$, with a tie to each of the first two tell rings; between the middle and third strands hangs the second bell, turned $s_2\pi$, tied to the middle and third tell rings. So the middle tell ring holds two ties and each outer tell ring holds one. The lead-out runs from the joining ring to the output. No two lines cross. Every ring is drawn whether its bit is 0 or 1, and every tie is a plain cord; there is no yellow box anywhere. We call the picture **The Storm Line**.



The picture has fifteen vertices: the input, the output, and thirteen spiders, five green and eight red. Every piece of it is exact with no number: the parting ring is Enc (Result 18.3), each weather and mending ring is X^{e_k} or X^{c_k}, the tell rings and bells are P_s (Result 18.5), and the joining ring is $\mathrm{Enc}^\dagger$. So the whole is $\mathrm{Enc}^\dagger X^c P_s X^e \mathrm{Enc}$. With the mending hand following the table, $c = c(s)$, it is the headline. With the mending bits left free, it says what every mending does.

> **Result 18.24 (The Storm Line).** For all bits e_1, e_2, e_3, s_1, s_2 and c_1, c_2, c_3, exactly,
>
> $$\text{The Storm Line} \;=\; [\, s = \sigma(e) \,] \cdot \begin{cases} I & \text{if } e \oplus c = 000, \\ X & \text{if } e \oplus c = 111, \\ 0 & \text{otherwise.} \end{cases}$$
>
> With $c = c(s)$ and $X^e \in \mathcal{E}_X$ it is Result 18.7; with $c = c(s)$ and any e it is Result 18.9.

Proof. **On the blackboard.** By Result 18.6, $P_s X^e \mathrm{Enc} = [s = \sigma(e)]\, X^e \mathrm{Enc}$. Then $X^c X^e = X^{e \oplus c}$, and $\mathrm{Enc}^\dagger X^f \mathrm{Enc}$ is I for $f = 000$, X for $f = 111$, and 0 otherwise (the proof of Result 18.16). **In the sketchbook.** Step (i) of §18.7 holds for every labelling: the bells become $(s_j + \sigma_j)\pi$ and the mending rings $(c_k + e_k)\pi$. If $s \neq \sigma(e)$, the last picture of §18.7 gives 0. If $s = \sigma(e)$, both bells are at rest, and there are three cases, by the weight of $e \oplus c$. If it is 000, steps (ii) to (v) give the bare wire. If it is 111, every mending ring is a half: slide the tell rings into the parting ring along the bare weather cords (row 57), and the bells come off with the number 1 (Result 18.8); push the first strand's half through the joining ring, which is at rest (row 120): it comes out on the other two strands, where it fuses with their halves to rest (row 57), and on the lead-out; the three strands are now bare, so the joining

ring slides into the parting ring, the two loops go (row 90), and what is left is a red half on a wire, X, number 1. If the weight of $e \oplus c$ is 1 or 2, push each mending ring's half back through its tell ring (row 120): it lands on that strand's weather cord and on the ties, and the bells become $\sigma_j(e \oplus c)\,\pi$, which is a half for at least one of them, since σ vanishes only on 000 and 111. The mending cords are now bare: slide the tell rings into the *joining* ring along them (row 57), and each bell is tied twice to it; a bell at a half gives 0 (Result 18.8). ■ See `Code/Ch18/BlindVote.py`, which builds this picture in PyZX, counts its vertices and cords, checks its value against the formula for every one of the 2^8 labellings, and lets `full_reduce` derive the bare wire, a single X with the number 1, or 0.

It is what the dialogue's storm line hangs as, the length of the Crab's front hall: rings and bells and cords, the message going in at the door and coming out at the kitchen, and nobody looking at it on the way.

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§18.9 What a Rewrite Cannot Carry

Take stock. Noise is a channel, and a channel is a sum of branches (Result 18.1). The classical vote of three fails with probability $F(p) = 3p^2 - 2p^3$ (Result 18.2). The one copier that linearity allows is a green spider, which copies dark and lit messages and spreads every other (Results 18.3 and 18.4). A bell asks two wires whether they agree (Result 18.5), and its answer depends on the gust and not on the message (Result 18.6), so a single gust is mended blind, exactly (Result 18.7, proved twice). A bell tied twice to one spider comes off with $(1 + e^{i\theta})/2$ (Result 18.8). Two gusts win the vote (Result 18.9); every mixture of single gusts is mended (Result 18.10); and independent flips become a logical flip of probability $F(p)$ (Result 18.11). A phase flip passes the bells unheard (Result 18.12), colour change gives the code that hears it (Result 18.13), and Shor's nine wires hear both (Result 18.14). A code may ask only questions whose answers do not depend on the message (Results 18.15 and 18.16); stabilizer codes (Result 18.17) ask them with checks of any weight (Result 18.18), and Kitaev's toric code asks them locally (Result 18.19). Below a threshold, codes within codes make

a computation as reliable as we like (Result 18.20). Splits and merges are spiders (Results 18.21 and 18.22), and a joiner is one of each, mended by a half (Result 18.23). *The Storm Line* says what every mending does (Result 18.24).

Now the honest part. **A rewrite proves that two processes are equal, and nothing else.** No picture in this chapter proves a threshold: Result 18.20 is Gottesman's, stated and not proved, and row 143 has no picture. No picture proves a distance, or that a code is the best of its kind: the quantum Hamming and Knill–Laflamme bounds, Theorem 5 and Kitaev's count are quoted, and the distances we checked, the 1 of $[[3, 1, 1]]$ in the text and the 3 of the nine-, seven- and five-wire codes in `Code/Ch18/WhatABellMayAsk.py`, were found by listing operators, not by drawing. The Knill–Laflamme theorem is stated; the pictures of Result 18.16 only evaluate its conditions on examples. De Beaudrap and Horsman's lemmas are claims about what surgery does to a patch of a real surface code; what we checked is that the matrices they name are spiders. Every probability in the chapter, $F(p)$, the fair coin of the joiner and every threshold figure, was computed or quoted on the blackboard. The joining ring was only ever the good branch of a decoder. And the headline itself went to the blackboard.

What the pictures do give is a way of seeing the whole construction as one thing. The checks hang from green spiders at rest, and a green spider at rest passes a red half from any leg to all the others: that one fact is why the bells hear the gust, why they hear it on exactly the strands they are tied to, and why they learn nothing of the message. Change the colour and the same fact makes the phase-flip code. Change the reading and it makes lattice surgery. Every picture in the chapter is exact, number included. The only "up to" is the overall number r with $|r| = 1$ that Shor's code may leave on a corrected state, and that is a global phase, which no reading can see.

The chapter's question has an answer. A message that must not be looked at survives a storm that turns it at random by being spread, not copied, over several wires by a green spider; by being questioned only about whether

its wires agree, never about what they hold; and by being mended by a table that reads only those answers. The storm's continuum of small turns collapses, under the questions, into a few whole ones, each undone exactly. Against both kinds of turn it takes nine wires, or seven, or five. And if the hardware's own mistakes are rare enough, below a threshold whose value is still argued over, codes inside codes make a whole computation as reliable as we please.

Every theorem in this book has been proved twice, once on the blackboard and once in the sketchbook, and this chapter has said plainly, more than once, where the second proof was the longer one. The last chapter turns to the pictures themselves. Chapter 19, *Logic of Pictures*, asks whether "proved twice" is always possible, and what the pictures' rules are when they are read as axioms.

Sinfonia in a Closed Box

閉ぢし箱
喉を鳴らすや
冬籠

tojishi hako
nodo o narasu ya
fuyugomori

The closed box, purring —
and all of us shut indoors
till the winter's done.

[*The Crab's music room, towards eleven on the night of the storm. The curtains are drawn and the door to the hall is shut. The Crab sits by the fire with a lute across his knees; its case stands shut on a low stool in the middle of the room, between the pianos and the fire. The Anteater, who came in an hour ago wet through, has the chair nearest the fire, and his coat hangs over another beside it. The Tortoise is at the table with her work-bag. Achilles has the chair nearest the case.*]

[*Nearly everything in the Crab's music room was bought as a pair and has stayed one, and at night, with the curtains drawn, the pairs close in round the fire: the two upright pianos side by side against the long wall, their lids down; the two music stands beside them; a lamp lit at either end of the table; and on the mantelpiece the two clocks by one maker, which do not keep step, their ticks falling anyhow under the noise of the weather. The one tall window has its heavy curtains drawn across it, a damask of leaves, and behind them the rain comes at the house in handfuls. Over*

the mantel, between the clocks, hangs Escher's Print Gallery *in a plain frame: a man in a gallery, looking at a print that holds the gallery he stands in. The firelight reaches no higher than the foot of the frame, and the lamps on the table leave the rest of it in a half-dark. It hung there before the storm got up, and it will hang there when the storm has blown itself out. The fire is new-made with wood that came in wet, and it hisses and sulks: a flame stands up at one end of a log and lies down again, another starts where nobody was looking, and which log has caught, nobody could yet say. The Anteater's coat steams on its chair, the red earth of Aunt Hillary's hill not quite washed out of its hem, and it hangs a little crooked, dragged down on one side by something heavy in an inner pocket. On its low stool the lute case stands shut, a maker's mark burned into the wood beside its clasp; the side of it towards the fire is warm, as anything standing so near that fire would be, and whether there is anything in it but its lining, the case does not say. Across the Crab's knees lies the lute that came out of it, pear-backed, with a rose cut in its belly as fine as lace, and in its bent-back pegbox a peg for every string: the top string alone, fine as a hair, and under it the pairs, each a shade heavier and darker than the pair above, down to the lowest, the two strings of every pair so alike and so close that only a hair's breadth of air between them says there are two.*]

Crab: Nobody could sleep through that wind, so I thought I'd do something useful with the damp. This is my lute, from a workshop in Füssen, on the Lech, where they have been making nothing else for longer than anyone remembers. The top string runs alone, and every course under it is a pair: two strings side by side, tuned to one note and struck together, so the pair sounds as one. A night like this puts them out, and puts each pair out with itself.

[*He shuts his eyes and sounds the top string by itself: one note, a shade flat. He turns its peg a hair and sounds it again, and the note comes up to pitch. What has changed: one string's note. What has held: every other string on the lute, untouched.*]

Crab: The first course, a single string.

Achilles: I never went up, you know. The bed looked lovely, and the wind said no.

Crab: Nor did I. [*Eyes still shut, he sounds the pair of strings just below the top*

string, both together.] The second course.

[*The two strings of the pair sound almost one note, and not quite: their two pitches beat against each other in a slow waver, like a wheel with a loose spoke. The Crab turns one peg of the pair a hair. The waver slows, slows, and stops, and the two strings sound as one note. What has changed: one string's pitch, and the waver gone out of the pair. What has held: its twin, which he never touched; the single string above; every pair below.*]

Anteater: You have a good ear, Mr Crab. I have been sitting in my own steam this last hour, and I could not have told you anything was wrong.

Crab: There's always something wrong with a lute. The trick is to have it all wrong together. [*The next pair, sounded together; one peg turned a hair; the waver gone.*] The third course. You came from Aunt Hillary's, Doctor?

Anteater: Straight from her, and wet through at every step. She has sealed every door in the hill against the storm: every chamber shut on its own, with its own ants in it and its own air. You could not get a message from one room of her to the next tonight if you begged.

[*Under the talk, the rain on the house, which a moment ago was only a noise, can be heard for what it is: a great many drops, each striking once, somewhere on the roof or the wall, and on its own, and out of all of them one steady hiss that no single drop makes. The Anteater turns his face to the fire and shuts his eyes, like somebody listening to a voice he knows.*]

Tortoise: And she's content with that?

Anteater: She is never more herself than when she is shut up in pieces. [*He reaches to his coat where it hangs, draws a small bundle from its deep inner pocket, and unwraps a brass lamp, tarnished and cold.*] Madam Tortoise, your lamp: the Genie's, from your top shelf. You asked me to bring it when I came, and I have kept it drier than I kept myself.

Tortoise: Thank you, Doctor. [*She takes it and sets it beside her work-bag.*]

Achilles: The Genie's lamp? What for?

Tortoise: Perhaps for nothing.

Crab: [*another pair sounded; a peg; the waver gone*] The fourth course.

[*From the shut case on its stool comes a purr.*]

Achilles: ...Did your case just purr?

Crab: [*opening his eyes*] I put the lute down an hour ago and shut the case, to keep the damp out of the lining. Empty. I'd swear to it.

Achilles: The Cat was on the hall settle at eight, and nobody saw it go anywhere. [*He leans towards the case.*] Cat? How are you getting on in there?

Cat: [*from inside the case*] Asleep, mostly. And a little awake, round the edges.

[*The case as it stands: shut, its brass clasp down. The Crab lays the lute across his knees, leans from his chair, and sets a claw-tip under the clasp; it lifts a hair's breadth. The Tortoise's hand comes down, gently, over his claw, and the clasp settles back. What has changed: nothing in the case, and nothing on it. What has held: the lid, down.*]

Tortoise: Open it and you've asked it.

Crab: [*drawing his claw back*] ...So I have. And one doesn't ask a guest whether it's asleep. [*He takes up the lute again.*] It may keep the lining; it's velvet from Lyon, and it was never meant for anything half so warm. [*A pair sounded; a peg; the waver gone.*] The fifth course.

Achilles: You tune them with your eyes shut. Is that for the ear?

Crab: It's for the strings. Stare at a string and it won't settle: it goes sharp from nerves. Try.

[*Achilles fixes his eyes on the next pair while the Crab sounds it. The waver comes and goes and will not stop; the Crab turns the peg, and the pitch slides past where it should rest, and back, and past again. Achilles looks away, into the fire. The Crab turns the peg a hair, and the pair settles into one note. What has changed: one string of the pair, brought to its twin. What has held: the lute, whole, and none the worse for being looked at.*]

Crab: The sixth course. You see.

Achilles: I see I'd better not.

Crab: [*eyes shut again; two more pairs, each a peg turned and a waver gone*] The seventh course. The eighth course. There's cocoa on the hob if anyone wants it. I've left the storm line up in the hall, and I'll not go out to it till this wind is done.

Tortoise: I'll pour. [*She pours the cocoa and hands the cups round.*]

[*The Anteater takes his cup in both hands and holds it under his nose for the steam. Achilles, his feet drawn up under him in the chair nearest the case, blows on his cocoa and watches the case over the rim; and the Tortoise, sitting down again with the last cup, glances across the table at Achilles.*]

Anteater: Near eleven, and not one of us abed.

Crab: [*another pair; a peg; the waver gone*] The ninth course. [*The last pair, the deepest, sounded together: a longer waver, slow as breathing, and then one low note.*] And the lowest course. There. [*He sweeps a claw once across the whole lute, the single top string and every pair beneath it, and they sound together as one chord.*] In tune, for as long as the storm lets it be.

[*The storm booms in the chimney, and goes on booming. Now that nobody is looking at it, the lute on the Crab's knee seems to answer the wind each time, a murmur somewhere among the pairs, too low to be sure of; the moment anyone turns to listen for it, there is only the storm.*]

First Challenge — *Every Card Balances*

Tortoise: [*bringing a squared pack of plain cards out of her work-bag*] While you all talked this evening, I wrote down every move we have made with pictures since the first evening we sat down together, one move to a card. On each card there's a picture on the left and a picture on the right, with "is" between them; and under the right-hand picture its number, when it has one, a size in words or a little dial for a turn of the whole. It's the rules of the house, Mr Crab, in one pack.

Crab: The rules of my own house, dealt to me. I've never been dealt my house before.

Achilles: And if one of them is wrong?

Anteater: At which level are you asking? A card by itself, or a card with a picture all round it, doing its work in company?

Achilles: By itself, first. [*He takes out his pad.*] My pad: I'll work both sides of every card in numbers, and give it a tick here when they come out the same. A tick on the pad, never on your cards. Deal.

[*The Tortoise sets the Genie's lamp on the mantel beside the left-hand clock, out of the way, and sits down again with the deck.*]

[*The deck as it stands: squared and face down in the Tortoise's hand; Achilles's pad blank but for a ruled margin. She turns the top card face up on the table, the right way up for Achilles, who sits across the table from her: a cord with a box and then another box on it is the two boxes run together; two cords, one above the other, each with its box, are the boxes side by side. Achilles bends over it, works the left side and the right side on his pad, and makes a small tick in the margin. What has changed: one card face up, one tick on the pad. What has held: the rest of the deck, face down in her hand.*]

Tortoise: One after another, and side by side.

Achilles: That's the least they could do. Tick.

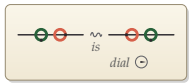
Tortoise: [*the next card face up*] The crossing. [*And the next.*] The bend. The bent cord is the pair, unweighted: both ends dark or both ends lit, with no number in front.

Achilles: [*working*] Dark with dark, lit with lit, and nothing to shrink it. And pulled straight it's a plain cord, with no number to pay. Tick, and tick.

Tortoise: Sliding together: two rings of one colour on a plain cord are one ring, and their beads add.

Achilles: An eighth and a quarter make three eighths, either side. Tick. You'll trip me somewhere; there's always one card that's wrong.

[*The deal goes on. Card by card she turns them up: the tube that changes a ring's colour; two tubes that are no tube; a point copied through a ring of the other colour, its number written under; a half passing a turn of the other colour, with a dial under the right-hand side; the cross undone, with its number; a loop that comes off. For each, Achilles works the two sides on his pad, frowning, and for each he makes his tick. The ticks go down the margin in a column; the face-up cards spread across the table; the pack in her hand grows thin.*]



Face up on the table

Tortoise: Two cords are none.

Achilles: A green ring and a red ring joined by two cords, the same as the two rings apart? [*He works it.*] No, they're not. They come to half as much. [*He looks up, pleased.*] There's your wrong card.

Tortoise: Look under the right-hand picture.

Achilles: "Halved." ...Oh. It's written there. Tick. Without that half it would be wrong by double, and a closed picture is nothing but its number.

Tortoise: Which is why every card carries its number.

Crab: [*with half an eye on the case*] Nothing's stirred in there.

[*The fire has taken at last, log by log, and burns now in many places at once, each flame going its own way on its own log and none of them like its neighbour. Achilles's eyes go from his pad to the case and back to his pad; across the table, between one card and the next, the Tortoise's go to Achilles. The case stands as it stood, warm on its fire side, and keeps its own counsel.*]

Anteater: You have ticked a great many cards alone, Achilles. But a card is never used alone. In a proof it sits in the middle of a picture, with things before it and after it and beside it.

Achilles: Then let me put one in company. [*He takes the sliding card and, on his pad, draws a box before it, a box after it, and a third box on a cord running alongside; he works the whole with the card's left side inside it, and again with its*

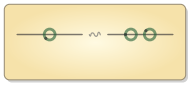
right side.] The same. Anything that stands round the card stands round both its sides alike, so it can't tell them apart. Tick, in company.

Tortoise: Now the mirror card. There's one more way to look at a card: hold it up to the lamp and read it from behind, through the paper, and left and right change places, and every turn reads the other way round. That is the mirror, and it costs nothing to look.

[*The mirror card as it stands, face up: on the left a box with its turns, then "is", then on the right the same box drawn the other way about, every turn reversed. She lifts it and holds it between the table lamp and Achilles, its inked face turned away from him, so that the lamp shines through the thin card and the pictures show through backwards. From his side the right-hand picture now stands on the left and the left on the right, and every bead that read a quarter round reads a quarter back. What has changed: which side is which, and which way every turn goes. What has held: the card itself, its "is", every cord and ring on it.*]

Achilles: It's the same card, with its two sides changed over.

Tortoise: [*holding the sliding card to the lamp the same way*] And this one?



Held to the lamp

Achilles: Two rings sliding into one, the beads turned the other way round and still adding. The sliding card again, with other turns.

Tortoise: [*holding up the card of the half passing a turn of the other colour*] And this?

[*From behind, each picture has crossed to the other side of the card, and in each the half now stands on the other side of the turn it passes; the little dial has crossed with its picture, and stands now under the left-hand one, pointing the other way round.*]



Read from behind

Achilles: The half on the other side of the turn, and the dial turned back. That's the same card read in the other order, and you wrote it both ways. All three are cards I've ticked already.

Tortoise: Hold any of them to the lamp, and what you see from behind is a card of the deck.

[*Where the two curtains meet across the window, the draught has turned the edge of one of them back a hand's breadth, and its back shows: the same damask leaves as on its face, turned the other way about, the pale and the dark changed over. It is a curtain from either side, and the Crab chose it for that.*]

[*She turns up the last card: letting go, and reading. Achilles works it and makes the last tick at the foot of the column; then he gathers the spread cards into one pack and knocks its edges square on the table. What has changed: every card has its tick, and the column on his pad is full. What has held: every card, just as she wrote it.*]

Achilles: [*handing her the squared deck*] Whatever the pictures say, the numbers say.

Tortoise: Every card balances.

Achilles: Then the numbers can check every picture. Can the pictures check every number?

Tortoise: Not in your kitchen yesterday, Mr Crab. Do you remember the hub with a quarter hanging from it?

Second Challenge — *The Hinge*

Crab: Remember it! I have a card in my recipe box for every dish that came out of that afternoon, and a blank one where that one goes.

Tortoise: A hub with a quarter hanging from it, standing on a quarter back on each line, is a tie in a sleeve. Yesterday I hadn't a move for it.

Achilles: My numbers settled it, though. Both pictures came to the same on every way the lines could start.

Tortoise: Your numbers checked it; my cards never reached it. Either the deck is a card short, or I wasn't clever enough. I would rather it were the deck.

Achilles: So what you're saying is the deck was wrong.

Tortoise: Short, not wrong. You've just ticked every card in it. [*She takes a blank card and a pencil.*]

[*The card as it starts: blank. On the left she draws one cord with three rings on it in a row: a green ring, its bead a quarter round; a red ring, a quarter round; a green ring, a quarter round. She writes "is". On the right she draws a plain cord with a single yellow tube on it, and under the tube a small dial, its arrow turned an eighth round. What has changed: a blank card holds a rule. What has held: the deck beside it, as Achilles ticked it.*]



No longer blank

Achilles: Green, red, green, a quarter each: glass, steel, glass! One click each, with the clock, and your ball stood just where the brass needle's two clicks had left it. The night the snow lay, Crab.

Crab: My lamp-black ball. I opened its shutter that night, and it has never turned true since. It sits in the cabinet still.

Achilles: But the ball never showed an eighth of anything. [*He works the card on his pad.*] There. The three quarters make the tube, and turn the whole an eighth round besides; a ball can't show a turn of the whole, and the numbers can. Tick.

[*Behind the drawn curtains and the cloud, the sky has wheeled a good eighth of the way round since eight o'clock; or the house has, with this room and everything in it, which comes to the same. Nothing in the room shows it: not the fire, not the cocoa in the cups, not one string of the lute.*]

Tortoise: I have never got that card out of the other cards, and those who have looked hardest say it can't be got out of theirs. So it goes in as a card

of its own: given, not got. [*She lays it on the deck.*]

Anteater: A new rule for the house, and at this hour of the night.

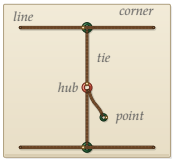
[*He says it fondly, as of a grandchild up long past its bedtime, and holds out his cup; the Crab fills it from the pot on the hob and sets the pot back to keep warm. Achilles stretches his feet towards the fire, his pad on his knee, and his eyes go to the case before they go back to the cards.*]

Tortoise: And three small ones come out of it. [*She draws three more cards quickly, and pencils a small cross in the top right-hand corner of each.*] The cross means taken from the others, not given. Three quarters back, green, red, green, is the tube with the whole an eighth back. A green turn passes a red point at rest untouched, and a red point at a half by turning the whole by the same turn. And a green point a quarter round is a red point a quarter back, with the whole an eighth round.

Achilles: [*working each on his pad*] Tick. Tick. And tick.

Tortoise: [*drawing her cloth out of her work-bag and spreading it on the table in front of her, its near edge on her side, so that what lies on it reads left to right, upper and lower, as she looks down at it from her chair; Achilles, across the table, sees it the other way up*] My drawing-cloth, with my rings and cords and yellow tubes, as ever: what lies on it is a picture of a thing, not the thing. [*She sets her little box of rings beside the cloth, and brings out a little packet of paper dials. Each ring opens at a small hinge, and each tube at a slit, so that either clips on or off a cord anywhere; on her cloth, as ever, a bead at rest sits at its ring's far side as she looks down, and a turn goes round clockwise.*] And these are new tonight: slips. A bead turns one ring; a slip laid in the corner of the cloth turns the whole picture, and two slips that cancel are taken away together.

[*She lays the hub as it stood in the Crab's kitchen just before she lifted it away whole, without the rest of that picture round it: two plain cords across the cloth, one above the other; on each a green ring, its bead a quarter back; from each of those a short tie running to one red ring between the cords, at rest, the hub; and from the hub a short cord running out and down to the right, to a green point lying on the cloth, its bead a quarter round. The corner of the cloth, its far right-hand corner, is empty.*]



Laid out flat

Tortoise: The hub, as it stood yesterday. Now the new cards.

[*The point as it stands, lying out to the right of the hub: green, a quarter round, on its short cord from the hub. The Tortoise lifts it off the end of the cord into her box and ties on in its place a red point, its bead a quarter back; then she sets a slip's arrow an eighth round and lays it in the corner. What has changed: the point's colour and its bead, and the corner, which holds one slip now. What has held: the hub, the two feet, both lines, and what the picture does; the crossed card says so.*]

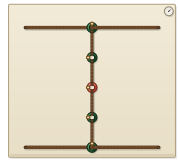
Achilles: A red point on a red hub. Red into red?

Tortoise: Red into red.

[*The red point, a quarter back, lies out to the right of the red hub on its short cord. She slides it up the cord into the hub: the cord is gone, the point is gone into the ring, and the hub's bead, which stood at rest, now stands a quarter back. What has changed: the hub has lost its short cord and taken the point's turn. What has held: the hub's two ties to the feet, the feet, and the one slip in the corner.*]

Achilles: So now the hub is only a red ring a quarter back, sitting on the tie between the two feet.

[*Each foot as it stands: a green ring a quarter back on its line, tied to the hub. She slides the upper foot apart along its own tie: a green ring at rest stays on the upper line, and a green ring a quarter back sits on the tie just below it. Then the lower foot, the same way: a green ring at rest on the lower line, and a green ring a quarter back on the tie just above it. What has changed: the feet stand at rest, and their turns have moved onto the tie. What has held: each line, the red ring in the middle, and the corner; nothing goes to the corner for this.*]



Feet at rest

Achilles: [*reading the tie from top to bottom*] Green a quarter back, red a quarter back, green a quarter back. That's your new card seen through.

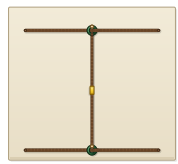
Tortoise: The crossed one.

[*The three rings on the tie as they stand: green, red, green, a quarter back each, in a row between the two rings at rest. She lifts the three off together into her box and slides a single yellow tube onto the tie in their place; then she sets a second slip's arrow an eighth back and lays it in the corner beside the first. What has changed: three rings have become one tube, and the corner holds two slips. What has held: the ring at rest on each line, and the tie's two ends.*]

Anteater: And your two slips, Madam? One says an eighth round, and one an eighth back.

Tortoise: Then they go together.

[*The corner as it stands: two slips, their arrows turned opposite ways. She picks them up one on the other and puts them in her bag. What has changed: the corner is empty. What has held: everything on the cloth, on each line a green ring at rest, and between them a tie with a tube on it.*]



The corner empty

Achilles: A sleeved tie. ...Oh. The eighth the ball couldn't show: the slips were carrying it the whole way, and it cancelled.

Crab: Yesterday's gap, shut. [*He reaches for the new card.*] I'm copying that into my recipe box tonight, and its three little crossed ones with it.

Tortoise: Whatever the numbers say, the pictures can say.

Achilles: With the right cards, you mean. Yesterday you were a card short. And one card mended one gap: how do you know there isn't another gap, waiting for another card?

Crab: Will they do it for something I've not seen?

Third Challenge — *Same Number, Same Picture*

Crab: A dare, then, and I'll hold the stakes. Achilles, you name two pictures whose numbers agree, anything you please, chosen with your pad and not with her cloth. Madam Tortoise gets from one to the other by her cards alone.

[*The Crab settles back in his chair, very pleased to be holding the stakes of anything. The fire has drawn itself up into a steady rank of flames along the logs, each standing where it stands and drawing evenly; and under them the embers have begun to glow in small cells, each brightening and dimming on its own, and the whole bed of them one steady red.*]

Achilles: [*working on his pad for a while, then turning it face down*] Here's one. A tube, then a quarter round, three times over, on one cord; and a bare cord with the whole turned an eighth round. My pad says they're the same. Get from one to the other with your cards, and don't look at my numbers.

Tortoise: Agreed. [*She clears the cloth: the rings of the sleeved tie into her box, and its tie, tube and all, into her work-bag with the cords of the lines.*]

[*She lays a single cord across the cloth, and on it, from left to right: a yellow tube, a green ring a quarter round, a tube, a green ring a quarter round, a tube, a green ring a quarter round. The corner is empty.*]



Achilles's dare

Achilles: Nothing in that picture says "an eighth". I chose it for that.

[*The cord as it stands. The Tortoise takes the first tube, the green ring after it and the tube after that, a ring with a tube on each of its cords, and draws the ring through both tubes: the tubes are gone into it, and it comes out red, its bead still a quarter round. What has changed: the ring's colour, and two tubes fewer on the cord. What has held: the ring's bead, and everything to the right of it.*]

Tortoise: A red quarter round. And this tube I read by the new card, backwards.

[*The one tube left on the cord sits between two green quarters. She lifts it off into her work-bag and threads in its place a green ring, a red ring and a green ring, each a quarter round, and lays a slip in the corner with its arrow an eighth back. The cord now reads: red quarter, green quarter, green quarter, red quarter, green quarter, green quarter. What has changed: one tube has become three rings, and the corner holds a slip. What has held: the rest of the cord, and what it does.*]



Read backwards

Anteater: You have made it longer.

Tortoise: To make it shorter. [*Green beside green, twice over: she slides each pair of green quarters together, the left-hand ring into the right-hand one, and each pair becomes one green ring at a half. The cord reads red quarter, green half, red quarter, green half.*]

[*The first green half as it stands, with a red quarter round just after it. She slides the half along the cord, past the red ring to just beyond it: as it passes, the red ring's bead swings from a quarter round to a quarter back, and she lays a second slip in the corner, its arrow a quarter round. The cord reads now: red quarter round, red quarter back, green half, green half. What has changed: the half has gone by, and the ring it passed turns the other way. What has held: the half itself, and every*

ring's colour.]



Slid past

Achilles: A red quarter round and a red quarter back, side by side.

[*She slides the two red rings together: a quarter round and a quarter back make rest, and the ring at rest, with only the cord through it, slips off into her box. Then the two green halves: a half and a half make a whole turn, which is rest, and that ring slips off into her box too. What lies there now is a bare cord, not a ring on it. In the corner lie a slip an eighth back and a slip a quarter round; she lays them one on the other, puts them in her work-bag, and lays a single slip in their place, its arrow an eighth round. What has changed: every ring and tube gone, and two slips become one. What has held: the cord's two ends.*]



A new slip

Tortoise: A bare cord, and the whole turned an eighth round. May I look at your pad now?

Achilles: [*turning it over*] A bare cord, the whole an eighth round. I lose, and I'm glad of it.

Crab: [*crowing*] Whatever the numbers say, the pictures can say.

Tortoise: Given the right cards, Mr Crab. And they aren't the same cards for every kind of picture.

[*On the nearer music stand lies a piece for the lute, written out in the letters a lute-player reads, which tell the fingers where to go; on the other stand lies the same piece in notes, which tell the ear what it will hear. The Crab copied the letters out of the notes himself, one winter, bar by bar, and has never found a bar that would not go.*]

Anteater: At which level, then? Are the right cards for small questions the right cards for large ones?

Tortoise: It goes by the kind of picture, not the size. For pictures that turn only by quarters, a right deck was found some years ago. For pictures with eighth turns the right deck was found only lately, and for any turn at all there's one too.

Achilles: And pictures that let go, like the one the Cat's in?

Tortoise: With quarter turns, yes. With eighths, the people who had found the eighth-turn deck left that one open. And none of those decks is quite my deck, so I'll not promise more than I've shown you tonight.

Achilles: [*looking at the case*] Then whatever I could work out about the Cat, you could draw. But I can't work out whether it's asleep.

Tortoise: No one can, from outside. So what can a closed box tell anyone?

[*Every eye in the room goes to the case, and the case gives nothing back: a shut lid, a clasp with no claw near it, and a warmth on its fire side that would do as well for something curled inside as for the fire alone. Asleep or awake, it has let the room wonder since eleven.*]

Fourth Challenge — *One Answer Out*

Achilles: It can tell me I've watched it all evening. I haven't taken my eyes off it since the purr.

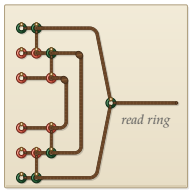
Tortoise: I know. I've been watching you.

Achilles: Me? What for?

Tortoise: Because you were watching the box. Let me lay the three of us out. [*She clears the cloth, the bare cord into her work-bag and its slip after it, and brings out more rings.*] The read ring you know: a green ring that takes a cord and its copy and gives out one single cord, the answer. And a cord let go is tied to its own copy, as ever; when I want a picture's odds and not only its shape, I lay it twice, as it is and seen through, one under the other.

[*She lays it out. At the top, the Cat's cord, starting from a green point at rest.*

Below it, Achilles's page: a cord starting from a red point at rest, blank. Below that, her own page, the same. On the Cat's cord she threads a green ring and ties it by a short tie to a red ring on Achilles's page: his look. Further along Achilles's page she threads a green ring and ties it to a red ring on her own page: her look. Then she lays the whole again beneath, seen through, but not as a card is held to the lamp: left and right stay as they were, and the rows come in the other order, so that her own page's copy lies nearest the middle, Achilles's below it and the Cat's lowest, with a clear row left between the layers; every bead keeps its rest at its ring's far side, and turns the other way round. Every bead is at rest, so bead for bead the copy looks just like the picture above it. She ties each page's cord, at its right-hand end, to its own copy's cord, with a knot where the cord turns down, her page's turn inside Achilles's: let go. And she brings the Cat's cord and its copy together into a read ring in the clear row, from which a cord runs off to the right-hand edge of the cloth.]



Pages and looks

Achilles: And which of those is me?

Tortoise: [*her finger on the upper layer*] The middle cord is your page. The Cat, as it was: evenly asleep and awake. Your look copies the Cat onto your page; my look copies your page onto mine. Both pages let go, and the Cat's cord read.

Achilles: That's us. That's this whole room.

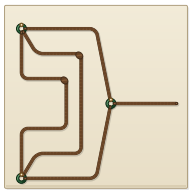
[*Achilles looks from the cloth to the case; the Tortoise, across the table, looks from the cloth to Achilles; the Crab looks at the pair of them; and the Anteater, warming his hands, looks at everybody. On the mantel the two clocks tick on out of step, as though neither had noticed the other all evening.*]

[*The cloth as it stands: on each layer, each page starts at a red point at rest, joined by plain cord to the red ring of a look. The Tortoise slides each red point along its*

cord into the red ring beside it, red into red, both at rest; and each ring, left at rest on its cord, slips off into her box. Four times over: on each page, as it is and seen through. What has changed: the red points and red rings are gone, and each look's short tie runs on as plain cord from its green ring to wherever the page went. What has held: the green rings, the let-go cords, the read ring.]

Achilles: My page is gone into your ring, and yours into nowhere.

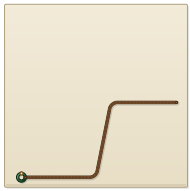
[*On each layer now: the Cat's green point, Achilles's green ring on the Cat's cord, and her green ring, all at rest and joined by plain cord. She slides them together, his look's ring and hers along the cord to the Cat's point, where they meet: the Cat's point, his look's ring and hers become one green ring at rest, with a cord to the read ring and the let-go cords running down to the same kind of ring on the copy below. The same on the copy. What has changed: the whole company on each layer has become one ring. What has held: the read ring, and the let-go cords between the layers.*]



Slid together

Anteater: Two rings now, one above and one below, and joined more ways than one.

[*The two green rings as they stand: joined by the let-go cords, and each running to the read ring. She slides the upper ring down a let-go cord into the lower: green into green, one ring; and what is left of the let-go cords now runs from that ring round and back into it, a loop. Then she slides the read ring in along the copy's cord, and the upper layer's cord, still running to it, becomes a loop too. She lifts the loops off into her work-bag. What lies on the cloth: one green ring at rest, just where the Cat's point lay on the copy, with a cord running from it to the cloth's edge, and no slip in the corner. What has changed: every look, every page, every copy, gone. What has held: the ring's rest, which was the Cat's point's, and the cord out to the edge.*]



Loops lifted off

Tortoise: There's the Cat's own point, on the answer's cord, just as it lay before anybody looked.

Achilles: But I looked! My page has it written down. That has to count for something.

Tortoise: It counts on your page. And your page is let go; it doesn't come out of the box.

[*Not a word said in this room tonight has got out of it: not through the shut door, not through the drawn curtains, not up the chimney against that wind. Anyone out in the lane, if anybody were fool enough to be out in the lane, would know of the whole room one thing only: a patch of warm light on the wet cobbles, where the curtains do not quite meet.*]

Achilles: [*working on his pad*] Evenly asleep and awake before anyone looked. After my look, and your look at me, and both pages let go: evenly asleep and awake. ...Oh. My watching changed nothing anyone outside could hear. Then no look can ever change a closed box's odds!

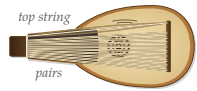
Tortoise: Not any look. Yours copied asleep-or-awake, and mine only watched you. A look that asked the Cat some other question could change them.

[*From the case, a purr.*]

Crab: [*taking up the lute*] Then here is what a closed box gives out.

[*The lute as it stands, lying across his knees belly up, its neck to his left, so that, seen from his chair, the top string is the farthest from him and the lowest pair the nearest: every pair in tune, the top string alone above them. Eyes shut, the Crab sounds the pairs one after another, from just under the top string down to the lowest,*

each pair's two strings sounding one note. Then he lays a claw flat across every string to still them, lifts it, and plucks the top string alone. What has changed: where there were many notes one after another, there is one, and it hangs in the room. What has held: every string, in tune, as he tuned it.]



Plucked alone

Tortoise: Those pairs of strings are like my cords and their copies: two, sounding as one. The top string is like the answer, which has no copy, as a record has none. Only a likeness, mind: a copy turns every bead the other way, and a twin string only agrees.

[*In the grate the logs, burned through where they crossed, settle into the embers together with a soft sound and a lift of sparks, and burn on, lower and quieter than before. The Anteater leans to the hearth and warms his hands at them.*]

Achilles: Then any machine's a closed box like that. It starts, it runs, and it gives out one answer, with odds.

Crab: And if the answer's wrong?

Achilles: Say it's right two times in three. Run it three times and take the majority. All three right, eight times in twenty-seven; just two right, in three ways, twelve times more. Twenty times in twenty-seven.

Crab: Better than two in three?

Achilles: Two in three is eighteen in twenty-seven, so better. And your cloth found the Cat's odds without a single number.

Anteater: The pad and the cloth, agreeing again.

Achilles: Whatever the numbers say, the pictures can say.

Crab: With the right cards.

Anteater: And if someone outside the room could undo the whole room, you and your page and all? Aunt Hillary's wardens can.

Fifth Challenge — *Watchers Watched*

Achilles: Undo a room? What wardens?

Anteater: Aunt Hillary keeps two sealed chambers for a game she is very fond of, and she plays it on nights like this. In one chamber an ant tosses a weighted coin: heads one time in three, tails the other two. On heads she sets a little token pointing down and sends it through a hatch into the other chamber. On tails she sets it crosswise, half down and half up.

Tortoise: And in the other chamber?

Anteater: An ant there looks at the token, and finds it lying one way or the other, down or up. Then both hatches are sealed. Outside each chamber sits a warden, who can undo the whole chamber, ant and token and memory and all, back into one thing, and ask it a single question: "ok, or not?"

Achilles: Undo an ant's memory? And the ant never knows?

Anteater: She saw what she saw; I don't doubt an ant for a moment. That is the difficulty. Each of them reasons from what she is certain of.

Tortoise: Such as?

Anteater: The looking ant, if she saw the token up, is certain the coin came down tails, for heads never sends it up. The tossing ant, on tails, is certain that the looker's warden will hear "not ok". And the tosser's warden, on hearing "ok", is certain the looking ant saw it up.

Achilles: Then chain them. The tosser's warden hears "ok", so the looking ant saw it up, so the coin was tails, so the looker's warden hears "not ok". They can never both hear "ok".

Anteater: Aunt Hillary has played it round after round. One round in twelve, both hear "ok".

Achilles: ...Then somebody's certainty is wrong. Whose?

Anteater: At which level are you asking? The ants', the wardens', or Aunt Hillary's own, who holds the lot? Three natural things were assumed: that her arithmetic holds in every chamber, the wardens' included; that those who are certain do not contradict one another; and that each ant saw one thing, and one thing only. The story does not say which to give up, and neither will she.

[*Round the case, the room; round the room, the shut house; round the house, the storm. Each is shut on the one inside it, and each has of the one inside only what gets through: the room, a purr now and then; the house, a murmur of voices behind a door; the storm, a patch of light it cannot get in at.*]

Achilles: Well, nobody in this room can undo anybody. You've only watched me, Tortoise.

Tortoise: Only watched. I haven't the hands to undo a whole Achilles.

Crab: Nor I a Cat, it seems.

Achilles: Then let me ask something bigger than a room. Tortoise, your lamp?

Tortoise: The Genie's lamp, on the mantel where I put it. Rub it once, and he comes.

[*Achilles takes the lamp down from the mantel. It is as it has been all evening: brass, tarnished, cold, its wick dry, never lit. He rubs its side once with his palm. The lamp does not warm; but from its spout comes a breath of pale mist that, instead of fading, gathers and rises and thickens, and there between the table and the fire stands the Genie, brushing a little damp from his sleeves. The lamp in Achilles's hands is as cold and dull as before. What is new is one more at the table.*]

Genie: The Office of Wishes. One query, the evening's only one: say it exactly, and say it once.

[*While he speaks the light in the room seems to draw in towards the table: the lamps at either end lean their flames a little inward, and the glow of the fire reaches further across the floor than it did, towards the old brass in Achilles's hands. The Crab sits very upright, as a host does when a guest has come in by some way other than the door.*]

Achilles: Two players, in two rooms that cannot speak to each other, sharing whatever pairs they like, and a referee who sets them any game he pleases. I want a way to tell, for every such game, whether they can win it every time or at best half the time.

Genie: A way to answer every question of a kind is a question about questions, and not mine to grant. Referred. [*He tilts his head, as though listening up a stairwell, and straightens at once.*]

[*In the chimney the wind goes up the flue with a single boom, and the echo of it comes back down before the boom has finished, so that the two are heard as one sound. The embers brighten and dim in the same instant.*]

Achilles: Referred where?

Genie: To the Meta-Genie, and above him, and above that, as high as it goes; and it goes up without end. The answer has come down already.

Crab: Already? From as high as it goes?

Genie: At once. It always does. The answer is no: there is no such way.

Achilles: No way at all? For every game?

Genie: For every machine, the Office can set a game that the pair win every time if the machine ever stops, and win at best half the time if it never does. So a way to tell every time from half the time, for every game, would tell of every machine whether it ever stops. Nothing tells that; the Office has it from very high up.

Achilles: Then my query's spent, and on a no.

Genie: A query answered is a query spent. The Office is exact.

[*The Genie thins to mist from the feet up, and the mist draws itself back into the spout of the lamp in Achilles's hands, the last wisp of it going in like a breath taken. The lamp is as it was: cold, dull, unlit. What has changed: one fewer at the table, and a query spent. What has held: the lamp. Achilles sets it down on the table.*]

[*The light goes back to its places: the lamps' flames stand straight again, and the firelight draws back to the hearth. Achilles sits down again in the chair nearest the*

case and holds his hands out to the embers; his eyes go back to the case, and across the table the Tortoise's go to him.]

Tortoise: You've watched that case all night, Achilles, and I've watched you watching it. Who has been watching me, I couldn't tell you.

[*Outside, the wind drops into a lull. In the quiet a flame stands up tall out of the embers, and for as long as it stands its light climbs the chimney-breast and lies full across the print over the mantel.*]

Achilles: [*looking up at the print over the mantel, between the clocks: Escher's* Print Gallery] Your print, Crab: a man in a gallery, looking at a print that holds the gallery he stands in. I've been sitting in it all night.

Crab: [*after a silence*] Then what have we got, out of the whole night?

Tortoise: Every card balances.

Anteater: That much we ticked.

Tortoise: With the right cards, the pictures can say anything the numbers can about a closed picture. And a closed box gives out one answer, with odds that no watcher who only takes note can change.

Achilles: And the Genie's kind of question?

Tortoise: No way, drawn or worked, answers every question of that kind. And the Doctor's story doesn't say which of the three to give up.

[*In the grate the logs have burned down into one bed of fire, and there is no telling any more where one of them lay and where another; the whole of it glows at once, and the room is warm into its corners. The storm has not let up, and the house leans into it and does not mind.*]

[*On the mantel the two clocks strike twelve, the one beginning before the other has done, so that the strokes fall in between one another.*]

Crab: Midnight, by both of them, more or less.

Anteater: [*feeling the sleeve of his coat where it hangs*] And my coat has dried. Aunt Hillary will be unsealing her doors by morning, one chamber at a time.

[*The Crab lays the lute on the table beside the stool where its case stands, every string still.*]

Crab: The last of the cocoa. [*He shares it out, and gives Achilles the fuller cup.*]

Achilles: Thank you. [*He holds the cup and does not drink; his eyes have gone back to the case.*]

Tortoise: You've watched it all night. Ask it.

Crab: Ask it, Achilles. I shan't touch the clasp.

[*For a moment the whole room looks at the case, and the case stands on its low stool as it has stood all night, warm on its fire side, shut as the Crab shut it. There may be something curled on the velvet from Lyon, or there may be only velvet; asleep or awake, the room has let it be both through all the storm.*]

Achilles: Cat — asleep or awake?

Cat: Awake.

[*The lid stays down. The purr stops; over the closed case a grin shows for a moment, the last of the Cat to go, and then is gone too.*]

[*The case is only a case now: a lute case on a low stool, shut, with velvet from Lyon inside it, asked its one question and done with being asked. Beside it on the table the lute lies as the Crab laid it, strung as he strung it: the top string alone, and beneath it the pairs, each two strings side by side, down to the lowest, all of them still. The Genie's lamp stands among the cups, a piece of old brass and nothing more. In the grate the one bed of fire has gone low and red under a skin of ash, and gives its warmth out steadily to the whole room. Outside the storm is up still, throwing the rain at the curtained window and booming in the chimney, and means to be till morning; inside, the two clocks tick on out of step into the first minutes of the day. Over the mantel the print hangs in its plain frame, the firelight no higher than the foot of it again, one more of the Crab's things. Nobody goes up to bed. Achilles sits in the chair nearest the case with his cup in both hands; the Anteater's coat hangs dry on its chair, and the Anteater dozes beside it; the Crab has his feet to the fire; the Tortoise, her work-bag on her knee, has stopped watching Achilles; and for the first time all night nobody in the room is watching anybody at all.*]

Chapter 19: Logic of Pictures

— Sinfonia in a Closed Box —

Part VI: Foundations

Escher: "Print Gallery" — a man in a gallery, looking at a print that holds the gallery he stands in

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If two pictures always give the same numbers, can the rules always turn one into the other?

Imagine two clerks who keep the accounts of one house. The first keeps them in figures. The second keeps them in drawings, and he has a short list of moves he is allowed to make on a drawing: slide this, join that, take a loop away. Whenever the draughtsman says "these two drawings come to the same thing", the clerk of figures can check him by adding up. So far, so safe. The question of this chapter runs the other way. Whenever the figures say that two drawings come to the same thing, can the draughtsman get from one drawing to the other *by his own moves alone*, without once borrowing the other clerk's sums?

It is not an idle question, because this book has been keeping its accounts in exactly this double way. Every chapter proved its main result twice, once in figures and once in drawings. Most of the time the drawings managed on their own. But not always. More than once the sketchbook reached a step it could not take by moves, and it crossed the gap by working out both sides in figures and noting that they agreed. Two chapters ago it said so plainly, and admitted that a skeptic who distrusted pictures would be right to point at that step.

So is the gap in the drawings, or only in the list of moves? If one more move would close it, is there always one more move? Could there be a list that never needs to borrow at all? And if there is, what does such a list buy, and what can no list of moves ever buy?

These questions have answers, and most of them are recent. People found lists of moves that never need to borrow, for one kind of drawing after another, in a burst of papers between 2013 and 2019; for one kind, the authors of the last of those papers left the question open. There is also a second part of the story: some questions about quantum systems cannot be settled by any procedure at all, drawn or computed; and some questions about watchers who watch other watchers turn out to have no consistent answer unless something natural is given up.

And at the very end there is a box, closed, with something asleep inside it, and a person watching the box, and another person watching him. We will ask what a closed box can tell anyone, and whether it matters who was watching.

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§19.1 Every Card Balances

Chapter 18 ended with this: "Every theorem in this book has been proved twice, once on the blackboard and once in the sketchbook, and this chapter has said plainly, more than once, where the second proof was the longer one. The last chapter turns to the pictures themselves. Chapter 19, *Logic of Pictures*, asks whether 'proved twice' is always possible, and what the pictures' rules are when they are read as axioms."

The chapter's question, in the form we can answer: **if two pictures have the same matrix, can the rules of the calculus always rewrite the one into the other?** We answer it in five steps. This section names what we have been drawing since Chapter 1, lists the rules as a deck of cards, and proves that every card is true. §19.2 finds one card the deck was missing, adds it, and uses it to close a gap that Chapter 17 left open; that is the chapter's headline. §19.3 states what is known about the general question, kind of picture by

kind of picture, and works one example with the cards alone. §19.4 asks what a closed picture of a whole computation tells us, and defines the class of problems quantum computers are said to solve. §19.5 lists, with sources, what pictures cannot do. The master diagram in §19.8 is a closed box with a Cat in it, and §19.9 watches the watchers.

What we have been drawing

Since Chapter 1 we have drawn two kinds of thing: **wires**, which stand for systems (a qubit, a pair of qubits, a classical bit), and **boxes**, which stand for processes that take some systems in and put some systems out. We had two ways of putting pictures together: *one after another*, joining the outputs of the first to the inputs of the second, and *side by side*, stacking two pictures with no wire between them (Chapter 4). A picture with no wires going in and none going out is a number (Chapter 2). A picture can be reflected left to right, which is its **mirror** (Chapter 3). From Chapter 6 a wire could bend: the **cup** is a state on two wires and the **cap** an effect on two wires, and an S-shaped bend pulls straight (row 42, yanking). And from Chapter 8 on, the boxes we cared most about were spiders, whose matrices depend only on which leg is joined to which (row 63).

Mathematicians have a name for any collection of systems and processes that can be put together in these ways. We will call it **a theory of processes**. A *category* is the part that composes one after another; a *monoidal* category adds side by side; *compact closed* adds the cups and caps with yanking; and *dagger* adds the mirror. So what we have been drawing is a **dagger compact closed category**. The name is long and the idea is the one the reader has been using since the first chapter: systems, processes, two ways to combine them, bends that yank, and a mirror.

On the blackboard. The theory we have used throughout is the one in which a system of k qubits is the space $\mathbb{C}^{2^k}$ and a process is a matrix. "One after another" is the matrix product, written right to left: the picture with g first and f after it is $f \circ g$. "Side by side" is the Kronecker product $\otimes$, top wire first. The mirror of a process f is its adjoint $f^\dagger$, the conjugate transpose,

and two laws tie it to the two ways of combining:

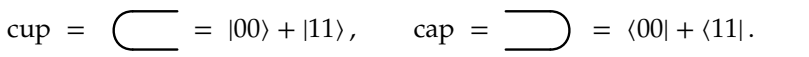
$$(f \circ g)^\dagger = g^\dagger \circ f^\dagger, \qquad (f \otimes g)^\dagger = f^\dagger \otimes g^\dagger.$$

The cup and the cap are, as always in this book, **unnormalized**:

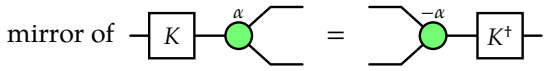
$$\text{cup} = |00\rangle + |11\rangle, \qquad \text{cap} = \langle 00| + \langle 11|, \qquad |\Phi^+\rangle = \frac{1}{\sqrt{2}}\,\text{cup},$$

so that yanking, $(\text{cap} \otimes I)(I \otimes \text{cup}) = I$, holds with no number. Selinger's survey states that "the category FdHilb of finite dimensional Hilbert spaces is dagger compact closed", "with $f^\dagger$ the usual linear algebra adjoint" (his Example 7.11). FdHilb is the grown-up name for "finite-dimensional complex vector spaces and the matrices between them", which is our blackboard.

In the sketchbook. Nothing new is drawn. The cup is a wire bent so that both ends point right; the cap, a wire bent so that both ends point left:



What is new is the claim that the mirror of a whole picture is always the adjoint of its matrix. For a single box this is how Chapter 3 defined the mirror. For a picture built of many pieces it is a fact to be proved, and here is an example of what it says. A box K, then a green spider turned α with one leg in and two out; in the mirror, the spider comes first, with two legs in and one out, turned $-\alpha$, and then the mirror of K:



The matrix of the left picture is $Z_{1\to 2}(\alpha)\,K$, where $Z_{1\to 2}(\alpha) = |00\rangle\langle 0| + e^{i\alpha}|11\rangle\langle 1|$ is the spider of row 53. Its adjoint is $K^\dagger(|0\rangle\langle 00| + e^{-i\alpha}|1\rangle\langle 11|)$, which is the matrix of the right picture, read left to right. The order has reversed, the legs have swapped sides, and the turn has changed sign. Does that happen for every picture, however large?

Result 19.1 (the mirror is the dagger). For every picture D

built from wires, boxes, spiders, yellow boxes, cups, caps and crossings, let the mirror of D be its reflection left to right, with every spider's turn α replaced by $-\alpha$ and every box K by its mirror $K^\dagger$. Then the matrix of the mirror of D is the adjoint of the matrix of D.

Proof. By the way D is built. For the single pieces: the adjoint of a green spider with m legs in, k out and turn α is $(|0\cdots0\rangle\langle0\cdots0| + e^{i\alpha}|1\cdots1\rangle\langle1\cdots1|)^\dagger$, which is the green spider with k legs in, m out and turn $-\alpha$; the same for red, with $|\pm\rangle$ in place of $|0\rangle, |1\rangle$ (rows 53, 54). $H^\dagger = H$. The adjoint of the cup is the cap, and a crossing is its own mirror, as a matrix and as a picture. A box's mirror is $K^\dagger$ by Chapter 3's definition. For the two ways of combining: the mirror of "g, then f" is "the mirror of f, then the mirror of g", and $(f \circ g)^\dagger = g^\dagger \circ f^\dagger$; the mirror of a side-by-side picture keeps top above bottom, and $(f \otimes g)^\dagger = f^\dagger \otimes g^\dagger$. ■ See `Code/Ch19/EveryCardBalances.py`, which takes pictures with random turns on every spider and checks that PyZX's mirror of each has the conjugate transpose as its matrix.

Selinger puts the same thing in one phrase: the graphical language of a dagger category comes "with the dagger of a diagram taken to be its mirror image".

Dirac's Blackboard.

Three reflections, kept apart. The book has three ways of turning a process over, and they are easy to confuse. The **transpose** A^T is what you get by bending both of a box's wires round (row 43): entries swapped, no conjugation. The **conjugate** $\bar{A}$ is what the mirror copy of a doubled wire carries (row 66): every entry conjugated, nothing swapped. The **adjoint** $A^\dagger = \bar{A}^T$ is the mirror, Result 19.1: both at once. On a spider they look like this. Take $Z_{1\to2}(\alpha) = |00\rangle\langle0| + e^{i\alpha}|11\rangle\langle1|$. Its transpose is $|0\rangle\langle00| + e^{i\alpha}|1\rangle\langle11|$, the same spider bent round, turn kept. Its conjugate is $|00\rangle\langle0| + e^{-i\alpha}|11\rangle\langle1|$, the same shape, turn reversed. Its adjoint is $|0\rangle\langle00| + e^{-i\alpha}|1\rangle\langle11|$: bent round *and* reversed. For a spider, bending is free (row 63), so the mirror of a spider is the conjugate of a spider. For a general box it is not: $\bar{A}$ and $A^\dagger$

are different matrices, and only the second is the mirror.

Why this definition. Why name the structure at all, when every example in the book is matrices? Because a name says which facts the pictures use. Every move with bends and mirrors that we have made depends only on the laws above, not on the entries of any matrix. Anything proved with those moves alone is true in *every* theory with those laws, not only in ours.

That last sentence is itself a theorem, and a different one from the chapter's question. Selinger states it for compact closed categories (his Theorem 4.33): an equation "follows from the axioms of compact closed categories if and only if it holds, up to isomorphism of diagrams, in the graphical language"; for the dagger version he says the same holds "up to isomorphism (for dagger compact closed categories)", "proved by the same method". He adds a caveat (4.34): "The special case of diagrams over a simple signature was proven by Kelly and Laplaza", and "The general case does not appear in the literature." That result is about *every* theory of processes at once, with the boxes left as unknowns. The chapter's question is about *one* theory, the qubits, and about particular boxes, the spiders, whose matrices we know. The two must be kept apart: the first is called coherence, and the second, the one this chapter is about, is called completeness.

Feynman's Notebook.

Who named what the reader was drawing. The idea that quantum protocols live in a category with bends and a mirror was put forward by Samson Abramsky and Bob Coecke in "A categorical semantics of quantum protocols", posted in February 2004. They called the structure a "strongly compact closed category". Peter Selinger's survey of graphical languages (2009) uses the name that has stuck, "dagger compact closed category", and credits the definition to that paper and to his own "Dagger compact closed categories and completely positive maps" of 2007. The pictures themselves are older than the quantum reading of them: Selinger credits the coherence of compact closed pictures, for the simplest case, to G. M. Kelly and M. L. Laplaza's "Coherence for compact closed categories" (1980). What was new was the

claim, in the abstract of Abramsky and Coecke's paper, that "the essential structures found in key quantum information protocols such as teleportation" "can be captured at this abstract level": the bends and a way of adding. Chapter 6 of this book made the same claim with a yank.

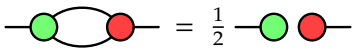
The deck

The rules of the calculus, as the book has used them, can be written one to a card, each card an equation between two pictures with its number. Here is the deck. Every card was dealt in an earlier chapter, and its row in the Rosetta Table is given.

- **Wiring.** One after another and side by side (Chapters 1 and 4); the crossing (Chapter 4); the bend and its yank, cup and cap unnormalized (rows 39–42); the mirror (Chapter 3, and Result 19.1).
- **Only the connectivity counts** (row 63): pieces may be slid, stretched and bent as long as the same legs stay joined.
- **Fusion** (row 57): two spiders of one colour joined by a plain wire slide together, and their turns add. Exact.
- **Identity** (Chapter 8): a spider at rest with two legs is a bare wire.
- **Colour change** (row 58): a spider with a yellow box on every leg is the other colour, same turn. **Two yellow boxes** in a row are none (row 56).
- **Copy** (rows 59, 81): a red point at rest or at a half, fed into a green spider at rest with one leg in and k out, comes out as k red points, times $2^{(1-k)/2}$; so $\frac{1}{\sqrt{2}}$ for two out. And the colours swapped.
- **A half passes the other colour** (row 60, with Chapter 14's number): $Z(\pi)X(\alpha) = e^{i\alpha}X(-\alpha)Z(\pi)$, and the colours swapped.
- **Bialgebra** (row 61), with the number $\frac{1}{\sqrt{2}}$, and **Hopf** (row 62), with the number $\frac{1}{2}$.
- **A loop comes off** (row 90): a plain wire from a spider back to itself may be removed, number 1.
- **Letting go and reading** (rows 67, 68), and **causality** (row 72), for doubled pictures (Chapter 9).

Here $Z(\alpha) = \mathrm{diag}(1, e^{i\alpha})$ and $X(\alpha) = HZ(\alpha)H$ are the two-legged green and red spiders, bare, and a *point* is a spider with one leg: a green point at rest is $|0\rangle + |1\rangle$, a red point at rest is $\sqrt{2}\,|0\rangle$ and a red point at a half is $\sqrt{2}\,|1\rangle$. One more card will join the deck in §19.2.

Every card carries its number. That matters, and here is why. The Hopf card, as Chapter 8 first dealt it, said that a green spider and a red spider joined by two wires are the same as the two spiders apart, "up to" a number. With its number written, it reads



On the blackboard: the green spider with one leg in and two out is $|00\rangle\langle 0| + |11\rangle\langle 1|$, and the red one with two in and one out sends $|00\rangle$ and $|11\rangle$ both to $\frac{1}{2}(|+\rangle + |-\rangle) = \frac{1}{\sqrt{2}}|0\rangle$. So the left side is $\frac{1}{\sqrt{2}}|0\rangle\,(\langle 0| + \langle 1|)$, and the right side is $\frac{1}{2}\cdot\sqrt{2}\,|0\rangle\,(\langle 0| + \langle 1|)$, the same. Now suppose the card had been dealt without its $\frac{1}{2}$, and feed a green point at rest into both sides. The left side gives $\frac{1}{\sqrt{2}}|0\rangle\cdot 2 = \sqrt{2}\,|0\rangle$. The right side, without the $\frac{1}{2}$, gives $\sqrt{2}\,|0\rangle\cdot 2 = 2\sqrt{2}\,|0\rangle$. A chain of moves using that card would "prove" that $\sqrt{2}\,|0\rangle = 2\sqrt{2}\,|0\rangle$. For shapes of processes that is harmless; for numbers it is fatal, and a closed picture *is* a number. So the cards must balance, number and all.

There is a second worry, which the Anteater of the dialogue puts as a question about levels. A card is checked by itself. But in a proof it is used inside a larger picture, with other pieces all round it. Is a card that is true alone still true in company?

> **Result 19.2 (soundness: every card balances).** Write $[\![D]\!]$ for the matrix of a picture D. (a) For every card of the deck, and every choice of the turns written on it, the two sides have the same matrix, number included. (b) If $[\![D]\!] = [\![D']\!]$, then for any pictures K, L and M that fit around them, the picture "K, then D beside L, then M" and the same picture with D' in place of D have the same matrix. So every chain of moves by the cards proves an equation of matrices.

Proof. (a) One card at a time. Each card was proved as an equation of matrices, with its number and for every turn, in the chapter that dealt it (the rows listed above), and the yank in Chapter 6. (b) By the way pictures are read, $[\![\cdot]\!]$ turns "one after another" into the matrix product and "side by side" into the Kronecker product. So the matrix of the larger picture is $[\![M]\!]\,([\![D]\!] \otimes [\![L]\!])\,[\![K]\!]$, and it does not change when $[\![D]\!]$ is replaced by the equal matrix $[\![D']\!]$. A chain of moves is a list of such replacements, so its first and last pictures have the same matrix. ■ See `Code/Ch19/EveryCardBalances.py`, which checks every card with random turns, and then again with each side wrapped in random non-real boxes on the left, on the right and alongside.

The sources say this the same way. Backens: "By considering the rewrite rules one-by-one, it is not too difficult to show that the ZX-calculus is sound". Jeandel, Perdrix and Vilmart: "It's routine to prove the soundness of the axioms". Van de Wetering's survey (§7.3) explains why checking the cards is enough: the matrix of a picture is defined piece by piece, and equal pieces make equal wholes.

In the sketchbook. Soundness says: *every card balances*. A card may be laid anywhere in a picture, and the picture keeps its value. That is what gives the sketchbook the right to call a chain of rewrites a proof. The strand's other half is Result 19.1 read as a card game: hold any card up to the light and read it from behind, and what you see, left and right exchanged and every turn reversed, is again a true card, because the mirror of both sides is the adjoint of both sides.

Why this definition. Why ask that every card hold *exactly*, number included, when many papers on the calculus drop the numbers? Because this book reads probabilities off closed pictures, and a closed picture is nothing but its number. A deck that is sound only "up to a number" can prove any closed picture equal to any other non-zero one. Where we quote such a paper we will say so.

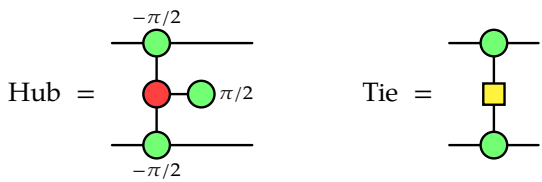
Soundness has a partner, recalled from Chapter 8. Theorem 8.7 there stated, with Coecke and Duncan, that the calculus is **universal**: every pure state, every unitary and every post-selected measurement on any number of

qubits can be drawn with spiders and yellow boxes. Van de Wetering (§7.3) calls this "fullness" of the map from pictures to matrices: every matrix of that kind is the matrix of some picture. Soundness says that the pictures never say anything false. Universality says that they can name anything. Neither says that the rules can prove every true equation. That is the third property, and it is the chapter's question.

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§19.2 The Hinge

Here is a place where the deck ran out. In Chapter 17's proof twice of the phase-polynomial circuit (§17.6), the sketchbook's step (iv) had to replace one small picture by another. The first was Chapter 14's controlled turn at $\alpha = -\pi$ (row 103): on each of two wires a green spider turned a quarter back, $-\frac{\pi}{2}$, each tied to one red hub at rest, and from the hub hangs a green point turned a quarter round, $\frac{\pi}{2}$. We call it Hub. The second is two green spiders at rest, one on each wire, joined by a wire with a yellow box on it (Chapter 16's Hadamard edge). We call it Tie:



Both have the same matrix, $\frac{1}{\sqrt{2}}\mathrm{CZ}$, where $\mathrm{CZ} = \mathrm{diag}(1, 1, 1, -1)$. Chapter 17 knew this, and wrote: "This step is a translation, checked by evaluating both sides, not a rewrite: we derive no chain of moves from the quarter hub to the Hadamard edge here". Its verdict added that a skeptic of pictures "may say the proof leans on the blackboard at its hinge, and the skeptic is right."

So here is a concrete case of the chapter's question: two pictures with the same matrix, and no chain of cards known to the book that joins them. Either the deck is missing a card, or we have not been clever enough. This

section shows that one card closes the gap.

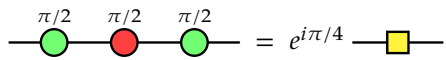
The Euler card

Chapter 3 found, in §3.6 ("Three turns"), that a quarter-turn about z, a quarter-turn about x and a quarter-turn about z make the Hadamard box: $SVS = e^{i\pi/4}H$, with $S = \mathrm{diag}(1, i)$ and $V = HSH$. In spiders, S is a green spider turned $\frac{\pi}{2}$ and V is a red one.

On the blackboard. $Z(\frac{\pi}{2}) = \mathrm{diag}(1, i)$ and $X(\frac{\pi}{2}) = \frac{1}{2}\left(\begin{smallmatrix} 1+i & 1-i \\ 1-i & 1+i \end{smallmatrix}\right)$. Multiplying $X(\frac{\pi}{2})$ on the right by $Z(\frac{\pi}{2})$ multiplies its second column by i, giving $\frac{1}{2}\left(\begin{smallmatrix} 1+i & 1+i \\ 1-i & -1+i \end{smallmatrix}\right)$; multiplying on the left by $Z(\frac{\pi}{2})$ multiplies the second row by i:

$$Z(\tfrac{\pi}{2})\,X(\tfrac{\pi}{2})\,Z(\tfrac{\pi}{2}) \;=\; \frac{1}{2}\begin{pmatrix} 1+i & 1+i \\ 1+i & -1-i \end{pmatrix} \;=\; \frac{1+i}{\sqrt{2}}\,H \;=\; e^{i\pi/4}\,H.$$

In the sketchbook. A green spider turned a quarter round, a red spider turned a quarter round and a green spider turned a quarter round, in a row on one wire, are the yellow box, with the whole turned an eighth round:



> **The Euler card (row 145).** $Z(\frac{\pi}{2})\,X(\frac{\pi}{2})\,Z(\frac{\pi}{2}) = e^{i\pi/4}H$, drawn as above, is **added to the deck as a rule**. It is true (the computation above), so the deck stays sound.

Why *add* it, when it is true and the blackboard proves it in a line? Because a card in the deck is a move the sketchbook may make without asking the blackboard, and until now this one was not in the deck. Could it have been got from the other cards? The book has derived no such chain, and the sources say that in Backens's calculus it cannot be: "The Euler decomposition rule cannot be derived from any combination of the other ZX-calculus rules", she writes (§3.3), citing Duncan and Perdrix's paper of 2009 on "the necessity of Euler decomposition". She also notes that the rule "is special in that it does not have a category-theoretical meaning, but follows from the fact that any unitary single qubit operator can be decomposed as a sequence

of three rotations around two orthogonal axes". Van de Wetering's survey goes further and *defines* the yellow box by this equation (his eq. (44)), and Jeandel, Perdrix and Vilmart's rule set contains it too. Backens states her calculus with numbers dropped; our card carries its number, $e^{i\pi/4}$.

Why this definition. Why this rule and not something simpler, such as "two yellow boxes are none"? We have that one already (row 56), and it only relates yellow boxes to each other. Colour change (row 58) relates a yellow box to spiders, but only when a spider is dressed in boxes on every leg. The Euler card is the one rule that says what a yellow box *is* in terms of spiders alone. Without it the yellow box is a separate piece that the spiders can push around but never take apart.

Three smaller cards follow from the new one. They are not new rules: each is derived from cards already in the deck, and each is marked so.

> **Result 19.3 (the mirror Euler; derived).** Three quarter turns back, green, red, green, are the yellow box with the whole turned an eighth back:
>
> $$Z(-\tfrac{\pi}{2})\,X(-\tfrac{\pi}{2})\,Z(-\tfrac{\pi}{2}) \;=\; e^{-i\pi/4}H.$$

Proof. On the blackboard, take the adjoint of both sides of the Euler card: $Z(\alpha)^\dagger = Z(-\alpha)$, $X(\alpha)^\dagger = X(-\alpha)$, $H^\dagger = H$, and the row of three reads the same backwards. In the sketchbook, by the cards alone, as the sidebar below does it. ■ See `Code/Ch19/TheHinge.py`.

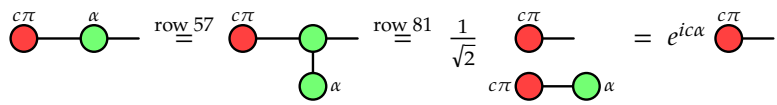
> **Coecke's Sketchbook.**
>
> *The mirror Euler, by the cards.* By Result 19.1 the mirror of the Euler card is a true equation, and one might simply hold the card up to the light. But a true equation is not yet a move: a card must be *derived* from cards in the deck. Here is the derivation. Start with the three quarters back on one wire: green, red, green, all $-\frac{\pi}{2}$. After them, put two yellow boxes in a row, which is no box at all (row 56). Now read the first of the two boxes by the Euler card backwards: it becomes $e^{-i\pi/4}$ times green, red, green, all $+\frac{\pi}{2}$. In the middle, a green quarter back meets a green

> quarter round: they slide together to a green spider at rest (row 57), which is bare wire (Chapter 8's identity card). That brings a red quarter back next to a red quarter round: the same again. That brings the first green quarter back next to the last green quarter round: the same again. What is left is one yellow box, and the number $e^{-i\pi/4}$. The pairs vanish from the inside out, like a zip.

Two lemmas about points come next. The first says what a green turn does to a red point at rest or at a half. For an example take a red point at a half, $\sqrt{2}\,|1\rangle$, followed by a green quarter round, $Z(\frac{\pi}{2}) = \mathrm{diag}(1, i)$. The result is $i\sqrt{2}\,|1\rangle$: the same point, with the whole turned a quarter round.

> **Result 19.4 (Lemma P: a green turn passes a basis point; derived).** For $c \in \{0, 1\}$ and every α, $Z(\alpha)\,\sqrt{2}\,|c\rangle = e^{ic\alpha}\,\sqrt{2}\,|c\rangle$. In pictures: a red point turned $c\pi$, followed by a green spider turned α, is the same red point with the whole turned by $c\alpha$.

Proof. On the blackboard, $Z(\alpha)$ is diagonal with entries 1 and $e^{i\alpha}$. In the sketchbook: slide the green spider apart into a green spider at rest on the wire with a green α point hanging from it (row 57 backwards); the red point copies through the green spider at rest (row 81), one copy on the wire and one on the hanging leg, with the number $\frac{1}{\sqrt{2}}$; the copy on the hanging leg now meets the green α point, and the two make a closed picture with no wire left open, which is a number (Chapter 2). It is $\sqrt{2}\,\langle c|\,(|0\rangle + e^{i\alpha}\,|1\rangle) = \sqrt{2}\,e^{ic\alpha}$, as in Chapter 14's "ledger of the lemma". The total is $\frac{1}{\sqrt{2}} \cdot \sqrt{2}\,e^{ic\alpha} = e^{ic\alpha}$:

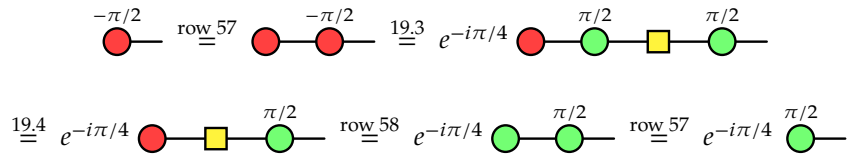


■ See `Code/Ch19/TheHinge.py`, which checks it for random α and both values of c.

The second lemma is about the point hanging from Hub: a green point turned a quarter round, $|0\rangle + i\,|1\rangle$. It has the wrong colour for anything we want to do with a red hub. Can it be traded for a red point?

> **Result 19.5 (Lemma A: a quarter point changes colour; derived).** A red point turned a quarter back is $e^{-i\pi/4}$ times a green point turned a quarter round: $X(-\frac{\pi}{2})\,\sqrt{2}\,|0\rangle = e^{-i\pi/4}(|0\rangle + i\,|1\rangle)$. Equivalently, a green $\frac{\pi}{2}$ point is $e^{i\pi/4}$ times a red $-\frac{\pi}{2}$ point.

Proof. On the blackboard, $X(-\frac{\pi}{2}) = \frac{1}{2}\left(\begin{smallmatrix} 1-i & 1+i \\ 1+i & 1-i \end{smallmatrix}\right)$, so $X(-\frac{\pi}{2})\sqrt{2}\,|0\rangle = \frac{1}{\sqrt{2}}((1-i)\,|0\rangle + (1+i)\,|1\rangle)$, and $\frac{1-i}{\sqrt{2}} = e^{-i\pi/4}$, $\frac{1+i}{\sqrt{2}} = e^{-i\pi/4}\,i$. In the sketchbook, five moves. Slide the red point apart into a red point at rest and a red spider turned $-\frac{\pi}{2}$ on its wire (row 57 backwards). Put a green quarter round and a green quarter back side by side at each end of that red spider (row 57 backwards and the identity card); the red spider now sits between two green quarters back, and by the mirror Euler (Result 19.3) the three are $e^{-i\pi/4}$ times a yellow box. So the wire reads: red point at rest, green $\frac{\pi}{2}$, yellow box, green $\frac{\pi}{2}$, with the number $e^{-i\pi/4}$. The first green quarter passes the red point at rest with the number 1 (Result 19.4, $c = 0$). The yellow box turns the red point at rest into a green point at rest (row 58). The last green quarter fuses into it (row 57):



■ See `Code/Ch19/TheHinge.py`.

In the sketchbook, then, the deck now holds one new card, given, and three derived from it, and the two lemmas are exactly what Hub needs: Lemma A turns its hanging point red, so that it can fuse into the red hub, and then the hub's tie reads as three quarter turns back, which the mirror Euler turns into a yellow box.

> **Result 19.6 (the hinge; the headline).** Hub = Tie, exactly, with the number 1. Both are $\frac{1}{\sqrt{2}}$ CZ.

It is proved twice in §19.7: on the blackboard in two lines, and in the sketchbook in five moves, each by a card of the deck. With it, Chapter 17's step (iv) is no longer a translation: it is a rewrite.

Dirac's Blackboard.

The hinge in two lines. Let $x = (x_1, x_2)$ be a basis word on the two wires. Each foot of Hub is a green spider, so it passes $|x_k\rangle$ along its wire and copies the bit x_k onto its tie, with the turn $e^{-i\frac{\pi}{2}x_k}$. The red hub at rest, fed x_1 and x_2 on two legs, puts out $2^{-1/2}\,|x_1 \oplus x_2\rangle$ on its third, and the hanging point closes that with $e^{i\frac{\pi}{2}(x_1 \oplus x_2)}$; this is row 129's hub, $2^{-1/2}\Gamma(\frac{\pi}{2})$. So

$$\mathrm{Hub}\,|x\rangle = 2^{-1/2}\, e^{\,i\frac{\pi}{2}\,(x_1 \oplus x_2 - x_1 - x_2)}\,|x\rangle = 2^{-1/2}(-1)^{x_1 x_2}\,|x\rangle\,,$$

because $x_1 + x_2 - (x_1 \oplus x_2) = 2x_1x_2$. And $\mathrm{Tie}\,|x\rangle = \langle x_2|\,H\,|x_1\rangle\,|x\rangle = 2^{-1/2}(-1)^{x_1x_2}\,|x\rangle$. Both are $\frac{1}{\sqrt{2}}\mathrm{CZ}$. The whole proof is the identity for the carry of a one-bit sum: add two bits, subtract their parity, and you are left with twice their product.

---

§19.3 Same Number, Same Picture

One card closed one gap. Is there always one more card? Before the general answer, a dare: name two pictures with the same matrix, chosen with nothing but the blackboard in mind, and see whether the deck can join them.

A dare

Take a yellow box followed by a green quarter round, three times over, on one wire. On the blackboard the picture is $SHSHSH = (SH)^3$, with $S = Z(\frac{\pi}{2}) = \mathrm{diag}(1, i)$ (the picture reads left to right, the product right to left). Now

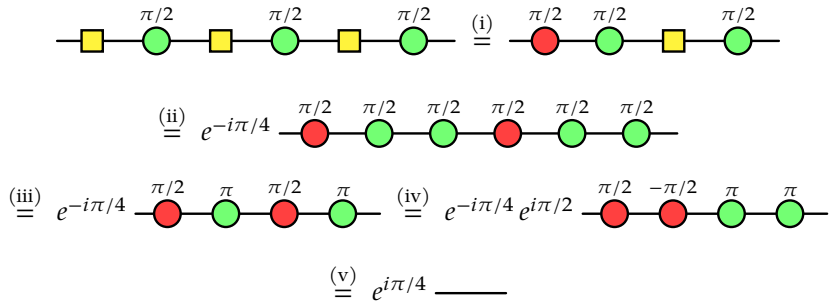
$$SH = \tfrac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ i & -i \end{pmatrix}, \qquad (SH)^2 = \tfrac{1}{2}\begin{pmatrix} 1+i & 1-i \\ 1+i & -1+i \end{pmatrix},$$

$$(SH)^3 = \tfrac{2+2i}{2\sqrt{2}}\, I = e^{i\pi/4}\, I.$$

The six pieces do nothing at all, except turn the whole by an eighth. Nothing in the picture suggests it. Can the cards find it?

> **Result 19.7 (a tube and a quarter, three times over).** $(SH)^3 = e^{i\pi/4}I$, and the deck derives it: the picture of a yellow box and a green quarter round, three times over, rewrites by the cards to a bare wire with the number $e^{i\pi/4}$.

Proof. The blackboard did it above. In the sketchbook, five moves, each by one card, with the pictures read left to right. **(i)** The first yellow box, green quarter and yellow box are a green spider with a box on each leg: colour change makes it a red quarter round (row 58). **(ii)** The next yellow box is replaced by the Euler card read backwards, $e^{-i\pi/4}$ times green, red, green quarters round (row 145). **(iii)** Neighbours of one colour fuse (row 57): the wire now reads red $\frac{\pi}{2}$, green π, red $\frac{\pi}{2}$, green π. **(iv)** The first green half passes the red quarter after it: the quarter is reversed and the whole turns a quarter round, $e^{i\pi/2}$ (row 60, with Chapter 14's number; on the blackboard, $X(\frac{\pi}{2})Z(\pi) = e^{i\pi/2}Z(\pi)X(-\frac{\pi}{2})$). **(v)** The two red quarters, round and back, fuse to rest, and so do the two green halves, which make a whole turn; spiders at rest with two legs are bare wire (row 57 and the identity card). The numbers: $e^{-i\pi/4}\,e^{i\pi/2} = e^{i\pi/4}$.



■ See `Code/Ch19/SameNumberSamePicture.py`, where PyZX's own simplifier takes the same picture to a bare wire with the number $e^{i\pi/4}$.

Step (ii) is the Euler card again. The chain needed it, and at no point did the sketchbook look at a matrix. That is one dare won. The general question is whether every dare can be won.

Completeness

On the blackboard. Fix a kind of picture: say, those whose spiders are turned only by multiples of $\frac{\pi}{2}$, or of $\frac{\pi}{4}$, or by anything at all. A deck of cards is **complete** for that kind if, for every two pictures D_1 and D_2 of that kind,

$$[\![D_1]\!] = [\![D_2]\!] \quad \Longrightarrow \quad D_1 \text{ rewrites to } D_2 \text{ by the cards of the deck.}$$

Soundness (Result 19.2) is the arrow the other way. Van de Wetering (§7.3) puts it in the language of §19.1: soundness says the map $[\![\cdot]\!]$ from pictures to matrices is well defined; universality says it is *full*, every matrix of the kind is hit; completeness says it is *faithful*, no two pictures that the cards cannot join are sent to the same matrix. He is plain about his own rule set: "The ruleset we presented in Figure 1 is not complete".

In the sketchbook. Completeness says: **same number, same picture**, *with the right cards*. If the blackboard says two pictures agree, the sketchbook can get from one to the other without borrowing anything from the blackboard. It says a chain *exists*; it does not say how long the chain is, or how to find it.

Why this definition. Why fix a *kind* of picture first, instead of asking for one deck that works for everything? Because the answer depends on the kind. Van de Wetering's Figure 1, which suffices for turns by quarters, is not enough for turns by eighths (§19.5 shows an equation that his survey says it cannot prove), and each kind has needed its own deck, found in its own paper. A definition that did not name the kind could not say which of those papers answered it.

Here is what is known, by kind, quoted from the papers that proved it. None of it is proved in this book.

> **Result 19.8 (completeness, by kind; stated, not proved).** (a) *Turns by multiples of* $\frac{\pi}{2}$ *(the Clifford pictures).* Miriam Backens (Oxford, 2013), Theorem 21: "The ZX-calculus is complete for stabilizer quantum mechanics." Her rule set includes the Euler rule, and it ignores non-zero numbers: "Since global phases have no physical effect and the normalization of pure states is fixed, we will ignore them" (her §3.3). (b) *Turns by multiples*

of $\frac{\pi}{4}$ (*Clifford*+T). Emmanuel Jeandel, Simon Perdrix and Renaud Vilmart (posted May 2017), Theorem 1: "The $\frac{\pi}{4}$-fragment of the ZX-Calculus as presented in Figure 1 is complete: for any two diagrams D_1, D_2 in the $\frac{\pi}{4}$-fragment of the ZX-Calculus, $[\![D_1]\!] = [\![D_2]\!]$ iff $ZX_{\pi/4} \vdash D_1 = D_2$." Their Figure 1 is a "Set of rules for the ZX-Calculus with scalars". (c) *Any turns.* Kang Feng Ng and Quanlong Wang (Oxford, 2017), Theorem 5.2: "The ZX-calculus is complete for universal pure qubit quantum mechanics: If $[\![D_1]\!] = [\![D_2]\!]$, then $ZX \vdash D_1 = D_2$". Their language is extended "with two new symbols (although in principle they could be eliminated ...)". (d) *With letting go* (*Chapter 9's doubled pictures*). Titouan Carette, Emmanuel Jeandel, Simon Perdrix and Renaud Vilmart (Nancy, 2019), Corollary 1 (of Theorem 1): "Given a †-CC C with enough isometries, if G is a †-CC universal complete graphical language for C then $G^\top$ is a universal complete language for CPM(C)." Their superscript is the ground symbol, which we write $\top$ as for the book's discard (row 67). They show that the turns by multiples of $\frac{\pi}{2}$, with letting go, meet the condition (their §4.2), and that the turns by multiples of $\frac{\pi}{4}$ do not; for these, "we leave open the complete axiomatisation of quantum operations for this fragment" (their §4.3).

Read the scope of each with care. In (b), Jeandel, Perdrix and Vilmart reached completeness "by adding two new axioms to the language", but *two* is counted against their own Figure 1, not against this book's deck, and we do not count cards against theirs. Whether the book's deck, with the Euler card, is the same as any of these rule sets we have not checked, and do not claim. In (d), "†-CC" is their abbreviation for a dagger compact closed category, the structure of §19.1, and CPM(C) is the doubled construction of Chapter 9. The words "enough isometries" name a technical condition (their Definition 10) that the kind of picture must meet for the discard to be added this simply.

This also corrects a sentence of Chapter 8. Theorem 8.7 there said that within the Clifford pictures "any two diagrams denoting the same map can

be turned into one another by the rules alone, with no return to matrices (Backens)." That is right, with two qualifications we can now state: Backens's rules include the Euler rule, which the book's deck lacked until §19.2; and her rules, like van de Wetering's Figure 1 ("these rules are only correct up to non-zero scalar"), ignore non-zero numbers, which the book's cards do not.

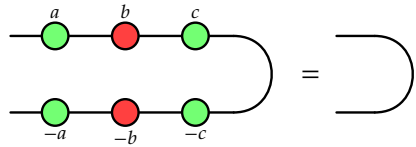
Feynman's Notebook.

The completeness race. Van de Wetering's survey (2020, §9.1) tells the story, dating each paper by its first posting on the arXiv. Coecke and Duncan introduced the calculus in a 2007 preprint, and more formally in 2008; the Hadamard box was then a piece of its own. Duncan and Perdrix saw in 2009 that its decomposition into turns was needed for completeness. Backens proved the Clifford case in 2013. The next step came through a different calculus, the ZW-calculus, for which Hadzihasanovic found a complete rule set "in the beginning of 2015". In May 2017 Jeandel, Perdrix and Vilmart used it to settle Clifford+T. In June 2017 Ng and Wang, at Oxford, settled the case of any turns, through a modified ZW-calculus from Hadzihasanovic's thesis. In January 2018 each side answered the other: Jeandel, Perdrix and Vilmart with any turns, Ng and Wang with Clifford+T. Van de Wetering calls it an "entangled" history, and the word is fair: within a year, two groups each did both, each in the other's wake.

An instance with letting go

Part (d) is about doubled pictures, with a wire and its mirror copy and a cap that lets them go (Chapter 9). Here is the simplest dare of that kind. On a wire, a green spider turned a, a red one turned b and a green one turned c; on its mirror copy, which carries the conjugate process, the same three turned $-a$, $-b$, $-c$; then let the wire go, which is a cap joining the wire to its copy. The blackboard says that a unitary, then letting go, is just letting go (causality, row 72): $\mathrm{tr}(U\rho U^{\dagger}) = \mathrm{tr}\,\rho$. Does the sketchbook agree without being told?

Result 19.9 (letting go after any turn). For all turns a, b, c, exactly: a green, a red and a green spider turned a, b, c on a wire, the same turned $-a$, $-b$, $-c$ on its mirror copy, and then the cap, equal the bare cap.



Proof. The cap is unnormalized, $\langle 00| + \langle 11|$, and as a picture it is a green spider at rest with two legs, bent (rows 40, 63). So the green c next to it slides into it (row 57), and so does the green $-c$ from the other side: the turns add to rest. Now the cap is bare again, and the red b and red $-b$ are joined through it by a plain wire, so they fuse to a red spider at rest, which is bare wire (row 57, identity). The same happens to the green a and $-a$. What is left is the bare cap. Every move is exact, with no number. On the blackboard: $(\text{cap})(U \otimes \bar{U}) = \text{cap}$ for every unitary U, since $\sum_i \langle ii| (U \otimes \bar{U}) = \sum_{i,j,k} U_{ij}\bar{U}_{ik} \langle jk| = \sum_{j,k} (U^\dagger U)_{kj} \langle jk|$, and $U^\dagger U = I$. ■ See `Code/Ch19/SameNumberSamePicture.py`, which checks it for random turns.

Here the pictures win outright: three slides through the cap against a sum over three indices. And the picture shows *why* the turns cancel. Each spider meets its own conjugate through the cap and the two turns add to nothing, from the inside out, as in the mirror Euler of §19.2.

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§19.4 One Answer Out

So far the pictures have had open wires: processes with inputs and outputs. A whole computation is different. It starts from states nobody chose freshly each time, runs a process, reads *one* answer, and lets everything else go. From the outside it is a closed box with one wire coming out. What can such a box tell anyone?

A run, and a class of problems

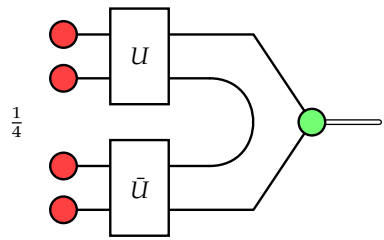
On the blackboard. A **run** of a quantum computer on n wires starts each wire at $|0\rangle$, applies a unitary U, reads the first wire, and ignores the rest. The answer is 1 with probability

$$p_1 \;=\; \langle 0\cdots 0|\, U^\dagger (|1\rangle\langle 1| \otimes I \otimes \cdots \otimes I)\, U\, |0\cdots 0\rangle\,,$$

and 0 with probability $p_0 = 1 - p_1$. A single run gives one bit, and it may be the wrong bit. So the useful notion is a machine that is right *often enough*.

A yes-or-no problem is in the class called **BQP** if there is a family of circuits, one for each size of input, with these three properties. The circuit for inputs of size n has a number of gates bounded by a polynomial in n. The circuits can be written down, one for each n, by an ordinary computer. And on every input, one run of its circuit answers correctly with probability at least $\frac{2}{3}$: on an input whose answer is "yes" it reads 1 with probability at least $\frac{2}{3}$, and on one whose answer is "no" it reads 1 with probability at most $\frac{1}{3}$. For the class and its place among the others, see Nielsen and Chuang's textbook. We use the class only to fix words; this book proves nothing about how BQP compares with any other class.

In the sketchbook. A run is a closed box with one wire out. Draw it doubled, as in Chapter 9: each quantum wire with its mirror copy below it, the copy carrying the conjugate process $\bar{U}$. Every wire and copy starts at a red point at rest, $\sqrt{2}\,|0\rangle$, so four such points need the number $\frac{1}{4}$ in front. The wire that is not read is let go: it is capped to its own copy (row 67). The wire that is read goes, with its copy, into the read-spider (row 68), a green spider at rest that takes a wire and its copy and gives out one classical wire, drawn doubled-stroke:



The copies are drawn in reflected order, the first wire's copy lowest, so that every cap is nested and no two lines cross; the box $\bar{U}$ takes its legs in that reflected order. The value of the whole picture is a classical vector, (p_0, p_1): the odds of the answer. That is all a closed box gives out: one answer, with odds.

Why this definition. Why a probability of $\frac{2}{3}$, and not certainty? Because a quantum computer's answer comes with odds, and demanding certainty would throw away most of what it can do. Why $\frac{2}{3}$ exactly? It is a convention: any fixed chance above one half can be pushed up by running the machine several times and taking a vote, and here is the smallest case of that, done exactly.

Three runs and a vote

Suppose one run is right two times in three. Run it three times, independently, and take the majority. The majority is right when all three runs are right, with probability $(\frac{2}{3})^3 = \frac{8}{27}$, or when exactly two are, in three ways, each with probability $(\frac{2}{3})^2 \cdot \frac{1}{3} = \frac{4}{27}$. In all, $\frac{8}{27} + \frac{12}{27} = \frac{20}{27}$, which is more than $\frac{2}{3} = \frac{18}{27}$. Is that a lucky number, or does a vote always help?

> **Result 19.10 (a majority of three).** If each of three independent runs is right with probability p, the majority is right with probability
>
> $$f(p) \;=\; 3p^2 - 2p^3.$$
>
> If $p \geq \frac{2}{3}$, then $f(p) \geq \frac{20}{27}$.

Proof. The majority is right when all three runs are, probability p^3, or exactly two are, three ways of probability $p^2(1-p)$ each: $p^3 + 3p^2(1-p) = 3p^2 - 2p^3$. Then $f'(p) = 6p - 6p^2 = 6p(1-p) \geq 0$ on $[0, 1]$, so f does not decrease there, and $p \geq \frac{2}{3}$ gives $f(p) \geq f(\frac{2}{3}) = \frac{20}{27}$. ■ See `Code/Ch19/OneAnswerOut.py`, which checks $f(\frac{2}{3}) = \frac{20}{27}$ exactly and $f(p) \geq \frac{20}{27}$ on a grid of p from $\frac{2}{3}$ to 1.

Dirac's Blackboard.

The same polynomial, read the other way. Chapter 18 counted the failure of a vote of three, each copy flipped with probability p: $F(p) = 3p^2 - 2p^3$. Here p is the chance a run is *right*, and the majority's success is the same polynomial. That is no accident. The majority of three is right exactly when the majority of their failures is not, so $f(p) = 1 - F(1-p)$, and

$$1-F(1-p) = 1-3(1-2p+p^2)+2(1-3p+3p^2-p^3) = 3p^2-2p^3.$$

So $F(p) + F(1-p) = 1$: the curve is symmetric about the point $(\frac{1}{2}, \frac{1}{2})$. And $f(p) - p = p(1-p)(2p-1)$, which is positive for $\frac{1}{2} < p < 1$. Votes can be stacked. Three groups of three runs, a majority taken in each group and then a majority of the three results, are right with probability

$$f(\tfrac{20}{27}) = 3 \cdot \tfrac{400}{729} - 2 \cdot \tfrac{8000}{19683} = \tfrac{32400-16000}{19683} = \tfrac{16400}{19683},$$

about 0.833. Each level raises a chance above one half, and the only resting points of f in $[0, 1]$ are 0, $\frac{1}{2}$ and 1, so stacked votes climb towards 1, as Chapter 18's toy showed.

In the sketchbook, three runs are three closed boxes side by side, three classical wires out, and the vote is a classical process on those three wires. The pictures could evaluate it, but the count of ways, and the inequality, are the blackboard's.

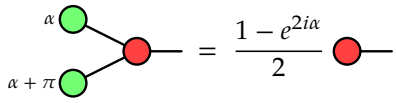
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§19.5 What Pictures Cannot Do

A book that has drawn pictures for many chapters owes its reader a list of their limits. Each item below is either sourced or said in the form "this book has derived no such chain". Nothing on the list is proved here, except one number.

Not every deck is complete. Van de Wetering: "The ruleset we presented in Figure 1 is not complete" (§7.3). His example opens his §9: an equation that, he writes, "is not provable using the rules of Figure 1 for any α that is not a multiple of $\frac{\pi}{2}$", citing Perdrix and Wang's paper of 2016 on why "supplementarity is necessary". He adds: "this equation only holds up to a global scalar". Here it is with its number worked out. Two green points, turned α and $\alpha + \pi$, feed a red spider with two legs in and one out:

Result 19.11 (an eighth-turn instance). For every α, exactly:



Proof. On the blackboard. The points are $|0\rangle + e^{i\alpha}|1\rangle$ and $|0\rangle - e^{i\alpha}|1\rangle$. The red spider sends $|ab\rangle$ to $|+\rangle\langle+|a\rangle\langle+|b\rangle + |-\rangle\langle-|a\rangle\langle-|b\rangle$. Here $\langle+|a\rangle\langle+|b\rangle = \frac{1}{2}(1+e^{i\alpha})(1-e^{i\alpha}) = \frac{1}{2}(1-e^{2i\alpha})$, and $\langle-|a\rangle\langle-|b\rangle = \frac{1}{2}(1-e^{i\alpha})(1+e^{i\alpha})$ is the same. So the output is $\frac{1}{2}(1-e^{2i\alpha})(|+\rangle + |-\rangle)$, and $|+\rangle + |-\rangle = \sqrt{2}\,|0\rangle$ is the red point at rest. ■ See `Code/Ch19/SameNumberSamePicture.py`, which checks the instance at $\alpha = \frac{\pi}{4}$, where the number is $\frac{1-i}{2}$.

At $\alpha = \frac{\pi}{4}$, the eighth turn, the number is $\frac{1-i}{2}$, which is not zero, so the equation says something. The blackboard proves it for every α in two lines. Whether this book's deck, with the Euler card, can prove it at $\alpha = \frac{\pi}{4}$ we do not know: the book has derived no such chain, and we have not compared the deck with van de Wetering's Figure 1, so his statement about Figure 1 is quoted as his and not claimed for the deck.

One rule was not enough for every turn. Ng and Wang, citing Schröder de Witt and Zamdzhiev: "the original version of the ZX-calculus plus the Euler decomposition of Hadamard gate is incomplete for the overall pure qubit quantum mechanics". The Euler card of §19.2 was what the Clifford pictures lacked; it is far from all that every turn needs.

A finite deck for all turns may be out of reach, or may not. Van de Wetering's §9.5 presents a complete rule set for any turns that needs one extra rule, and says that its "complicated side condition" betrays the benefits of

graphical reasoning "a bit"; he adds, citing others, that "it however seems unlikely that all the relations between phases that are not rational multiples of π can be captured by a finite set of rules". That is a judgement, not a theorem, and we quote it as one.

Some processes have no known single picture. Van de Wetering (§10.1, 2020): "there is currently no known representation of partial decoherence in the ZX-calculus". Chapter 18's noise (row 136) was drawn as a weighted *sum* of two pictures, which is consistent with this: a sum is not one picture.

For one kind, completeness with letting go was left open. Carette, Jeandel, Perdrix and Vilmart (2019, §4.3), for turns by multiples of $\frac{\pi}{4}$ with discard: "we leave open the complete axiomatisation of quantum operations for this fragment." We report what the paper says; we do not claim to know whether the question is still open.

A rewrite proves an equality of processes, and nothing else. This is the book's own rule, and it is the most important item on the list. A chain of cards proves that two pictures have the same matrix. It does not prove that a circuit is short, that an algorithm is fast, that a code is good, or that a problem is in any class. Completeness says that a chain *exists*; it says nothing about how long the chain is or how to find it, and the book makes no claim about either. And no rewrite decides a question about *all* pictures of a kind at once: §19.9 meets a theorem that says, for one such question, that no procedure of any kind can.

In the sketchbook. Row 152 of the Rosetta Table is the only row of the book whose right-hand column is empty on purpose. Its entry is: no picture. The equation of Result 19.11 has a picture on each side, and so does every equation a deck can prove; the claims on this list are claims *about* decks, and a claim about all pictures of a kind is not itself a picture.

§19.6 The Rosetta Table

The rows this chapter adds to the running dictionary, numbered on from Chapter 18's row 144. The chapter introduces no new string-diagram element. Row 145 is a new *rule* over old elements; rows 146 and 147 are derived from it; rows 148 to 151 name what the book has been doing; and row 152 is the list of §19.5.

#	Algebra (Dirac)	Picture (Coecke)
145	$Z(\frac{\pi}{2})X(\frac{\pi}{2})Z(\frac{\pi}{2}) = e^{i\pi/4}H$ (Chapter 3's SVS); the mirror, $Z(-\frac{\pi}{2})X(-\frac{\pi}{2})Z(-\frac{\pi}{2}) = e^{-i\pi/4}H$	**the Euler card**: green, red, green quarters round = the yellow box, the whole an eighth round; a *rule* of the calculus from now on (not derivable from the other rules of Backens's calculus, her §3.3, citing Duncan and Perdrix; the book has derived no such chain from its own deck); the mirror derived (rows 56, 57)
146	$Z(\alpha)\sqrt{2}\,\lvert c\rangle = e^{ic\alpha}\sqrt{2}\,\lvert c\rangle$; $X(-\frac{\pi}{2})\sqrt{2}\,\lvert 0\rangle = e^{-i\pi/4}(\lvert 0\rangle + i\,\lvert 1\rangle)$	**a green turn passes a basis point**, leaving $e^{ic\alpha}$ (rows 57, 81); **a quarter point changes colour**: a green $\frac{\pi}{2}$ point $= e^{i\pi/4}\times$ a red $-\frac{\pi}{2}$ point (derived, row 145)
147	Hub = Tie = $\frac{1}{\sqrt{2}}$CZ	**the hinge**: the quarter hub is the tie with a yellow box, number 1 (rows 57, 145, 146); Chapter 17's step (iv) is now a rewrite

#	Algebra (Dirac)	Picture (Coecke)
148	$[\![\cdot]\!]$ turns $\circ$ into the matrix product and $\otimes$ into the Kronecker product; every card holds exactly	**soundness**: every card balances; a card used inside any picture keeps it equal
149	FdHilb is dagger compact closed (Selinger, Example 7.11); $(f \circ g)^\dagger = g^\dagger \circ f^\dagger$, $(f \otimes g)^\dagger = f^\dagger \otimes g^\dagger$	**the pictures are a theory of processes**: the mirror is the dagger (Result 19.1); bends yank; only the connectivity counts (rows 42, 63)
150	Backens, Theorem 21; Jeandel–Perdrix–Vilmart, Theorem 1; Ng–Wang, Theorem 5.2; Carette et al., Corollary 1	**completeness, by kind**: same matrix $\Rightarrow$ same picture, *by that kind's deck*; stated, not drawn
151	BQP: right with probability $\geq \frac{2}{3}$ on every input; majority of three, $f(p) = 3p^2 - 2p^3$	**a run is a closed box**: points in, a process, one read wire out, the rest let go; its value is the answer's odds
152	what pictures cannot do (§19.5)	**no picture**: a rewrite proves an equality, never a bound, a class or a decision procedure

Every picture row is exact, number included. Row 145's card holds with its number $e^{i\pi/4}$, where Backens's and van de Wetering's versions drop it; row 150 quotes theorems whose own scope, with or without numbers, is stated in §19.3; row 151's value is a vector of probabilities; and row 152 has no picture at all.

§19.7 The Theorem, Twice

> **Result 19.6 (the hinge), restated.** Let Hub be two wires, each carrying a green spider turned $-\frac{\pi}{2}$, the two tied to one red spider at rest, from which hangs a green point turned $\frac{\pi}{2}$. Let Tie be two wires, each carrying a green spider at rest, the two joined by a wire with a yellow box on it. Then, exactly, with the number 1,
>
> $$\text{Hub} \;=\; \text{Tie} \;=\; \frac{1}{\sqrt{2}}\,\text{CZ}.$$

Dirac's Blackboard

Write a basis word on the two wires as $|x_1x_2\rangle$, top wire first.

Step 1: the feet. Each foot is a green spider with one leg in and two out (one along the wire, one to the hub), turned $-\frac{\pi}{2}$: $|00\rangle\langle 0| + e^{-i\pi/2}\,|11\rangle\langle 1|$. So on $|x_1x_2\rangle$ the two feet give

$$e^{-i\frac{\pi}{2}x_1}\,e^{-i\frac{\pi}{2}x_2}\;\;|x_1x_2\rangle \otimes |x_1x_2\rangle_{\text{ties}}\,.$$

Step 2: the hub. The red spider at rest, with the two ties as inputs and one output, is $|+\rangle\langle ++| + |-\rangle\langle --|$. On $|y_1y_2\rangle$ it gives $\frac{1}{2}(|+\rangle + (-1)^{y_1+y_2}\,|-\rangle) = \frac{1}{\sqrt{2}}\,|y_1 \oplus y_2\rangle$. The hanging green point, as an effect, is $\langle 0| + e^{i\pi/2}\,\langle 1|$, so it closes the output with $e^{i\frac{\pi}{2}(y_1\oplus y_2)}$. Hub and point together send the ties $|x_1x_2\rangle$ to the number $2^{-1/2}\,e^{i\frac{\pi}{2}(x_1\oplus x_2)}$.

Step 3: the product.

$$\text{Hub}\,|x_1x_2\rangle \;=\; 2^{-1/2}\,e^{i\frac{\pi}{2}(x_1\oplus x_2 - x_1 - x_2)}\,|x_1x_2\rangle\,.$$

Since $x_1 + x_2 - (x_1 \oplus x_2) = 2x_1x_2$ for bits, the exponent is $-i\pi x_1x_2$, and $e^{-i\pi x_1x_2} = (-1)^{x_1x_2}$. Word by word: $|00\rangle$, $|01\rangle$, $|10\rangle$ get $2^{-1/2}$, and $|11\rangle$ gets $2^{-1/2}e^{-i\pi}e^{0} = -2^{-1/2}$. So Hub $= \frac{1}{\sqrt{2}}\,\text{diag}(1,1,1,-1) = \frac{1}{\sqrt{2}}\text{CZ}$.

Step 4: the tie. Each green spider at rest copies its wire's bit onto the joining wire, $|x_k\rangle \;\mapsto\; |x_k\rangle\,|x_k\rangle$. The joining wire carries x_1 from the top, through

H, into the bottom spider's leg, where it must meet x_2: the amplitude is $\langle x_2|H|x_1\rangle = 2^{-1/2}(-1)^{x_1 x_2}$. So Tie $= \frac{1}{\sqrt{2}}$CZ as well. ■

Coecke's Sketchbook

Start from Hub. Every step names its card, and every number is written.

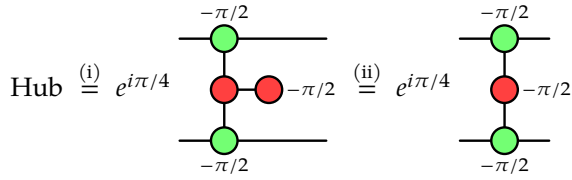
Step (i): the hanging point changes colour (Result 19.5, Lemma A). The green point turned a quarter round is $e^{i\pi/4}$ times a red point turned a quarter back.

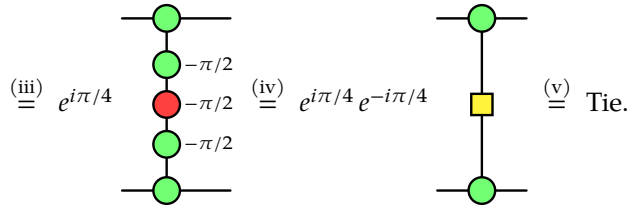
Step (ii): the point joins the hub (row 57). The red point and the red hub at rest are joined by a plain wire, so they fuse. The hub now has two legs, the two ties, and is turned $-\frac{\pi}{2}$.

Step (iii): each foot slides apart along its own tie (row 57 backwards). The upper foot becomes a green spider at rest on the wire and a green spider turned $-\frac{\pi}{2}$ on the tie; the same for the lower foot. Number 1. The tie now runs, from top to bottom: a green quarter back, a red quarter back, a green quarter back. By row 63 it may be read as one wire, straight down.

Step (iv): three quarters back are a yellow box (Result 19.3, the mirror Euler). They are replaced by a yellow box and the number $e^{-i\pi/4}$.

Step (v): the numbers. $e^{i\pi/4} \cdot e^{-i\pi/4} = 1$. What is left is two green spiders at rest joined by a wire with a yellow box on it: Tie.





■ See `Code/Ch19/TheHinge.py`, which builds every picture of this chain and checks each against the one before, with its number; there PyZX's own simplifier, which is not this book's deck, also takes Hub to Tie.

The verdict

The blackboard wins, and the chapter says so. Its proof is two lines and one identity about bits, $x_1 + x_2 - (x_1 \oplus x_2) = 2x_1x_2$. The sketchbook needs five moves, and before it could make them it needed a new card, the Euler card, and three cards derived from it. By any count of ink the algebra is shorter.

But the point of the sketchbook here is not that it is shorter. It is that it *exists*. Chapter 17 reached this step and could only check it; now the step is a chain of cards, and a skeptic of pictures has one fewer place to point. What the picture shows, and the algebra hides, is where the quarter turns go. The hub's hanging quarter changes colour and slides into the tie, the feet give up their quarters to the tie, and the three quarters on the tie are one yellow box: a controlled half-turn *is* a Hadamard edge because an Euler decomposition was hiding inside the gadget. What the algebra shows, and the picture hides, is why the number is -1 on exactly one word: the carry of a one-bit sum. Elsewhere in the chapter the pictures win outright: letting go after any turn (Result 19.9) is three slides through a cap, and in *The Closed Box* the watchers slide away by fusion alone, where the blackboard needs a partial trace of a three-system state. The numbers of §19.4 and §19.9, $\frac{20}{27}$ and $\frac{1}{12}$, are the blackboard's alone.

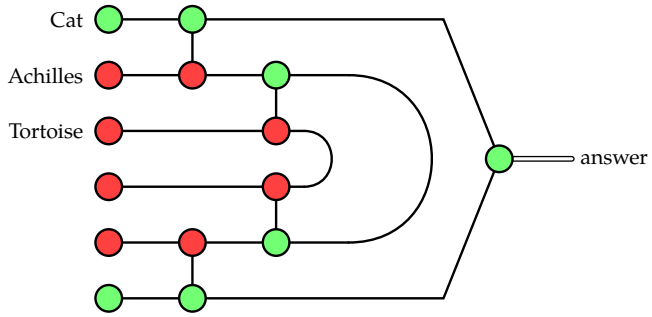
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§19.8 The Closed Box

Here is the chapter's master diagram. It is the watching scene of the dialogue, drawn: a Cat asleep in a closed box, Achilles watching the box, the Tortoise watching Achilles, the answer read from the Cat, and both watchers' pages let go. It is also the shape of every run of §19.4: points in, a process, one read wire out, the rest let go.

There are three systems, each started by a point. **The Cat's wire** starts at a green point at rest, $|0\rangle + |1\rangle$: its two basis states stand for asleep and awake, in even measure. **Achilles's page** starts at a red point at rest, $\sqrt{2}\,|0\rangle$: blank. **The Tortoise's page** starts the same way. There are two looks, each a joiner (row 64): **Achilles's look** is a green spider on the Cat's wire tied to a red spider on his page, and **the Tortoise's look** is a green spider on Achilles's page tied to a red spider on hers. At the end, the Cat's wire is read by the read-spider (row 68), which gives out one classical wire, the answer, and both pages are let go (row 67).

The picture is drawn doubled and unfolded, as in Chapter 9: each quantum line has its mirror copy below, in reflected order, so that the Tortoise's copy lies nearest the middle and the Cat's lowest. Every spider is at rest, so each mirror copy is drawn the same as its original. Walk it by its lines. On the Cat's wire: from its point to Achilles's look, and from the look to the read-spider; and the same two on the copy. On Achilles's page: from its point to his look's red spider, from there to the Tortoise's look's green spider, and from there to the cap that lets the page go; the same three on the copy, the last of them being the cap's other leg. On the Tortoise's page: from its point to her look's red spider, and from there to the cap; and the same two on the copy. The two looks' ties, and their copies. And, alone, the classical answer, drawn with a double stroke. No two lines cross, and every cap is the unnormalized one, $\langle 00| + \langle 11|$.

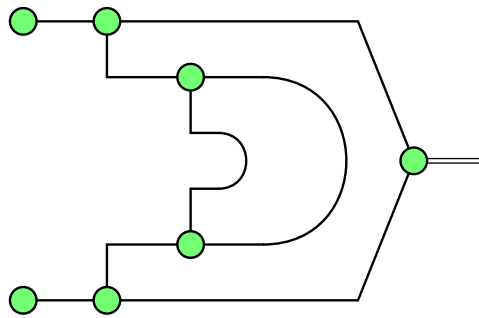


We call the picture **The Closed Box**. Before evaluating it, guess: does it matter that anyone watched? Achilles's page, after his look, records the Cat; the Tortoise's page records Achilles's page. The Cat's odds were even before anyone looked. Are they still?

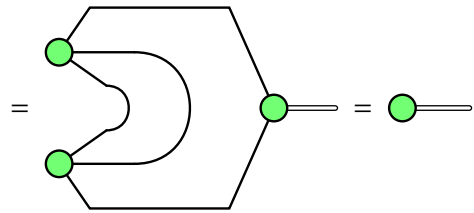
> **Result 19.12 (The Closed Box).** The Closed Box equals the Cat's own green point at rest, on the answer wire, with the number 1: as a classical vector, $(1, 1)$. Normalized, the answer is asleep or awake with odds $\frac{1}{2}$ and $\frac{1}{2}$, the Cat's own.

Proof. **On the blackboard.** Undoubled, the three points give $(|0\rangle + |1\rangle) \otimes \sqrt{2}\,|0\rangle \otimes \sqrt{2}\,|0\rangle$, top wire first. Each look is CNOT$/\sqrt{2}$ (row 64): Achilles's look gives $\sqrt{2}\,(|00\rangle + |11\rangle) \otimes |0\rangle$, and the Tortoise's look gives $|000\rangle + |111\rangle$, so all three records agree. Doubled, the state is $|\Psi\rangle \otimes |\bar{\Psi}\rangle$ with $|\Psi\rangle = |000\rangle + |111\rangle$. Letting go of both pages is the partial trace over them: the Cat is left with $|0\rangle\langle 0| + |1\rangle\langle 1|$. The read-spider, $\sum_i |i\rangle\langle ii|$, takes the diagonal: $(1, 1)$.

In the sketchbook, by fusion and one loop. Each red point at rest is joined by a plain wire to a red spider, so it fuses into it (row 57); each of those red spiders is left at rest with two legs, which is bare wire (the identity card). So each look becomes a plain wire from the green spider above it:



Now the Cat's point, the green spider of Achilles's look and the green spider of the Tortoise's look are joined by plain wires, and all three are green at rest: they fuse into one green spider (row 57), with three legs, one to the read-spider and one to each cap; the same on the mirror copy. The caps are plain wires between the two green spiders (only the connectivity counts, row 63). Fuse the two along one cap (row 57); the other cap becomes a loop. The read-spider is green at rest, so it fuses in along one of its two legs, and its other leg becomes a loop too. Both loops come off, with the number 1 each (row 90). What is left is one green spider at rest with one leg, the classical wire: a green point at rest, the Cat's own.



Every move is exact, with no number but 1. ■

See `Code/Ch19/OneAnswerOut.py`. It builds *The Closed Box* line by line as drawn, checks its value, and lets PyZX's simplifier reduce it to one green point at rest with the number 1.

A warning about that number. The Cat's point is unnormalized, $|0\rangle+|1\rangle$. The picture's value is the vector $(1, 1)$, and if the answer wire were closed as well, summing over both answers, the number would be 2, not 1. Normalized, the odds are $\frac{1}{2}$ and $\frac{1}{2}$.

The pictures made the watchers vanish, but only these watchers. Would any watching do the same? Not any: a look that copied the Cat in another basis could change its odds. If the Cat were surely asleep, $|0\rangle$, and a look copied its $|\pm\rangle$ instead, the joint state would be $\frac{1}{\sqrt{2}}(|+\rangle|0\rangle + |-\rangle|1\rangle)$, and the Cat alone would be left with $\frac{1}{2}(|+\rangle\langle+| + |-\rangle\langle-|) = \frac{1}{2}I$: asleep or awake at even odds. But the looks in *The Closed Box* are of two harmless kinds.

> **Result 19.13 (whoever watched).** Let the Cat's wire, together with any watchers' pages, be in any state. Apply any sequence of looks of two kinds: (a) a look that copies the Cat's asleep-or-awake, $\sum_c |c\rangle\langle c| \otimes U_c$, with $|c\rangle\langle c|$ on the Cat's wire and any unitaries U_c on the pages; (b) any unitary on the pages alone, which includes a look by one watcher at another. Then the Cat's odds of asleep and awake are unchanged.

Proof. Let ρ be the state and $\rho_C = \mathrm{tr}_{\text{pages}}\,\rho$ the Cat's part. Its odds are the diagonal entries $\langle c|\rho_C|c\rangle$. For (b), $\mathrm{tr}_{\text{pages}}((I \otimes V)\rho(I \otimes V^\dagger)) = \rho_C$, since the trace does not change under V. For (a), write $W = \sum_c |c\rangle\langle c| \otimes U_c$. Then $\langle c|\,\mathrm{tr}_{\text{pages}}(W\rho W^\dagger)\,|c\rangle = \mathrm{tr}(U_c\,\langle c|\rho|c\rangle\,U_c^\dagger) = \mathrm{tr}\,\langle c|\rho|c\rangle = \langle c|\rho_C|c\rangle$, where $\langle c|\rho|c\rangle$ is the block of ρ on the pages for the Cat's value c. ■ See `Code/Ch19/OneAnswerOut.py`, which checks it with random non-real states for the Cat and both pages and random unitaries in both looks.

In *The Closed Box* Achilles's look is of kind (a) and the Tortoise's of kind (b). So the answer's odds are the box's own, whoever watched, by a look of these kinds. That is all a closed box gives out, and nobody who only takes note of it changes it.

In the dialogue the same picture is the Crab's lute. Each paired course, a string and its unison twin, stands for a line and its mirror copy; the single top string stands for the answer, which has no twin, as a record has none. It is an analogy and no more: a mirror copy carries the conjugate of its line, and a unison twin carries nothing of the kind.

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§19.9 Watchers Watched

Take stock. The pictures we have drawn since Chapter 1 form a theory of processes, a dagger compact closed category, in which the mirror is the dagger (Result 19.1). Every card of the deck balances, number and all, in any surroundings (Result 19.2). One card was missing, the Euler card; with it, three more cards follow (Results 19.3 to 19.5), and the hinge that Chapter 17 could only check is a rewrite (Result 19.6, proved twice). A dare was won with the cards alone (Result 19.7). For each of four kinds of picture a complete deck is known, quoted with its source and its scope (Result 19.8), and letting go after any turn is letting go (Result 19.9). A vote of three runs turns two in three into twenty in twenty-seven (Result 19.10). One equation with an eighth turn shows that a deck can be sound and still not enough (Result 19.11). And a closed box gives out one answer, with the odds of what is inside it, whoever watched, by a look that copies the Cat's asleep-or-awake or watches only the watchers (Results 19.12 and 19.13).

The last result had a condition, and the condition matters. Its watchers only took note. What if a watcher could do more: undo a whole room, watcher and all, and ask it a question? And what if the question is not about one box but about every box of a kind? Two recent results answer these, and we report them at the level of a survey: stated, quoted, and not proved.

Watchers who can undo a room

Frauchiger and Renner begin from Wigner. In their words: "In 1967, Wigner proposed an argument, known as the Wigner's friend paradox, which should show that 'quantum mechanics cannot have unlimited validity'." A friend F in a sealed laboratory L measures a spin and sees one result. A watcher W outside, who has not seen it, may treat the whole laboratory, friend and all, as one quantum system in a superposition. Frauchiger and Renner then build a thought experiment with two sealed laboratories and two outside watchers.

In laboratory $\bar{L}$, agent $\bar{F}$ runs a random generator that gives $r =$ heads with probability $\frac{1}{3}$ and tails with $\frac{2}{3}$. On heads she prepares a spin in the state $|{\downarrow}\rangle$;

on tails, in $|{\rightarrow}\rangle = \sqrt{1/2}\,(|{\downarrow}\rangle + |{\uparrow}\rangle)$; and she sends it to agent F in laboratory L, who measures it and records $z = -\frac{1}{2}$ for $|{\downarrow}\rangle$ or $z = +\frac{1}{2}$ for $|{\uparrow}\rangle$. Then watcher $\bar{W}$ measures the whole of $\bar{L}$, and watcher W the whole of L, each in a basis that contains a state called $|\text{ok}\rangle$, and each announces "ok" or "fail". The rounds are repeated until both announce "ok". Frauchiger and Renner credit the preparation and the choice of measurements to "a construction due to Hardy".

On the blackboard. Write heads as 0 and tails as 1, and $z = -\frac{1}{2}$ as 0 and $z = +\frac{1}{2}$ as 1, and let each laboratory's state stand for its record. Before the watchers act, the two laboratories are in

$$\psi \;=\; \tfrac{1}{\sqrt{3}}(|00\rangle + |10\rangle + |11\rangle),$$

laboratory $\bar{L}$ first: heads with $|{\downarrow}\rangle$, or tails with $|{\rightarrow}\rangle$ spread over both records. For both laboratories $|\text{ok}\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$: Frauchiger and Renner's $\sqrt{1/2}(|\bar{h}\rangle - |\bar{t}\rangle)$ for $\bar{L}$ and $\sqrt{1/2}(|{-\tfrac{1}{2}}\rangle - |{+\tfrac{1}{2}}\rangle)$ for L.

Result 19.14 (the numbers of the sealed rooms). Exactly,

$$(\langle 1| \otimes \langle \text{ok}|)\,\psi = 0, \qquad (\langle 0| \otimes \langle 1|)\,\psi = 0, \qquad (\langle \text{ok}| \otimes \langle 0|)\,\psi = 0,$$

and $(\langle \text{ok}| \otimes \langle \text{ok}|)\,\psi = \frac{1}{2\sqrt{3}}$, so both watchers announce "ok" with probability $\frac{1}{12}$.

Proof. The first: the tails part of ψ is $\frac{1}{\sqrt{3}}\,|1\rangle\,(|0\rangle + |1\rangle)$, and $\langle \text{ok}|\,(|0\rangle + |1\rangle) = 0$. The second: the heads part is $\frac{1}{\sqrt{3}}\,|0\rangle\,|0\rangle$, which has no $|1\rangle$ in L. The third: the part with L at 0 is $\frac{1}{\sqrt{3}}(|0\rangle + |1\rangle)\,|0\rangle$, and again $\langle \text{ok}|\,(|0\rangle + |1\rangle) = 0$. The last: $\langle \text{ok}| \otimes \langle \text{ok}| = \frac{1}{2}(\langle 00| - \langle 01| - \langle 10| + \langle 11|)$, which meets ψ in $\frac{1}{2\sqrt{3}}(1 - 1 + 1)$; squared, $\frac{1}{12}$. ■ See `Code/Ch19/WatchersWatched.py`.

Each zero is a certainty held by an agent, in Frauchiger and Renner's analysis. $\bar{F}$, having seen tails, is certain that W will announce "fail" (the first zero). F, having seen $z = +\frac{1}{2}$, is certain that r was not heads (the second). $\bar{W}$, having announced "ok", is certain that F saw $z = +\frac{1}{2}$ (the third). Chained, the

certainties say: if $\bar{W}$ announces "ok", then F saw $z = +\frac{1}{2}$, so r was tails, so W will announce "fail". Yet one round in twelve, both announce "ok", and the experiment halts on exactly that round.

Frauchiger and Renner name three assumptions that together give the contradiction: quantum theory everywhere (Q), that an agent "can be certain that a given proposition holds whenever the quantum-mechanical Born rule assigns probability 1 to it"; consistency (C), that "the different agents' predictions are not contradictory"; and single outcomes (S), that "from the viewpoint of an agent who carries out a particular measurement, this measurement has one single outcome".

> **Result 19.15 (Frauchiger and Renner's theorem; quoted, not proved).** "Any theory that satisfies assumptions (Q), (C), and (S) yields contradictory statements when applied to the Gedankenexperiment described in Section 2.1."

And, in their introduction: "The theorem itself is neutral in the sense that it does not tell us which of these three assumptions is wrong." This book takes no side either. The numbers of Result 19.14 are all we have checked.

In the sketchbook. The state ψ is a sum of two pictures:

$$\sqrt{3}\,\psi \;=$$  $$-\;\tfrac{1}{2}$$ 

Two green points at rest make $|00\rangle + |01\rangle + |10\rangle + |11\rangle$; a red point at rest beside a red point at a half makes $2\,|01\rangle$; half of that, taken away, removes the one corner, $|01\rangle$, that ψ lacks.

> **Coecke's Sketchbook.**
>
> *Four closed pictures and a sum.* Every number of Result 19.14 is a closed picture, or rather a sum of two. The effect $\langle \text{ok}|$ is $\frac{1}{\sqrt{2}}$ times a green spider turned π with one leg; the effects $\langle 0|$ and $\langle 1|$ are $\frac{1}{\sqrt{2}}$ times red effects at rest and at a half. Close each of the two pictures above with a pair of these, and each closed picture reduces, by copy and fusion, to a number; subtract, as

the sum says. Take the first zero. In the first term, $\langle \text{ok}|$ on the lower wire meets a green point at rest; the two fuse into a green spider with no legs, turned π, whose number is $1 + e^{i\pi} = 0$. In the second term, the effect $\langle 1|$ on the upper wire meets a red point at rest, and they fuse the same way, to 0. The zeros are exact cancellations, and the $\frac{1}{12}$ is one term surviving. What the pictures do *not* do is add: a sum of pictures is not one picture, so the subtraction is the blackboard's, and so is the paradox. The pictures agree with every number and settle nothing about which assumption to drop.

Feynman's Notebook.

From Wigner's friend to Frauchiger and Renner. Frauchiger and Renner's paper, "Quantum theory cannot consistently describe the use of itself", was posted in 2016. Their first figure shows two older arguments. In Wigner's, as they tell it, the outside watcher, "having no access to" the friend's result, would assign a superposition to the whole laboratory; Frauchiger and Renner point out that this is no contradiction, since the friend and the watcher simply know different things. Deutsch "later noted that agent W could in principle test this state assignment by applying a carefully designed measurement" to the laboratory. Frauchiger and Renner's step was to let agents use quantum theory to reason about other agents who also use quantum theory: "such a self-referential use of the theory yields contradictory claims". They are careful about the laboratory itself: "we do not claim that the experiment is technologically feasible". A watcher who can undo a room is a thought experiment, and its purpose, they write, is "to scrutinise the consistency of our currently best available theories".

The Closed Box is consistent with all of this. Nobody in it undoes anybody: the Tortoise watches Achilles, and never unmixes him. Result 19.13 needs exactly that, and it is exactly what Frauchiger and Renner's watchers do not do.

A question about all games

The second result is about questions of a different size. A **nonlocal game** is played by two or more players in separate rooms. A referee sends each a question; each answers without hearing the others; the referee declares a win or a loss by a fixed rule. The players may agree on a strategy beforehand and may share an entangled state. The best winning probability over all such quantum strategies is the game's *entangled value*, written val^*. Chapter 7 played two such games: the CHSH game, whose best quantum strategy wins with probability $\cos^2(\pi/8)$ (Tsirelson's bound, stated there), and a game for three players who, sharing one three-legged state, win every time.

Ji, Natarajan, Vidick, Wright and Yuen (posted January 2020) proved "that the class MIP* of languages that can be decided by a classical verifier interacting with multiple all-powerful quantum provers sharing entanglement is equal to the class RE of recursively enumerable languages". RE contains problems no computer can decide, among them "the Halting problem, which is to decide whether a given Turing machine eventually halts". The inclusion $\mathrm{MIP}^* \subseteq \mathrm{RE}$ is the easy direction, which they note first; their main result is the reverse, $\mathrm{RE} \subseteq \mathrm{MIP}^*$, and they obtain it by building, for every Turing machine M, a nonlocal game G_M such that $\mathrm{val}^*(G_M) = 1$ if M halts and $\mathrm{val}^*(G_M) \leq \frac{1}{2}$ if it does not.

> **Result 19.16** ($\mathrm{MIP}^* = \mathrm{RE}$**; Ji, Natarajan, Vidick, Wright and Yuen; quoted, not proved**)**.** $\mathrm{MIP}^* = \mathrm{RE}$. "In particular for any $0 < \varepsilon < 1$, it is an undecidable problem to determine whether a given nonlocal game has entangled value 1 or at most $1 - \varepsilon$ (promised that one is the case)."

"Using a known connection", they add, the result gives a negative answer to Tsirelson's problem and a refutation of Connes' embedding conjecture, one a question about quantum correlations, the other a conjecture "from the theory of von Neumann algebras". We name them and do not explain them.

The games of the proof refer to themselves. The authors describe a procedure that compresses games, and write: "Intuitively, one would expect that iterating this procedure and 'taking the limit' gives" the family of games

they need. "Formally, we do not take such a limit but instead define directly the family of games ... as a fixed point of the Turing machine that implements the compression procedure." A footnote names the tool: "Kleene's recursion theorem". The dialogue's Genie, whose answer to a question about all questions goes up an endless chain of Meta-Genies and comes back at once, is the intuitive picture, the iteration and its limit. The paper's proof is not a limit but a fixed point.

Coecke's Sketchbook.

A strategy is a picture; the question is not. Draw a strategy for a nonlocal game: a state on two wires, a cup if you like, then on each side a box chosen by that side's question, then a read on each side. Each winning probability is a closed doubled picture, and the pictures of this book evaluate it exactly. Chapter 7 drew one such strategy and read off its value. So for any one strategy the sketchbook can say, with the right cards, everything the blackboard can. What Result 19.16 is about is different: whether the *best* of all strategies, over boxes of every size and states on wires of every number, wins every time or loses at least some fixed fraction of the time. That is a question about all pictures of a kind at once. No rewrite of one picture into another can answer it, and the theorem says that no procedure of any kind, drawn or computed, answers it for every game.

The book's rule, honestly

This book has proved each chapter's headline twice, and the chapter's question asked whether that is always possible. Here is the honest answer.

Where a complete deck is known, yes. For pictures whose turns are multiples of $\frac{\pi}{2}$, or of $\frac{\pi}{4}$, or anything at all, and for the doubled pictures of the $\frac{\pi}{2}$ kind with letting go, every equation the blackboard proves between two such pictures has a chain of rewrites by that kind's deck, with the scope its authors give it, Backens's up to non-zero numbers (Result 19.8). Whatever the numbers say, the pictures can say, *with the right cards*. The book's second proof is licensed there.

But the book's own deck is not shown to be any of those decks, and the book's own chapters did not always reach a chain. The hinge closes one recorded gap: Chapter 17's step (iv) is now a rewrite, from the Euler card, the rule the sources say cannot be derived from the others of Backens's calculus. Other gaps stay open, and we say so. In Chapter 14 the sketchbook borrowed the kickback of a general box from the blackboard; a box that stands for an unknown unitary is not a picture of any of the kinds above, and no completeness theorem speaks of it. In Chapter 15 a drawing was checked by its matrix. In Chapter 16 re-tying and the pivot were stated, not derived; the Euler card is now in the deck, but the derivation is not given here. The rule "proved twice" is a rule of this book's writing, and completeness does not make it a theorem about this book.

And some questions were never the pictures' to answer: the number of steps a chain needs, the class a problem belongs to, the value of the best strategy in every game, and which of three natural assumptions the watchers of the sealed rooms must give up.

The chapter's question has an answer. If two pictures always give the same numbers, the rules can turn one into the other, *if the rules are the right ones for the kind of picture*: for quarter turns, for eighth turns, for any turns, and for quarter turns with letting go, such rules are known, and for eighth turns with letting go they were left open in 2019. This book's own deck lacked one card, the Euler card, and with it the step Chapter 17 could only check became a rewrite. Every card balances, so every rewrite is true; a closed box gives out one answer, with odds that no watcher who only takes note can change; and no procedure, drawn or computed, decides every question of the Genie's kind.

The chapters are done, but there is one more rewrite. In the Epilogue, *The Empty Diagram*, the last rewrite removes the last wire, and what is left is the empty diagram: the number 1, where the Prelude began.

Postlude on the Same Sheet

冬晴に
見ず仕舞ひけり
白き紙

fuyubare ni
mizu shimaikeri
shiroki kami

In the clear winter
put away, and never seen —
the white sheet is left.

[*Achilles's table, a clear morning after a storm. Achilles and the Tortoise come in together; he carries a stone bottle.*]

[*Outside, the storm has left the sky washed clear and the wet roofs shining; drops still fall quick from the eaves into the brimming water-butt in the yard. The sun is up, low and pale, and lies on the floor by the window, not yet on the table. The grate is swept and cold, no fire laid, and the wall above the hearth is bare. The table is bare too, with nothing on it to write with; Achilles's shawl hangs over the back of his chair.*]

Achilles: Home at last. The lane is all puddles from the Crab's door to mine.

Tortoise: The storm blew itself out before dawn, and the Crab sent us off with his cocoa, still warm.

Achilles: [*He sets the bottle on the table, fetches two cups to stand beside it, and sits down.*] A whole night and not a wink. I asked the Cat my one question, and it said awake; this morning I mean to ask nothing.

Tortoise: [*She takes the chair beside him.*] Good. Did you keep what I gave you in the autumn?

[*He opens the drawer on his side of the table, takes out a sheet of plain paper, blank on both sides, and lays it flat before them, leaving the drawer open.*]

Achilles: Your picture from the autumn. You told me to keep it, and I kept it, in there beside my old coin.

Tortoise: May I have the coin too? The one I kept under my foot from the middle of one evening to the end of the next, and you never saw how it lay. One toss, and this time neither of us will see.

Achilles: [*He takes the coin from the drawer, puts it in her hand, and rests his hands on his cup.*] Toss away.

[*The coin lies in the Tortoise's fingers; the sheet lies blank; the drawer stands open. She tosses the coin, it turns in the air, and she claps it flat under her palm on the sheet. Now her palm lies on the sheet over the coin, one face up, and neither of them has seen which. What has changed: the coin is down, its face unknown. What has held still: the sheet, blank round her hand; Achilles's hands on his cup.*]

Tortoise: What is it worth, the whole of it, if we never look?

Achilles: A half. Heads, a half.

[*The Tortoise lets the answer stand. Her palm lies flat over the coin; the drawer is open below the table's edge. Without lifting it she slides her palm, and the coin beneath, off the sheet and over the edge; the coin drops into the open drawer with one small sound, and with her other hand she pushes the drawer shut at once, neither of them looking down. What has changed: her palm is empty; the coin is in the drawer, its face never seen by anyone. What has held still: the sheet, exactly as it lay before the coin came down on it.*]

[*In the yard a last drop gathers at the eaves, lets go, and falls into the brimming butt. It is one with the water at once, and the surface holds the washed sky just as it held it before. Achilles looks from the window to his cup, and says nothing yet.*]

Tortoise: I didn't ask about heads.

Achilles: Heads a half, tails a half, and it lies one way or the other for certain. A half and a half: once. Had it been loaded five to one, five sixths and one sixth: once still. Whatever way it was set, once.

Tortoise: And what is left on the table?

Achilles: Your sheet, as it was. Worth once, as we found it in the autumn: the very picture you showed me then. ...Oh.

[*For a while neither of them says anything. The eaves have gone quiet, and the sun has come round onto the table and lies there, pale and warm. Achilles sits back and rubs his eyes; the Tortoise rests her hands against the stone bottle, still warm.*]

Tortoise: In the autumn you asked me why a picture of nothing should come to one, and not to nothing.

Achilles: Because it is what is left when a thing is let go unseen. Whatever the thing was, let go with all its chances together, it comes to once, not to nothing. So your sheet is a picture after all: the picture of what is left when the last thing is let go.

Tortoise: Then the question kept, as I said it would. [*She pours the cocoa.*] Drink it while it's warm; the sun is on the table.

Achilles: [*He puts his shawl round his shoulders.*] Awake all night, and only now awake. I shan't open the drawer to see how the coin lies.

Tortoise: Then leave it where it lies, and the sheet where you laid it. The question has its answer, and what is left is what was here in the autumn: my sheet, as blank as it came.

[*Achilles drinks his cocoa in his shawl, and the Tortoise stays where she is. Outside, the butt stands full and still under the dry eaves, holding the whole of the sky, and the puddles in the lane are going down. Indoors the sun lies on the table, on the cups and the stone bottle and the sheet where he laid it, and nobody opens the drawer. Neither of them moves; there is nothing left to do, and no hurry.*]

Epilogue: The Empty Diagram

— Postlude on the Same Sheet —

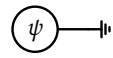
What is left when the last wire goes?

Every picture since the Prelude has had something on it, and wires have come and gone. This last page follows one wire to its end, and asks what remains, and what that is worth.

The Last Wire

A wire can leave a picture in three ways. It can be **yanked**: an S-bend pulled straight into one plain wire (Theorem 6.2, row 42). It can be **fused** away: two spiders of one colour slide together along the wire between them (row 57), and a plain wire from a spider back to itself comes off with the number 1 (Chapter 19's card *A loop comes off*). And it can be **let go** into the discard, the ground symbol of Chapter 9, which asks it no question at all (row 67). Earlier chapters closed pictures into numbers (Theorem 2.1's 1, §9.3's 2) and wrote the number down; the last rewrite ends on the blank itself.

Here is the picture this page undoes: a state ψ of one wire, let go.



The one line stands for two: the wire of a state is doubled, the system and its conjugate copy (row 66), and the ground symbol is a cap joining the copies (row 67). On the blackboard the discard, written $\top$, is the trace: $\top : \rho \mapsto \operatorname{tr} \rho$.

An example first. Take $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, whose matrix Chapter 9 com-

puted in §9.2: $|+\rangle\langle+| = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$. Let it go and the trace adds the diagonal: $\operatorname{tr}|+\rangle\langle+| = \frac{1}{2} + \frac{1}{2} = 1$.

Letting go is not looking. A look closes the wire with an effect (Chapter 2), which picks one answer and weighs it: the effect 0 on the doubled $|+\rangle$, a $\langle 0|$ on each copy, gives $|\langle 0|+\rangle|^2 = \frac{1}{2}$, the square of the amplitude $1/\sqrt{2}$. Letting go asks nothing, and weighs every answer at once. Reading an answer and then forgetting it is letting go (Result 9.5).

A wire does not always go for nothing. A bare loop, a cup closed by a cap, is 2: the doubled bare cup, discarded, is $\operatorname{tr} I = 2$ (§9.3). An unnormalized point let go is 2 as well, since $|0\rangle + |1\rangle$ meets itself in $1 + 1$. With $\frac{1}{2}$ beside it the cup is the maximally mixed state (row 69), a system cupped to its own mirror copy; let go, it closes into the loop, and $\frac{1}{2} \times 2 = 1$. That is the dialogue's covered fair coin: a half and a half. So the last wire goes with the number 1 only because what it carried was a state, with $\langle\psi|\psi\rangle = 1$.

The Rosetta Table

Two rows, following Chapter 19's row 152. Row 153 gives the Prelude's picture the row it was left without; its right-hand column is a picture, the empty diagram, not row 152's "no picture". Neither adds a new picture element.

#	Algebra (Dirac)	Picture (Coecke)
153	the number 1: a process from no wire to no wire, the 1×1 matrix (1)	**the empty diagram**: nothing drawn, no wire in and none out; a picture, not row 152's "no picture"; the Prelude's page, given its row
154	$\operatorname{tr}\|\psi\rangle\langle\psi| = \langle\psi|\psi\rangle = 1$ for every state $\|\psi\rangle$ (row 72 with no wire in)	**a state let go is the empty diagram**: a state box, a wire and the discard = nothing drawn; the last rewrite

The Theorem, Twice

> **Theorem (the last rewrite).** For every state ψ of one wire, the picture "ψ, then let go" is the empty diagram: $\operatorname{tr}|\psi\rangle\langle\psi| = 1$, exactly, scalar included.

Proof idea. To let a state go is to meet it with its own mirror image, and that meeting is the number 1.

Dirac's Blackboard

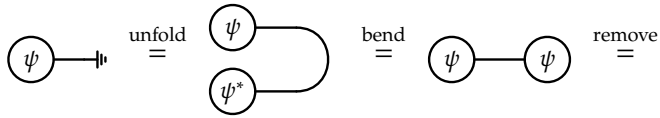
Letting go is taking the trace (row 67). Write $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Then

$$\operatorname{tr}|\psi\rangle\langle\psi| \;=\; \operatorname{tr}\begin{pmatrix} \alpha\bar{\alpha} & \alpha\bar{\beta} \\ \beta\bar{\alpha} & \beta\bar{\beta} \end{pmatrix} \;=\; \alpha\bar{\alpha} + \beta\bar{\beta} \;=\; \langle\psi|\psi\rangle \;=\; 1$$

by Theorem 2.1. The answer is the 1×1 matrix (1), a process from no wire to no wire: the number 1, which is the empty diagram (Theorem 0.1). ■

Coecke's Sketchbook

Three moves:



Unfold the discard. The ground symbol is shorthand for the cap on a system's two copies (§9.3; rows 66, 67), and a state is its two copies side by side, ψ above and its conjugate copy ψ^* below (§9.2). The cap is the unnormalized $\langle 00| + \langle 11|$ of Chapter 6, and its scalar is carried, not dropped. One line has become two.

Carry the copy round the bend. A box slid round a bend arrives transposed (Theorem 6.4 in its cap form, rows 43–44), as Chapter 9 slid the conjugate

copy of its isometry round a cap to prove causality (§9.6). A state is a box with no wire in, and states slide too: neither proof of Theorem 6.4 needs the box to have a wire in (Chapter 6, §6.4). So ψ^* arrives on the upper line as $(\psi^*)^T = \psi^\dagger$: the effect ψ, the state's mirror image, because the mirror conjugates (rows 10, 16). Two lines have become one.

Remove the closed picture. A state meets its own mirror image in the number 1 (Theorem 2.1, row 14); a closed picture is its number (row 11); and the number 1 is the empty diagram (Theorem 0.1). The last term is the empty diagram: nothing is drawn there. The last wire has gone. ■

The theorem is Result 9.4, causality, with nothing going in. There, a doubled isometry capped by a discard left a bare discard, because its mirror image undid it. Here the isometry takes no wire in, so it is a state; its mirror undoes it, $\langle\psi|\psi\rangle = 1$; and no discard is left over, because no wire is. Chapter 9 already let its fresh pointer go this way (proof of Result 9.6). Nothing new is proved; what is new is where it lands.

A mixed state is a weighted list of pure ones whose weights sum to 1 (§9.2), so it too lets go to 1; that weighted sum belongs to the blackboard, since a sum of pictures is not one picture.

The verdict

The blackboard wins on length, honestly. Its proof is one line: the trace of $|\psi\rangle\langle\psi|$ is $\langle\psi|\psi\rangle$, which is 1. The sketchbook needs three moves, and its last move *is* the blackboard's fact, Theorem 2.1, read as a picture.

What the sketchbook shows, and the blackboard cannot, is a wire going: one line drawn, two unfolded, one bent, then none. The blackboard's answer was a 1×1 matrix from the start, and it never had a wire to lose. The rewrite proves an equality of processes and nothing more; here, that was all there was to prove.

The Empty Diagram

The Prelude would not say what a diagram is; now it can be said. A diagram is pieces joined and read from left to right. One with no wire coming in and none going out is closed, and a closed diagram is a number (Chapter 2). The last term of the chain is the closed diagram with nothing in it: what the last rewrite leaves. Its number is 1, and it is the Prelude's picture, drawn again as it was drawn there:



The Empty Diagram.

The Prelude showed this page before anything had been drawn, and said the book would come back to such a page only at its end. Here it is, reached from the other side: a state, let go unseen, leaves exactly this. It is worth what it was worth then, one, and nothing is drawn after it.

Appendix: References

The works each chapter leans on, chapter by chapter, with the sections where they are cited. Where a work is freely available its arXiv number is given. A work cited in several chapters is listed under each.

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